Al-Balqa Applied University
Faculty of Engineering Technology
Communication Technology Department

Chapter 1 Semiconductors Diodes
Lecture 1

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1.2 SEMICONDUCTOR MATERIALS: Ge, Si, AND GaAs

The construction of every discrete (individual) solid-state (hard crystal structure) electronic device or integrated circuit begins with a semiconductor material of the highest quality.

Semiconductors are a special class of elements having a conductivity between that of a good conductor and that of an insulator.

In general, semiconductor materials fall into one of two classes: *single-crystal* and *compound*. Single-crystal semiconductors such as germanium (Ge) and silicon (Si) have a repetitive crystal structure, whereas compound semiconductors such as gallium arsenide (GaAs), cadmium sulfide (CdS), gallium nitride (GaN), and gallium arsenide phosphide (GaAsP) are constructed of two or more semiconductor materials of different atomic structures.

The three semiconductors used most frequently in the construction of electronic devices are Ge, Si, and GaAs.

As time moved on, however, the field of electronics became increasingly sensitive to issues of speed. Computers were operating at higher and higher speeds, and communication systems were operating at higher levels of performance. A semiconductor material capable of meeting these new needs had to be found. The result was the development of the first GaAs transistor in the early 1970s. This new transistor had speeds of operation up to five times that of Si. The problem, however, was that because of the years of intense design efforts and manufacturing improvements using Si, Si transistor networks for most applications were cheaper to manufacture and had the advantage of highly efficient design strategies. GaAs was more difficult to manufacture at high levels of purity, was more expensive, and had little design support in the early years of development. However, in time the demand for increased speed resulted in more funding for GaAs research, to the point that today it is often used as the base material for new high-speed, very large scale integrated (VLSI) circuit designs.

1.3 COVALENT BONDING AND INTRINSIC MATERIALS

To fully appreciate why Si, Ge, and GaAs are the semiconductors of choice for the electronics industry requires some understanding of the atomic structure of each and how the atoms are bound together to form a crystalline structure. The fundamental components of an atom are the electron, proton, and neutron. In the lattice structure, neutrons and protons form the nucleus and electrons appear in fixed orbits around the nucleus. The Bohr model for the three materials is provided in Fig. 1.3.

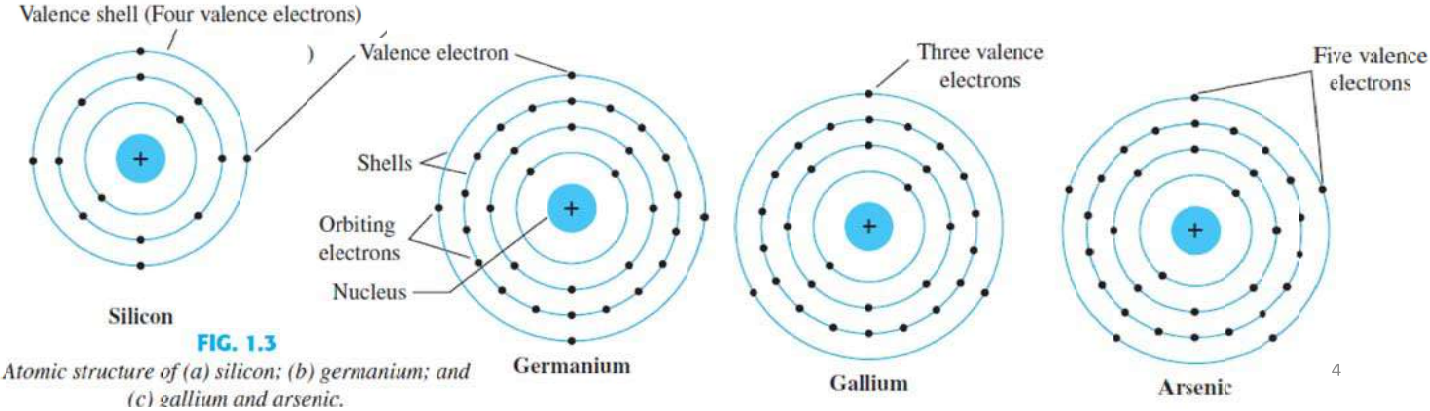

FIG. 1.3
Atomic structure of (a) silicon; (b) germanium; and (c) gallium and arsenic.

As indicated in Fig. 1.3, silicon has 14 orbiting electrons, germanium has 32 electrons, gallium has 31 electrons, and arsenic has 33 orbiting electrons (the same arsenic that is a very poisonous chemical agent). For germanium and silicon there are four electrons in the outermost shell, which are referred to as *valence electrons*. Gallium has three valence electrons and arsenic has five valence electrons. Atoms that have four valence electrons are called *tetravalent*, those with three are called *trivalent*, and those with five are called *pentavalent*. The term *valence* is used to indicate that the potential (ionization potential) required to remove any one of these electrons from the atomic structure is significantly lower than that required for any other electron in the structure.

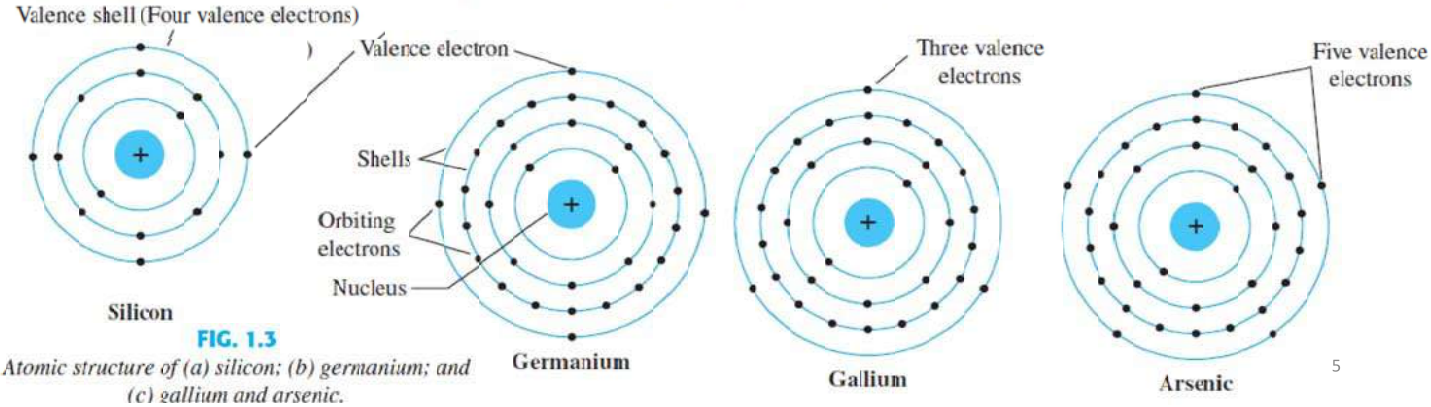

FIG. 1.3
Atomic structure of (a) silicon; (b) germanium; and (c) gallium and arsenic.

In a pure silicon or germanium crystal the four valence electrons of one atom form a bonding arrangement with four adjoining atoms, as shown in Fig. 1.4.

This bonding of atoms, strengthened by the sharing of electrons, is called covalent bonding.

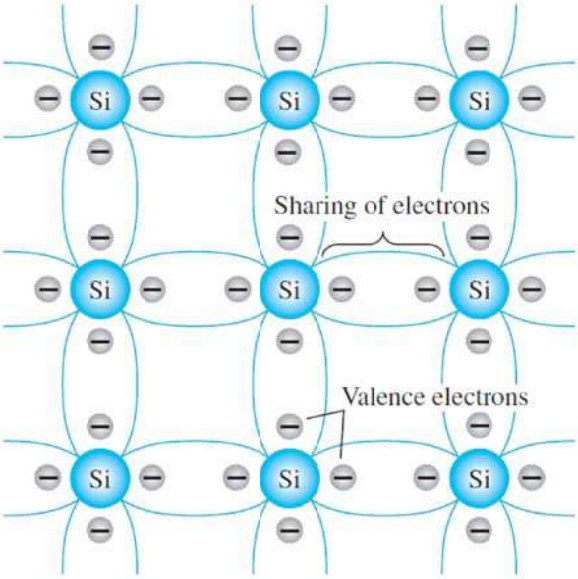

FIG. 1.4
Covalent bonding of the silicon atom.

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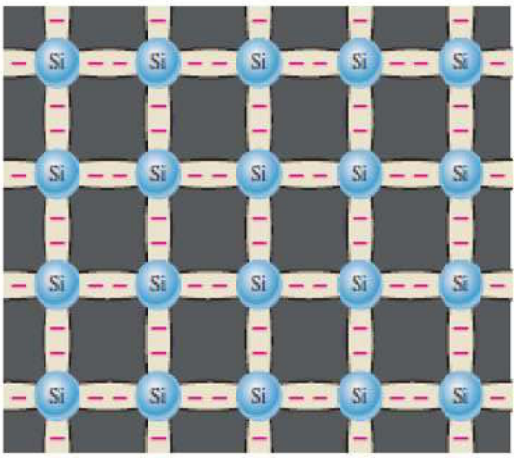
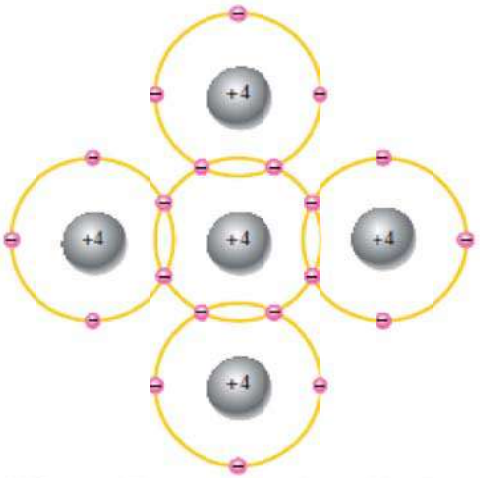
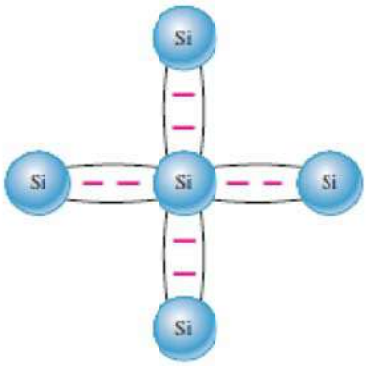
This bonding of atoms, strengthened by the sharing of electrons, is called covalent bonding.



Covalent bonds in a silicon crystal.



(a) The center silicon atom shares an electron with each of the four surrounding silicon atoms, creating a covalent bond with each. The surrounding atoms are in turn bonded to other atoms, and soon.



(b) Bonding diagram. The red negative signs represent the shared valence electrons.

Because GaAs is a compound semiconductor, there is sharing between the two different atoms, as shown in Fig. 1.5. Each atom, gallium or arsenic, is surrounded by atoms of the complementary type. There is still a sharing of electrons similar in structure to that of Ge and Si, but now five electrons are provided by the As atom and three by the Ga atom.

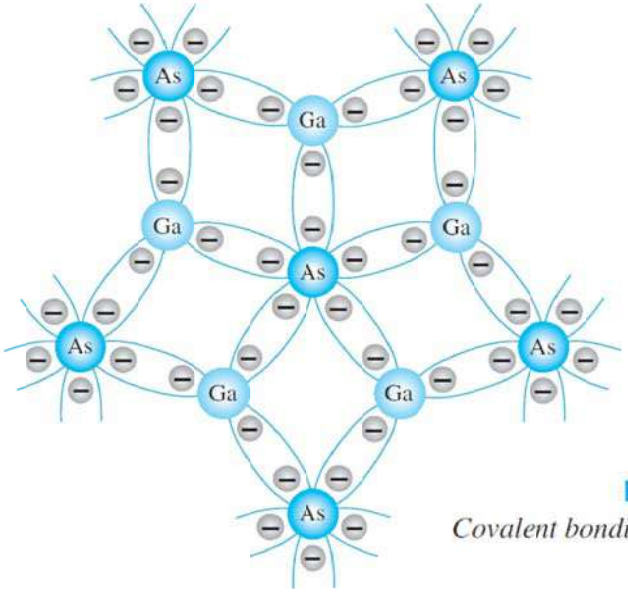


FIG. 1.5
Covalent bonding of the GaAs crystal.

The term* intrinsic *is applied to any semiconductor material that has been carefully refined to reduce the number of impurities to a very low level—essentially as pure as can be made available through modern technology.

The free electrons in a material due only to external causes are referred to as *intrinsic carriers*. Table 1.1 compares the number of intrinsic carriers per cubic centimeter (abbreviated n_i) for Ge, Si, and GaAs. It is interesting to note that Ge has the highest number and GaAs the lowest. In fact, Ge has more than twice the number as GaAs. The number of carriers in the intrinsic form is important, but other characteristics of the material are more significant in determining its use in the field. One such factor is the *relative mobility* (μ_n) of the free carriers in the material, that is, the ability of the free carriers to move throughout the material. Table 1.2 clearly reveals that the free carriers in GaAs have more than five times the mobility of free carriers in Si, a factor that results in response times using GaAs electronic devices that can be up to five times those of the same devices made from Si. Note also that free carriers in Ge have more than twice the mobility of electrons in Si, a factor that results in the continued use of Ge in high-speed radio frequency applications.

TABLE 1.1
Intrinsic Carriers n_i

Semiconductor	Intrinsic Carriers (per cubic centimeter)
GaAs	1.7×10^6
Si	1.5×10^{10}
Ge	2.5×10^{13}

TABLE 1.2
Relative Mobility Factor μ_n

Semiconductor	μ_n ($cm^2/V \cdot s$)
Si	1500
Ge	3900
GaAs	8500

We obviously have to deal with a whole new level of comparison when we deal with the semiconductor medium. The ability to change the characteristics of a material through this process is called *doping*, something that germanium, silicon, and gallium arsenide readily and easily accept. The doping process is discussed in detail in Sections 1.5 and 1.6.

Temperature Coefficient

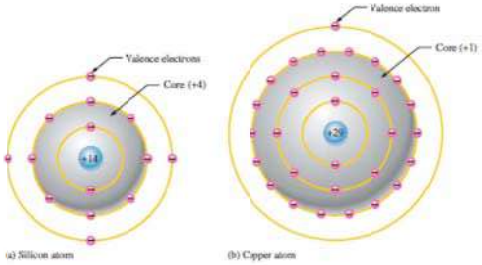
One important and interesting difference between semiconductors and conductors is their reaction to the application of heat. For conductors, the resistance increases with an increase in heat. This is because the numbers of carriers in a conductor do not increase significantly with temperature, but their vibration pattern about a relatively fixed location makes it increasingly difficult for a sustained flow of carriers through the material. Materials that react in this manner are said to have a *positive temperature coefficient*. Semiconductor materials, however, exhibit an increased level of conductivity with the application of heat. As the temperature rises, an increasing number of valence electrons absorb sufficient thermal energy to break the covalent bond and to contribute to the number of free carriers. Therefore:

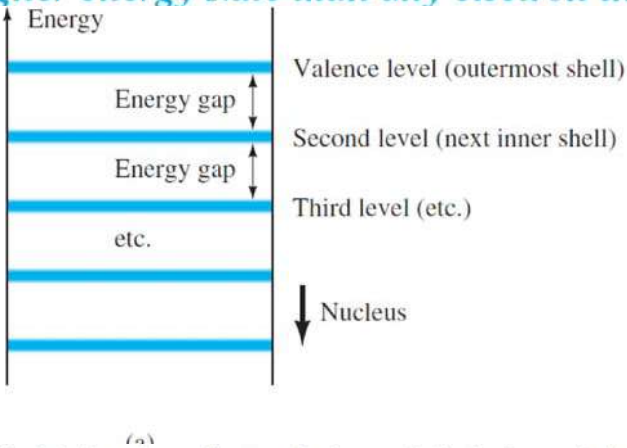
Semiconductor materials have a negative temperature coefficient.

1.4 ENERGY LEVELS

Within the atomic structure of each and every *isolated* atom there are specific energy levels associated with each shell and orbiting electron, as shown in Fig. 1.6. The energy levels associated with each shell will be different for every element. However, in general:

The farther an electron is from the nucleus, the higher is the energy state, and any electron that has left its parent atom has a higher energy state than any electron in the atomic structure.





(a)

Note in Fig. 1.6a that only specific energy levels can exist for the electrons in the atomic structure of an isolated atom. The result is a series of gaps between allowed energy levels

1.4 ENERGY LEVELS

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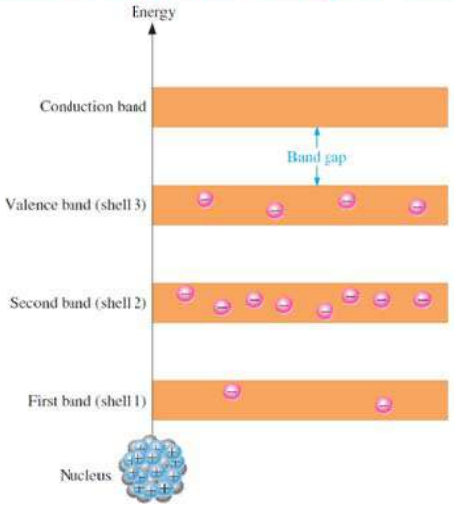
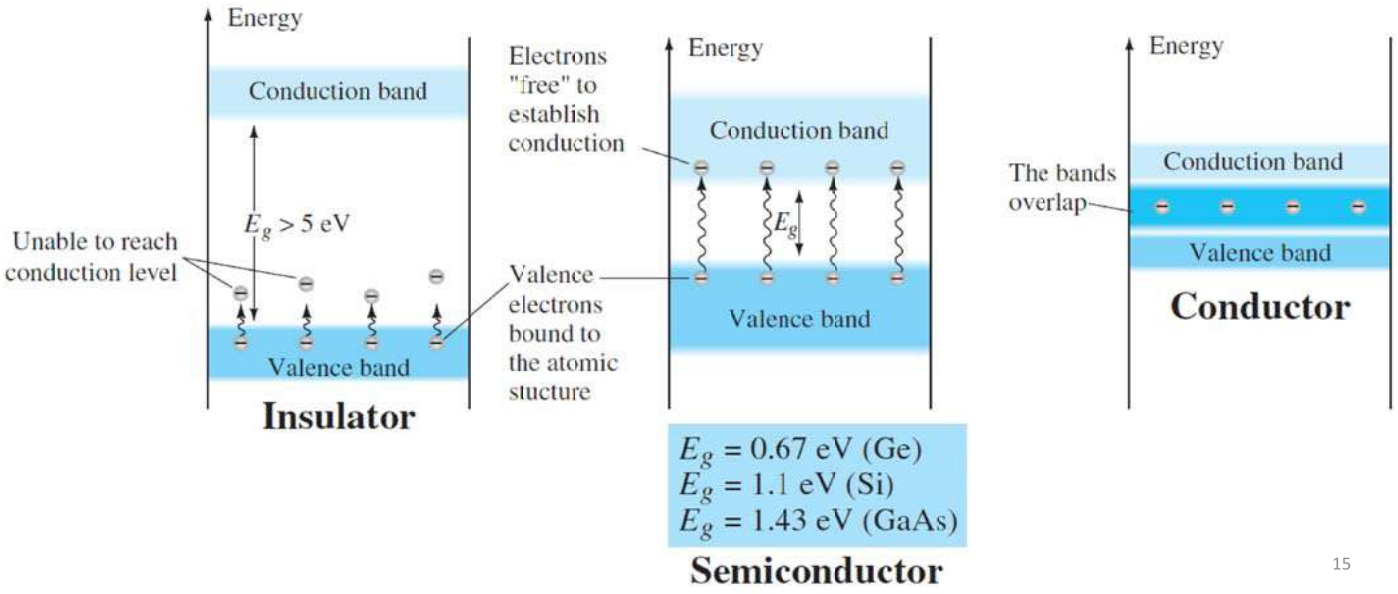
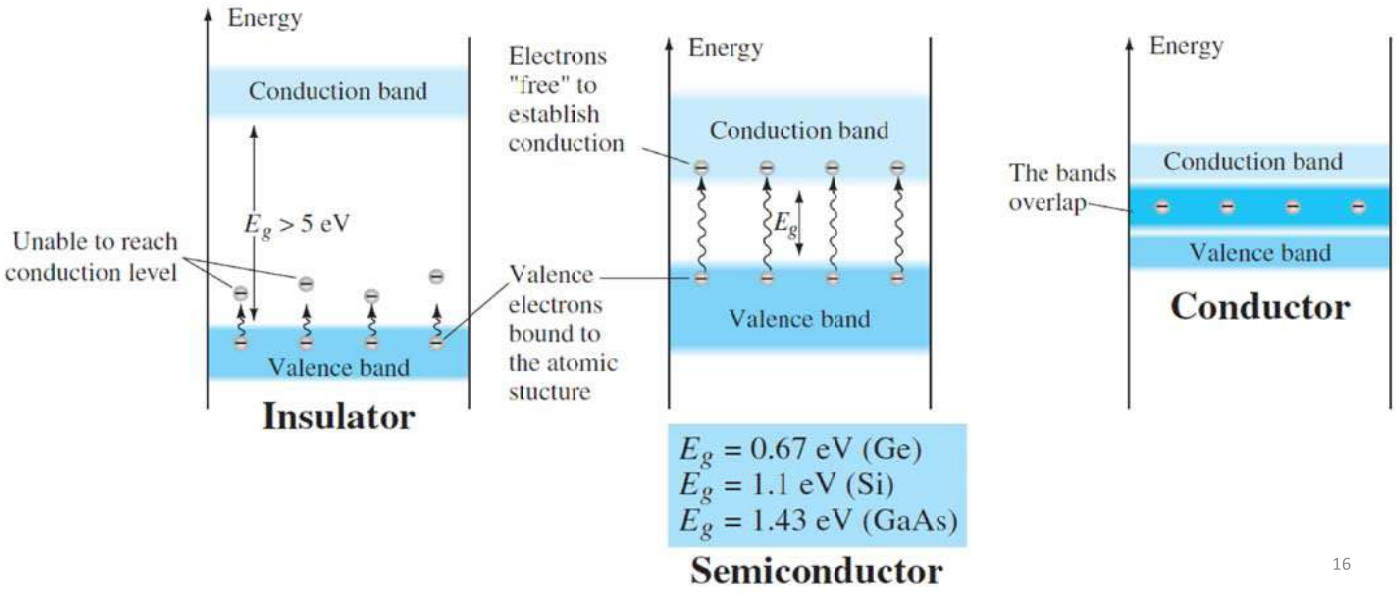


An electron in the valence band of silicon must absorb more energy than one in the valence band of germanium to become a free carrier. Similarly, an electron in the valence band of gallium arsenide must gain more energy than one in silicon or germanium to enter the conduction band.



This difference in energy gap requirements reveals the sensitivity of each type of semiconductor to changes in temperature. For instance, as the temperature of a Ge sample increases, the number of electrons that can pick up thermal energy and enter the conduction band will increase quite rapidly because the energy gap is quite small. However, the number of electrons entering the conduction band for Si or GaAs would be a great deal less.



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The energy gap also reveals which elements are useful in the construction of light-emitting devices such as light-emitting diodes (LEDs), which will be introduced shortly. The wider the energy gap, the greater is the possibility of energy being released in the form of visible or invisible (infrared) light waves. For conductors, the overlapping of valence and conduction bands essentially results in all the additional energy picked up by the electrons being dissipated in the form of heat. Similarly, for Ge and Si, because the energy gap is so small, most of the electrons that pick up sufficient energy to leave the valence band end up in the conduction band, and the energy is dissipated in the form of heat. However, for GaAs the gap is sufficiently large to result in significant light radiation. For LEDs (Section 1.9) the level of doping and the materials chosen determine the resulting color.

Before we leave this subject, it is important to underscore the importance of understanding the units used for a quantity. In Fig. 1.6 the units of measurement are *electron volts* (eV). The unit of measure is appropriate because W (energy) $= QV$ (as derived from the defining equation for voltage: $V = W/Q$). Substituting the charge of one electron and a potential difference of 1 V results in an energy level referred to as one *electron volt*.

That is,

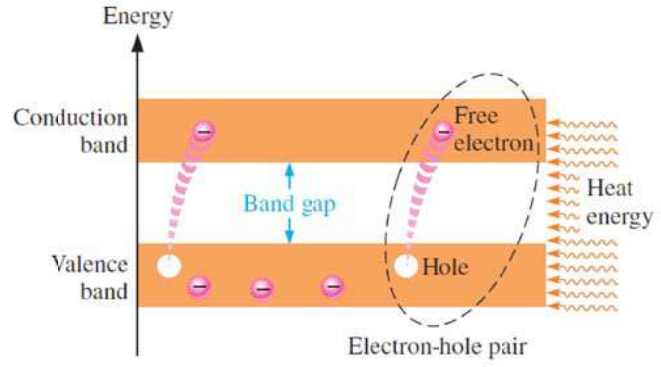
$$\begin{aligned} W &= QV \\ &= (1.6 \times 10^{-19}\,\text{C})(1\,\text{V}) \\ &= 1.6 \times 10^{-19}\,\text{J} \end{aligned}$$

and

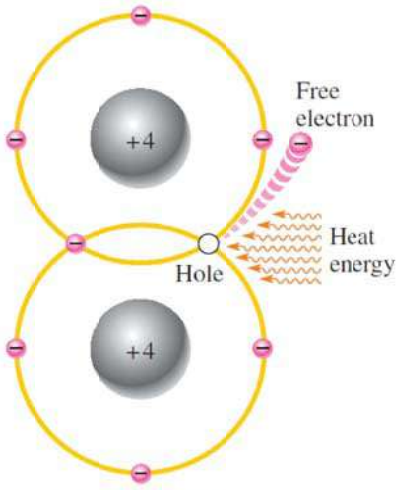
$$\boxed{1\,\text{eV} = 1.6 \times 10^{-19}\,\text{J}} \qquad \textbf{(1.1)}$$

Conduction Electrons and Holes

An intrinsic (pure) silicon crystal at room temperature has sufficient heat (thermal) energy for some valence electrons to jump the gap from the valence band into the conduction band, becoming free electrons. Free electrons are also called **conduction electrons.** This is illus-

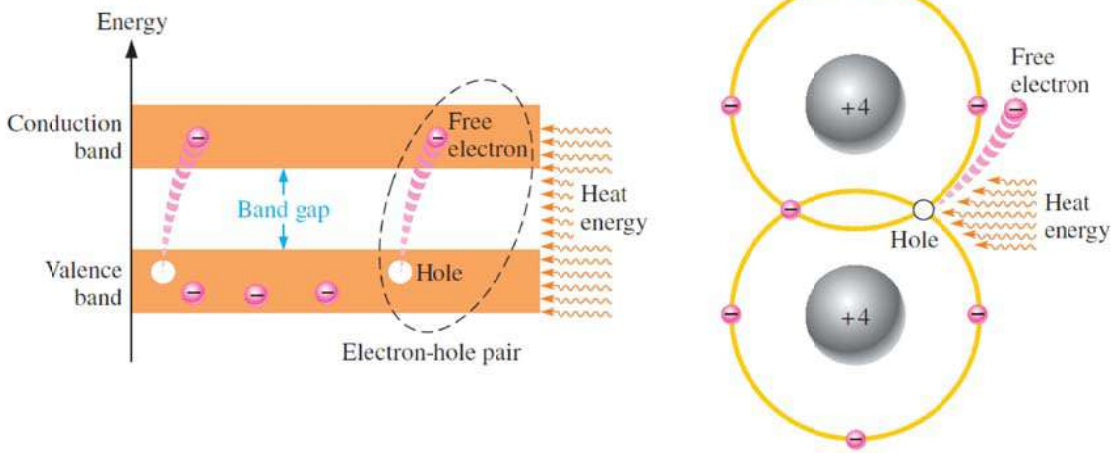


(a) Energy diagram



(b) Bonding diagram

When an electron jumps to the conduction band, a vacancy is left in the valence band within the crystal. This vacancy is called a **hole**. For every electron raised to the conduction band by external energy, there is one hole left in the valence band, creating what is called an **electron-hole pair. Recombination** occurs when a conduction-band electron loses energy and falls back into a hole in the valence band.



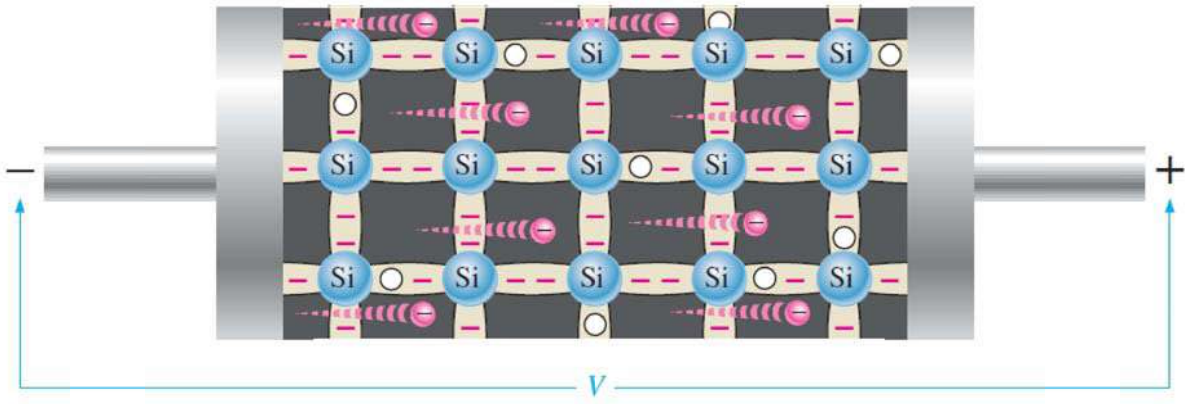
(a) Energy diagram

(b) Bonding diagram

21

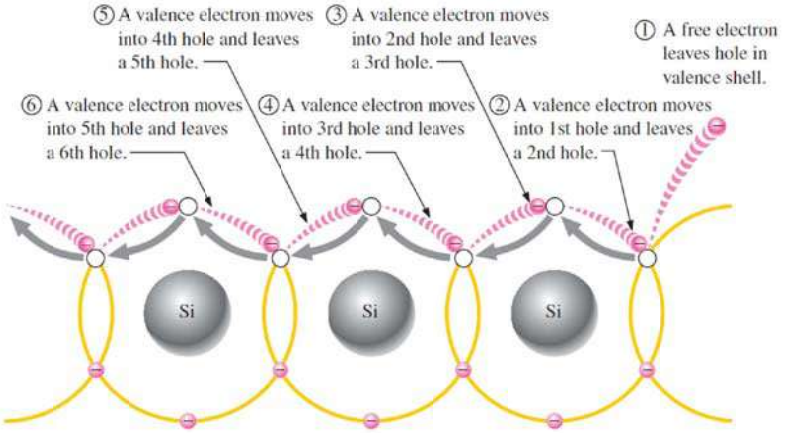
Electron and Hole Current

When a voltage is applied across a piece of intrinsic silicon, as shown in Figure 1-15, the thermally generated free electrons in the conduction band, which are free to move randomly in the crystal structure, are now easily attracted toward the positive end. This movement of free electrons is one type of **current** in a semiconductive material and is called *electron current*.



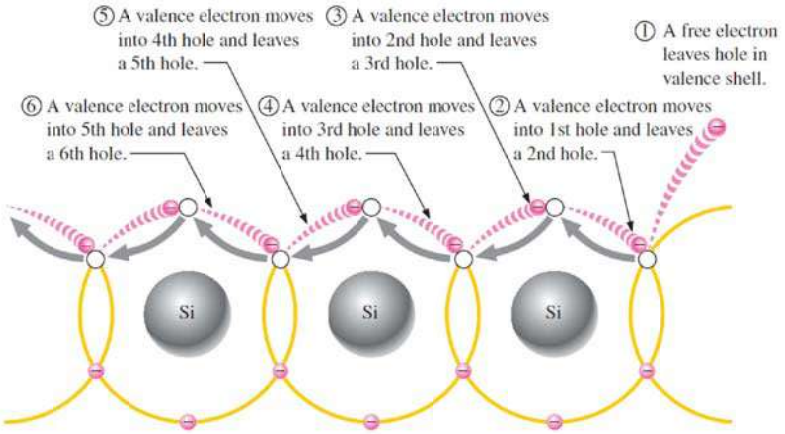
22

Another type of current occurs in the valence band, where the holes created by the free electrons exist. Electrons remaining in the valence band are still attached to their atoms and are not free to move randomly in the crystal structure as are the free electrons. However, a valence electron can move into a nearby hole with little change in its energy level, thus leaving another hole where it came from. Effectively the hole has moved from one place to another in the crystal structure, as illustrated in Figure . Although current in the valence band is produced by valence electrons, it is called *hole current* to distinguish it from electron current in the conduction band.



As you have seen, conduction in semiconductors is considered to be either the movement of free electrons in the conduction band or the movement of holes in the valence band, which is actually the movement of valence electrons to nearby atoms, creating hole current in the opposite direction.

When a valence electron moves left to right to fill a hole while leaving another hole behind, the hole has effectively moved from right to left. Gray arrows indicate effective movement of a hole.



1.5 *n*-TYPE AND *p*-TYPE MATERIALS

Because Si is the material used most frequently as the base (substrate) material in the construction of solid-state electronic devices, the discussion to follow in this and the next few sections deals solely with Si semiconductors. Because Ge, Si, and GaAs share a similar covalent bonding, the discussion can easily be extended to include the use of the other materials in the manufacturing process.

As indicated earlier, the characteristics of a semiconductor material can be altered significantly by the addition of specific impurity atoms to the relatively pure semiconductor material. These impurities, although only added at 1 part in 10 million, can alter the band structure sufficiently to totally change the electrical properties of the material.

A semiconductor material that has been subjected to the doping process is called an extrinsic material.

There are two extrinsic materials of immeasureable importance to semiconductor device fabrication: *n*-type and *p*-type materials. Each is described in some detail in the following subsections.

n-Type Material

Both *n*-type and *p*-type materials are formed by adding a predetermined number of impurity atoms to a silicon base. An *n*-type material is created by introducing impurity elements that have *five* valence electrons (*pentavalent*), such as *antimony*, *arsenic*, and *phosphorus*. Each is a member of a subset group of elements in the Periodic Table of Elements referred to as Group V because each has five valence electrons.

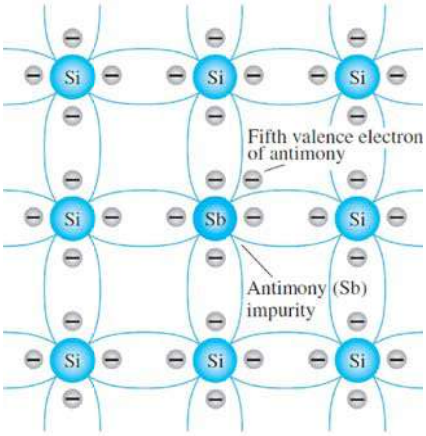
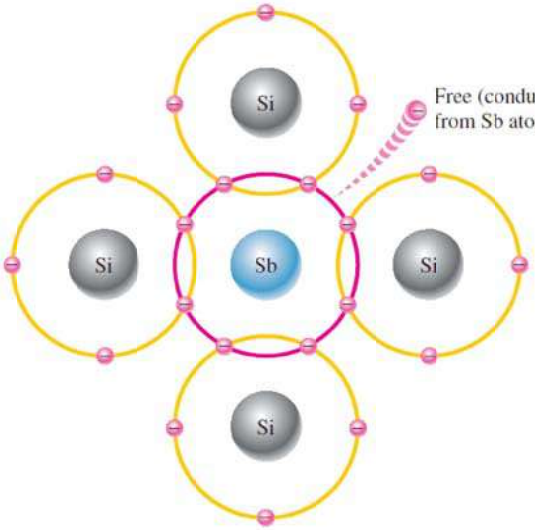
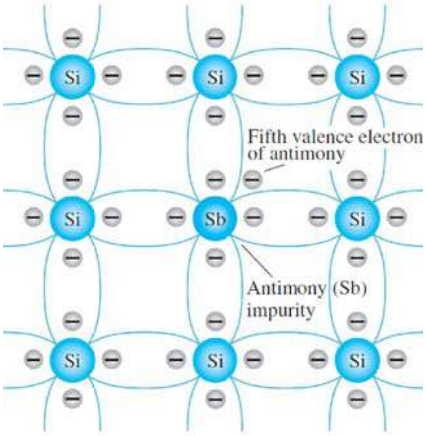


FIG. 1.7
Antimony impurity in n-type material.



n-Type Material

The effect of such impurity elements is indicated in
Fig. 1.7 (using antimony as the impurity in a silicon base). Note that the four covalent bonds are still present. There is, however, an additional fifth electron due to the impurity atom, which is *unassociated* with any particular covalent bond. This remaining electron, loosely bound to its parent (antimony) atom, is relatively free to move within the newly formed *n*-type material. Since the inserted impurity atom has donated a relatively "free" electron to the structure:



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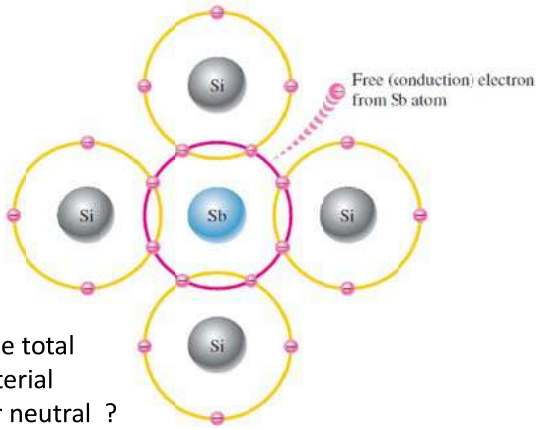
Antimony impurity in n-type material.

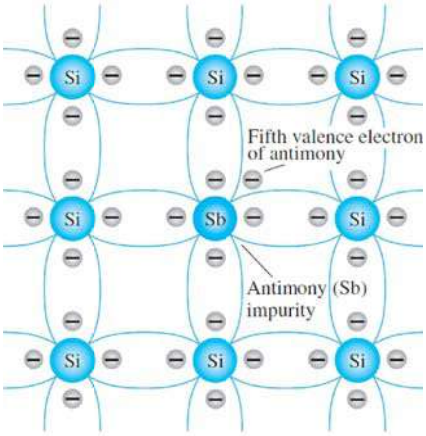


Question : what is the total charge of n-type material negative ,positive, or neutral ?

n-Type Material

Diffused impurities with five valence electrons are called donor atoms.

It is important to realize that even though a large number of free carriers have been established in the *n*-type material, it is still electrically *neutral* since ideally the number of positively charged protons in the nuclei is still equal to the number of free and orbiting negatively charged electrons in the structure.



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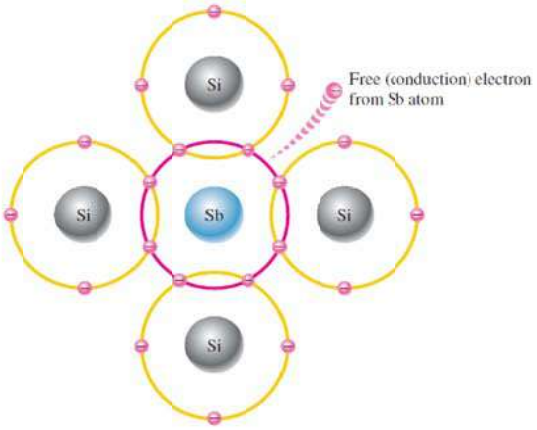


The effect of this doping process on the relative conductivity can best be described through the use of the energy-band diagram of Fig. 1.8. Note that a discrete energy level (called the *donor level*) appears in the forbidden band with an E_g significantly less than that of the intrinsic material. Those free electrons due to the added impurity sit at this energy level and have less difficulty absorbing a sufficient measure of thermal energy to move into the conduction band at room temperature. The result is that at room temperature, there are a large number of carriers (electrons) in the conduction level, and the conductivity of the material increases significantly. At room temperature in an intrinsic Si material there is about one free electron for every 10^{12} atoms. If the dosage level is 1 in 10 million (10^7), the ratio $10^{12}/10^7 = 10^5$ indicates that the carrier concentration has increased by a ratio of 100,000:1.

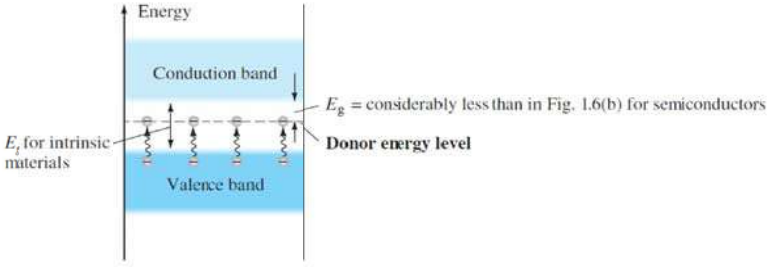


FIG. 1.8
Effect of donor impurities on the energy band structure.

p-Type Material

The *p*-type material is formed by doping a pure germanium or silicon crystal with impurity atoms having *three* valence electrons. The elements most frequently used for this purpose are *boron*, *gallium*, and *indium*. Each is a member of a subset group of elements in the Periodic Table of Elements referred to as Group III because each has three valence electrons.

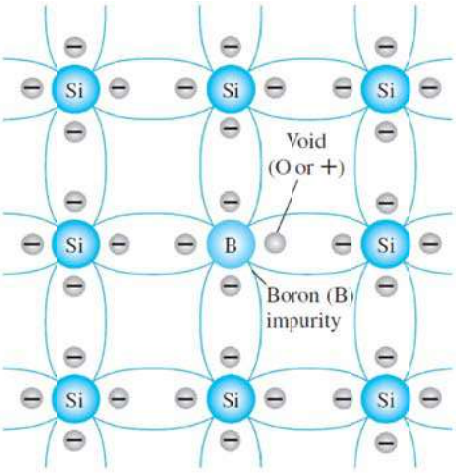
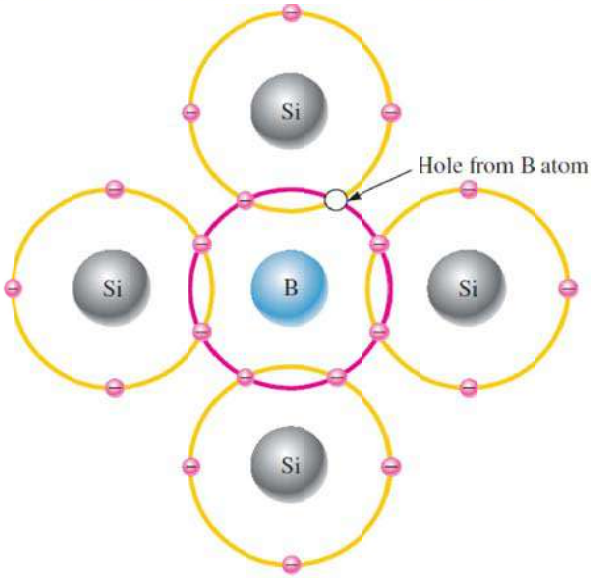


FIG. 1.9
Boron impurity in p-type material.



p-Type Material

The effect of one of these elements, boron, on a base of silicon is indicated in Fig. 1.9.

Note that there is now an insufficient number of electrons to complete the covalent bonds of the newly formed lattice. The resulting vacancy is called a *hole* and is represented by a small circle or a plus sign, indicating the absence of a negative charge. Since the resulting vacancy will readily *accept* a free electron:

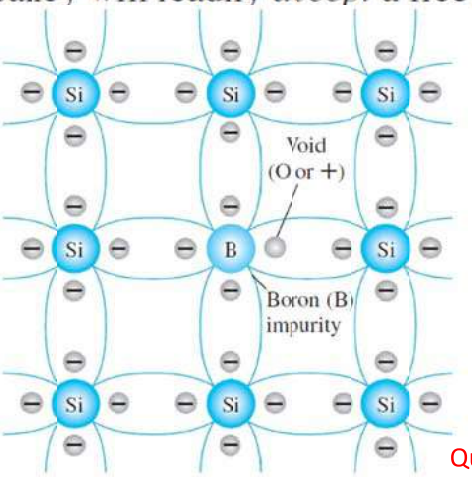
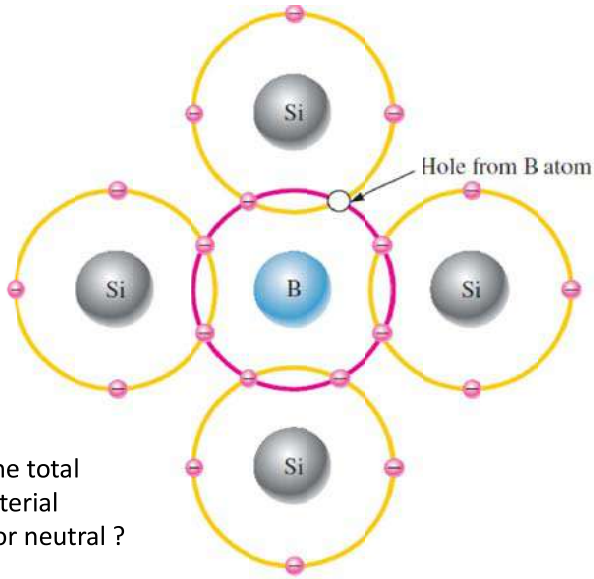


FIG. 1.9
Boron impurity in p-type material.



Question : what is the total charge of p-type material
Negative , positive, or neutral ?

p-Type Material

The diffused impurities with three valence electrons are called acceptor atoms.

The resulting *p*-type material is electrically neutral, for the same reasons described for the *n*-type material.

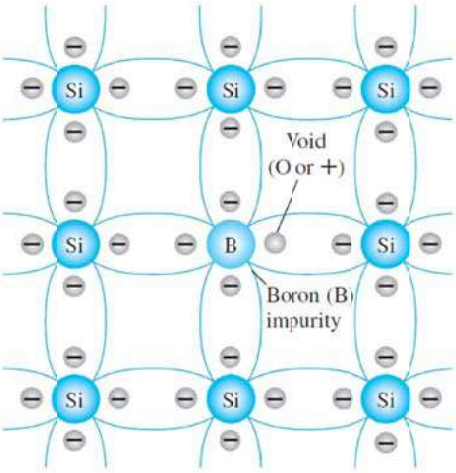
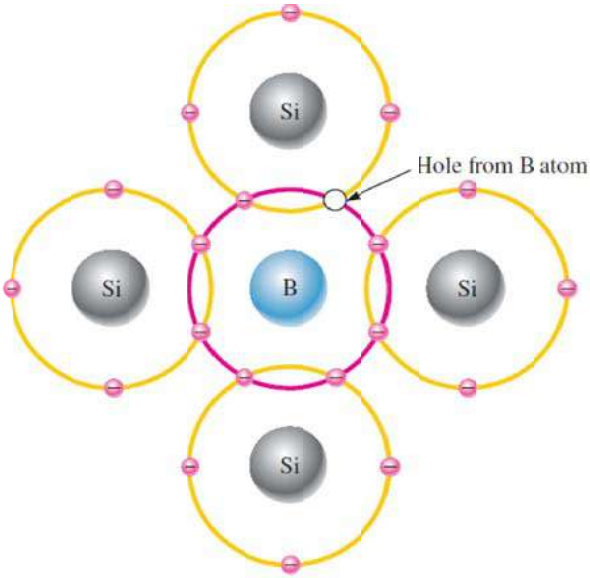


FIG. 1.9
Boron impurity in p-type material.



Electron versus Hole Flow

The effect of the hole on conduction is shown in Fig. 1.10. If a valence electron acquires sufficient kinetic energy to break its covalent bond and fills the void created by a hole, then a vacancy, or hole, will be created in the covalent bond that released the electron. There is, therefore, a transfer of holes to the left and electrons to the right, as shown in Fig. 1.10. The direction to be used in this text is that of *conventional flow*, which is indicated by the direction of hole flow.

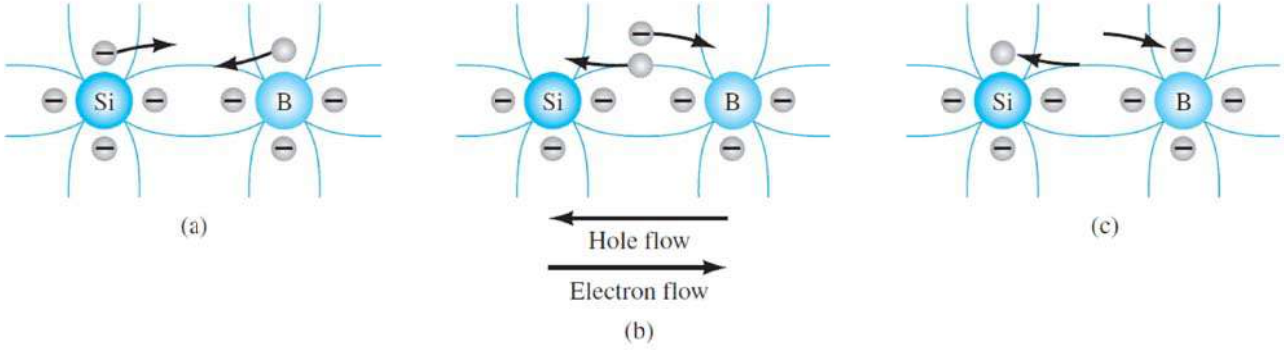


FIG. 1.10
Electron versus hole flow.

Majority and Minority Carriers

In the intrinsic state, the number of free electrons in Ge or Si is due only to those few electrons in the valence band that have acquired sufficient energy from thermal or light sources to break the covalent bond or to the few impurities that could not be removed. The vacancies left behind in the covalent bonding structure represent our very limited supply of holes. In an *n*-type material, the number of holes has not changed significantly from this intrinsic level. The net result, therefore, is that the number of electrons far outweighs the number of holes. For this reason:

In an n-type material (Fig. 1.11a) the electron is called the majority carrier and the hole the minority carrier.

For the *p*-type material the number of holes far outweighs the number of electrons, as shown in Fig. 1.11b. Therefore:

In a p-type material the hole is the majority carrier and the electron is the minority carrier.

When the fifth electron of a donor atom leaves the parent atom, the atom remaining acquires a net positive charge: hence the plus sign in the donor-ion representation. For similar reasons, the minus sign appears in the acceptor ion.

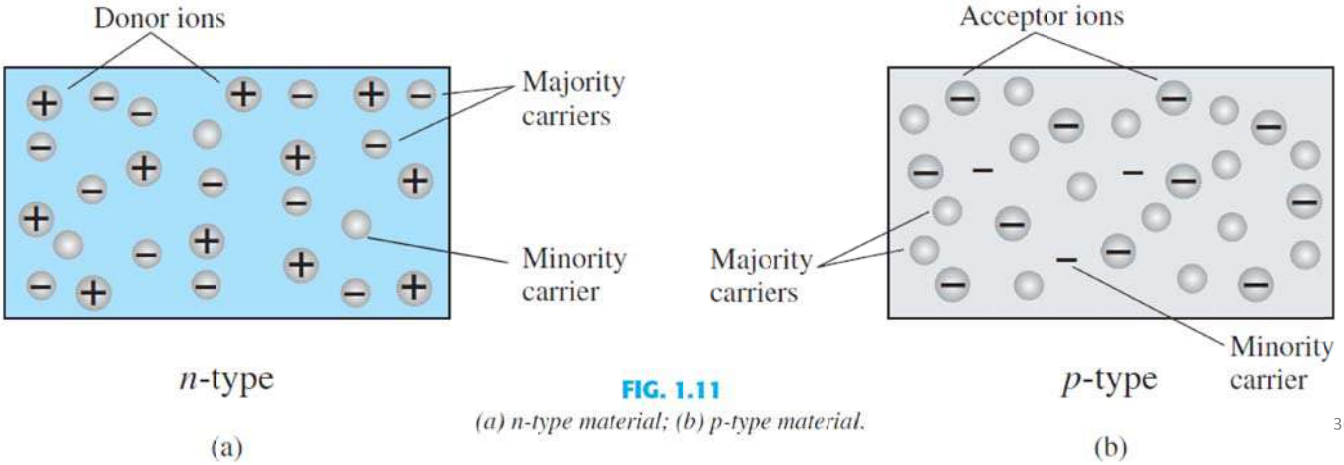


FIG. 1.11
(a) n-type material; (b) p-type material.

The *n*- and *p*-type materials represent the basic building blocks of semiconductor devices. We will find in the next section that the "joining" of a single *n*-type material with a *p*-type material will result in a semiconductor element of considerable importance in electronic systems.

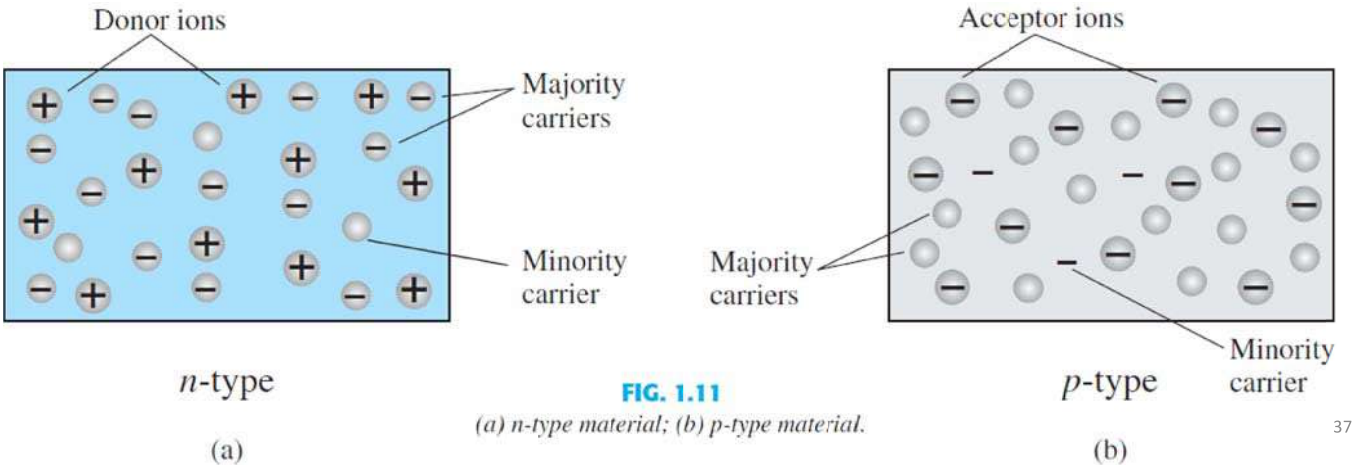
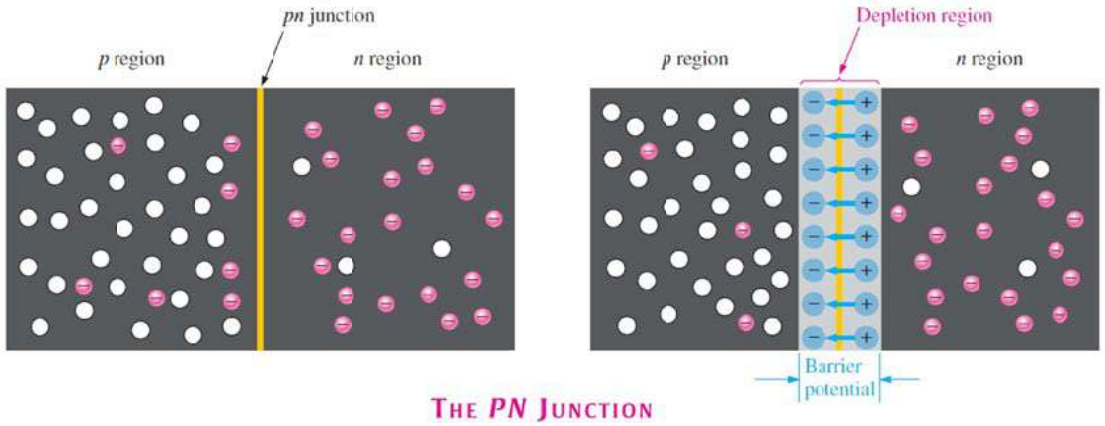


FIG. 1.11
(a) n-type material; (b) p-type material.

1.6 SEMICONDUCTOR DIODE

Now that both *n*- and *p*-type materials are available, we can construct our first solid-state electronic device: The *semiconductor diode*, with applications too numerous to mention, is created by simply joining an *n*-type and a *p*-type material together, nothing more, just the joining of one material with a majority carrier of electrons to one with a majority carrier of holes. The basic simplicity of its construction simply reinforces the importance of the development of this solid-state era.



THE *PN* JUNCTION

No Applied Bias ($V = 0$ V)

At the instant the two materials are "joined" the electrons and the holes in the region of the junction will combine, resulting in a lack of free carriers in the region near the junction, as shown in Fig. 1.12a. Note in Fig. 1.12a that the only particles displayed in this region are the positive and the negative ions remaining once the free carriers have been absorbed.

This region of uncovered positive and negative ions is called the depletion region due to the "depletion" of free carriers in the region.

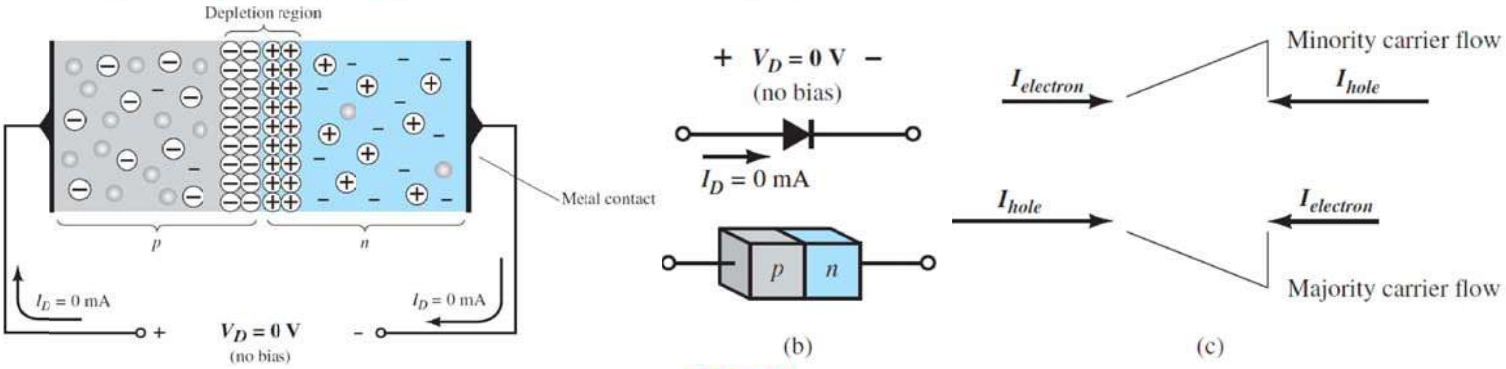


FIG. 1.12

A p–n junction with no external bias: (a) an internal distribution of charge; (b) a diode symbol, with the defined polarity and the current direction; (c) demonstration that the net carrier flow is zero at the external terminal of the device when $V_D = 0$ V.

If leads are connected to the ends of each material, a *two-terminal device* results, as shown in Figs. 1.12a and 1.12b. Three options then become available: *no bias*, *forward bias*, and *reverse bias*. The term *bias* refers to the application of an external voltage across the two terminals of the device to extract a response. The condition shown in Figs. 1.12a and 1.12b is the no-bias situation because there is no external voltage applied.

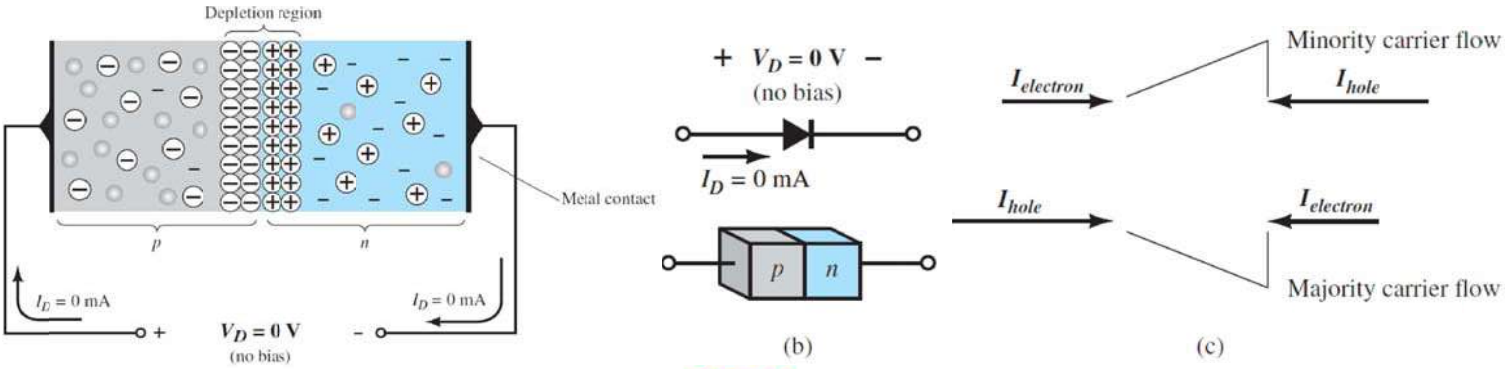


FIG. 1.12

A p–n junction with no external bias: (a) an internal distribution of charge; (b) a diode symbol, with the defined polarity and the current direction; (c) demonstration that the net carrier flow is zero at the external terminal of the device when $V_D = 0$ V.

Under no-bias conditions, any minority carriers (holes) in the *n*-type material that find themselves within the depletion region for any reason whatsoever will pass quickly into the *p*-type material. The closer the minority carrier is to the junction, the greater is the attraction for the layer of negative ions and the less is the opposition offered by the positive ions in the depletion region of the *n*-type material. We will conclude, therefore, for future discussions, that any minority carriers of the *n*-type material that find themselves in the depletion region will pass directly into the *p*-type material. This carrier flow is indicated at the top of Fig. 1.12c for the minority carriers of each material.

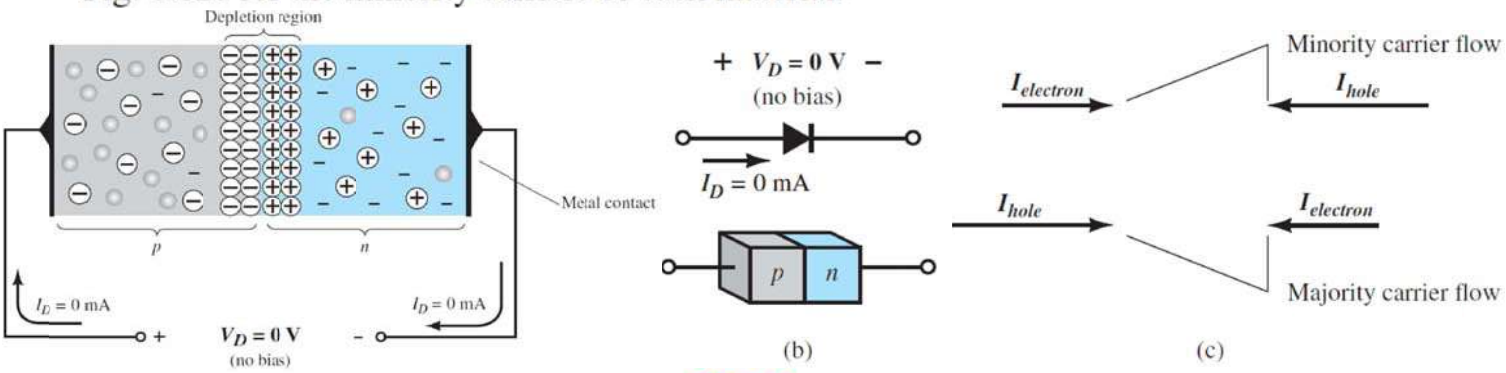


FIG. 1.12

A p–n junction with no external bias: (a) an internal distribution of charge; (b) a diode symbol, with the defined polarity and the current direction; (c) demonstration that the net carrier flow is zero at the external terminal of the device when $V_D = 0$ V.

The majority carriers (electrons) of the *n*-type material must overcome the attractive forces of the layer of positive ions in the *n*-type material and the shield of negative ions in the *p*-type material to migrate into the area beyond the depletion region of the *p*-type material. However, the number of majority carriers is so large in the *n*-type material that there will invariably be a small number of majority carriers with sufficient kinetic energy to pass through the depletion region into the *p*-type material. Again, the same type of discussion can be applied to the majority carriers (holes) of the *p*-type material. The resulting flow due to the majority carriers is shown at the bottom of Fig. 1.12c.

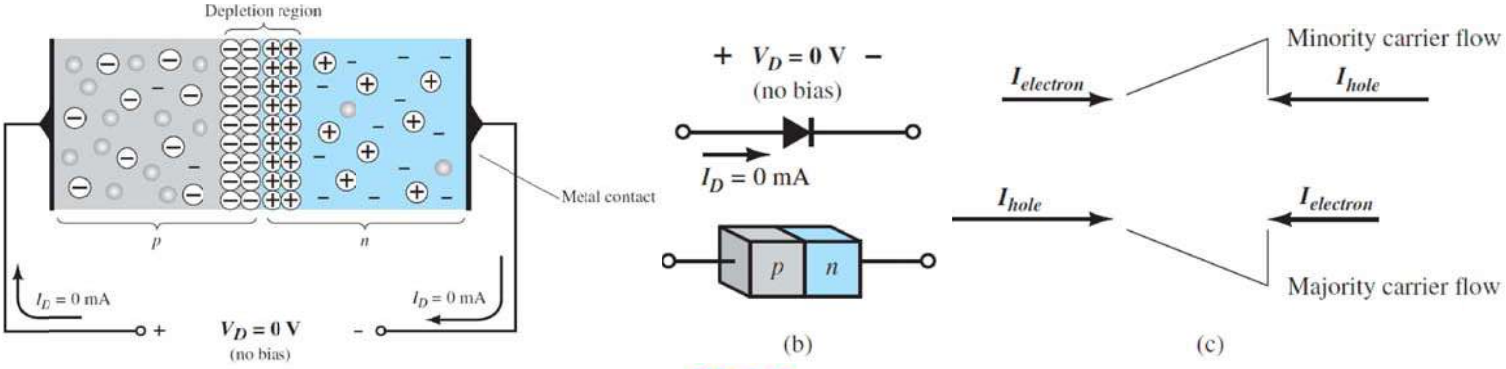


FIG. 1.12

A p–n junction with no external bias: (a) an internal distribution of charge; (b) a diode symbol, with the defined polarity and the current direction; (c) demonstration that the net carrier flow is zero at the external terminal of the device when $V_D = 0$ V.

A close examination of Fig. 1.12c will reveal that the relative magnitudes of the flow vectors are such that the net flow in either direction is zero. This cancellation of vectors for each type of carrier flow is indicated by the crossed lines. The length of the vector representing hole flow is drawn longer than that of electron flow to demonstrate that the two magnitudes need not be the same for cancellation and that the doping levels for each material may result in an unequal carrier flow of holes and electrons. In summary, therefore:

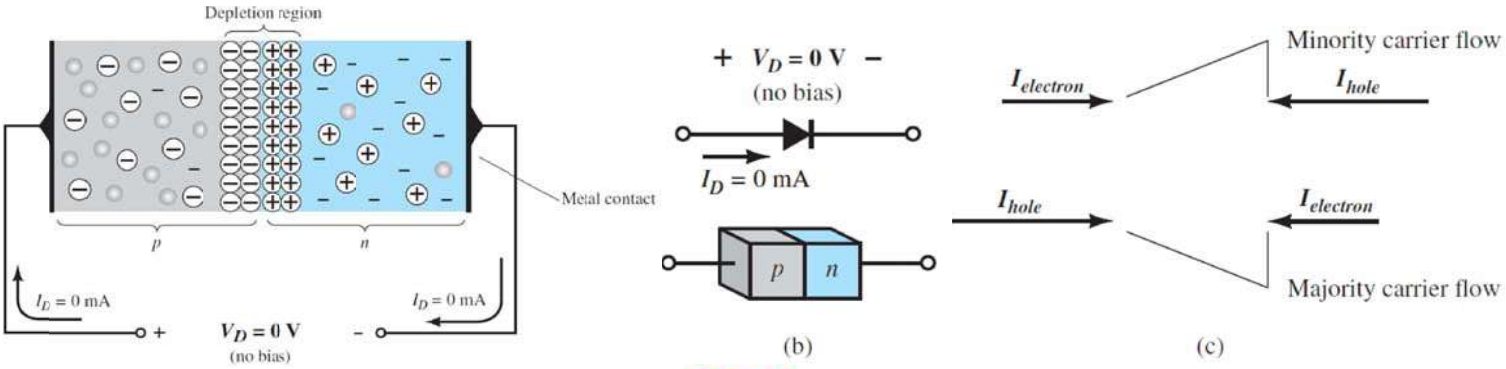


FIG. 1.12

A p–n junction with no external bias: (a) an internal distribution of charge; (b) a diode symbol, with the defined polarity and the current direction; (c) demonstration that the net carrier flow is zero at the external terminal of the device when $V_D = 0$ V.

It is simply

a diode with two leads sitting isolated on a laboratory bench. In Fig. 1.12b the symbol for a semiconductor diode is provided to show its correspondence with the *p–n* junction. In each figure it is clear that the applied voltage is 0 V (no bias) and the resulting current is 0 A, much like an isolated resistor. The absence of a voltage across a resistor results in zero current through it. Even at this early point in the discussion it is important to note the polarity of the voltage across the diode in Fig. 1.12b and the direction given to the current.

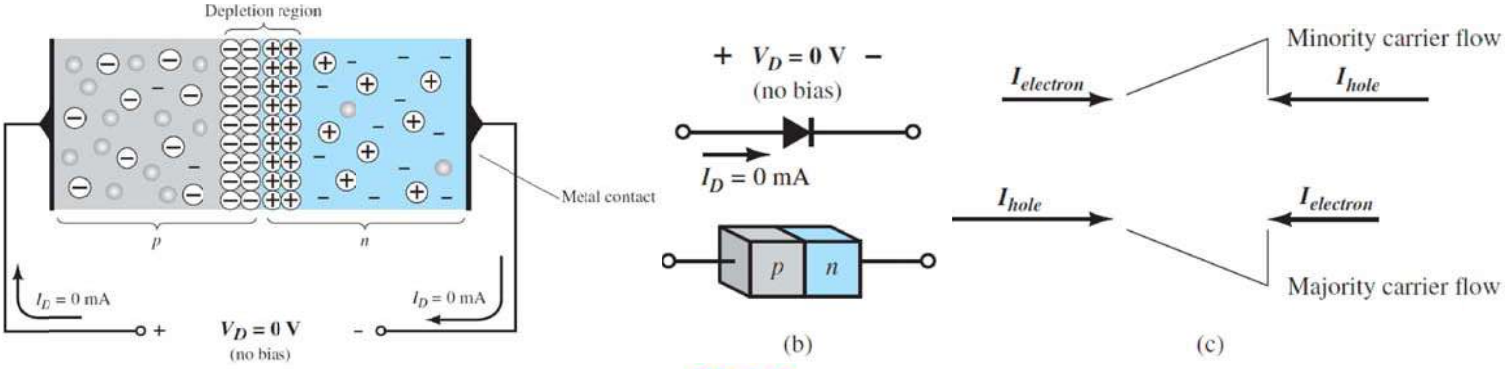


FIG. 1.12

A p–n junction with no external bias: (a) an internal distribution of charge; (b) a diode symbol, with the defined polarity and the current direction; (c) demonstration that the net carrier flow is zero at the external terminal of the device when $V_D = 0$ V.

Those polarities will be recognized as the *defined polarities* for the semiconductor diode. If a voltage applied across the diode has the same polarity across the diode as in Fig. 1.12b, it will be considered a positive voltage. If the reverse, it is a negative voltage. The same standards can be applied to the defined direction of current in Fig. 1.12b.

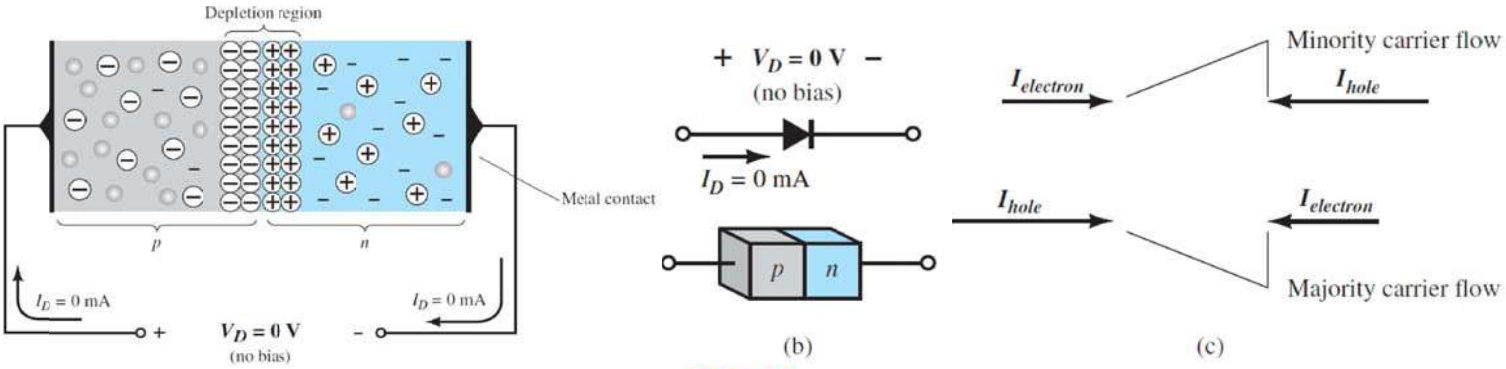


FIG. 1.12

A p–n junction with no external bias: (a) an internal distribution of charge; (b) a diode symbol, with the defined polarity and the current direction; (c) demonstration that the net carrier flow is zero at the external terminal of the device when $V_D = 0$ V.

In the absence of an applied bias across a semiconductor diode, the net flow of charge in one direction is zero.

In other words, the current under no-bias conditions is zero, as shown in Figs. 1.12a and 1.12b.

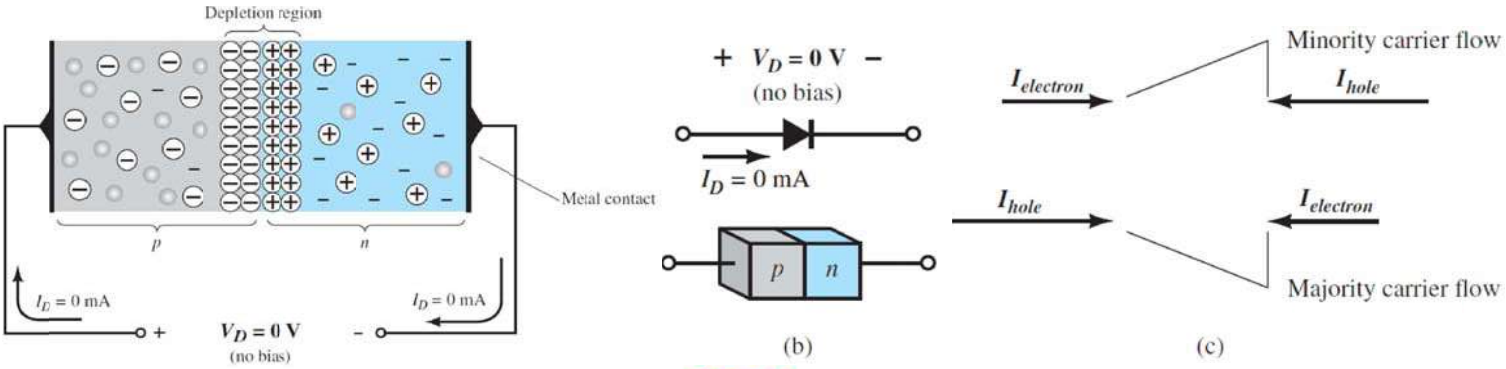
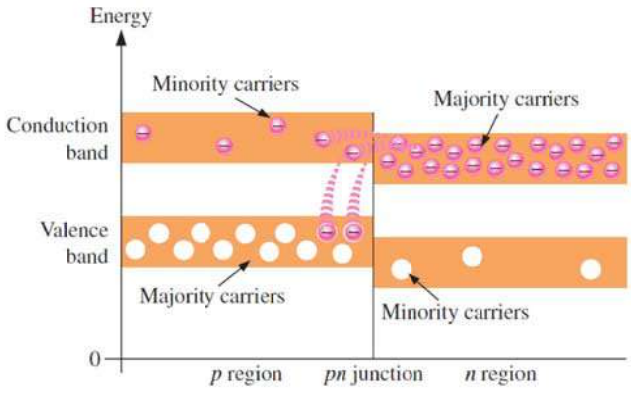


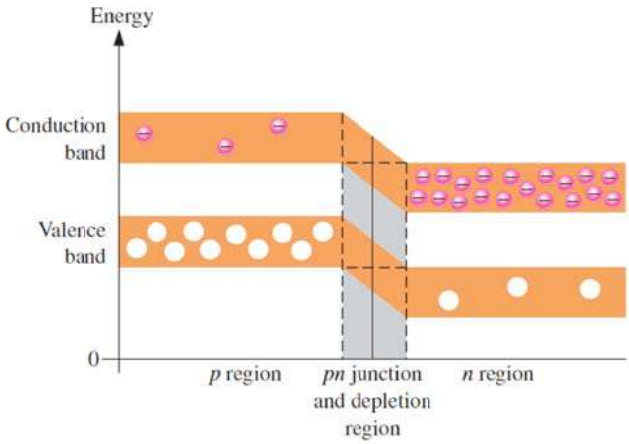
FIG. 1.12

A p–n junction with no external bias: (a) an internal distribution of charge; (b) a diode symbol, with the defined polarity and the current direction; (c) demonstration that the net carrier flow is zero at the external terminal of the device when $V_D = 0$ V.

Energy diagrams illustrating the formation of the *pn* junction and depletion region.



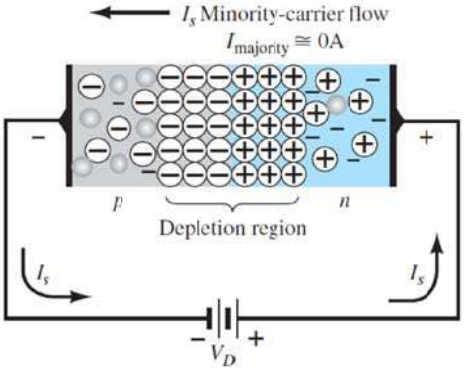
(a) At the instant of junction formation



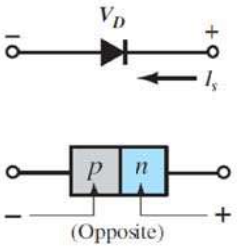
(b) At equilibrium

Reverse-Bias Condition ($V_D < 0$ V)

If an external potential of V volts is applied across the p–n junction such that the positive terminal is connected to the n-type material and the negative terminal is connected to the p-type material as shown in Fig. 1.13, the number of uncovered positive ions in the depletion region of the n-type material will increase due to the large number of free electrons drawn to the positive potential of the applied voltage. For similar reasons, the number of uncovered negative ions will increase in the p-type material. The net effect, therefore, is a



(a)



(b)

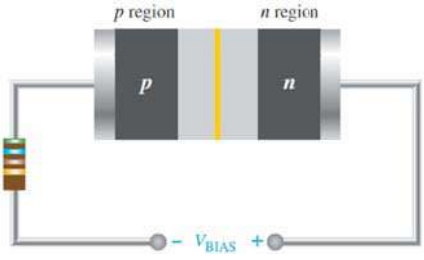


FIG. 1.13

Reverse-biased p–n junction: (a) internal distribution of charge under reverse-bias conditions; (b) reverse-bias polarity and direction of reverse saturation current.

widening of the depletion region. This widening of the depletion region will establish too great a barrier for the majority carriers to overcome, effectively reducing the majority carrier flow to zero, as shown in Fig. 1.13a.

The number of minority carriers, however, entering the depletion region will not change, resulting in minority-carrier flow vectors of the same magnitude indicated in Fig. 1.12c with no applied voltage.

The current that exists under reverse-bias conditions is called the reverse saturation current and is represented by I_s.

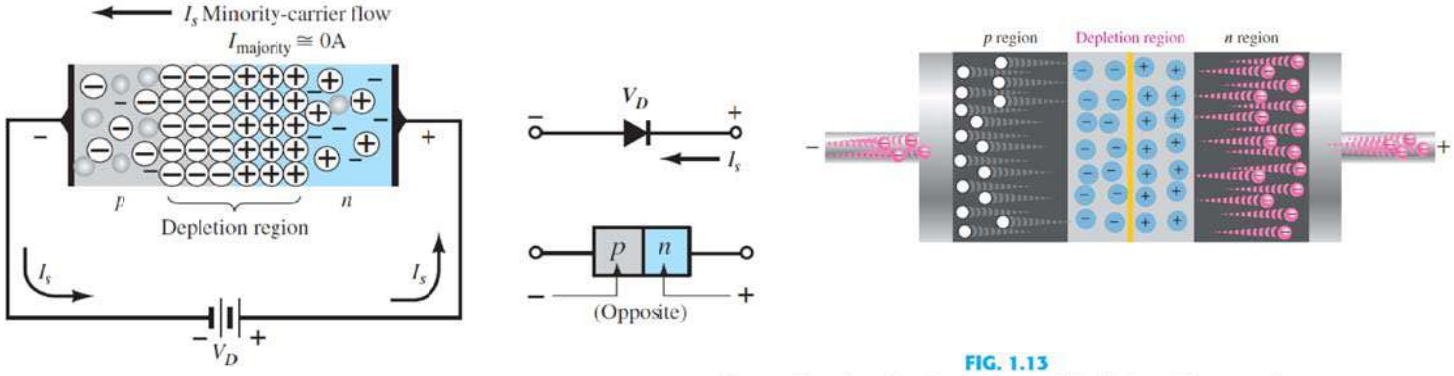


FIG. 1.13

Reverse-biased p–n junction: (a) internal distribution of charge under reverse-bias conditions; (b) reverse-bias polarity and direction of reverse saturation current.

The reverse saturation current is seldom more than a few microamperes and typically in nA, except for high-power devices. The term *saturation* comes from the fact that it reaches its maximum level quickly and does not change significantly with increases in the reverse-bias potential, as shown on the diode characteristics of Fig. 1.15 for $V_D < 0$ V. The reverse-biased conditions are depicted in Fig. 1.13b for the diode symbol and *p–n* junction. Note, in particular, that the direction of I_s is against the arrow of the symbol. Note also that the negative side of the applied voltage is connected to the *p*-type material and the positive side to the *n*-type material, the difference in underlined letters for each region revealing a reverse-bias condition.

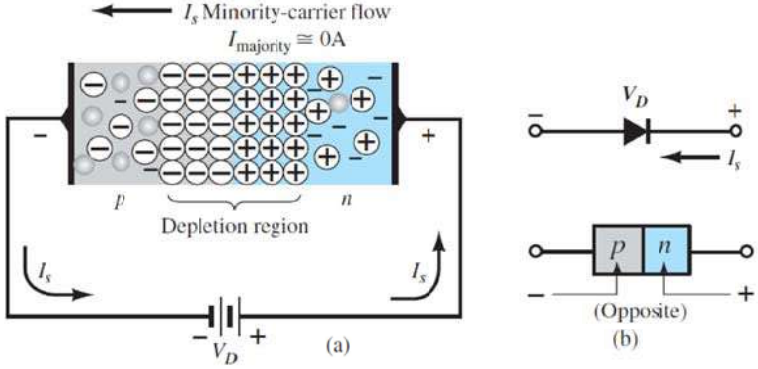


FIG. 1.13

Reverse-biased p–n junction: (a) internal distribution of charge under reverse-bias conditions; (b) reverse-bias polarity and direction of reverse saturation current.

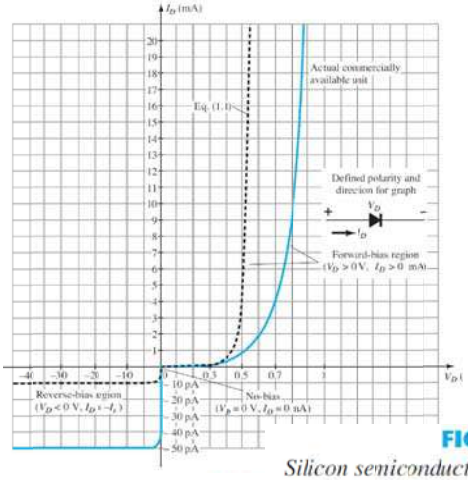


FIG. 1.15

Silicon semiconductor diode characteristics.

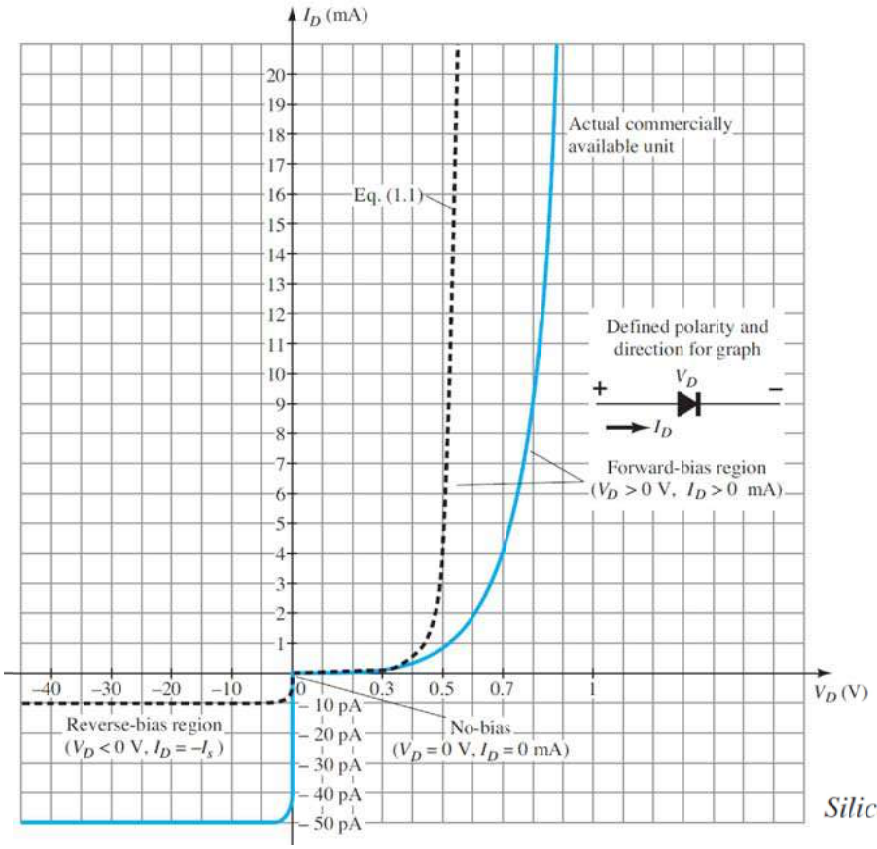


FIG. 1.15
Silicon semiconductor diode characteristics.

Forward-Bias Condition ($V_D > 0$ V)

A *forward-bias* or "on" condition is established by applying the positive potential to the *p*-type material and the negative potential to the *n*-type material as shown in Fig. 1.14.

The application of a forward-bias potential V_D will "pressure" electrons in the *n*-type material and holes in the *p*-type material to recombine with the ions near the boundary and reduce the width of the depletion region as shown in Fig. 1.14a.

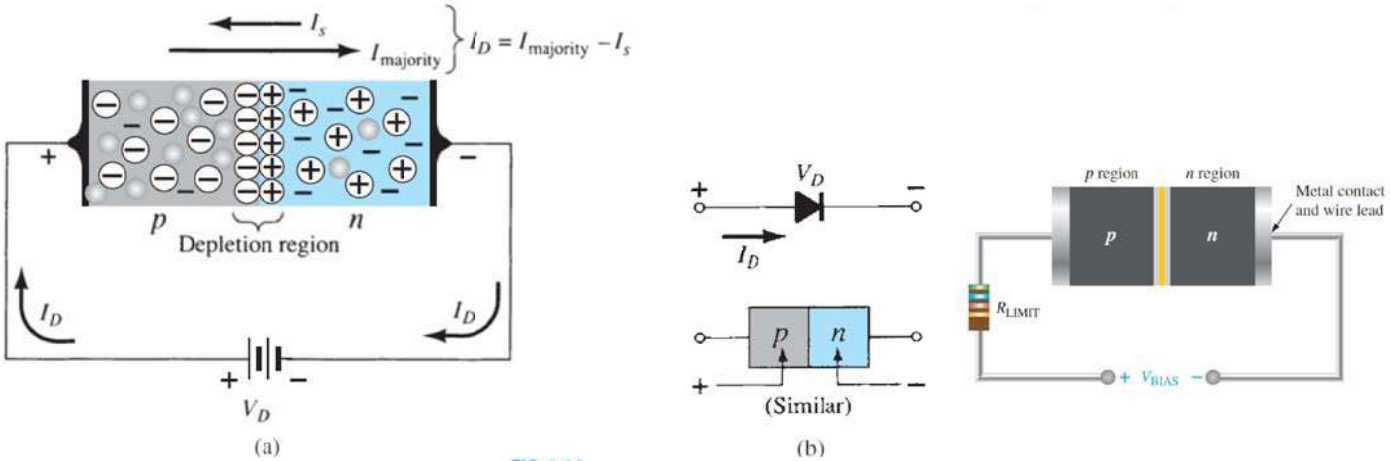
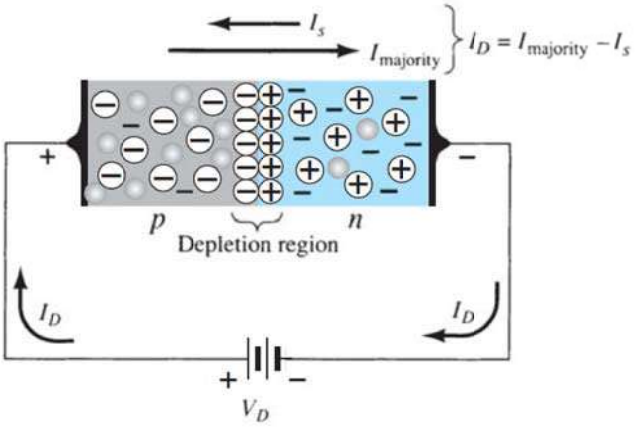


FIG. 1.14
Forward-biased p–n junction: (a) internal distribution of charge under forward-bias conditions; (b) forward-bias polarity and direction of resulting current.

The resulting minority-carrier flow of electrons from the *p*-type material to the *n*-type material (and of holes from the *n*-type material to the *p*-type material) has not changed in magnitude (since the conduction level is controlled primarily by the limited number of impurities in the material), but the reduction in the width of the depletion region has resulted in a heavy majority flow across the junction. An electron of the *n*-type material now "sees" a reduced barrier at the junction due to the reduced depletion region and a strong attraction for the positive potential applied to the *p*-type material.

As the applied bias increases in magnitude, the depletion region will continue to decrease in width until a flood of electrons can pass through the junction, resulting in an exponential rise in current as shown in the forward-bias region of the characteristics of Fig. 1.15. Note that the vertical scale of Fig. 1.15 is measured in milliamperes (although some semiconductor diodes have a vertical scale measured in amperes), and the horizontal scale in the forward-bias region has a maximum of 1 V. Typically, therefore, the voltage across a forward-biased diode will be less than 1 V. Note also how quickly the current rises beyond the knee of the curve.



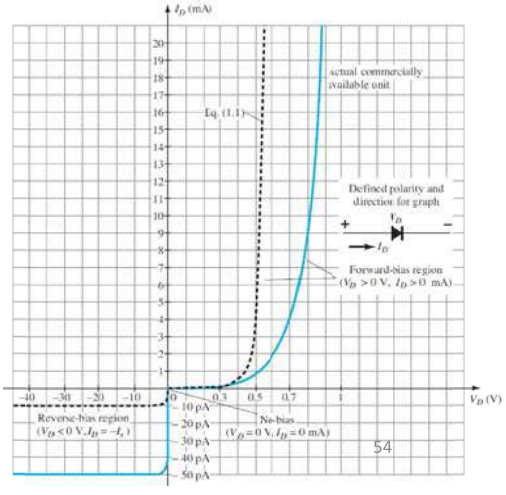


FIG. 1.15
nductor diode characteristics.

It can be demonstrated through the use of solid-state physics that the general characteristics of a semiconductor diode can be defined by the following equation, referred to as Shockley's equation, for the forward- and reverse-bias regions:

$$I_D = I_s(e^{V_D/nV_T} - 1) \quad \text{(A)} \tag{1.2}$$

where I_s is the reverse saturation current
V_D is the applied forward-bias voltage across the diode
n is an ideality factor, which is a function of the operating conditions and physical construction; it has a range between 1 and 2 depending on a wide variety of factors ($n = 1$ will be assumed throughout this text unless otherwise noted).

The voltage V_T in Eq. (1.1) is called the *thermal voltage* and is determined by

$$V_T = \frac{kT_K}{q} \quad \text{(V)} \tag{1.3}$$

where k is Boltzmann's constant $= 1.38 \times 10^{-23}$ J/K
T_K is the absolute temperature in kelvins = 273 + the temperature in °C
q is the magnitude of electronic charge $= 1.6 \times 10^{-19}$ C

EXAMPLE 1.1 At a temperature of 27°C (common temperature for components in an enclosed operating system), determine the thermal voltage V_T.

Solution: Substituting into Eq. (1.3), we obtain

$$T = 273 + °C = 273 + 27 = 300 \text{ K}$$

$$V_T = \frac{kT_K}{q} = \frac{(1.38 \times 10^{-23} \text{ J/K})(30 \text{ K})}{1.6 \times 10^{-19} \text{ C}}$$

$$= 25.875 \text{ mV} \cong 26 \text{ mV}$$

The thermal voltage will become an important parameter in the analysis to follow in this chapter and a number of those to follow.

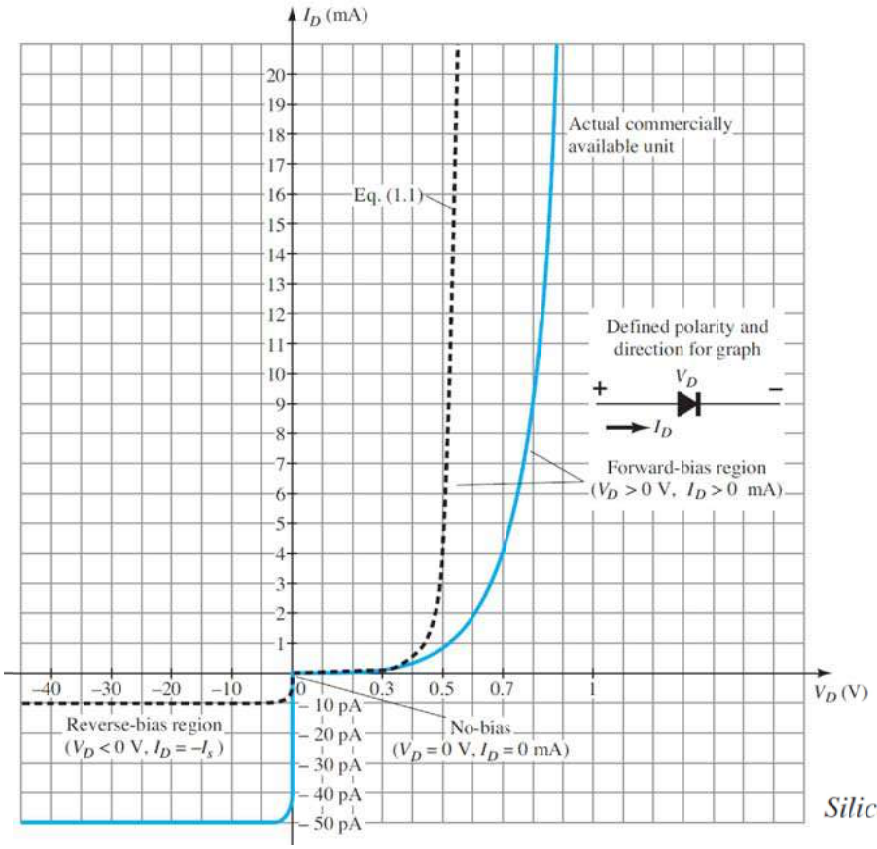


FIG. 1.15
Silicon semiconductor diode characteristics.

The exponential curve of Fig. 1.16 increases very rapidly with increasing values of x. At $x = 0$, $e^0 = 1$, whereas at $x = 5$, it jumps to greater than 148. If we continued to $x = 10$, the curve jumps to greater than 22,000. Clearly, therefore, as the value of x increases, the curve becomes almost vertical, an important conclusion to keep in mind when we examine the change in current with increasing values of applied voltage.

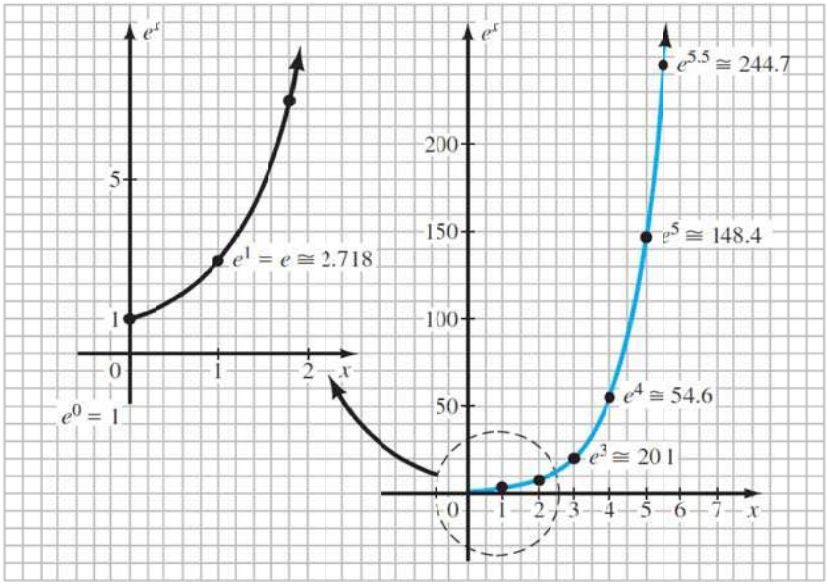


FIG. 1.16
Plot of e^x.

$$I_D = I_s(e^{V_D/nV_T} - 1)$$

For negative values of V_D the exponential term drops very quickly below the level of I, and the resulting equation for I_D is simply

$$I_D \cong -I_s \qquad (V_D \text{ negative})$$

Note in Fig. 1.15 that for negative values of V_D the current is essentially horizontal at the level of $-I_s$.

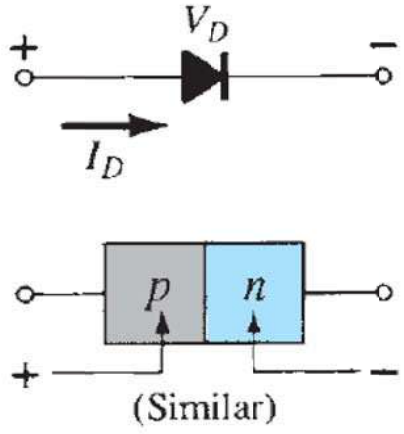
At $V = 0$ V, Eq. (1.2) becomes

$$I_D = I_s(e^0 - 1) = I_s(1 - 1) = 0 \text{ mA}$$

as confirmed by Fig. 1.15.

It is important to note in Fig. 1.14b how:

The defined direction of conventional current for the positive voltage region matches the arrowhead in the diode symbol.



Breakdown Region

Even though the scale of Fig. 1.15 is in tens of volts in the negative region, there is a point where the application of too negative a voltage with the reverse polarity will result in a sharp change in the characteristics, as shown in Fig. 1.17. The current increases at a very rapid rate in a direction opposite to that of the positive voltage region. The reverse-bias potential that results in this dramatic change in characteristics is called the *breakdown potential* and is given the label V_{BV}.

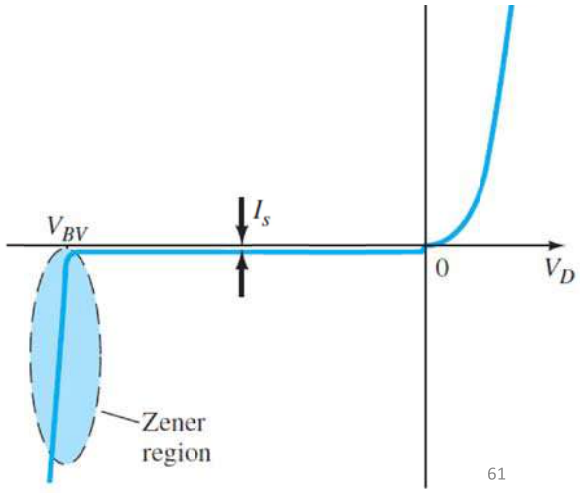


FIG. 1.17
Breakdown region.

The breakdown region of the semiconductor diode described must be avoided if the response of a system is not to be completely altered by the sharp change in characteristics in this reverse-voltage region.

The maximum reverse-bias potential that can be applied before entering the breakdown region is called the peak inverse voltage (referred to simply as the PIV rating) or the peak reverse voltage (denoted the PRV rating).

If an application requires a PIV rating greater than that of a single unit, a number of diodes of the same characteristics can be connected in series. Diodes are also connected in parallel to increase the current-carrying capacity.

Important

Ge, Si, and GaAs

The discussion thus far has solely used Si as the base semiconductor material. It is now important to compare it to the other two materials of importance: GaAs and Ge. A plot comparing the characteristics of Si, GaAs, and Ge diodes is provided in Fig. 1.18. The curves are not

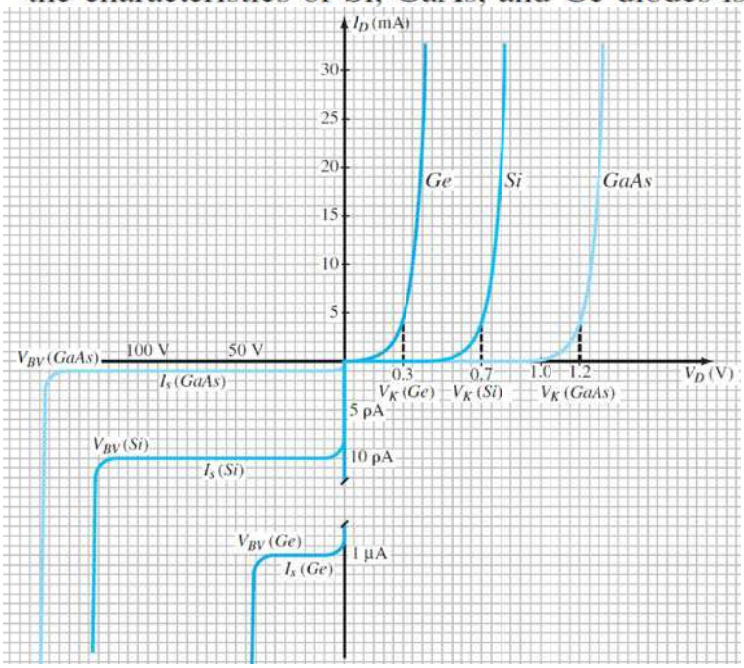


TABLE 1.3
Knee Voltages V_K

Semiconductor	V_K(V)
Ge	0.3
Si	0.7
GaAs	1.2

FIG. 1.18
Comparison of Ge, Si, and GaAs commercial diodes.

Temperature Effects

Temperature can have a marked effect on the characteristics of a semiconductor diode, as demonstrated by the characteristics of a silicon diode shown in Fig. 1.19:

In the forward-bias region the characteristics of a silicon diode shift to the left at a rate of 2.5 mV per centigrade degree increase in temperature.

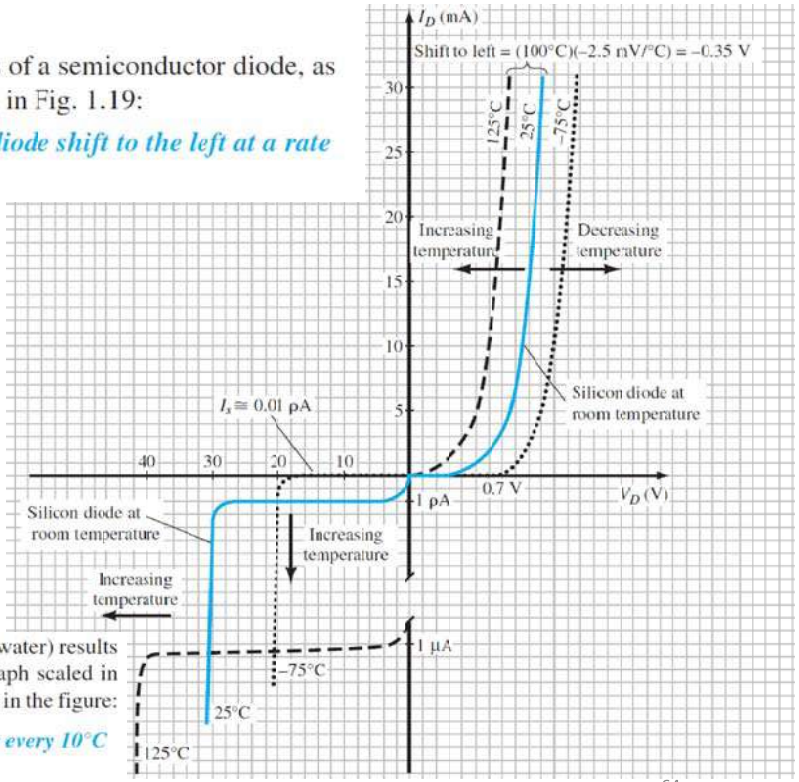


FIG. 1.19
Variation in Si diode characteristics with temperature change.

An increase from room temperature (20°C) to 100°C (the boiling point of water) results in a drop of 80(2.5 mV) = 200 mV, or 0.2 V, which is significant on a graph scaled in tenths of volts. A decrease in temperature has the reverse effect, as also shown in the figure:

In the reverse-bias region the reverse current of a silicon diode doubles for every 10°C rise in temperature.

TABLE 1.5

The Current Commercial Use of Ge, Si, and GaAs

Ge:	Germanium is in limited production due to its temperature sensitivity and high reverse saturation current. It is still commercially available but is limited to some high-speed applications (due to a relatively high mobility factor) and applications that use its sensitivity to light and heat such as photodetectors and security systems.
Si:	Without question the semiconductor used most frequently for the full range of electronic devices. It has the advantage of being readily available at low cost and has relatively low reverse saturation currents, good temperature characteristics, and excellent breakdown voltage levels. It also benefits from decades of enormous attention to the design of large-scale integrated circuits and processing technology.
GaAs:	Since the early 1990s the interest in GaAs has grown in leaps and bounds, and it will eventually take a good share of the development from silicon devices, especially in very large scale integrated circuits. Its high-speed characteristics are in more demand every day, with the added features of low reverse saturation currents, excellent temperature sensitivities, and high breakdown voltages. More than 80% of its applications are in optoelectronics with the development of light-emitting diodes, solar cells, and other photodetector devices, but that will probably change dramatically as its manufacturing costs drop and its use in integrated circuit design continues to grow; perhaps the semiconductor material of the future.

1.7 IDEAL VERSUS PRACTICAL

In the previous section we found that a *p–n* junction will permit a generous flow of charge when forward-biased and a very small level of current when reverse-biased. Both conditions are reviewed in Fig. 1.21, with the heavy current vector in Fig. 1.21a matching the direction of the arrow in the diode symbol and the significantly smaller vector in the opposite direction in Fig. 1.21b representing the reverse saturation current.

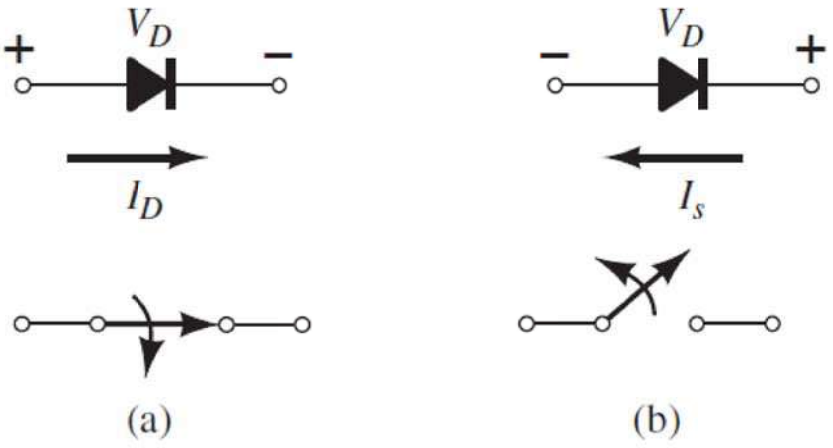


FIG. 1.21

Ideal semiconductor diode: (a) forward-biased; (b) reverse-biased.

In other words:

The semiconductor diode behaves in a manner similar to a mechanical switch in that it can control whether current will flow between its two terminals.

However, it is important to also be aware that:

The semiconductor diode is different from a mechanical switch in the sense that when the switch is closed it will only permit current to flow in one direction.

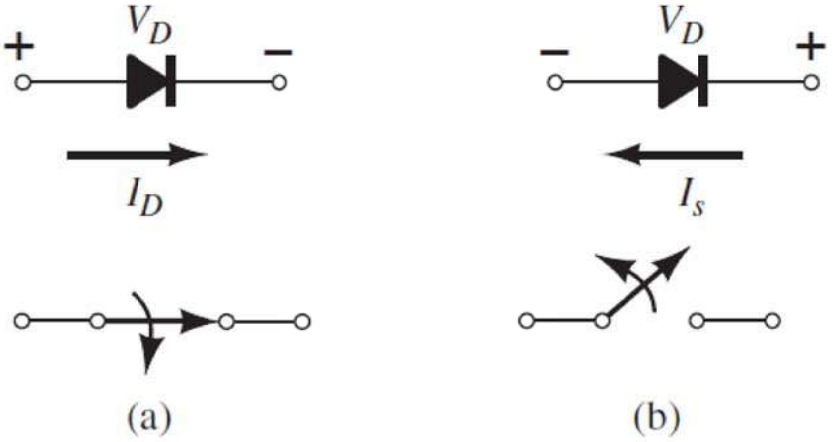


FIG. 1.21
Ideal semiconductor diode: (a) forward-biased; (b) reverse-biased.

Ideally, if the semiconductor diode is to behave like a closed switch in the forward-bias region, the resistance of the diode should be 0 Ω. In the reverse-bias region its resistance should be ∞Ω to represent the open-circuit equivalent. Such levels of resistance in the forward- and reverse-bias regions result in the characteristics of Fig. 1.22.

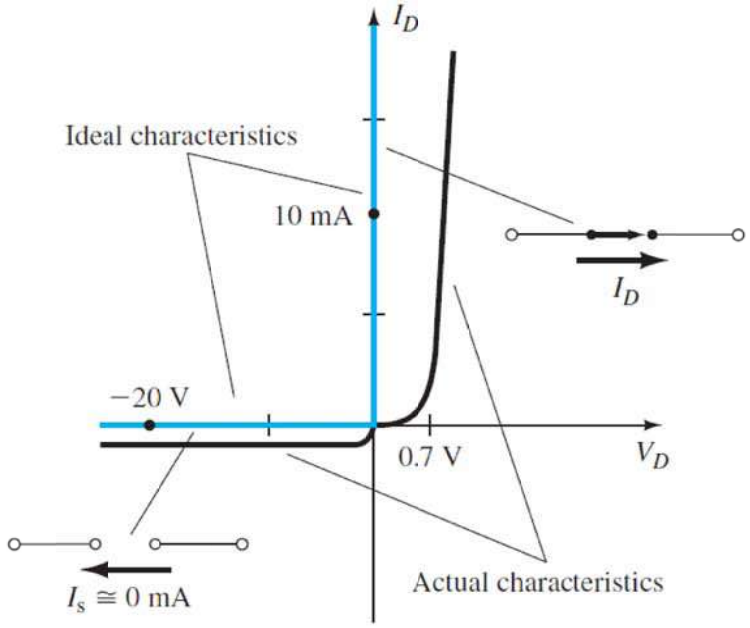
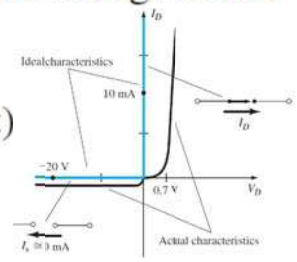


FIG. 1.22
Ideal versus actual semiconductor characteristics.

The characteristics have been superimposed to compare the ideal Si diode to a real-world Si diode. First impressions might suggest that the commercial unit is a poor impression of the ideal switch. However, when one considers that the only major difference is that the commercial diode rises at a level of 0.7 V rather than 0 V, there are a number of similarities between the two plots.

When a switch is closed the resistance between the contacts is assumed to be 0 Ω. At the plot point chosen on the vertical axis the diode current is 5 mA and the voltage across the diode is 0 V. Substituting into Ohm's law results in

$$R_F = \frac{V_D}{I_D} = \frac{0\text{ V}}{5\text{ mA}} = \mathbf{0\ \Omega} \qquad \text{(short-circuit equivalent)}$$



In fact:

At any current level on the vertical line, the voltage across the ideal diode is 0 V and the resistance is 0 Ω.

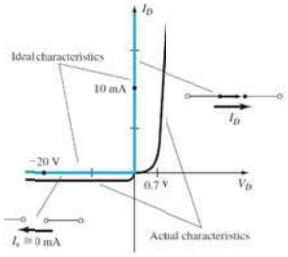
For the horizontal section, if we again apply Ohm's law, we find

$$R_R = \frac{V_D}{I_D} = \frac{20\text{ V}}{0\text{ mA}} \cong \infty\,\Omega \qquad \text{(open-circuit equivalent)}$$

Again:

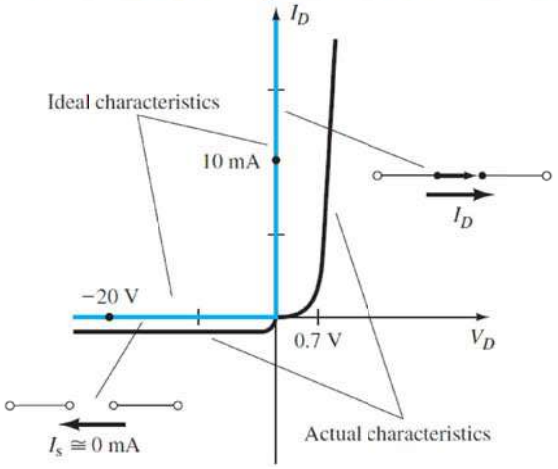
Because the current is 0 mA anywhere on the horizontal line, the resistance is considered to be infinite ohms (an open-circuit) at any point on the axis.

The result, therefore, is that there are sufficient similarities between the ideal switch and the semiconductor diode to make it an effective electronic device. In the next section the various resistance levels of importance are determined for use in the next chapter, where the response of diodes in an actual network is examined.

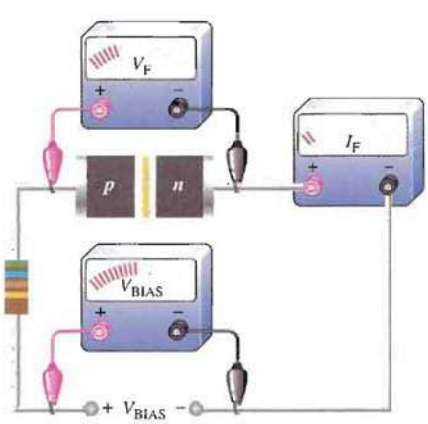


1.8 RESISTANCE LEVELS

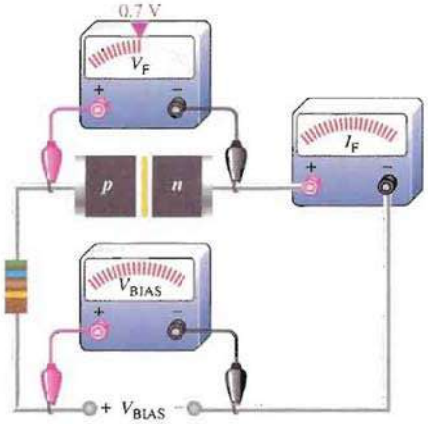
As the operating point of a diode moves from one region to another the resistance of the diode will also change due to the nonlinear shape of the characteristic curve. It will be demonstrated in the next few paragraphs that the type of applied voltage or signal will define the resistance level of interest. Three different levels will be introduced in this section, which will appear again as we examine other devices. It is therefore paramount that their determination be clearly understood.



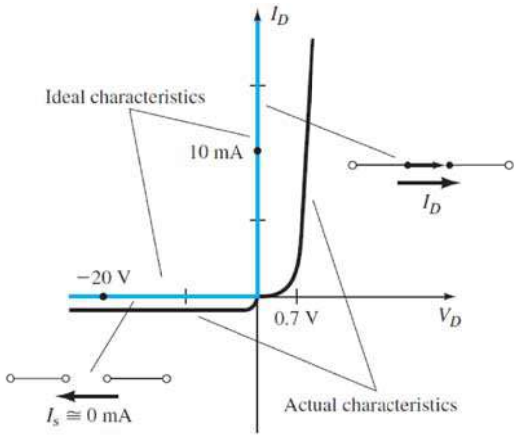
71



(a) Small forward-bias voltage ($V_F < 0.7$ V), very small forward current.



(b) Forward voltage reaches and remains at approximately 0.7 V. Forward current continues to increase as the bias voltage is increased.



Forward-bias measurements show general changes in V_F and I_F as V_{BIAS} is increased.

72

DC or Static Resistance

The application of a dc voltage to a circuit containing a semiconductor diode will result in an operating point on the characteristic curve that will not change with time. The resistance of the diode at the operating point can be found simply by finding the corresponding levels of V_D and I_D as shown in Fig. 1.23 and applying the following equation:

$$R_D = \frac{V_D}{I_D} \qquad (1.4)$$

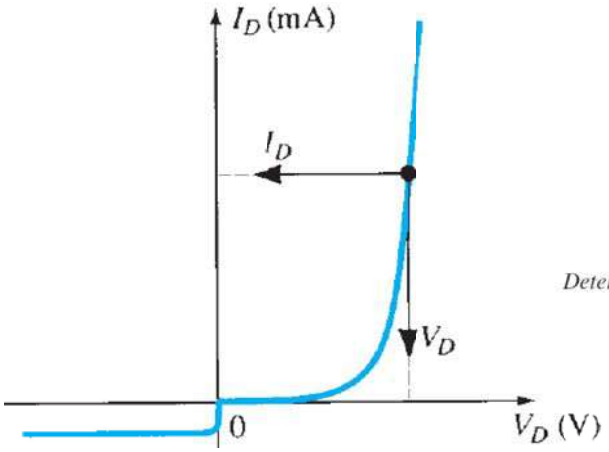
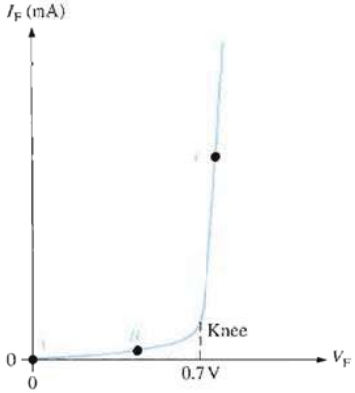


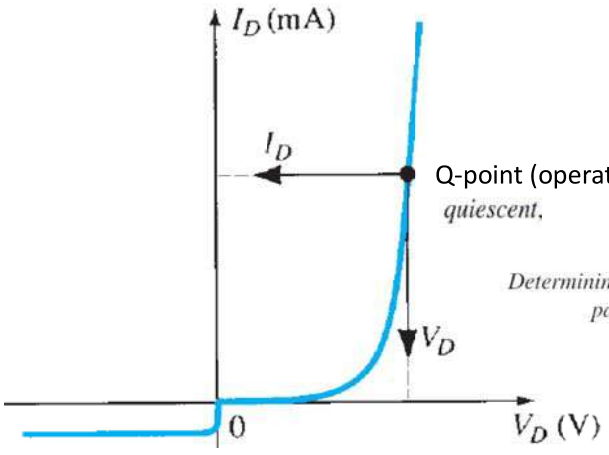
FIG. 1.23
Determining the dc resistance of a diode at a particular operating point.

The dc resistance levels at the knee and below will be greater than the resistance levels obtained for the vertical rise section of the characteristics. The resistance levels in the reverse-bias region will naturally be quite high. Since ohmmeters typically employ a relatively constant-current source, the resistance determined will be at a preset current level (typically, a few milliamperes).

In general, therefore, the higher the current through a diode, the lower is the dc resistance level.

Typically, the dc resistance of a diode in the active (most utilized) will range from about 10 Ω to 80 Ω.



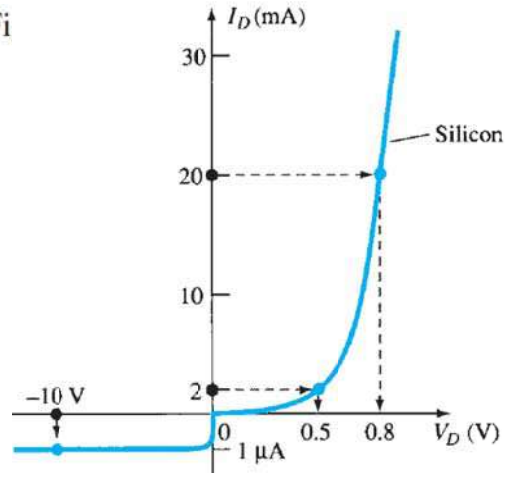


Q-point (operating point)
quiescent,

FIG. 1.23
Determining the dc resistance of a diode at a particular operating point.

EXAMPLE 1.3 Determine the dc resistance levels for the diode of Fi

a. $I_D = 2$ mA (low level)
b. $I_D = 20$ mA (high level)
c. $V_D = -10$ V (reverse-biased)



Solution:

a. At $I_D = 2$ mA, $V_D = 0.5$ V (from the curve) and

$$R_D = \frac{V_D}{I_D} = \frac{0.5\text{ V}}{2\text{ mA}} = \mathbf{250\ \Omega}$$

b. At $I_D = 20$ mA, $V_D = 0.8$ V (from the curve) and

$$R_D = \frac{V_D}{I_D} = \frac{0.8\text{ V}}{20\text{ mA}} = \mathbf{40\ \Omega}$$

c. At $V_D = -10$ V, $I_D = -I_s = -1\ \mu$A (from the curve) and

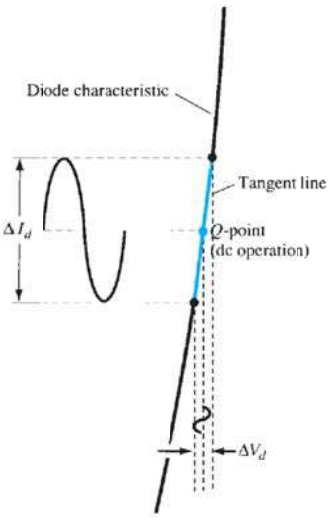
$$R_D = \frac{V_D}{I_D} = \frac{10\text{ V}}{1\ \mu\text{A}} = \mathbf{10\ M\Omega}$$

clearly supporting some of the earlier comments regarding the dc resistance levels of a diode.

AC or Dynamic Resistance

Eq. (1.4) and Example 1.3 reveal that

the dc resistance of a diode is independent of the shape of the characteristic in the region surrounding the point of interest.



AC or Dynamic Resistance

If a sinusoidal rather than a dc input is applied, the situation will change completely. The varying input will move the instantaneous operating point up and down a region of the characteristics and thus defines a specific change in current and voltage as shown in Fig. 1.25. With no applied varying signal, the point of operation would be the *Q*-point appearing on Fig. 1.25, determined by the applied dc levels. The designation *Q-point* is derived from the word *quiescent*, which means "still or unvarying."

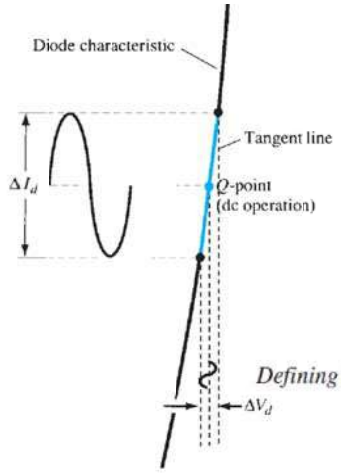


FIG. 1.25
Defining the dynamic or ac resistance.

A straight line drawn tangent to the curve through the *Q*-point as shown in Fig. 1.26 will define a particular change in voltage and current that can be used to determine the *ac* or *dynamic* resistance for this region of the diode characteristics. An effort should be made to keep the change in voltage and current as small as possible and equidistant to either side of the *Q*-point. In equation form,

$$r_d = \frac{\Delta V_d}{\Delta I_d} \tag{1.5}$$

where Δ signifies a finite change in the quantity.

The steeper the slope, the lower is the value of ΔV_d for the same change in ΔI_d and the lower is the resistance. The ac resistance in the vertical-rise region of the characteristic is therefore quite small, whereas the ac resistance is much higher at low current levels.

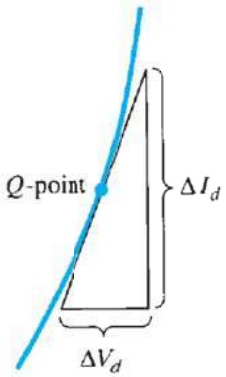
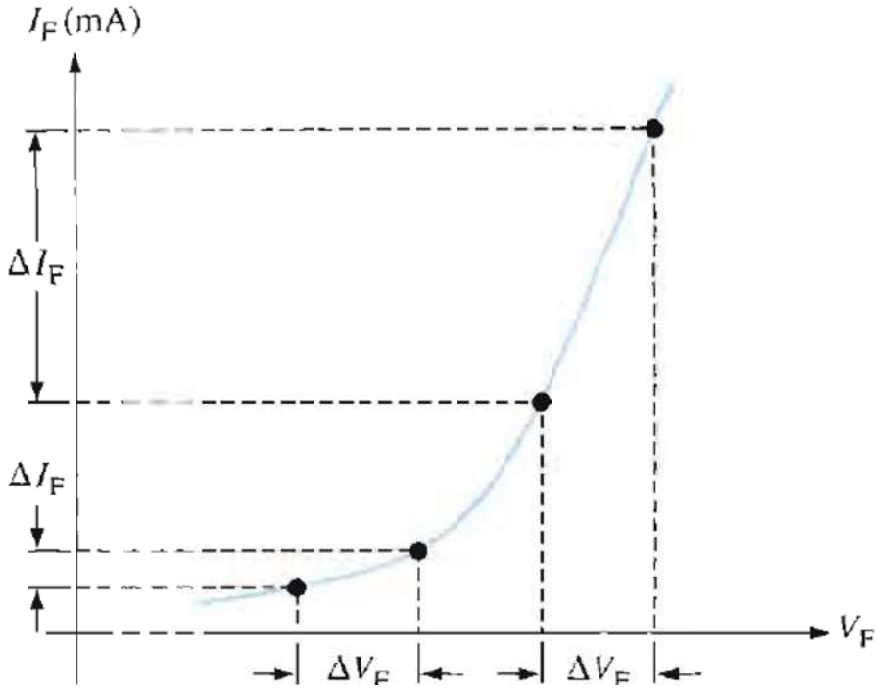


FIG. 1.26
Determining the ac resistance at a Q-point.

In general, therefore, the lower the Q-point of operation (smaller current or lower voltage), the higher is the ac resistance.



EXAMPLE 1.4 For the characteristics of Fig. 1.27:

a. Determine the ac resistance at $I_D = 2$ mA.
b. Determine the ac resistance at $I_D = 25$ mA.
c. Compare the results of parts (a) and (b) to the dc resistances at each current level.

Solution:

a. For $I_D = 2$ mA, the tangent line at $I_D = 2$ mA was drawn as shown in Fig. 1.27 and a swing of 2 mA above and below the specified diode current was chosen. At $I_D = 4$ mA, $V_D = 0.76$ V, and at $I_D = 0$ mA, $V_D = 0.65$ V. The resulting changes in current and voltage are, respectively,

$$\Delta I_d = 4\text{ mA} - 0\text{ mA} = 4\text{ mA}$$

and

$$\Delta V_d = 0.76\text{ V} - 0.65\text{ V} = 0.11\text{ V}$$

and the ac resistance is

$$r_d = \frac{\Delta V_d}{\Delta I_d} = \frac{0.11\text{ V}}{4\text{ mA}} = \mathbf{27.5\ \Omega}$$

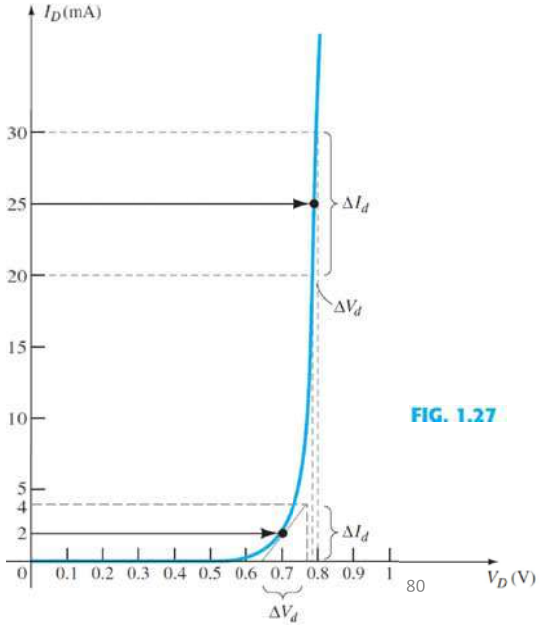


FIG. 1.27

b. For $I_D = 25$ mA, the tangent line at $I_D = 25$ mA was drawn as shown in Fig. 1.27 and a swing of 5 mA above and below the specified diode current was chosen. At $I_D = 30$ mA, $V_D = 0.8$ V, and at $I_D = 20$ mA, $V_D = 0.78$ V. The resulting changes in current and voltage are, respectively,

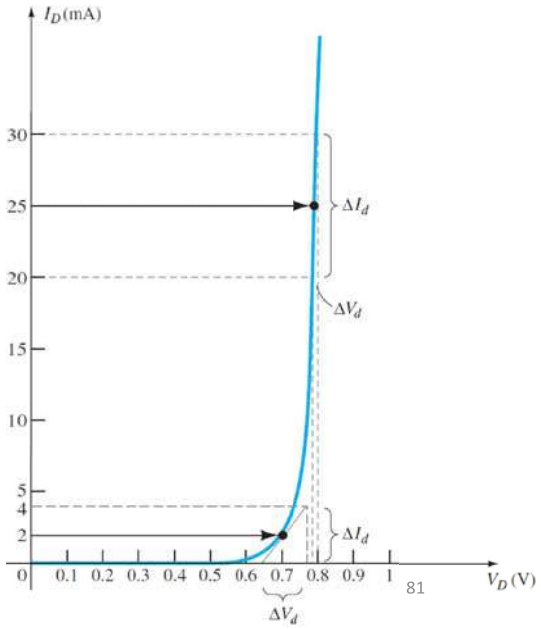
$$\Delta I_d = 30\text{ mA} - 20\text{ mA} = 10\text{ mA}$$

and

$$\Delta V_d = 0.8\text{ V} - 0.78\text{ V} = 0.02\text{ V}$$

and the ac resistance is

$$r_d = \frac{\Delta V_d}{\Delta I_d} = \frac{0.02\text{ V}}{10\text{ mA}} = \mathbf{2\ \Omega}$$



c. For $I_D = 2$ mA, $V_D = 0.7$ V and

$$R_D = \frac{V_D}{I_D} = \frac{0.7\text{ V}}{2\text{ mA}} = \mathbf{350\ \Omega}$$

which far exceeds the r_d of 27.5 Ω.

For $I_D = 25$ mA, $V_D = 0.79$ V and

$$R_D = \frac{V_D}{I_D} = \frac{0.79\text{ V}}{25\text{ mA}} = \mathbf{31.62\ \Omega}$$

which far exceeds the r_d of 2 Ω.

$$r_d = \frac{26\text{ mV}}{I_D} \qquad \textbf{(1.6)}$$

26 mV is the thermal voltage

The significance of Eq. (1.6) must be clearly understood. It implies that

the dynamic resistance can be found simply by substituting the quiescent value of the diode current into the equation.

There is no need to have the characteristics available or to worry about sketching tangent lines as defined by Eq. (1.5). It is important to keep in mind, however, that Eq. (1.6) is accurate only for values of I_D in the vertical-rise section of the curve. For lesser values of I_D, $n = 2$ (silicon) and the value of r_d obtained must be multiplied by a factor of 2. For small values of I_D below the knee of the curve, Eq. (1.6) becomes inappropriate.

All the resistance levels determined thus far have been defined by the p–n junction and do not include the resistance of the semiconductor material itself (called *body* resistance) and the resistance introduced by the connection between the semiconductor material and the external metallic conductor (called *contact* resistance). These additional resistance levels can be included in Eq. (1.6) by adding a resistance denoted r_B:

$$r'_d = \frac{26\text{ mV}}{I_D} + r_B \text{ ohms} \qquad \textbf{(1.7)}$$

The resistance r'_d, therefore, includes the dynamic resistance defined by Eq. (1.6) and the resistance r_B just introduced. The factor r_B can range from typically 0.1 Ω for high-power devices to 2 Ω for some low-power, general-purpose diodes. For Example 1.4 the ac resistance at 25 mA was calculated to be 2 Ω. Using Eq. (1.6), we have

$$r_d = \frac{26\text{ mV}}{I_D} = \frac{26\text{ mV}}{25\text{ mA}} = \mathbf{1.04\ \Omega}$$

The difference of about 1 Ω could be treated as the contribution of r_B.

For Example 1.4 the ac resistance at 2 mA was calculated to be 27.5 Ω. Using Eq. (1.6) but multiplying by a factor of 2 for this region (in the knee of the curve $n = 2$),

$$r_d = 2\left(\frac{26\text{ mV}}{I_D}\right) = 2\left(\frac{26\text{ mV}}{2\text{ mA}}\right) = 2(13\ \Omega) = \mathbf{26\ \Omega}$$

The difference of 1.5 Ω could be treated as the contribution due to r_B.

Average AC Resistance

If the input signal is sufficiently large to produce a broad swing such as indicated in Fig. 1.28, the resistance associated with the device for this region is called the *average ac resistance*. The average ac resistance is, by definition, the resistance determined by a straight line drawn between the two intersections established by the maximum and minimum values of input voltage. In equation form (note Fig. 1.28),

$$r_{av} = \left.\frac{\Delta V_d}{\Delta I_d}\right|_{\text{pt. to pt.}} \qquad \textbf{(1.8)}$$

For the situation indicated by Fig. 1.28,

$$\Delta I_d = 17\,\text{mA} - 2\,\text{mA} = 15\,\text{mA}$$

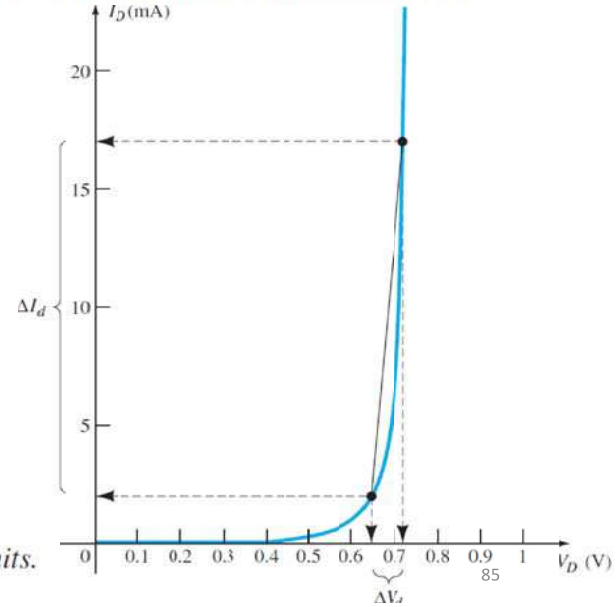


FIG. 1.28
Determining the average ac resistance between indicated limits.

and
$$\Delta V_d = 0.725\,\text{V} - 0.65\,\text{V} = 0.075\,\text{V}$$

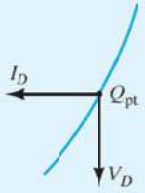
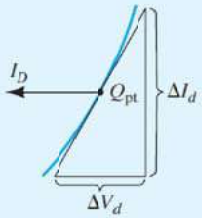
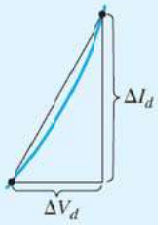
with
$$r_{av} = \frac{\Delta V_d}{\Delta I_d} = \frac{0.075\,\text{V}}{15\,\text{mA}} = \mathbf{5\ \Omega}$$

If the ac resistance (r_d) were determined at $I_D = 2$ mA, its value would be more than 5 Ω, and if determined at 17 mA, it would be less. In between, the ac resistance would make the transition from the high value at 2 mA to the lower value at 17 mA. Equation (1.7) defines a value that is considered the average of the ac values from 2 mA to 17 mA. The fact that one resistance level can be used for such a wide range of the characteristics will prove quite useful in the definition of equivalent circuits for a diode in a later section.

As with the dc and ac resistance levels, the lower the level of currents used to determine the average resistance, the higher is the resistance level.

$$r'_d = \frac{26\,\text{mV}}{I_D} + r_B \ \text{ohms} \qquad \textbf{(1.7)}$$

TABLE 1.6
Resistance Levels

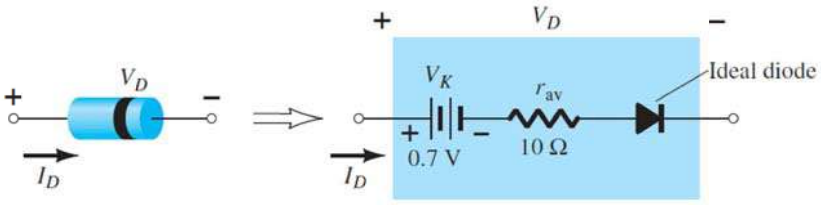
Type	Equation	Special Characteristics	Graphical Determination
DC or static	$R_D = \frac{V_D}{I_D}$	Defined as a point on the characteristics	
AC or dynamic	$r_d = \frac{\Delta V_d}{\Delta I_d} = \frac{26\text{ mV}}{I_D}$	Defined by a tangent line at the Q-point	
Average ac	$r_{av} = \frac{\Delta V_d}{\Delta I_d}\Big\|_{\text{pt. to pt.}}$	Defined by a straight line between limits of operation	

1.9 DIODE EQUIVALENT CIRCUITS

An equivalent circuit is a combination of elements properly chosen to best represent the actual terminal characteristics of a device or system in a particular operating region.

In other words, once the equivalent circuit is defined, the device symbol can be removed from a schematic and the equivalent circuit inserted in its place without severely affecting the actual behavior of the system. The result is often a network that can be solved using traditional circuit analysis techniques.

Piecewise-Linear Equivalent Circuit



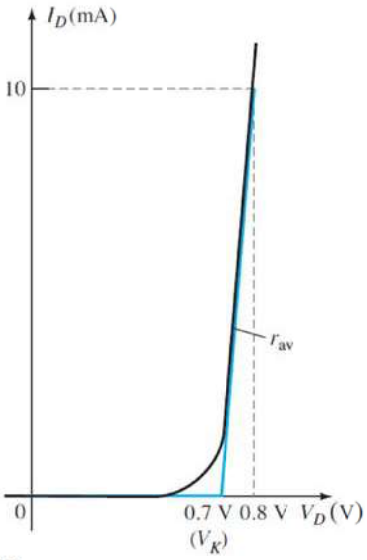


FIG. 1.30

Components of the piecewise-linear equivalent circuit.

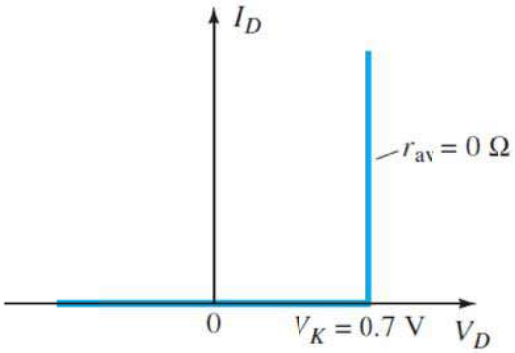
$V_D = 0.8$ V, we know that for silicon a shift of 0.7 V is required before the characteristics rise, and we obtain

$$r_{av} = \left.\frac{\Delta V_d}{\Delta I_d}\right|_{\text{pt. to pt.}} = \frac{0.8\text{ V} - 0.7\text{ V}}{10\text{ mA} - 0\text{ mA}} = \frac{0.1\text{ V}}{10\text{ mA}} = \mathbf{10\ \Omega}$$

as obtained for Fig. 1.29.

If the characteristics or specification sheet for a diode is not available the resistance r_{av} can be approximated by the ac resistance r_d.

Simplified Equivalent Circuit



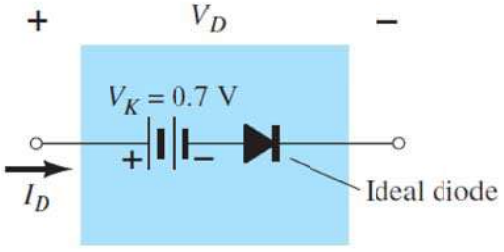


FIG. 1.31

Simplified equivalent circuit for the silicon semiconductor diode.

Ideal Equivalent Circuit

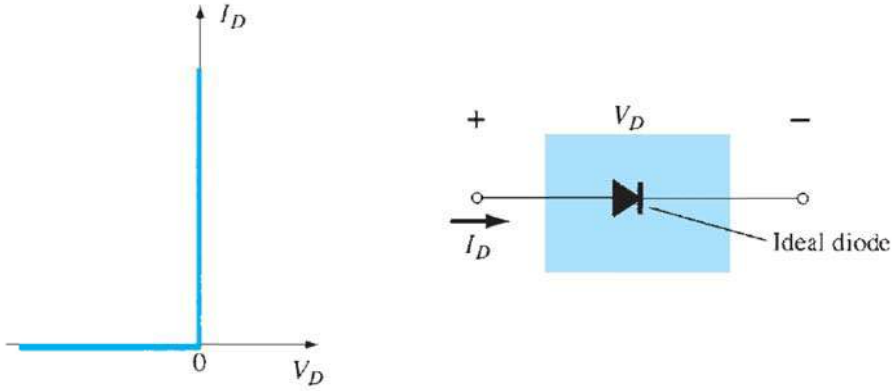
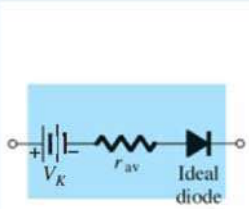
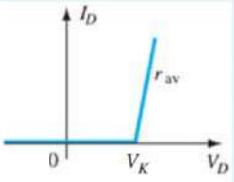
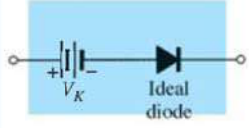
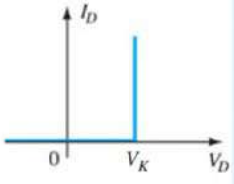
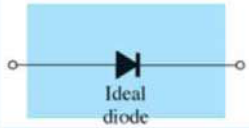
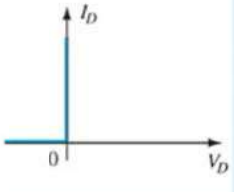


FIG. 1.32
Ideal diode and its characteristics.

In industry a popular substitution for the phrase "diode equivalent circuit" is diode *model*—a model by definition being a representation of an existing device, object, system, and so on. In fact, this substitute terminology will be used almost exclusively in the chapters to follow.

TABLE 1.7
Diode Equivalent Circuits (Models)

Type	Conditions	Model	Characteristics
Piecewise-linear model			
Simplified model	$R_{network} \gg r_{av}$		
Ideal device	$R_{network} \gg r_{av}$ $E_{network} \gg V_K$		

1.11 REVERSE RECOVERY TIME

There are certain pieces of data that are normally provided on diode specification sheets provided by manufacturers. One such quantity that has not been considered yet is the reverse recovery time, denoted by t_{rr}. In the forward-bias state it was shown earlier that there are a large number of electrons from the n-type material progressing through the p-type material and a large number of holes in the n-type material—a requirement for conduction. The electrons in the p-type material and holes progressing through the n-type material establish a large number of minority carriers in each material. If the applied voltage should be reversed to establish a reverse-bias situation, we would ideally like to see the diode change instantaneously from the conduction state to the nonconduction state.

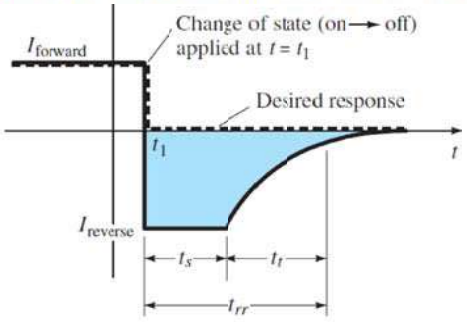


FIG. 1.35
Defining the reverse recovery time.

93

However, because of the large number of minority carriers in each material, the diode current will simply reverse as shown in Fig. 1.35 and stay at this measurable level for the period of time t_s (storage time) required for the minority carriers to return to their majority-carrier state in the opposite material. In essence, the diode will remain in the short-circuit state with a current $I_{reverse}$ determined by the network parameters. Eventually, when this storage phase has passed, the current will be reduced in level to that associated with the nonconduction state. This second period of time is denoted by t_t (transition interval). The reverse recovery time is the sum of these two intervals: $t_{rr} = t_s + t_t$. This is an important consideration in

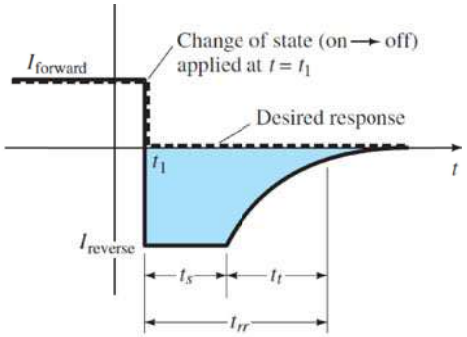


FIG. 1.35
Defining the reverse recovery time.

94

high-speed switching applications. Most commercially available switching diodes have a t_{rr} in the range of a few nanoseconds to 1 μs. Units are available, however, with a t_{rr} of only a few hundred picoseconds (10^{-12} s).

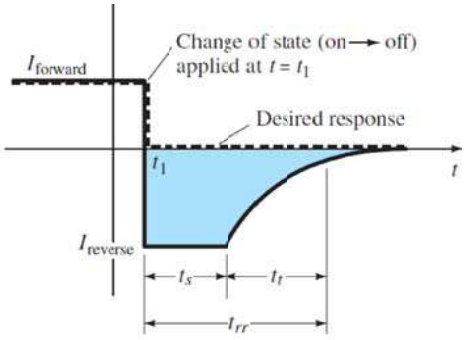


FIG. 1.35
Defining the reverse recovery time.

95

1.12 DIODE SPECIFICATION SHEETS

Data on specific semiconductor devices are normally provided by the manufacturer in one of two forms. Most frequently, they give a very brief description limited to perhaps one page. At other times, they give a thorough examination of the characteristics using graphs, artwork, tables, and so on. In either case, there are specific pieces of data that must be included for proper use of the device. They include:

1. The forward voltage V_F (at a specified current and temperature)
2. The maximum forward current I_F (at a specified temperature)
3. The reverse saturation current I_R (at a specified voltage and temperature)
4. The reverse-voltage rating [PIV or PRV or V(BR), where BR comes from the term "breakdown" (at a specified temperature)]
5. The maximum power dissipation level at a particular temperature
6. Capacitance levels
7. Reverse recovery time t_{rr}
8. Operating temperature range

96

Depending on the type of diode being considered, additional data may also be provided, such as frequency range, noise level, switching time, thermal resistance levels, and peak repetitive values. For the application in mind, the significance of the data will usually be self-apparent. If the maximum power or dissipation rating is also provided, it is understood to be equal to the following product:

$$P_{D\max} = V_D I_D \tag{1.11}$$

where I_D and V_D are the diode current and voltage, respectively, at a particular point of operation.

If we apply the simplified model for a particular application (a common occurrence), we can substitute $V_D = V_T = 0.7$ V for a silicon diode in Eq. (1.11) and determine the resulting power dissipation for comparison against the maximum power rating. That is,

$$P_{\text{dissipated}} \cong (0.7\text{ V})I_D \tag{1.12}$$

The data provided for a high-voltage/low-leakage diode appear in Figs. 1.36 and 1.37. This example would represent the expanded list of data and characteristics. The term *rectifier* is applied to a diode when it is frequently used in a *rectification* process, described in Chapter 2.

Specific areas of the specification sheet are highlighted in blue, with letters corresponding to the following description:

A The data sheet highlights the fact that the silicon high-voltage diode has a *minimum* reverse-bias voltage of 125 V at a specified reverse-bias current.

B Note the wide range of temperature operation. Always be aware that data sheets typically use the centigrade scale, with 200°C = 392°F and −65°C = −85°F.

C The maximum power dissipation level is given by $P_D = V_D I_D = 500\ \text{mW} = 0.5\ \text{W}$. The effect of the linear derating factor of 3.33 mW/°C is demonstrated in Fig. 1.37a. Once the temperature exceeds 25°C the maximum power rating will drop by 3.33 mW for each 1°C increase in temperature. At a temperature of 100°C, which is the boiling point of water, the maximum power rating has dropped to one half of its original value. An initial temperature of 25°C is typical inside a cabinet containing operating electronic equipment in a low-power situation.

D The maximum sustainable current is 500 mA. The plot of Fig. 1.37b reveals that the forward current at 0.5 V is about 0.01 mA, but jumps to 1 mA (100 times greater) at about 0.65 V. At 0.8 V the current is more than 10 mA, and just above 0.9 V it is close

DIFFUSED SILICON PLANAR

A — • BV . . . 125 V (MIN) @ 100 μA (BAY73)

ABSOLUTE MAXIMUM RATINGS (Note 1)

Temperatures

B — Storage Temperature Range	-65°C to +200°C
B — Maximum Junction Operating Temperature	+175°C
B — Lead Temperature	+260°C

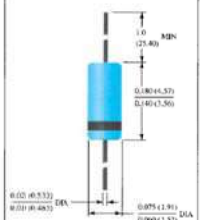
Power Dissipation (Note 2)

C — Maximum Total Power Dissipation at 25°C Ambient	500 mW
C — Linear Power Derating Factor (from 25°C)	3.33 mW/°C

Maximum Voltage and Currents

WIV	Working Inverse Voltage	BAY73	100 V
I_O	Average Rectified Current		200 mA
D — I_F	Continuous Forward Current		500 mA
i_f	Peak Repetitive Forward Current		600 mA
$i_{f(surge)}$	Peak Forward Surge Current		
	Pulse Width = 1 s		1.0 A
	Pulse Width = 1 μs		4.0 A

DO-35 OUTLINE



NOTES:
Copperclad steel rods, tin plated
Gold plated leads available
Hermetically sealed glass package
Package weight 0.14 gram

ELECTRICAL CHARACTERISTICS (25°C Ambient Temperature unless otherwise noted)

SYMBOL	CHARACTERISTIC	BAY73 MIN	BAY73 MAX	UNITS	TEST CONDITIONS
E — V_F	Forward Voltage	0.85	1.00	V	I_F = 200 mA
		0.81	0.94	V	I_F = 100 mA
		0.78	0.88	V	I_F = 50 mA
		0.69	0.80	V	I_F = 10 mA
		0.67	0.75	V	I_F = 5.0 mA
		0.60	0.68	V	I_F = 1.0 mA
F — I_R	Reverse Current		500	nA	V_R = 20 V, T_A = 125°C
			1.0	μA	V_R = 100 V, T_A = 125°C
			0.2	nA	V_R = 20 V, T_A = 25°C
			0.5	nA	V_R = 100 V, T_A = 25°C
BV	Breakdown Voltage	125		V	I_R = 100 μA
G — C	Capacitance		5.0	pF	V_R = 0, f = 1.0 MHz
H — t_{rr}	Reverse Recovery Time		3.0	μs	I_F = 10 mA, V_t = 35 V, R_L = 1.0 to 100 kΩ, C_L = 10 pF, JAN 256

NOTES
1. These ratings are limiting values above which the serviceability of the diode may be impaired.
2. These are steady state limits. The factory should be consulted on applications involving pulses or low duty-cycle operation.

FIG. 1.36
Electrical characteristics of a high-voltage, low-leakage diode

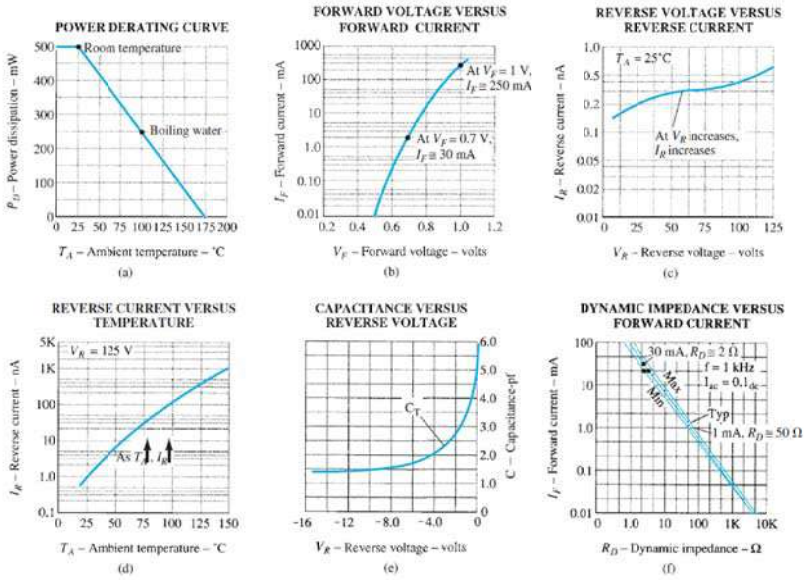


FIG. 1.37
Terminal characteristics of a high-voltage diode.

See pages 32-35

1.13 SEMICONDUCTOR DIODE NOTATION

The notation most frequently used for semiconductor diodes is provided in Fig. 1.38. For most diodes any marking such as a dot or band, as shown in Fig. 1.38, appears at the cathode end. The terminology anode and cathode is a carryover from vacuum-tube notation. The anode refers to the higher or positive potential, and the cathode refers to the lower or negative terminal. This combination of bias levels will result in a forward-bias or "on" condition for the diode. A number of commercially available semiconductor diodes appear in Fig. 1.39.

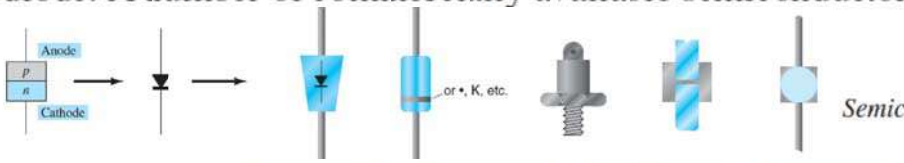


FIG. 1.38
Semiconductor diode notation.

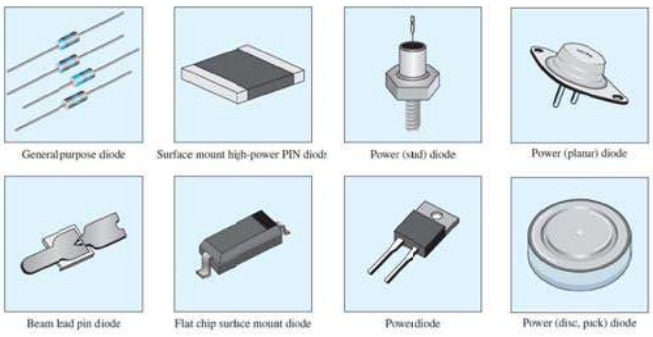


FIG. 1.39
Various types of junction diodes.

1.14 DIODE TESTING

The condition of a semiconductor diode can be determined quickly using (1) a digital display meter (DDM) with a *diode checking function*, (2) the *ohmmeter section* of a multimeter, or (3) a *curve tracer*.

Diode Checking Function

A digital display meter with a diode checking capability appears in Fig. 1.40. Note the small diode symbol at the top right of the rotating dial. When set in this position and hooked up as shown in Fig. 1.41a, the diode should be in the "on" state and the display will provide an indication of the forward-bias voltage such as 0.67 V (for Si). The meter has an internal constant-current source (about 2 mA) that will define the voltage level as indicated in Fig. 1.41b. An OL indication with the hookup of Fig. 1.41a reveals an open (defective) diode. If the leads are reversed, an OL indication should result due to the expected open-circuit equivalence for the diode. In general, therefore, an OL indication in both directions is an indication of an open or defective diode.

FIG. 1.40
Digital display meter. (Courtesy of

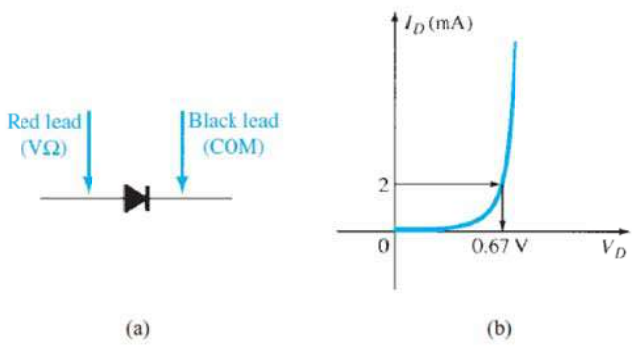


FIG. 1.41
Checking a diode in the forward-bias state.

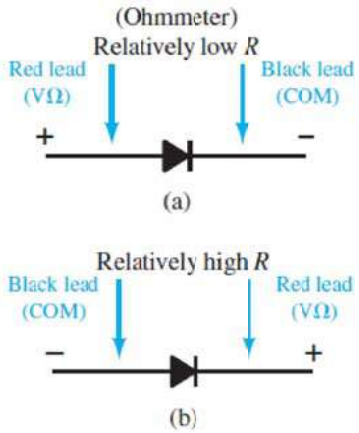


FIG. 1.42
Checking a diode with an ohmmeter.

1.15 ZENER DIODES

For the Zener diode the direction of conduction is opposite to that of the arrow in the symbol, as pointed out in the introduction to this section. Note also that the polarity of V_D and V_Z are the same as would be obtained if each were a resistive element as shown in Fig. 1.46c.

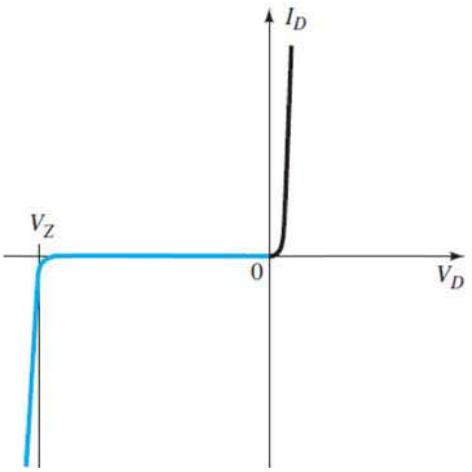


FIG. 1.45
Reviewing the Zener region.

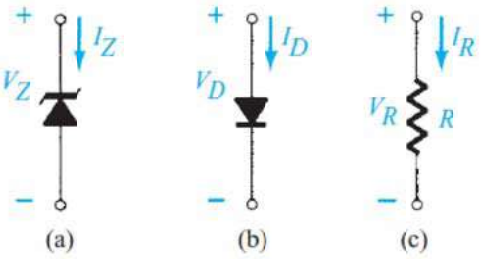
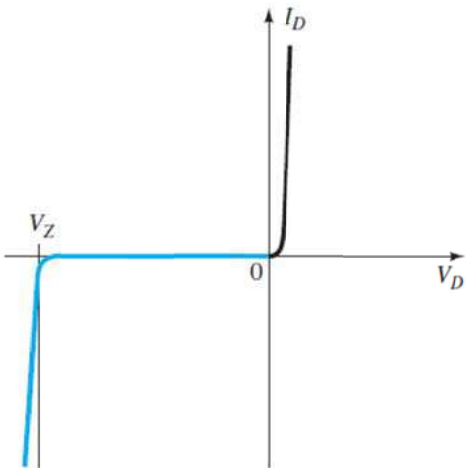


FIG. 1.46
Conduction direction: (a) Zener diode; (b) semiconductor diode; (c) resistive element.

The location of the Zener region can be controlled by varying the doping levels. An increase in doping that produces an increase in the number of added impurities, will decrease the Zener potential. Zener diodes are available having Zener potentials of 1.8 V to 200 V with power ratings from ¼ W to 50 W. Because of its excellent temperature and current capabilities, silicon is the preferred material in the manufacture of Zener diodes.



It would be nice to assume the Zener diode is ideal with a straight vertical line at the Zener potential. However, there is a slight slope to the characteristics requiring the piecewise equivalent model appearing in Fig. 1.47 for that region. For most of the applications appearing in this text the series resistive element can be ignored and the reduced equivalent model of just a dc battery of V_Z volts employed.

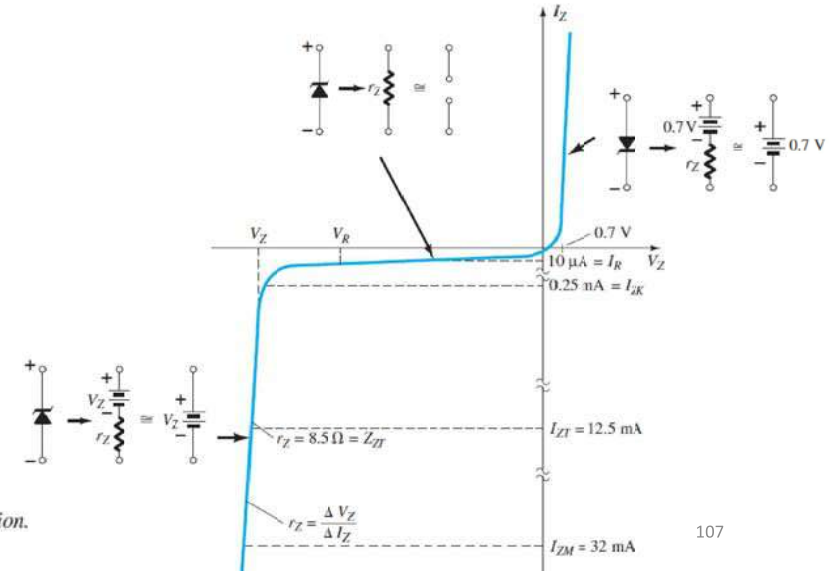


FIG. 1.47
Zener diode characteristics with the equivalent model for each region.

Since some applications of Zener diodes swing between the Zener region and the forward-bias region, it is important to understand the operation of the Zener diode in all regions. As shown in Fig. 1.47, the equivalent model for a Zener diode in the reverse-bias region below V_Z is a very large resistor (as for the standard diode). For most applications this resistance is so large it can be ignored and the open-circuit equivalent employed. For the forward-bias region the piecewise equivalent is the same as described in earlier sections

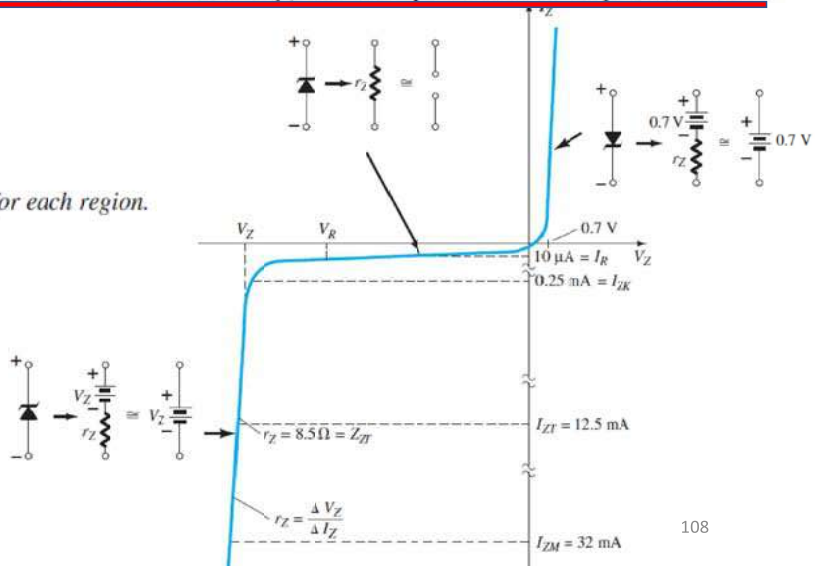


FIG. 1.47
Zener diode characteristics with the equivalent model for each region.

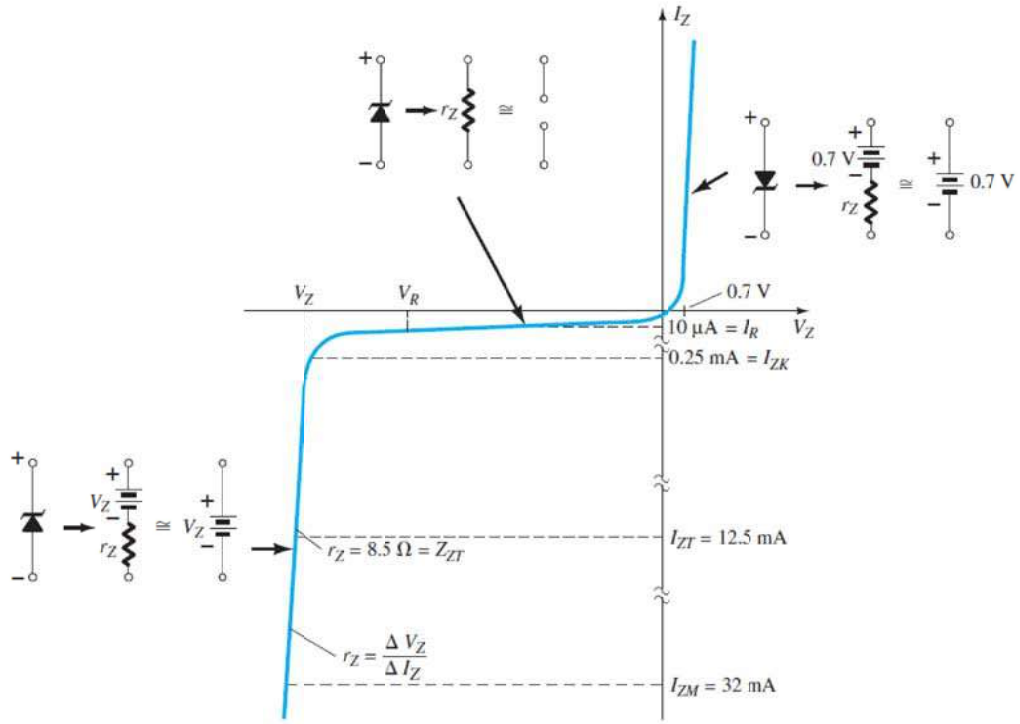


FIG. 1.47
Zener diode characteristics with the equivalent model for each region.

The terminal identification and the casing for a variety of Zener diodes appear in Fig. 1.49. Their appearance is similar in many ways to that of the standard diode. Some areas of application for the Zener diode will be examined in Chapter 2.

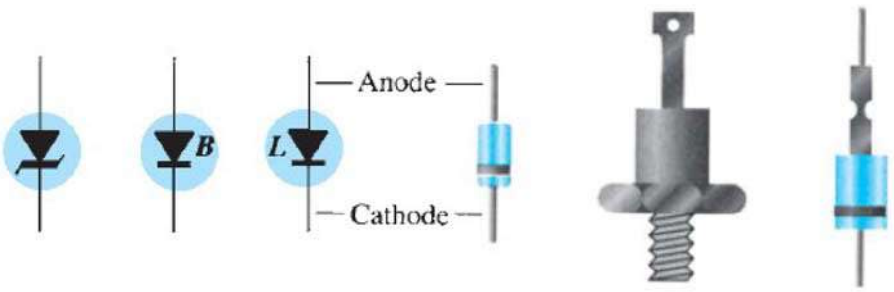


FIG. 1.49
Zener terminal identification and symbols.

1.16 LIGHT-EMITTING DIODES

The increasing use of digital displays in calculators, watches, and all forms of instrumentation has contributed to an extensive interest in structures that emit light when properly biased. The two types in common use to perform this function are the light-emitting diode (LED) and the liquid-crystal display (LCD). Since the LED falls within the family of *p–n* junction devices and will appear in some of the networks of the next few chapters, it will be introduced in this chapter.

1.16 LIGHT-EMITTING DIODES

As the name implies, the light-emitting diode is a diode that gives off visible or invisible (infrared) light when energized. In any forward-biased *p–n* junction there is, within the structure and primarily close to the junction, a recombination of holes and electrons. This recombination requires that the energy possessed by the unbound free electrons be transferred to another state. In all semiconductor *p–n* junctions some of this energy is given off in the form of heat and some in the form of photons.

In Si and Ge diodes the greater percentage of the energy converted during recombination at the junction is dissipated in the form of heat within the structure, and the emitted light is insignificant.

For this reason, silicon and germanium are not used in the construction of LED devices. On the other hand:

Diodes constructed of GaAs emit light in the infrared (invisible) zone during the recombination process at the p–n junction.

Even though the light is not visible, infrared LEDs have numerous applications where visible light is not a desirable effect. These include security systems, industrial processing, optical coupling, safety controls such as on garage door openers, and in home entertainment centers, where the infrared light of the remote control is the controlling element.

Through other combinations of elements a coherent visible light can be generated. Table 1.9 provides a list of common compound semiconductors and the light they generate. In addition, the typical range of forward bias potentials for each is listed.

TABLE 1.9
Light-Emitting Diodes

Color	Construction	Typical Forward Voltage (V)
Amber	AlInGaP	2.1
Blue	GaN	5.0
Green	GaP	2.2
Orange	GaAsP	2.0
Red	GaAsP	1.8
White	GaN	4.1
Yellow	AlInGaP	2.1

The basic construction of an LED appears in Fig. 1.50 with the standard symbol used for the device. The external metallic conducting surface connected to the *p*-type material is smaller to permit the emergence of the maximum number of photons of light energy when the device is forward-biased. Note in the figure that the recombination of the injected carriers due to the forward-biased junction results in emitted light at the site of the recombination.

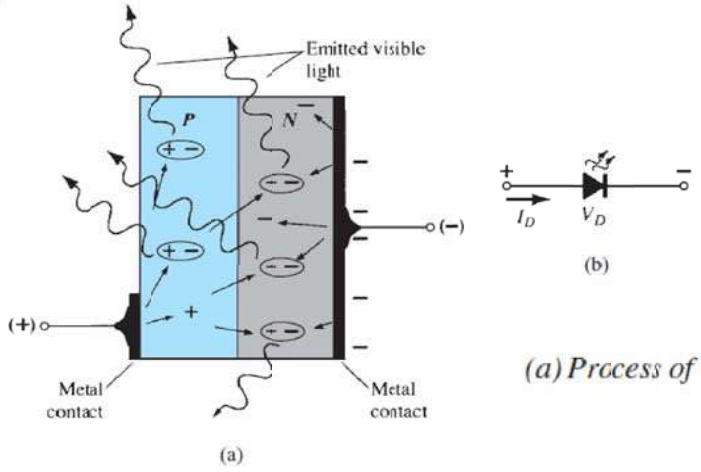


FIG. 1.50
(a) Process of electroluminescence in the LED; (b) graphic symbol.

There will, of course, be some absorption of the packages of photon energy in the structure itself, but a very large percentage can leave, as shown in the figure.

Just as different sounds have different frequency spectra (high-pitched sounds generally have high-frequency components, and low sounds have a variety of low-frequency components), the same is true for different light emissions.

The frequency spectrum for infrared light extends from about 100 THz (T = tera = 10^{12}) to 400 THz, with the visible light spectrum extending from about 400 to 750 THz.

It is interesting to note that invisible light has a lower frequency spectrum than visible light.

In general, when one talks about the response of electroluminescent devices, one references their wavelength rather than their frequency.

The two quantities are related by the following equation:

$$\lambda = \frac{c}{f} \quad \text{(m)} \tag{1.15}$$

where $c = 3 \times 10^8$ m/s (the speed of light in a vacuum)
f = frequency in Hertz
λ = wavelength in meters.

EXAMPLE 1.6 Using Eq. (1.15), find the range of wavelength for the frequency range of visible light (400 THz–750 THz).

Solution:

$$c = 3 \times 10^8 \frac{\cancel{m}}{s}\left[\frac{10^9 \text{ nm}}{\cancel{m}}\right] = 3 \times 10^{17} \text{ nm/s}$$

$$\lambda = \frac{c}{f} = \frac{3 \times 10^{17} \text{ nm/s}}{400 \text{ THz}} = \frac{3 \times 10^{17} \text{ nm/s}}{400 \times 10^{12} \text{ Hz}} = 750 \text{ nm}$$

$$\lambda = \frac{c}{f} = \frac{3 \times 10^{17} \text{ nm/s}}{750 \text{ THz}} = \frac{3 \times 10^{17} \text{ nm/s}}{750 \times 10^{12} \text{ Hz}} = 400 \text{ nm}$$

400 nm to 750 nm

Note in the above example the resulting inversion from higher frequency to smaller wavelength. That is, the higher frequency results in the smaller wavelength. Also, most charts use either nanometers (nm) or angstrom (Å) units. One angstrom unit is equal to 10^{-10} m.

The response of the average human eye as provided in Fig. 1.51 extends from about 350 nm to 800 nm with a peak near 550 nm.

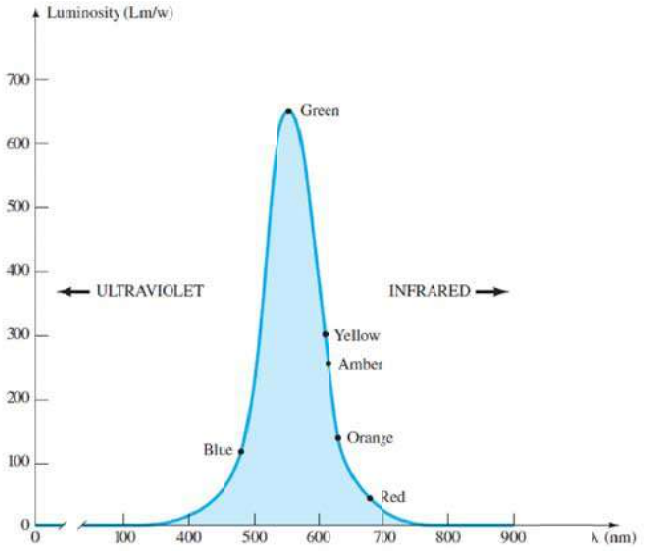


FIG. 1.51
Standard response curve of the human eye, showing the eye's response to light energy peaks at green and falls off for blue and red.

It is interesting to note that the peak response of the eye is to the color green, with red and blue at the lower ends of the bell curve. The curve reveals that a red or a blue LED must have a much stronger efficiency than a green one to be visible at the same intensity. In other words, the eye is more sensitive to the color green than to other colors. Keep in mind that the wavelengths shown are for the peak response of each color. All the colors indicated on the plot will have a bell-shaped curve response, so green, for example, is still visible at 600 nm, but at a lower intensity level.

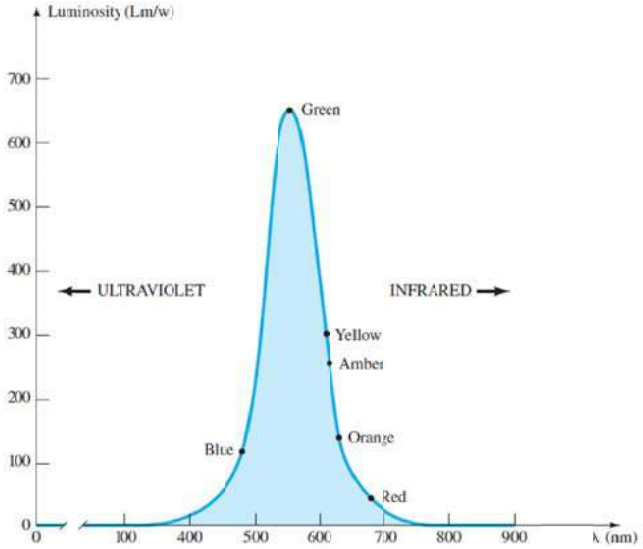


FIG. 1.51
Standard response curve of the human eye, showing the eye's response to light energy peaks at green and falls off for blue and red.

In Section 1.4 it was mentioned briefly that GaAs with its higher energy gap of 1.43 eV made it suitable for electromagnetic radiation of visible light, whereas Si at 1.1 eV resulted primarily in heat dissipation on recombination. The effect of this difference in energy gaps can be explained to some degree by realizing that to move an electron from one discrete energy level to another requires a specific amount of energy. The amount of energy involved is given by

$$E_g = \frac{hc}{\lambda} \qquad \textbf{(1.16)}$$

with E_g = joules (J) [1 eV = 1.6×10^{-19} J]
h = Planck's constant = 6.626×10^{-34} J · s.
$c = 3 \times 10^8$ m/s
λ = wavelength in meters

If we substitute the energy gap level of 1.43 eV for GaAs into the equation, we obtain the following wavelength:

$$1.43 \text{ eV}\left[\frac{1.6 \times 10^{-19} \text{ J}}{1 \text{ eV}}\right] = 2.288 \times 10^{-19} \text{ J}$$

and

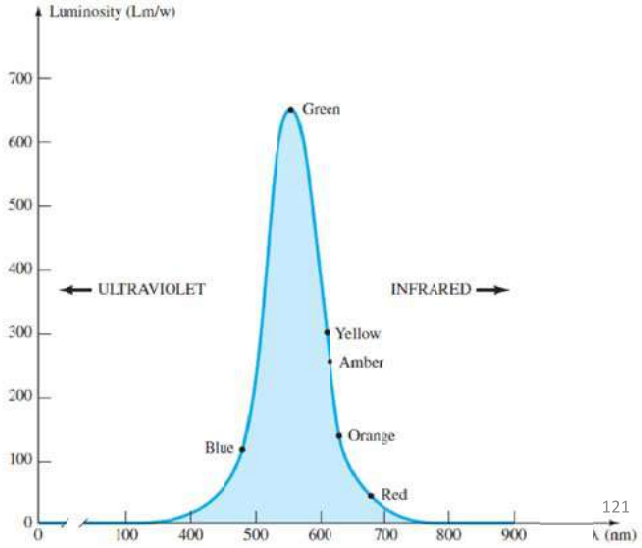
$$\lambda = \frac{hc}{E_g} = \frac{(6.626 \times 10^{-34} \text{ J} \cdot \text{s})(3 \times 10^{8} \text{ m/s})}{2.288 \times 10^{-19} \text{ J}}$$

$$= 869 \textbf{ nm}$$

For silicon, with $E_g = 1.1$ eV

$$\lambda = \textbf{1130 nm}$$

which is well beyond the visible range of Fig. 1.51.



The wavelength of 869 nm places GaAs in the wavelength zone typically used in infrared devices. For a compound material such as GaAsP with a band gap of 1.9 eV the resulting wavelength is 654 nm, which is in the center of the red zone, making it an excellent compound semiconductor for LED production. In general, therefore:

The wavelength and frequency of light of a specific color are directly related to the energy band gap of the material.

A first step, therefore, in the production of a compound semiconductor that can be used to generate light is to come up with a combination of elements that will generate the desired energy band gap.

The light intensity of an LED will increase with forward current until a point of saturation arrives where any further increase in current will not effectively increase the level of illumination.

One of the major concerns when using an LED is the reverse-bias breakdown voltage, which is typically between 3 V and 5 V (an occasional device has a 10-V level).

For many years the only colors available were green, yellow, orange, and red, permitting the use of the average values of $V_F = 2$ V and $I_F = 20$ mA for obtaining an approximate operating level.

However, with the introduction of blue in the early 1990s and white in the late 1990s the magnitude of these two parameters has changed. For blue the average forward bias voltage can be as high as 5 V, and for white about 4.1 V, although both have a typical operating current of 20 mA or more. In general, therefore:

Assume an average forward-bias voltage of 5 V for blue and 4 V for white LEDs at currents of 20 mA to initiate an analysis of networks with these types of LEDs.

FIG. 1.53
LED residential and commercial lighting.

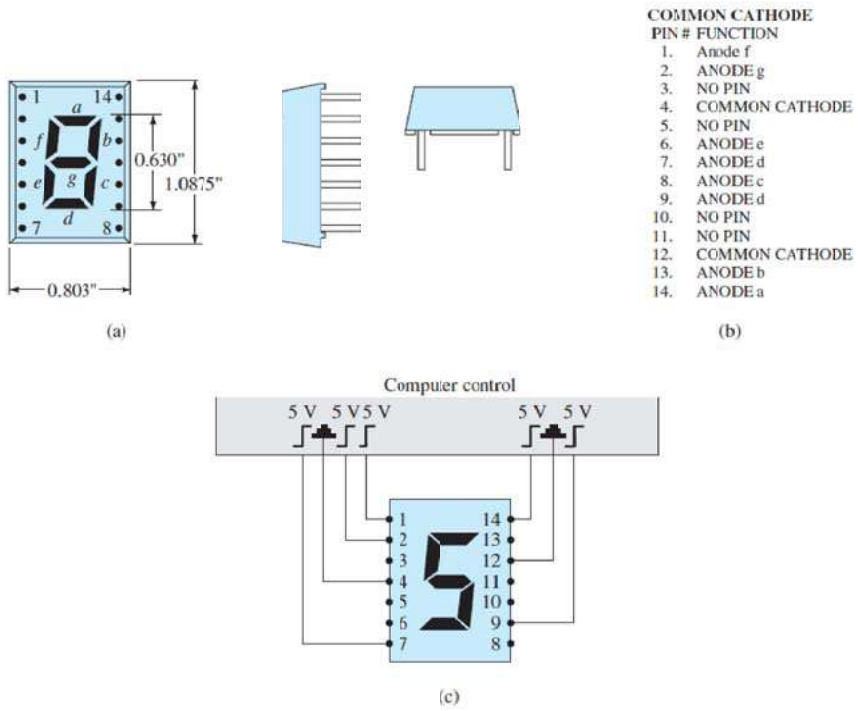


FIG. 1.54
Seven-segment display: (a) face with pin identification; (b) pin function; (c) displaying the numeral 5.

125

Equations

$$I_D = I_s(e^{V_D/nV_T} - 1) \qquad V_T = \frac{kT}{q} \qquad T_K = T_C + 273° \qquad k = 1.38 \times 10^{-23}\ \text{J/K}$$

$$V_K \cong 0.7\ \text{V (Si)}$$

$$V_K \cong 1.2\ \text{V (GaAs)}$$

$$V_K \cong 0.3\ \text{V (Ge)}$$

$$R_D = \frac{V_D}{I_D}$$

$$r_d = \frac{\Delta V_d}{\Delta I_d} = \frac{26\ \text{mV}}{I_D}$$

$$r_{av} = \left.\frac{\Delta V_d}{\Delta I_d}\right|_{\text{pt. to pt.}}$$

$$P_{D_{max}} = V_D I_D$$

130

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Chapter 2 Diode Applications
Lecture 1

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Chapter 2

Diode Applications

CHAPTER OBJECTIVES

- Understand the concept of load-line analysis and how it is applied to diode networks.
- Become familiar with the use of equivalent circuits to analyze series, parallel, and series-parallel diode networks.
- Understand the process of rectification to establish a dc level from a sinusoidal ac input.
- Be able to predict the output response of a clipper and clamper diode configuration.
- Become familiar with the analysis of and the range of applications for Zener diodes.

2.2 LOAD-LINE ANALYSIS

The circuit of Fig. 2.1 is the simplest of diode configurations. It will be used to describe the analysis of a diode circuit using its actual characteristics. In the next section we will replace the characteristics by an approximate model for the diode and compare solutions. Solving the circuit of Fig. 2.1 is all about finding the current and voltage levels that will satisfy both the characteristics of the diode and the chosen network parameters at the same time.

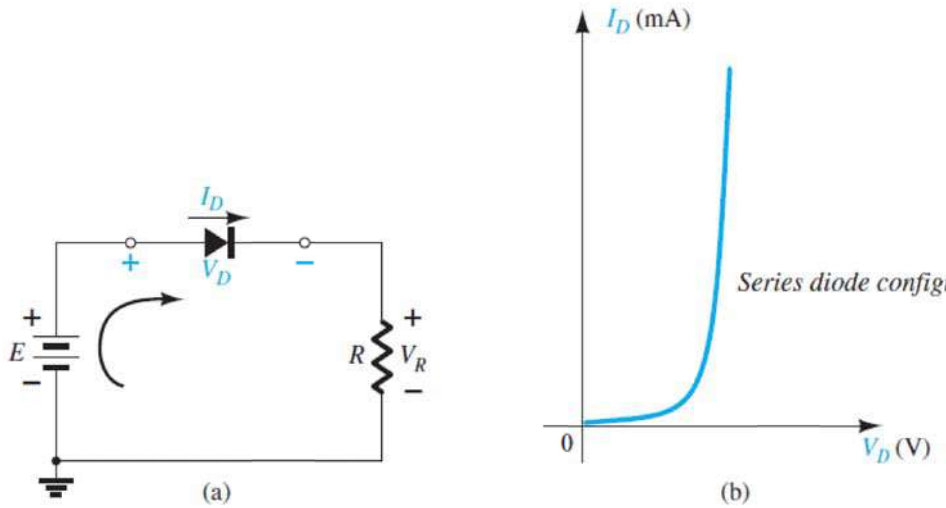


FIG. 2.1
Series diode configuration: (a) circuit; (b) characteristics.

In Fig. 2.2 the diode characteristics are placed on the same set of axes as a straight line defined by the parameters of the network. The straight line is called a *load line* because the intersection on the vertical axis is defined by the applied load R. The analysis to follow is therefore called *load-line analysis.* The intersection of the two curves will define the solution for the network and define the current and voltage levels for the network.

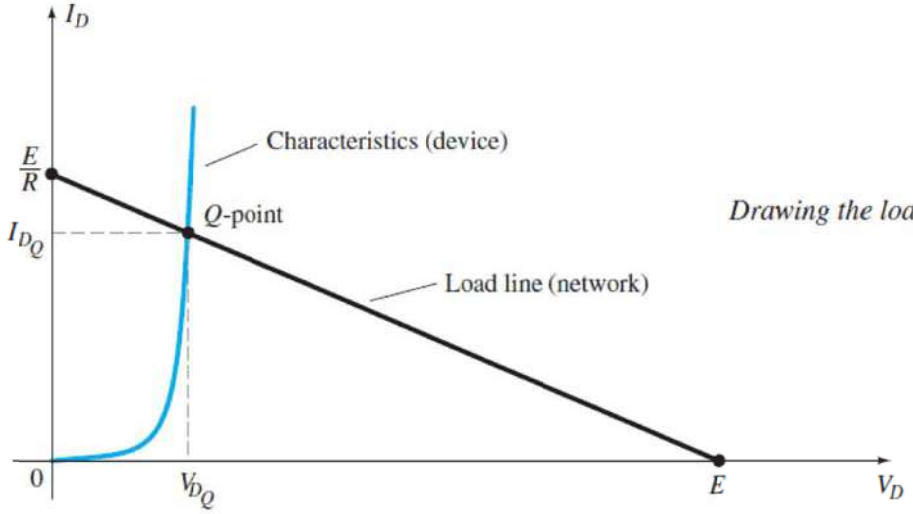


FIG. 2.2
Drawing the load line and finding the point of operation.

Before reviewing the details of drawing the load line on the characteristics, we need to determine the expected response of the simple circuit of Fig. 2.1. Note in Fig. 2.1 that the effect of the "pressure" established by the dc supply is to establish a conventional current in the direction indicated by the clockwise arrow. The fact that the direction of this current has the same direction as the arrow in the diode symbol reveals that the diode is in the "on" state and will conduct a high level of current. The polarity of the applied voltage has resulted in a forward-bias situation.

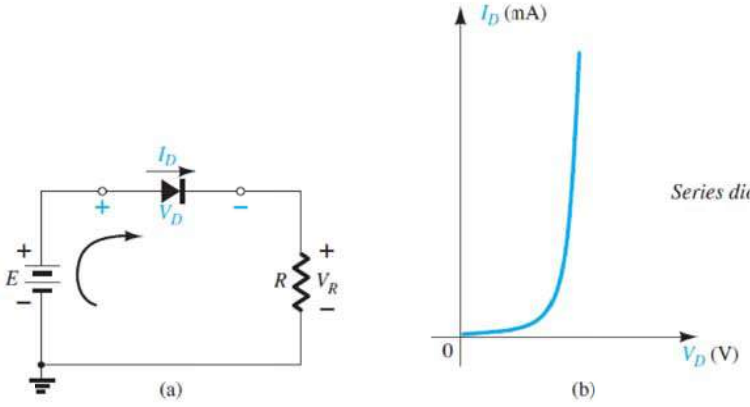


FIG. 2.1
Series diode configuration: (a) circuit; (b) characteristics.

With the current direction established, the polarities for the voltage across the diode and resistor can be superimposed. The polarity of V_D and the direction of I_D clearly reveal that the diode is indeed in the forward-bias state, resulting in a voltage across the diode in the neighborhood of 0.7 V and a current on the order of 10 mA or more.

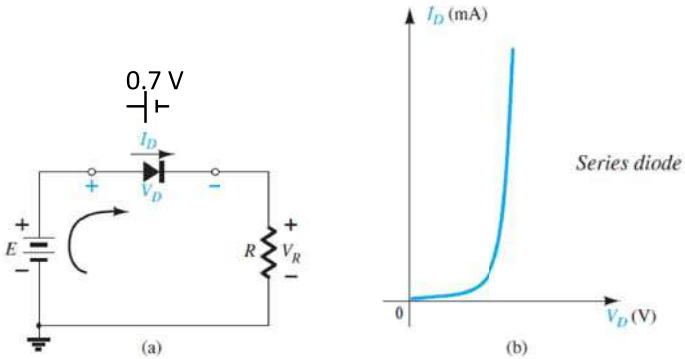


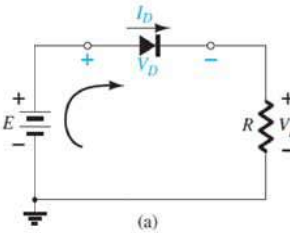
FIG. 2.1
Series diode configuration: (a) circuit; (b) characteristics.

The intersections of the load line on the characteristics of Fig. 2.2 can be determined by first applying Kirchhoff's voltage law in the clockwise direction, which results in

$$+E - V_D - V_R = 0$$

or

$$E = V_D + I_D R \tag{2.1}$$



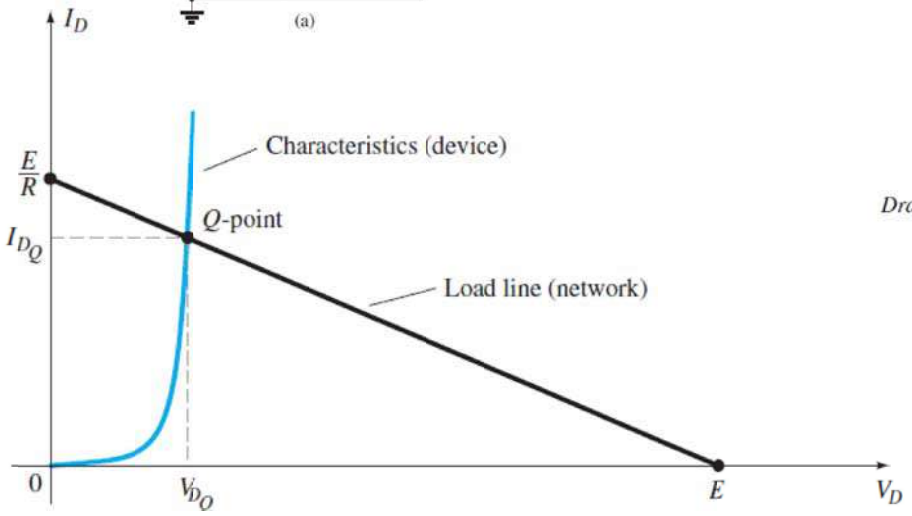


FIG. 2.2
Drawing the load line and finding the point of operation.

The two variables of Eq. (2.1), V_D and I_D, are the same as the diode axis variables of Fig. 2.2. This similarity permits plotting Eq. (2.1) on the same characteristics of Fig. 2.2.

The intersections of the load line on the characteristics can easily be determined if one simply employs the fact that anywhere on the horizontal axis $I_D = 0$ A and anywhere on the vertical axis $V_D = 0$ V.

$$E = V_D + I_D R \tag{2.1}$$

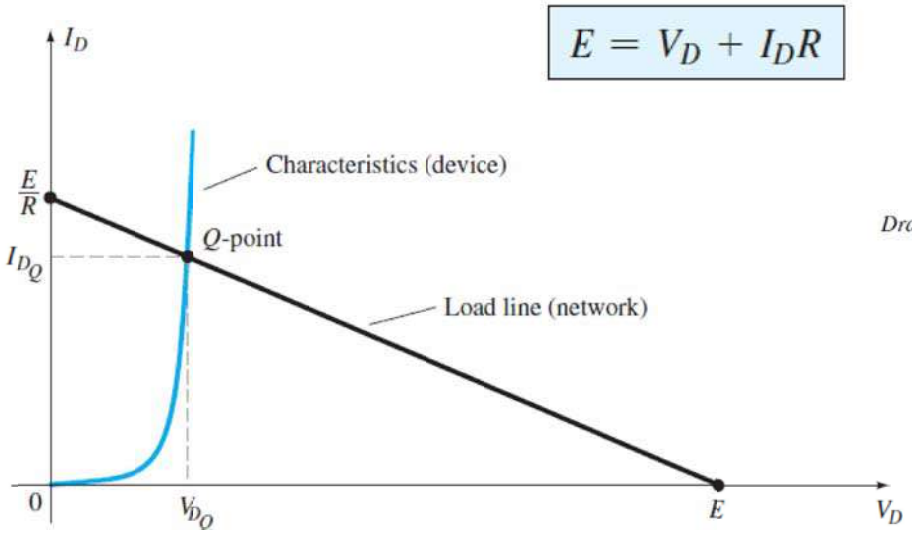


FIG. 2.2
Drawing the load line and finding the point of operation.

If we *set* $V_D = 0$ V in Eq. (2.1) and solve for I_D, we have the magnitude of I_D *on* the vertical axis. Therefore, with $V_D = 0$ V, Eq. (2.1) becomes

$$E = V_D + I_D R$$
$$= 0\,\text{V} + I_D R$$

and

$$\boxed{I_D = \frac{E}{R}\bigg|_{V_D = 0\,\text{V}}} \tag{2.2}$$

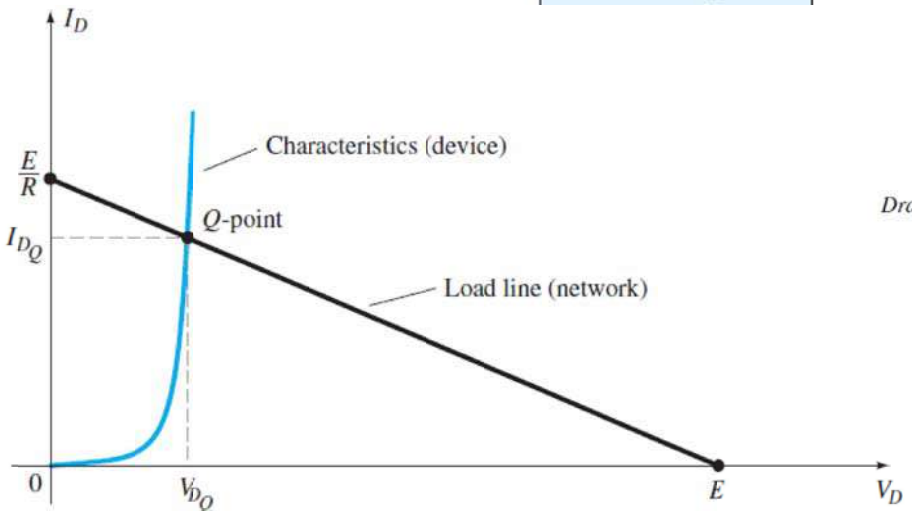


FIG. 2.2
Drawing the load line and finding the point of operation.

as shown in Fig. 2.2. If we *set* $I_D = 0$ A in Eq. (2.1) and solve for V_D, we have the magnitude of V_D *on* the horizontal axis. Therefore, with $I_D = 0$ A, Eq. (2.1) becomes

$$E = V_D + I_D R$$
$$= V_D + (0\,\text{A})R$$

and

$$\boxed{V_D = E|_{I_D = 0\,\text{A}}} \tag{2.3}$$

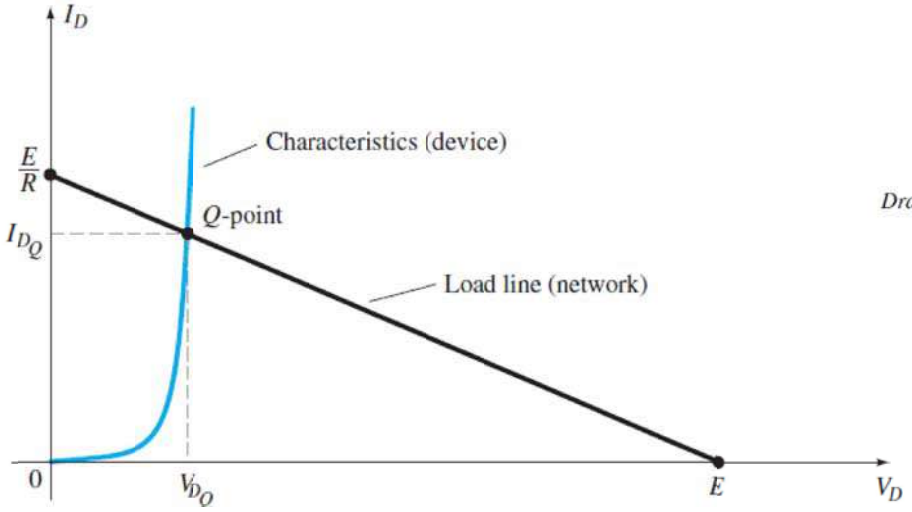


FIG. 2.2
Drawing the load line and finding the point of operation.

as shown in Fig. 2.2. A straight line drawn between the two points will define the load line as depicted in Fig. 2.2. Change the level of R (the load) and the intersection on the vertical axis will change. The result will be a change in the slope of the load line and a different point of intersection between the load line and the device characteristics.

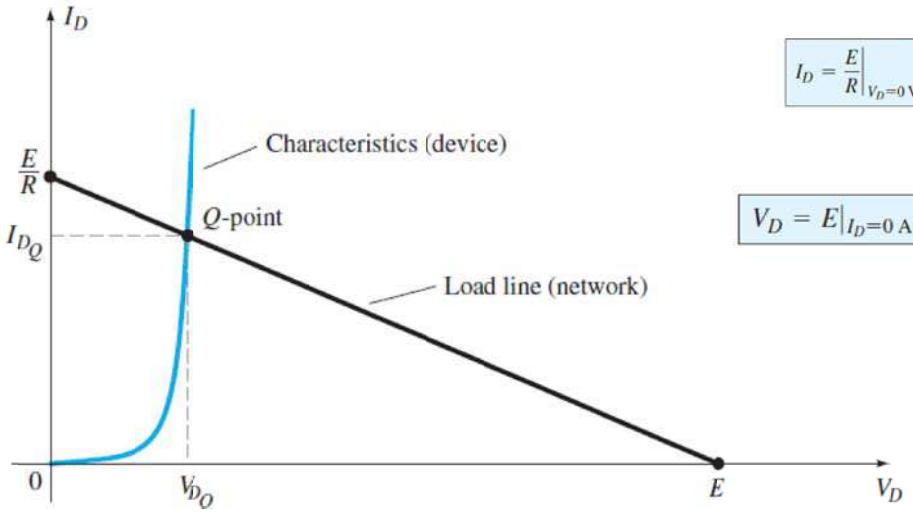


FIG. 2.2
Drawing the load line and finding the point of operation.

$$I_D = \left.\frac{E}{R}\right|_{V_D=0\,\text{V}} \tag{2.2}$$

$$V_D = E|_{I_D=0\,\text{A}} \tag{2.3}$$

The solution obtained at the intersection of the two curves is the same as would be obtained by a simultaneous mathematical solution of

$$I_D = \frac{E}{R} - \frac{V_D}{R} \qquad [\text{derived from Eq. (2.1)}]$$

and

$$I_D = I_s(e^{V_D/nV_T} - 1)$$

Since the curve for a diode has nonlinear characteristics, the mathematics involved would require the use of nonlinear techniques that are beyond the needs and scope of this book.

EXAMPLE 2.1 For the series diode configuration of Fig. 2.3a, employing the diode characteristics of Fig. 2.3b, determine:

a. V_{D_Q} and I_{D_Q}.
b. V_R.

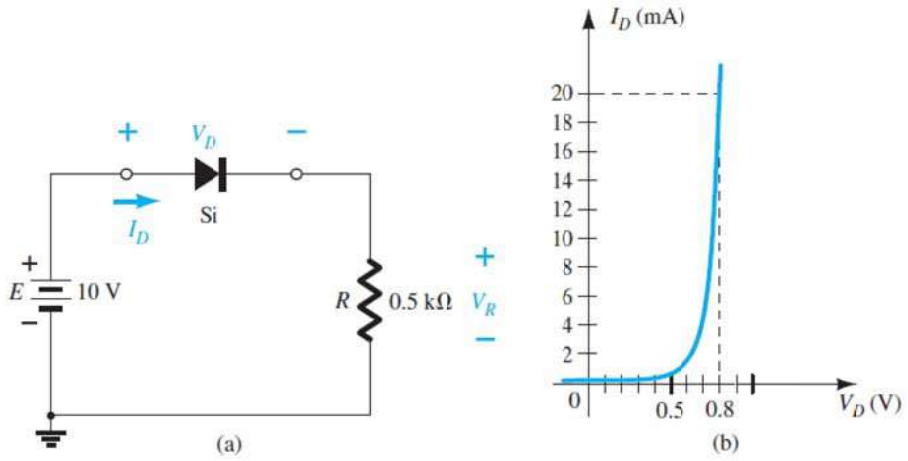


FIG. 2.3
(a) Circuit; (b) characteristics.

Solution:

a. Eq. (2.2): $$I_D = \frac{E}{R}\bigg|_{V_D=0\text{ V}} = \frac{10\text{ V}}{0.5\text{ k}\Omega} = 20\text{ mA}$$

Eq. (2.3): $$V_D = E|_{I_D=0\text{ A}} = 10\text{ V}$$

The resulting load line appears in Fig. 2.4. The intersection between the load line and the characteristic curve defines the Q-point as

$$V_{D_Q} \cong \mathbf{0.78\text{ V}}$$
$$I_{D_Q} \cong \mathbf{18.5\text{ mA}}$$

The level of V_D is certainly an estimate, and the accuracy of I_D is limited by the chosen scale. A higher degree of accuracy would require a plot that would be much larger and perhaps unwieldy.

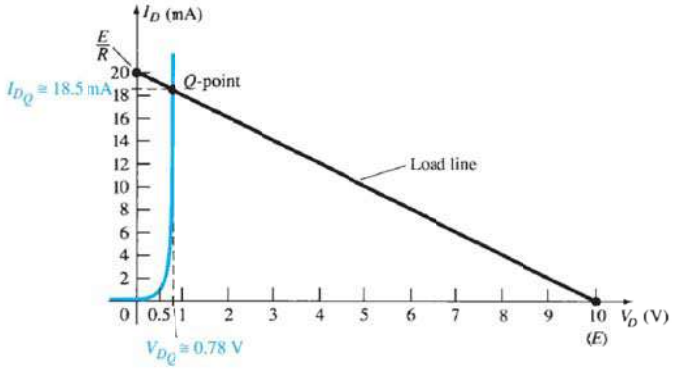
b. $V_R = E - V_D = 10\text{ V} - 0.78\text{ V} = \mathbf{9.22\text{ V}}$

As noted in the example above,

the load line is determined solely by the applied network, whereas the characteristics are defined by the chosen device.

Changing the model we use for the diode will not disturb the network so the load line to be drawn will be exactly the same as appearing in the example above.

Since the network of Example 2.1 is a dc network the Q-point of Fig.2.4 will remain fixed with $V_{D_Q} = 0.78$ V and $I_{D_Q} = 18.5$ mA. In Chapter 1 a dc resistance was defined at any point on the characteristics by $R_{DC} = V_D/I_D$.



Using the Q-point values, the dc resistance for Example 2.1 is

$$R_D = \frac{V_{D_Q}}{I_{D_Q}} = \frac{0.78\text{ V}}{18.5\text{ mA}} = 42.16\ \Omega$$

An equivalent network (for these operating conditions only) can then be drawn as shown in Fig. 2.5.

The current

$$I_D = \frac{E}{R_D + R} = \frac{10\text{ V}}{42.16\ \Omega + 500\ \Omega} = \frac{10\text{ V}}{542.16\ \Omega} \cong \mathbf{18.5\text{ mA}}$$

and

$$V_R = \frac{RE}{R_D + R} = \frac{(500\ \Omega)(10\text{ V})}{42.16\ \Omega + 500\ \Omega} = \mathbf{9.22\text{ V}}$$

matching the results of Example 2.1.

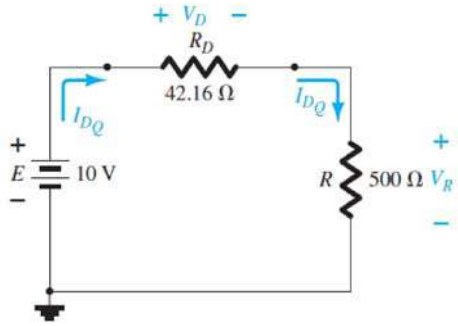
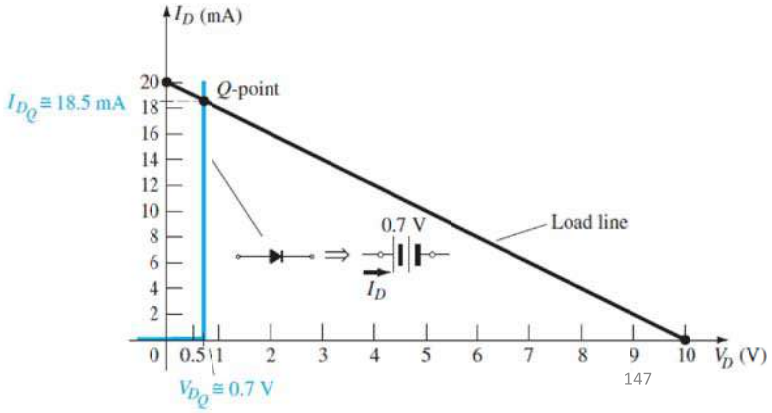


FIG. 2.5
Network quivalent to Fig. 2.4.

EXAMPLE 2.2 Repeat Example 2.1 using the approximate equivalent model for the silicon semiconductor diode.

Solution: The load line is redrawn as shown in Fig. 2.6 with the same intersections as defined in Example 2.1. The characteristics of the approximate equivalent circuit for the diode have also been sketched on the same graph. The resulting Q-point is

$$V_{D_Q} = \mathbf{0.7\ V}$$
$$I_{D_Q} = \mathbf{18.5\ mA}$$



For this situation the dc resistance of the Q-point is

$$R_D = \frac{V_{D_Q}}{I_{D_Q}} = \frac{0.7\ \text{V}}{18.5\ \text{mA}} = 37.84\ \Omega$$

which is still relatively close to that obtained for the full characteristics.

In the next example we go a step further and substitute the ideal model. The results will reveal the conditions that must be satisfied to apply the ideal equivalent properly.

EXAMPLE 2.3 Repeat Example 2.1 using the ideal diode model.

Solution: As shown in Fig. 2.7, the load line is the same, but the ideal characteristics now intersect the load line on the vertical axis. The *Q*-point is therefore defined by

$$V_{D_Q} = \mathbf{0\ V}$$
$$I_{D_Q} = \mathbf{20\ mA}$$

In this case,

$$R_D = \frac{V_{D_Q}}{I_{D_Q}} = \frac{0\ \text{V}}{20\ \text{mA}} = 0\ \Omega$$

(or a short-circuit equivalent)

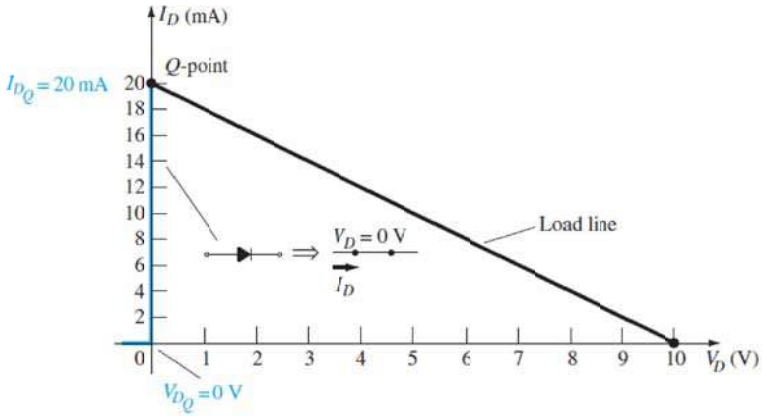


FIG. 2.7
Solution to Example 2.1 using the ideal diode model.

Chapter 2

Diode Applications

CHAPTER OBJECTIVES

- Understand the concept of load-line analysis and how it is applied to diode networks.
- Become familiar with the use of equivalent circuits to analyze series, parallel, and series-parallel diode networks.
- Understand the process of rectification to establish a dc level from a sinusoidal ac input.
- Be able to predict the output response of a clipper and clamper diode configuration.
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The circuit of Fig. 2.1 is the simplest of diode configurations. It will be used to describe the analysis of a diode circuit using its actual characteristics. In the next section we will replace the characteristics by an approximate model for the diode and compare solutions. Solving the circuit of Fig. 2.1 is all about finding the current and voltage levels that will satisfy both the characteristics of the diode and the chosen network parameters at the same time.

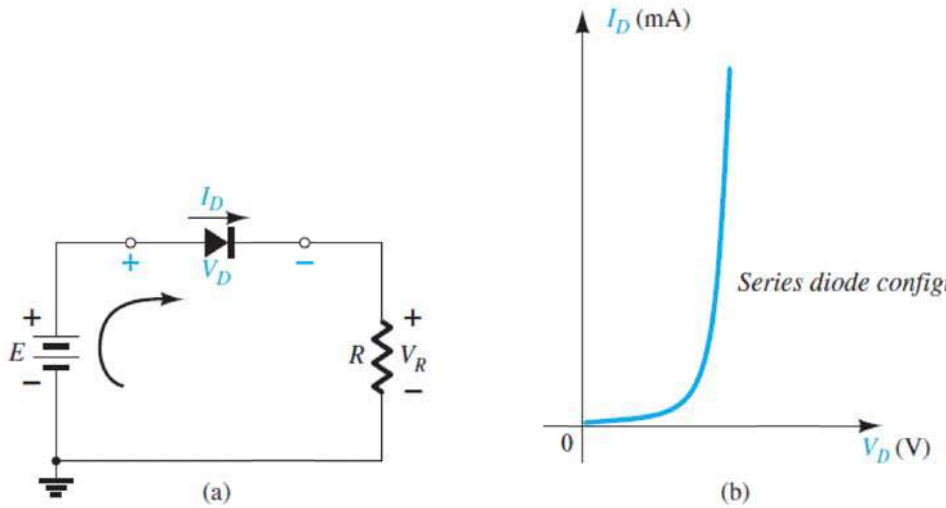


FIG. 2.1
Series diode configuration: (a) circuit; (b) characteristics.

In Fig. 2.2 the diode characteristics are placed on the same set of axes as a straight line defined by the parameters of the network. The straight line is called a *load line* because the intersection on the vertical axis is defined by the applied load R. The analysis to follow is therefore called *load-line analysis.* The intersection of the two curves will define the solution for the network and define the current and voltage levels for the network.

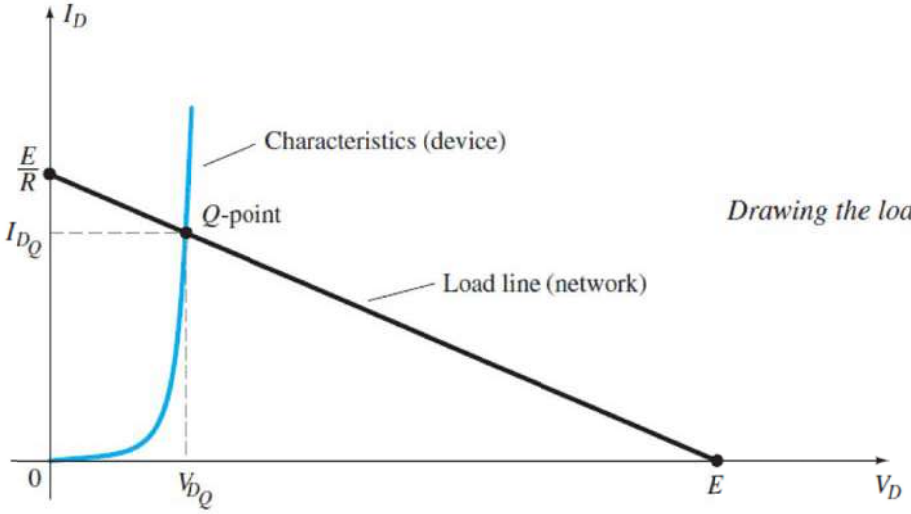


FIG. 2.2
Drawing the load line and finding the point of operation.

Before reviewing the details of drawing the load line on the characteristics, we need to determine the expected response of the simple circuit of Fig. 2.1. Note in Fig. 2.1 that the effect of the "pressure" established by the dc supply is to establish a conventional current in the direction indicated by the clockwise arrow. The fact that the direction of this current has the same direction as the arrow in the diode symbol reveals that the diode is in the "on" state and will conduct a high level of current. The polarity of the applied voltage has resulted in a forward-bias situation.

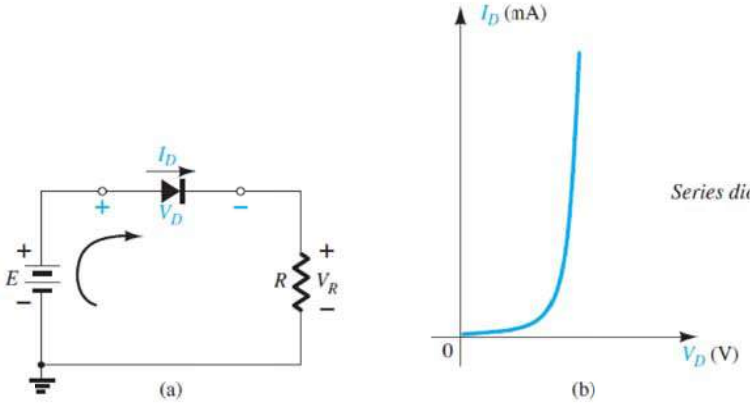


FIG. 2.1
Series diode configuration: (a) circuit; (b) characteristics.

With the current direction established, the polarities for the voltage across the diode and resistor can be superimposed. The polarity of V_D and the direction of I_D clearly reveal that the diode is indeed in the forward-bias state, resulting in a voltage across the diode in the neighborhood of 0.7 V and a current on the order of 10 mA or more.

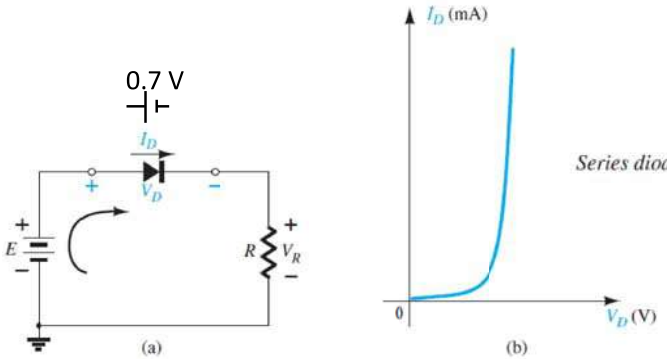


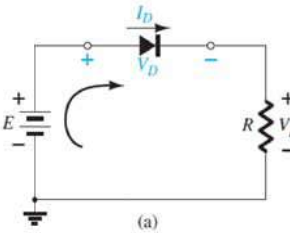
FIG. 2.1
Series diode configuration: (a) circuit; (b) characteristics.

The intersections of the load line on the characteristics of Fig. 2.2 can be determined by first applying Kirchhoff's voltage law in the clockwise direction, which results in

$$+E - V_D - V_R = 0$$

or

$$E = V_D + I_D R \tag{2.1}$$



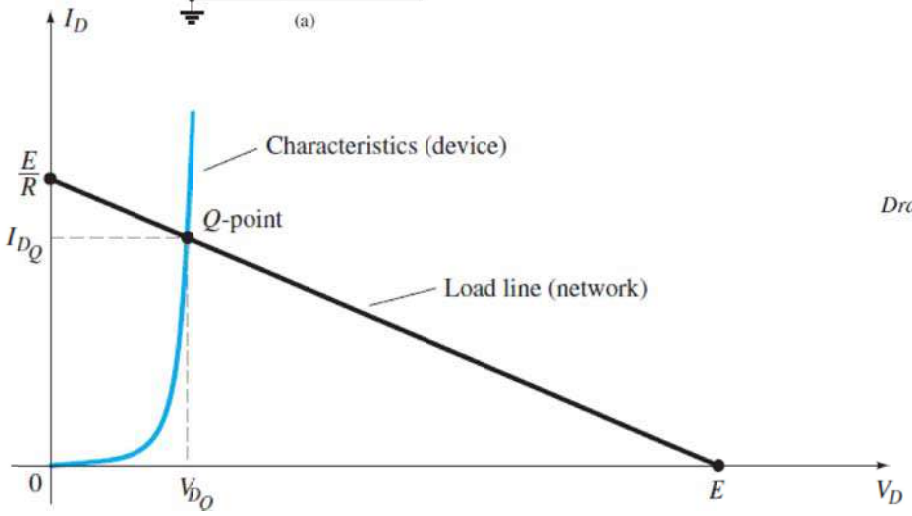


FIG. 2.2
Drawing the load line and finding the point of operation.

The two variables of Eq. (2.1), V_D and I_D, are the same as the diode axis variables of Fig. 2.2. This similarity permits plotting Eq. (2.1) on the same characteristics of Fig. 2.2.

The intersections of the load line on the characteristics can easily be determined if one simply employs the fact that anywhere on the horizontal axis $I_D = 0$ A and anywhere on the vertical axis $V_D = 0$ V.

$$E = V_D + I_D R \tag{2.1}$$

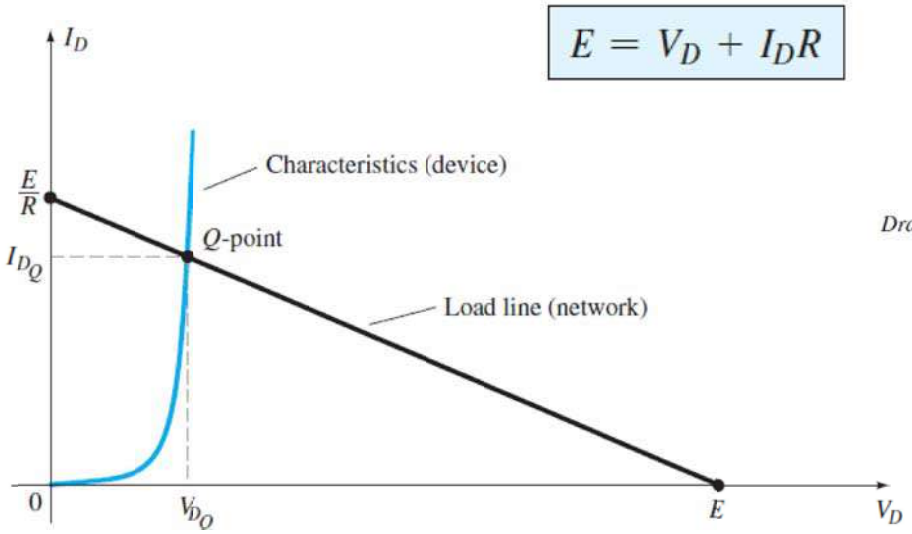


FIG. 2.2
Drawing the load line and finding the point of operation.

If we *set* $V_D = 0$ V in Eq. (2.1) and solve for I_D, we have the magnitude of I_D *on* the vertical axis. Therefore, with $V_D = 0$ V, Eq. (2.1) becomes

$$E = V_D + I_D R$$
$$= 0\,\text{V} + I_D R$$

and

$$\boxed{I_D = \frac{E}{R}\bigg|_{V_D = 0\,\text{V}}} \tag{2.2}$$

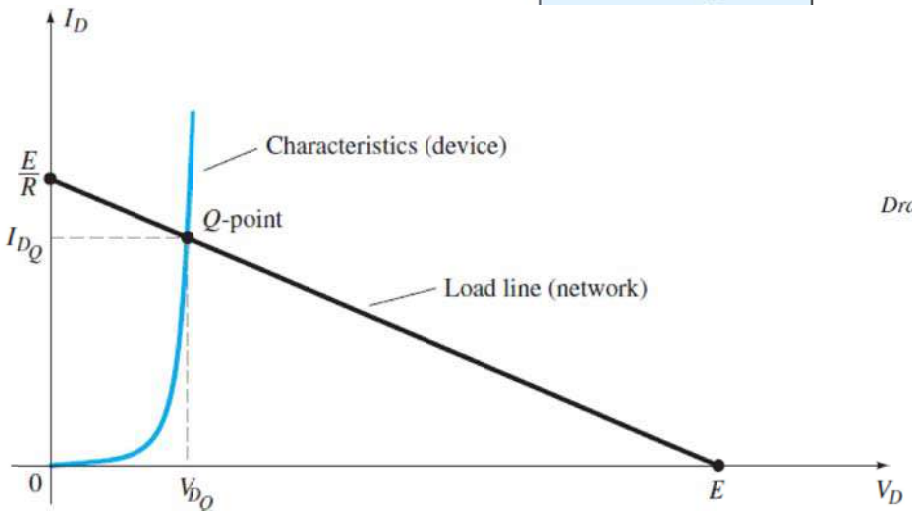


FIG. 2.2
Drawing the load line and finding the point of operation.

as shown in Fig. 2.2. If we *set* $I_D = 0$ A in Eq. (2.1) and solve for V_D, we have the magnitude of V_D *on* the horizontal axis. Therefore, with $I_D = 0$ A, Eq. (2.1) becomes

$$E = V_D + I_D R$$
$$= V_D + (0\,\text{A})R$$

and

$$\boxed{V_D = E|_{I_D = 0\,\text{A}}} \tag{2.3}$$

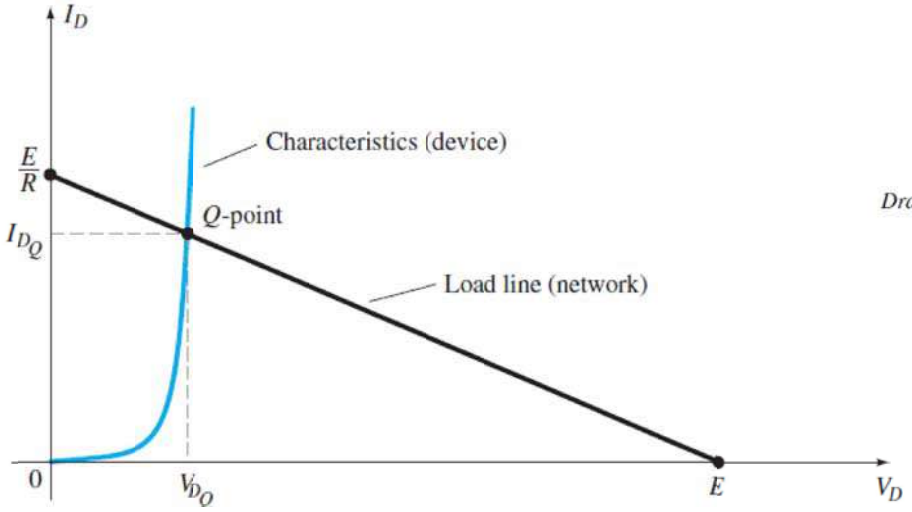


FIG. 2.2
Drawing the load line and finding the point of operation.

as shown in Fig. 2.2. A straight line drawn between the two points will define the load line as depicted in Fig. 2.2. Change the level of R (the load) and the intersection on the vertical axis will change. The result will be a change in the slope of the load line and a different point of intersection between the load line and the device characteristics.

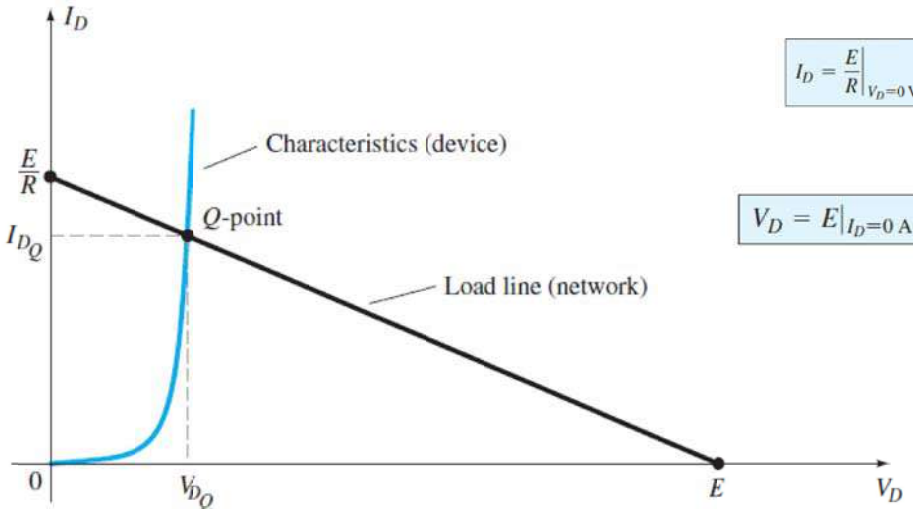


FIG. 2.2
Drawing the load line and finding the point of operation.

$$I_D = \left.\frac{E}{R}\right|_{V_D=0\,\text{V}} \qquad (2.2)$$

$$V_D = E|_{I_D=0\,\text{A}} \qquad (2.3)$$

The solution obtained at the intersection of the two curves is the same as would be obtained by a simultaneous mathematical solution of

$$I_D = \frac{E}{R} - \frac{V_D}{R} \qquad [\text{derived from Eq. (2.1)}]$$

and

$$I_D = I_s(e^{V_D/nV_T} - 1)$$

Since the curve for a diode has nonlinear characteristics, the mathematics involved would require the use of nonlinear techniques that are beyond the needs and scope of this book.

EXAMPLE 2.1 For the series diode configuration of Fig. 2.3a, employing the diode characteristics of Fig. 2.3b, determine:

a. V_{D_Q} and I_{D_Q}.
b. V_R.

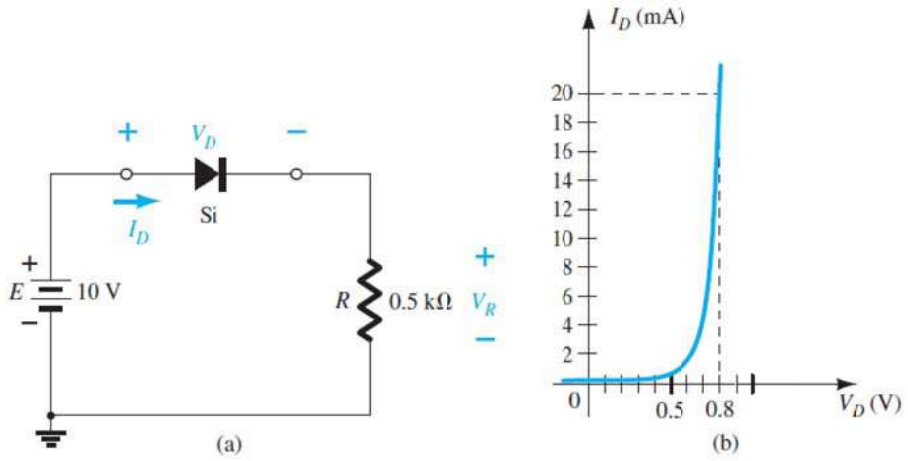


FIG. 2.3
(a) Circuit; (b) characteristics.

Solution:

a. Eq. (2.2): $$I_D = \frac{E}{R}\bigg|_{V_D=0\text{ V}} = \frac{10\text{ V}}{0.5\text{ k}\Omega} = 20\text{ mA}$$

Eq. (2.3): $$V_D = E|_{I_D=0\text{ A}} = 10\text{ V}$$

The resulting load line appears in Fig. 2.4. The intersection between the load line and the characteristic curve defines the Q-point as

$$V_{D_Q} \cong \mathbf{0.78\text{ V}}$$
$$I_{D_Q} \cong \mathbf{18.5\text{ mA}}$$

The level of V_D is certainly an estimate, and the accuracy of I_D is limited by the chosen scale. A higher degree of accuracy would require a plot that would be much larger and perhaps unwieldy.

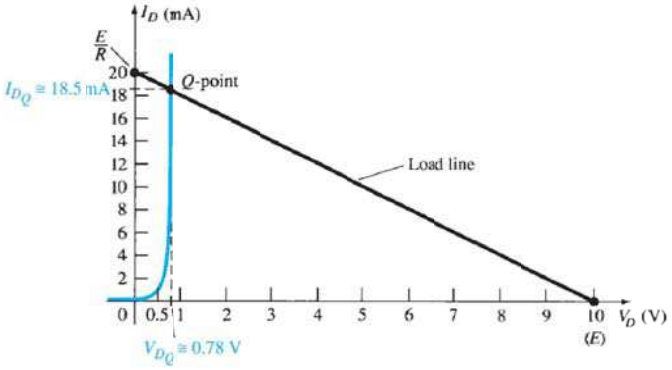
b. $V_R = E - V_D = 10\text{ V} - 0.78\text{ V} = \mathbf{9.22\text{ V}}$

As noted in the example above,

the load line is determined solely by the applied network, whereas the characteristics are defined by the chosen device.

Changing the model we use for the diode will not disturb the network so the load line to be drawn will be exactly the same as appearing in the example above.

Since the network of Example 2.1 is a dc network the Q-point of Fig.2.4 will remain fixed with $V_{D_Q} = 0.78$ V and $I_{D_Q} = 18.5$ mA. In Chapter 1 a dc resistance was defined at any point on the characteristics by $R_{DC} = V_D/I_D$.



Using the Q-point values, the dc resistance for Example 2.1 is

$$R_D = \frac{V_{D_Q}}{I_{D_Q}} = \frac{0.78\text{ V}}{18.5\text{ mA}} = 42.16\ \Omega$$

An equivalent network (for these operating conditions only) can then be drawn as shown in Fig. 2.5.

The current

$$I_D = \frac{E}{R_D + R} = \frac{10\text{ V}}{42.16\ \Omega + 500\ \Omega} = \frac{10\text{ V}}{542.16\ \Omega} \cong \mathbf{18.5\text{ mA}}$$

and

$$V_R = \frac{RE}{R_D + R} = \frac{(500\ \Omega)(10\text{ V})}{42.16\ \Omega + 500\ \Omega} = \mathbf{9.22\text{ V}}$$

matching the results of Example 2.1.

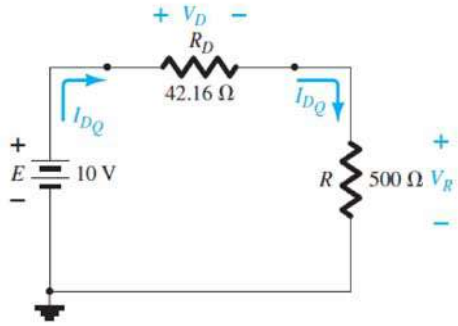
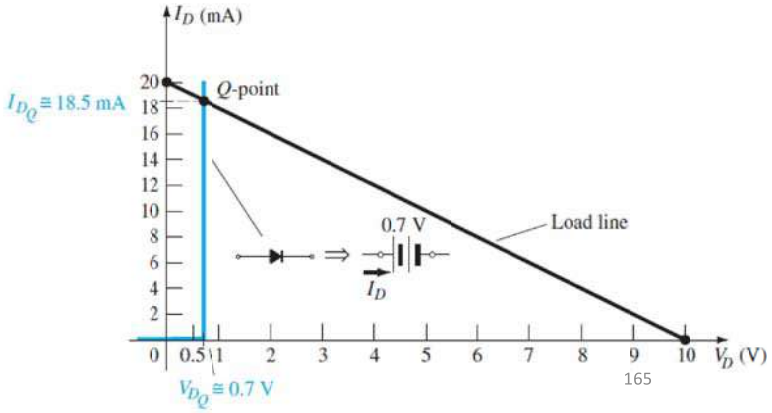


FIG. 2.5
Network quivalent to Fig. 2.4

EXAMPLE 2.2 Repeat Example 2.1 using the approximate equivalent model for the silicon semiconductor diode.

Solution: The load line is redrawn as shown in Fig. 2.6 with the same intersections as defined in Example 2.1. The characteristics of the approximate equivalent circuit for the diode have also been sketched on the same graph. The resulting Q-point is

$$V_{D_Q} = \mathbf{0.7\ V}$$
$$I_{D_Q} = \mathbf{18.5\ mA}$$



For this situation the dc resistance of the Q-point is

$$R_D = \frac{V_{D_Q}}{I_{D_Q}} = \frac{0.7\ \text{V}}{18.5\ \text{mA}} = 37.84\ \Omega$$

which is still relatively close to that obtained for the full characteristics.

In the next example we go a step further and substitute the ideal model. The results will reveal the conditions that must be satisfied to apply the ideal equivalent properly.

EXAMPLE 2.3 Repeat Example 2.1 using the ideal diode model.

Solution: As shown in Fig. 2.7, the load line is the same, but the ideal characteristics now intersect the load line on the vertical axis. The Q-point is therefore defined by

$$V_{D_Q} = \mathbf{0\ V}$$
$$I_{D_Q} = \mathbf{20\ mA}$$

In this case,

$$R_D = \frac{V_{D_Q}}{I_{D_Q}} = \frac{0\ \text{V}}{20\ \text{mA}} = 0\ \Omega$$

(or a short-circuit equivalent)

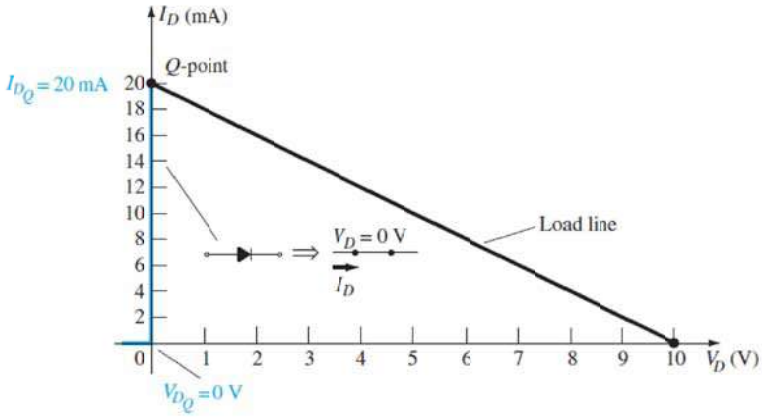


FIG. 2.7
Solution to Example 2.1 using the ideal diode model.

2.3 SERIES DIODE CONFIGURATIONS

In the last section we found that the results obtained using the approximate piecewise-linear equivalent model were quite close, if not equal, to the response obtained using the full characteristics.

The primary purpose of this text is to develop a general knowledge of the behavior, capabilities, and possible areas of application of a device in a manner that will minimize the need for extensive mathematical developments.

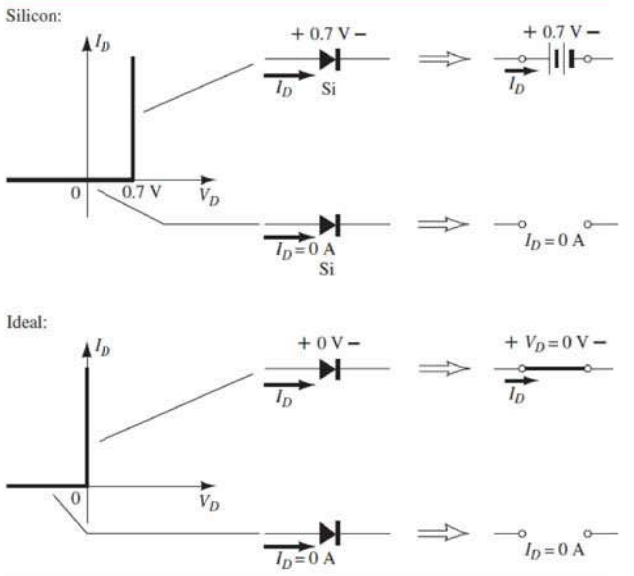
For all the analysis to follow in this chapter it is assumed that

The forward resistance of the diode is usually so small compared to the other series elements of the network that it can be ignored.

This is a valid approximation for the vast majority of applications that employ diodes. Using this fact will result in the approximate equivalents for a silicon diode and an ideal diode that appear in Table 2.1.

TABLE 2.1

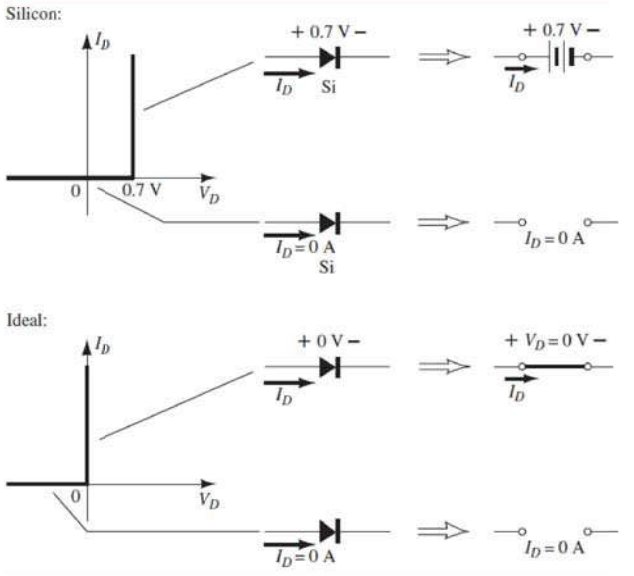
Approximate and Ideal Semiconductor Diode Models.



For a Ge diode the offset voltage is 0.3 V and for a GaAs diode it is 1.2 V. Otherwise the equivalent networks are the same. For each diode the label Si, Ge, or GaAs will appear along with the diode symbol. For networks with ideal diodes the diode symbol will appear as shown in Table 2.1 without any labels.

TABLE 2.1

Approximate and Ideal Semiconductor Diode Models.



In general, a diode is in the "on" state if the current established by the applied sources is such that its direction matches that of the arrow in the diode symbol, and $V_D \geq 0.7$ V for silicon, $V_D \geq 0.3$ V for germanium, and $V_D \geq 1.2$ V for gallium arsenide.

For each configuration, *mentally* replace the diodes with resistive elements and note the resulting current direction as established by the applied voltages ("pressure"). If the resulting direction is a "match" with the arrow in the diode symbol, conduction through the diode will occur and the device is in the "on" state. The description above is, of course, contingent on the supply having a voltage greater than the "turn-on" voltage (V_K) of each diode.

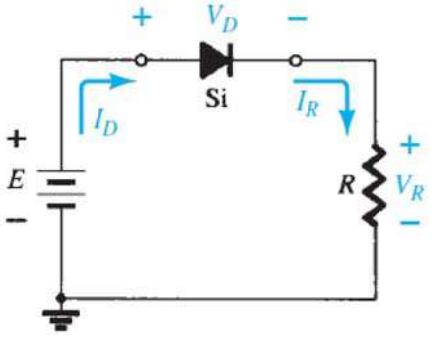


FIG. 2.8
Series diode configuration.

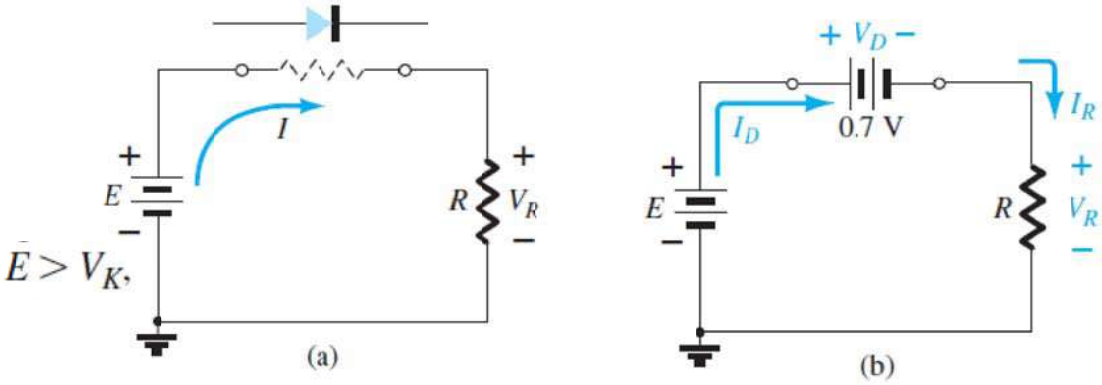


FIG. 2.9
(a) Determining the state of the diode of Fig. 2.8; (b) substituting the equivalent model for the "on" diode of Fig. 2.9a.

$$V_R = E - V_K \qquad (2.5)$$

$$I_D = I_R = \frac{V_R}{R} \qquad (2.6)$$

In Reverse Bias

The diode is in the "off" state, resulting in the equivalent circuit of Fig. 2.12. Due to the open circuit, the diode current is 0 A and the voltage across the resistor R is the following:

$$V_R = I_R R = I_D R = (0\text{ A})R = \mathbf{0\text{ V}}$$

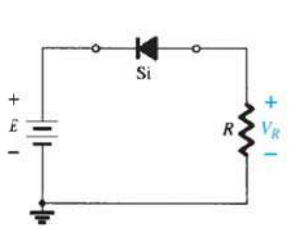


FIG. 2.10
Reversing the diode of Fig. 2.8.

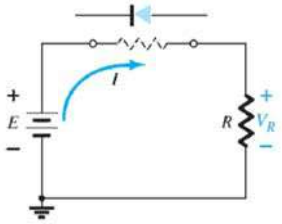


FIG. 2.11
Determining the state of the diode of Fig. 2.10.

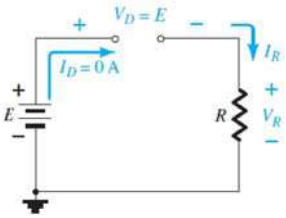


FIG. 2.12
Substituting the equivalent model for the "off" diode of Fig. 2.10.

The fact that $V_R = 0$ V will establish E volts across the open circuit as defined by Kirchhoff's voltage law. Always keep in mind that under any circumstances—dc, ac instantaneous values, pulses, and so on—Kirchhoff's voltage law must be satisfied!

EXAMPLE 2.4 For the series diode configuration of Fig. 2.13, determine V_D, V_R, and I_D.

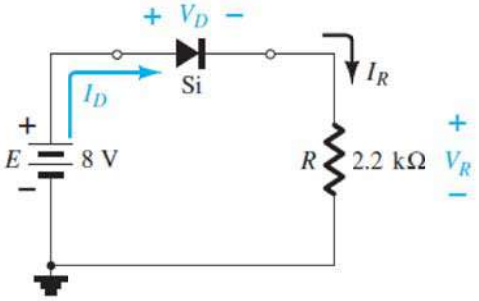


FIG. 2.13
Circuit for Example 2.4.

Solution: Since the applied voltage establishes a current in the clockwise direction to match the arrow of the symbol and the diode is in the "on" state,

$$V_D = \mathbf{0.7\text{ V}}$$

$$V_R = E - V_D = 8\text{ V} - 0.7\text{ V} = \mathbf{7.3\text{ V}}$$

$$I_D = I_R = \frac{V_R}{R} = \frac{7.3\text{ V}}{2.2\text{ k}\Omega} \cong \mathbf{3.32\text{ mA}}$$

EXAMPLE 2.5 Repeat Example 2.4 with the diode reversed.

Solution: Removing the diode, we find that the direction of I is opposite to the arrow in the diode symbol and the diode equivalent is the open circuit no matter which model is employed. The result is the network of Fig. 2.14, where $I_D = \mathbf{0\ A}$ due to the open circuit. Since $V_R = I_R R$, we have $V_R = (0)R = \mathbf{0\ V}$. Applying Kirchhoff's voltage law around the closed loop yields $E - V_D - V_R = 0$

$$V_D = E - V_R = E - 0 = E = 8\ \text{V}$$

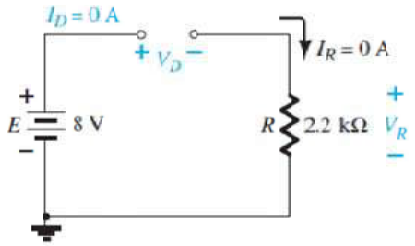


FIG. 2.14
Determining the unknown quantities for

For ideal diode

An open circuit can have any voltage across its terminals, but the current is always 0 A.

A short circuit has a 0-V drop across its terminals, but the current is limited only by the surrounding network.

2.3 SERIES DIODE CONFIGURATIONS Examples

In the next example the notation of Fig. 2.15 will be employed for the applied voltage. It is a common industry notation and one with which the reader should become very familiar. Such notation and other defined voltage levels are treated further in Chapter 4.

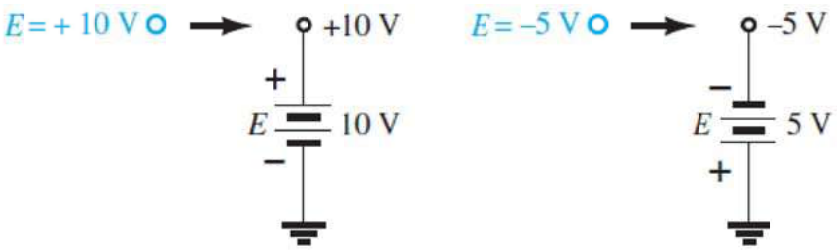


FIG. 2.15
Source notation.

EXAMPLE 2.6 For the series diode configuration of Fig. 2.16, determine V_D, V_R, and I_D.

Solution: Although the "pressure" establishes a current with the same direction as the arrow symbol, the level of applied voltage is insufficient to turn the silicon diode "on." The point of operation on the characteristics is shown in Fig. 2.17, establishing the open-circuit equivalent as the appropriate approximation, as shown in Fig. 2.18. The resulting voltage and current levels are therefore the following:

$$I_D = \mathbf{0\ A}$$

$$V_R = I_R R = I_D R = (0\ \text{A})\,1.2\ \text{k}\Omega = \mathbf{0\ V}$$

and

$$V_D = E = \mathbf{0.5\ V}$$

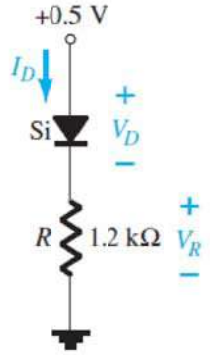


FIG. 2.16
Series diode circuit for Example 2.6.

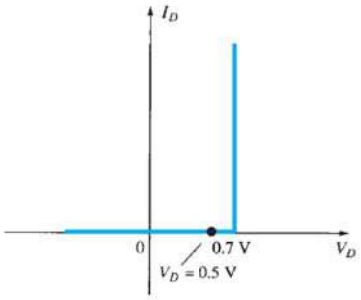


FIG. 2.17
Operating point with E = 0.5 V.

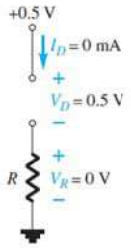


FIG. 2.18
Determining I_D, V_R, and V_D for the circuit of Fig. 2.16.

EXAMPLE 2.7 Determine V_o and I_D for the series circuit of Fig. 2.19.

Solution: An attack similar to that applied in Example 2.4 will reveal that the resulting current has the same direction as the arrowheads of the symbols of both diodes, and the network of Fig. 2.20 results because $E = 12\ \text{V} > (0.7\ \text{V} + 1.8\ \text{V}\ [\text{Table 1.8}]) = 2.5\ \text{V}$. Note the redrawn supply of 12 V and the polarity of V_o across the 680-Ω resistor. The resulting voltage is

$$V_o = E - V_{K_1} - V_{K_2} = 12\ \text{V} - 2.5\ \text{V} = \mathbf{9.5\ V}$$

and

$$I_D = I_R = \frac{V_R}{R} = \frac{V_o}{R} = \frac{9.5\ \text{V}}{680\ \Omega} = \mathbf{13.97\ mA}$$

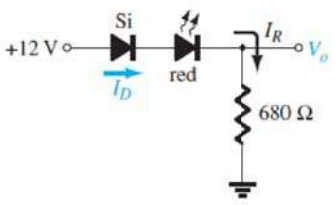


FIG. 2.19
Circuit for Example 2.7.

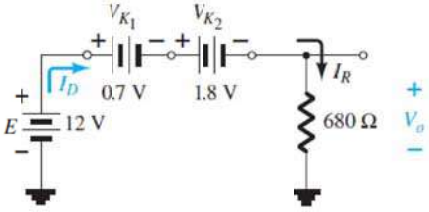


FIG. 2.20
Determining the unknown quantities for Example 2.7.

EXAMPLE 2.8 Determine I_D, V_{D_2}, and V_o for the circuit of Fig. 2.21.

Solution: Removing the diodes and determining the direction of the resulting current I result in the circuit of Fig. 2.22. There is a match in current direction for one silicon diode but not for the other silicon diode. The combination of a short circuit in series with an open circuit always results in an open circuit and $I_D = \mathbf{0\ A}$, as shown in Fig. 2.23.

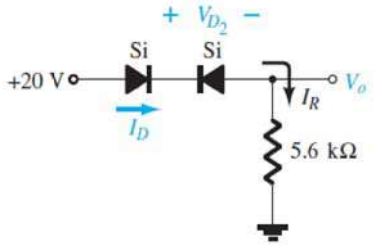


FIG. 2.21
Circuit for Example 2.8.

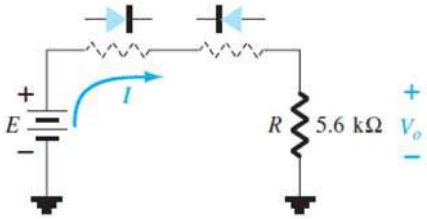


FIG. 2.22
Determining the state of the diodes of Fig. 2.21.

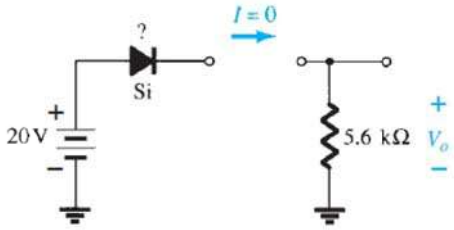


FIG. 2.23
Substituting the equivalent state for the open diode.

The question remains as to what to substitute for the silicon diode. For the analysis to follow in this and succeeding chapters, simply recall for the actual practical diode that when $I_D = 0$ A, $V_D = 0$ V (and vice versa), as described for the no-bias situation in Chapter 1. The conditions described by $I_D = 0$ A and $V_{D_1} = 0$ V are indicated in Fig. 2.24. We have

$$V_o = I_R R = I_D R = (0\ \text{A})R = \mathbf{0\ V}$$

and

$$V_{D_2} = V_{\text{open circuit}} = E = \mathbf{20\ V}$$

Applying Kirchhoff's voltage law in a clockwise direction gives

$$E - V_{D_1} - V_{D_2} - V_o = 0$$

and

$$V_{D_2} = E - V_{D_1} - V_o = 20\ \text{V} - 0 - 0$$
$$= \mathbf{20\ V}$$

with

$$V_o = \mathbf{0\ V}$$

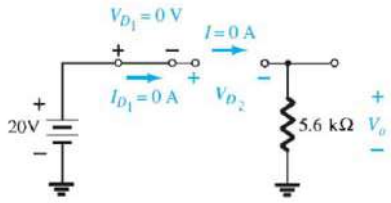


FIG. 2.24
Determining the unknown quantities for the circuit of Example 2.8.

EXAMPLE 2.9 Determine I, V_1, V_2, and V_o for the series dc configuration of Fig. 2.25.

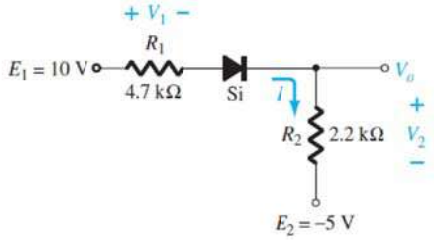


FIG. 2.25

Solution: The sources are drawn and the current direction indicated as shown in Fig. 2.26. The diode is in the "on" state and the notation appearing in Fig. 2.27 is included to indicate this state. Note that the "on" state is noted simply by the additional $V_D = 0.7$ V on the figure. This eliminates the need to redraw the network and avoids any confusion that may

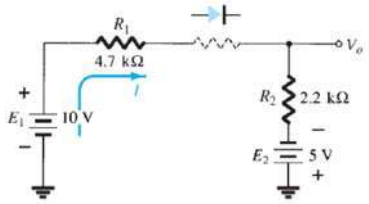


FIG. 2.26
Determining the state of the diode for the network of Fig. 2.25.

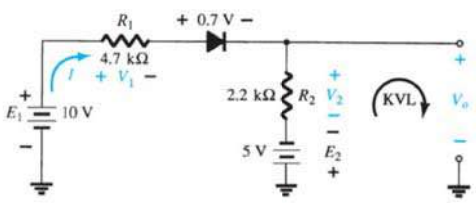


FIG. 2.27
Determining the unknown quantities for the network of Fig. 2.25. KVL, Kirchhoff voltage loop.

The resulting current through the circuit is

$$I = \frac{E_1 + E_2 - V_D}{R_1 + R_2} = \frac{10\text{ V} + 5\text{ V} - 0.7\text{ V}}{4.7\text{ k}\Omega + 2.2\text{ k}\Omega} = \frac{14.3\text{ V}}{6.9\text{ k}\Omega}$$
$$\cong \mathbf{2.07\text{ mA}}$$

and the voltages are

$$V_1 = IR_1 = (2.07\text{ mA})(4.7\text{ k}\Omega) = \mathbf{9.73\text{ V}}$$
$$V_2 = IR_2 = (2.07\text{ mA})(2.2\text{ k}\Omega) = \mathbf{4.55\text{ V}}$$

Applying Kirchhoff's voltage law to the output section in the clockwise direction results in

$$-E_2 + V_2 - V_o = 0$$

and

$$V_o = V_2 - E_2 = 4.55\text{ V} - 5\text{ V} = \mathbf{-0.45\text{ V}}$$

The minus sign indicates that V_o has a polarity opposite to that appearing in Fig. 2.25.

EXAMPLE 2.9 Determine I, V_1, V_2, and V_o for the series dc configuration of Fig. 2.25.

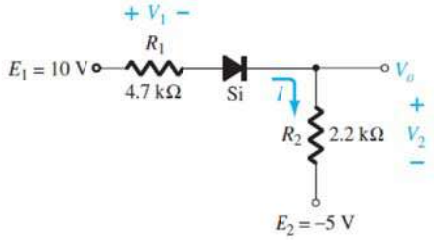


FIG. 2.25

Solution: The sources are drawn and the current direction indicated as shown in Fig. 2.26. The diode is in the "on" state and the notation appearing in Fig. 2.27 is included to indicate this state. Note that the "on" state is noted simply by the additional $V_D = 0.7$ V on the figure. This eliminates the need to redraw the network and avoids any confusion that may

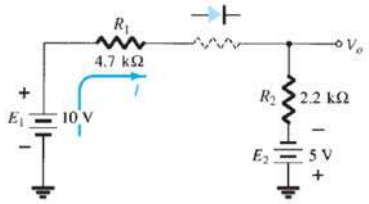


FIG. 2.26
Determining the state of the diode for the network of Fig. 2.25.

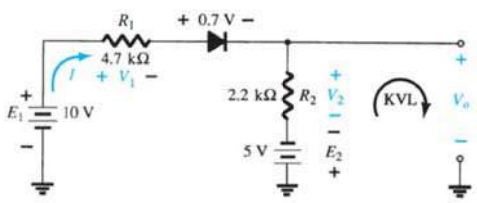


FIG. 2.27
Determining the unknown quantities for the network of Fig. 2.25. KVL, Kirchhoff voltage loop.

183

The resulting current through the circuit is

$$I = \frac{E_1 + E_2 - V_D}{R_1 + R_2} = \frac{10\text{ V} + 5\text{ V} - 0.7\text{ V}}{4.7\text{ k}\Omega + 2.2\text{ k}\Omega} = \frac{14.3\text{ V}}{6.9\text{ k}\Omega}$$
$$\cong \mathbf{2.07\text{ mA}}$$

and the voltages are

$$V_1 = IR_1 = (2.07\text{ mA})(4.7\text{ k}\Omega) = \mathbf{9.73\text{ V}}$$
$$V_2 = IR_2 = (2.07\text{ mA})(2.2\text{ k}\Omega) = \mathbf{4.55\text{ V}}$$

Applying Kirchhoff's voltage law to the output section in the clockwise direction results in

$$-E_2 + V_2 - V_o = 0$$

and

$$V_o = V_2 - E_2 = 4.55\text{ V} - 5\text{ V} = \mathbf{-0.45\text{ V}}$$

The minus sign indicates that V_o has a polarity opposite to that appearing in Fig. 2.25.

184

2.4 PARALLEL AND SERIES–PARALLEL CONFIGURATIONS

The methods applied in Section 2.3 can be extended to the analysis of parallel and series–parallel configurations. For each area of application, simply match the sequential series of steps applied to series diode configurations.

EXAMPLE 2.10 Determine V_o, I_1, I_{D_1}, and I_{D_2} for the parallel diode configuration of Fig. 2.28.

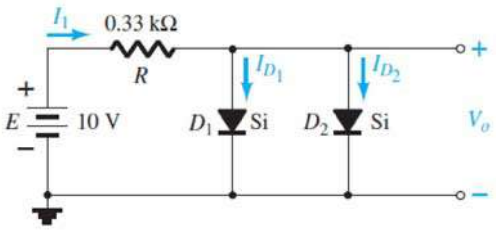


FIG. 2.28
Network for Example 2.10.

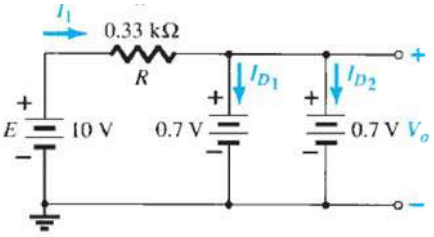


FIG. 2.29
Determining the unknown quantities for the network of Example 2.10.

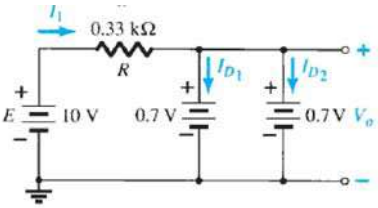
185

Solution: For the applied voltage the "pressure" of the source acts to establish a current through each diode in the same direction as shown in Fig. 2.29. Since the resulting current direction matches that of the arrow in each diode symbol and the applied voltage is greater than 0.7 V, both diodes are in the "on" state. The voltage across parallel elements is always the same and

$$V_o = \mathbf{0.7\ V}$$

The current is

$$I_1 = \frac{V_R}{R} = \frac{E - V_D}{R} = \frac{10\ \text{V} - 0.7\ \text{V}}{0.33\ \text{k}\Omega} = \mathbf{28.18\ mA}$$



Assuming diodes of similar characteristics, we have

$$I_{D_1} = I_{D_2} = \frac{I_1}{2} = \frac{28.18\ \text{mA}}{2} = \mathbf{14.09\ mA}$$

This example demonstrates one reason for placing diodes in parallel. If the current rating of the diodes of Fig. 2.28 is only 20 mA, a current of 28.18 mA would damage the device if it appeared alone in Fig. 2.28. By placing two in parallel, we limit the current to a safe value of 14.09 mA with the same terminal voltage.

186

EXAMPLE 2.11 In this example there are two LEDs that can be used as a polarity detector. Apply a positive source voltage and a green light results. Negative supplies result in a red light. Packages of such combinations are commercially available.

Find the resistor R to ensure a current of 20 mA through the "on" diode for the configuration of Fig. 2.30. Both diodes have a reverse breakdown voltage of 3 V and an average turn-on voltage of 2 V.

Solution: The application of a positive supply voltage results in a conventional current that matches the arrow of the green diode and turns it on.

The polarity of the voltage across the green diode is such that it reverse biases the red diode by the same amount. The result is the equivalent network of Fig. 2.31.

Applying Ohm's law, we obtain

$$I = 20\text{ mA} = \frac{E - V_{LED}}{R} = \frac{8\text{ V} - 2\text{ V}}{R}$$

and

$$R = \frac{6\text{ V}}{20\text{ mA}} = \mathbf{300\ \Omega}$$

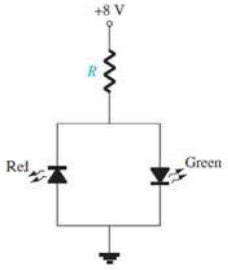


FIG. 2.30
Network for Example 2.11.

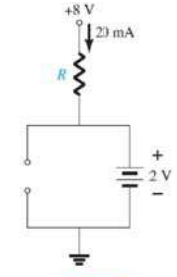


FIG. 2.31
Operating conditions for the network of Fig. 2.30.

Note that the reverse breakdown voltage across the red diode is 2 V, which is fine for an LED with a reverse breakdown voltage of 3 V.

However, if the green diode were to be replaced by a blue diode, problems would develop, as shown in Fig. 2.32. Recall that the forward bias required to turn on a blue diode is about 5 V. The result would appear to require a smaller resistor R to establish the current of 20 mA. However, note that the reverse bias voltage of the red LED is 5 V, but the reverse breakdown voltage of the diode is only 3 V. The result is the voltage across the red LED would lock in at 3 V as shown in Fig. 2.33. The voltage across R would be 5 V and the current limited to 20 mA with a 250 Ω resistor but neither LED would be on.

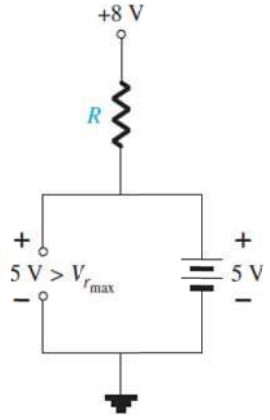


FIG. 2.32
Network of Fig. 2.31 with a blue diode.

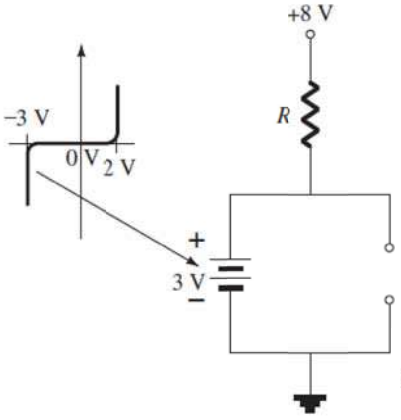


FIG. 2.33
Demonstrating damage to the red LED if the reverse breakdown voltage is exceeded.

A simple solution to the above is to add the appropriate resistance level in series with each diode to establish the desired 20 mA and to include another diode to add to the reverse-bias total reverse breakdown voltage rating, as shown in Fig. 2.34. When the blue LED is on, the diode in series with the blue LED will also be on, causing a total voltage drop of 5.7 V across the two series diodes and a voltage of 2.3 V across the resistor R_1, establishing a high emission current of 19.17 mA. At the same time the red LED diode and

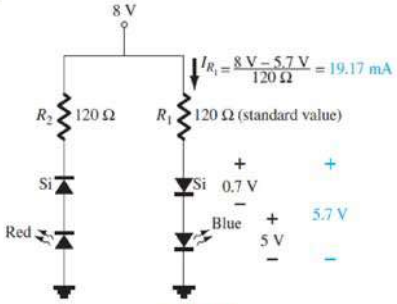


FIG. 2.34
Protective measure for the red LED of Fig. 2.33.

its series diode will also be reverse biased, but now the standard diode with a reverse breakdown voltage of 20 V will prevent the full reverse-bias voltage of 8 V from appearing across the red LED. When forward biased, the resistor R_2 will establish a current of 19.63 mA to ensure a high level of intensity for the red LED.

EXAMPLE 2.12 Determine the voltage V_o for the network of Fig. 2.35.

Solution: Initially, it might appear that the applied voltage will turn both diodes "on" because the applied voltage ("pressure") is trying to establish a conventional current through each diode that would suggest the "on" state. However, if both were on, there would be more than one voltage across the parallel diodes, violating one of the basic rules of network analysis: The voltage must be the same across parallel elements.

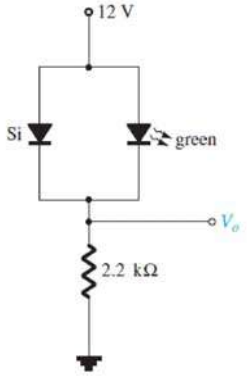


FIG. 2.35
Network for Example 2.12.

The resulting action can best be explained by remembering that there is a period of build-up of the supply voltage from 0 V to 12 V even though it may take milliseconds or microseconds. At the instant the increasing supply voltage reaches 0.7 V the silicon diode will turn "on" and maintain the level of 0.7 V since the characteristic is vertical at this voltage—the current of the silicon diode will simply rise to the defined level. The result is that the voltage across the green LED will never rise above 0.7 V and will remain in the equivalent open-circuit state as shown in Fig. 2.36.

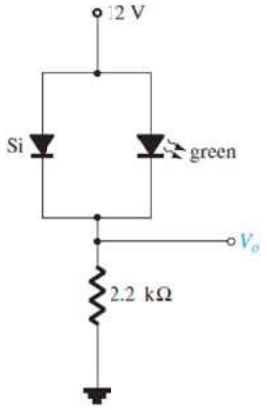
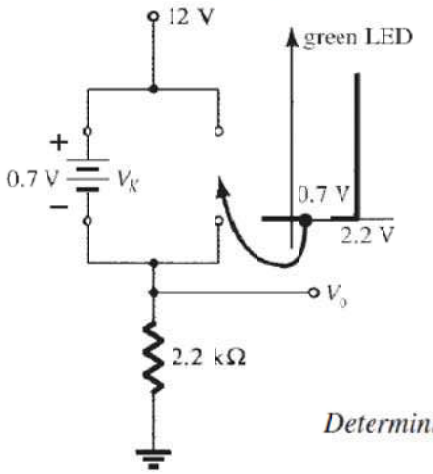


FIG. 2.35
Network for Example 2.12.

The result is



$$V_o = 12\,\text{V} - 0.7\,\text{V} = \mathbf{11.3\,V}$$

FIG. 2.36
Determining V_o for the network of Fig. 2.35.

EXAMPLE 2.13 Determine the currents I_1, I_2, and I_{D_2} for the network of Fig. 2.37.

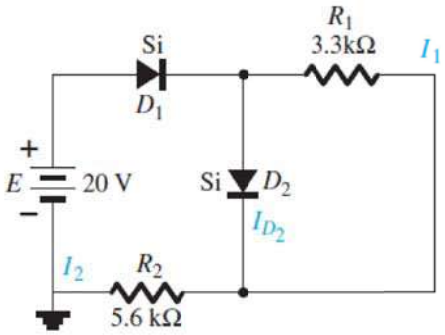


FIG. 2.37
Network for Example 2.13.

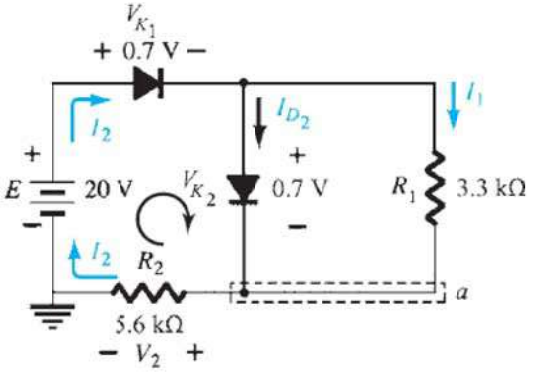


FIG. 2.38
Determining the unknown quantities for Example 2.13

Solution: The applied voltage (pressure) is such as to turn both diodes on, as indicated by the resulting current directions in the network of Fig. 2.38. Note the use of the abbreviated notation for "on" diodes and that the solution is obtained through an application of techniques applied to dc series–parallel networks. We have

$$I_1 = \frac{V_{K_2}}{R_1} = \frac{0.7\text{ V}}{3.3\text{ k}\Omega} = \mathbf{0.212\text{ mA}}$$

Applying Kirchhoff's voltage law around the indicated loop in the clockwise direction yields

$$-V_2 + E - V_{K_1} - V_{K_2} = 0$$

and

$$V_2 = E - V_{K_1} - V_{K_2} = 20\text{ V} - 0.7\text{ V} - 0.7\text{ V} = \mathbf{18.6\text{ V}}$$

with

$$I_2 = \frac{V_2}{R_2} = \frac{18.6\text{ V}}{5.6\text{ k}\Omega} = \mathbf{3.32\text{ mA}}$$

At the bottom node *a*,

$$I_{D_2} + I_1 = I_2$$

and

$$I_{D_2} = I_2 - I_1 = 3.32\text{ mA} - 0.212\text{ mA} \cong \mathbf{3.11\text{ mA}}$$

193

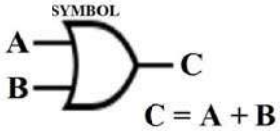
2.5 AND/OR GATES

The tools of analysis are now at our disposal, and the opportunity to investigate a computer configuration is one that will demonstrate the range of applications of this relatively simple device. Our analysis will be limited to determining the voltage levels and will not include a detailed discussion of Boolean algebra or positive and negative logic.

The network to be analyzed in Example 2.14 is an OR gate for positive logic. That is, the 10-V level of Fig. 2.39 is assigned a "1" for Boolean algebra and the 0-V input is assigned a "0." An OR gate is such that the output voltage level will be a 1 if either *or* both inputs is a 1. The output is a 0 if both inputs are at the 0 level.

OR Gate

SYMBOL



C = A + B

TRUTH TABLE

INPUT		OUTPUT
A	B	A OR B
0	0	0
0	1	1
1	0	1
1	1	1

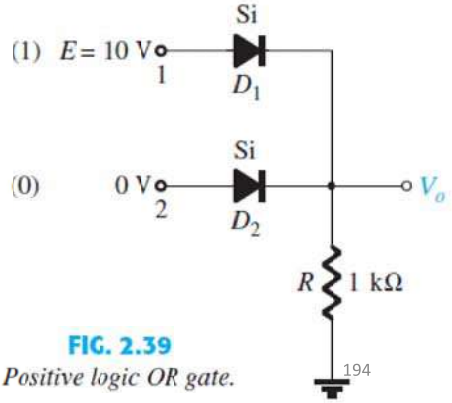


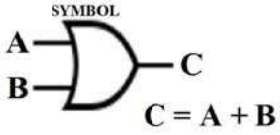
FIG. 2.39
Positive logic OR gate.

194

2.5 AND/OR GATES

The analysis of AND/OR gates is made easier by using the approximate equivalent for a diode rather than the ideal because we can stipulate that the voltage across the diode must be 0.7 V positive for the silicon diode to switch to the "on" state.

OR Gate



TRUTH TABLE

INPUT		OUTPUT
A	B	A OR B
0	0	0
0	1	1
1	0	1
1	1	1

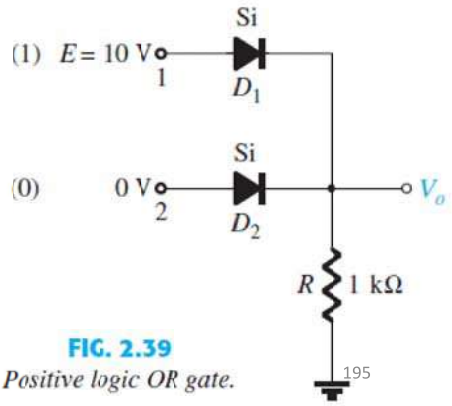


FIG. 2.39
Positive logic OR gate.

EXAMPLE 2.14 Determine V_o for the network of Fig. 2.39.

Solution: First note that there is only one applied potential; 10 V at terminal 1. Terminal 2 with a 0-V input is essentially at ground potential, as shown in the redrawn network of Fig. 2.40. Figure 2.40 "suggests" that D_1 is probably in the "on" state due to the applied 10 V, whereas D_2 with its "positive" side at 0 V is probably "off." Assuming these states will result in the configuration of Fig. 2.41.

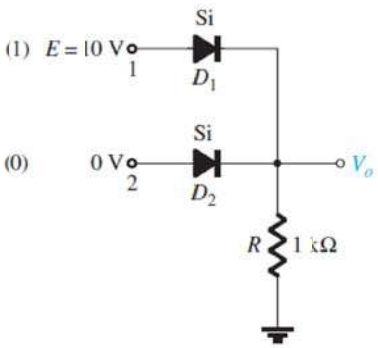


FIG. 2.39
Positive logic OR gate.

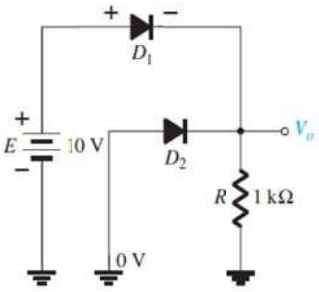


FIG. 2.40
Redrawn network of Fig. 2.39.

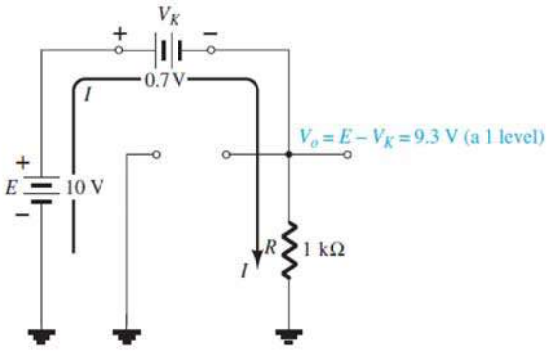
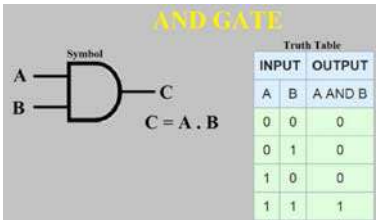


FIG. 2.41
Assumed diode states for Fig. 2.40.

EXAMPLE 2.15 Determine the output level for the positive logic AND gate of Fig. 2.42. An AND gate is one where a 1 output is only obtained when a 1 input appears at each and every input.

Solution: Note in this case that an independent source appears in the grounded leg of the network. For reasons soon to become obvious, it is chosen at the same level as the input logic level. The network is redrawn in Fig. 2.43 with our initial assumptions regarding the state of the diodes. With 10 V at the cathode side of D_1 it is assumed that D_1 is in the "off" state even though there is a 10-V source connected to the anode of D_1 through the resistor.



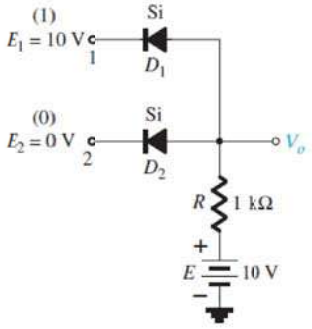


FIG. 2.42
Positive logic AND gate.

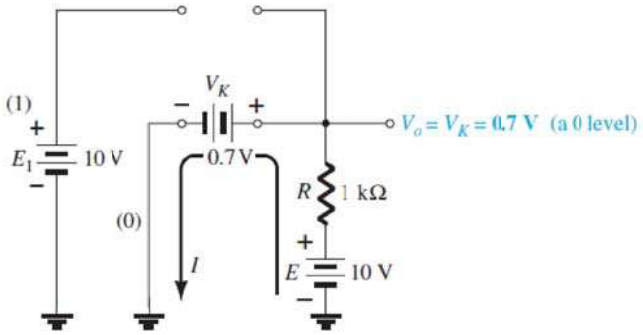
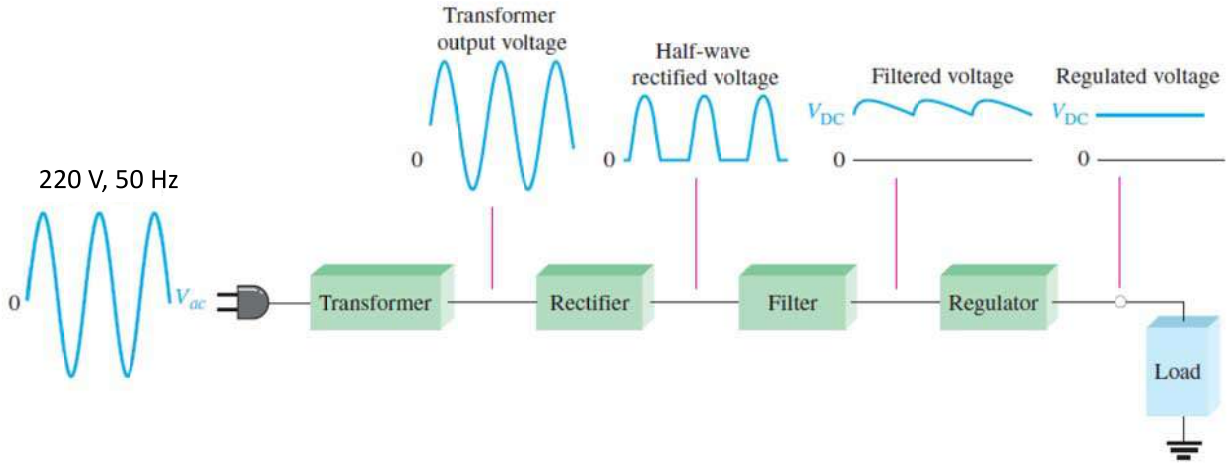


FIG. 2.43
Substituting the assumed states for the diodes of Fig. 2.42.

2.6 SINUSOIDAL INPUTS; HALF-WAVE RECTIFICATION



(a) Complete power supply with transformer, rectifier, filter, and regulator

2.6 SINUSOIDAL INPUTS; HALF-WAVE RECTIFICATION

The diode analysis will now be expanded to include time-varying functions such as the sinusoidal waveform and the square wave. There is no question that the degree of difficulty will increase, but once a few fundamental maneuvers are understood, the analysis will be fairly direct and follow a common thread.

The simplest of networks to examine with a time-varying signal appears in Fig. 2.44. For the moment we will use the ideal model (note the absence of the Si, Ge, or GaAs label) to ensure that the approach is not clouded by additional mathematical complexity.

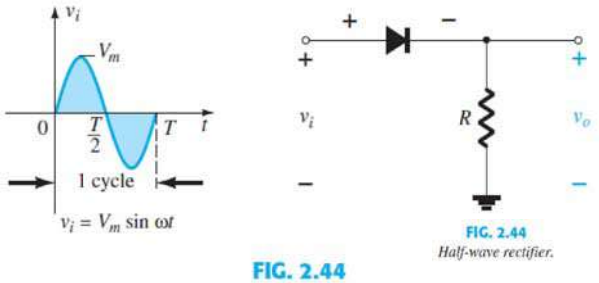


FIG. 2.44
Half-wave rectifier.

FIG. 2.44
Half-wave rectifier.

2.6 SINUSOIDAL INPUTS; HALF-WAVE RECTIFICATION

Over one full cycle, defined by the period T of Fig. 2.44, the average value (the algebraic sum of the areas above and below the axis) is zero. The circuit of Fig. 2.44, called a *half-wave rectifier*, will generate a waveform v_o that will have an average value of particular use in the ac-to-dc conversion process. When employed in the rectification process, a diode is typically referred to as a *rectifier*. Its power and current ratings are typically much higher than those of diodes employed in other applications, such as computers and communication systems.

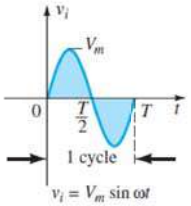


FIG. 2.44
Half-wave rectifier.

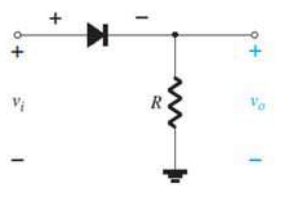
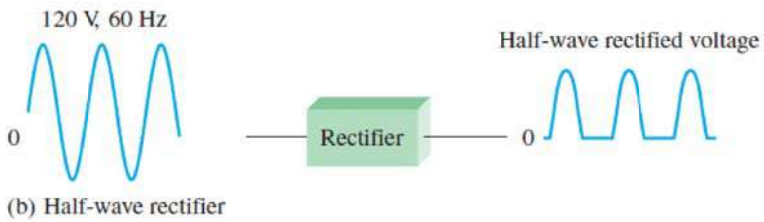


FIG. 2.44
Half-wave rectifier.



(b) Half-wave rectifier

During the interval $t = 0 \rightarrow T/2$ in Fig. 2.44 the polarity of the applied voltage v_i is such as to establish "pressure" in the direction indicated and turn on the diode with the polarity appearing above the diode. Substituting the short-circuit equivalence for the ideal diode will result in the equivalent circuit of Fig. 2.45, where it is fairly obvious that the output signal is an exact replica of the applied signal. The two terminals defining the output voltage are connected directly to the applied signal via the short-circuit equivalence of the diode.

For the period $T/2 \rightarrow T$, the polarity of the input v_i is as shown in Fig. 2.46, and the resulting polarity across the ideal diode produces an "off" state with an open-circuit equivalent. The result is the absence of a path for charge to flow, and $v_o = iR = (0)R = 0$ V for the period $T/2 \rightarrow T$. for

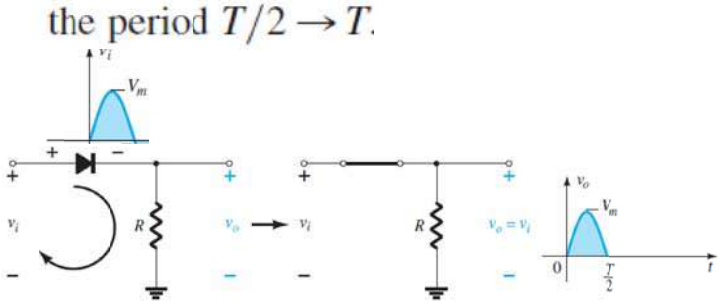


FIG. 2.45
Conduction region ($0 \rightarrow T/2$).

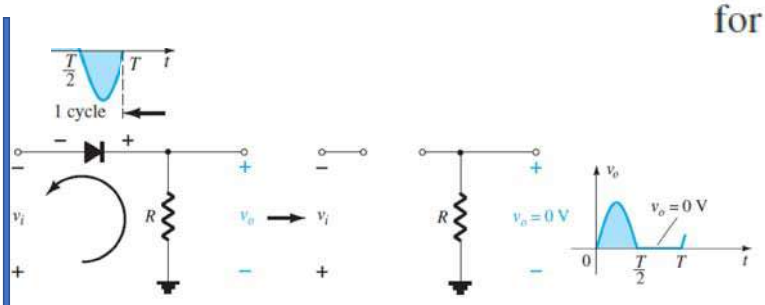


FIG. 2.46
Nonconduction region ($T/2 \rightarrow T$).

The input v_i and the output v_o are sketched together in Fig. 2.47 for comparison purposes. The output signal v_o now has a net positive area above the axis over

$f_{in} = 1/T$
$f_{out} = 1/T$
$f_{out} = f_{in}$

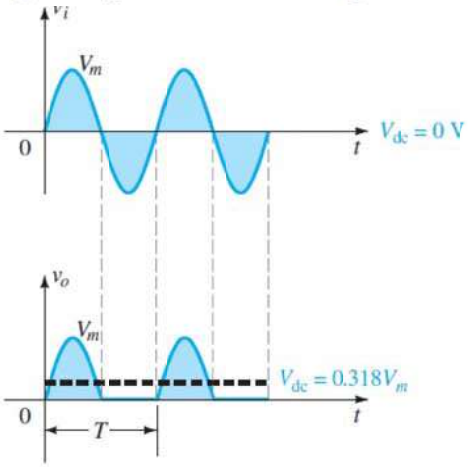


FIG. 2.47

a full period and an average value determined by

$V_{DC} = V_m/\pi$

$$V_{dc} = 0.318\,V_m \quad \text{half-wave} \tag{2.7}$$

The process of removing one-half the input signal to establish a dc level is called *half-wave rectification*.

The effect of using a silicon diode with $V_K = 0.7$ V is demonstrated in Fig. 2.48 for the forward-bias region. The applied signal must now be at least 0.7 V before the diode can turn "on." For levels of v_i less than 0.7 V, the diode is still in an open-circuit state and $v_o = 0$ V, as shown in the same figure. When conducting, the difference between v_o and v_i is a fixed

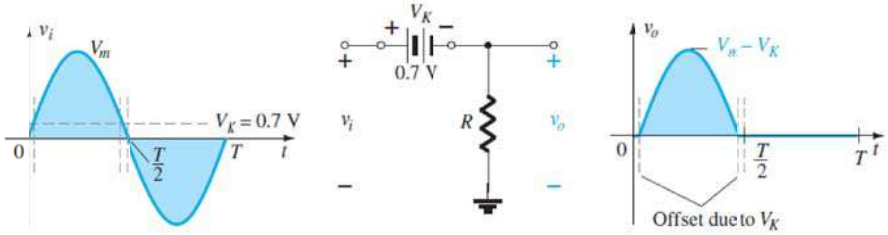


FIG. 2.48
Effect of V_K on half-wave rectified signal.

level of $V_K = 0.7$ V and $v_o = v_i - V_K$, as shown in the figure. The net effect is a reduction in area above the axis, which reduces the resulting dc voltage level. For situations where $V_m \gg V_K$, the following equation can be applied to determine the average value with a relatively high level of accuracy.

$V_{DC} = (V_m - 0.7)/\pi$

$$V_{dc} \cong 0.318(V_m - V_K) \qquad \textbf{(2.8)}$$

In fact, if V_m is sufficiently greater than V_K, Eq. (2.7) is often applied as a first approximation for V_{dc}.

EXAMPLE 2.16

a. Sketch the output v_o and determine the dc level of the output for the network of Fig. 2.49.
b. Repeat part (a) if the ideal diode is replaced by a silicon diode.
c. Repeat parts (a) and (b) if V_m is increased to 200 V, and compare solutions using Eqs. (2.7) and (2.8).

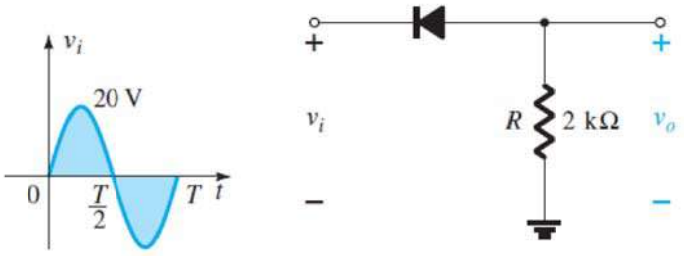


FIG. 2.49
Network for Example 2.16.

Solution:

a. In this situation the diode will conduct during the negative part of the input as shown in Fig. 2.50, and v_o will appear as shown in the same figure. For the full period, the dc level is

$$V_{dc} = -0.318V_m = -0.318(20\text{ V}) = \mathbf{-6.36\text{ V}}$$

The negative sign indicates that the polarity of the output is opposite to the defined polarity of Fig. 2.49.

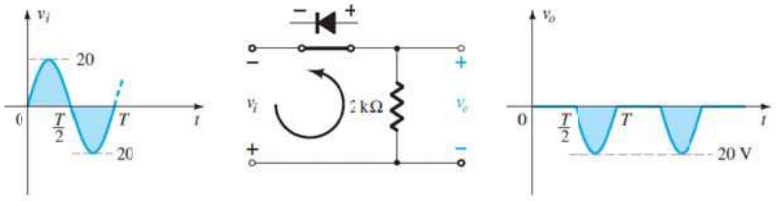


FIG. 2.50
Resulting v_o for the circuit of Example 2.16.

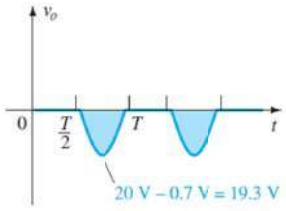


FIG. 2.51
Effect of V_K on output of Fig. 2.50.

Solution:

b. For a silicon diode, the output has the appearance of Fig. 2.51, and

$$V_{dc} \cong -0.318(V_m - 0.7\text{ V}) = -0.318(19.3\text{ V}) \cong \mathbf{-6.14\text{ V}}$$

The resulting drop in dc level is 0.22 V, or about 3.5%.

c. Eq. (2.7): $V_{dc} = -0.318\,V_m = -0.318(200\text{ V}) = \mathbf{-63.6\text{ V}}$

Eq. (2.8): $V_{dc} = -0.318(V_m - V_K) = -0.318(200\text{ V} - 0.7\text{ V})$

$$= -(0.318)(199.3\text{ V}) = \mathbf{-63.38\text{ V}}$$

which is a difference that can certainly be ignored for most applications. For part (c) the offset and drop in amplitude due to V_K would not be discernible on a typical oscilloscope if the full pattern is displayed.

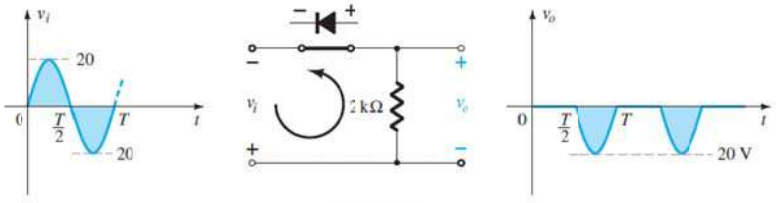


FIG. 2.50
Resulting v_o for the circuit of Example 2.16.

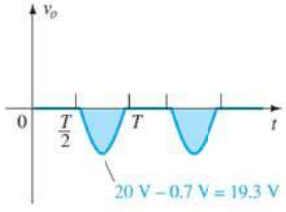


FIG. 2.51
Effect of V_K on output of Fig. 2.50.

PIV (PRV)

The peak inverse voltage (PIV) [or PRV (peak reverse voltage)] rating of the diode is of primary importance in the design of rectification systems. Recall that it is the voltage rating that must not be exceeded in the reverse-bias region or the diode will enter the Zener avalanche region. The required PIV rating for the half-wave rectifier can be determined from Fig. 2.52, which displays the reverse-biased diode of Fig. 2.44 with maximum applied voltage. Applying Kirchhoff's voltage law, it is fairly obvious that the PIV rating of the diode must equal or exceed the peak value of the applied voltage. Therefore,

$$\text{PIV rating} \geqq V_m \quad \text{half-wave rectifier} \tag{2.9}$$

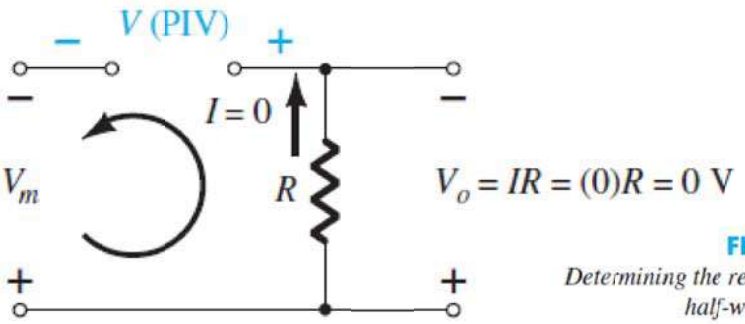


FIG. 2.52
Determining the required PIV rating for the half-wave rectifier.

2.7 FULL-WAVE RECTIFICATION

Bridge Network

The dc level obtained from a sinusoidal input can be improved 100% using a process called *full-wave rectification*. The most familiar network for performing such a function appears in Fig. 2.53 with its four diodes in a *bridge* configuration.

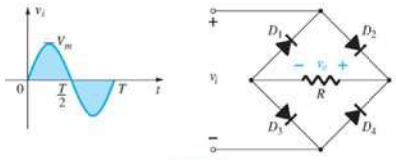


FIG. 2.53
Full-wave bridge rectifier.

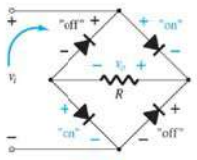
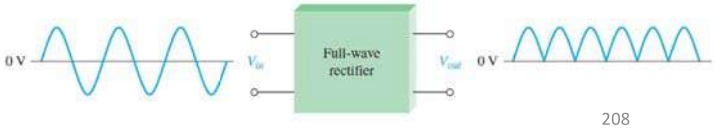


FIG. 2.54
Network of Fig. 2.53 for the period $0 \rightarrow T/2$ of the input voltage v_i.



2.7 FULL-WAVE RECTIFICATION

Bridge Network

During the period $t = 0$ to $T/2$ the polarity of the input is as shown in Fig. 2.54. The resulting polarities across the ideal diodes are also shown in Fig. 2.54 to reveal that D_2 and D_3 are conducting, whereas D_1 and D_4 are in the "off" state. The net result is the configuration of Fig. 2.55, with its indicated current and polarity across R. Since the diodes are ideal, the load voltage is $v_o = v_i$, as shown in the same figure.

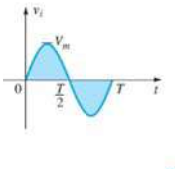


FIG. 2.53
Full-wave bridge rectifier.

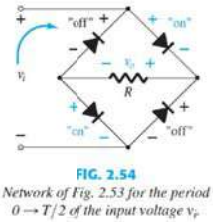


FIG. 2.54
Network of Fig. 2.53 for the period $0 \rightarrow T/2$ of the input voltage v_i.

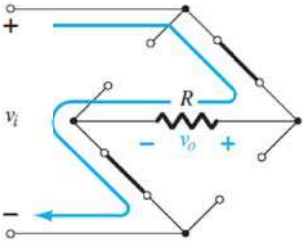
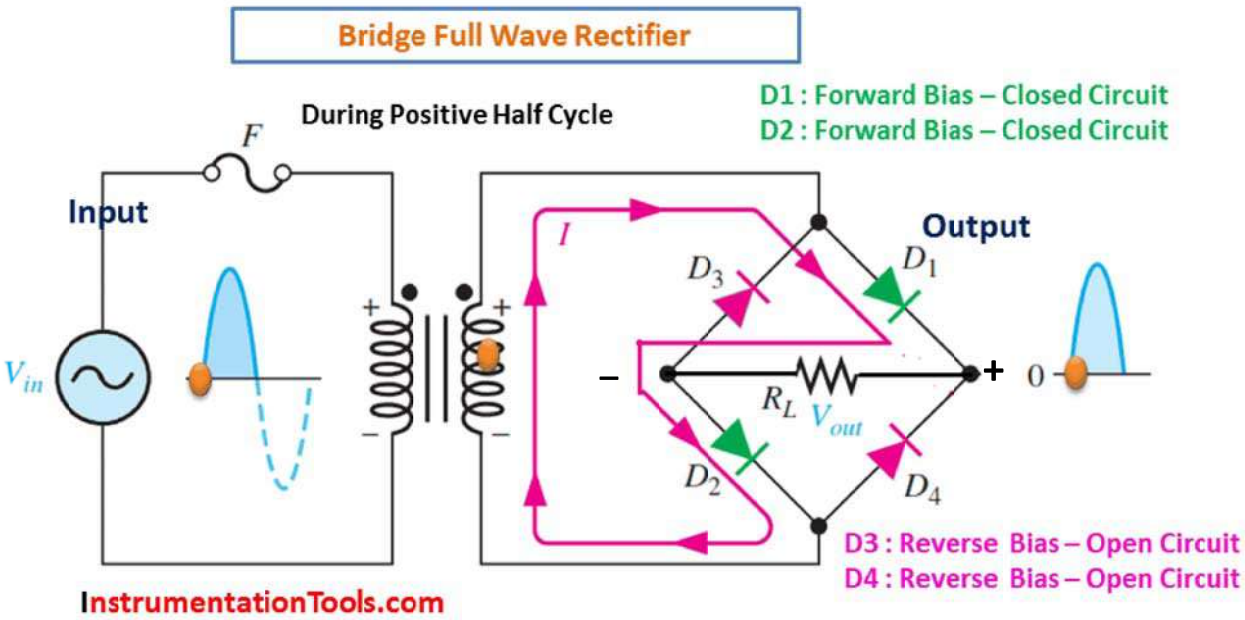


FIG. 2.55



For the negative region of the input the conducting diodes are D_1 and D_4, resulting in the configuration of Fig. 2.56. The important result is that the polarity across the load resistor R is the same as in Fig. 2.54, establishing a second positive pulse, as shown in Fig. 2.56. Over one full cycle the input and output voltages will appear as shown in Fig. 2.57.

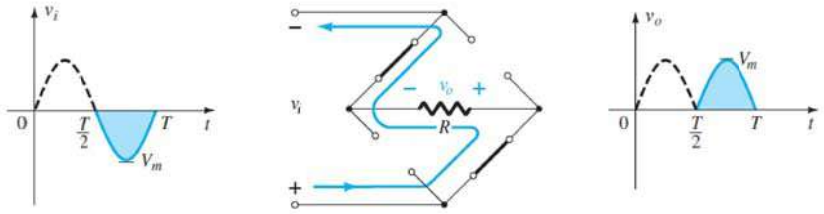
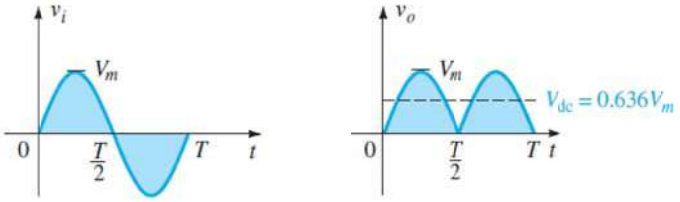
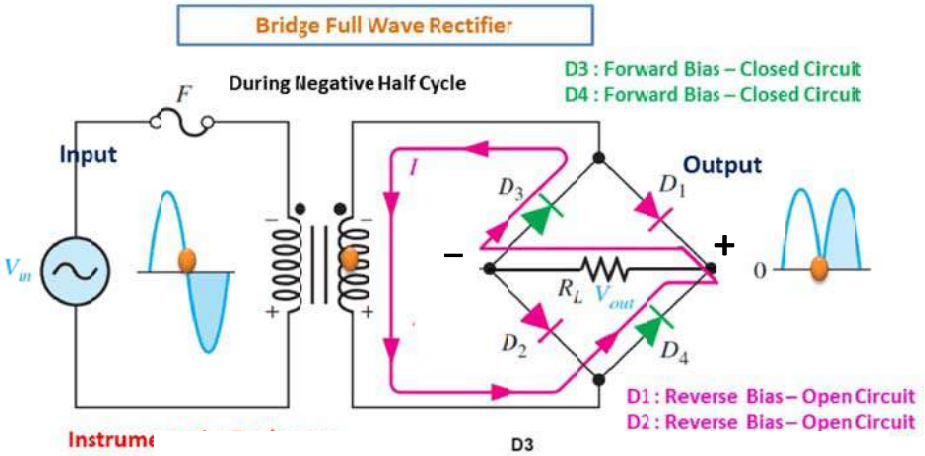


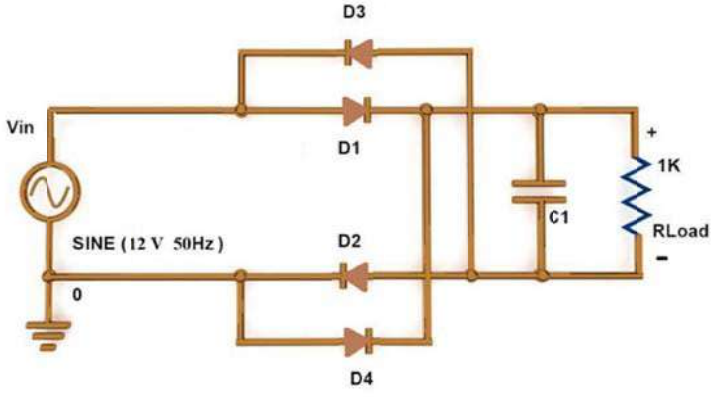
FIG. 2.56
Conduction path for the negative region of v_i.



f_{in} =1/T
f_{out} = 1/(T/2)
f_{out}=2f_{in}

FIG. 2.57
Input and output waveforms for a full-wave rectifier.





Since the area above the axis for one full cycle is now twice that obtained for a half-wave system, the dc level has also been doubled and

$V_{AVE} = \frac{2V_p}{\pi} = 0.637V_p$ $\qquad V_{dc} = 2[\text{Eq. (2.7)}] = 2(0.318V_m)$

or

$$\boxed{V_{dc} = 0.636\,V_m} \quad \text{full-wave} \qquad \textbf{(2.10)}$$

If silicon rather than ideal diodes are employed as shown in Fig. 2.58, the application of Kirchhoff's voltage law around the conduction path results in

$$v_i - V_K - v_o - V_K = 0$$

and

$$v_o = v_i - 2V_K$$

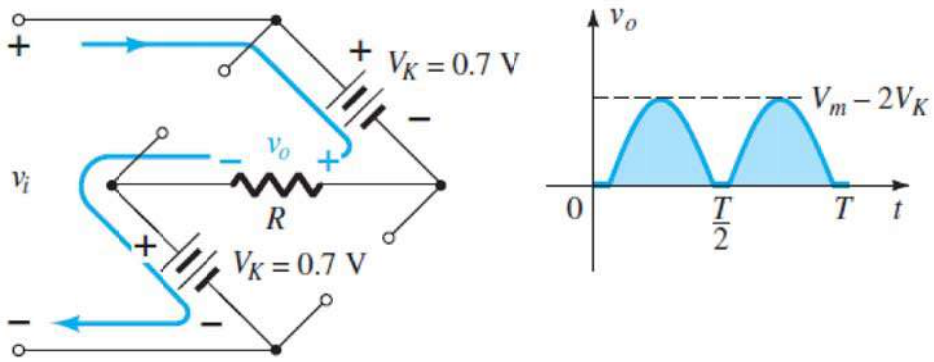


FIG. 2.58

Determining $V_{o_{max}}$ for silicon diodes in the bridge configuration.

The peak value of the output voltage v_o is therefore

$$V_{o_{\max}} = V_m - 2V_K$$

For situations where $V_m \gg 2V_K$, the following equation can be applied for the average value with a relatively high level of accuracy:

$$V_{dc} \cong 0.636(V_m - 2V_K) \qquad \textbf{(2.11)}$$

Then again, if V_m is sufficiently greater than $2V_K$, then Eq. (2.10) is often applied as a first approximation for V_{dc}.

PIV The required PIV of each diode (ideal) can be determined from Fig. 2.59 obtained at the peak of the positive region of the input signal. For the indicated loop the maximum voltage across R is V_m and the PIV rating is defined by

$$\text{PIV} \geqq V_m$$

full-wave bridge rectifier

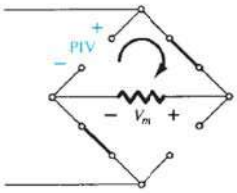
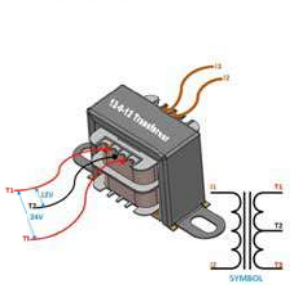


FIG. 2.59
Determining the required PIV for the bridge configuration.

For practical diodes
PIV >= Vm +V_k or
Vi -0.7 V (Vi -1.4 +0.7 V)

Center-Tapped Transformer

A second popular full-wave rectifier appears in Fig. 2.60 with only two diodes but requiring a center-tapped (CT) transformer to establish the input signal across each section of the secondary of the transformer. During the positive portion of v_i applied to the primary of the transformer, the network will appear as shown in Fig. 2.61 with a positive pulse across each section of the secondary coil. D_1 assumes the short-circuit equivalent and D_2 the open-circuit equivalent, as determined by the secondary voltages and the resulting current directions. The output voltage appears as shown in Fig. 2.61.



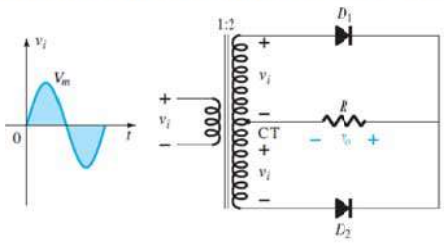


FIG. 2.60
Center-tapped transformer full-wave rectifier.

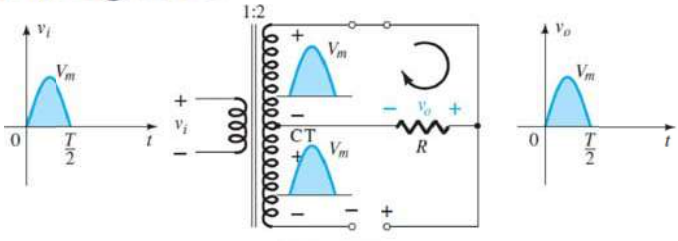


FIG. 2.61
Network conditions for the positive region of v_i.

During the negative portion of the input the network appears as shown in Fig. 2.62, reversing the roles of the diodes but maintaining the same polarity for the voltage across the load resistor R. The net effect is the same output as that appearing in Fig. 2.57 with the same dc levels.

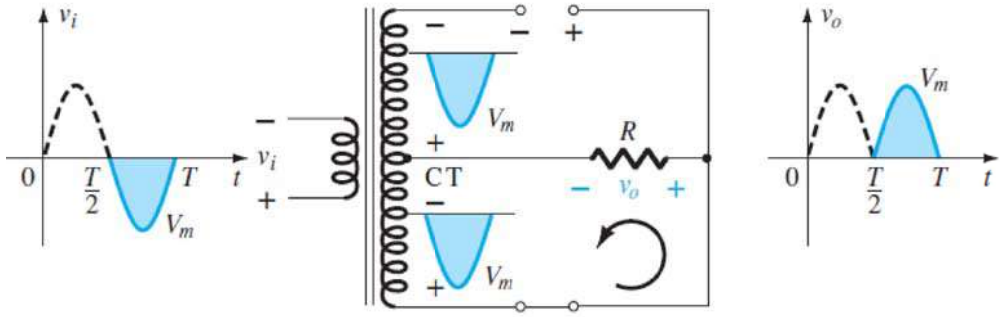


FIG. 2.62
Network conditions for the negative region of v_i.

PIV The network of Fig. 2.63 will help us determine the net PIV for each diode for this full-wave rectifier. Inserting the maximum voltage for the secondary voltage and V_m as established by the adjoining loop results in

$$\text{PIV} = V_{\text{secondary}} + V_R$$
$$= V_m + V_m$$

and

$$\boxed{\text{PIV} \geqq 2V_m} \quad \text{CT transformer, full-wave rectifier} \qquad (2.13)$$

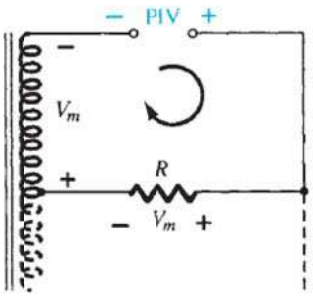
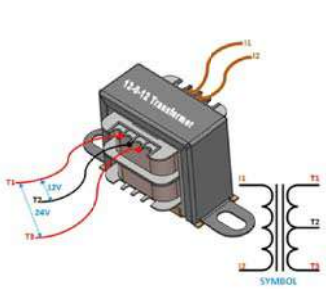
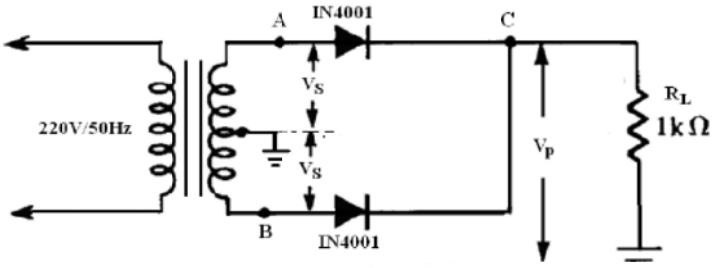


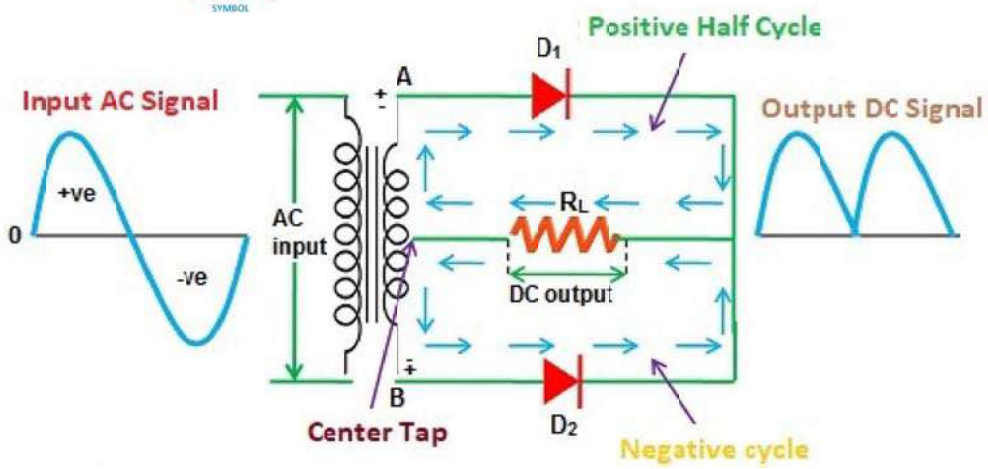
FIG. 2.63
Determining the PIV level for the diodes of the CT transformer full-wave rectifier.

For practical diodes
PIV >= 2Vm +V_k or
2Vi -0.7 V ([2(Vi -0.7)] +0.7) V

- **Center-Tapped Full-wave Rectifier**

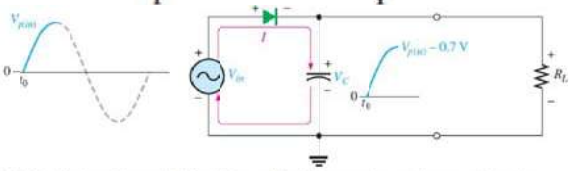




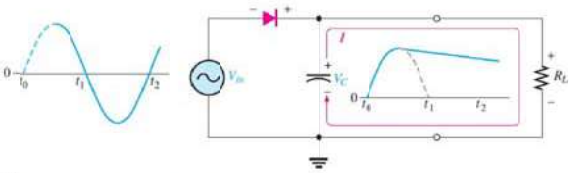


Capacitor-Input Filter

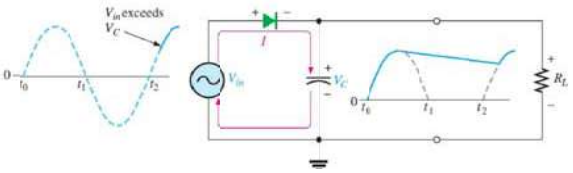
A half-wave rectifier with a capacitor-input filter is shown in Figure 2–43. The filter is simply a capacitor connected from the rectifier output to ground. R_L represents the equivalent resistance of a load. We will use the half-wave rectifier to illustrate the basic principle and then expand the concept to full-wave rectification.

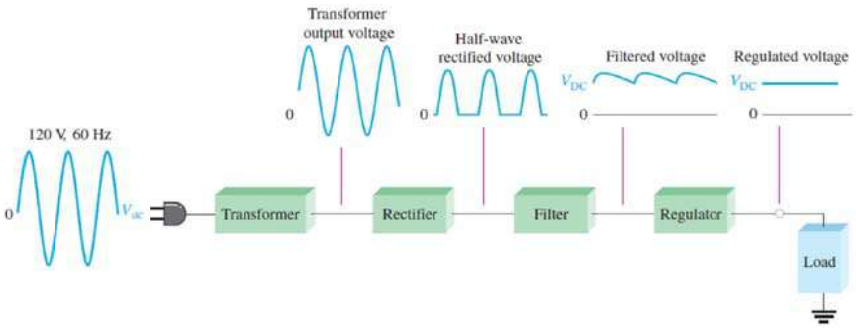


(a) Initial charging of the capacitor (diode is forward-biased) happens only once when power is turned on.



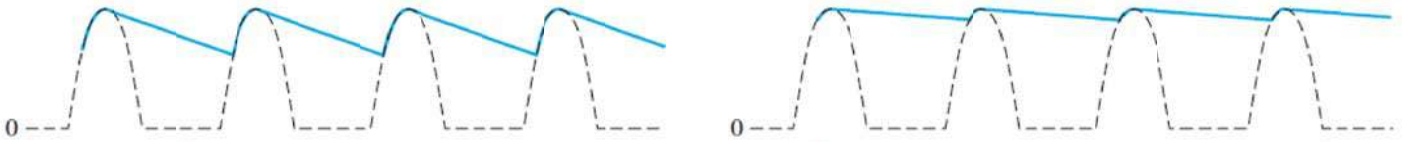
(b) The capacitor discharges through R_L after peak of positive alternation when the diode is reverse-biased. This discharging occurs during the portion of the input voltage indicated by the solid dark blue curve.





(a) Complete power supply with transformer, rectifier, filter, and regulator

Ripple Voltage As you have seen, the capacitor quickly charges at the beginning of a cycle and slowly discharges through R_L after the positive peak of the input voltage (when the diode is reverse-biased). The variation in the capacitor voltage due to the charging and discharging is called the **ripple voltage**. Generally, ripple is undesirable; thus, the smaller the ripple, the better the filtering action, as illustrated in Figure 2–44.

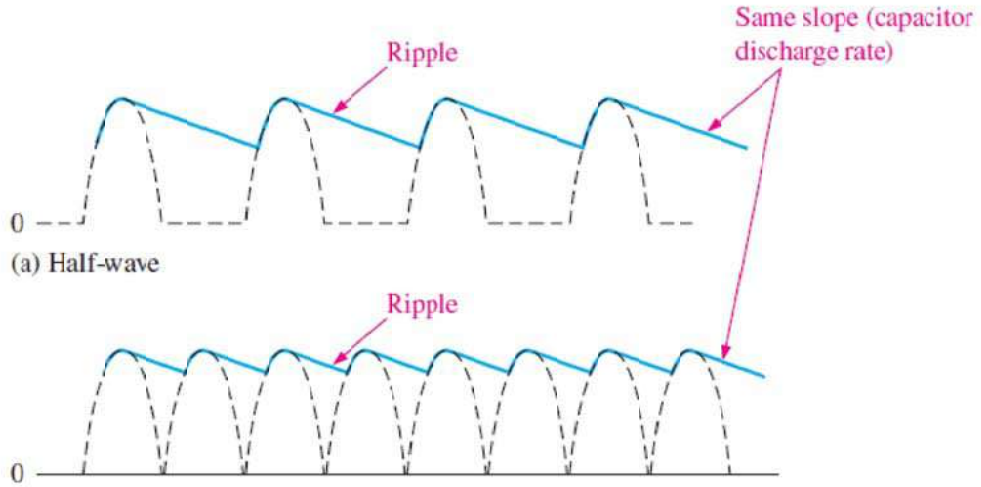


(a) Larger ripple (blue) means less effective filtering.

(b) Smaller ripple means more effective filtering. Generally, the larger the capacitor value, the smaller the ripple for the same input and load.

▲ FIGURE 2–44

Half-wave ripple voltage (blue line).

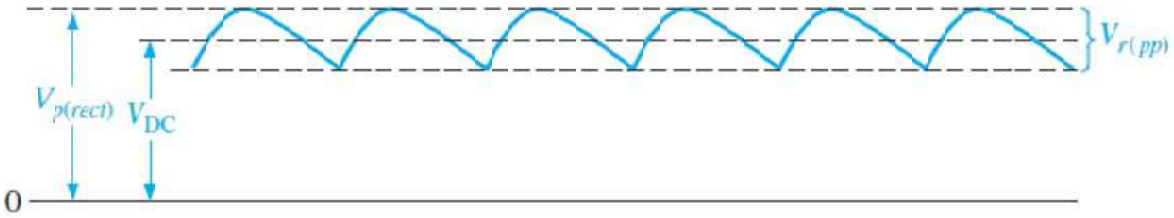


Ripple Factor The **ripple factor** (r) is an indication of the effectiveness of the filter and is defined as

$$r = \frac{V_{r(pp)}}{V_{DC}}$$

$$r = \frac{V_{r(pp)}}{V_{DC}}$$

where $V_{r(pp)}$ is the peak-to-peak ripple voltage and V_{DC} is the dc (average) value of the filter's output voltage, as illustrated in Figure 2–47. The lower the ripple factor, the better the filter. The ripple factor can be lowered by increasing the value of the filter capacitor or increasing the load resistance.



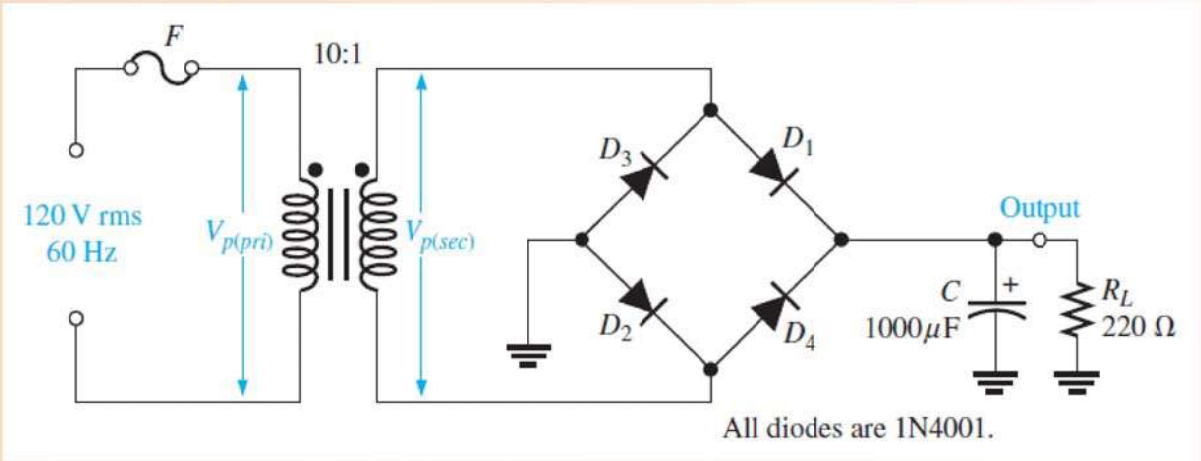
For a full-wave rectifier with a capacitor-input filter, approximations for the peak-to-peak ripple voltage, $V_{r(pp)}$, and the dc value of the filter output voltage, V_{DC}, are given in the following equations. The variable $V_{p(rect)}$ is the unfiltered peak rectified voltage. Notice that if R_L or C increases, the ripple voltage decreases and the dc voltage increases.

$$V_{r(pp)} \cong \left(\frac{1}{fR_LC}\right)V_{p(rect)}$$

$$V_{DC} \cong \left(1 - \frac{1}{2fR_LC}\right)V_{p(rect)}$$

Example

Determine the ripple factor for the filtered bridge rectifier with a load as indicated in Figure 2–48.



Solution The transformer turns ratio is $n = 0.1$. The peak primary voltage is

$$V_{p(pri)} = 1.414V_{rms} = 1.414(120\text{ V}) = 170\text{ V}$$

The peak secondary voltage is

$$V_{p(sec)} = nV_{p(pri)} = 0.1(170\text{ V}) = 17.0\text{ V}$$

The unfiltered peak full-wave rectified voltage is

$$V_{p(rect)} = V_{p(sec)} - 1.4\text{ V} = 17.0\text{ V} - 1.4\text{ V} = 15.6\text{ V}$$

The frequency of a full-wave rectified voltage is 120 Hz. The approximate peak-to-peak ripple voltage at the output is

$$V_{r(pp)} \cong \left(\frac{1}{fR_LC}\right)V_{p(rect)} = \left(\frac{1}{(120\text{ Hz})(220\ \Omega)(1000\ \mu\text{F})}\right)15.6\text{ V} = 0.591\text{ V}$$

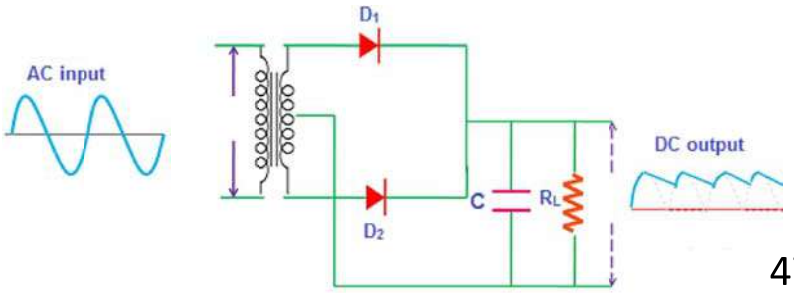
The approximate dc value of the output voltage is determined as follows:

$$V_{DC} = \left(1 - \frac{1}{2fR_LC}\right)V_{p(rect)} = \left(1 - \frac{1}{(240\text{ Hz})(220\ \Omega)(1000\ \mu\text{F})}\right)15.6\text{ V} = 15.3\text{ V}$$

The resulting ripple factor is

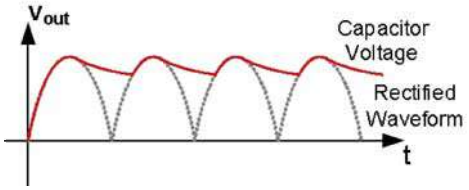
$$r = \frac{V_{r(pp)}}{V_{DC}} = \frac{0.591\text{ V}}{15.3\text{ V}} = \mathbf{0.039}$$

The percent ripple is 3.9%.

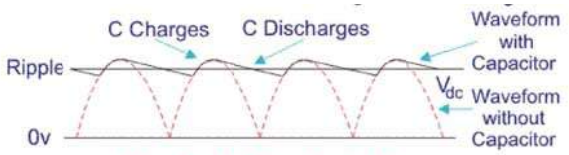


Full wave rectifier with capacitor filter

47µF

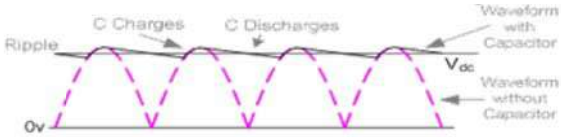


100µF



Diode Bridge Rectifier

200µF



EXAMPLE 2.17 Determine the output waveform for the network of Fig. 2.64 and calculate the output dc level and the required PIV of each diode.

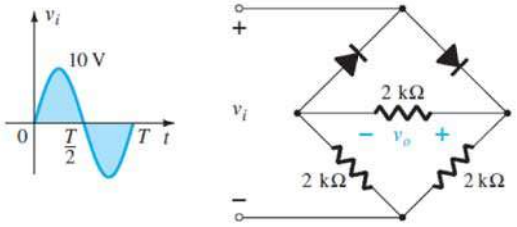


FIG. 2.64
Bridge network for Example 2.17.

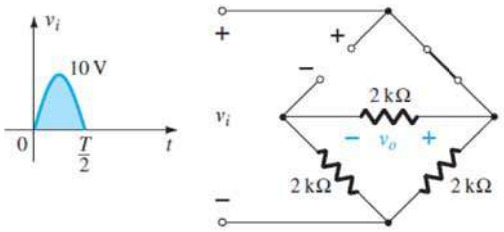


FIG. 2.65
Network of Fig. 2.64 for the positive region of v_i.

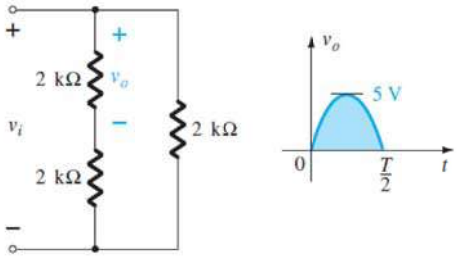


FIG. 2.66
Redrawn network of Fig. 2.65.

EXAMPLE 2.17 Determine the output waveform for the network of Fig. 2.64 and calculate the output dc level and the required PIV of each diode.

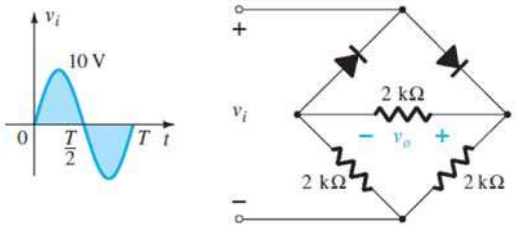


FIG. 2.64
Bridge network for Example 2.17.

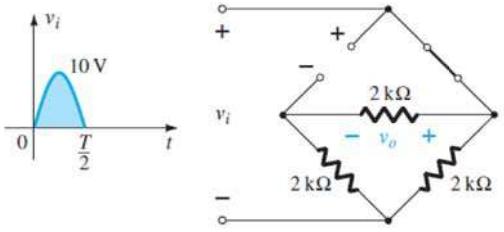


FIG. 2.65
Network of Fig. 2.64 for the positive region of v_i.

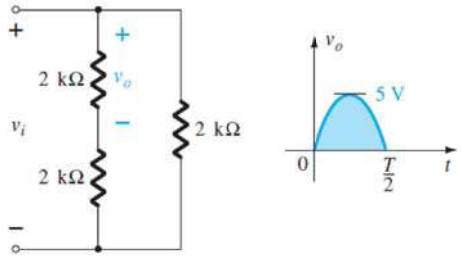


FIG. 2.66
Redrawn network of Fig. 2.65.

Solution: The network appears as shown in Fig. 2.65 for the positive region of the input voltage. Redrawing the network results in the configuration of Fig. 2.66, where $v_o = \frac{1}{2}v_i$ or $V_{o_{max}} = \frac{1}{2}V_{i_{max}} = \frac{1}{2}(10\text{ V}) = 5\text{ V}$, as shown in Fig. 2.66. For the negative part of the input, the roles of the diodes are interchanged and v_o appears as shown in Fig. 2.67.

The effect of removing two diodes from the bridge configuration is therefore to reduce the available dc level to the following:

$$V_{dc} = 0.636(5\text{ V}) = \mathbf{3.18\text{ V}}$$

the available dc level to the following:

$$V_{dc} = 0.636(5\text{ V}) = \mathbf{3.18\text{ V}}$$

or that available from a half-wave rectifier with the same input. However, the PIV as determined from Fig. 2.59 is equal to the maximum voltage across R, which is 5 V, or half of that required for a half-wave rectifier with the same input.

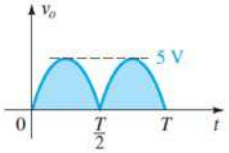


FIG. 2.67
Resulting output for Example 2.17.

2.8 CLIPPERS

The previous section on rectification gives clear evidence that diodes can be used to change the appearance of an applied waveform. This section on clippers and the next on clampers will expand on the wave-shaping abilities of diodes.

Clippers are networks that employ diodes to "clip" away a portion of an input signal without distorting the remaining part of the applied waveform.

The half-wave rectifier of Section 2.6 is an example of the simplest form of diode clipper—one resistor and a diode. Depending on the orientation of the diode, the positive or negative region of the applied signal is "clipped" off.

There are two general categories of clippers: *series* and *parallel*. The series configuration is defined as one where the diode is in series with the load, whereas the parallel variety has the diode in a branch parallel to the load.

Series

The response of the series configuration of Fig. 2.68a to a variety of alternating waveforms is provided in Fig. 2.68b. Although first introduced as a half-wave rectifier (for sinusoidal waveforms), there are no boundaries on the type of signals that can be applied to a clipper.

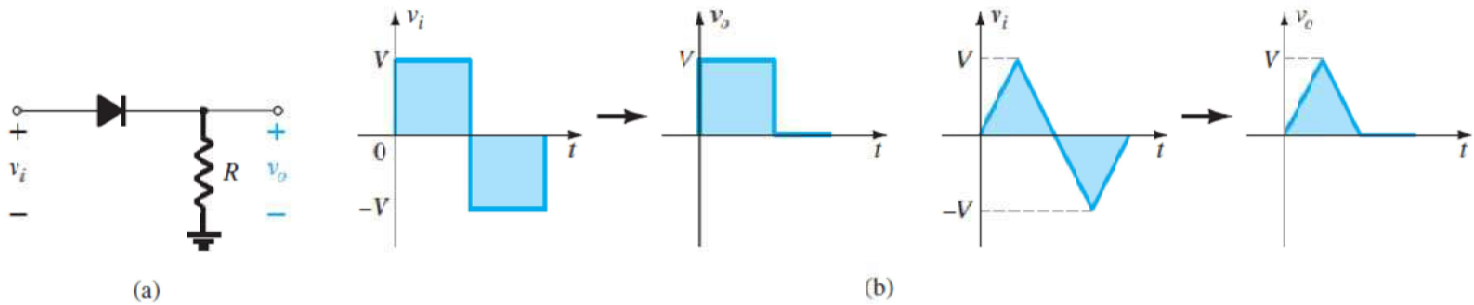


FIG. 2.68
Series clipper.

The addition of a dc supply to the network as shown in Fig. 2.69 can have a pronounced effect on the analysis of the series clipper configuration. The response is not as obvious because the dc supply can aid or work against the source voltage, and the dc supply can be in the leg between the supply and output or in the branch parallel to the output.

There is no general procedure for analyzing networks such as the type in Fig. 2.69, but there are some things one can do to give the analysis some direction.

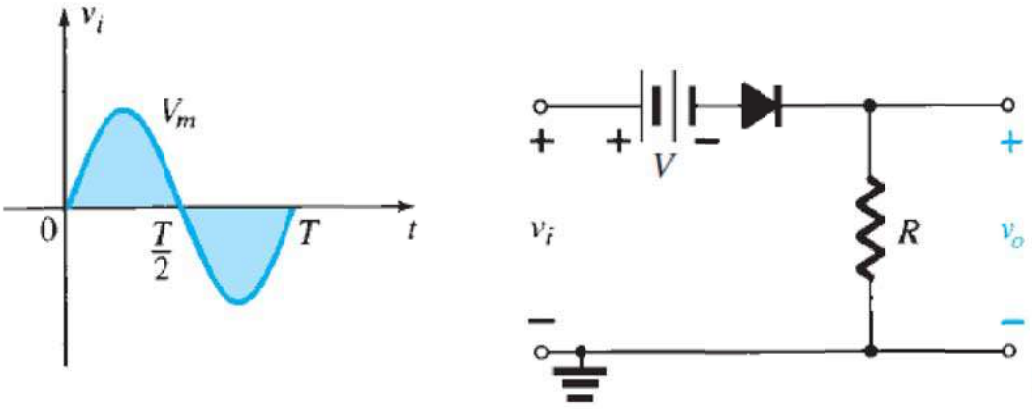


FIG. 2.69
Series clipper with a dc supply.

First and most important:

1. **Take careful note of where the output voltage is defined.**

In Fig. 2.69 it is directly across the resistor R. In some cases it may be across a combination of series elements.

Next:

2. **Try to develop an overall sense of the response by simply noting the "pressure" established by each supply and the effect it will have on the conventional current direction through the diode.**

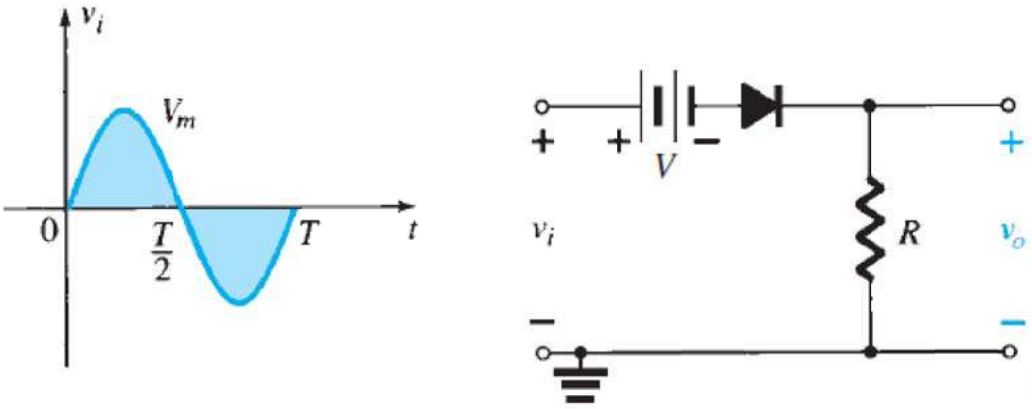


FIG. 2.69
Series clipper with a dc supply.

In Fig. 2.69, for instance, any positive voltage of the supply will try to turn the diode on by establishing a conventional current through the diode that matches the arrow in the diode symbol. However, the added dc supply V will oppose that applied voltage and try to keep the diode in the "off" state. The result is that any supply voltage greater than V volts will turn the diode on and conduction can be established through the load resistor. Keep in mind that we are dealing with an ideal diode for the moment, so the turn-on voltage is simply 0 V. In general, therefore, for the network of Fig. 2.69 we can conclude that the diode will be on for any voltage v_i that is greater than V volts and off for any lesser voltage. For the "off" condition, the output would be 0 V due to the lack of current, and for the "on" condition it would simply be $v_o = v_i - V$ as determined by Kirchhoff's voltage law.

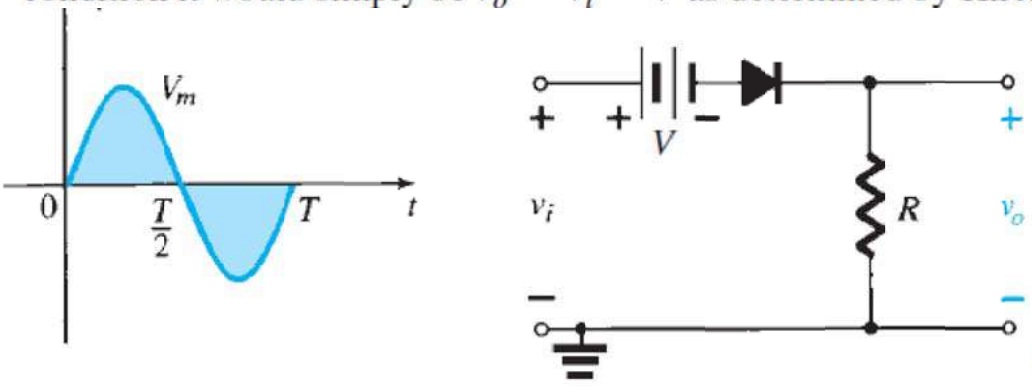
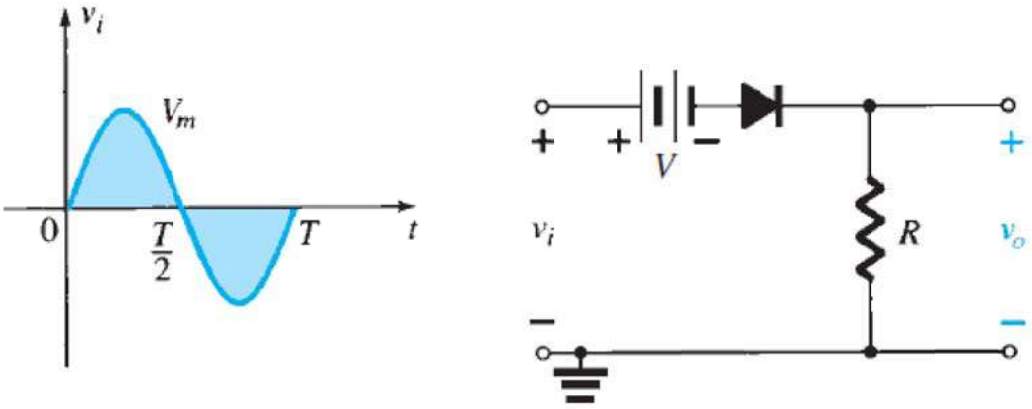


FIG. 2.69
Series clipper with a dc supply.

3. Determine the applied voltage (transition voltage) that will result in a change of state for the diode from the "off" to the "on" state.

This step will help to define a region of the applied voltage when the diode is on and when it is off. On the characteristics of an ideal diode this will occur when $V_D = 0$ V and $I_D = 0$ mA. For the approximate equivalent this is determined by finding the applied voltage when the diode has a drop of 0.7 V across it (for silicon) and $I_D = 0$ mA.



This exercise was applied to the network of Fig. 2.69 as shown in Fig. 2.70. Note the substitution of the short-circuit equivalent for the diode and the fact that the voltage across the resistor is 0 V because the diode current is 0 mA. The result is $v_i - V = 0$, and so

$$\boxed{v_i = V} \qquad \textbf{(2.14)}$$

is the transition voltage.

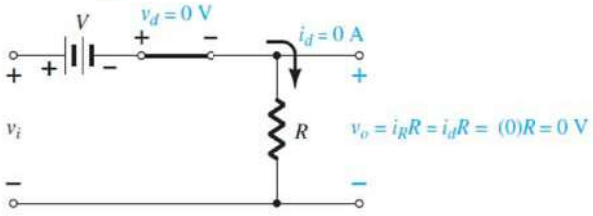


FIG. 2.70
Determining the transition level for the circuit of Fig. 2.69.

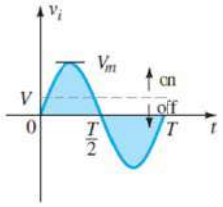


FIG. 2.71
Using the transition voltage to define the "on" and "off" regions.

This permits drawing a line on the sinusoidal supply voltage as shown in Fig. 2.71 to define the regions where the diode is on and off.

For the "on" region, as shown in Fig. 2.72, the diode is replaced by a short-circuit equivalent, and the output voltage is defined by

$$\boxed{v_o = v_i - V} \qquad \textbf{(2.15)}$$

For the "off" region, the diode is an open circuit, $I_D = 0$ mA, and the output voltage is

$$\boxed{v_o = 0 \text{ V}}$$

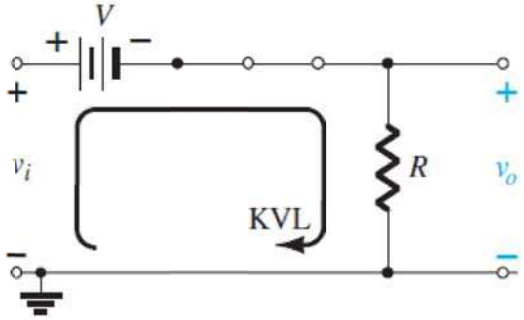


FIG. 2.72
Determining v_o for the diode in the "on" state.

4. **It is often helpful to draw the output waveform directly below the applied voltage using the same scales for the horizontal axis and the vertical axis.**

Using this last piece of information, we can establish the 0-V level on the plot of Fig. 2.73 for the region indicated. For the "on" condition, Eq. (2.15) can be used to find the output voltage when the applied voltage has its peak value:

$$v_{o_{peak}} = V_m - V$$

and this can be added to the plot of Fig. 2.73. It is then simple to fill in the missing section of the output curve.

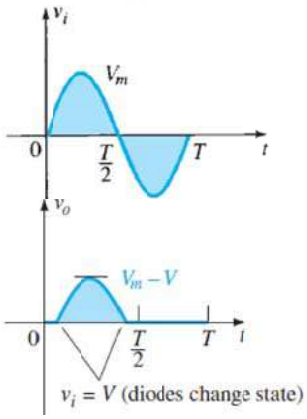


FIG. 2.73
Sketching the waveform of v_o using the results obtained for v_o above and below the transition level.

EXAMPLE 2.18 Determine the output waveform for the sinusoidal input of Fig. 2.74.

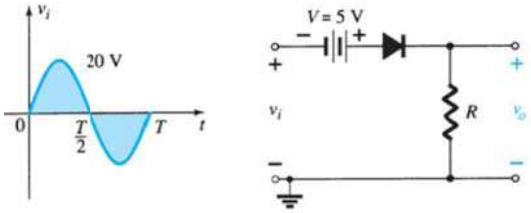


FIG. 2.74
Series clipper for Example 2.18.

Solution:

Step 1: The output is again directly across the resistor R.

Step 2: The positive region of v_i and the dc supply are both applying "pressure" to turn the diode on. The result is that we can safely assume the diode is in the "on" state for the entire range of positive voltages for v_i. Once the supply goes negative, it would have to exceed the dc supply voltage of 5 V before it could turn the diode off.

Step 3: The transition model is substituted in Fig. 2.75, and we find that the transition from one state to the other will occur when

$$v_i + 5\,\text{V} = 0\,\text{V}$$

or

$$v_i = -5\,\text{V}$$

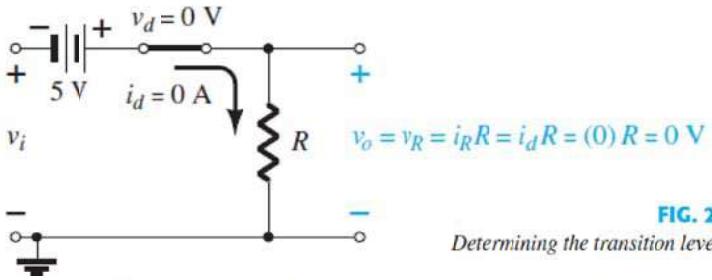


FIG. 2.75
Determining the transition level for the clipper of Fig. 2.74.

Step 4: In Fig. 2.76 a horizontal line is drawn through the applied voltage at the transition level. For voltages less than −5 V the diode is in the open-circuit state and the output is 0 V, as shown in the sketch of v_o. Using Fig. 2.76, we find that for conditions when the diode is on and the diode current is established the output voltage will be the following, as determined using Kirchhoff's voltage law:

$$v_o = v_i + 5\text{ V}$$

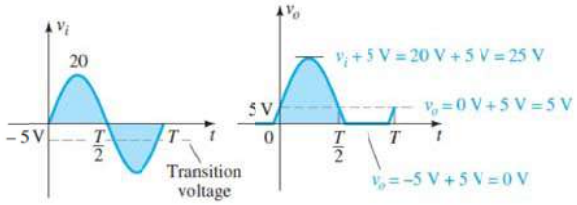


FIG. 2.76
Sketching v_o for Example 2.18.

The analysis of clipper networks with square-wave inputs is actually easier than with sinusoidal inputs because only two levels have to be considered. In other words, the network can be analyzed as if it had two dc level inputs with the resulting v_o plotted in the proper time frame. The next example demonstrates the procedure.

EXAMPLE 2.19 Find the output voltage for the network examined in Example 2.18 if the applied signal is the square wave of Fig. 2.77.

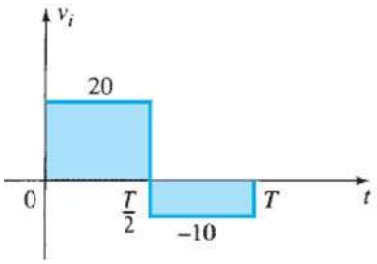
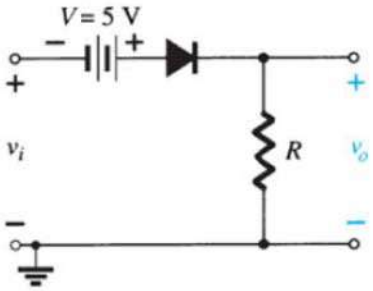


FIG. 2.77
Applied signal for Example 2.19.



Solution: For $v_i = 20\text{ V}$ $(0 \rightarrow T/2)$ the network of Fig. 2.78 results. The diode is in the short-circuit state, and $v_o = 20\text{ V} + 5\text{ V} = 25\text{ V}$. For $v_i = -10\text{ V}$ the network of Fig. 2.79

results, placing the diode in the "off" state, and $v_o = i_R R = (0)R = 0\text{ V}$. The resulting output voltage appears in Fig. 2.80.

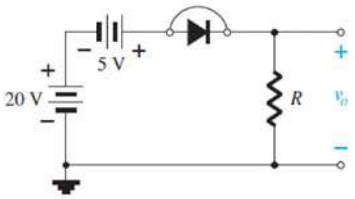


FIG. 2.78
v_o at $v_i = +20$ V.

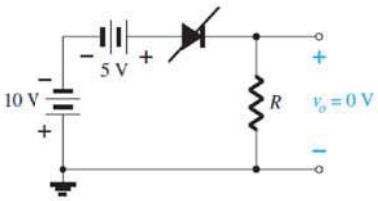


FIG. 2.79
v_o at $v_i = -10$ V.

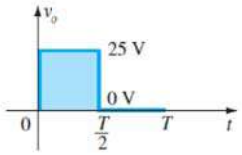


FIG. 2.80
Sketching v_o for Example 2.19.

Note in Example 2.19 that the clipper not only clipped off 5 V from the total swing, but also raised the dc level of the signal by 5 V.

Parallel

The network of Fig. 2.81 is the simplest of parallel diode configurations with the output for the same inputs of Fig. 2.68. The analysis of parallel configurations is very similar to that applied to series configurations, as demonstrated in the next example.

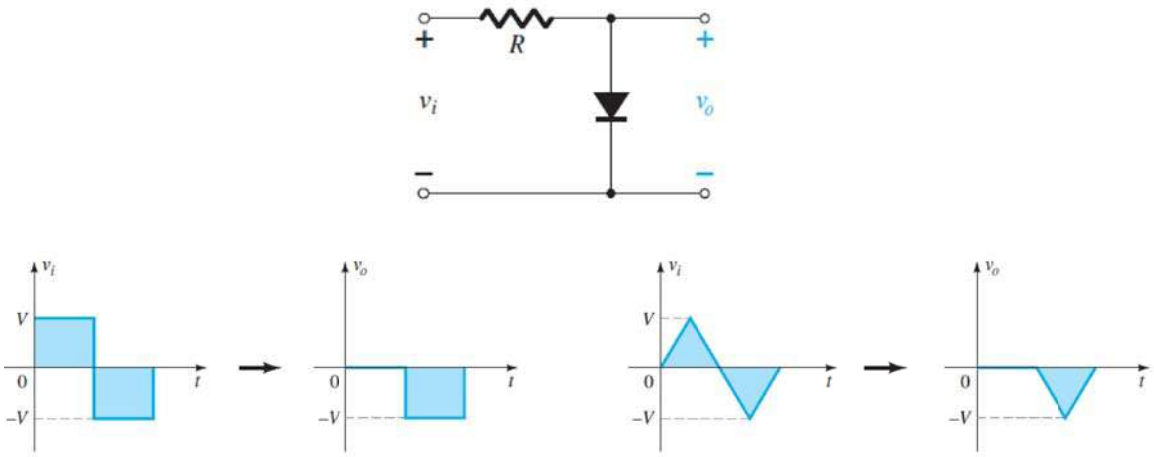


FIG. 2.81
Response to a parallel clipper.

EXAMPLE 2.20 Determine v_o for the network of Fig. 2.82.

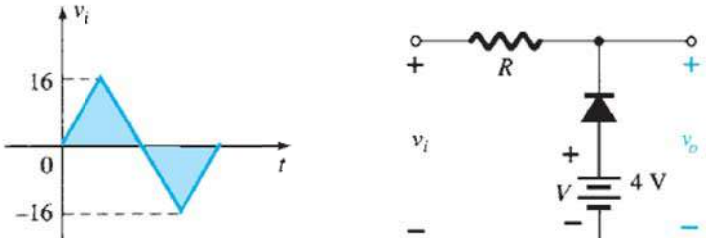


FIG. 2.82
Example 2.20.

Solution:

Step 1: In this example the output is defined across the series combination of the 4-V supply and the diode, not across the resistor R.

Step 2: The polarity of the dc supply and the direction of the diode strongly suggest that the diode will be in the "on" state for a good portion of the negative region of the input signal. In fact, it is interesting to note that since the output is directly across the series combination, when the diode is in its short-circuit state the output voltage will be directly across the 4-V dc supply, requiring that the output be fixed at 4 V. In other words, when the diode is on the output will be 4 V. Other than that, when the diode is an open circuit, the current through the series network will be 0 mA and the voltage drop across the resistor will be 0 V. That will result in $v_o = v_i$ whenever the diode is off.

Step 3: The transition level of the input voltage can be found from Fig. 2.83 by substituting the short-circuit equivalent and remembering the diode current is 0 mA at the instant of transition. The result is a change in state when

$$v_i = 4 \text{ V}$$

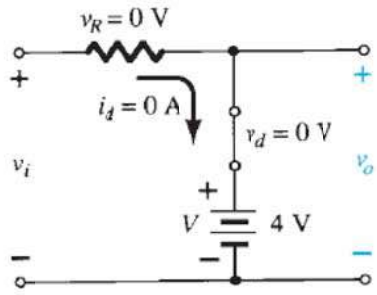


FIG. 2.83
Determining the transition level for Example 2.20.

Step 4: In Fig. 2.84 the transition level is drawn along with $v_o = 4$ V when the diode on. For $v_i \geq 4$ V, $v_o = 4$ V, and the waveform is simply repeated on the output plot.

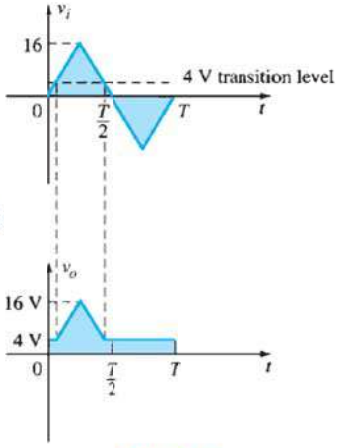


FIG. 2.84
Sketching v_o for Example 2.20.

To examine the effects of the knee voltage V_K of a silicon diode on the output response, the next example will specify a silicon diode rather than the ideal diode equivalent.

EXAMPLE 2.21 Repeat Example 2.20 using a silicon diode with $V_K = 0.7$ V.

Solution: The transition voltage can first be determined by applying the condition $i_d = 0$ A at $v_d = V_D = 0.7$ V and obtaining the network of Fig. 2.85. Applying Kirchhoff's voltage law around the output loop in the clockwise direction, we find that

$$v_i + V_K - V = 0$$

and

$$v_i = V - V_K = 4\text{ V} - 0.7\text{ V} = \mathbf{3.3\text{ V}}$$

For input voltages greater than 3.3 V, the diode will be an open circuit and $v_o = v_i$. For input voltages less than 3.3 V, the diode will be in the "on" state and the network of Fig. 2.86 results, where

$$v_o = 4\text{ V} - 0.7\text{ V} = \mathbf{3.3\text{ V}}$$

The resulting output waveform appears in Fig. 2.87. Note that the only effect of V_K was to drop the transition level to 3.3 from 4 V.

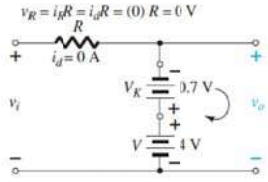


FIG. 2.85
Determining the transition level for the network of Fig. 2.82.

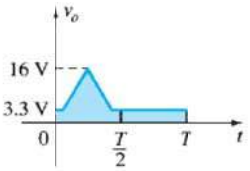
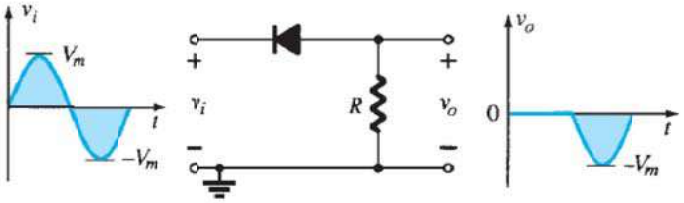


FIG. 2.87
Sketching v_o for Example 2.21.

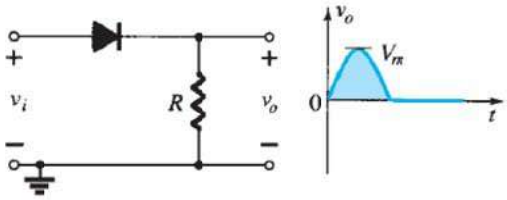
There is no question that including the effects of V_K will complicate the analysis somewhat, but once the analysis is understood with the ideal diode, the procedure, including the effects of V_K, will not be that difficult.

Simple Series Clippers (Ideal Diodes)

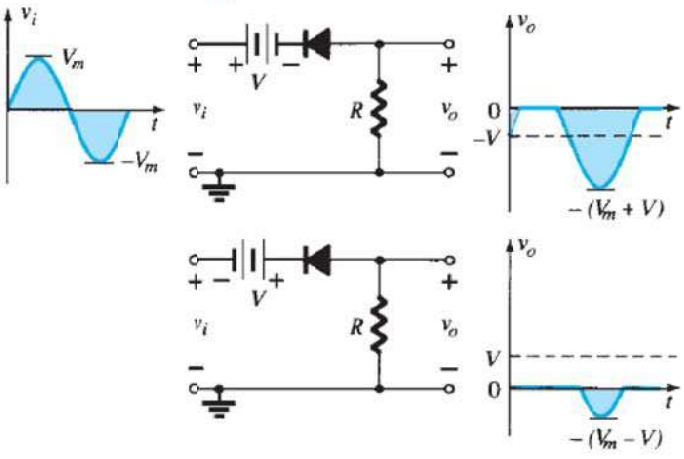
POSITIVE

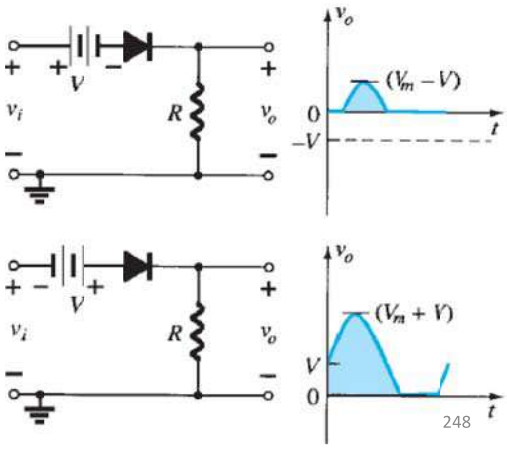


NEGATIVE

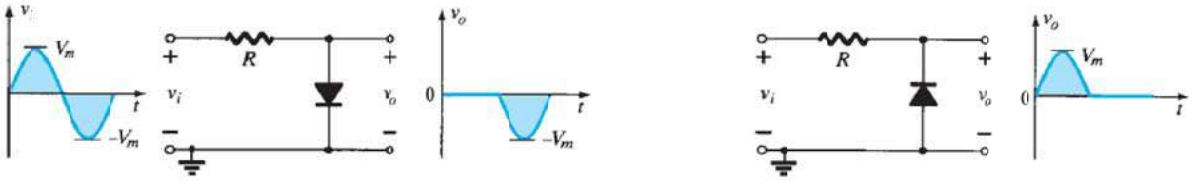


Biased Series Clippers (Ideal Diodes)

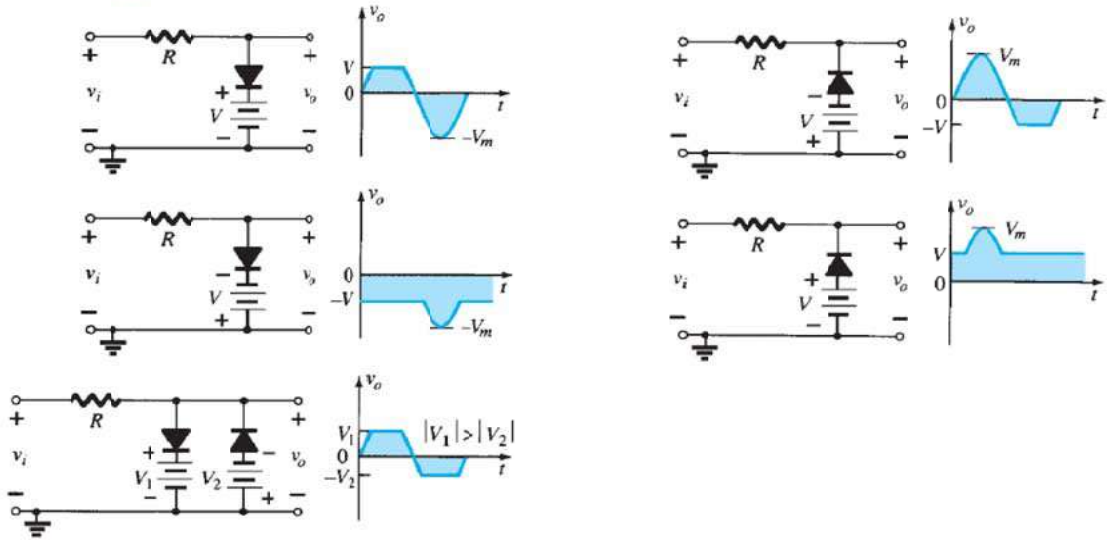




Simple Parallel Clippers (Ideal Diodes)



Biased Parallel Clippers (Ideal Diodes)



2.9 CLAMPERS

The previous section investigated a number of diode configurations that clipped off a portion of the applied signal without changing the remaining part of the waveform.

This section will examine a variety of diode configurations that shift the applied signal to a different level.

A clamper is a network constructed of a diode, a resistor, and a capacitor that shifts a waveform to a different dc level without changing the appearance of the applied signal.

2.9 CLAMPERS

Additional shifts can also be obtained by introducing a dc supply to the basic structure. The chosen resistor and capacitor of the network must be chosen such that the time constant determined by $\tau = RC$ is sufficiently large to ensure that the voltage across the capacitor does not discharge significantly during the interval the diode is nonconducting. Throughout the analysis we assume that for all practical purposes the capacitor fully charges or discharges in five time constants.

The simplest of clamper networks is provided in Fig. 2.89. It is important to note that the capacitor is connected directly between input and output signals and the resistor and the diode are connected in parallel with the output signal.

Clamping networks have a capacitor connected directly from input to output with a resistive element in parallel with the output signal. The diode is also in parallel with the output signal but may or may not have a series dc supply as an added element.

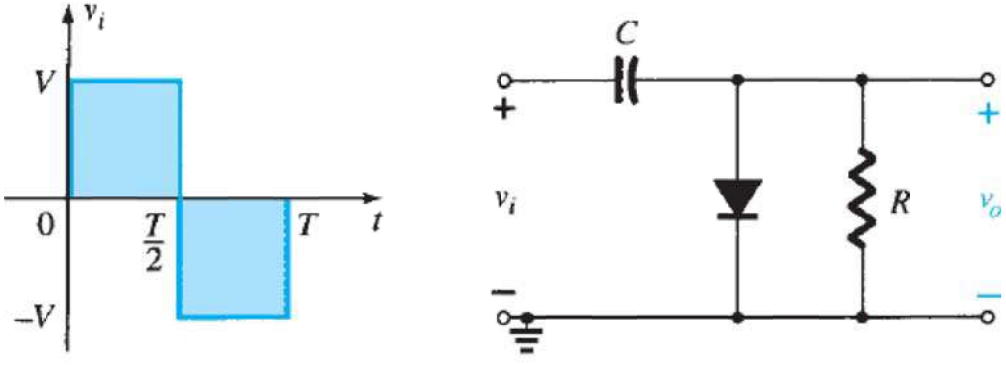


FIG. 2.89
Clamper.

There is a sequence of steps that can be applied to help make the analysis straightforward. It is not the only approach to examining clampers, but it does offer an option if difficulties surface.

Step 1: Start the analysis by examining the response of the portion of the input signal that will forward bias the diode.

Step 2: During the period that the diode is in the "on" state, assume that the capacitor will charge up instantaneously to a voltage level determined by the surrounding network.

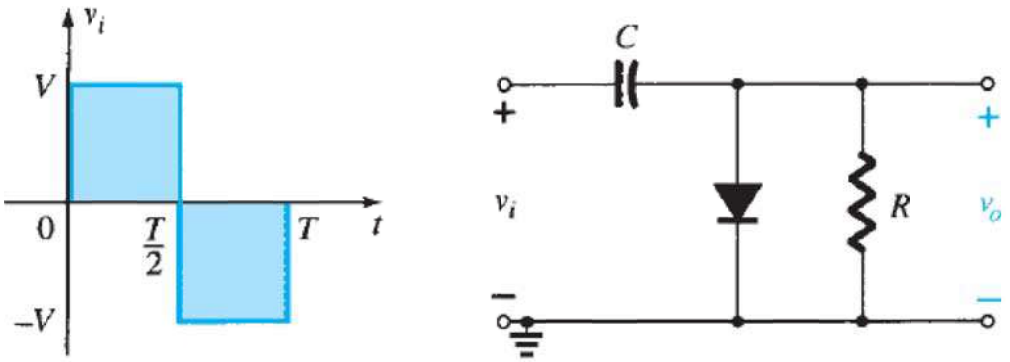


FIG. 2.89
Clamper.

For the network of Fig. 2.89 the diode will be forward biased for the positive portion of the applied signal. For the interval 0 to $T/2$ the network will appear as shown in Fig. 2.90. The short-circuit equivalent for the diode will result in $v_o = 0$ V for this time interval, as shown in the sketch of v_o in Fig. 2.92. During this same interval of time, the time constant determined by $\tau = RC$ is very small because the resistor R has been effectively "shorted out" by the conducting diode and the only resistance present is the inherent (contact, wire) resistance of the network. The result is that the capacitor will quickly charge to the peak value of V volts as shown in Fig. 2.90 with the polarity indicated.

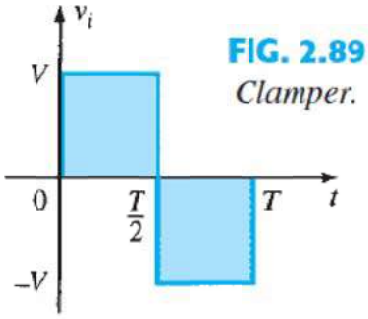
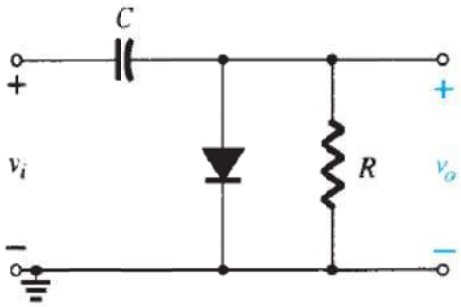


FIG. 2.89
Clamper.



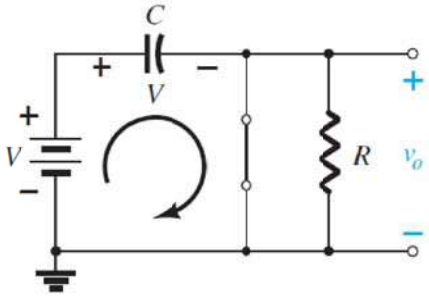


FIG. 2.90
Diode "on" and the capacitor charging to V volts.

Step 3: Assume that during the period when the diode is in the "off" state the capacitor holds on to its established voltage level.

Step 4: Throughout the analysis, maintain a continual awareness of the location and defined polarity for v_o to ensure that the proper levels are obtained.

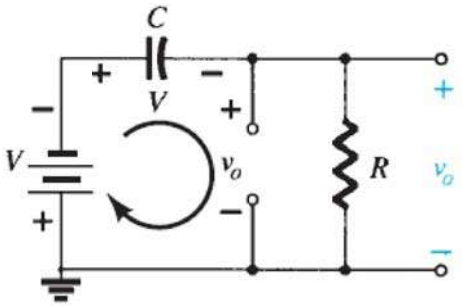


FIG. 2.91
Determining v_o with the diode "off."

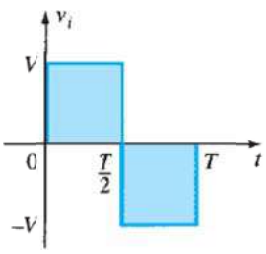
When the input switches to the $-V$ state, the network will appear as shown in Fig. 2.91, with the open-circuit equivalent for the diode determined by the applied signal and stored voltage across the capacitor—both "pressuring" current through the diode from cathode to anode. Now that R is back in the network the time constant determined by the RC product is sufficiently large to establish a discharge period 5τ, much greater than the period $T/2 \rightarrow T$, and it can be assumed on an approximate basis that the capacitor holds onto all its charge and, therefore, voltage (since $V = Q/C$) during this period.

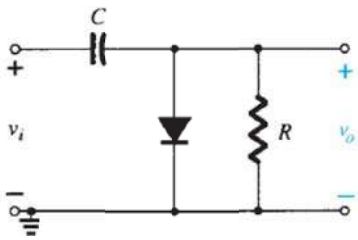
Since v_o is in parallel with the diode and resistor, it can also be drawn in the alternative position shown in Fig. 2.91. Applying Kirchhoff's voltage law around the input loop results in

$$-V - V - v_o = 0$$

and

$$v_o = -2V$$





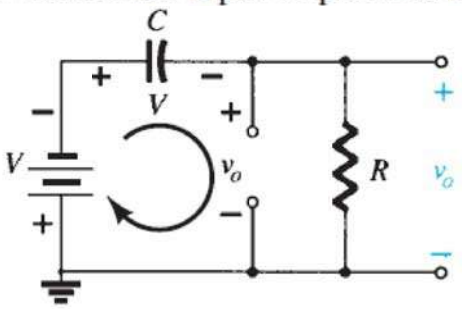


FIG. 2.91
Determining v_o with the diode "off."

The negative sign results from the fact that the polarity of $2V$ is opposite to the polarity defined for v_o. The resulting output waveform appears in Fig. 2.92 with the input signal. The output signal is clamped to 0 V for the interval 0 to $T/2$ but maintains the same total swing ($2V$) as the input.

Step 5: Check that the total swing of the output matches that of the input.

This is a property that applies for all clamping networks, giving an excellent check on the results obtained.

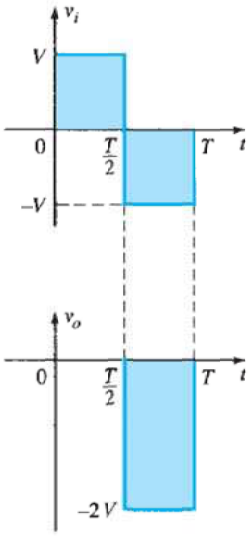


FIG. 2.92
Sketching v_o for the network of Fig. 2.91.

EXAMPLE 2.22 Determine v_o for the network of Fig. 2.93 for the input indicated.

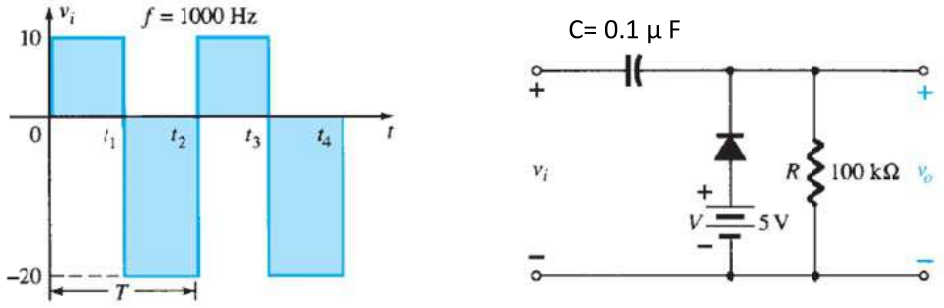


FIG. 2.93
Applied signal and network for Example 2.22.

Solution: Note that the frequency is 1000 Hz, resulting in a period of 1 ms and an interval of 0.5 ms between levels. The analysis will begin with the period $t_1 \rightarrow t_2$ of the input signal since the diode is in its short-circuit state

T=1/f =1/1000 =1ms

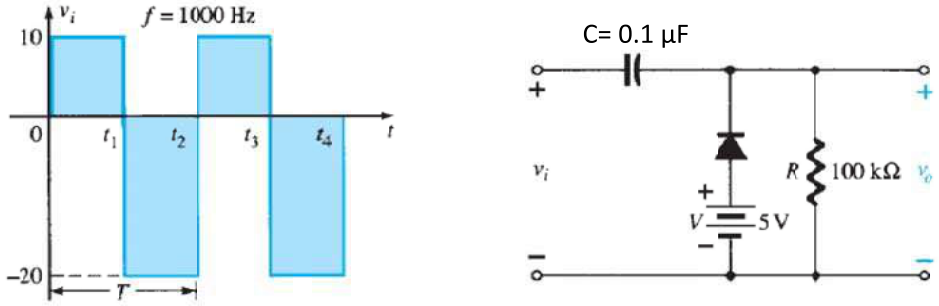


FIG. 2.93
Applied signal and network for Example 2.22.

Solution:

For this interval the network will appear as shown in Fig. 2.94. The output is across R, but it is also directly across the 5-V battery if one follows the direct connection between the defined terminals for v_o and the battery terminals. The result is $v_o = 5$ V for this interval. Applying Kirchhoff's voltage law around the input loop results in

$$-20\text{ V} + V_C - 5\text{ V} = 0$$

and

$$V_C = 25\text{ V}$$

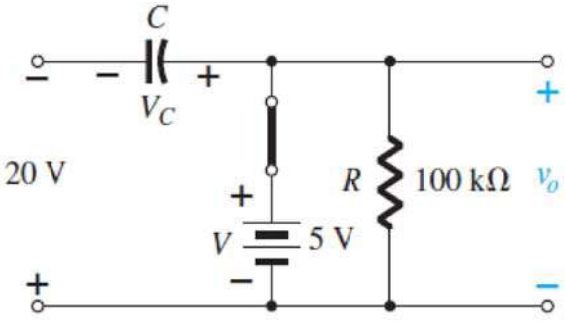


FIG. 2.94
Determining v_o and V_C with the diode in the "on" state.

The capacitor will therefore charge up to 25 V. In this case the resistor R is not shorted out by the diode, but a Thévenin equivalent circuit of that portion of the network that includes the battery and the resistor will result in $R_{Th} = 0\,\Omega$ with $E_{Th} = V = 5$ V. For the period $t_2 \rightarrow t_3$ the network will appear as shown in Fig. 2.95.

The open-circuit equivalent for the diode removes the 5-V battery from having any effect on v_o, and applying Kirchhoff's voltage law around the outside loop of the network results in

$$+10\text{ V} + 25\text{ V} - v_o = 0$$

and

$$v_o = 35\text{ V}$$

The time constant of the discharging network of Fig. 2.95 is determined by the product RC and has the magnitude

$$\tau = RC = (100\,\text{k}\Omega)(0.1\,\mu\text{F}) = 0.01\text{ s} = 10\text{ ms}$$

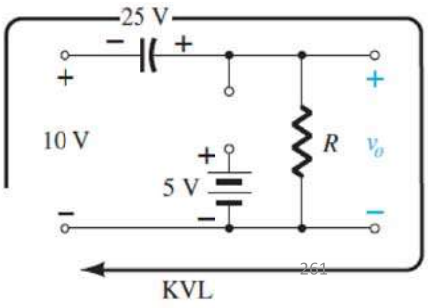


FIG. 2.95
Determining v_o with the diode in the "off" state.

The total discharge time is therefore $5\tau = 5(10\text{ ms}) = 50$ ms.

Since the interval $t_2 \rightarrow t_3$ will only last for 0.5 ms, it is certainly a good approximation that the capacitor will hold its voltage during the discharge period between pulses of the input signal. The resulting output appears in Fig. 2.96 with the input signal. Note that the output swing of 30 V matches the input swing as noted in step 5.

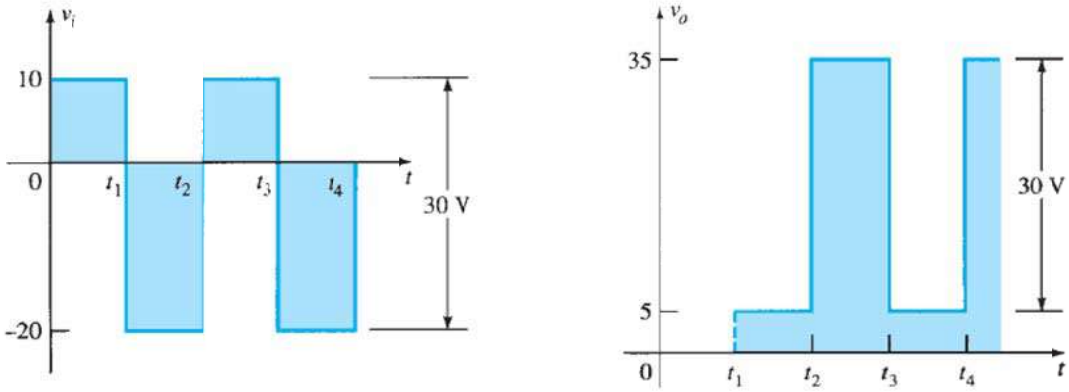


FIG. 2.96
v_i and v_o for the clamper of Fig. 2.93.

EXAMPLE 2.23 Repeat Example 2.22 using a silicon diode with $V_K = 0.7$ V.

Solution: For the short-circuit state the network now takes on the appearance of Fig. 2.97, and v_o can be determined by Kirchhoff's voltage law in the output section:

$$+5\text{ V} - 0.7\text{ V} - v_o = 0$$

and

$$v_o = 5\text{ V} - 0.7\text{ V} = 4.3\text{ V}$$

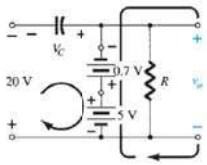


FIG. 2.97
Determining v_o and V_C with the diode in the "on" state.

For the input section Kirchhoff's voltage law results in

$$-20\text{ V} + V_C + 0.7\text{ V} - 5\text{ V} = 0$$

and

$$V_C = 25\text{ V} - 0.7\text{ V} = 24.3\text{ V}$$

For the period $t_2 \rightarrow t_3$ the network will now appear as in Fig. 2.98, with the only change being the voltage across the capacitor. Applying Kirchhoff's voltage law yields

$$+10\text{ V} + 24.3\text{ V} - v_o = 0$$

and

$$v_o = 34.3\text{ V}$$

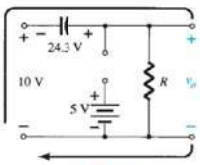


FIG. 2.98
Determining v_o with the diode in the open state.

The resulting output appears in Fig. 2.99, verifying the statement that the input and output swings are the same.

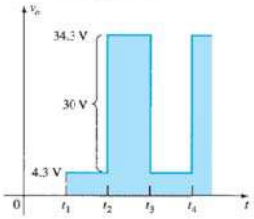


FIG. 2.99
Sketching v_o for the clamper of Fig. 2.93 with a silicon diode.

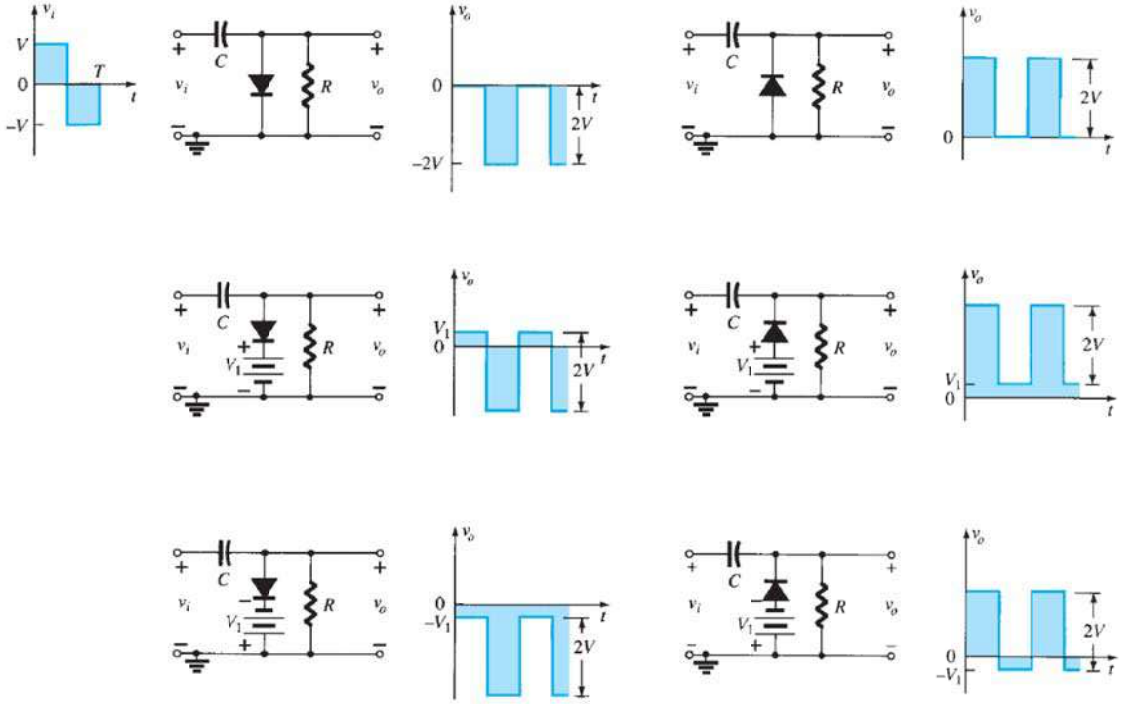


FIG. 2.100
Clamping circuits with ideal diodes ($5\tau = 5RC \gg T/2$).

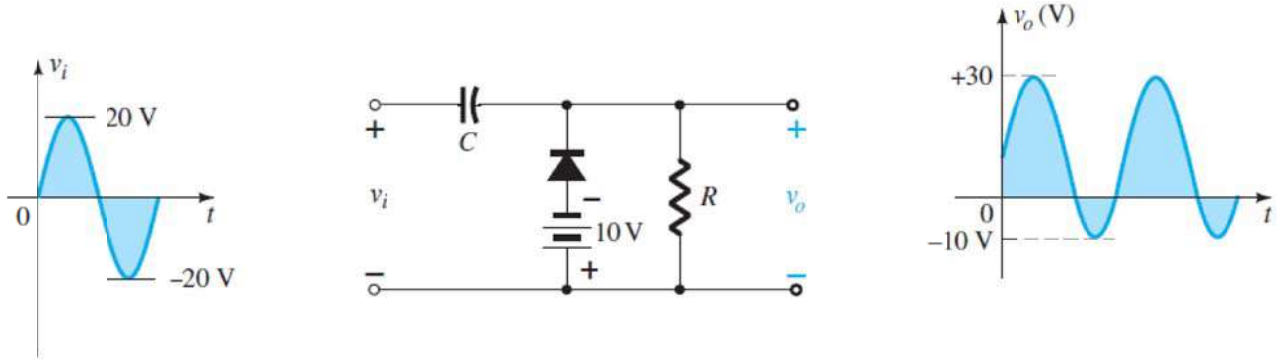


FIG. 2.101
Clamping network with a sinusoidal input.

2.10 NETWORKS WITH A DC AND AC SOURCE

The analysis thus far has been limited to circuits with a single dc, ac, or square wave input. This section will expand that analysis to include both an ac and a dc source in the same configuration. In Fig. 2.102 the simplest of two-source networks has been constructed.

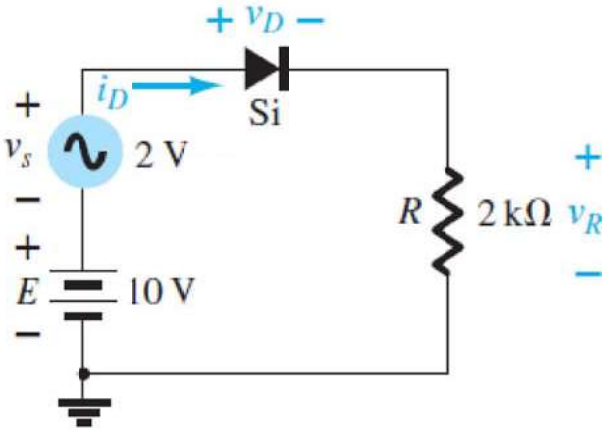


FIG. 2.102
Network with a dc and ac supply.

For such a system it is especially important that the Superposition Theorem can be applied. That is,

The response of any network with both an ac and a dc source can be found by finding the response to each source independently and then combining the results.

DC Source

The network is redrawn as shown in Fig. 2.103 for the dc source. Note that the ac source was removed by simply replacing it with a short-circuit equivalent to the condition $v_s = 0$ V.

Using the approximate equivalent circuit for the diode, the output voltage is

$$V_R = E - V_D = 10\text{ V} - 0.7\text{ V} = 9.3\text{ V}$$

and the currents are

$$I_D = I_R = \frac{9.3\text{ V}}{2\text{ k}\Omega} = 4.65\text{ mA}$$

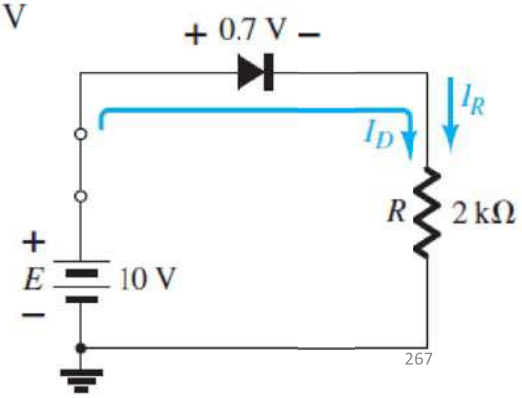


FIG. 2.103
Applying superposition to determine effects of the dc source.

AC Source

The dc source is also replaced by a short-circuit equivalent, as shown in Fig. 2.104. The diode will be replaced by the ac resistance, as determined by Eq. 1.5 in Chapter 1—the current in the equation being the quiescent or dc value. For this case,

$$r_d = \frac{26\text{ mV}}{I_D} = \frac{26\text{ mV}}{4.65\text{ mA}} = 5.59\ \Omega$$

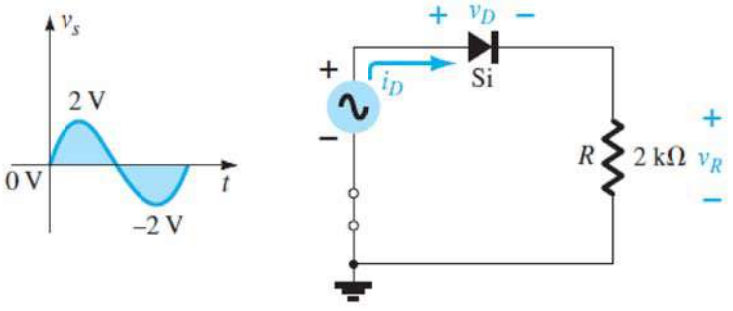


FIG. 2.104
Determing the response of v_R to the applied ac source.

Replacing the diode by this resistance will result in the circuit of Fig. 2.105. For the peak value of the applied voltage, the peak values of v_R and v_D will be

$$v_{R_{peak}} = \frac{2\,k\Omega\,(2\,V)}{2\,k\Omega + 5.59\,\Omega} \cong 1.99\,V$$

and

$$v_{D_{peak}} = v_{s_{peak}} - v_{R_{peak}} = 2\,V - 1.99\,V = 0.01\,V = 10\,mV$$

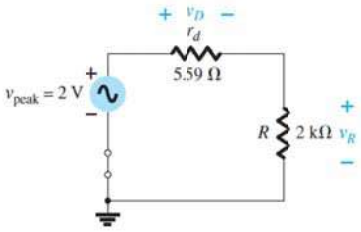
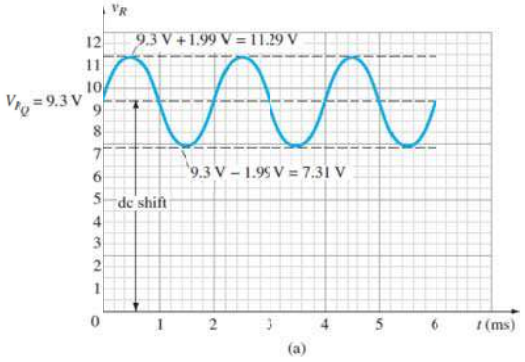
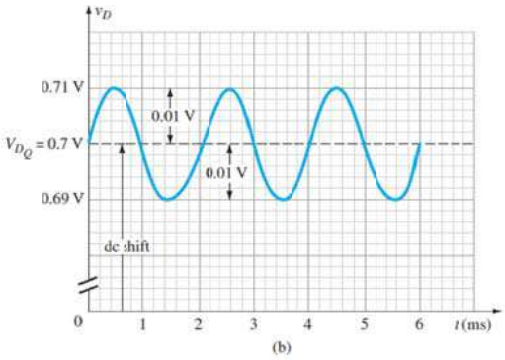


FIG. 2.105
Replacing the diode of Fig. 2.104 by its equivalent ac resistance.

Combining the results of the dc and ac analysis will result in the waveforms of Fig. 2.106 for v_R and v_D.





Note that the diode has an important impact on the resulting output voltage v_R but very little impact on the ac swing.

For comparison purposes the same system will now be analyzed using the actual characteristics and a load-line analysis. In Fig. 2.107 the dc load line has been drawn as described in Section 2.2. The resulting dc current is now slightly less due to a voltage drop across the diode that is slightly more than the approximate value of 0.7 V. For the peak value of the input voltage the load line will have intersections of $E = 12$ V and $I = \frac{E}{R} = \frac{12\,V}{2\,k\Omega} = 6$ mA. For the negative peak the intersections are at 8 V and 4 mA. Take particular note of the region of the diode characteristics traversed by the ac swing. It defines the region for which the diode resistance was determined in the analysis above. In this case, however, the quiescent value of dc current is ≅4.6 mA so the new ac resistance is

$$r_d = \frac{26\,mV}{4.6\,mA} = 5.65\,\Omega$$

which is very close to the above value.

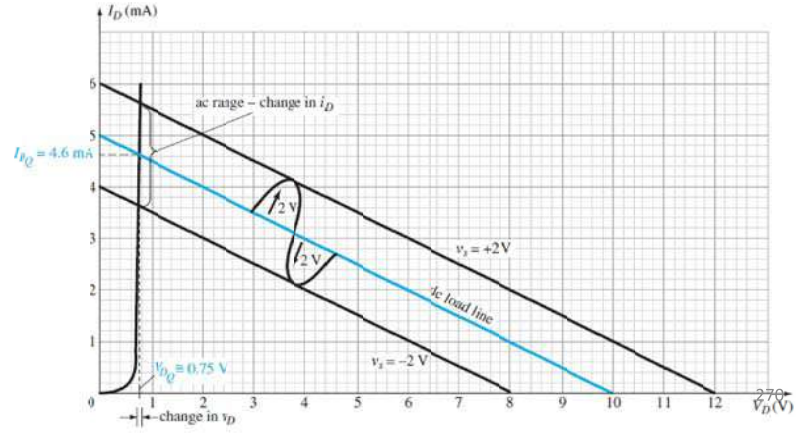


FIG. 2.107
Shifting loadline due to v_s source.

2.12 VOLTAGE-MULTIPLIER CIRCUITS

Voltage-multiplier circuits are employed to maintain a relatively low transformer peak voltage while stepping up the peak output voltage to two, three, four, or more times the peak rectified voltage.

Voltage Doubler

The network of Fig. 2.123 is a half-wave voltage doubler. During the positive voltage half-cycle across the transformer, secondary diode D_1 conducts (and diode D_2 is cut off), charging capacitor C_1 up to the peak rectified voltage (V_m). Diode D_1 is ideally a short during this half-cycle, and the input voltage charges capacitor C_1 to V_m with the polarity shown in Fig. 2.124a. During the negative half-cycle of the secondary voltage, diode D_1 is cut off and diode D_2 conducts charging capacitor C_2. Since diode D_2 acts as a short during the negative half-cycle (and diode D_1 is open), we can sum the voltages around the outside loop (see Fig. 2.124b):

$$-V_m - V_{C_1} + V_{C_2} = 0$$
$$-V_m - V_m + V_{C_2} = 0$$

from which we obtain $V_{C_2} = 2V_m$

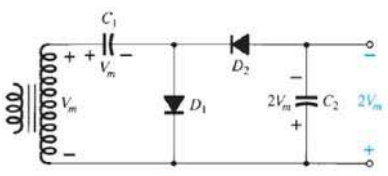


FIG. 2.123
Half-wave voltage doubler.

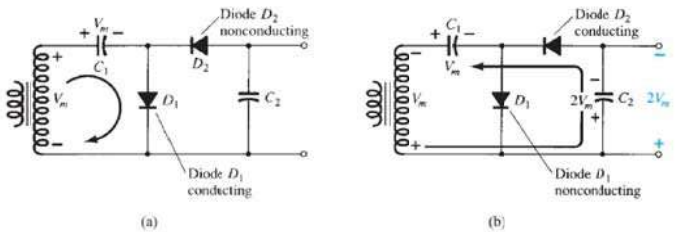


FIG. 2.124
Double operation, showing each half-cycle of operation: (a) positive half-cycle; (b) negative half-cycle

On the next positive half-cycle, diode D_2 is nonconducting and capacitor C_2 will discharge through the load. If no load is connected across capacitor C_2, both capacitors stay charged—C_1 to V_m and C_2 to $2V_m$. If, as would be expected, there is a load connected to the output of the voltage doubler, the voltage across capacitor C_2 drops during the positive half-cycle (at the input) and the capacitor is recharged up to $2V_m$ during the negative half-cycle. The output waveform across capacitor C_2 is that of a half-wave signal filtered by a capacitor filter. The peak inverse voltage across each diode is $2V_m$.

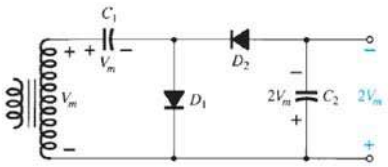


FIG. 2.123
Half-wave voltage doubler.

Half-wave voltage doubler.

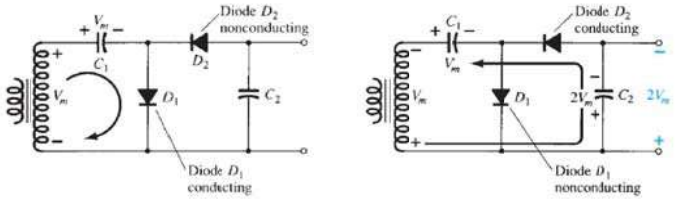


FIG. 2.124
Double operation, showing each half-cycle of operation: (a) positive half-cycle; (b) negative half-cycle

Another doubler circuit is the full-wave doubler of Fig. 2.125. During the positive half-cycle of transformer secondary voltage (see Fig. 2.126a) diode D_1 conducts, charging capacitor C_1 to a peak voltage V_m. Diode D_2 is nonconducting at this time.

During the negative half-cycle (see Fig. 2.126b) diode D_2 conducts, charging capacitor C_2, while diode D_1 is nonconducting. If no load current is drawn from the circuit, the voltage across capacitors C_1 and C_2 is $2V_m$. If load current is drawn from the circuit, the voltage across capacitors C_1 and C_2 is the same as that across a capacitor fed by a full-wave rectifier circuit. One difference is that the effective capacitance is that of C_1 and C_2 in series, which is less than the capacitance of either C_1 or C_2 alone. The lower capacitor value will provide poorer filtering action than the single-capacitor filter circuit.

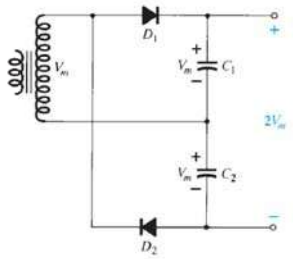


FIG. 2.125
Full-wave voltage doubler.

The peak inverse voltage across each diode is $2V_m$, as it is for the filter capacitor circuit. In summary, the half-wave or full-wave voltage-doubler circuits provide twice the peak voltage of the transformer secondary while requiring no center-tapped transformer and only $2V_m$ PIV rating for the diodes.

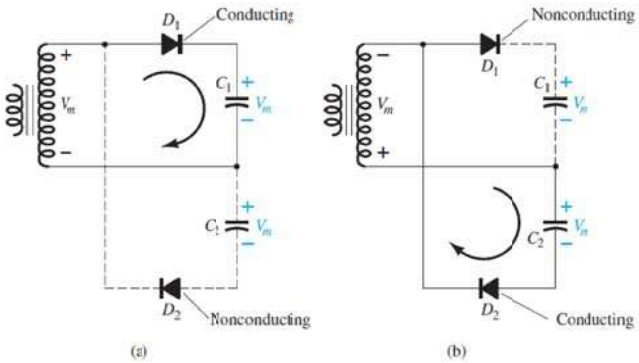


FIG. 2.126
Alternate half-cycles of operation for full-wave voltage doubler.

Voltage Tripler and Quadrupler

Figure 2.127 shows an extension of the half-wave voltage doubler, which develops three and four times the peak input voltage. It should be obvious from the pattern of the circuit

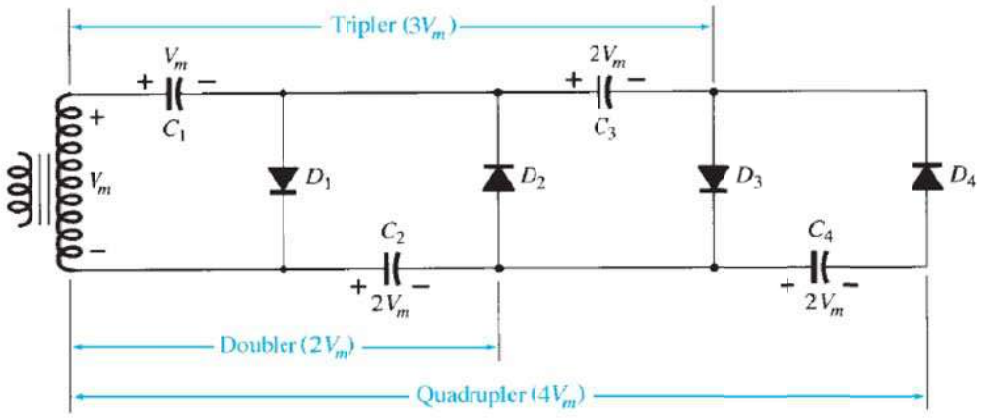


FIG. 2.127
Voltage tripler and quadrupler.

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from which we obtain $V_{C_2} = 2V_m$

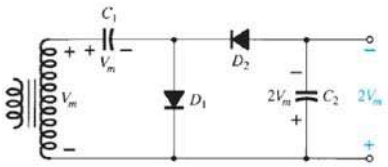

FIG. 2.123
Half-wave voltage doubler.

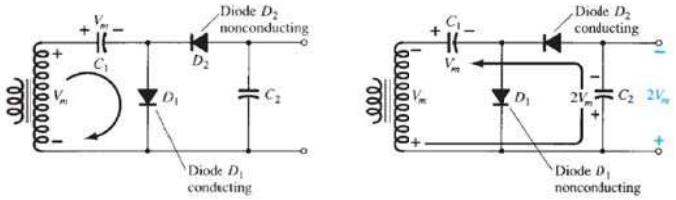

FIG. 2.124
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Half-wave voltage doubler.

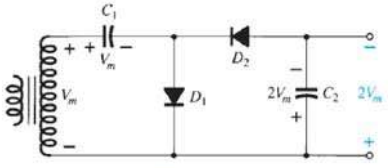

FIG. 2.123
Half-wave voltage doubler.

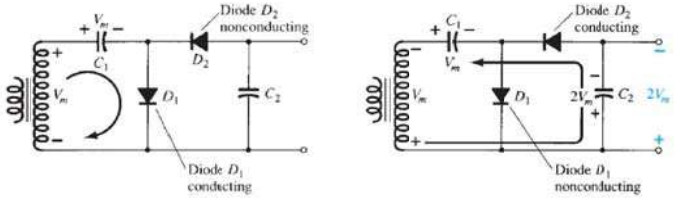

FIG. 2.124
Double operation, showing each half-cycle of operation: (a) positive half-cycle; (b) negative half-cycle

Another doubler circuit is the full-wave doubler of Fig. 2.125. During the positive half-cycle of transformer secondary voltage (see Fig. 2.126a) diode D_1 conducts, charging capacitor C_1 to a peak voltage V_m. Diode D_2 is nonconducting at this time.

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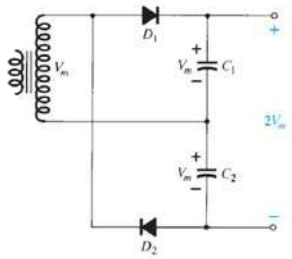


FIG. 2.125
Full-wave voltage doubler.

The peak inverse voltage across each diode is $2V_m$, as it is for the filter capacitor circuit. In summary, the half-wave or full-wave voltage-doubler circuits provide twice the peak voltage of the transformer secondary while requiring no center-tapped transformer and only $2V_m$ PIV rating for the diodes.

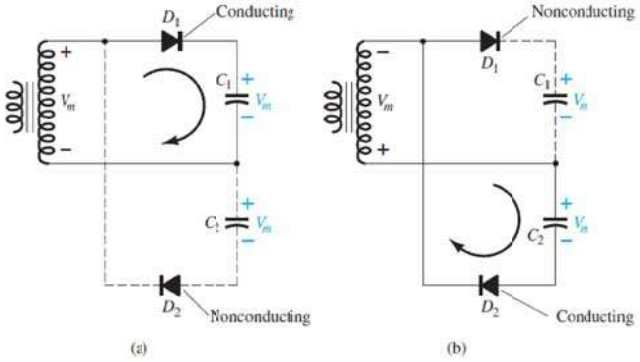


FIG. 2.126
Alternate half-cycles of operation for full-wave voltage doubler.

Voltage Tripler and Quadrupler

Figure 2.127 shows an extension of the half-wave voltage doubler, which develops three and four times the peak input voltage. It should be obvious from the pattern of the circuit

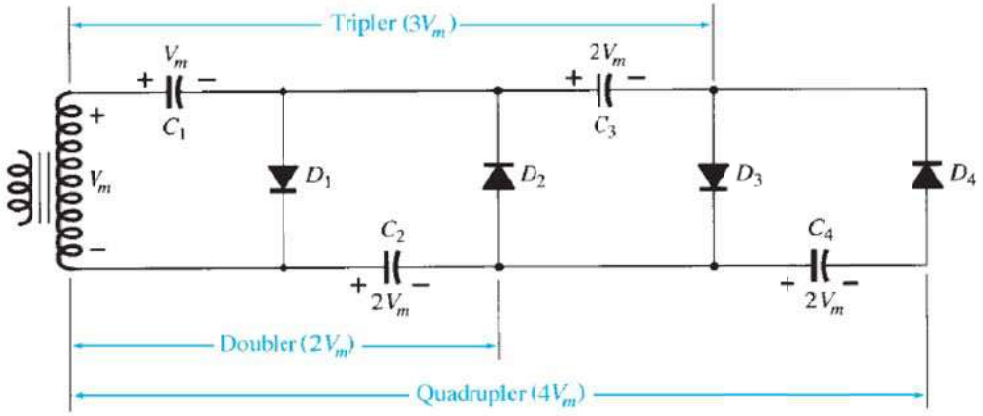


FIG. 2.127
Voltage tripler and quadrupler.

2.11 ZENER DIODES

The analysis of networks employing Zener diodes is quite similar to the analysis of semiconductor diodes in previous sections. First the state of the diode must be determined, followed by a substitution of the appropriate model and a determination of the other unknown quantities of the network. Figure 2.108 reviews the approximate equivalent circuits for each region of a Zener diode assuming the straight-line approximations at each break point. Note that the forward-bias region is included because occasionally an application will skip into this region also.

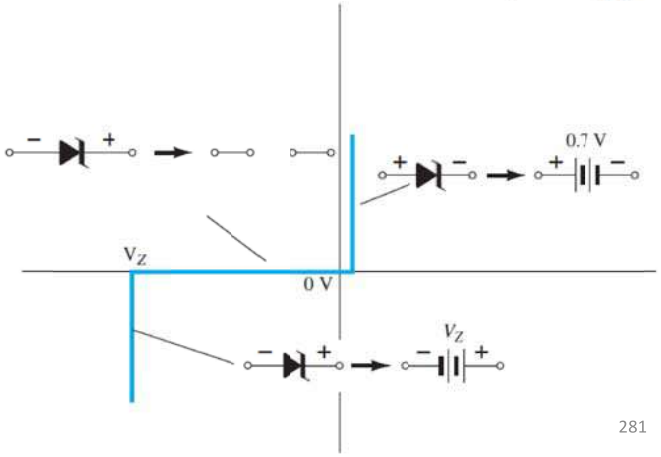


FIG. 2.108
Approximate equivalent circuits for the Zener diode in the three possible regions of application.

The first two examples will demonstrate how a Zener diode can be used to establish reference voltage levels and act as a protection device. The use of a Zener diode as a *regulator* will then be described in detail because it is one of its major areas of application. A regulator is a combination of elements designed to ensure that the output voltage of a supply remains fairly constant.

EXAMPLE 2.24 Determine the reference voltages provided by the network of Fig. 2.109, which uses a white LED to indicate that the power is on. What is the level of current through the LED and the power delivered by the supply? How does the power absorbed by the LED compare to that of the 6-V Zener diode?

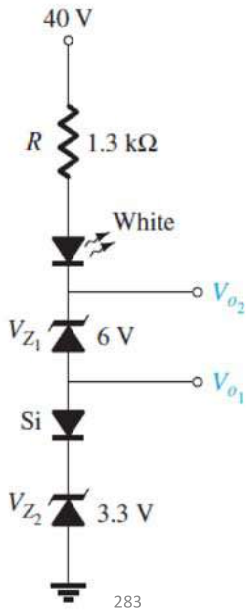


FIG. 2.109
Reference setting circuit for Example 2.24.

Solution: First we have to check that there is sufficient applied voltage to turn on all the series diode elements. The white LED will have a drop of about 4 V across it, the 6-V and 3.3-V Zener diodes have a total of 9.3 V, and the forward-biased silicon diode has 0.7 V, for a total of 14 V. The applied 40 V is then sufficient to turn on all the elements and, one hopes, establish a proper operating current.

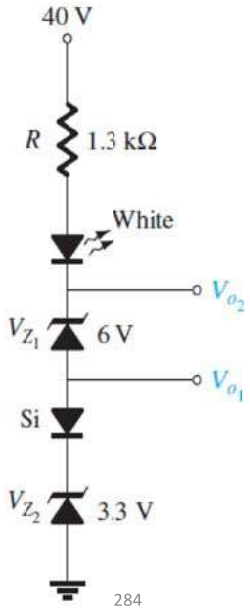


FIG. 2.109
Reference setting circuit for Example 2.24.

Note that the silicon diode was used to create a reference voltage of 4 V because

$$V_{o_1} = V_{Z_2} + V_K = 3.3\text{ V} + 0.7\text{ V} = \mathbf{4.0\text{ V}}$$

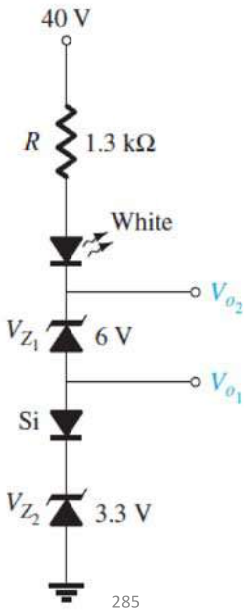
Combining the voltage of the 6-V Zener diode with the 4 V results in

$$V_{o_2} = V_{o_1} + V_{Z_1} = 4\text{ V} + 6\text{ V} = \mathbf{10\text{ V}}$$

Finally, the 4 V across the white LED will leave a voltage of 40 V – 14 V = 26 V across the resistor, and

$$I_R = I_{\text{LED}} = \frac{V_R}{R} = \frac{40\text{ V} - V_{o_2} - V_{\text{LED}}}{1.3\text{ k}\Omega} = \frac{40\text{ V} - 10\text{ V} - 4\text{ V}}{1.3\text{ k}\Omega} = \frac{26\text{ V}}{1.3\text{ k}\Omega} = \mathbf{20\text{ mA}}$$

which should establish the proper brightness for the LED.



The power delivered by the supply is simply the product of the supply voltage and current drain as follows:

$$P_s = EI_s = EI_R = (40\text{ V})(20\text{ mA}) = \mathbf{800\text{ mW}}$$

The power absorbed by the LED is

$$P_{\text{LED}} = V_{\text{LED}}I_{\text{LED}} = (4\text{ V})(20\text{ mA}) = \mathbf{80\text{ mW}}$$

and the power absorbed by the 6-V Zener diode is

$$P_Z = V_Z I_Z = (6\text{ V})(20\text{ mA}) = \mathbf{120\text{ mW}}$$

The power absorbed by the Zener diode exceeds that of the LED by 40 mW.

EXAMPLE 2.25 The network of Fig. 2.110 is designed to limit the voltage to 20 V during the positive portion of the applied voltage and to 0 V for a negative excursion of the applied voltage. Check its operation and plot the waveform of the voltage across the system for the applied signal. Assume the system has a very high input resistance so it will not affect the behavior of the network.

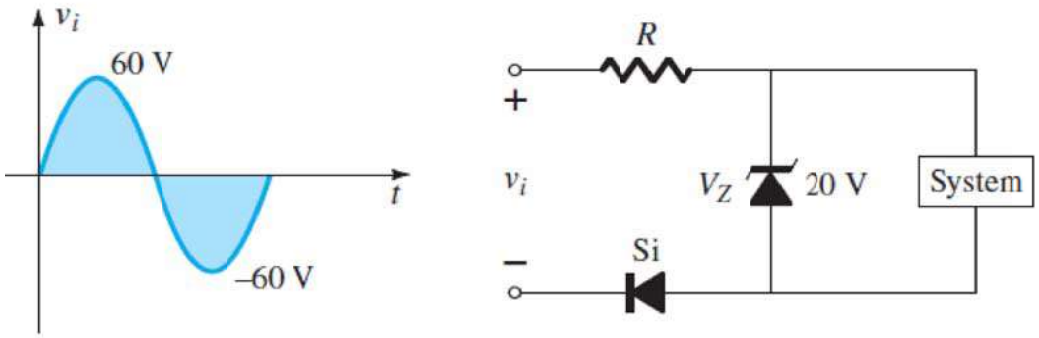
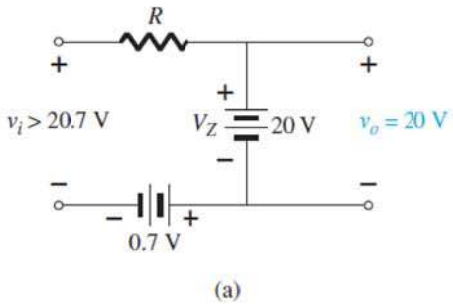


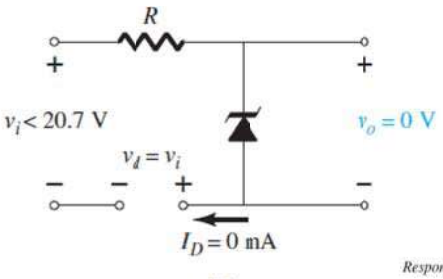
FIG. 2.110
Controlling network for Example 2.25.

Solution: For positive applied voltages less than the Zener potential of 20 V the Zener diode will be in its approximate open-circuit state, and the input signal will simply distribute itself across the elements, with the majority going to the system because it has such a high resistance level.

Once the voltage across the Zener diode reaches 20 V the Zener diode will turn on as shown in Fig. 2.111a and the voltage across the system will lock in at 20 V. Further increases in the applied voltage will simply appear across the series resistor with the voltage across the system and the forward-biased diode remaining fixed at 20 V and 0.7 V, respectively. The voltage across the system is fixed at 20 V, as shown in Fig. 2.111a, because the 0.7 V of the diode is not between the defined output terminals. The system is therefore safe from any further increases in applied voltage.



(a)



(b)

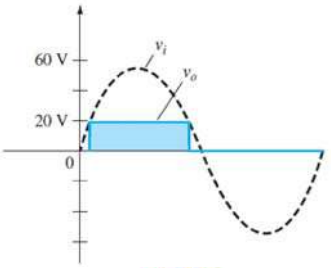
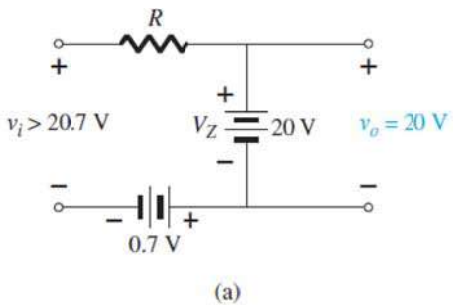


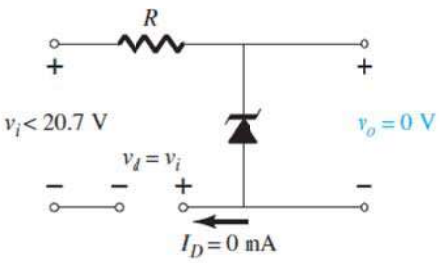
FIG. 2.111
Response of the network of Fig. 2.110 to the application of a 60-V sinusoidal signal.

For the negative region of the applied signal the silicon diode is reverse biased and presents an open circuit to the series combination of elements. The result is that the full negatively applied signal will appear across the open-circuited diode and the negative voltage across the system locked in at 0 V, as shown in Fig. 2.111b.

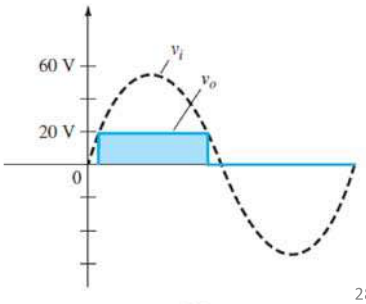
The voltage across the system will therefore appear as shown in Fig. 2.111c.



(a)



(b)



(c)

The use of the Zener diode as a regulator is so common that three conditions surrounding the analysis of the basic Zener regulator are considered. The analysis provides an excellent opportunity to become better acquainted with the response of the Zener diode to different operating conditions. The basic configuration appears in Fig. 2.112. The analysis is first for fixed quantities, followed by a fixed supply voltage and a variable load, and finally a fixed load and a variable supply.

V_i and R Fixed

Fixed V_i, Variable R_L

Fixed R_L, Variable V_i

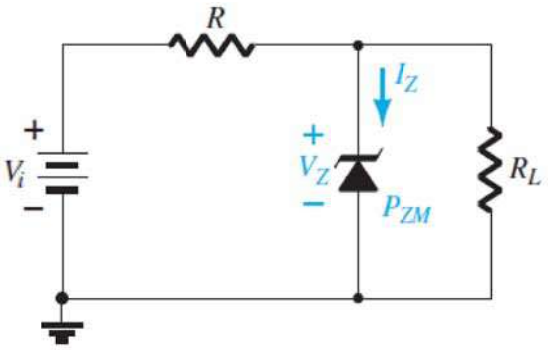


FIG. 2.112
Basic Zener regulator.

V_i and R Fixed

The simplest of Zener diode regulator networks appears in Fig. 2.112. The applied dc voltage is fixed, as is the load resistor. The analysis can fundamentally be broken down into two steps.

1. *Determine the state of the Zener diode by removing it from the network and calculating the voltage across the resulting open circuit.*

Applying step 1 to the network of Fig. 2.112 results in the network of Fig. 2.113, where an application of the voltage divider rule results in

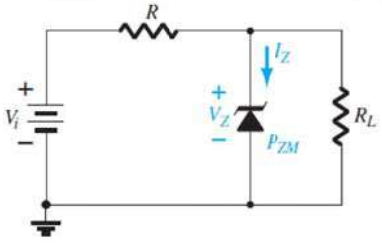


FIG. 2.112
Basic Zener regulator.

$$V = V_L = \frac{R_L V_i}{R + R_L}$$

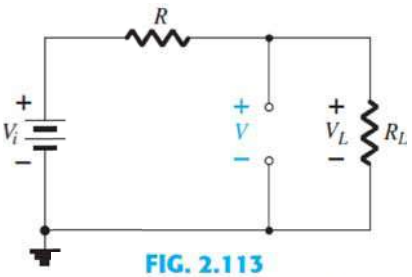


FIG. 2.113
Determining the state of the Zener diode.

If $V \geq V_Z$, the Zener diode is on, and the appropriate equivalent model can be substituted.
If $V < V_Z$, the diode is off, and the open-circuit equivalence is substituted.

2. *Substitute the appropriate equivalent circuit and solve for the desired unknowns.*

For the network of Fig. 2.112, the "on" state will result in the equivalent network of Fig. 2.114. Since voltages across parallel elements must be the same, we find that

$$V_L = V_Z \tag{2.17}$$

If $V \geq V_Z$, the Zener diode is on, and the appropriate equivalent model can be substituted.

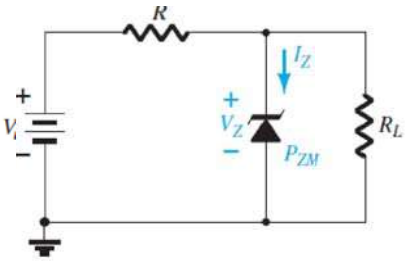


FIG. 2.112
Basic Zener regulator.

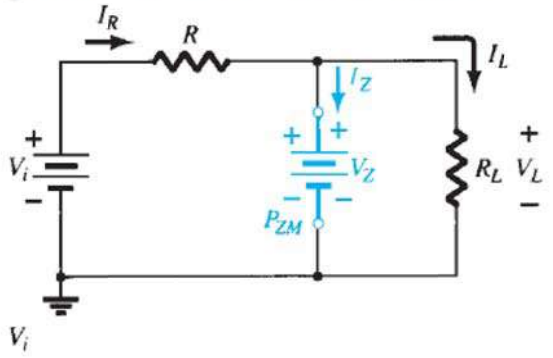


FIG. 2.114
Substituting the Zener equivalent for the "on" situation.

291

The Zener diode current must be determined by an application of Kirchhoff's current law. That is,

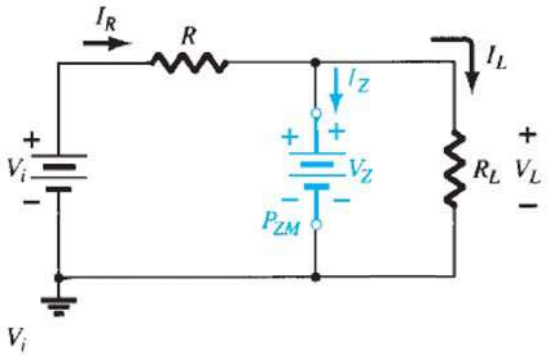
$$I_R = I_Z + I_L$$

and

$$I_Z = I_R - I_L$$

where

$$I_L = \frac{V_L}{R_L} \quad \text{and} \quad I_R = \frac{V_R}{R} = \frac{V_i - V_L}{R}$$



The power dissipated by the Zener diode is determined by

$$P_Z = V_Z I_Z \tag{2.19}$$

that must be less than the P_{ZM} specified for the device.

Before continuing, it is particularly important to realize that the first step was employed only to determine the *state of the Zener diode.* If the Zener diode is in the "on" state, the voltage across the diode is not V volts. When the system is turned on, the Zener diode will turn on as soon as the voltage across the Zener diode is V_Z volts. It will then "lock in" at this level and never reach the higher level of V volts.

292

EXAMPLE 2.26

a. For the Zener diode network of Fig. 2.115, determine V_L, V_R, I_Z, and P_Z.
b. Repeat part (a) with $R_L = 3\,\text{k}\Omega$.

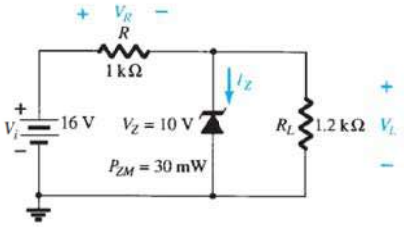


FIG. 2.115
Zener diode regulator for Example 2.26.

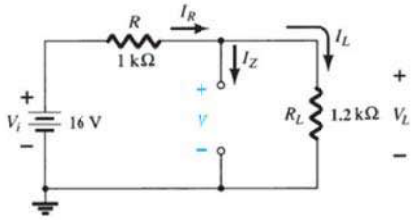


FIG. 2.116
Determining V for the regulator of Fig. 2.115.

Solution:

a. Following the suggested procedure, we redraw the network as shown in Fig. 2.116. Applying Eq. (2.16) gives

$$V = \frac{R_L V_i}{R + R_L} = \frac{1.2\,\text{k}\Omega(16\text{ V})}{1\,\text{k}\Omega + 1.2\,\text{k}\Omega} = 8.73\text{ V}$$

Since $V = 8.73$ V is less than $V_Z = 10$ V, the diode is in the "off" state, as shown on the characteristics of Fig. 2.117. Substituting the open-circuit equivalent results in the same network as in Fig. 2.116, where we find that

$$V_L = V = \mathbf{8.73\text{ V}}$$
$$V_R = V_i - V_L = 16\text{ V} - 8.73\text{ V} = \mathbf{7.27\text{ V}}$$
$$I_Z = \mathbf{0\text{ A}}$$

and

$$P_Z = V_Z I_Z = V_Z(0\text{ A}) = \mathbf{0\text{ W}}$$

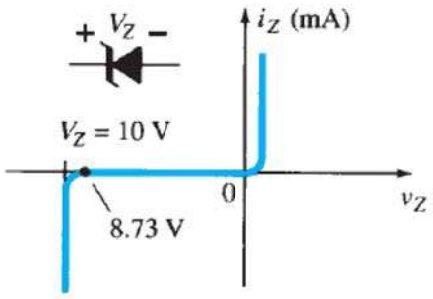


FIG. 2.117
Resulting operating point for the network of Fig. 2.115.

b. Applying Eq. (2.16) results in

$$V = \frac{R_L V_i}{R + R_L} = \frac{3\,\text{k}\Omega(16\,\text{V})}{1\,\text{k}\Omega + 3\,\text{k}\Omega} = 12\,\text{V}$$

Since $V = 12$ V is greater than $V_Z = 10$ V, the diode is in the "on" state and the network of Fig. 2.118 results. Applying Eq. (2.17) yields

$$V_L = V_Z = \mathbf{10\,V}$$

and

$$V_R = V_i - V_L = 16\,\text{V} - 10\,\text{V} = \mathbf{6\,V}$$

with

$$I_L = \frac{V_L}{R_L} = \frac{10\,\text{V}}{3\,\text{k}\Omega} = 3.33\,\text{mA}$$

and

$$I_R = \frac{V_R}{R} = \frac{6\,\text{V}}{1\,\text{k}\Omega} = 6\,\text{mA}$$

so that

$$I_Z = I_R - I_L\,[\text{Eq. (2.18)}]$$
$$= 6\,\text{mA} - 3.33\,\text{mA}$$
$$= \mathbf{2.67\,mA}$$

The power dissipated is

$$P_Z = V_Z I_Z = (10\,\text{V})(2.67\,\text{mA}) = \mathbf{26.7\,mW}$$

which is less than the specified $P_{ZM} = 30$ mW.

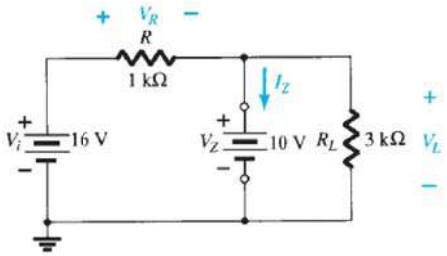


FIG. 2.118
Network of Fig. 2.115 in the "on" state.

Fixed V_i, Variable R_L

Due to the offset voltage V_Z, there is a specific range of resistor values (and therefore load current) that will ensure that the Zener is in the "on" state. Too small a load resistance R_L will result in a voltage V_L across the load resistor less than V_Z, and the Zener device will be in the "off" state.

To determine the minimum load resistance of Fig. 2.112 that will turn the Zener diode on, simply calculate the value of R_L that will result in a load voltage $V_L = V_Z$. That is,

$$V_L = V_Z = \frac{R_L V_i}{R_L + R}$$

Solving for R_L, we have

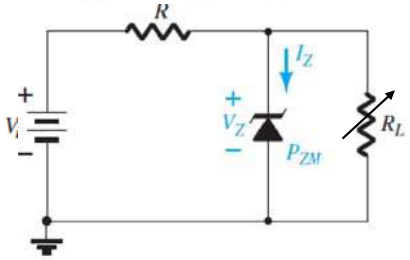


FIG. 2.112
Basic Zener regulator.

$$R_{L_{\text{min}}} = \frac{RV_Z}{V_i - V_Z} \qquad \textbf{(2.20)}$$

For example , R_{Lmin} =1*10/16-10=1.66 kΩ

Any load resistance value greater than the R_L obtained from Eq. (2.20) will ensure that the Zener diode is in the "on" state and the diode can be replaced by its V_Z source equivalent.

The condition defined by Eq. (2.20) establishes the minimum R_L, but in turn specifies the maximum I_L as

$$I_{L_{max}} = \frac{V_L}{R_L} = \frac{V_Z}{R_{L_{min}}} \tag{2.21}$$

Once the diode is in the "on" state, the voltage across R remains fixed at

$$V_R = V_i - V_Z \tag{2.22}$$

and I_R remains fixed at

$$I_R = \frac{V_R}{R} \tag{2.23}$$

The Zener current

$$I_Z = I_R - I_L \tag{2.24}$$

297

resulting in a minimum I_Z when I_L is a maximum and a maximum I_Z when I_L is a minimum value, since I_R is constant.

Since I_Z is limited to I_{ZM} as provided on the data sheet, it does affect the range of R_L and therefore I_L. Substituting I_{ZM} for I_Z establishes the minimum I_L as

$$I_{L_{min}} = I_R - I_{ZM} \tag{2.25}$$

and the maximum load resistance as

$$R_{L_{max}} = \frac{V_Z}{I_{L_{min}}} \tag{2.26}$$

298

EXAMPLE 2.27

a. For the network of Fig. 2.119, determine the range of R_L and I_L that will result in V_{RL} being maintained at 10 V.
b. Determine the maximum wattage rating of the diode.

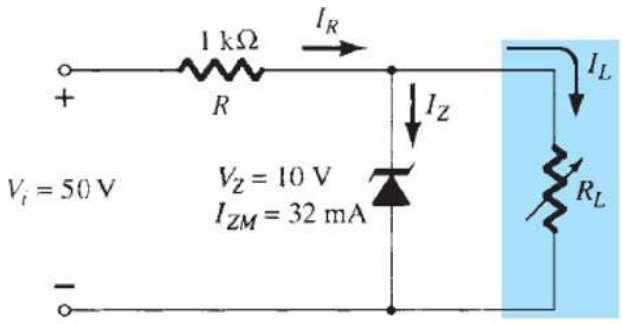


FIG. 2.119
Voltage regulator for Example 2.27.

Solution:

a. To determine the value of R_L that will turn the Zener diode on, apply Eq. (2.20

$$R_{L_{min}} = \frac{RV_Z}{V_i - V_Z} = \frac{(1\text{ k}\Omega)(10\text{ V})}{50\text{ V} - 10\text{ V}} = \frac{10\text{ k}\Omega}{40} = \mathbf{250\ \Omega}$$

The voltage across the resistor R is then determined by Eq. (2.22):

$$V_R = V_i - V_Z = 50\text{ V} - 10\text{ V} = \mathbf{40\ V}$$

and Eq. (2.23) provides the magnitude of I_R:

$$I_R = \frac{V_R}{R} = \frac{40\text{ V}}{1\text{ k}\Omega} = \mathbf{40\ mA}$$

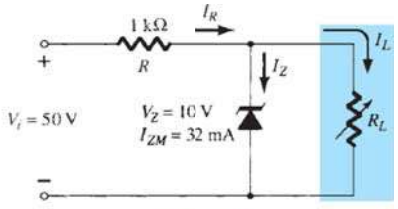


FIG. 2.119
Voltage regulator for Example 2.27.

The minimum level of I_L is then determined by Eq. (2.25):

$$I_{L_{min}} = I_R - I_{ZM} = 40\text{ mA} - 32\text{ mA} = \mathbf{8\ mA}$$

with Eq. (2.26) determining the maximum value of R_L:

$$R_{L_{max}} = \frac{V_Z}{I_{L_{min}}} = \frac{10\text{ V}}{8\text{ mA}} = \mathbf{1.25\ k\Omega}$$

A plot of V_L versus R_L appears in Fig. 2.120a and for V_L versus I_L in Fig. 2.120b.

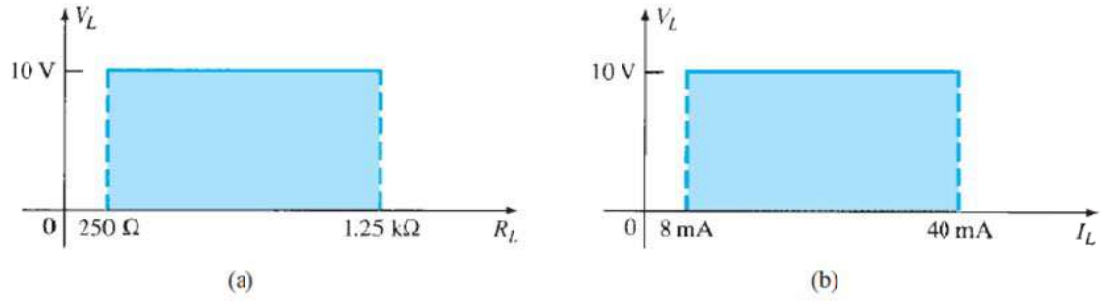


FIG. 2.120
V_L versus R_L and I_L for the regulator of Fig. 2.119.

b. $P_{max} = V_Z I_{ZM}$
$= (10\text{ V})(32\text{ mA}) = \mathbf{320\text{ mW}}$

Fixed R_L, Variable V_i

For fixed values of R_L in Fig. 2.112, the voltage V_i must be sufficiently large to turn the Zener diode on. The minimum turn-on voltage $V_i = V_{i_{min}}$ is determined by

$$V_L = V_Z = \frac{R_L V_i}{R_L + R}$$

and

$$\boxed{V_{i_{min}} = \frac{(R_L + R)V_Z}{R_L}} \qquad \textbf{(2.27)}$$

The maximum value of V_i is limited by the maximum Zener current I_{ZM}. Since $I_{ZM} = I_R - I_L$,

$$\boxed{I_{R_{max}} = I_{ZM} + I_L} \qquad \textbf{(2.28)}$$

Since I_L is fixed at V_Z/R_L and I_{ZM} is the maximum value of I_Z, the maximum V_i is defined by

$$V_{i_{max}} = V_{R_{max}} + V_Z$$

$$\boxed{V_{i_{max}} = I_{R_{max}} R + V_Z} \qquad \textbf{(2.29)}$$

EXAMPLE 2.28 Determine the range of values of V_i that will maintain the Zener diode of Fig. 2.121 in the "on" state.

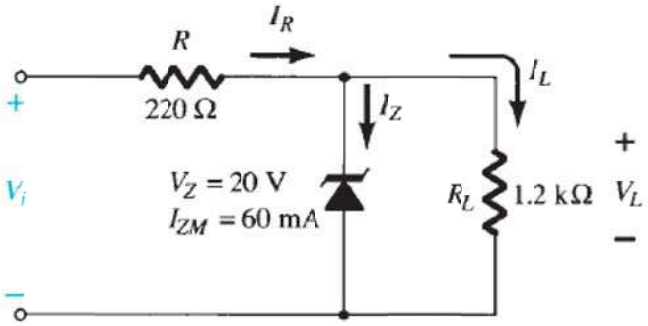


FIG. 2.121
Regulator for Example 2.28.

Eq. (2.27):
$$V_{i_{\min}} = \frac{(R_L + R)V_Z}{R_L} = \frac{(1200\ \Omega + 220\ \Omega)(20\ \text{V})}{1200\ \Omega} = \mathbf{23.67\ V}$$

$$I_L = \frac{V_L}{R_L} = \frac{V_Z}{R_L} = \frac{20\ \text{V}}{1.2\ \text{k}\Omega} = 16.67\ \text{mA}$$

Eq. (2.28):
$$I_{R_{\max}} = I_{ZM} + I_L = 60\ \text{mA} + 16.67\ \text{mA}$$
$$= 76.67\ \text{mA}$$

Eq. (2.29):
$$V_{i_{\max}} = I_{R_{\max}}R + V_Z$$
$$= (76.67\ \text{mA})(0.22\ \text{k}\Omega) + 20\ \text{V}$$
$$= 16.87\ \text{V} + 20\ \text{V}$$
$$\mathbf{36.87\ V}$$

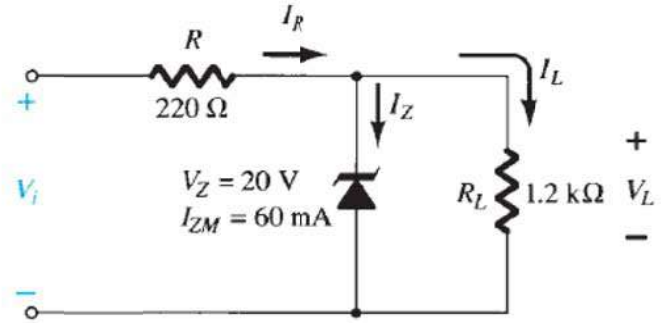


FIG. 2.121
Regulator for Example 2.28.

A plot of V_L versus V_i is provided in Fig. 2.122.

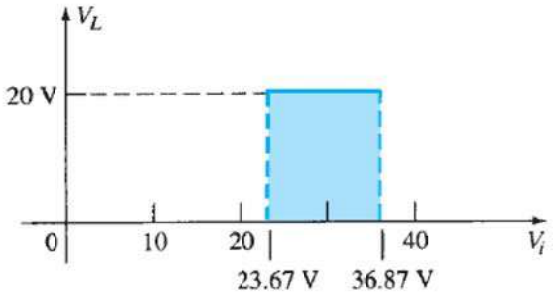


FIG. 2.122
V_L versus V_i for the regulator of Fig. 2.121.

The results of Example 2.28 reveal that for the network of Fig. 2.121 with a fixed R_L, the output voltage will remain fixed at 20 V for a range of input voltage that extends from 23.67 V to 36.87 V.

Conduction in Semiconductors

1.1 What happens when heat is added to silicon?

1.2 Name the two energy bands where current is produced in silicon.

1.1 When heat is added to silicon, more free electrons and holes are produced.

1.2 Current is produced in silicon at the **conduction** band and the **valence** band.

N-Type and P-Type Semiconductors

1.3 Describe the process of doping and explain how it alters the atomic structure if silicon.

N-Type and P-Type Semiconductors

1.3

Doping is the carefully controlled addition of trivalent or pentavalent atoms to pure (intrinsic) semiconductor material for the purpose of increasing the number of majority carriers (free electrons or holes).

The Diode

1.4 How is the electric field in the *pn* junction created?

1.5 Because of its barrier, can a diode be used as a voltage source? Explain.

The Diode

1.4

The electric field across the *pn* junction of a diode is created by donor atoms in the *n* region losing free electrons to acceptor atoms in the *p* region. This creates positive ions in the *n* region near the junction and negative ions in the *p* region near the junction. A field is then established between the ions.

1.5

The barrier potential of a diode represents an energy gradient that must be overcome by conduction electrons and produces a voltage drop, not a source of energy.

Biasing a Diode

1.6 Explain why a series resistance is necessary when a diode is forward biased.

Biasing a Diode

1.6

A series resistor is needed to **limit the current** through a forward-biased diode to a value that will not damage the diode because the diode itself has very little resistance.

Diode Models

1.7 Determine whether each diode in Figure 1.1 is forward-biased or reversed-biased.

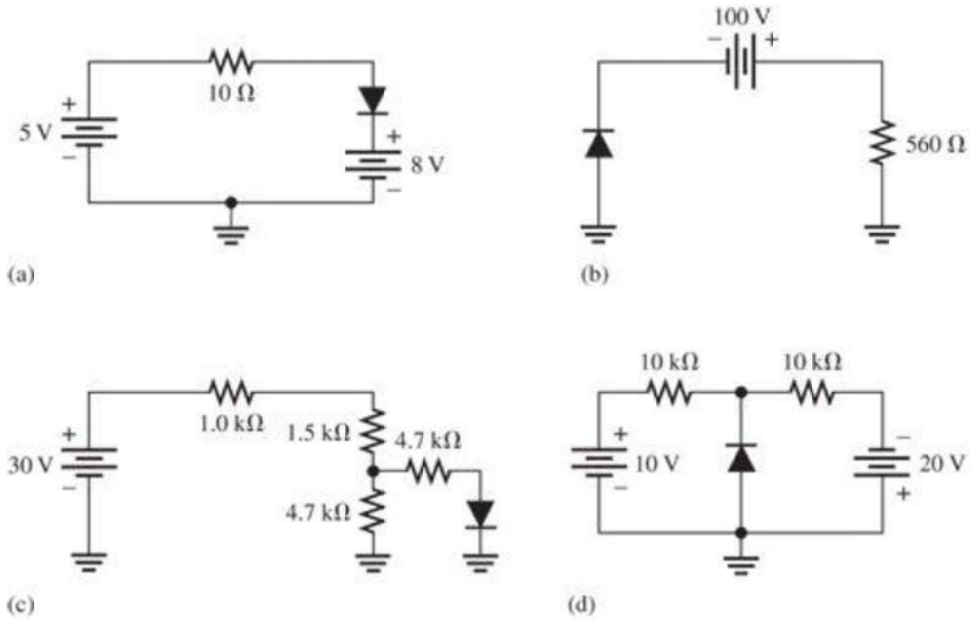
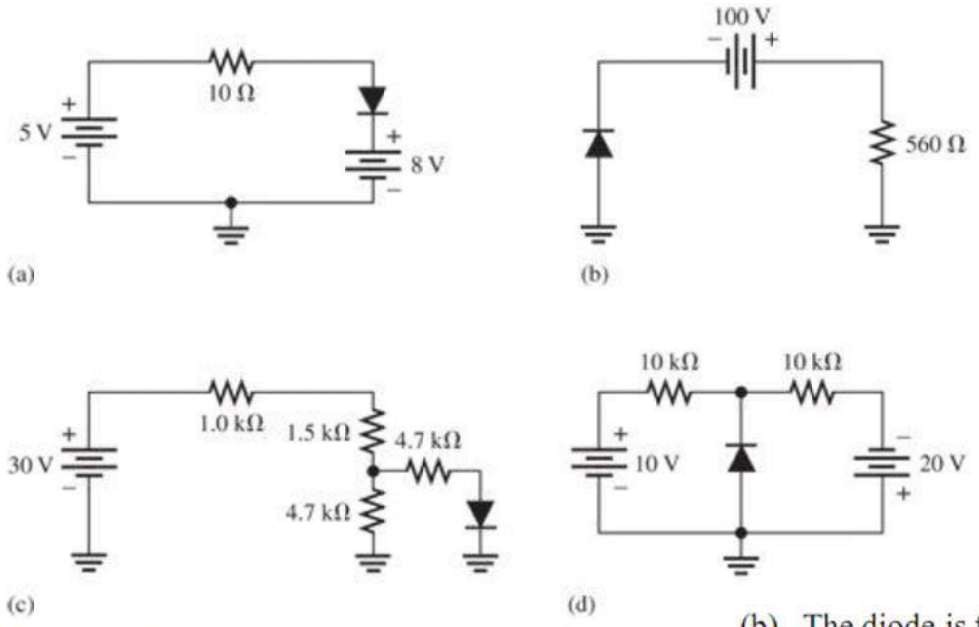


Figure 1.1

Diode Models

1.7 Determine whether each diode in Figure 1.1 is forward-biased or reversed-biased.

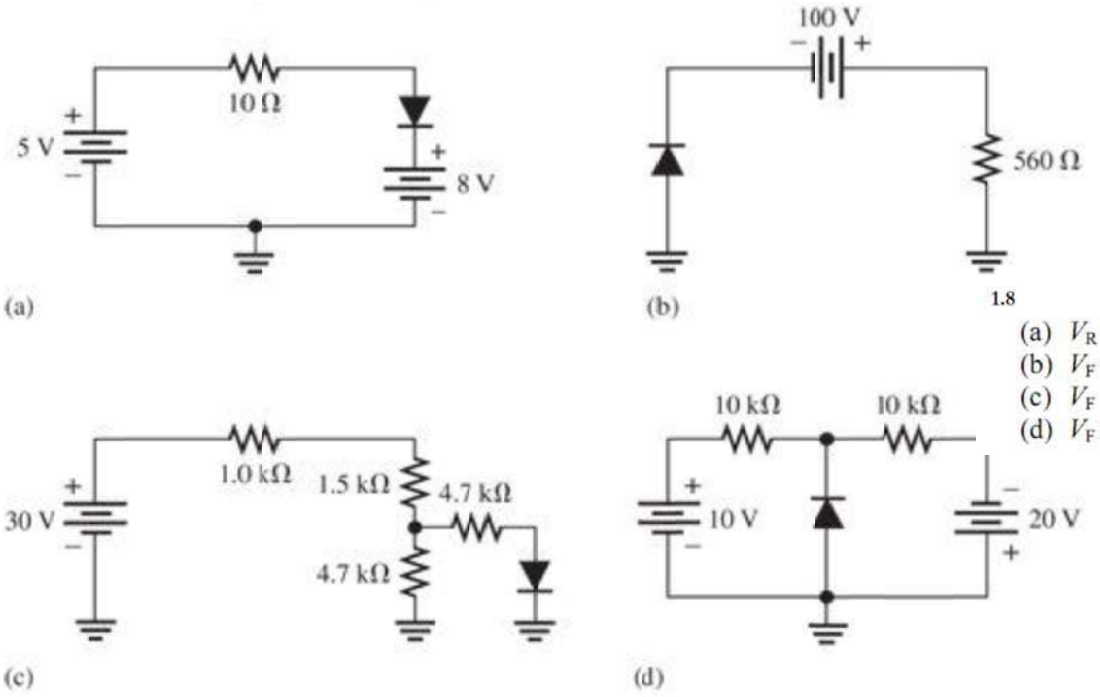


(a) The diode is reverse-biased.
(c) The diode is forward-biased.

Figure 1.1

(b) The diode is forward-biased.
(d) The diode is forward-biased.

1.8 Determine the voltage across each diode in Figure 1.1, assuming the practical model.



1.8

(a) $V_R = 5\text{ V} - 8\text{ V} = \mathbf{-3\text{ V}}$
(b) $V_F = \mathbf{0.7\text{ V}}$
(c) $V_F = \mathbf{0.7\text{ V}}$
(d) $V_F = \mathbf{0.7\text{ V}}$

Figure 1.1

1.9 Determine the voltage across each diode in Figure 1.1, assuming the ideal model.

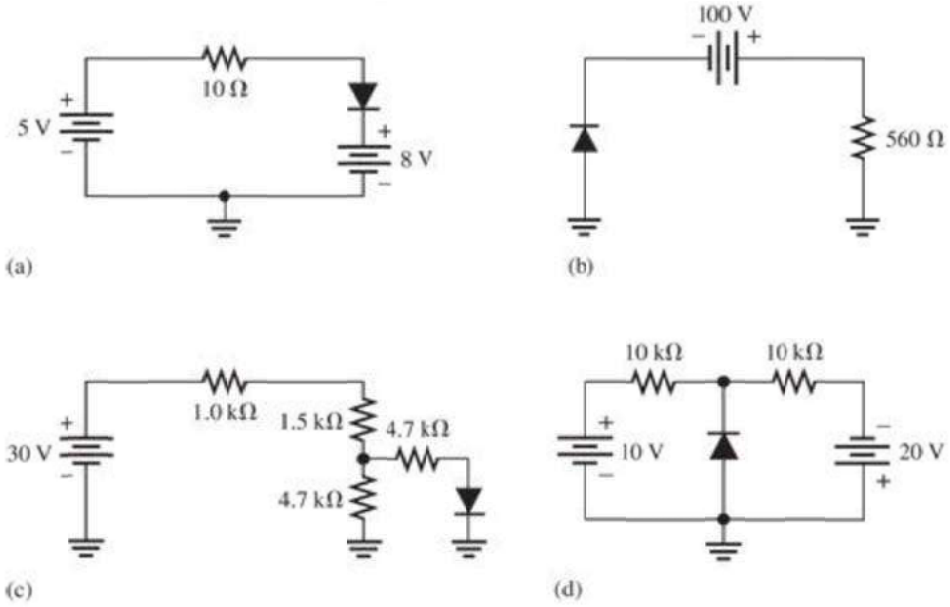
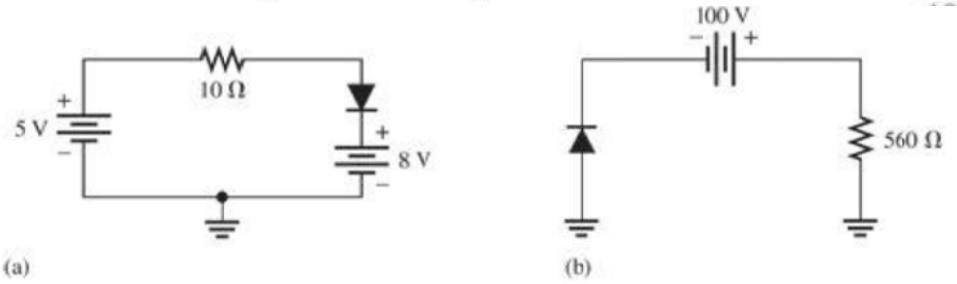


Figure 1.1

1.9

(a) $V_R = 5\text{ V} - 8\text{ V} = \mathbf{-3\text{ V}}$
(b) $V_F = \mathbf{0\text{ V}}$
(c) $V_F = \mathbf{0\text{ V}}$
(d) $V_F = \mathbf{0\text{ V}}$

1.10 Determine the voltage across each diode in Figure 1.1, using the complete diode model with $r'_d = 7\Omega$ and $r'_R = 100M\Omega$.



1.10

Ignoring r'_R

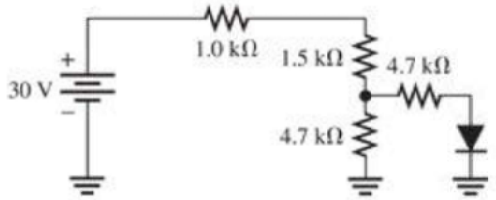
(a) $V_R \approx 5V - 8V = -3V$

(b)

$$I_F = \frac{100V - 0.7V}{560\Omega + 7\Omega} = 175.1mA$$

$$V_F = I_F r'_d + V_B = (175.1mA)(7\Omega) + 0.7V = 1.9V$$

1.10 Determine the voltage across each diode in Figure 1.1, using the complete diode model with $r'_d = 7\Omega$ and $r'_R = 100M\Omega$.



(c)

Figure 1.1

(c)

$$I_{tot} = \frac{30V}{R_{tot}}$$

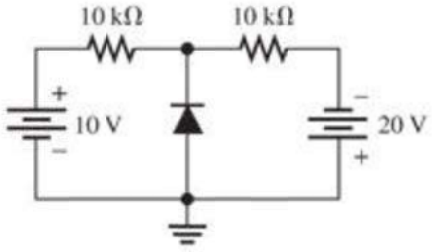
$$R_{tot} = 1K\Omega + 1.5K\Omega + \frac{4.7K\Omega}{2} = 2.5K\Omega + 2.35K\Omega = 4.85K\Omega$$

$$I_{tot} = \frac{30V}{4.85K\Omega} = 6.19mA$$

$$I_F = \frac{6.19mA}{2} = 3.1mA$$

$$V_F = I_F r'_d + V_B = (3.1mA)(7\Omega) + 0.7V = 0.722V$$

1.10 Determine the voltage across each diode in Figure 1.1, using the complete diode



(d)

(d)

Approximately all of the current from the 20 V source is through the diode. No current from the 10 V source is through the diode.

$$I_F = \frac{20V - 0.7V}{10K\Omega + 7\Omega} = 1.93mA$$

$$V_F = I_F r'_d + V_B = (1.93mA)(7\Omega) + 0.7V = 0.71V$$

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Chapter 3 Bipolar Junction Transistor

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CHAPTER OBJECTIVES ●

- Become familiar with the basic construction and operation of the Bipolar Junction Transistor.
- Be able to apply the proper biasing to insure operation in the active region.
- Recognize and be able to explain the characteristics of an *npn* or *pnp* transistor.
- Become familiar with the important parameters that define the response of a transistor.
- Be able to test a transistor and identify the three terminals.

On December 23, 1947, however, the electronics industry was to experience the advent of a completely new direction of interest and development. It was on the afternoon of this day that Dr. S. William Shockley, Walter H. Brattain, and John Bardeen demonstrated the amplifying action of the first transistor at the Bell Telephone Laboratories as shown inFig. 3.1 . The original transistor (a point-contact transistor) is shown in Fig. 3.2 . The advantages of this three-terminal solid-state device over the tube were immediately obvious:

- It was smaller and lightweight;
- it had no heater requirement or heater loss;
- it had a rugged construction;
- it was more efficient since less power was absorbed by the device itself;
- it was instantly available for use, requiring no warm-up period; and lower operating voltages were possible.

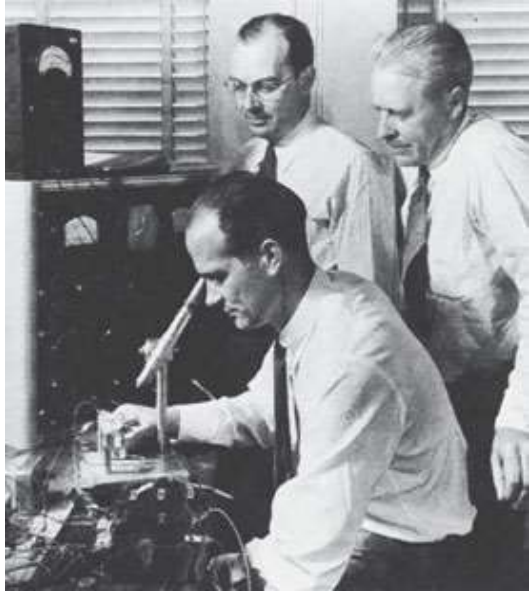


FIG. 3.1
Coinventors of the first transistor at Bell Laboratories.

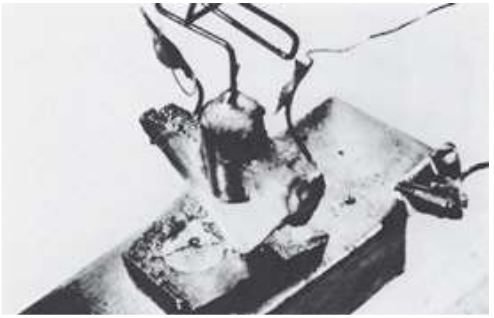
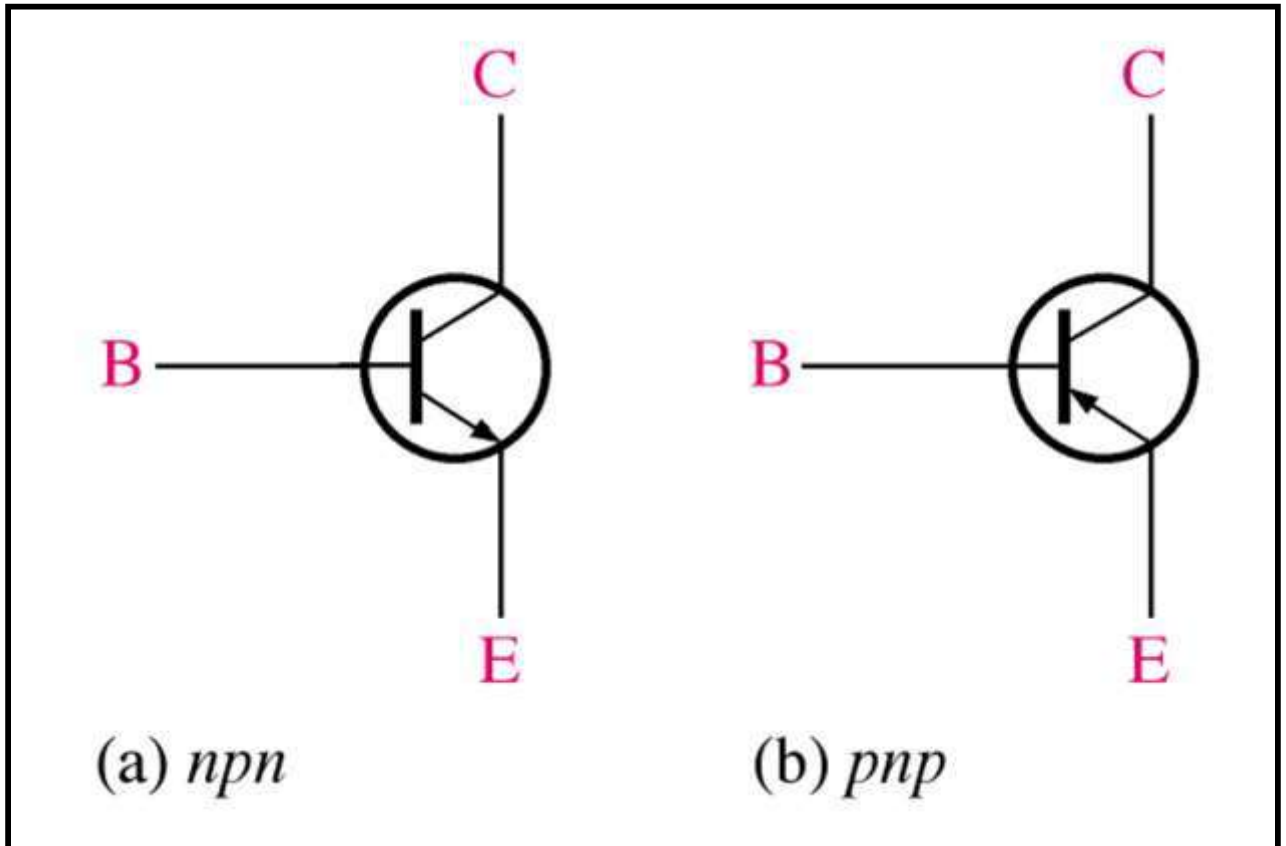


FIG. 3.2
The first transistor. (Courtesy of AT&T Archives and History Center.)

Note that this chapter is our first discussion of devices with three or more terminals. You will find that all amplifiers (devices that increase the voltage, current, or power level) have at least three terminals, with one controlling the flow or potential between the other two.

A **transistor** is a device which can be used as either an **amplifier** or a **switch**. Let's first consider its operation in a more simple view as a current controlling device.



TRANSISTOR CONSTRUCTION

The transistor is a three-layer semiconductor device consisting of either two *n* - and one *p* -type layers of material or two *p* - and one *n* -type layers of material. The former is called an *npn transistor* , and the latter is called a *pnp transistor* . Both are shown in Fig. 3.3 with the proper dc biasing. The emitter layer is heavily doped, with the base and collector only lightly doped. The outer layers have widths much greater than the sandwiched p - or n -type material.

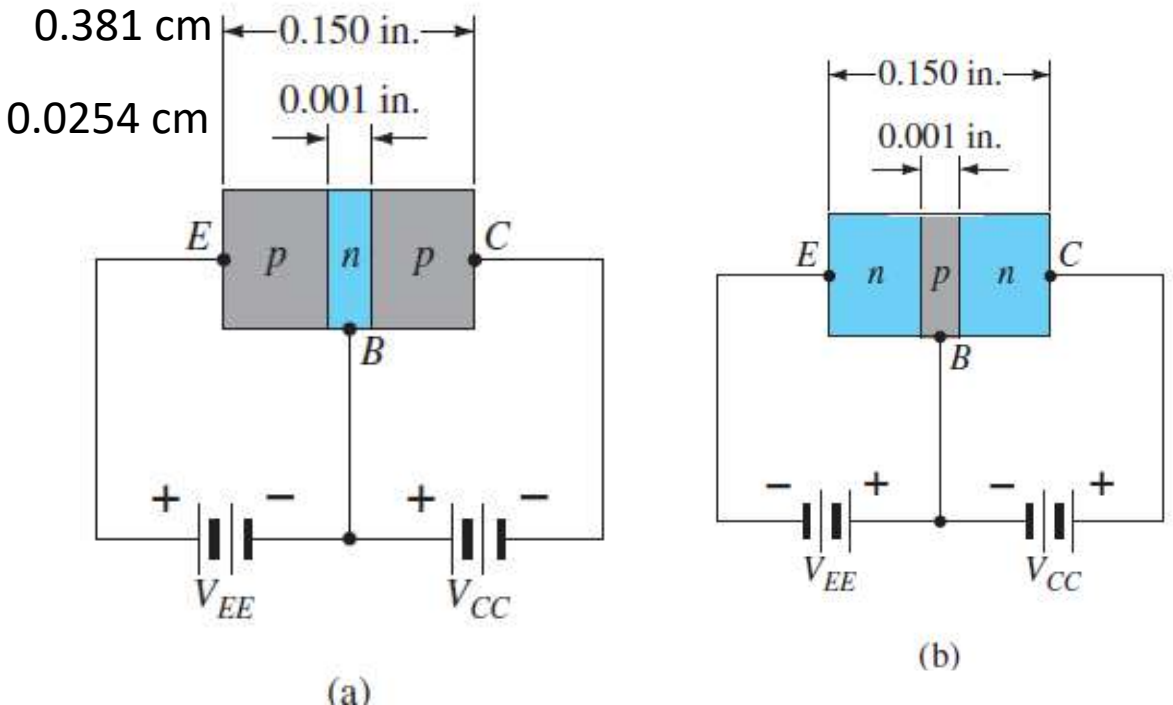


FIG. 3.3
Types of transistors: (a) pnp; (b) npn.

TRANSISTOR OPERATION

The basic operation of the transistor will now be described using the *pnp* transistor of Fig. 3.3a . The operation of the *npn* transistor is exactly the same if the roles played by the electron and hole are interchanged. In Fig. 3.4a the *pnp* transistor has been redrawn without the base-to collector bias. Note the similarities between this situation and that of the *forward-biased* diode in Chapter 1 . The depletion region has been reduced in width due to the applied bias, resulting in a heavy flow of majority carriers from the *p* - to the *n* -type material.

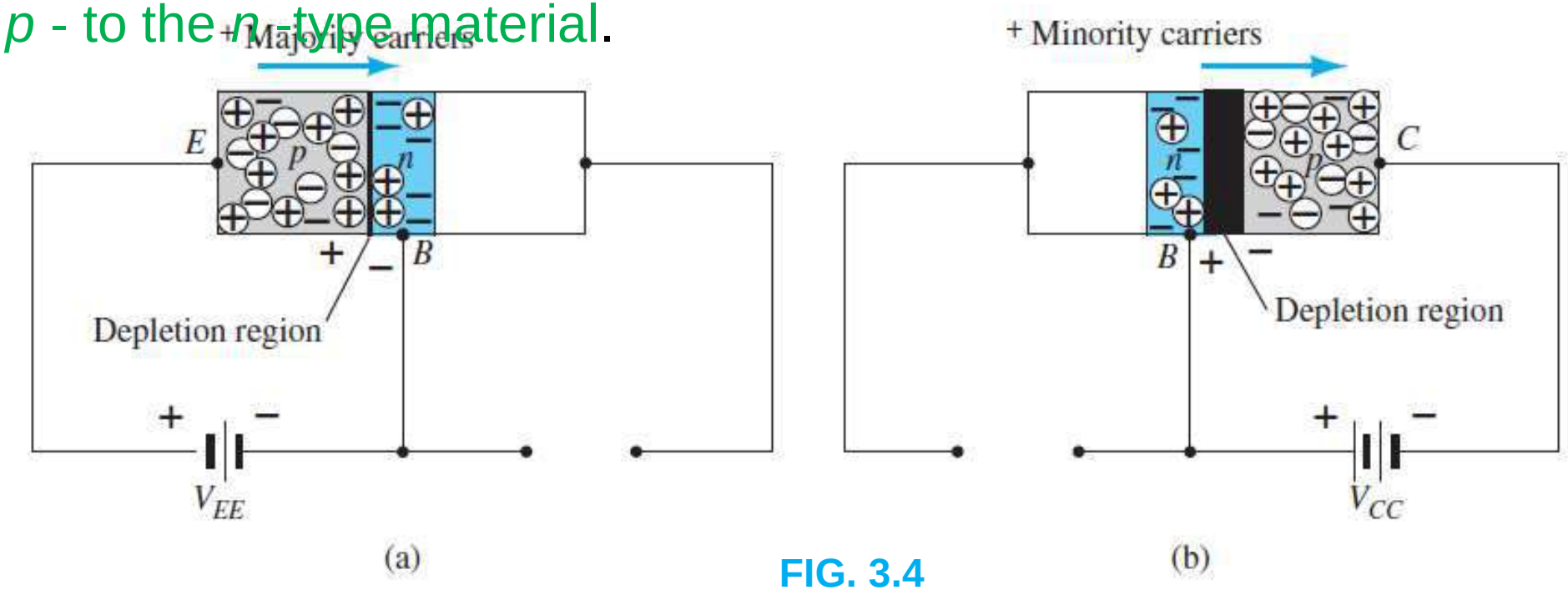


FIG. 3.4

Biasing a transistor: (a) forward-bias; (b) reverse-bias.

TRANSISTOR OPERATION

Let us now remove the base-to-emitter bias of the *pnp* transistor of Fig. 3.3a as shown in Fig. 3.4b . Consider the similarities between this situation and that of the *reverse-biased* diode of Section 1.6 . Recall that the flow of majority carriers is zero, resulting in only a minority-carrier flow, as indicated in Fig. 3.4b. In summary, therefore:

One p–n junction of a transistor is reverse-biased, whereas the other is forward·

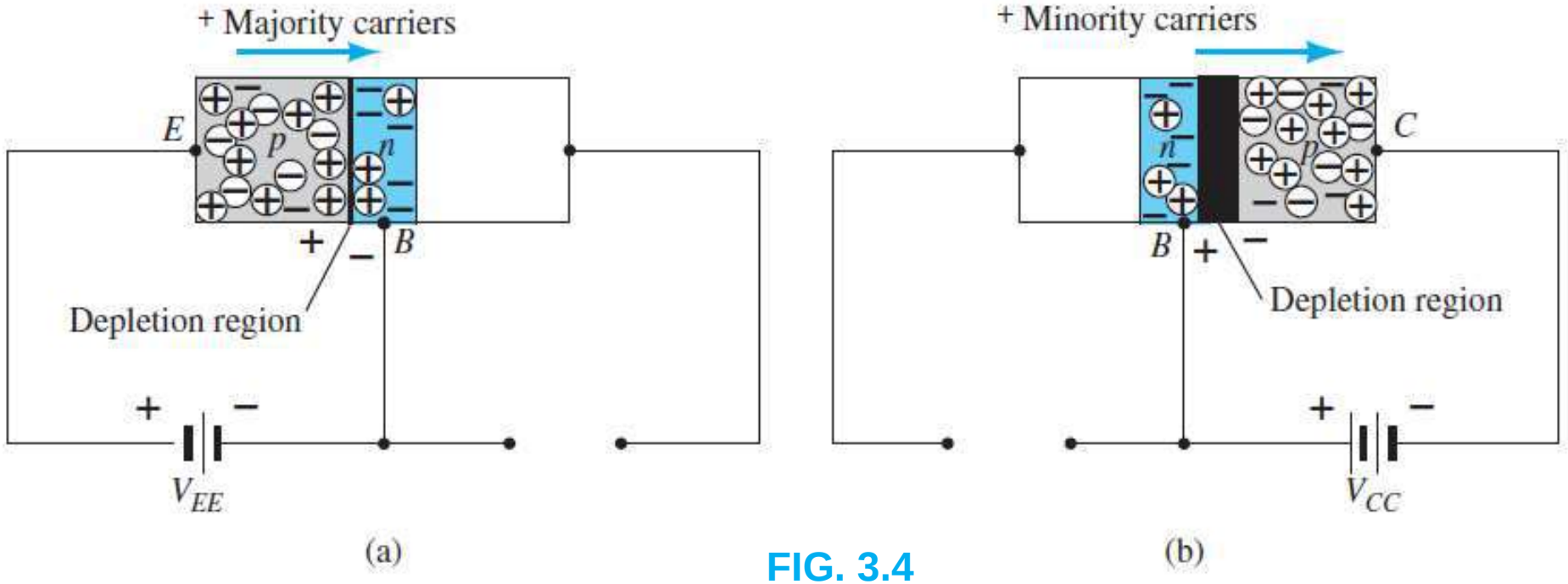
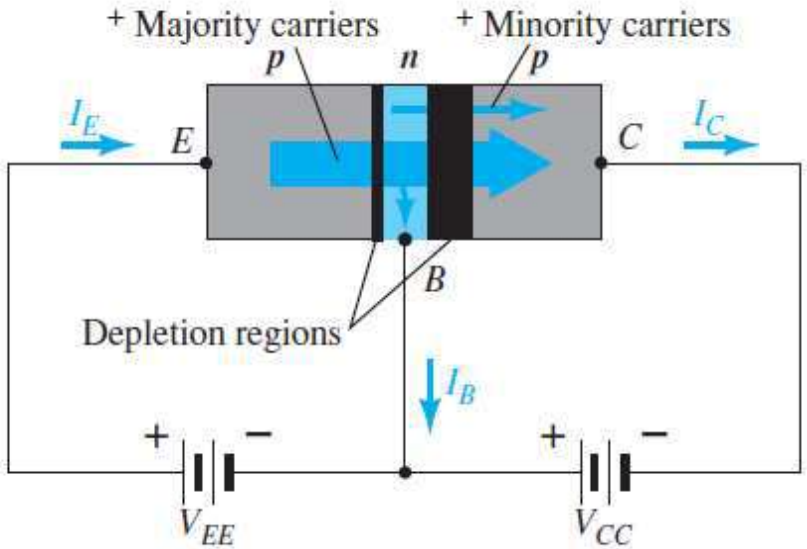


FIG. 3.4

Biasing a transistor: (a) forward-bias; (b) reverse-bias.

In Fig. 3.5 both biasing potentials have been applied to a *pnp* transistor, with the resulting majority- and minority-carrier flows indicated. Note in Fig. 3.5 the widths of the depletion regions, indicating clearly which junction is forward-biased and which is reverse-biased. As indicated in Fig. 3.5 , a large number of majority carriers will diffuse across the forward biased *p–n* junction into the *n* -type material. The question then is whether these carriers will contribute directly to the base current I_B or pass directly into the *p*-type material. Since the sandwiched *n* -type material is very thin and has a low conductivity, a very small number of these carriers will take this path of high resistance to the base terminal. The magnitude of the base current is typically on the order of microamperes, as compared to milliamperes for the emitter and collector currents. The larger number of these majority carriers will diffuse across the reverse-biased junction into the *p* -type material connected to the collector terminal as indicated in Fig. 3.5



Applying Kirchhoff's current law to the transistor of Fig. 3.5 as if it were a single node, we obtain

$$I_E = I_C + I_B$$

and find that the emitter current is the sum of the collector and base currents. The collector current, however, comprises two components—the majority and the minority carriers as indicated in Fig. 3.5 . The minority-current component is called the *leakage current* and is given the symbol I_{CO} (I_C current with emitter terminal Open). The collector current, therefore, is determined in total by

$$I_C = I_{C\,\text{majority}} + I_{CO\text{minority}}$$

For general-purpose transistors, I_C is measured in milliamperes and I_{CO} is measured in microamperes or nanoamperes. I_{CO} , like I_S for a reverse-biased diode, is temperature sensitive and must be examined carefully when applications of wide temperature ranges are considered . It can severely affect the stability of a system at high temperature if not considered properly. Improvements in construction techniques have resulted in significantly lower levels of I_{CO} , to the point where its effect can often be ignored

COMMON-BASE CONFIGURATION

The notation and symbols are indicated in Fig. 3.6 for the common-base configuration with *pnp* and *npn* transistors. The common-base terminology is derived from the fact that the base is common to both the int and output sides of the configuration.

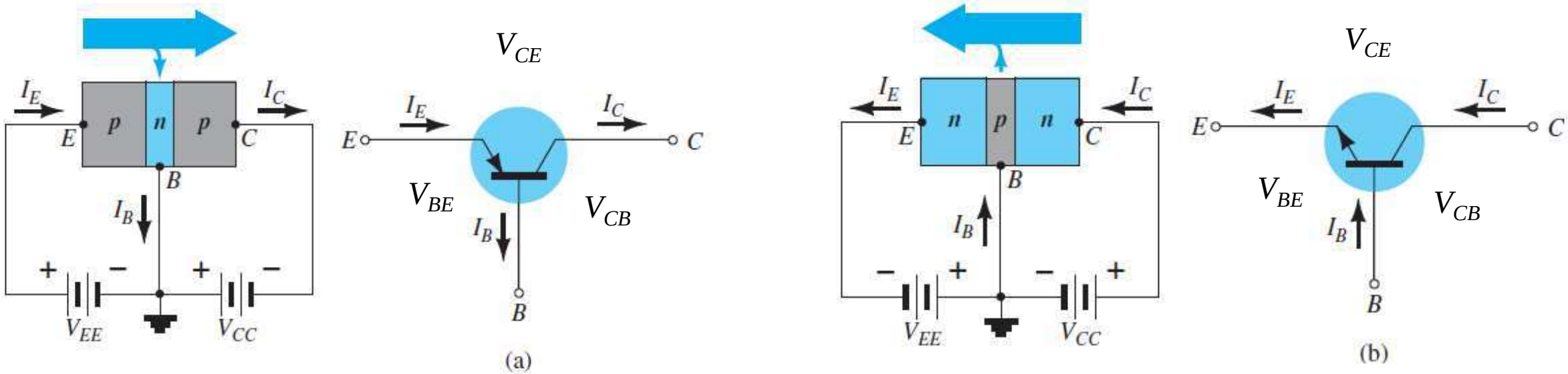


FIG. 3.6 *Notation and symbols used with the common-base configuration: (a) pnp transistor; (b) npn transistor.*

For the transistor:

The arrow in the graphic symbol defines the direction of emitter current (conventional flow) through the device.

All the current directions appearing in Fig. 3.6 are the actual directions as defined by the choice of conventional flow. Note in each case that $I_E = I_C + I_B$. Note also that the applied biasing (voltage sources) are such as to establish current in the direction indicated for each branch. That is, compare the direction of I_E to the polarity of V_{EE} for each configuration and the direction of I_C to the polarity of V_{CC} .

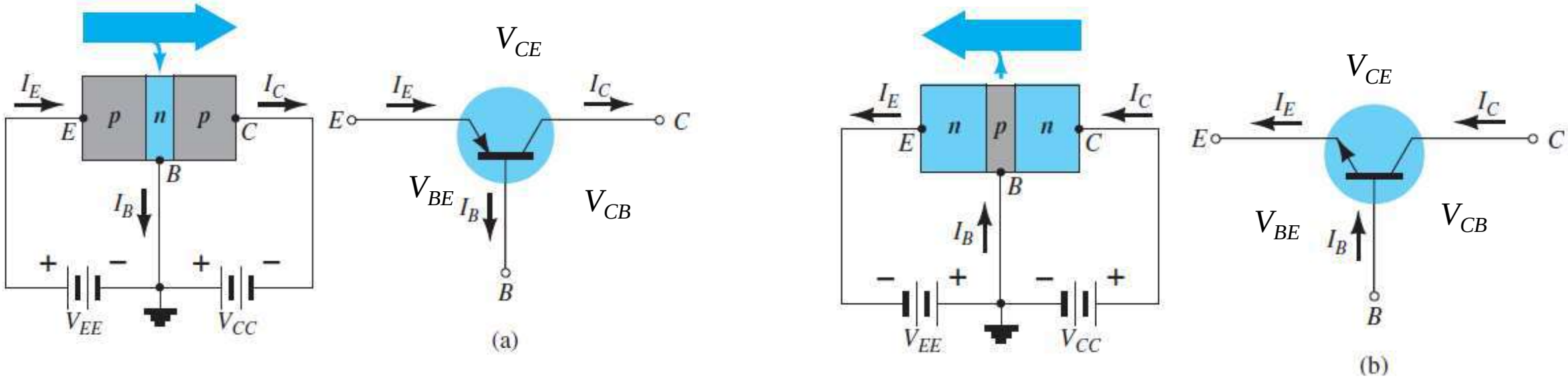
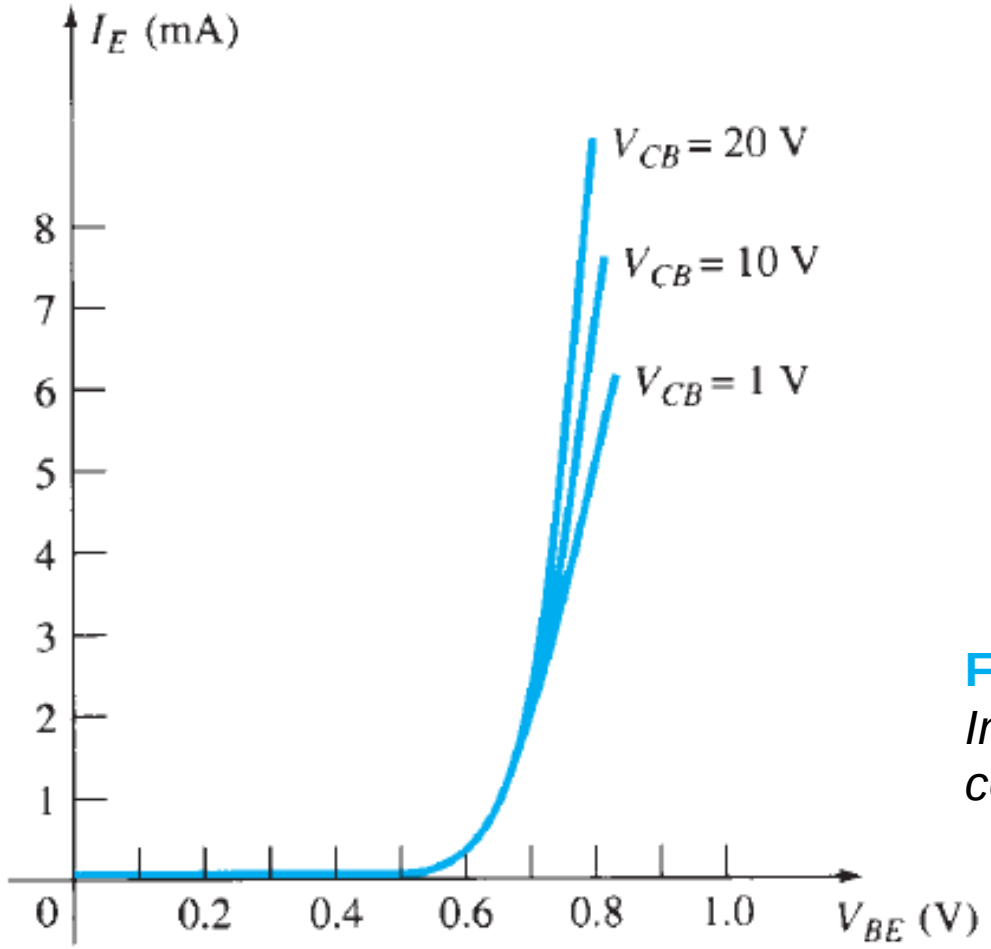


FIG. 3.6 *Notation and symbols used with the common-base configuration: (a) pnp transistor; (b) npn transistor.*

The input set for the common-base amplifier as shown in Fig. 3.7 relates an input current (I_E) to an input voltage (V_{BE}) for various levels of output voltage (V_{CB}).



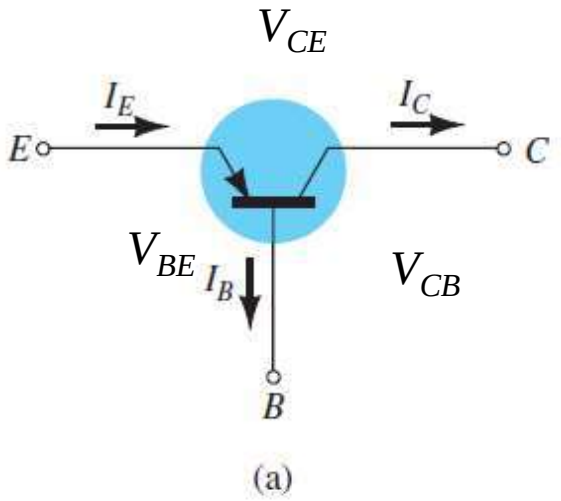


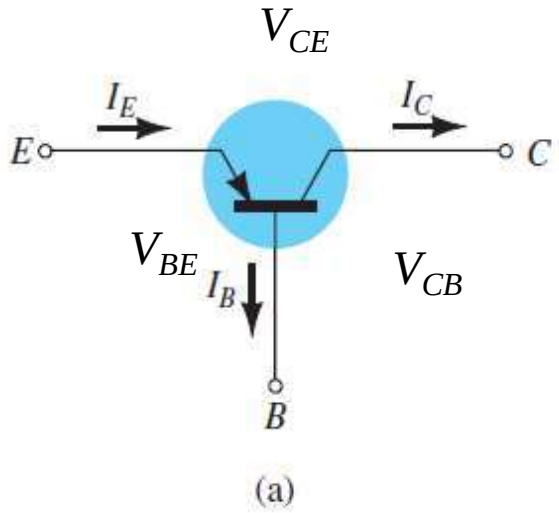
FIG. 3.7
Input or driving point characteristics for a common-base silicon transistor amplifier.

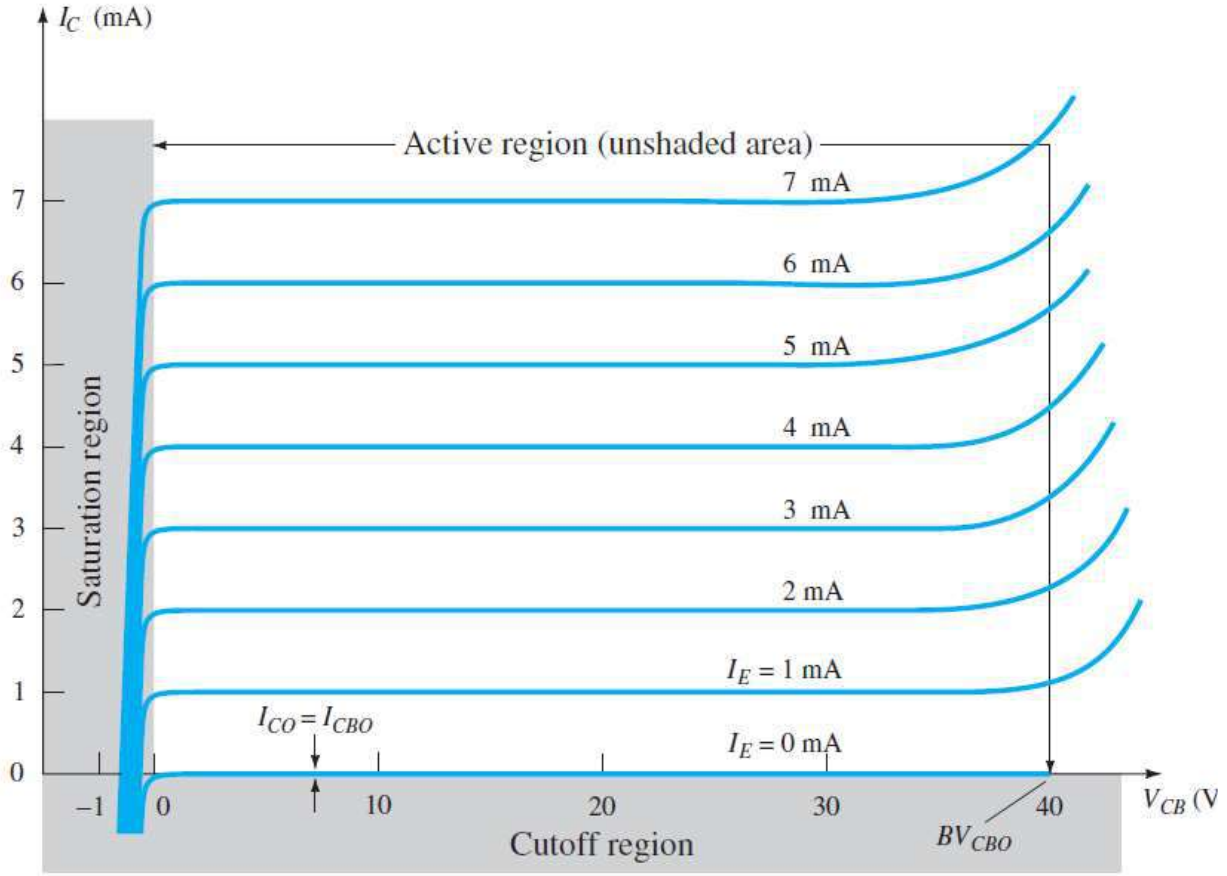
The output set relates an output current (I_C) to an output voltage (V_{CB}) for various levels of input current (I_E) as shown in Fig. 3.8 . The output or *collector* set of characteristics has three basic regions of interest, as indicated in Fig. 3.8 *: the active* , *cutoff* , and *saturation regions*. The active region is the region normally employed for linear (undistorted) amplifiers. In particular:

In the active region the base–emitter junction is forward-biased, whereas the collector–base junction is reverse-biased.

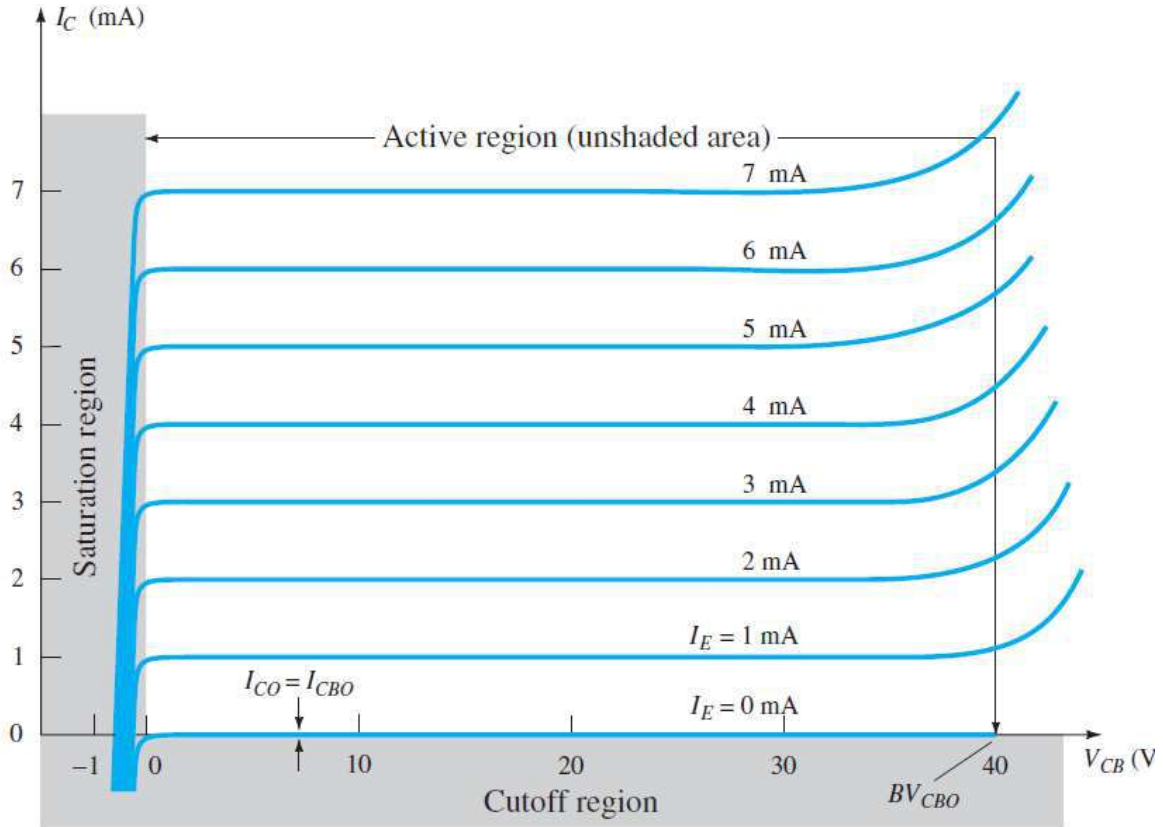
FIG. 3.8

Output or collector characteristics for a common-base transistor amplifier.





The active region is defined by the biasing arrangements of Fig. 3.6 . At the lower end of the active region the emitter current (I_E) is zero, and the collector current is simply that due to the reverse saturation current I_{CO} , as indicated in Fig. 3.9 . The current I_{CO} is so small (microamperes) in magnitude compared to the vertical scale of I_C (milliamperes) that it appears on virtually the same horizontal line as I_C 0. The circuit conditions that exist when I_E 0 for the common-base configuration are shown in Fig. 3.9 . The notation most frequently used for I_{CO} on data and specification sheets is,



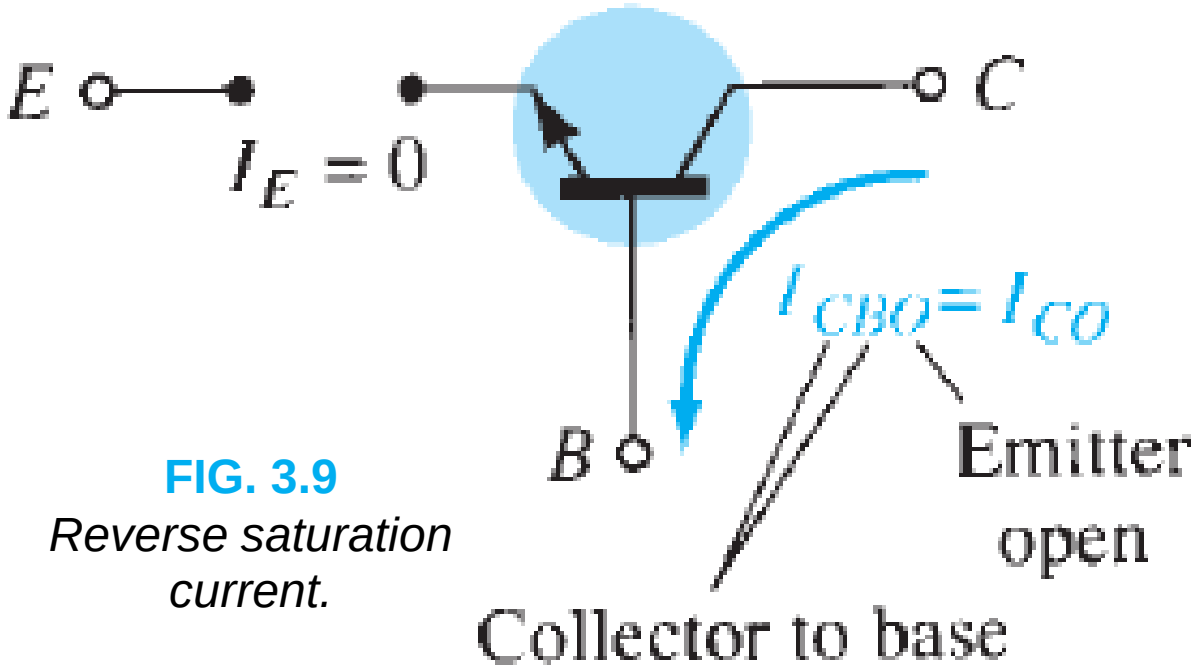


FIG. 3.9
Reverse saturation current.

As indicated in Fig. 3.9 , I_{CBO} (the collector-to-base current with the emitter leg open). its effect can be ignored. However, for higher power units I_{CBO} will still appear in the microampere range. In addition, keep in mind that I_{CBO} , like I_S , for the diode (both reverse leakage currents) is temperature sensitive. At higher temperatures, the effect of I_{CBO} may become an important factor since it increases so rapidly with temperature.

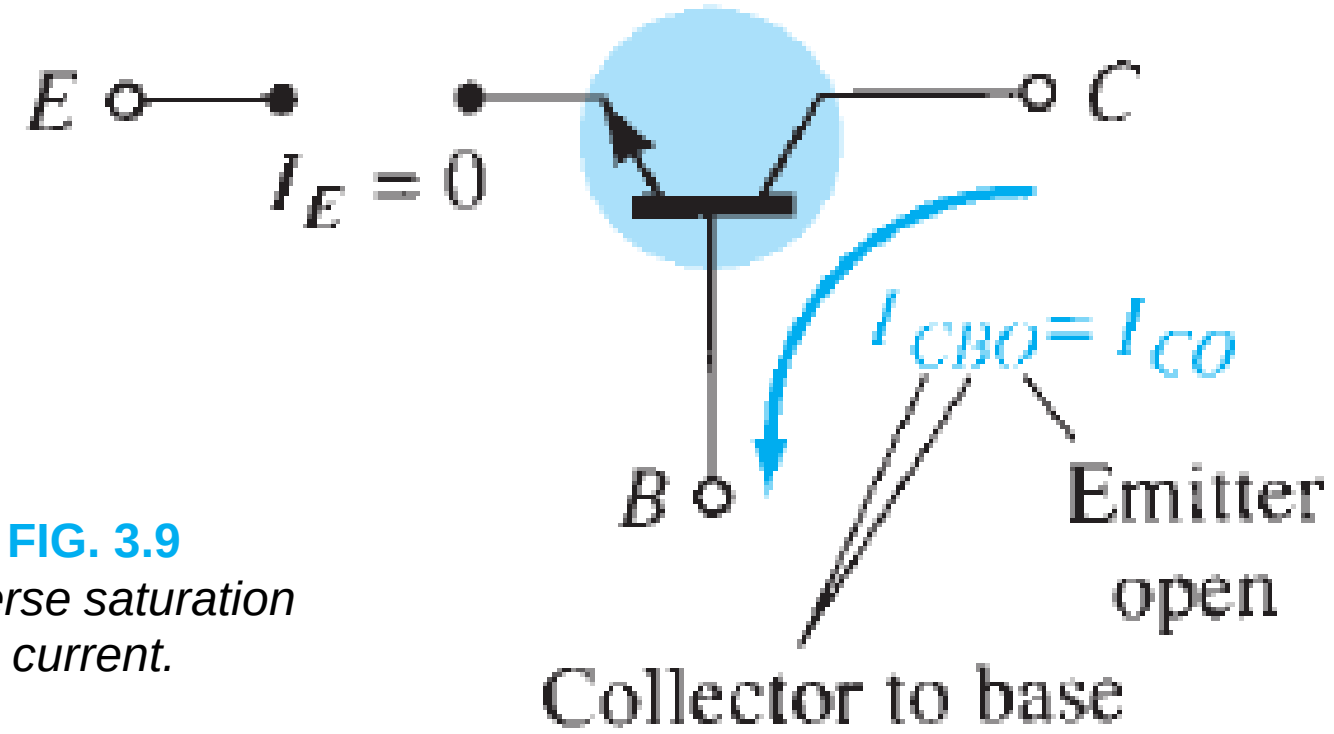


FIG. 3.9
Reverse saturation current.

Note in Fig. 3.8 that as the emitter current increases above zero, the collector current increases to a magnitude essentially equal to that of the emitter current as determined by the basic transistor-current relations. Note also the almost negligible effect of V_{CB} on the collector current for the active region. The curves clearly indicate that *a first approximation to the relationship between I_E and I_C in the active region is given by*

$$I_C \cong I_E$$

As inferred by its name, the **cutoff region** is defined as that region where the collector current is 0 A, as revealed on Fig. 3.8 .

In addition: In the cutoff region the base–emitter collector–base junctions of a transistor are both reverse-biased.

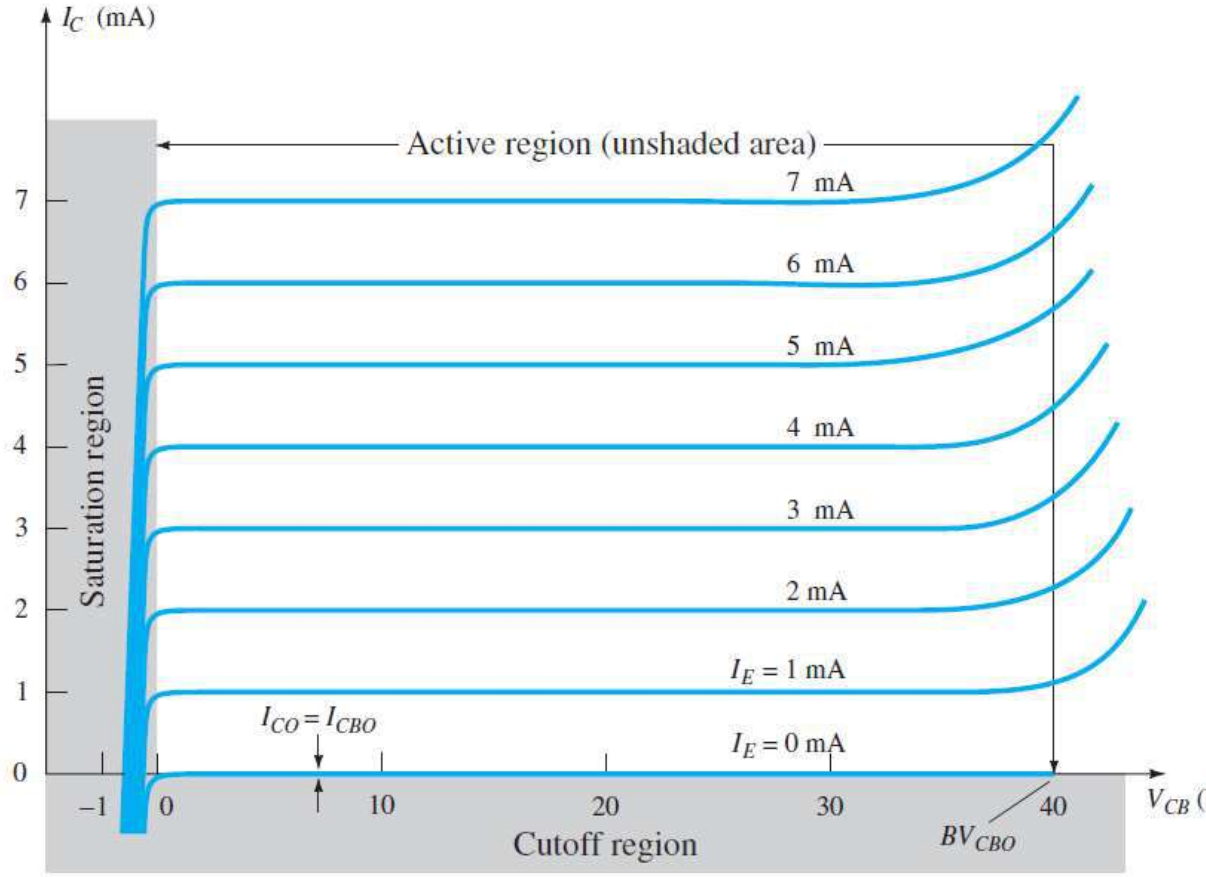


FIG. 3.8
Output or collector characteristics for a common-base transistor amplifier.

The saturation region is defined as that region of the characteristics to the left of V_{CB} 0 V. The horizontal scale in this region was expanded to clearly show the dramatic change in characteristics in this region. Note the exponential increase in collector current as the voltage V_{CB} increases toward 0 V.

In the saturation region the base–emitter and collector–base junctions are forward-biased.

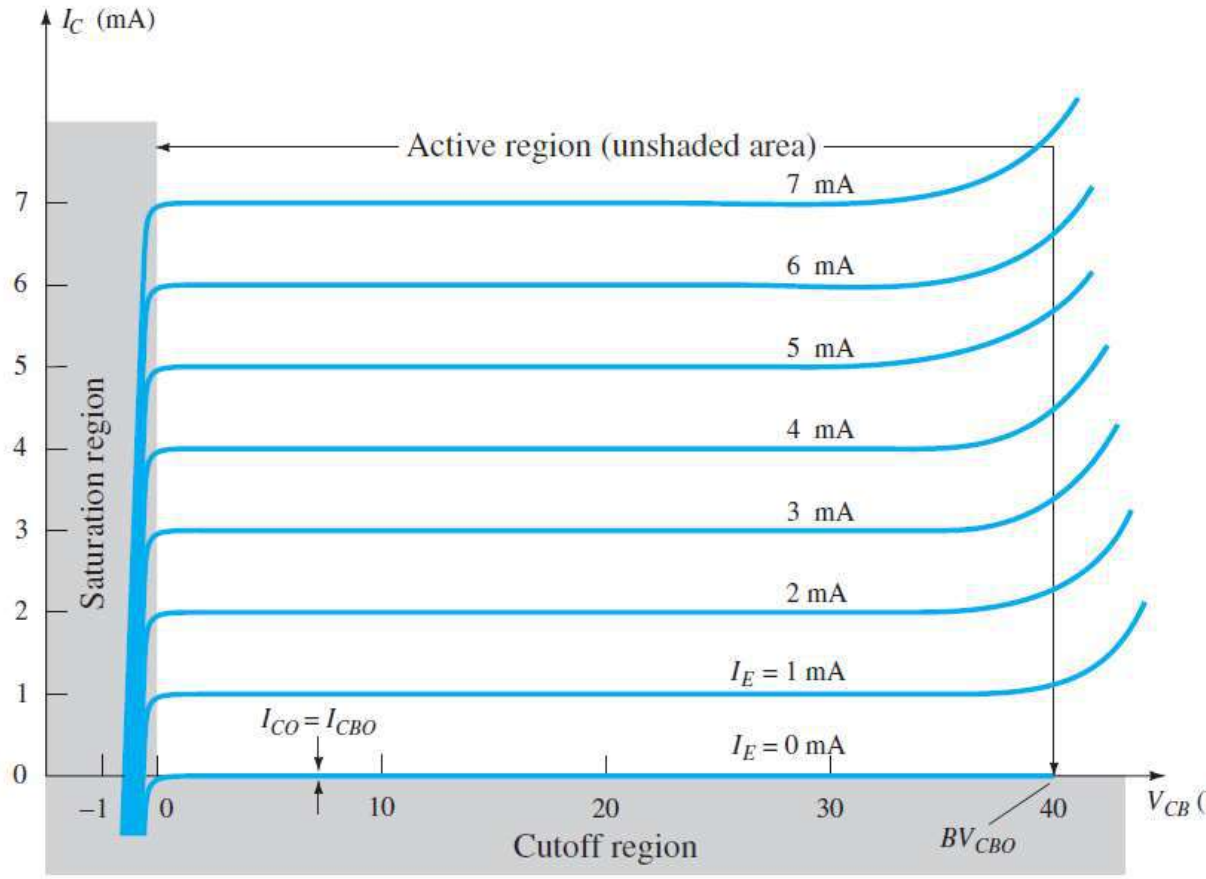


FIG. 3.8
Output or collector characteristics for a common-base transistor amplifier.

The input characteristics of Fig. 3.7 reveal that for fixed values of collector voltage (V_{CB}), as the base-to-emitter voltage increases, the emitter current increases in a manner that closely resembles the diode characteristics. In fact, increasing levels of V_{CB} have such a small effect on the characteristics that as a first approximation the change due to changes in ***V_{CB}* can be ignored** and the characteristics drawn assumed to be the following:

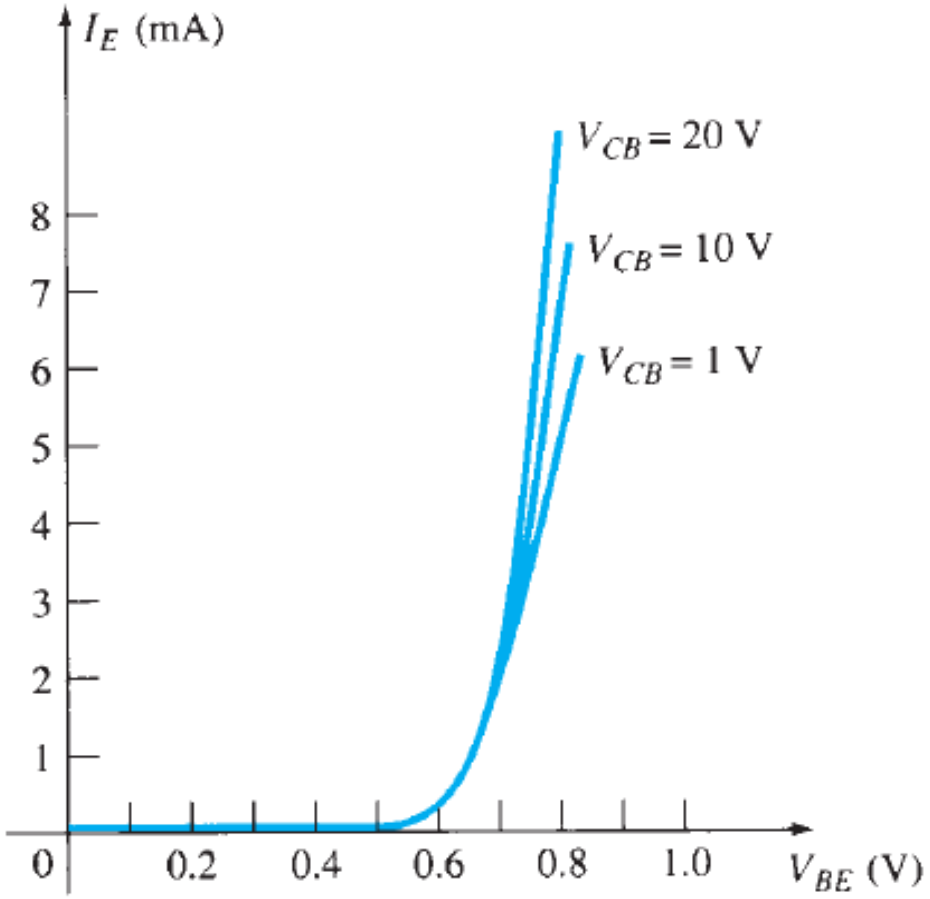


FIG. 3.7
Input or driving point characteristics for a common-base silicon transistor amplifier.

As shown in Fig. 3.10a . If we then apply the piecewise linear approach, the characteristics of Fig. 3.10b result. Taking it a step further and ignoring the slope of the curve and therefore the resistance associated with the forward-biased junction results in the characteristics of Fig. 3.10c . For the analysis to follow in this book the equivalent model of Fig. 3.10c will be employed for all dc analysis of transistor networks. **That is, once a transistor is in the "on" state, the base-to-emitter voltage will be assumed to be the following**: $V_{BE} \cong 0.7$ V

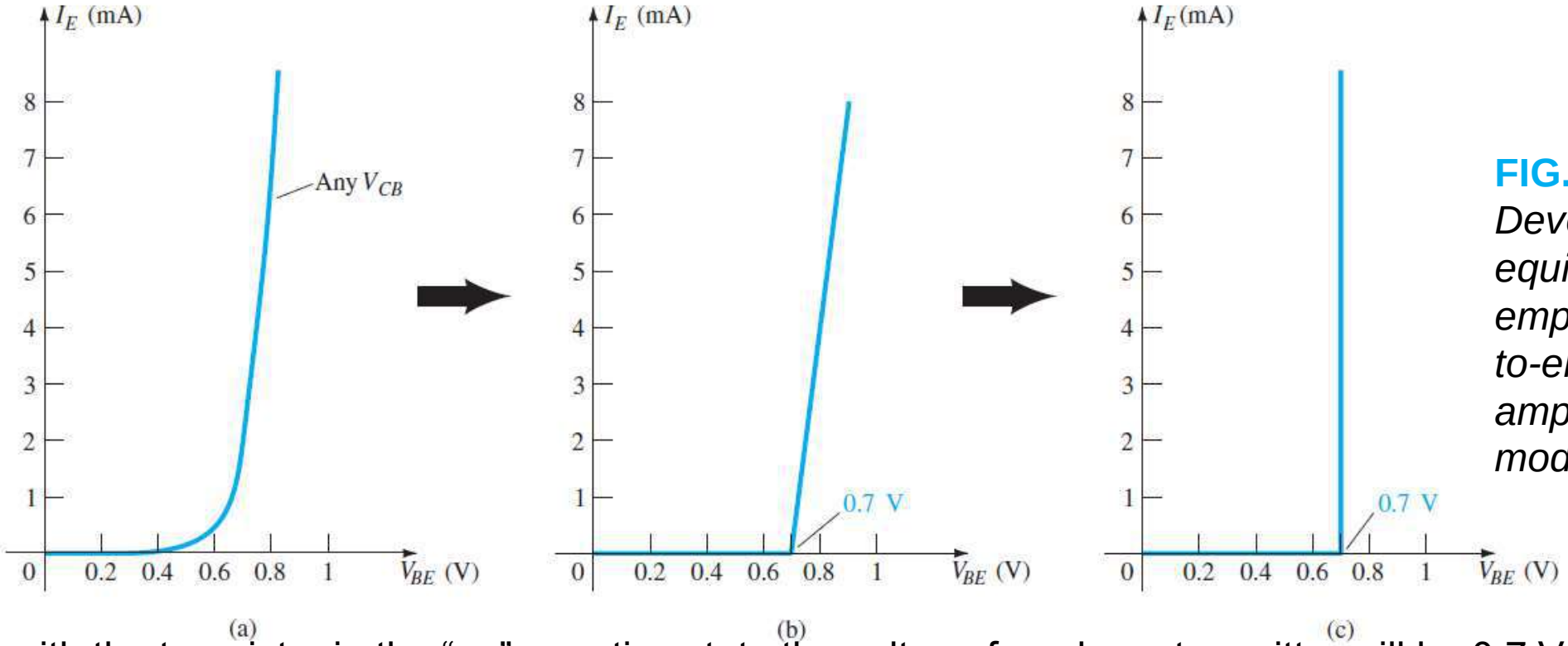


FIG. 3.10
Developing the equivalent model to be employed for the base-to-emitter region of an amplifier in the dc mode.

with the transistor in the "on" or active state the voltage from base to emitter will be 0.7 V at *any* level of emitter current as controlled by the external network

Alpha (A)

DC Mode In the dc mode the levels of I_C and I_E due to the majority carriers are related by a quantity called *alpha* and defined by the following equation:

$$\alpha = \frac{I_C}{I_E}$$

where I_C and I_E are the levels of current at the point of operation. For practical devices alpha typically extends from 0.90 to 0.998, with most values approaching the high end of the range. Since alpha is
defined solely for the majority carriers, Eq. (3.2) becomes

$$I_C = \alpha I_E + I_{CBO}$$

For the characteristics of Fig. 3.8 when I_E 0 mA, I_C is therefore equal to I_{CBO} ,. In other words, when I_E 0 mA, I_C also appears to be 0 mA for the range of V_{CB} values.

AC Mode For ac situations where the point of operation moves on the characteristic curve, an ac alpha is defined by

$$\alpha_{ac} = \left.\frac{\Delta I_C}{\Delta I_E}\right|_{V_{CB}=\text{constant}}$$

Biasing

The proper biasing of the common-base configuration in the active region can be determined quickly using the approximation $I_C \cong I_E$ and assuming for the moment that $I_B \cong 0\ \mu A$. The result is the configuration of Fig. 3.11 for the *pnp* transistor. The arrow of the symbol defines the direction of conventional flow for $I_E \cong I_C$. The dc supplies are then inserted with a polarity that will support the resulting current direction. For the *npn* transistor the polarities will be reversed.

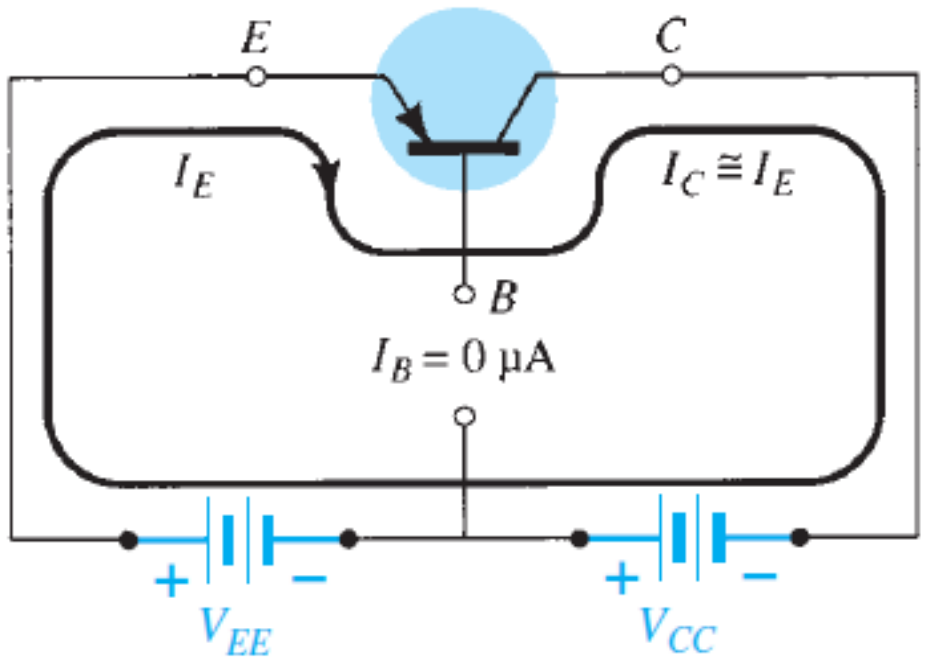


FIG. 3.11
Establishing the proper biasing management for a common-base pnp transistor in the active region.

Breakdown Region

As the applied voltage V_{CB} increases there is a point where the curves take a dramatic upswing in Fig. 3.8. This is due primarily to an avalanche effect similar to that described for the diode in Chapter 1 when the reverse-bias voltage reached the breakdown region.

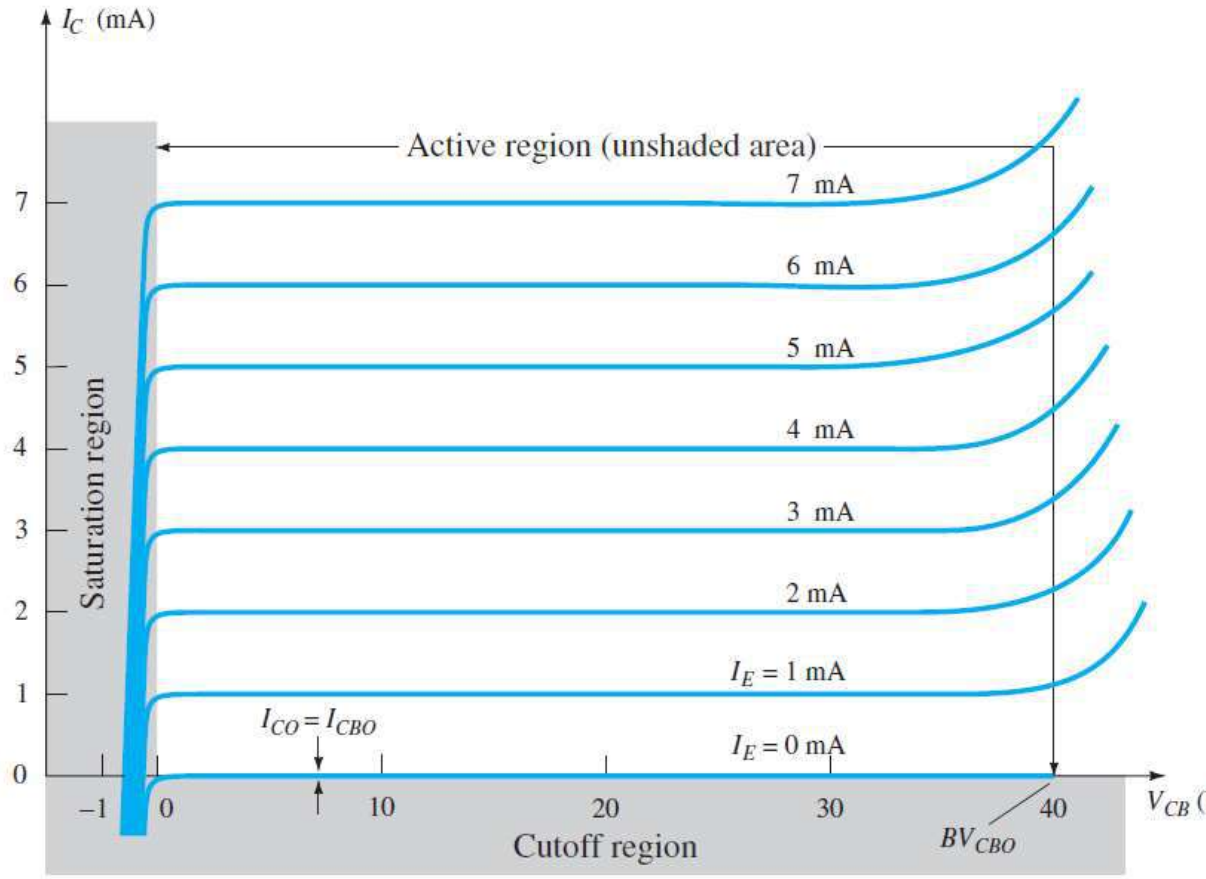


FIG. 3.8
Output or collector characteristics for a common-base transistor amplifier.

As stated earlier the base-to-collector junction is reversed biased in the active region, but there is a point where too large a reverse-bias voltage will lead to the avalanche effect. The result is a large increase in current for small increases in the base-to-collector voltage. The largest permissible base-to-collector voltage is labeled BV_{CBO} as shown in Fig. 3.8. It is also referred to as $V_{(BR)CBO}$ as shown on the characteristics of Fig. 3.23 to be discussed later.

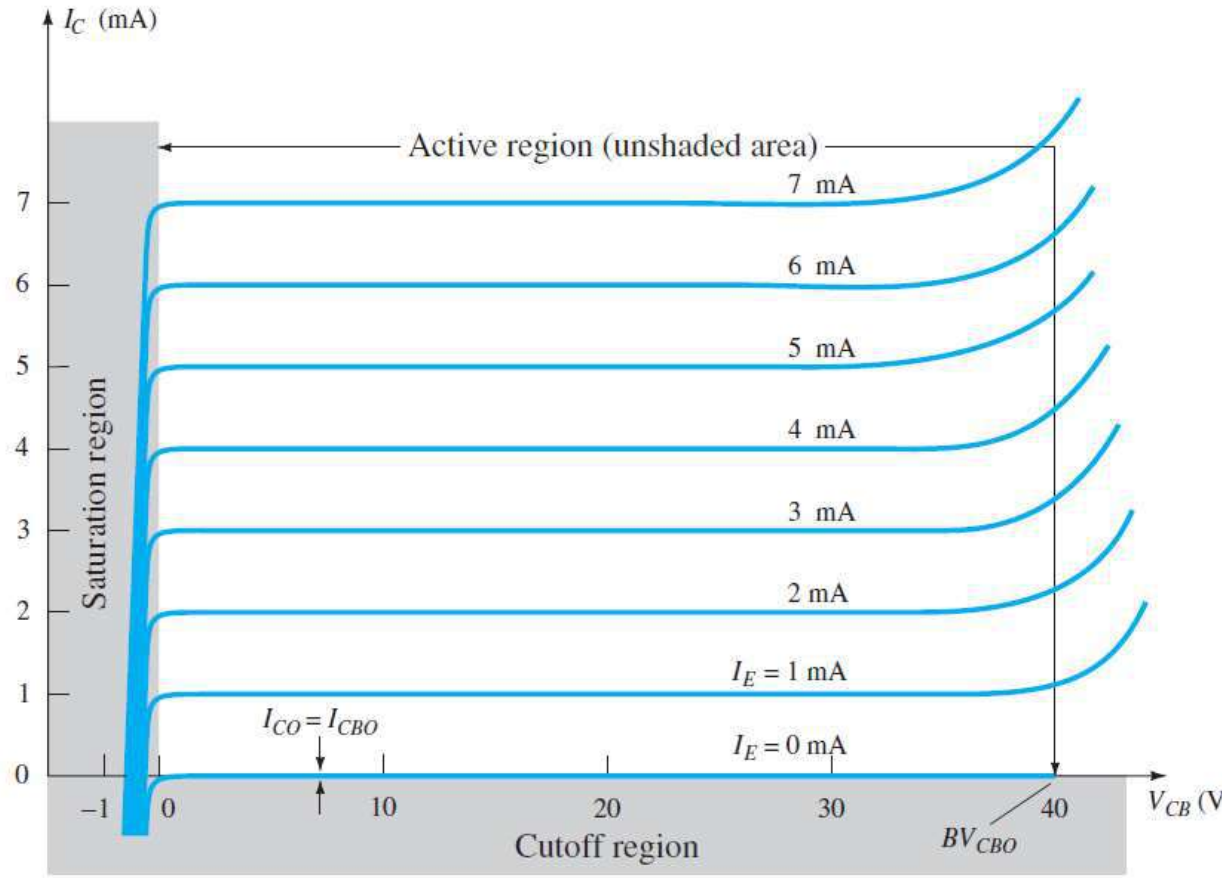


FIG. 3.8

Output or collector characteristics for a common-base transistor amplifier.

TRANSISTOR AMPLIFYING ACTION

Now that the relationship between I_C and I_E has been established in Section 3.4, the basic amplifying action of the transistor can be introduced on a surface level using the network of Fig. 3.12. The dc biasing does not appear in the figure since our interest will be limited to the ac response. For the common-base configuration the ac input resistance determined by the characteristics of Fig. 3.7 is quite small and typically varies from 10 to 100 Ω. The output resistance as determined by the curves of Fig. 3.8 is quite high (the more horizontal the curves the higher the resistance) and typically varies from 50 kΩ to 1 MΩ (100 kΩ for the transistor of Fig. 3.12). The difference in resistance is due to the forward-biased junction at the input (base to emitter) and the reverse-biased junction at the output (base to collector). Using a common value of 20 Ω for the input resistance, we find that

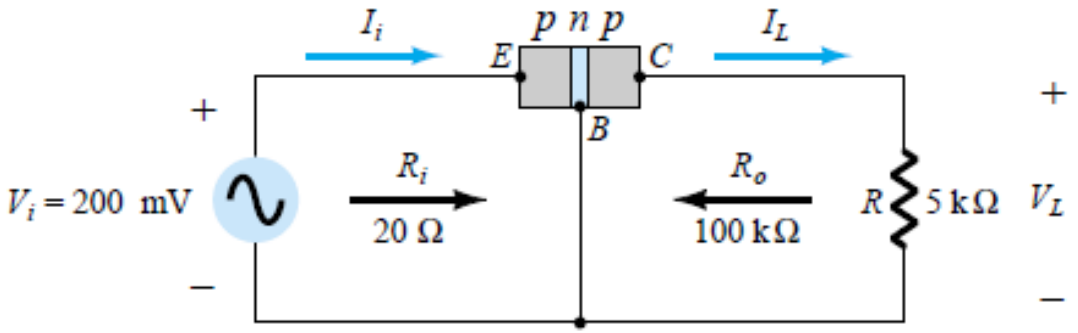
$$I_i = \frac{V_i}{R_i} = \frac{200\text{ mV}}{20\ \Omega} = \mathbf{10\ mA}$$

If we assume for the moment that $\alpha_{ac} = 1$ ($I_c = I_e$),

$$I_L = I_i = 10\text{ mA}$$

and

$$V_L = I_L R$$
$$= (10\text{ mA})(5\text{ k}\Omega)$$
$$= \mathbf{50\ V}$$



12 Basic voltage amplification action of the common-base configuration.

The voltage amplification is

$$A_v = \frac{V_L}{V_i} = \frac{50\text{ V}}{200\text{ mV}} = \mathbf{250}$$

Typical values of voltage amplification for the common-base configuration vary from 50 to 300. The current amplification (I_C/I_E) is always less than 1 for the common-base configuration. This latter characteristic should be obvious since $I_C = \alpha I_E$ and α is always less than 1.

The basic amplifying action was produced by transferring a current I from a low- to a high-*resistance* circuit. The combination of the two terms in italics results in the label *transistor;* that is,

$$\textit{trans}\text{fer} + \text{re}\textit{sistor} \rightarrow \textit{transistor}$$

3.5 COMMON-EMITTER CONFIGURATION

The most frequently encountered transistor configuration appears in Fig. 3.12 for the *pnp* and *npn* transistors. It is called the *common-emitter configuration* because the emitter is common to both the input and output terminals (in this case common to both the base and collector terminals).

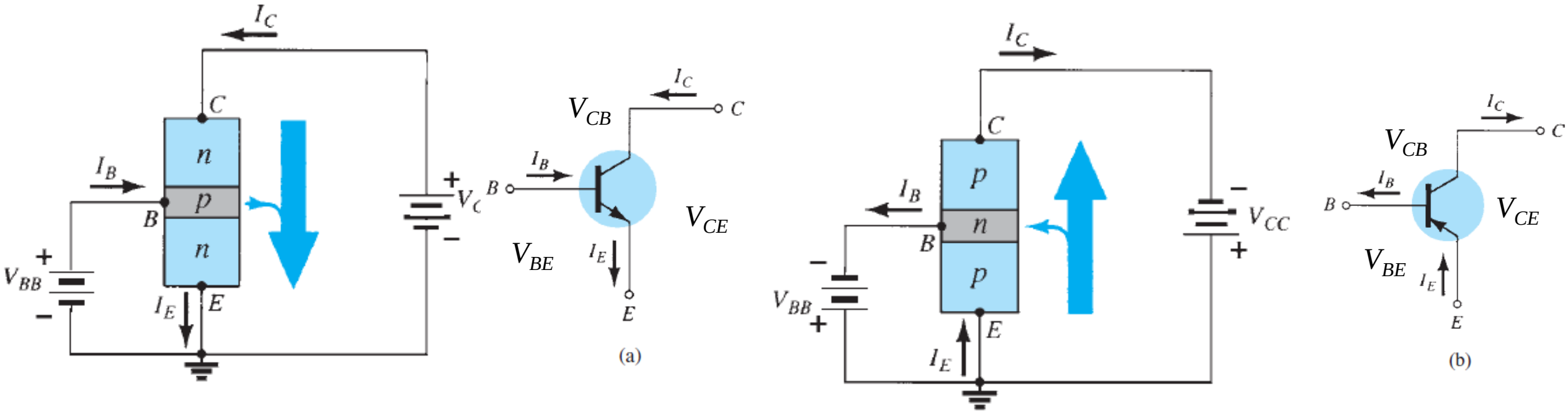


FIG. 3.12

Notation and symbols used with the common-emitter configuration: (a) npn transistor; (b) pnp transistor.

Two sets of characteristics are again necessary to describe fully the behavior of the common-emitter configuration: one for the *input* or *base–emitter* circuit and one for the *output* or *collector–emitter* circuit. Both are shown in Fig. 3.13.

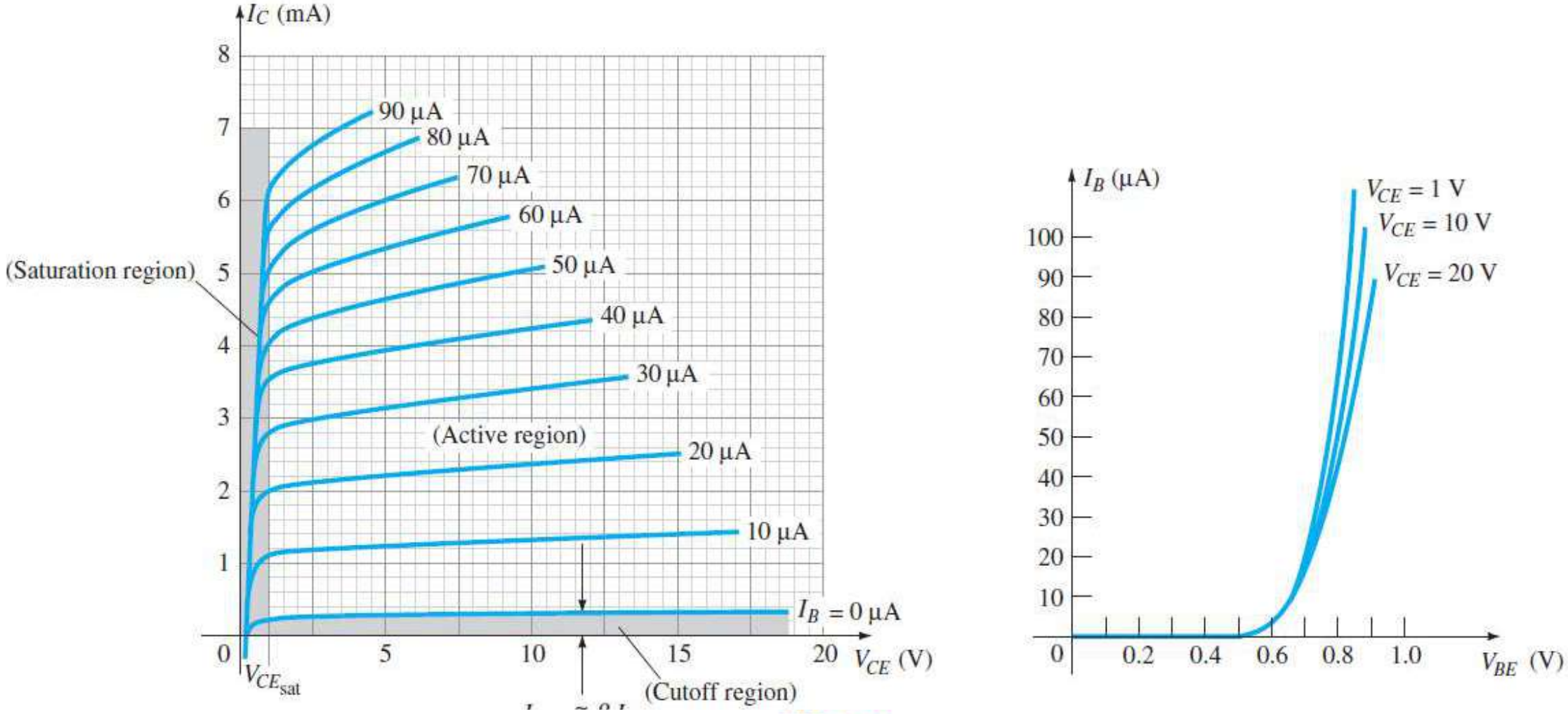
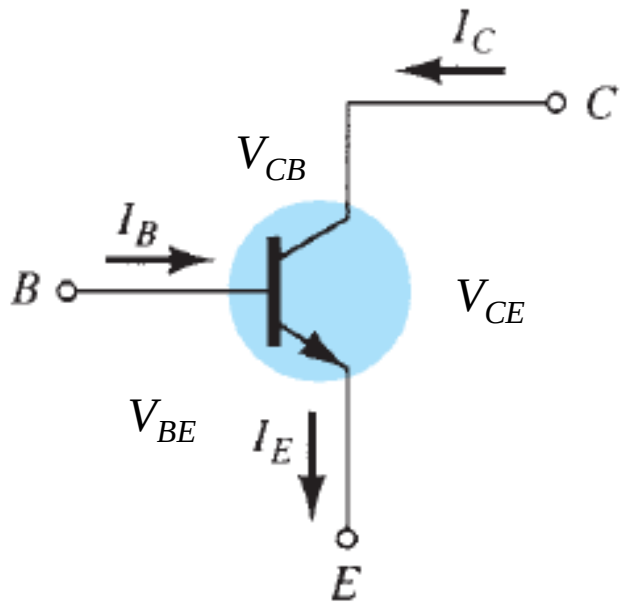


FIG. 3.13

Characteristics of a silicon transistor in the common-emitter configuration: (a) collector characteristics; (b) base characteristics.

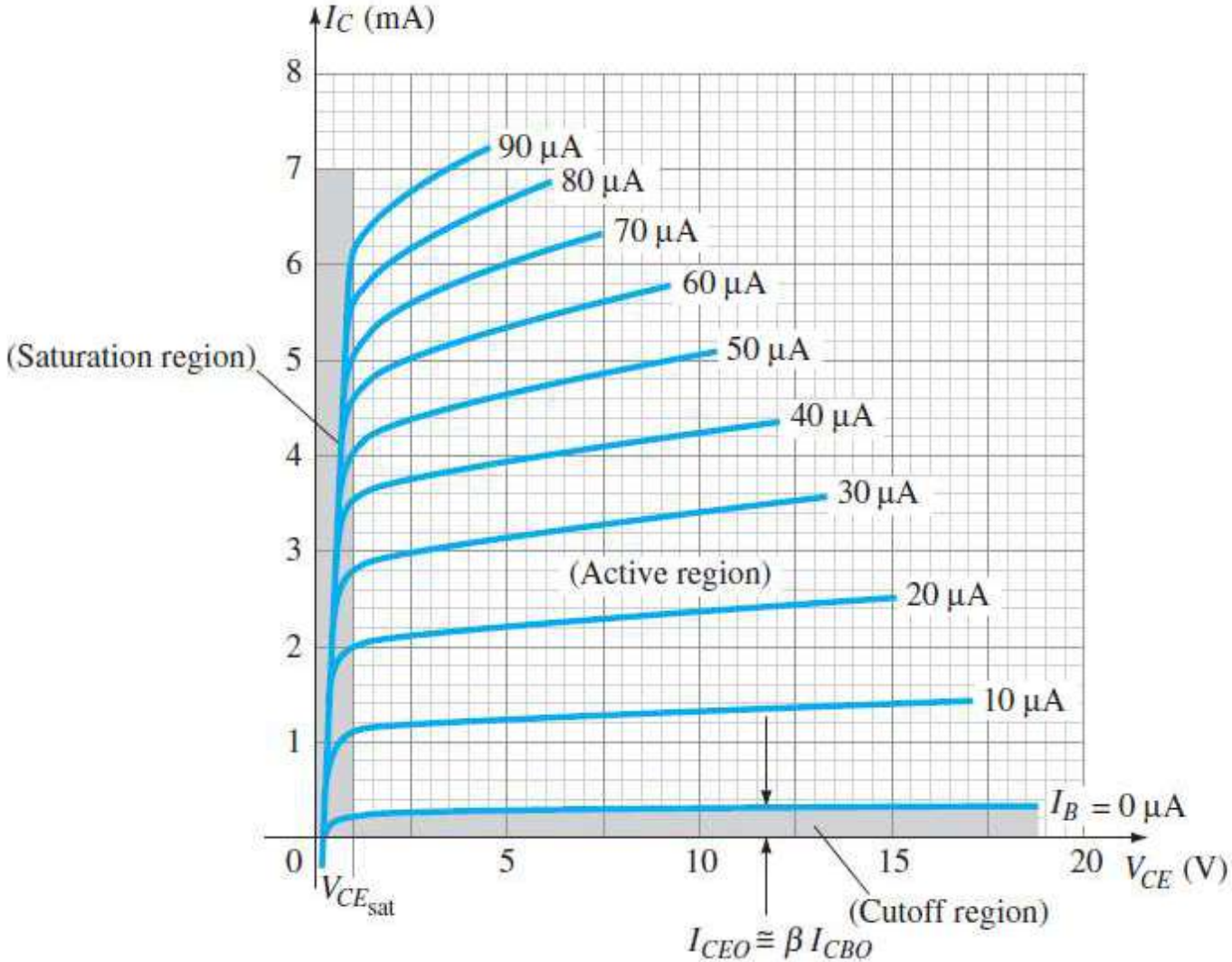
The emitter, collector, and base currents are shown in their actual conventional current direction. Even though the transistor configuration has changed, the current relations developed earlier for the common-base configuration are still applicable. That is, $I_E = I_C + I_B$ and $I_C = \alpha I_E$.

For the common-emitter configuration the output characteristics are a plot of the output current (I_C) versus output voltage (V_{CE}) for a range of values of input current (I_B). The input characteristics are a plot of the input current (I_B) versus the input voltage (V_{BE}) for a range of values of output voltage (V_{CE}).



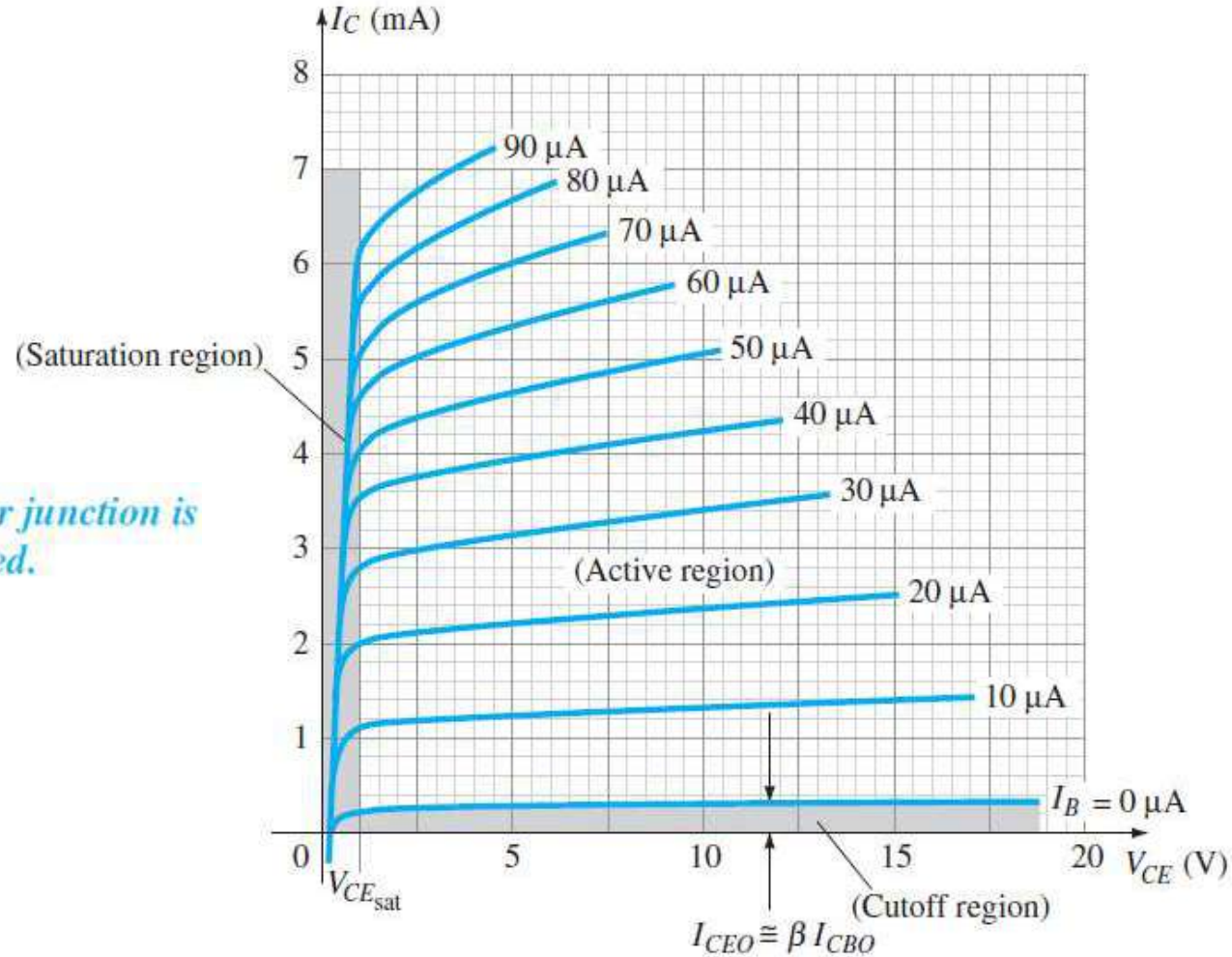
(a)

Note that on the characteristics of Fig. the magnitude of I_B is in microamperes, compared to milliamperes of I_C. Consider also that the curves of I_B are not as horizontal as those obtained for I_E in the common-base configuration, indicating that the collector-to-emitter voltage will influence the magnitude of the collector current.



The active region for the common-emitter configuration is that portion of the upper-right quadrant that has the greatest linearity, that is, that region in which the curves for I_B are nearly straight and equally spaced. In Fig. this region exists to the right of the vertical dashed line at $V_{CE_{sat}}$ and above the curve for I_B equal to zero. The region to the left of $V_{CE_{sat}}$ is called the saturation region.

In the active region of a common-emitter amplifier, the base–emitter junction is forward-biased, whereas the collector–base junction is reverse-biased.



(a)

The cutoff region for the common-emitter configuration is not as well defined as for the common-base configuration. Note on the collector characteristics of Fig. 3.14 that I_C is not equal to zero when I_B is zero. For the common-base configuration, when the input current I_E was equal to zero, the collector current was equal only to the reverse saturation current I_{CO}, so that the curve $I_E = 0$ and the voltage axis were, for all practical purposes, one.

The reason for this difference in collector characteristics can be derived through the proper manipulation of Eqs. (3.3) and (3.6). That is,

$$\text{Eq. (3.6):} \quad I_C = \alpha I_E + I_{CBO}$$

Substitution gives

$$\text{Eq. (3.3):} \quad I_C = \alpha(I_C + I_B) + I_{CBO}$$

Rearranging yields

$$I_C = \frac{\alpha I_B}{1 - \alpha} + \frac{I_{CBO}}{1 - \alpha} \tag{3.8}$$

If we consider the case discussed above, where $I_B = 0$ A, and substitute a typical value of α such as 0.996, the resulting collector current is the following:

$$I_C = \frac{\alpha(0\text{ A})}{1 - \alpha} + \frac{I_{CBO}}{1 - 0.996}$$

$$= \frac{I_{CBO}}{0.004} = 250I_{CBO}$$

If I_{CBO} were 1 μA, the resulting collector current with $I_B = 0$ A would be 250(1 μA) = 0.25 mA, as reflected in the characteristics of Fig. 3.14.

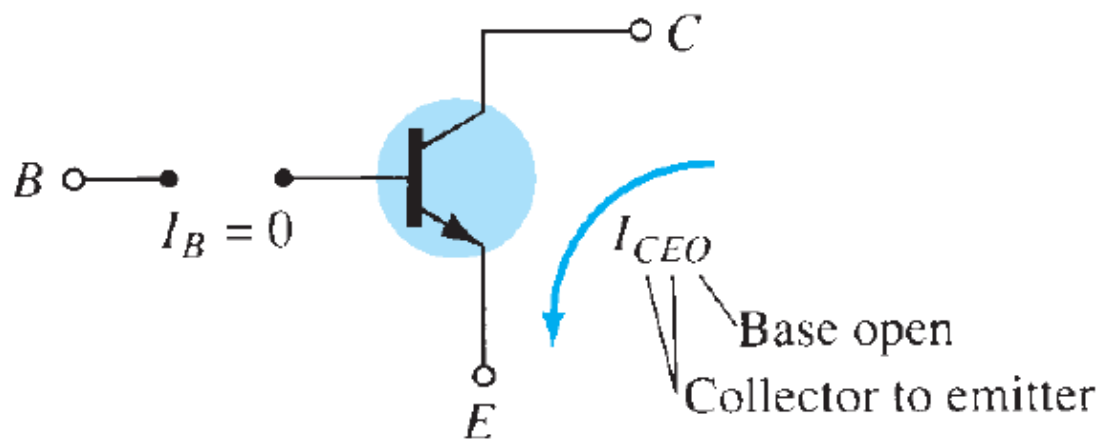
For future reference, the collector current defined by the condition $I_B = 0$ μA will be assigned the notation indicated by the following equation:

$$I_{CEO} = \left.\frac{I_{CBO}}{1 - \alpha}\right|_{I_B = 0\,\mu\text{A}} \tag{3.9}$$

For linear (least distortion) amplification purposes, cutoff for the common-emitter configuration will be defined by $I_C = I_{CEO}$.

In other words, the region below $I_B = 0\ \mu A$ is to be avoided if an undistorted output signal is required.

When employed as a switch in the logic circuitry of a computer, a transistor will have two points of operation of interest: one in the cutoff and one in the saturation region. The cutoff condition should ideally be $I_C = 0$ mA for the chosen V_{CE} voltage. Since I_{CEO} is typically low in magnitude for silicon materials, *cutoff will exist for switching purposes when $I_B = 0\ \mu A$ or $I_C = I_{CEO}$ for silicon transistors only. For germanium transistors, however, cutoff for switching purposes will be defined as those conditions that exist when $I_C = I_{CBO}$.* This condition can normally be obtained for germanium transistors by reverse-biasing the base-to-emitter junction a few tenths of a volt.



Recall for the common-base configuration that the input set of characteristics was approximated by a straight-line equivalent that resulted in $V_{BE} = 0.7$ V for any level of I_E greater than 0 mA. For the common-emitter configuration the same approach can be taken, resulting in the approximate equivalent of Fig. 3.15. The result supports our earlier conclusion that for a transistor in the "on" or active region the base-to-emitter voltage is 0.7 V. In this case the voltage is fixed for any level of base current.

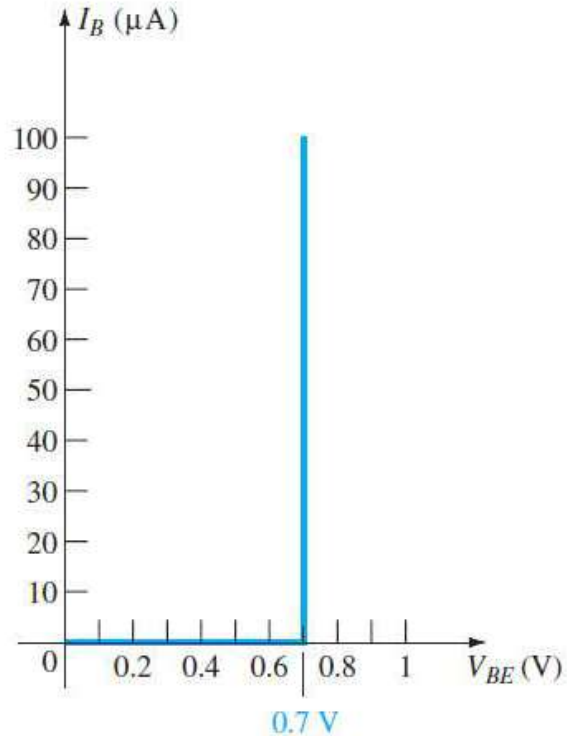


FIG. 3.15
Piecewise-linear equivalent for the diode characteristics of Fig. 3.13b.

Beta (β)

DC Mode In the dc mode the levels of I_C and I_B are related by a quantity called *beta* and defined by the following equation:

$$\beta_{dc} = \frac{I_C}{I_B} \tag{3.10}$$

where I_C and I_B are determined at a particular operating point on the characteristics. For practical devices the level of β typically ranges from about 50 to over 400, with most in the midrange. As for α, the parameter β reveals the relative magnitude of one current with respect to the other. For a device with a β of 200, the collector current is 200 times the magnitude of the base current.

On specification sheets β_{dc} is usually included as h_{FE} with the italic letter h derived from an ac *h*ybrid equivalent circuit to be introduced in Chapter 5. The subscript FE is derived from *f*orward-current amplification and common-*e*mitter configuration, respectively.

AC Mode For ac situations an ac beta is defined as follows:

$$\beta_{ac} = \left.\frac{\Delta I_C}{\Delta I_B}\right|_{V_{CE}=\text{constant}} \tag{3.11}$$

The formal name for β_{ac} is *common-emitter*, *forward-current*, *amplification factor*. Since the collector current is usually the output current for a common-emitter configuration and the base current is the input current, the term *amplification* is included in the nomenclature above.

On specification sheets β_{ac} is normally referred to as h_{fe}. Note that the only difference between the notation used for the dc beta, specifically, $\beta_{dc} = h_{FE}$, is the type of lettering for each subscript quantity.

The use of Eq. (3.11) is best described by a numerical example using an actual set of characteristics such as appearing in Fig. 3.13a and repeated in Fig. 3.17. Let us determine β_{ac} for a region of the characteristics defined by an operating point of $I_B = 25\ \mu A$ and $V_{CE} = 7.5$ V as indicated on Fig. 3.16. The restriction of V_{CE} = constant requires that a vertical line be drawn through the operating point at $V_{CE} = 7.5$ V. At any location on this vertical line the voltage V_{CE} is 7.5 V, a constant. The change in $I_B(\Delta I_B)$ as appearing in Eq. (3.11) is then defined by choosing two points on either side of the Q-point along the vertical axis of about equal distances to either side of the Q-point. For this situation the $I_B = 20\ \mu A$ and $30\ \mu A$ curves meet the requirement without extending too far from the Q-point. They also

$$\beta_{ac} = \left.\frac{\Delta I_C}{\Delta I_B}\right|_{V_{CE}=\text{constant}} \tag{3.11}$$

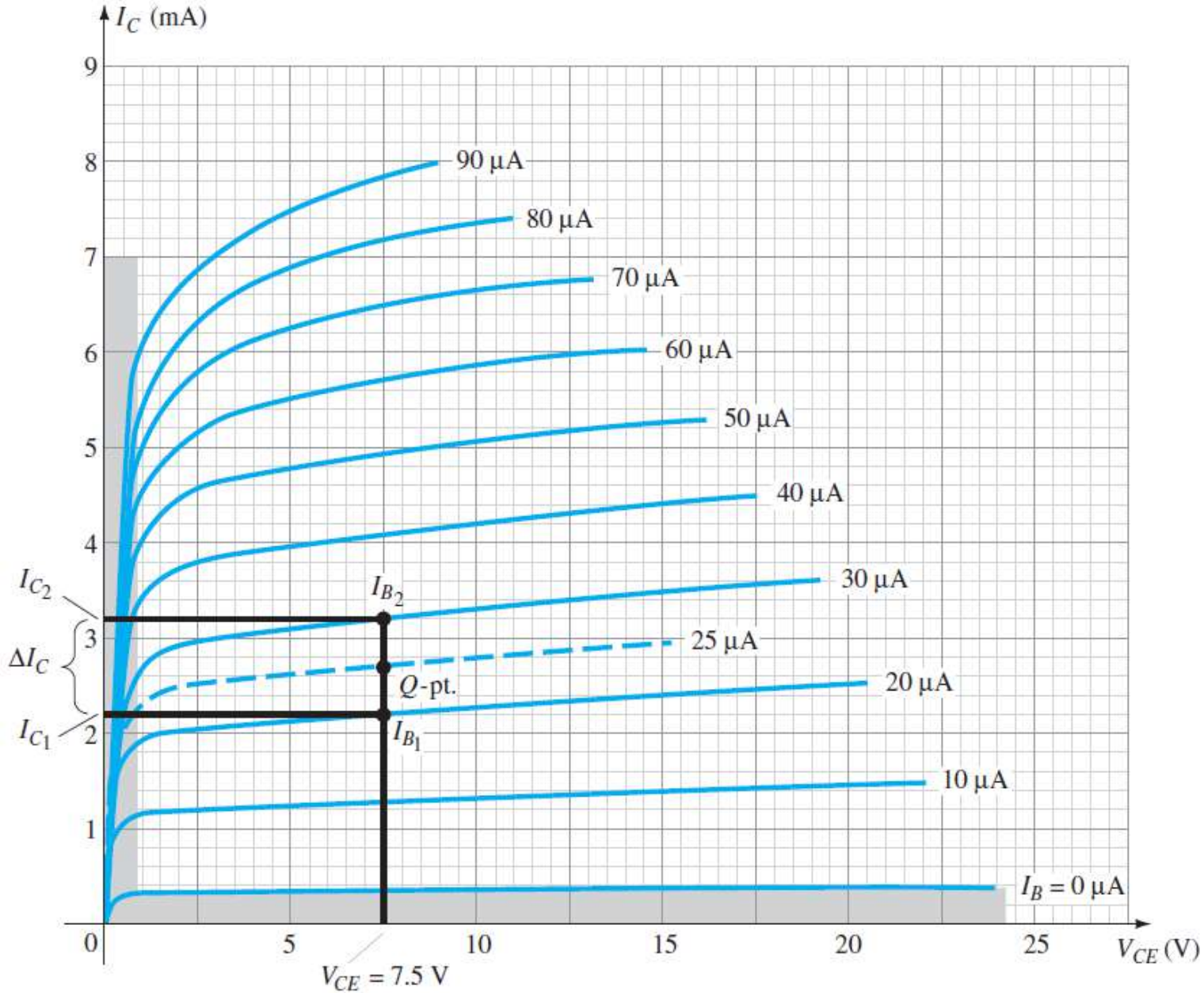


FIG. 3.16

Determining β_{ac} and β_{dc} from the collector characteristics.

define levels of I_B that are easily defined rather than require interpolation of the level of I_B between the curves. It should be mentioned that the best determination is usually made by keeping the chosen ΔI_B as small as possible. At the two intersections of I_B and the vertical axis, the two levels of I_C can be determined by drawing a horizontal line over to the vertical axis and reading the resulting values of I_C. The resulting β_{ac} for the region can then be determined by

$$\beta_{ac} = \left.\frac{\Delta I_C}{\Delta I_B}\right|_{V_{CE}=\text{constant}} = \frac{I_{C_2} - I_{C_1}}{I_{B_2} - I_{B_1}}$$

$$= \frac{3.2\text{ mA} - 2.2\text{ mA}}{30\ \mu\text{A} - 20\ \mu\text{A}} = \frac{1\text{ mA}}{10\ \mu\text{A}}$$

$$= \mathbf{100}$$

The solution above reveals that for an ac input at the base, the collector current will be about 100 times the magnitude of the base current.

If we determine the dc beta at the Q-point, we obtain

$$\beta_{dc} = \frac{I_C}{I_B} = \frac{2.7\text{ mA}}{25\ \mu\text{A}} = \mathbf{108}$$

If the characteristics of a transistor are approximated by those appearing in Fig. 3.17, the level of β_{ac} would be the same in every region of the characteristics. Note that the step in I_B is fixed at 10 μA and the vertical spacing between curves is the same at every point in the characteristics—namely, 2 mA. Calculating the β_{ac} at the Q-point indicated results in

$$\beta_{ac} = \left.\frac{\Delta I_C}{\Delta I_B}\right|_{V_{CE}=\text{constant}} = \frac{9\text{ mA} - 7\text{ mA}}{45\ \mu\text{A} - 35\ \mu\text{A}} = \frac{2\text{ mA}}{10\ \mu\text{A}} = \mathbf{200}$$

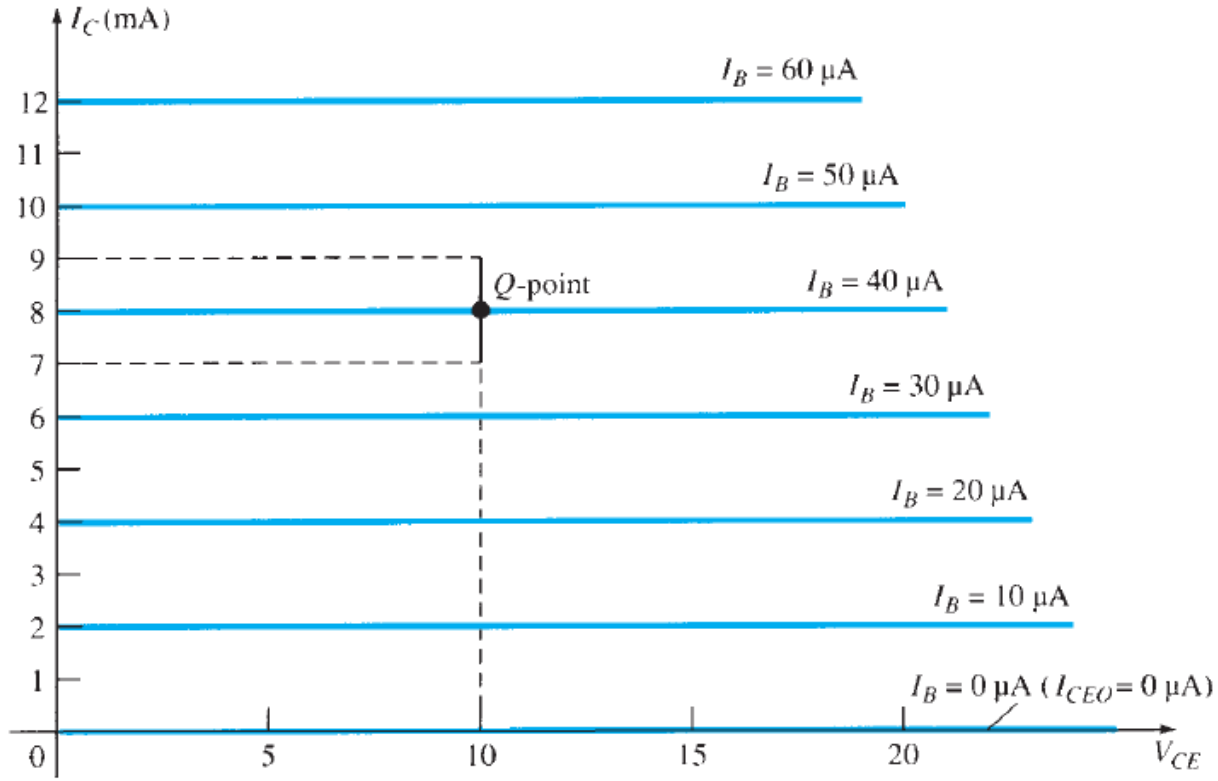


FIG. 3.17

Characteristics in which β_{ac} is the same everywhere and $\beta_{ac} = \beta_{dc}$.

Determining the dc beta at the same Q-point results in

$$\beta_{dc} = \frac{I_C}{I_B} = \frac{8\text{ mA}}{40\ \mu\text{A}} = \mathbf{200}$$

revealing that if the characteristics have the appearance of Fig. 3.17, the magnitudes of β_{ac} and β_{dc} *will be the same* at every point on the characteristics. In particular, note that $I_{CEO} = 0\ \mu\text{A}$.

A relationship can be developed between β and α using the basic relationships introduced thus far. Using $\beta = I_C/I_B$, we have $I_B = I_C/\beta$, and from $\alpha = I_C/I_E$ we have $I_E = I_C/\alpha$. Substituting into

$$I_E = I_C + I_B$$

we have

$$\frac{I_C}{\alpha} = I_C + \frac{I_C}{\beta}$$

and dividing both sides of the equation by I_C results in

$$\frac{1}{\alpha} = 1 + \frac{1}{\beta}$$

or

$$\beta = \alpha\beta + \alpha = (\beta + 1)\alpha$$

so that

$$\boxed{\alpha = \frac{\beta}{\beta + 1}} \tag{3.12}$$

or

$$\boxed{\beta = \frac{\alpha}{1 - \alpha}} \tag{3.13}$$

In addition, recall that

$$I_{CEO} = \frac{I_{CBO}}{1 - \alpha}$$

but using an equivalence of

$$\frac{1}{1 - \alpha} = \beta + 1$$

derived from the above, we find that

$$I_{CEO} = (\beta + 1)I_{CBO}$$

or

$$\boxed{I_{CEO} \cong \beta I_{CBO}} \tag{3.14}$$

as indicated on Fig. 3.13a. Beta is a particularly important parameter because it provides a direct link between current levels of the input and output circuits for a common-emitter configuration. That is,

$$\boxed{I_C = \beta I_B} \tag{3.15}$$

and since

$$\begin{aligned} I_E &= I_C + I_B \\ &= \beta I_B + I_B \end{aligned}$$

we have

$$\boxed{I_E = (\beta + 1) I_B} \tag{3.16}$$

Both of the equations above play a major role in the analysis in Chapter 4.

Biasing

The proper biasing of a common-emitter amplifier can be determined in a manner similar to that introduced for the common-base configuration. Let us assume that we are presented with an *npn* transistor such as shown in Fig. 3.18a and asked to apply the proper biasing to place the device in the active region.

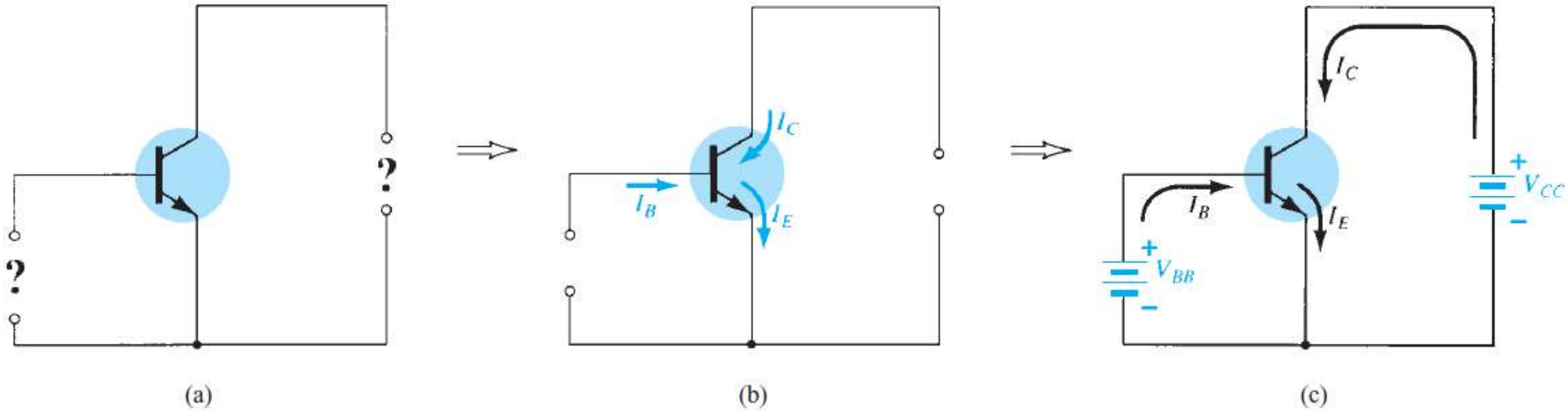


FIG. 3.18

Determining the proper biasing arrangement for a common-emitter npn transistor configuration.

The first step is to indicate the direction of I_E as established by the arrow in the transistor symbol as shown in Fig. 3.18b. Next, the other currents are introduced as shown, keeping in mind Kirchhoff's current law relationship: $I_C + I_B = I_E$. That is, I_E is the sum of I_C and I_B and both I_C and I_B must enter the transistor structure. Finally, the supplies are introduced with polarities that will support the resulting directions of I_B and I_C as shown in Fig. 3.18c to complete the picture. The same approach can be applied to *pnp* transistors. If the transistor of Fig. 3.18 was a *pnp* transistor, all the currents and polarities of Fig. 3.18c would be reversed.

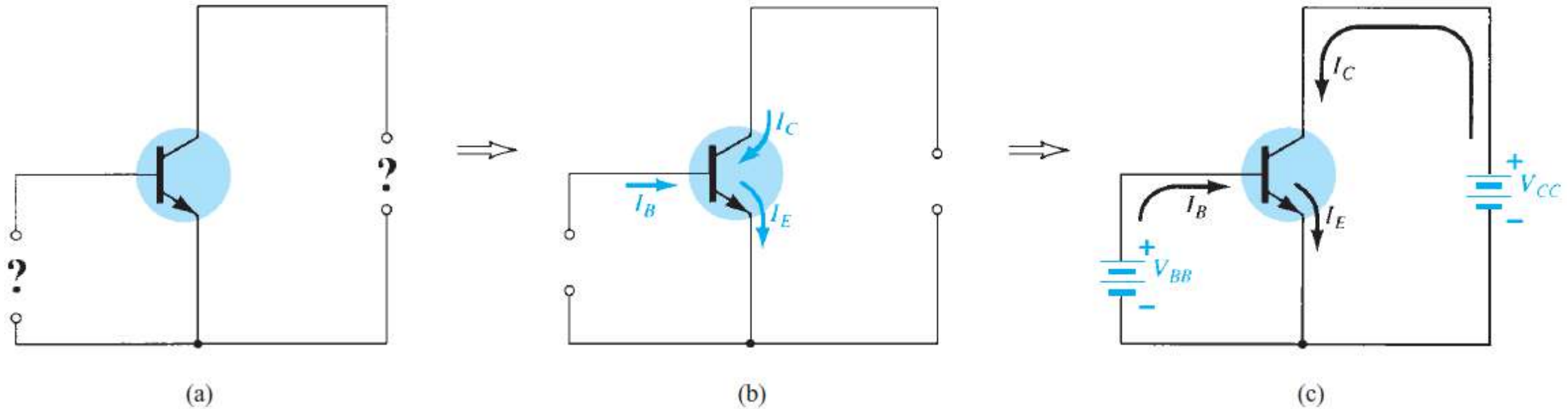
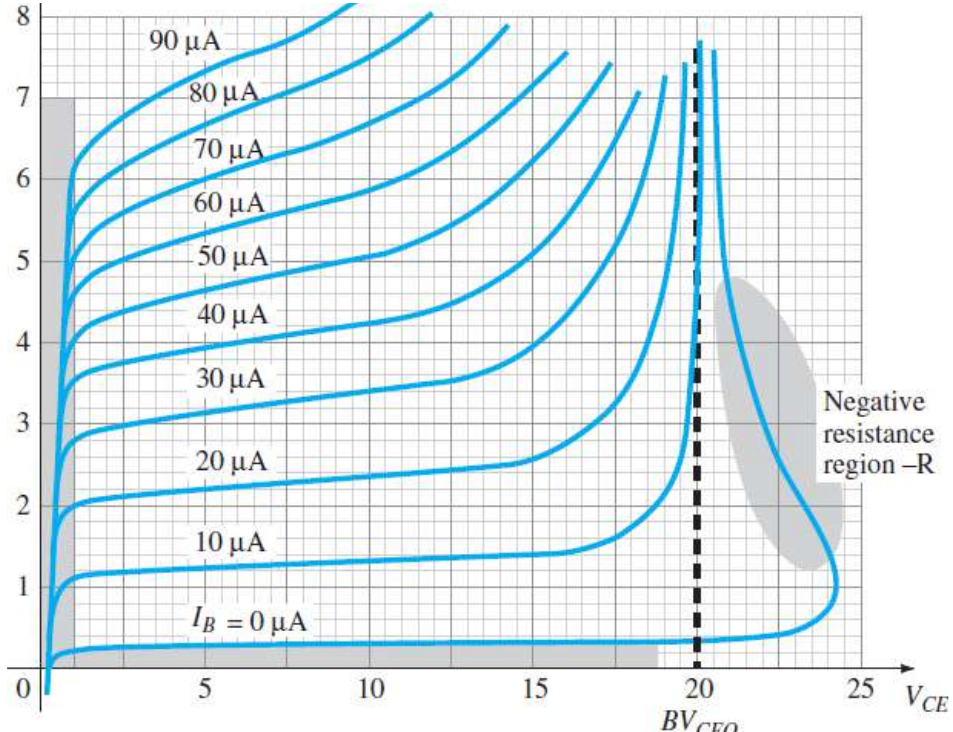


FIG. 3.18

Determining the proper biasing arrangement for a common-emitter npn transistor configuration.

Breakdown Region

As with the common-base configuration, there is a maximum collector-emitter voltage that can be applied and still remain in the active stable region of operation. In Fig. 3.19 the characteristics of Fig. 3.8 have been extended to demonstrate the impact on the characteristics at high levels of V_{CE}. At high levels of base current the currents almost climb vertically, whereas at lower levels a region develops that seems to back up on itself. This region is particularly noteworthy because an increase in current is resulting in a drop in voltage—totally different from that of any resistive element where an increase in current results in an increase in potential drop across the resistor. Regions of this nature are said to have a



negative-resistance characteristic. Although the concept of a negative resistance may seem strange at this point, this text will introduce devices and systems that rely on this type of characteristic to perform their desired task.

The recommended maximum value for a transistor under normal operating conditions is labeled BV_{CEO} as shown in Fig. 3.19 or $V_{(BR)CEO}$ as shown in Fig. 3.23. It is less than BV_{CBO}

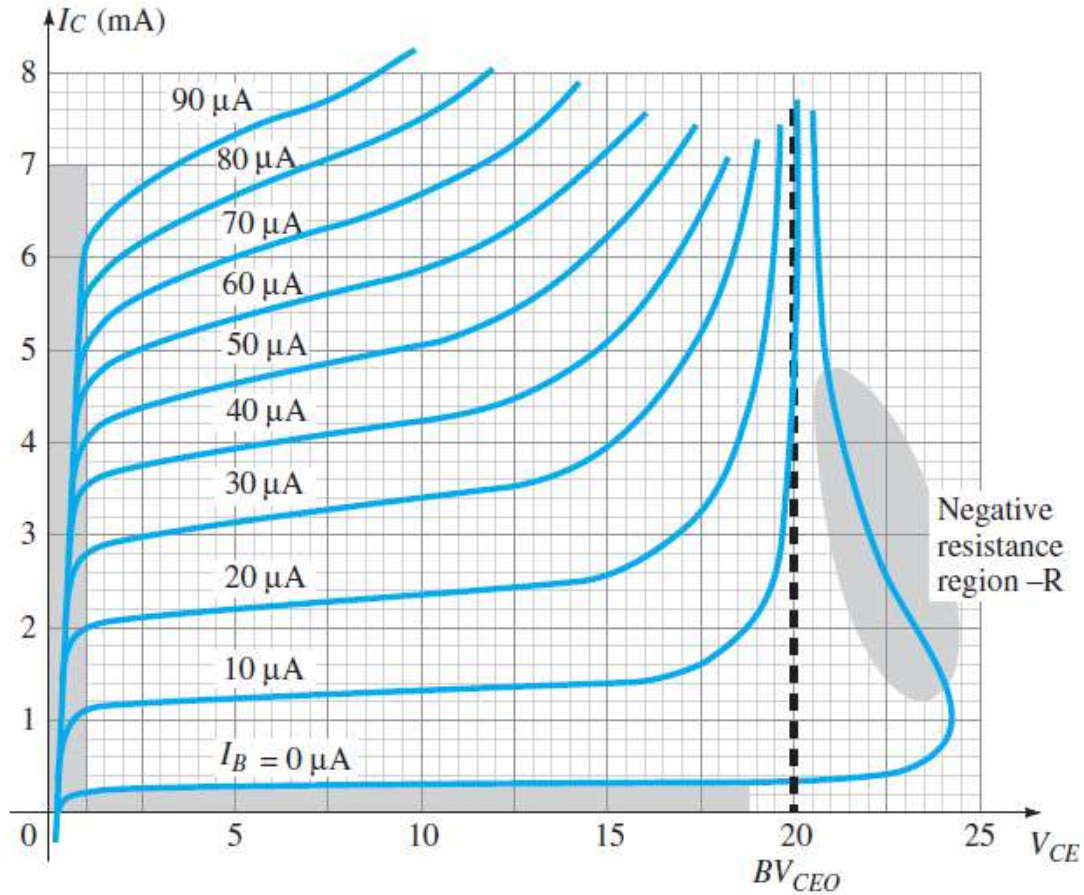


FIG. 3.19

Examining the breakdown region of a transistor in the common-emitter configuration.

3.6 COMMON-COLLECTOR CONFIGURATION

The third and final transistor configuration is the *common-collector configuration*, shown in Fig. 3.20 with the proper current directions and voltage notation. The common-collector configuration is used primarily for impedance-matching purposes since it has a high input impedance and low output impedance, opposite to that of the common-base and common-emitter configurations.

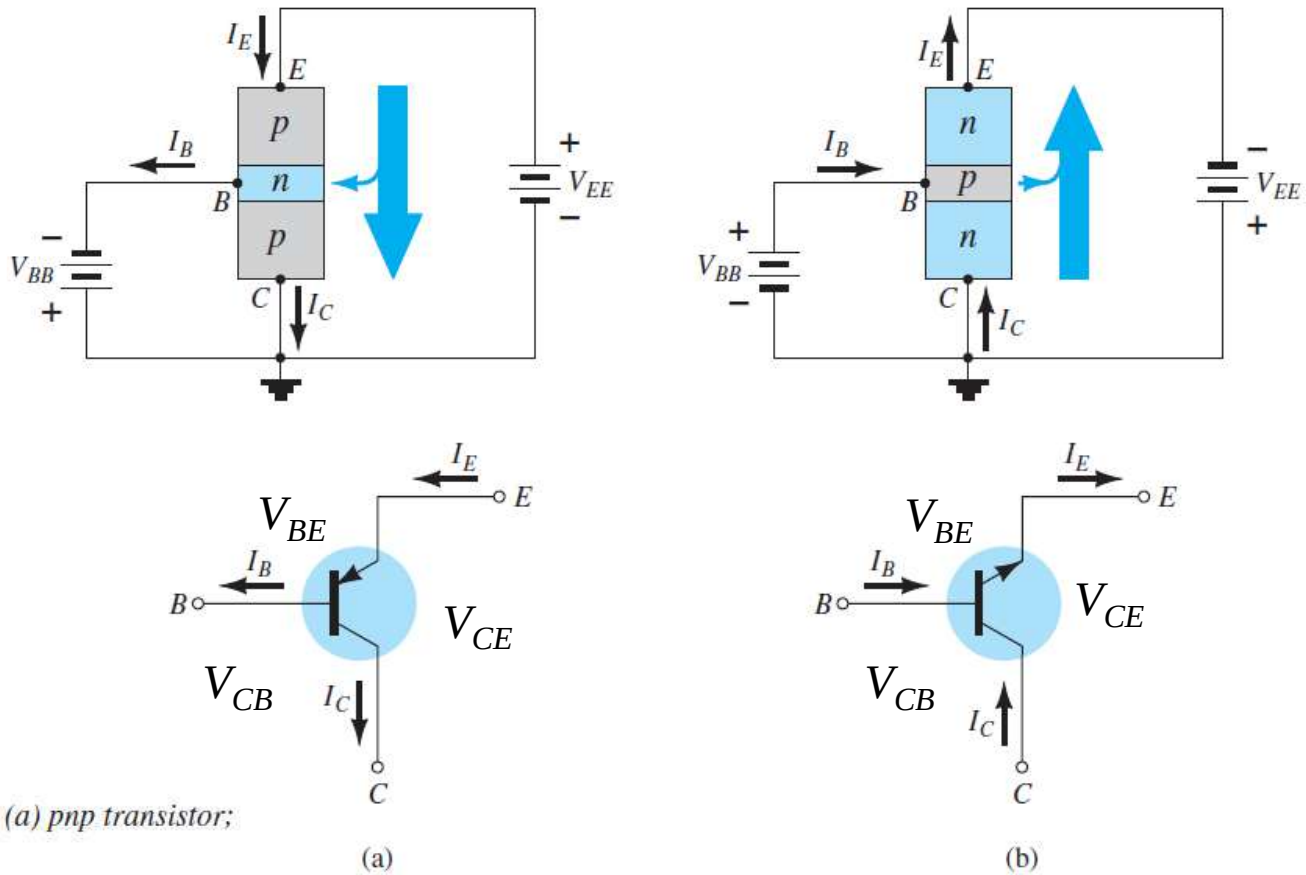


FIG. 3.20
Notation and symbols used with the common-collector configuration: (a) pnp transistor; (b) npn transistor.

A common-collector circuit configuration is provided in Fig. 3.21 with the load resistor connected from emitter to ground. Note that the collector is tied to ground even though the transistor is connected in a manner similar to the common-emitter configuration. From a design viewpoint, there is no need for a set of common-collector characteristics to choose the parameters of the circuit of Fig. 3.21. It can be designed using the common-emitter characteristics of Section 3.5. For all practical purposes, the output characteristics of the common-collector configuration are the same as for the common-emitter configuration. For the common-collector configuration the output characteristics are a plot of I_E versus V_{CE} for a range of values of I_B. The input current, therefore, is the same for both the common-emitter and common-collector characteristics. The horizontal voltage axis for the common-collector configuration is obtained by simply changing the sign of the collector-to-emitter voltage of the common-emitter characteristics. Finally, there is an almost unnoticeable

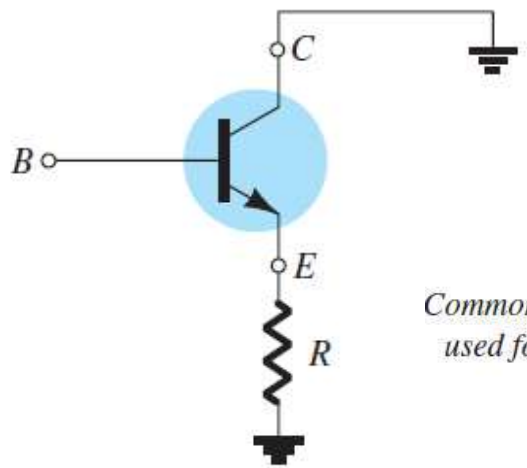


FIG. 3.21
Common-collector configuration used for impedance-matching purposes.

Finally, there is an almost unnoticeable change in the vertical scale of I_C of the common-emitter characteristics if I_C is replaced by I_E for the common-collector characteristics (since $\alpha \cong 1$). For the input circuit of the common-collector configuration the common-emitter base characteristics are sufficient for obtaining the required information.

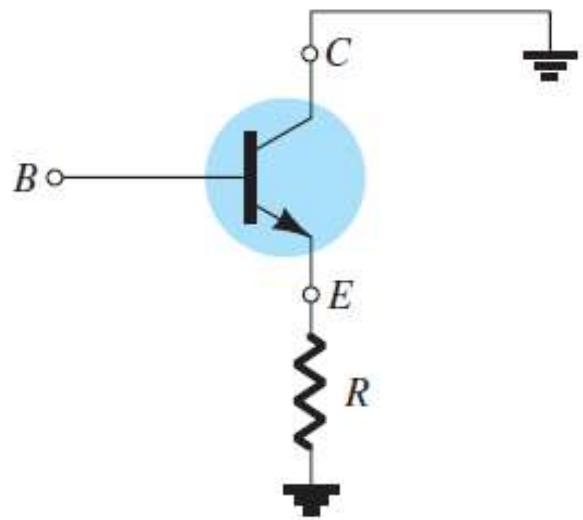


FIG. 3.21
Common-collector configuration used for impedance-matching purposes.

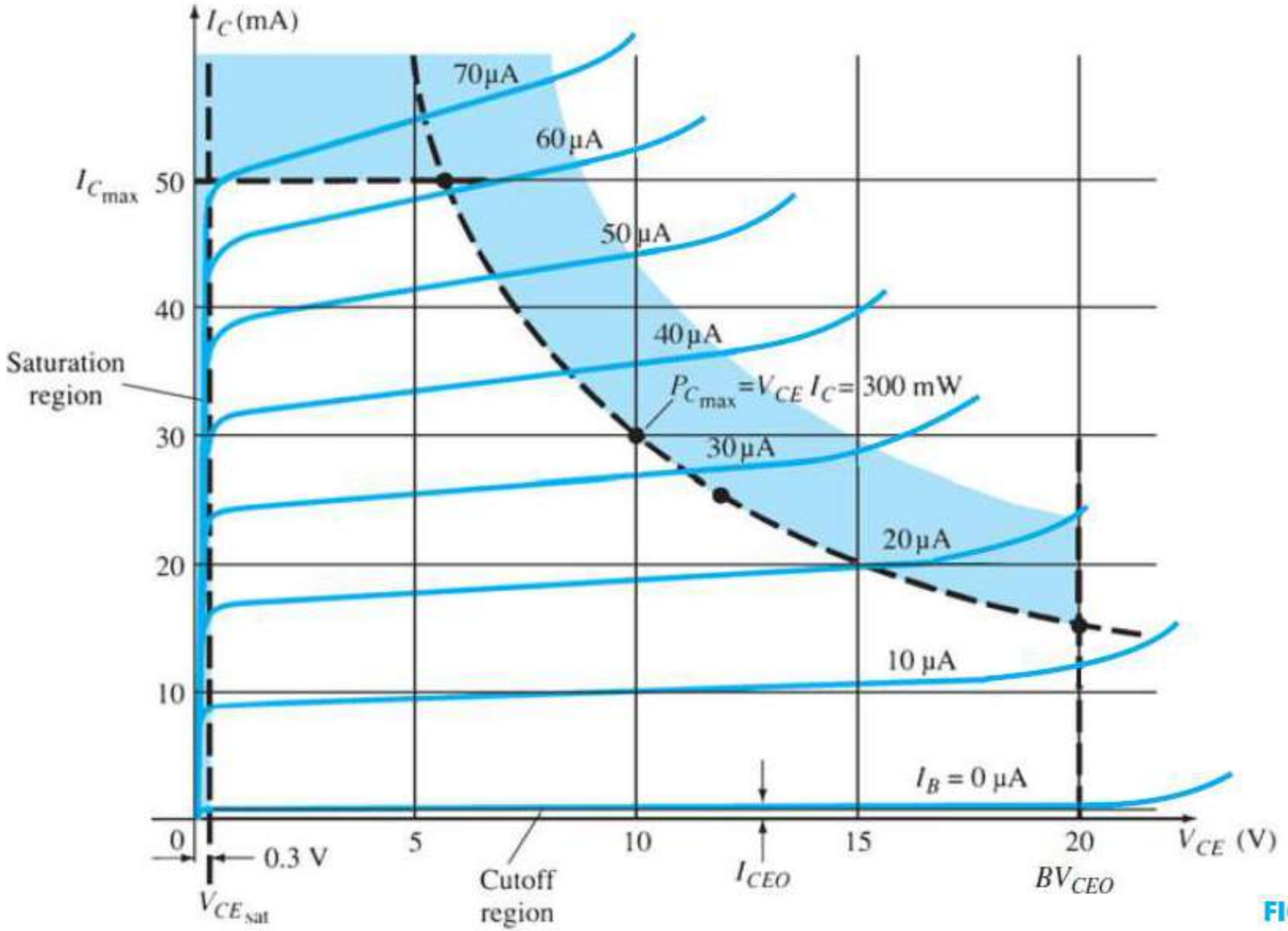


FIG. 3.22

Defining the linear (undistorted) region of operation for a transistor.

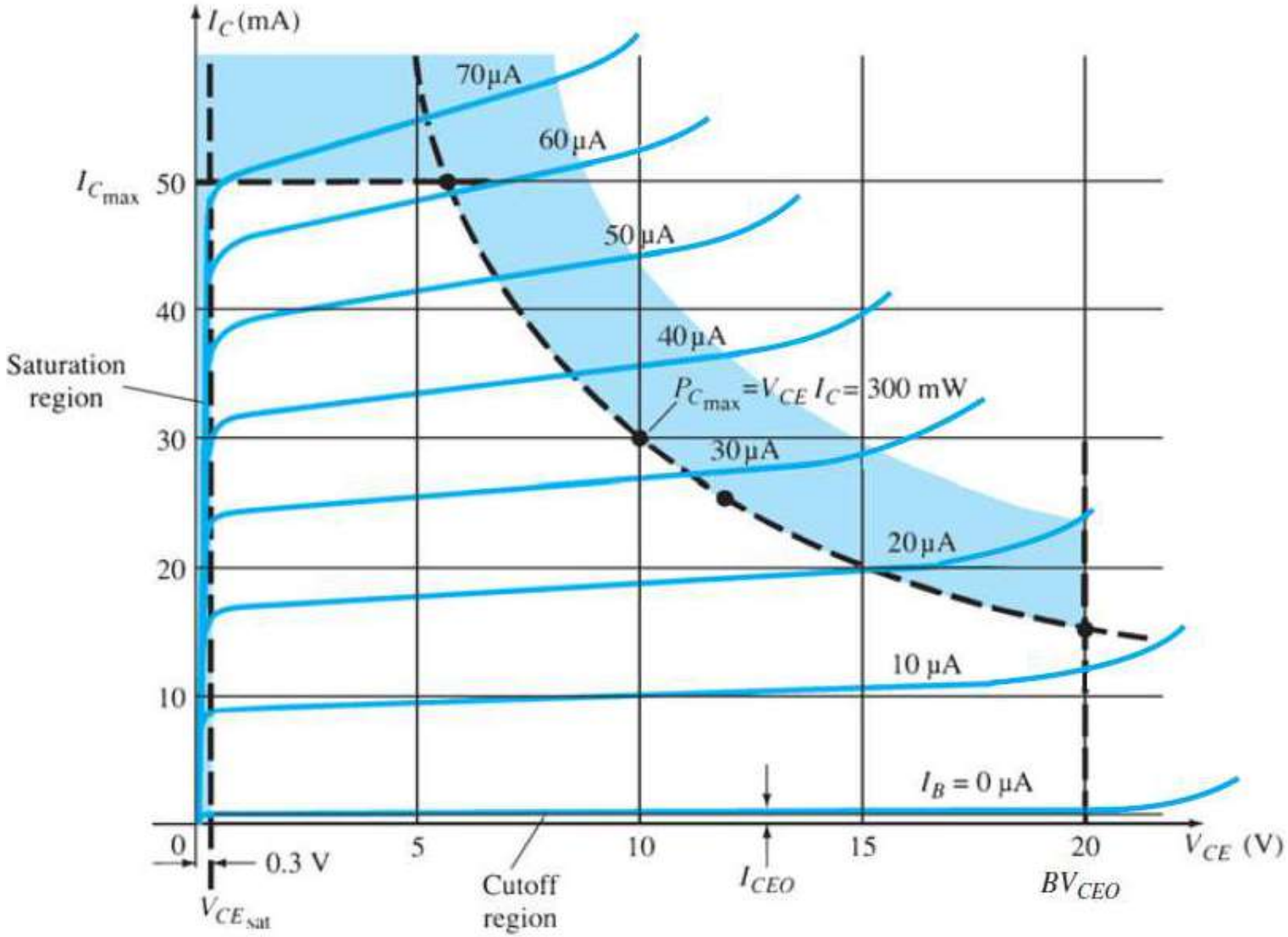
For the device of Fig. 3.22, the collector power dissipation was specified as 300 mW. The question then arises of how to plot the collector power dissipation curve specified by the fact that

$$P_{C_{\max}} = V_{CE}I_C = 300 \text{ mW}$$

or

$$V_{CE}I_C = 300 \text{ mW}$$

At $I_{C_{max}}$ At any point on the characteristics the product of V_{CE} and I_C must be equal to 300 mW. If we choose I_C to be the maximum value of 50 mA and substitute into the relationship above, we obtain

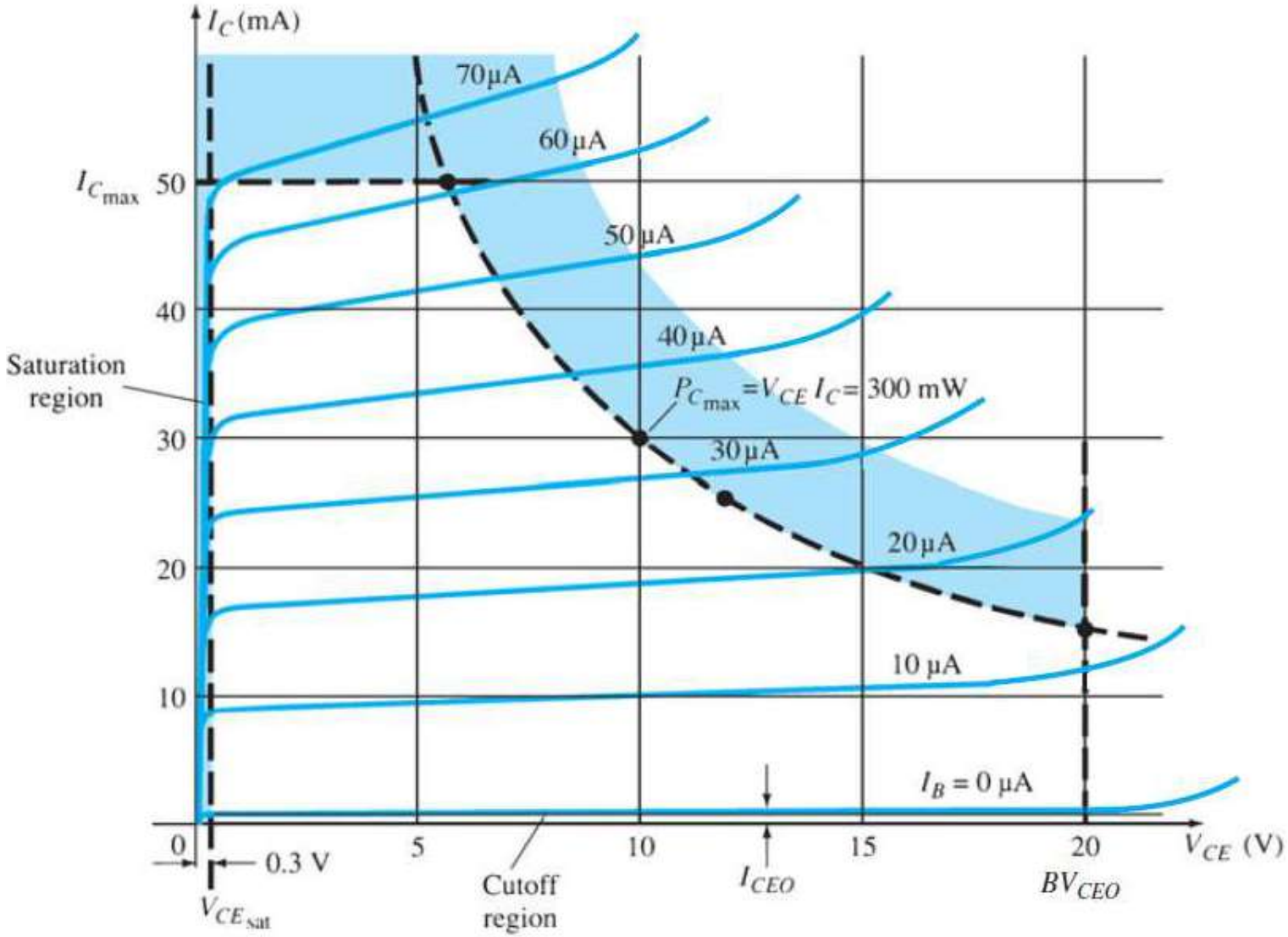


$$V_{CE}I_C = 300 \text{ mW}$$

$$V_{CE}(50 \text{ mA}) = 300 \text{ mW}$$

$$V_{CE} = \frac{300 \text{ mW}}{50 \text{ mA}} = \mathbf{6\ V}$$

At $V_{CE_{max}}$ As a result we find that if $I_C = 50$ mA, then $V_{CE} = 6$ V on the power dissipation curve as indicated in Fig. 3.22. If we now choose V_{CE} to be its maximum value of 20 V, the level of I_C is the following:

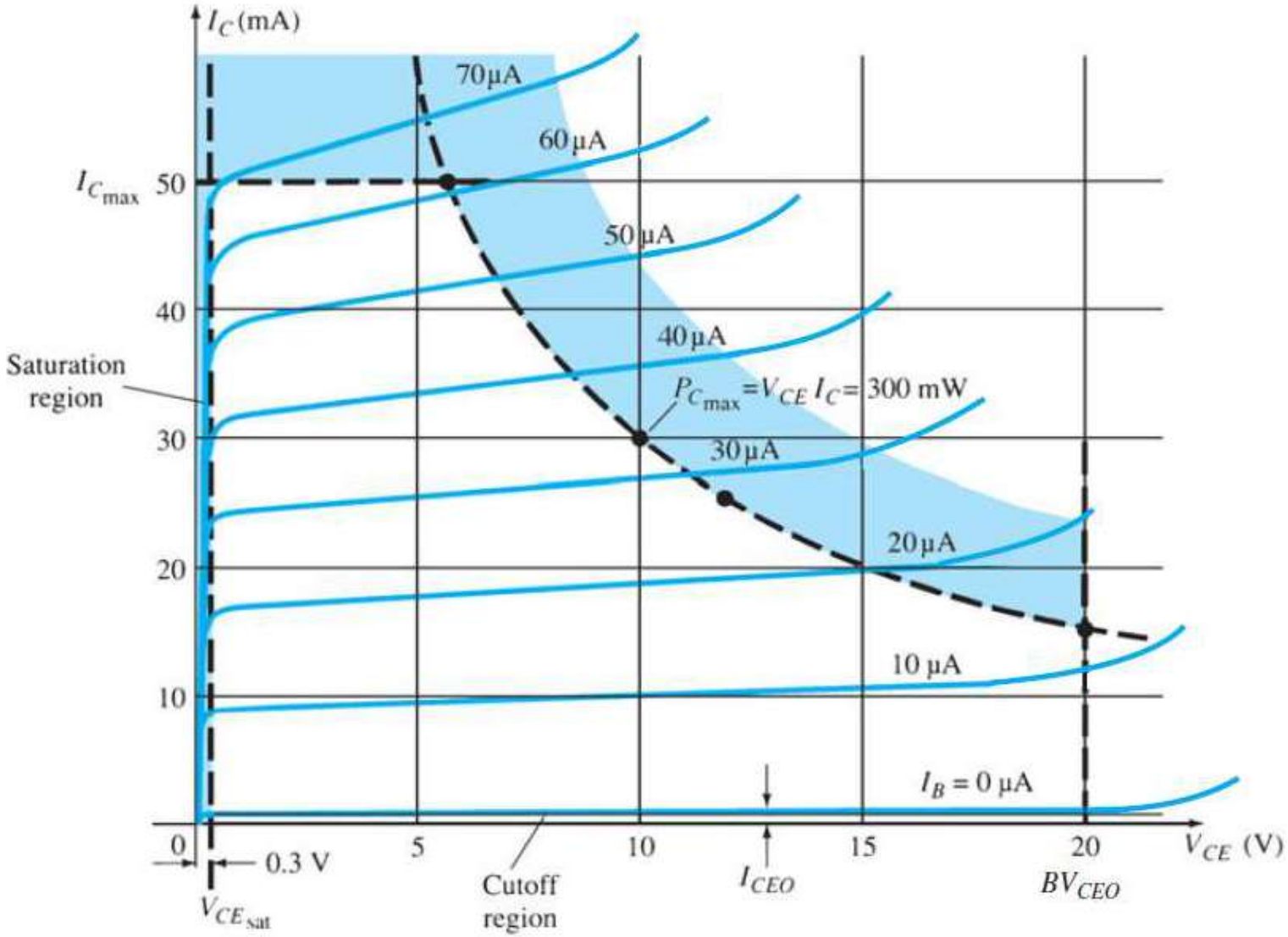


$$(20\text{ V})I_C = 300\text{ mW}$$

$$I_C = \frac{300\text{ mW}}{20\text{ V}} = \mathbf{15\text{ mA}}$$

defining a second point on the power curve.

At $I_C = \frac{1}{2}I_{C_{max}}$ If we now choose a level of I_C in the midrange such as 25 mA and solve for the resulting level of V_{CE}, we obtain



$$V_{CE}(25\text{ mA}) = 300\text{ mW}$$

$$V_{CE} = \frac{300\text{ mW}}{25\text{ mA}} = \mathbf{12\text{ V}}$$

If the characteristic curves are unavailable or do not appear on the specification sheet (as is often the case), one must simply be sure that I_C, V_{CE}, and their product $V_{CE}I_C$ fall into the following range:

$$\begin{aligned} I_{CEO} \leqq I_C &\leqq I_{C_{max}} \\ V_{CE_{sat}} \leqq V_{CE} &\leqq V_{CE_{max}} \\ V_{CE}I_C &\leqq P_{C_{max}} \end{aligned} \tag{3.18}$$

For the common-base characteristics the maximum power curve is defined by the following product of output quantities:

$$P_{C_{max}} = V_{CB}I_C \tag{3.19}$$

3.8 TRANSISTOR SPECIFICATION SHEET

Since the specification sheet is the communication link between the manufacturer and user, it is particularly important that the information provided be recognized and correctly understood. Although all the parameters have not been introduced, a broad number will now be familiar. The remaining parameters will be introduced in the chapters that follow. Reference will then be made to this specification sheet to review the manner in which the parameter is presented.

The information provided as Fig. 3.23 is provided by the Fairchild Semiconductor Corporation. The 2N4123 is a general-purpose *npn* transistor with the casing and terminal

MAXIMUM RATINGS

Rating	Symbol	2N4123	Unit
Collector-Emitter Voltage	V_{CEO}	30	Vdc
Collector-Base Voltage	V_{CBO}	40	Vdc
Emitter-Base Voltage	V_{EBO}	5.0	Vdc
Collector Current – Continuous	I_C	200	mAdc
Total Device Dissipation @ $T_A = 25°C$ Derate above 25°C	P_D	625 5.0	mW mW°C
Operating and Storage Junction Temperature Range	T_J, T_{stg}	–55 to +150	°C

THERMAL CHARACTERISTICS

Characteristic	Symbol	Max	Unit
Thermal Resistance, Junction to Case	$R_{\theta JC}$	83.3	°C W
Thermal Resistance, Junction to Ambient	$R_{\theta JA}$	200	°C W

ELECTRICAL CHARACTERISTICS ($T_A = 25°C$ unless otherwise noted)

FAIRCHILD
SEMICONDUCTOR ™

2N4123

General Purpose Transistor NPN Silicon

Characteristic	Symbol	Min	Max	Unit
OFF CHARACTERISTICS				
Collector-Emitter Breakdown Voltage (1) ($I_C = 1.0$ mAdc, $I_E = 0$)	$V_{(BR)CEO}$	30		Vdc
Collector-Base Breakdown Voltage ($I_C = 10$ µAdc, $I_E = 0$)	$V_{(BR)CBO}$	40		Vdc
Emitter-Base Breakdown Voltage ($I_E = 10$ µAdc, $I_C = 0$)	$V_{(BR)EBO}$	5.0	–	Vdc
Collector Cutoff Current ($V_{CB} = 20$ Vdc, $I_E = 0$)	I_{CBO}	–	50	nAdc
Emitter Cutoff Current ($V_{BE} = 3.0$ Vdc, $I_C = 0$)	I_{EBO}	–	50	nAdc
ON CHARACTERISTICS				
DC Current Gain(1) ($I_C = 2.0$ mAdc, $V_{CE} = 1.0$ Vdc) ($I_C = 50$ mAdc, $V_{CE} = 1.0$ Vdc)	h_{FE}	50 25	150 –	–
Collector-Emitter Saturation Voltage(1) ($I_C = 50$ mAdc, $I_B = 5.0$ mAdc)	$V_{CE(sat)}$	–	0.3	Vdc
Base-Emitter Saturation Voltage(1) ($I_C = 50$ mAdc, $I_B = 5.0$ mAdc)	$V_{BE(sat)}$	–	0.95	Vdc
SMALL-SIGNAL CHARACTERISTICS				
Current-Gain – Bandwidth Product ($I_C = 10$ mAdc, $V_{CE} = 20$ Vdc, f = 100 MHz)	f_T	250		MHz
Output Capacitance ($V_{CB} = 5.0$ Vdc, $I_E = 0$, f = 100 MHz)	C_{obo}	–	4.0	pF
Input Capacitance ($V_{BE} = 0.5$ Vdc, $I_C = 0$, f = 100 kHz)	C_{ibo}	–	8.0	pF
Collector-Base Capacitance ($I_E = 0$, $V_{CB} = 5.0$ V, f = 100 kHz)	C_{cb}	–	4.0	pF
Small-Signal Current Gain ($I_C = 2.0$ mAdc, $V_{CE} = 10$ Vdc, f = 1.0 kHz)	h_{fe}	50	200	–
Current Gain – High Frequency ($I_C = 10$ mAdc, $V_{CE} = 20$ Vdc, f = 100 MHz) ($I_C = 2.0$ mAdc, $V_{CE} = 10$ V, f = 1.0 kHz)	h_{fe}	2.5 50	– 200	–
Noise Figure ($I_C = 100$ µAdc, $V_{CE} = 5.0$ Vdc, $R_S = 1.0$ k ohm, f = 1.0 kHz)	NF	–	6.0	dB

(1) Pulse Test: Pulse Width = 300 µs. Duty Cycle = 2.0%

For more information
Textbook 146-152

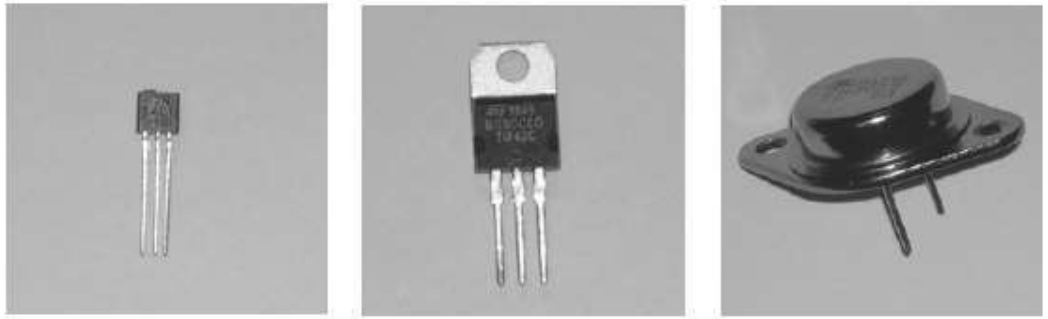


FIG. 3.29
Various types of general-purpose or switching transistors: (a) low power; (b) medium power; (c) medium to high power.

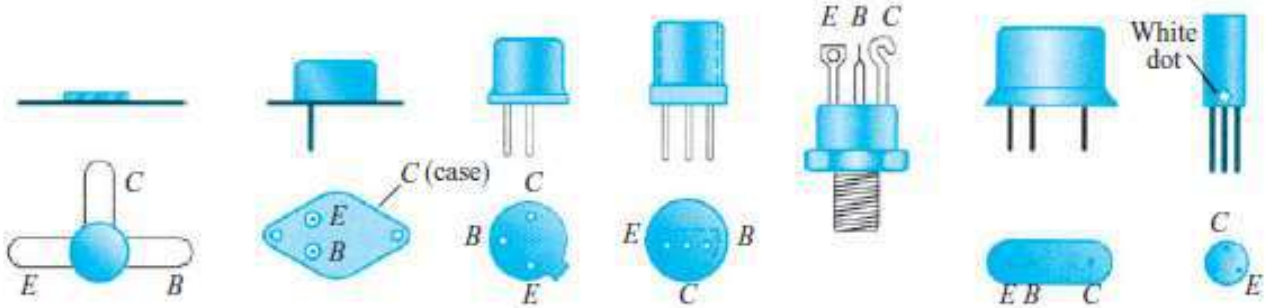


FIG. 3.30
Transistor terminal identification.

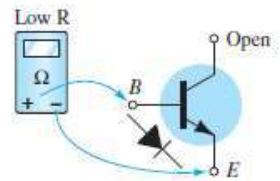


FIG. 3.27
Checking the forward-biased base-to-emitter junction of an npn transistor.

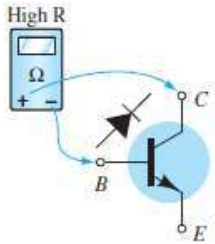


FIG. 3.28
Checking the reverse-biased base-to-collector junction of an npn transistor.

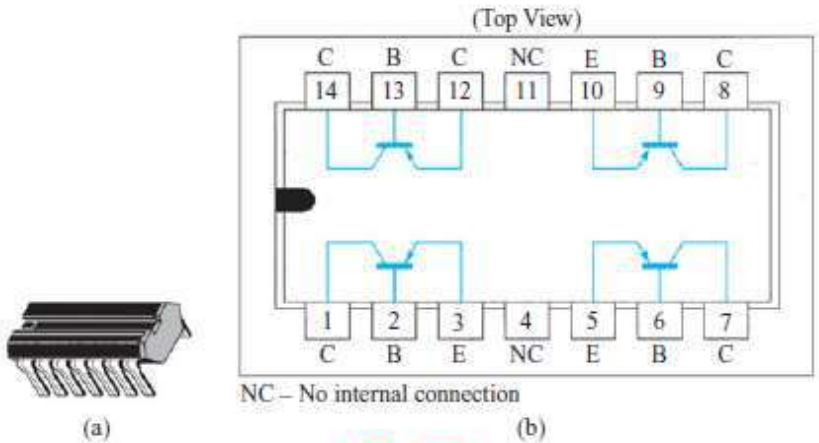


FIG. 3.32
Type Q2T2905 Texas Instruments quad pnp silicon transistor: (a) appearance; (b) pin connections.

Al-Balqa Applied University
Faculty of Engineering Technology
Communication Technology Department

Chapter 4: DC Biasing—BJTs

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CHAPTER OBJECTIVES ●

- Be able to determine the dc levels for the variety of important BJT configurations.
- Understand how to measure the important voltage levels of a BJT transistor configuration and use them to determine whether the network is operating properly.
- Become aware of the saturation and cutoff conditions of a BJT network and the
expected voltage and current levels established by each condition.
- Be able to perform a load-line analysis of the most common BJT configurations.
- Become acquainted with the design process for BJT amplifiers.
- Understand the basic operation of transistor switching networks.
- Begin to understand the troubleshooting process as applied to BJT configurations.

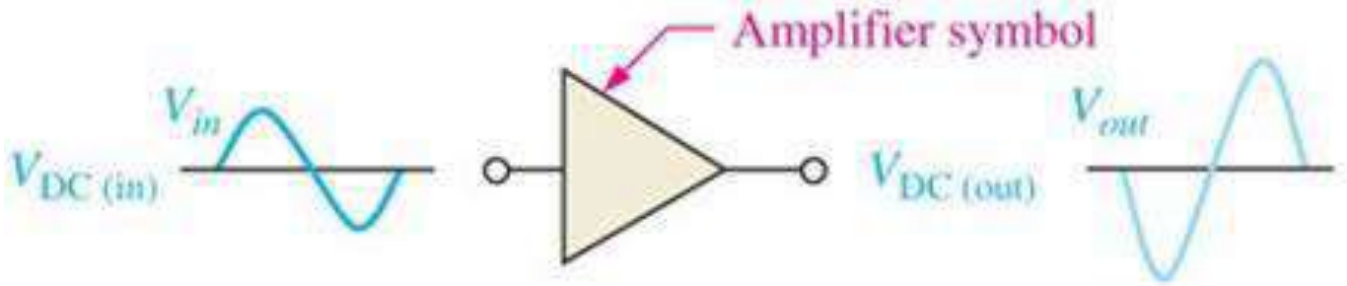
The analysis or design of a transistor amplifier requires a knowledge of both the **dc** and the **ac** response of the system. Too often it is assumed that the transistor is a magical device that can raise the level of the applied ac input without the assistance of an external energy source. In actuality, ***any increase in ac voltage, current, or power is the result of a transfer of energy from the applied dc supplies.***

The analysis or design of any electronic amplifier therefore has two components: **a dc and an ac portion**. Fortunately, the superposition theorem is applicable, and the investigation of the dc conditions can be totally separated from the ac response. However, one must keep in mind that during the design or synthesis stage the choice of parameters for the required dc levels will affect the ac response, and vice versa.

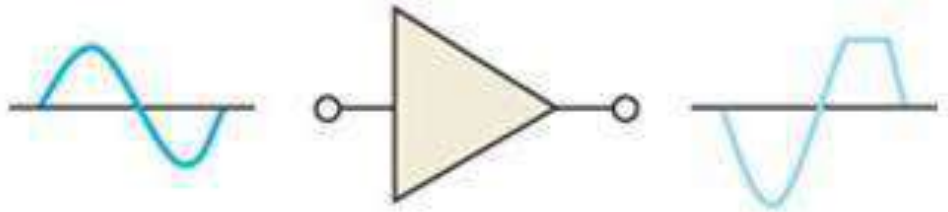
The dc level of operation of a transistor is controlled by a number of factors, including the range of possible operating points on the device characteristics
A transistor must be properly biased with a dc voltage in order to operate as a linear amplifier.

A dc operating point must be set so that signal variations at the input terminal are amplified and accurately reproduced at the output terminal.

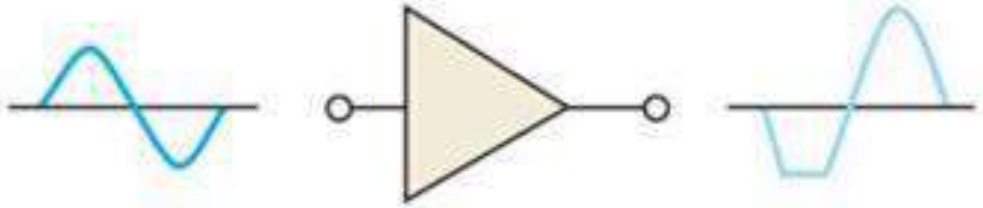
- If an amplifier is not biased with the correct DC voltages, it can go into saturation and cutoff.
- Figure 1(a) shows the correct Linear operation with amplified output.
- Figure 1(b) shows nonlinear operation where the amplifier is in cutoff. The clipping in the positive cycle is always due to cutoff.
- Figure 1(c) shows nonlinear operation where the amplifier is in saturation. The clipping in the negative cycle is always due to saturation.



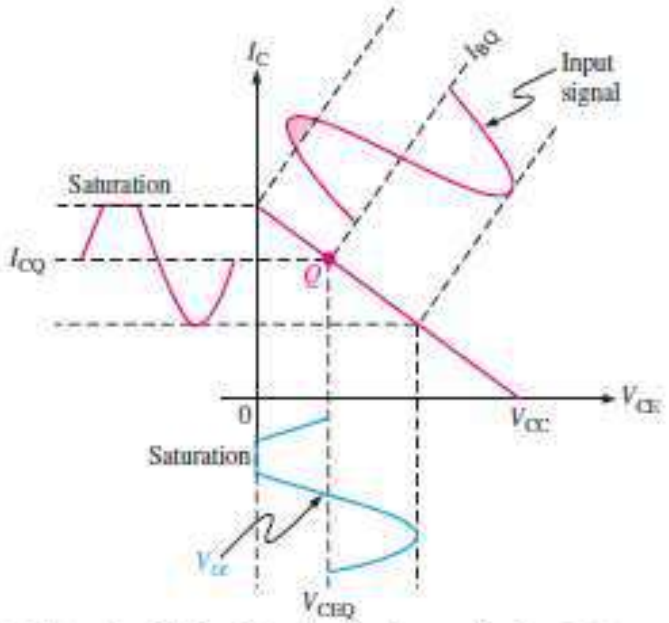
(a) Linear operation: larger output has same shape as input except that it is inverted



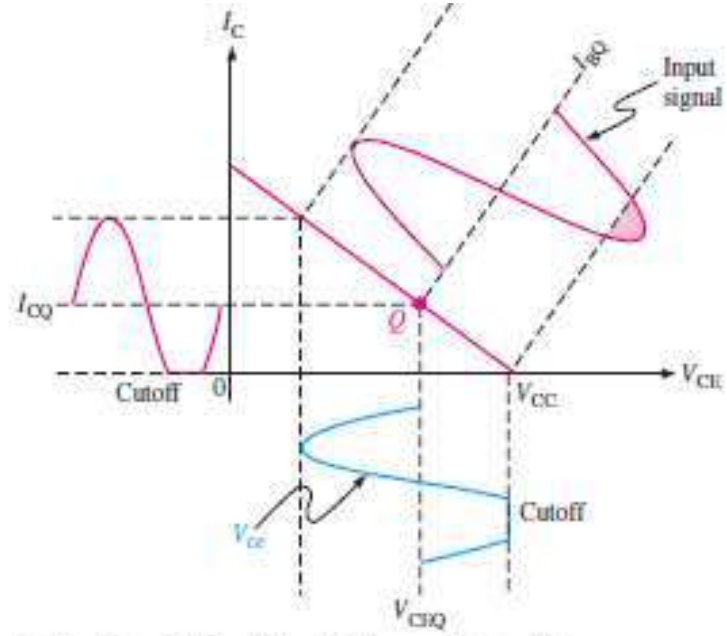
(b) Nonlinear operation: output voltage limited (clipped) by cutoff



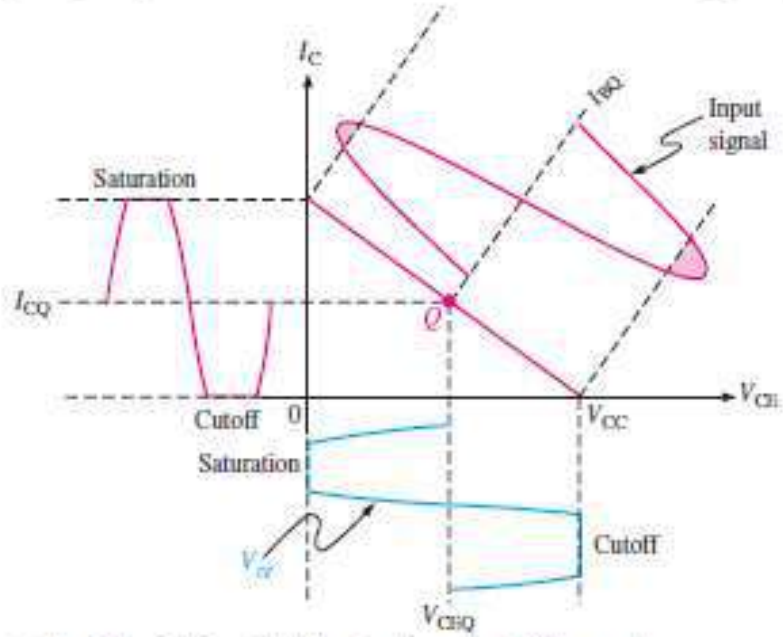
(c) Nonlinear operation: output voltage limited (clipped) by saturation



(a) Transistor is driven into saturation because the Q-point is too close to saturation for the given input signal.



(b) Transistor is driven into cutoff because the Q-point is too close to cutoff for the given input signal.



(c) Transistor is driven into both saturation and cutoff because the input signal is too large.

Although a number of networks are analyzed in this chapter, there is an underlying similarity in the analysis of each configuration due to the recurring use of the following important basic relationships for a transistor:

$$V_{BE} \cong 0.7\ \text{V} \tag{4.1}$$

$$I_E = (\beta + 1)I_B \cong I_C \tag{4.2}$$

$$I_C = \beta I_B \tag{4.3}$$

The term *biasing* appearing in the title of this chapter is an all-inclusive term for the application of dc voltages to establish a fixed level of current and voltage. For transistor amplifiers the resulting dc current and voltage establish an *operating point* on the characteristics that define the region that will be employed for amplification of the applied signal. Because the operating point is a fixed point on the characteristics, it is also called the *quiescent point* (abbreviated *Q* -point). By definition, *quiescent* means quiet, still, inactive. Figure.1 shows a general output device characteristic with four operating points indicated.

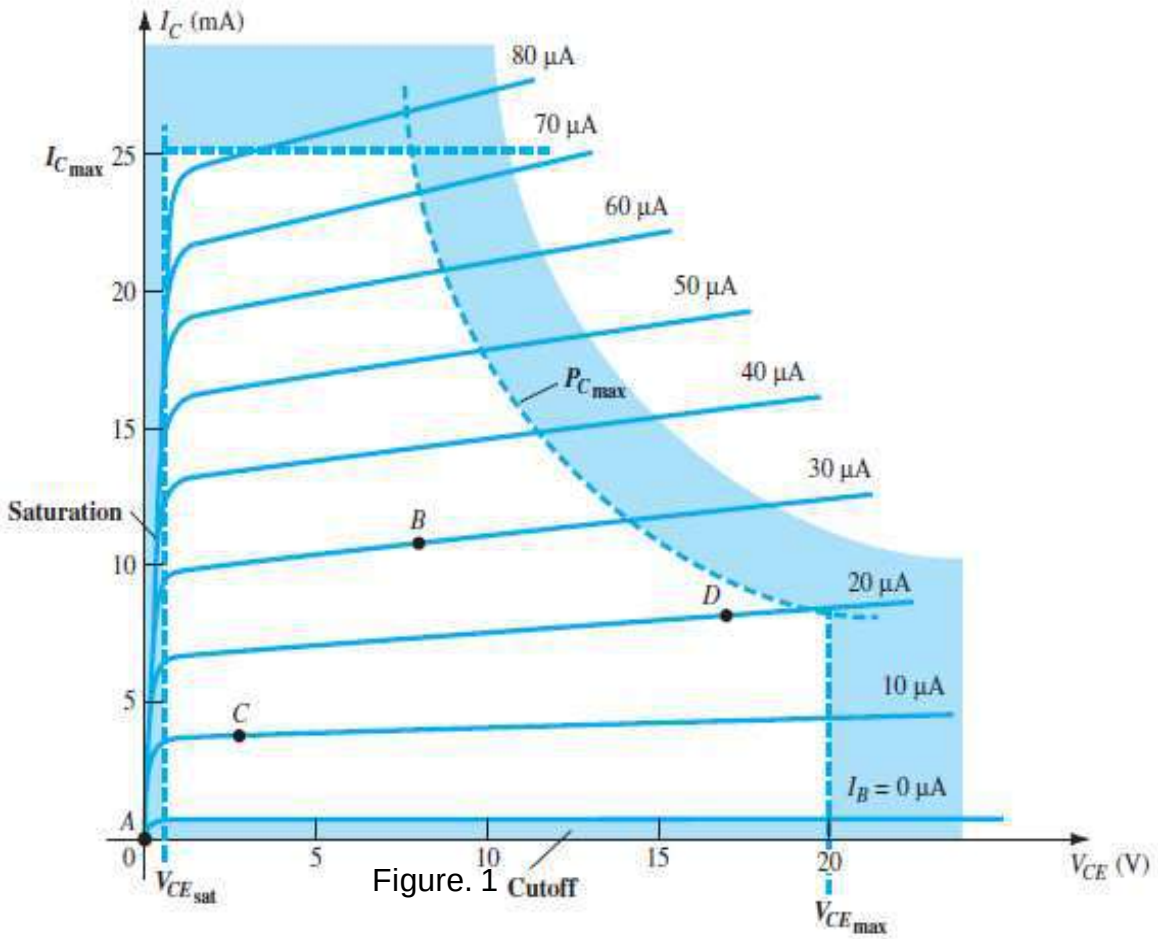
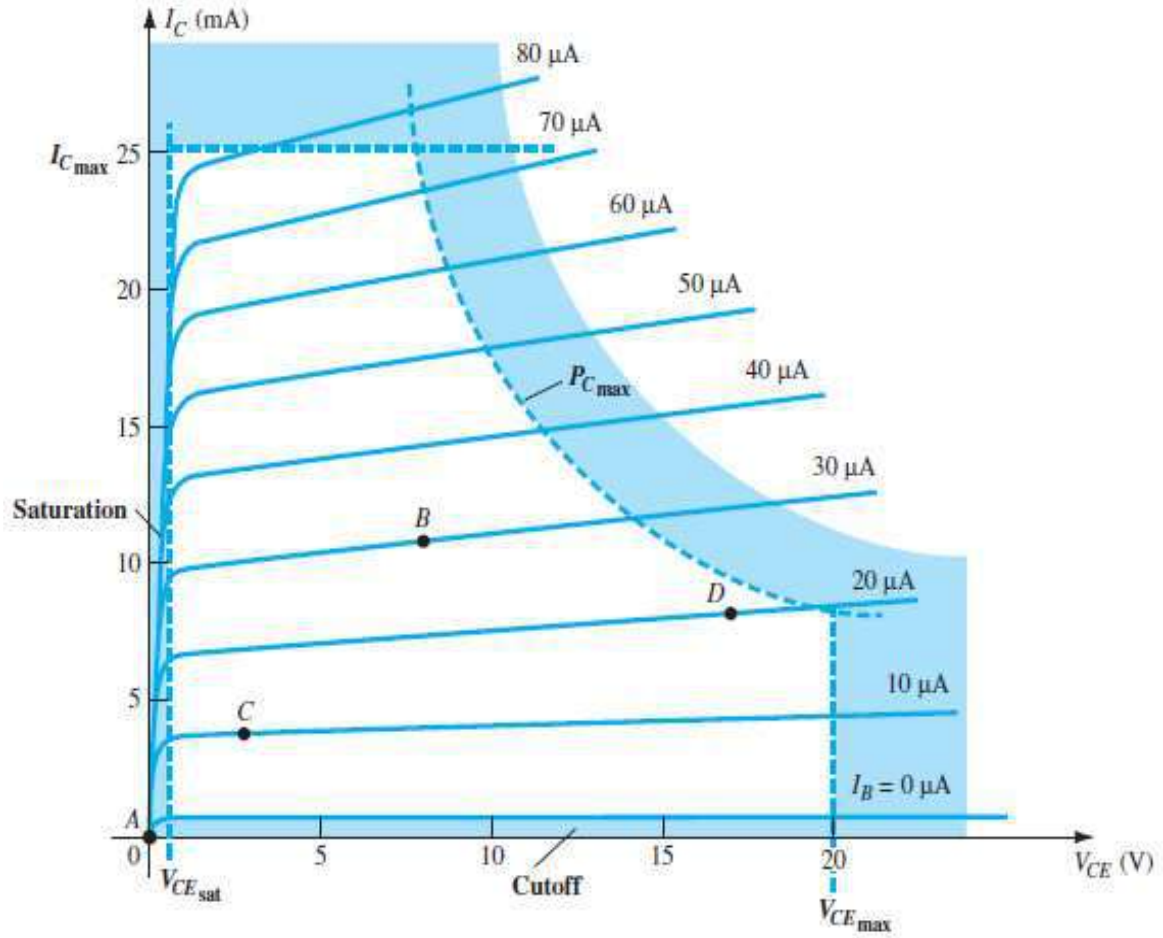


Figure. 1

Various operating points within the limits of operation of a transistor

The biasing circuit can be designed to set the device operation at any of these points or others within the *active region.* The maximum power constraint is defined by the curve P_{Cmax} in the same figure. At the lower end of the scales are the *cutoff region* , defined by $I_{\text{B}} \leq 0$ mA, and the *saturation region*, defined by $V_{\text{CE}} \leq V_{CE\ \text{sat}}$

Confining ourselves to the *active* region, we can select many different operating areas or points. The chosen Q -point often depends on the intended use of the circuit.



The chosen *Q* -point often depends on the intended use of the circuit.
For the BJT to be biased in its linear or active operating region the following must be true:

1. ***The base–emitter junction must be forward-biased (p-region voltage more positive), with a resulting forward-bias voltage of about 0.6 V to 0.7 V.***

2. ***The base–collector junction must be reverse-biased (n-region more positive), with the reverse-bias voltage being any value within the maximum limits of the device.***

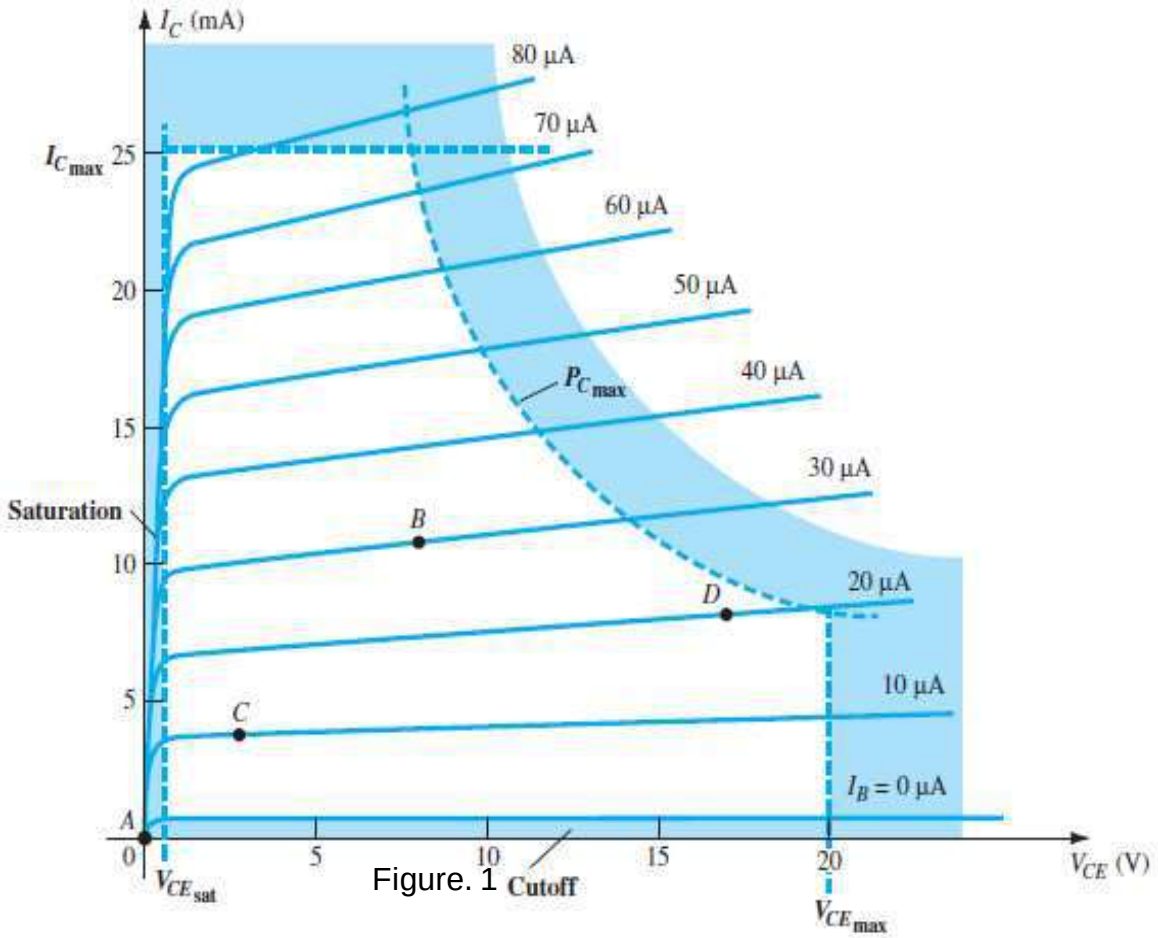


Figure. 1 **Various operating points within the limits of operation of a transistor**

Operation in the cutoff, saturation, and linear regions of the BJT characteristic are provided
as follows:

1. *Linear-region operation:*

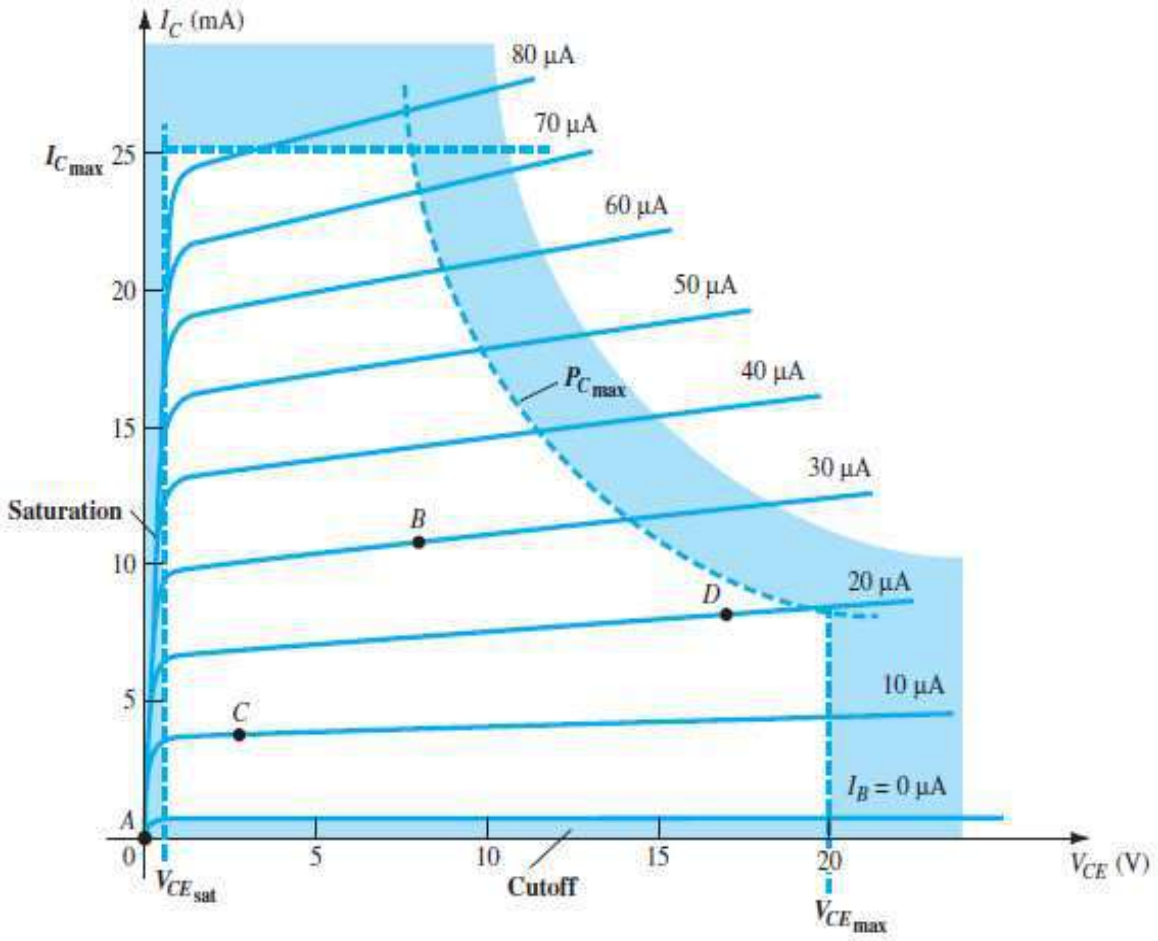
Base–emitter junction forward-biased
Base–collector junction reverse-biased

2. *Cutoff-region operation:*

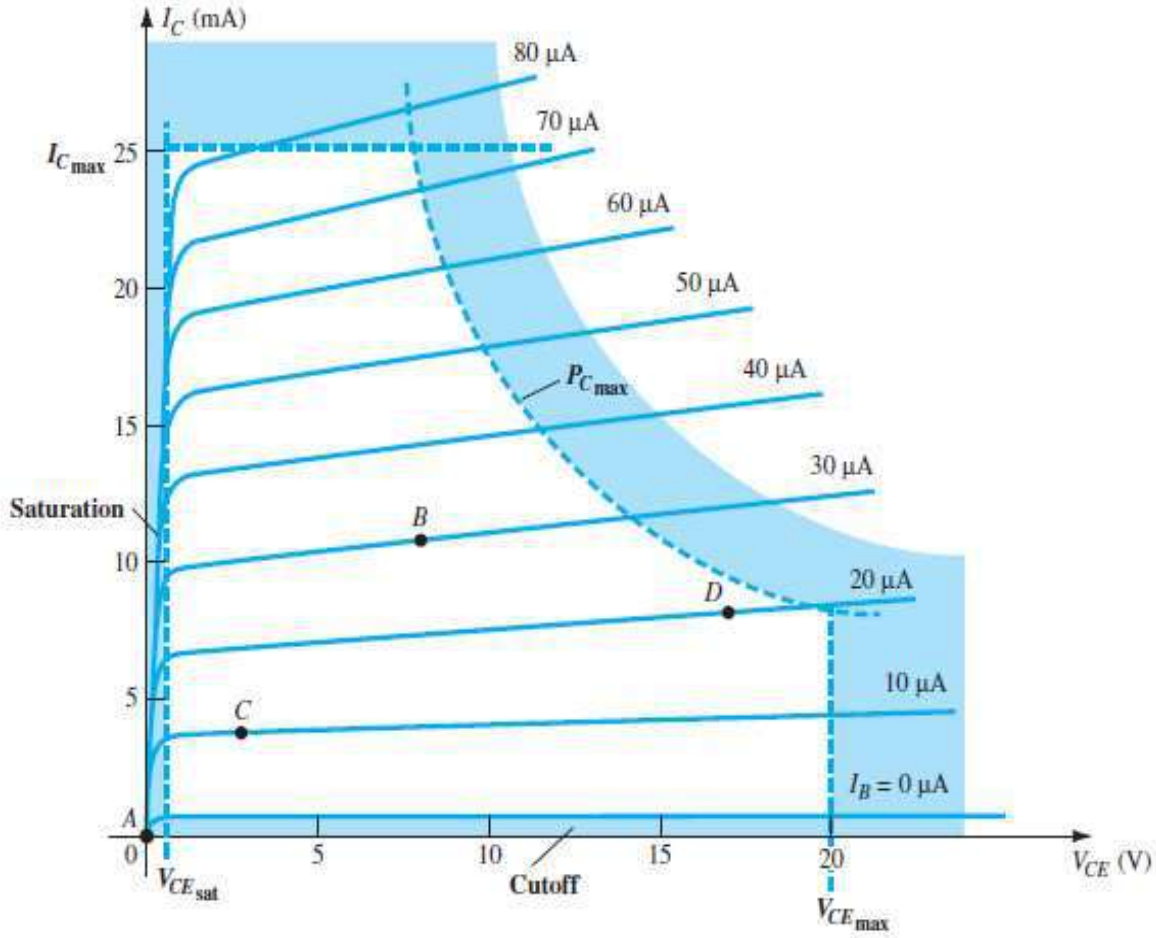
Base–emitter junction reverse-biased
Base–collector junction reverse-biased

3. *Saturation-region operation:*

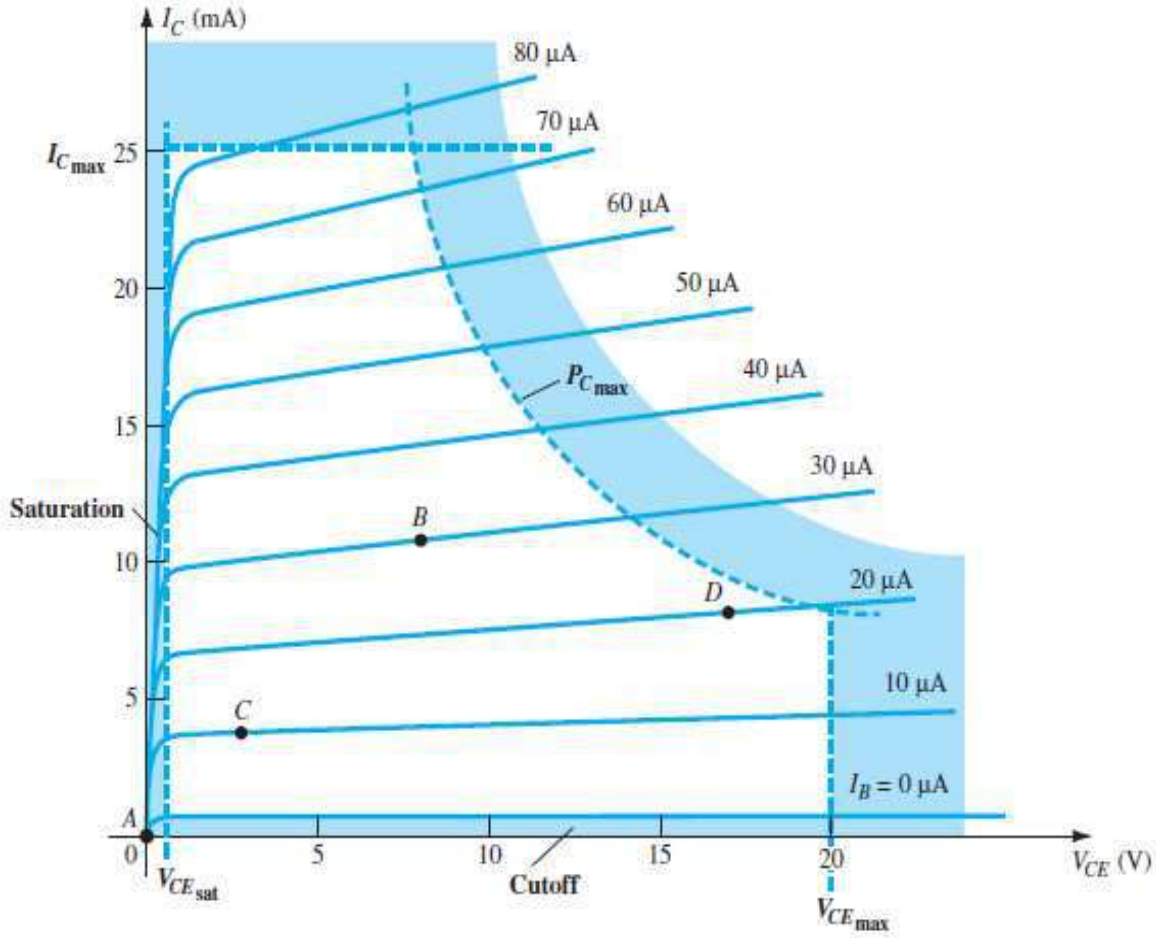
Base–emitter junction forward-biased
Base–collector junction forward-biased



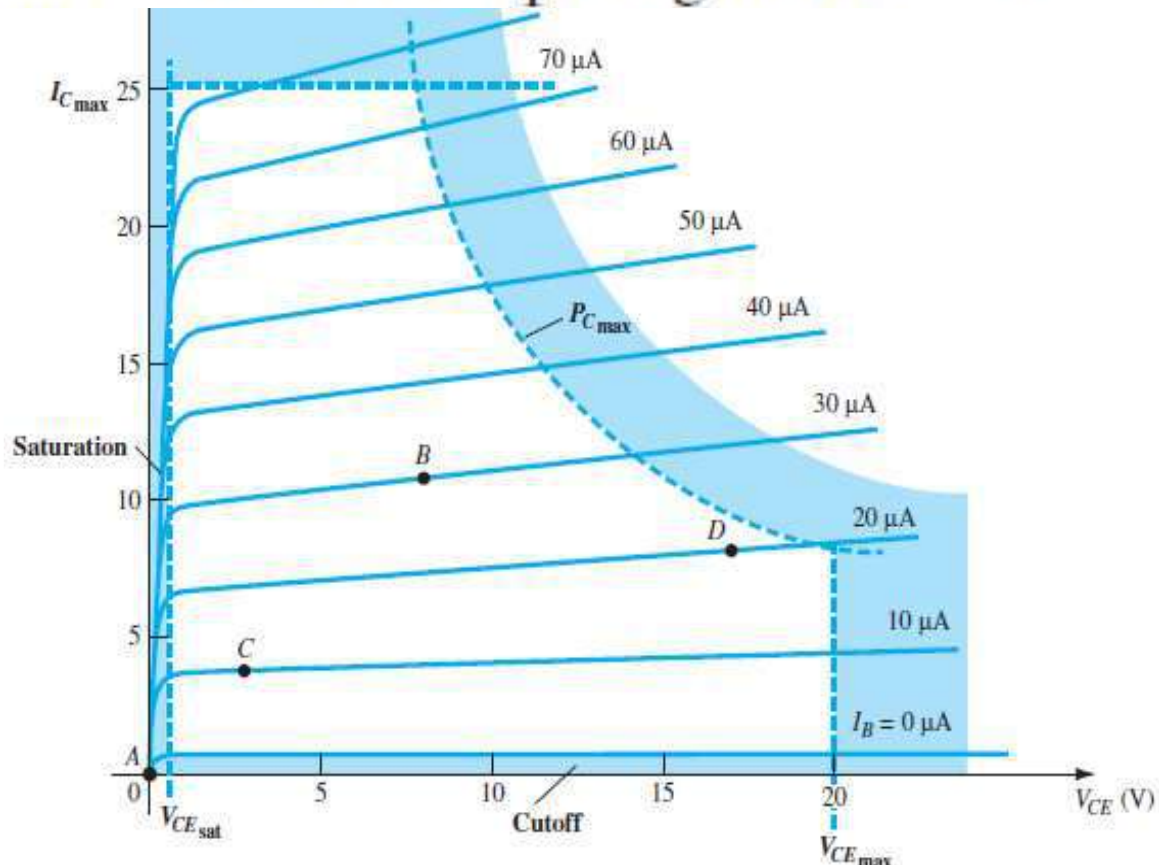
If no bias were used, the device would initially be completely off, resulting in a *Q*-point at *A*—namely, zero current through the device (and zero voltage across it). Because it is necessary to bias a device so that it can respond to the entire range of an input signal, point *A* would not be suitable.



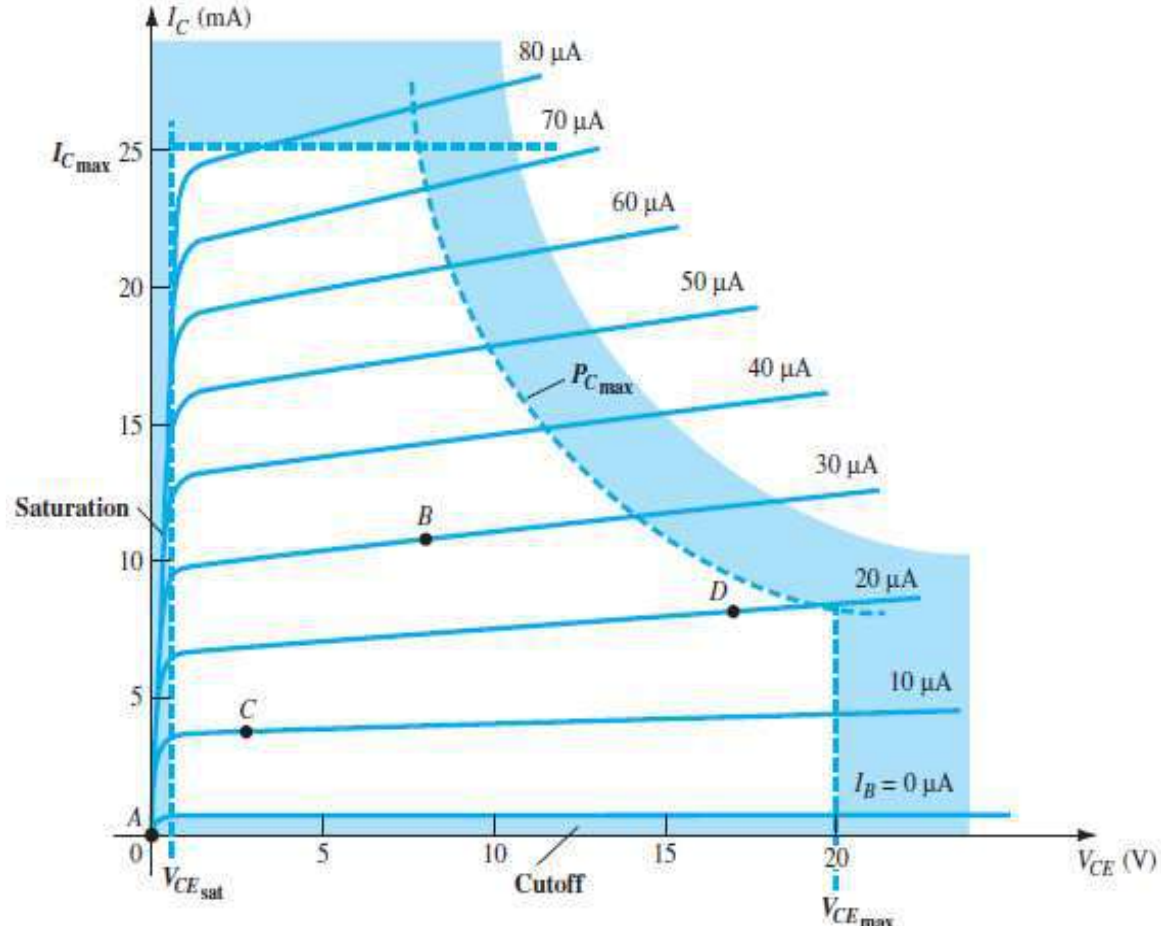
For point *B*, if a signal is applied to the circuit, the device will vary in current and voltage from the operating point, allowing the device to react to (and possibly amplify) both the positive and negative excursions of the input signal. If the input signal is properly chosen, the voltage and current of the device will vary, but not enough to drive the device into *cutoff* or *saturation*.



Point C would allow some positive and negative variation of the output signal, but the peak-to-peak value would be limited by the proximity of $V_{CE} = 0$ V and $I_C = 0$ mA. Operating at point C also raises some concern about the nonlinearities introduced by the fact that the spacing between I_B curves is rapidly changing in this region. In general, it is preferable to operate where the gain of the device is fairly constant (or linear) to ensure that the amplification over the entire swing of input signal is the same. Point B is a region of more linear spacing and therefore more linear operation, as shown in Fig. 4.1.



Point *D* sets the device operating point near the maximum voltage and power level. The output voltage swing in the positive direction is thus limited if the maximum voltage is not to be exceeded. Point *B* therefore seems the best operating point in terms of linear gain and largest possible voltage and current swing.



The biasing circuit can be designed to set the device operation at any of these points or others within the *active region*. The maximum power constraint is defined by the curve P_{Cmax} in the same figure. At the lower end of the scales are the *cutoff region* , defined by I_B … 0 mA, and the *saturation region*, defined by V_{CE} … $V_{CE\,sat}$

Confining ourselves to the *active* region, we can select many different operating areas or points. The chosen *Q* -point often depends on the intended use of the circuit.

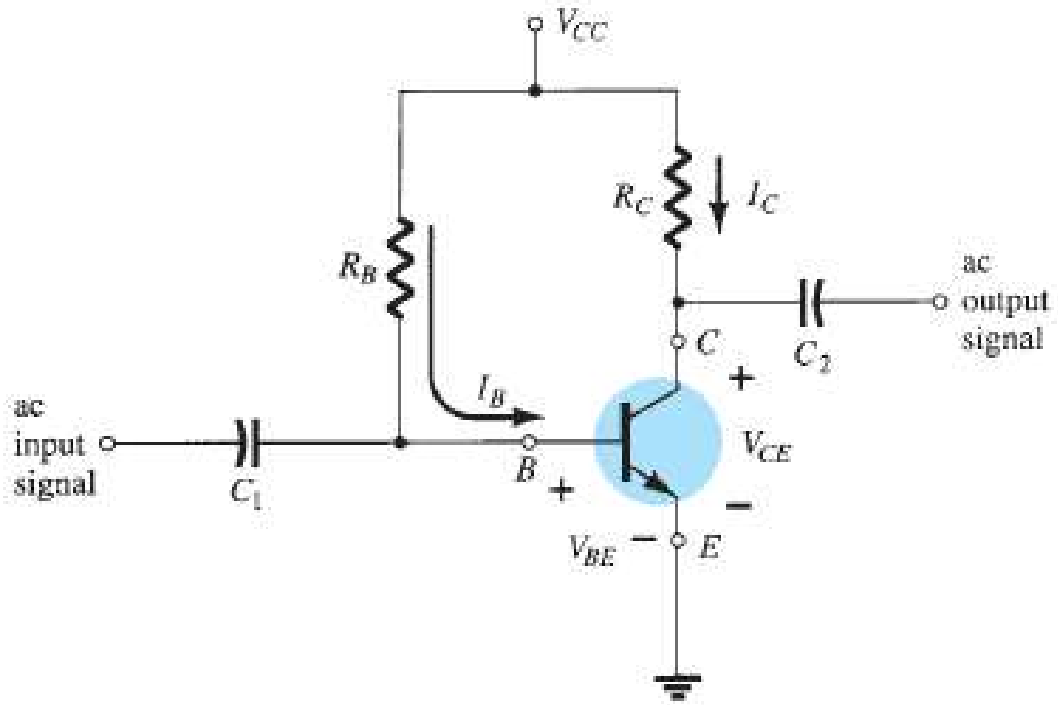
For the BJT to be biased in its linear or active operating region the following must be true:

1. ***The base–emitter junction must be forward-biased (p-region voltage more positive), with a resulting forward-bias voltage of about 0.6 V to 0.7 V.***

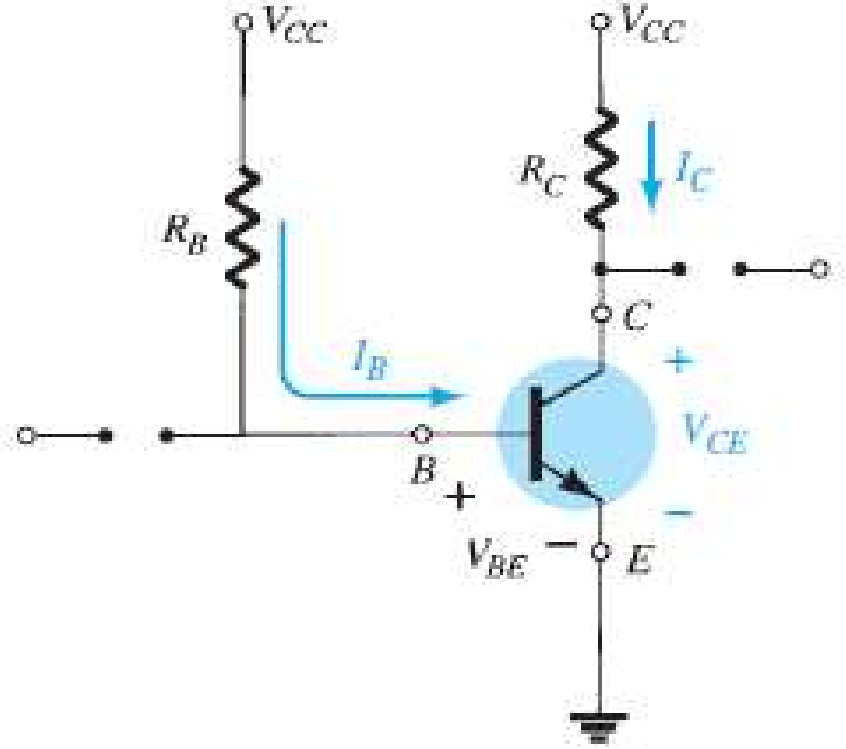
2. ***The base–collector junction must be reverse-biased (n-region more positive), with the reverse-bias voltage being any value within the maximum limits of the device.***

FIXED-BIAS CONFIGURATION

The network employs an *npn* transistor, the equations and calculations apply equally well to a *pnp* transistor configuration merely by changing all current directions and voltage polarities.

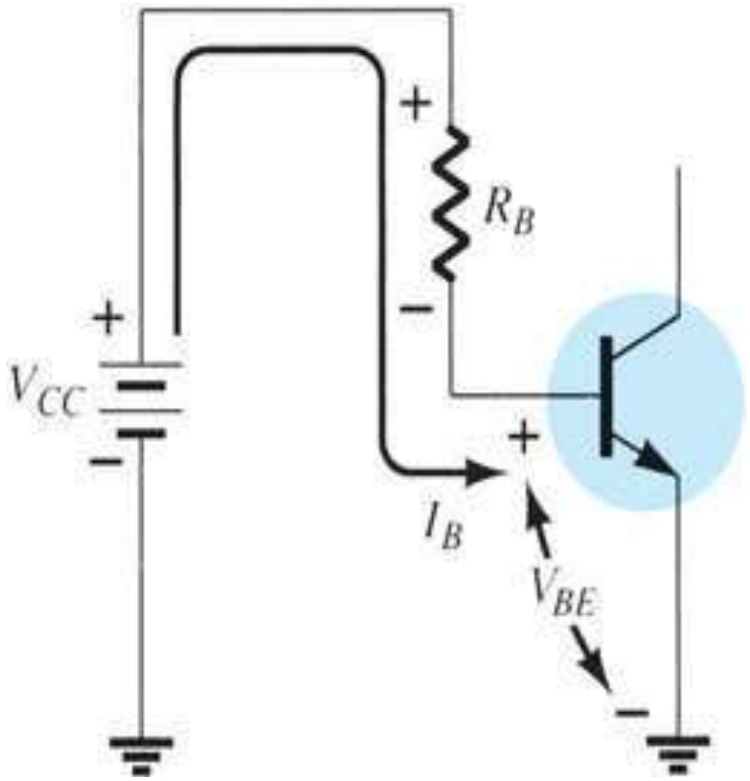


Fixed-bias circuit.



DC equivalent

Forward Bias of Base–Emitter



Writing Kirchhoff's voltage equation , in the clockwise direction for the loop, we obtain

$$+V_{CC} - I_B R_B - V_{BE} = 0$$

$$I_B = \frac{V_{CC} - V_{BE}}{R_B}$$

Collector–Emitter Loop

The collector–emitter section of the network appears in Fig. 4.5 with the indicated direction of current I_C and the resulting polarity across R_C. The magnitude of the collector current is related directly to I_B through

$$I_C = \beta I_B \tag{4.5}$$

Applying Kirchhoff's voltage law in the clockwise direction around the indicated closed loop of Fig. 4.5 results in the following:

$$V_{CE} + I_C R_C - V_{CC} = 0$$

and

$$V_{CE} = V_{CC} - I_C R_C \tag{4.6}$$

which states that the voltage across the collector–emitter region of a transistor in the fixed-bias configuration is the supply voltage less the drop across R_C.

As a brief review of single- and double-subscript notation recall that

$$V_{CE} = V_C - V_E \tag{4.7}$$

where V_{CE} is the voltage from collector to emitter and V_C and V_E are the voltages from collector and emitter to ground, respectively. *In this case,* since $V_E = 0$ V, we have

$$V_{CE} = V_C \tag{4.8}$$

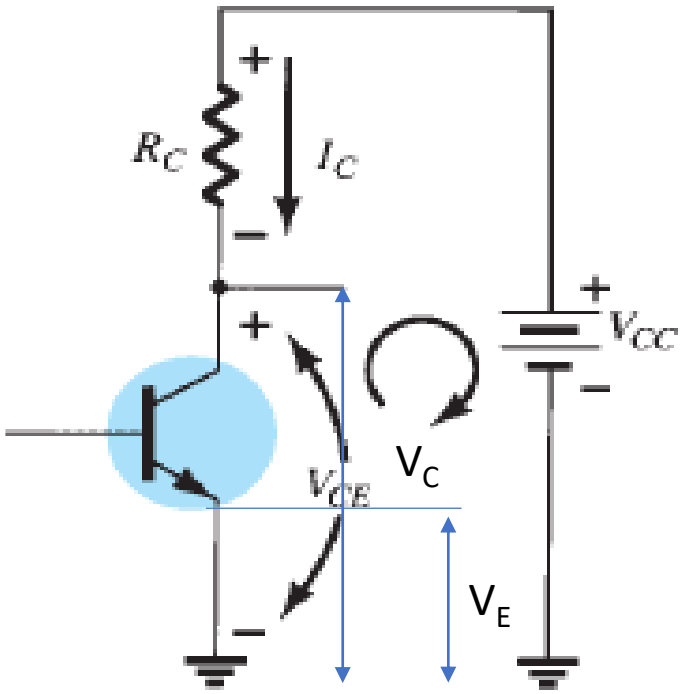


Fig. 4.5

In addition, because

$$V_{BE} = V_B - V_E$$

and $V_E = 0$ V, then

$$V_{BE} = V_B$$

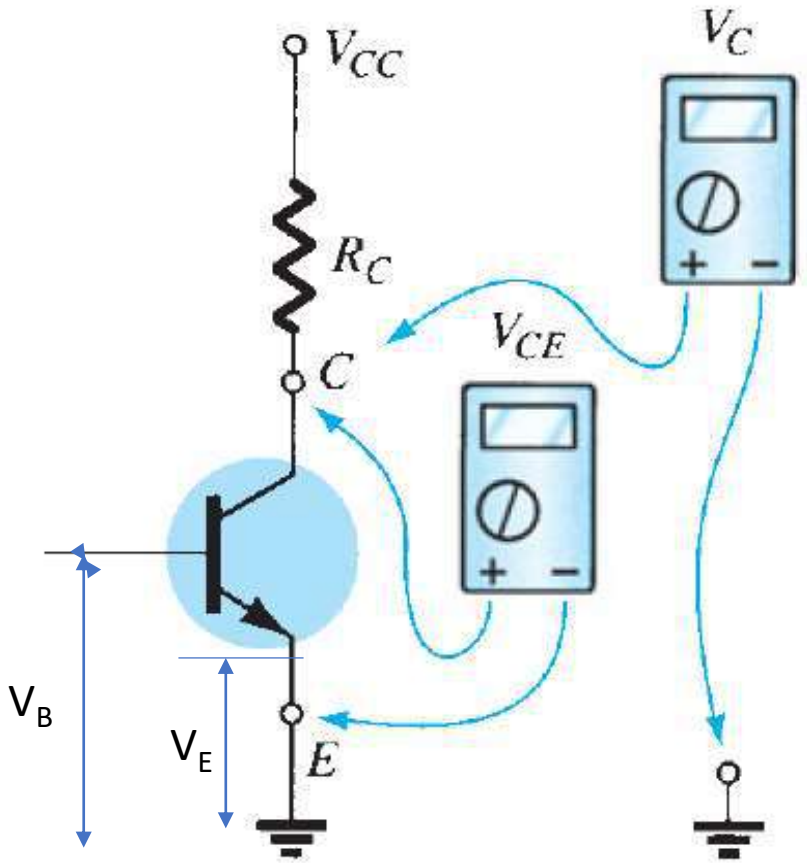
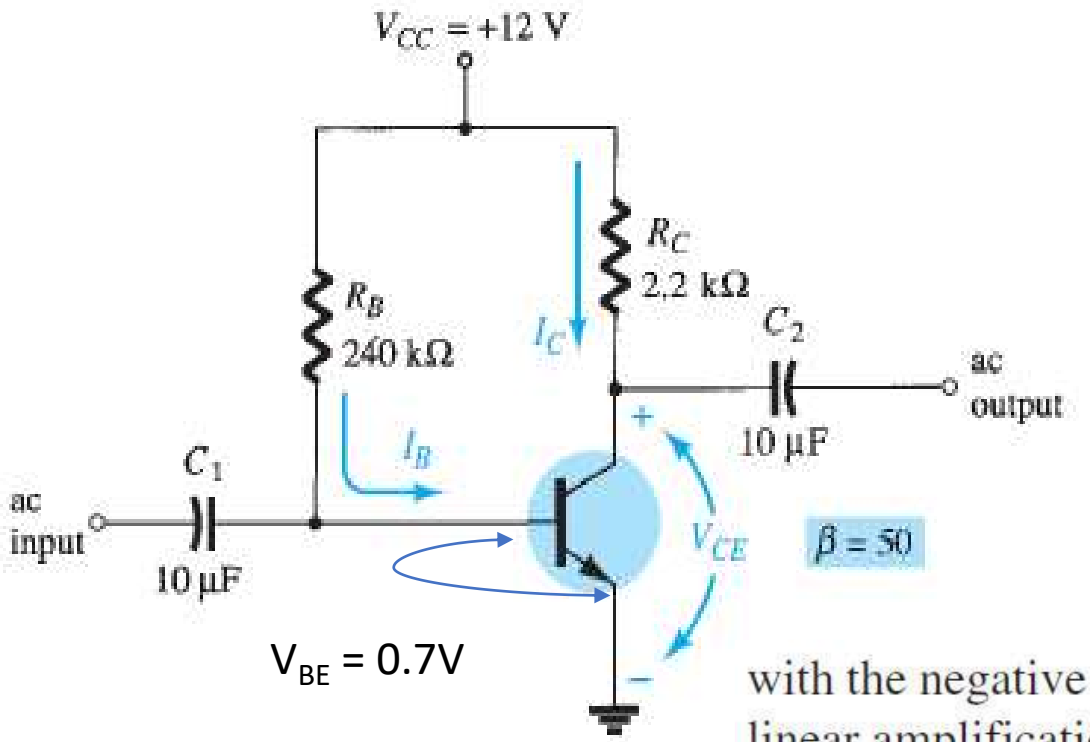


FIG. 4.6
Measuring V_{CE} and V_C.

EXAMPLE.1 Determine the following for the fixed-bias configuration of following Figure:

a. I_{BQ} and $I_{CQ.}$
b. V_{CEQ}.
c. V_B and V_C.
d. V_{BC}.



Solution:

a. Eq. (4.4): $$I_{B_Q} = \frac{V_{CC} - V_{BE}}{R_B} = \frac{12\text{ V} - 0.7\text{ V}}{240\text{ k}\Omega} = \mathbf{47.08\ \mu A}$$

Eq. (4.5): $$I_{C_Q} = \beta I_{BQ} = (50)(47.08\ \mu\text{A}) = \mathbf{2.35\ mA}$$

b. Eq. (4.6): $$V_{CE_Q} = V_{CC} - I_C R_C$$
$$= 12\text{ V} - (2.35\text{ mA})(2.2\text{ k}\Omega)$$
$$= \mathbf{6.83\ V}$$

c. $V_B = V_{BE} = \mathbf{0.7\ V}$
$V_C = V_{CE} = \mathbf{6.83\ V}$

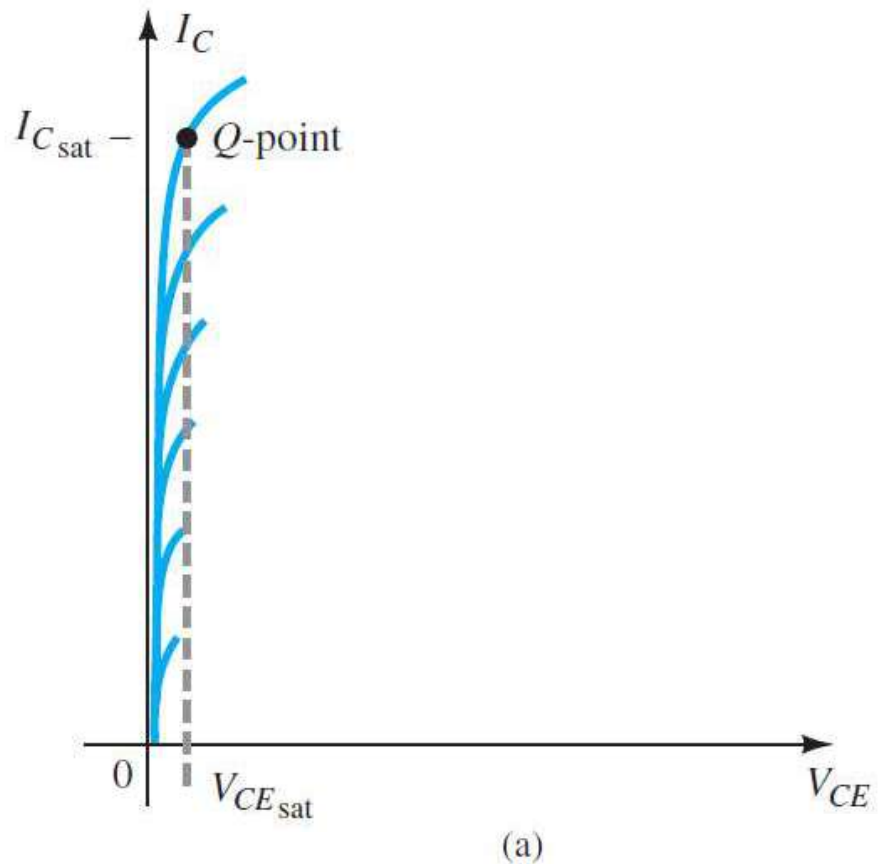
d. Using double-subscript notation yields

$$V_{BC} = V_B - V_C = 0.7\text{ V} - 6.83\text{ V}$$
$$= \mathbf{-6.13\ V}$$

with the negative sign revealing that the junction is reversed-biased, as it should be for linear amplification.

Transistor Saturation

The term *saturation* is applied to any system where levels have reached their maximum values. A saturated sponge is one that cannot hold another drop of water. For a transistor operating in the saturation region, the current is a maximum value *for the particular design.* Change the design and the corresponding saturation level may rise or drop. Of course, the highest saturation level is defined by the maximum collector current as provided by the specification sheet.



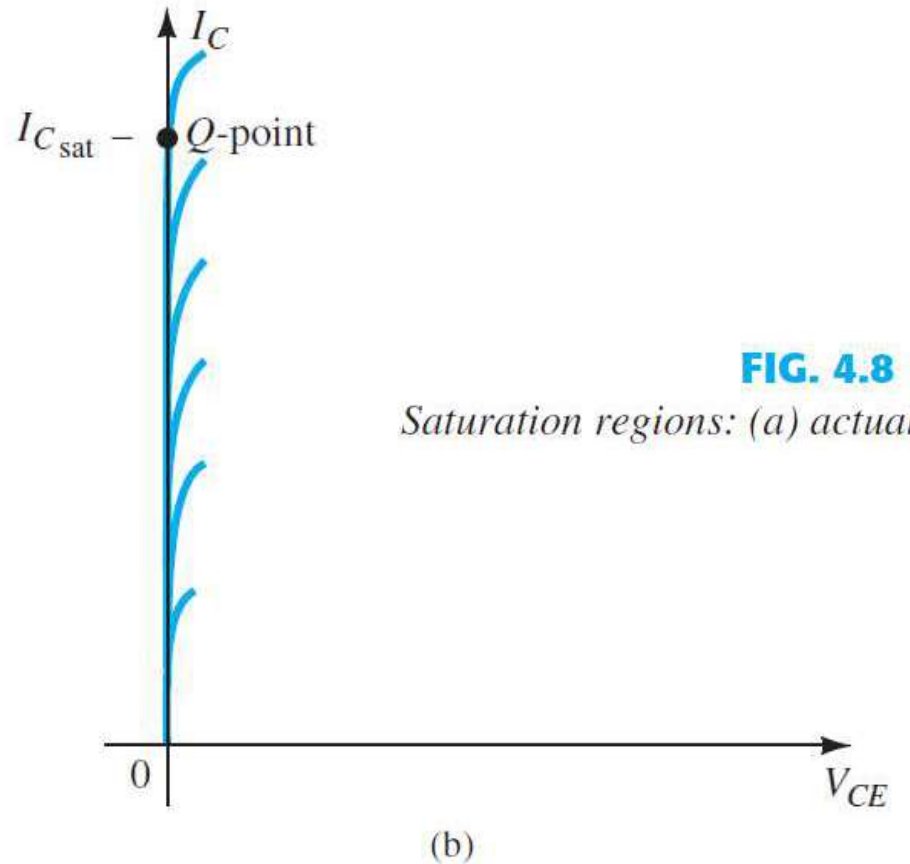


FIG. 4.8
Saturation regions: (a) actual; (b) approximate.

If we approximate the curves of Fig. 4.8a by those appearing in Fig. 4.8b, a quick, direct method for determining the saturation level becomes apparent. In Fig. 4.8b, the current is relatively high, and the voltage V_{CE} is assumed to be 0 V. Applying Ohm's law, we can determine the resistance between collector and emitter terminals as follows:

$$R_{CE} = \frac{V_{CE}}{I_C} = \frac{0\text{ V}}{I_{C_{sat}}} = 0\ \Omega$$

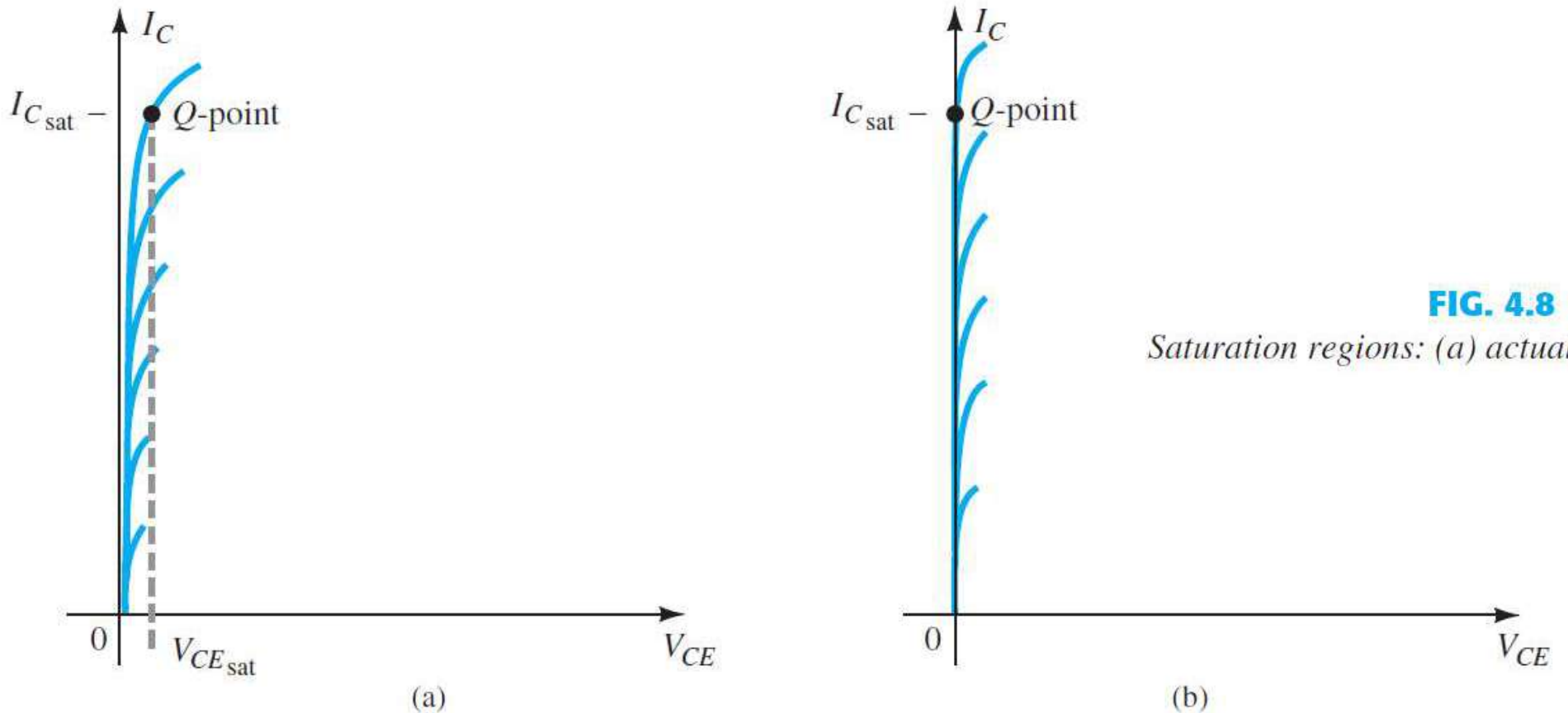
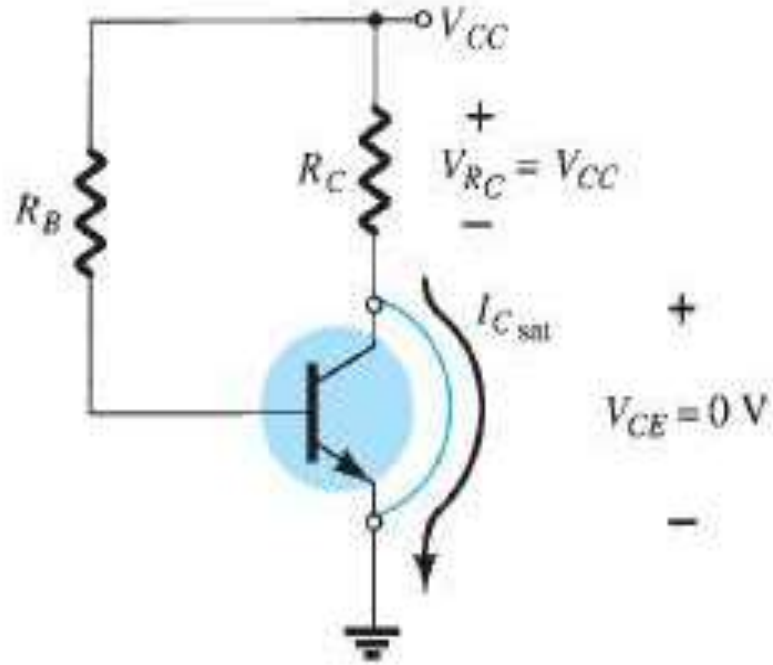


FIG. 4.8
Saturation regions: (a) actual; (b) approximate.

The resulting saturation current for the fixed-bias configuration is

$$I_{C_{sat}} = \frac{V_{CC}}{R_C}$$



Once $I_{C_{sat}}$ is known, we have some idea of the maximum possible collector current for the chosen design and the level to stay below if we expect linear amplification.

EXAMPLE 4.2 Determine the saturation level for the network of Fig. 4.7.

Solution:

$$I_{C_{sat}} = \frac{V_{CC}}{R_C} = \frac{12\text{ V}}{2.2\text{ k}\Omega} = \mathbf{5.45\text{ mA}}$$

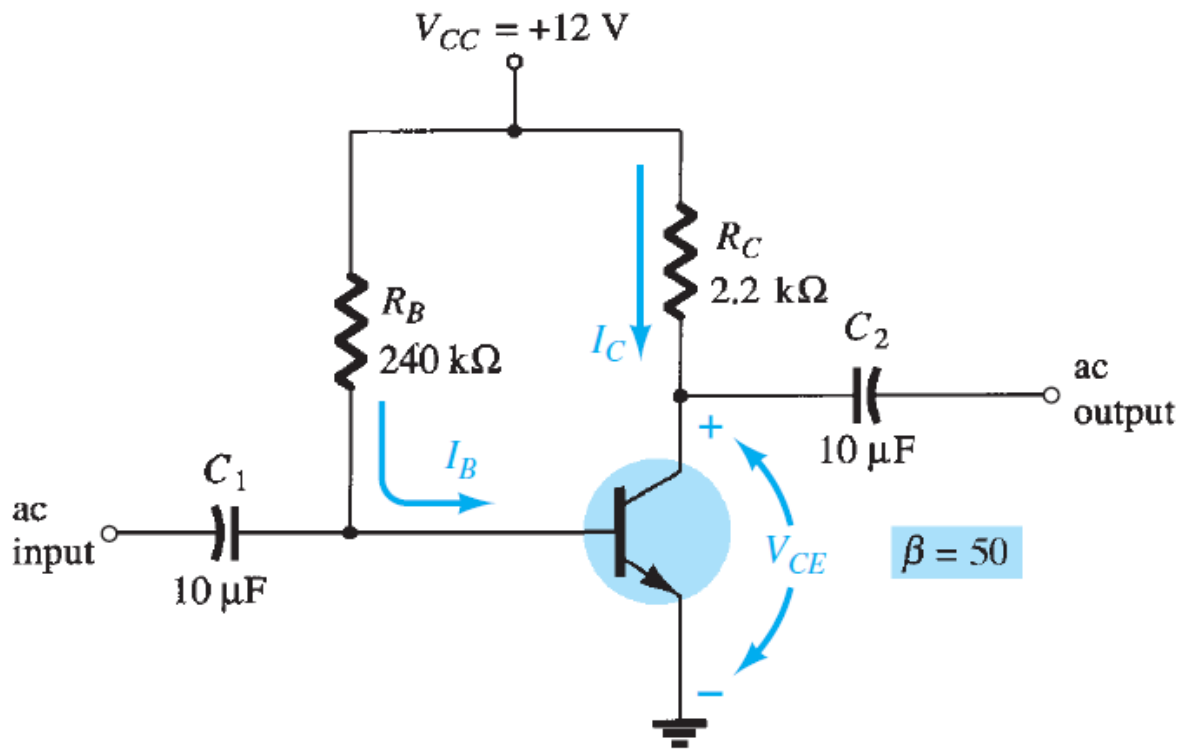
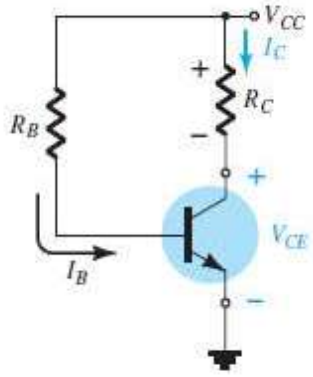


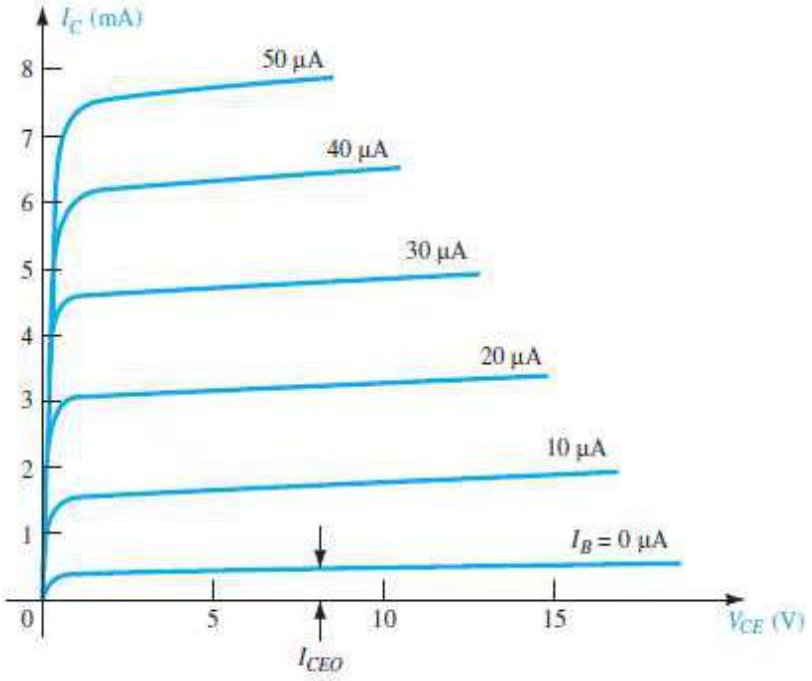
FIG. 4.7

Load-Line Analysis

The characteristics of the BJT are superimposed on a plot of the network equation defined by the same axis parameters. The load resistor R_C for the fixed-bias configuration will define the slope of the network equation and the resulting intersection between the two plots. The smaller the load resistance, the steeper the slope of the network load line



(a)



(b)

The network of above Fig. a establishes an output equation that relates the variables I_C and V_{CE} in the following manner:

$$V_{CE} = V_{CC} - I_C R_C$$

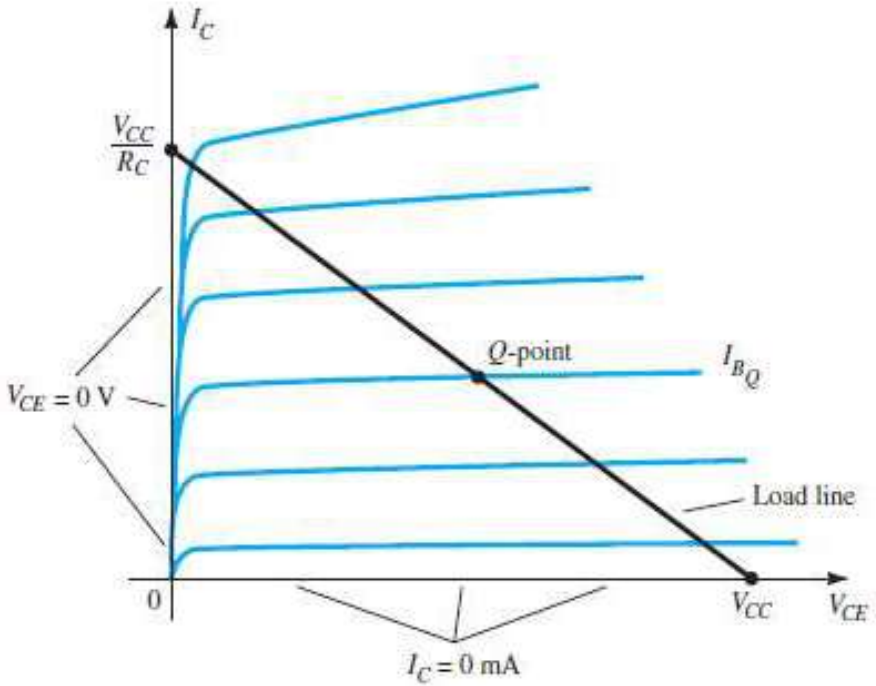
The output characteristics of the transistor also relate the same two variables I_C and V_{CE} as shown in Fig. We must now superimpose the straight line defined by the above equation on the characteristics.

$$V_{CE} = V_{CC} - (0)R_C$$

$$V_{CE} = V_{CC}|_{I_C = 0\ \text{mA}}$$

$$0 = V_{CC} - I_C R_C$$

$$I_C = \left.\frac{V_{CC}}{R_C}\right|_{V_{CE} = 0\ \text{V}}$$



If the level of I_B is changed by varying the value of R_B, the Q-point moves up or down the load line as shown in Fig. 4.13 for increasing values of I_B. If V_{CC} is held fixed and R_C increased, the load line will shift as shown in Fig. 4.14. If I_B is held fixed, the Q-point will move as shown in the same figure. If R_C is fixed and V_{CC} decreased, the load line shifts as shown in Fig. 4.15.

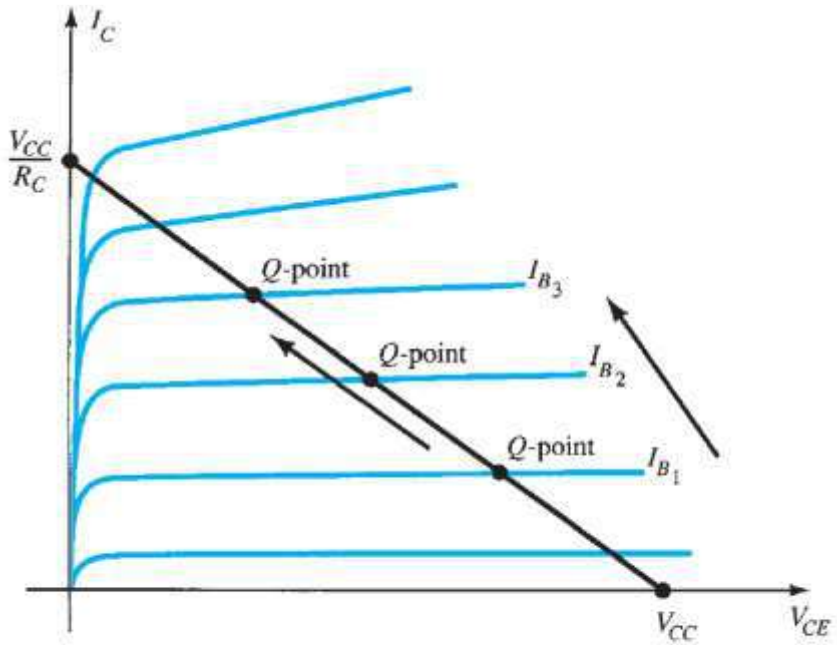


FIG. 4.13
Movement of the Q-point with increasing level of I_B.

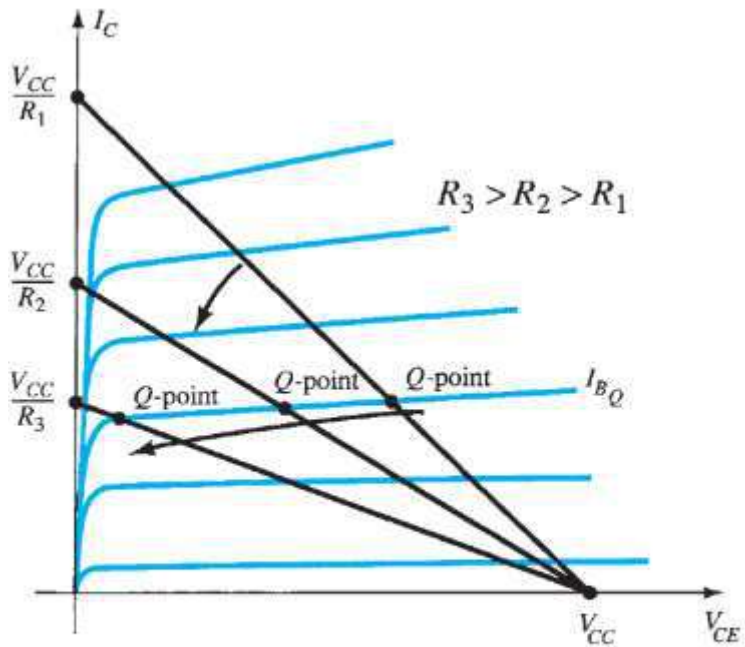


FIG. 4.14
Effect of an increasing level of R_C on the load line and the Q-point.

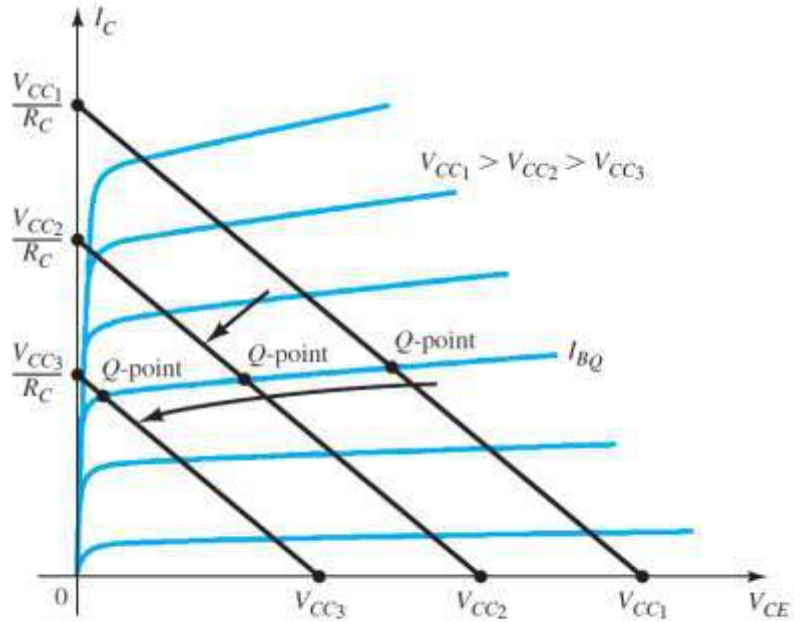
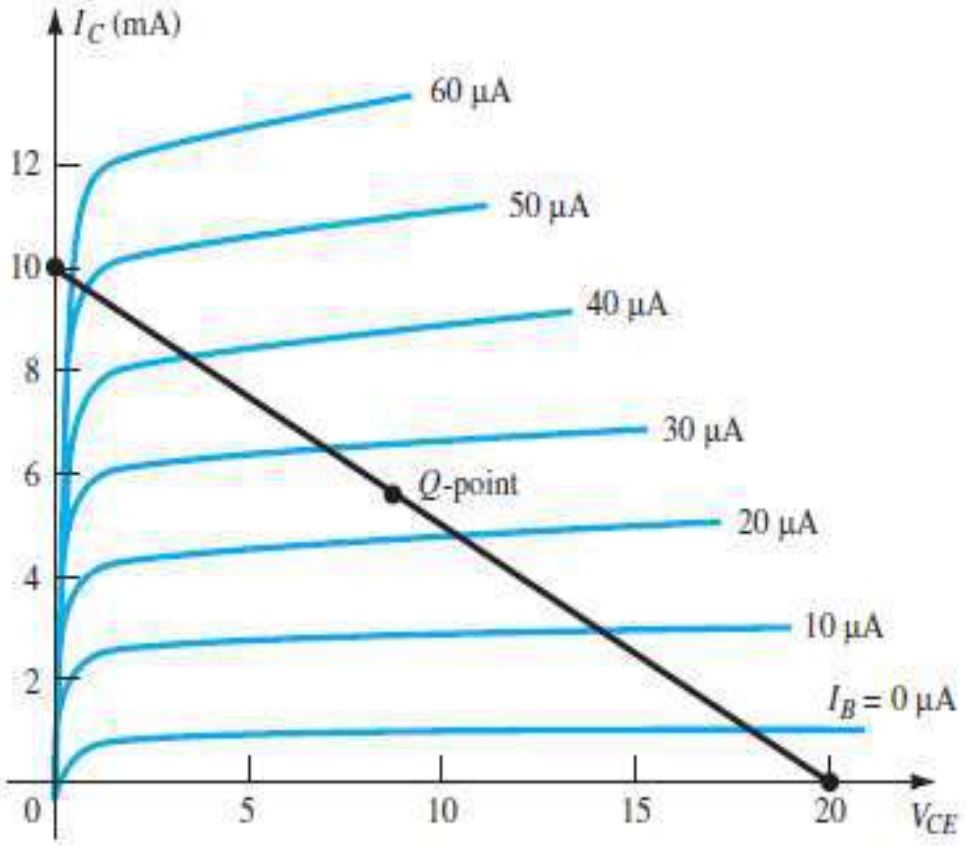


FIG. 4.15
Effect of lower values of V_{CC} on the load line and the Q-point.

EXAMPLE 4.3 Given the load line of the following characteristics and the defined Q-point, determine the required values of V_{CC}, R_C, and R_B for a fixed-bias configuration.



Solution:

$$V_{CE} = V_{CC} = \mathbf{20\ V} \text{ at } I_C = 0 \text{ mA}$$

$$I_C = \frac{V_{CC}}{R_C} \text{ at } V_{CE} = 0 \text{ V}$$

$$R_C = \frac{V_{CC}}{I_C} = \frac{20 \text{ V}}{10 \text{ mA}} = \mathbf{2\ k\Omega}$$

$$I_B = \frac{V_{CC} - V_{BE}}{R_B}$$

$$R_B = \frac{V_{CC} - V_{BE}}{I_B} = \frac{20 \text{ V} - 0.7 \text{ V}}{25\ \mu\text{A}} = \mathbf{772\ k\Omega}$$

EMITTER-BIAS CONFIGURATION

The dc bias network of the following Figure 4.17 contains an emitter resistor to improve the stability level over that of the fixed-bias configuration. The more stable a configuration, the less its response will change due to undesirable changes in temperature and parameter variations.

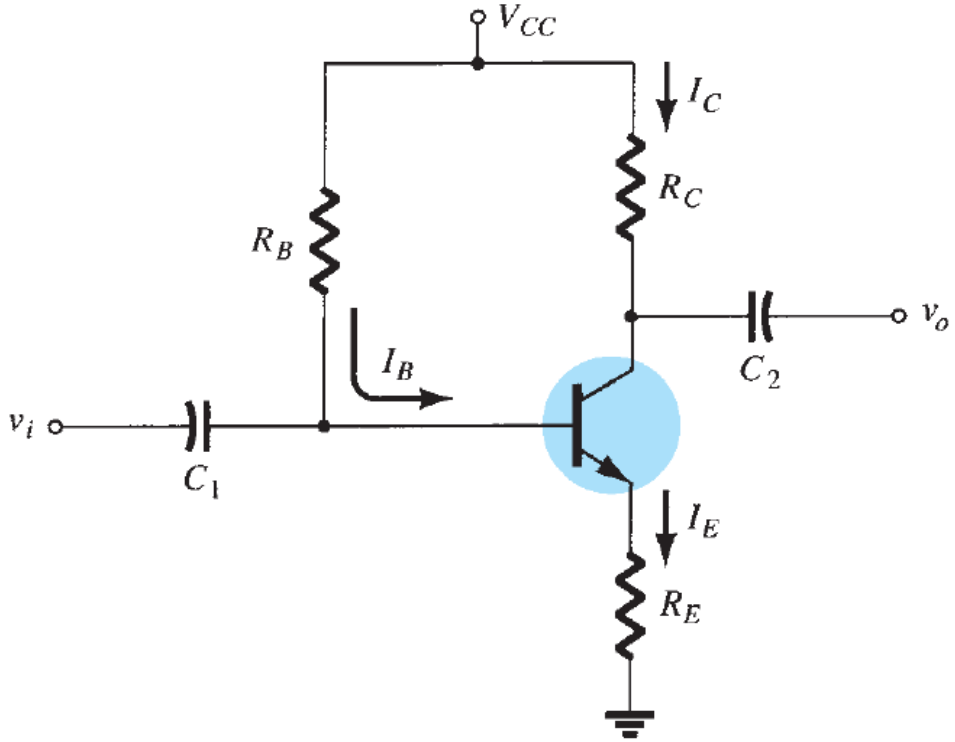


FIG. 4.17

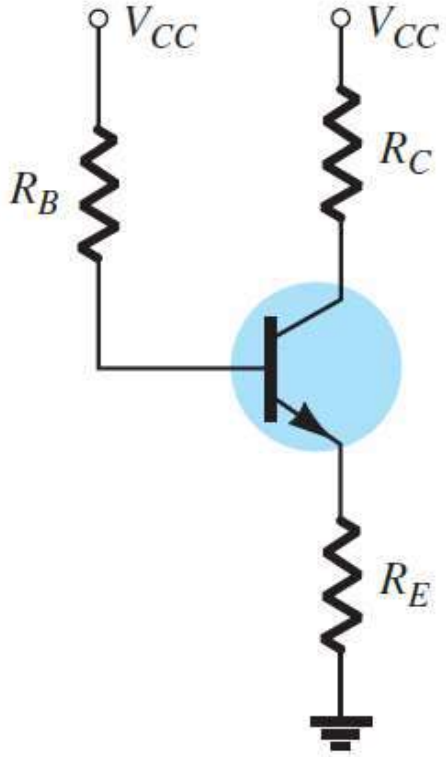


FIG. 4.18
DC equivalent of Fig. 4.17.

Base–Emitter Loop:

The base–emitter loop of the network of Fig. 4.18 can be redrawn as shown in Fig. 4.19. Writing Kirchhoff's voltage law around the indicated loop in the clockwise direction results in the following equation:

$$+V_{CC} - I_B R_B - V_{BE} - I_E R_E = 0 \qquad \textbf{(4.15)}$$

Recall from Chapter 3 that

$$I_E = (\beta + 1) I_B \qquad \textbf{(4.16)}$$

Substituting for I_E in Eq. (4.15) results in

$$V_{CC} - I_B R_B - V_{BE} - (\beta + I) I_B R_E = 0$$

Grouping terms then provides the following:

$$-I_B(R_B + (\beta + 1)R_E) + V_{CC} - V_{BE} = 0$$

Multiplying through by (-1), we have

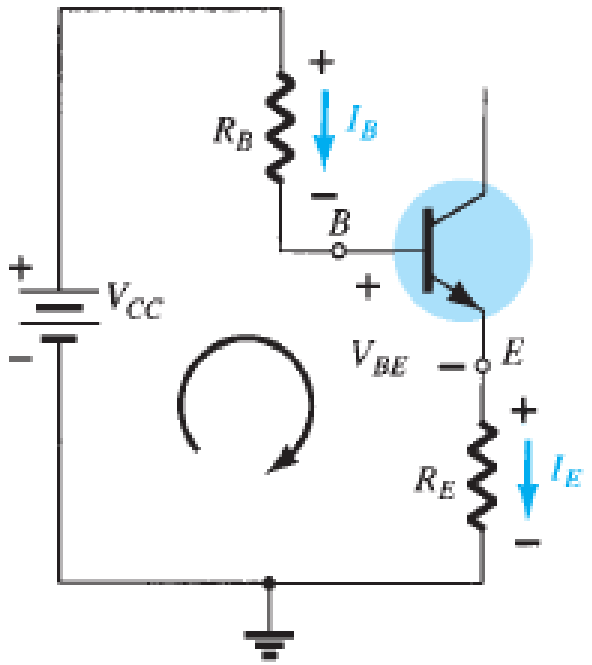
$$I_B(R_B + (\beta + 1)R_E) - V_{CC} + V_{BE} = 0$$

with

$$I_B(R_B + (\beta + 1)R_E) = V_{CC} - V_{BE}$$

and solving for I_B gives

$$I_B = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1)R_E} \qquad \textbf{(4.17)}$$



Note that the only difference between this equation for I_B and that obtained for the fixed-bias configuration is the term $(\beta + 1)R_E$.

There is an interesting result that can be derived from Eq. (4.17) if the equation is used to sketch a series network that would result in the same equation. Such is the case for the network of Fig. 4.20. Solving for the current I_B results in the same equation as obtained above. Note that aside from the base-to-emitter voltage V_{BE}, the resistor R_E is *reflected* back to the input base circuit by a factor $(\beta + 1)$. In other words, the emitter resistor, which is part of the collector–emitter loop, "appears as" $(\beta + 1)R_E$ in the base–emitter loop. Because β is typically 50 or more, the emitter resistor appears to be a great deal larger in the base circuit. In general, therefore, for the configuration of Fig. 4.21,

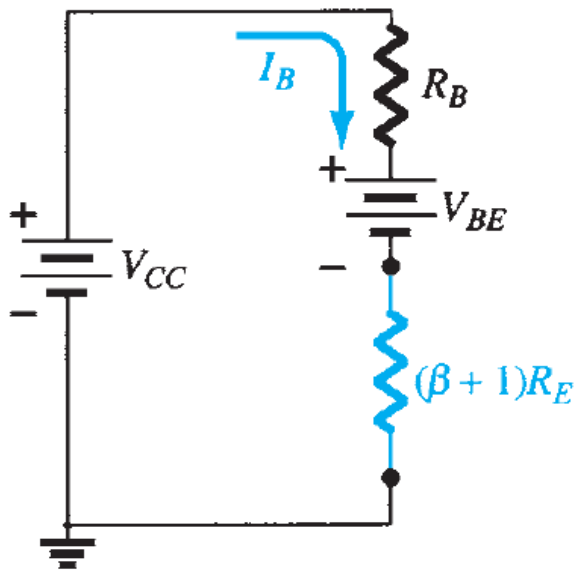


FIG. 4.20
Network derived from Eq. (4.17).

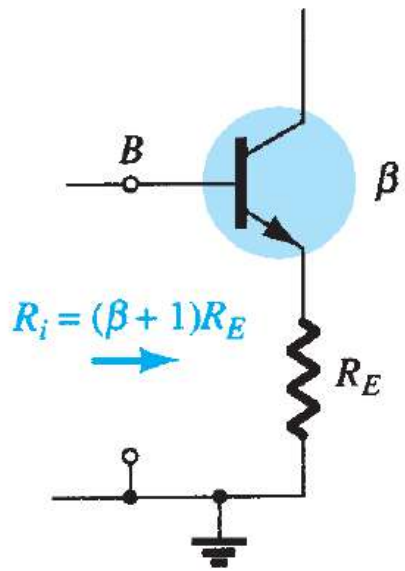


FIG. 4.21
Reflected impedance level of R_E.

$$R_i = (\beta + 1)R_E \tag{4.18}$$

Equation (4.18) will prove useful in the analysis to follow. In fact, it provides a fairly easy way to remember Eq. (4.17). Using Ohm's law, we know that the current through a system is the voltage divided by the resistance of the circuit. For the base–emitter circuit the net voltage is $V_{CC} - V_{BE}$. The resistance levels are R_B plus R_E reflected by $(\beta + 1)$. The result is Eq. (4.17).

$$I_B = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1)R_E} \tag{4.17}$$

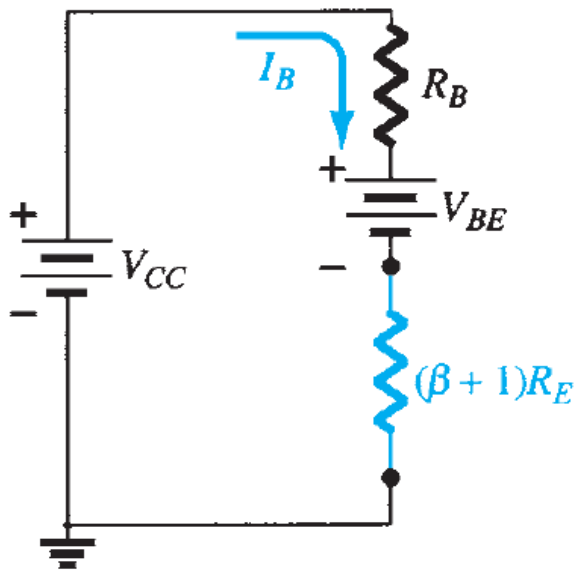


FIG. 4.20
Network derived from Eq. (4.17).

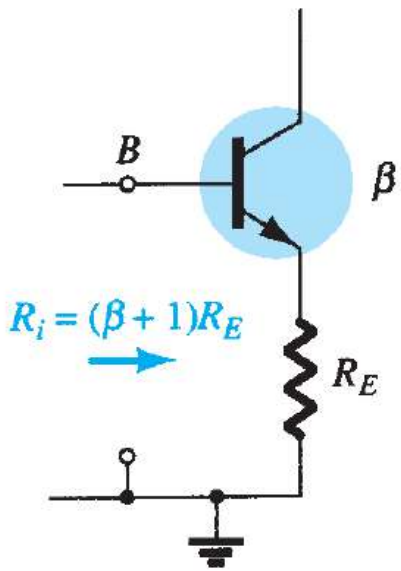


FIG. 4.21
Reflected impedance level of R_E.

$$R_i = (\beta + 1)R_E \tag{4.18}$$

Collector–Emitter Loop

The collector–emitter loop appears in Fig. 4.22. Writing Kirchhoff's voltage law for the indicated loop in the clockwise direction results in

$$+I_E R_E + V_{CE} + I_C R_C - V_{CC} = 0$$

Substituting $I_E \cong I_C$ and grouping terms gives

$$V_{CE} - V_{CC} + I_C(R_C + R_E) = 0$$

and

$$\boxed{V_{CE} = V_{CC} - I_C(R_C + R_E)}$$

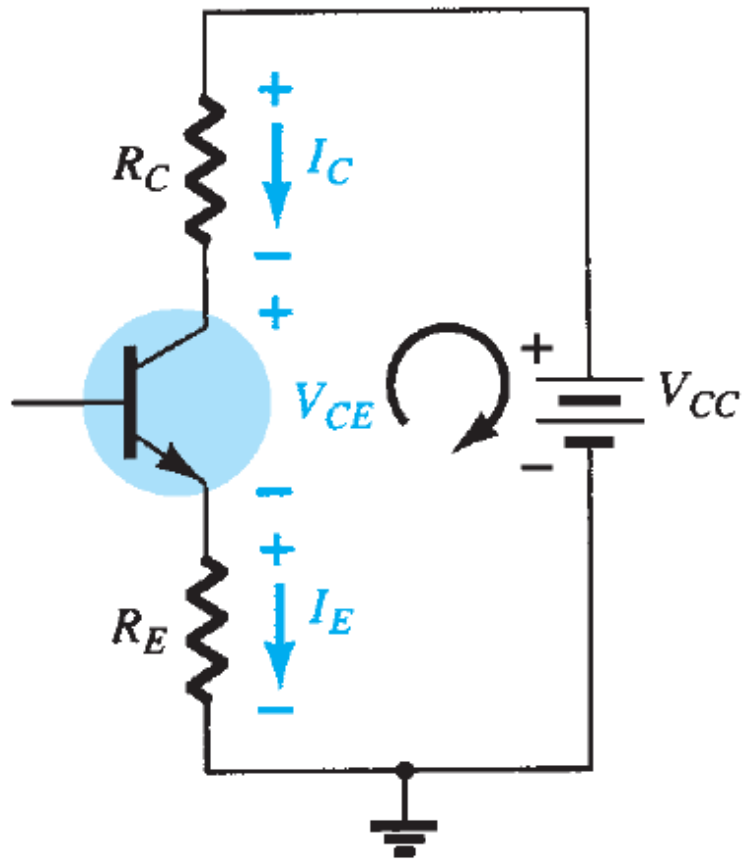


FIG. 4.22
Collector–emitter loop.

The single-subscript voltage V_E is the voltage from emitter to ground and is determined by

$$V_E = I_E R_E \tag{4.20}$$

whereas the voltage from collector to ground can be determined from

$$V_{CE} = V_C - V_E$$

and

$$V_C = V_{CE} + V_E \tag{4.21}$$

or

$$V_C = V_{CC} - I_C R_C \tag{4.22}$$

The voltage at the base with respect to ground can be determined using Fig. 4.18

$$V_B = V_{CC} - I_B R_B \tag{4.23}$$

or

$$V_B = V_{BE} + V_E \tag{4.24}$$

EXAMPLE 4.4 For the emitter-bias network of Fig. 4.23, determine:

a. I_B.
b. I_C.
c. V_{CE}.
d. V_C.
e. V_E.
f. V_B.
g. V_{BC}.

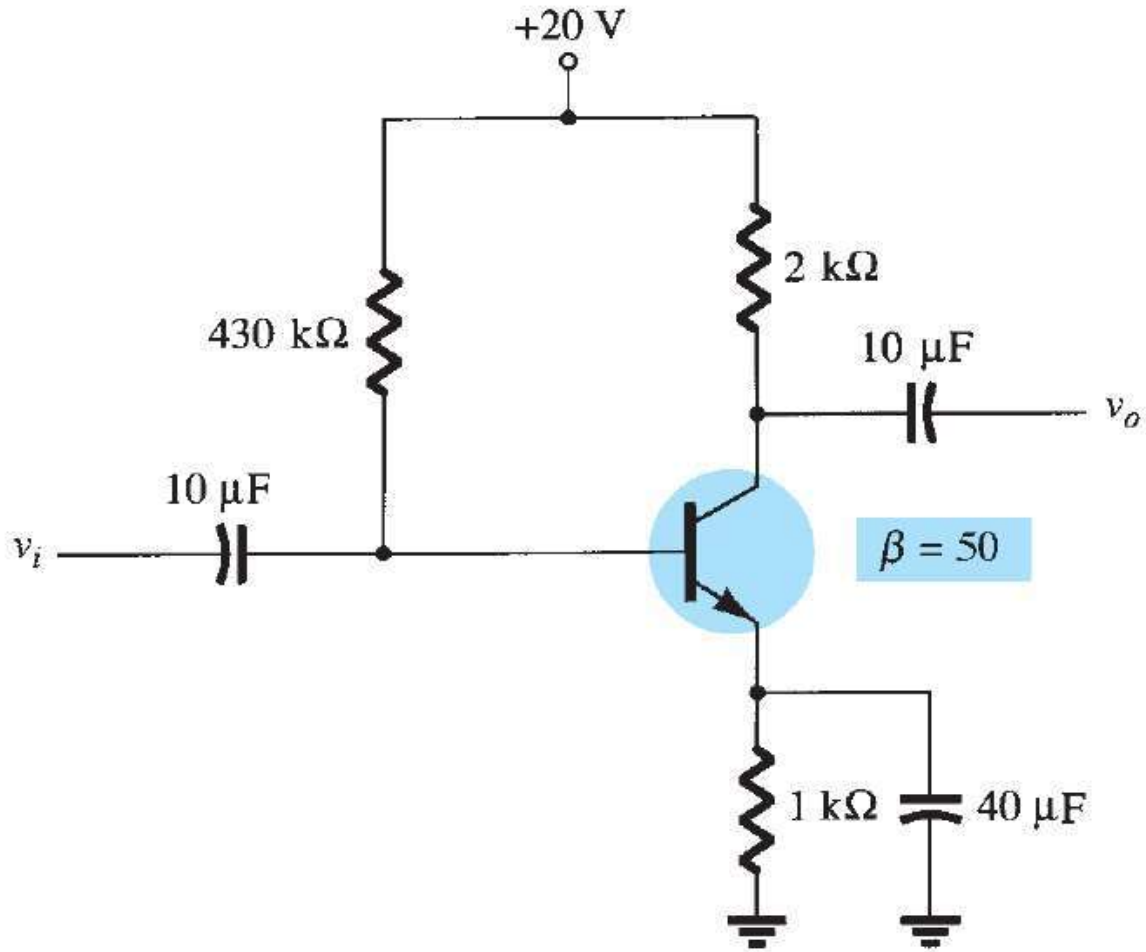


FIG. 4.23
Emitter-stabilized bias circuit for Example 4.4.

Solution:

a. Eq. (4.17): $$I_B = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1)R_E} = \frac{20\text{ V} - 0.7\text{ V}}{430\text{ k}\Omega + (51)(1\text{ k}\Omega)}$$

$$= \frac{19.3\text{ V}}{481\text{ k}\Omega} = \mathbf{40.1\ \mu A}$$

b. $I_C = \beta I_B$

$= (50)(40.1\ \mu\text{A})$

$\cong$ **2.01 mA**

c. Eq. (4.19): $V_{CE} = V_{CC} - I_C(R_C + R_E)$

$= 20\text{ V} - (2.01\text{ mA})(2\text{ k}\Omega + 1\text{ k}\Omega) = 20\text{ V} - 6.03\text{ V}$

$=$ **13.97 V**

d. $V_C = V_{CC} - I_C R_C$

$= 20\text{ V} - (2.01\text{ mA})(2\text{ k}\Omega) = 20\text{ V} - 4.02\text{ V}$

$=$ **15.98 V**

e. $V_E = V_C - V_{CE}$

$= 15.98\text{ V} - 13.97\text{ V}$

$=$ **2.01 V**

or $V_E = I_E R_E \cong I_C R_E$

$= (2.01\text{ mA})(1\text{ k}\Omega)$

$=$ **2.01 V**

f. $V_B = V_{BE} + V_E$

$= 0.7\text{ V} + 2.01\text{ V}$

$=$ **2.71 V**

g. $V_{BC} = V_B - V_C$

Improved Bias Stability

The addition of the emitter resistor to the dc bias of the BJT provides improved stability, that is, the dc bias currents and voltages remain closer to where they were set by the circuit when outside conditions, such as temperature and transistor beta, change. Although a mathematical analysis is provided in Section 4.12, some comparison of the improvement can be obtained as demonstrated by Example 4.5.

EXAMPLE 4.5 Prepare a table and compare the bias voltage and currents of the circuits of Fig. 4.7 and Fig. 4.23 for the given value of $\beta = 50$ and for a new value of $\beta = 100$. Compare the changes in I_C and V_{CE} for the same increase in β.

Solution: Using the results calculated in Example 4.1 and then repeating for a value of $\beta = 100$ yields the following:

Effect of β variation on the response of the fixed-bias configuration of Fig. 4.7.

β	I_B (μA)	I_C (mA)	V_{CE} (V)
50	47.08	2.35	6.83
100	47.08	4.71	1.64

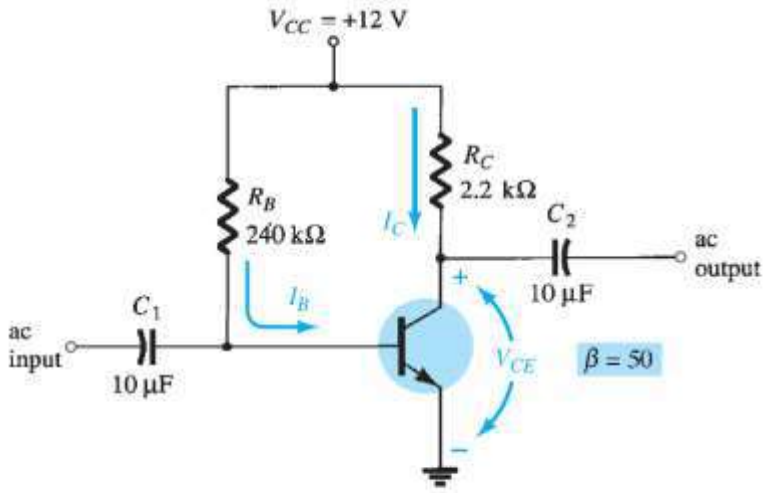


FIG. 4.7

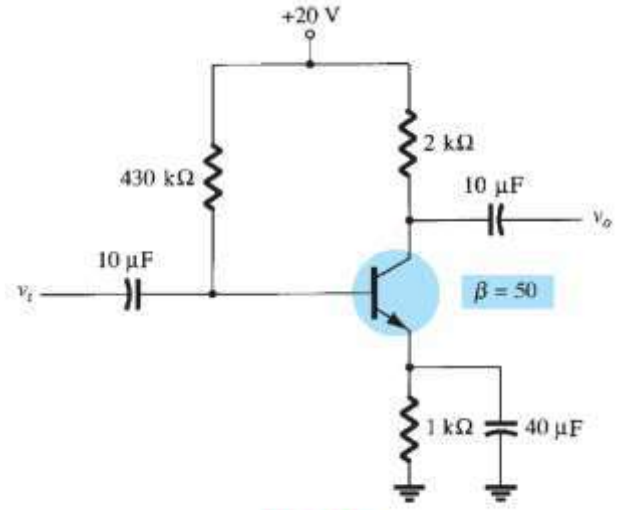


FIG. 4.23

The BJT collector current is seen to change by 100% due to the 100% change in the value of β. The value of I_B is the same, and V_{CE} decreased by 76%.

Using the results calculated in Example 4.4 and then repeating for a value of $\beta = 100$, we have the following:

Effect of β variation on the response of the emitter-bias configuration of Fig. 4.23.

β	I_B (μA)	I_C (mA)	V_{CE} (V)
50	40.1	2.01	13.97
100	36.3	3.63	9.11

Now the BJT collector current increases by about 81% due to the 100% increase in β. Notice that I_B decreased, helping maintain the value of I_C—or at least reducing the overall change in I_C due to the change in β. The change in V_{CE} has dropped to about 35%. The network of Fig. 4.23 is therefore more stable than that of Fig. 4.7 for the same change in β.

Saturation Level

The collector saturation level or maximum collector current for an emitter-bias design can be determined using the same approach applied to the fixed-bias configuration: Apply a short circuit between the collector–emitter terminals as shown in Fig. 4.24 and calculate the resulting collector current. For Fig. 4.24

$$I_{C_{sat}} = \frac{V_{CC}}{R_C + R_E} \tag{4.25}$$

The addition of the emitter resistor reduces the collector saturation level below tha obtained with a fixed-bias configuration using the same collector resistor.

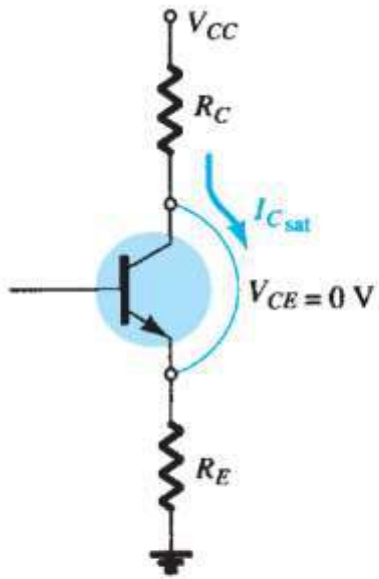


FIG. 4.24
Determining $I_{C_{sat}}$ for the emitter-stabilized bias circuit.

EXAMPLE 4.6 Determine the saturation current for the network of Example 4.4.

Solution:

$$I_{C_{sat}} = \frac{V_{CC}}{R_C + R_E}$$

$$= \frac{20\text{ V}}{2\text{ k}\Omega + 1\text{ k}\Omega} = \frac{20\text{ V}}{3\text{ k}\Omega}$$

$$= \mathbf{6.67\text{ mA}}$$

which is about three times the level of $I_{C_{sat}}$ for Example 4.4.

Load-Line Analysis

The load-line analysis of the emitter-bias network is only slightly different from that encountered for the fixed-bias configuration. The level of I_B as determined by Eq. (4.17) defines the level of I_B on the characteristics of Fig. 4.25 (denoted I_{B_Q}).

The collector–emitter loop equation that defines the load line is

$$V_{CE} = V_{CC} - I_C(R_C + R_E)$$

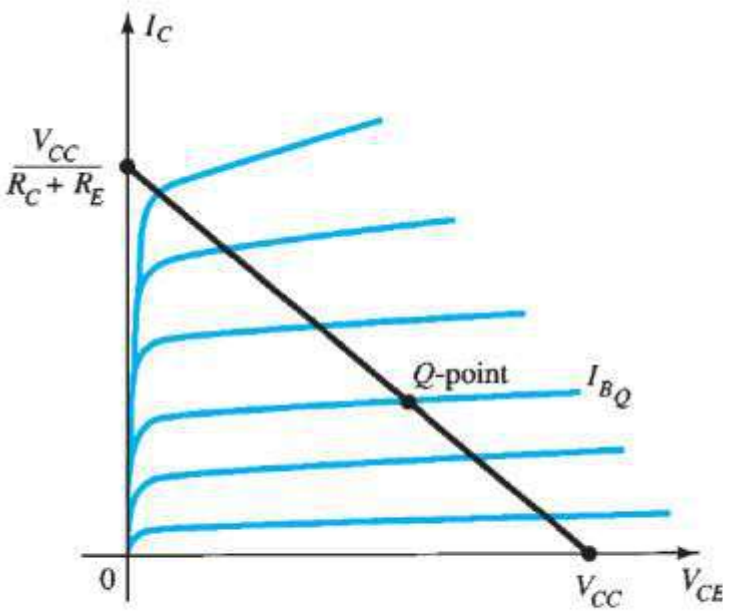


FIG. 4.25
Load line for the emitter-bias configuration.

Choosing $I_C = 0$ mA gives

$$\boxed{V_{CE} = V_{CC}\big|_{I_C=0\text{ mA}}} \tag{4.26}$$

as obtained for the fixed-bias configuration. Choosing $V_{CE} = 0$ V gives

$$\boxed{I_C = \frac{V_{CC}}{R_C + R_E}\bigg|_{V_{CE}=0\text{ V}}} \tag{4.27}$$

as shown in Fig. 4.25. Different levels of I_{B_Q} will, of course, move the Q-point up or down the load line.

EXAMPLE 4.7

a. Draw the load line for the network of Fig. 4.26a on the characteristics for the transistor appearing in Fig. 4.26b.
b. For a Q-point at the intersection of the load line with a base current of 15 μA, find the values of I_{C_Q} and V_{CE_Q}.
c. Determine the dc beta at the Q-point.
d. Using the beta for the network determined in part c, calculate the required value of R_B and suggest a possible standard value.

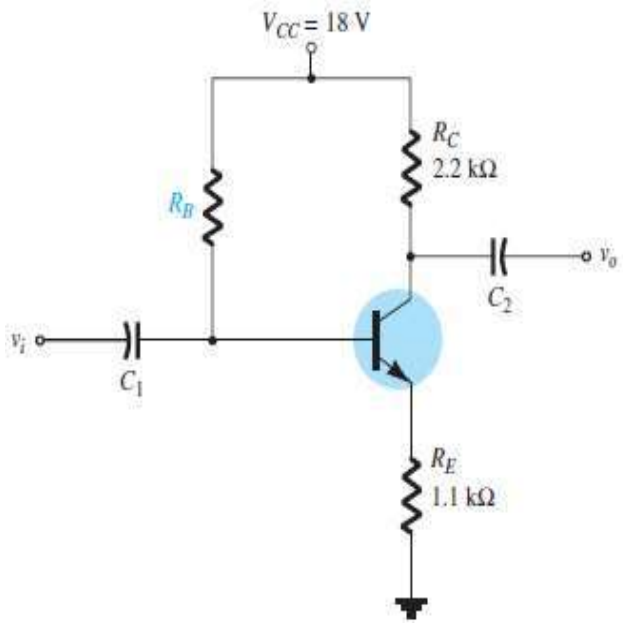


FIG. 4.26a

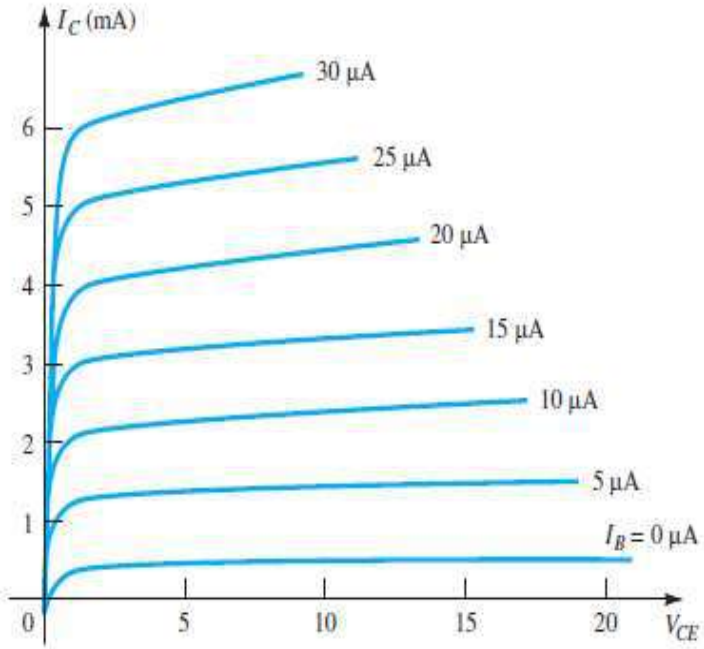


FIG. 4.26b

Solution:

a. Two points on the characteristics are required to draw the load line.

$$\text{At } V_{CE} = 0\text{ V}: \quad I_C = \frac{V_{CC}}{R_C + R_E} = \frac{18\text{ V}}{2.2\text{ k}\Omega + 1.1\text{ k}\Omega} = \frac{18\text{ V}}{3.3\text{ k}\Omega} = 5.45\text{ mA}$$

$$\text{At } I_C = 0\text{ mA}: V_{CE} = V_{CC} = 18\text{ V}$$

The resulting load line appears in Fig. 4.27.

b. From the characteristics of Fig. 4.27 we find

$$V_{CE_Q} \cong \mathbf{7.5\text{ V}}, I_{C_Q} \cong \mathbf{3.3\text{ mA}}$$

c. The resulting dc beta is:

$$\beta = \frac{I_{C_Q}}{I_{B_Q}} = \frac{3.3\text{ mA}}{15\ \mu\text{A}} = \mathbf{220}$$

d. Applying Eq. 4.17:

$$I_B = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1)R_E} = \frac{18\text{ V} - 0.7\text{ V}}{R_B + (220 + 1)(1.1\text{ k}\Omega)}$$

$$\text{and } 15\ \mu\text{A} = \frac{17.3\text{ V}}{R_B + (221)(1.1\text{ k}\Omega)} = \frac{17.3\text{ V}}{R_B + 243.1\text{ k}\Omega}$$

so that $(15\ \mu\text{A})(R_B) + (15\ \mu\text{A})(243.1\text{ k}\Omega) = 17.3\text{ V}$

and $(15\ \mu\text{A})(R_B) = 17.3\text{ V} - 3.65\text{ V} = 13.65\text{ V}$

$$\text{resulting in } R_B = \frac{13.65\text{ V}}{15\ \mu\text{A}} = \mathbf{910\text{ k}\Omega}$$

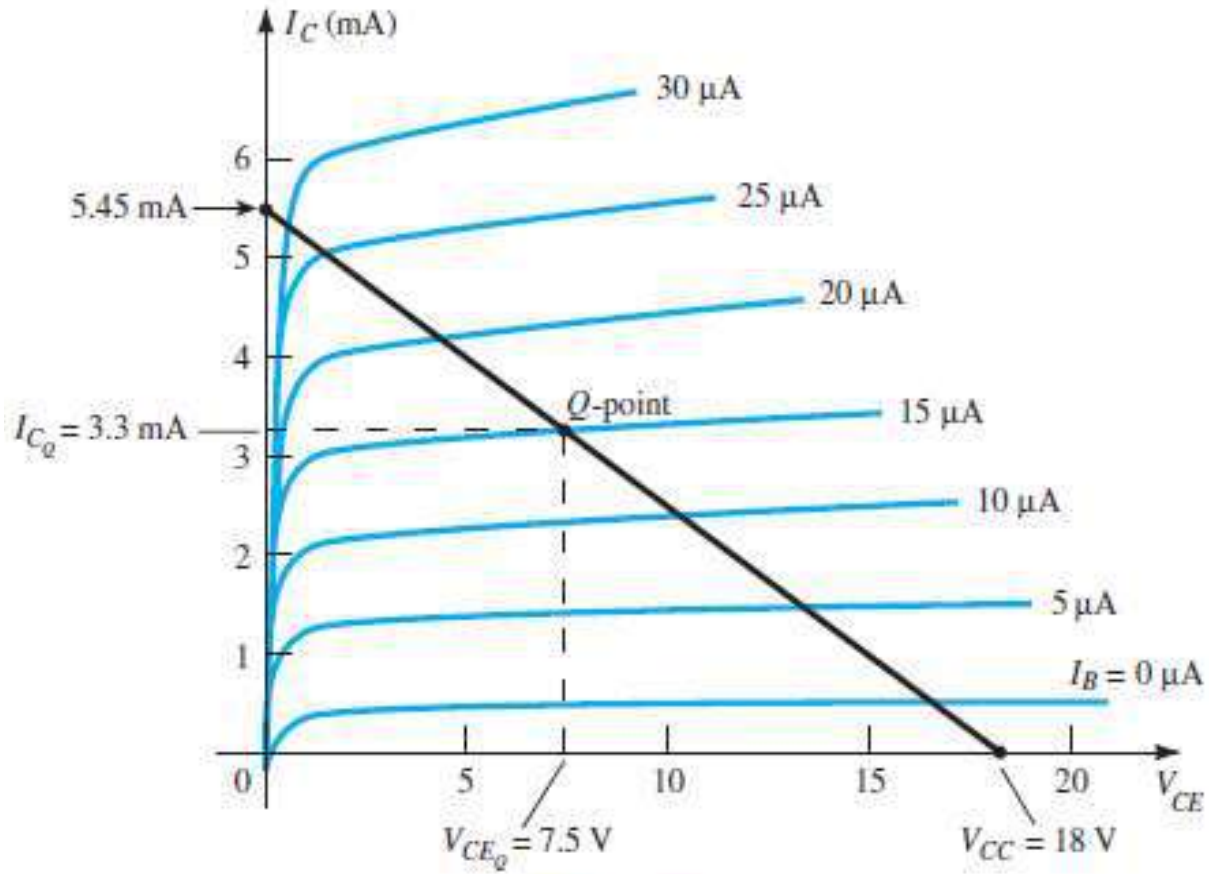


FIG. 4.27

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Chapter 4: DC Biasing—BJTs

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CHAPTER OBJECTIVES ●

- Be able to determine the dc levels for the variety of important BJT configurations.
- Understand how to measure the important voltage levels of a BJT transistor configuration and use them to determine whether the network is operating properly.
- Become aware of the saturation and cutoff conditions of a BJT network and the expected voltage and current levels established by each condition.
- Be able to perform a load-line analysis of the most common BJT configurations.
- Become acquainted with the design process for BJT amplifiers.
- Understand the basic operation of transistor switching networks.
- Begin to understand the troubleshooting process as applied to BJT configurations.
- Develop a sense for the stability factors of a BJT configuration and how they affect its operation due to changes in specific characteristics and environmental changes.

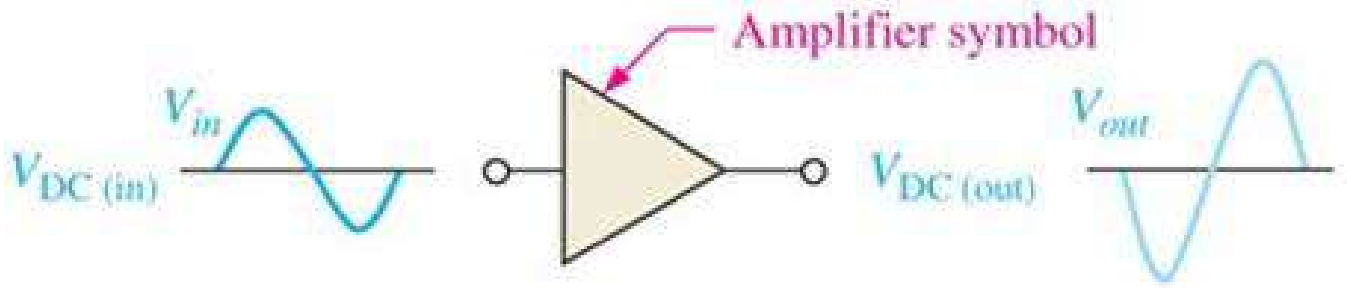
The analysis or design of a transistor amplifier requires a knowledge of both the dc and the ac response of the system. Too often it is assumed that the transistor is a magical device that can raise the level of the applied ac input without the assistance of an external energy source. In actuality, ***any increase in ac voltage, current, or power is the result of a transfer of energy from the applied dc supplies.***

The analysis or design of any electronic amplifier therefore has two components: a dc and an ac portion. Fortunately, the superposition theorem is applicable, and the investigation of the dc conditions can be totally separated from the ac response. However, one must keep in mind that during the design or synthesis stage the choice of parameters for the required dc levels will affect the ac response, and vice versa.

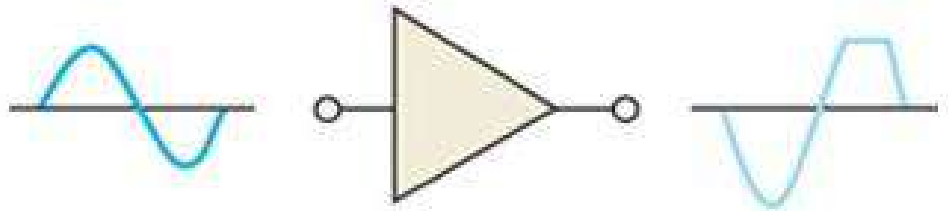
The dc level of operation of a transistor is controlled by a number of factors, including the range of possible operating points on the device characteristics
A transistor must be properly biased with a dc voltage in order to operate as a linear amplifier.

A dc operating point must be set so that signal variations at the input terminal are amplified and accurately reproduced at the output terminal.

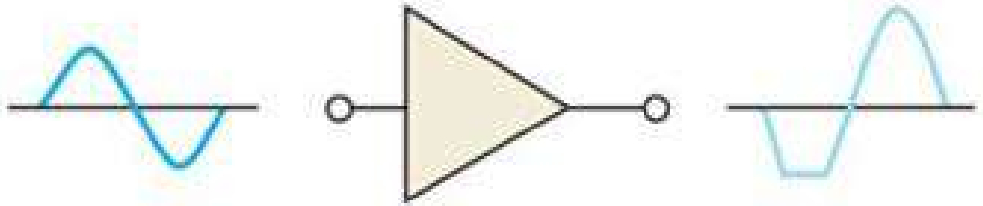
- If an amplifier is not biased with the correct DC voltages, it can go into saturation and cutoff.
- Figure 1(a) shows the correct Linear operation with amplified output.
- Figure 1(b) shows nonlinear operation where the amplifier is in cutoff. The clipping in the positive cycle is always due to cutoff.
- Figure 1(c) shows nonlinear operation where the amplifier is in saturation. The clipping in the negative cycle is always due to saturation.



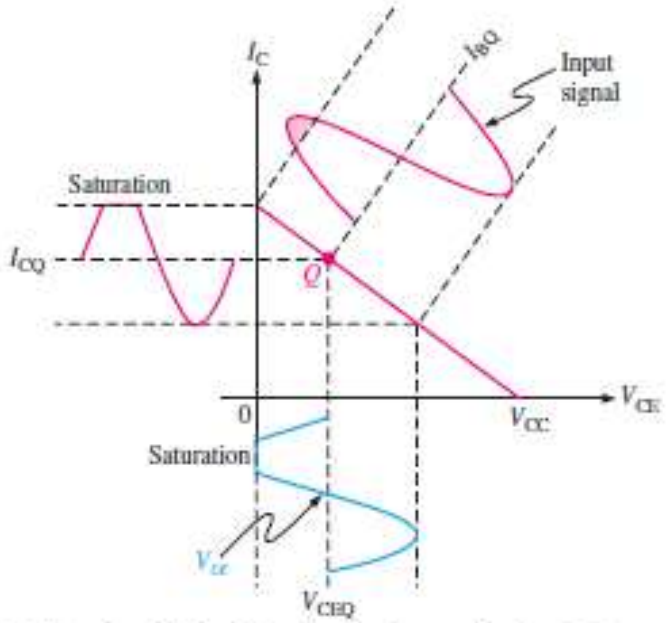
(a) Linear operation: larger output has same shape as input except that it is inverted



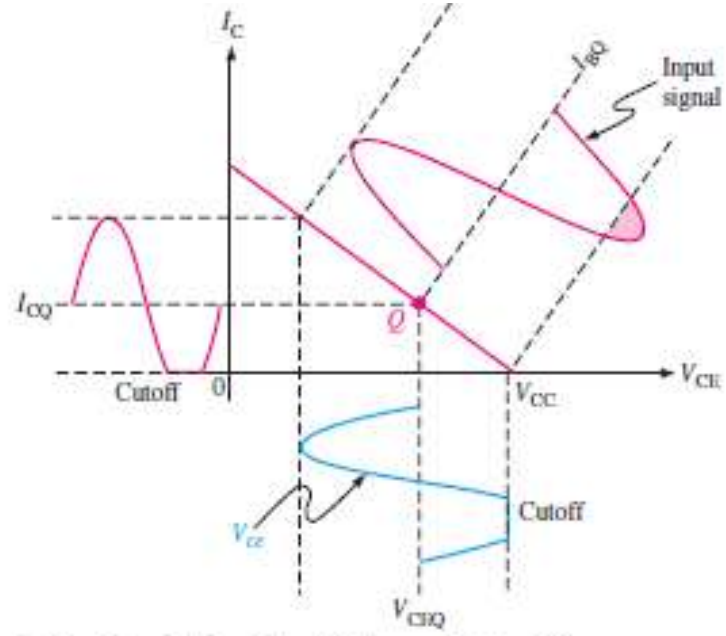
(b) Nonlinear operation: output voltage limited (clipped) by cutoff



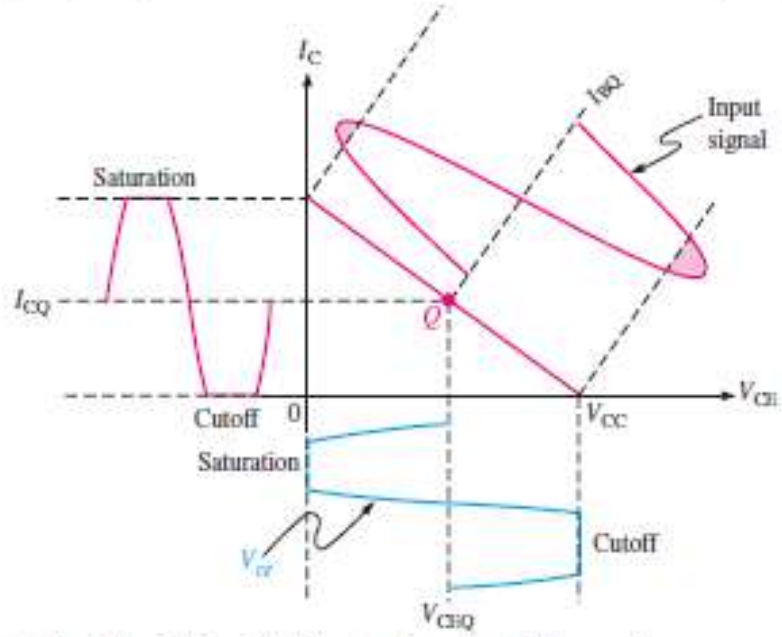
(c) Nonlinear operation: output voltage limited (clipped) by saturation



(a) Transistor is driven into saturation because the Q-point is too close to saturation for the given input signal.



(b) Transistor is driven into cutoff because the Q-point is too close to cutoff for the given input signal.



(c) Transistor is driven into both saturation and cutoff because the input signal is too large.

Although a number of networks are analyzed in this chapter, there is an underlying similarity in the analysis of each configuration due to the recurring use of the following important basic relationships for a transistor:

$$V_{BE} \cong 0.7\ \text{V} \tag{4.1}$$

$$I_E = (\beta + 1)I_B \cong I_C \tag{4.2}$$

$$I_C = \beta I_B \tag{4.3}$$

The term *biasing* appearing in the title of this chapter is an all-inclusive term for the application of dc voltages to establish a fixed level of current and voltage. For transistor amplifiers the resulting dc current and voltage establish an *operating point* on the characteristics that define the region that will be employed for amplification of the applied signal. Because the operating point is a fixed point on the characteristics, it is also called the *quiescent point* (abbreviated *Q* -point). By definition, *quiescent* means quiet, still, inactive. Figure.1 shows a general output device characteristic with four operating points indicated.

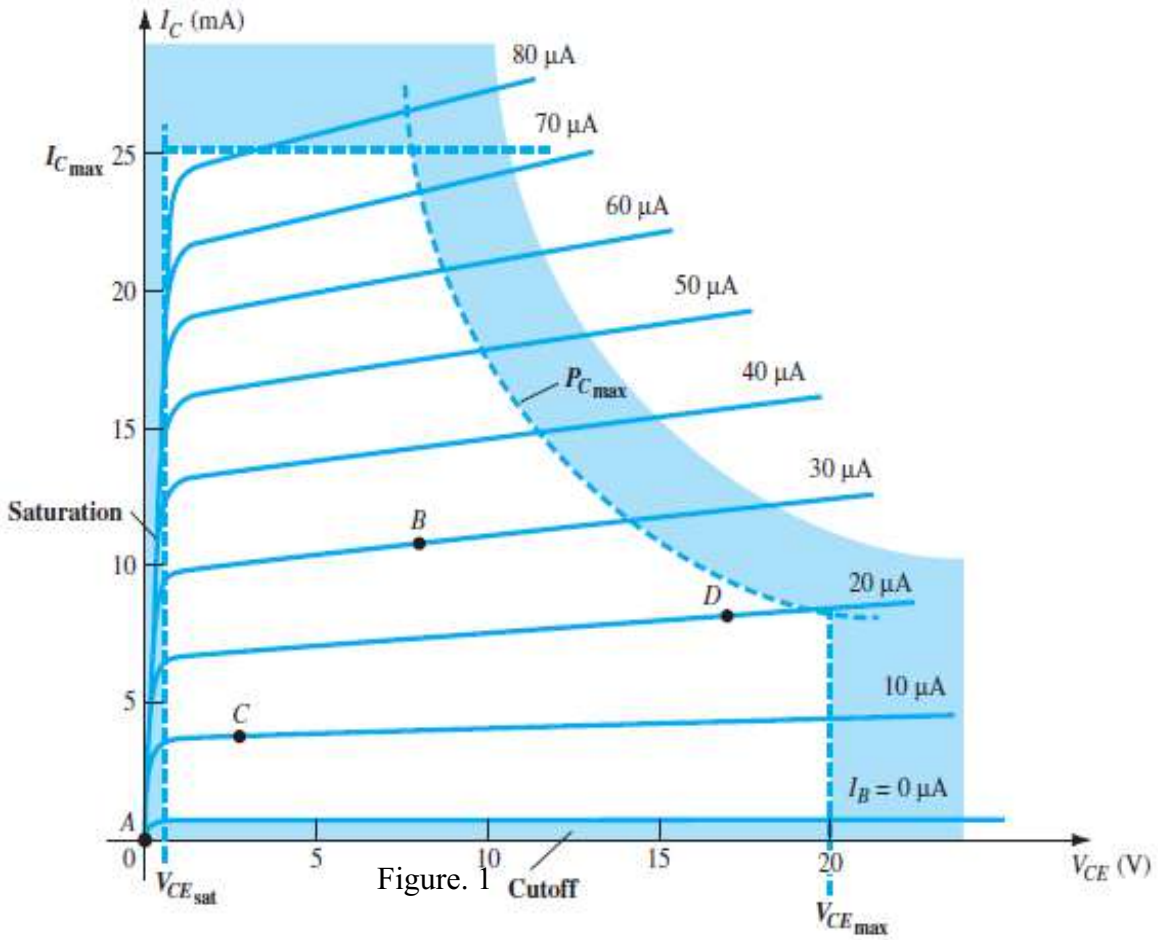
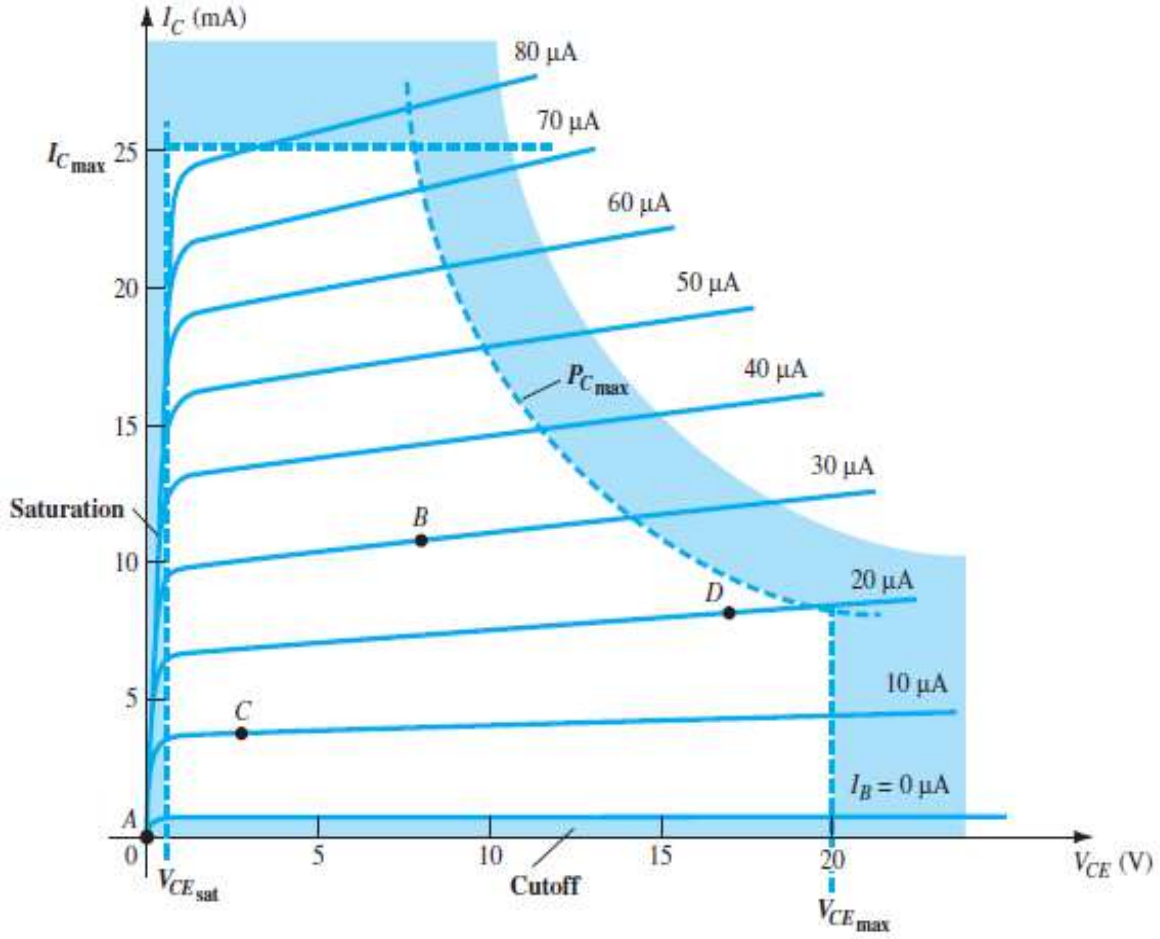


Figure. 1 **Various operating points within the limits of operation of a transistor**

The biasing circuit can be designed to set the device operation at any of these points or others within the *active region.* The maximum power constraint is defined by the curve $P_{C\max}$ in the same figure. At the lower end of the scales are the *cutoff region* , defined by $I_B \leq 0$ mA, and the *saturation region,* defined by $V_{CE} \leq V_{CE\ \text{sat}}$

Confining ourselves to the *active* region, we can select many different operating areas or points. The chosen Q -point often depends on the intended use of the circuit.



The chosen Q -point often depends on the intended use of the circuit.
For the BJT to be biased in its linear or active operating region the following must be true:

1. ***The base–emitter junction must be forward-biased (p-region voltage more positive), with a resulting forward-bias voltage of about 0.6 V to 0.7 V.***

2. ***The base–collector junction must be reverse-biased (n-region more positive), with the reverse-bias voltage being any value within the maximum limits of the device.***

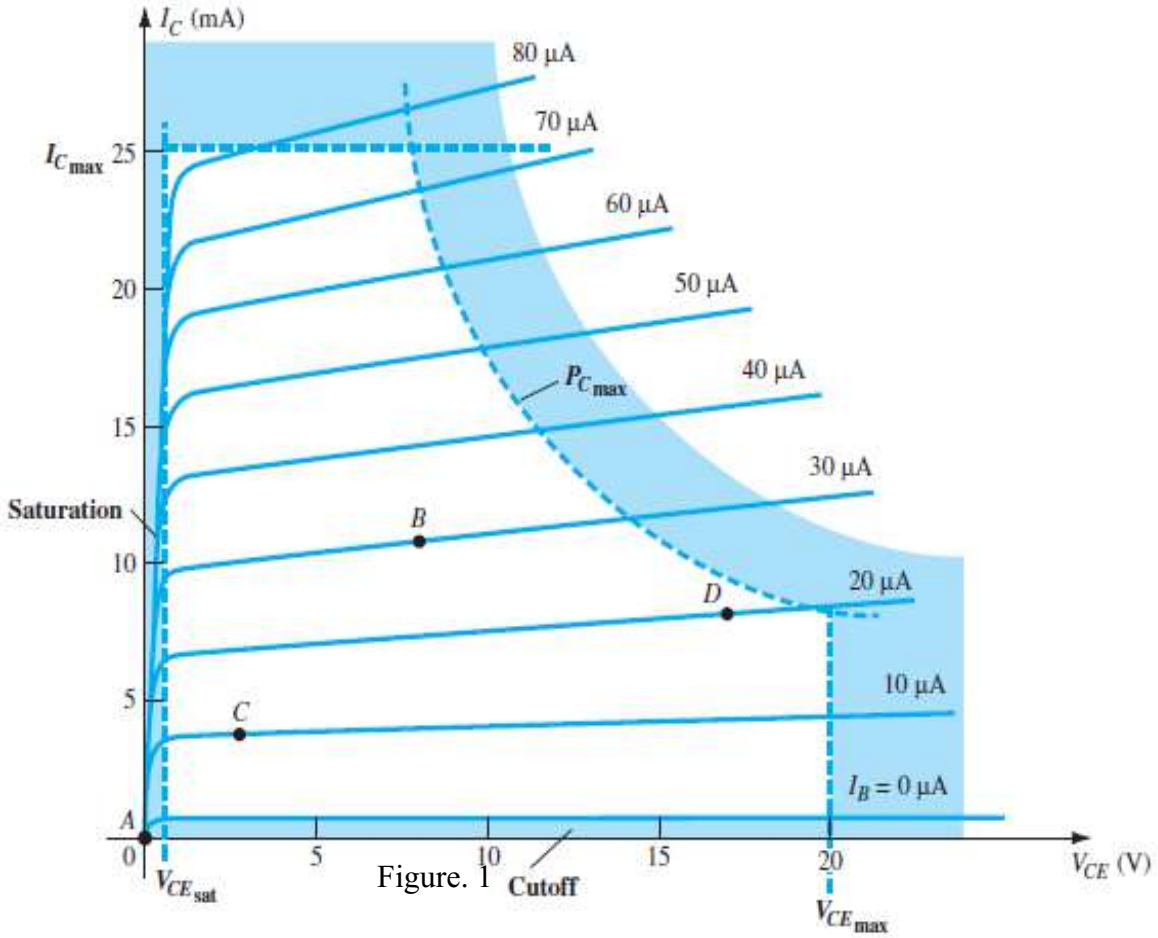


Figure. 1

Various operating points within the limits of operation of a transistor

Operation in the cutoff, saturation, and linear regions of the BJT characteristic are provided
as follows:

1. *Linear-region operation:*

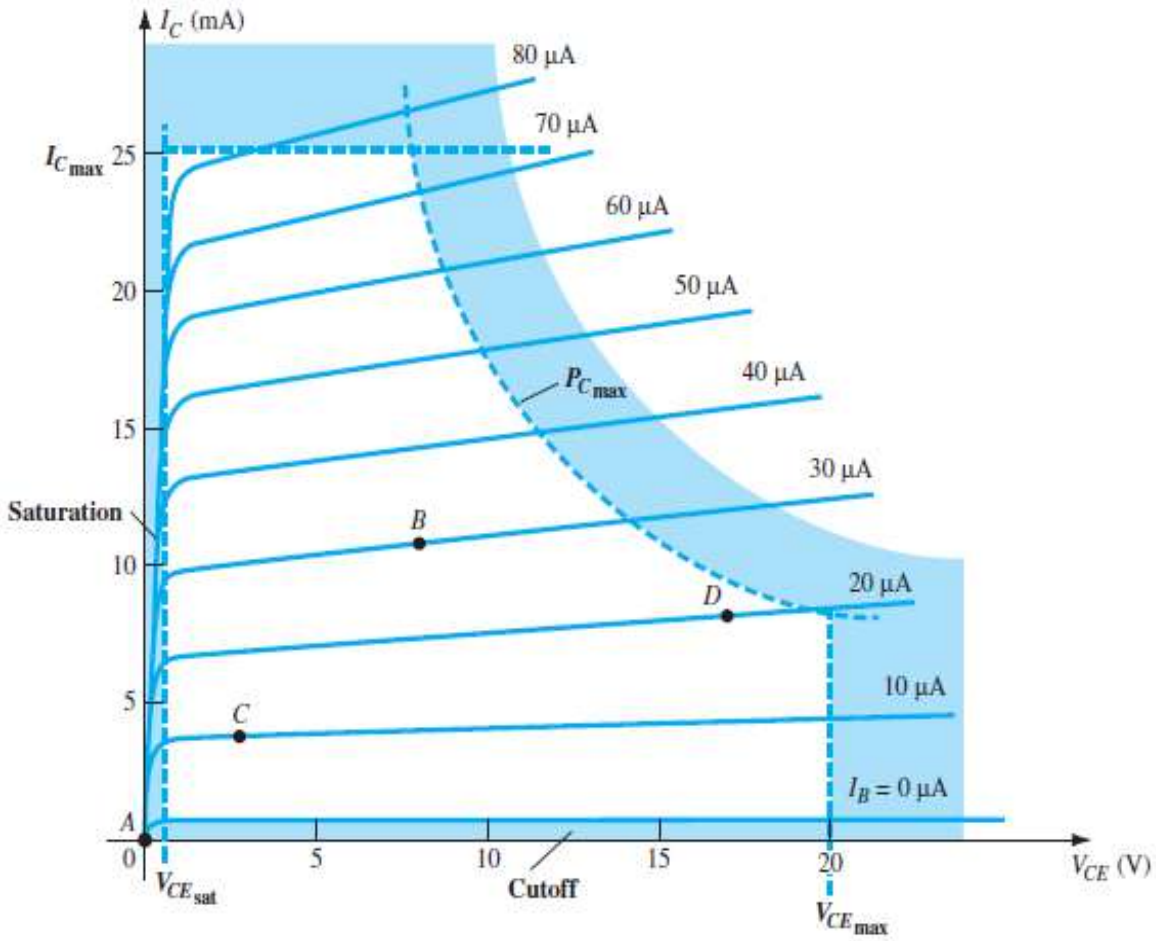
Base–emitter junction forward-biased
Base–collector junction reverse-biased

2. *Cutoff-region operation:*

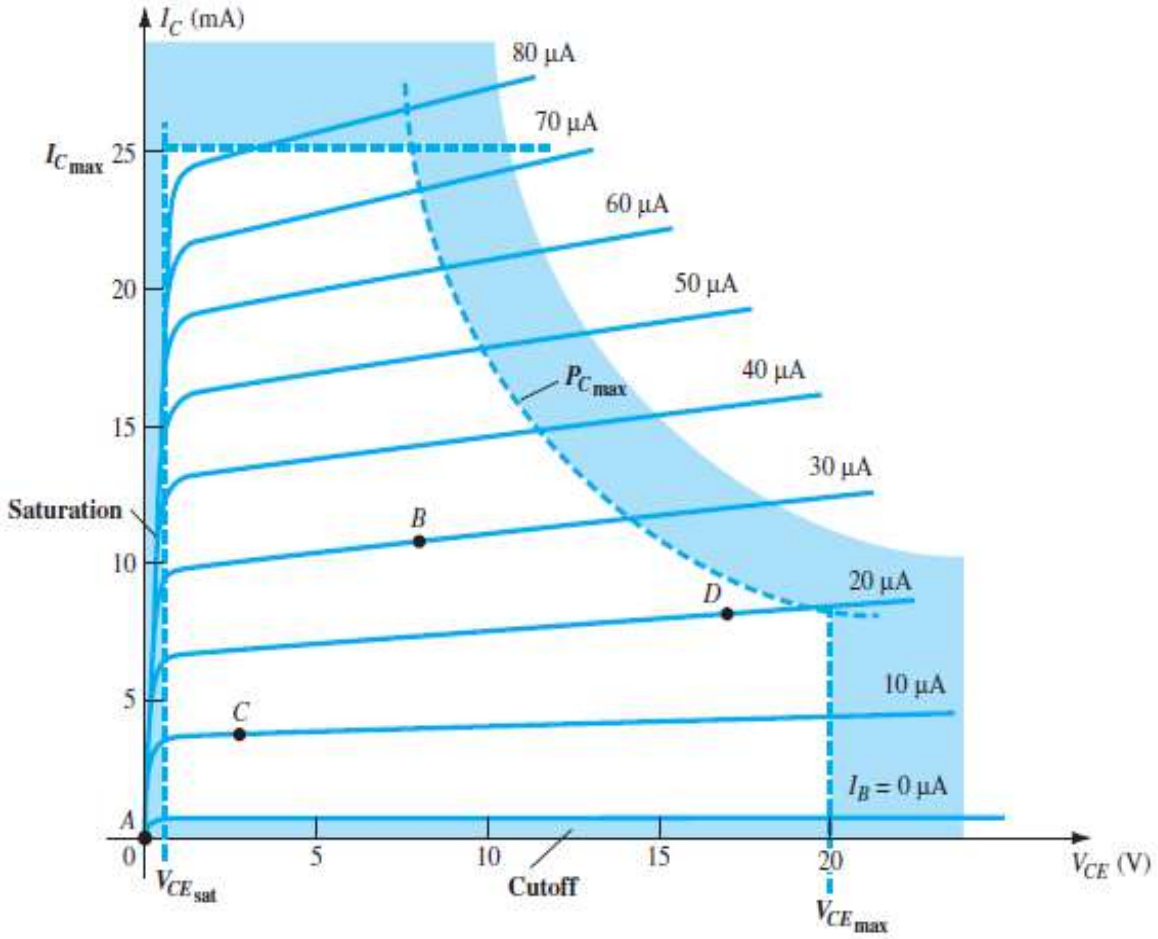
Base–emitter junction reverse-biased
Base–collector junction reverse-biased

3. *Saturation-region operation:*

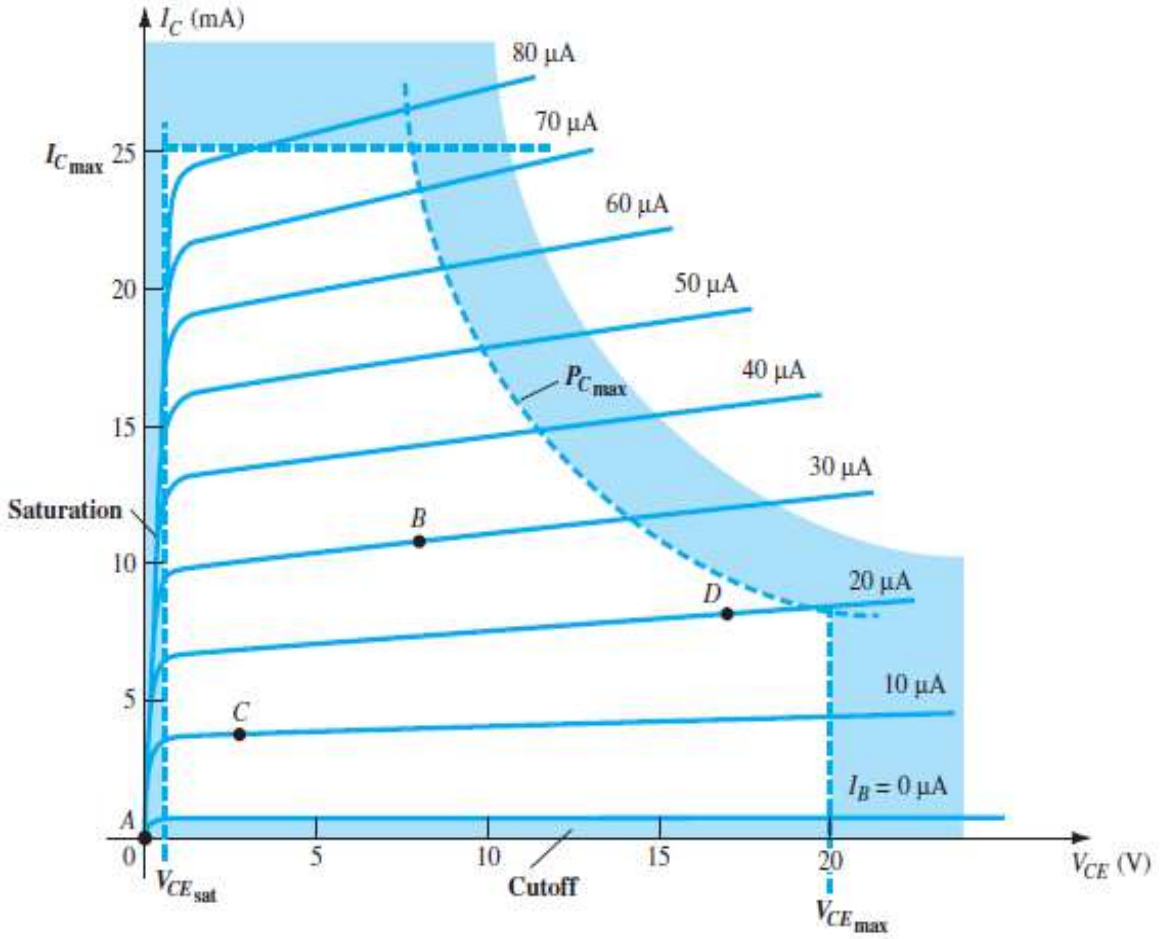
Base–emitter junction forward-biased
Base–collector junction forward-biased



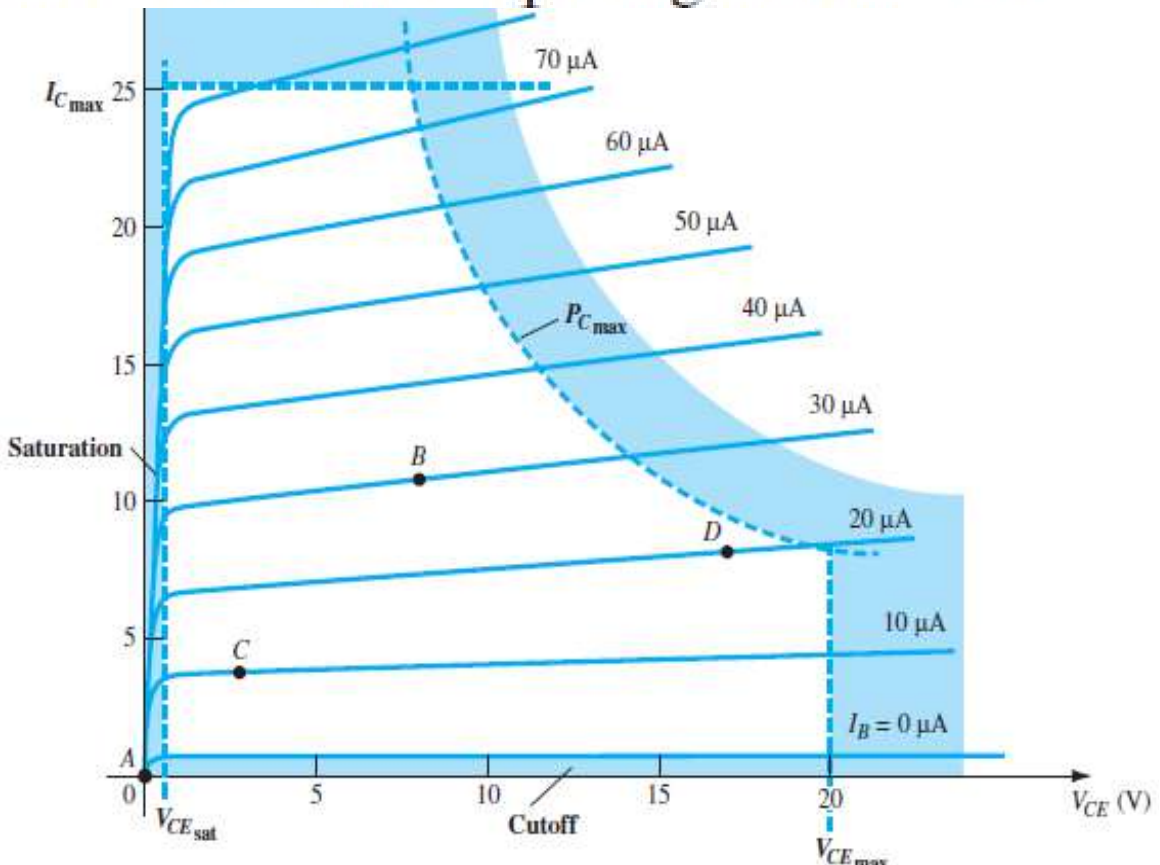
If no bias were used, the device would initially be completely off, resulting in a *Q*-point at *A*—namely, zero current through the device (and zero voltage across it). Because it is necessary to bias a device so that it can respond to the entire range of an input signal, point *A* would not be suitable.



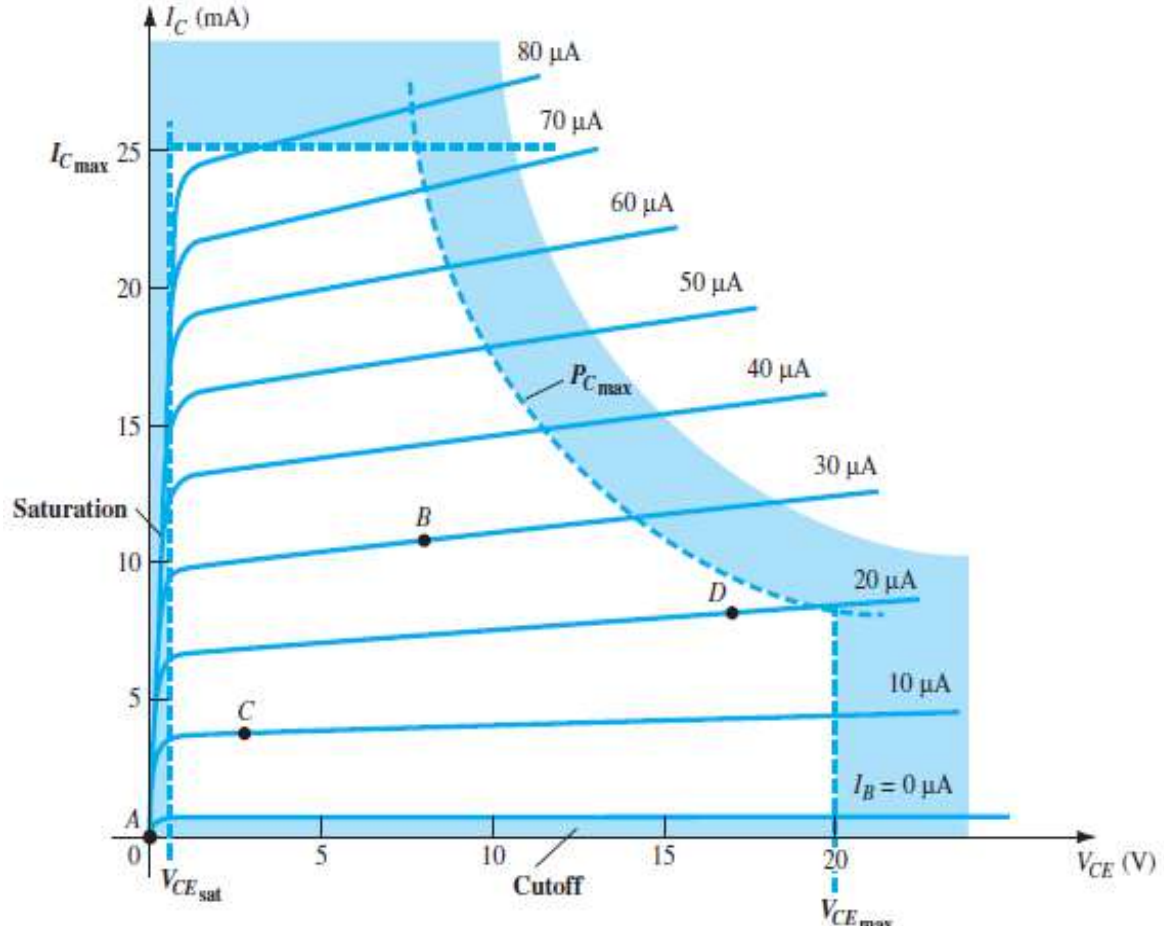
For point *B*, if a signal is applied to the circuit, the device will vary in current and voltage from the operating point, allowing the device to react to (and possibly amplify) both the positive and negative excursions of the input signal. If the input signal is properly chosen, the voltage and current of the device will vary, but not enough to drive the device into *cutoff* or *saturation*.



Point C would allow some positive and negative variation of the output signal, but the peak-to-peak value would be limited by the proximity of $V_{CE} = 0$ V and $I_C = 0$ mA. Operating at point C also raises some concern about the nonlinearities introduced by the fact that the spacing between I_B curves is rapidly changing in this region. In general, it is preferable to operate where the gain of the device is fairly constant (or linear) to ensure that the amplification over the entire swing of input signal is the same. Point B is a region of more linear spacing and therefore more linear operation, as shown in Fig. 4.1.



Point *D* sets the device operating point near the maximum voltage and power level. The output voltage swing in the positive direction is thus limited if the maximum voltage is not to be exceeded. Point *B* therefore seems the best operating point in terms of linear gain and largest possible voltage and current swing.



The biasing circuit can be designed to set the device operation at any of these points or others within the *active region.* The maximum power constraint is defined by the curve $P_{C\max}$ in the same figure. At the lower end of the scales are the *cutoff region* , defined by I_B … 0 mA, and the *saturation region,* defined by V_{CE} … $V_{CE\ \text{sat}}$
Confining ourselves to the *active* region, we can select many different operating areas or points. The chosen Q -point often depends on the intended use of the circuit.
For the BJT to be biased in its linear or active operating region the following must be true:

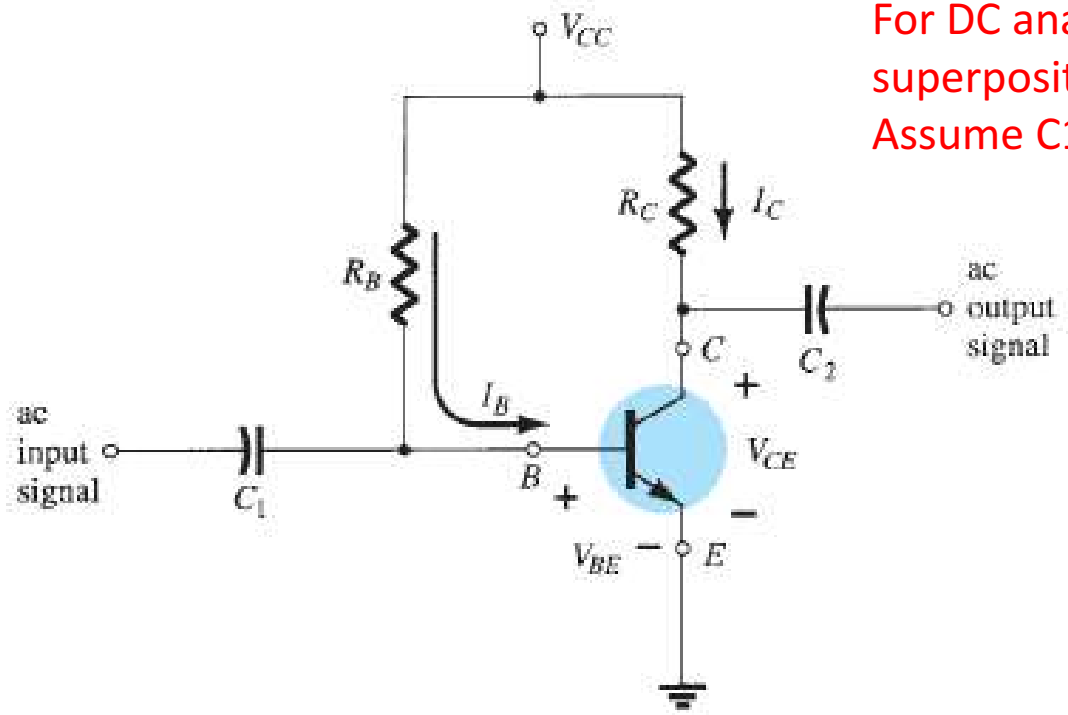
1. ***The base–emitter junction must be forward-biased (p-region voltage more positive), with a resulting forward-bias voltage of about 0.6 V to 0.7 V.***

2. The base–collector junction must be reverse-biased (n-region more positive), with the reverse-bias voltage being any value within the maximum limits of the device.

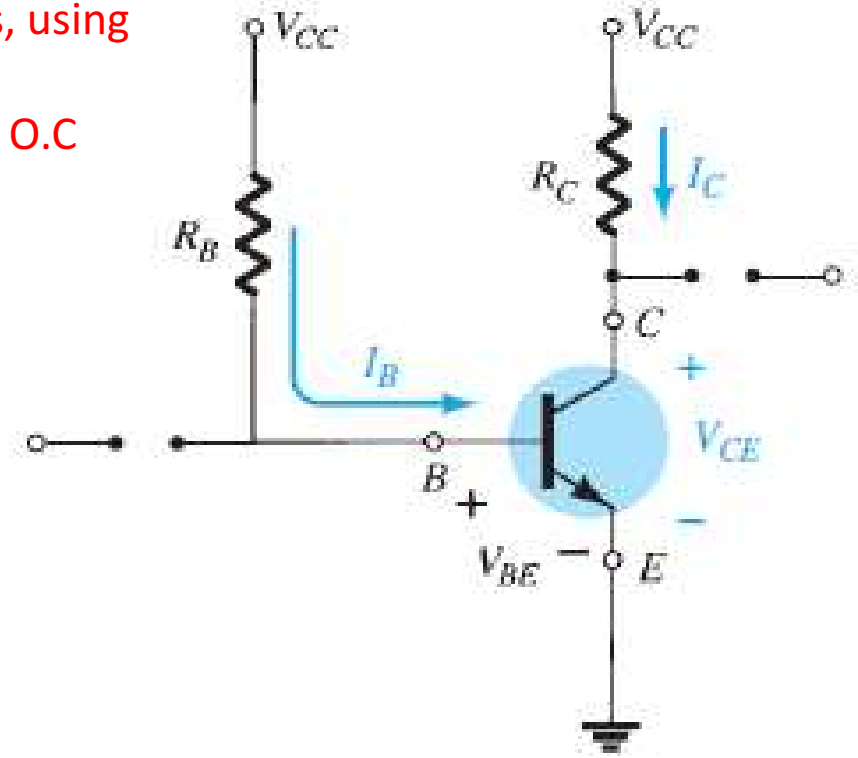
FIXED-BIAS CONFIGURATION

The network employs a *npn* transistor, the equations and calculations apply equally well to a *pnp* transistor configuration merely by changing all current directions and voltage polarities.

For DC analysis, using superposition
Assume C1, C2 O.C

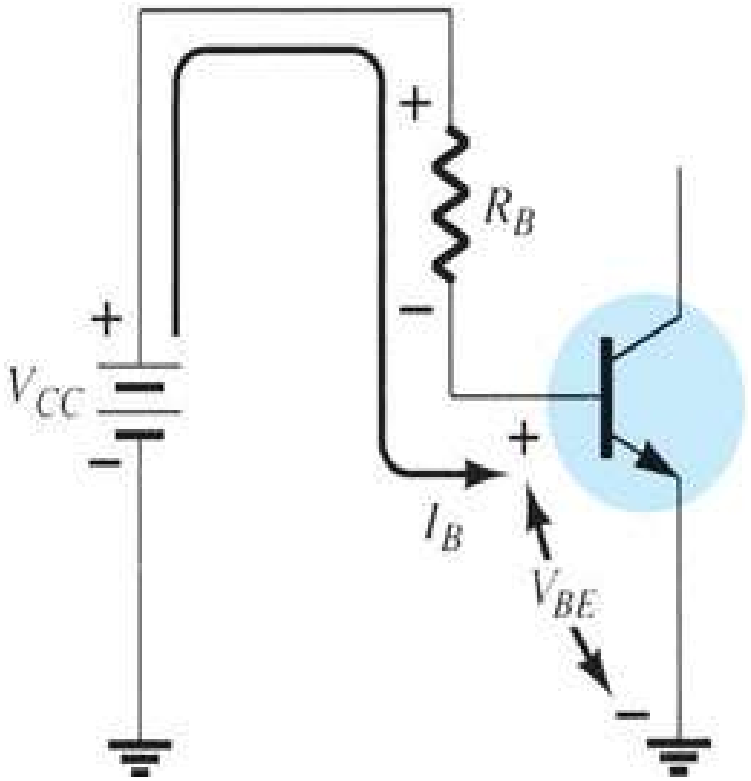


Fixed-bias circuit.



DC equivalent

Forward Bias of Base–Emitter



Writing Kirchhoff's voltage equation , in the clockwise direction for the loop, we obtain

$$+V_{CC} - I_B R_B - V_{BE} = 0$$

$$I_B = \frac{V_{CC} - V_{BE}}{R_B}$$

Collector–Emitter Loop

The collector–emitter section of the network appears in Fig. 4.5 with the indicated direction of current I_C and the resulting polarity across R_C. The magnitude of the collector current is related directly to I_B through

$$I_C = \beta I_B \tag{4.5}$$

Applying Kirchhoff's voltage law in the clockwise direction around the indicated closed loop of Fig. 4.5 results in the following:

$$V_{CE} + I_C R_C - V_{CC} = 0$$

and

$$V_{CE} = V_{CC} - I_C R_C \tag{4.6}$$

which states that the voltage across the collector–emitter region of a transistor in the fixed-bias configuration is the supply voltage less the drop across R_C.

As a brief review of single- and double-subscript notation recall that

$$V_{CE} = V_C - V_E \tag{4.7}$$

where V_{CE} is the voltage from collector to emitter and V_C and V_E are the voltages from collector and emitter to ground, respectively. *In this case,* since $V_E = 0$ V, we have

$$V_{CE} = V_C \tag{4.8}$$

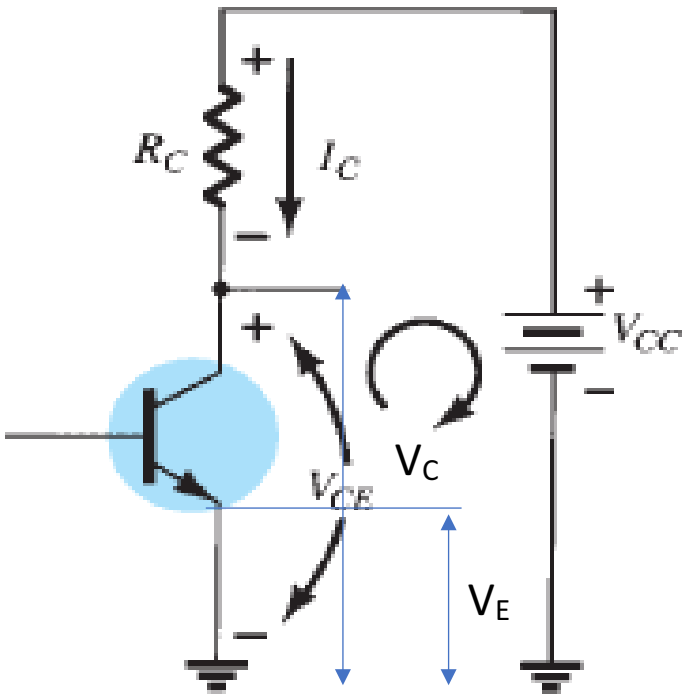


Fig. 4.5

In addition, because

$$V_{BE} = V_B - V_E$$

and $V_E = 0$ V, then

$$V_{BE} = V_B$$

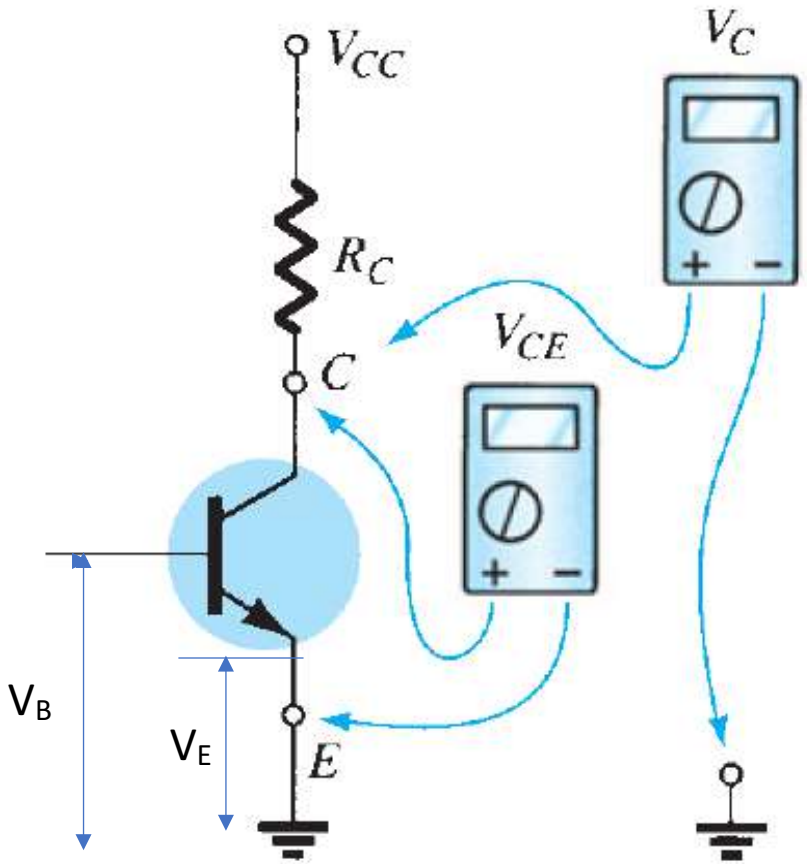
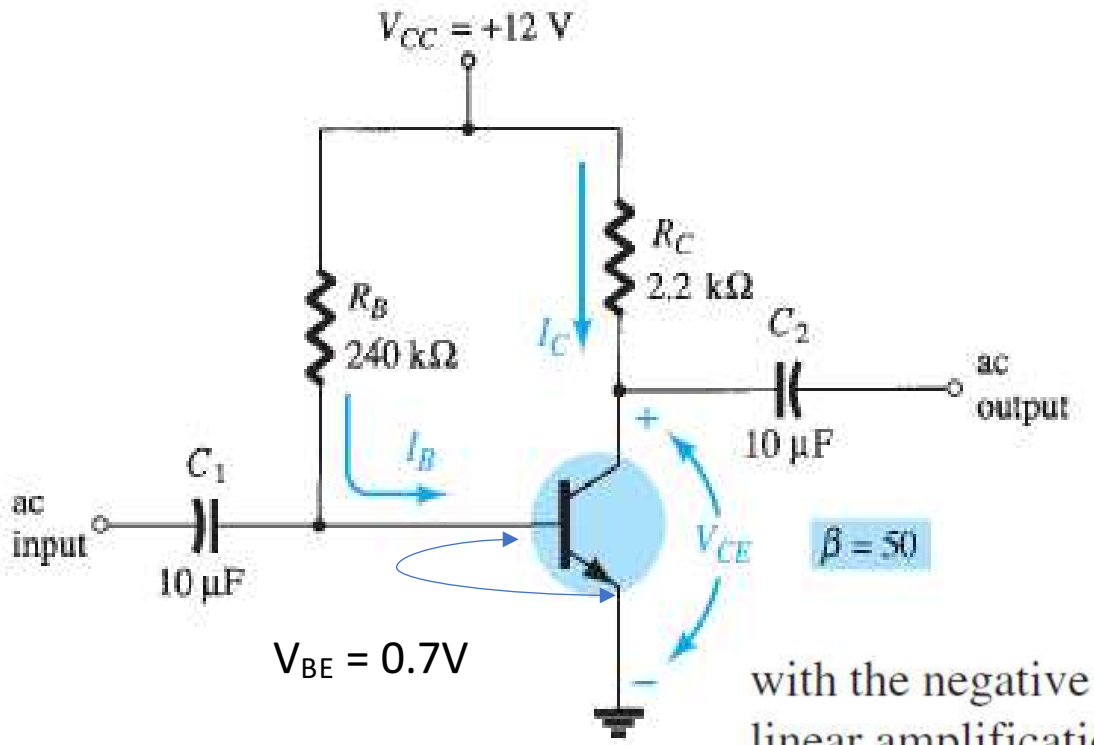


FIG. 4.6

Measuring V_{CE} and V_C.

EXAMPLE 1 Determine the following for the fixed–bias configuration of following Figure:

a. I_{BQ} and I_{CQ}.
b. V_{CEQ}.
c. V_B and V_C.
d. V_{BC}.



Solution:

a. Eq. (4.4): $$I_{B_Q} = \frac{V_{CC} - V_{BE}}{R_B} = \frac{12\text{ V} - 0.7\text{ V}}{240\text{ k}\Omega} = \mathbf{47.08\ \mu A}$$

Eq. (4.5): $$I_{C_Q} = \beta I_{BQ} = (50)(47.08\ \mu\text{A}) = \mathbf{2.35\ mA}$$

b. Eq. (4.6): $$V_{CE_Q} = V_{CC} - I_C R_C$$
$$= 12\text{ V} - (2.35\text{ mA})(2.2\text{ k}\Omega)$$
$$= \mathbf{6.83\ V}$$

c. $V_B = V_{BE} = \mathbf{0.7\ V}$
$V_C = V_{CE} = \mathbf{6.83\ V}$

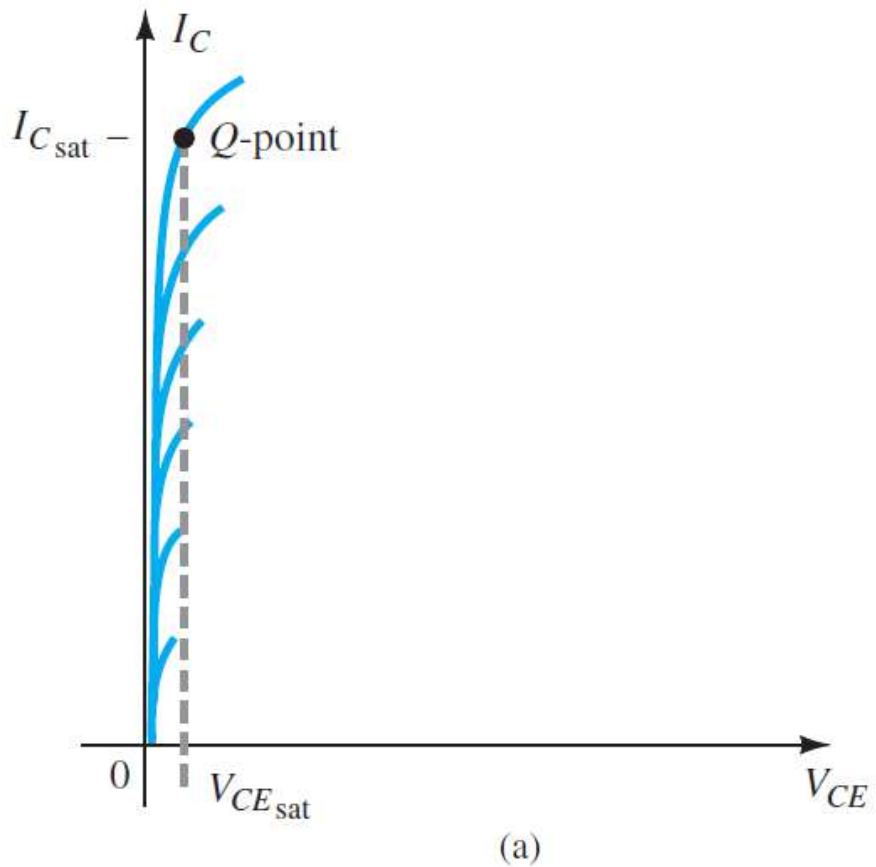
d. Using double-subscript notation yields

$$V_{BC} = V_B - V_C = 0.7\text{ V} - 6.83\text{ V}$$
$$= \mathbf{-6.13\ V}$$

with the negative sign revealing that the junction is reversed-biased, as it should be for linear amplification.

Transistor Saturation

The term *saturation* is applied to any system where levels have reached their maximum values. A saturated sponge is one that cannot hold another drop of water. For a transistor operating in the saturation region, the current is a maximum value *for the particular design.* Change the design and the corresponding saturation level may rise or drop. Of course, the highest saturation level is defined by the maximum collector current as provided by the specification sheet.



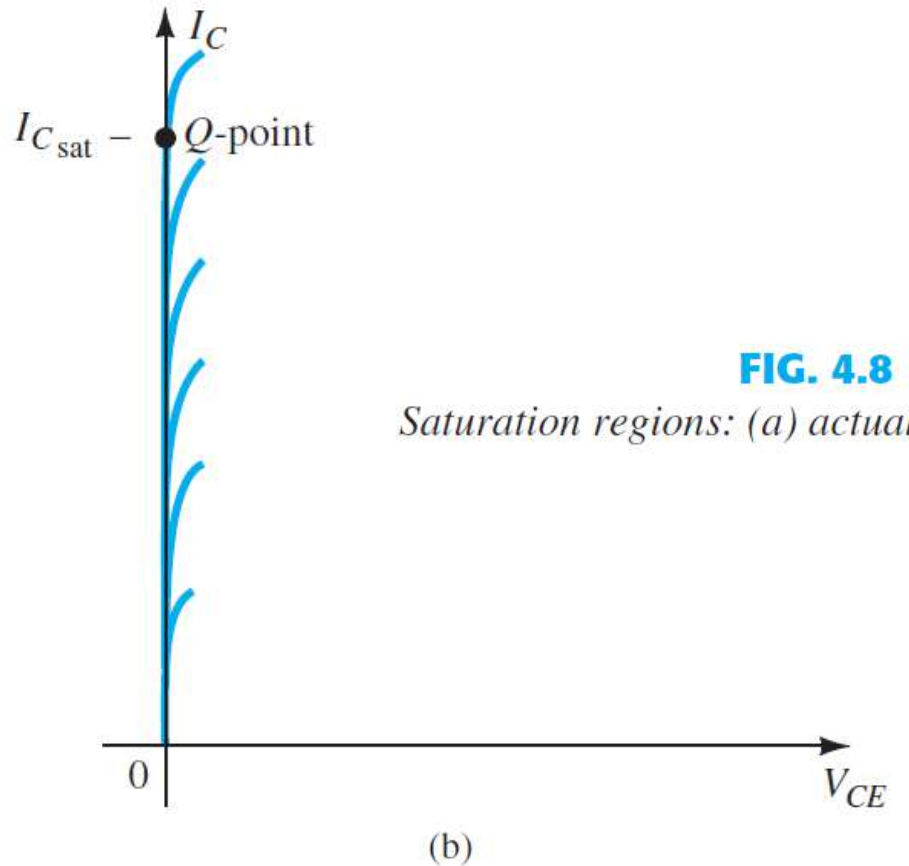


FIG. 4.8
Saturation regions: (a) actual; (b) approximate.

If we approximate the curves of Fig. 4.8a by those appearing in Fig. 4.8b, a quick, direct method for determining the saturation level becomes apparent. In Fig. 4.8b, the current is relatively high, and the voltage V_{CE} is assumed to be 0 V. Applying Ohm's law, we can determine the resistance between collector and emitter terminals as follows:

$$R_{CE} = \frac{V_{CE}}{I_C} = \frac{0\text{ V}}{I_{C_{sat}}} = 0\ \Omega$$

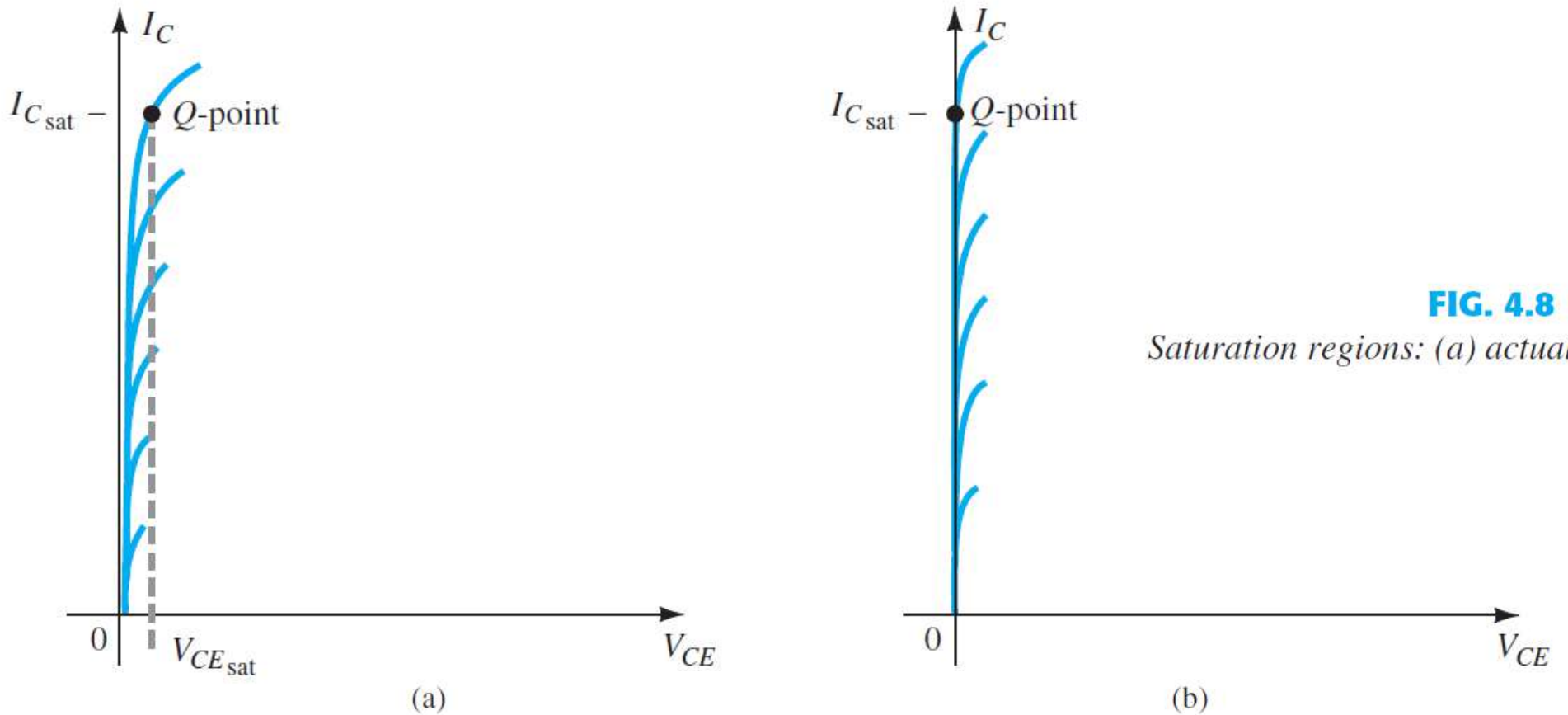
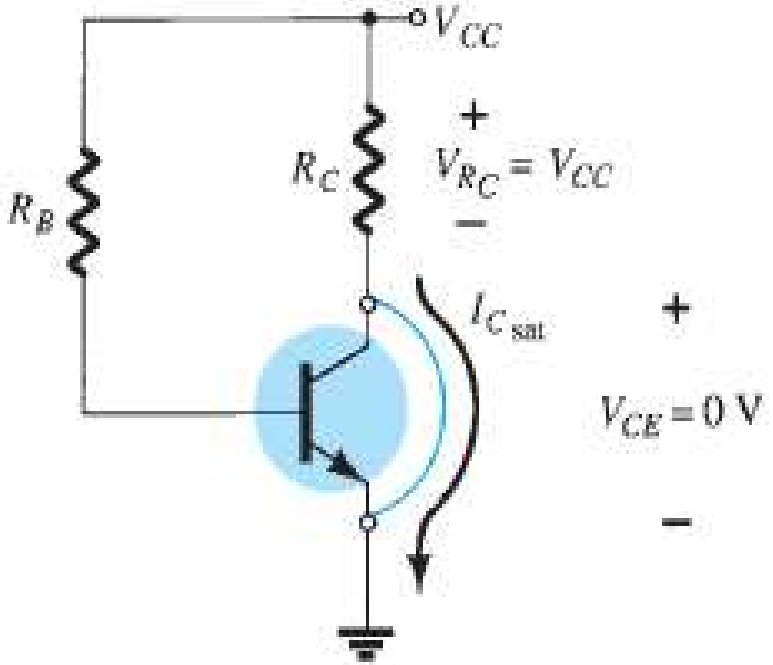


FIG. 4.8
Saturation regions: (a) actual; (b) approximate.

The resulting saturation current for the fixed-bias configuration is

$$I_{C_{sat}} = \frac{V_{CC}}{R_C}$$



Once $I_{C_{sat}}$ is known, we have some idea of the maximum possible collector current for the chosen design and the level to stay below if we expect linear amplification.

EXAMPLE 4.2 Determine the saturation level for the network of Fig. 4.7.

Solution:

$$I_{C_{sat}} = \frac{V_{CC}}{R_C} = \frac{12\text{ V}}{2.2\text{ k}\Omega} = \mathbf{5.45\text{ mA}}$$

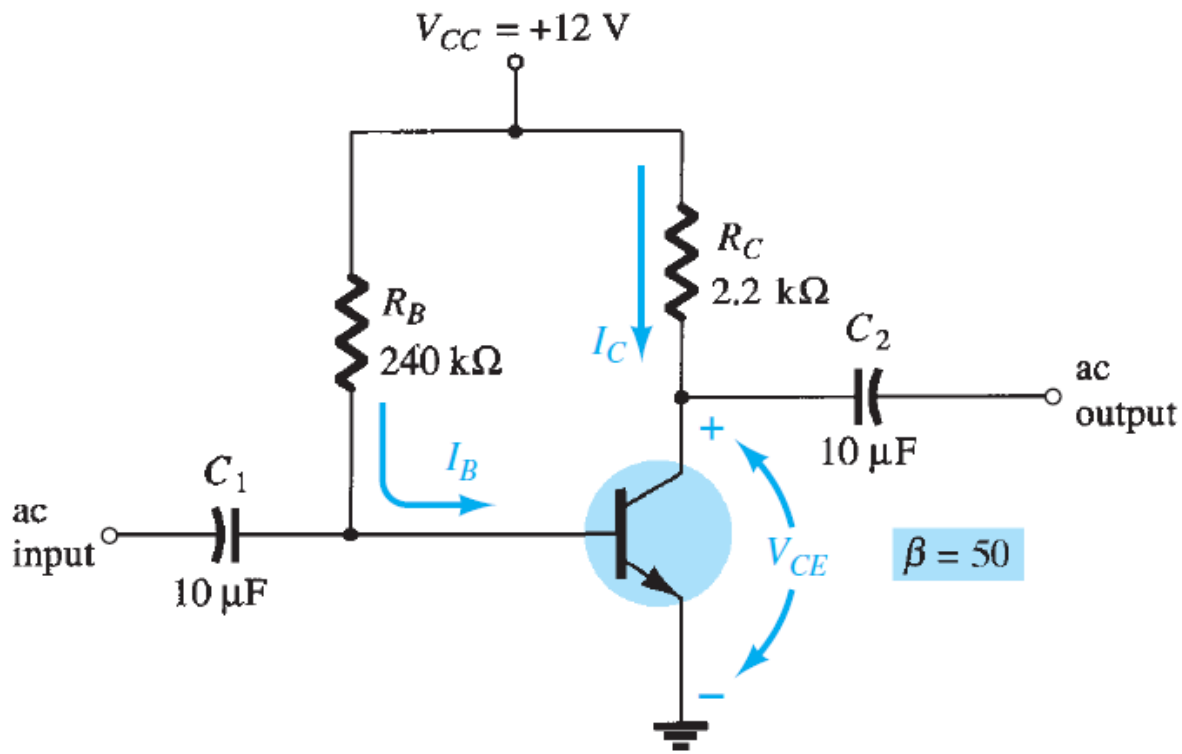
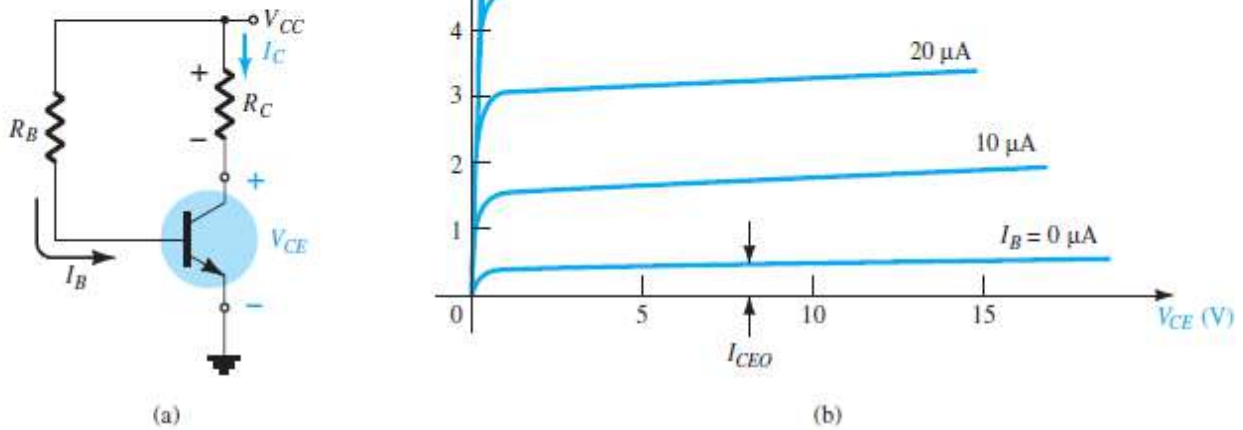


FIG. 4.7

Load–Line Analysis

The characteristics of the BJT are superimposed on a plot of the network equation defined by the same axis parameters. The load resistor R_C for the fixed-bias configuration will define the slope of the network equation and the resulting intersection between the two plots. The smaller the load resistance, the steeper the slope of the network load line



The network of above Fig. a establishes an output equation that relates the variables I_C and V_{CE} in the following manner:

$$V_{CE} = V_{CC} - I_C R_C$$

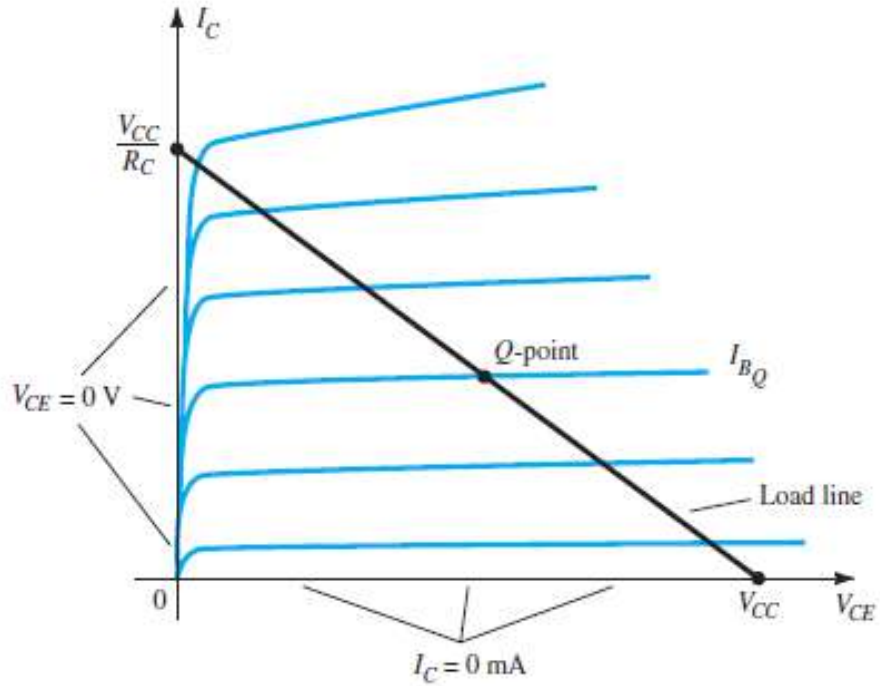
The output characteristics of the transistor also relate the same two variables I_C and V_{CE} as shown in Fig. We must now superimpose the straight line defined by the above equation on the characteristics.

$$V_{CE} = V_{CC} - (0)R_C$$

$$V_{CE} = V_{CC}|_{I_C = 0\ \text{mA}}$$

$$0 = V_{CC} - I_C R_C$$

$$I_C = \frac{V_{CC}}{R_C}\bigg|_{V_{CE} = 0\ \text{V}}$$



If the level of I_B is changed by varying the value of R_B, the Q-point moves up or down the load line as shown in Fig. 4.13 for increasing values of I_B. If V_{CC} is held fixed and R_C increased, the load line will shift as shown in Fig. 4.14. If I_B is held fixed, the Q-point will move as shown in the same figure. If R_C is fixed and V_{CC} decreased, the load line shifts as shown in Fig. 4.15.

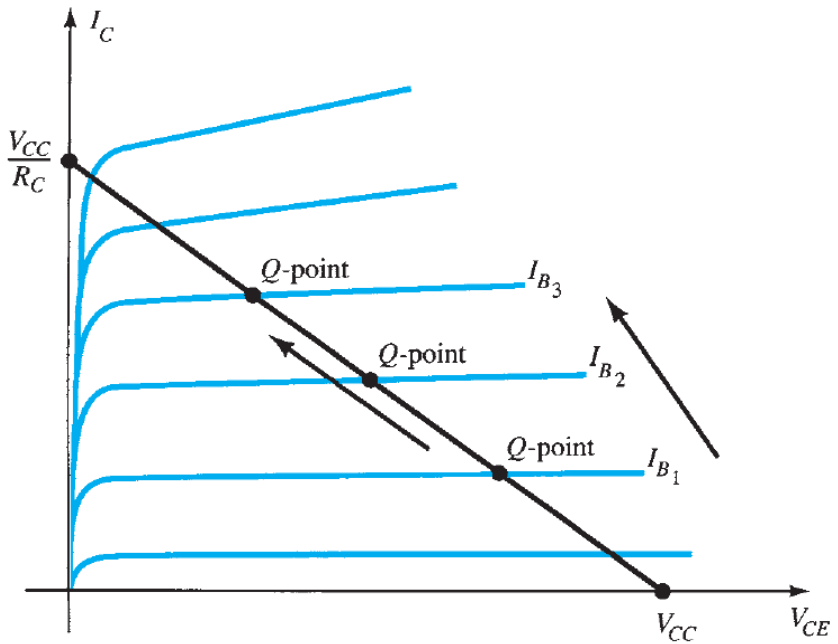


FIG. 4.13
Movement of the Q-point with increasing level of I_B.

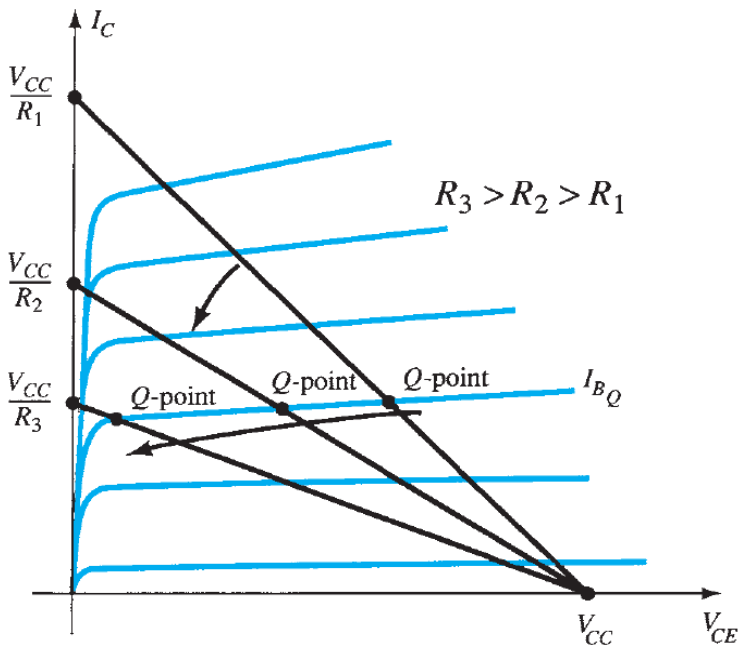


FIG. 4.14
Effect of an increasing level of R_C on the load line and the Q-point.

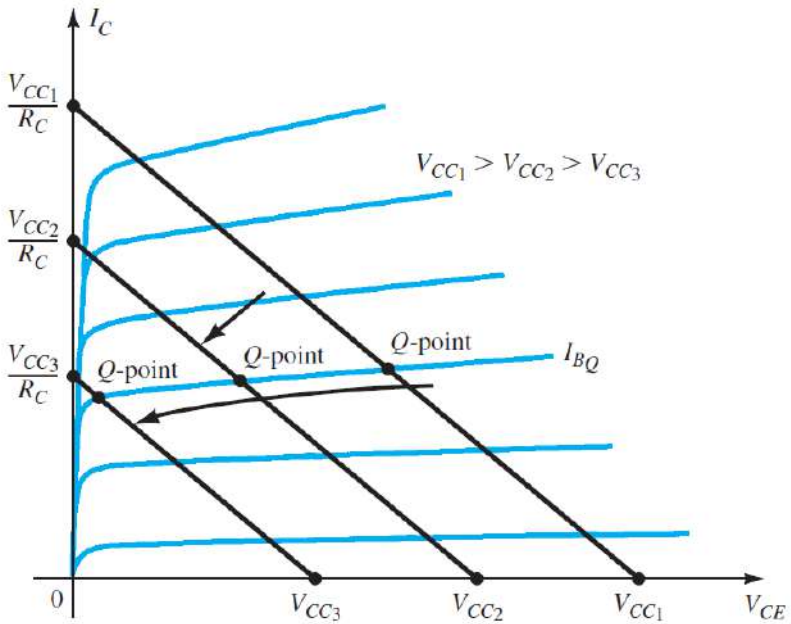
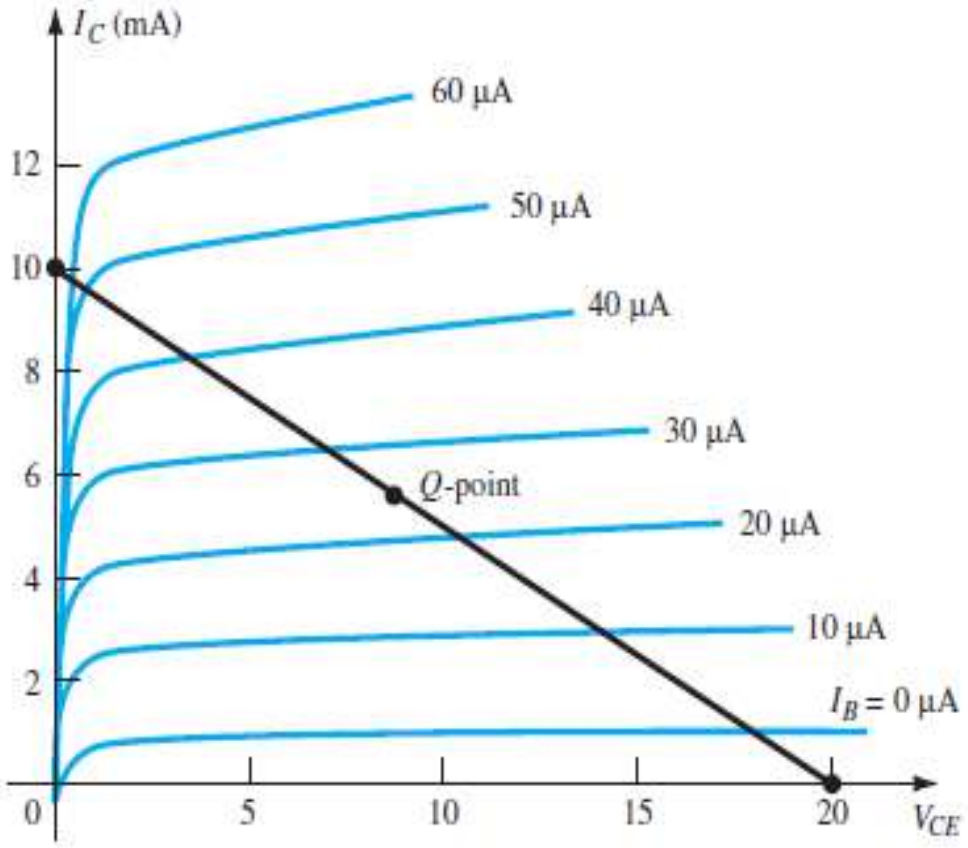


FIG. 4.15
Effect of lower values of V_{CC} on the load line and the Q-point.

EXAMPLE 4.3 Given the load line of the following characteristics and the defined Q-point, determine the required values of V_{CC}, R_C, and R_B for a fixed-bias configuration.



Solution:

$$V_{CE} = V_{CC} = \mathbf{20\ V} \text{ at } I_C = 0\ \text{mA}$$

$$I_C = \frac{V_{CC}}{R_C} \text{ at } V_{CE} = 0\ \text{V}$$

$$R_C = \frac{V_{CC}}{I_C} = \frac{20\ \text{V}}{10\ \text{mA}} = \mathbf{2\ k\Omega}$$

$$I_B = \frac{V_{CC} - V_{BE}}{R_B}$$

$$R_B = \frac{V_{CC} - V_{BE}}{I_B} = \frac{20\ \text{V} - 0.7\ \text{V}}{25\ \mu\text{A}} = \mathbf{772\ k\Omega}$$

EMITTER–BIAS CONFIGURATION

The dc bias network of the following Figure 4.17 contains an emitter resistor to improve the stability level over that of the fixed-bias configuration. The more stable a configuration, the less its response will change due to undesirable changes in temperature and parameter variations.

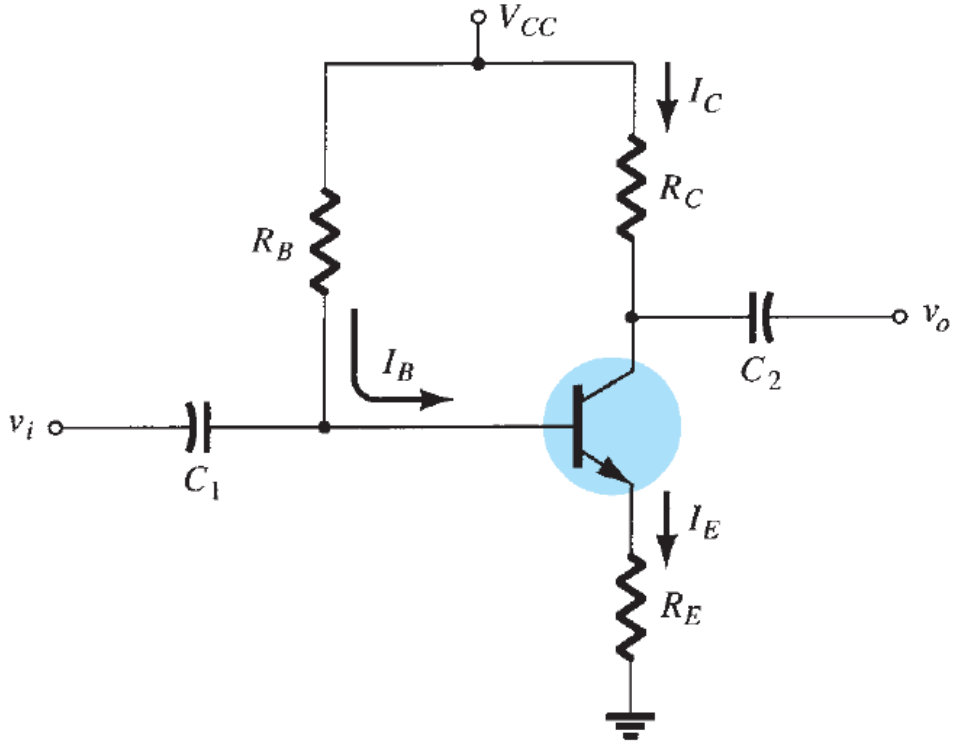


FIG. 4.17

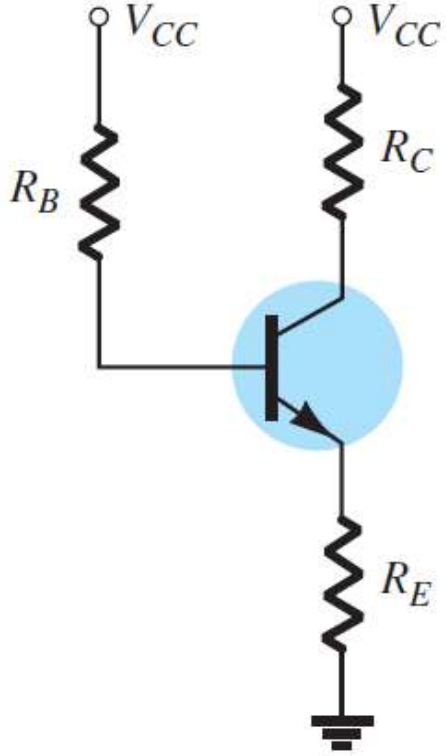


FIG. 4.18
DC equivalent of Fig. 4.17.

Base–Emitter Loop:

The base–emitter loop of the network of Fig. 4.18 can be redrawn as shown in Fig. 4.19. Writing Kirchhoff's voltage law around the indicated loop in the clockwise direction results in the following equation:

$$+V_{CC} - I_B R_B - V_{BE} - I_E R_E = 0 \qquad \textbf{(4.15)}$$

Recall from Chapter 3 that

$$I_E = (\beta + 1) I_B \qquad \textbf{(4.16)}$$

Substituting for I_E in Eq. (4.15) results in

$$V_{CC} - I_B R_B - V_{BE} - (\beta + I) I_B R_E = 0$$

Grouping terms then provides the following:

$$-I_B(R_B + (\beta + 1)R_E) + V_{CC} - V_{BE} = 0$$

Multiplying through by (-1), we have

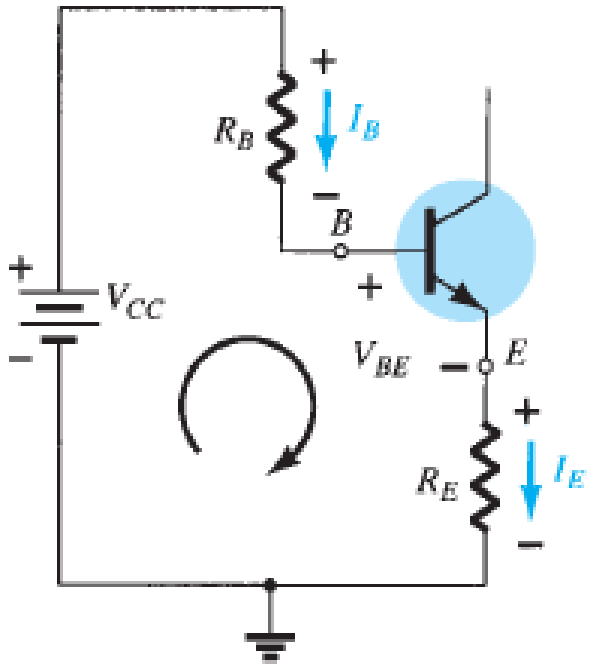
$$I_B(R_B + (\beta + 1)R_E) - V_{CC} + V_{BE} = 0$$

with

$$I_B(R_B + (\beta + 1)R_E) = V_{CC} - V_{BE}$$

and solving for I_B gives

$$I_B = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1)R_E} \qquad \textbf{(4.17)}$$



Note that the only difference between this equation for I_B and that obtained for the fixed-bias configuration is the term $(\beta + 1)R_E$.

There is an interesting result that can be derived from Eq. (4.17) if the equation is used to sketch a series network that would result in the same equation. Such is the case for the network of Fig. 4.20. Solving for the current I_B results in the same equation as obtained above. Note that aside from the base-to-emitter voltage V_{BE}, the resistor R_E is *reflected* back to the input base circuit by a factor $(\beta + 1)$. In other words, the emitter resistor, which is part of the collector–emitter loop, "appears as" $(\beta + 1)R_E$ in the base–emitter loop. Because β is typically 50 or more, the emitter resistor appears to be a great deal larger in the base circuit. In general, therefore, for the configuration of Fig. 4.21,

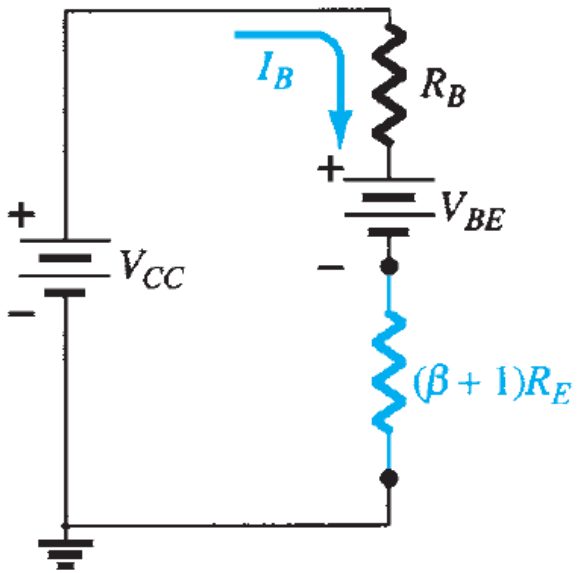


FIG. 4.20
Network derived from Eq. (4.17).

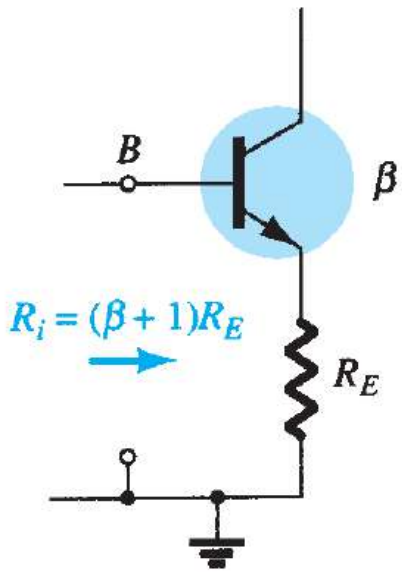


FIG. 4.21
Reflected impedance level of R_E.

$$R_i = (\beta + 1)R_E \tag{4.18}$$

Equation (4.18) will prove useful in the analysis to follow. In fact, it provides a fairly easy way to remember Eq. (4.17). Using Ohm's law, we know that the current through a system is the voltage divided by the resistance of the circuit. For the base–emitter circuit the net voltage is $V_{CC} - V_{BE}$. The resistance levels are R_B plus R_E reflected by $(\beta + 1)$. The result is Eq. (4.17).

$$I_B = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1)R_E} \tag{4.17}$$

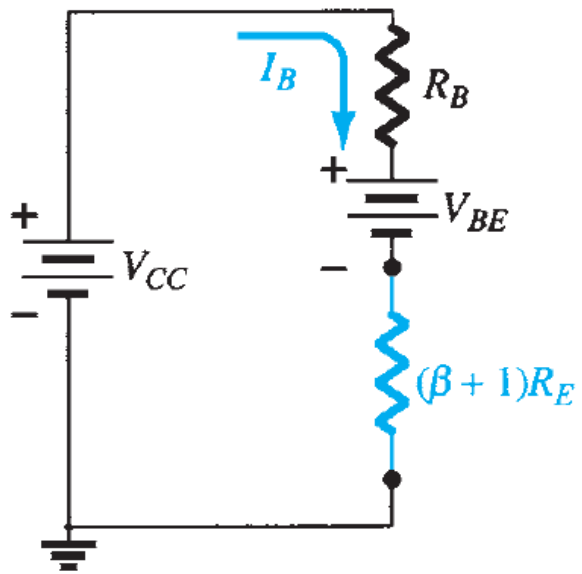


FIG. 4.20
Network derived from Eq. (4.17).

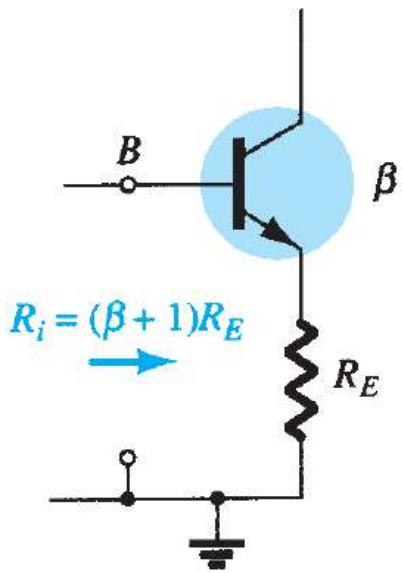


FIG. 4.21
Reflected impedance level of R_E.

$$R_i = (\beta + 1)R_E \tag{4.18}$$

Collector–Emitter Loop

The collector–emitter loop appears in Fig. 4.22. Writing Kirchhoff's voltage law for the indicated loop in the clockwise direction results in

$$+I_E R_E + V_{CE} + I_C R_C - V_{CC} = 0$$

Substituting $I_E \cong I_C$ and grouping terms gives

$$V_{CE} - V_{CC} + I_C(R_C + R_E) = 0$$

and

$$\boxed{V_{CE} = V_{CC} - I_C(R_C + R_E)}$$

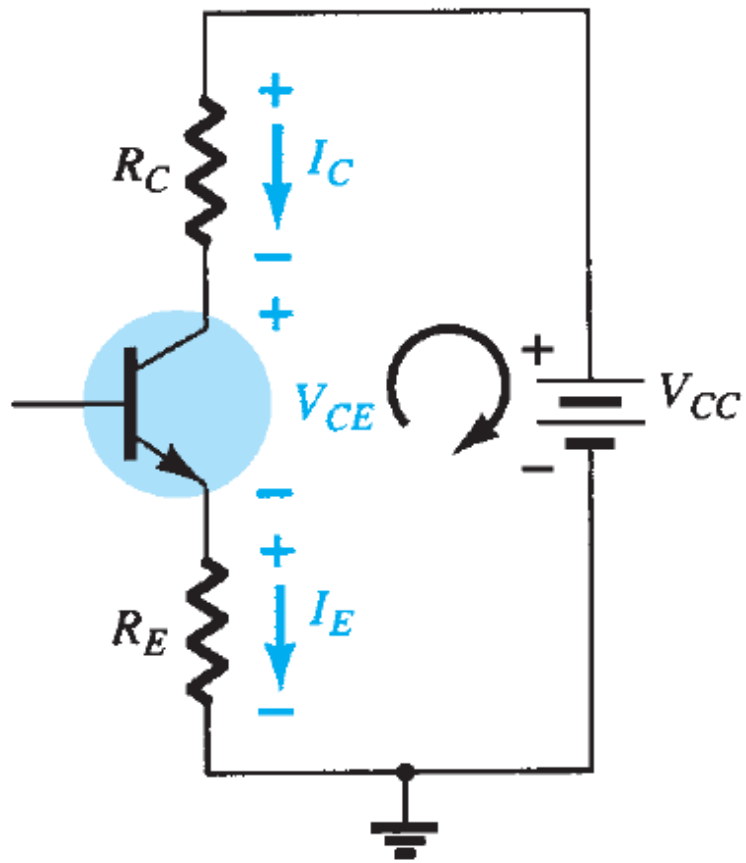


FIG. 4.22
Collector–emitter loop.

The single-subscript voltage V_E is the voltage from emitter to ground and is determined by

$$V_E = I_E R_E \tag{4.20}$$

whereas the voltage from collector to ground can be determined from

$$V_{CE} = V_C - V_E$$

and

$$V_C = V_{CE} + V_E \tag{4.21}$$

or

$$V_C = V_{CC} - I_C R_C \tag{4.22}$$

The voltage at the base with respect to ground can be determined using Fig. 4.18

$$V_B = V_{CC} - I_B R_B \tag{4.23}$$

or

$$V_B = V_{BE} + V_E \tag{4.24}$$

EXAMPLE 4.4 For the emitter-bias network of Fig. 4.23, determine:

a. I_B.
b. I_C.
c. V_{CE}.
d. V_C.
e. V_E.
f. V_B.
g. V_{BC}.

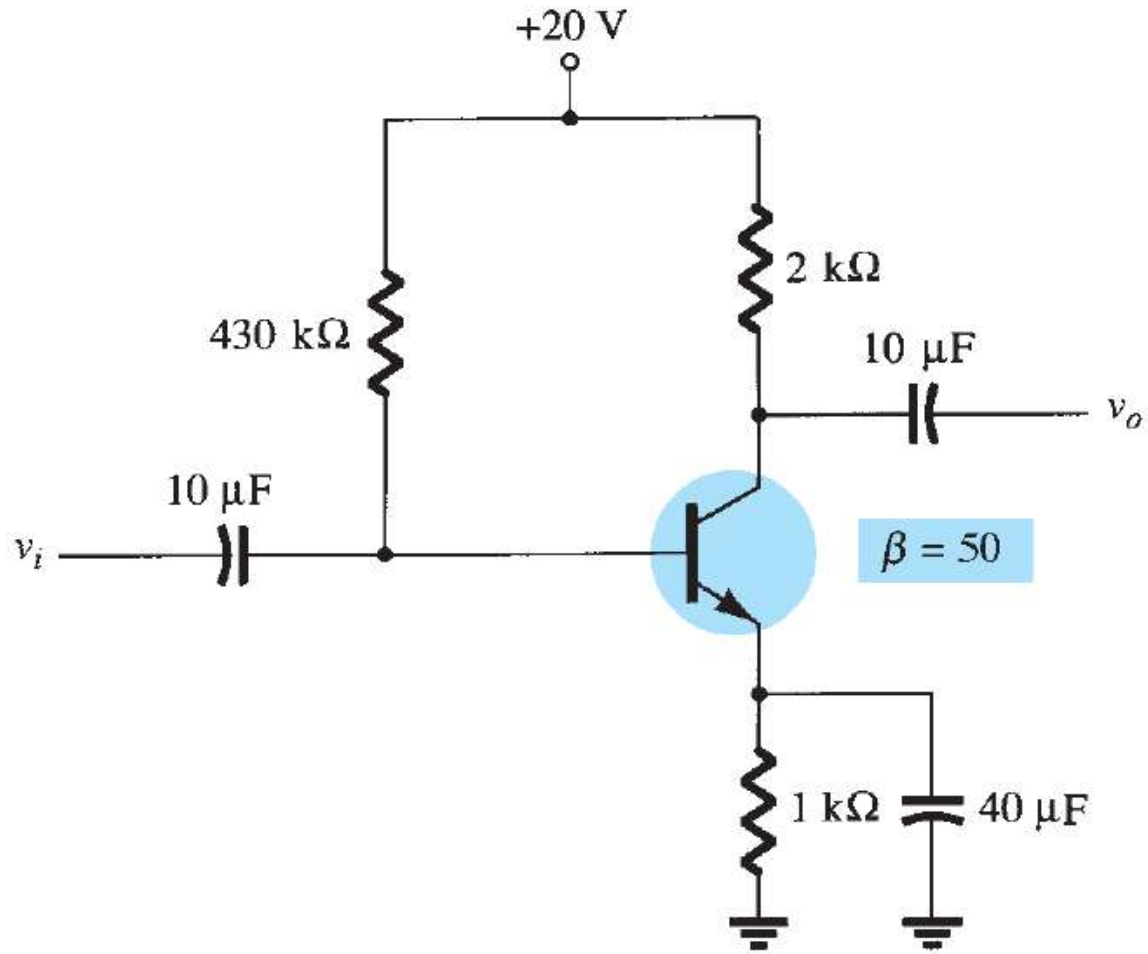


FIG. 4.23
Emitter-stabilized bias circuit for Example 4.4.

Solution:

a. Eq. (4.17): $$I_B = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1)R_E} = \frac{20\text{ V} - 0.7\text{ V}}{430\text{ k}\Omega + (51)(1\text{ k}\Omega)}$$

$$= \frac{19.3\text{ V}}{481\text{ k}\Omega} = \mathbf{40.1\ \mu A}$$

b. $I_C = \beta I_B$

$= (50)(40.1\ \mu\text{A})$

$\cong$ **2.01 mA**

c. Eq. (4.19): $V_{CE} = V_{CC} - I_C(R_C + R_E)$

$= 20\text{ V} - (2.01\text{ mA})(2\text{ k}\Omega + 1\text{ k}\Omega) = 20\text{ V} - 6.03\text{ V}$

$=$ **13.97 V**

d. $V_C = V_{CC} - I_C R_C$

$= 20\text{ V} - (2.01\text{ mA})(2\text{ k}\Omega) = 20\text{ V} - 4.02\text{ V}$

$=$ **15.98 V**

e. $V_E = V_C - V_{CE}$

$= 15.98\text{ V} - 13.97\text{ V}$

$=$ **2.01 V**

or $V_E = I_E R_E \cong I_C R_E$

$= (2.01\text{ mA})(1\text{ k}\Omega)$

$=$ **2.01 V**

f. $V_B = V_{BE} + V_E$

$= 0.7\text{ V} + 2.01\text{ V}$

$=$ **2.71 V**

g. $V_{BC} = V_B - V_C$

Improved Bias Stability

The addition of the emitter resistor to the dc bias of the BJT provides improved stability, that is, the dc bias currents and voltages remain closer to where they were set by the circuit when outside conditions, such as temperature and transistor beta, change. Although a mathematical analysis is provided in Section 4.12, some comparison of the improvement can be obtained as demonstrated by Example 4.5.

EXAMPLE 4.5 Prepare a table and compare the bias voltage and currents of the circuits of Fig. 4.7 and Fig. 4.23 for the given value of $\beta = 50$ and for a new value of $\beta = 100$. Compare the changes in I_C and V_{CE} for the same increase in β.

Solution: Using the results calculated in Example 4.1 and then repeating for a value of $\beta = 100$ yields the following:

Effect of β variation on the response of the fixed-bias configuration of Fig. 4.7.

β	I_B (μA)	I_C (mA)	V_{CE} (V)
50	47.08	2.35	6.83
100	47.08	4.71	1.64

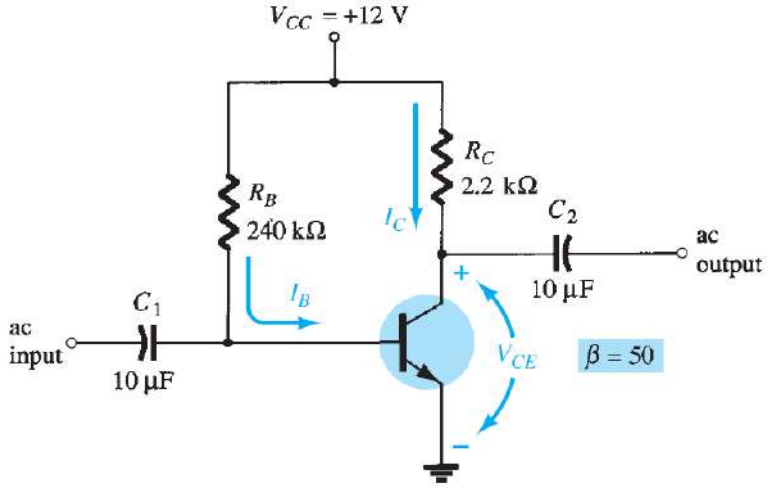


FIG. 4.7

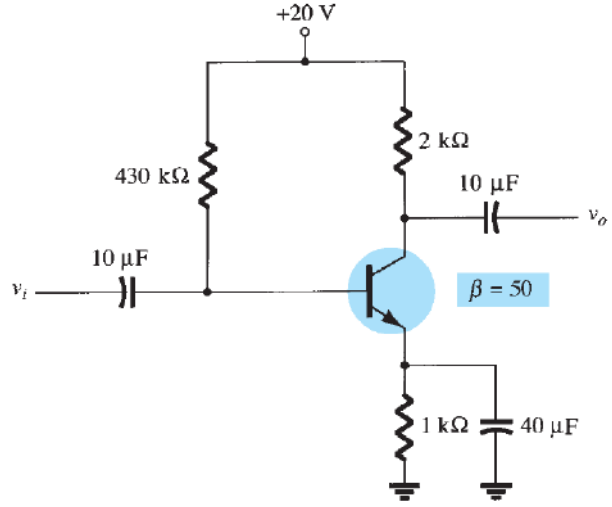


FIG. 4.23

The BJT collector current is seen to change by 100% due to the 100% change in the value of β. The value of I_B is the same, and V_{CE} decreased by 76%.

Using the results calculated in Example 4.4 and then repeating for a value of $\beta = 100$, we have the following:

Effect of β variation on the response of the emitter-bias configuration of Fig. 4.23.

β	I_B (μA)	I_C (mA)	V_{CE} (V)
50	40.1	2.01	13.97
100	36.3	3.63	9.11

Now the BJT collector current increases by about 81% due to the 100% increase in β. Notice that I_B decreased, helping maintain the value of I_C—or at least reducing the overall change in I_C due to the change in β. The change in V_{CE} has dropped to about 35%. The network of Fig. 4.23 is therefore more stable than that of Fig. 4.7 for the same change in β.

Saturation Level

The collector saturation level or maximum collector current for an emitter-bias design can be determined using the same approach applied to the fixed-bias configuration: Apply a short circuit between the collector–emitter terminals as shown in Fig. 4.24 and calculate the resulting collector current. For Fig. 4.24

$$I_{C_{sat}} = \frac{V_{CC}}{R_C + R_E} \tag{4.25}$$

The addition of the emitter resistor reduces the collector saturation level below tha obtained with a fixed-bias configuration using the same collector resistor.

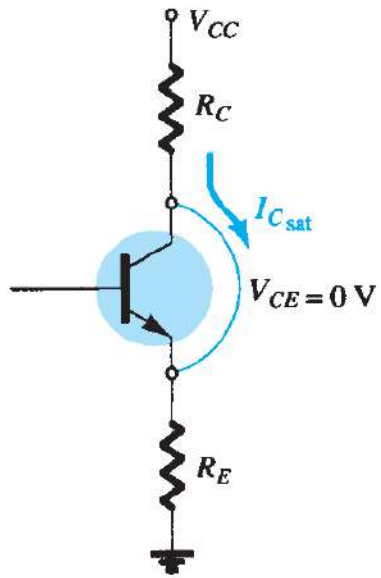


FIG. 4.24
Determining $I_{C_{sat}}$ for the emitter-stabilized bias circuit.

EXAMPLE 4.6 Determine the saturation current for the network of Example 4.4.

Solution:

$$I_{C_{sat}} = \frac{V_{CC}}{R_C + R_E}$$

$$= \frac{20\text{ V}}{2\text{ k}\Omega + 1\text{ k}\Omega} = \frac{20\text{ V}}{3\text{ k}\Omega}$$

$$= \mathbf{6.67\text{ mA}}$$

which is about three times the level of $I_{C_{sat}}$ for Example 4.4.

Load-Line Analysis

The load-line analysis of the emitter-bias network is only slightly different from that encountered for the fixed-bias configuration. The level of I_B as determined by Eq. (4.17) defines the level of I_B on the characteristics of Fig. 4.25 (denoted I_{B_Q}).

The collector–emitter loop equation that defines the load line is

$$V_{CE} = V_{CC} - I_C(R_C + R_E)$$

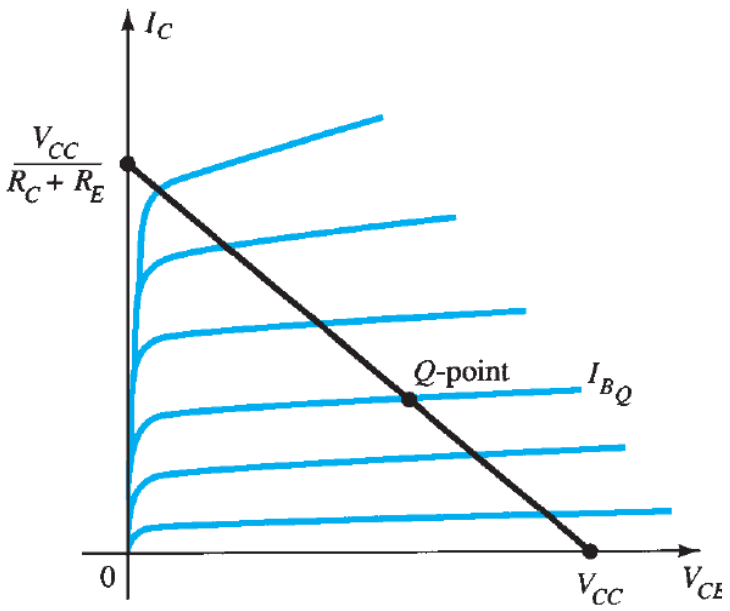


FIG. 4.25
Load line for the emitter-bias configuration.

Choosing $I_C = 0$ mA gives

$$\boxed{V_{CE} = V_{CC}\big|_{I_C=0\text{ mA}}} \tag{4.26}$$

as obtained for the fixed-bias configuration. Choosing $V_{CE} = 0$ V gives

$$\boxed{I_C = \frac{V_{CC}}{R_C + R_E}\bigg|_{V_{CE}=0\text{ V}}} \tag{4.27}$$

as shown in Fig. 4.25. Different levels of I_{B_Q} will, of course, move the Q-point up or down the load line.

EXAMPLE 4.7

a. Draw the load line for the network of Fig. 4.26a on the characteristics for the transistor appearing in Fig. 4.26b.
b. For a Q-point at the intersection of the load line with a base current of 15 μA, find the values of I_{C_Q} and V_{CE_Q}.
c. Determine the dc beta at the Q-point.
d. Using the beta for the network determined in part c, calculate the required value of R_B and suggest a possible standard value.

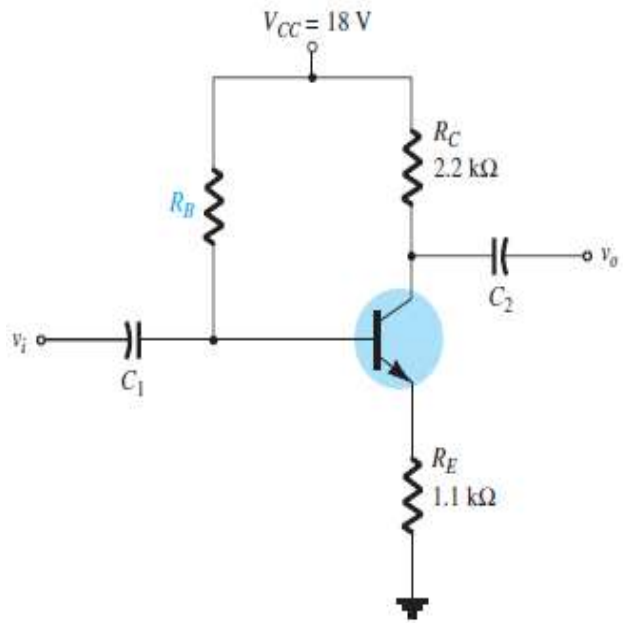


FIG. 4.26a

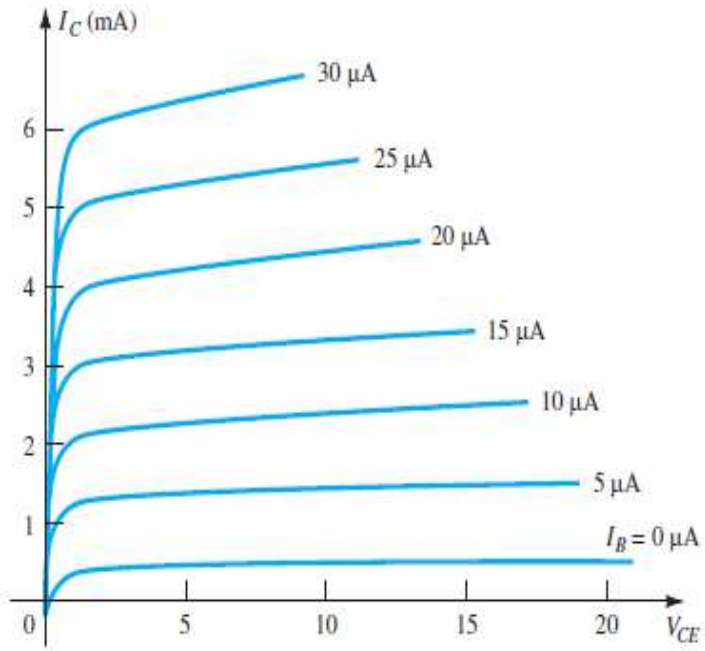


FIG. 4.26b

Solution:

a. Two points on the characteristics are required to draw the load line.

$$\text{At } V_{CE} = 0\text{ V}: \quad I_C = \frac{V_{CC}}{R_C + R_E} = \frac{18\text{ V}}{2.2\text{ k}\Omega + 1.1\text{ k}\Omega} = \frac{18\text{ V}}{3.3\text{ k}\Omega} = 5.45\text{ mA}$$

$$\text{At } I_C = 0\text{ mA}: V_{CE} = V_{CC} = 18\text{ V}$$

The resulting load line appears in Fig. 4.27.

b. From the characteristics of Fig. 4.27 we find

$$V_{CE_Q} \cong 7.5\text{ V}, I_{C_Q} \cong 3.3\text{ mA}$$

c. The resulting dc beta is:

$$\beta = \frac{I_{C_Q}}{I_{B_Q}} = \frac{3.3\text{ mA}}{15\ \mu\text{A}} = 220$$

d. Applying Eq. 4.17:

$$I_B = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1)R_E} = \frac{18\text{ V} - 0.7\text{ V}}{R_B + (220 + 1)(1.1\text{ k}\Omega)}$$

$$\text{and } 15\ \mu\text{A} = \frac{17.3\text{ V}}{R_B + (221)(1.1\text{ k}\Omega)} = \frac{17.3\text{ V}}{R_B + 243.1\text{ k}\Omega}$$

$$\text{so that } (15\ \mu\text{A})(R_B) + (15\ \mu\text{A})(243.1\text{ k}\Omega) = 17.3\text{ V}$$

$$\text{and } (15\ \mu\text{A})(R_B) = 17.3\text{ V} - 3.65\text{ V} = 13.65\text{ V}$$

$$\text{resulting in } R_B + \frac{13.65\text{ V}}{15\ \mu\text{A}} = 910\text{ k}\Omega$$

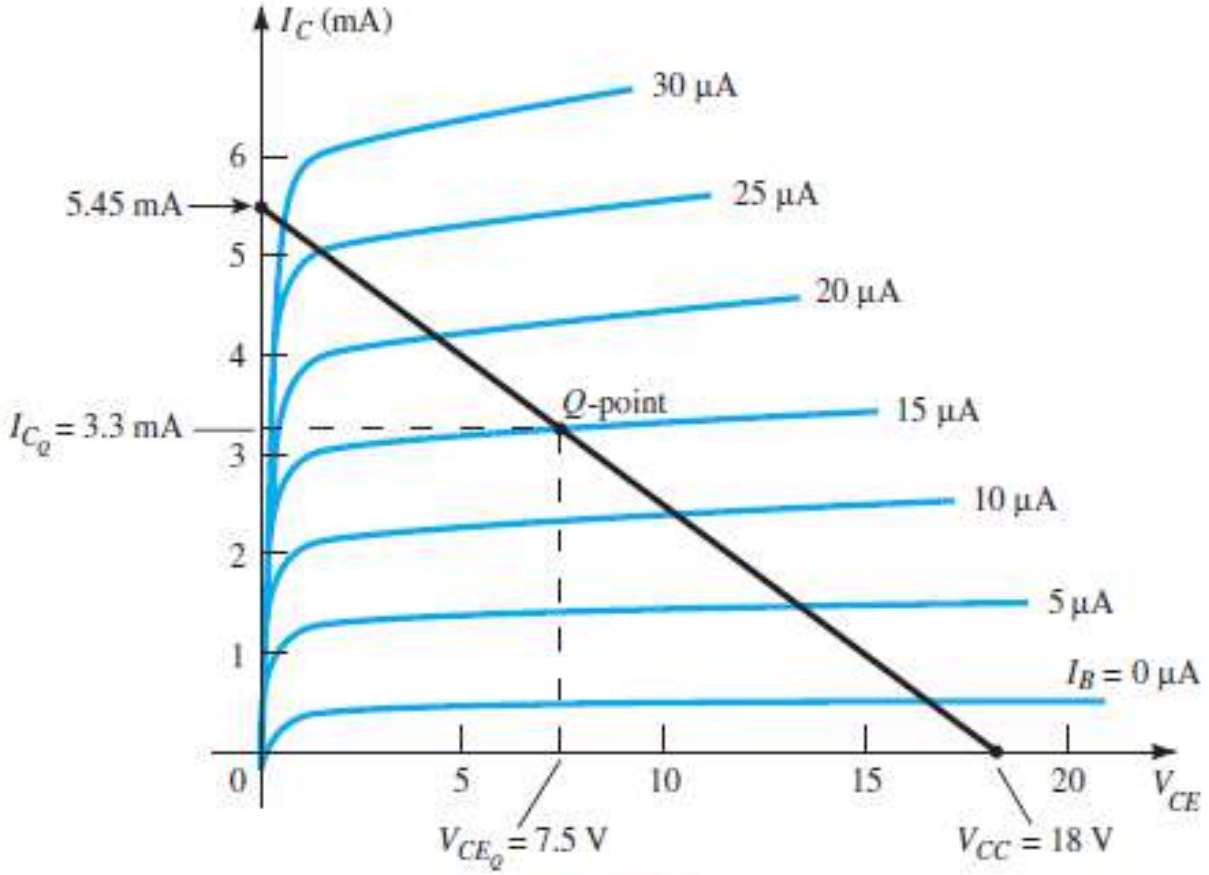


FIG. 4.27

4.5 VOLTAGE-DIVIDER BIAS CONFIGURATION

In the previous bias configurations the bias current I_{C_Q} and voltage V_{CE_Q} were a function of the current gain β of the transistor. However, because β is temperature sensitive, especially for silicon transistors, and the actual value of beta is usually not well defined, it would be desirable to develop a bias circuit that is less dependent on, or in fact is independent of, the transistor beta. The voltage-divider bias configuration of Fig. 4.28 is such a network. If analyzed on an exact basis, the sensitivity to changes in beta is quite small. If the circuit parameters are properly chosen, the resulting levels of I_{C_Q} and V_{CE_Q} can be almost totally independent of beta. Recall from previous discussions that a Q-point is defined by a fixed level of I_{C_Q} and V_{CE_Q} as shown in Fig. 4.29. The level of I_{B_Q} will change with the change in beta, but the operating point on the characteristics defined by I_{C_Q} and V_{CE_Q} can remain fixed if the proper circuit parameters are employed.

As noted earlier, there are two methods that can be applied to analyze the voltage-divider configuration. The reason for the choice of names for this configuration will become obvious in the analysis to follow. The first to be demonstrated is the *exact method*, which can be applied to *any* voltage-divider configuration. The second is referred to as the *approximate method* and can be applied only if specific conditions are satisfied. The approximate approach permits a more direct analysis with a savings in time and energy. It is also particularly helpful in the design mode to be described in a later section. All in all, the approximate approach can be applied to the majority of situations and therefore should be examined with the same interest as the exact method.

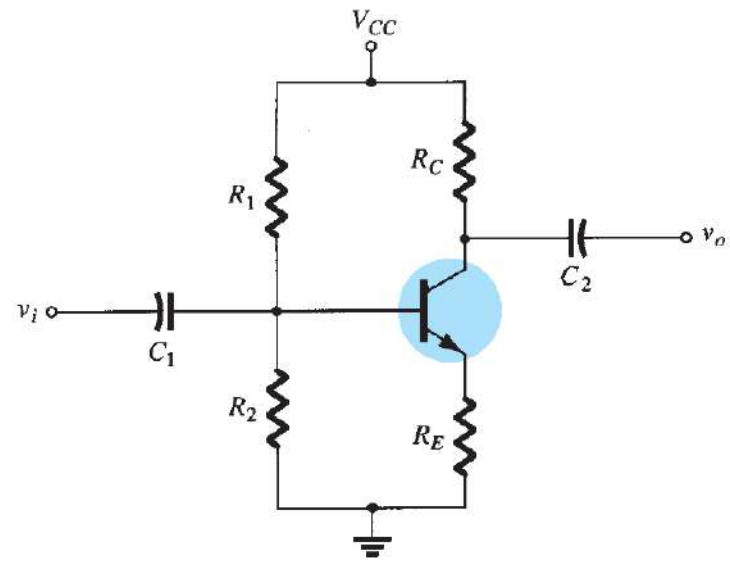


FIG. 4.28
Voltage-divider bias configuration.

Exact Analysis

For the dc analysis the network of Fig. 4.28 can be redrawn as shown in Fig. 4.30. The input side of the network can then be redrawn as shown in Fig. 4.31 for the dc analysis. The Thévenin equivalent network for the network to the left of the base terminal can then be found in the following manner:

R_{Th} The voltage source is replaced by a short-circuit equivalent as shown in Fig. 4.32:

$$R_{Th} = R_1 \| R_2 \tag{4.28}$$

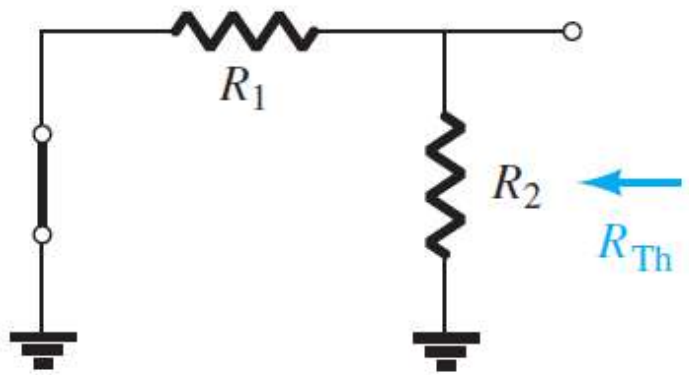


FIG. 4.32
Determining R_{Th}.

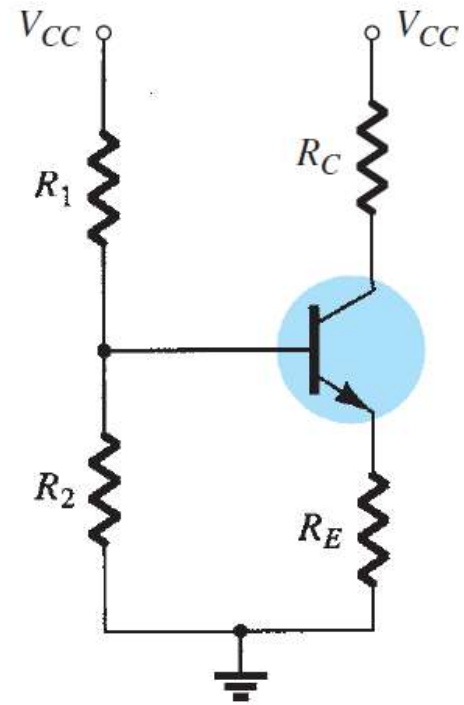


FIG. 4.30
DC components of the voltage-divider configuration.

E_{Th} The voltage source V_{CC} is returned to the network and the open-circuit Thévenin voltage of Fig. 4.33 determined as follows:

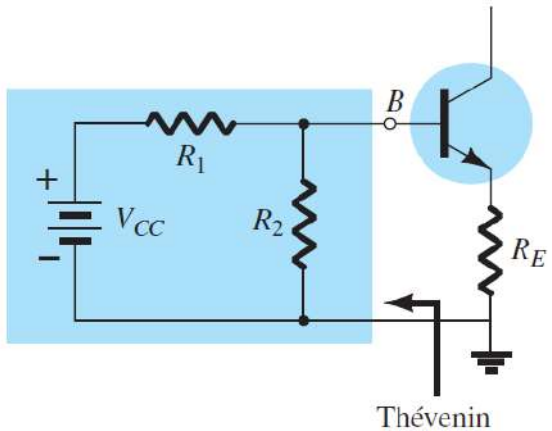


FIG. 4.31
Redrawing the input side of the network of Fig. 4.28.

Applying the voltage-divider rule gives

$$E_{\text{Th}} = V_{R_2} = \frac{R_2 V_{CC}}{R_1 + R_2} \tag{4.29}$$

The Thévenin network is then redrawn as shown in Fig. 4.34, and I_{B_Q} can be determined by first applying Kirchhoff's voltage law in the clockwise direction for the loop indicated:

$$E_{\text{Th}} - I_B R_{\text{Th}} - V_{BE} - I_E R_E = 0$$

Substituting $I_E = (\beta + 1)I_B$ and solving for I_B yields

$$I_B = \frac{E_{\text{Th}} - V_{BE}}{R_{\text{Th}} + (\beta + 1)R_E} \tag{4.30}$$

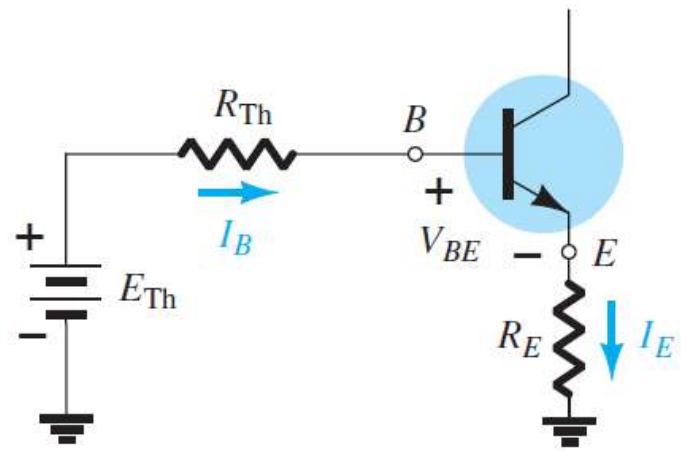


FIG. 4.34
Inserting the Thévenin equivalent circuit.

Although Eq. (4.30) initially appears to be different from those developed earlier, note that the numerator is again a difference of two voltage levels and the denominator is the base resistance plus the emitter resistor reflected by $(\beta + 1)$—certainly very similar to Eq. (4.17).

Once I_B is known, the remaining quantities of the network can be found in the same manner as developed for the emitter-bias configuration. That is,

$$V_{CE} = V_{CC} - I_C(R_C + R_E) \tag{4.31}$$

which is exactly the same as Eq. (4.19). The remaining equations for V_E, V_C, and V_B are also the same as obtained for the emitter-bias configuration.

EXAMPLE 4.8 Determine the dc bias voltage V_{CE} and the current I_C for the voltage-divider configuration of Fig. 4.35

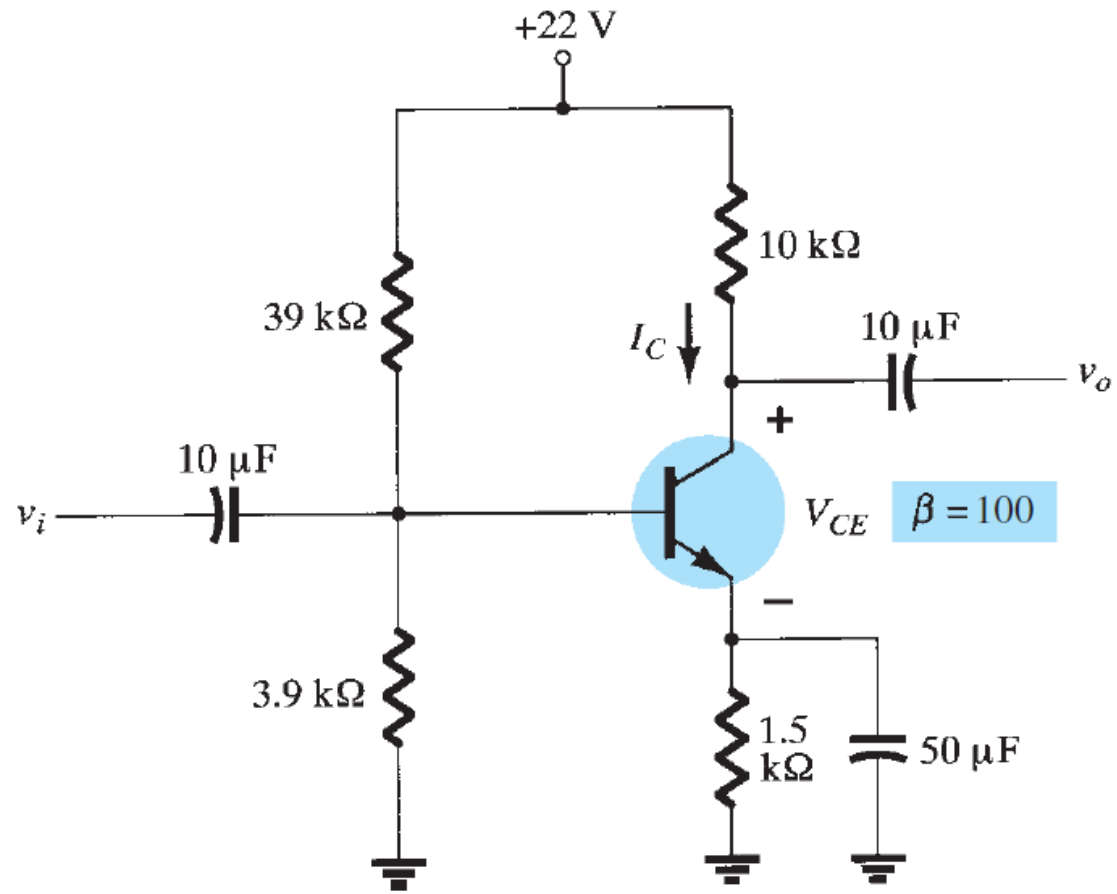


FIG. 4.35
Beta-stabilized circuit for Example 4.8.

Solution:

Eq. (4.28): $$R_{Th} = R_1 \| R_2$$

$$= \frac{(39\ \text{k}\Omega)(3.9\ \text{k}\Omega)}{39\ \text{k}\Omega + 3.9\ \text{k}\Omega} = 3.55\ \text{k}\Omega$$

Eq. (4.29): $$E_{Th} = \frac{R_2 V_{CC}}{R_1 + R_2}$$

$$= \frac{(3.9\ \text{k}\Omega)(22\ \text{V})}{39\ \text{k}\Omega + 3.9\ \text{k}\Omega} = 2\ \text{V}$$

Eq. (4.30): $$I_B = \frac{E_{Th} - V_{BE}}{R_{Th} + (\beta + 1)R_E}$$

$$= \frac{2\ \text{V} - 0.7\ \text{V}}{3.55\ \text{k}\Omega + (101)(1.5\ \text{k}\Omega)} = \frac{1.3\ \text{V}}{3.55\ \text{k}\Omega + 151.5\ \text{k}\Omega}$$

$$= 8.38\ \mu\text{A}$$

$$I_C = \beta I_B$$

$$= (100)(8.38\ \mu\text{A})$$

$$= \mathbf{0.84\ mA}$$

Eq. (4.31): $$V_{CE} = V_{CC} - I_C(R_C + R_E)$$

$$= 22\ \text{V} - (0.84\ \text{mA})(10\ \text{k}\Omega + 1.5\ \text{k}\Omega)$$

$$= 22\ \text{V} - 9.66\ \text{V}$$

$$= \mathbf{12.34\ V}$$

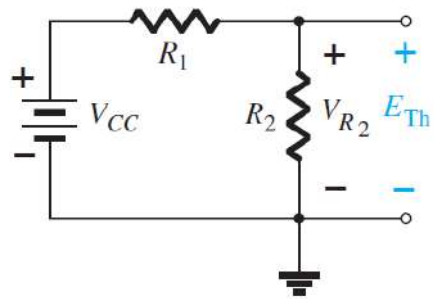


FIG. 4.33
Determining E_{Th}.

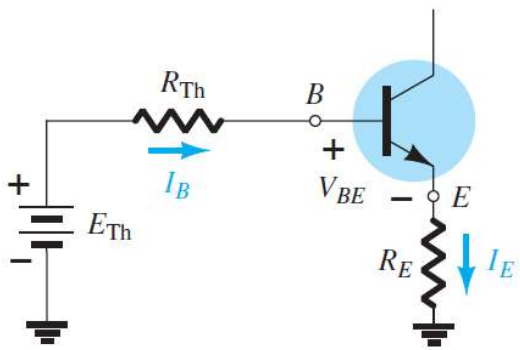


FIG. 4.34
Inserting the Thévenin equivalent circuit.

Approximate Analysis

The input section of the voltage-divider configuration can be represented by the network of Fig. 4.36. The resistance R_i is the equivalent resistance between base and ground for the transistor with an emitter resistor R_E. Recall from Section 4.4 [Eq. (4.18)] that the reflected resistance between base and emitter is defined by $R_i = (\beta + 1)R_E$. If R_i is much larger than the resistance R_2, the current I_B will be much smaller than I_2 (current always seeks the path of least resistance) and I_2 will be approximately equal to I_1. If we accept the approximation that I_B is essentially 0 A compared to I_1 or I_2, then $I_1 = I_2$, and R_1 and R_2 can be considered series elements. The voltage across R_2, which is actually the base voltage, can be

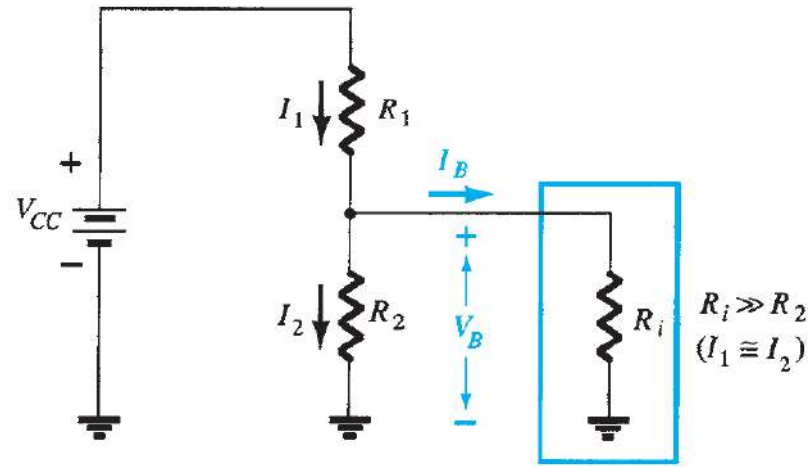


FIG. 4.36
Partial-bias circuit for calculating the approximate base voltage V_B.

determined using the voltage-divider rule (hence the name for the configuration). That is,

$$V_B = \frac{R_2 V_{CC}}{R_1 + R_2} \tag{4.32}$$

Because $R_i = (\beta + 1)R_E \cong \beta R_E$ the condition that will define whether the approximate approach can be applied is

$$\beta R_E \geq 10R_2 \tag{4.33}$$

In other words, if β times the value of R_E is at least 10 times the value of R_2, the approximate approach can be applied with a high degree of accuracy.

Once V_B is determined, the level of V_E can be calculated from

$$V_E = V_B - V_{BE} \tag{4.34}$$

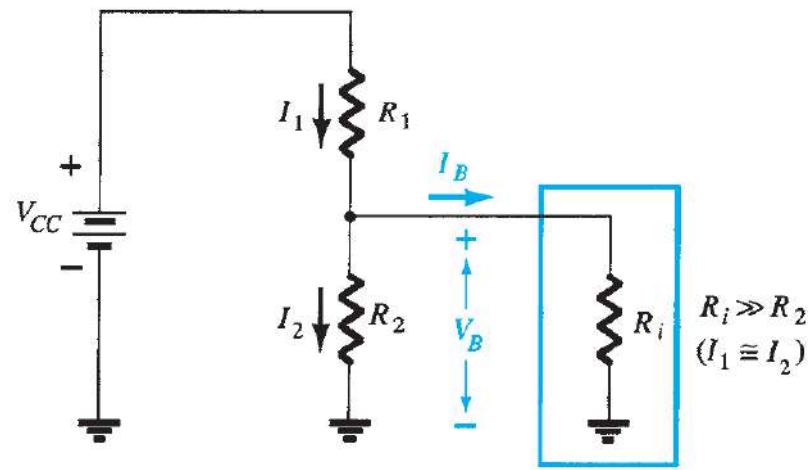


FIG. 4.36
Partial-bias circuit for calculating the approximate base voltage V_B.

and the emitter current can be determined from

$$I_E = \frac{V_E}{R_E} \tag{4.35}$$

and

$$I_{C_Q} \cong I_E \tag{4.36}$$

The collector-to-emitter voltage is determined by

$$V_{CE} = V_{CC} - I_C R_C - I_E R_E$$

but because $I_E \cong I_C$,

$$V_{CE_Q} = V_{CC} - I_C(R_C + R_E) \tag{4.37}$$

Note in the sequence of calculations from Eq. (4.33) through Eq. (4.37) that β does not appear and I_B was not calculated. The Q-point (as determined by I_{C_Q} and V_{CE_Q}) is therefore independent of the value of β.

EXAMPLE 4.9 Repeat the analysis of Fig. 4.35 using the approximate technique, and compare solutions for I_{C_Q} and V_{CE_Q}.

Solution: Testing:

$$\beta R_E \geq 10R_2$$
$$(100)(1.5\,\text{k}\Omega) \geq 10(3.9\,\text{k}\Omega)$$
$$150\,\text{k}\Omega \geq 39\,\text{k}\Omega \ (\textit{satisfied})$$

$$V_B = \frac{R_2 V_{CC}}{R_1 + R_2}$$
$$= \frac{(3.9\,\text{k}\Omega)(22\,\text{V})}{39\,\text{k}\Omega + 3.9\,\text{k}\Omega}$$
$$= 2\,\text{V}$$

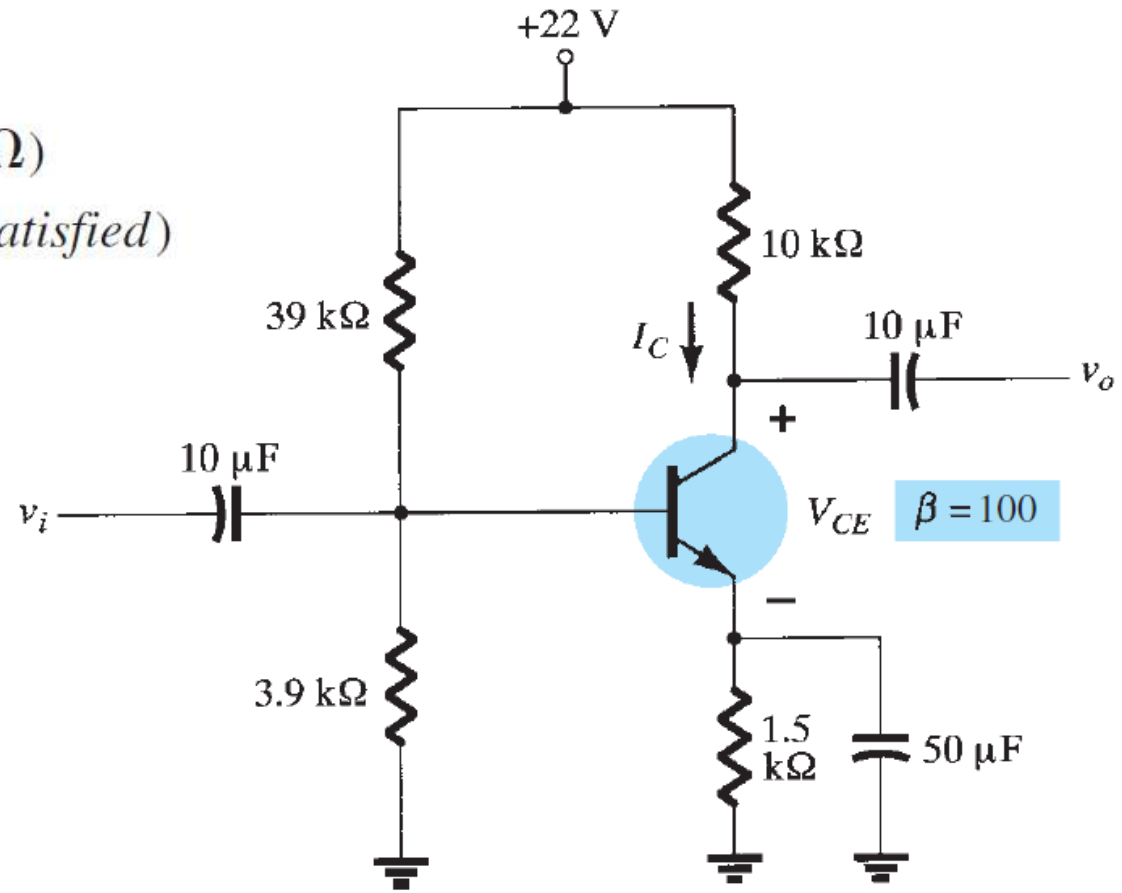


FIG. 4.35
Beta-stabilized circuit for Example 4.8.

Note that the level of V_B is the same as E_{Th} determined in Example 4.7. Essentially, therefore, the primary difference between the exact and approximate techniques is the effect of R_{Th} in the exact analysis that separates E_{Th} and V_B.

Eq. (4.34):
$$\begin{aligned} V_E &= V_B - V_{BE} \\ &= 2\text{ V} - 0.7\text{ V} \\ &= 1.3\text{ V} \end{aligned}$$

$$I_{CQ} \cong I_E = \frac{V_E}{R_E} = \frac{1.3\text{ V}}{1.5\text{ k}\Omega} = \mathbf{0.867\text{ mA}}$$

compared to 0.84 mA with the exact analysis. Finally,

$$\begin{aligned} V_{CE_Q} &= V_{CC} - I_C(R_C + R_E) \\ &= 22\text{ V} - (0.867\text{ mA})(10\text{ kV} + 1.5\text{ k}\Omega) \\ &= 22\text{ V} - 9.97\text{ V} \\ &= \mathbf{12.03\text{ V}} \end{aligned}$$

versus 12.34 V obtained in Example 4.8.

The results for I_{C_Q} and V_{CE_Q} are certainly close, and considering the actual variation in parameter values, one can certainly be considered as accurate as the other. The larger the level of R_i compared to R_2, the closer is the approximate to the exact solution. Example 4.11 will compare solutions at a level well below the condition established by Eq. (4.33).

EXAMPLE 4.10 Repeat the exact analysis of Example 4.8 if β is reduced to 50, and compare solutions for I_{C_Q} and V_{CE_Q}.

Solution: This example is not a comparison of exact versus approximate methods, but a testing of how much the Q-point will move if the level of β is cut in half. R_{Th} and E_{Th} are the same:

$$R_{\text{Th}} = 3.55\ \text{k}\Omega, \qquad E_{\text{Th}} = 2\ \text{V}$$

$$I_B = \frac{E_{\text{Th}} - V_{BE}}{R_{\text{Th}} + (\beta + 1)R_E}$$

$$= \frac{2\ \text{V} - 0.7\ \text{V}}{3.55\ \text{k}\Omega + (51)(1.5\ \text{k}\Omega)} = \frac{1.3\ \text{V}}{3.55\ \text{k}\Omega + 76.5\ \text{k}\Omega}$$

$$= 16.24\ \mu\text{A}$$

$$I_{C_Q} = \beta I_B$$

$$= (50)(16.24\ \mu\text{A})$$

$$= \mathbf{0.81\ mA}$$

$$V_{CE_Q} = V_{CC} - I_C(R_C + R_E)$$

$$= 22\ \text{V} - (0.81\ \text{mA})(10\ \text{k}\Omega + 1.5\ \text{k}\Omega)$$

$$= \mathbf{12.69\ V}$$

Tabulating the results, we have:

Effect of β variation on the response of the voltage-divider configuration of Fig. 4.35.

β	I_{C_Q} (mA)	V_{CE_Q} (V)
100	0.84 mA	12.34 V
50	0.81 mA	12.69 V

The results clearly show the relative insensitivity of the circuit to the change in β. Even though β is drastically cut in half, from 100 to 50, the levels of I_{C_Q} and V_{CE_Q} are essentially the same.

Important Note: Looking back on the results for the fixed-bias configuration, we find the current decreased from 4.71 mA to 2.35 mA when beta dropped from 100 to 50. For the voltage-divider configuration, the same change in beta only resulted in a change in current from 0.84 mA to 0.81 mA. Even more noticeable is the change in V_{CE_Q} for the fixed-bias configuration. Dropping beta from 100 to 50 resulted in an increase in voltage from 1.64 to 6.83 V (a change of over 300%). For the voltage-divider configuration, the increase in voltage was only from 12.34 V to 12.69 V, which is a change of less than 3%. In summary, therefore, changing beta by 50% resulted in a change in an important network parameter of over 300% for the fixed-bias configuration and less than 3% for the voltage-divider configuration—a significant difference.

EXAMPLE 4.11 Determine the levels of I_{C_Q} and V_{CE_Q} for the voltage-divider configuration of Fig. 4.37 using the exact and approximate techniques and compare solutions. In this case, the conditions of Eq. (4.33) *will not be satisfied* and the results will reveal the difference in solution if the criterion of Eq. (4.33) is ignored.

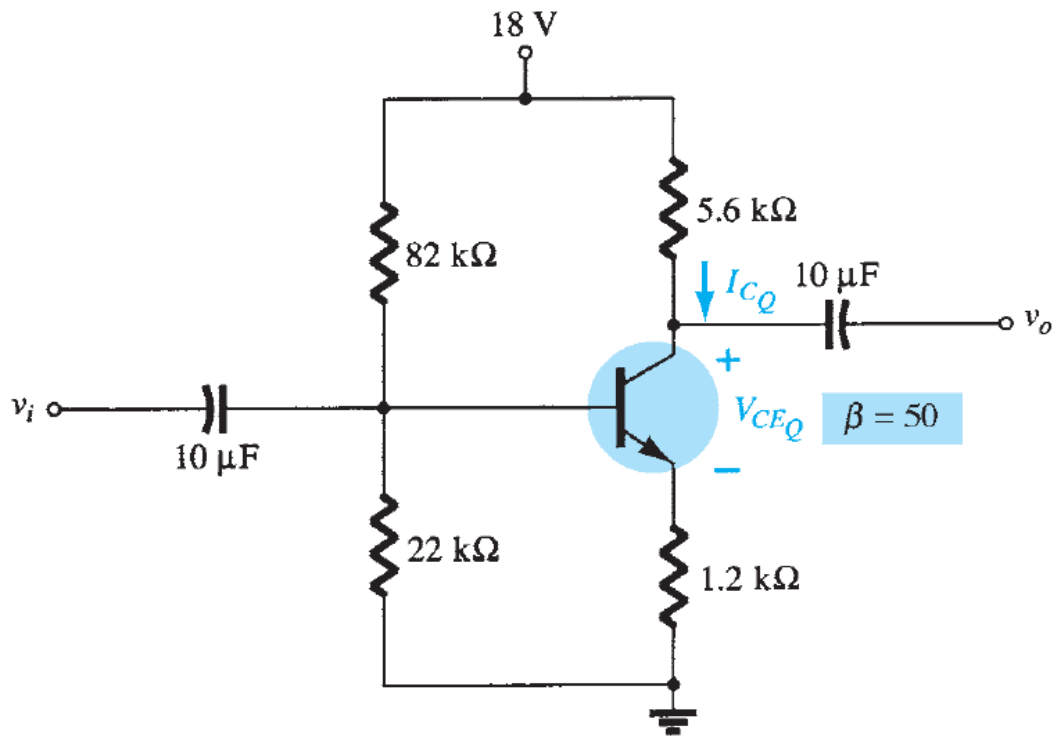


FIG. 4.37
Voltage-divider configuration for Example 4.11.

Solution: Exact analysis:

Eq. (4.33):

$$\beta R_E \geq 10R_2$$

$$(50)(1.2\,\text{k}\Omega) \geq 10(22\,\text{k}\Omega)$$

$$60\,\text{k}\Omega \not\geq 220\,\text{k}\Omega \ (\textit{not satisfied})$$

$$R_{\text{Th}} = R_1 \| R_2 = 82\,\text{k}\Omega \| 22\,\text{k}\Omega = 17.35\,\text{k}\Omega$$

$$E_{\text{Th}} = \frac{R_2 V_{CC}}{R_1 + R_2} = \frac{22\,\text{k}\Omega(18\,\text{V})}{82\,\text{k}\Omega + 22\,\text{k}\Omega} = 3.81\,\text{V}$$

$$I_B = \frac{E_{\text{Th}} - V_{BE}}{R_{\text{Th}} + (\beta + 1)R_E} = \frac{3.81\,\text{V} - 0.7\,\text{V}}{17.35\,\text{k}\Omega + (51)(1.2\,\text{k}\Omega)} = \frac{3.11\,\text{V}}{78.55\,\text{k}\Omega} = 39.6\,\mu\text{A}$$

$$I_{C_Q} = \beta I_B = (50)(39.6\,\mu\text{A}) = \mathbf{1.98\,mA}$$

$$V_{CE_Q} = V_{CC} - I_C(R_C + R_E)$$

$$= 18\,\text{V} - (1.98\,\text{mA})(5.6\,\text{k}\Omega + 1.2\,\text{k}\Omega)$$

$$= \mathbf{4.54\,V}$$

Approximate analysis:

$$V_B = E_{\text{Th}} = 3.81\ \text{V}$$

$$V_E = V_B - V_{BE} = 3.81\ \text{V} - 0.7\ \text{V} = 3.11\ \text{V}$$

$$I_{C_Q} \cong I_E = \frac{V_E}{R_E} = \frac{3.11\ \text{V}}{1.2\ \text{k}\Omega} = \mathbf{2.59\ mA}$$

$$V_{CE_Q} = V_{CC} - I_C(R_C + R_E)$$

$$= 18\ \text{V} - (2.59\ \text{mA})(5.6\ \text{k}\Omega + 1.2\ \text{k}\Omega)$$

$$= \mathbf{3.88\ V}$$

Tabulating the results, we have:

Comparing the exact and approximate approaches.

	I_{C_Q} (mA)	V_{CE_Q} (V)
Exact	1.98	4.54
Approximate	2.59	3.88

The results reveal the difference between exact and approximate solutions. I_{C_Q} is about 30% greater with the approximate solution, whereas V_{CE_Q} is about 10% less. The results are notably different in magnitude, but even though βR_E is only about three times larger than R_2, the results are still relatively close to each other. For the future, however, our analysis will be dictated by Eq. (4.33) to ensure a close similarity between exact and approximate solutions.

Transistor Saturation

The output collector–emitter circuit for the voltage-divider configuration has the same appearance as the emitter-biased circuit analyzed in Section 4.4. The resulting equation for the saturation current (when V_{CE} is set to 0 V on the schematic) is therefore the same as obtained for the emitter-biased configuration. That is,

$$I_{C_{sat}} = I_{C_{max}} = \frac{V_{CC}}{R_C + R_E} \tag{4.38}$$

Load-Line Analysis

The similarities with the output circuit of the emitter-biased configuration result in the same intersections for the load line of the voltage-divider configuration. The load line will therefore have the same appearance as that of Fig. 4.25, with

$$I_C = \left.\frac{V_{CC}}{R_C + R_E}\right|_{V_{CE}=0\text{ V}} \tag{4.39}$$

and

$$V_{CE} = V_{CC}\big|_{I_C=0\text{ mA}} \tag{4.40}$$

The level of I_B is of course determined by a different equation for the voltage-divider bias and the emitter-bias configurations.

4.6 COLLECTOR FEEDBACK CONFIGURATION

An improved level of stability can also be obtained by introducing a feedback path from collector to base as shown in Fig. 4.38. Although the Q-point is not totally independent of beta (even under approximate conditions), the sensitivity to changes in beta or temperature variations is normally less than encountered for the fixed-bias or emitter-biased configurations. The analysis will again be performed by first analyzing the base–emitter loop, with the results then applied to the collector–emitter loop.

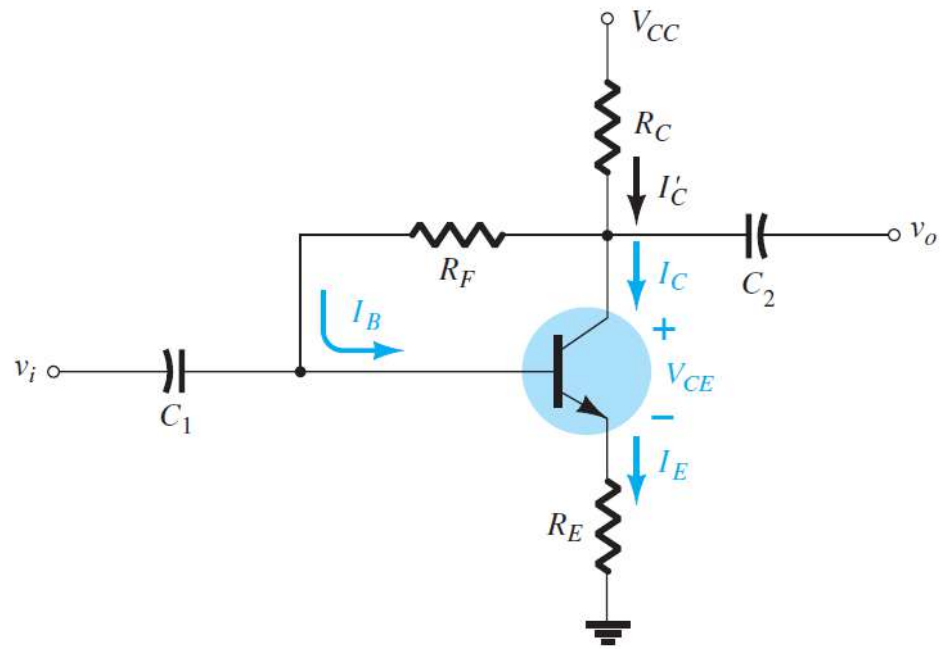


FIG. 4.38
DC bias circuit with voltage feedback.

Base–Emitter Loop

Figure 4.39 shows the base–emitter loop for the voltage feedback configuration. Writing Kirchhoff's voltage law around the indicated loop in the clockwise direction will result in

$$V_{CC} - I'_C R_C - I_B R_F - V_{BE} - I_E R_E = 0$$

It is important to note that the current through R_C is not I_C, but I'_C (where $I'_C = I_C + I_B$). However, the level of I_C and I'_C far exceeds the usual level of I_B, and the approximation $I'_C \cong I_C$ is normally employed. Substituting $I'_C \cong I_C = \beta I_B$ and $I_E \cong I_C$

$$V_{CC} - \beta I_B R_C - I_B R_F - V_{BE} - \beta I_B R_E = 0$$

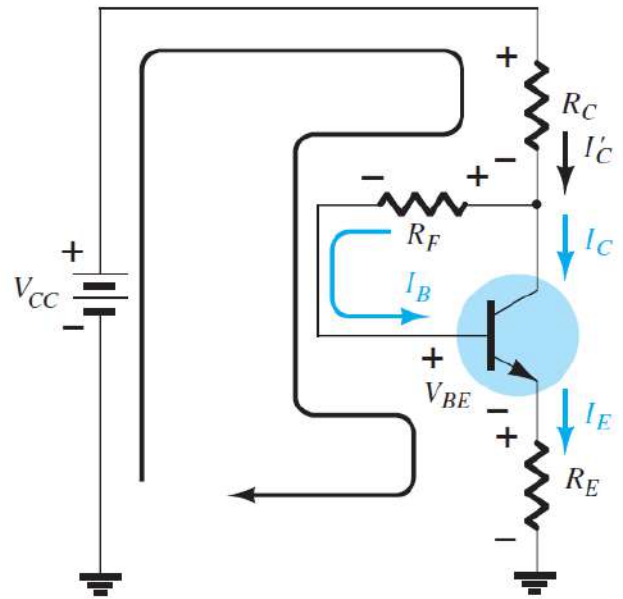


FIG. 4.39

Base–emitter loop for the network of Fig. 4.38.

Gathering terms, we have

$$V_{CC} - V_{BE} - \beta I_B(R_C + R_E) - I_B R_F = 0$$

and solving for I_B yields

$$I_B = \frac{V_{CC} - V_{BE}}{R_F + \beta(R_C + R_E)} \tag{4.41}$$

The result is quite interesting in that the format is very similar to equations for I_B obtained for earlier configurations. The numerator is again the difference of available voltage levels, whereas the denominator is the base resistance plus the collector and emitter resistors reflected by beta. In general, therefore, the feedback path results in a reflection of the resistance R_C back to the input circuit, much like the reflection of R_E.

In general, the equation for I_B has the following format, which can be compared with the result for the fixed-bias and emitter-bias configurations.

$$I_B = \frac{V'}{R_F + \beta R'}$$

$$I_B = \frac{V'}{R_F + \beta R'}$$

For the fixed-bias configuration $\beta R'$ does not exist. For the emitter-bias setup (with $\beta + 1 \cong \beta$), $R' = R_E + R_C$

Because $I_C = \beta I_B$,

$$I_{C_Q} = \frac{\beta V'}{R_F + \beta R'} = \frac{V'}{\frac{R_F}{\beta} + R'}$$

In general, the larger R' is compared with $\frac{R_F}{\beta}$, the more accurate the approximation that

$$I_{C_Q} \cong \frac{V'}{R'}$$

The result is an equation absent of β, which would be very stable for variations in β. Because R' is typically larger for the voltage-feedback configuration than for the emitter-bias configuration, the sensitivity to variations in beta is less. Of course, R' is 0 Ω for the fixed-bias configuration and is therefore quite sensitive to variations in beta.

Collector–Emitter Loop

The collector–emitter loop for the network of Fig. 4.38 is provided in Fig. 4.40. Applying Kirchhoff's voltage law around the indicated loop in the clockwise direction results in

$$I_E R_E + V_{CE} + I'_C R_C - V_{CC} = 0$$

Because $I'_C \cong I_C$ and $I_E \cong I_C$, we have

$$I_C(R_C + R_E) + V_{CE} - V_{CC} = 0$$

and

$$\boxed{V_{CE} = V_{CC} - I_C(R_C + R_E)} \qquad \textbf{(4.42)}$$

which is exactly as obtained for the emitter-bias and voltage-divider bias configurations.

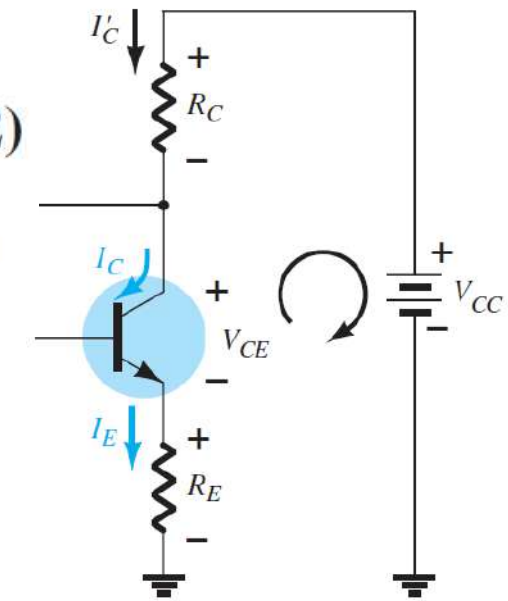


FIG. 4.40
Collector–emitter loop for the network of Fig. 4.38.

EXAMPLE 4.12 Determine the quiescent levels of I_{C_Q} and V_{CE_Q} for the network of Fig. 4.41.

Solution: Eq. (4.41):

$$I_B = \frac{V_{CC} - V_{BE}}{R_F + \beta(R_C + R_E)}$$

$$= \frac{10\text{ V} - 0.7\text{ V}}{250\text{ k}\Omega + (90)(4.7\text{ k}\Omega + 1.2\text{ k}\Omega)}$$

$$= \frac{9.3\text{ V}}{250\text{ k}\Omega + 531\text{ k}\Omega} = \frac{9.3\text{ V}}{781\text{ k}\Omega}$$

$$= 11.91\ \mu\text{A}$$

$$I_{C_Q} = \beta I_B = (90)(11.91\ \mu\text{A})$$

$$= \mathbf{1.07\text{ mA}}$$

$$V_{CE_Q} = V_{CC} - I_C(R_C + R_E)$$

$$= 10\text{ V} - (1.07\text{ mA})(4.7\text{ k}\Omega + 1.2\text{ k}\Omega)$$

$$= 10\text{ V} - 6.31\text{ V}$$

$$= \mathbf{3.69\text{ V}}$$

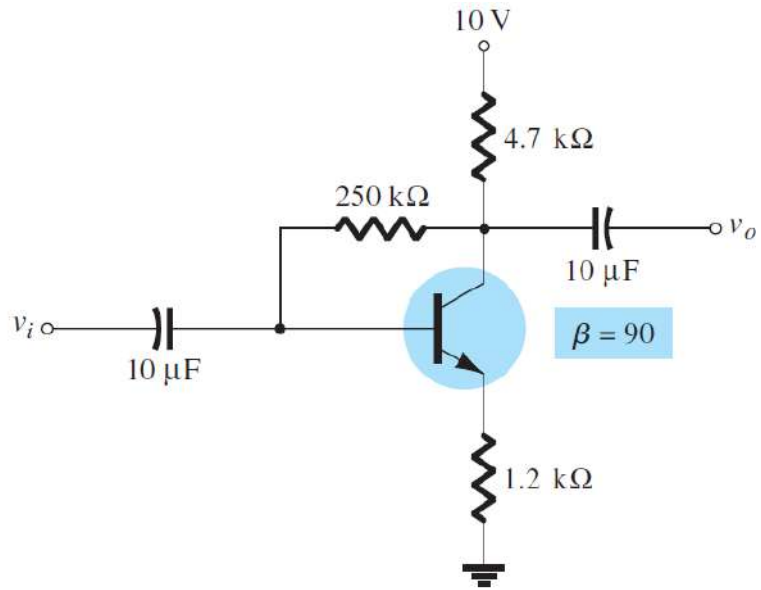


FIG. 4.41
Network for Example 4.12.

EXAMPLE 4.13 Repeat Example 4.12 using a beta of 135 (50% greater than in Example 4.12).

Solution: It is important to note in the solution for I_B in Example 4.12 that the second term in the denominator of the equation is much larger than the first. Recall in a recent discussion that the larger this second term is compared to the first, the less is the sensitivity to changes in beta. In this example, the level of beta is increased by 50%, which will increase the magnitude of this second term even more compared to the first. It is more important to note in these examples, however, that once the second term is relatively large compared to the first, the sensitivity to changes in beta is significantly less.

$$I_B = \frac{V_{CC} - V_{BE}}{R_F + \beta(R_C + R_E)}$$

Solving for I_B gives

$$I_B = \frac{V_{CC} - V_{BE}}{R_B + \beta(R_C + R_E)}$$

$$= \frac{10\text{ V} - 0.7\text{ V}}{250\text{ k}\Omega + (135)(4.7\text{ k}\Omega + 1.2\text{ k}\Omega)}$$

$$= \frac{9.3\text{ V}}{250\text{ k}\Omega + 796.5\text{ k}\Omega} = \frac{9.3\text{ V}}{1046.5\text{ k}\Omega}$$

$$= 8.89\ \mu\text{A}$$

and

$$I_{C_Q} = \beta I_B$$

$$= (135)(8.89\ \mu\text{A})$$

$$= \mathbf{1.2\text{ mA}}$$

and

$$V_{CE_Q} = V_{CC} - I_C(R_C + R_E)$$

$$= 10\text{ V} - (1.2\text{ mA})(4.7\text{ k}\Omega + 1.2\text{ k}\Omega)$$

$$= 10\text{ V} - 7.08\text{ V}$$

$$= \mathbf{2.92\text{ V}}$$

Even though the level of β increased 50%, the level of I_{C_Q} only increased 12.1%, whereas the level of V_{CE_Q} decreased about 20.9%. If the network were a fixed-bias design, a 50% increase in β would have resulted in a 50% increase in I_{C_Q} and a dramatic change in the location of the Q-point.

EXAMPLE 4.14 Determine the dc level of I_B and V_C for the network of Fig. 4.42.

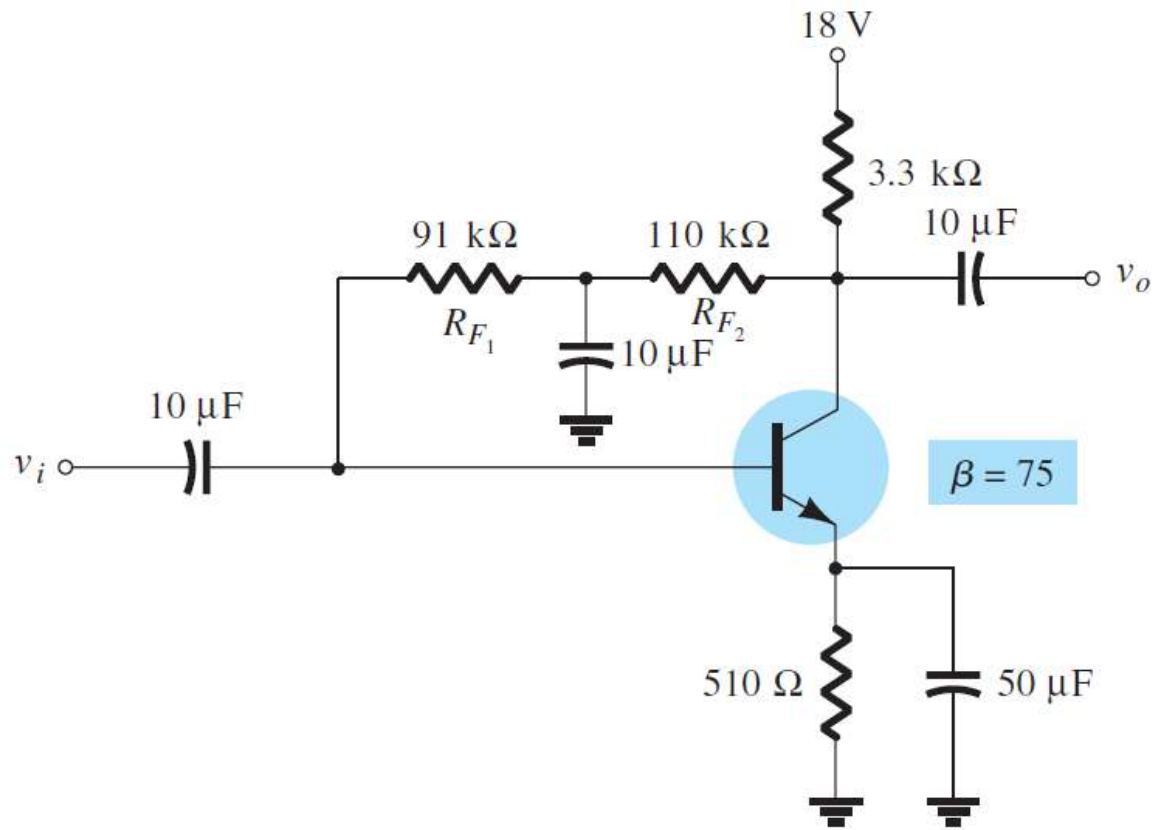


FIG. 4.42
Network for Example 4.14.

Solution: In this case, the base resistance for the dc analysis is composed of two resistors with a capacitor connected from their junction to ground. For the dc mode, the capacitor assumes the open-circuit equivalence, and $R_B = R_{F_1} + R_{F_2}$.

Solving for I_B gives

$$I_B = \frac{V_{CC} - V_{BE}}{R_B + \beta(R_C + R_E)}$$

$$= \frac{18\text{ V} - 0.7\text{ V}}{(91\text{ k}\Omega + 110\text{ k}\Omega) + (75)(3.3\text{ k}\Omega + 0.51\text{ k}\Omega)}$$

$$= \frac{17.3\text{ V}}{201\text{ k}\Omega + 285.75\text{ k}\Omega} = \frac{17.3\text{ V}}{486.75\text{ k}\Omega}$$

$$= \mathbf{35.5\ \mu A}$$

$$I_C = \beta I_B$$

$$= (75)(35.5\ \mu\text{A})$$

$$= 2.66\text{ mA}$$

$$V_C = V_{CC} - I'_C R_C \cong V_{CC} - I_C R_C$$

$$= 18\text{ V} - (2.66\text{ mA})(3.3\text{ k}\Omega)$$

$$= 18\text{ V} - 8.78\text{ V}$$

$$= \mathbf{9.22\text{ V}}$$

Saturation Conditions

Using the approximation $I_C' = I_C$, we find that the equation for the saturation current is the same as obtained for the voltage-divider and emitter-bias configurations. That is,

$$I_{C_{sat}} = I_{C_{max}} = \frac{V_{CC}}{R_C + R_E} \tag{4.43}$$

Load-Line Analysis

Continuing with the approximation $I_C' = I_C$ results in the same load line defined for the voltage-divider and emitter-biased configurations. The level of I_{B_Q} is defined by the chosen bias configuration.

EXAMPLE 4.15 Given the network of Fig. 4.43 and the BJT characteristics of Fig. 4.44.

a. Draw the load line for the network on the characteristics.
b. Determine the dc beta in the center region of the characteristics. Define the choser point as the Q-point.
c. Using the dc beta calculated in part b, find the dc value of I_B.
d. Find I_{C_Q} and I_{CE_Q}.

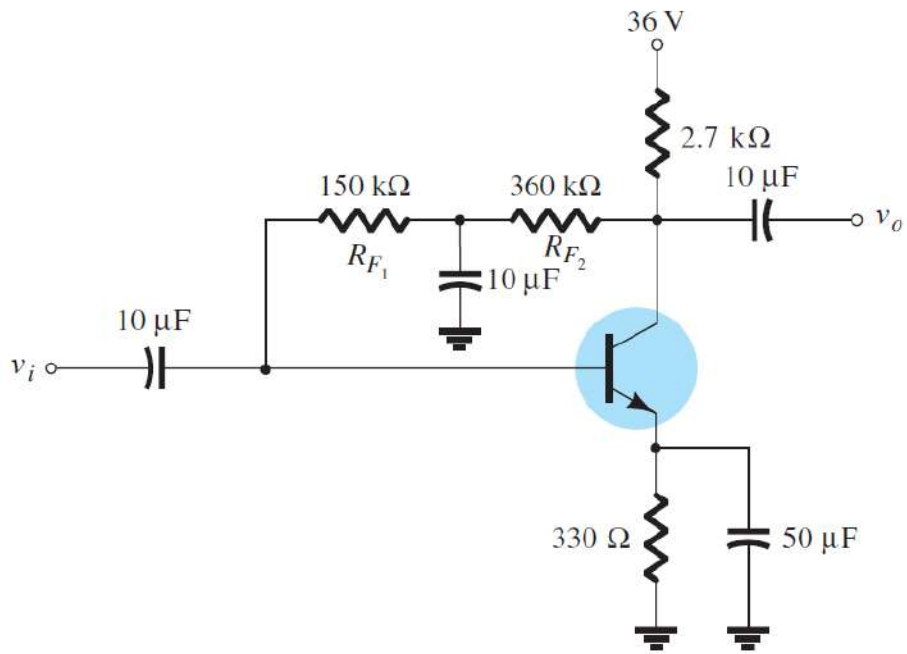


FIG. 4.43
Network for Example 4.15.

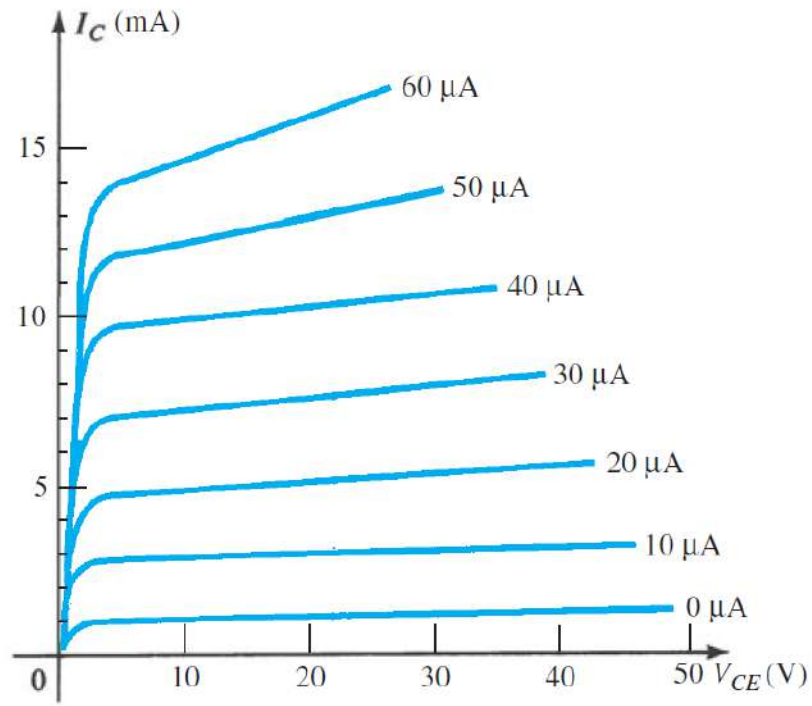


FIG. 4.44
BJT characteristics.

Solution:

a. The load line is drawn on Fig. 4.45 as determined by the following intersections:

$$V_{CE} = 0\text{ V}: I_C = \frac{V_{CC}}{R_C + R_E} = \frac{36\text{ V}}{2.7\text{ k}\Omega + 330\ \Omega} = \mathbf{11.88\ mA}$$

$$I_C = 0\text{ mA}: V_{CE} = V_{CC} = \mathbf{36\ V}$$

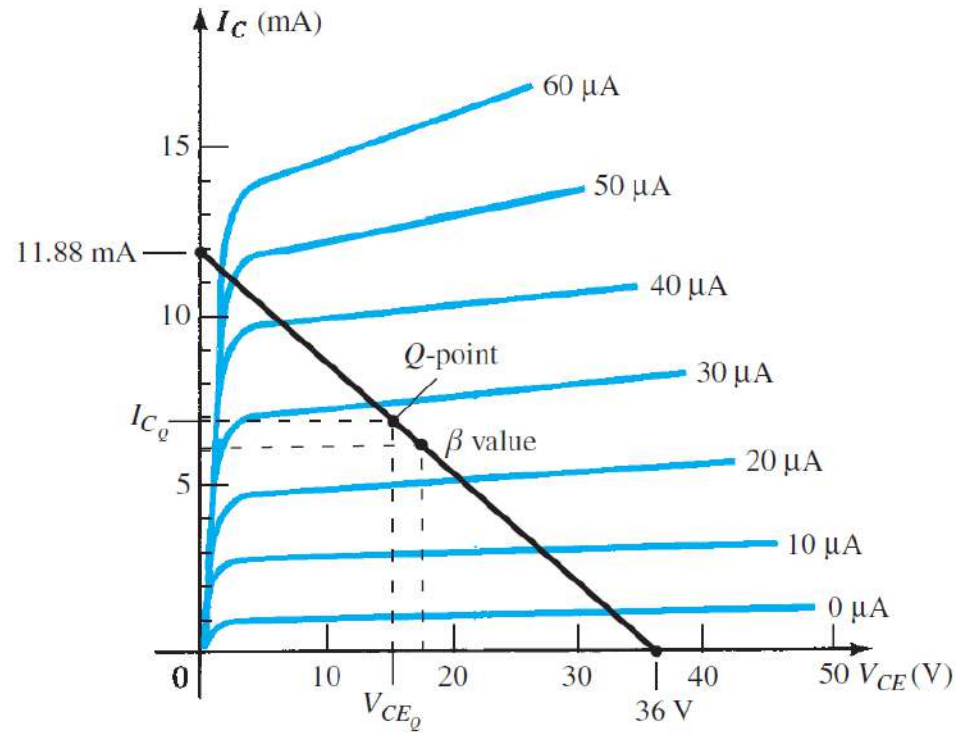


FIG. 4.45

Defining the Q-point for the voltage-divider bias configuration of Fig. 4.43.

b. The dc beta was determined using $I_B = 25\ \mu\text{A}$ and V_{CE} about 17 V.

$$\beta \cong \frac{I_{C_Q}}{I_{B_Q}} = \frac{6.2\ \text{mA}}{25\ \mu\text{A}} = \mathbf{248}$$

c. Using Eq. 4.41:

$$I_B = \frac{V_{CC} - V_{BE}}{R_B + \beta(R_C + R_E)} = \frac{36\ \text{V} - 0.7\ \text{V}}{510\ \text{k}\Omega + 248(2.7\ \text{k}\Omega + 330\ \Omega)}$$

$$= \frac{35.3\ \text{V}}{510\ \text{k}\Omega + 751.44\ \text{k}\Omega}$$

and $I_B = \dfrac{35.3\ \text{V}}{1.261\ \text{M}\Omega} = \mathbf{28\ \mu A}$

d. From Fig. 4.45 the quiescent values are

$I_{C_Q} \cong \mathbf{6.9\ mA}$ and $V_{CE_Q} \cong \mathbf{15\ V}$

4.7 EMITTER-FOLLOWER CONFIGURATION

The previous sections introduced configurations in which the output voltage is typically taken off the collector terminal of the BJT. This section will examine a configuration where the output is taken off the emitter terminal as shown in Fig. 4.46. The configuration of Fig. 4.46 is not the only one where the output can be taken off the emitter terminal. In fact, any of the configurations just described can be used so long as there is a resistor in the emitter leg.

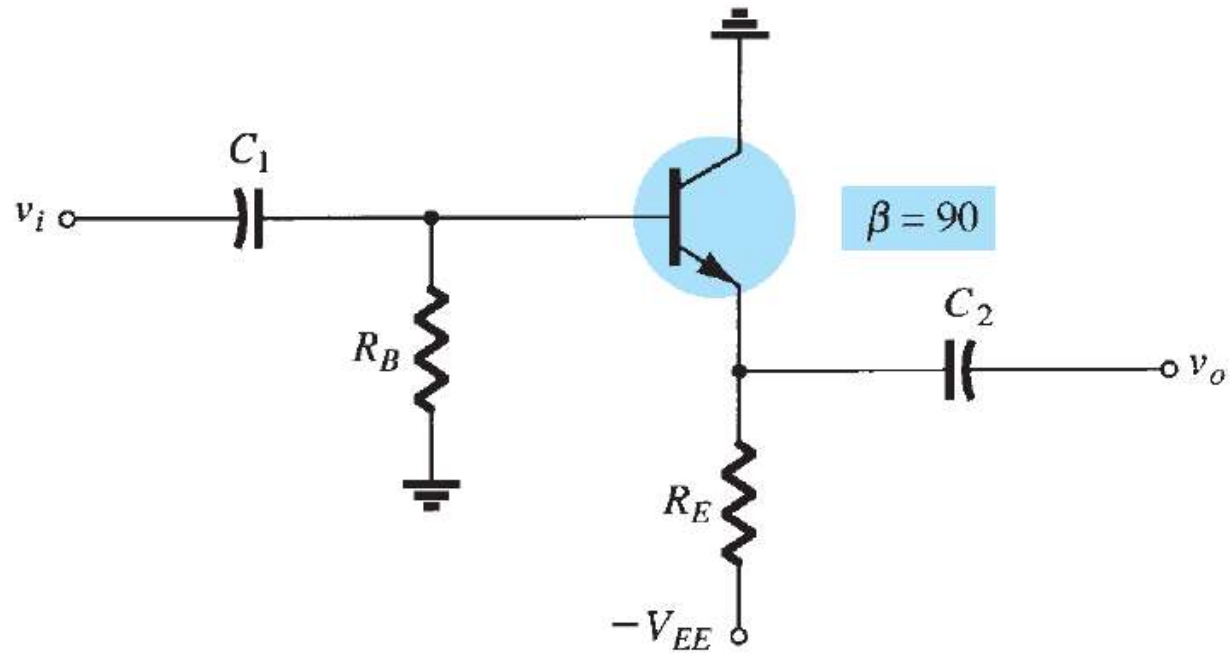


FIG. 4.46
Common-collecter (emitter-follower) configuration.

The dc equivalent of the network of Fig. 4.46 appears in Fig. 4.47

Applying Kirchhoff's voltage rule to the input circuit will result in

$$-I_B R_B - V_{BE} - I_E R_E + V_{EE} = 0$$

and using $I_E = (\beta + 1)I_B$

$$I_B R_B + (\beta + 1)I_B R_E = V_{EE} - V_{BE}$$

so that

$$\boxed{I_B = \frac{V_{EE} - V_{BE}}{R_B + (\beta + 1)R_E}} \qquad \textbf{(4.44)}$$

For the output network, an application of Kirchhoff's voltage law will result in

$$-V_{CE} - I_E R_E + V_{EE} = 0$$

and

$$\boxed{V_{CE} = V_{EE} - I_E R_E} \qquad (4$$

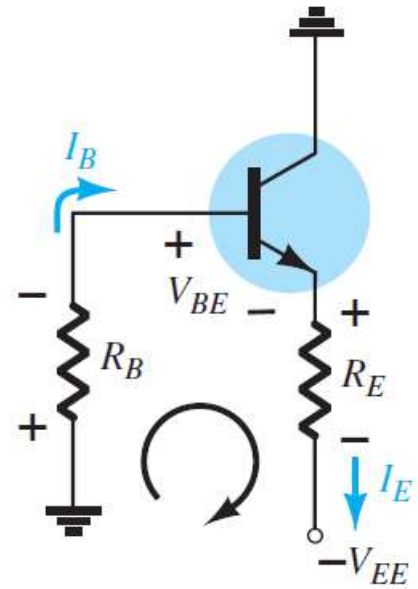


FIG. 4.47
dc equivalent of Fig. 4.46.

EXAMPLE 4.16 Determine V_{CE_Q} and I_{E_Q} for the network of Fig. 4.48.

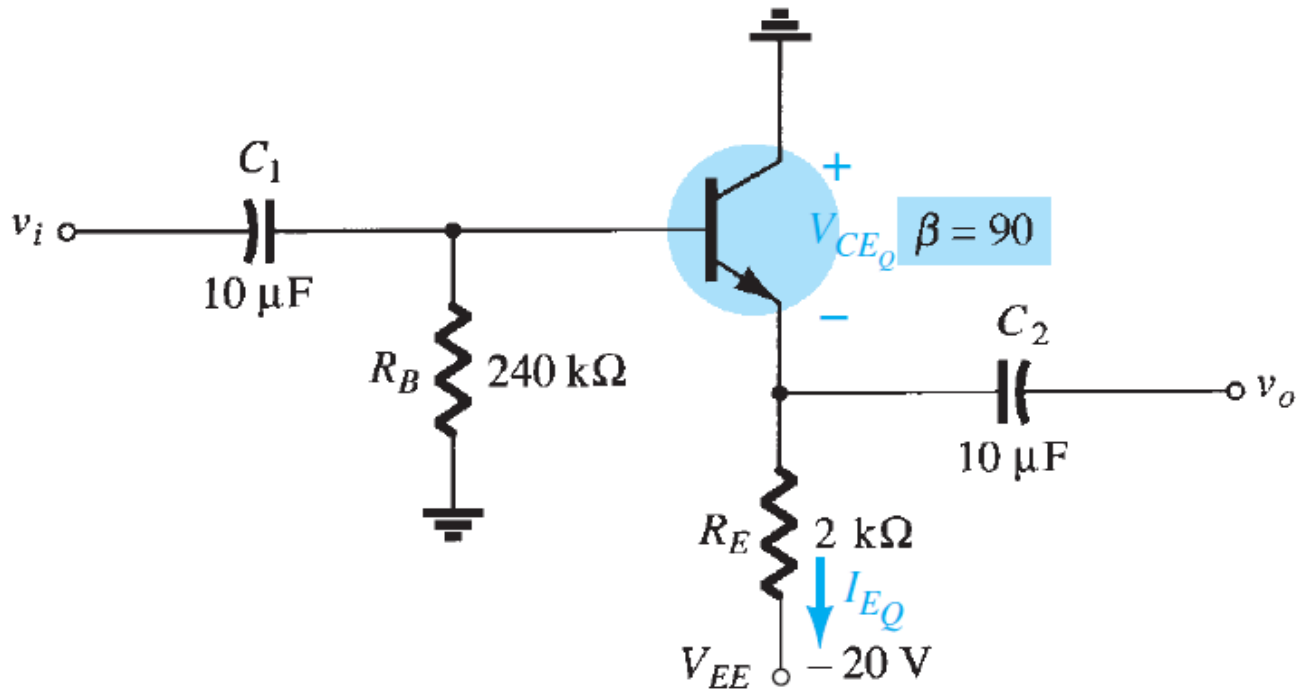


FIG. 4.48
Example 4.16.

Solution:

Eq. 4.44:
$$I_B = \frac{V_{EE} - V_{BE}}{R_B + (\beta + 1)R_E}$$
$$= \frac{20\text{ V} - 0.7\text{ V}}{240\text{ k}\Omega + (90 + 1)2\text{ k}\Omega} = \frac{19.3\text{ V}}{240\text{ k}\Omega + 182\text{ k}\Omega}$$
$$= \frac{19.3\text{ V}}{422\text{ k}\Omega} = 45.73\ \mu\text{A}$$

and Eq. 4.45:
$$V_{CE_Q} = V_{EE} - I_E R_E$$
$$= V_{EE} - (\beta + 1)I_B R_E$$
$$= 20\text{ V} - (90 + 1)(45.73\ \mu\text{A})(2\text{ k}\Omega)$$
$$= 20\text{ V} - 8.32\text{ V}$$
$$= \mathbf{11.68\text{ V}}$$
$$I_{E_Q} = (\beta + 1)I_B = (91)(45.73\ \mu\text{A})$$
$$= 4.16\text{ mA}$$

4.8 COMMON-BASE CONFIGURATION

The common-base configuration is unique in that the applied signal is connected to the emitter terminal and the base is at, or just above, ground potential. It is a fairly popular configuration because in the ac domain it has a very low input impedance, high output impedance, and good gain.

A typical common-base configuration appears in Fig. 4.49. Note that two supplies are used in this configuration and the base is the common terminal between the input emitter terminal and output collector terminal.

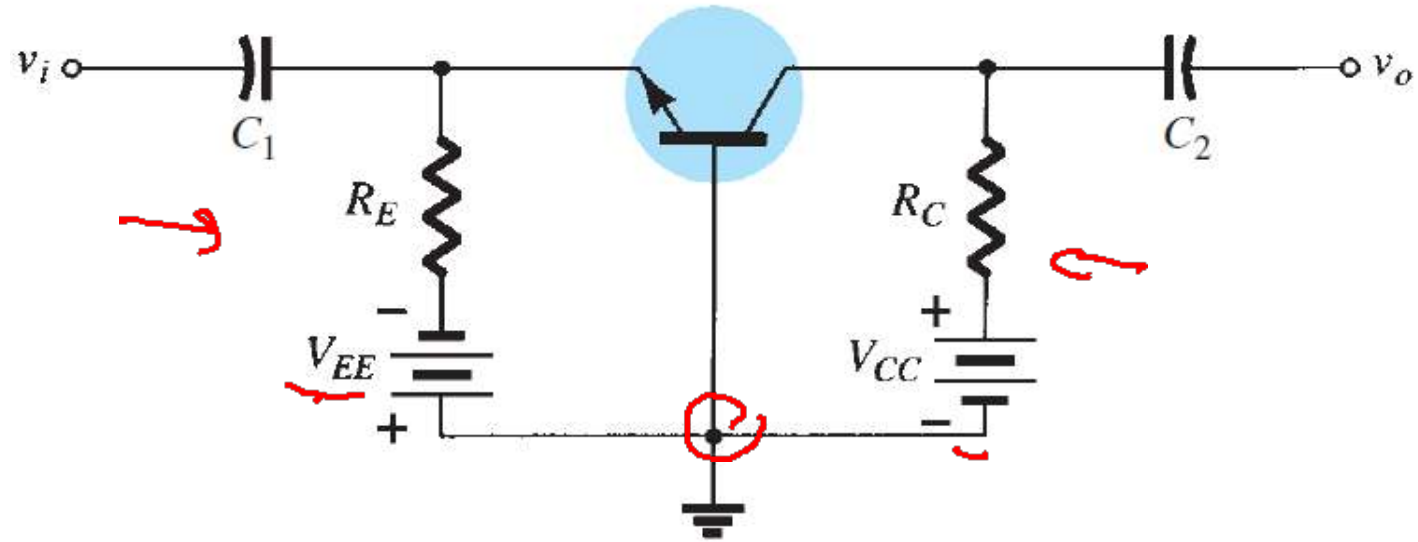


FIG. 4.49
Common-base configuration.

The dc equivalent of the input side of Fig. 4.49 appears in Fig. 4.50.

Applying Kirchhoff's voltage law will result in

$$-V_{EE} + I_E R_E + V_{BE} = 0$$

$$I_E = \frac{V_{EE} - V_{BE}}{R_E} \qquad \textbf{(4.46)}$$

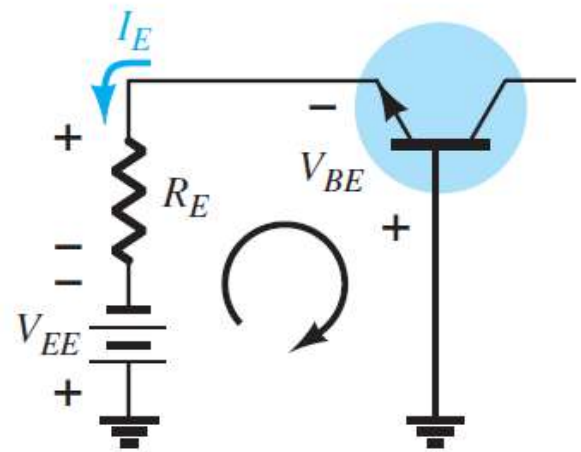


FIG. 4.50
Input dc equivalent of Fig. 4.49.

Applying Kirchhoff's voltage law to the entire outside perimeter of the network of Fig. 4.51 will result in

$$-V_{EE} + I_E R_E + V_{CE} + I_C R_C - V_{CC} = 0$$

and solving for V_{CE}:

$$V_{CE} = V_{EE} + V_{CC} - I_E R_E - I_C R_C$$

Because

$$I_E \cong I_C$$

$$\boxed{V_{CE} = V_{EE} + V_{CC} - I_E(R_C + R_E)} \qquad \textbf{(4.47)}$$

The voltage V_{CB} of Fig. 4.51 can be found by applying Kirchhoff's voltage law to the output loop of Fig 4.51 to obtain:

$$V_{CB} + I_C R_C - V_{CC} = 0$$

or

$$V_{CB} = V_{CC} - I_C R_C$$

Using

$$I_C \cong I_E$$

we have

$$\boxed{V_{CB} = V_{CC} - I_C R_C}$$

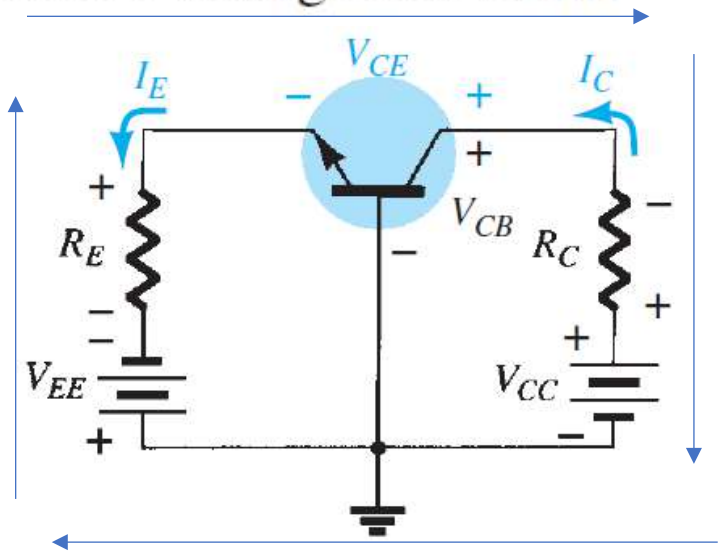


FIG. 4.51
Determining V_{CE} and V_{CB}.

EXAMPLE 4.17 Determine the currents I_E and I_B and the voltages V_{CE} and V_{CB} for the common-base configuration of Fig. 4.52.

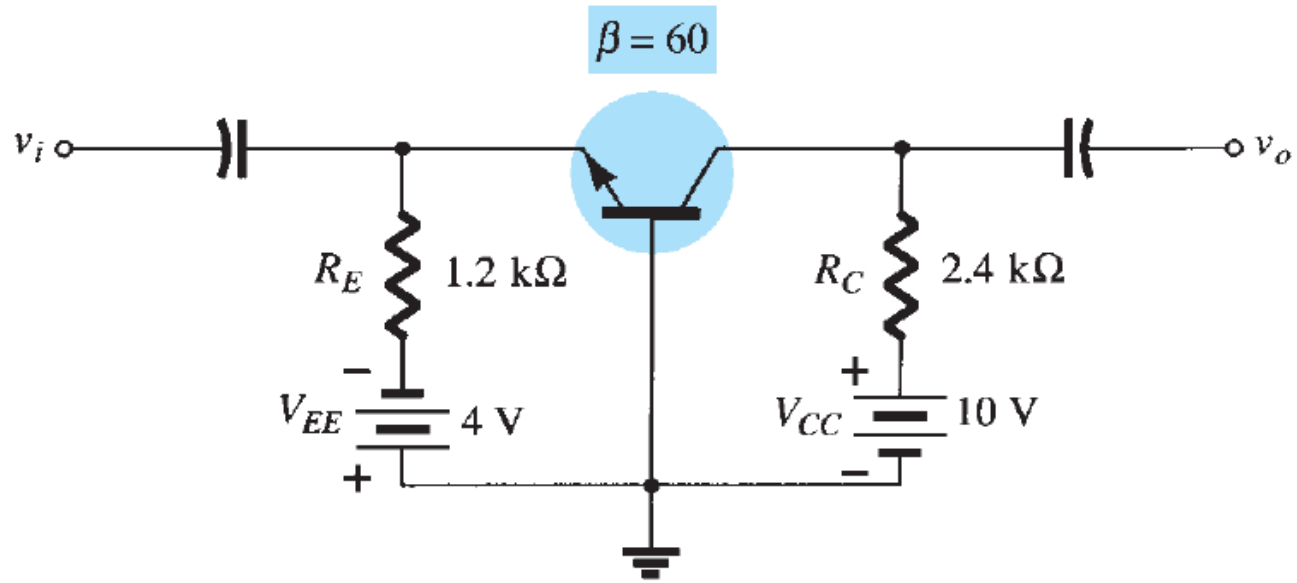


FIG. 4.52
Example 4.17.

Solution: Eq. 4.46:

$$I_E = \frac{V_{EE} - V_{BE}}{R_E}$$

$$= \frac{4\text{ V} - 0.7\text{ V}}{1.2\text{ k}\Omega} = \mathbf{2.75\text{ mA}}$$

$$I_B = \frac{I_E}{\beta + 1} = \frac{2.75\text{ mA}}{60 + 1} = \frac{2.75\text{ mA}}{61}$$

$$= \mathbf{45.08\ \mu A}$$

Eq. 4.47:

$$V_{CE} = V_{EE} + V_{CC} - I_E(R_C + R_E)$$

$$= 4\text{ V} + 10\text{ V} - (2.75\text{ mA})(2.4\text{ k}\Omega + 1.2\text{ k}\Omega)$$

$$= 14\text{ V} - (2.75\text{ mA})(3.6\text{ k}\Omega)$$

$$= 14\text{ V} - 9.9\text{ V}$$

$$= \mathbf{4.1\text{ V}}$$

Eq. 4.48:

$$V_{CB} = V_{CC} - I_C R_C = V_{CC} - \beta I_B R_C$$

$$= 10\text{ V} - (60)(45.08\ \mu\text{A})(24\text{ k}\Omega)$$

$$= 10\text{ V} - 6.49\text{ V}$$

$$= \mathbf{3.51\text{ V}}$$

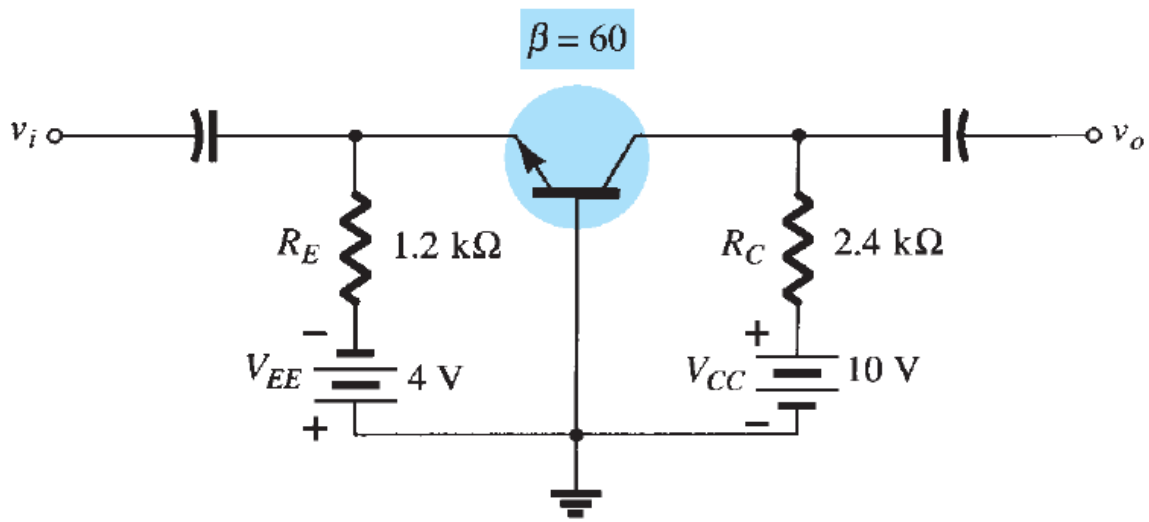
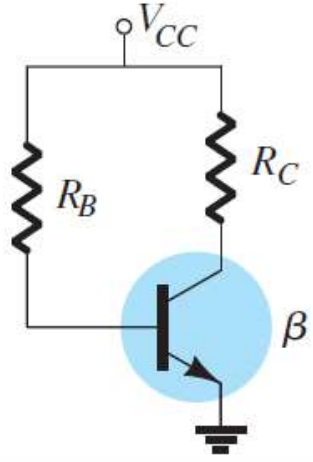
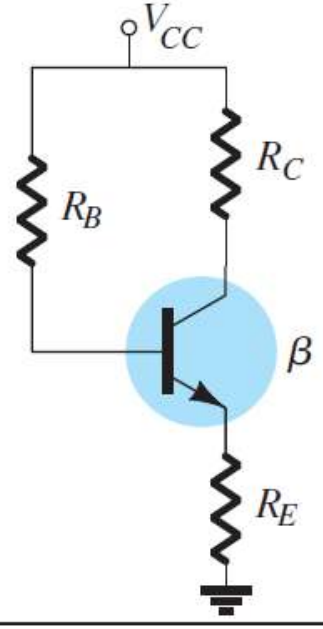
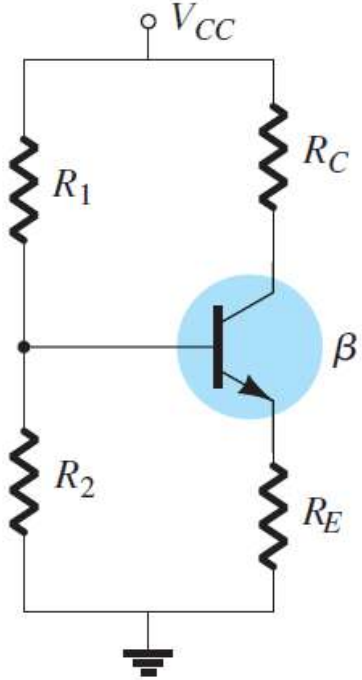


TABLE 4.1

BJT Bias Configurations

Type	Configuration	Pertinent Equations
Fixed-bias		$I_B = \frac{V_{CC} - V_{BE}}{R_B}$ $I_C = \beta I_B, I_E = (\beta + 1)I_B$ $V_{CE} = V_{CC} - I_C R_C$
Emitter-bias		$I_B = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1)R_E}$ $I_C = \beta I_B, I_E = (\beta + 1)I_B$ $R_i = (\beta + 1)R_E$ $V_{CE} = V_{CC} - I_C (R_C + R_E)$

Voltage-divider bias



EXACT: $R_{Th} = R_1 \| R_2, E_{Th} = \dfrac{R_2 V_{CC}}{R_1 + R_2}$

$$I_B = \frac{E_{Th} - V_{BE}}{R_{Th} + (\beta + 1)R_E}$$

$$I_C = \beta I_B, I_E = (\beta + 1) I_B$$

$$V_{CE} = V_{CC} - I_C (R_C + R_E)$$

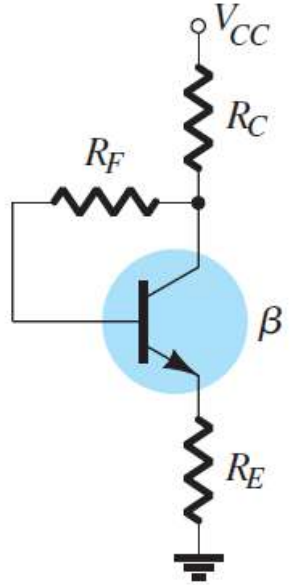
APPROXIMATE: $\beta R_E \geq 10 R_2$

$$V_B = \frac{R_2 V_{CC}}{R_1 + R_2}, V_E = V_B - V_{BE}$$

$$I_E = \frac{V_E}{R_E}, I_B = \frac{I_E}{\beta + 1}$$

$$V_{CE} = V_{CC} - I_C (R_C + R_E)$$

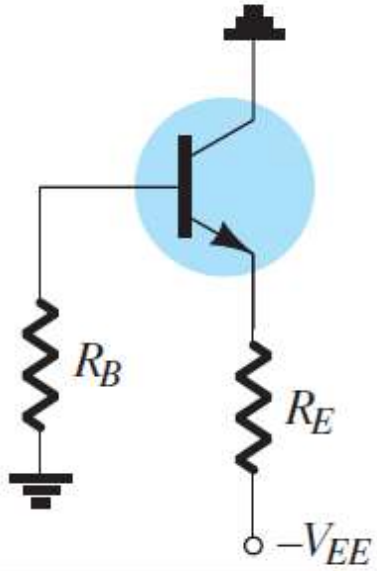
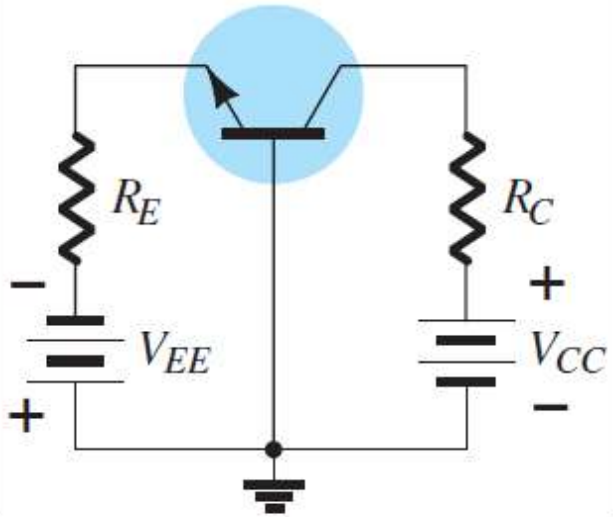
Collector-feedback



$$I_B = \frac{V_{CC} - V_{BE}}{R_F + \beta(R_C + R_E)}$$

$$I_C = \beta I_B, I_E = (\beta + 1) I_B$$

$$V_{CE} = V_{CC} - I_C (R_C + R_E)$$

Emitter-follower		$I_B = \dfrac{V_{EE} - V_{BE}}{R_B + (\beta + 1)R_E}$ $I_C = \beta I_B, I_E = (\beta + 1)I_B$ $V_{CE} = V_{EE} - I_E R_E$
Common-base		$I_E = \dfrac{V_{EE} - V_{BE}}{R_E}$ $I_B = \dfrac{I_E}{\beta + 1}, I_C = \beta I_B$ $V_{CE} = V_{EE} + V_{CC} - I_E(R_C + R_E)$ $V_{CB} = V_{CC} - I_C R_C$

4.9 MISCELLANEOUS BIAS CONFIGURATIONS

There are a number of BJT bias configurations that do not match the basic mold of those analyzed in the previous sections. In fact, there are variations in design that would require many more pages than is possible in a single publication. However, the primary purpose here is to emphasize those characteristics of the device that permit a dc analysis of the configuration and to establish a general procedure toward the desired solution. For each configuration discussed thus far, the first step has been the derivation of an expression for the base current. Once the base current is known, the collector current and voltage levels of the output circuit can be determined quite directly. This is not to imply that all solutions will take this path, but it does suggest a possible route to follow if a new configuration is encountered.

The first example is simply one where the emitter resistor has been dropped from the voltage-feedback configuration of Fig. 4.38. The analysis is quite similar, but does require dropping R_E from the applied equation.

EXAMPLE 4.18 For the network of Fig. 4.53:

a. Determine I_{C_Q} and V_{CE_Q}.
b. Find V_B, V_C, V_E, and V_{BC}.

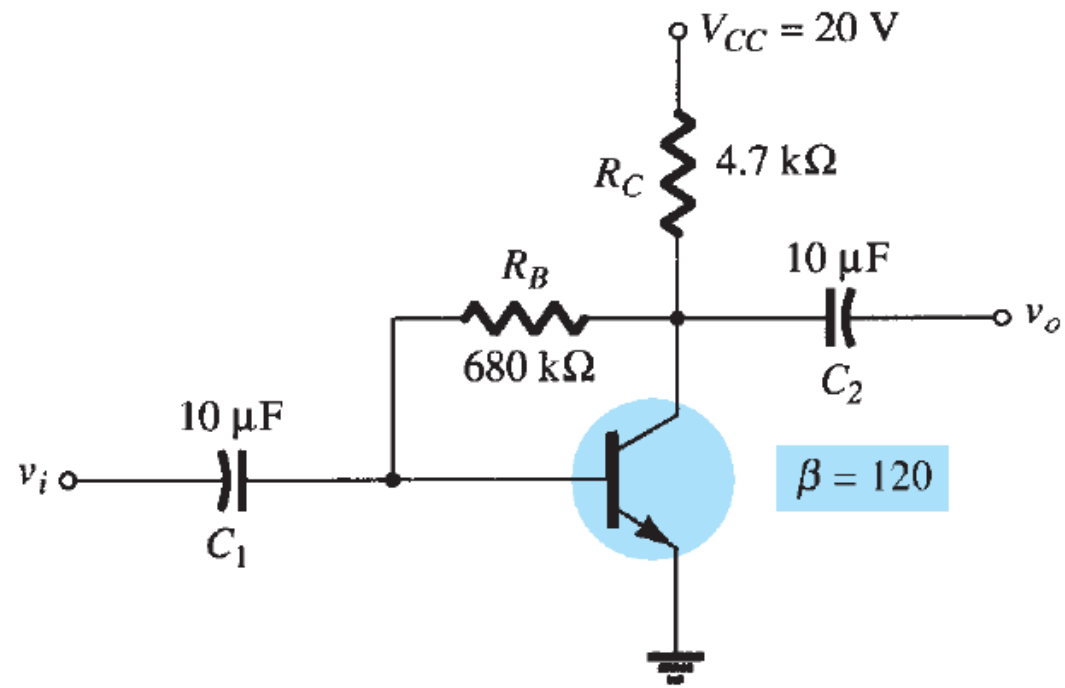


FIG. 4.53
Collector feedback with $R_E = 0\ \Omega$.

Solution:

a. The absence of R_E reduces the reflection of resistive levels to simply that of R_C, and the equation for I_B reduces to

$$I_B = \frac{V_{CC} - V_{BE}}{R_B + \beta R_C}$$

$$= \frac{20\text{ V} - 0.7\text{ V}}{680\text{ k}\Omega + (120)(4.7\text{ k}\Omega)} = \frac{19.3\text{ V}}{1.244\text{ M}\Omega}$$

$$= \mathbf{15.51\ \mu A}$$

$$I_{C_Q} = \beta I_B = (120)(15.51\ \mu\text{A})$$

$$= \mathbf{1.86\text{ mA}}$$

$$V_{CE_Q} = V_{CC} - I_C R_C$$

$$= 20\text{ V} - (1.86\text{ mA})(4.7\text{ k}\Omega)$$

$$= \mathbf{11.26\text{ V}}$$

b.

$$V_B = V_{BE} = \mathbf{0.7\text{ V}}$$

$$V_C = V_{CE} = \mathbf{11.26\text{ V}}$$

$$V_E = \mathbf{0\text{ V}}$$

$$V_{BC} = V_B - V_C = 0.7\text{ V} - 11.26\text{ V}$$

$$= \mathbf{-10.56\text{ V}}$$

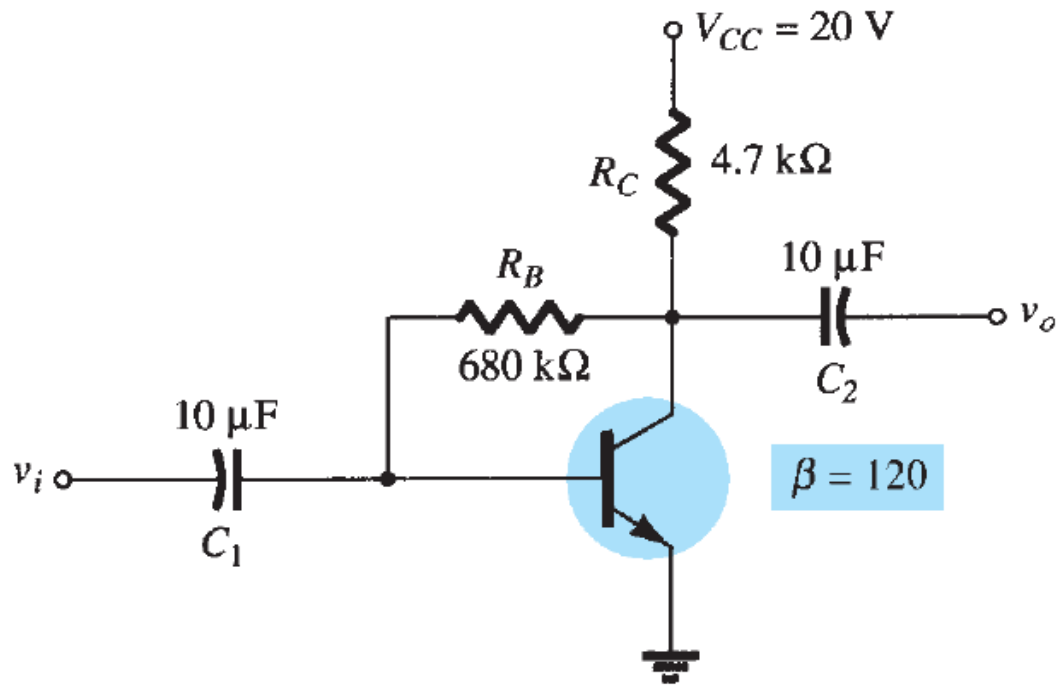


FIG. 4.53
Collector feedback with $R_E = 0\ \Omega$.

In the next example, the applied voltage is connected to the emitter leg and R_C is connected directly to ground. Initially, it appears somewhat unorthodox and quite different from those encountered thus far, but one application of Kirchhoff's voltage law to the base circuit will result in the desired base current.

EXAMPLE 4.19 Determine V_C and V_B for the network of Fig. 4.54.

Solution: Applying Kirchhoff's voltage law in the clockwise direction for the base–emitter loop results in

$$-I_B R_B - V_{BE} + V_{EE} = 0$$

$$I_B = \frac{V_{EE} - V_{BE}}{R_B}$$

$$I_B = \frac{9\text{ V} - 0.7\text{ V}}{100\text{ k}\Omega}$$

$$= \frac{8.3\text{ V}}{100\text{ k}\Omega}$$

$$= 83\ \mu\text{A}$$

$$I_C = \beta I_B$$

$$= (45)(83\ \mu\text{A})$$

$$= 3.735\text{ mA}$$

$$V_C = -I_C R_C$$

$$= -(3.735\text{ mA})(1.2\text{ k}\Omega)$$

$$= \mathbf{-4.48\text{ V}}$$

$$V_B = -I_B R_B$$

$$= -(83\ \mu\text{A})(100\text{ k}\Omega)$$

$$= \mathbf{-8.3\text{ V}}$$

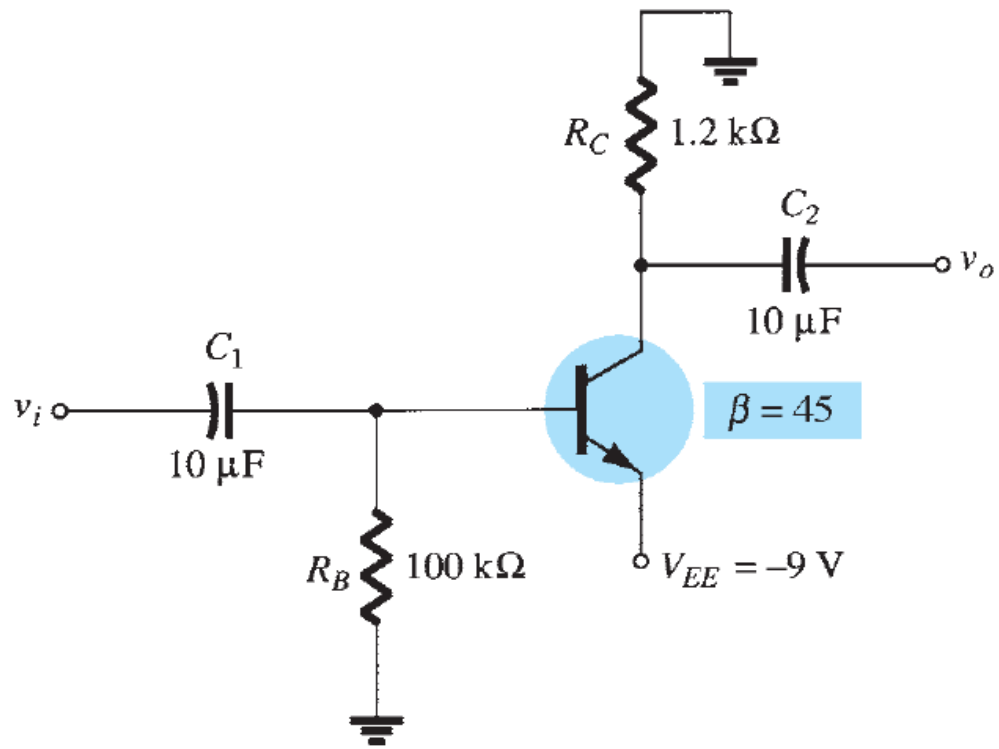


FIG. 4.54

Example 4.20 employs a split supply and will require the application of Thévenin's theorem to determine the desired unknowns.

EXAMPLE 4.20 Determine V_C and V_B for the network of Fig. 4.55.

Solution: The Thévenin resistance and voltage are determined for the network to the left of the base terminal as shown in Figs. 4.56 and 4.57.

R_{Th}

$$R_{Th} = 8.2\,\text{k}\Omega \,\|\, 2.2\,\text{k}\Omega = 1.73\,\text{k}\Omega$$

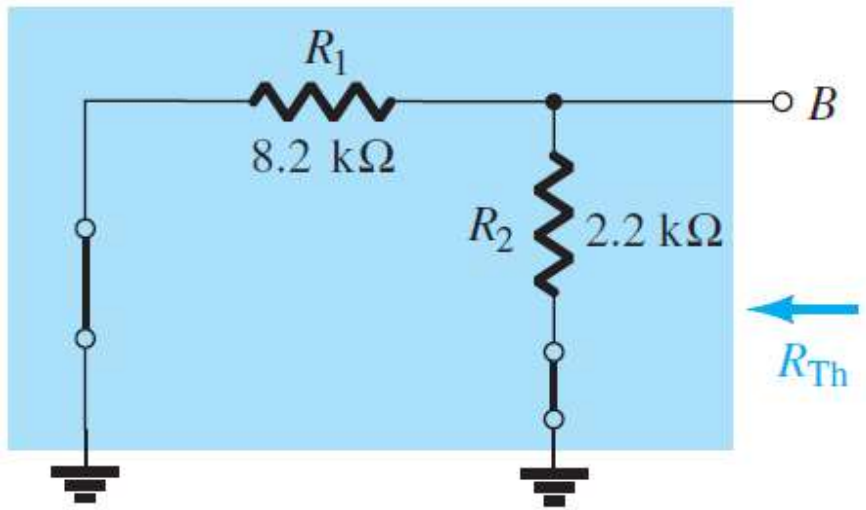


FIG. 4.56
Determining R_{Th}.

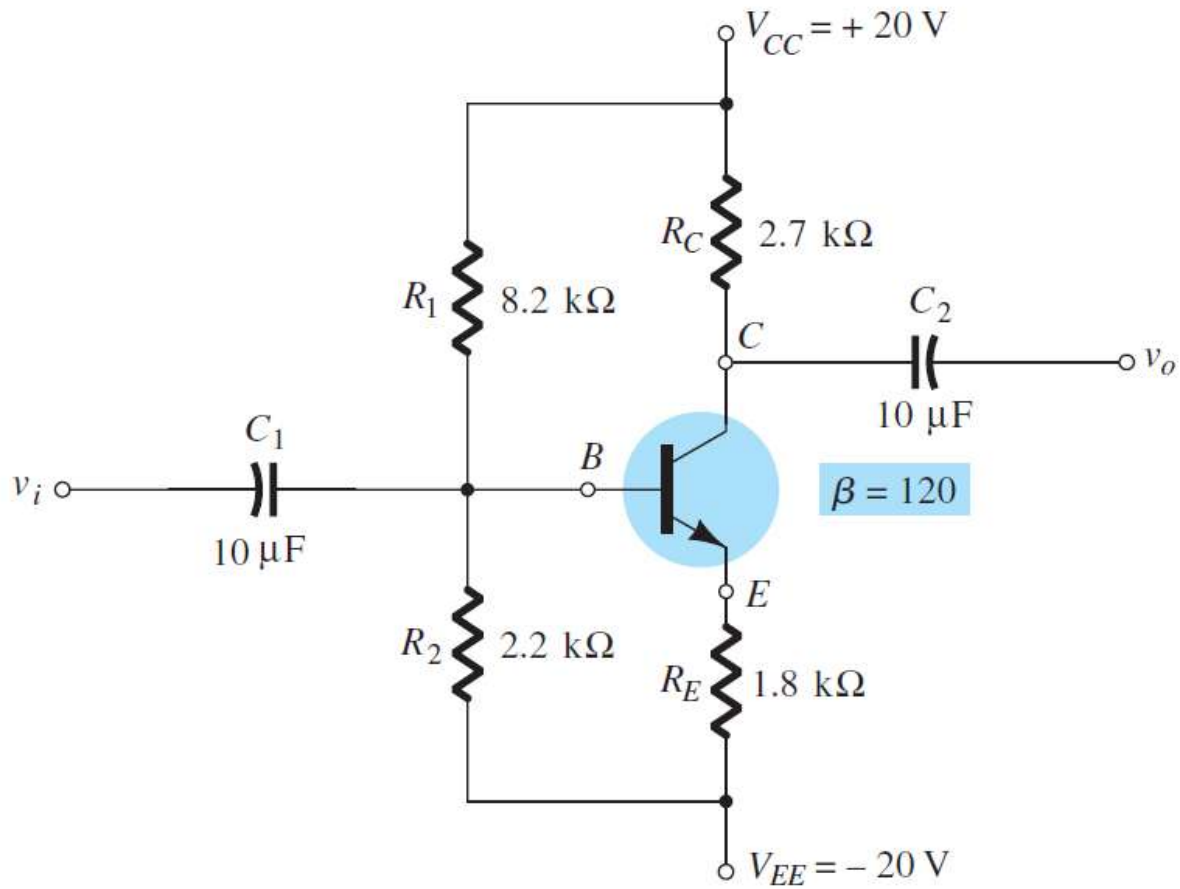


FIG. 4.55

E_{Th}

$$I = \frac{V_{CC} + V_{EE}}{R_1 + R_2} = \frac{20\text{ V} + 20\text{ V}}{8.2\text{ k}\Omega + 2.2\text{ k}\Omega} = \frac{40\text{ V}}{10.4\text{ k}\Omega}$$
$$= 3.85\text{ mA}$$
$$E_{Th} = IR_2 - V_{EE}$$
$$= (3.85\text{ mA})(2.2\text{ k}\Omega) - 20\text{ V}$$
$$= -11.53\text{ V}$$

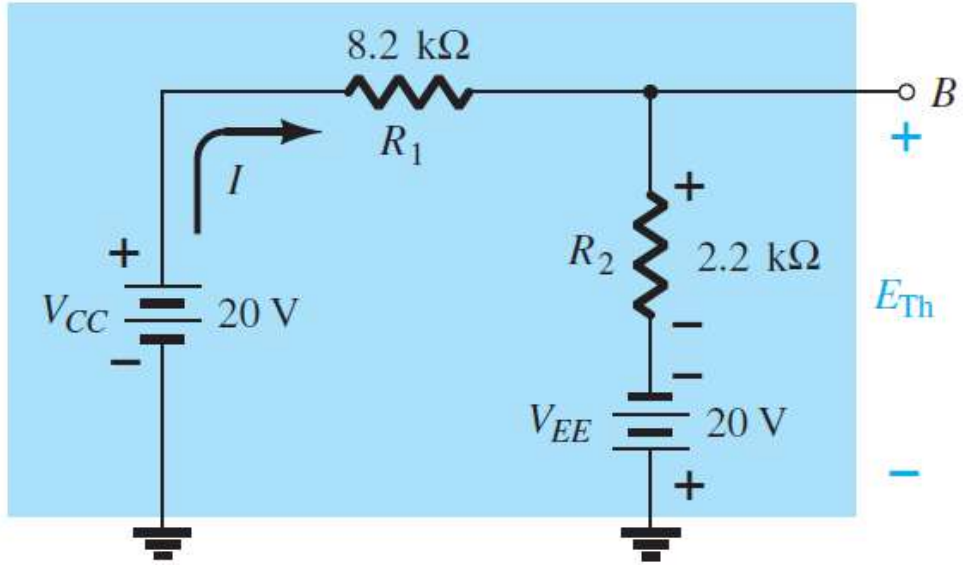


FIG. 4.57
Determining E_{Th}.

The network can then be redrawn as shown in Fig. 4.58, where the application of Kirchhoff's voltage law results in

$$-E_{\text{Th}} - I_B R_{\text{Th}} - V_{BE} - I_E R_E + V_{EE} = 0$$

Substituting $I_E = (\beta + 1)I_B$ gives

$$V_{EE} - E_{\text{Th}} - V_{BE} - (\beta + 1)I_B R_E - I_B R_{\text{Th}} = 0$$

$$I_B = \frac{V_{EE} - E_{\text{Th}} - V_{BE}}{R_{\text{Th}} + (\beta + 1)R_E}$$

$$= \frac{20\text{ V} - 11.53\text{ V} - 0.7\text{ V}}{1.73\text{ k}\Omega + (121)(1.8\text{ k}\Omega)}$$

$$= \frac{7.77\text{ V}}{219.53\text{ k}\Omega}$$

$$= 35.39\ \mu\text{A}$$

$$I_C = \beta I_B$$

$$= (120)(35.39\ \mu\text{A})$$

$$= 4.25\text{ mA}$$

$$V_C = V_{CC} - I_C R_C$$

$$= 20\text{ V} - (4.25\text{ mA})(2.7\text{ k}\Omega)$$

$$= \mathbf{8.53\text{ V}}$$

$$V_B = -E_{\text{Th}} - I_B R_{\text{Th}}$$

$$= -(11.53\text{ V}) - (35.39\ \mu\text{A})(1.73\text{ k}\Omega)$$

$$= \mathbf{-11.59\text{ V}}$$

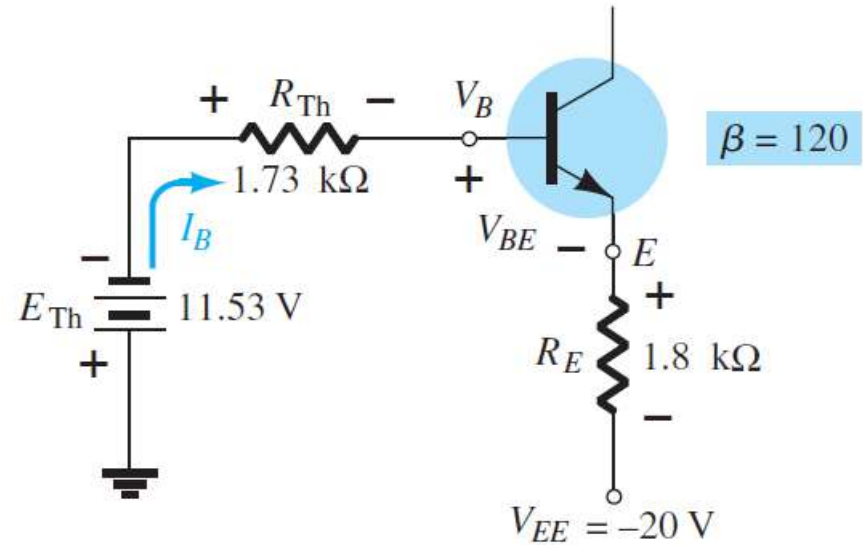


FIG. 4.58

4.11 DESIGN OPERATIONS

Discussions thus far have focused on the analysis of existing networks. All the elements are in place, and it is simply a matter of solving for the current and voltage levels of the configuration. The design process is one where a current and/or voltage may be specified and the elements required to establish the designated levels must be determined. This synthesis process requires a clear understanding of the characteristics of the device, the basic equations for the network, and a firm understanding of the basic laws of circuit analysis, such as Ohm's law, Kirchhoff's voltage law, and so on. In most situations the thinking process is challenged to a higher degree in the design process than in the analysis sequence. The path toward a solution is less defined and in fact may require a number of basic assumptions that do not have to be made when simply analyzing a network.

The design sequence is obviously sensitive to the components that are already specified and the elements to be determined. If the transistor and supplies are specified, the design process will simply determine the required resistors for a particular design. Once the theoretical values of the resistors are determined, the nearest standard commercial values are normally chosen and any variations due to not using the exact resistance values are accepted as part of the design. This is certainly a valid approximation considering the tolerances normally associated with resistive elements and the transistor parameters.

If resistive values are to be determined, one of the most powerful equations is simply Ohm's law in the following form:

$$R_{\text{unknown}} = \frac{V_R}{I_R} \tag{4.49}$$

In a particular design the voltage across a resistor can often be determined from specified levels. If additional specifications define the current level, Eq. (4.49) can then be used to calculate the required resistance level. The first few examples will demonstrate how particular elements can be determined from the design specifications. A complete design procedure will then be introduced for two popular configurations.

EXAMPLE 4.21 Given the device characteristics of Fig. 4.59a, determine V_{CC}, R_B, and R_C for the fixed-bias configuration of Fig. 4.59b. ***Solution:*** From the load line

$$V_{CC} = \mathbf{20\ V}$$

$$I_C = \left.\frac{V_{CC}}{R_C}\right|_{V_{CE}=0\ V}$$

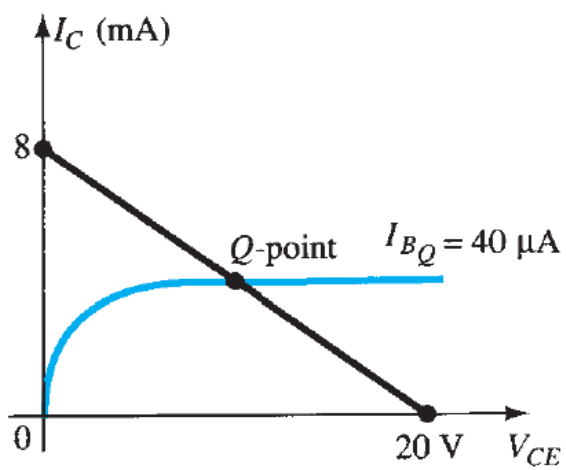
$$R_C = \frac{V_{CC}}{I_C} = \frac{20\ V}{8\ mA} = \mathbf{2.5\ k\Omega}$$

$$I_B = \frac{V_{CC} - V_{BE}}{R_B}$$

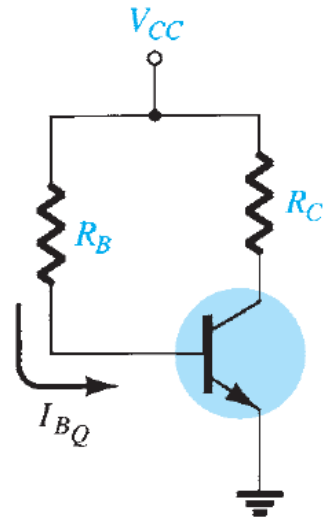
$$R_B = \frac{V_{CC} - V_{BE}}{I_B}$$

$$= \frac{20\ V - 0.7\ V}{40\ \mu A} = \frac{19.3\ V}{40\ \mu A}$$

$$= \mathbf{482.5\ k\Omega}$$



(a)



(b)

FIG. 4.59

Standard resistor values are

$$R_C = 2.4\ k\Omega$$

$$R_B = 470\ k\Omega$$

Using standard resistor values gives

$$I_B = 41.1\ \mu A$$

which is well within 5% of the value specified.

EXAMPLE 4.22 Given that $I_{C_Q} = 2\text{ mA}$ and $V_{CE_Q} = 10\text{ V}$, determine R_1 and R_C for the network of Fig. 4.60.

Solution:

$$V_E = I_E R_E \cong I_C R_E$$
$$= (2\text{ mA})(1.2\text{ k}\Omega) = 2.4\text{ V}$$
$$V_B = V_{BE} + V_E = 0.7\text{ V} + 2.4\text{ V} = 3.1\text{ V}$$
$$V_B = \frac{R_2 V_{CC}}{R_1 + R_2} = 3.1\text{ V}$$
$$\frac{(18\text{ k}\Omega)(18\text{ V})}{R_1 + 18\text{ k}\Omega} = 3.1\text{ V}$$
$$324\text{ k}\Omega = 3.1R_1 + 55.8\text{ k}\Omega$$
$$3.1R_1 = 268.2\text{ k}\Omega$$
$$R_1 = \frac{268.2\text{ k}\Omega}{3.1} = \mathbf{86.52\text{ k}\Omega}$$

$$R_C = \frac{V_{R_C}}{I_C} = \frac{V_{CC} - V_C}{I_C}$$
$$V_C = V_{CE} + V_E = 10\text{ V} + 2.4\text{ V} = 12.4\text{ V}$$
$$R_C = \frac{18\text{ V} - 12.4\text{ V}}{2\text{ mA}}$$
$$= \mathbf{2.8\text{ k}\Omega}$$

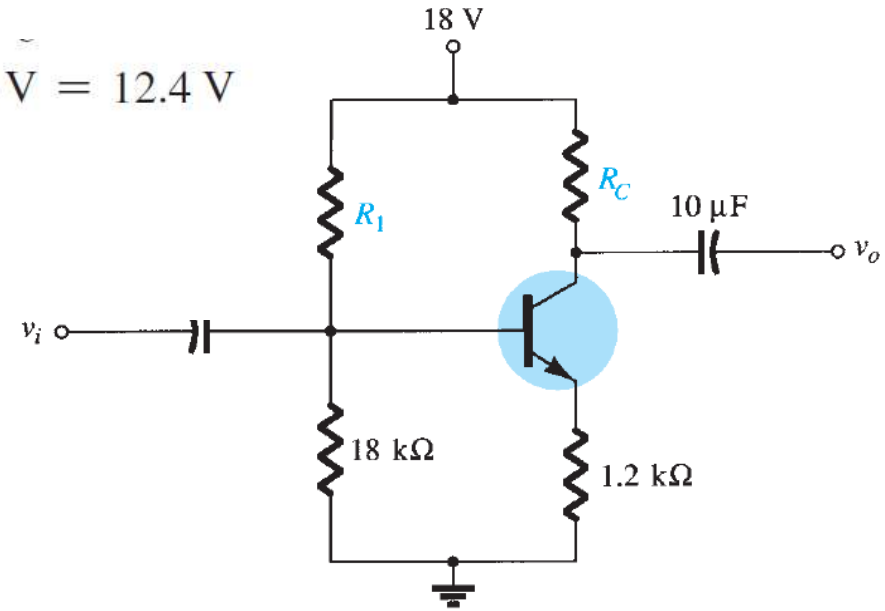


FIG. 4.60

The nearest standard commercial values to R_1 are 82 kΩ and 91 kΩ. However, using the series combination of standard values of 82 kΩ and 4.7 kΩ = 86.7 kΩ would result in a value very close to the design level.

EXAMPLE 4.23 The emitter-bias configuration of Fig. 4.61 has the following specifications: $I_{C_Q} = \frac{1}{2}I_{sat}$, $I_{C_{sat}} = 8\text{ mA}$, $V_C = 18\text{ V}$, and $\beta = 110$. Determine R_C, R_E, and R_B.

Solution:

$$I_{C_Q} = \tfrac{1}{2}I_{C_{sat}} = 4\text{ mA}$$

$$R_C = \frac{V_{R_C}}{I_{C_Q}} = \frac{V_{CC} - V_C}{I_{C_Q}}$$

$$= \frac{28\text{ V} - 18\text{ V}}{4\text{ mA}} = \mathbf{2.5\ k\Omega}$$

$$I_{C_{sat}} = \frac{V_{CC}}{R_C + R_E}$$

$$R_C + R_E = \frac{V_{CC}}{I_{C_{sat}}} = \frac{28\text{ V}}{8\text{ mA}} = 3.5\text{ k}\Omega$$

$$R_E = 3.5\text{ k}\Omega - R_C$$

$$= 3.5\text{ k}\Omega - 2.5\text{ k}\Omega$$

$$= \mathbf{1\ k\Omega}$$

$$I_{B_Q} = \frac{I_{C_Q}}{\beta} = \frac{4\text{ mA}}{110} = 36.36\ \mu\text{A}$$

$$I_{B_Q} = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1)R_E}$$

$$R_B + (\beta + 1)R_E = \frac{V_{CC} - V_{BE}}{I_{B_Q}}$$

$$R_B = \frac{V_{CC} - V_{BE}}{I_{B_Q}} - (\beta + 1)R_E$$

$$= \frac{28\text{ V} - 0.7\text{ V}}{36.36\ \mu\text{A}} - (111)(1\text{ k}\Omega)$$

$$= \frac{27.3\text{ V}}{36.36\ \mu\text{A}} - 111\text{ k}\Omega$$

$$= \mathbf{639.8\ k\Omega}$$

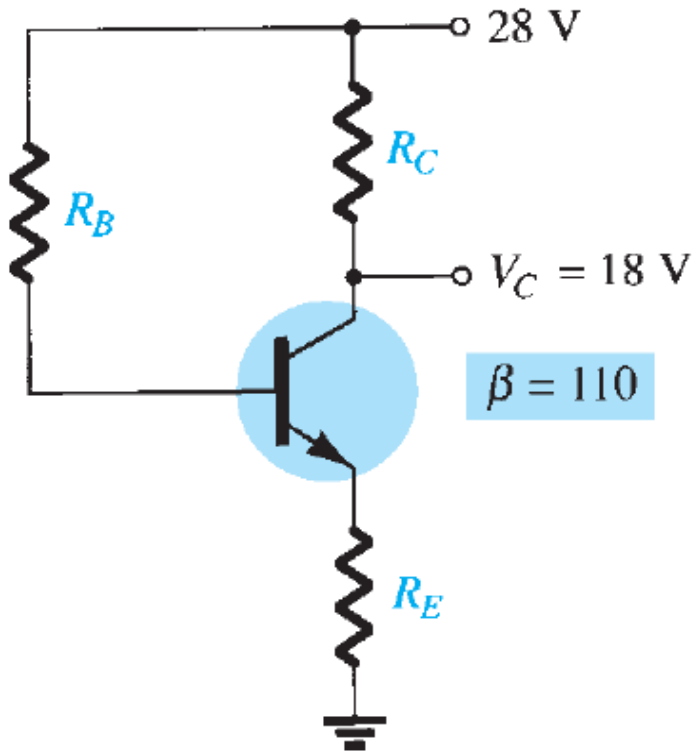


FIG. 4.61

$$R_C = 2.4\text{ k}\Omega$$

$$R_E = 1\text{ k}\Omega$$

$$R_B = 620\text{ k}\Omega$$

The discussion to follow will introduce one technique for designing an entire circuit to operate at a specified bias point. Often the manufacturer's specification (spec) sheets provide information on a suggested operating point (or operating region) for a particular transistor. In addition, other system components connected to the given amplifier stage may also define the current swing, voltage swing, value of common supply voltage, and so on, for the design.

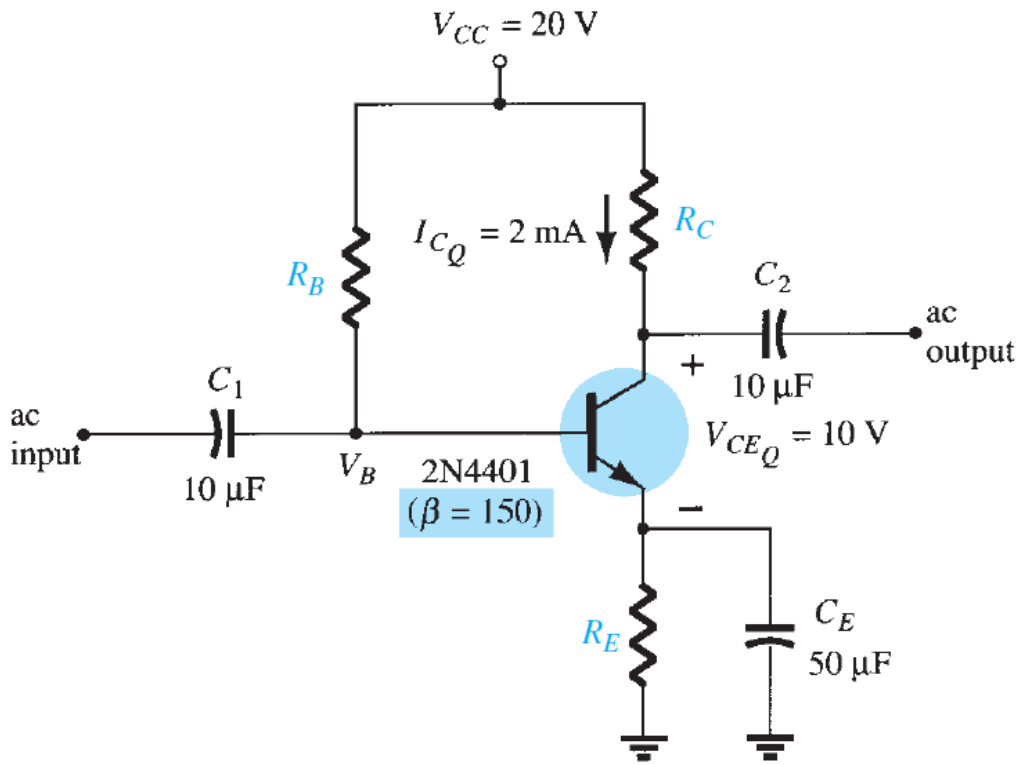
In actual practice, many other factors may have to be considered that may affect the selection of the desired operating point. For the moment we concentrate on determining the component values to obtain a specified operating point. The discussion will be limited to the emitter-bias and voltage-divider bias configurations, although the same procedure can be applied to a variety of other transistor circuits.

Design of a Bias Circuit with an Emitter Feedback Resistor

Consider first the design of the dc bias components of an amplifier circuit having emitter-resistor bias stabilization as shown in Fig. 4.62. The supply voltage and operating point were selected from the manufacturer's information on the transistor used in the amplifier.

The selection of collector and emitter resistors cannot proceed directly from the information just specified. The equation that relates the voltages around the collector–emitter loop has two unknown quantities present—the resistors R_C and R_E. At this point some engineering judgment must be made, such as the level of the emitter voltage compared to the applied supply voltage. Recall that the need for including a resistor from emitter to ground was to provide a means of dc bias stabilization so that the change of collector current due to leakage currents in the transistor and the transistor beta would not cause a large shift in the operating point. The emitter resistor cannot be unreasonably large because the voltage across it limits the range of swing of the voltage from collector to emitter (to be noted when the ac response is discussed).

The examples examined in this chapter reveal that the voltage from emitter to ground is typically around one-fourth to one-tenth of the supply voltage. Selecting the conservative case of one-tenth will permit calculating the emitter resistor R_E and the resistor R_C in a manner similar to the examples just completed. In the next example we perform a complete design of the network of Fig. 4.62 using the criteria just introduced for the emitter voltage.



Emitter-stabilized bias circuit for design consideration.

FIG. 4.62

EXAMPLE 4.24 Determine the resistor values for the network of Fig. 4.62 for the indicated operating point and supply voltage.

Solution:

$$V_E = \frac{1}{10}V_{CC} = \frac{1}{10}(20\text{ V}) = 2\text{ V}$$

$$R_E = \frac{V_E}{I_E} \cong \frac{V_E}{I_C} = \frac{2\text{ V}}{2\text{ mA}} = \mathbf{1\,k\Omega}$$

$$R_C = \frac{V_{R_C}}{I_C} = \frac{V_{CC} - V_{CE} - V_E}{I_C} = \frac{20\text{ V} - 10\text{ V} - 2\text{ V}}{2\text{ mA}} = \frac{8\text{ V}}{2\text{ mA}}$$

$$= \mathbf{4\,k\Omega}$$

$$I_B = \frac{I_C}{\beta} = \frac{2\text{ mA}}{150} = 13.33\ \mu\text{A}$$

$$R_B = \frac{V_{R_B}}{I_B} = \frac{V_{CC} - V_{BE} - V_E}{I_B} = \frac{20\text{ V} - 0.7\text{ V} - 2\text{ V}}{13.33\ \mu\text{A}}$$

$$\cong \mathbf{1.3\,M\Omega}$$

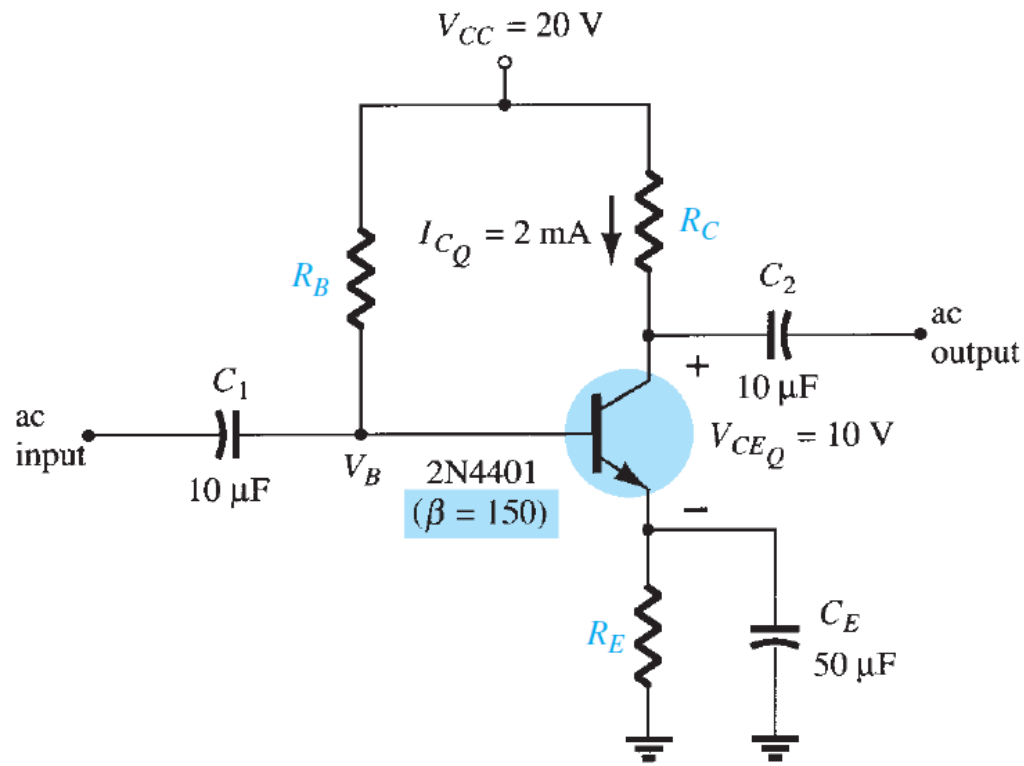
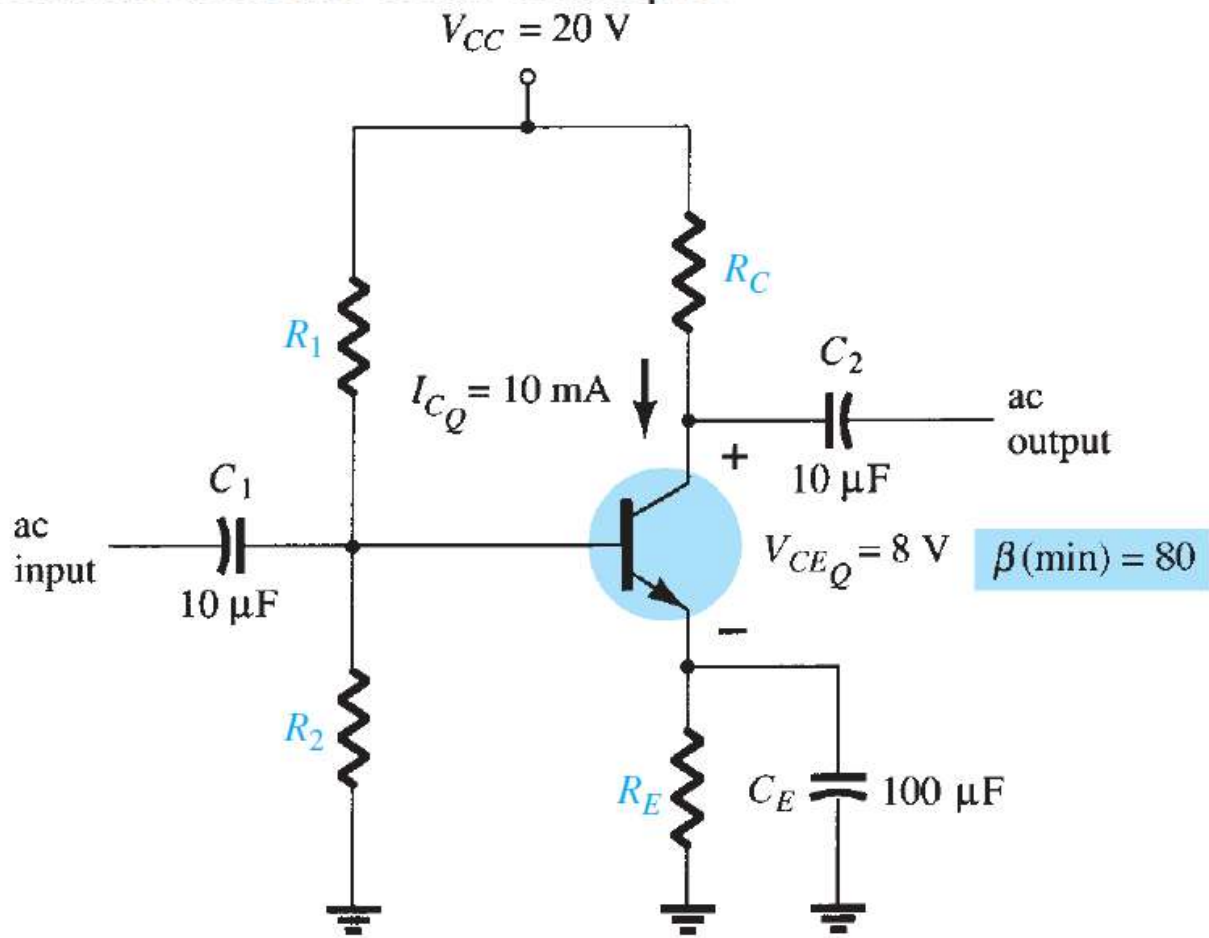


FIG. 4.62

Emitter-stabilized bias circuit for design consideration.

Design of a Current-Gain-Stabilized (Beta-Independent) Circuit

The circuit of Fig. 4.63 provides stabilization both for leakage and current gain (beta) changes. The four resistor values shown must be obtained for the specified operating point. Engineering judgment in selecting a value of emitter voltage V_E, as in the previous design consideration, leads to a direct, straightforward solution for all the resistor values. The design steps are all demonstrated in the next example.



EXAMPLE 4.25 Determine the levels of R_C, R_E, R_1, and R_2 for the network of Fig. 4.63 for the operating point indicated.

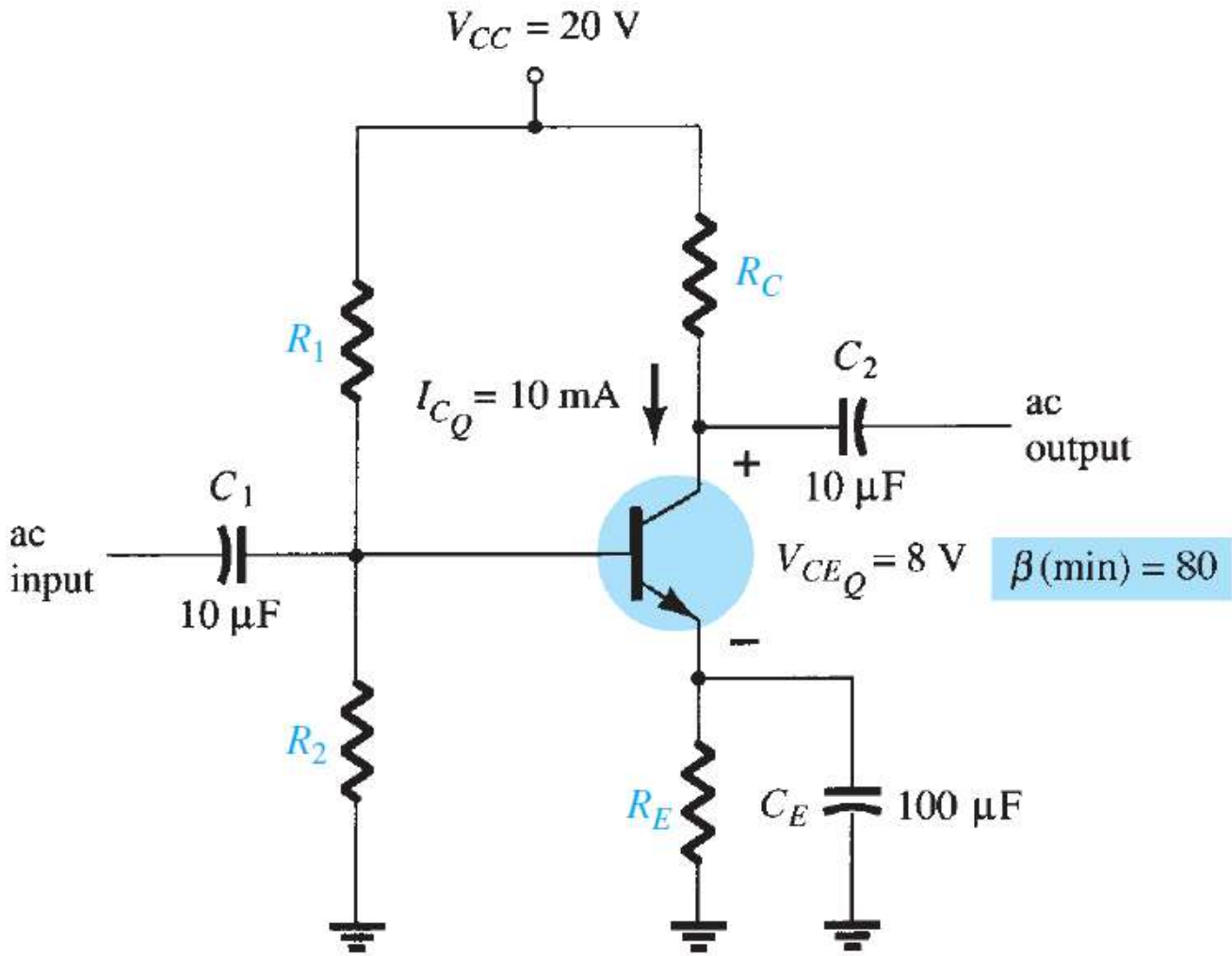


FIG. 4.63
Current-gain-stabilized circuit for design considerations.

Solution:

$$V_E = \frac{1}{10}V_{CC} = \frac{1}{10}(20\text{ V}) = 2\text{ V}$$

$$R_E = \frac{V_E}{I_E} \cong \frac{V_E}{I_C} = \frac{2\text{ V}}{10\text{ mA}} = \mathbf{200\ \Omega}$$

$$R_C = \frac{V_{R_C}}{I_C} = \frac{V_{CC} - V_{CE} - V_E}{I_C} = \frac{20\text{ V} - 8\text{ V} - 2\text{ V}}{10\text{ mA}} = \frac{10\text{ V}}{10\text{ mA}}$$

$$= \mathbf{1\ k\Omega}$$

$$V_B = V_{BE} + V_E = 0.7\text{ V} + 2\text{ V} = 2.7\text{ V}$$

The equations for the calculation of the base resistors R_1 and R_2 will require a little thought. Using the value of base voltage calculated above and the value of the supply voltage will provide one equation—but there are two unknowns, R_1 and R_2. An additional equation can be obtained from an understanding of the operation of these two resistors in providing the necessary base voltage. For the circuit to operate efficiently, it is assumed that the current through R_1 and R_2 should be approximately equal to and much larger than the base current (at least 10:1). This fact and the voltage-divider equation for the base voltage provide the two relationships necessary to determine the base resistors. That is,

$$R_2 \leq \frac{1}{10}\beta R_E$$

$$V_B = \frac{R_2}{R_1 + R_2} V_{CC}$$

$$R_2 \leq \tfrac{1}{10}(80)(0.2\,\text{k}\Omega)$$
$$= \mathbf{1.6\,k\Omega}$$

$$V_B = 2.7\,\text{V} = \frac{(1.6\,\text{k}\Omega)(20\,\text{V})}{R_1 + 1.6\,\text{k}\Omega}$$

$$2.7R_1 + 4.32\,\text{k}\Omega = 32\,\text{k}\Omega$$
$$2.7R_1 = 27.68\,\text{k}\Omega$$
$$R_1 = \mathbf{10.25\,k\Omega} \quad (\text{use } 10\,\text{k}\Omega)$$

Important Equations

$$V_{BE} \cong 0.7\ \text{V}, \qquad I_E = (\beta + 1)I_B \cong I_C, \qquad I_C = \beta I_B$$

Fixed bias:

$$I_B = \frac{V_{CC} - V_{BE}}{R_B}, \qquad I_C = \beta I_B$$

Emitter stabilized:

$$I_B = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1)R_E}, \qquad R_i = (\beta + 1)R_E$$

Voltage-divider bias:

$$\textit{Exact:} \quad R_{\text{Th}} = R_1 \| R_2, \qquad E_{\text{Th}} = V_{R_2} = \frac{R_2 V_{CC}}{R_1 + R_2}, \qquad I_B = \frac{E_{\text{Th}} - V_{BE}}{R_{\text{Th}} + (\beta + 1)R_E}$$

$$\textit{Approximate: Test} \quad \beta R_E \geq 10R_2$$

$$V_B = \frac{R_2 V_{CC}}{R_1 + R_2}, \qquad V_E = V_B - V_{BE}, \qquad I_E = \frac{V_E}{R_E} \cong I_C$$

DC bias with voltage feedback:

$$I_B = \frac{V_{CC} - V_{BE}}{R_B + \beta(R_C + R_E)}, \qquad I'_C \cong I_C \cong I_E$$

Common base:

$$I_E = \frac{V_{EE} - V_{BE}}{R_E}, \qquad I_C \cong I_E$$

4.12 MULTIPLE BJT NETWORKS

The BJT networks introduced thus far have only been single-stage configurations. This section will cover some of the most popular networks using multiple transistors. It will demonstrate how the methods introduced thus far in this chapter can be applied to networks with any number of components.

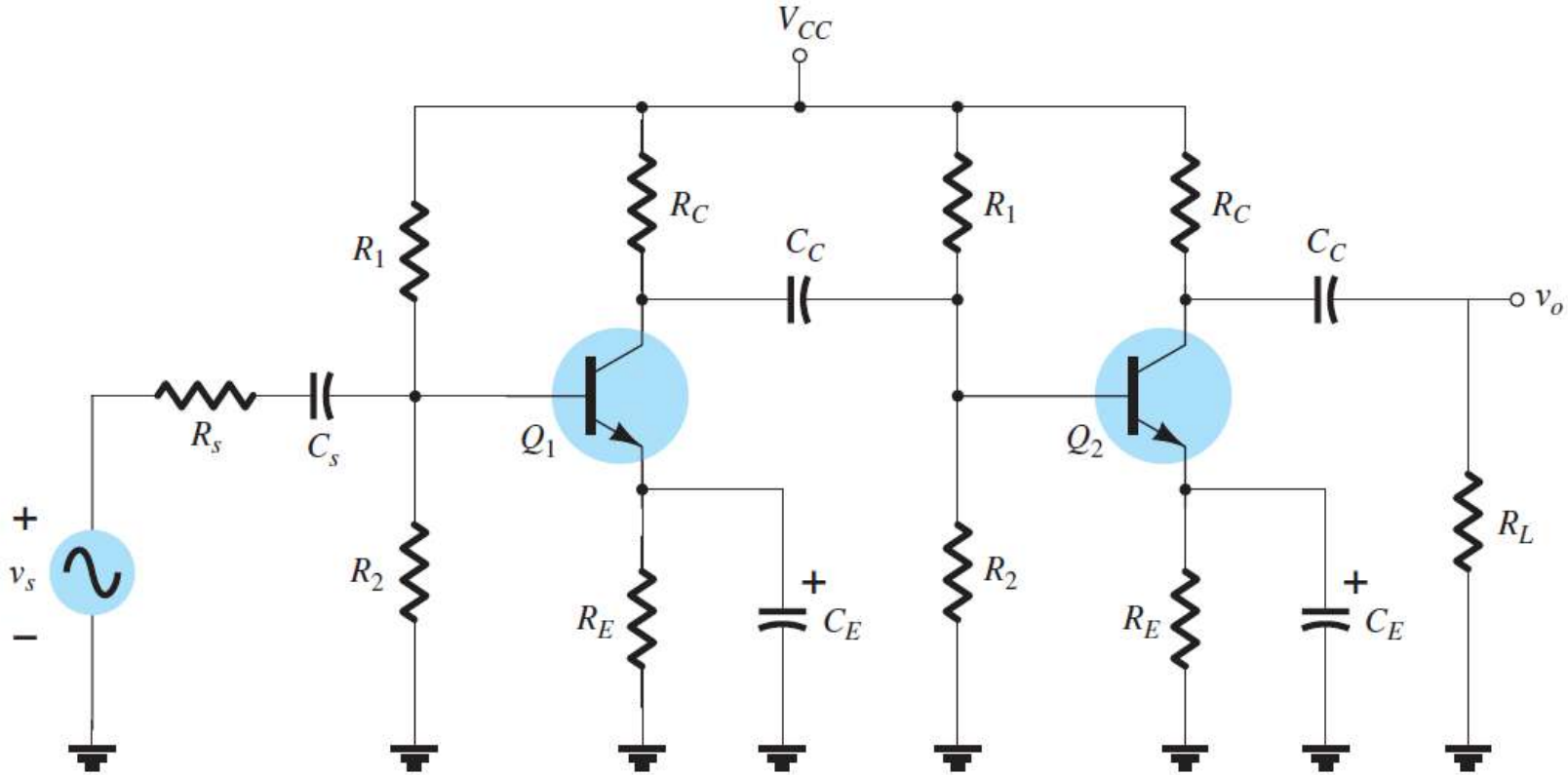


FIG. 4.64
R–C coupled BJT amplifiers.

The **R–C coupling** of Fig. 4.64 is probably the most common. The collector output of one stage is fed directly into the base of the next stage using a coupling capacitor C_C. The capacitor is chosen to ensure that it will block dc between the stages and act like a short circuit to any ac signal. The network of Fig. 4.64 has two voltage-divider stages, but the same coupling can be used between any combination of networks such as the fixed-bias or emitter-follower configurations.

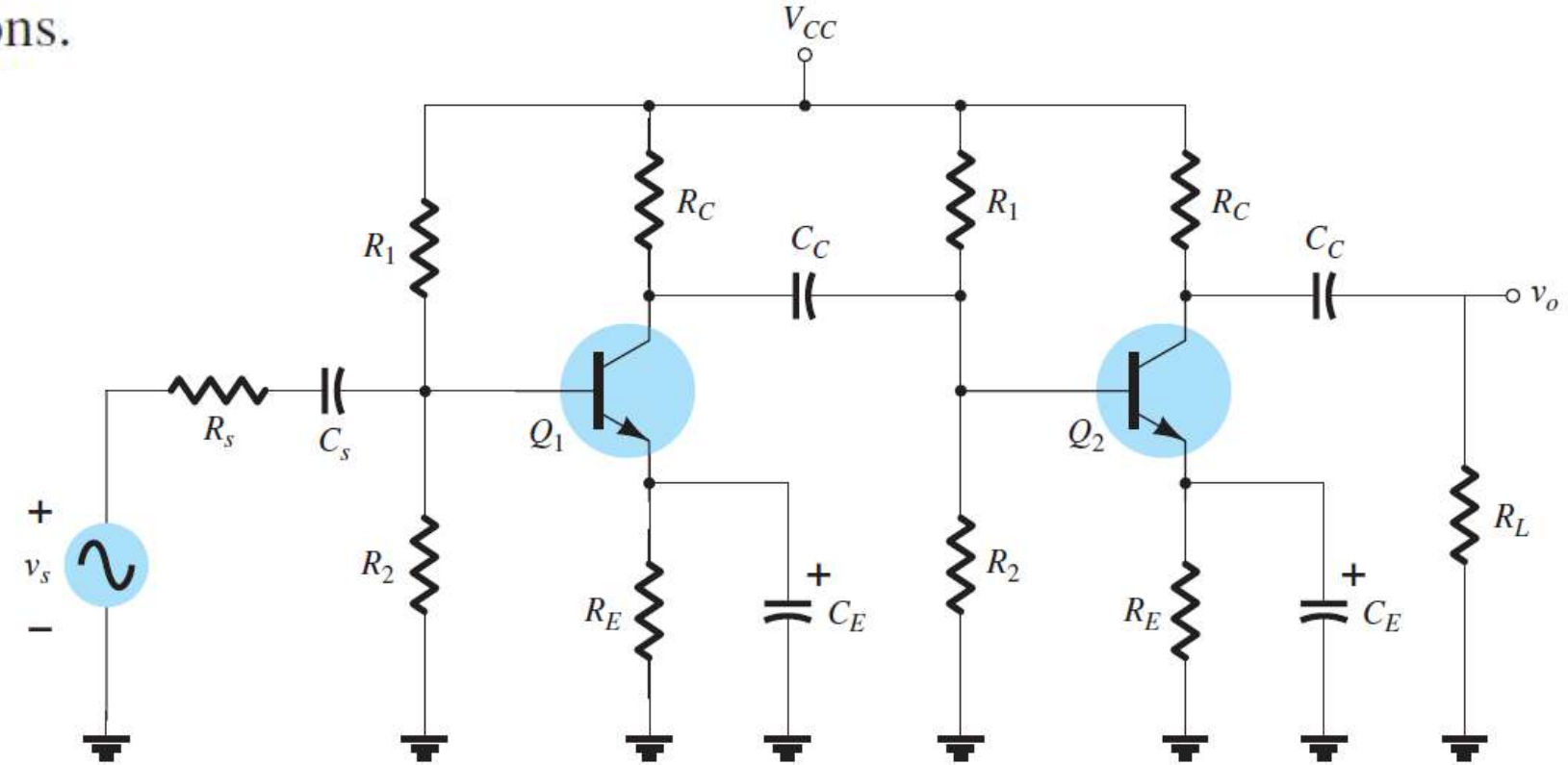


FIG. 4.64
R–C coupled BJT amplifiers.

Substituting an open-circuit equivalent for C_C and the other capacitors of the network will result in the two bias arrangements shown in Fig. 4.65. The methods of analysis introduced in this chapter can then be applied to each stage separately since one stage will not affect the other. Of course, the 20 V dc supply must be applied to each isolated component.

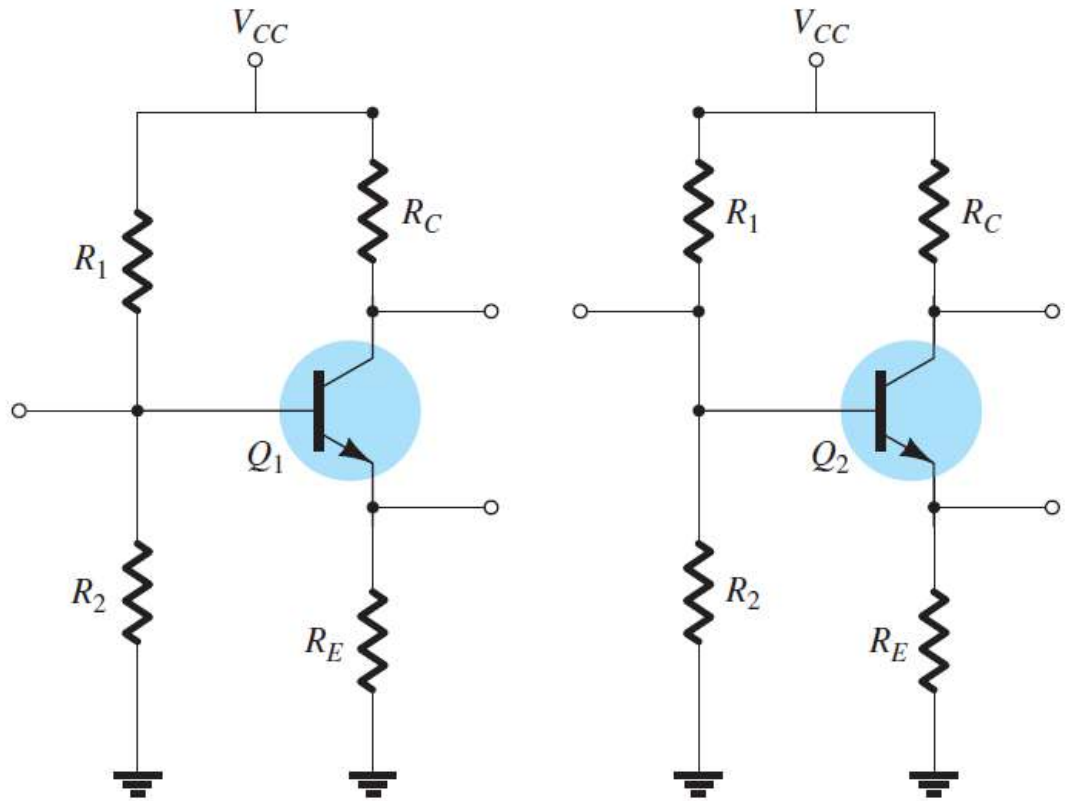


FIG. 4.65
DC equivalent of Fig. 4.64.

The **Darlington** configuration of Fig. 4.66 feeds the output of one stage directly into the input of the succeeding stage. Since the output of Fig. 4.66 is taken directly off the emitter terminal, you will find in the next chapter that the ac gain is very close to 1 but the input impedance is very high, making it attractive for use in amplifiers operating off sources that have a relatively high internal resistance. If a load resistor were added to the collector leg and the output taken off the collector terminal, the configuration would provide a very high gain.

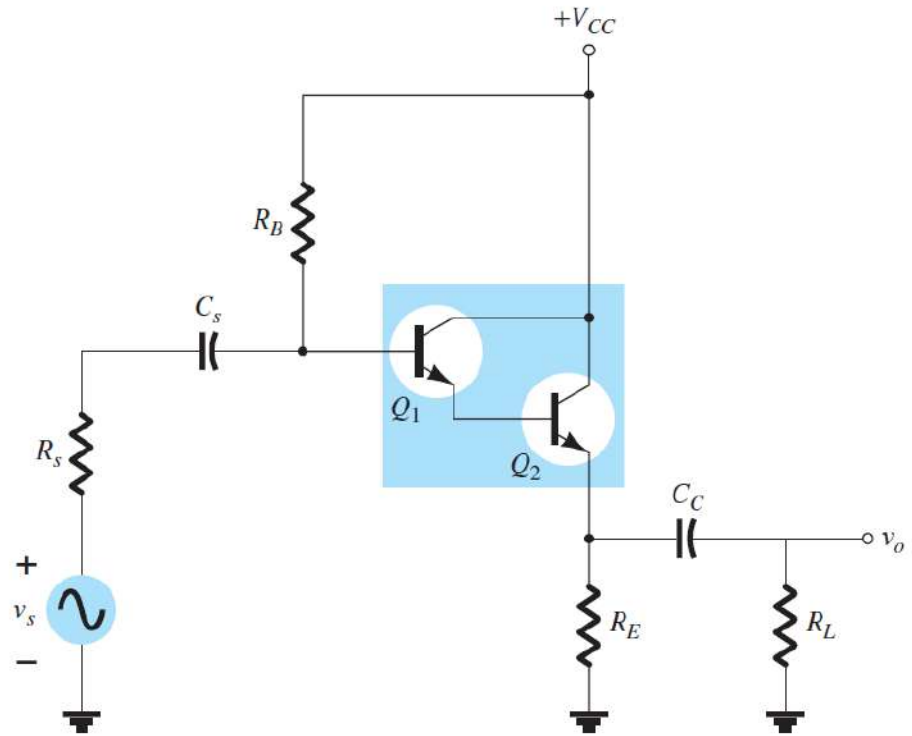


FIG. 4.66

For the dc analysis of Fig. 4.67 assuming a beta β_1 for the first transistor and β_2 for the second, the base current for the second transistor is

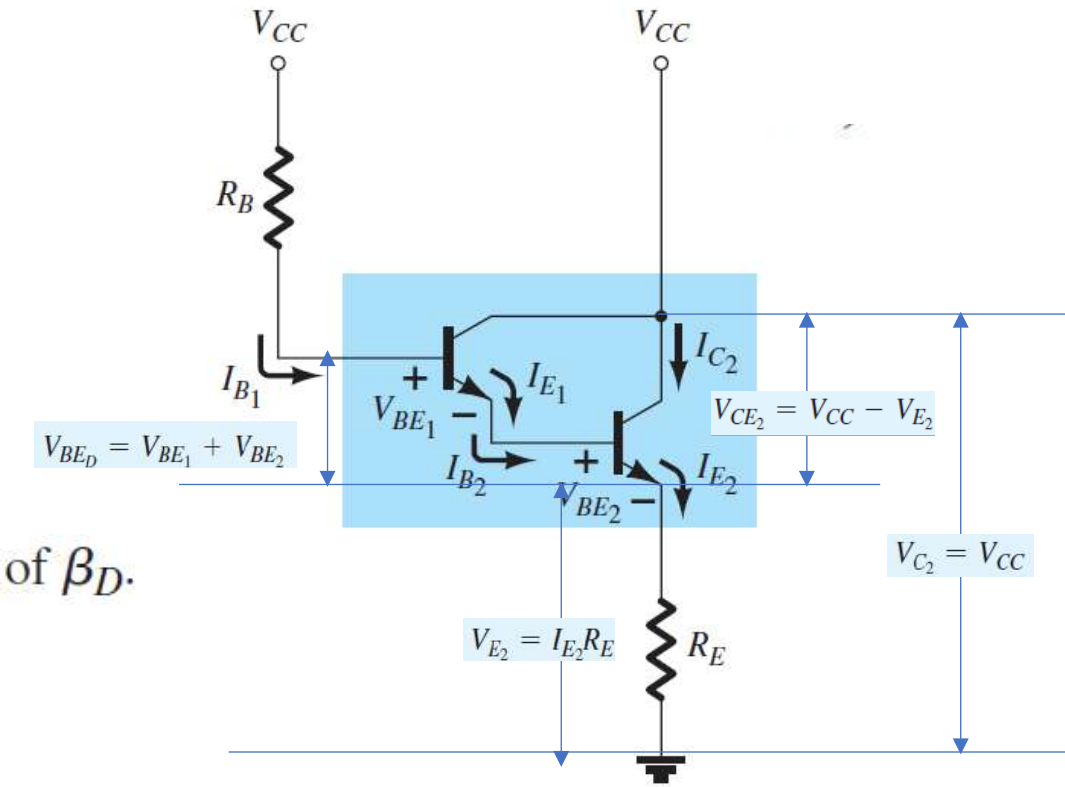
$$I_{B_2} = I_{E_1} = (\beta_1 + 1)I_{B_1}$$

and the emitter current for the second transistor is

$$I_{E_2} = (\beta_2 + 1)I_{B_2} = (\beta_2 + 1)(\beta_1 + 1)I_{B_1}$$

Assuming $\beta \gg 1$ for each transistor, we find the net beta for the configuration is

$$\boxed{\beta_D = \beta_1\beta_2}$$



which compares directly with a single-stage amplifier having a gain of β_D.

Applying an analysis similar to that of Section 4.4 will result in the following equation for the base current:

$$I_{B_1} = \frac{V_{CC} - V_{BE_1} - V_{BE_2}}{R_B + (\beta_D + 1)R_E}$$

Defining

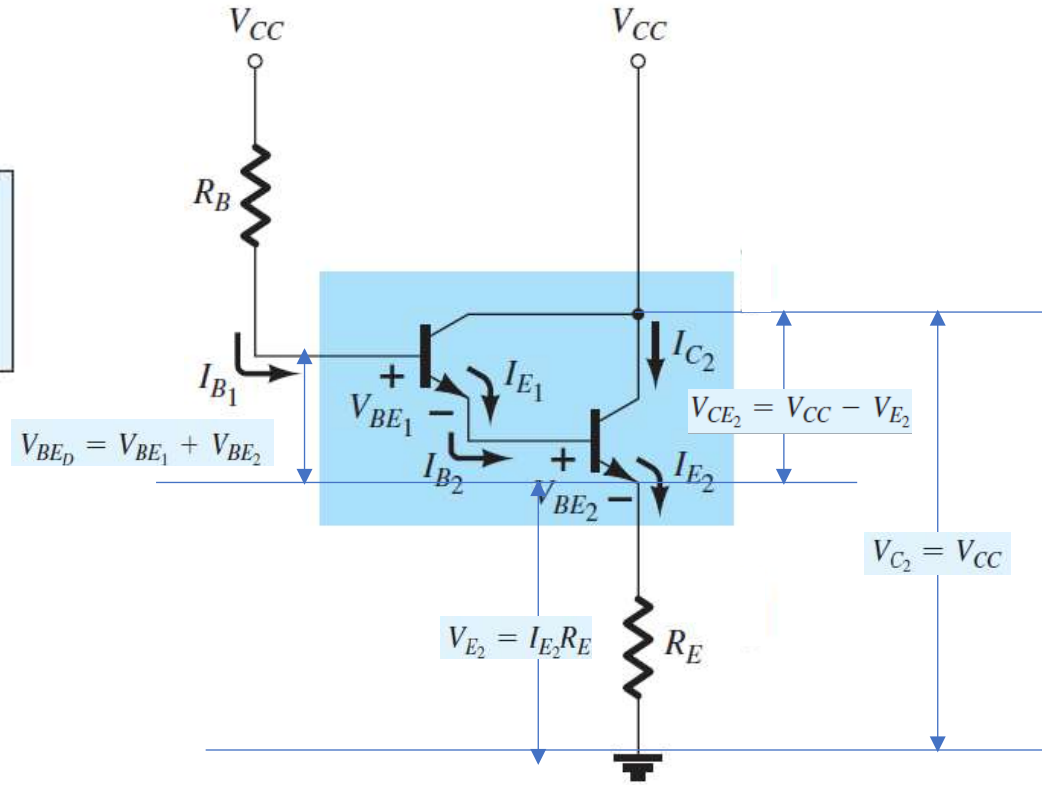
$$\boxed{V_{BE_D} = V_{BE_1} + V_{BE_2}}$$

we have

$$\boxed{I_{B_1} = \frac{V_{CC} - V_{BE_D}}{R_B + (\beta_D + 1)R_E}}$$

The currents

$$\boxed{I_{C_2} \cong I_{E_2} = \beta_D I_{B_1}}$$



and the dc voltage at the emitter terminal is

$$V_{E_2} = I_{E_2} R_E \tag{4.54}$$

The collector voltage for this configuration is obviously equal to that of the source V.

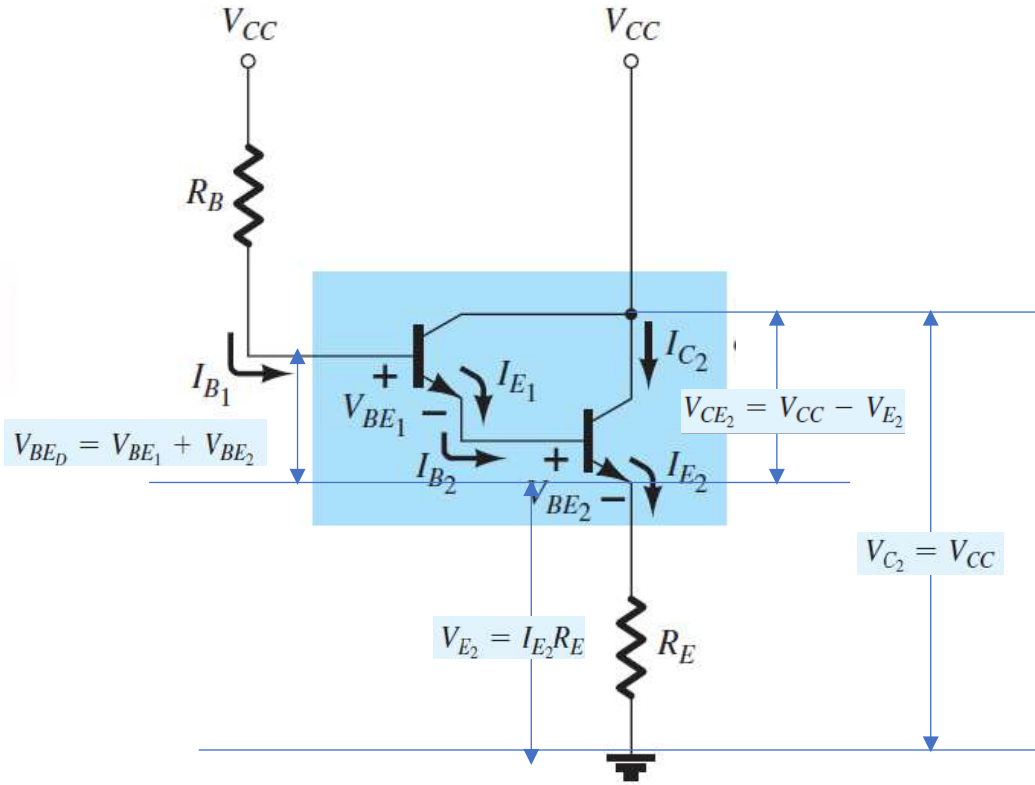
$$V_{C_2} = V_{CC} \tag{4.55}$$

and the voltage across the output of the transistor is

$$V_{CE_2} = V_{C_2} - V_{E_2}$$

and

$$V_{CE_2} = V_{CC} - V_{E_2}$$



The **Cascode** configuration of Fig. 4.68 ties the collector of one transistor to the emitter of the other. In essence it is a voltage-divider network with a common-base configuration at the collector. The result is a network with a high gain and a reduced Miller capacitance—a topic to be examined in Section 9.9.

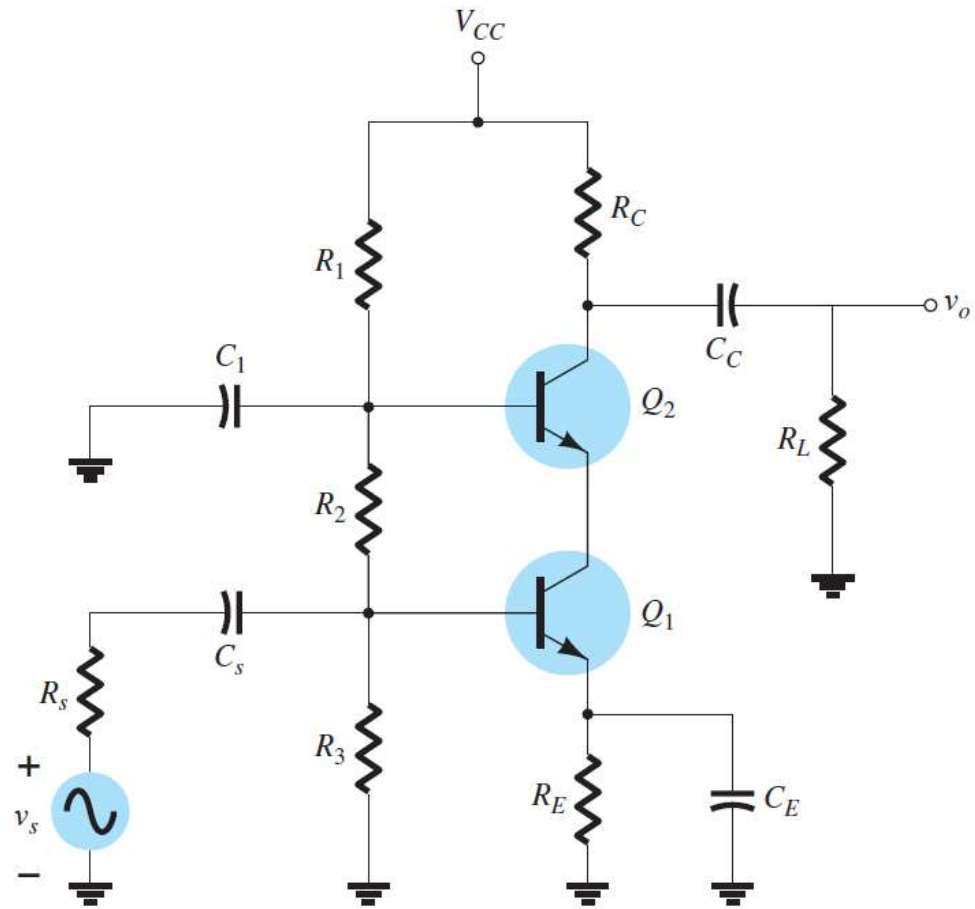


FIG. 4.68

The dc analysis is initiated by assuming the current through the bias resistors R_1, R_2, and R_3 of Fig. 4.69 is much larger than the base current of each transistor. That is,

$$I_{R_1} \cong I_{R_2} \cong I_{R_3} \gg I_{B_1} \text{ or } I_{B_2}$$

The result is that the voltage at the base of the transistor Q_1 is simply determined by an application of the voltage-divider rule:

$$V_{B_1} = \frac{R_3}{R_1 + R_2 + R_3} V_{CC}$$

The voltage at the base of the transistor Q_2 is found in the same man

$$V_{B_2} = \frac{(R_2 + R_3)}{R_1 + R_2 + R_3} V_{CC}$$

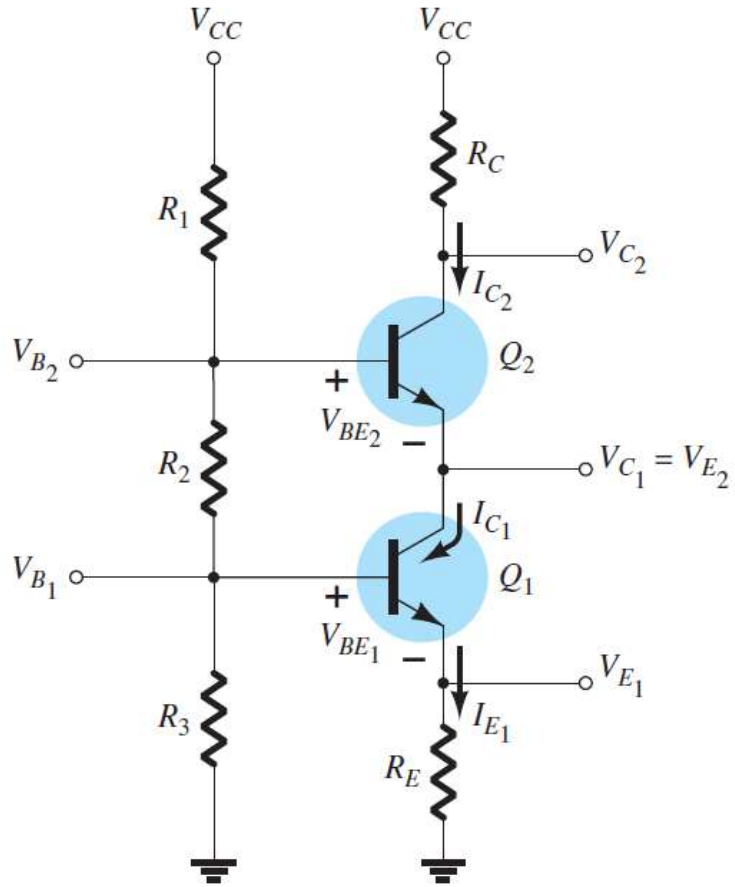


FIG. 4.69

The emitter voltages are then determined by

$$V_{E_1} = V_{B_1} - V_{BE_1}$$

and

$$V_{E_2} = V_{B_2} - V_{BE_2}$$

with the emitter and collector currents determined by:

$$I_{C_2} \cong I_{E_2} \cong I_{C_1} \cong I_{E_1} = \frac{V_{B_1} - V_{BE_1}}{R_{E_1} + R_{E_2}}$$

The collector voltage V_{C_1}:

$$V_{C_1} = V_{B_2} - V_{BE_2}$$

and the collector voltage V_{C_2}:

$$V_{C_2} = V_{CC} - I_{C_2}R_C$$

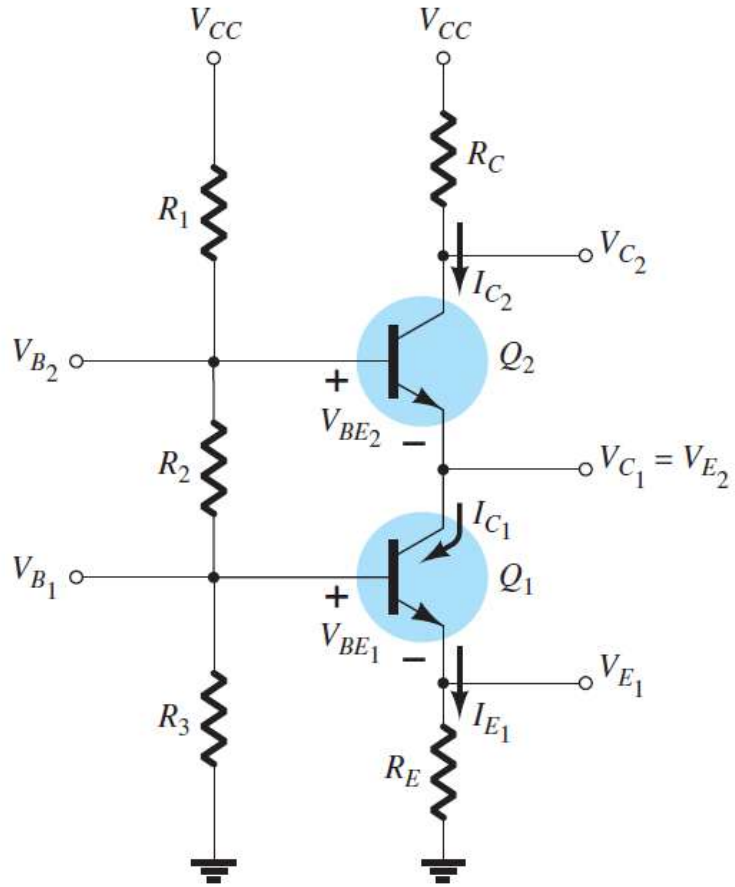


FIG. 4.69

The current through the biasing resistors is

$$I_{R_1} \cong I_{R_2} \cong I_{R_3} = \frac{V_{CC}}{R_1 + R_2 + R_3}$$

and each base current is determined by

$$I_{B_1} = \frac{I_{C_1}}{\beta_1}$$

with

$$I_{B_2} = \frac{I_{C_2}}{\beta_2}$$

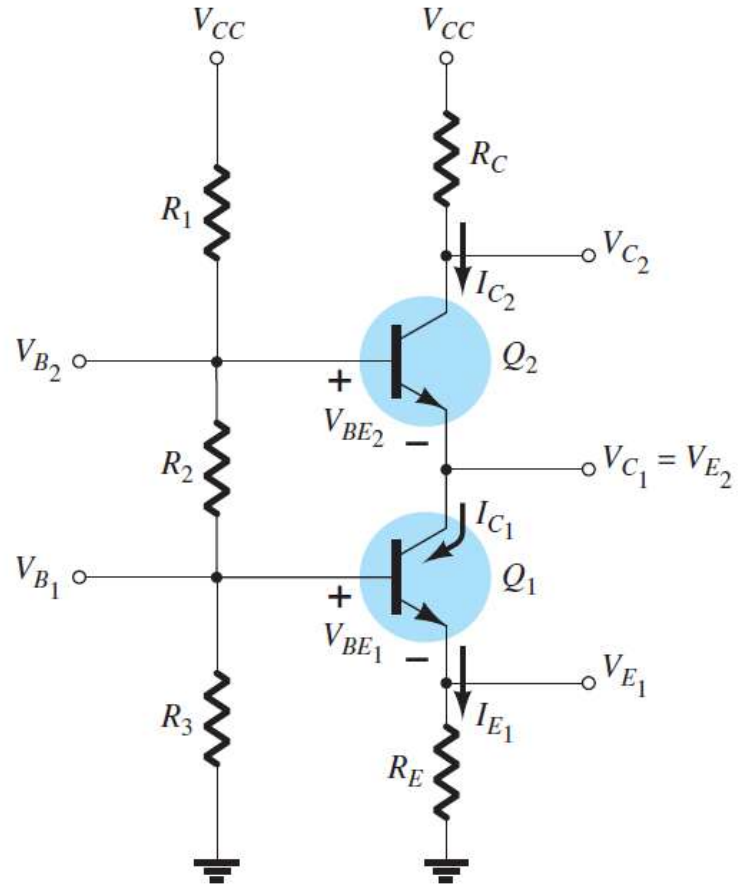


FIG. 4.69

The next multistage configuration to be introduced is the **Feedback Pair** of Fig. 4.70, which employs both an *npn* and *pnp* transistor. The result is a configuration that provides high gain with increased stability.

The dc version with all the currents labeled appears in Fig. 4.71.

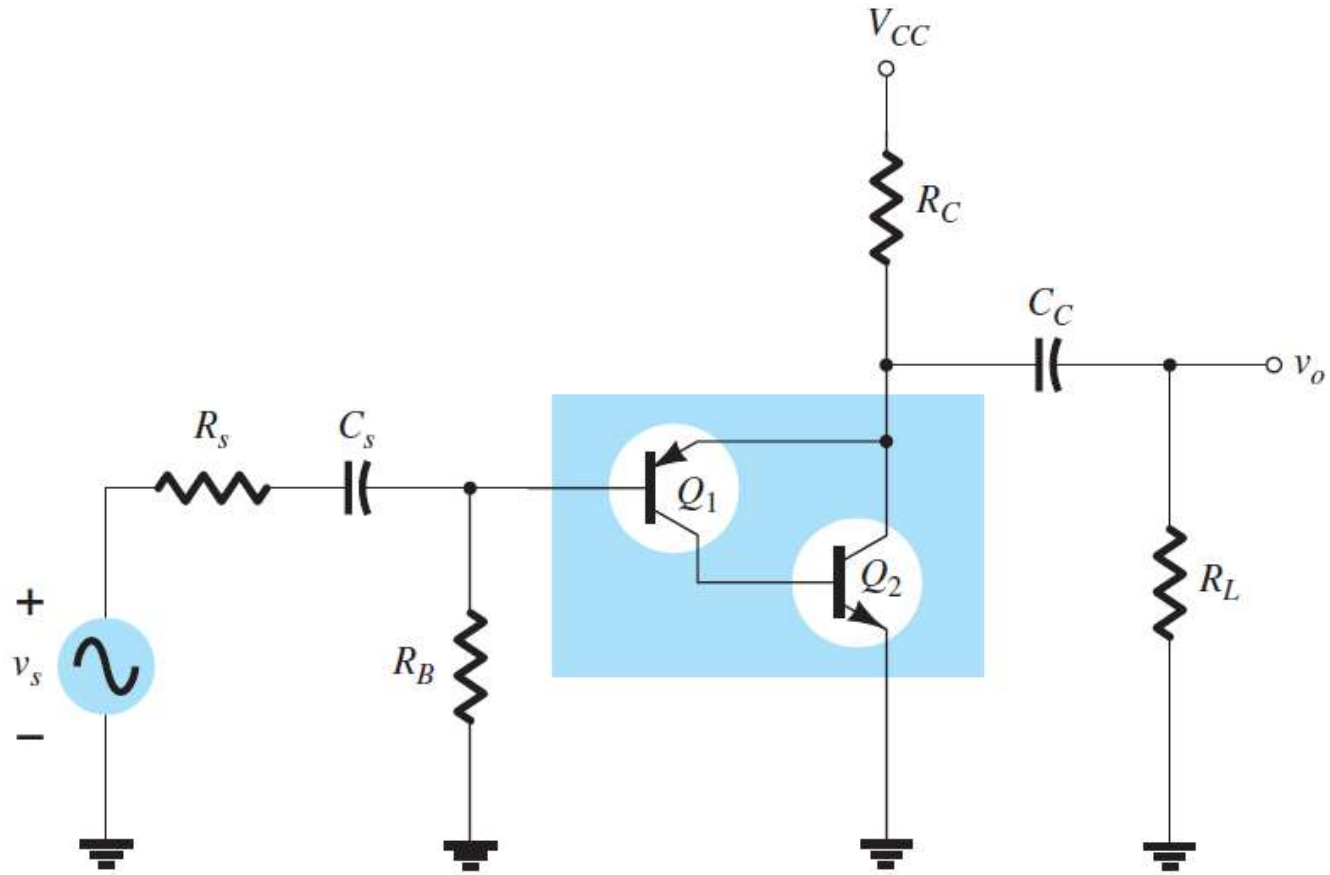


FIG. 4.70
Feedback Pair amplifier.

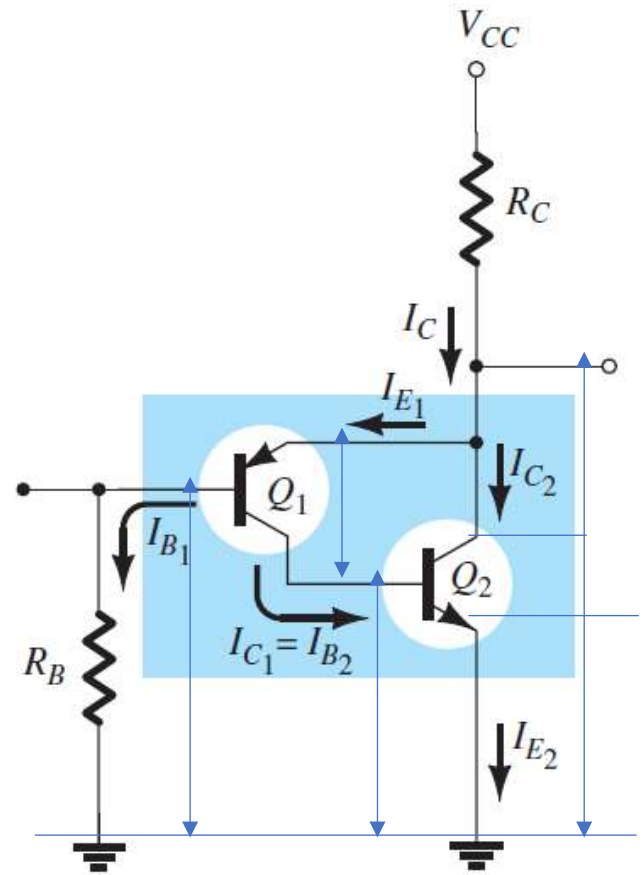


FIG. 4.71
DC equivalent of Fig. 4.70.

The base current

$$I_{B_2} = I_{C_1} = \beta_1 I_{B_1}$$

and

$$I_{C_2} = \beta_2 I_{B_2}$$

so that

$$\boxed{I_{C_2} \cong I_{E_2} = \beta_1 \beta_2 I_{B_1}}$$

The collector current

$$\begin{aligned} I_C &= I_{E_1} + I_{E_2} \\ &\cong \beta_1 I_{B_1} + \beta_1 \beta_2 I_{B_1} \\ &= \beta_1 (1 + \beta_2) I_{B_1} \end{aligned}$$

so that

$$\boxed{I_C \cong \beta_1 \beta_2 I_{B_1}}$$

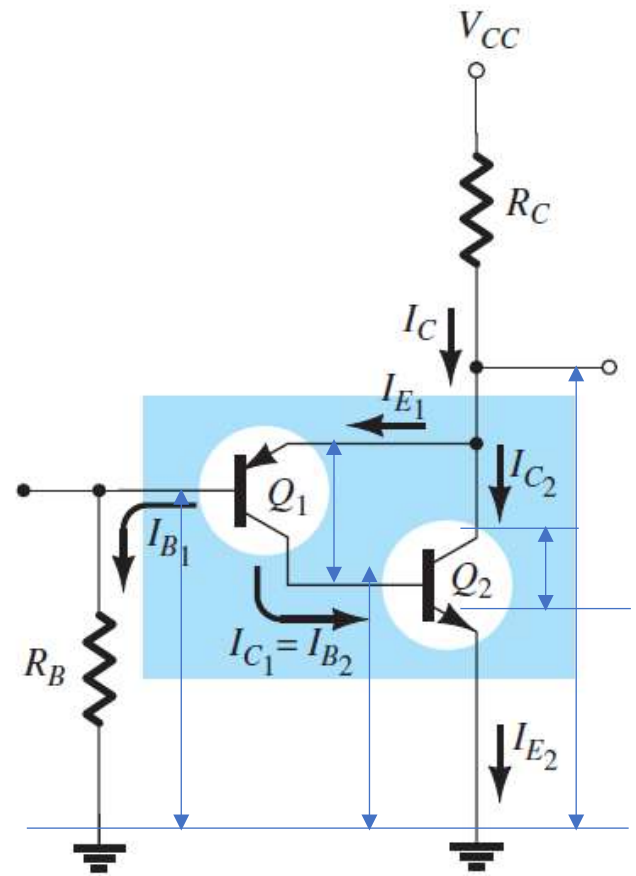


FIG. 4.71
DC equivalent of Fig. 4.70.

Applying Kirchhoff's voltage law down from the source to ground will result in

$$V_{CC} - I_C R_C - V_{EB_1} - I_{B_1} R_B = 0$$

or

$$V_{CC} - V_{EB_1} - \beta_1 \beta_2 I_{B_1} R_C - I_{B_1} R_B = 0$$

and

$$\boxed{I_{B_1} = \frac{V_{CC} - V_{EB_1}}{R_B + \beta_1 \beta_2 R_C}}$$

The base voltage V_{B_1} is

$$\boxed{V_{B_1} = I_{B_1} R_B}$$

and

$$\boxed{V_{B_2} = V_{BE_2}}$$

The collector voltage $V_{C_2} = V_{E_1}$ is

$$\boxed{V_{C_2} = V_{CC} - I_C R_C}$$

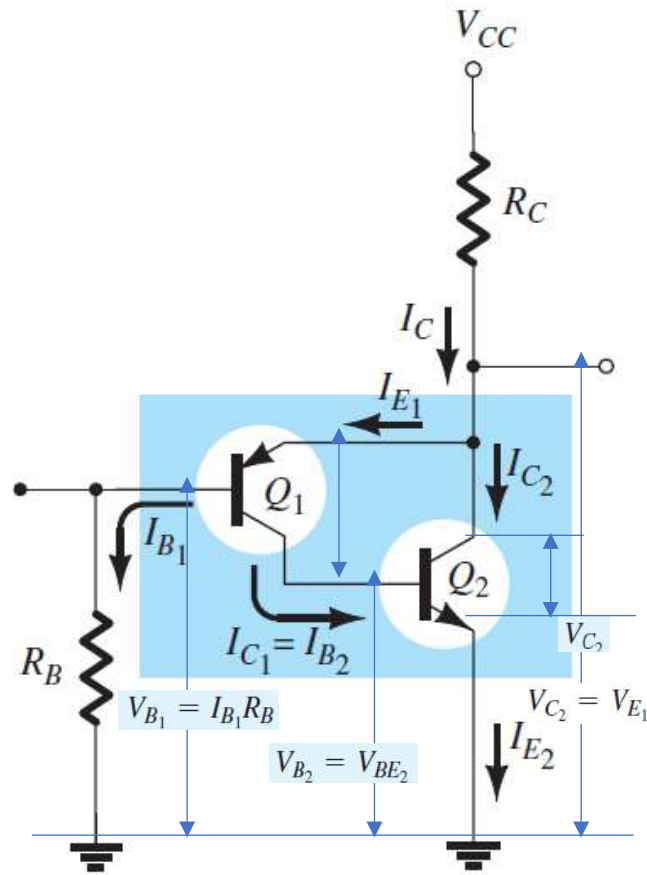


FIG. 4.71
DC equivalent of Fig. 4.70.

and

$$V_{C_1} = V_{BE_2}$$

In this case

$$V_{CE_2} = V_{C_2}$$

and

$$V_{EC_1} = V_{E_1} - V_{C_1}$$

so that

$$V_{EC_1} = V_{C_2} - V_{BE_2}$$

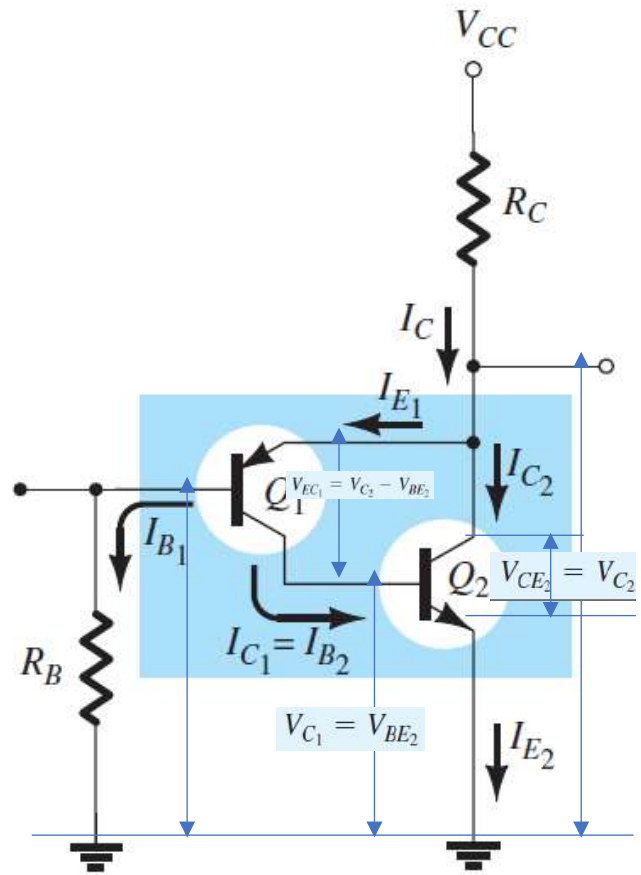


FIG. 4.71

DC equivalent of Fig. 4.70.

The last multistage configuration to be introduced is the **Direct Coupled** amplifier such as appearing in Example 4.26. Note the absence of a coupling capacitor to isolate the dc levels of each stage. The dc levels in one stage will directly affect the dc levels in succeeding stages. The benefit is that the coupling capacitor typically limits the low-frequency response of the amplifier. Without coupling capacitors, the amplifier can amplify signals of very low frequency—in fact down to dc. The disadvantage is that any variation in dc levels due to a variety of reasons in one stage can affect the dc levels in the succeeding stages of the amplifier.

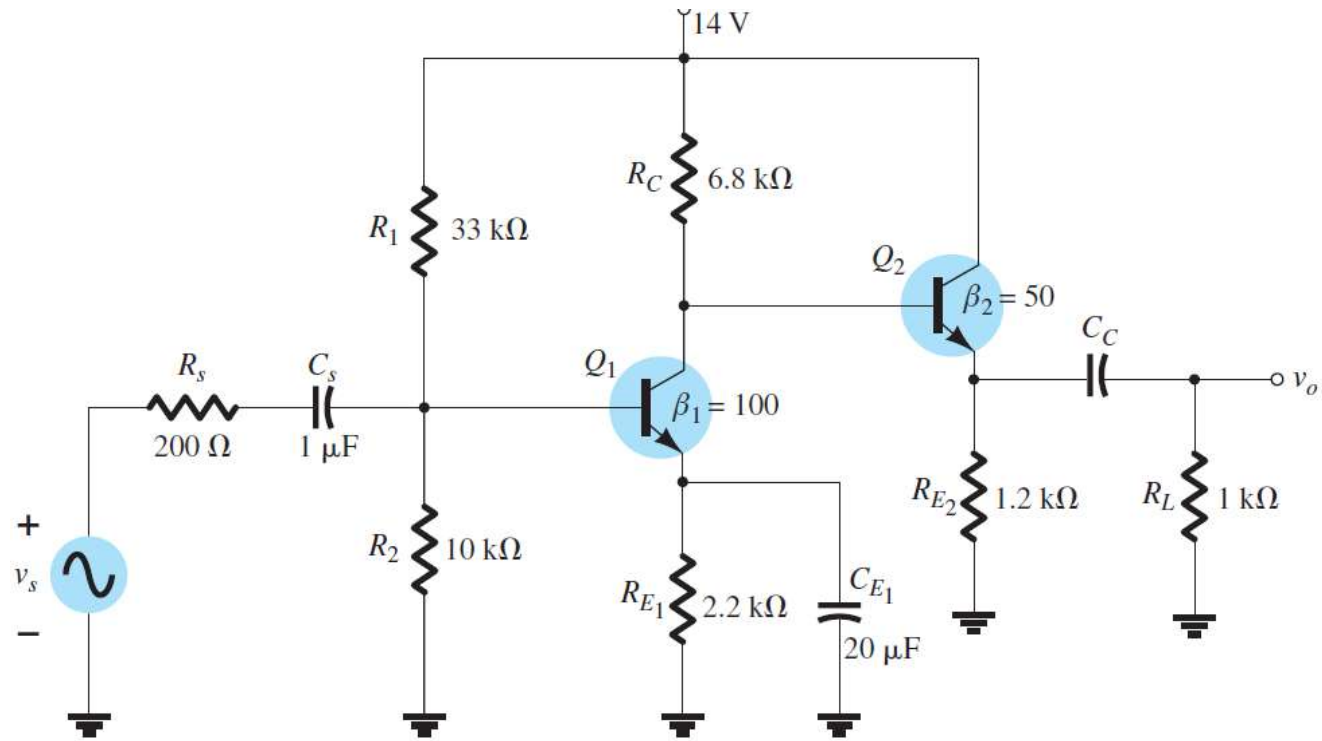


FIG. 4.72

4.14 CURRENT SOURCE CIRCUITS

The concept of a power supply provides the starting point in our consideration of current source circuits. A practical voltage source (Fig. 4.80a) is a voltage supply in series with a resistance. An ideal voltage source has $R = 0$, whereas a practical source includes some small resistance. A practical current source (Fig. 4.80b) is a current supply in parallel with a resistance. An ideal current source has $R = \infty\,\Omega$, whereas a practical current source includes some very large resistance.

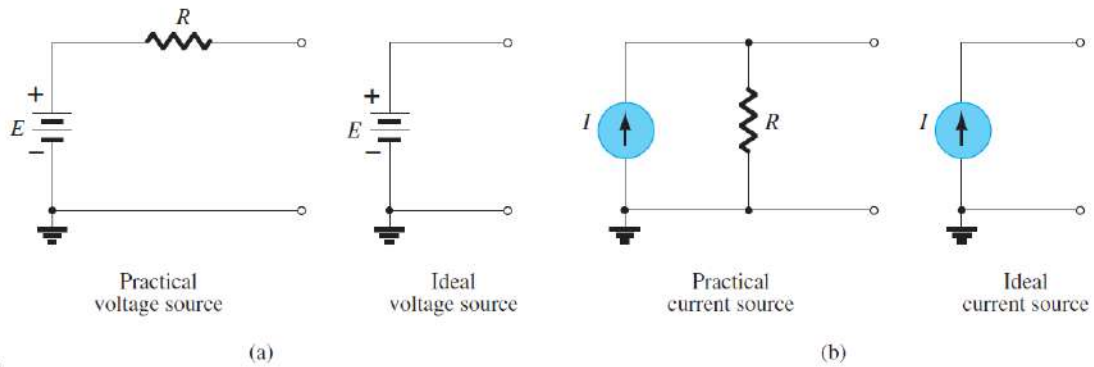


FIG. 4.80
Voltage and current sources.

An ideal current source provides a constant current regardless of the load connected to it. There are many uses in electronics for a circuit providing a constant current at a very high impedance. Constant-current circuits can be built using bipolar devices, FET devices, and a combination of these components. There are circuits used in discrete form and others more suitable for operation in integrated circuits.

Bipolar Transistor Constant-Current Source

Bipolar transistors can be connected in a circuit that acts as a constant-current source in a number of ways. Figure 4.81 shows a circuit using a few resistors and an *npn* transistor for operation as a constant-current circuit. The current through I_E can be determined as follows. Assuming that the base input impedance is much larger than R_1 or R_2, we have

$$V_B = \frac{R_1}{R_1 + R_2}(-V_{EE})$$

and

$$V_E = V_B - 0.7\ \text{V}$$

with

$$I_E = \frac{V_E - (-V_{EE})}{R_E} \approx I_C$$

where I_C is the constant current provided by the circuit of Fig. 4.81.

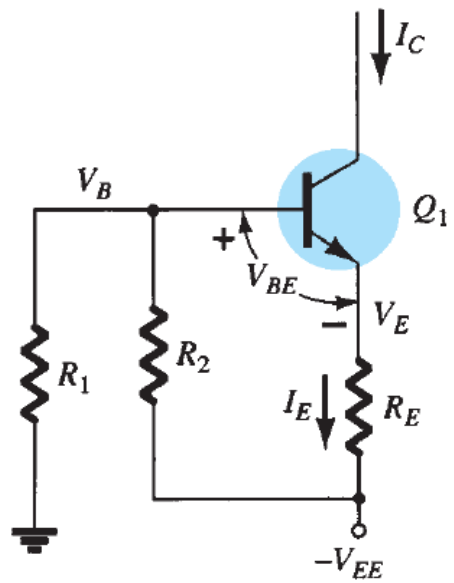


FIG. 4.81
Discrete constant-current source.

EXAMPLE 4.29 Calculate the constant current I in the circuit of Fig. 4.82.

Solution:

$$V_B = \frac{R_1}{R_1 + R_2}(-V_{EE}) = \frac{5.1\ \text{k}\Omega}{5.1\ \text{k}\Omega + 5.1\ \text{k}\Omega}(-20\ \text{V}) = -10\ \text{V}$$

$$V_E = V_B - 0.7\ \text{V} = 10\ \text{V} - 0.7\ \text{V} = -10.7\ \text{V}$$

$$I = I_E = \frac{V_E - (-V_{EE})}{R_E} = \frac{-10.7\ \text{V} - (-20\ \text{V})}{2\ \text{k}\Omega}$$

$$= \frac{9.3\ \text{V}}{2\ \text{k}\Omega} = \mathbf{4.65\ mA}$$

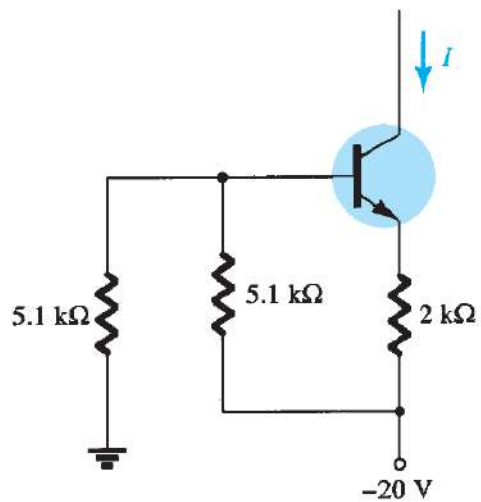


FIG. 4.82
Constant-current source for Example 4.29.

Transistor/Zener Constant-Current Source

Replacing resistor R_2 with a Zener diode, as shown in Fig. 4.83, provides an improved constant-current source over that of Fig. 4.81. The Zener diode results in a constant current calculated using the base-emitter KVL (Kirchhoff voltage loop) equation. The value of I can be calculated using

$$I \approx I_E = \frac{V_Z - V_{BE}}{R_E} \tag{4.83}$$

A major point to consider is that the constant current depends on the Zener diode voltage, which remains quite constant, and the emitter resistor R_E. The voltage supply V_{EE} has no effect on the value of I.

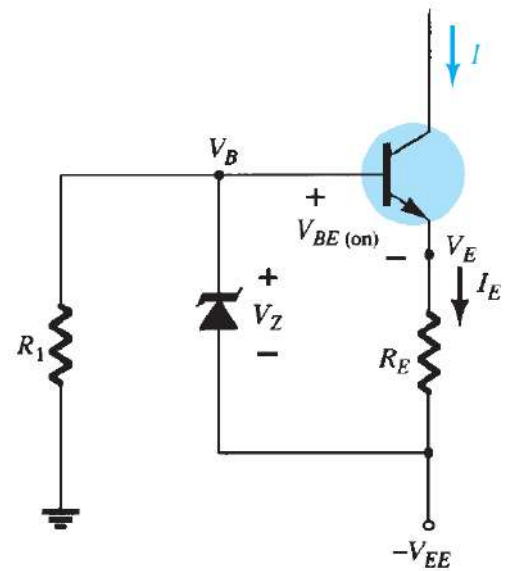


FIG. 4.83
Constant-current circuit using Zener diode.

EXAMPLE 4.30 Calculate the constant current I in the circuit of Fig. 4.84.

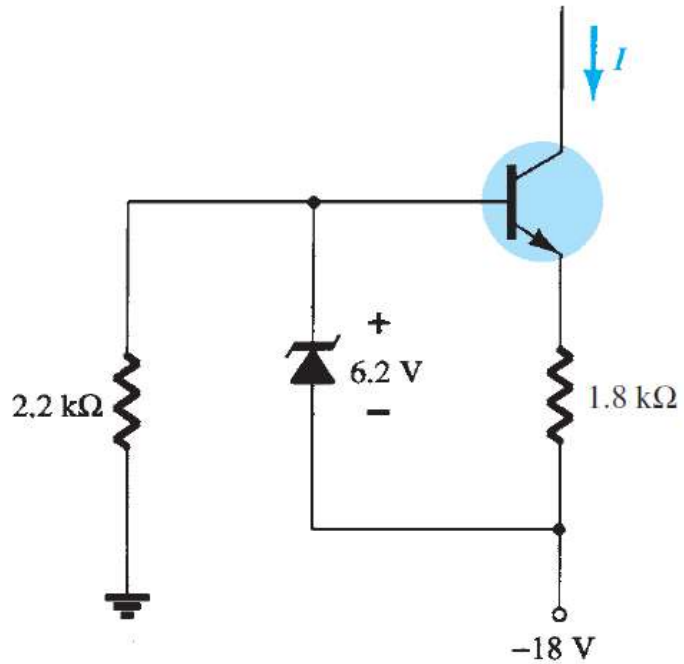


FIG. 4.84
Constant-current circuit for Example 4.30.

Solution:

$$\text{Eq. (4.83):} \quad I = \frac{V_Z - V_{BE}}{R_E} = \frac{6.2\text{ V} - 0.7\text{ V}}{1.8\text{ k}\Omega} = 3.06\text{ mA} \approx \mathbf{3\text{ mA}}$$

4.15 *pnp* TRANSISTORS

The analysis thus far has been limited totally to *npn* transistors to ensure that the initial analysis of the basic configurations was as clear as possible and uncomplicated by switching between types of transistors. Fortunately, the analysis of *pnp* transistors follows the same pattern established for *npn* transistors. The level of I_B is first determined, followed by the application of the appropriate transistor relationships to determine the list of unknown quantities. In fact, the only difference between the resulting equations for a network in which an *npn* transistor has been replaced by a *pnp* transistor is the sign associated with particular quantities.

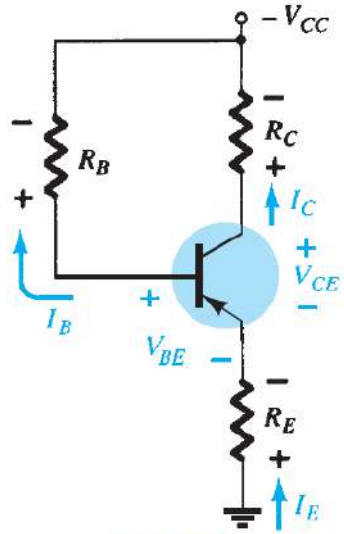


FIG. 4.85
pnp transistor in an emitter-stabilized configuration.

As noted in Fig. 4.85, the double-subscript notation continues as normally defined. The current directions, however, have been reversed to reflect the actual conduction directions. Using the defined polarities of Fig. 4.85, both V_{BE} and V_{CE} will be negative quantities.

Applying Kirchhoff's voltage law to the base–emitter loop results in the following equation for the network of Fig. 4.85:

$$-I_E R_E + V_{BE} - I_B R_B + V_{CC} = 0$$

Substituting $I_E = (\beta + 1)I_B$ and solving for I_B yields

$$I_B = \frac{V_{CC} + V_{BE}}{R_B + (\beta + 1)R_E}$$

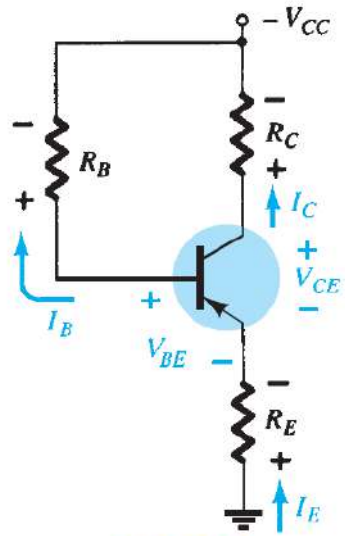


FIG. 4.85
pnp transistor in an emitter-stabilized configuration.

The resulting equation is the same as Eq. (4.17) except for the sign for V_{BE}. However, in this case $V_{BE} = -0.7$ V and the substitution of values results in the same sign for each term of Eq. (4.84) as Eq. (4.17). Keep in mind that the direction of I_B is now defined opposite of that for a *pnp* transistor as shown in Fig. 4.85.

For V_{CE} Kirchhoff's voltage law is applied to the collector–emitter loop, resulting in the following equation:

$$-I_E R_E + V_{CE} - I_C R_C + V_{CC} = 0$$

Substituting $I_E \cong I_C$ gives

$$\boxed{V_{CE} = -V_{CC} + I_C(R_C + R_E)} \qquad \textbf{(4.85)}$$

The resulting equation has the same format as Eq. (4.19), but the sign in front of each term on the right of the equal sign has changed. Because V_{CC} will be larger than the magnitude of the succeeding term, the voltage V_{CE} will have a negative sign, as noted in an earlier paragraph.

EXAMPLE 4.31 Determine V_{CE} for the voltage-divider bias configuration of Fig. 4.86.

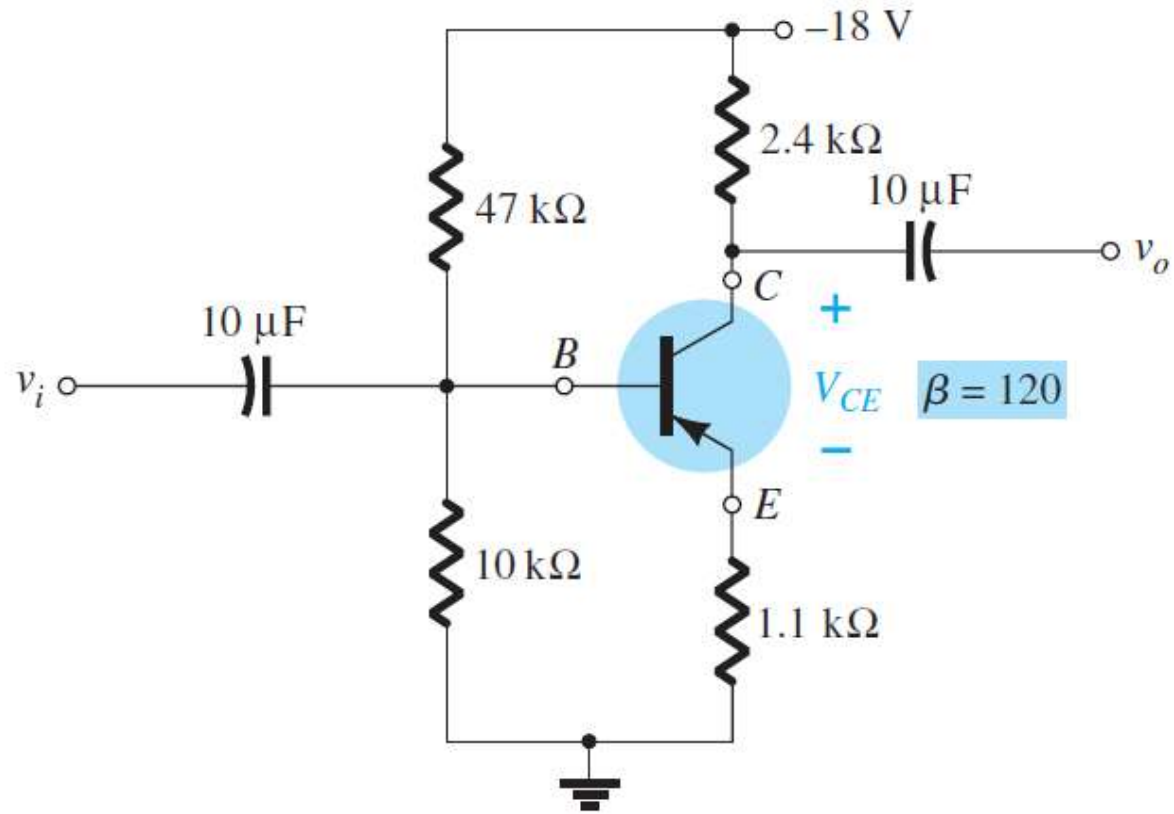


FIG. 4.86
pnp transistor in a voltage-divider bias configuration.

Solution: Testing the condition

$$\beta R_E \geq 10R_2$$

results in

$$(120)(1.1\ \text{k}\Omega) \geq 10(10\ \text{k}\Omega)$$
$$132\ \text{k}\Omega \geq 100\ \text{k}\Omega \quad (\textit{satisfied})$$

Solving for V_B, we have

$$V_B = \frac{R_2 V_{CC}}{R_1 + R_2} = \frac{(10\ \text{k}\Omega)(-18\ \text{V})}{47\ \text{k}\Omega + 10\ \text{k}\Omega} = -3.16\ \text{V}$$

Note the similarity in format of the equation with the resulting negative voltage for V_B.

Applying Kirchhoff's voltage law around the base–emitter loop yields

$$+V_B - V_{BE} - V_E = 0$$

and

$$V_E = V_B - V_{BE}$$

Substituting values, we obtain

$$V_E = -3.16\ \text{V} - (-0.7\ \text{V})$$
$$= -3.16\ \text{V} + 0.7\ \text{V}$$
$$= -2.46\ \text{V}$$

Note in the equation above that the standard single- and double-subscript notation is employed. For an *npn* transistor the equation $V_E = V_B - V_{BE}$ would be exactly the same. The only difference surfaces when the values are substituted.

The current is

$$I_E = \frac{V_E}{R_E} = \frac{2.46\ \text{V}}{1.1\ \text{k}\Omega} = 2.24\ \text{mA}$$

For the collector–emitter loop,

$$-I_E R_E + V_{CE} - I_C R_C + V_{CC} = 0$$

Substituting $I_E \cong I_C$ and gathering terms, we have

$$V_{CE} = -V_{CC} + I_C(R_C + R_E)$$

Substituting values gives

$$\begin{aligned} V_{CE} &= -18\ \text{V} + (2.24\ \text{mA})(2.4\ \text{k}\Omega + 1.1\ \text{k}\Omega) \\ &= -18\ \text{V} + 7.84\ \text{V} \\ &= \mathbf{-10.16\ V} \end{aligned}$$

4.16 TRANSISTOR SWITCHING NETWORKS

The application of transistors is not limited solely to the amplification of signals. Through proper design, transistors can be used as switches for computer and control applications. The network of Fig. 4.87a can be employed as an *inverter* in computer logic circuitry. Note that the output voltage V_C is opposite to that applied to the base or input terminal. In addition, note the absence of a dc supply connected to the base circuit. The only dc source is connected to the collector or output side, and for computer applications is typically equal to the magnitude of the "high" side of the applied signal—in this case 5 V. The resistor R_B will ensure that the full applied voltage of 5 V will not appear across the base-to-emitter junction. It will also set the I_B level for the "on" condition.

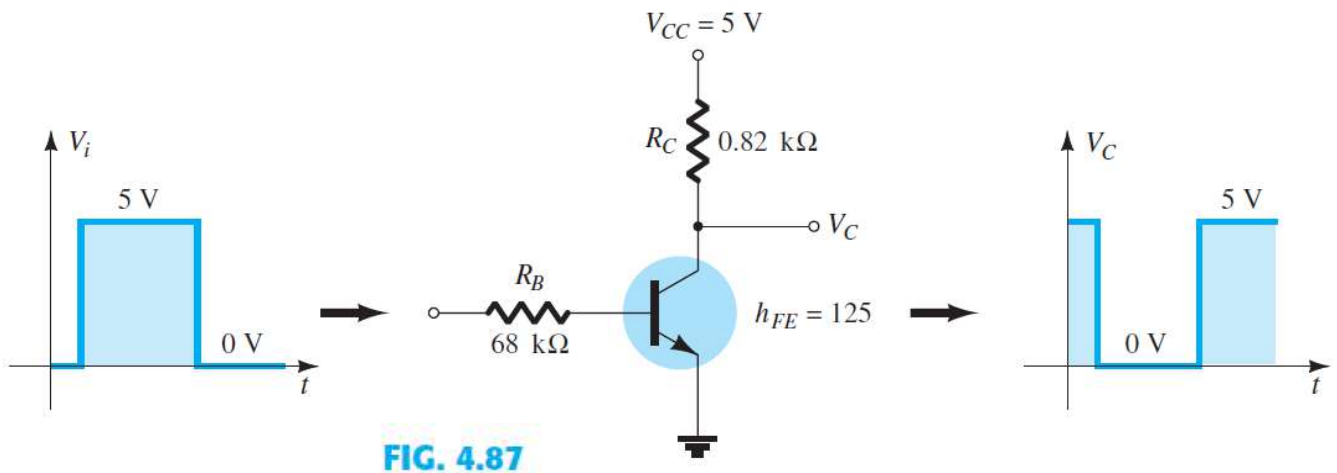
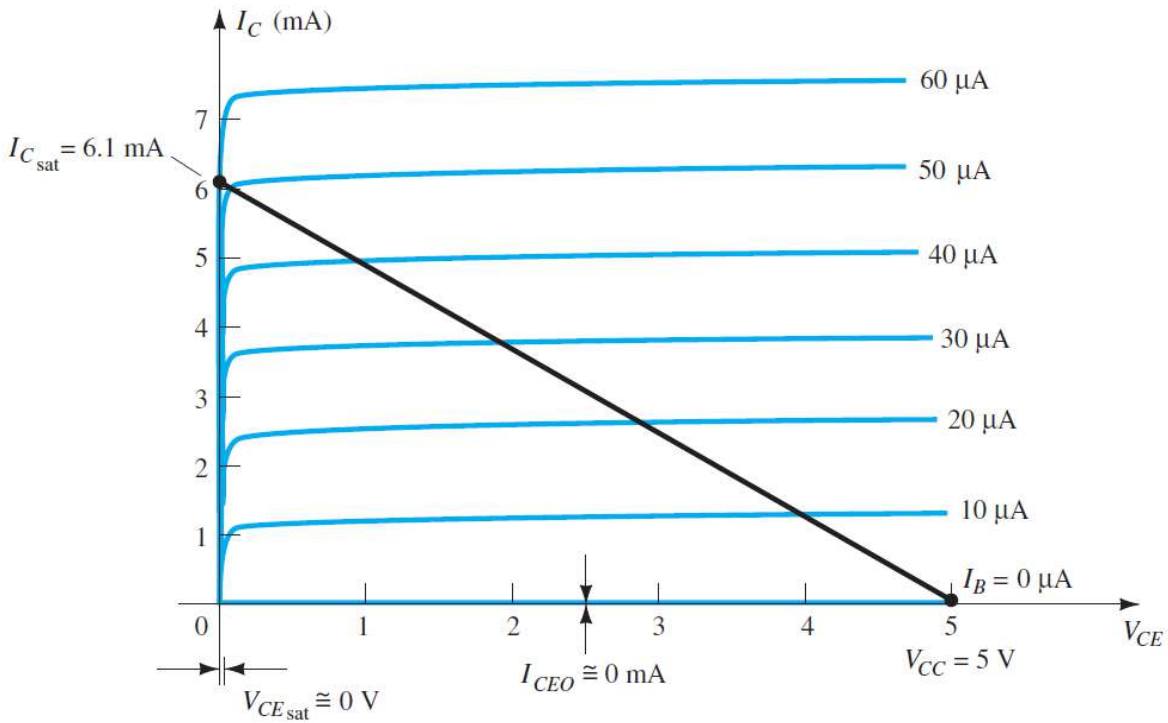


FIG. 4.87
Transistor inverter. (a)

4.16 TRANSISTOR SWITCHING NETWORKS

Proper design for the inversion process requires that the operating point switch from cutoff to saturation along the load line depicted in Fig. 4.87b. For our purposes we will assume that $I_C = I_{CEO} \cong 0$ mA when $I_B = 0\ \mu$A (an excellent approximation in light of improving construction techniques), as shown in Fig. 4.87b. In addition, we will assume that $V_{CE} = V_{CE_{sat}} \cong 0$ V rather than the typical 0.1-V to 0.3-V level.

When $V_i = 5$ V, the transistor will be "on" and the design must ensure that the network is heavily saturated by a level of I_B greater than that associated with the I_B curve appearing



(b)

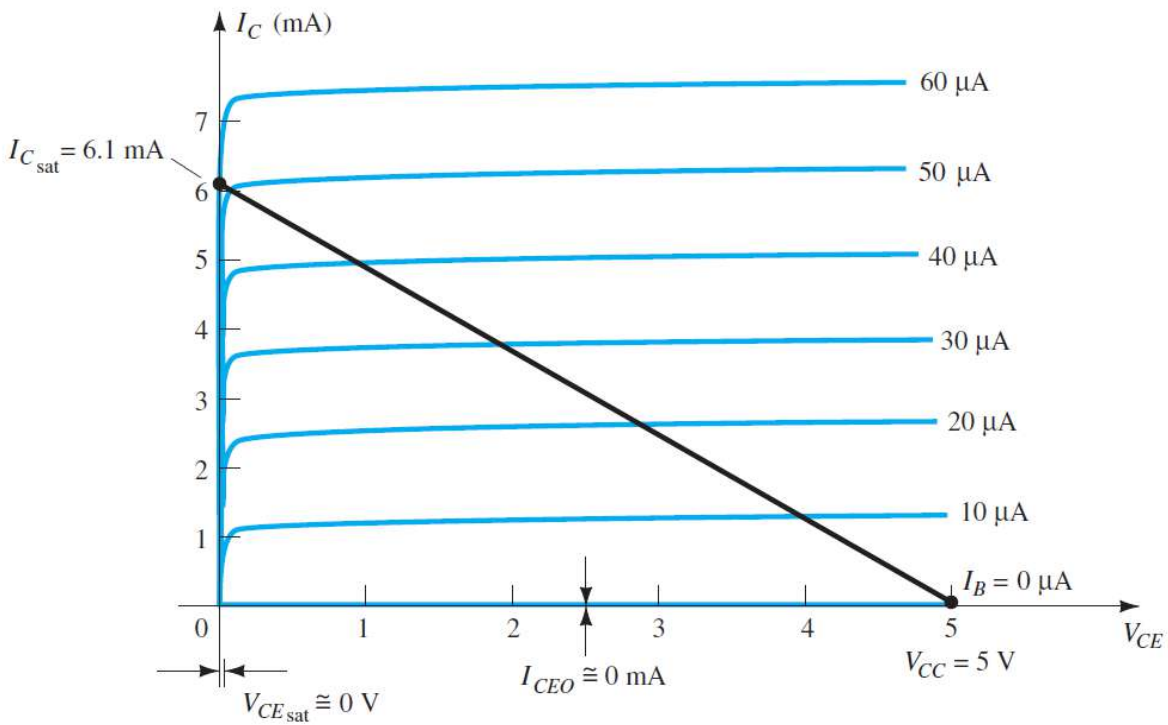
FIG. 4.87
Transistor inverter.

near the saturation level. In Fig. 4.87b, this requires that $I_B > 50\ \mu A$. The saturation level for the collector current for the circuit of Fig. 4.87a is defined by

$$I_{C_{sat}} = \frac{V_{CC}}{R_C}$$

The level of I_B in the active region just before saturation results can be approximated by the following equation:

$$I_{B_{max}} \cong \frac{I_{C_{sat}}}{\beta_{dc}}$$



(b)

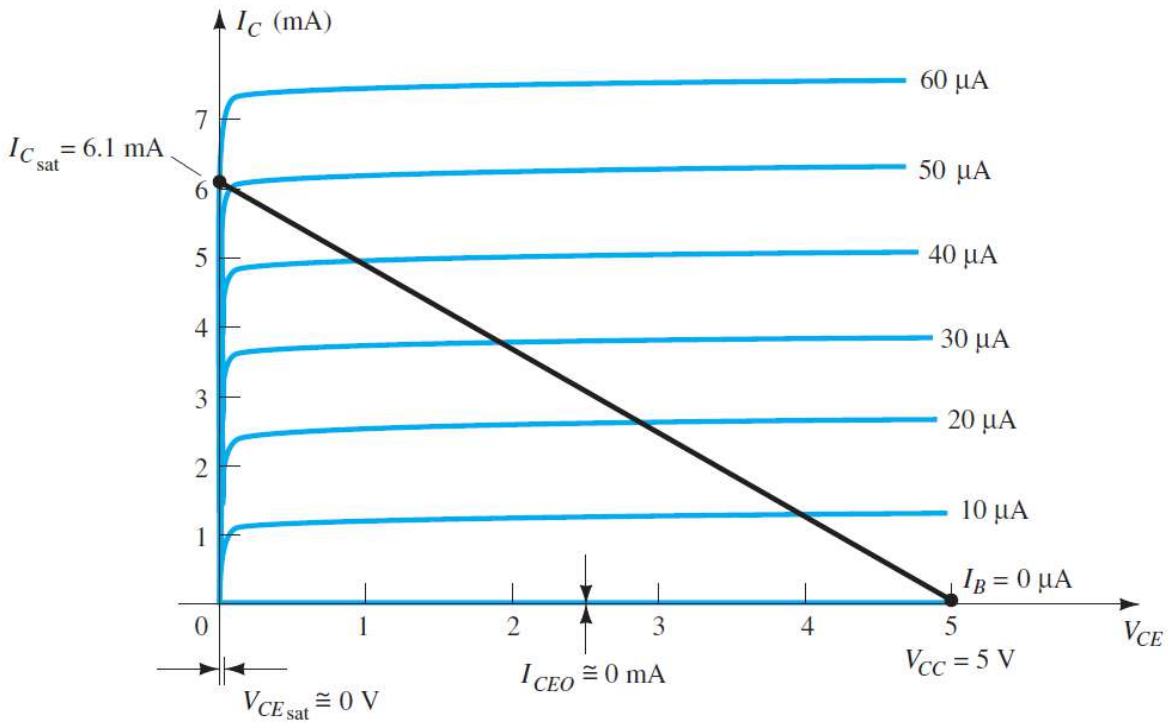
FIG. 4.87
Transistor inverter.

For the network of Fig. 4.87b, when $V_i = 5$ V, the resulting level of I_B is

$$I_B = \frac{V_i - 0.7\text{ V}}{R_B} = \frac{5\text{ V} - 0.7\text{ V}}{68\text{ k}\Omega} = 63\ \mu\text{A}$$

and

$$I_{C_{\text{sat}}} = \frac{V_{CC}}{R_C} = \frac{5\text{ V}}{0.82\text{ k}\Omega} \cong 6.1\text{ mA}$$



(b)

FIG. 4.87
Transistor inverter.

Testing Eq. (4.87) gives

$$I_B = 63\,\mu\text{A} > \frac{I_{C_{\text{sat}}}}{\beta_{\text{dc}}} = \frac{6.1\text{ mA}}{125} = 48.8\,\mu\text{A}$$

which is satisfied. Certainly, any level of I_B greater than 60 μA will pass through a Q-point on the load line that is very close to the vertical axis.

For $V_i = 0$ V, $I_B = 0\,\mu$A, and because we are assuming that $I_C = I_{CEO} = 0$ mA, the voltage drop across R_C as determined by $V_{RC} = I_C R_C = 0$ V, resulting in $V_C = +5$ V for the response indicated in Fig. 4.87a.

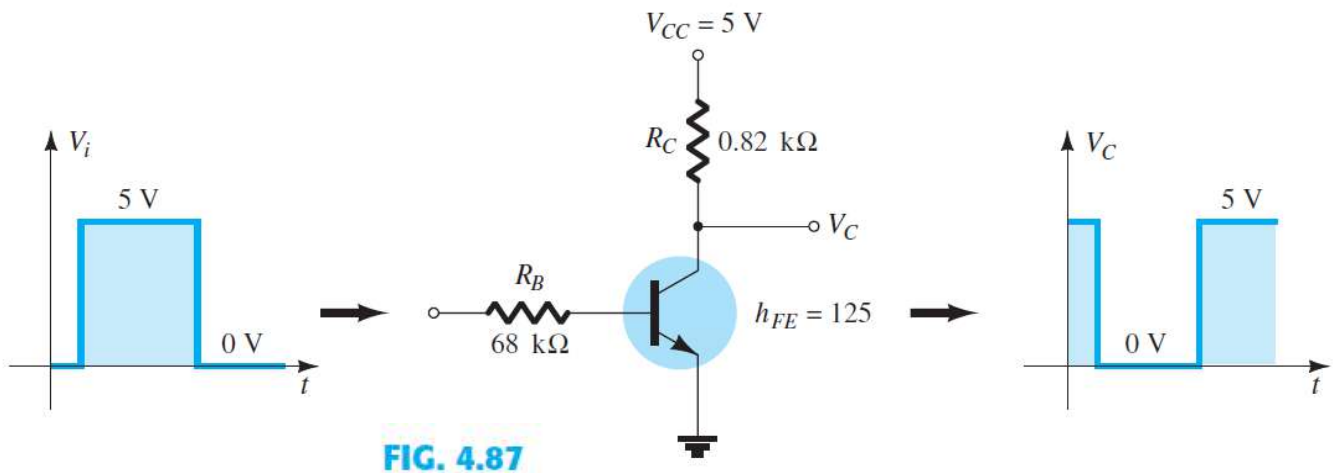


FIG. 4.87
Transistor inverter. (a)

In addition to its contribution to computer logic, the transistor can also be employed as a switch using the same extremities of the load line. At saturation, the current I_C is quite high and the voltage V_{CE} very low. The result is a resistance level between the two terminals determined by

$$R_{\text{sat}} = \frac{V_{CE_{\text{sat}}}}{I_{C_{\text{sat}}}}$$

and is depicted in Fig. 4.88.

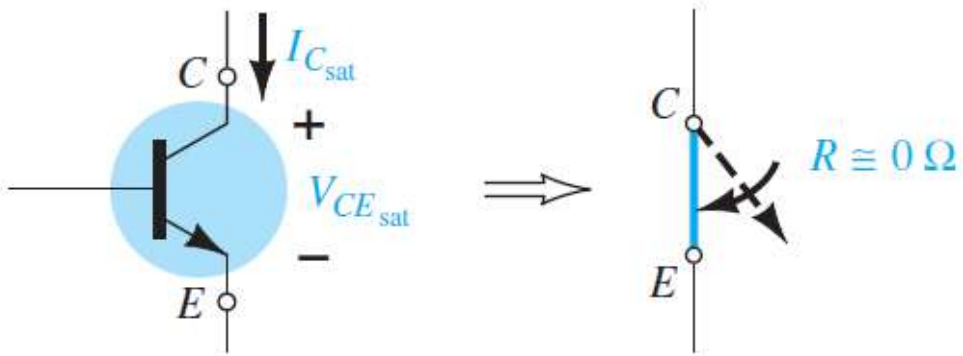


FIG. 4.88
Saturation conditions and the resulting terminal resistance.

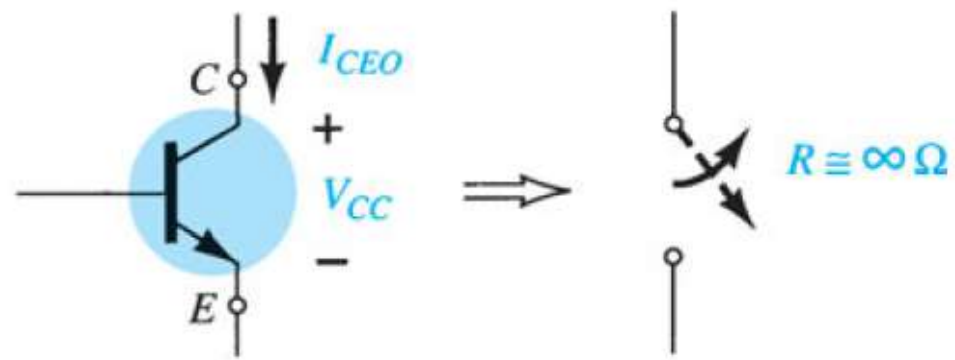


FIG. 4.89
Cutoff conditions and the resulting terminal resistance.

Using a typical average value of $V_{CE_{sat}}$ such as 0.15 V gives

$$R_{sat} = \frac{V_{CE_{sat}}}{I_{C_{sat}}} = \frac{0.15\ \text{V}}{6.1\ \text{mA}} = 24.6\ \Omega$$

which is a relatively low value and can be considered as approximately 0 Ω when placed in series with resistors in the kilohm range.

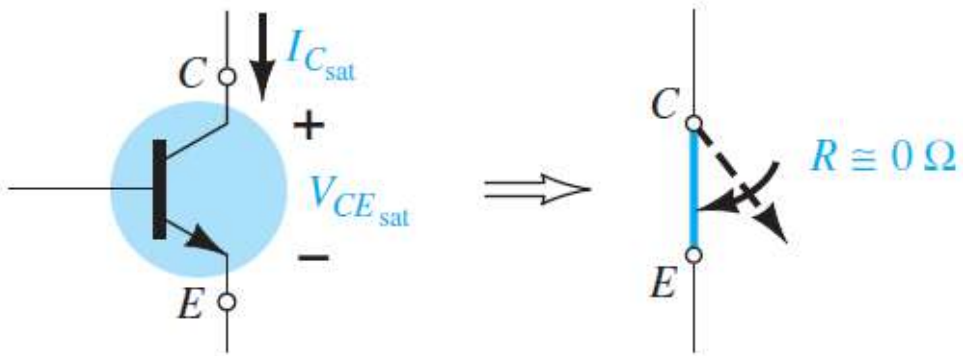


FIG. 4.88
Saturation conditions and the resulting terminal resistance.

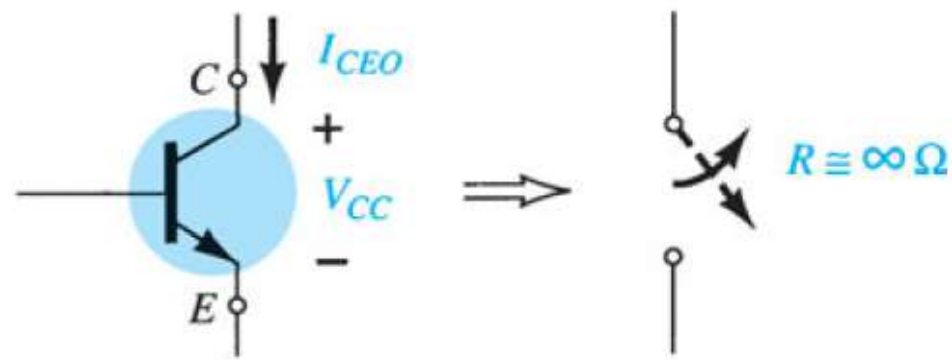


FIG. 4.89
Cutoff conditions and the resulting terminal resistance.

For $V_i = 0$ V, as shown in Fig. 4.89, the cutoff condition results in a resistance level of the following magnitude:

$$R_{\text{cutoff}} = \frac{V_{CC}}{I_{CEO}} = \frac{5\text{ V}}{0\text{ mA}} = \boldsymbol{\infty\,\Omega}$$

resulting in the open-circuit equivalence. For a typical value of $I_{CEO} = 10\,\mu$A, the magnitude of the cutoff resistance is

$$R_{\text{cutoff}} = \frac{V_{CC}}{I_{CEO}} = \frac{5\text{ V}}{10\,\mu\text{A}} = \boldsymbol{500\text{ k}\Omega}$$

which certainly approaches an open-circuit equivalence for many situations.

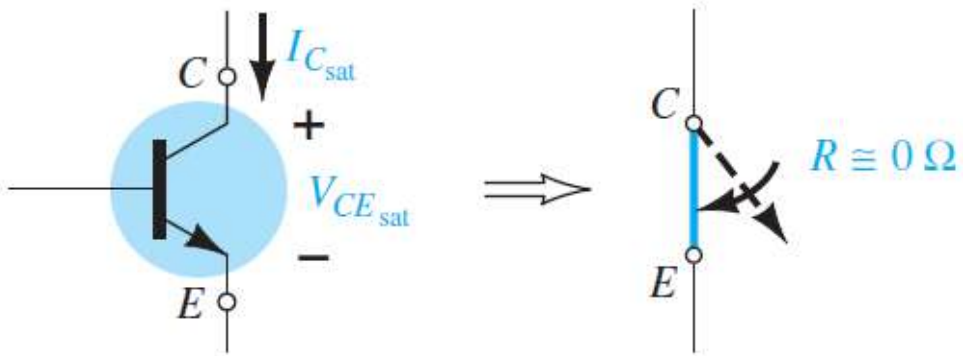


FIG. 4.88
Saturation conditions and the resulting terminal resistance.

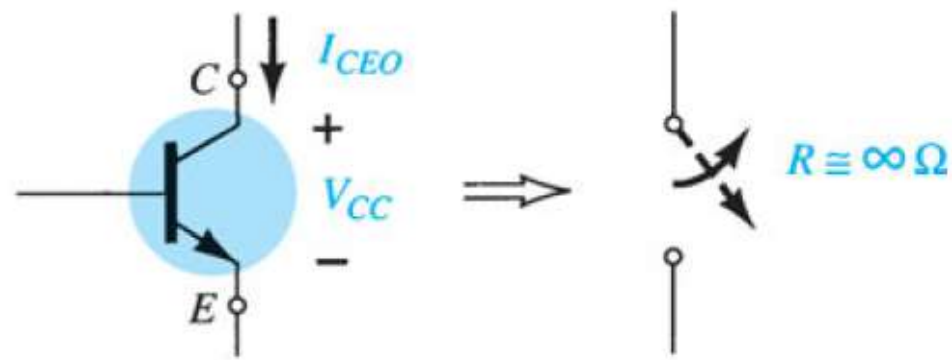


FIG. 4.89
Cutoff conditions and the resulting terminal resistance.

EXAMPLE 4.32 Determine R_B and R_C for the transistor inverter of Fig. 4.90 if $I_{C_{sat}} = 10$ mA.

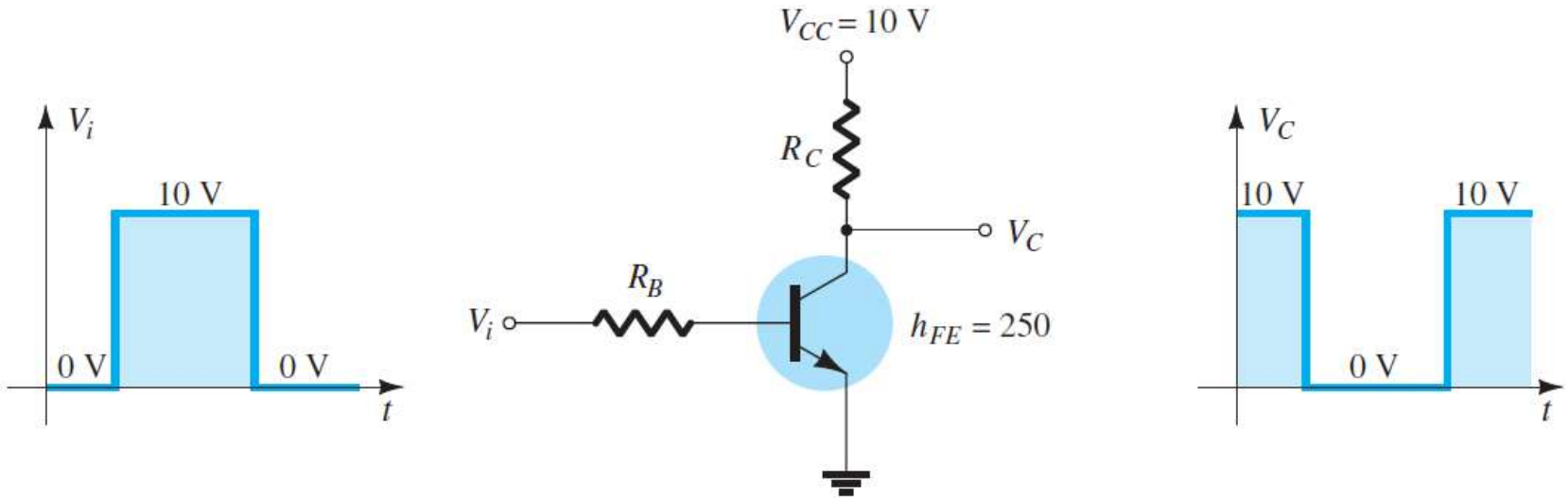


FIG. 4.90

Solution: At saturation,

$$I_{C_{sat}} = \frac{V_{CC}}{R_C}$$

$$10\text{ mA} = \frac{10\text{ V}}{R_C}$$

$$R_C = \frac{10\text{ V}}{10\text{ mA}} = 1\text{ k}\Omega$$

At saturation,

$$I_B \cong \frac{I_{C_{sat}}}{\beta_{dc}} = \frac{10\text{ mA}}{250} = 40\ \mu\text{A}$$

Choosing $I_B = 60\ \mu\text{A}$ to ensure saturation and using $I_B = \dfrac{V_i - 0.7\text{ V}}{R_B}$ ➡ $R_B = \dfrac{V_i - 0.7\text{ V}}{I_B} = \dfrac{10\text{ V} - 0.7\text{ V}}{60\ \mu\text{A}} = 155\text{ k}\Omega$

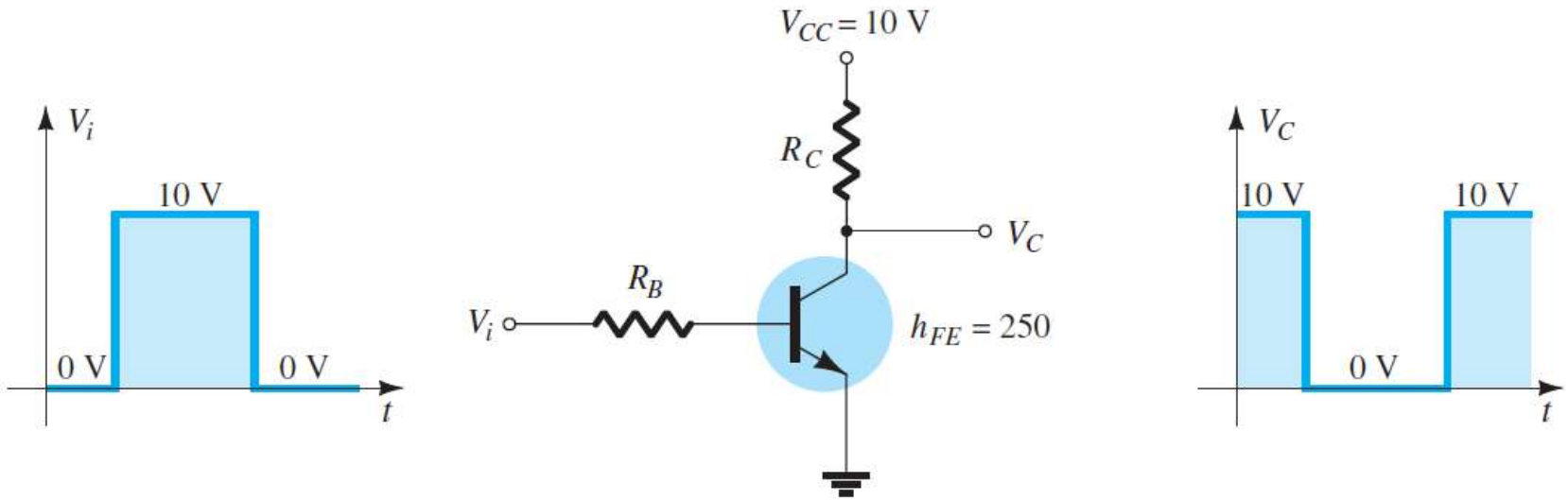


FIG. 4.90

Choose $R_B = 150\,\text{k}\Omega$, which is a standard value. Then

$$I_B = \frac{V_i - 0.7\,\text{V}}{R_B} = \frac{10\,\text{V} - 0.7\,\text{V}}{150\,\text{k}\Omega} = 62\,\mu\text{A}$$

and

$$I_B = 62\,\mu\text{A} > \frac{I_{C_{\text{sat}}}}{\beta_{\text{dc}}} = 40\,\mu\text{A}$$

Therefore, use $R_B = \mathbf{150\,k\Omega}$ and $R_C = \mathbf{1\,k\Omega}$.

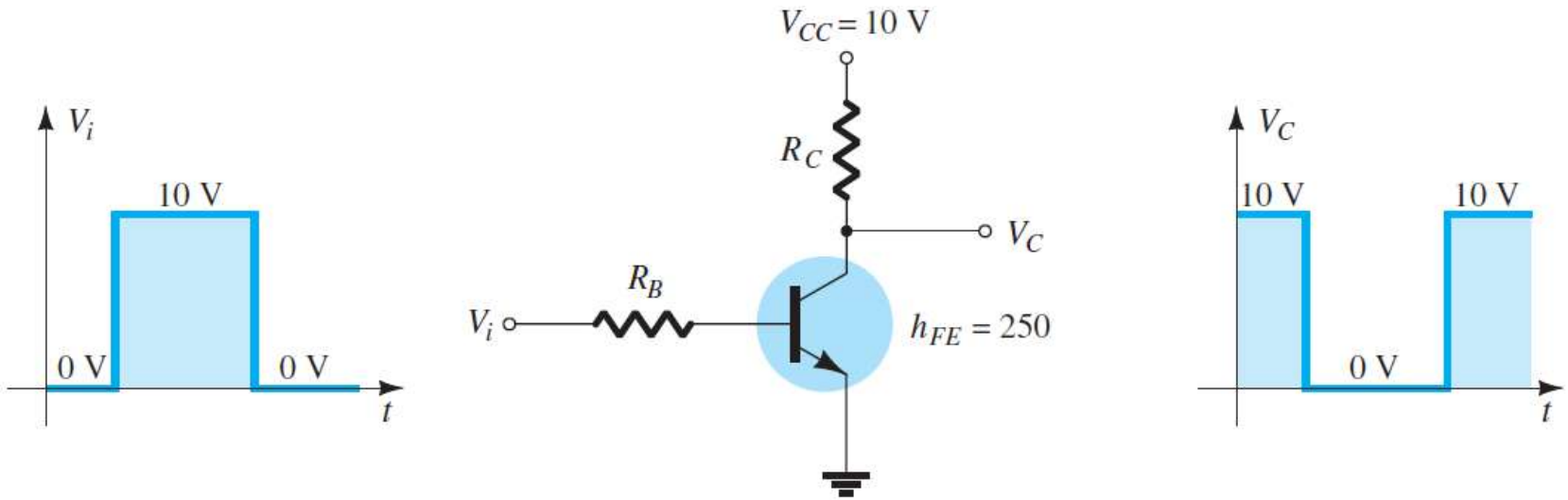


FIG. 4.90

4.17 TROUBLESHOOTING TECHNIQUES

The art of troubleshooting is such a broad topic that a full range of possibilities and techniques cannot be covered in a few sections of a book. However, the practitioner should be aware of a few basic maneuvers and measurements that can isolate the problem area and possibly identify a solution.

Quite obviously, the first step in being able to troubleshoot a network is to fully understand the behavior of the network and to have some idea of the expected voltage and current levels. For the transistor in the active region, the most important measurable dc level is the base-to-emitter voltage.

For an "on" transistor, the voltage V_{BE} should be in the neighborhood of 0.7 V.

For an "on" transistor, the voltage V_{BE} should be in the neighborhood of 0.7 V.

The proper connections for measuring V_{BE} appear in Fig. 4.92. Note that the positive (red) lead is connected to the base terminal for an *npn* transistor and the negative (black) lead to the emitter terminal. Any reading totally different from the expected level of about 0.7 V, such as 0, 4, or 12 V or a negative value, would be suspect and the device or network connections should be checked. For a *pnp* transistor, the same connections can be used, but a negative reading should be expected.

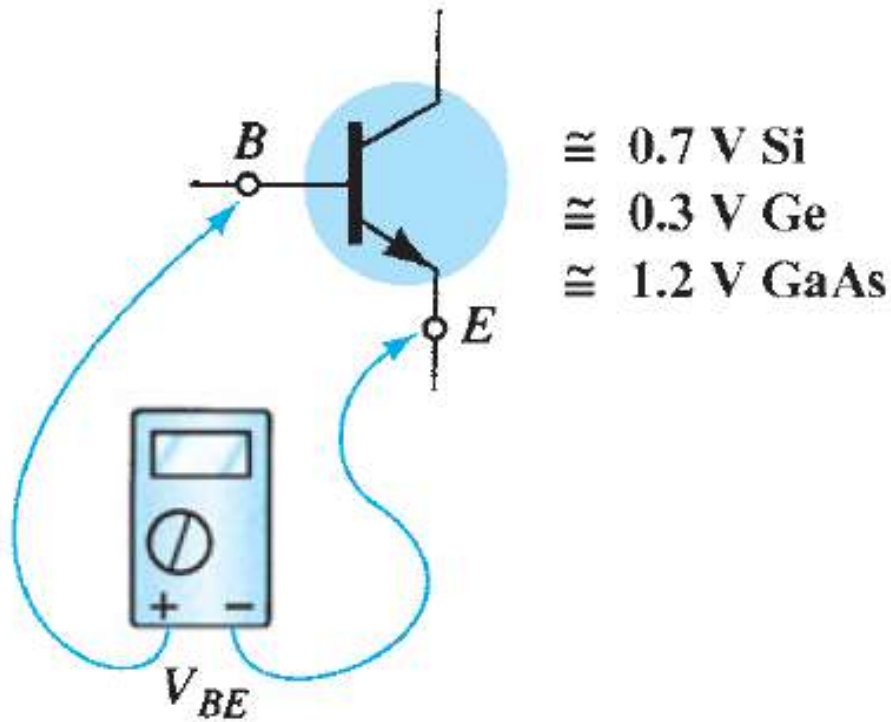


FIG. 4.92

A voltage level of equal importance is the collector-to-emitter voltage. Recall from the general characteristics of a BJT that levels of V_{CE} in the neighborhood of 0.3 V suggest a saturated device—a condition that should not exist unless it is being employed in a switching mode. However:

For the typical transistor amplifier in the active region, V_{CE} is usually about 25% to 75% of V_{CC}.

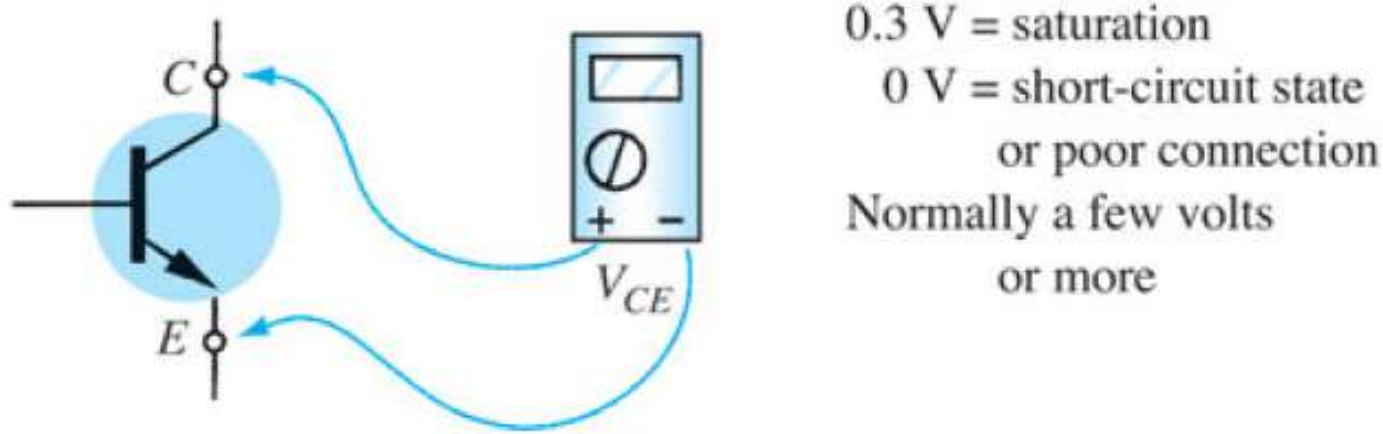


FIG. 4.93
Checking the dc level of V_{CE}.

For $V_{CC} = 20$ V, a reading of V_{CE} of 1 V to 2 V or from 18 V to 20 V as measured in Fig. 4.93 is certainly an uncommon result, and unless the device was knowingly designed for this response, the design and operation should be investigated. If $V_{CE} = 20$ V (with $V_{CC} = 20$ V) at least two possibilities exist—either the device (BJT) is damaged and has the characteristics of an open circuit between collector and emitter terminals or a connection in the collector–emitter or base–emitter circuit loop is open as shown in Fig. 4.94, establishing I_C at 0 mA and $V_{R_C} = 0$ V.

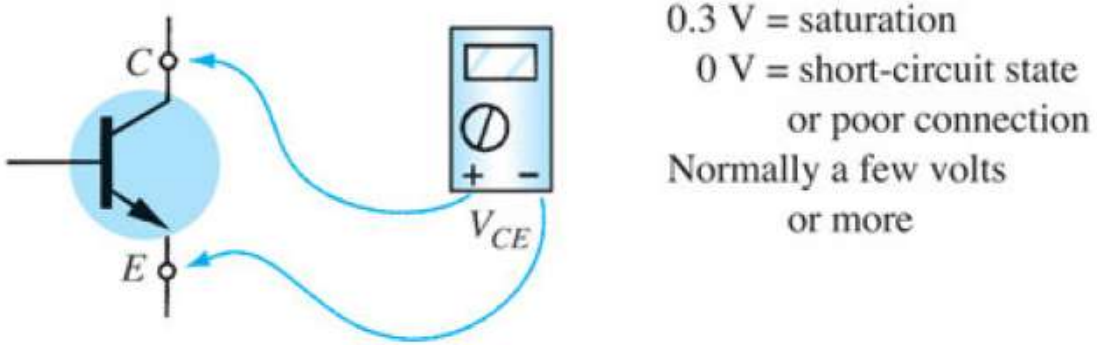


FIG. 4.93
Checking the dc level of V_{CE}.

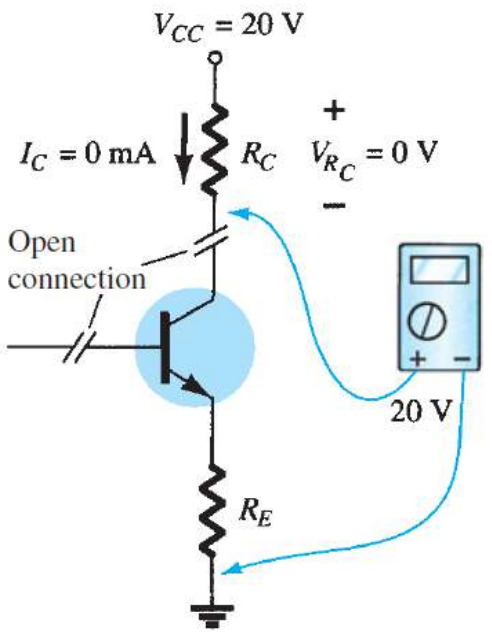


FIG. 4.94

In Fig. 4.94, the black lead of the voltmeter is connected to the common ground of the supply and the red lead to the bottom terminal of the resistor. The absence of a collector current and a consequent zero voltage drop across R_C will result in a reading of 20 V. If the meter is connected between the collector terminal and ground of the BJT, the reading will be 0 V because V_{CC} is blocked from the active device by the open circuit. One of the most common errors in the laboratory is the use of the wrong resistance value for a given design. Imagine the impact of using a 680-Ω resistor for R_B rather than the design value of 680 kΩ. For $V_{CC} = 20$ V and a fixed-bias configuration, the resulting base current would be

$$I_B = \frac{20\text{ V} - 0.7\text{ V}}{680\ \Omega} = 28.4\text{ mA}$$

rather than the desired 28.4 μA—a significant difference!

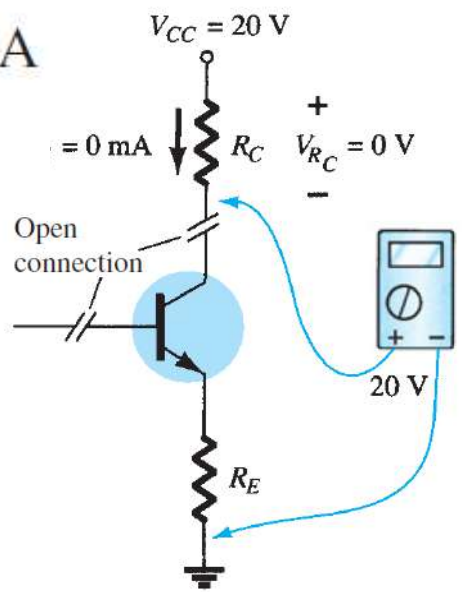


FIG. 4.94

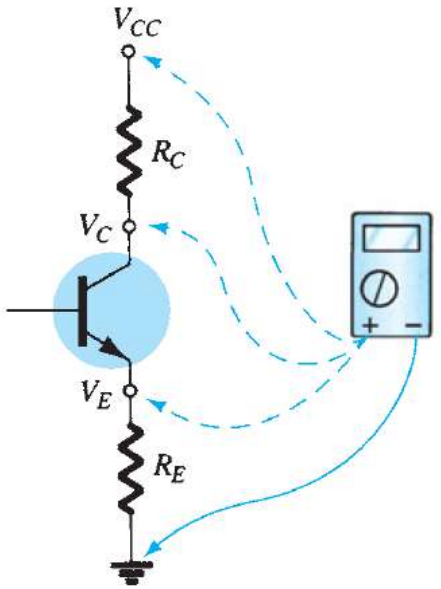


FIG. 4.95

EXAMPLE 4.33 Based on the readings provided in Fig. 4.96, determine whether the network is operating properly and, if not, the probable cause.

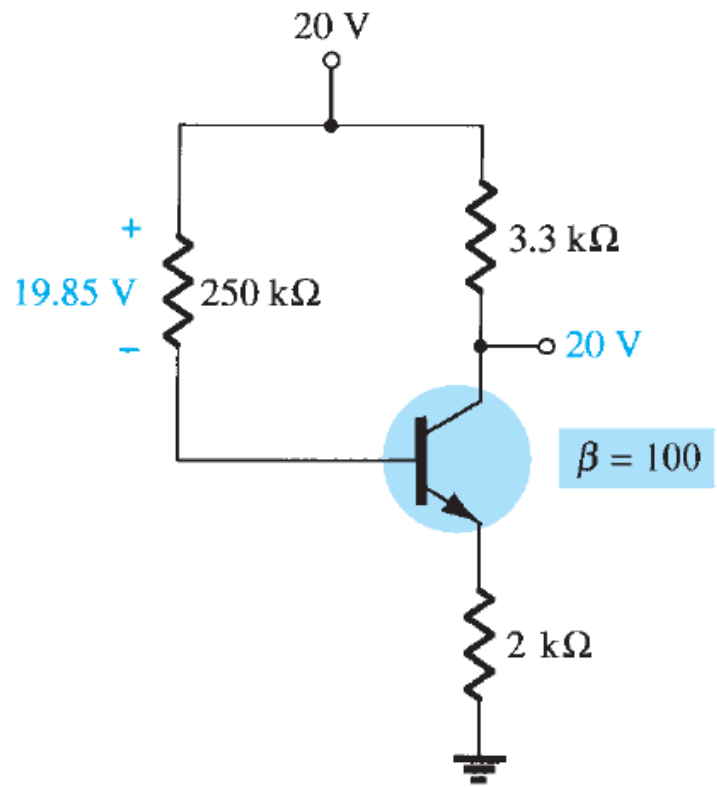
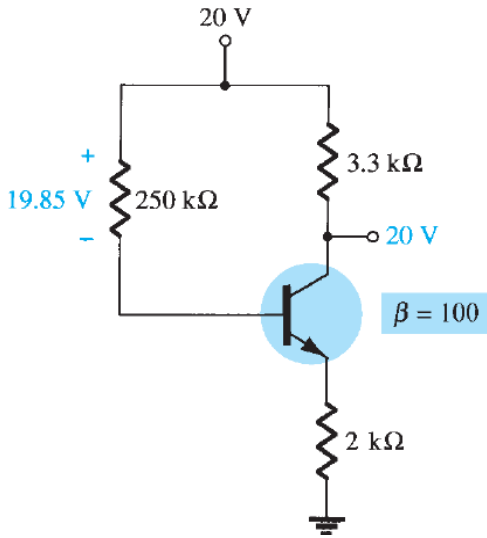


FIG. 4.96
Network for Example 4.33.

Solution: The 20 V at the collector immediately reveals that $I_C = 0$ mA, due to an open circuit or a nonoperating transistor. The level of $V_{R_B} = 19.85$ V also reveals that the transistor is "off" because the difference of $V_{CC} - V_{R_B} = 0.15$ V is less than that required to turn "on" the transistor and provide some voltage for V_E. In fact, if we assume a short-circuit condition from base to emitter, we obtain the following current through R_B:

$$I_{R_B} = \frac{V_{CC}}{R_B + R_E} = \frac{20\text{ V}}{252\text{ k}\Omega} = 79.4\ \mu\text{A}$$



which matches that obtained from

$$I_{R_B} = \frac{V_{R_B}}{R_B} = \frac{19.85\text{ V}}{250\text{ k}\Omega} = 79.4\ \mu\text{A}$$

If the network were operating properly, the base current should be

$$I_B = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1)R_E} = \frac{20\text{ V} - 0.7\text{ V}}{250\text{ k}\Omega + (101)(2\text{ k}\Omega)} = \frac{19.3\text{ V}}{452\text{ k}\Omega} = 42.7\ \mu\text{A}$$

The result, therefore, is that the transistor is in a damaged state, with a short-circuit condition between base and emitter.

EXAMPLE 4.34 Based on the readings appearing in Fig. 4.97, determine whether the transistor is "on" and the network is operating properly.

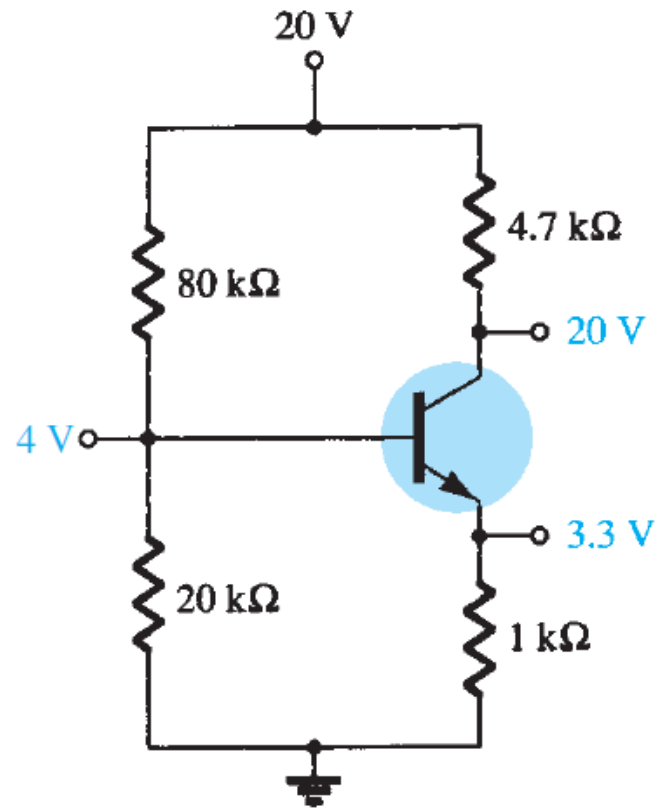
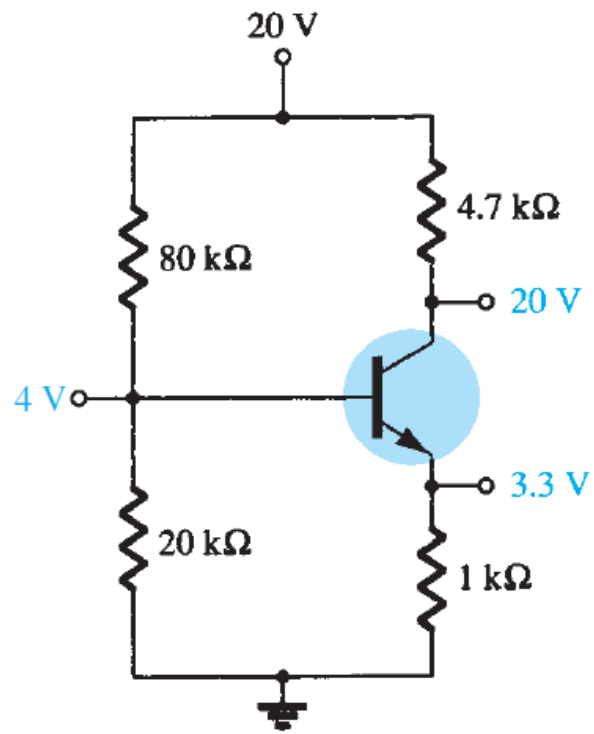


FIG. 4.97

Solution: Based on the resistor values of R_1 and R_2 and the magnitude of V_{CC}, the voltage $V_B = 4$ V seems appropriate (and in fact it is). The 3.3 V at the emitter results in a 0.7-V drop across the base-to-emitter junction of the transistor, suggesting an "on" transistor. However, the 20 V at the collector reveals that $I_C = 0$ mA, although the connection to the supply must be "solid" or the 20 V would not appear at the collector of the device. Two possibilities exist—there can be a poor connection between R_C and the collector terminal of the transistor or the transistor has an open base-to-collector junction. First, check the continuity at the collector junction using an ohm-meter, and if it is okay, check the transistor using one of the methods described in Chapter 3.



4.3 Fixed-Bias Configuration

1. For the fixed-bias configuration of Fig. 4.118, determine:
   a. I_{B_Q}.
   b. I_{C_Q}.
   c. V_{CE_Q}.
   d. V_C.
   e. V_B.
   f. V_E.

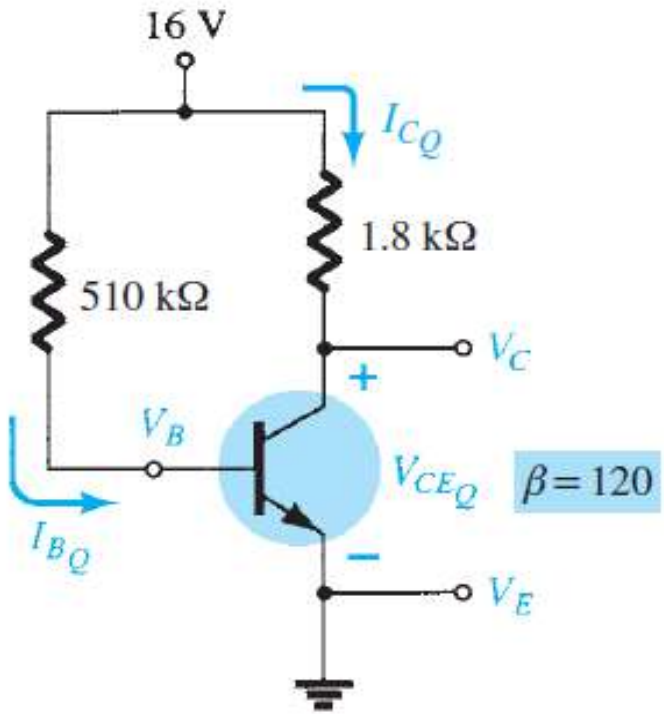


FIG. 4.118
Problems 1, 4, 6, 7, 14, 65, 69, 71, and 75.

(a) $I_{B_Q} = \dfrac{V_{CC} - V_{BE}}{R_B} = \dfrac{16\text{ V} - 0.7\text{ V}}{470\text{ k}\Omega} = \dfrac{15.3\text{ V}}{470\text{ k}\Omega} = \mathbf{32.55\ \mu A}$

(b) $I_{C_Q} = \beta I_{B_Q} = (90)(32.55\ \mu\text{A}) = \mathbf{2.93\text{ mA}}$

(c) $V_{CE_Q} = V_{CC} - I_{C_Q}R_C = 16\text{ V} - (2.93\text{ mA})(2.7\text{ k}\Omega) = \mathbf{8.09\text{ V}}$

(d) $V_C = V_{CE_Q} = \mathbf{8.09\text{ V}}$

(e) $V_B = V_{BE} = \mathbf{0.7\text{ V}}$

(f) $V_E = \mathbf{0\text{ V}}$

2. Given the information appearing in Fig. 4.119, determine:

a. I_C.

b. R_C.

c. R_B.

d. V_{CE}.

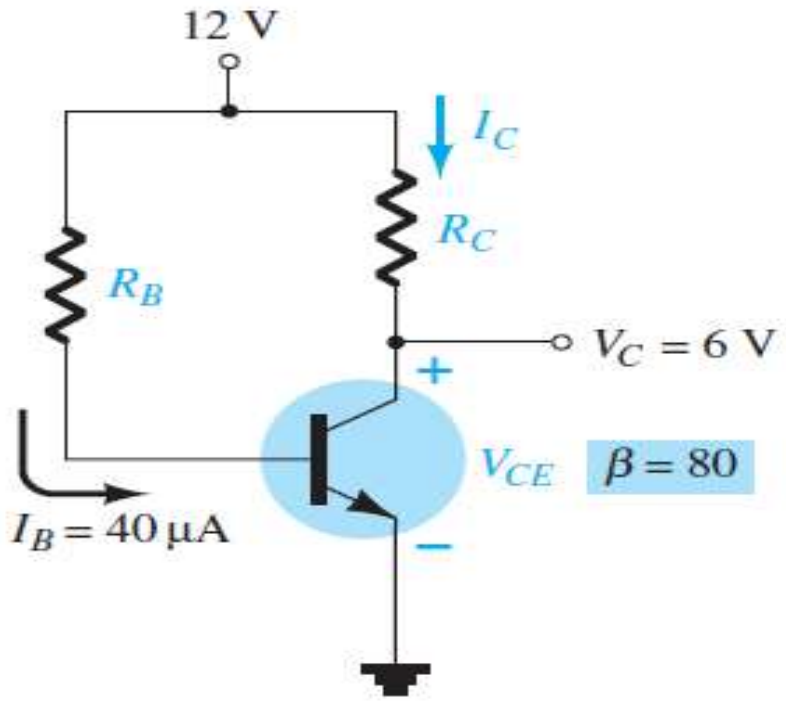


FIG. 4.119
Problem 2.

(a) $I_C = \beta I_B = 80(40\ \mu A) = \mathbf{3.2\ mA}$

(b) $R_C = \dfrac{V_{R_C}}{I_C} = \dfrac{V_{CC} - V_C}{I_C} = \dfrac{12\ V - 6\ V}{3.2\ mA} = \dfrac{6\ V}{3.2\ mA} = \mathbf{1.875\ k\Omega}$

(c) $R_B = \dfrac{V_{R_B}}{I_B} = \dfrac{12\ V - 0.7\ V}{40\ \mu A} = \dfrac{11.3\ V}{40\ \mu A} = \mathbf{282.5\ k\Omega}$

(d) $V_{CE} = V_C = \mathbf{6\ V}$

3. Given the information appearing in Fig. 4.120, determine:

a. I_C.

b. V_{CC}.

c. β.

d. R_B.

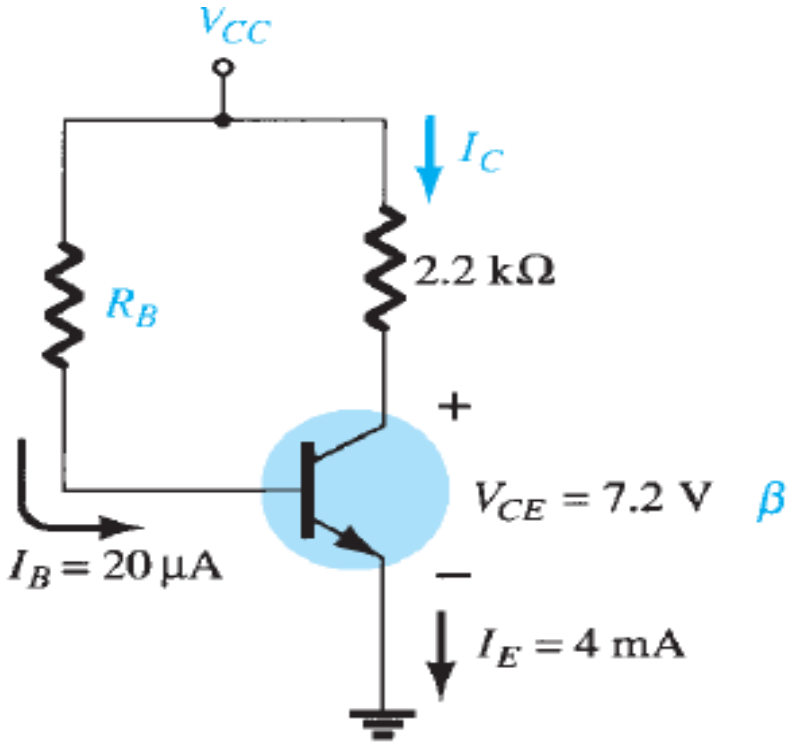


FIG. 4.120
Problem 3.

(a) $I_C = I_E - I_B = 4\text{ mA} - 20\ \mu\text{A} = \mathbf{3.98\ mA} \cong 4\text{ mA}$

(b) $V_{CC} = V_{CE} + I_C R_C = 7.2\text{ V} + (3.98\text{ mA})(2.2\text{ k}\Omega)$

$= \mathbf{15.96\ V} \cong 16\text{ V}$

(c) $\beta = \dfrac{I_C}{I_B} = \dfrac{3.98\text{ mA}}{20\ \mu\text{A}} = \mathbf{199} \cong 200$

(d) $R_B = \dfrac{V_{R_B}}{I_B} = \dfrac{V_{CC} - V_{BE}}{I_B} = \dfrac{15.96\text{ V} - 0.7\text{ V}}{20\ \mu\text{A}} = \mathbf{763\ k\Omega}$

4.4 Emitter-Bias Configuration

8. For the emitter-stabilized bias circuit of Fig. 4.122, determin
   a. I_{B_Q}.
   b. I_{C_Q}.
   c. V_{CE_Q}.
   d. V_C.
   e. V_B.
   f. V_E.

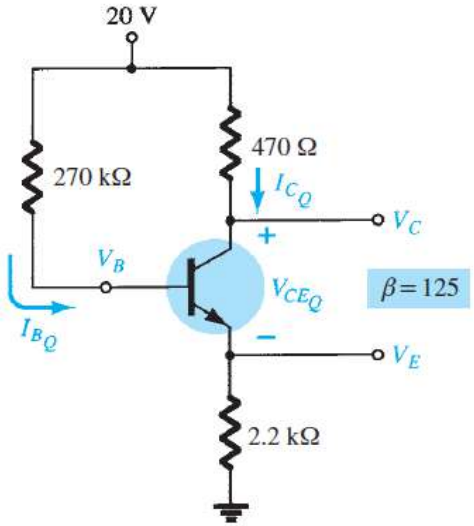
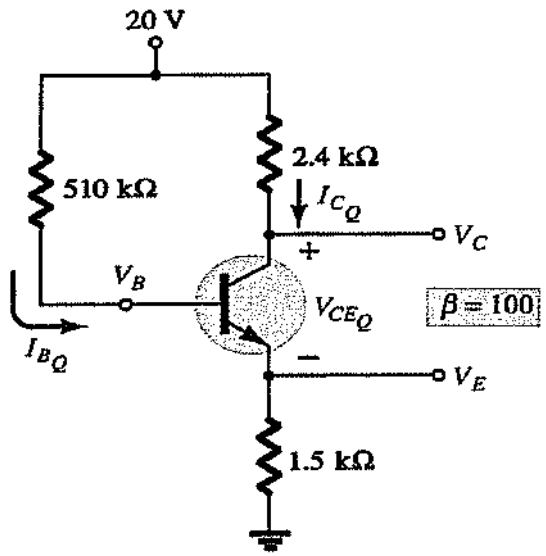


FIG. 4.122



(a) $$I_{B_Q} = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1)R_E} = \frac{20\text{ V} - 0.7\text{ V}}{510\text{ k}\Omega + (101)1.5\text{ k}\Omega} = \frac{19.3\text{ V}}{661.5\text{ k}\Omega}$$
$$= \mathbf{29.18\ \mu A}$$

(b) $$I_{C_Q} = \beta I_{B_Q} = (100)(29.18\ \mu\text{A}) = \mathbf{2.92\ mA}$$

(c) $$V_{CE_Q} = V_{CC} - I_C(R_C + R_E) = 20\text{ V} - (2.92\text{ mA})(2.4\text{ k}\Omega + 1.5\text{ k}\Omega)$$
$$= 20\text{ V} - 11.388\text{ V}$$
$$= \mathbf{8.61\ V}$$

(d) $$V_C = V_{CC} - I_C R_C = 20\text{ V} - (2.92\text{ mA})(2.4\text{ k}\Omega) = 20\text{ V} - 7.008\text{ V}$$
$$= \mathbf{13\ V}$$

(e) $$V_B = V_{CC} - I_B R_B = 20\text{ V} - (29.18\ \mu\text{A})(510\text{ k}\Omega)$$
$$= 20\text{ V} - 14.882\text{ V} = \mathbf{5.12\ V}$$

(f) $$V_E = V_C - V_{CE} = 13\text{ V} - 8.61\text{ V} = \mathbf{4.39\ V}$$

10. Given the information provided in Fig. 4.123, determine:

a. R_C.

b. R_E.

c. R_B.

d. V_{CE}.

e. V_B.

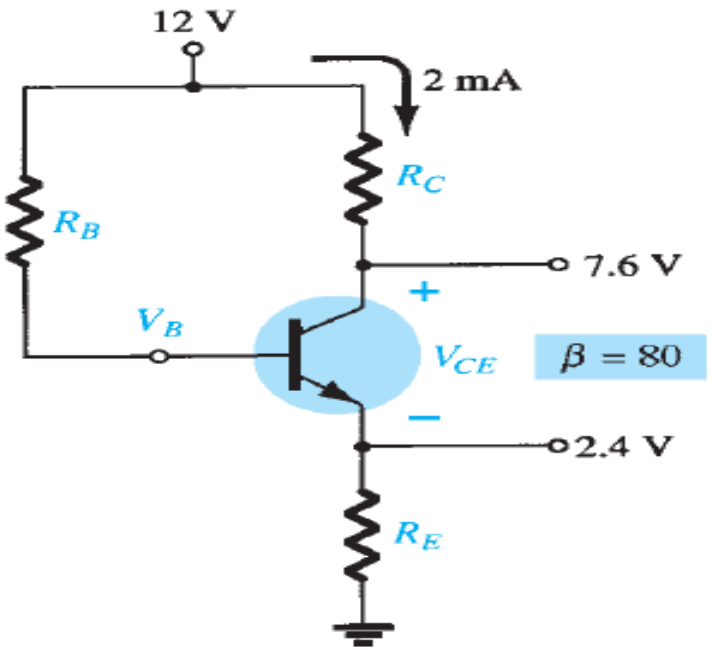


FIG. 4.123
Problem 10.

(a) $$R_C = \frac{V_{CC} - V_C}{I_C} = \frac{12\text{ V} - 7.6\text{ V}}{2\text{ mA}} = \frac{4.4\text{ V}}{2\text{ mA}} = \mathbf{2.2\ k\Omega}$$

(b) $$I_E \cong I_C:\ R_E = \frac{V_E}{I_E} = \frac{2.4\text{ V}}{2\text{ mA}} = \mathbf{1.2\ k\Omega}$$

(c) $$R_B = \frac{V_{R_B}}{I_B} = \frac{V_{CC} - V_{BE} - V_E}{I_B} = \frac{12\text{ V} - 0.7\text{ V} - 2.4\text{ V}}{2\text{ mA}/80} = \frac{8.9\text{ V}}{25\ \mu\text{A}} = \mathbf{356\ k\Omega}$$

(d) $$V_{CE} = V_C - V_E = 7.6\text{ V} - 2.4\text{ V} = \mathbf{5.2\text{ V}}$$

(e) $$V_B = V_{BE} + V_E = 0.7\text{ V} + 2.4\text{ V} = \mathbf{3.1\text{ V}}$$

11. Given the information provided in Fig. 4.124, determine:
    a. β.
    b. V_{CC}.
    c. R_B.

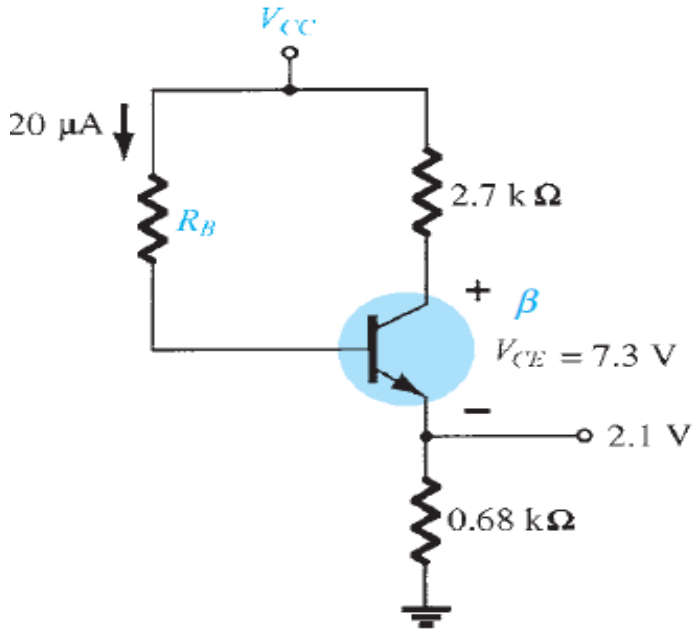


FIG. 4.124
Problem 11.

4.5 Voltage-Divider Bias Configuration

15. For the voltage-divider bias configuration of Fig. 4.125, determine:
    a. I_{B_Q}.
    b. I_{C_Q}.
    c. V_{CE_Q}.
    d. V_C.
    e. V_E.
    f. V_B.

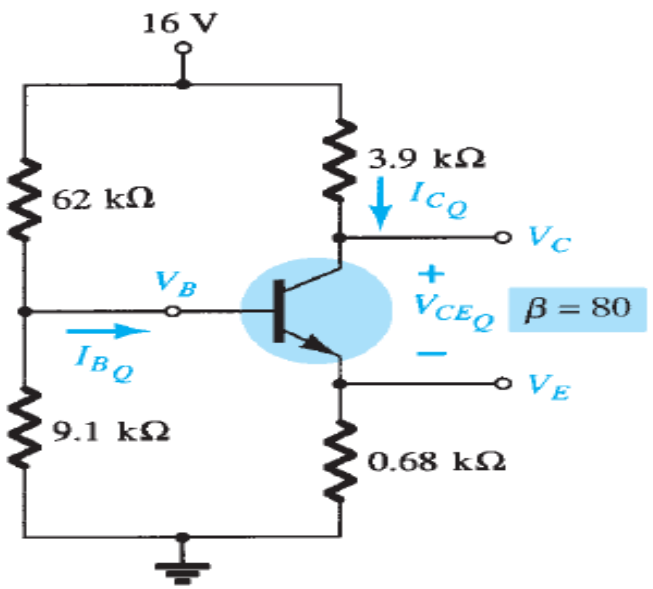


FIG. 4.125
Problems 15, 16, 20, 23, 25, 67, 69, 70, 73, and 77.

17. Given the information provided in Fig. 4.126, determine:

a. I_C.

b. V_E.

c. V_B.

d. R_1.

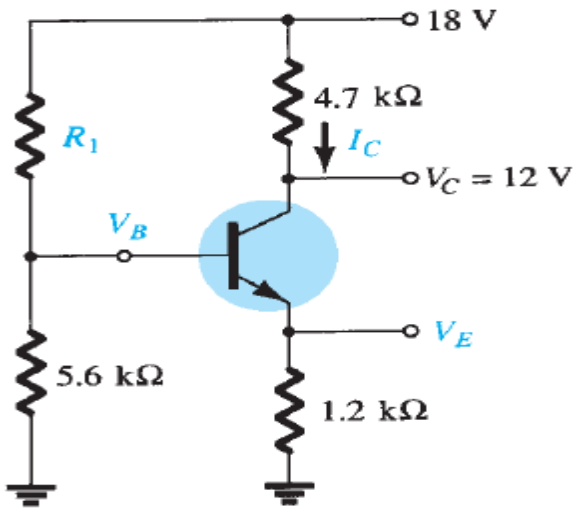


FIG. 4.126

18. Given the information appearing in Fig. 4.127, determine:

a. I_C.

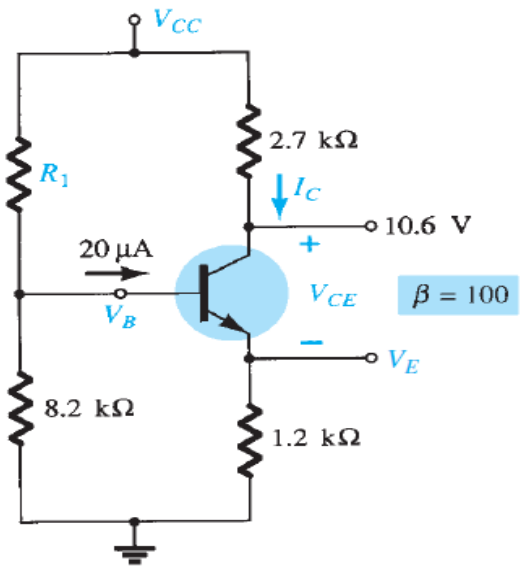
b. V_E.

c. V_{CC}.

d. V_{CE}.

e. V_B.

f. R_1.



*21. Determine the following for the voltage-divider configuration of Fig. 4.128 using the approximate approach if the condition established by Eq. (4.33) is satisfied.

a. I_C.
b. V_{CE}.
c. I_B.
d. V_E.
e. V_B.

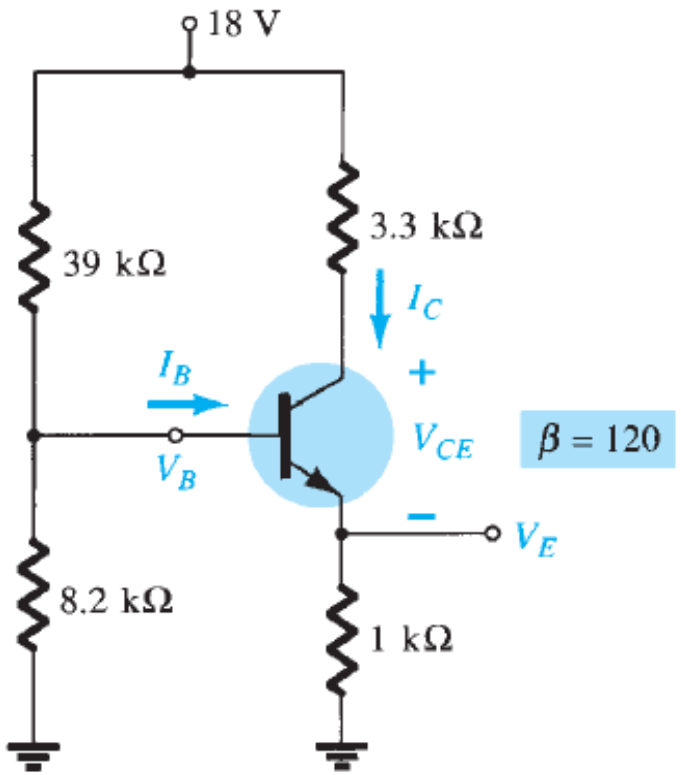


FIG. 4.128

4.6 Collector-Feedback Configuration

27. For the collector-feedback configuration of Fig. 4.129, determine:
    a. I_B.
    b. I_C.
    c. V_C.

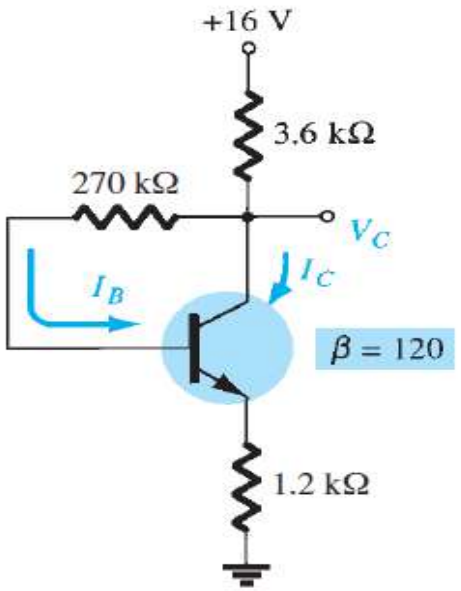
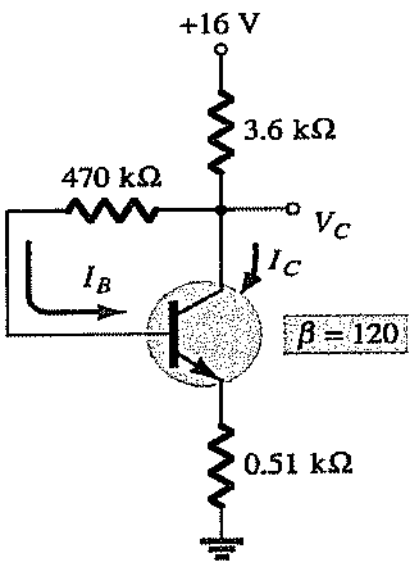


FIG. 4.129



29. For the voltage feedback network of Fig. 4.130, determine:

a. I_C.

b. V_C.

c. V_E.

d. V_{CE}.

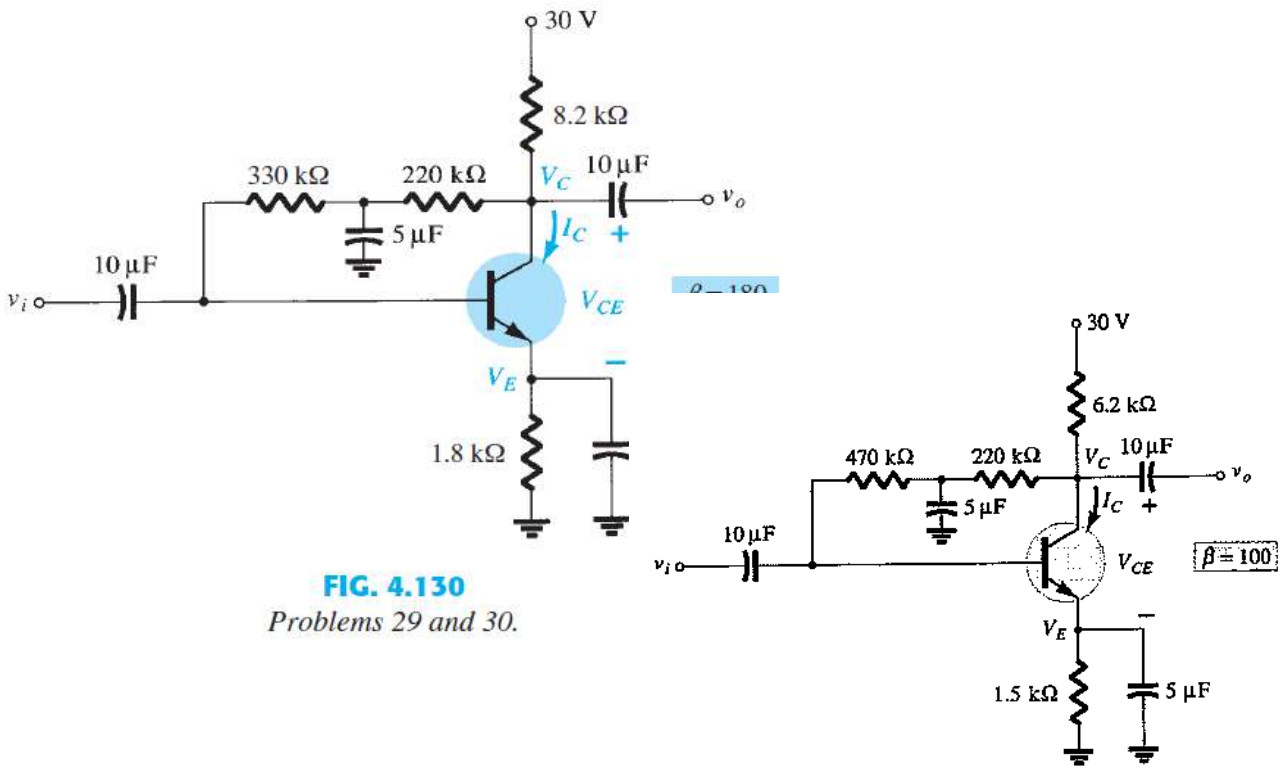
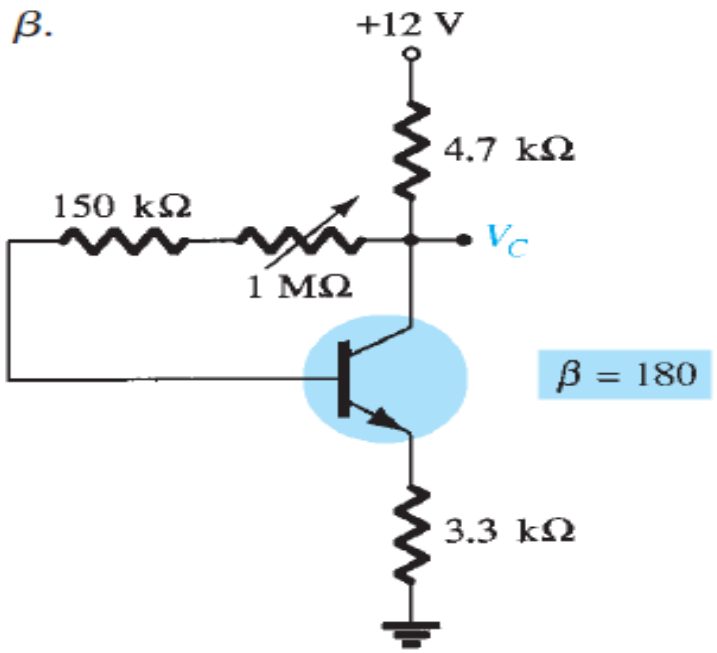


FIG. 4.130
Problems 29 and 30.

32. Determine the range of possible values for V_C for the network of Fig. 4.132 using the 1-MΩ potentiometer.

f. β.



$1\ \text{M}\Omega = 0\ \Omega,\ R_B = 150\ \text{k}\Omega$

$$I_B = \frac{V_{CC} - V_{BE}}{R_B + \beta(R_C + R_E)} = \frac{12\ \text{V} - 0.7\ \text{V}}{150\ \text{k}\Omega + (180)(4.7\ \text{k}\Omega + 3.3\ \text{k}\Omega)}$$

$$= 7.11\ \mu\text{A}$$

$$I_C = \beta I_B = (180)(7.11\ \mu\text{A}) = 1.28\ \text{mA}$$

$$V_C = V_{CC} - I_C R_C = 12\ \text{V} - (1.28\ \text{mA})(4.7\ \text{k}\Omega)$$

$$= \mathbf{5.98\ V}$$

Full 1 MΩ: $R_B = 1{,}000\ \text{k}\Omega + 150\ \text{k}\Omega = 1{,}150\ \text{k}\Omega = 1.15\ \text{M}\Omega$

$$I_B = \frac{V_{CC} - V_{BE}}{R_B + \beta(R_C + R_E)} = \frac{12\ \text{V} - 0.7\ \text{V}}{1.15\ \text{M}\Omega + (180)(4.7\ \text{k}\Omega + 3.3\ \text{k}\Omega)}$$

$$= 4.36\ \mu\text{A}$$

$$I_C = \beta I_B = (180)(4.36\ \mu\text{A}) = 0.785\ \text{mA}$$

$$V_C = V_{CC} - I_C R_C = 12\ \text{V} - (0.785\ \text{mA})(4.7\ \text{k}\Omega)$$

$$= \mathbf{8.31\ V}$$

V_C ranges from **5.98 V to 8.31 V**

*33. Given $V_B = 4$ V for the network of Fig. 4.133, determine:

a. V_E.
b. I_C.
c. V_C.
d. V_{CE}.
e. I_B.
f. β.

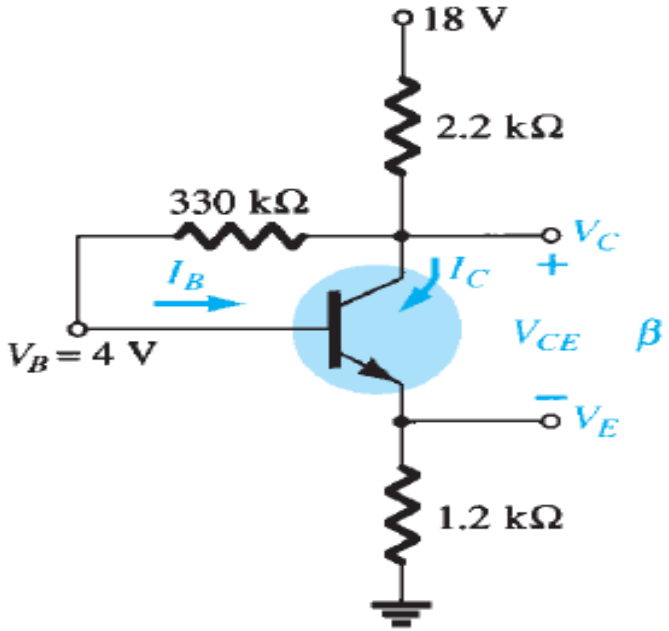


FIG. 4.133
Problem 33.

(a) $V_E = V_B - V_{BE} = 4\text{ V} - 0.7\text{ V} = \mathbf{3.3\text{ V}}$

(b) $I_C \cong I_E = \dfrac{V_E}{R_E} = \dfrac{3.3\text{ V}}{1.2\text{ k}\Omega} = \mathbf{2.75\text{ mA}}$

(c) $V_C = V_{CC} - I_C R_C = 18\text{ V} - (2.75\text{ mA})(2.2\text{ k}\Omega)$
$= \mathbf{11.95\text{ V}}$

(d) $V_{CE} = V_C - V_E = 11.95\text{ V} - 3.3\text{ V} = \mathbf{8.65\text{ V}}$

(e) $I_B = \dfrac{V_{R_B}}{R_B} = \dfrac{V_C - V_B}{R_B} = \dfrac{11.95\text{ V} - 4\text{ V}}{330\text{ k}\Omega} = \mathbf{24.09\ \mu A}$

(f) $\beta = \dfrac{I_C}{I_B} = \dfrac{2.75\text{ mA}}{24.09\ \mu\text{A}} = \mathbf{114.16}$

4.7 Emitter-Follower Configuration

*34. Determine the level of V_E and I_E for the network of Fig. 4.134.

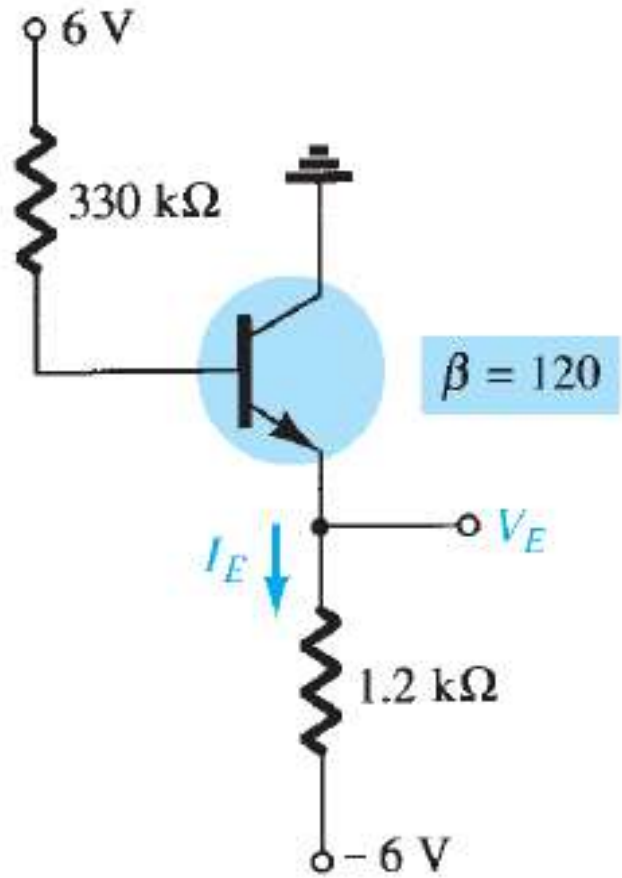


FIG. 4.134
Problem 34.

35. For the emitter follower network of Fig. 4.135

a. Find I_B, I_C, and I_E.

b. Determine V_B, V_C, and V_E.

c. Calculate V_{BC} and V_{CE}.

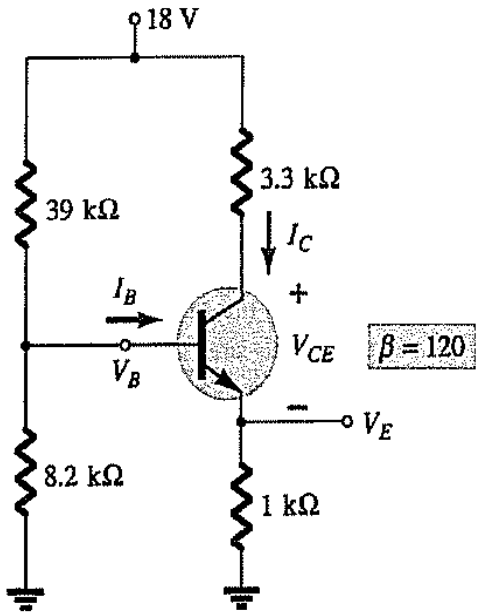


FIG. 4.96

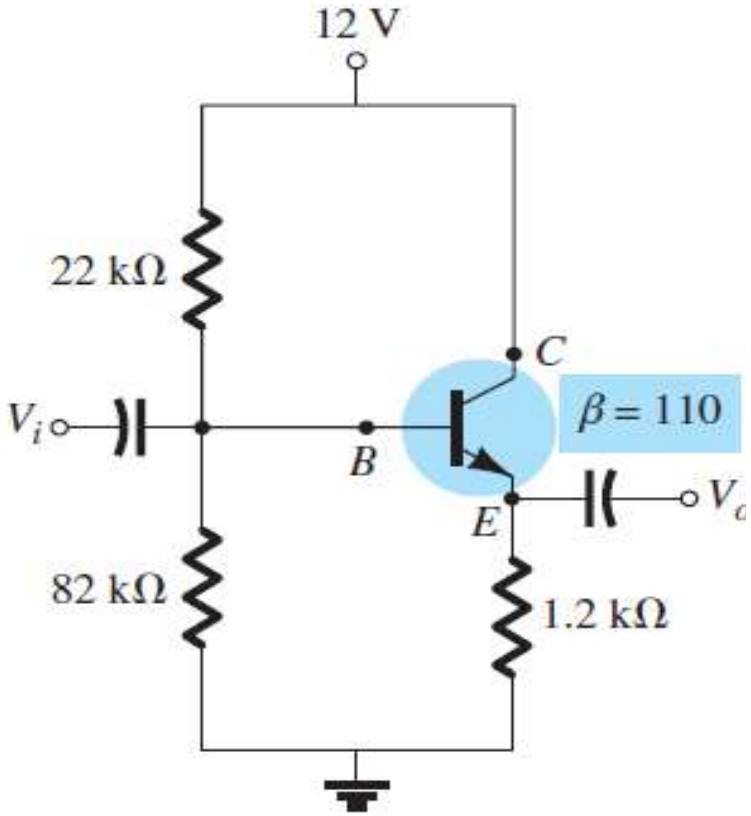


FIG. 4.135
Problem 35.

(a) $\beta R_E \geq 10R_2$

$(120)(1\ \text{k}\Omega) \geq 10(8.2\ \text{k}\Omega)$

$120\ \text{k}\Omega \geq 82\ \text{k}\Omega$ (checks)

$$\therefore V_B = \frac{R_2 V_{CC}}{R_1 + R_2} = \frac{(8.2\ \text{k}\Omega)(18\ \text{V})}{39\ \text{k}\Omega + 8.2\ \text{k}\Omega} = 3.13\ \text{V}$$

$$V_E = V_B - V_{BE} = 3.13\ \text{V} - 0.7\ \text{V} = 2.43\ \text{V}$$

$$I_C \cong I_E = \frac{V_E}{R_E} = \frac{2.43\ \text{V}}{1\ \text{k}\Omega} = \mathbf{2.43\ mA}$$

(b) $V_{CE} = V_{CC} - I_C(R_C + R_E)$

$= 18\ \text{V} - (2.43\ \text{mA})(3.3\ \text{k}\Omega + 1\ \text{k}\Omega)$

$= \mathbf{7.55\ V}$

(c) $$I_B = \frac{I_C}{\beta} = \frac{2.43\ \text{mA}}{120} = \mathbf{20.25\ \mu A}$$

(d) $V_E = I_E R_E \cong I_C R_E = (2.43\ \text{mA})(1\ \text{k}\Omega) = \mathbf{2.43\ V}$

(e) $V_B = \mathbf{3.13\ V}$

Repeat Problem 16 using the exact (Thévenin) approach and compare solutions. Based on the results, is the approximate approach a valid analysis technique if Eq. (4.33) is satisfied?

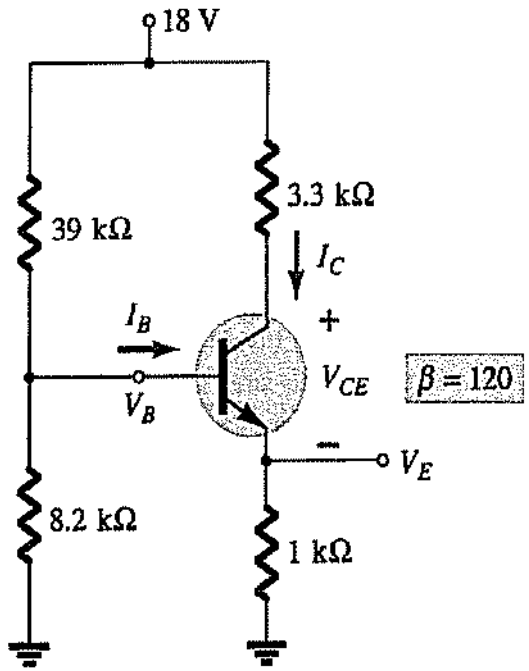


FIG. 4.96

(a) $R_{Th} = R_1 \parallel R_2 = 39\ \text{k}\Omega \parallel 8.2\ \text{k}\Omega = 6.78\ \text{k}\Omega$

$$E_{Th} = \frac{R_C V_{CC}}{R_1 + R_2} = \frac{8.2\ \text{k}\Omega(18\ \text{V})}{39\ \text{k}\Omega + 8.2\ \text{k}\Omega} = 3.13\ \text{V}$$

$$I_B = \frac{E_{Th} - V_{BE}}{R_{Th} + (\beta + 1)R_E} = \frac{3.13\ \text{V} - 0.7\ \text{V}}{6.78\ \text{k}\Omega + (121)(1\ \text{k}\Omega)}$$

$$= \frac{2.43\ \text{V}}{127.78\ \text{k}\Omega} = 19.02\ \mu\text{A}$$

$I_C = \beta I_B = (120)(19.02\ \mu\text{A}) = \mathbf{2.28\ mA}$ (vs. 2.43 mA #16)

(b) $V_{CE} = V_{CC} - I_C(R_C + R_E) = 18\ \text{V} - (2.28\ \text{mA})(3.3\ \text{k}\Omega + 1\ \text{k}\Omega)$

$= 18\ \text{V} - 9.8\ \text{V} = \mathbf{8.2\ V}$ (vs. 7.55 V #16)

(c) **19.02 μA** (vs. 20.25 μA #16)

(d) $V_E = I_E R_E \cong I_C R_E = (2.28\ \text{mA})(1\ \text{k}\Omega) = \mathbf{2.28\ V}$ (vs. 2.43 V #16)

(e) $V_B = V_{BE} + V_E = 0.7\ \text{V} + 2.28\ \text{V} = \mathbf{2.98\ V}$ (vs. 3.13 V #16)

The results suggest that the approximate approach is valid if Eq. 4.33 is satisfied.

4.8 Common-Base Configuration

*36. For the network of Fig. 4.136, determine:

a. I_B.

b. I_C.

c. V_{CE}.

d. V_C.

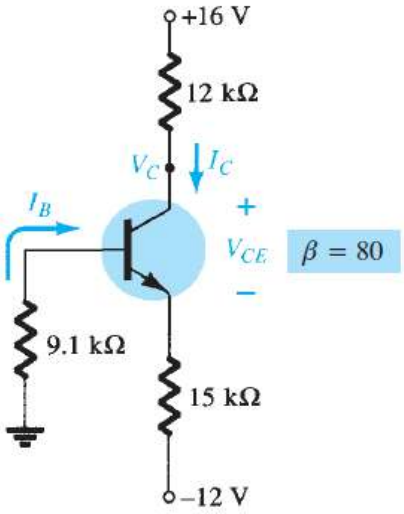
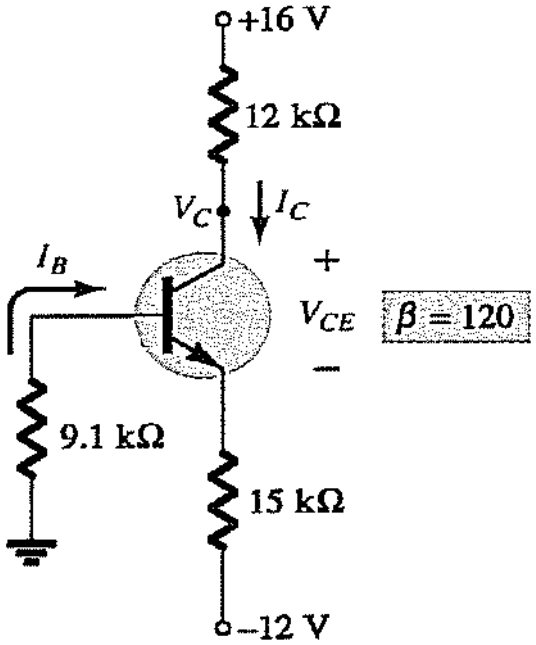


FIG. 4.136
Problem 36.



(a) $$I_B = \frac{V_{EE} - V_{BE}}{R_B + (\beta + 1)R_E} = \frac{12\text{ V} - 0.7\text{ V}}{9.1\text{ k}\Omega + (120 + 1)15\text{ k}\Omega}$$
$$= \mathbf{6.2\ \mu A}$$

(b) $$I_C = \beta I_B = (120)(6.2\ \mu\text{A}) = \mathbf{0.744\ mA}$$

(c) $$V_{CE} = V_{CC} + V_{EE} - I_C(R_C + R_E)$$
$$= 16\text{ V} + 12\text{ V} - (0.744\text{ mA})(27\text{ k}\Omega)$$
$$= \mathbf{7.91\ V}$$

(d) $$V_C = V_{CC} - I_C R_C = 16\text{ V} - (0.744\text{ mA})(12\text{ k}\Omega) = \mathbf{7.07\ V}$$

*37. For the network of Fig. 4.137, determine:

a. I_E.

b. V_C.

c. V_{CE}.

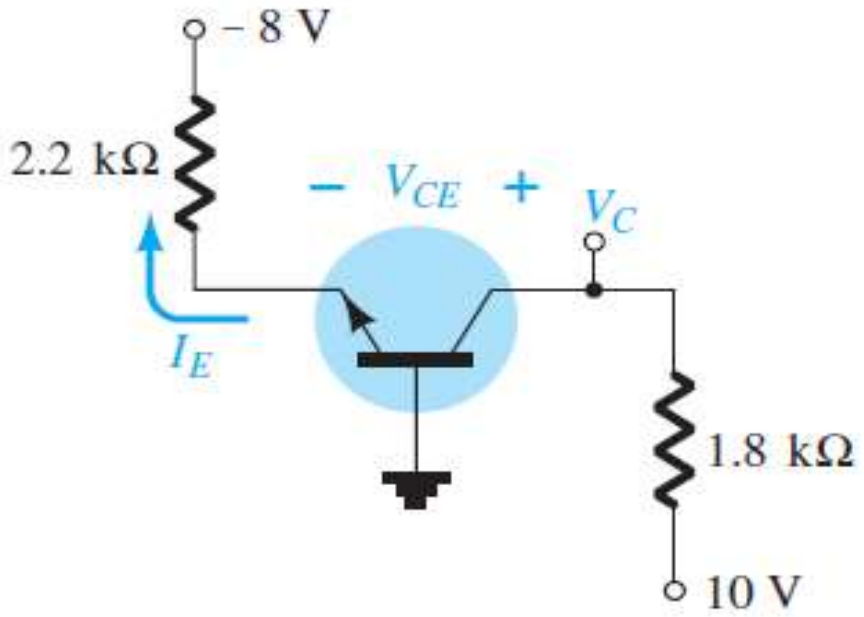


FIG. 4.137
Problem 37.

(a) $$I_E = \frac{8\text{ V} - 0.7\text{ V}}{2.2\text{ k}\Omega} = \frac{7.3\text{ V}}{2.2\text{ k}\Omega} = \mathbf{3.32\text{ mA}}$$

(b) $$V_C = 10\text{ V} - (3.32\text{ mA})(1.8\text{ k}\Omega) = 10\text{ V} - 5.976$$
$$= \mathbf{4.02\text{ V}}$$

(c) $$V_{CE} = 10\text{ V} + 8\text{ V} - (3.32\text{ mA})(2.2\text{ k}\Omega + 1.8\text{ k}\Omega)$$
$$= 18\text{ V} - 13.28\text{ V}$$
$$= \mathbf{4.72\text{ V}}$$

38. For the common-base network of Fig. 4.138
    a. Using the information provided determine the value of R_C.
    b. Find the currents I_B and I_E.
    c. Determine the voltages V_{BC} and V_{CE}.

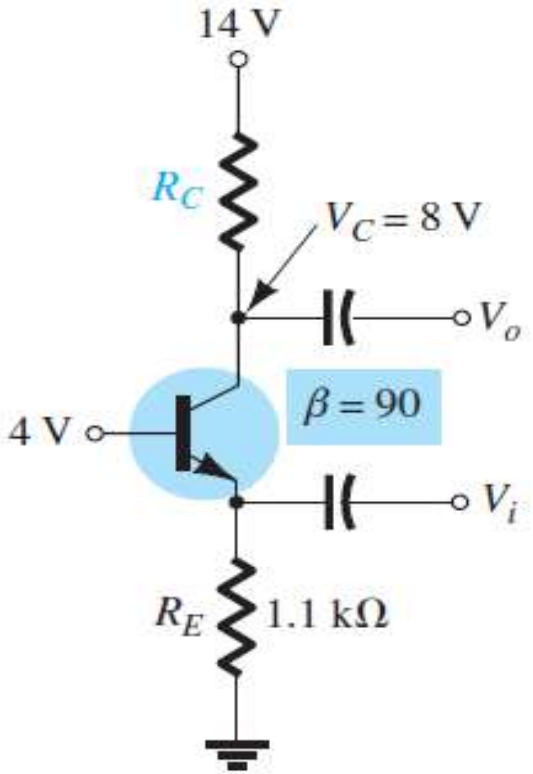


FIG. 4.138

4.9 Miscellaneous Bias Configurations

*39. For the network of Fig. 4.139, determine:

a. I_B.

b. I_C.

c. V_E.

d. V_{CE}.

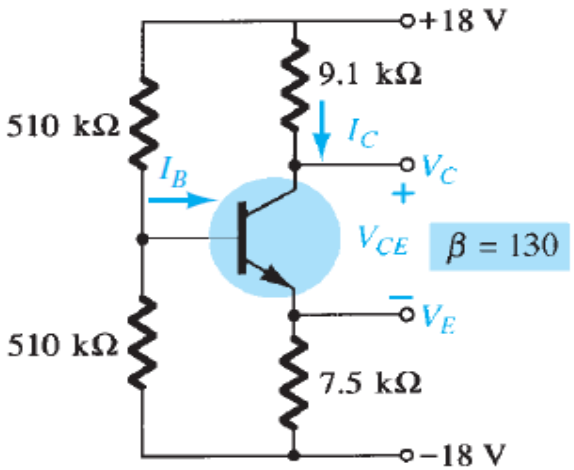
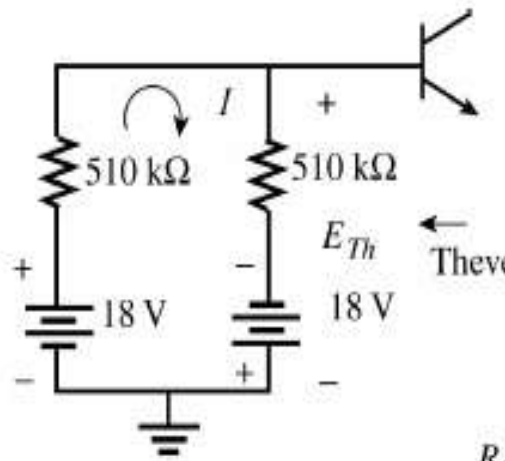
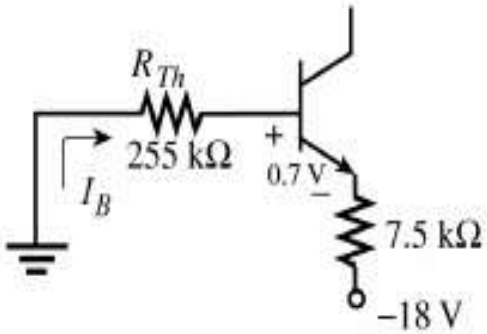


FIG. 4.139
Problem 39.

(a) $\beta R_E > 10R_2$ not satisfied $\therefore$ Use exact approach:
Network redrawn to determine the Thevenin equivalent:





$$R_{Th} = \frac{510\ \text{k}\Omega}{2} = \mathbf{255\ k\Omega}$$

$$I = \frac{18\ \text{V} + 18\ \text{V}}{510\ \text{k}\Omega + 510\ \text{k}\Omega} = 35.29\ \mu\text{A}$$

$$E_{Th} = -18\ \text{V} + (35.29\ \mu\text{A})(510\ \text{k}\Omega) = \mathbf{0\ V}$$

$$I_B = \frac{18\ \text{V} - 0.7\ \text{V}}{255\ \text{k}\Omega + (130+1)(7.5\ \text{k}\Omega)} = \mathbf{13.95\ \mu A}$$

(b) $I_C = \beta I_B = (130)(13.95\ \mu\text{A}) = \mathbf{1.81\ mA}$

(c) $V_E = -18\ \text{V} + (1.81\ \text{mA})(7.5\ \text{k}\Omega)$
$= -18\ \text{V} + 13.58\ \text{V}$
$= \mathbf{-4.42\ V}$

(d) $V_{CE} = 18\ \text{V} + 18\ \text{V} - (1.81\ \text{mA})(9.1\ \text{k}\Omega + 7.5\ \text{k}\Omega)$
$= 36\ \text{V} - 30.05\ \text{V} = \mathbf{5.95\ V}$

40. Given $V_C = 8$ V for the network of Fig. 4.140, determine:

a. I_B.

b. I_C.

c. β.

d. V_{CE}.

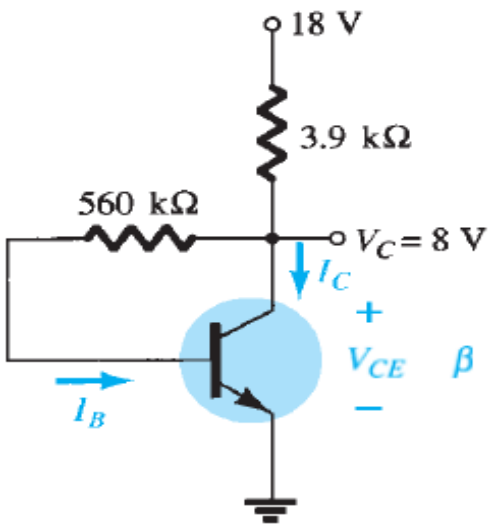


FIG. 4.140

4.11 Design Operations

41. Determine R_C and R_B for a fixed-bias configuration if $V_{CC} = 12\text{ V}$, $\beta = 80$, and $I_{C_Q} = 2.5\text{ mA}$ with $V_{CE_Q} = 6\text{ V}$. Use standard values.

$$I_B = \frac{I_C}{\beta} = \frac{2.5\text{ mA}}{80} = 31.25\ \mu\text{A}$$

$$R_B = \frac{V_{R_B}}{I_B} = \frac{V_{CC} - V_{BE}}{I_B} = \frac{12\text{ V} - 0.7\text{ V}}{31.25\ \mu\text{A}} = \mathbf{361.6\ k\Omega}$$

$$R_C = \frac{V_{R_C}}{I_C} = \frac{V_{CC} - V_C}{I_C} = \frac{V_{CC} - V_{CE_Q}}{I_{C_Q}} = \frac{12\text{ V} - 6\text{ V}}{2.5\text{ mA}} = \frac{6\text{ V}}{2.5\text{ mA}}$$

$$= \mathbf{2.4\ k\Omega}$$

Standard values:

$R_B = \mathbf{360\ k\Omega}$

$R_C = \mathbf{2.4\ k\Omega}$

42. Design an emitter-stabilized network at $I_{C_Q} = \frac{1}{2}I_{C_{sat}}$ and $V_{CE_Q} = \frac{1}{2}V_{CC}$. Use $V_{CC} = 20$ V, $I_{C_{sat}} = 10$ mA, $\beta = 120$, and $R_C = 4R_E$. Use standard values.

$$I_{C_{sat}} = \frac{V_{CC}}{R_C + R_E} = 10 \text{ mA}$$

$$\frac{20 \text{ V}}{4R_E + R_E} = 10 \text{ mA} \Rightarrow \frac{20 \text{ V}}{5R_E} = 10 \text{ mA} \Rightarrow 5R_E = \frac{20 \text{ V}}{10 \text{ mA}} = 2 \text{ k}\Omega$$

$$R_E = \frac{2 \text{ k}\Omega}{5} = \mathbf{400\ \Omega}$$

$$R_C = 4R_E = \mathbf{1.6\ k\Omega}$$

$$I_B = \frac{I_C}{\beta} = \frac{5 \text{ mA}}{120} = 41.67\ \mu\text{A}$$

$$R_B = V_{RB}/I_B = \frac{20 \text{ V} - 0.7 \text{ V} - 5 \text{ mA}(0.4 \text{ k}\Omega)}{41.67\ \mu\text{A}} = \frac{19.3 - 2 \text{ V}}{41.67\ \mu\text{A}}$$

$$= \mathbf{415.17\ k\Omega}$$

Standard values: $R_E = \mathbf{390\ \Omega}$, $R_C = \mathbf{1.6\ k\Omega}$, $R_B = \mathbf{430\ k\Omega}$

43. Design a voltage-divider bias network using a supply of 24 V, a transistor with a beta of 110, and an operating point of $I_{C_Q} = 4$ mA and $V_{CE_Q} = 8$ V. Choose $V_E = \frac{1}{8}V_{CC}$. Use standard values.

$$R_E = \frac{V_E}{I_E} \cong \frac{V_E}{I_C} = \frac{3\text{ V}}{4\text{ mA}} = \mathbf{0.75\ k\Omega}$$

$$R_C = \frac{V_{R_C}}{I_C} = \frac{V_{CC} - V_C}{I_C} = \frac{V_{CC} - (V_{CE_Q} + V_E)}{I_C}$$

$$= \frac{24\text{ V} - (8\text{ V} + 3\text{ V})}{4\text{ mA}} = \frac{24\text{ V} - 11\text{ V}}{4\text{ mA}} = \frac{13\text{ V}}{4\text{ mA}} = \mathbf{3.25\ k\Omega}$$

$$V_B = V_E + V_{BE} = 3\text{ V} + 0.7\text{ V} = 3.7\text{ V}$$

$$\left.\begin{aligned} V_B &= \frac{R_2 V_{CC}}{R_2 + R_1} \\ \Rightarrow 3.7\text{ V} &= \frac{R_2(24\text{ V})}{R_2 + R_1} \end{aligned}\right\} \text{ 2 unknowns!}$$

$\therefore$ use $\beta R_E \geq 10R_2$ for increased stability

$(110)(0.75\text{ k}\Omega) = 10R_2$

$R_2 = 8.25\text{ k}\Omega$

Choose $R_2 = \mathbf{7.5\ k\Omega}$

Substituting in the above equation:

$$3.7\text{ V} = \frac{7.5\text{ k}\Omega(24\text{ V})}{7.5\text{ k}\Omega + R_1}$$

$R_1 = \mathbf{41.15\ k\Omega}$

Standard values:

$R_E = \mathbf{0.75\ k\Omega}$, $R_C = \mathbf{3.3\ k\Omega}$, $R_2 = \mathbf{7.5\ k\Omega}$, $R_1 = \mathbf{43\ k\Omega}$

*44. Using the characteristics of Fig. 4.121, design a voltage-divider configuration to have a saturation level of 10 mA and a Q-point one-half the distance between cutoff and saturation. The available supply is 28 V, and V_E is to be one-fifth of V_{CC}. The condition established by Eq. (4.33) should also be met to provide a high stability factor. Use standard values.

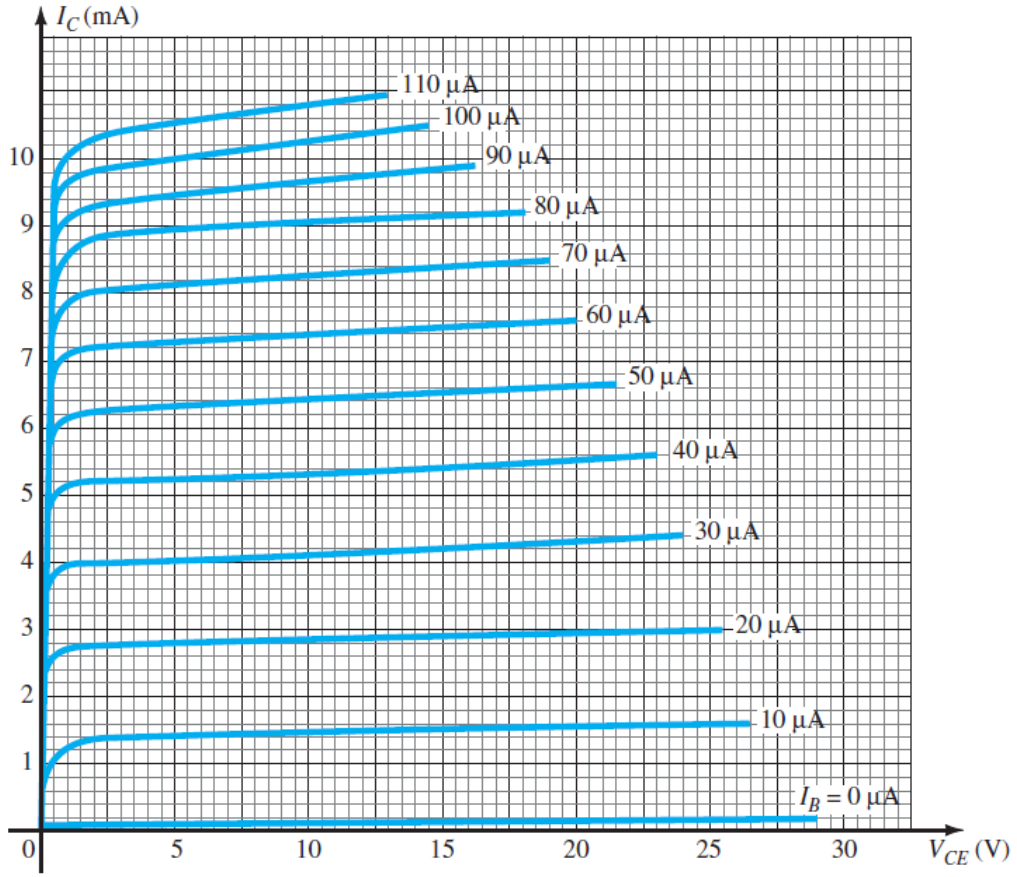


FIG. 4.121

$$V_E = \frac{1}{5}V_{CC} = \frac{1}{5}(28\text{ V}) = 5.6\text{ V}$$

$$R_E = \frac{V_E}{I_E} = \frac{5.6\text{ V}}{5\text{ mA}} = \mathbf{1.12\ k\Omega}\ (\text{use } \mathbf{1.1\ k\Omega})$$

$$V_C = \frac{V_{CC}}{2} + V_E = \frac{28\text{ V}}{2} + 5.6\text{ V} = 14\text{ V} + 5.6\text{ V} = 19.6\text{ V}$$

$$V_{R_C} = V_{CC} - V_C = 28\text{ V} - 19.6\text{ V} = 8.4\text{ V}$$

$$R_C = \frac{V_{R_C}}{I_C} = \frac{8.4\text{ V}}{5\text{ mA}} = \mathbf{1.68\ k\Omega}\ (\text{use } 1.6\text{ k}\Omega)$$

$$V_B = V_{BE} + V_E = 0.7\text{ V} + 5.6\text{ V} = 6.3\text{ V}$$

$$V_B = \frac{R_2 V_{CC}}{R_2 + R_1} \Rightarrow 6.3\text{ V} = \frac{R_2(28\text{ V})}{R_2 + R_1}\ (2\text{ unknowns})$$

$$\beta = \frac{I_C}{I_B} = \frac{5\text{ mA}}{37\ \mu\text{A}} = 135.14$$

$$\beta R_E = 10R_2$$

$$(135.14)(1.12\text{ k}\Omega) = 10(R_2)$$

$$R_2 = 15.14\text{ k}\Omega\ (\text{use } 15\text{ k}\Omega)$$

Substituting: $6.3\text{ V} = \dfrac{(15.14\text{ k}\Omega)(28\text{ V})}{15.14\text{ k}\Omega + R_1}$

Solving, $R_1 = 52.15\text{ k}\Omega\ (\text{use } 51\text{ k}\Omega)$

Standard values:

$R_E = \mathbf{1.1\ k\Omega}$

$R_C = \mathbf{1.6\ k\Omega}$

$R_1 = \mathbf{51\ k\Omega}$

$R_2 = \mathbf{15\ k\Omega}$

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Chapter 5: field Effect Transistor FET

Lecture 1

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CHAPTER OBJECTIVES

- Become familiar with the construction and operating characteristics of Junction Field Effect (JFET), Metal-Oxide Semiconductor FET (MOSFET), [redacted].
- Be able to sketch the transfer characteristics from the drain characteristics of a JFET, MOSFET, [redacted]
- Understand the vast amount of information provided on the specification sheet for each type of FET.
- Be aware of the differences between the dc analysis of the various types of FETs.

6.1 INTRODUCTION

The field-effect transistor (FET) is a three-terminal device used for a variety of applications that match, to a large extent, those of the BJT transistor described in Chapters 3 through 5. Although there are important differences between the two types of devices, there are also many similarities, which will be pointed out in the sections to follow.

The primary difference between the two types of transistors is the fact that:

The BJT transistor is a current-controlled device as depicted in Fig. 6.1a, whereas the JFET transistor is a voltage-controlled device as shown in Fig. 6.1b.

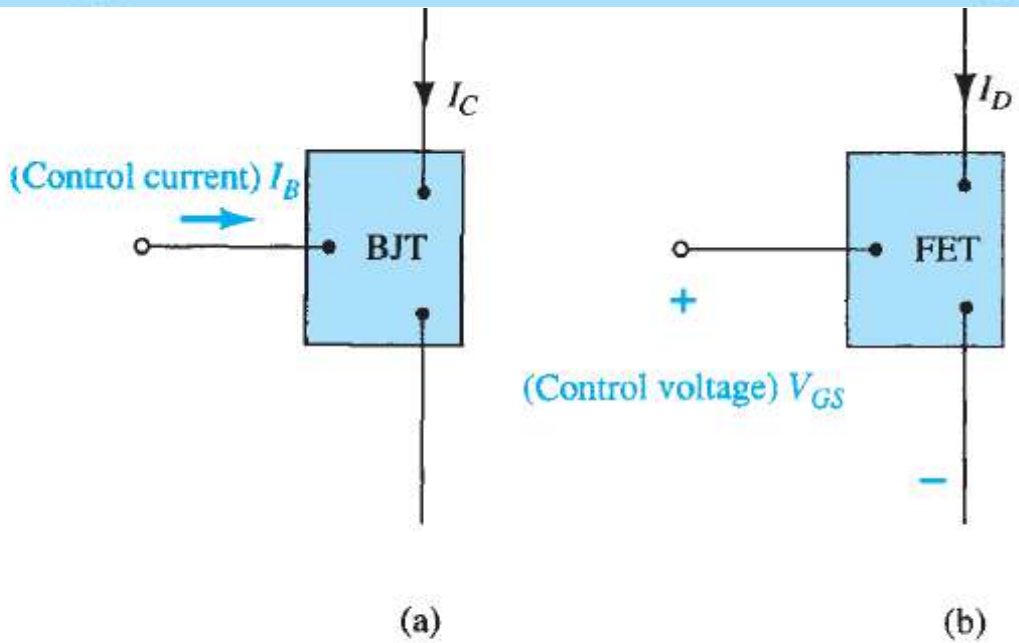


FIG. 6.1

(a) Current-controlled and (b) voltage-controlled amplifiers.

In other words, the current I_C in Fig. 6.1a is a direct function of the level of I_B. For the FET the current I_D will be a function of the voltage V_{GS} applied to the input circuit as shown in Fig. 6.1b. In each case the current of the output circuit is controlled by a parameter of the input circuit—in one case a current level and in the other an applied voltage.

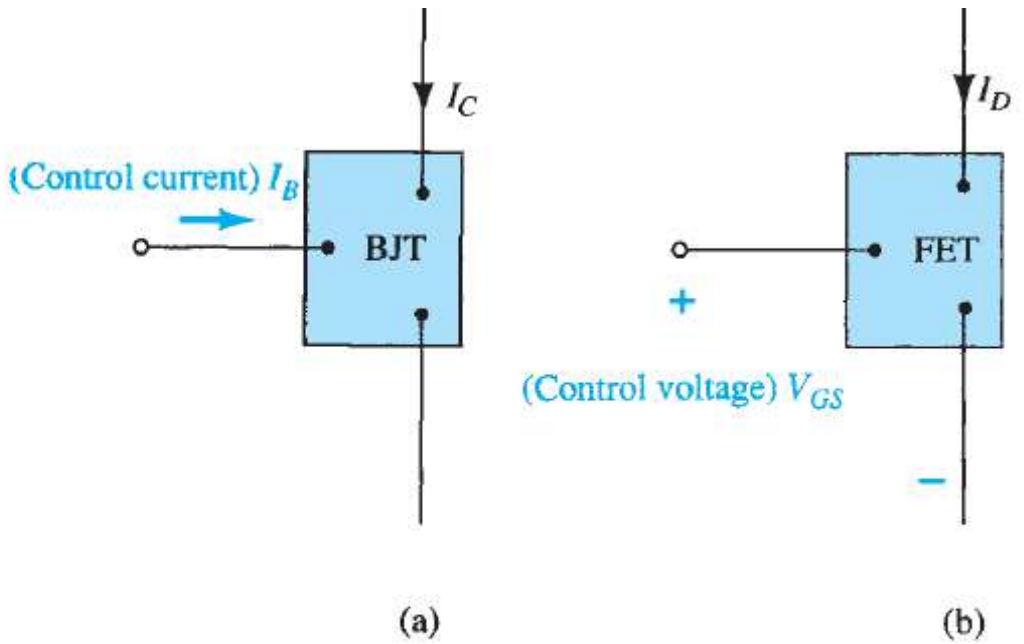
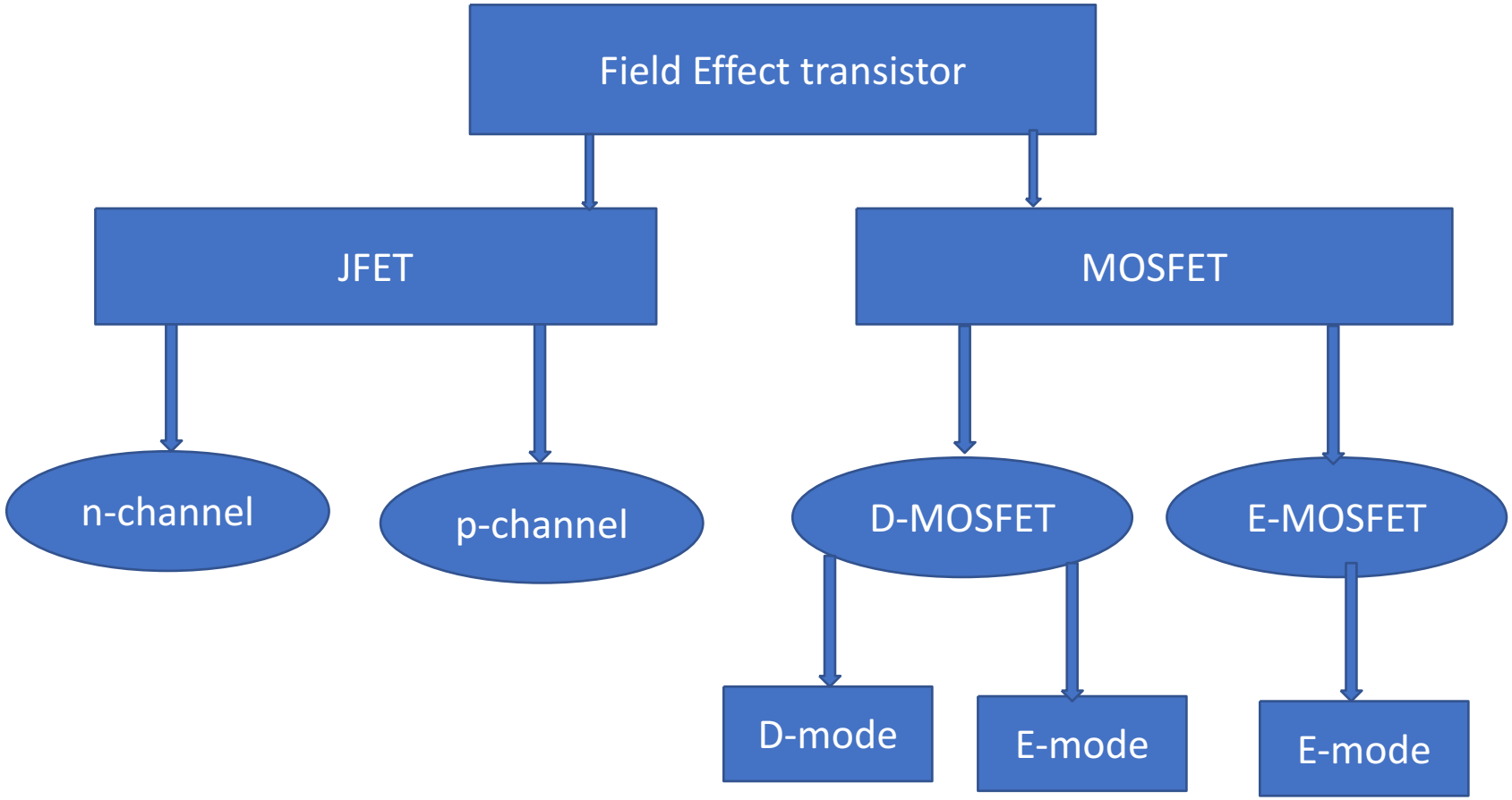


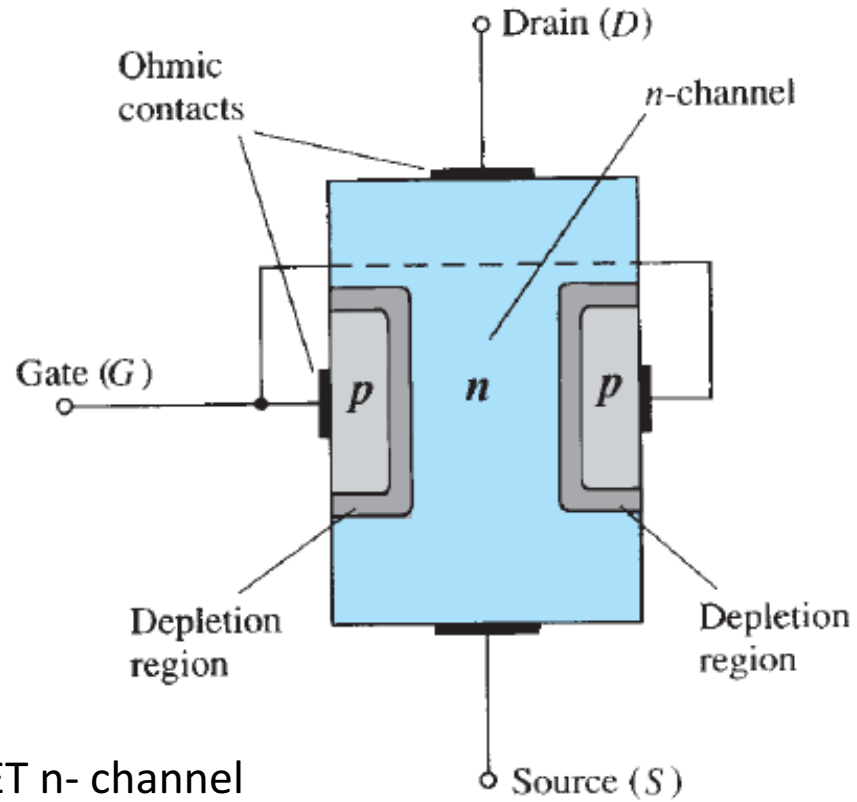
FIG. 6.1

(a) Current-controlled and (b) voltage-controlled amplifiers.

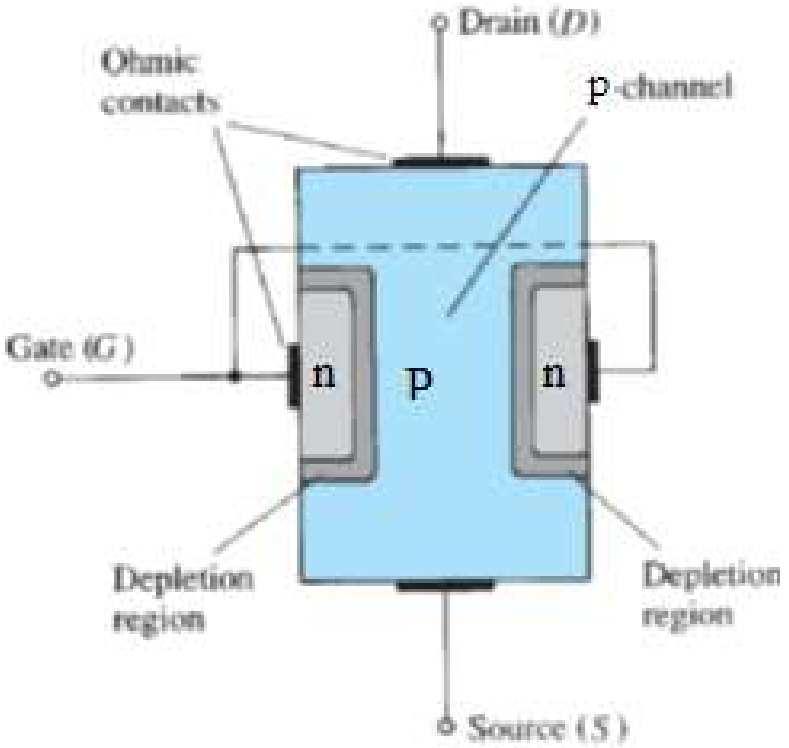

Field Effect transistor
JFET
MOSFET
n-channel
p-channel
D-MOSFET
E-MOSFET
D-mode
E-mode
E-mode

JFET Junction Field Effect Transistor

Just as there are *npn* and *pnp* bipolar transistors, there are *n-channel* and *p-channel* field-effect transistors. However, it is important to keep in mind that the BJT transistor is a *bipolar* device—the prefix *bi* indicates that the conduction level is a function of two charge carriers, electrons and holes. The FET is a *unipolar* device depending solely on either electron (*n*-channel) or hole (*p*-channel) conduction.



JFET n- channel



JFET p- channel

One of the most important characteristics of the FET is its high input impedance.

At a level of 1 MΩ to several hundred megohms it far exceeds the typical input resistance levels of the BJT transistor configurations—a very important characteristic in the design of linear ac amplifier systems. On the other hand, the BJT transistor has a much higher sensitivity to changes in the applied signal. In other words, the variation in output current is typically a great deal more for BJTs than for FETs for the same change in the applied voltage.

For this reason:

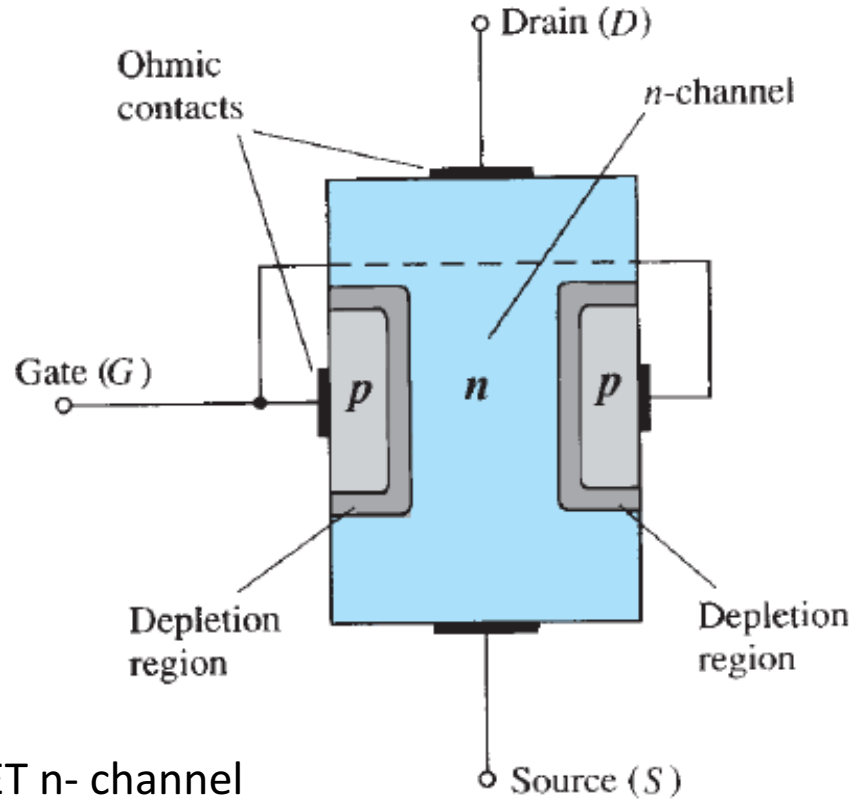
Typical ac voltage gains for BJT amplifiers are a great deal more than for FETs.

However,

FETs are more temperature stable than BJTs, and FETs are usually smaller than BJTs, making them particularly useful in integrated-circuit (IC) chips.

6.2 CONSTRUCTION AND CHARACTERISTICS OF JFETs

As indicated earlier, the JFET is a three-terminal device with one terminal capable of controlling the current between the other two. In our discussion of the BJT transistor the *npn* transistor was employed through the major part of the analysis and design sections, with a section devoted to the effect of using a *pnp* transistor. For the JFET transistor the *n*-channel device will be the prominent device, with paragraphs and sections devoted to the effect of using a *p*-channel JFET.



JFET n- channel

Analogies are seldom perfect and at times can be misleading, but the water analogy of Fig. 6.4 does provide a sense for the JFET control at the gate terminal and the appropriateness of the terminology applied to the terminals of the device. The source of water pressure can be likened to the applied voltage from drain to source, which establishes a flow of water (electrons) from the spigot (source). The "gate," through an applied signal (potential), controls the flow of water (charge) to the "drain." The drain and source terminals are at opposite ends of the *n*-channel as introduced in Fig. 6.3 because the terminology is defined for electron flow.

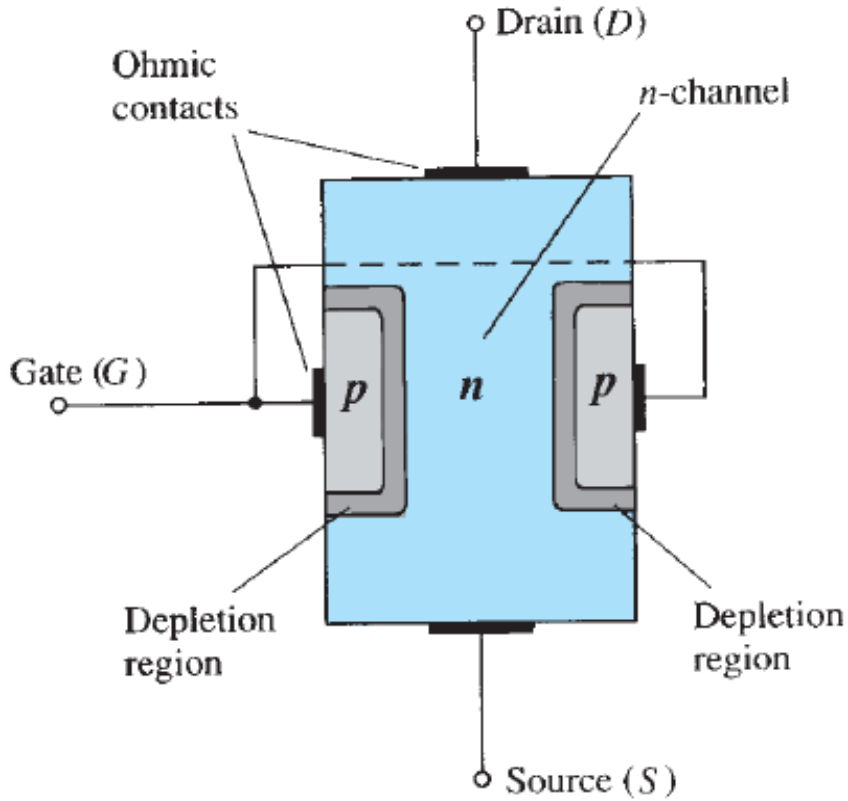


FIG. 6.3
Junction field-effect transistor (JFET

FIG. 6.4
Water analogy for the JFET control mechanism.

$V_{GS} = 0$ V, V_{DS} Some Positive Value

In Fig. 6.5, a positive voltage V_{DS} is applied across the channel and the gate is connected directly to the source to establish the condition $V_{GS} = 0$ V. The result is a gate and a source terminal at the same potential and a depletion region in the low end of each *p*-material similar to the distribution of the no-bias conditions of Fig. 6.3. The instant the voltage V_{DD} ($=V_{DS}$) is applied, the electrons are drawn to the drain terminal, establishing the conventional current I_D with the defined direction of Fig. 6.5. The path of charge flow clearly reveals that the drain and source currents are equivalent ($I_D = I_S$). Under the conditions in Fig. 6.5, the flow of charge is relatively uninhibited and is limited solely by the resistance of the *n*-channel between drain and source.

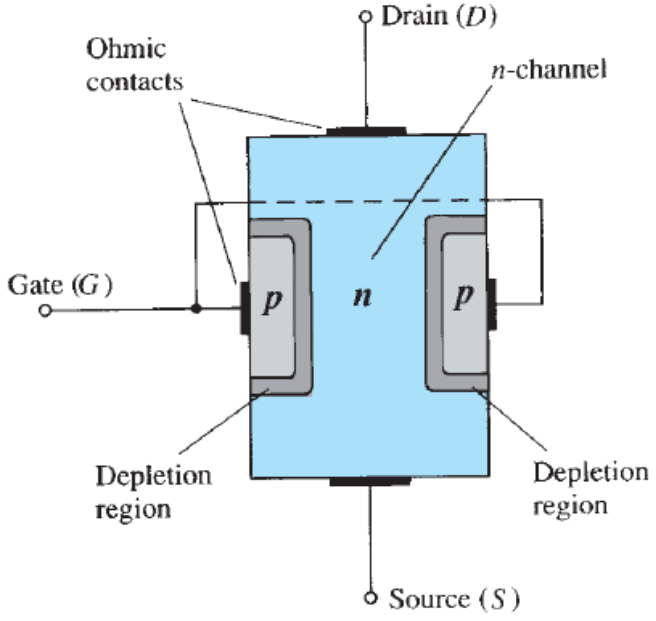


FIG. 6.3
Junction field-effect transistor (JFET).

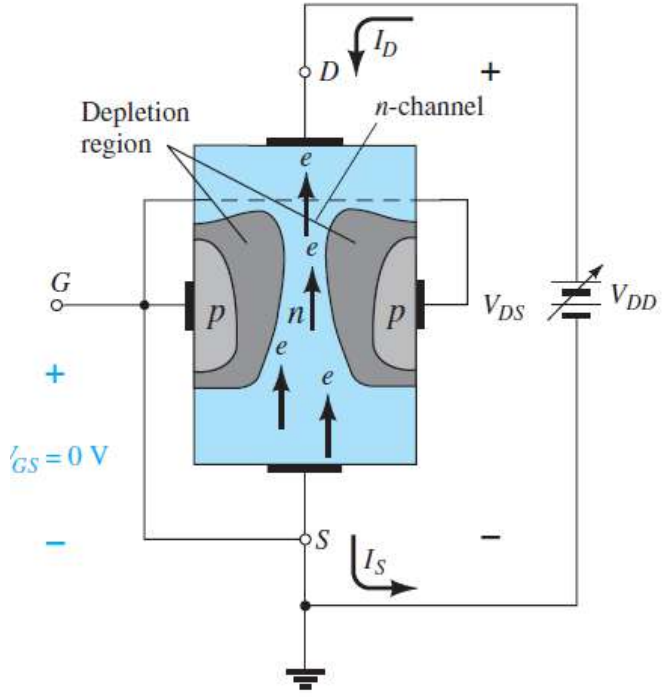


FIG. 6.5
JFET at $V_{GS} = 0$ V and $V_{DS} > 0$ V.

It is important to note that the depletion region is wider near the top of both *p*-type materials. The reason for the change in width of the region is best described through the help of Fig. 6.6.

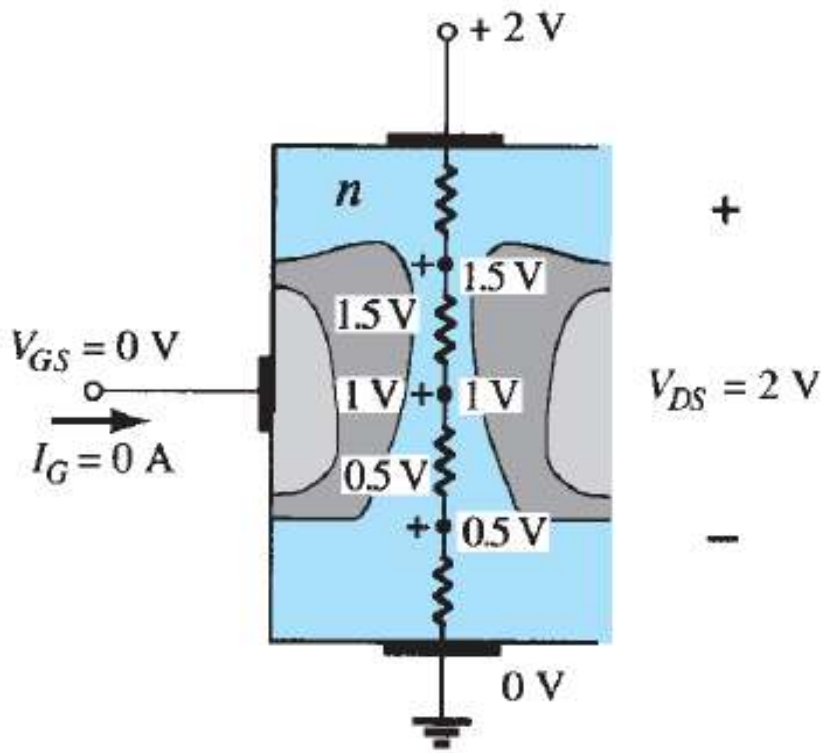


FIG. 6.6

Assuming a uniform resistance in the *n*-channel, we can break down the resistance of the channel into the divisions appearing in Fig. 6.6. The current I_D will establish the voltage levels through the channel as indicated on the same figure. The result is that the upper region of the *p*-type material will be reverse-biased by about 1.5 V, with the lower region only reverse-biased by 0.5 V. Recall from the discussion of the diode operation that the greater the applied reverse bias, the wider is the depletion region—hence the distribution of the depletion region as shown in Fig. 6.6.

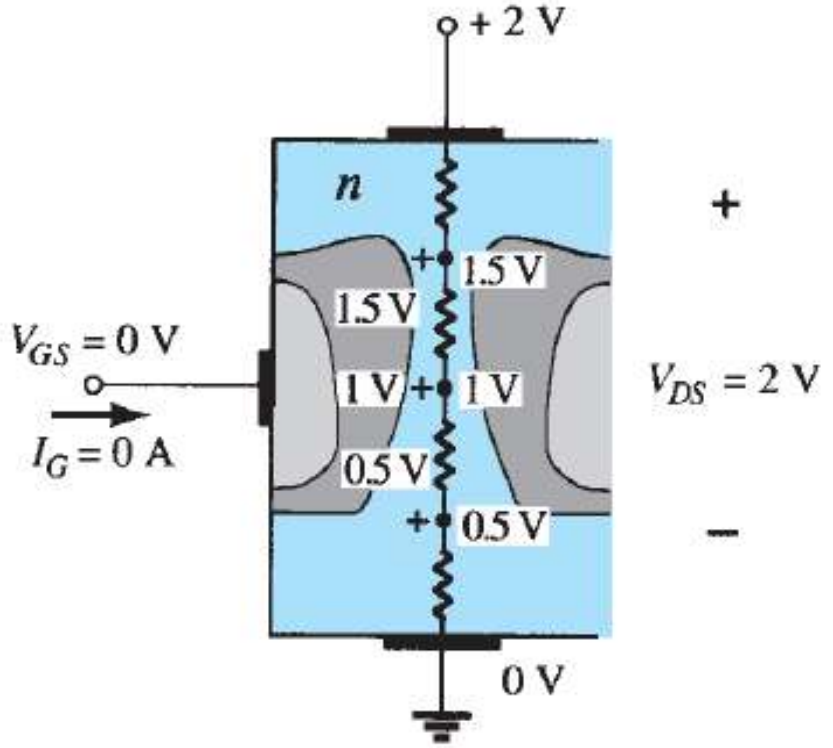


FIG. 6.6

The fact that
the p–n junction is reverse-biased for the length of the channel results in a gate current of zero amperes, as shown in the same figure. The fact that $I_G = 0$ A is an important characteristic of the JFET.

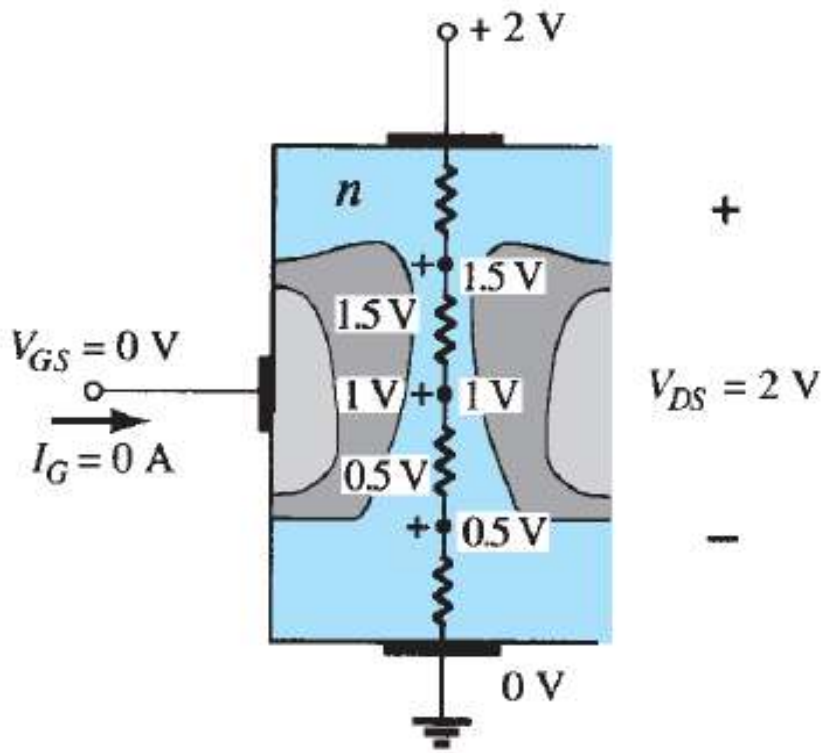


FIG. 6.6

As the voltage V_{DS} is increased from 0 V to a few volts, the current will increase as determined by Ohm's law and the plot of I_D versus V_{DS} will appear as shown in Fig. 6.7. The relative straightness of the plot reveals that for the region of low values of V_{DS}, the resistance is essentially constant. As V_{DS} increases and approaches a level referred to as V_P in Fig. 6.7, the depletion regions of Fig. 6.5 will widen, causing a noticeable reduction in the channel width. The reduced path of conduction causes the resistance to increase and the curve in the graph of Fig. 6.7 to occur.

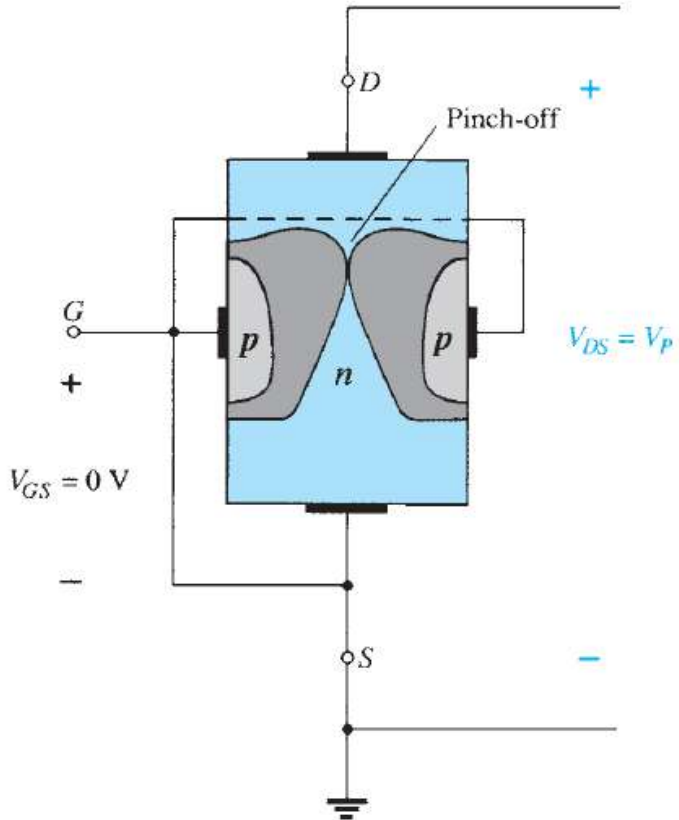


FIG. 6.8
Pinch-off ($V_{GS} = 0$ V, $V_{DS} = V_P$).

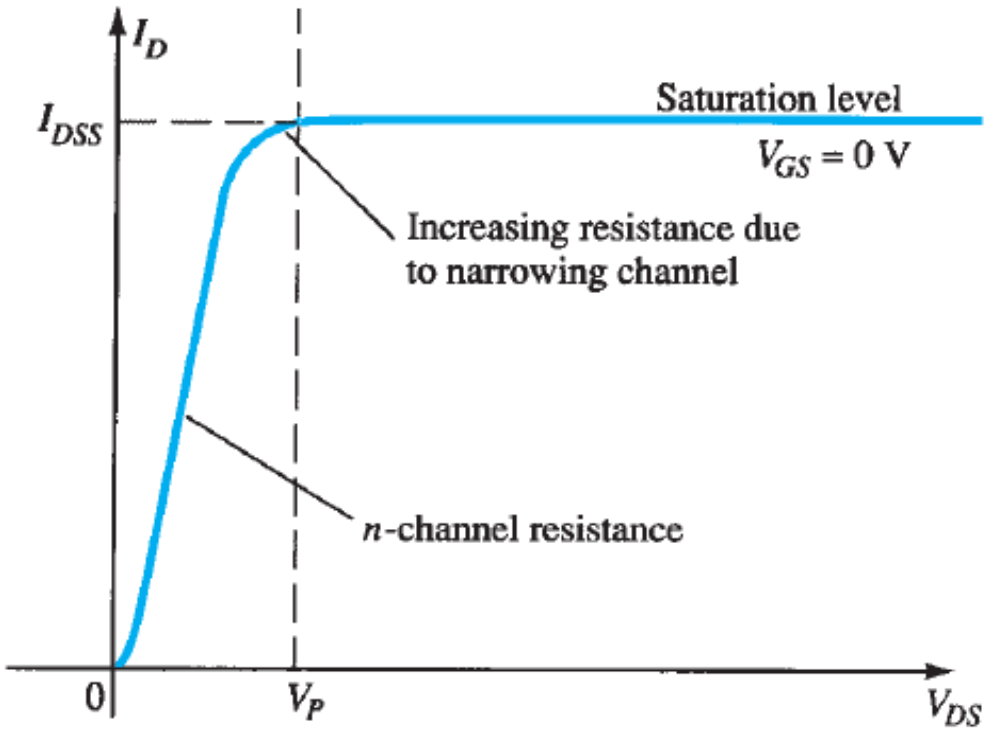


FIG. 6.7

The more horizontal the curve, the higher the resistance, suggesting that the resistance is approaching "infinite" ohms in the horizontal region. If V_{DS} is increased to a level where it appears that the two depletion regions would "touch" as shown in Fig. 6.8, a condition referred to as *pinch-off* will result. The level of V_{DS} that establishes this condition is referred to as the *pinch-off voltage* and is denoted by V_P, as shown in Fig. 6.7.

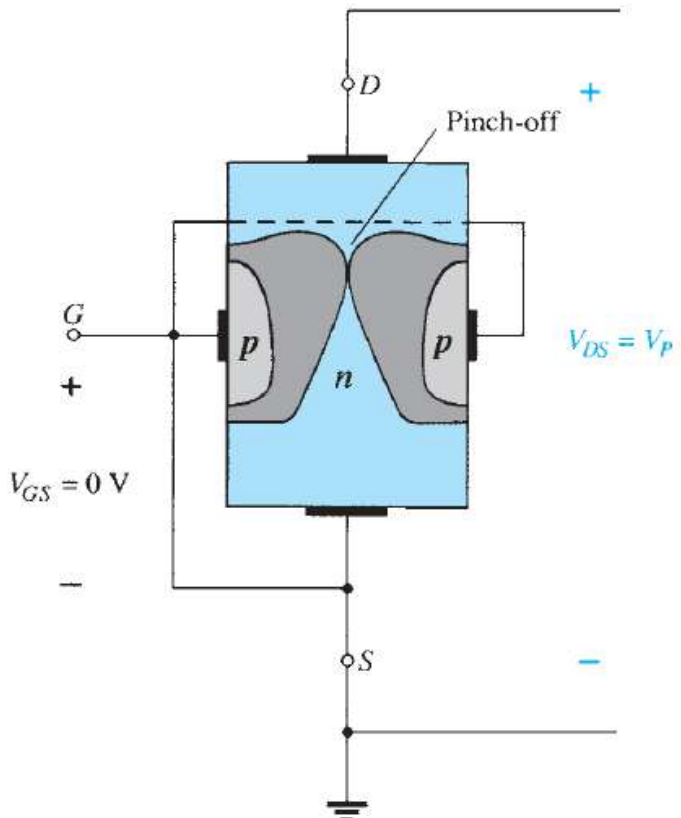


FIG. 6.8
Pinch-off ($V_{GS} = 0$ V, $V_{DS} = V_P$)

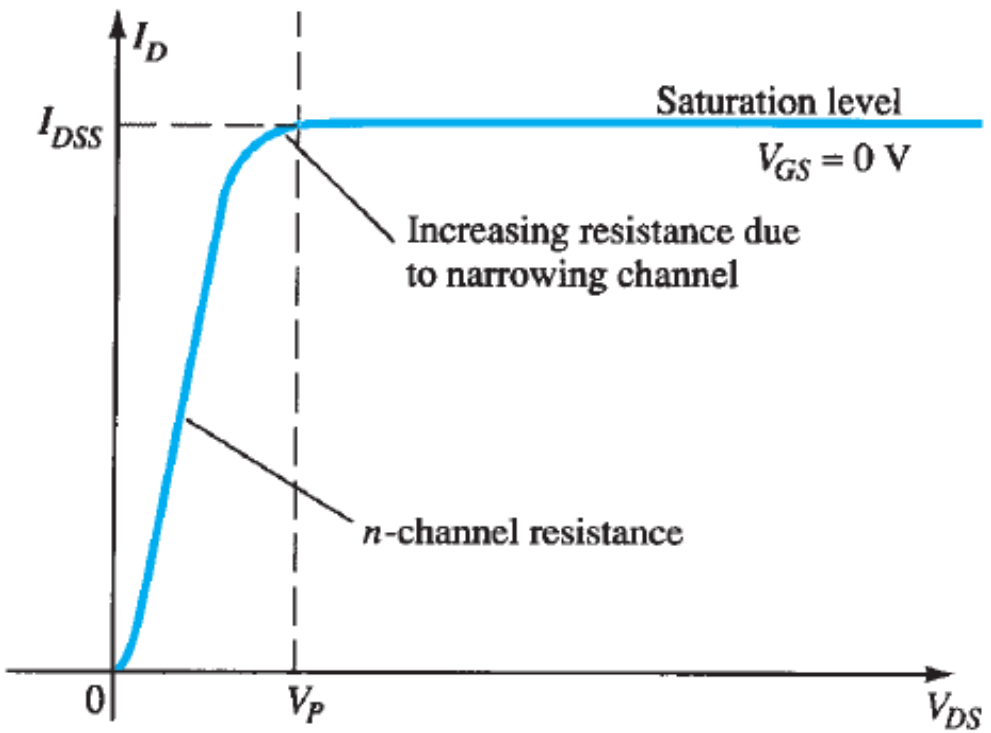


FIG. 6.7

In actuality, the term *pinch-off* is a misnomer in that it suggests the current I_D is pinched off and drops to 0 A. As shown in Fig. 6.7, however, this is hardly the case— I_D maintains a saturation level defined as I_{DSS} in Fig. 6.7. In reality a very small channel still exists, with a current of very high density.

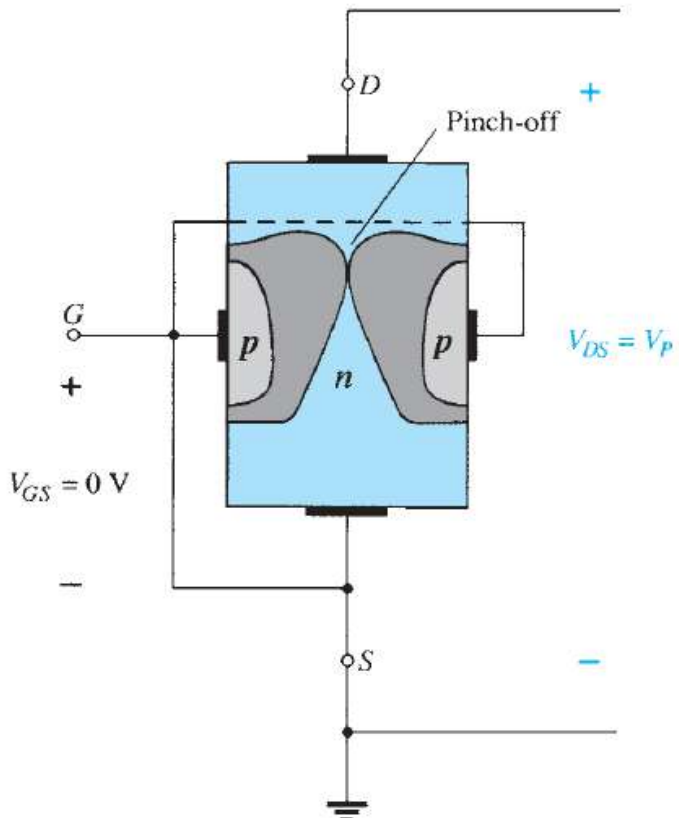


FIG. 6.8
Pinch-off ($V_{GS} = 0$ V, $V_{DS} = V_P$)

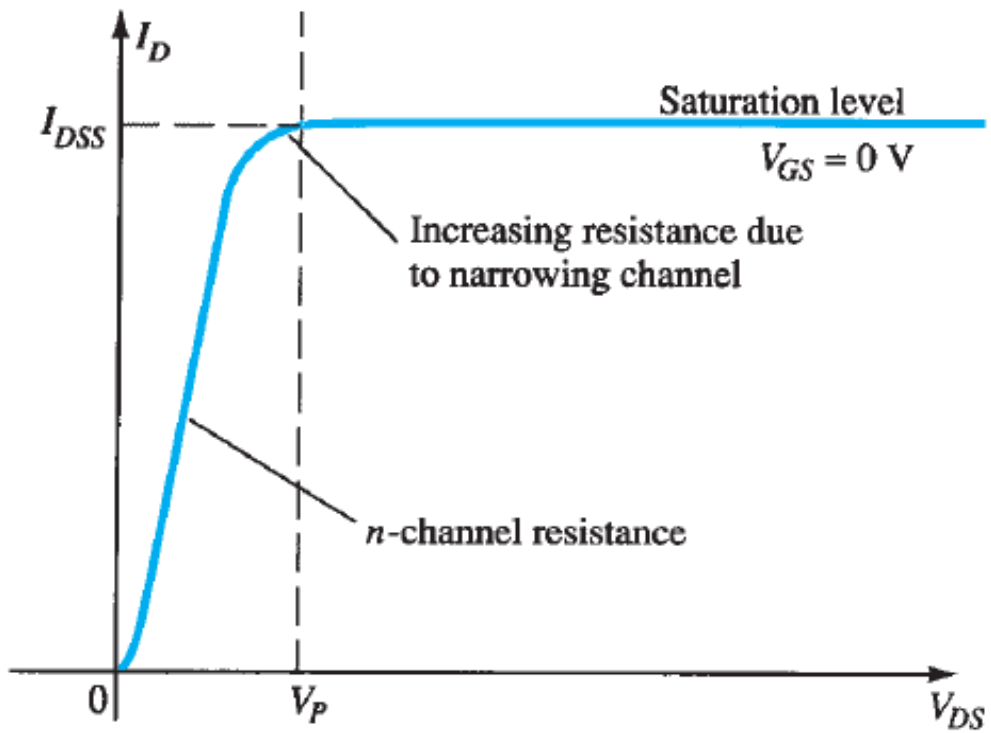


FIG. 6.7

As V_{DS} is increased beyond V_P, the region of close encounter between the two depletion regions increases in length along the channel, but the level of I_D remains essentially the same. In essence, therefore, once $V_{DS} > V_P$ the JFET has the characteristics of a current source. As shown in Fig. 6.9, the current is fixed at $I_D = I_{DSS}$, but the voltage V_{DS} (for levels $> V_P$) is determined by the applied load.

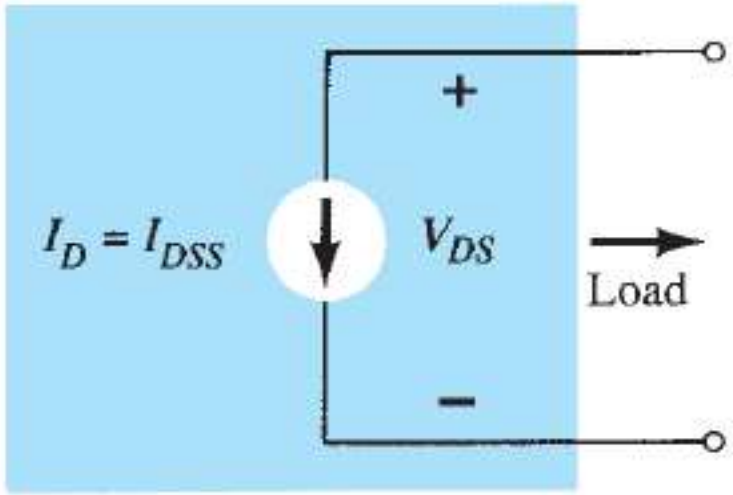


FIG. 6.9
Current source equivalent for $V_{GS} = 0$ V, $V_{DS} > V_P$.

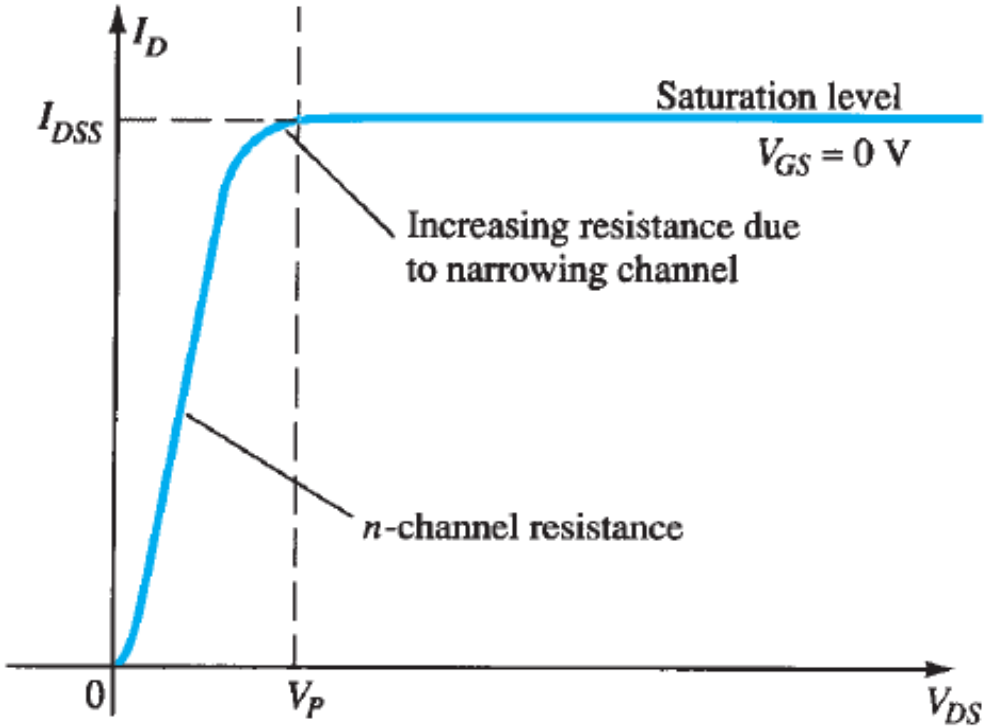


FIG. 6.7

The choice of notation I_{DSS} is derived from the fact that it is the *d*rain-to-*s*ource current with a *s*hort-circuit connection from gate to source. As we continue to investigate the characteristics of the device we will find that:

I_{DSS} is the maximum drain current for a JFET and is defined by the conditions $V_{GS} = 0\ V$ and $V_{DS} > |V_P|$.

Note in Fig. 6.7 that $V_{GS} = 0$ V for the entire length of the curve. The next few paragraphs will describe how the characteristics of Fig. 6.7 are affected by changes in the level of V_{GS}.

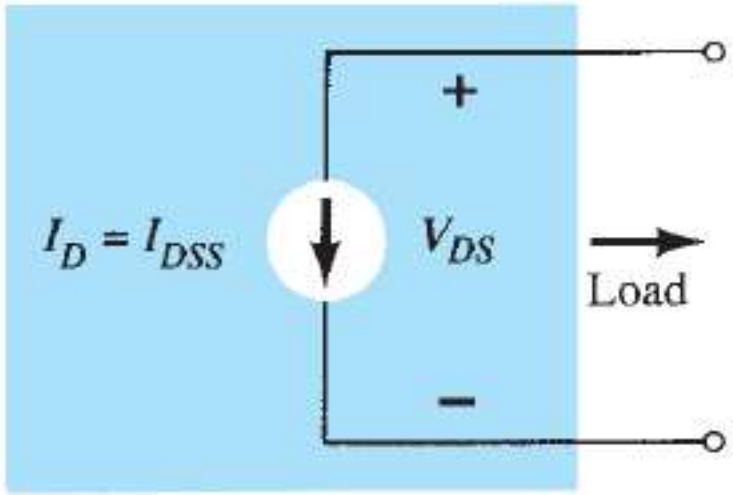


FIG. 6.9
Current source equivalent for $V_{GS} = 0\ V$, $V_{DS} > V_P$.

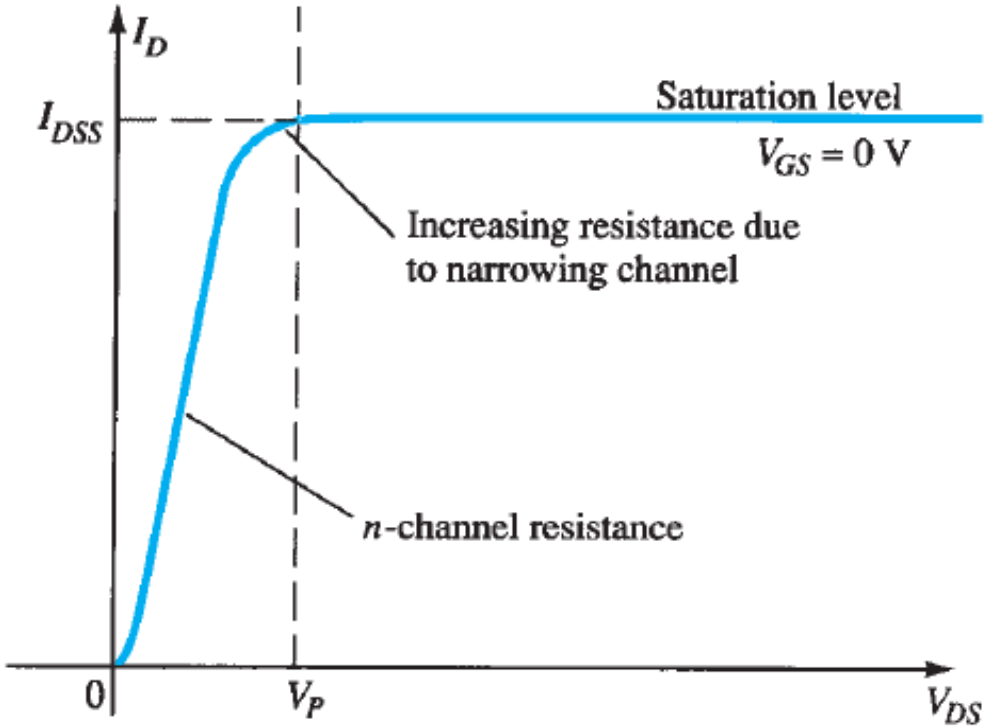


FIG. 6.7

$V_{GS} < 0$ V

The voltage from gate to source, denoted V_{GS}, is the controlling voltage of the JFET. Just as various curves for I_C versus V_{CE} were established for different levels of I_B for the BJT transistor, curves of I_D versus V_{DS} for various levels of V_{GS} can be developed for the JFET. For the *n*-channel device the controlling voltage V_{GS} is made more and more negative from its $V_{GS} = 0$ V level. In other words, the gate terminal will be set at lower and lower potential levels as compared to the source.

In Fig. 6.10 a negative voltage of -1 V is applied between the gate and source terminals for a low level of V_{DS}. The effect of the applied negative-bias V_{GS} is to establish depletion regions similar to those obtained with $V_{GS} = 0$ V, but at lower levels of V_{DS}. Therefore, the result of applying a negative bias to the gate is to reach the saturation level at a lower level of V_{DS}, as shown in Fig. 6.11 for $V_{GS} = -1$ V. The resulting saturation level for I_D has been reduced and in fact will continue to decrease as V_{GS} is made more and more negative. Note also in Fig. 6.11 how the pinch-off voltage continues to drop in a parabolic manner as V_{GS} becomes more and more negative. Eventually, V_{GS} when $V_{GS} = -V_P$ will be sufficiently negative to establish a saturation level that is essentially 0 mA, and for all practical purposes the device has been "turned off." In summary:

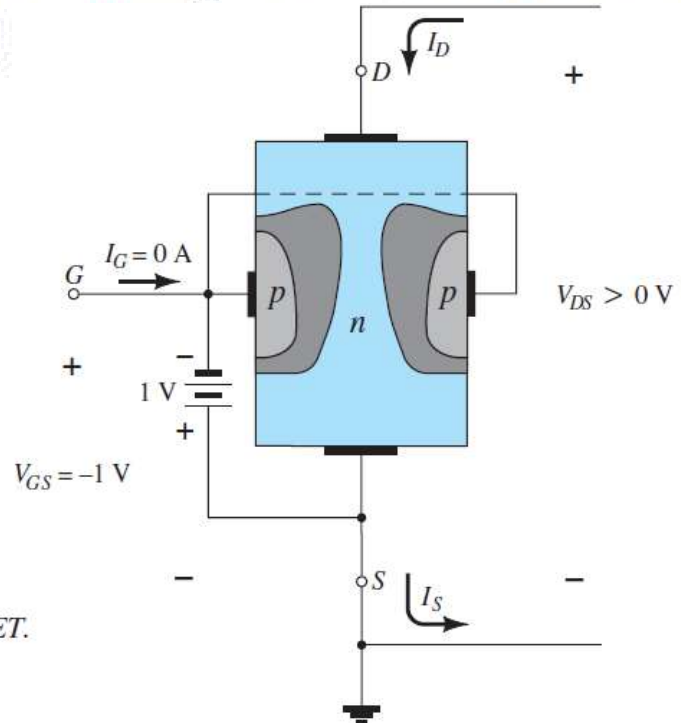


FIG. 6.10
Application of a negative voltage to the gate of a JFET.

In summary:

The level of V_{GS} that results in $I_D = 0$ mA is defined by $V_{GS} = V_P$, with V_P being a negative voltage for n-channel devices and a positive voltage for p-channel JFETs.

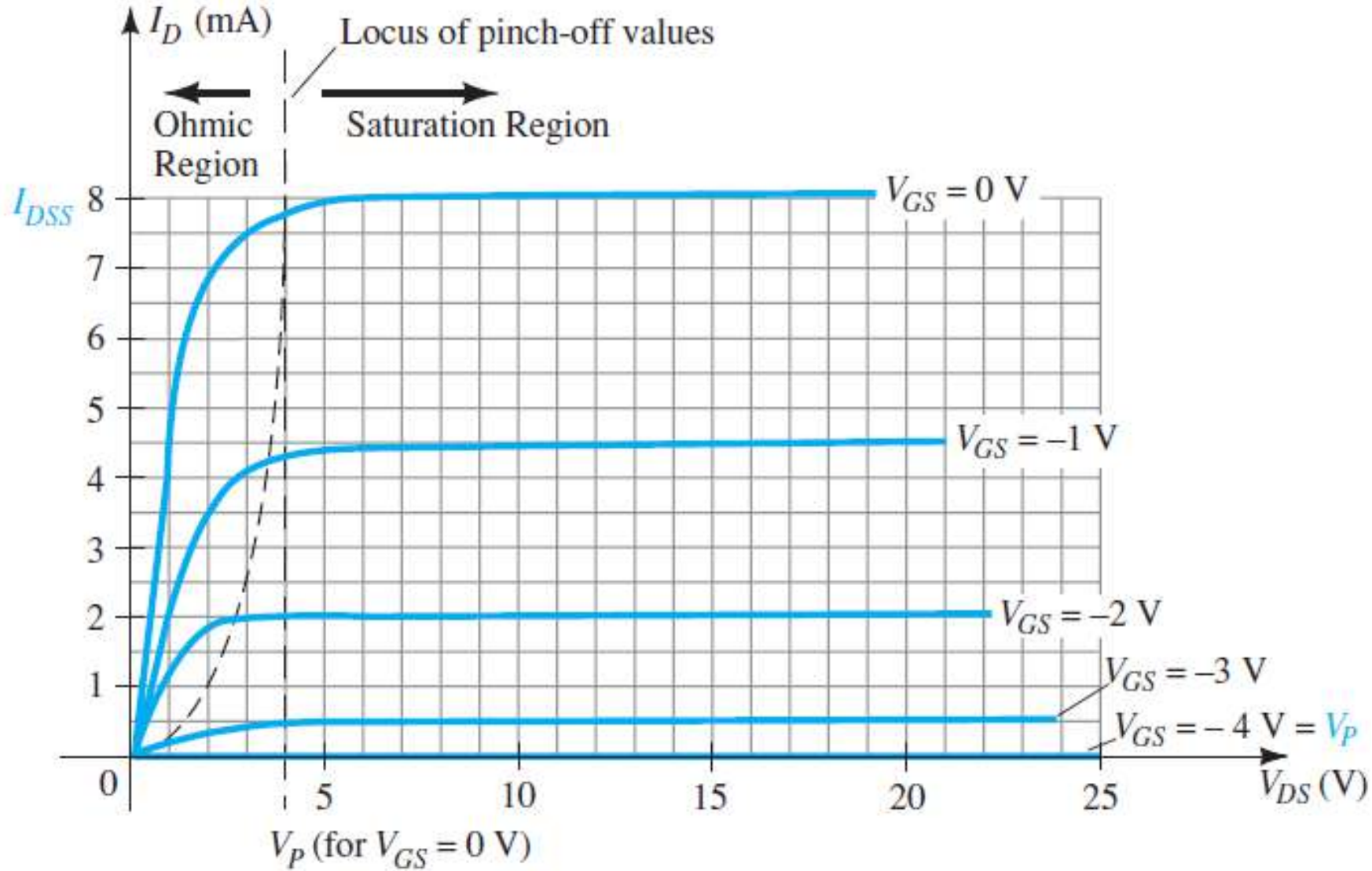


FIG. 6.11

On most specification sheets the pinch-off voltage is specified as $V_{GS(off)}$ rather than V_P. A specification sheet will be reviewed later in the chapter when the majority of the controlling elements have been introduced.

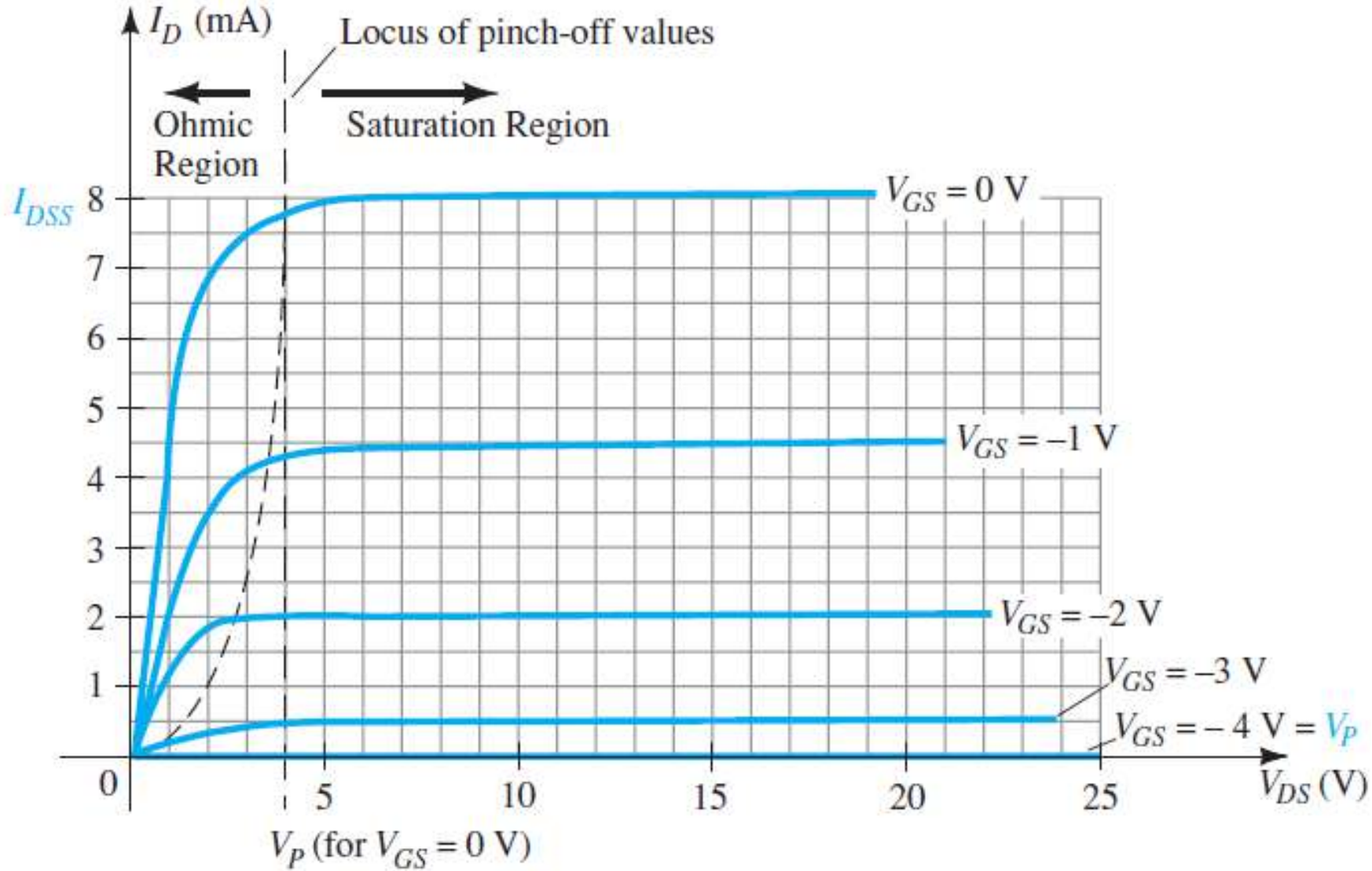


FIG. 6.11

The region to the right of the pinch-off locus of Fig. 6.11 is the region typically employed in linear amplifiers (amplifiers with minimum distortion of the applied signal) and is commonly referred to as the *constant-current, saturation,* or *linear amplification region.*

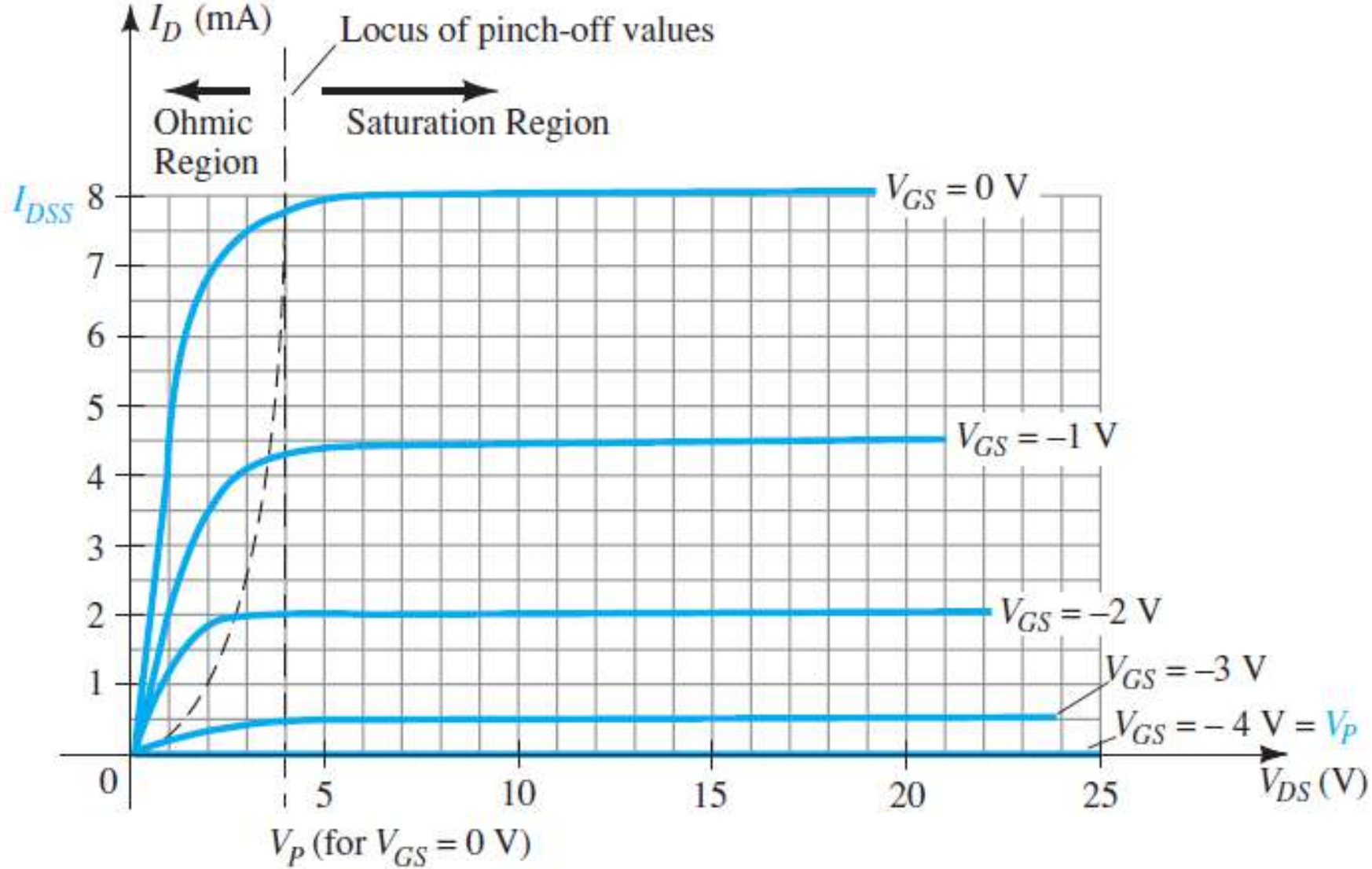


FIG. 6.11

Voltage-Controlled Resistor

The region to the left of the pinch-off locus of Fig. 6.11 is referred to as the *ohmic* or *voltage-controlled resistance region.* In this region the JFET can actually be employed as a variable resistor (possibly for an automatic gain control system) whose resistance is controlled by the applied gate-to-source voltage. Note in Fig. 6.11 that the slope of each curve and therefore the resistance of the device between drain and source for $V_{DS} < V_P$ are a function of the applied voltage V_{GS}.

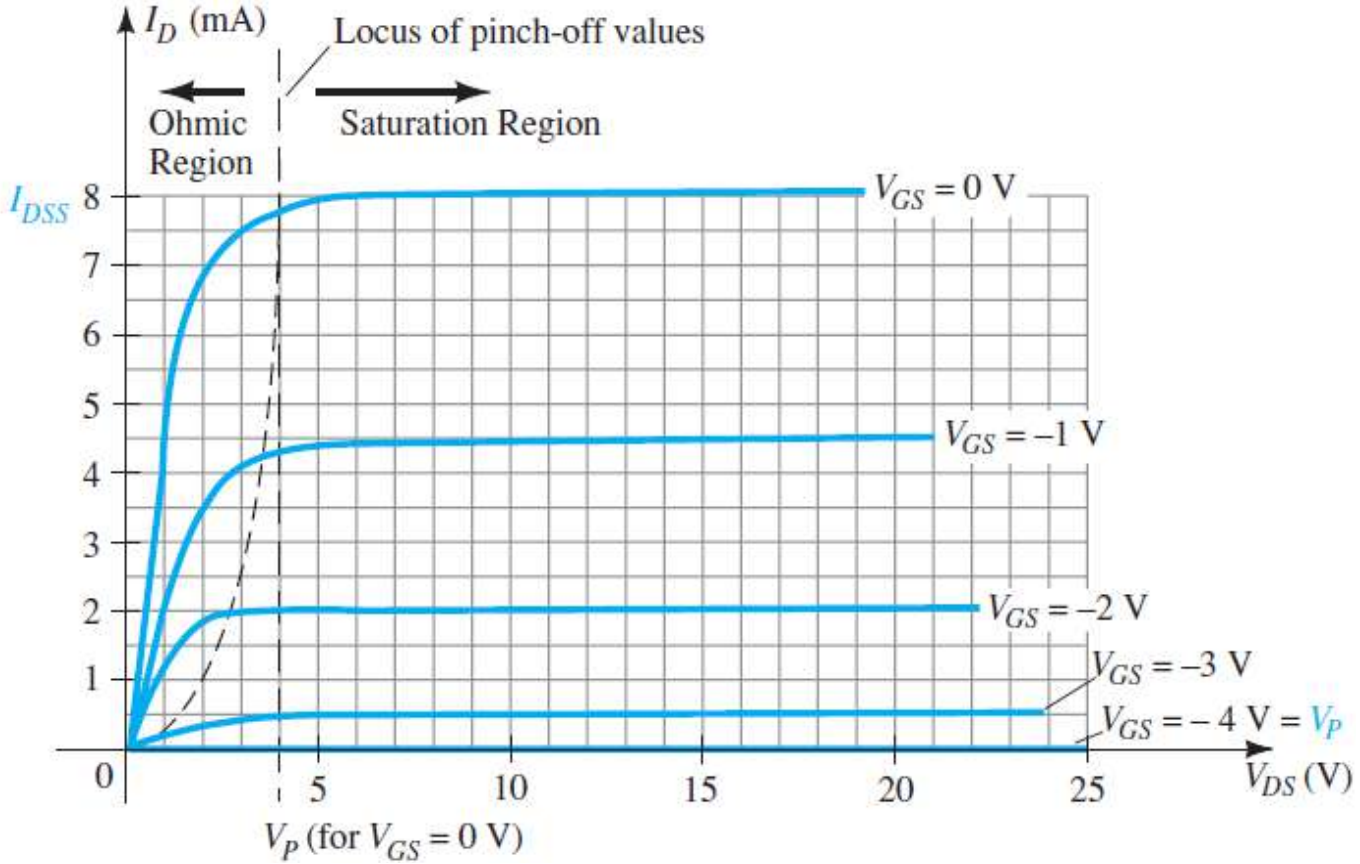


FIG. 6.11

As V_{GS} becomes more and more negative, the slope of each curve becomes more and more horizontal, corresponding to an increasing resistance level. The following equation provides a good first approximation to the resistance level in terms of the applied voltage V_{GS}:

$$r_d = \frac{r_o}{(1 - V_{GS}/V_P)^2} \tag{6.1}$$

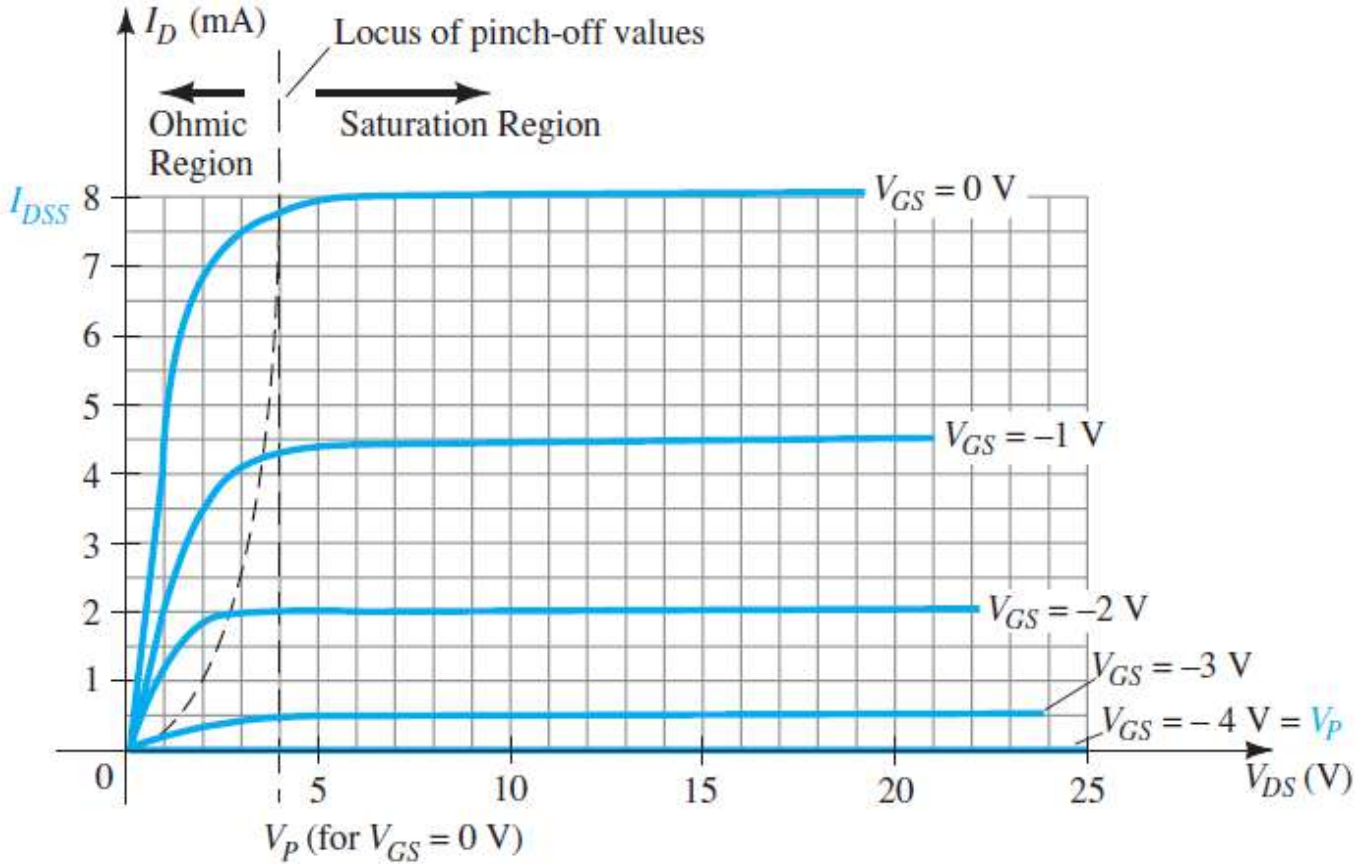


FIG. 6.11

$$r_d = \frac{r_o}{(1 - V_{GS}/V_P)^2} \tag{6.1}$$

where r_o is the resistance with $V_{GS} = 0$ V and r_d is the resistance at a particular level of V_{GS}.

For an *n*-channel JFET with $r_o = 10\ \text{k}\Omega$ ($V_{GS} = 0$ V, $V_P = -6$ V), Eq. (6.1) results in 40 kΩ at $V_{GS} = -3$ V.

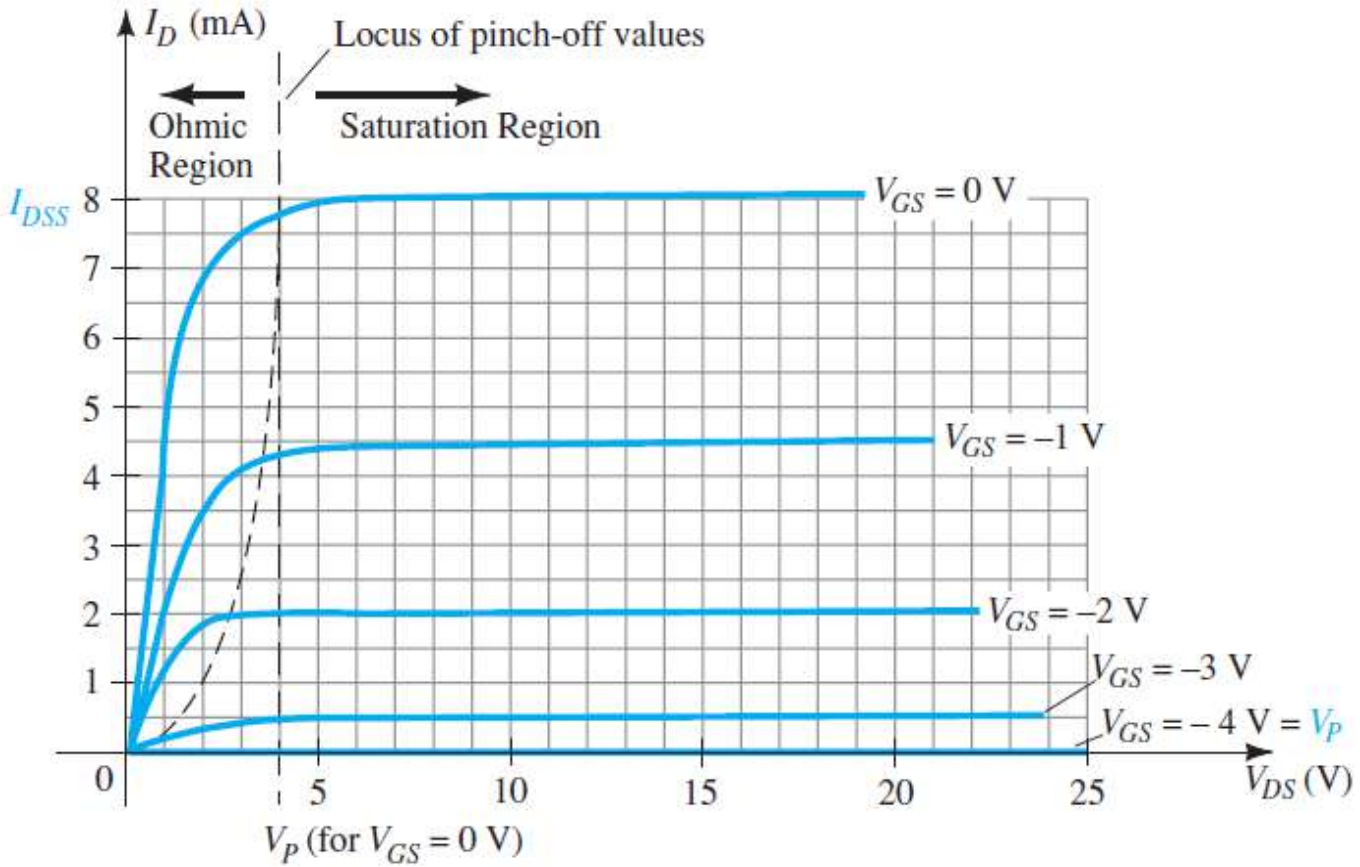


FIG. 6.11

p-Channel Devices

The *p*-channel JFET is constructed in exactly the same manner as the *n*-channel device of Fig. 6.3 but with a reversal of the *p*- and *n*-type materials as shown in Fig. 6.12. The defined current directions are reversed, as are the actual polarities for the voltages V_{GS} and V_{DS}. For the *p*-channel device, the channel will be constricted by increasing positive voltages from gate to source and the double-subscript notation for V_{DS} will result in negative voltages for V_{DS} on the characteristics of Fig. 6.13, which has an I_{DSS} of 6 mA and a pinch-off voltage of $V_{GS} = +6$ V. Do not let the minus signs for V_{DS} confuse you. They simply indicate that the source is at a higher potential than the drain.

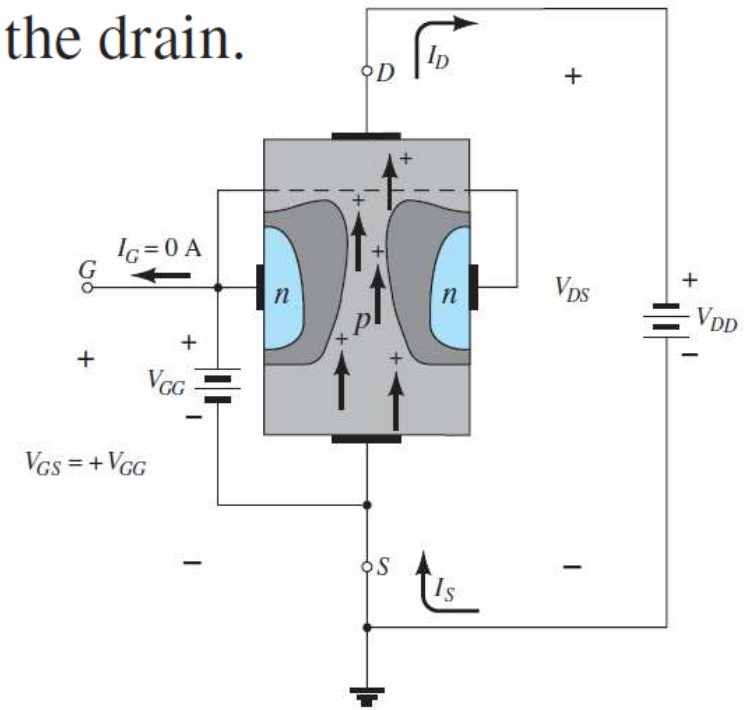


FIG. 6.12
p-Channel JFET.

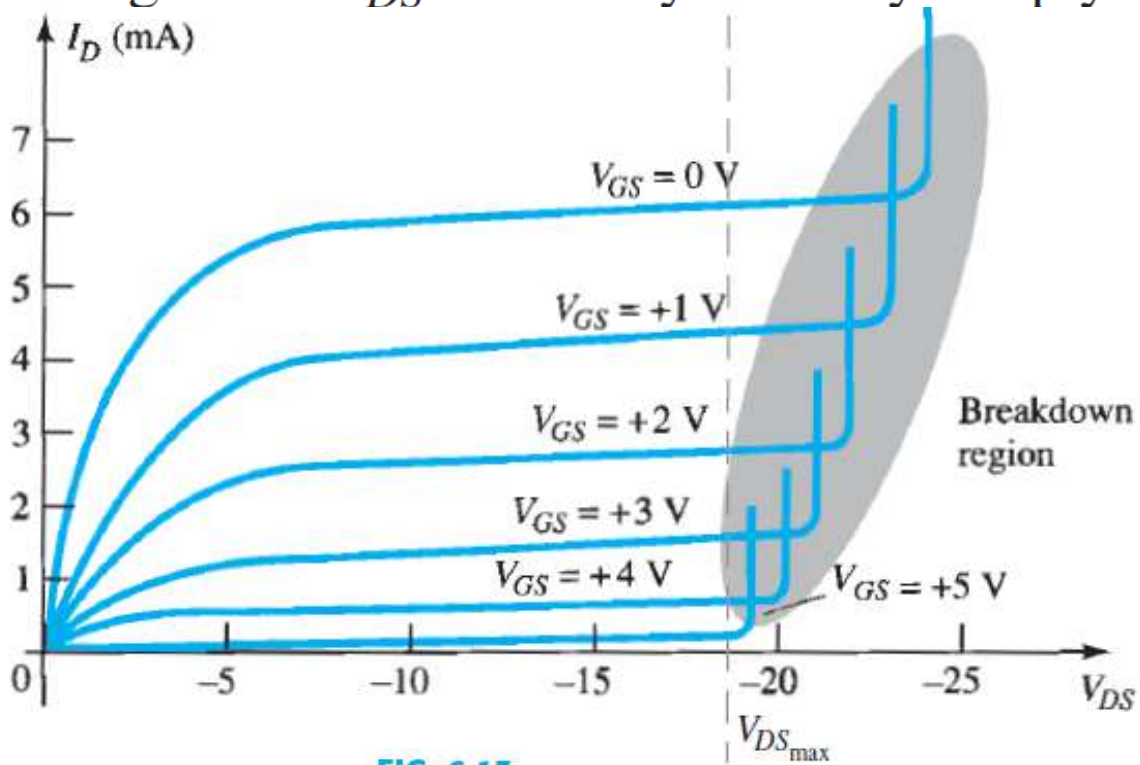
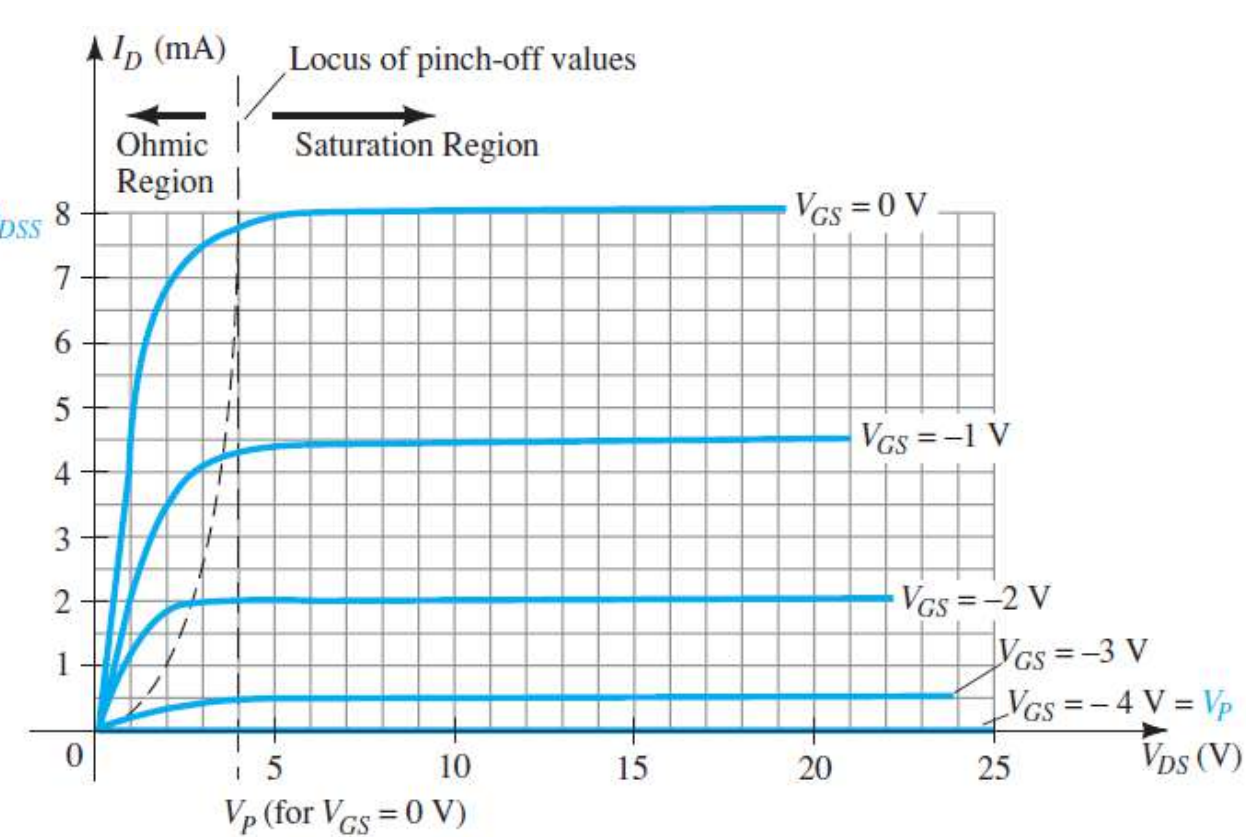


FIG. 6.13
p-Channel JFET characteristics with $I_{DSS} = 6$ mA and $V_P = +6$ V.

Note at high levels of V_{DS} that the curves suddenly rise to levels that seem unbounded. The vertical rise is an indication that breakdown has occurred and the current through the channel (in the same direction as normally encountered) is now limited solely by the external circuit. Although not appearing in Fig. 6.11 for the *n*-channel device, they do occur for the *n*-channel device if sufficient voltage is applied. This region can be avoided if the level of $V_{DS_{max}}$ is noted on the specification sheet and the design is such that the actual level of V_{DS} is less than this value for *all* values of V_{GS}.



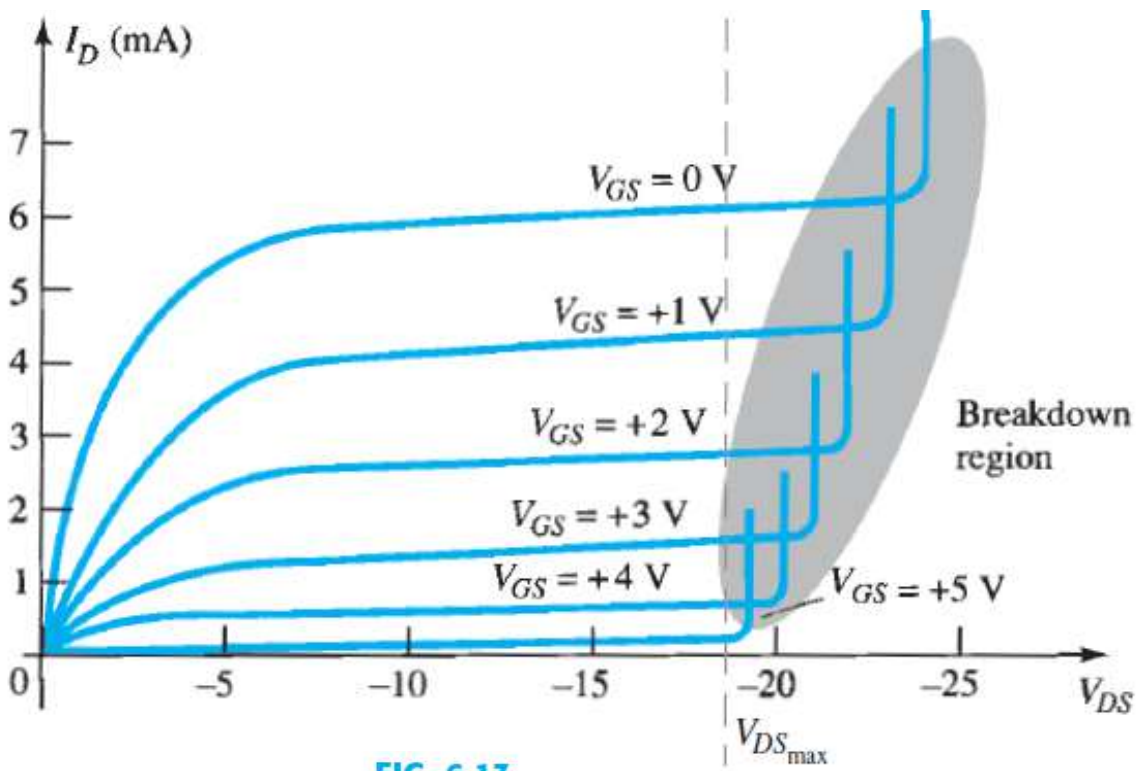


FIG. 6.13

p-Channel JFET characteristics with $I_{DSS} = 6$ mA and $V_P = +6$ V.

Symbols

The graphic symbols for the *n*-channel and *p*-channel JFETs are provided in Fig. 6.14. Note that the arrow is pointing in for the *n*-channel device of Fig. 6.14a to represent the direction in which I_G would flow if the *p*–*n* junction were forward-biased. For the *p*-channel device (Fig. 6.14b) the only difference in the symbol is the direction of the arrow in the symbol.

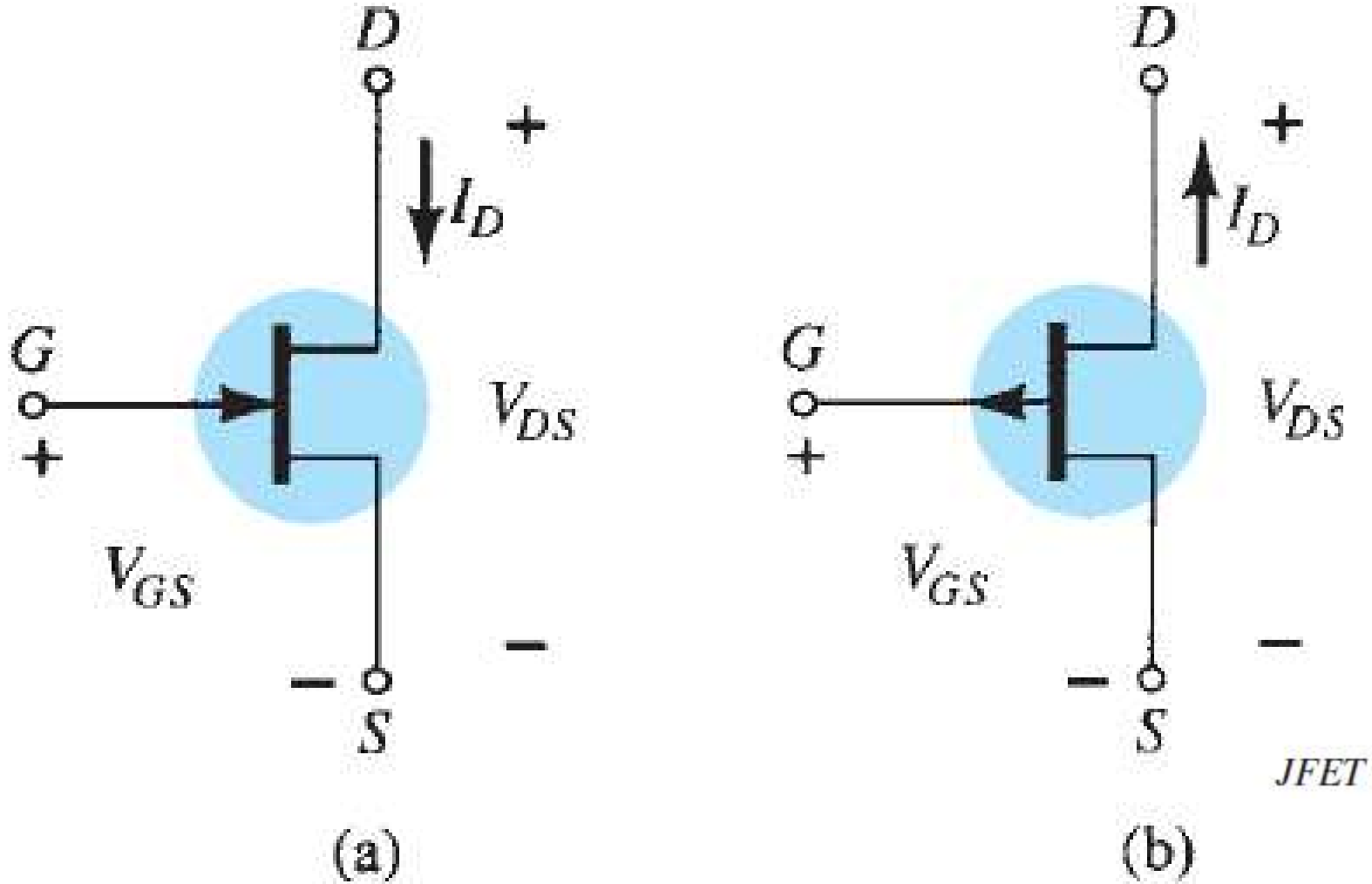


FIG. 6.14
JFET symbols: (a) n-channel; (b) p-channel.

Summary

A number of important parameters and relationships were introduced in this section. A few that will surface frequently in the analysis to follow in this chapter and the next for *n*-channel JFETs include the following:

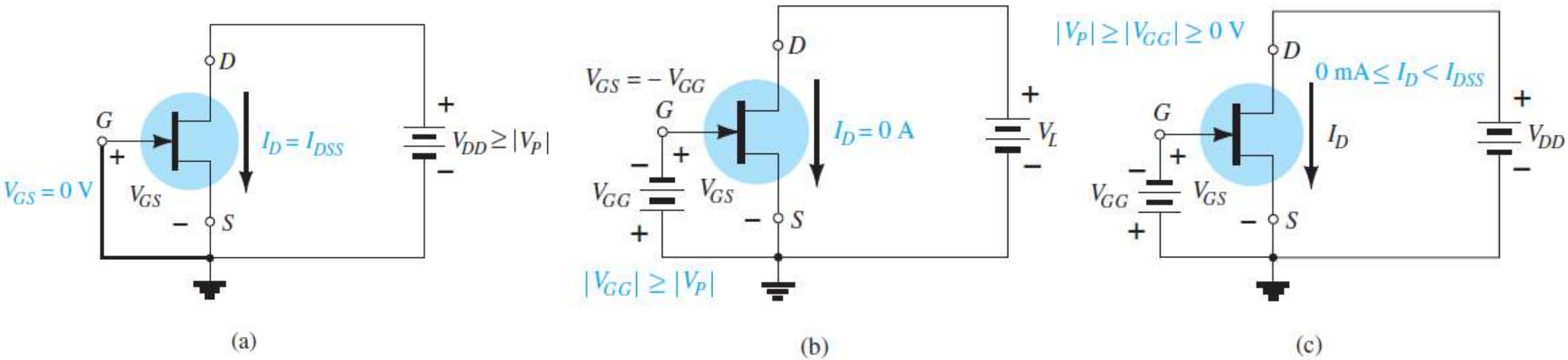


FIG. 6.15

(a) $V_{GS} = 0$ V, $I_D = I_{DSS}$; (b) cutoff ($I_D = 0$ A) V_{GS} less than the pinch-off level; (c) I_D is between 0 A and I_{DSS} for $V_{GS} \leq 0$ V and greater than the pinch-off level.

The maximum current is defined as I_{DSS} and occurs when $V_{GS} = 0$ V and $V_{DS} \geq |V_P|$, as shown in Fig. 6.15a.

For gate-to-source voltages V_{GS} is less than (more negative than) the pinch-off level, the drain current is 0 A ($I_D = 0$ A), as in Fig. 6.15b.

For all levels of V_{GS} between 0 V and the pinch-off level, the current I_D will range between I_{DSS} and 0 A, respectively, as in Fig. 6.15c.

A similar list can be developed for p-channel JFETs.

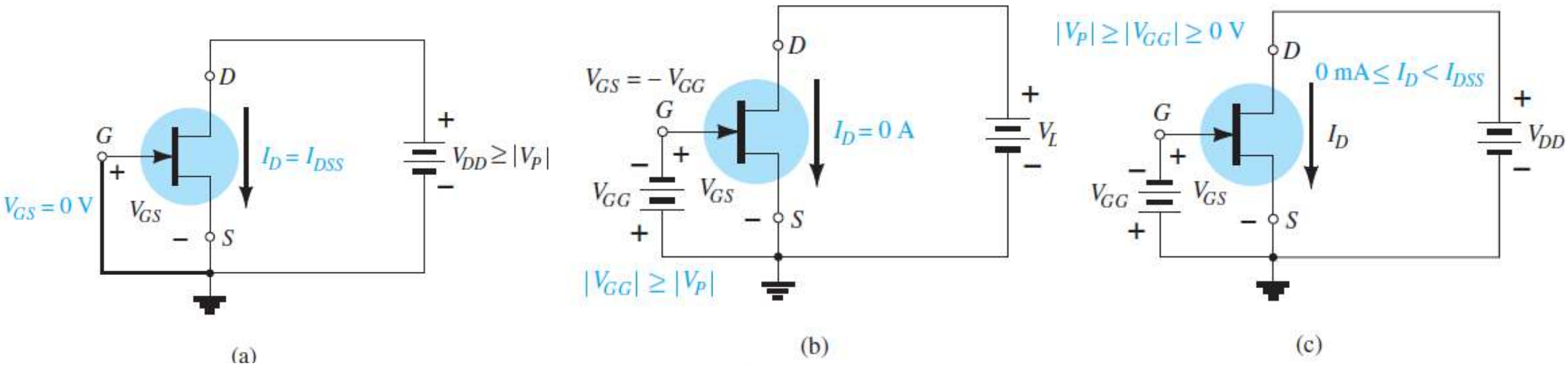


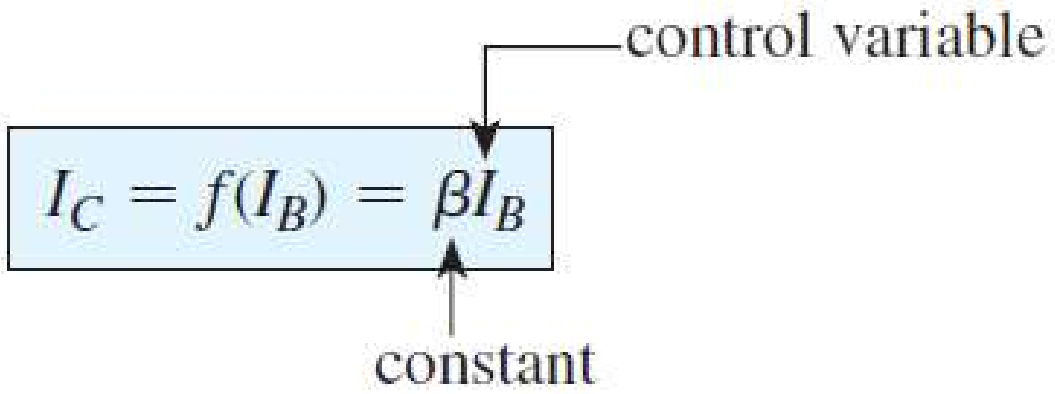
FIG. 6.15

(a) $V_{GS} = 0$ V, $I_D = I_{DSS}$; (b) cutoff ($I_D = 0$ A) V_{GS} less than the pinch-off level; (c) I_D is between 0 A and I_{DSS} for $V_{GS} \leq 0$ V and greater than the pinch-off level.

6.3 TRANSFER CHARACTERISTICS

Derivation

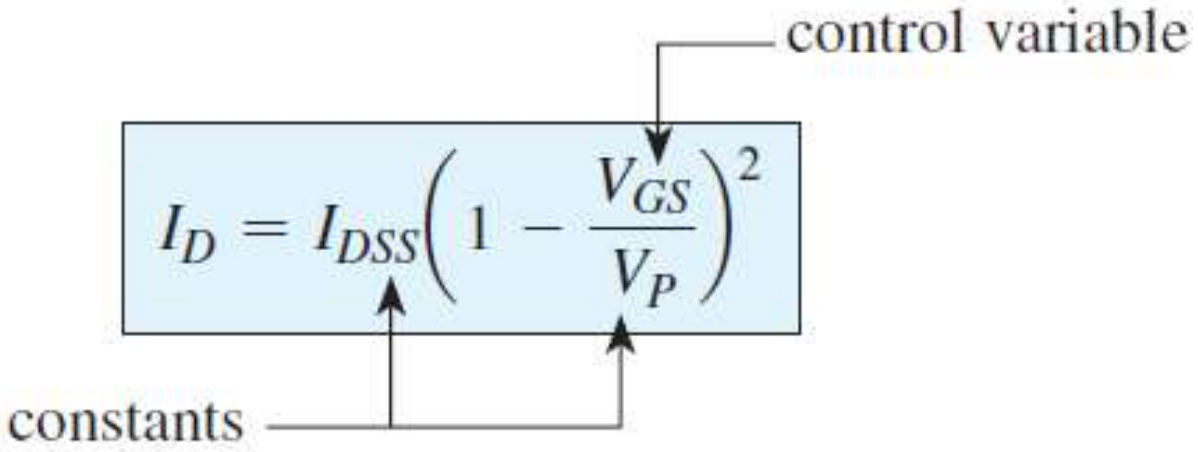
For the BJT transistor the output current I_C and the input controlling current I_B are related by beta, which was considered constant for the analysis to be performed. In equation form,



(6.2)

In Eq. (6.2) a linear relationship exists between I_C and I_B. Double the level of I_B and I_C will increase by a factor of two also.

Unfortunately, this linear relationship does not exist between the output and input quantities of a JFET. The relationship between I_D and V_{GS} is defined by *Shockley's equation* (see Fig. 6.16):



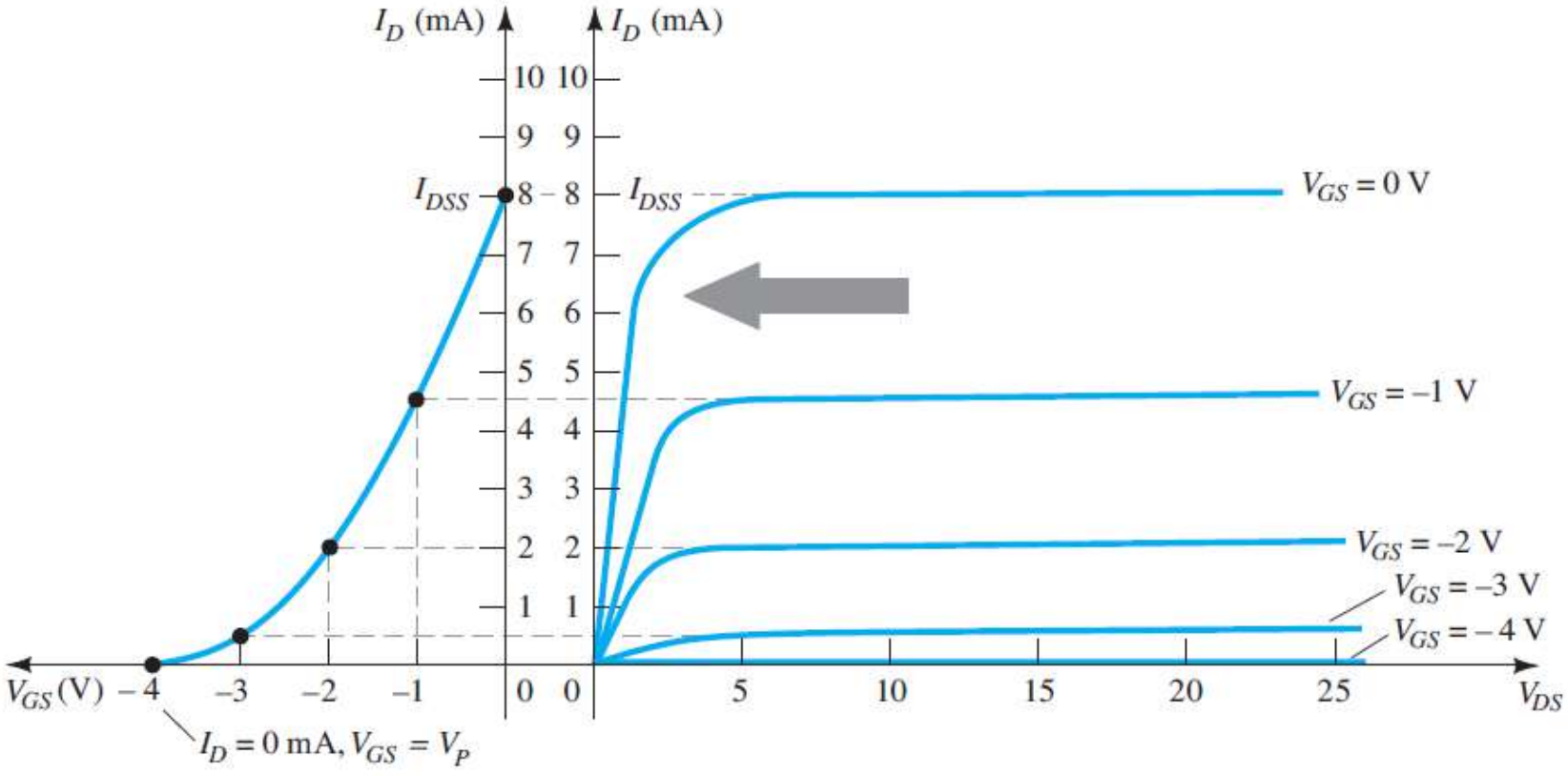
$$I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_{GS_{off}}}\right)^2$$

The squared term in the equation results in a nonlinear relationship between I_D and V_{GS}, producing a curve that grows exponentially with decreasing magnitude of V_{GS}.

Note : For n-channel V_{GSoff} and V_{GS} always negative, for p-channel V_{GSoff} and V_{GS} always positive ,so I_D always less than or equal I_{DSS}

The network equation may change along with the intersection between the two curves, but the transfer curve defined by Eq. (6.3) is unaffected. In general, therefore:

The transfer characteristics defined by Shockley's equation are unaffected by the network in which the device is employed.

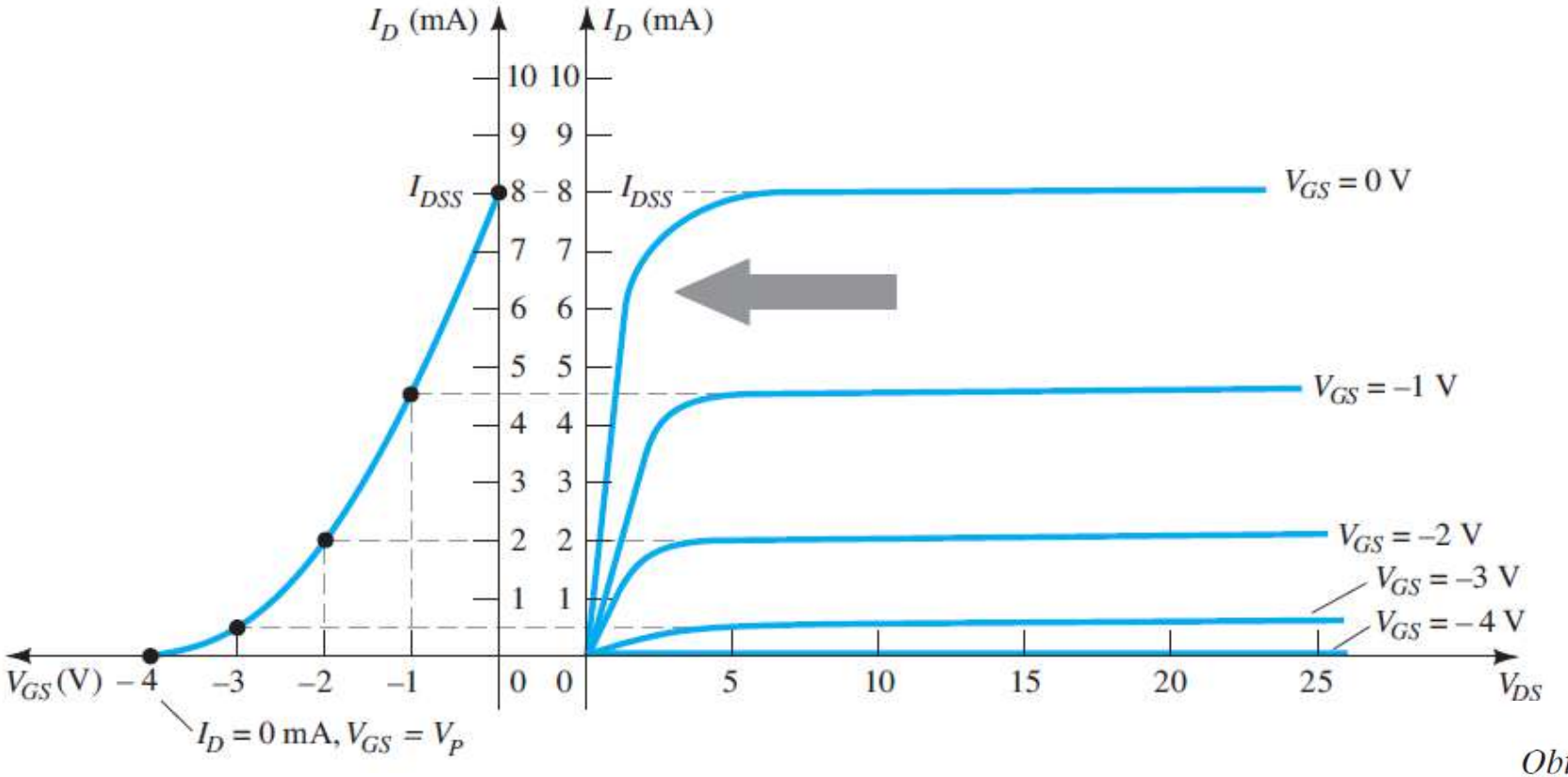


$$I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_{GS_{off}}}\right)^2$$

FIG. 6.17

btaining the transfer curve from the drain characteristics.

The transfer curve can be obtained using Shockley's equation or from the output characteristics of Fig. 6.11. In Fig. 6.17 two graphs are provided, with the vertical scaling in milliamperes for each graph. One is a plot of I_D versus V_{DS}, whereas the other is I_D versus V_{GS}. Using the drain characteristics on the right of the "y" axis, we can draw a horizontal line from the saturation region of the curve denoted $V_{GS} = 0$ V to the I_D axis. The resulting current level for both graphs is I_{DSS}. The point of intersection on the I_D versus V_{GS} curve will be as shown since the vertical axis is defined as $V_{GS} = 0$ V.



$$I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_{GS_{off}}}\right)^2$$

FIG. 6.17

Obtaining the transfer curve from the drain characteristics.

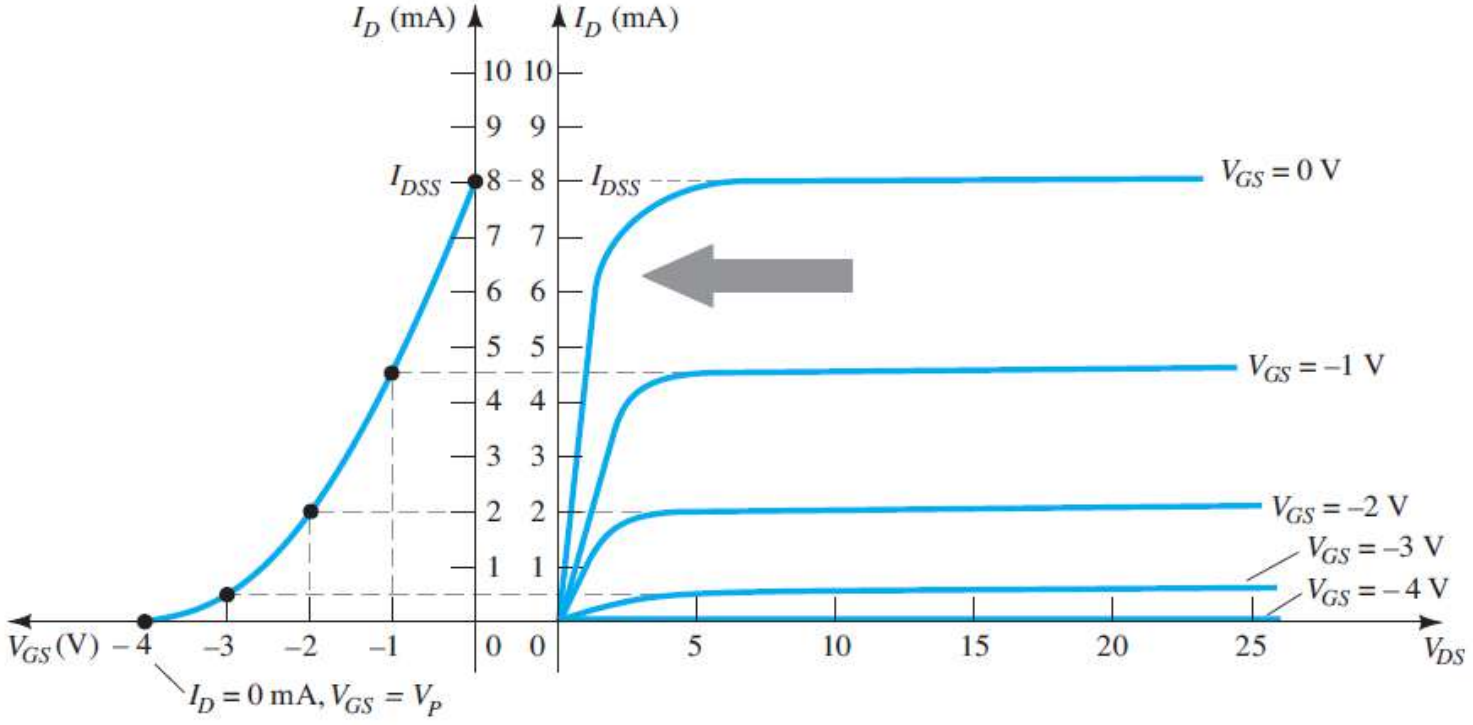
In review:

$$\text{When } V_{GS} = 0\text{ V}, \quad I_D = I_{DSS} \tag{6.4}$$

When $V_{GS} = V_P = -4$ V, the drain current is 0 mA, defining another point on the transfer curve. That is:

$$\text{When } V_{GS} = V_P, \quad I_D = 0\text{ mA} \tag{6.5}$$

Before continuing, it is important to realize that the drain characteristics relate one output (or drain) quantity to another output (or drain) quantity—both axes are defined by variables in the same region of the device characteristics. The transfer characteristics are a plot of an output (or drain) current versus an input-controlling quantity

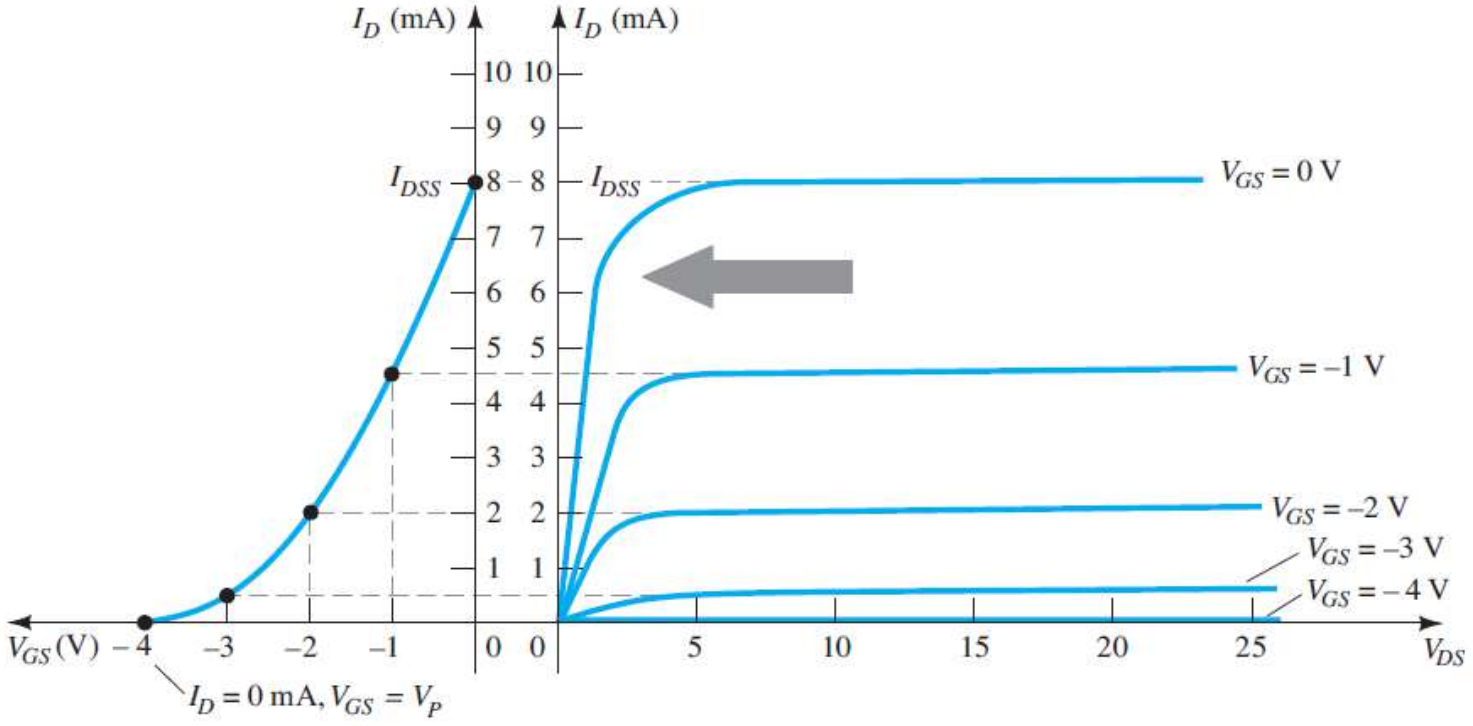


$$I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_{GS_{off}}}\right)^2$$

FIG. 6.17

Obtaining the transfer curve from the drain characteristics.

There is therefore a direct "transfer" from input to output variables when employing the curve to the left of Fig. 6.17. If the relationship were linear, the plot of I_D versus V_{GS} would result in a straight line between I_{DSS} and V_P. However, a parabolic curve will result because the vertical spacing between steps of V_{GS} on the drain characteristics of Fig. 6.17 decreases noticeably as V_{GS} becomes more and more negative. Compare the spacing between $V_{GS} = 0$ V and $V_{GS} = -1$ V to that between $V_{GS} = -3$ V and pinch-off. The change in V_{GS} is the same, but the resulting change in I_D is quite different.



$$I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_{GS_{off}}}\right)^2$$

FIG. 6.17

Obtaining the transfer curve from the drain characteristics.

Applying Shockley's Equation

The transfer curve of Fig. 6.17 can also be obtained directly from Shockley's equation (6.3) given simply the values of I_{DSS} and V_P. The levels of I_{DSS} and V_P define the limits of the curve on both axes and leave only the necessity of finding a few intermediate plot points. The validity of Eq. (6.3) as a source of the transfer curve of Fig. 6.17 is best demonstrated by examining a few specific levels of one variable and finding the resulting level of the other as follows:

Substituting $V_{GS} = 0$ V gives

$$\text{Eq. (6.3):} \quad I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_P}\right)^2 \qquad I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_{GS_{off}}}\right)^2$$

$$= I_{DSS}\left(1 - \frac{0}{V_P}\right)^2 = I_{DSS}(1 - 0)^2$$

and

$$\boxed{I_D = I_{DSS}\,|_{V_{GS}=0\,\text{V}}} \tag{6.6}$$

Substituting $V_{GS} = V_P$ yields

$$I_D = I_{DSS}\left(1 - \frac{V_P}{V_P}\right)^2$$
$$= I_{DSS}(1 - 1)^2 = I_{DSS}(0)$$

$$\boxed{I_D = 0\ \text{A}\big|_{V_{GS}=V_P}} \tag{6.7}$$

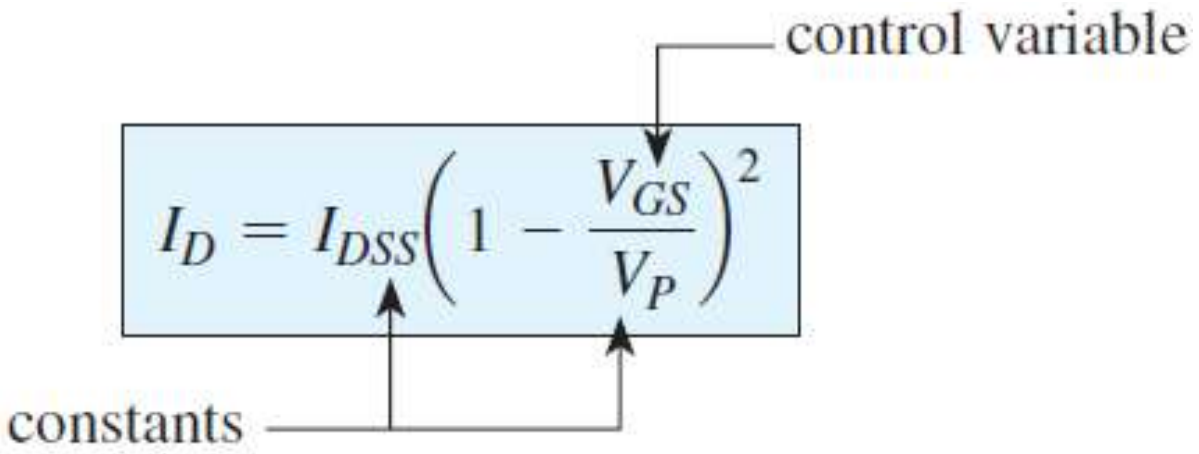
For the drain characteristics of Fig. 6.17, if we substitute $V_{GS} = -1$ V,

$$I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_P}\right)^2$$
$$= 8\ \text{mA}\left(1 - \frac{-1\ \text{V}}{-4\ \text{V}}\right)^2 = 8\ \text{mA}\left(1 - \frac{1}{4}\right)^2 = 8\ \text{mA}\,(0.75)^2$$
$$= 8\ \text{mA}\,(0.5625)$$
$$= \mathbf{4.5\ mA}$$

as shown in Fig. 6.17. Note the care taken with the negative signs for V_{GS} and V_P in the calculations above. The loss of one sign would result in a totally erroneous result.

It should be obvious from the above that given I_{DSS} and V_P (as is normally provided on specification sheets), the level of I_D can be found for any level of V_{GS}. Conversely, by using basic algebra we can obtain [from Eq. (6.3)] an equation for the resulting level of V_{GS} for a given level of I_D. The derivation is quite straightforward and results in

$$V_{GS} = V_P\left(1 - \sqrt{\frac{I_D}{I_{DSS}}}\right) \tag{6.8}$$



Let us test Eq. (6.8) by finding the level of V_{GS} that will result in a drain current of 4.5 mA for the device with the characteristics of Fig. 6.17. We find

$$V_{GS} = -4\,\text{V}\left(1 - \sqrt{\frac{4.5\,\text{mA}}{8\,\text{mA}}}\right)$$
$$= -4\,\text{V}(1 - \sqrt{0.5625}) = -4\,\text{V}(1 - 0.75)$$
$$= -4\,\text{V}(0.25)$$
$$= \mathbf{-1\,V}$$

as substituted in the above calculation and verified by Fig. 6.17.

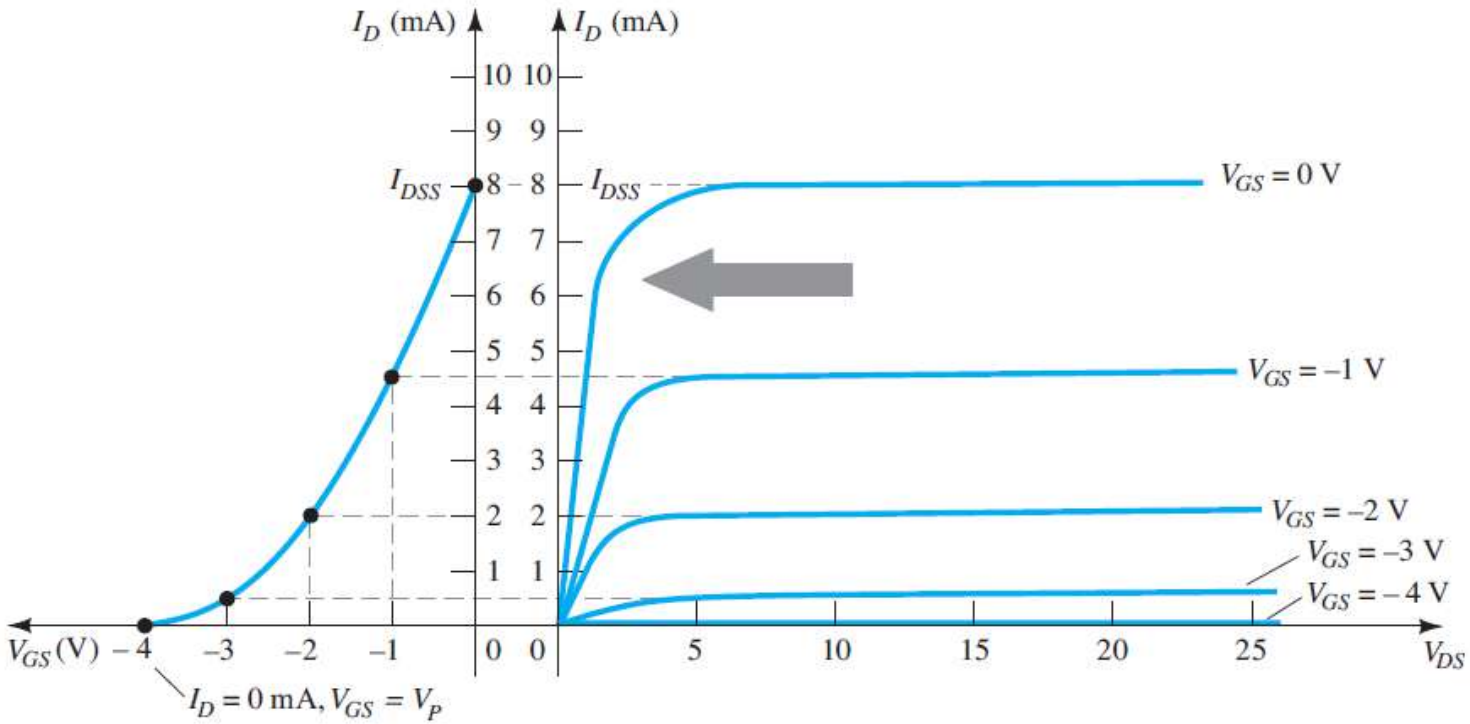


FIG. 6.17

Obtaining the transfer curve from the drain characteristics.

Shorthand Method

Since the transfer curve must be plotted so frequently, it would be quite advantageous to have a shorthand method for plotting the curve in the quickest, most efficient manner while maintaining an acceptable degree of accuracy. The format of Eq. (6.3) is such that specific levels of V_{GS} will result in levels of I_D that can be memorized to provide the plot points needed to sketch the transfer curve. If we specify V_{GS} to be one-half the pinch-off value V_P, the resulting level of I_D will be the following, as determined by Shockley's equation:

$$I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_P}\right)^2$$

$$= I_{DSS}\left(\frac{1 - V_P/2}{V_P}\right)^2 = I_{DSS}\left(1 - \frac{1}{2}\right)^2 = I_{DSS}(0.5)^2$$

$$= I_{DSS}(0.25)$$

and

$$\boxed{I_D = \frac{I_{DSS}}{4}\Big|_{V_{GS} = V_P/2}} \qquad \textbf{(6.9)}$$

If we choose $I_D = I_{DSS}/2$ and substitute into Eq. (6.8), we find that

$$V_{GS} = V_P\left(1 - \sqrt{\frac{I_D}{I_{DSS}}}\right)$$

$$= V_P\left(1 - \sqrt{\frac{I_{DSS}/2}{I_{DSS}}}\right) = V_P(1 - \sqrt{0.5}) = V_P(0.293)$$

and

$$\boxed{V_{GS} \cong 0.3V_P|_{I_D=I_{DSS}/2}} \quad \textbf{(6.10)}$$

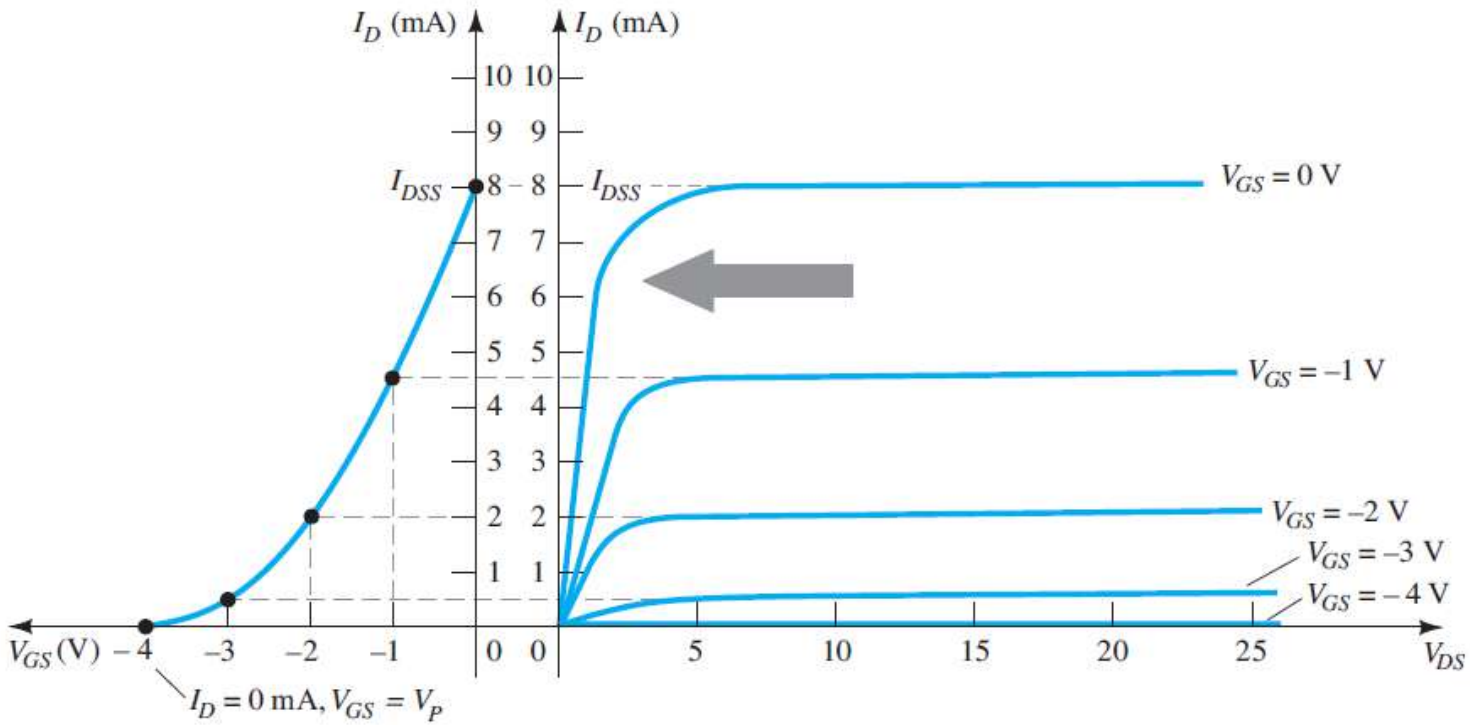


FIG. 6.17

Obtaining the transfer curve from the drain characteristics.

Additional points can be determined, but the transfer curve can be sketched to a satisfactory level of accuracy simply using the four plot points defined above and reviewed in Table 6.1. In fact, in the analysis of Chapter 7, a maximum of four plot points are used to sketch the transfer curves. On most occasions using just the plot point defined by $V_{GS} = V_P/2$ and the axis intersections at I_{DSS} and V_P will provide a curve accurate enough for most calculations.

TABLE 6.1

V_{GS} versus I_D Using Shockley's Equation

V_{GS}	I_D
0	I_{DSS}
$0.3V_P$	$I_{DSS}/2$
$0.5V_P$	$I_{DSS}/4$
V_P	0 mA

EXAMPLE 6.1 Sketch the transfer curve defined by $I_{DSS} = 12\text{ mA}$ and $V_P = -6\text{ V}$.

Solution: Two plot points are defined by

$$I_{DSS} = \mathbf{12\text{ mA}} \quad \text{and} \quad V_{GS} = \mathbf{0\text{ V}}$$

and

$$I_D = \mathbf{0\text{ mA}} \quad \text{and} \quad V_{GS} = V_P$$

At $V_{GS} = V_P/2 = -6\text{ V}/2 = \mathbf{-3\text{ V}}$ the drain current is determined by $I_D = I_{DSS}/4 = 12\text{ mA}/4 = \mathbf{3\text{ mA}}$. At $I_D = I_{DSS}/2 = 12\text{ mA}/2 = \mathbf{6\text{ mA}}$ the gate-to-source voltage is determined by $V_{GS} \cong 0.3V_P = 0.3(-6\text{ V}) = \mathbf{-1.8\text{ V}}$. All four plot points are well defined on Fig. 6.18 with the complete transfer curve.

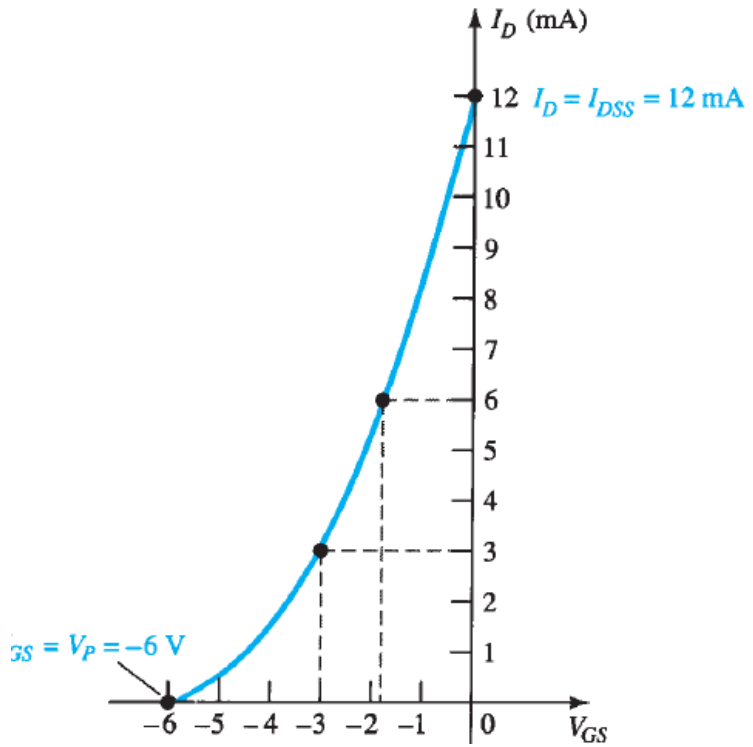


FIG. 6.18
Transfer curve for Example 6.1.

For *p*-channel devices Shockley's equation (6.3) can still be applied exactly as it appears. In this case, both V_P and V_{GS} will be positive and the curve will be the mirror image of the transfer curve obtained with an *n*-channel and the same limiting values.

EXAMPLE 6.2 Sketch the transfer curve for a *p*-channel device with $I_{DSS} = 4\,\text{mA}$ and $V_P = 3\,\text{V}$.

Solution: At $V_{GS} = V_P/2 = 3\,\text{V}/2 = \mathbf{1.5\,V}$, $I_D = I_{DSS}/4 = 4\,\text{mA}/4 = \mathbf{1\,mA}$. At $I_D = I_{DSS}/2 = 4\,\text{mA}/2 = \mathbf{2\,mA}$, $V_{GS} = 0.3V_P = 0.3(3\,\text{V}) = \mathbf{0.9\,V}$. Both plot points appear in Fig. 6.19 along with the points defined by I_{DSS} and V_P.

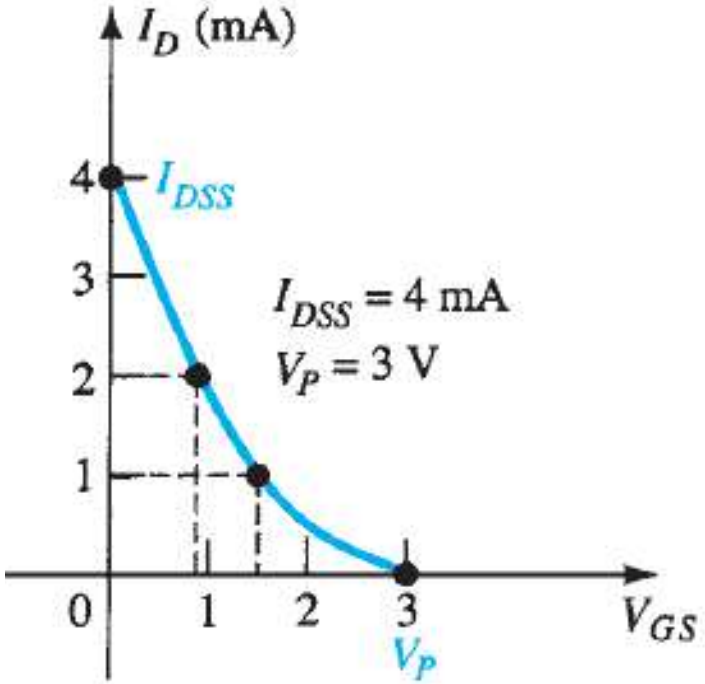


FIG. 6.19

6.4 SPECIFICATION SHEETS (JFETs)

As with any electronic device it is important to be able to understand the data provided on a specification sheet. Often times the notation used is different than we normally apply so a measure of translation may have to be applied. In general, however, the headings for the data are uniform and include **Maximum Ratings, Thermal Characteristics, Electrical Characteristics**, and sets of **Typical Characteristics**. In Fig. 6.20 the specification sheets for a Fairchild Semiconductor 2N5457 *n*-channel JFET appears with two types of packaging techniques. The TO-92 package is for a higher power device than the surface mount SOT-23 unit.

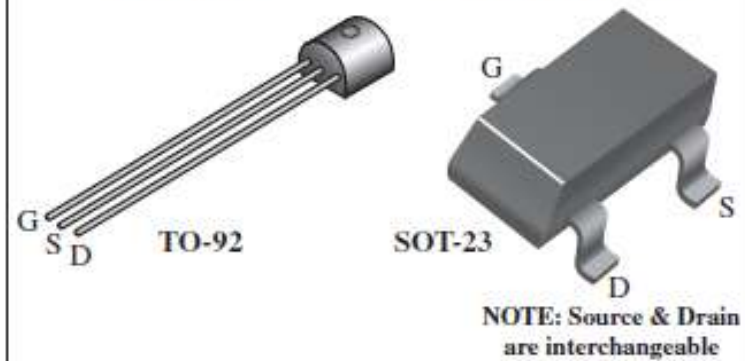
More information page 391-392

ABSOLUTE MAXIMUM RATINGS

Symbol	Parameter	Value	Units
V_{DS}	Drain-Source Voltage	25	V
V_{DG}	Drain-Gate Voltage	25	V
V_{GS}	Gate-Source Voltage	–25	V
I_{GF}	Forward Gate Current	10	mA
T_j, T_{stg}	Operating and Storage Junction Temperature Range	–55 to +150	°C

FAIRCHILD
SEMICONDUCTOR ™

2N5457 **MMBF5457**



NOTE: Source & Drain are interchangeable

N-Channel General Purpose Amplifier

This device is a low-level audio amplifier and switching transistor, and can be used for analog switching applications.

Operating Region

The specification sheet and the curve defined by the pinch-off levels at each level of V_{GS} define the region of operation for linear amplification on the drain characteristics as shown in Fig. 6.21. The ohmic region defines the minimum permissible values of V_{DS} at each level of V_{GS}, and $V_{DS_{max}}$ specifies the maximum value for this parameter. The saturation

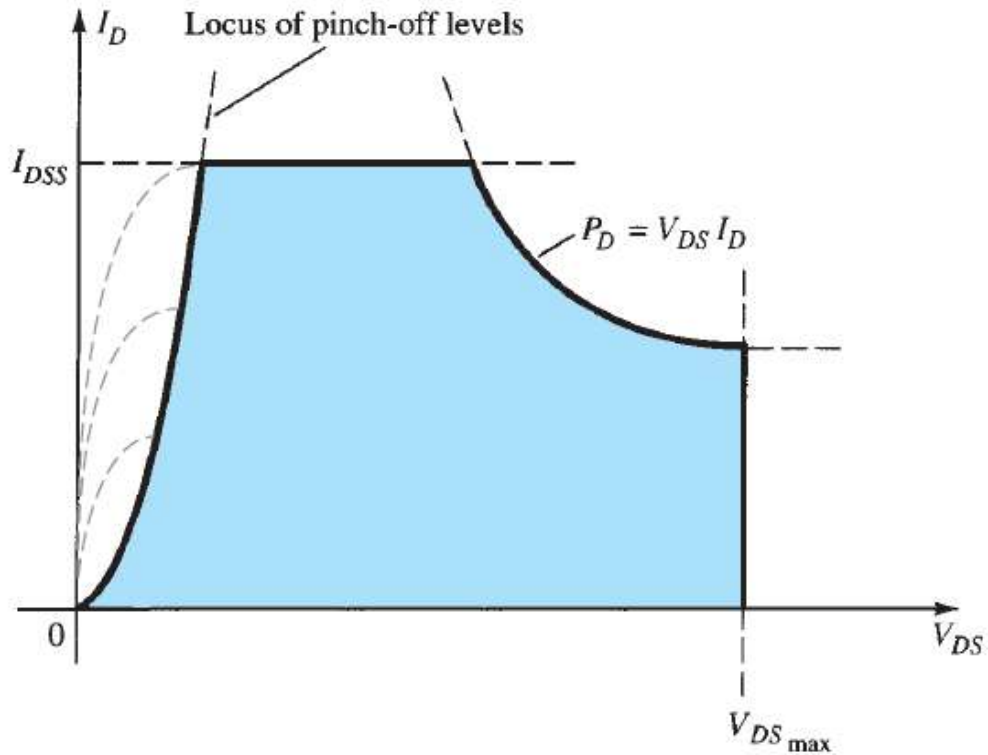


FIG. 6.21
Normal operating region for linear amplifier design.

current I_{DSS} is the maximum drain current, and the maximum power dissipation level defines the curve drawn in the same manner as described for BJT transistors. The resulting shaded region is the normal operating region for amplifier design.

6.6 IMPORTANT RELATIONSHIPS

A number of important equations and operating characteristics for the JFET have been introduced that are of particular importance for the analysis of dc and ac configurations that will follow. To isolate and emphasize their importance, they are repeated in Table 6.2 next to corresponding equations for the BJT transistor. The JFET equations are defined for the configuration of Fig. 6.23a, whereas the BJT equations relate to Fig. 6.23b.

TABLE 6.2

JFET		*BJT*
$I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_P}\right)^2$	$\Leftrightarrow$	$I_C = \beta I_B$
$I_D = I_S$	$\Leftrightarrow$	$I_C \cong I_E$
$I_G \cong 0\ \text{A}$	$\Leftrightarrow$	$V_{BE} \cong 0.7\ \text{V}$

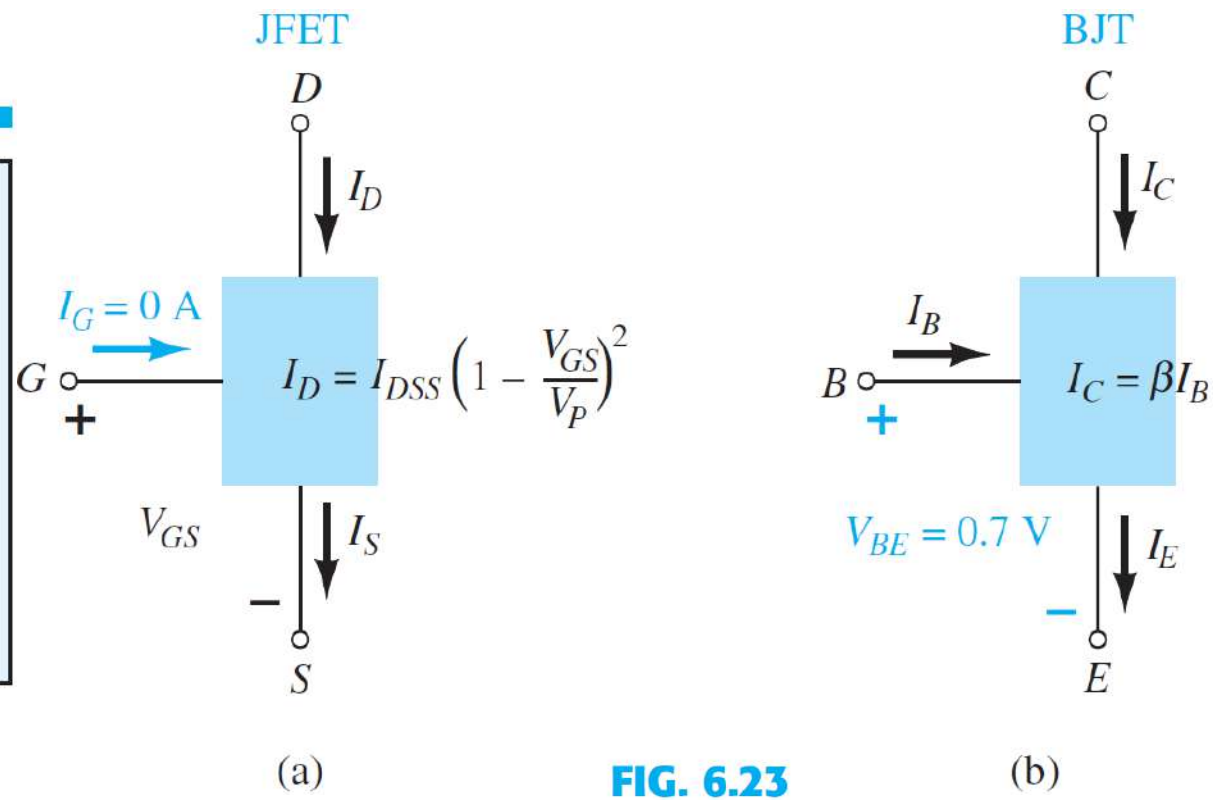


FIG. 6.23
(a) JFET versus (b) BJT.

A clear understanding of the effect of each of the equations above is sufficient background to approach the most complex of dc configurations. Recall that $V_{BE} = 0.7\ \text{V}$ was often the key to initiating an analysis of a BJT configuration. Similarly, the condition $I_G = 0\ \text{A}$ is often the starting point for the analysis of a JFET configuration. For the BJT configuration, I_B is normally the first parameter to be determined. For the JFET, it is normally V_{GS}. The number of similarities between the analysis of BJT and JFET dc configurations will become quite apparent in Chapter 7.

TABLE 6.2

JFET		*BJT*
$I_D = I_{DSS}\left(1 - \dfrac{V_{GS}}{V_P}\right)^2$	$\Leftrightarrow$	$I_C = \beta I_B$
$I_D = I_S$	$\Leftrightarrow$	$I_C \cong I_E$
$I_G \cong 0\ \text{A}$	$\Leftrightarrow$	$V_{BE} \cong 0.7\ \text{V}$

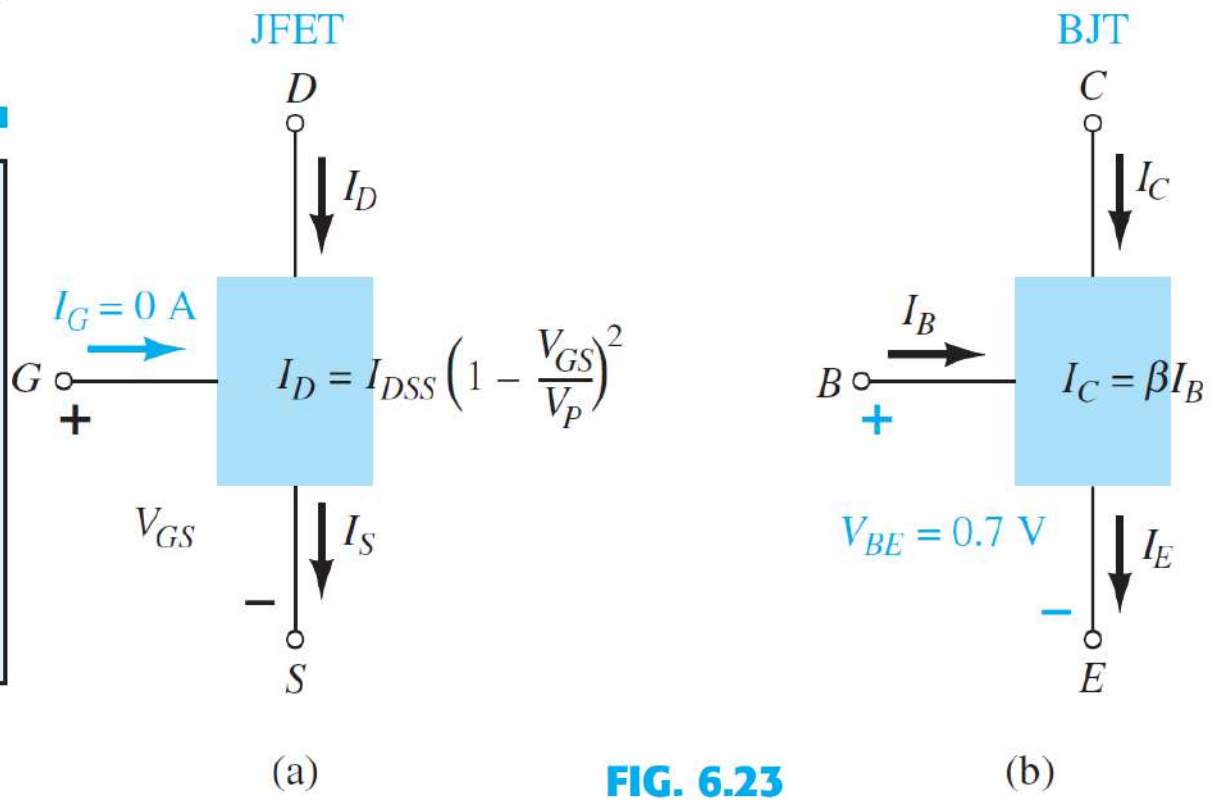
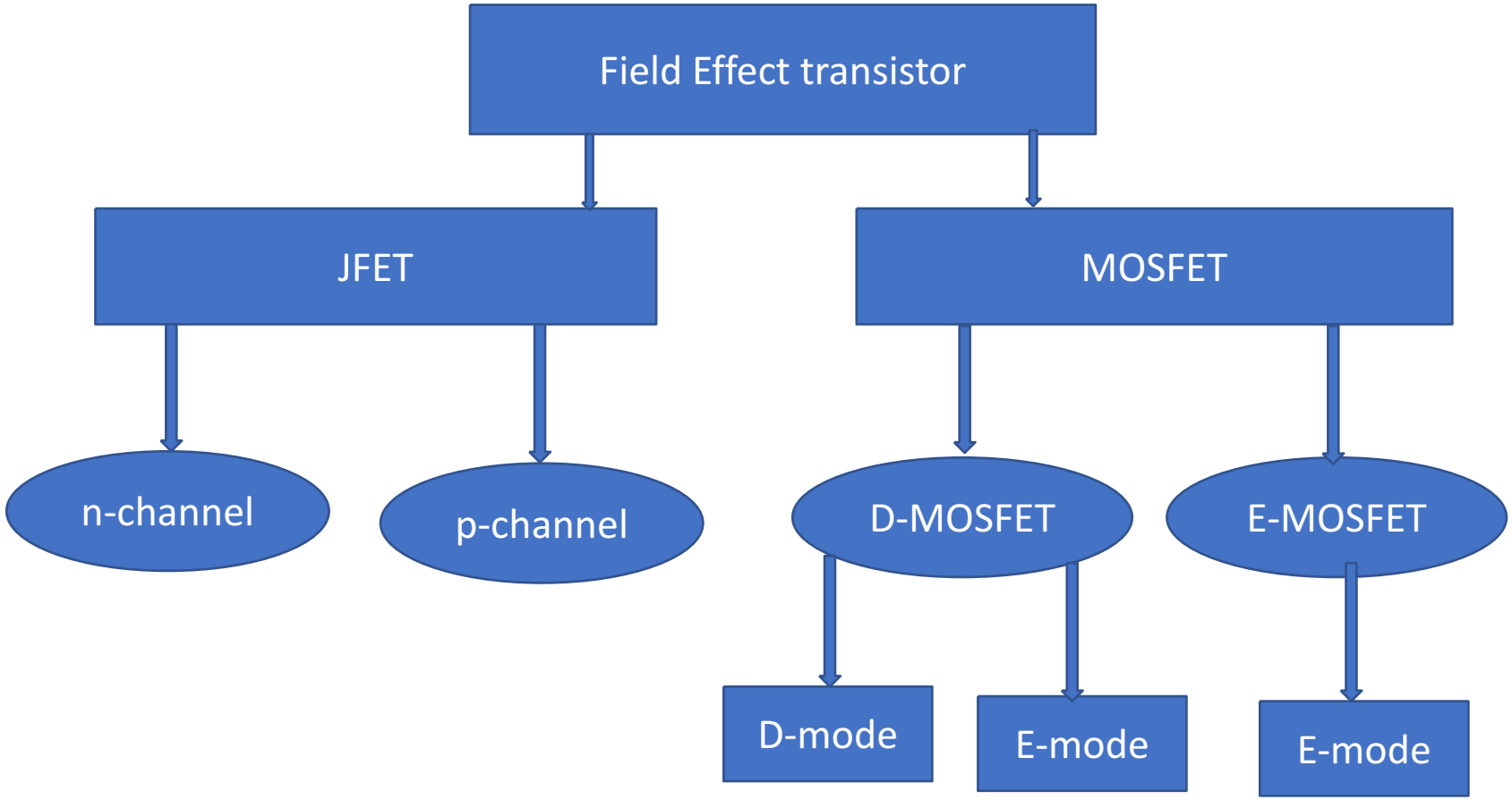


FIG. 6.23
(a) JFET versus (b) BJT.

Just as there are *npn* and *pnp* bipolar transistors, there are *n-channel* and *p-channel* field-effect transistors. However, it is important to keep in mind that the BJT transistor is a *bipolar* device—the prefix *bi* indicates that the conduction level is a function of two charge carriers, electrons and holes. The FET is a *unipolar* device depending solely on either electron (*n*-channel) or hole (*p*-channel) conduction.


Field Effect transistor
JFET
MOSFET
n-channel
p-channel
D-MOSFET
E-MOSFET
D-mode
E-mode
E-mode

6.7 DEPLETION-TYPE MOSFET

As noted in the introduction, there are three types of FETs: JFETs, MOSFETs, and MESFETs. MOSFETs are further broken down into *depletion type* and *enhancement type*. The terms *depletion* and *enhancement* define their basic mode of operation; the name MOSFET stands for *m*etal–*o*xide–*s*emiconductor *f*ield-*e*ffect *t*ransistor. Since there are differences in the characteristics and operation of different types of MOSFET, they are covered in separate sections. In this section we examine the depletion-type MOSFET, which has characteristics similar to those of a JFET between cutoff and saturation at I_{DSS}, and also has the added feature of characteristics that extend into the region of opposite polarity for V_{GS}.

Basic Construction

The basic construction of the *n*-channel depletion-type MOSFET is provided in Fig. 6.24. A slab of *p*-type material is formed from a silicon base and is referred to as the *substrate*. It is the foundation on which the device is constructed. In some cases the substrate is internally connected to the source terminal. However, many discrete devices provide an additional terminal labeled *SS*, resulting in a four-terminal device, such as that in Fig. 6.24. The

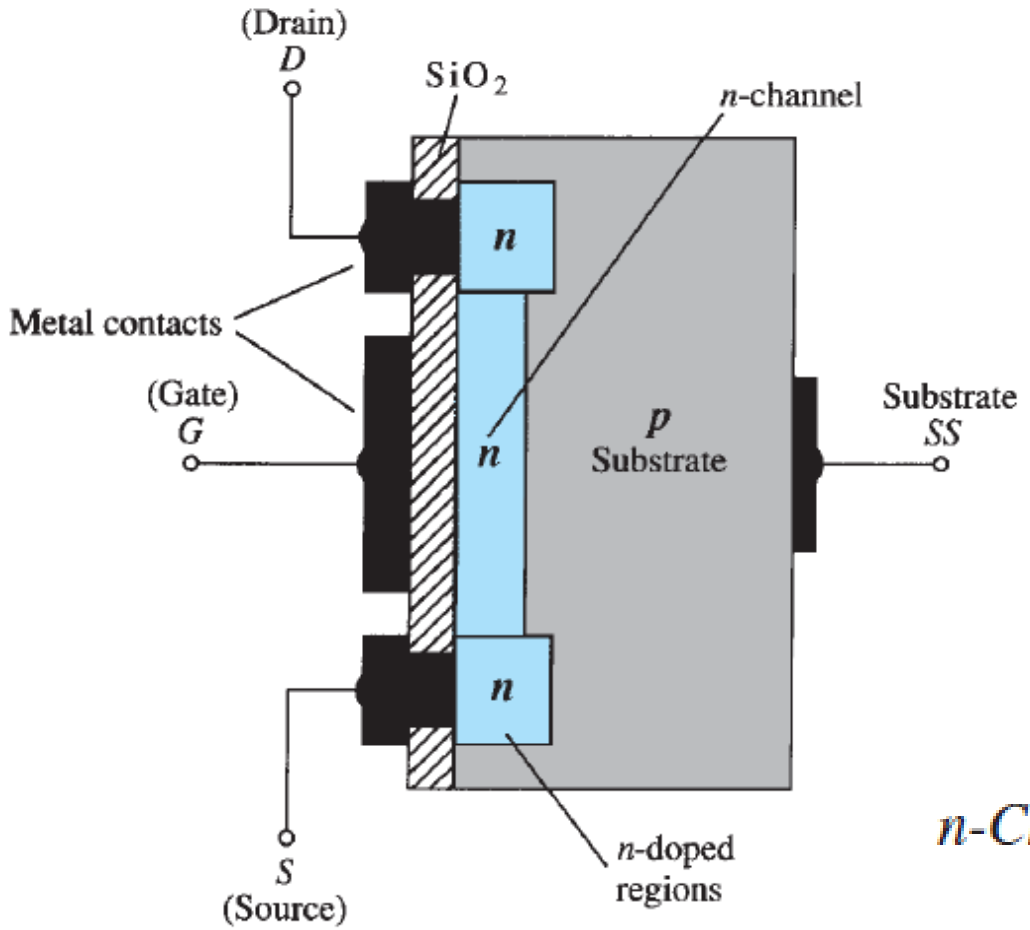
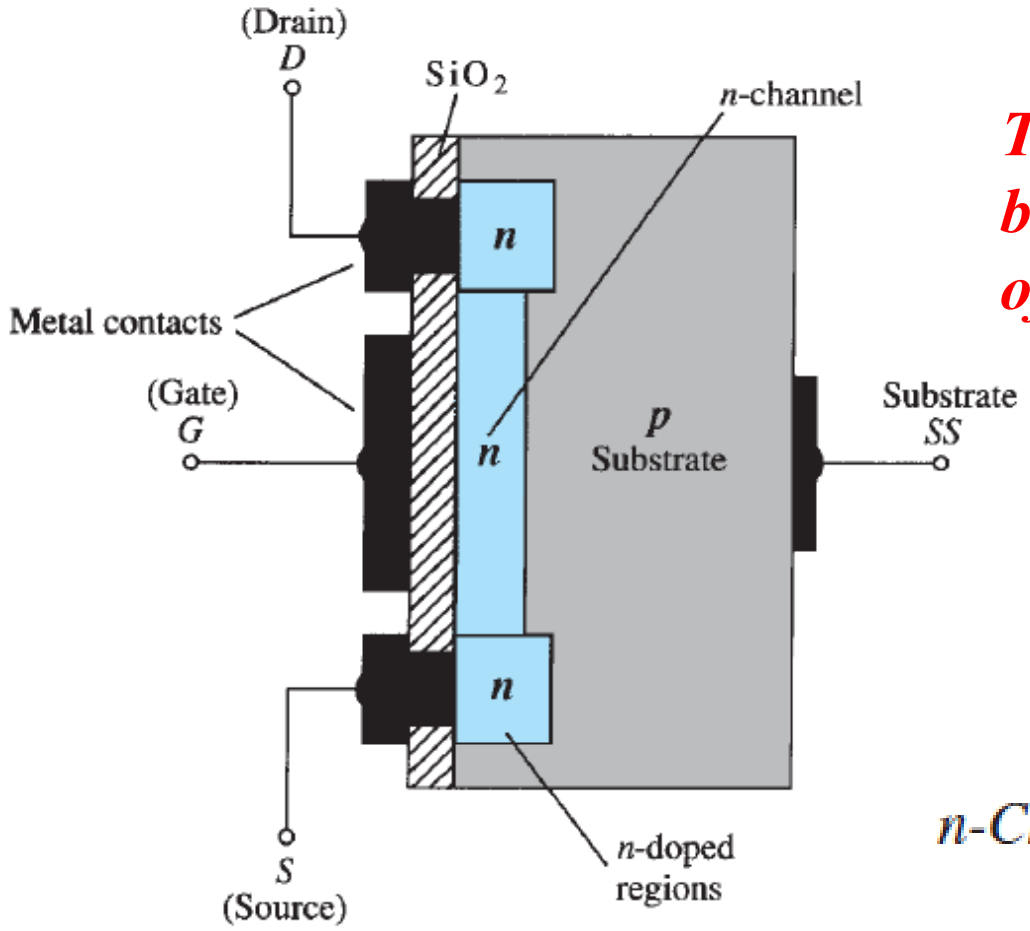


FIG. 6.24
n-Channel depletion-type MOSFET.

The source and drain terminals are connected through metallic contacts to *n*-doped regions linked by an *n*-channel as shown in the figure. The gate is also connected to a metal contact surface but remains insulated from the *n*-channel by a very thin silicon dioxide (SiO_2) layer. SiO_2 is a type of insulator referred to as a *dielectric*, which sets up opposing (as indicated by the prefix *di-*) electric fields within the dielectric when exposed to an externally applied field. The fact that the SiO_2 layer is an insulating layer means that:



There is no direct electrical connection between the gate terminal and the channel of a MOSFET

FIG. 6.24
n-Channel depletion-type MOSFET.

In addition:

It is the insulating layer of SiO_2 in the MOSFET construction that accounts for the very desirable high input impedance of the device.

In fact, the input resistance of a MOSFET is usually more than that of a typical JFET, even though the input impedance of most JFETs is sufficiently high for most applications. Because of the very high input impedance, the gate current I_G is essentially 0 A for dc-biased configurations.

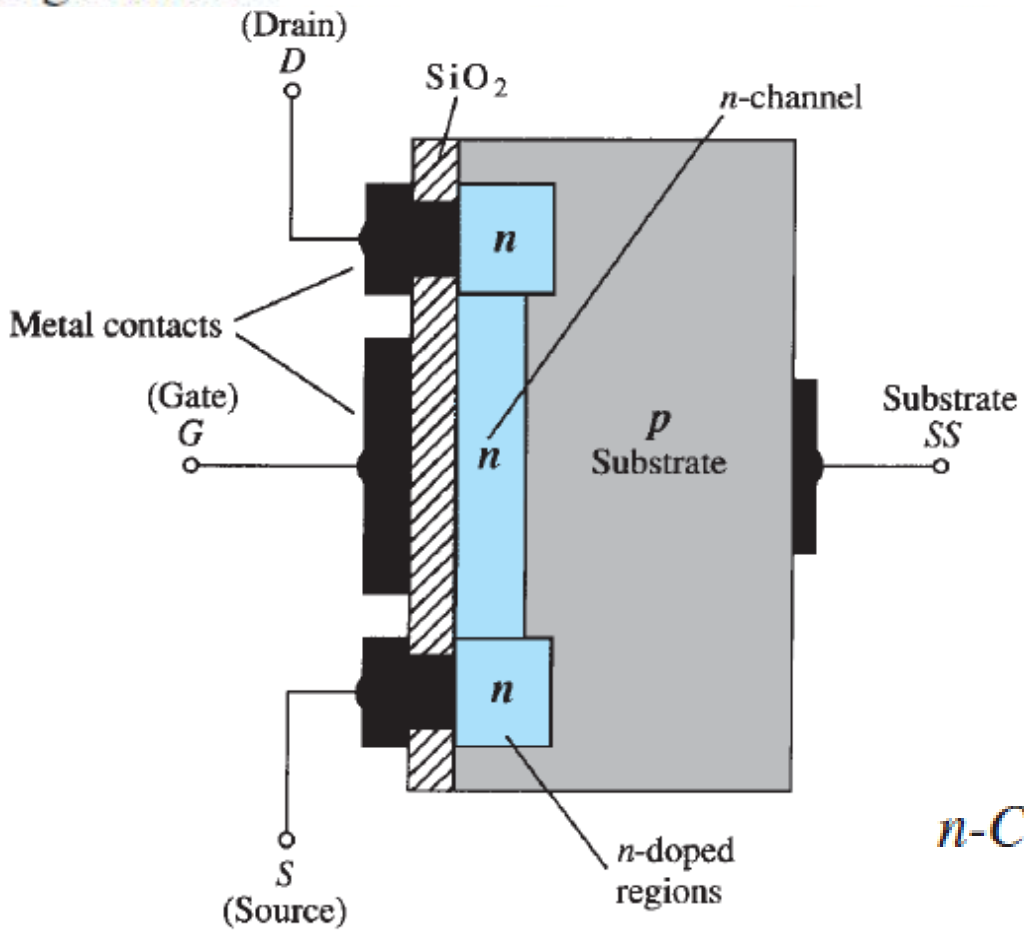


FIG. 6.24
n-Channel depletion-type MOSFET.

The reason for the label metal–oxide–semiconductor FET is now fairly obvious: *metal* for the drain, source, and gate connections; *oxide* for the silicon dioxide insulating layer; and *semiconductor* for the basic structure on which the *n*- and *p*-type regions are diffused. The insulating layer between the gate and the channel has resulted in another name for the device: *insulated-gate FET*, or *IGFET*, although this label is used less and less in the literature.

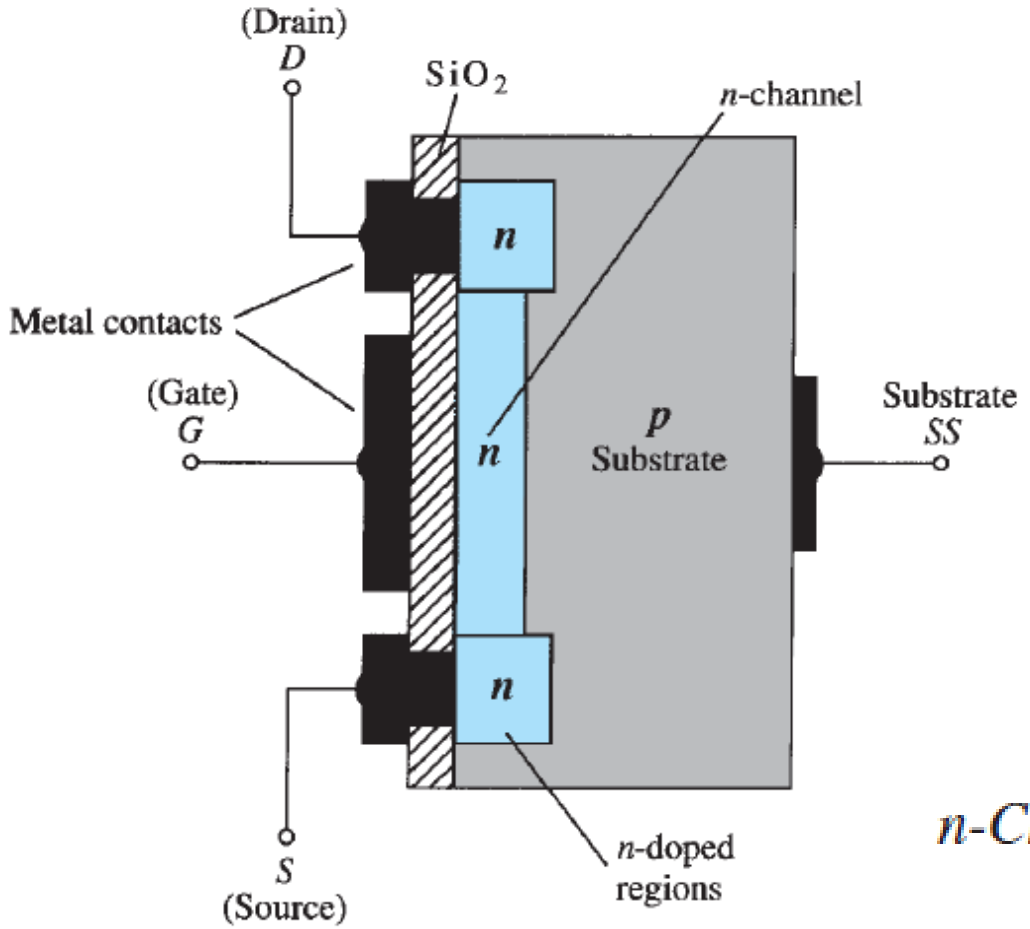


FIG. 6.24
n-Channel depletion-type MOSFET.

Basic Operation and Characteristics

In Fig. 6.25 the gate-to-source voltage is set to 0 V by the direct connection from one terminal to the other, and a voltage V_{DD} is applied across the drain-to-source terminals. The result is an attraction of the free electrons of the *n*-channel for the positive voltage at the drain. The result is a current similar to that flowing in the channel of the JFET. In fact, the resulting current with $V_{GS} = 0$ V continues to be labeled I_{DSS}, as shown in Fig. 6.26.

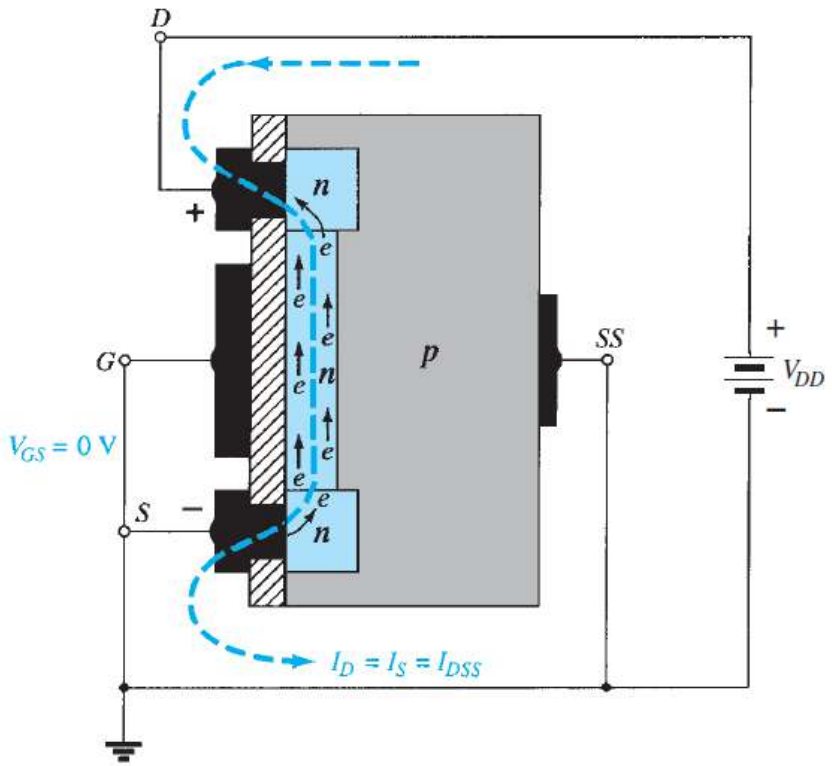


FIG. 6.25

n-Channel depletion-type MOSFET with $V_{GS} = 0$ V and applied voltage V_{DD}.

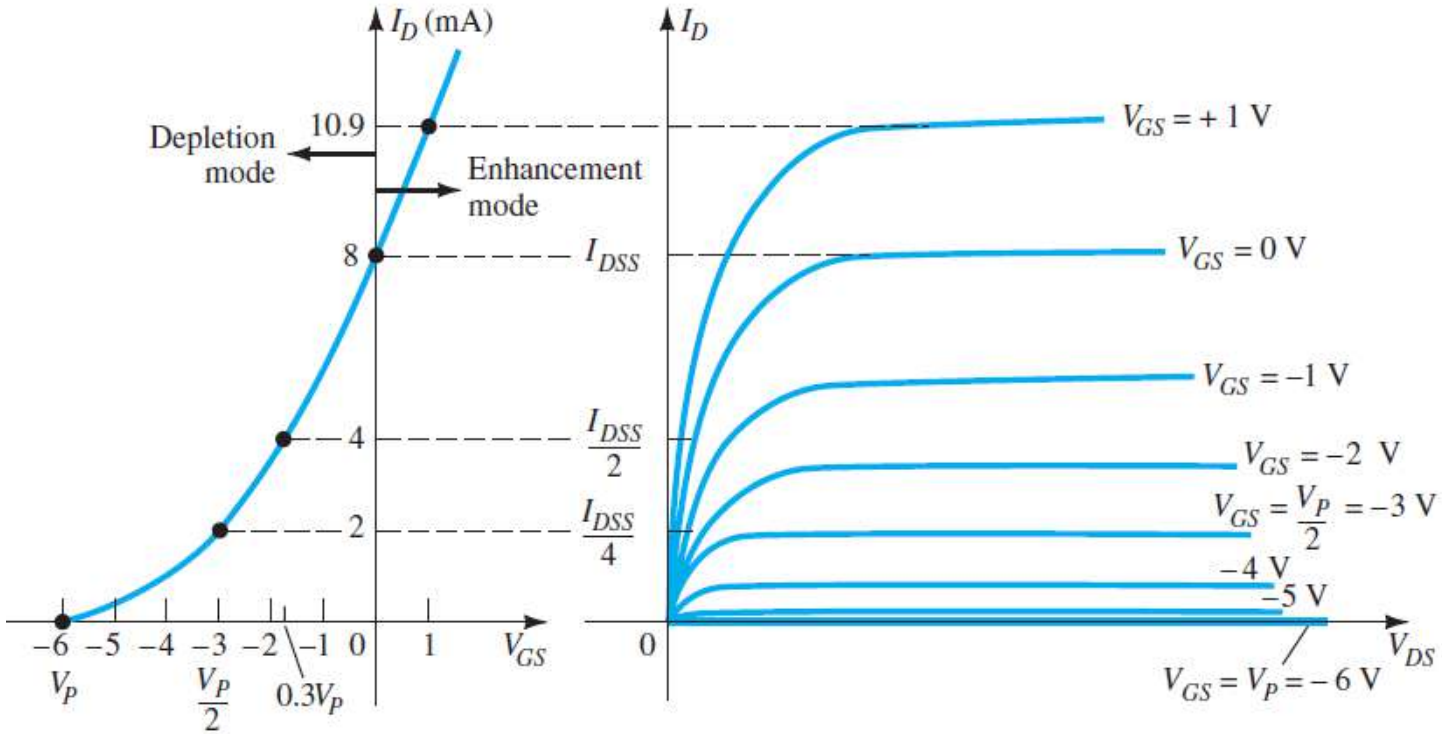


FIG. 6.26

Drain and transfer characteristics for an n-channel depletion-type MOSFET.

In Fig. 6.27, V_{GS} is set at a negative voltage such as −1 V. The negative potential at the gate will tend to pressure electrons toward the *p*-type substrate (like charges repel) and attract holes from the *p*-type substrate (opposite charges attract) as shown in Fig. 6.27.

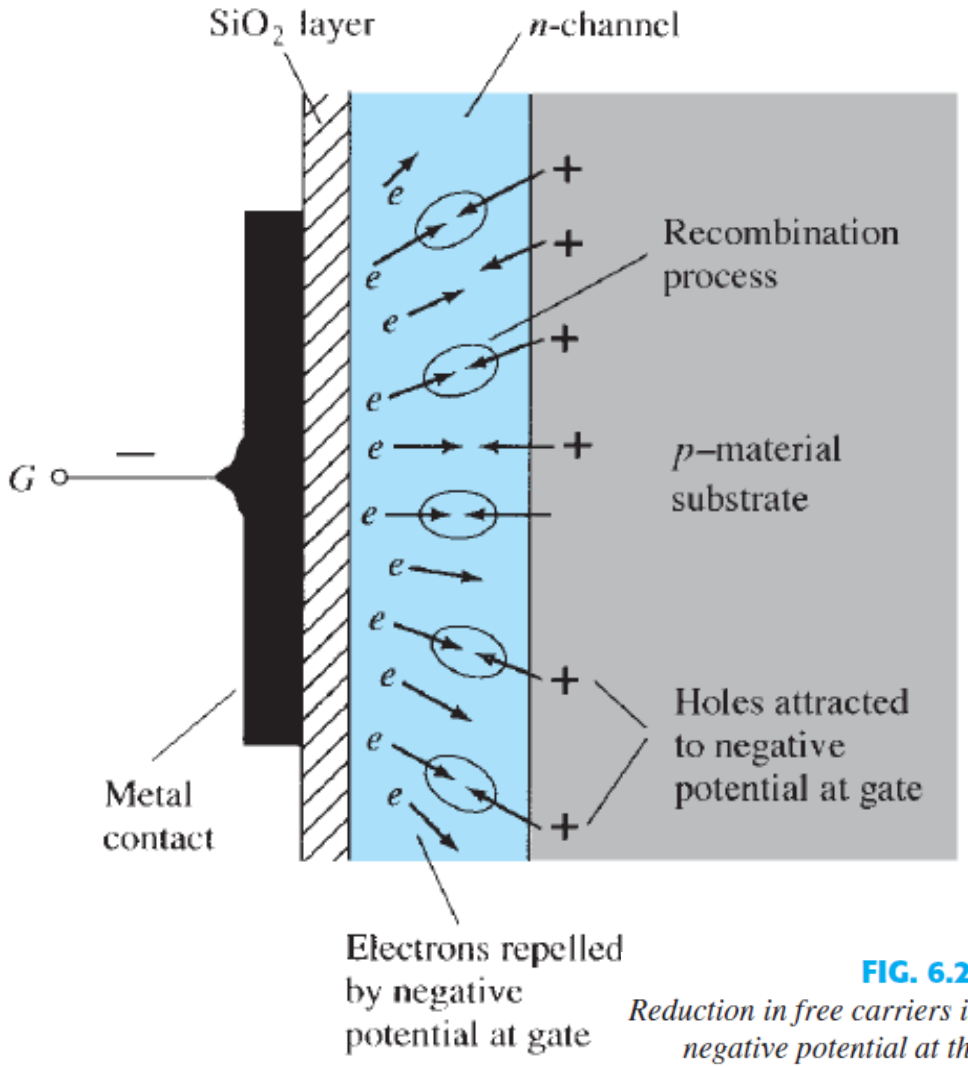


FIG. 6.27
Reduction in free carriers in a channel due to a negative potential at the gate terminal.

Depending on the magnitude of the negative bias established by V_{GS}, a level of recombination between electrons and holes will occur that will reduce the number of free electrons in the *n*-channel available for conduction. The more negative the bias, the higher is the rate of recombination.

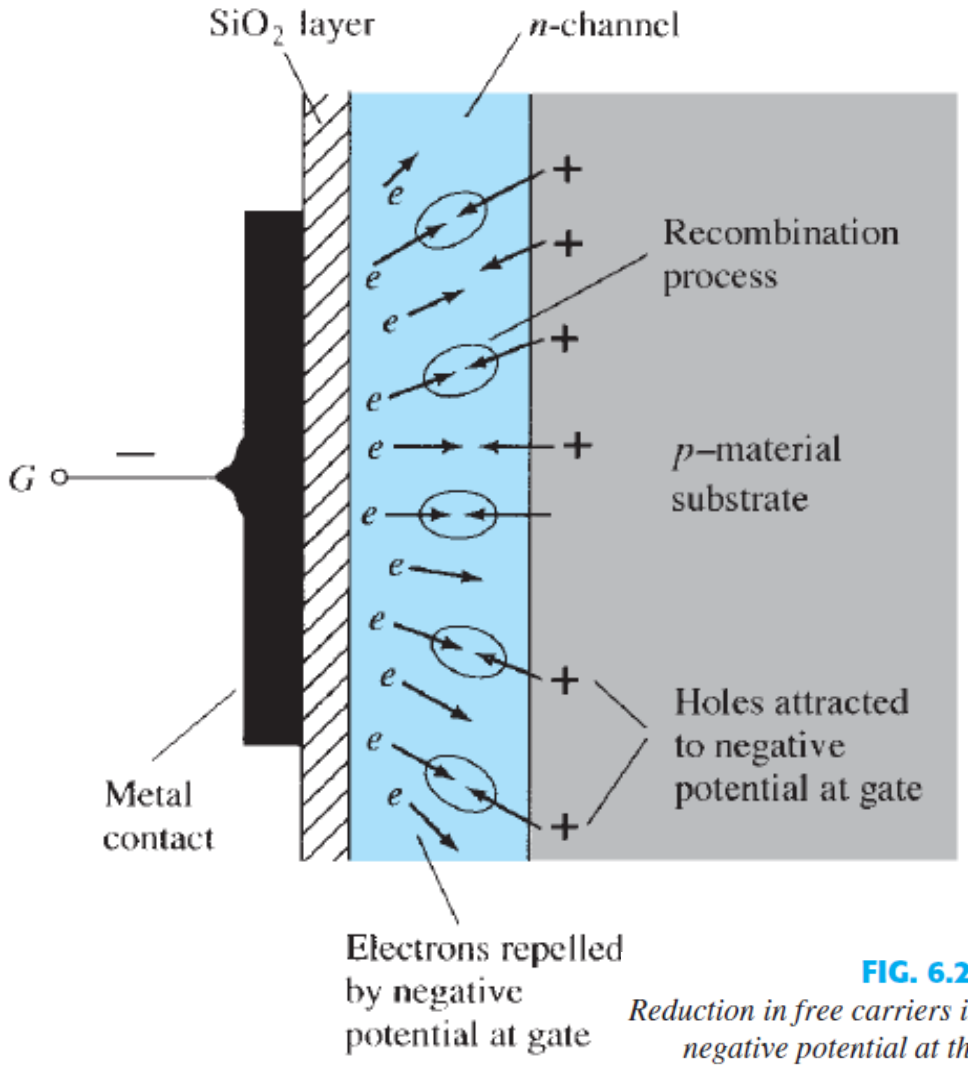


FIG. 6.27
Reduction in free carriers in a channel due to a negative potential at the gate terminal.

The resulting level of drain current is therefore reduced with increasing negative bias for V_{GS}, as shown in Fig. 6.26 for $V_{GS} = -1$ V, -2 V, and so on, to the pinch-off level of -6 V. The resulting levels of drain current and the plotting of the transfer curve proceed exactly as described for the JFET.

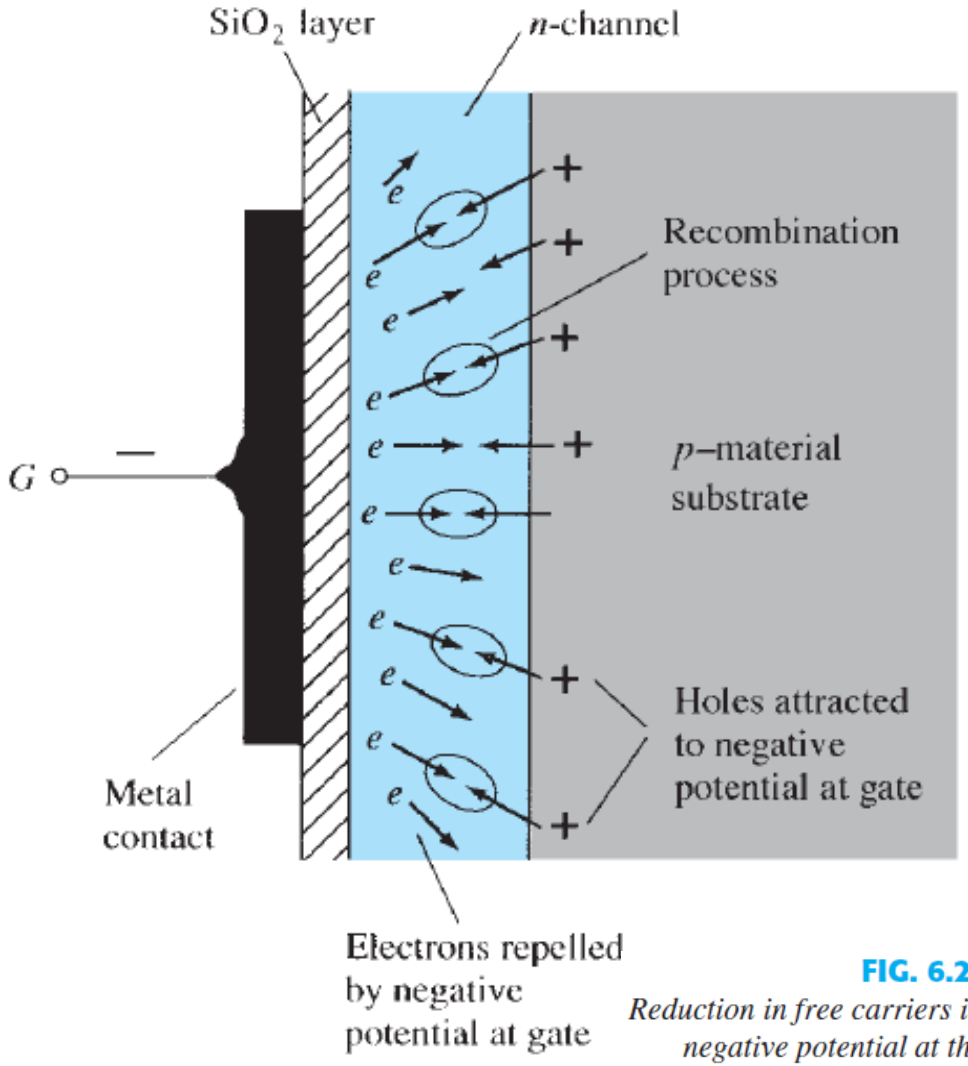


FIG. 6.27
Reduction in free carriers in a channel due to a negative potential at the gate terminal.

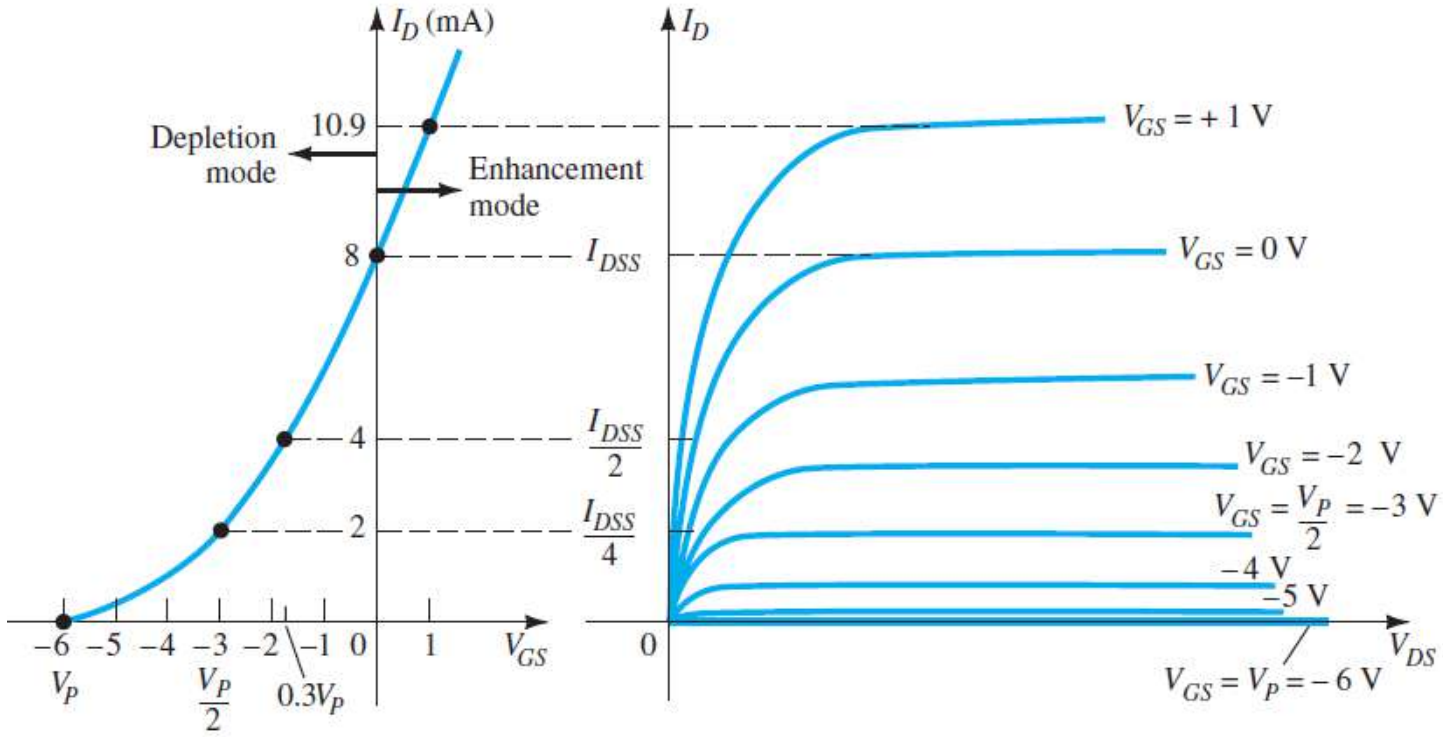


FIG. 6.26
Drain and transfer characteristics for an n-channel depletion-type MOSFET.

For positive values of V_{GS}, the positive gate will draw additional electrons (free carriers) from the *p*-type substrate due to the reverse leakage current and establish new carriers through the collisions resulting between accelerating particles. As the gate-to-source voltage continues to increase in the positive direction, Fig. 6.26 reveals that the drain current will increase at a rapid rate for the reasons listed above. The vertical spacing between the $V_{GS} = 0$ V and $V_{GS} = +1$ V curves of Fig. 6.26 is a clear indication of how much the current has increased for the 1-V change in V_{GS}. Due to the rapid rise, the user must be aware of the maximum drain current rating since it could be exceeded with a positive gate voltage.

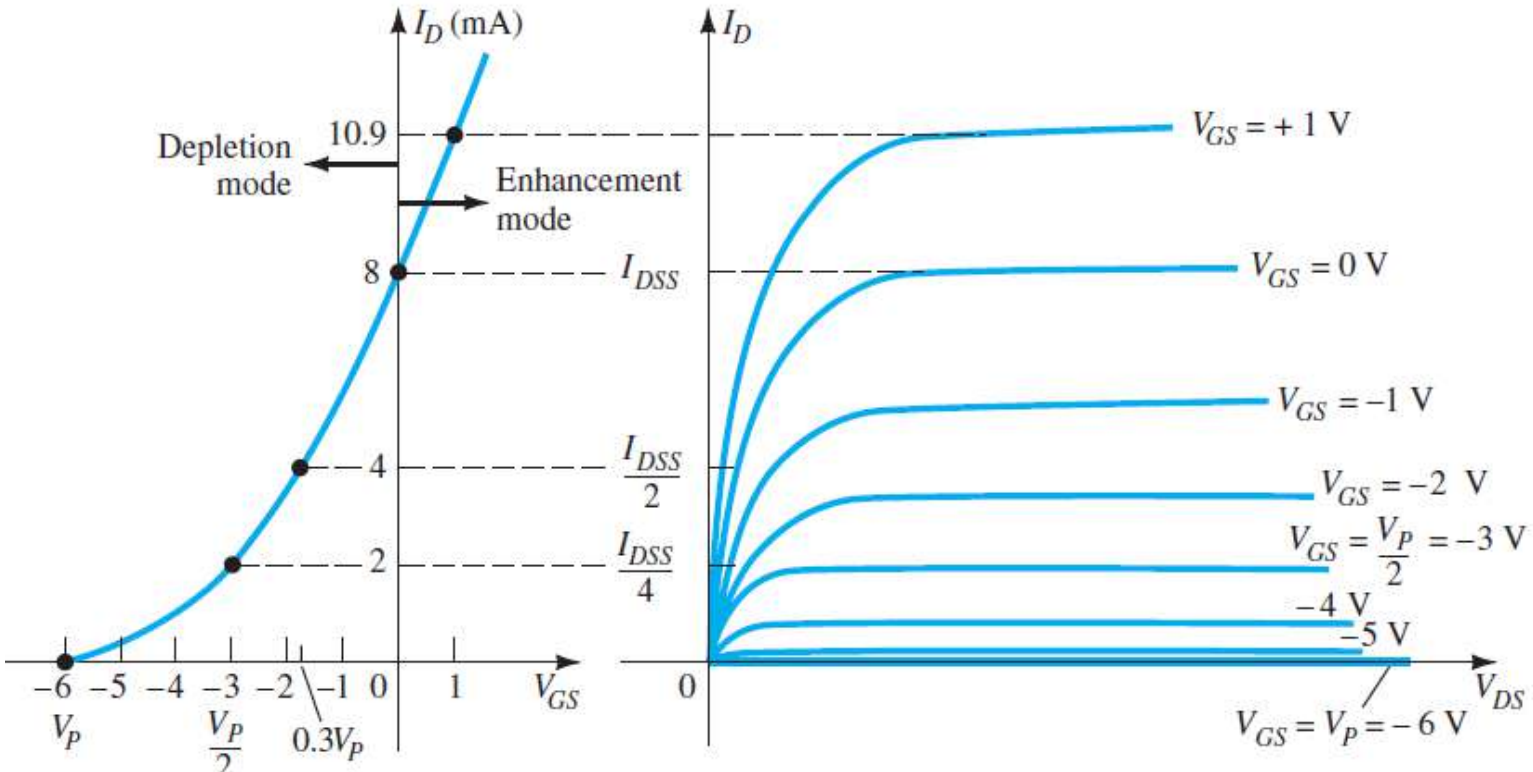


FIG. 6.26
Drain and transfer characteristics for an n-channel depletion-type MOSFET.

That is, for the device of Fig. 6.26, the application of a voltage $V_{GS} = +4$ V would result in a drain current of 22.2 mA, which could possibly exceed the maximum rating (current or power) for the device. As revealed above, the application of a positive gate-to-source voltage has "enhanced" the level of free carriers in the channel compared to that encountered with $V_{GS} = 0$ V. For this reason the region of positive gate voltages on the drain or transfer characteristics is often referred to as the *enhancement region*, with the region between cutoff and the saturation level of I_{DSS} referred to as the *depletion region*.

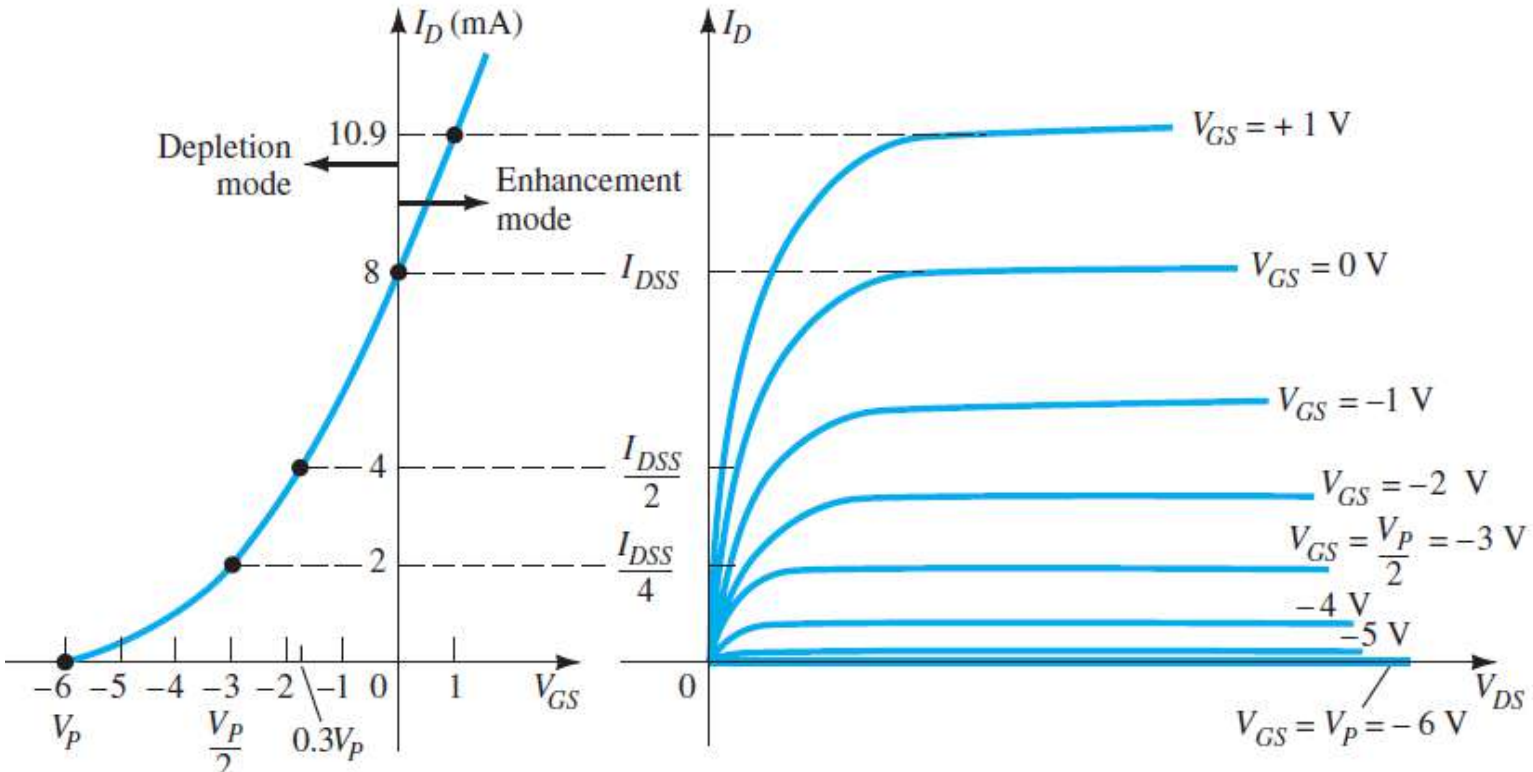
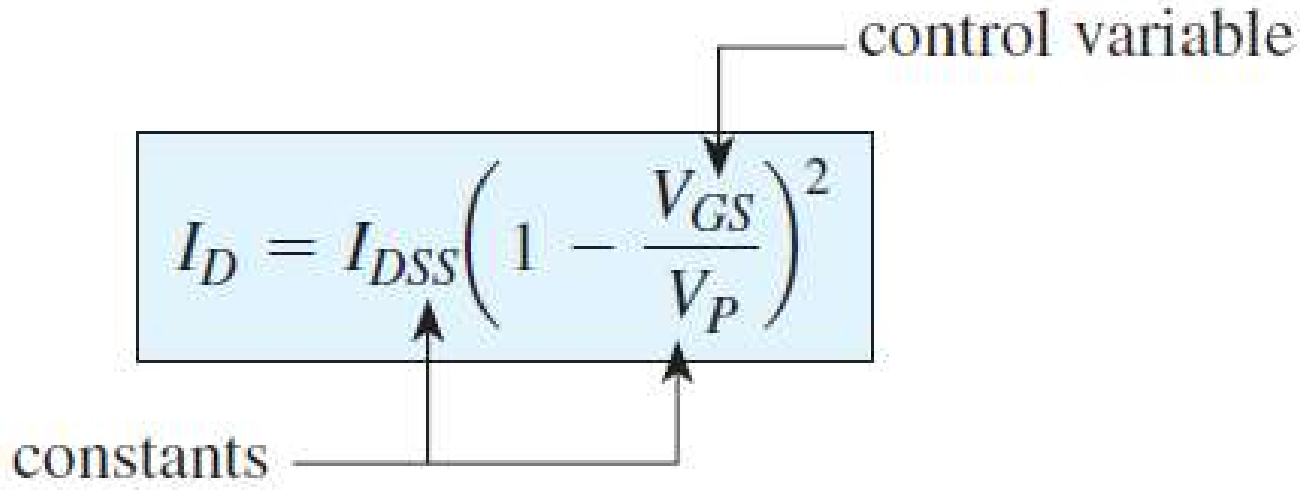


FIG. 6.26
Drain and transfer characteristics for an n-channel depletion-type MOSFET.

It is particularly interesting and helpful that Shockley's equation will continue to be applicable for the depletion-type MOSFET characteristics in both the depletion and enhancement regions. For both regions, it is simply necessary that the proper sign be included with V_{GS} in the equation and the sign be carefully monitored in the mathematical operations.



(6.3)

Important note : For Enhancement mode I_D more than d I_{DSS} ,

EXAMPLE 6.3 Sketch the transfer characteristics for an *n*-channel depletion-type MOSFET with $I_{DSS} = 10$ mA and $V_P = -4$ V.

Solution:

$$\text{At } V_{GS} = 0 \text{ V}, \quad I_D = I_{DSS} = 10 \text{ mA}$$

$$V_{GS} = V_P = -4 \text{ V}, \quad I_D = 0 \text{ mA}$$

$$V_{GS} = \frac{V_P}{2} = \frac{-4 \text{ V}}{2} = -2 \text{ V}, \quad I_D = \frac{I_{DSS}}{4} = \frac{10 \text{ mA}}{4} = 2.5 \text{ mA}$$

and at $I_D = \dfrac{I_{DSS}}{2}$,

$$V_{GS} = 0.3V_P = 0.3(-4 \text{ V}) = -1.2 \text{ V}$$

all of which appear in Fig. 6.28.

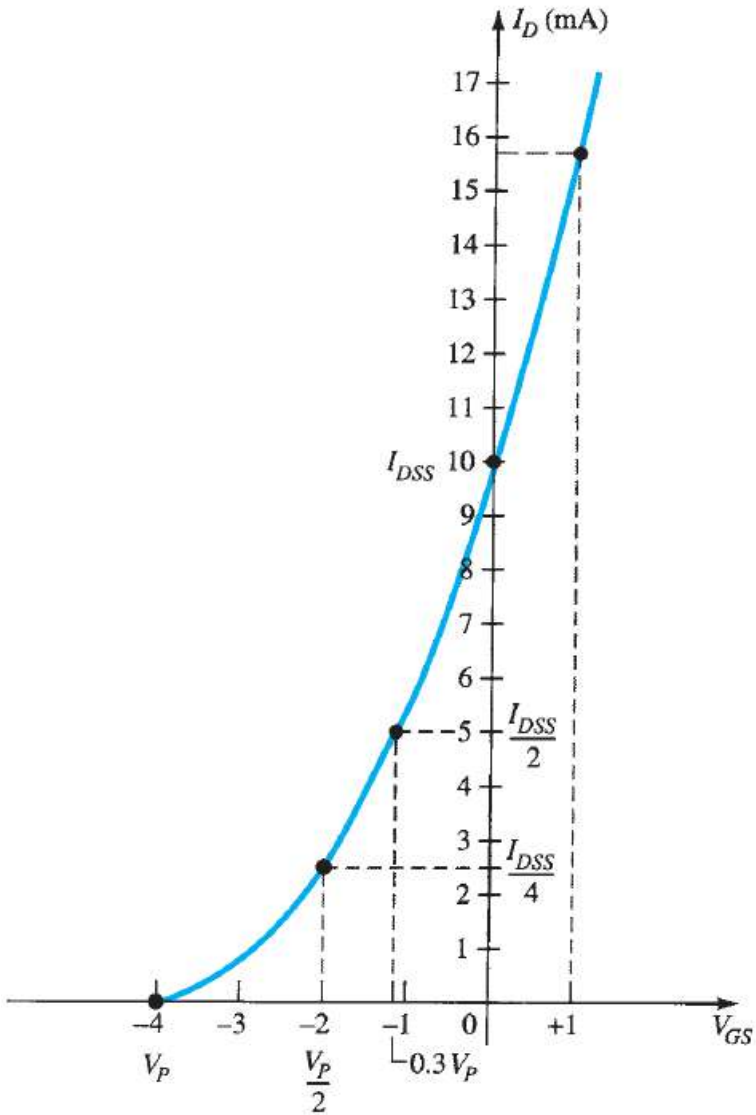


FIG. 6.28
Transfer characteristics for an n-channel depletion-type MOSFET with $I_{DSS} = 10$ mA and $V_P = -4$ V.

Before plotting the positive region of V_{GS}, keep in mind that I_D increases very rapidly with increasing positive values of V_{GS}. In other words, be conservative with the choice of values to be substituted into Shockley's equation. In this case, we try +1 V as follows:

$$I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_P}\right)^2$$

$$= (10\text{ mA})\left(1 - \frac{+1\text{ V}}{-4\text{ V}}\right)^2 = (10\text{ mA})(1 + 0.25)^2 = (10\text{ mA})(1.5625)$$

$$\cong 15.63\text{ mA}$$

which is sufficiently high to finish the plot.

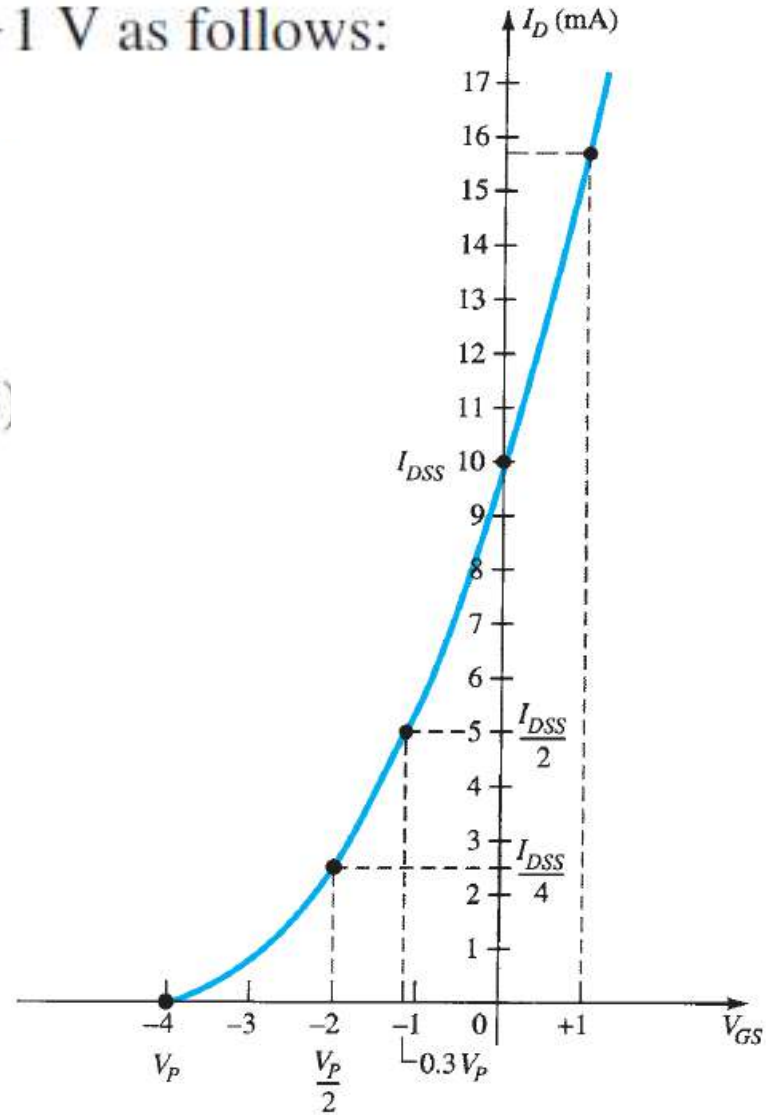


FIG. 6.28

Transfer characteristics for an n-channel depletion-type MOSFET with $I_{DSS} = 10$ mA and $V_P = -4$ V.

p-Channel Depletion-Type MOSFET

The construction of a *p*-channel depletion-type MOSFET is exactly the reverse of that appearing in Fig. 6.24. That is, there is now an *n*-type substrate and a *p*-type channel, as shown in Fig. 6.29a. The terminals remain as identified, but all the voltage polarities and the current directions are reversed, as shown in the same figure. The drain characteristics would appear exactly as in Fig. 6.26, but with V_{DS} having negative values, I_D having positive values as indicated (since the defined direction is now reversed), and V_{GS} having the opposite polarities as shown in Fig. 6.29c. The reversal in V_{GS} will result in a mirror image (about the I_D axis) for the transfer characteristics as shown in Fig. 6.29b. In other words,

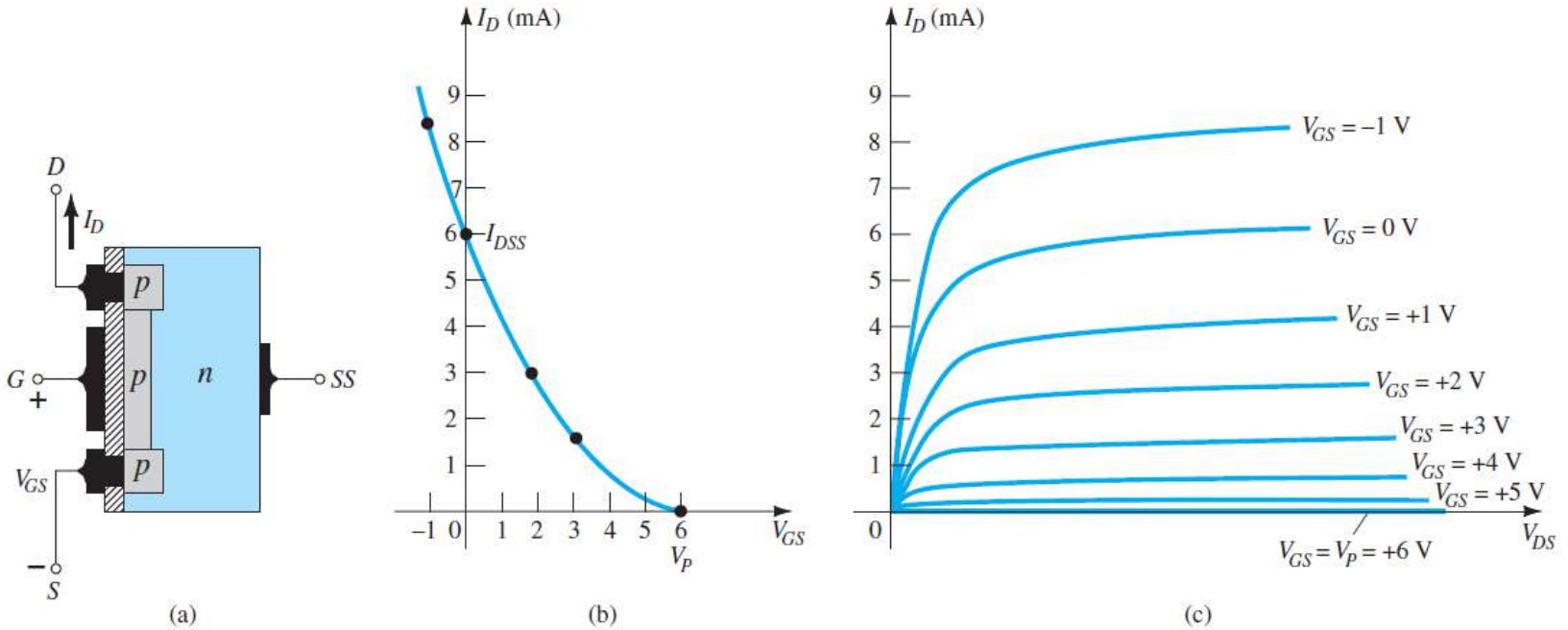


FIG. 6.29
p-Channel depletion-type MOSFET with $I_{DSS} = 6$ mA and $V_P = +6$ V.

p-Channel Depletion-Type MOSFET

the drain current will increase from cutoff at $V_{GS} = V_P$ in the positive V_{GS} region to I_{DSS} and then continue to increase for increasingly negative values of V_{GS}. Shockley's equation is still applicable and requires simply placing the correct sign for both V_{GS} and V_P in the equation.

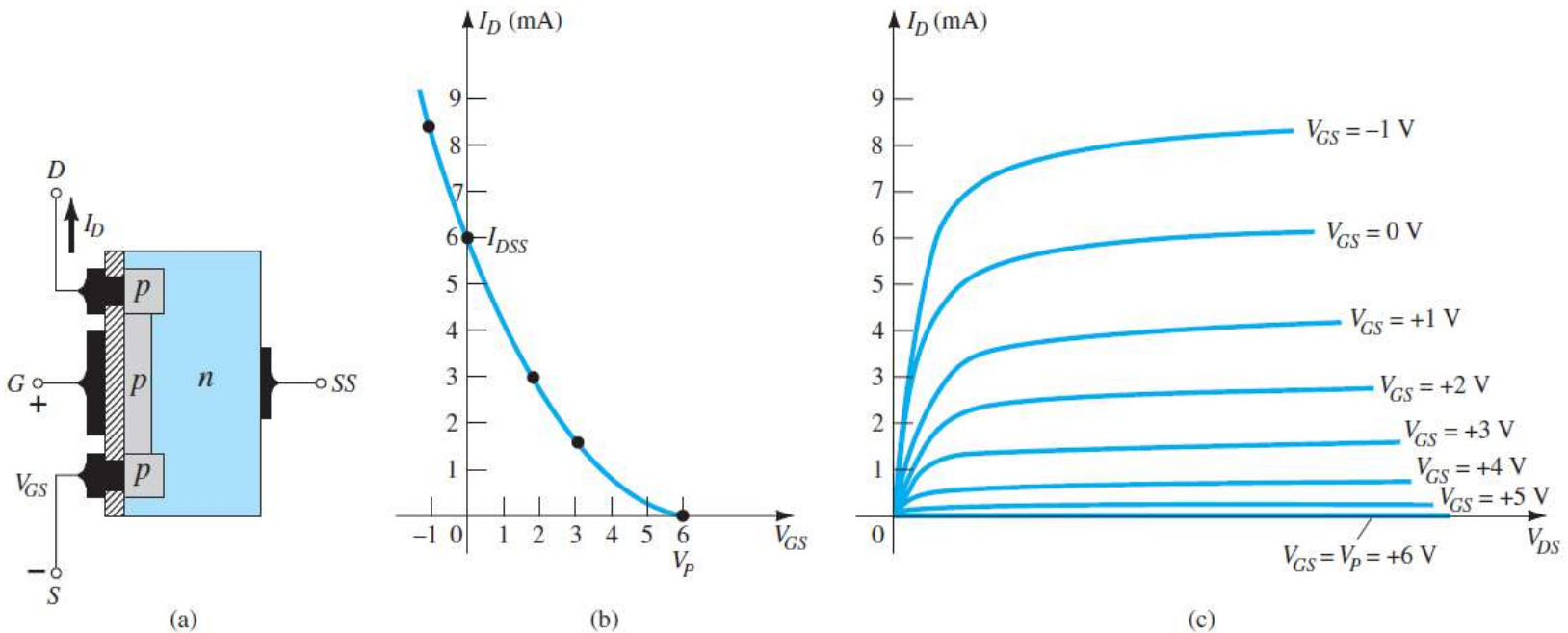


FIG. 6.29
p-Channel depletion-type MOSFET with $I_{DSS} = 6$ mA and $V_P = +6$ V.

Symbols, Specification Sheets, and Case Construction

The graphic symbols for an *n*- and *p*-channel depletion-type MOSFET are provided in Fig. 6.30. Note how the symbols chosen try to reflect the actual construction of the device. The lack of a direct connection (due to the gate insulation) between the gate and the channel is represented by a space between the gate and the other terminals of the symbol. The vertical line representing the channel is connected between the drain and the source and is "supported" by the substrate. Two symbols are provided for each type of channel to reflect the fact that in some cases the substrate is externally available, whereas in others it is not. For most of the analysis to follow in Chapter 7, the substrate and the source will be connected and the lower symbols will be employed.

FIG. 6.30
Graphic symbols for: (a) n-channel depletion-type MOSFETs and (b) p-channel depletion-type MOSFETs.

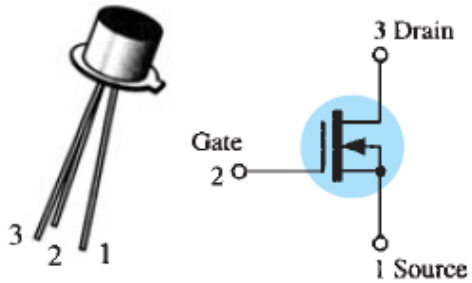
The device appearing in Fig. 6.31 has three terminals, with the terminal identification appearing in the same figure. The specification sheet for a depletion-type MOSFET is similar to that of a JFET. The levels of V_P and I_{DSS} are provided along with a list of maximum values and typical "on" and "off" characteristics. In addition, however, since I_D can extend beyond the I_{DSS} level, another point is normally provided that reflects a typical value of I_D for some positive voltage (for an *n*-channel device). For the unit of Fig. 6.31, I_D is specified as $I_{D(\text{on})} = 9$ mA dc, with $V_{DS} = 10$ V and $V_{GS} = 3.5$ V.

MAXIMUM RATINGS

Rating	Symbol	Value	Unit
Drain–Source Voltage 2N3797	V_{DS}	20	Vdc
Gate–Source Voltage	V_{GS}	±10	Vdc
Drain Current	I_D	20	mAdc
Total Device Dissipation @ $T_A = 25°C$ Derate above 25°C	P_D	200 1.14	mW mW/°C
Junction Temperature Range	T_J	+175	°C
Storage Channel Temperature Range	T_{stg}	–65 to +200	°C

2N3797

MOSFET
LOW–POWER AUDIO



N-CHANNEL – DEPLETION

FIG. 6.31
2N3797 Motorola n-channel depletion-type MOSFET.

ELECTRICAL CHARACTERISTICS (T_A = 25°C unless otherwise noted)

Characteristic		Symbol	Min	Typ	Max	Unit
OFF CHARACTERISTICS						
Drain Source Breakdown Voltage (V_{GS} = –7.0 V, I_D = 5.0 μA)	2N3797	$V_{(BR)DSX}$	20	25	–	Vdc
Gate Reverse Current (1) (V_{GS} = –10 V, V_{DS} = 0) (V_{GS} = –10 V, V_{DS} = 0, T_A = 150°C)		I_{GSS}	– –	– –	1.0 200	pAdc
Gate Source Cutoff Voltage (I_D = 2.0 μA, V_{DS} = 10 V)	2N3797	$V_{GS(off)}$	–	–5.0	–7.0	Vdc
Drain-Gate Reverse Current (1) (V_{DG} = 10 V, I_S = 0)		I_{DGO}	–	–	1.0	pAdc
ON CHARACTERISTICS						
Zero-Gate-Voltage Drain Current (V_{DS} = 10 V, V_{GS} = 0)	2N3797	I_{DSS}	2.0	2.9	6.0	mAdc
On-State Drain Current (V_{DS} = 10 V, V_{GS} = +3.5 V)	2N3797	$I_{D(on)}$	9.0	14	18	mAdc

FIG. 6.31

2N3797 Motorola n-channel depletion-type MOSFET.

6.8 ENHANCEMENT-TYPE MOSFET

Although there are some similarities in construction and mode of operation between depletion-type and enhancement-type MOSFETs, the characteristics of the enhancement-type MOSFET are quite different from anything obtained thus far. The transfer curve is not defined by Shockley's equation, and the drain current is now cut off until the gate-to-source voltage reaches a specific magnitude. In particular, current control in an *n*-channel device is now effected by a positive gate-to-source voltage rather than the range of negative voltages encountered for *n*-channel JFETs and *n*-channel depletion-type MOSFETs.

Basic Construction

The basic construction of the *n*-channel enhancement-type MOSFET is provided in Fig. 6.32. A slab of *p*-type material is formed from a silicon base and is again referred to as the substrate. As with the depletion-type MOSFET, the substrate is sometimes internally connected to the source terminal, whereas in other cases a fourth lead (labeled SS) is made available for external control of its potential level. The source and drain terminals are again connected through metallic contacts to *n*-doped regions, but note in Fig. 6.32 the absence of a channel between the two *n*-doped regions

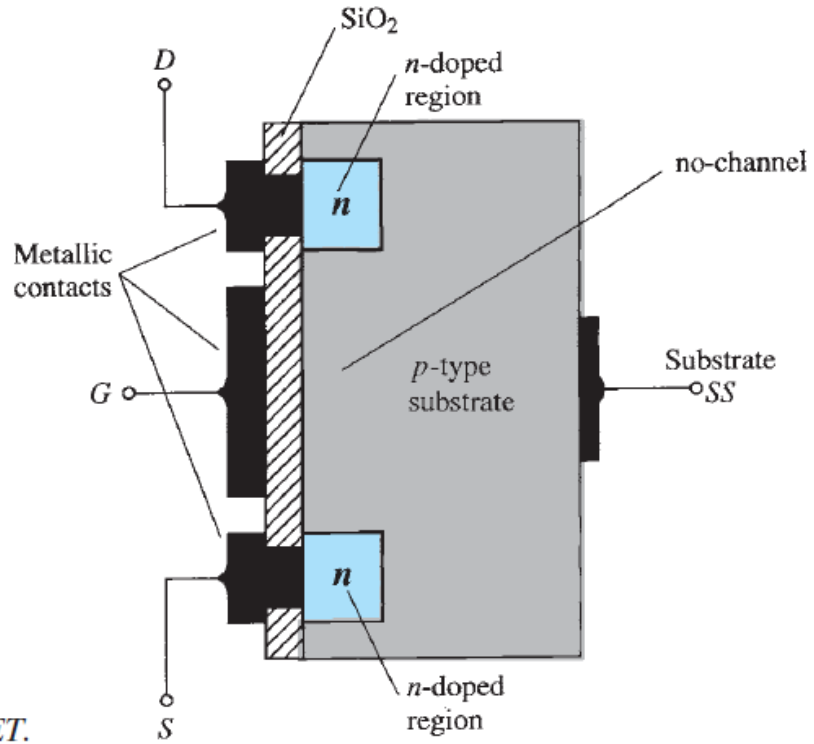


FIG. 6.32
n-Channel enhancement-type MOSFET.

Basic Construction

This is the primary difference between the construction of depletion-type and enhancement-type MOSFETs—the absence of a channel as a constructed component of the device. The SiO_2 layer is still present to isolate the gate metallic platform from the region between the drain and source, but now it is simply separated from a section of the *p*-type material. In summary, therefore, the construction of an enhancement-type MOSFET is quite similar to that of the depletion-type MOSFET, except for the absence of a channel between the drain and source terminals.

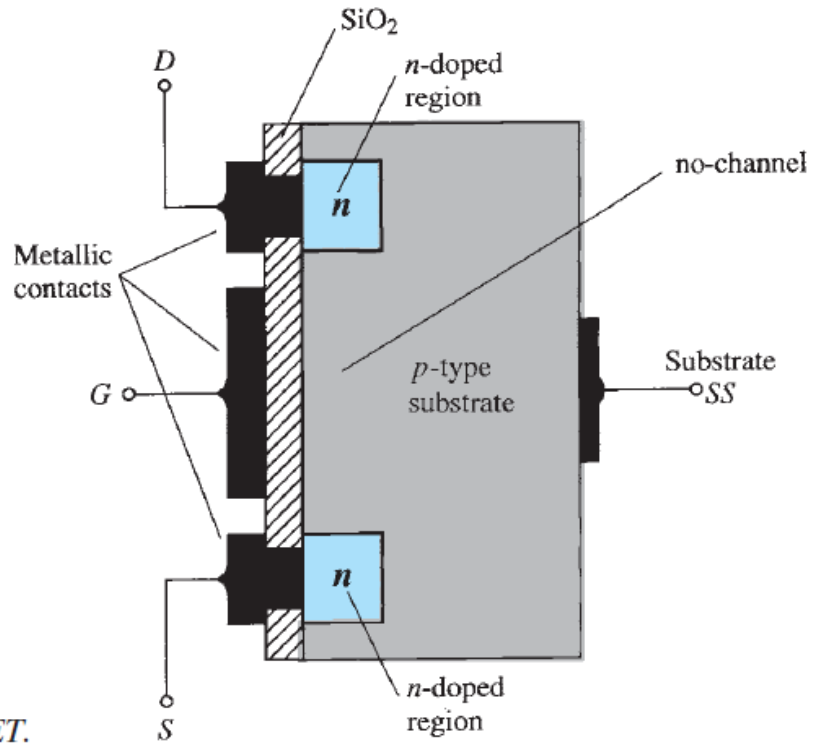


FIG. 6.32
n-Channel enhancement-type MOSFET.

Basic Operation and Characteristics

If V_{GS} is set at 0 V and a voltage applied between the drain and the source of the device of Fig. 6.32, the absence of an *n*-channel (with its generous number of free carriers) will result in a current of effectively 0 A—quite different from the depletion-type MOSFET and JFET, where $I_D = I_{DSS}$. It is not sufficient to have a large accumulation of carriers (electrons) at the drain and the source (due to the *n*-doped regions) if a path fails to exist between the two. With V_{DS} some positive voltage, V_{GS} at 0 V, and terminal *SS* directly connected to the source, there are in fact two reverse-biased *p*–*n* junctions between the *n*-doped regions and the *p*-substrate to oppose any significant flow between drain and source.

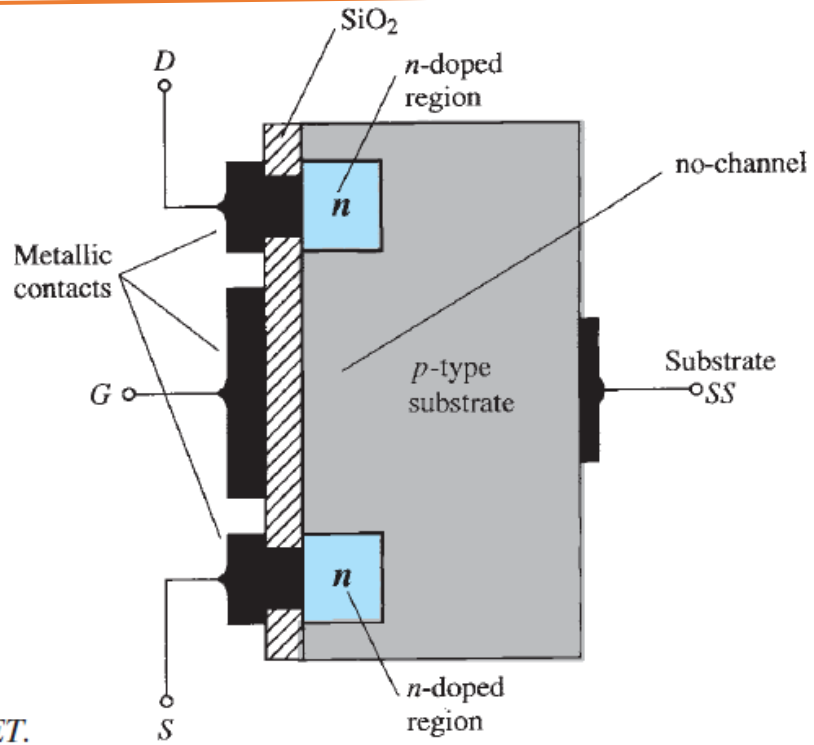


FIG. 6.32
n-Channel enhancement-type MOSFET.

Basic Operation and Characteristics

In Fig. 6.33, both V_{DS} and V_{GS} have been set at some positive voltage greater than 0 V, establishing the drain and the gate at a positive potential with respect to the source.

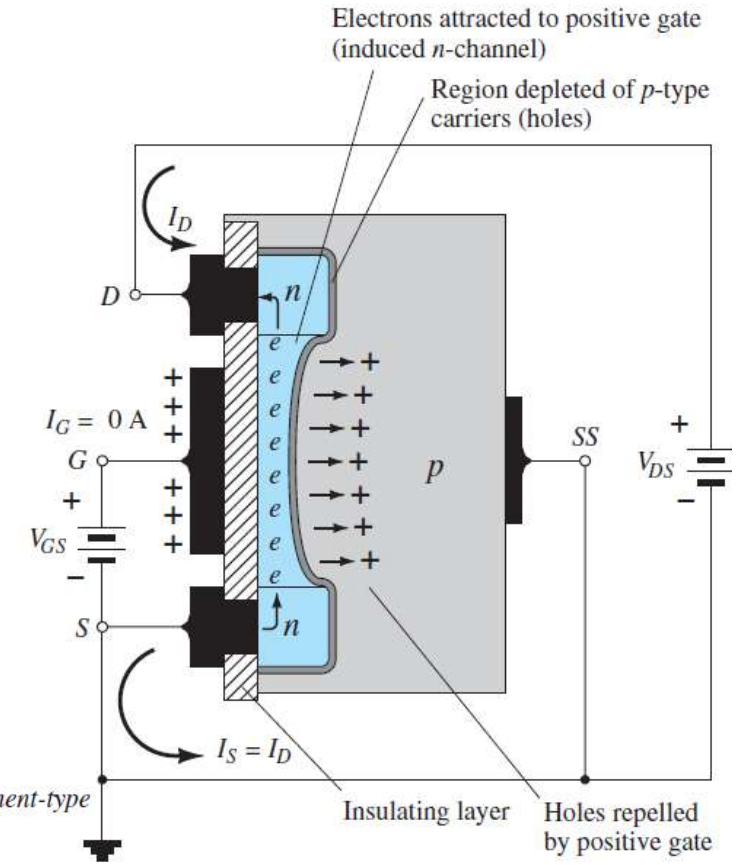


FIG. 6.33
Channel formation in the n-channel enhancement-type MOSFET.

The positive potential at the gate will pressure the holes (since like charges repel) in the *p*-substrate along the edge of the SiO_2 layer to leave the area and enter deeper regions of the *p*-substrate, as shown in the figure. The result is a depletion region near the SiO_2 insulating layer void of holes. However, the electrons in the *p*-substrate (the minority carriers of the material) will be attracted to the positive gate and accumulate in the region near the surface of the SiO_2 layer. The SiO_2 layer and its insulating qualities will prevent the negative carriers from being absorbed at the gate terminal.

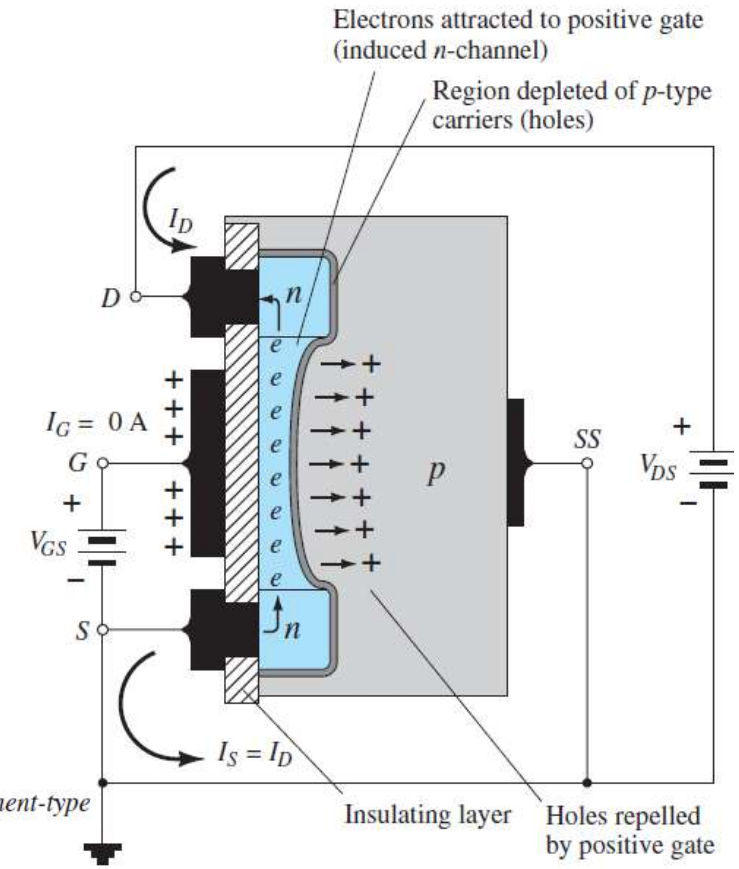


FIG. 6.33
Channel formation in the n-channel enhancement-type MOSFET.

As V_{GS} increases in magnitude, the concentration of electrons near the SiO_2 surface increases until eventually the induced *n*-type region can support a measurable flow between drain and source. The level of V_{GS} that results in the significant increase in drain current is called the *threshold voltage* and is given the symbol V_T. On specification sheets it is referred to as $V_{GS(Th)}$, although V_T is less unwieldy and will be used in the analysis to follow. Since the channel is nonexistent with $V_{GS} = 0$ V and "enhanced" by the application of a positive gate-to-source voltage, this type of MOSFET is called an

enhancement-type MOSFET.

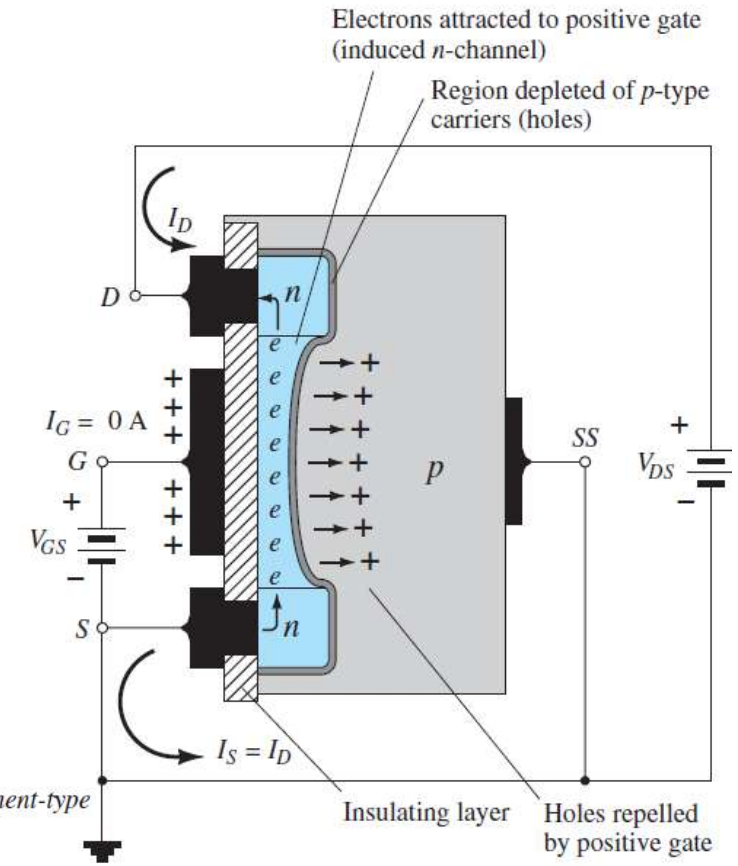
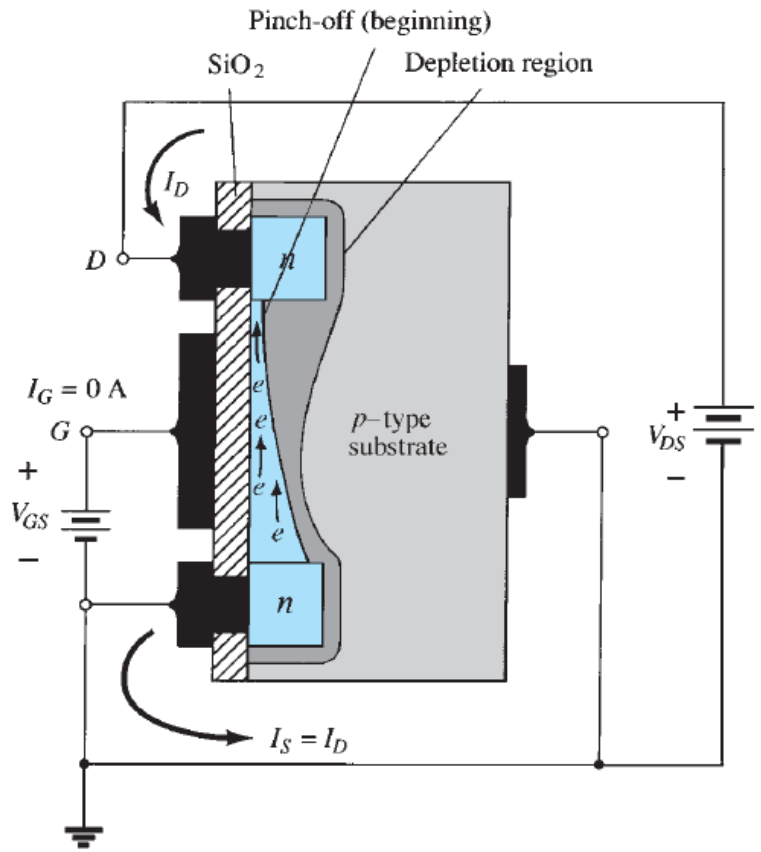


FIG. 6.33
Channel formation in the n-channel enhancement-type MOSFET.

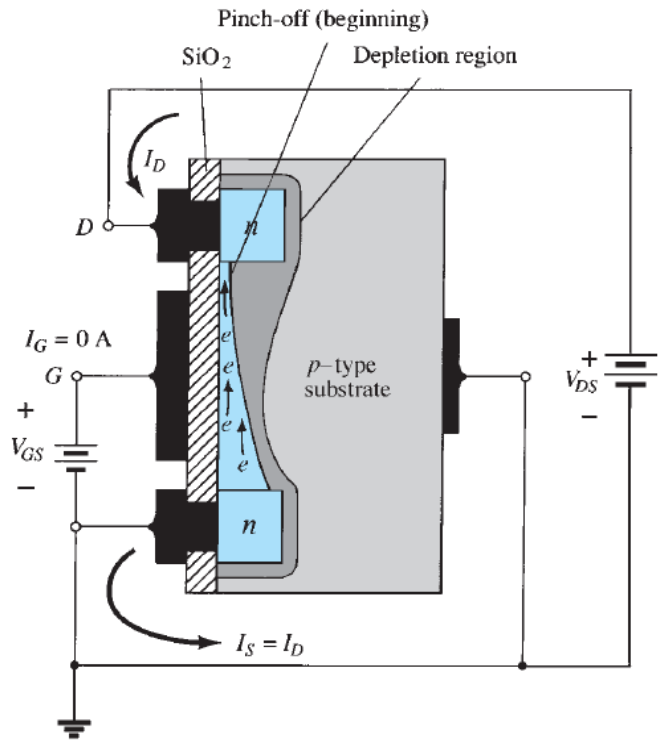
As V_{GS} is increased beyond the threshold level, the density of free carriers in the induced channel will increase, resulting in an increased level of drain current. However, if we hold V_{GS} constant and increase the level of V_{DS}, the drain current will eventually reach a saturation level as occurred for the JFET and depletion-type MOSFET. The leveling off of I_D is due to a pinching-off process depicted by the narrower channel at the drain end of the induced channel as shown in Fig. 6.34. Applying Kirchhoff's voltage law to the terminal voltages of the MOSFET of Fig. 6.34, we find that



$$V_{DG} = V_{DS} - V_{GS} \tag{6.13}$$

FIG. 6.34
Change in channel and depletion region with increasing level of V_{DS} for a fixed value of V_{GS}.

If V_{GS} is held fixed at some value such as 8 V and V_{DS} is increased from 2 V to 5 V, the voltage V_{DG} [by Eq. (6.13)] will increase from −6 V to −3 V and the gate will become less and less positive with respect to the drain. This reduction in gate-to-drain voltage will in turn reduce the attractive forces for free carriers (electrons) in this region of the induced channel, causing a reduction in the effective channel width. Eventually, the channel will be reduced to the point of pinch-off and a saturation condition will be established as described earlier for the JFET and depletion-type MOSFET.



$$V_{DG} = V_{DS} - V_{GS} \tag{6.13}$$

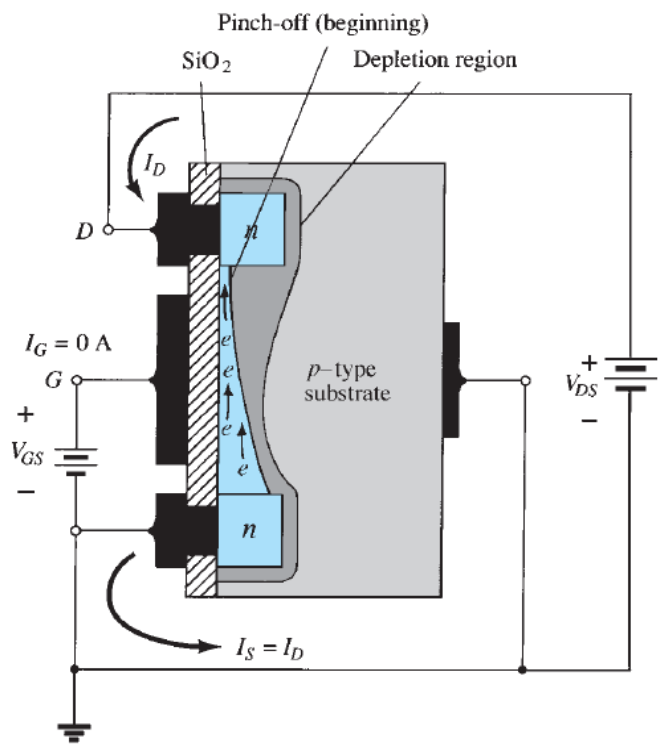
In other words, any further increase in V_{DS} at the fixed value of V_{GS} will not affect the saturation level of I_D until breakdown conditions are encountered

FIG. 6.34
Change in channel and depletion region with increasing level of V_{DS} for a fixed value of V_{GS}.

The drain characteristics of Fig. 6.35 reveal that for the device of Fig. 6.34 with $V_{GS} = 8$ V, saturation occurs at a level of $V_{DS} = 6$ V. In fact, the saturation level for V_{DS} is related to the level of applied V_{GS} by

$$V_{DS_{sat}} = V_{GS} - V_T \tag{6.14}$$

Obviously, therefore, for a fixed value of V_T, the higher the level of V_{GS}, the greater is the saturation level for V_{DS}, as shown in Fig. 6.34 by the locus of saturation levels.



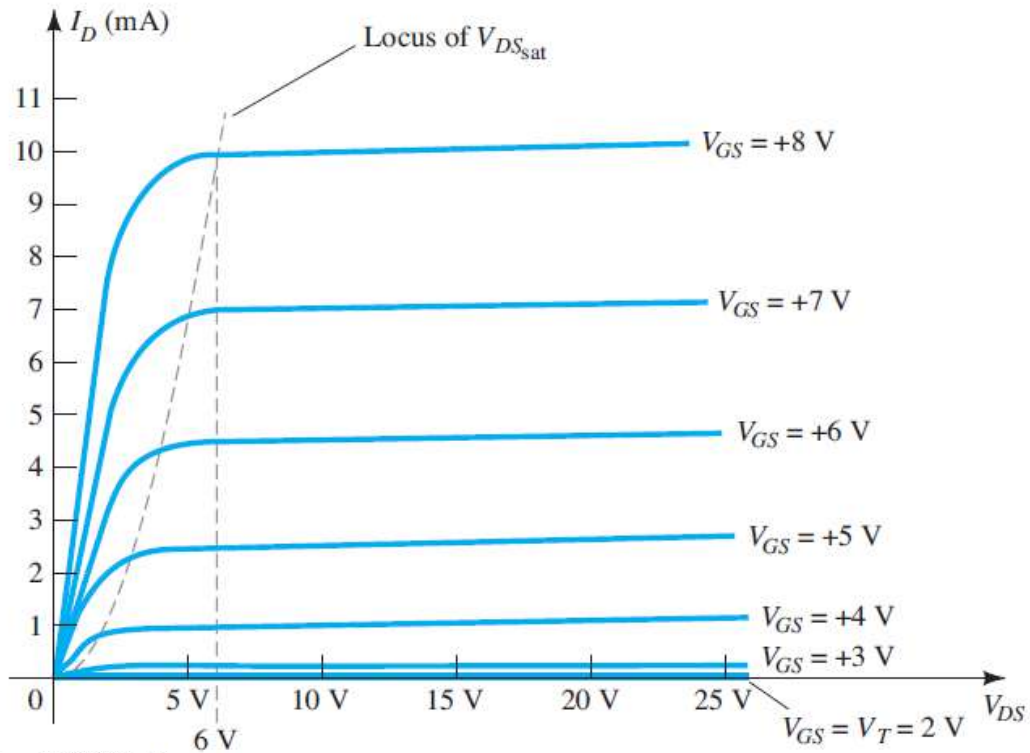


FIG. 6.35
Drain characteristics of an n-channel enhancement-type MOSFET with $V_T = 2$ V and $k = 0.278 \times 10^{-3}$ A/V^2.

For the characteristics of Fig. 6.34, the level of V_T is 2 V, as revealed by the fact that the drain current has dropped to 0 mA. In general, therefore:

For values of V_{GS} less than the threshold level, the drain current of an enhancement-type MOSFET is 0 mA.

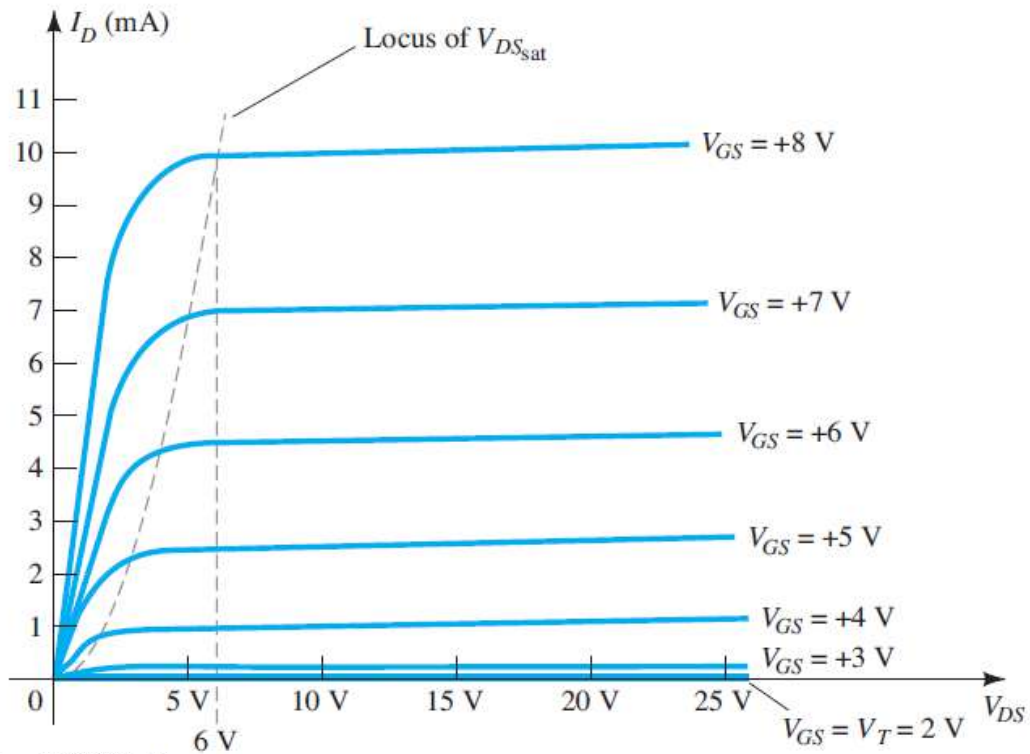


FIG. 6.35
Drain characteristics of an n-channel enhancement-type MOSFET with $V_T = 2$ V and $k = 0.278 \times 10^{-3}$ A/V^2.

Figure 6.35 clearly reveals that as the level of V_{GS} increases from V_T to 8 V, the resulting saturation level for I_D also increases from a level of 0 mA to 10 mA. In addition, it is quite noticeable that the spacing between the levels of V_{GS} increases as the magnitude of V_{GS} increases, resulting in ever-increasing increments in drain current.

For levels of $V_{GS} > V_T$, the drain current is related to the applied gate-to-source voltage by the following nonlinear relationship:

$$I_D = k(V_{GS} - V_T)^2 \qquad \textbf{(6.15)}$$

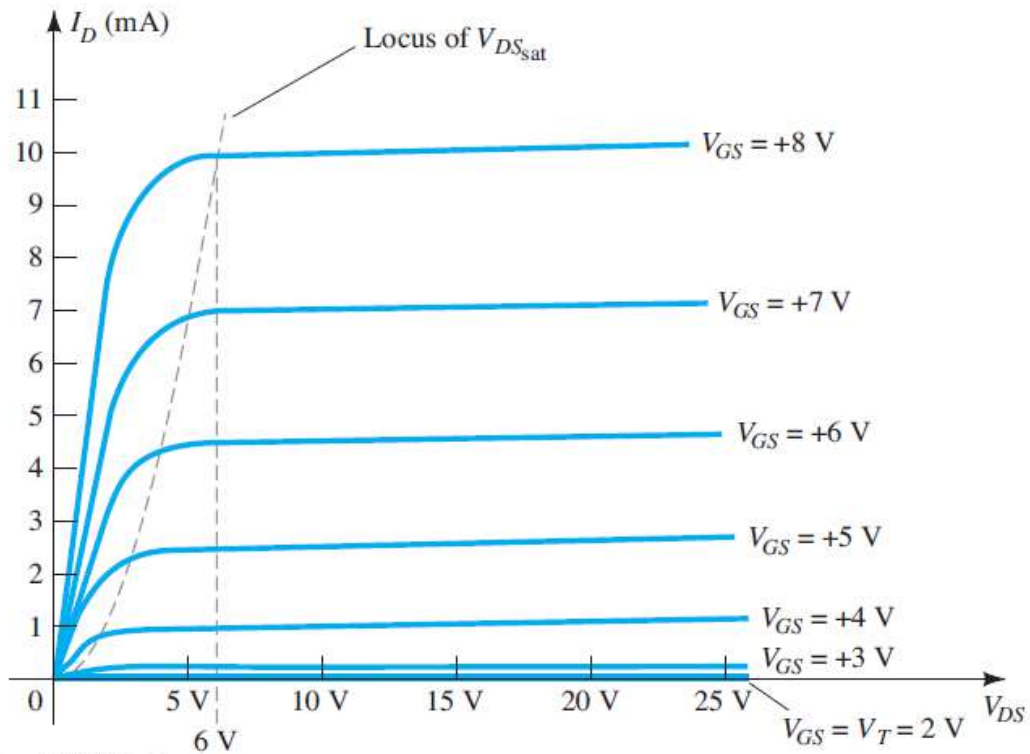


FIG. 6.35
Drain characteristics of an n-channel enhancement-type MOSFET with $V_T = 2$ V and $k = 0.278 \times 10^{-3}$ A/V^2.

$$I_D = k(V_{GS} - V_T)^2 \qquad \textbf{(6.15)}$$

Again, it is the squared term that results in the nonlinear (curved) relationship between I_D and V_{GS}. The k term is a constant that is a function of the construction of the device. The value of k can be determined from the following equation [derived from Eq. (6.15)], where $I_{D(\text{on})}$ and $V_{GS(\text{on})}$ are the values for each at a particular point on the characteristics of the device.

$$k = \frac{I_{D(\text{on})}}{(V_{GS(\text{on})} - V_T)^2} \qquad \textbf{(6.16)}$$

Substituting $I_{D(on)} = 10\text{ mA}$ when $V_{GS(on)} = 8\text{ V}$ from the characteristics of Fig. 6.35 yields

$$k = \frac{10\text{ mA}}{(8\text{ V} - 2\text{ V})^2} = \frac{10\text{ mA}}{(6\text{ V})^2} = \frac{10\text{ mA}}{36\text{ V}^2}$$
$$= \mathbf{0.278 \times 10^{-3}\ A/V^2}$$

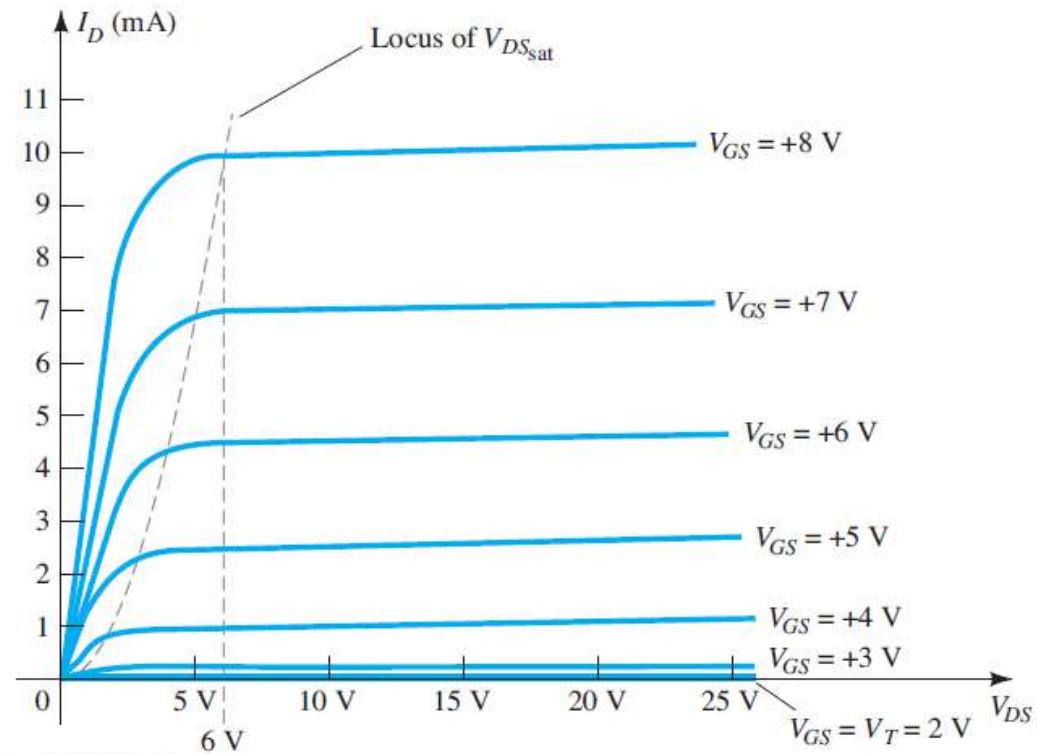


FIG. 6.35
Drain characteristics of an n-channel enhancement-type MOSFET with $V_T = 2$ V and $k = 0.278 \times 10^{-3}$ A/V^2.

and a general equation for I_D for the characteristics of Fig. 6.35 results in

$$I_D = 0.278 \times 10^{-3}(V_{GS} - 2\,\text{V})^2$$

Substituting $V_{GS} = 4$ V, we find that

$$I_D = 0.278 \times 10^{-3}(4\,\text{V} - 2\,\text{V})^2 = 0.278 \times 10^{-3}(2)^2$$
$$= 0.278 \times 10^{-3}(4) = \mathbf{1.11\ mA}$$

as verified by Fig. 6.35. At $V_{GS} = V_T$, the squared term is 0, and $I_D = 0$ mA.

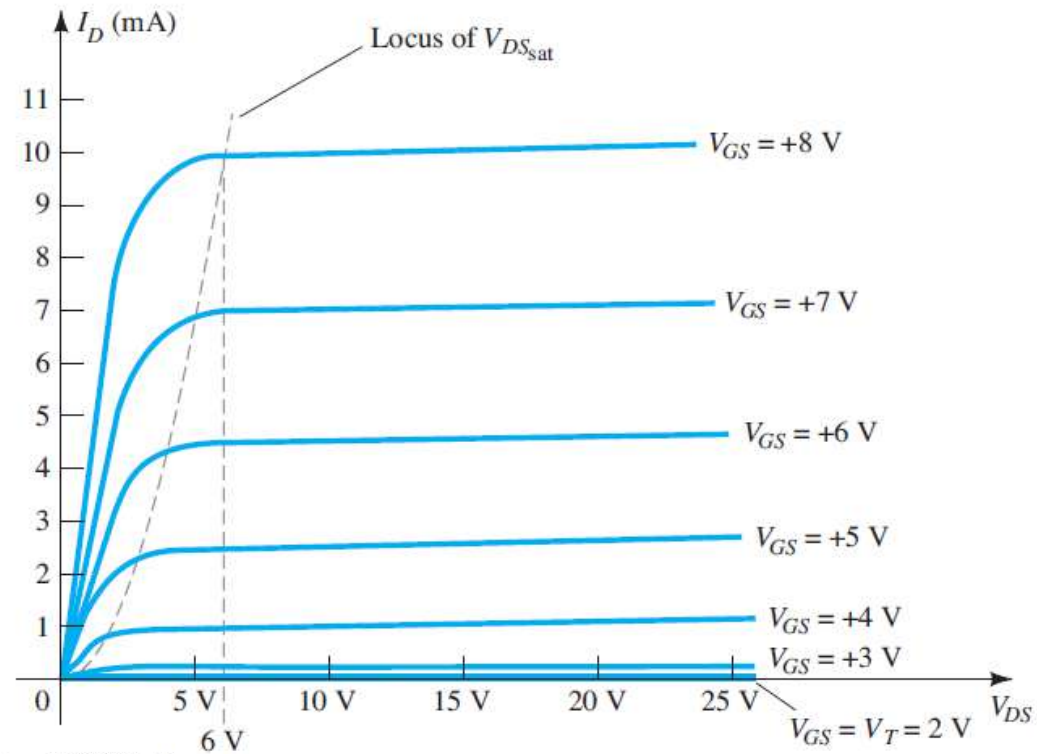


FIG. 6.35
Drain characteristics of an n-channel enhancement-type MOSFET with $V_T = 2$ V and $k = 0.278 \times 10^{-3}$ A/V^2.

The transfer characteristics

In Fig. 6.36 , the drain and transfer characteristics have been set side by side to describe the transfer process from one to the other. Essentially, it proceeds as introduced earlier for the JFET and depletion-type MOSFETs. In this case, however, **it must be remembered that the drain current is 0 mA for $V_{GS} \leq V_T$**. As V_{GS} is increased beyond V_T , the drain current I_D will begin to flow at an increasing rate in accordance with Eq. (6.15). Note that in defining **the points on the transfer characteristics from the drain characteristics, only the saturation levels are employed, thereby limiting the region of operation to levels of V_{DS} greater than the saturation levels as defined by Eq. (6.14).**

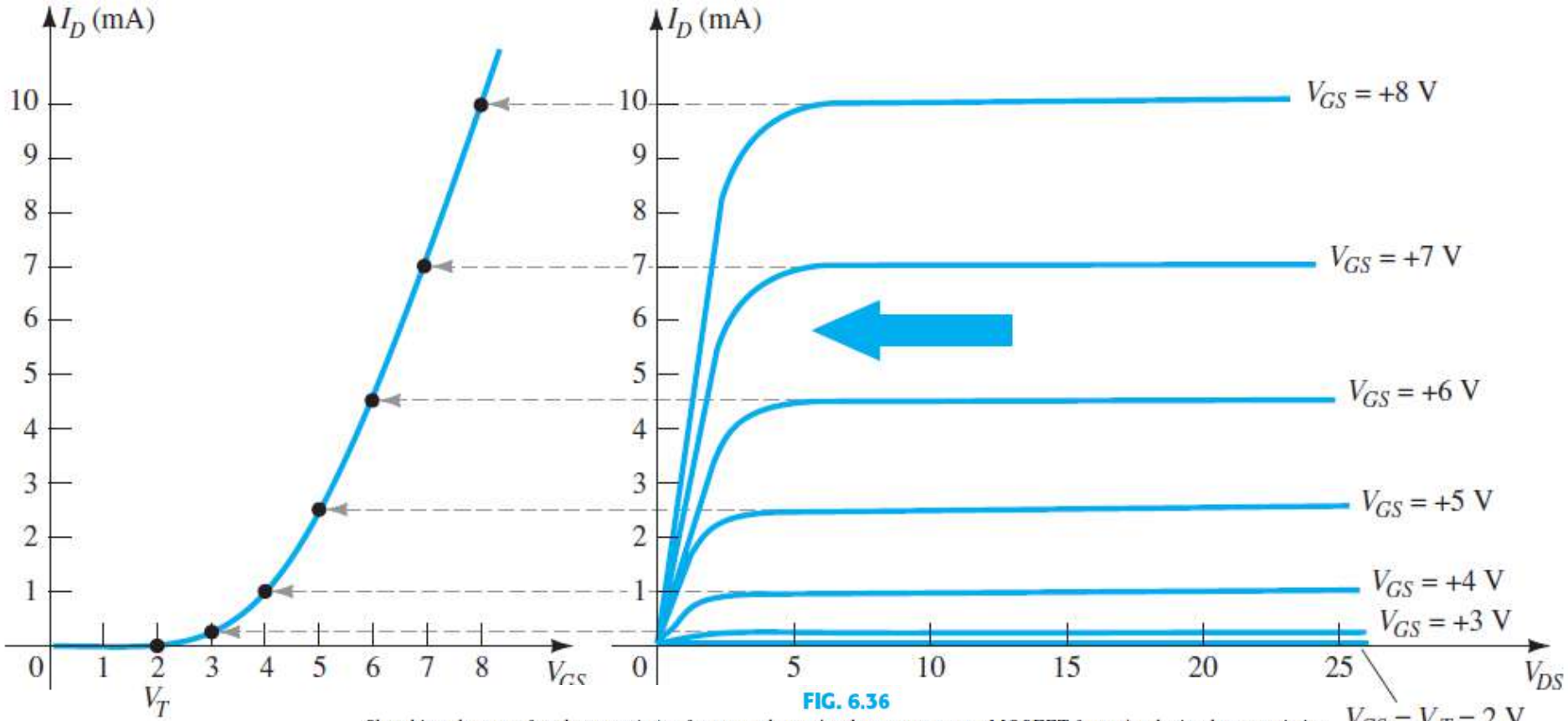


FIG. 6.36
Sketching the transfer characteristics for an n-channel enhancement-type MOSFET from the drain characteristics.

$$I_D = k(V_{GS} - V_T)^2 \qquad (6.15)$$

$$V_{DS_{sat}} = V_{GS} - V_T \qquad (6.14)$$

The transfer curve of Fig. 6.36 is certainly quite different from those obtained earlier. For an *n*-channel (induced) device, it is now totally in the positive V_{GS} region and does not rise until $V_{GS} = V_T$. The question now surfaces as to how to plot the transfer characteristics given the levels of k and V_T as included below for a particular MOSFET:

$$I_D = 0.5 \times 10^{-3}(V_{GS} - 4\text{ V})^2$$

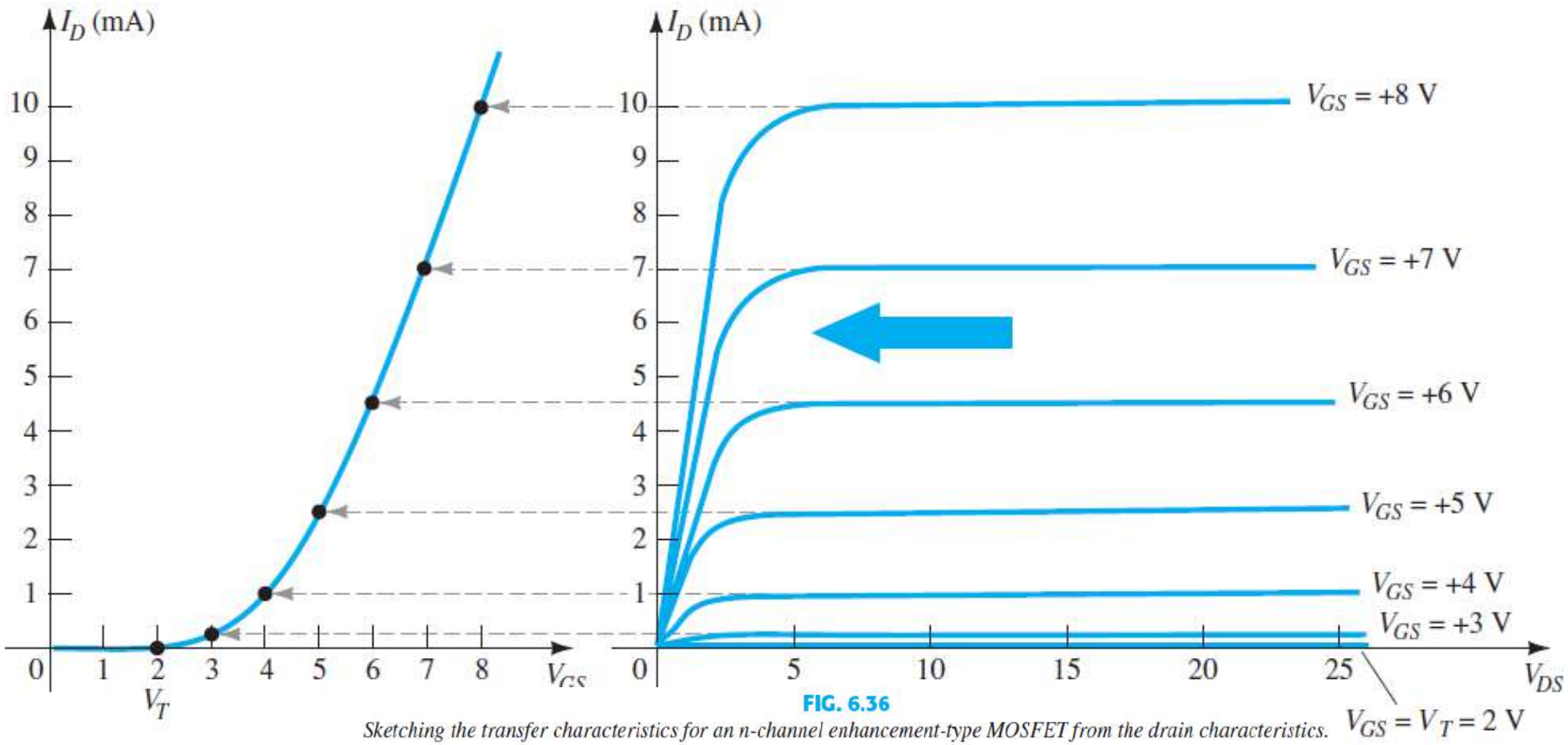


FIG. 6.36
Sketching the transfer characteristics for an n-channel enhancement-type MOSFET from the drain characteristics.

$$I_D = k(V_{GS} - V_T)^2 \qquad \textbf{(6.15)}$$

$$V_{DS_{sat}} = V_{GS} - V_T \qquad \textbf{(6.14)}$$

First, a horizontal line is drawn at $I_D = 0$ mA from $V_{GS} = 0$ V to $V_{GS} = 4$ V as shown in Fig. 6.37a. Next, a level of V_{GS} greater than V_T such as 5 V is chosen and substituted into Eq. (6.15) to determine the resulting level of I_D as follows:

$$\begin{aligned} I_D &= 0.5 \times 10^{-3}(V_{GS} - 4\text{ V})^2 \\ &= 0.5 \times 10^{-3}(5\text{ V} - 4\text{ V})^2 = 0.5 \times 10^{-3}(1)^2 \\ &= \mathbf{0.5\text{ mA}} \end{aligned}$$

and a point on the plot is obtained as shown in Fig. 6.37b. Finally, additional levels of V_{GS} are chosen and the resulting levels of I_D obtained. In particular, at $V_{GS} = 6$, 7, and 8 V, the level of I_D is 2, 4.5, and 8 mA, respectively, as shown on the resulting plot of Fig. 6.37c.

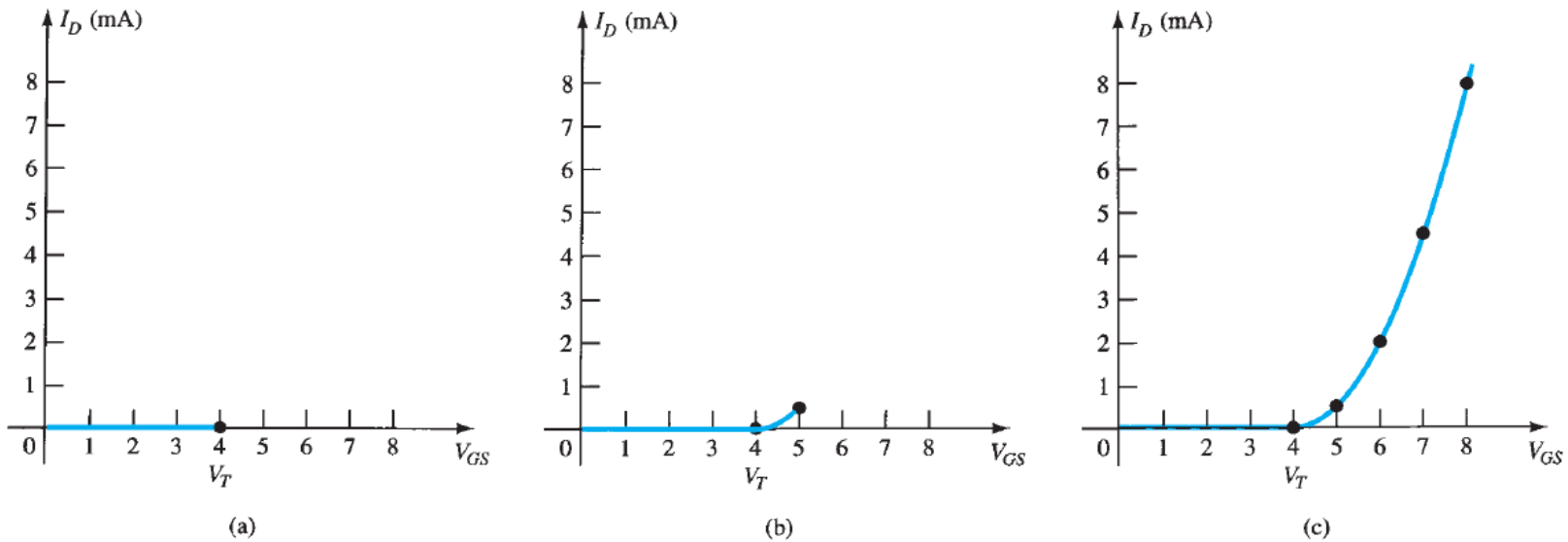


FIG. 6.37

Plotting the transfer characteristics of an n-channel enhancement-type MOSFET with $k = 0.5 \times 10^{-3}$ A/V^2 and $V_T = 4$ V.

p-Channel Enhancement-Type MOSFETs

The construction of a *p*-channel enhancement-type MOSFET is exactly the reverse of that appearing in Fig. 6.32, as shown in Fig. 6.38a. That is, there is now an *n*-type substrate and *p*-doped regions under the drain and source connections. The terminals remain as identified, but all the voltage polarities and the current directions are reversed. The drain characteristics will appear as shown in Fig. 6.38c, with increasing levels of current resulting from increasingly negative values of V_{GS}. The transfer characteristics of Fig. 6.38b will be the mirror image (about the I_D axis) of the transfer curve of Fig. 6.36, with I_D increasing with increasingly negative values of V_{GS} beyond V_T, as shown in Fig. 6.38c. Equations (6.13) through (6.16) are equally applicable to *p*-channel devices.

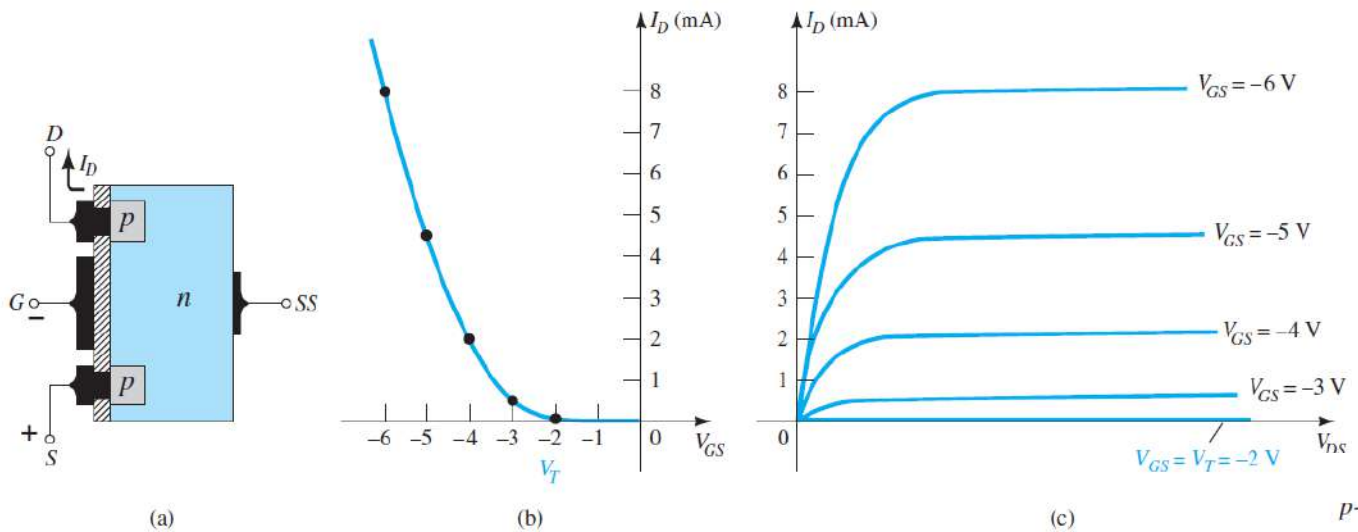


FIG. 6.38
p-Channel enhancement-type MOSFET with $V_T = 2$ V and $k = 0.5 \times 10^{-3}$ A/V^2.

Symbols, Specification Sheets, and Case Construction

The graphic symbols for the *n*- and *p*-channel enhancement-type MOSFETs are provided as Fig. 6.39. Again note how the symbols try to reflect the actual construction of the device. The dashed line between drain and source is chosen to reflect the fact that a channel does not exist between the two under no-bias conditions. It is, in fact, the only difference between the symbols for the depletion-type and enhancement-type MOSFETs.

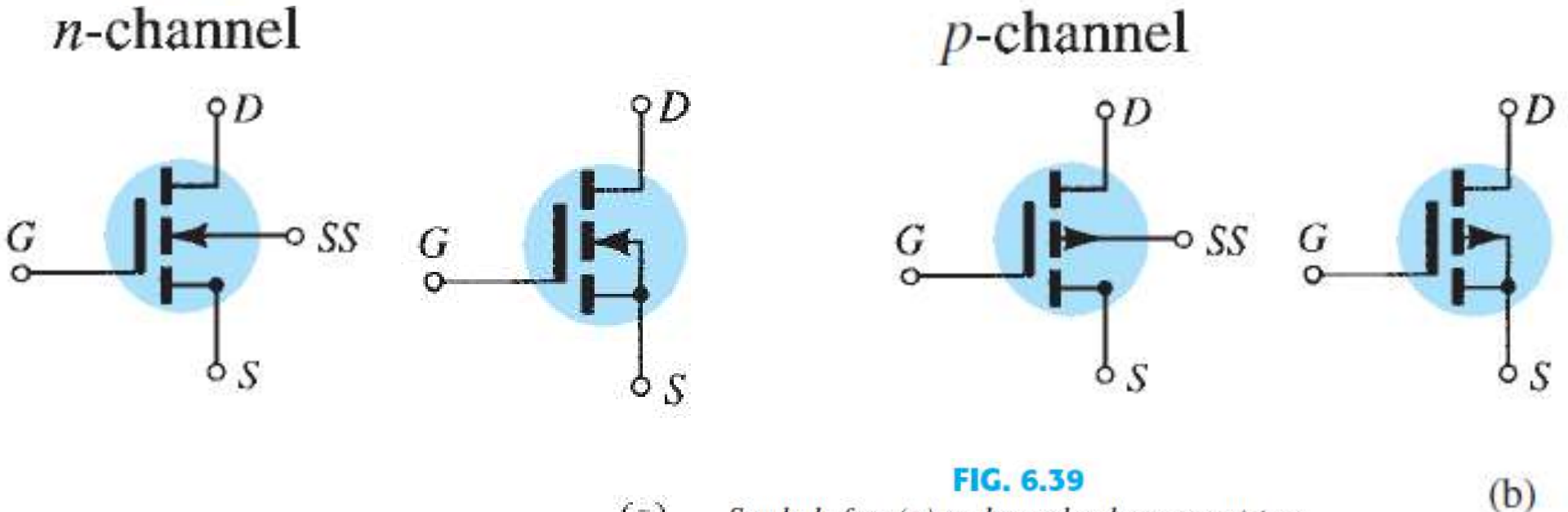


FIG. 6.39
Symbols for: (a) n-channel enhancement-type MOSFETs and (b) p-channel enhancement-type MOSFETs.

The specification sheet for a Motorola *n*-channel enhancement-type MOSFET is provided as Fig. 6.40. The case construction and the terminal identification are provided next to the maximum ratings, which now include a maximum drain current of 30 mA dc. The specification sheet provides the level of I_{DSS} under "off" conditions, which is now simply 10 nA dc (at $V_{DS} = 10$ V and $V_{GS} = 0$ V), compared to the milliampere range for the JFET and the depletion-type MOSFET.

MAXIMUM RATINGS

Rating	Symbol	Value	Unit
Drain–Source Voltage	V_{DS}	25	Vdc
Drain–Gate Voltage	V_{DG}	30	Vdc
Gate–Source Voltage*	V_{GS}	30	Vdc
Drain Current	I_D	30	mAdc
Total Device Dissipation @ $T_A = 25°C$ Derate above 25°C	P_D	300 1.7	mW mW/°C
Junction Temperature Range	T_J	175	°C
Storage Temperature Range	T_{stg}	–65 to +175	°C

* Transient potentials of ± 75 Volt will not cause gate-oxide failure.

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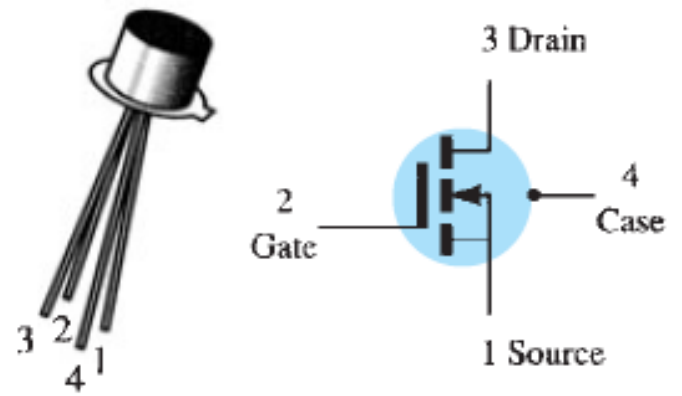
MOSFET SWITCHING



N-CHANNEL – ENHANCEMENT

The threshold voltage is specified as $V_{GS(Th)}$ and has a range of 1 to 5 V dc, depending on the device employed. Rather than provide a range of k in Eq. (6.15), a typical level of $I_{D(on)}$ (3 mA in this case) is specified at a particular level of $V_{GS(on)}$ (10 V for the specified I_D level). In other words, when $V_{GS} = 10$ V, $I_D = 3$ mA. The given levels of $V_{GS(Th)}$, $I_{D(on)}$, and $V_{GS(on)}$ permit a determination of k from Eq. (6.16) and a writing of the general equation for the transfer characteristics. The handling requirements of MOSFETs are reviewed in Section 6.9.

ELECTRICAL CHARACTERISTICS ($T_A = 25°C$ unless otherwise noted.)

Characteristic	Symbol	Min	Max	Unit
OFF CHARACTERISTICS				
Drain-Source Breakdown Voltage ($I_D = 10\ \mu A$, $V_{GS} = 0$)	$V_{(BR)DSX}$	25	–	Vdc
Zero-Gate-Voltage Drain Current ($V_{DS} = 10$ V, $V_{GS} = 0$) $T_A = 25°C$ $T_A = 150°C$	I_{DSS}	– –	10 10	nAdc μAdc
Gate Reverse Current ($V_{GS} = \pm 15$ Vdc, $V_{DS} = 0$)	I_{GSS}	–	± 10	pAdc
ON CHARACTERISTICS				
Gate Threshold Voltage ($V_{DS} = 10$ V, $I_D = 10\mu A$)	$V_{GS(Th)}$	1.0	5	Vdc
Drain-Source On-Voltage ($I_D = 2.0$ mA, $V_{GS} = 10$V)	$V_{DS(on)}$	–	1.0	V
On-State Drain Current ($V_{GS} = 10$ V, $V_{DS} = 10$ V)	$I_{D(on)}$	3.0	–	mAdc

ELECTRICAL CHARACTERISTICS (T_A = 25°C unless otherwise noted.)

Characteristic	Symbol	Min	Max	Unit
OFF CHARACTERISTICS				
Drain-Source Breakdown Voltage (I_D = 10 μA, V_{GS} = 0)	$V_{(BR)DSX}$	25	–	Vdc
Zero-Gate-Voltage Drain Current (V_{DS} = 10 V, V_{GS} = 0) T_A = 25°C T_A = 150°C	I_{DSS}	– –	10 10	nAdc μAdc
Gate Reverse Current (V_{GS} = ± 15 Vdc, V_{DS} = 0)	I_{GSS}	–	± 10	pAdc
ON CHARACTERISTICS				
Gate Threshold Voltage (V_{DS} = 10 V, I_D = 10 μA)	$V_{GS(Th)}$	1.0	5	Vdc
Drain-Source On-Voltage (I_D = 2.0 mA, V_{GS} = 10V)	$V_{DS(on)}$	–	1.0	V
On-State Drain Current (V_{GS} = 10 V, V_{DS} = 10 V)	$I_{D(on)}$	3.0	–	mAdc

EXAMPLE 6.4 Using the data provided on the specification sheet of Fig. 6.40 and an average threshold voltage of $V_{GS(Th)} = 3$ V, determine:

a. The resulting value of k for the MOSFET.
b. The transfer characteristics.

ON CHARACTERISTICS

Gate Threshold Voltage (V_{DS} = 10 V, I_D = 10 μA)	$V_{GS(Th)}$	1.0	5	Vdc
Drain-Source On-Voltage (I_D = 2.0 mA, V_{GS} = 10V)	$V_{DS(on)}$	–	1.0	V
On-State Drain Current (V_{GS} = 10 V, V_{DS} = 10 V)	$I_{D(on)}$	3.0	–	mAdc

Solution:

a. Eq. (6.16): $$k = \frac{I_{D(on)}}{(V_{GS(on)} - V_{GS(Th)})^2}$$

$$= \frac{3 \text{ mA}}{(10 \text{ V} - 3 \text{ V})^2} = \frac{3 \text{ mA}}{(7 \text{ V})^2} = \frac{3 \times 10^{-3}}{49} \text{A/V}^2$$

$$= \mathbf{0.061 \times 10^{-3} \text{ A/V}^2}$$

b. Eq. (6.15):
$$I_D = k(V_{GS} - V_T)^2$$
$$= 0.061 \times 10^{-3}(V_{GS} - 3\text{ V})^2$$

For $V_{GS} = 5$ V,

$$I_D = 0.061 \times 10^{-3}(5\text{ V} - 3\text{ V})^2 = 0.061 \times 10^{-3}(2)^2$$
$$= 0.061 \times 10^{-3}(4) = 0.244\text{ mA}$$

For $V_{GS} = 8$, 10, 12, and 14 V, I_D will be 1.525, 3 (as defined), 4.94, and 7.38 mA, respectively. The transfer characteristics are sketched in Fig. 6.41.

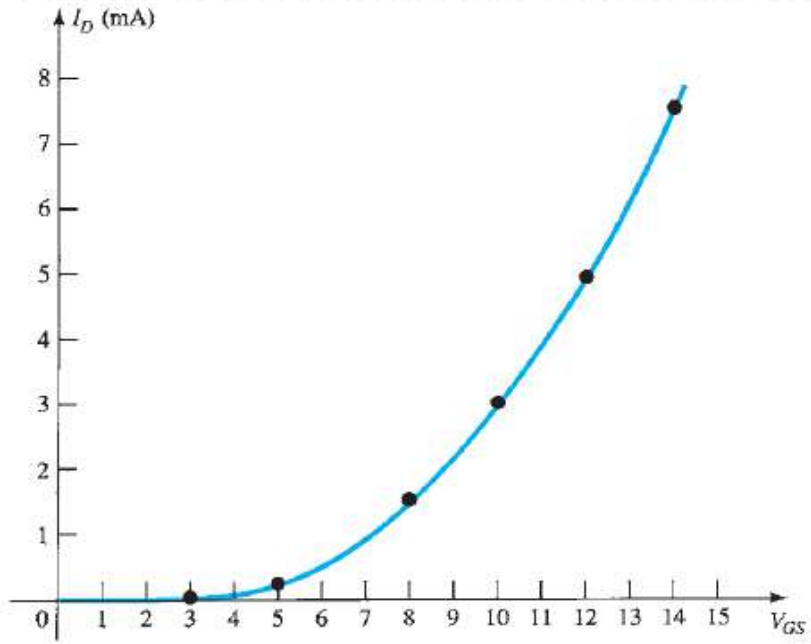
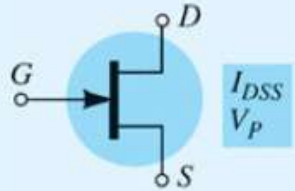
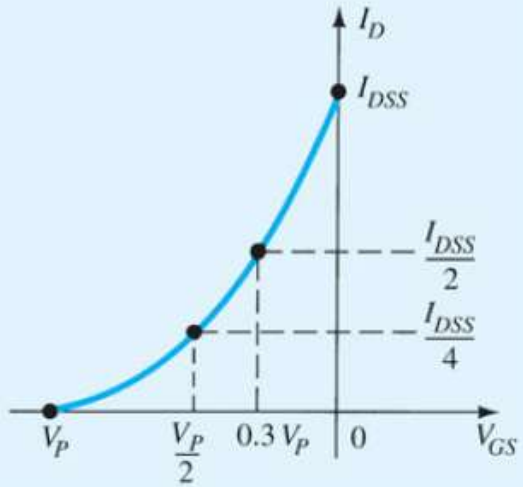
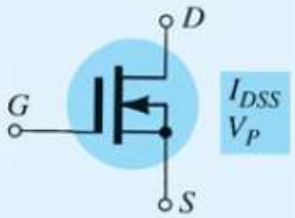
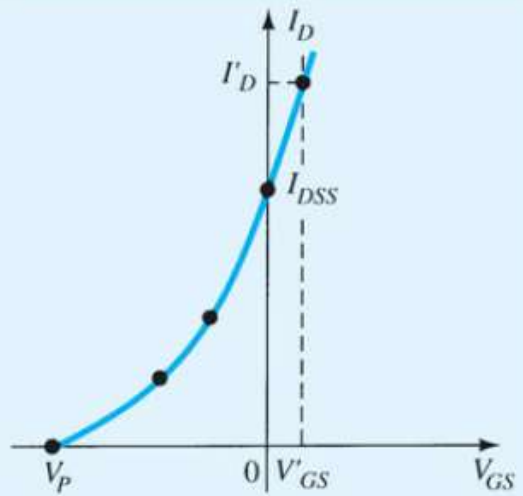
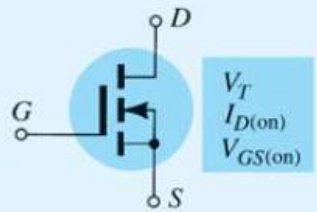
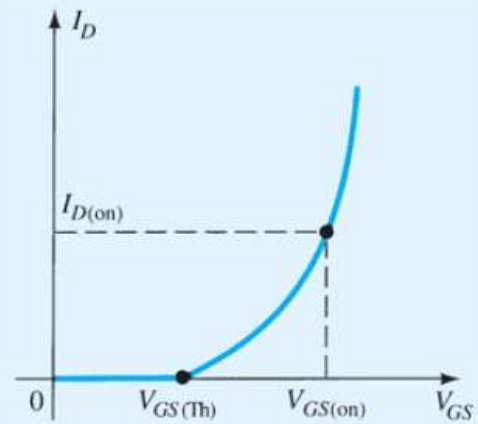


FIG. 6.41
Solution to Example 6.4.

TABLE 6.3

Field Effect Transistors

Type	Symbol and Basic Relationships	Transfer Curve	Input Resistance and Capacitance
JFET (*n*-channel)	$I_G = 0\ \text{A}, I_D = I_S$  $I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_P}\right)^2$	 	$R_i > 100\ \text{M}\Omega$ C_i: (1 − 10) pF
MOSFET depletion type (*n*-channel)	$I_G = 0\ \text{A}, I_D = I_S$  $I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_P}\right)^2$	 	$R_i > 10^{10}\ \Omega$ C_i: (1 − 10) pF

MOSFET enhancement type (*n*-channel)	$I_G = 0$ A, $I_D = I_S$  $I_D = k\,(V_{GS} - V_{GS\,(Th)})^2$ $k = \dfrac{I_{D(on)}}{(V_{GS(on)} - V_{GS\,(Th)})^2}$	 	$R_i > 10^{10}\ \Omega$ C_i: (1 − 10) pF

Equations

JFET:

$$I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_P}\right)^2$$

$$I_D = I_{DSS}\big|_{V_{GS}=0\ V}, \quad I_D = 0\ \text{mA}\big|_{V_{GS}=V_P}, \quad I_D = \left.\frac{I_{DSS}}{4}\right|_{V_{GS}=V_P/2}, \quad V_{GS} \cong 0.3V_P\big|_{I_D=I_{DSS}/2}$$

$$V_{GS} = V_P\left(1 - \sqrt{\frac{I_D}{I_{DSS}}}\right)$$

$$P_D = V_{DS}I_D$$

$$r_d = \frac{r_o}{(1 - V_{GS}/V_P)^2}$$

MOSFET (enhancement):

$$I_D = k(V_{GS} - V_T)^2$$

$$k = \frac{I_{D(\text{on})}}{(V_{GS(\text{on})} - V_T)^2}$$

6.9 MOSFET HANDLING

The thin SiO_2 layer between the gate and the channel of MOSFETs has the positive effect of providing a high-input-impedance characteristic for the device, but because of its extremely thin layer, it introduces a concern for its handling that was not present for the BJT or JFET transistors. There is often sufficient accumulation of static charge (picked up from the surroundings) to establish a potential difference across the thin layer that can break down the layer and establish conduction through it. It is therefore imperative to leave the shorting (or conduction) shipping foil (or ring) connecting the leads of the device together until the device is to be inserted in the system. The shorting ring prevents the possibility of applying a potential across any two terminals of the device. With the ring, the potential difference between any two terminals is maintained at 0 V. At the very least always touch ground to permit discharge of the accumulated static charge before handling the device, and always pick up the transistor by the casing.

There are often transients (sharp changes in voltage or current) in a network when elements are removed or inserted if the power is on. The transient levels can often be more than the device can handle, and therefore the power should always be off when network changes are made.

The maximum gate-to-source voltage is normally provided in the list of maximum ratings of the device. One method of ensuring that this voltage is not exceeded (perhaps by transient effects) for either polarity is to introduce two Zener diodes, as shown in Fig. 6.42. The Zeners are back to back to ensure protection for either polarity. If both are 30-V Zeners and a positive transient of 40 V appears, the lower Zener will "fire" at 30 V and the upper will turn on with a 0-V drop (ideally—for the positive "on" region of a semiconductor diode) across the other diode. The result is a maximum of 30 V for the gate-to-source voltage. One disadvantage introduced by the Zener protection is that the off resistance of a Zener diode is less than the input impedance established by the SiO_2 layer. The result is a reduction in input resistance, but even so, it is still high enough for most applications. So many of the discrete devices now have the Zener protection that some of the concerns listed above are not as troublesome. However, it is still best to be somewhat cautious when handling discrete MOSFET devices.

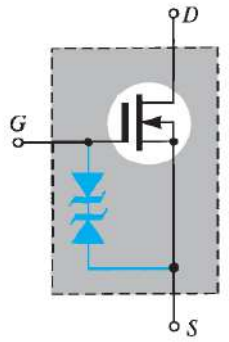


FIG. 6.42
Zener-protected MOSFET.

6.10 VMOS AND UMOS POWER MOSFETs

One of the disadvantages of the typical planar MOSFET is the reduced power handling (typically less than 1 W) and current levels compared with the broad range of bipolar transistors. However, through a vertical design such as shown for the VMOS MOSFET in Fig. 6.43a and the UMOS MOSFET in Fig. 6.43b, power and current levels have been increased along with higher switching speeds and reduced operating dissipation. All the elements of the planar MOSFET are present in the VMOS or UMOS MOSFETs—the metallic surface connection to the terminals of the device, the SiO_2 layer between the gate, and the *p*-type region between the drain and the source for the growth of the induced *n*-channel (enhancement-mode operation). The term *vertical* is due primarily to the fact that the channel is now formed in the *vertical* direction resulting in a vertical current direction rather than the horizontal direction for the planar device. However, the channel of Fig. 6.43a also has the appearance of a "V" cut in the semiconductor base, which often stands out as the reason for the name for the device. The construction of Fig. 6.43a is somewhat simplistic in nature, leaving out some of the transition levels of doping, but it does permit a description of the most important facets of its operation.

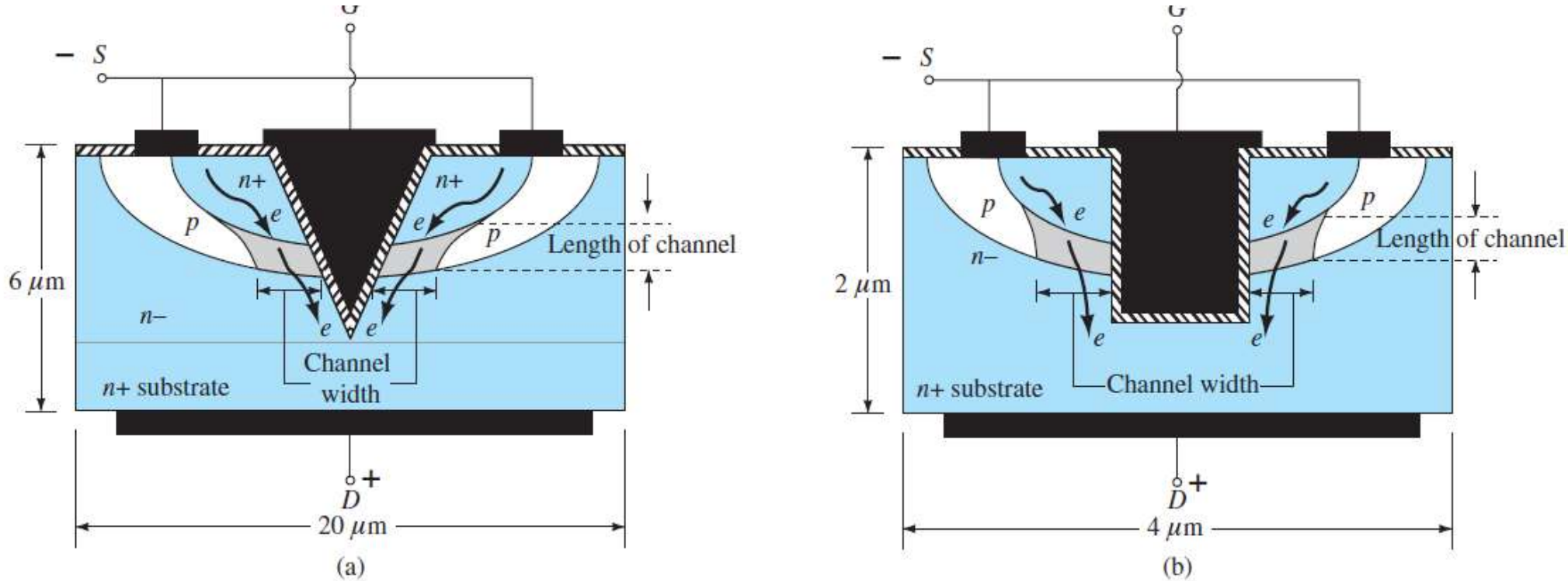


FIG. 6.43

(a) VMOS MOSFET; (b) UMOS MOSFET.

The application of a positive voltage to the drain and a negative voltage to the source with the gate at 0 V or some typical positive "on" level as shown in Fig. 6.43a results in the induced *n*-channel in the narrow *p*-type region of the device. The length of the channel is now defined by the vertical height of the *p*-region, which can be made significantly less than that of a channel using planar construction. On a horizontal plane the length of the channel is limited to 1 μm to 2 μm (1 μm = 10^{-6} m). Diffusion layers (such as the *p*-region of Fig. 6.43) can be controlled to small fractions of a micrometer. Since decreasing channel lengths result in reduced resistance levels, the power dissipation level of the device (power lost in the form of heat) at operating current levels will be reduced. In addition, the contact area between the channel and the n^+ region is greatly increased by the vertical mode construction, contributing to a further decrease in the resistance level and an increased area for current between the doping layers. There is also the existence of two conduction paths between drain and source, as shown in Fig. 6.43, to further contribute to a higher current rating. The net result is a device with drain currents that can reach the ampere levels with power levels exceeding 10 W.

The VMOS MOSFET was the first in line of vertical MOSFETs designed primarily to be used as power switches to control the operation of power supplies, low-voltage motor controllers, DC- to DC-convertors, flat-panel displays, and a host of applications in today's automobiles. Fundamentally, a good power switch should work at relatively low voltages (less than 200 V), has excellent high-speed characteristics, and low levels of "on" resistance to ensure minimum power losses during operation. Over time, a variety of other vertical designs began to surface to improve on the "V" construction of Fig. 6.43a. The delicate

etching required to establish the V groove resulted in difficulties establishing a consistent threshold voltage, and the sharp tip at the end of the channel created high electric fields, which affected the breakdown voltage of the MOSFET. The breakdown voltage is important because it is directly related to the "on" resistance. Increase the breakdown voltage and the "on" resistance begins to increase.

One improvement over the "V" design is the "U" groove or channel as appearing in Fig. 6.43b. The operation of this UMOS MOSFET (also called Trench MOSFET) is very similar to that of the VMOS MOSFET but with improved characteristics. First the fabrication process is preferred because the trench-etching process developed for memory cells in DRAMs can be utilized. The result is reduced widths in the neighborhood of 2–10 μm compared with the VMOS construction with widths in the 20–30 μm range. The channel width itself may be only 1 μm with a height of 2 μm. The "on" resistance is less using the trench approach because the channel length is decreased and the width of the current path is increased near the bottom of the trench. However, due to the large surface area required for the heavy current flow, there are capacitive effects that must be considered at frequencies beyond 100 kHz. The three that have to be considered are C_{GS}, C_{GD}, and C_{DS} (respectively referred to as C_{iss}, C_{rss}, and C_{oss} on specification sheets). For the UMOS MOSFET the gate-to-source capacitance at the input is the largest and typically thousands of pF.

6.11 CMOS

A very effective logic circuit can be established by constructing a *p*-channel and an *n*-channel MOSFET on the same substrate as shown in Fig. 6.44. Note the induced *p*-channel on the left and the induced *n*-channel on the right for the *p*- and *n*-channel devices, respectively. The configuration is referred to as a *complementary MOSFET* arrangement (CMOS); it has extensive applications in computer logic design. The relatively high input impedance, fast switching speeds, and lower operating power levels of the CMOS configuration have resulted in a whole new discipline referred to as *CMOS logic design.*

One very effective use of the complementary arrangement is as an inverter, as shown in Fig. 6.45. As introduced for switching transistors, an inverter is a logic element that "inverts" the applied signal. That is, if the logic levels of operation are 0 V (0-state) and 5 V (1-state), an input level of 0 V will result in an output level of 5 V, and vice versa. Note in Fig. 6.45 that both gates are connected to the applied signal and both drain to the output V_o. The source of the *p*-channel MOSFET (Q_2) is connected directly to the applied voltage V_{SS}, whereas the source of the *n*-channel MOSFET (Q) is connected to ground. For the logic levels defined above, the application of 5 V at the i V

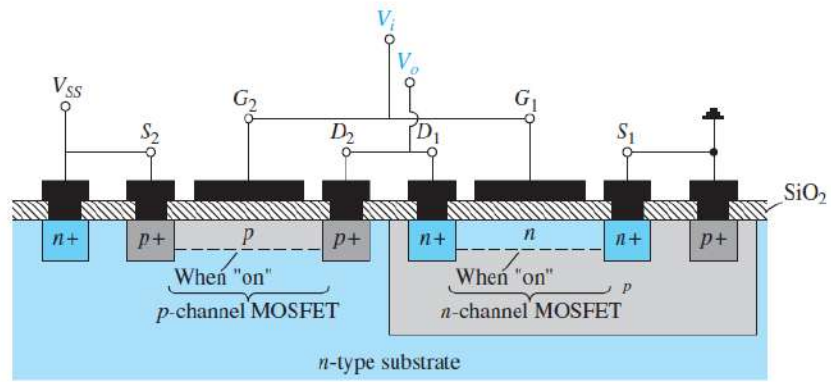


FIG. 6.44

at the output. With 5 V at V_i (with respect to ground), $V_{GS_1} = V_i$, and Q_1 is "on," resulting in a relatively low resistance between drain and source as shown in Fig. 6.46. Since V_i and V_{SS} are at 5 V, $V_{GS_2} = 0$ V, which is less than the required V_T for the device, resulting in an "off" state. The resulting resistance level between drain and source is quite high for Q_2, as shown in Fig. 6.46. A simple application of the voltage-divider rule will reveal that V_o is very close to 0 V, or the 0-state, establishing the desired inversion process. For an applied voltage V_i of 0 V (0-state), $V_{GS_1} = 0$ V, and Q_1 will be "off" with $V_{SS_2} = -5$ V, turning on the *p*-channel MOSFET. The result is that Q_2 will present a small resistance level, Q_1 a high resistance, and $V_o = V_{SS} = 5$ V (the 1-state). Since the drain current that flows for either case is limited by the "off" transistor to the leakage value, the power dissipated by the device in either state is very low. Additional comment on the application of CMOS logic is presented in Chapter 13.

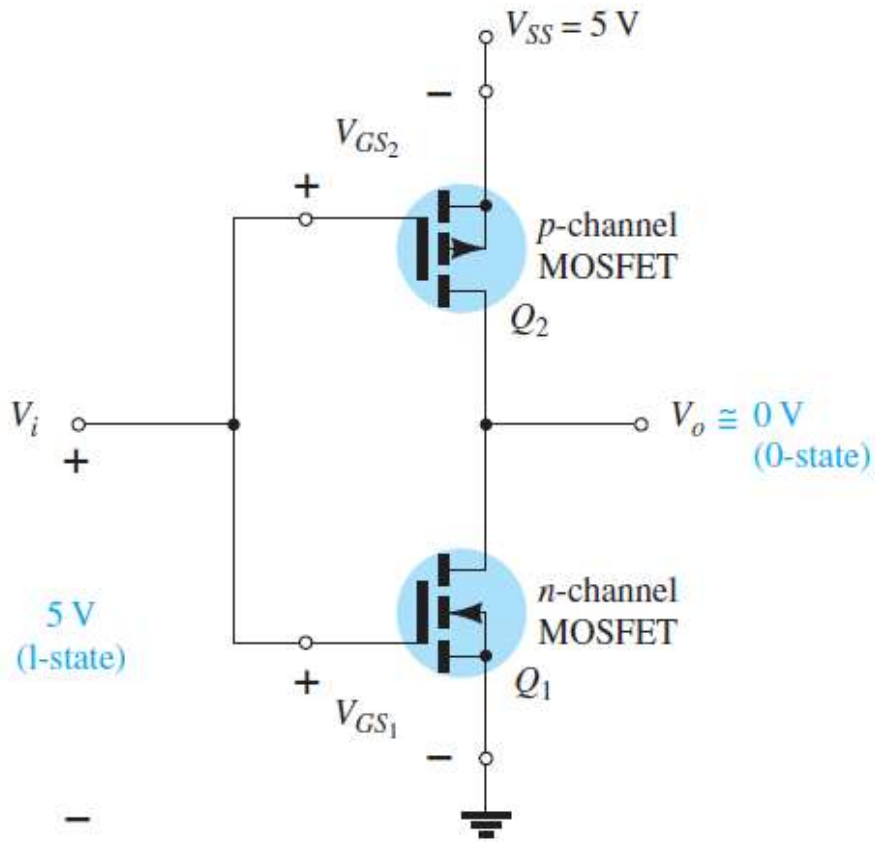


FIG. 6.45
CMOS inverter.

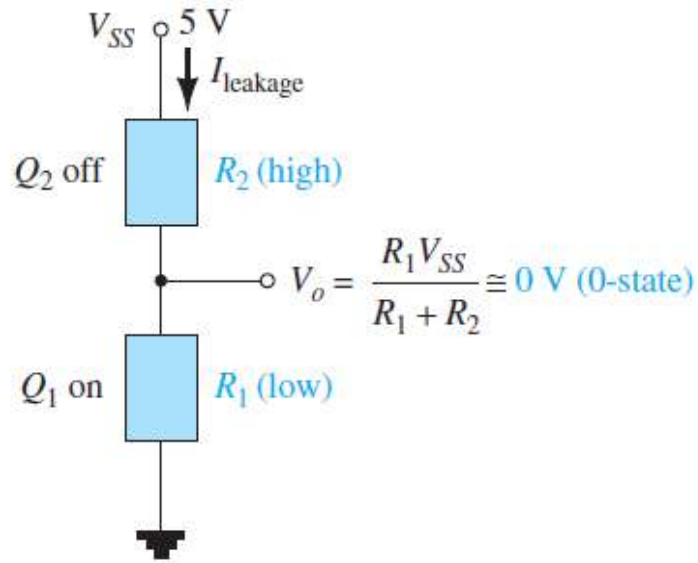
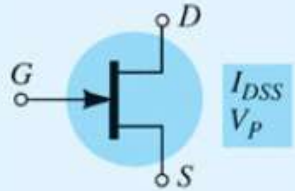
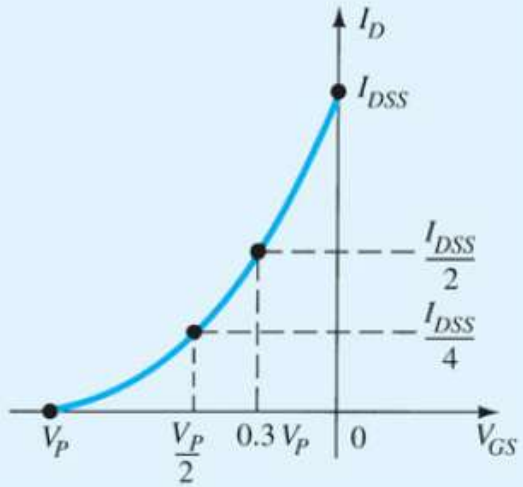
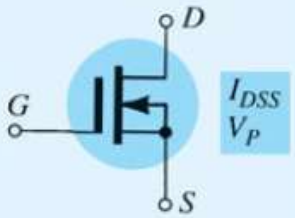
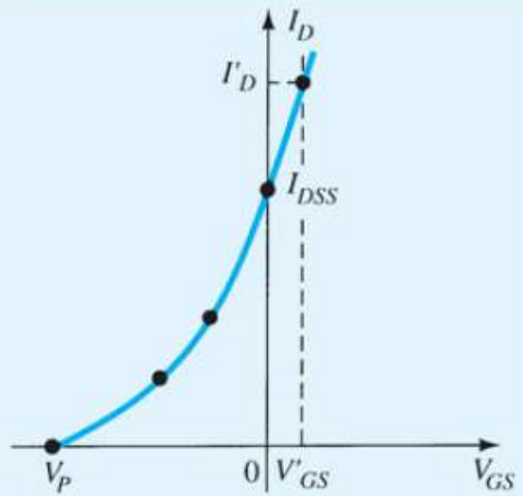
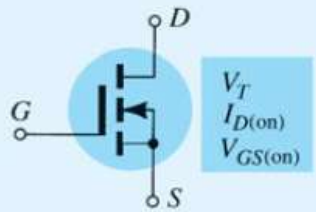
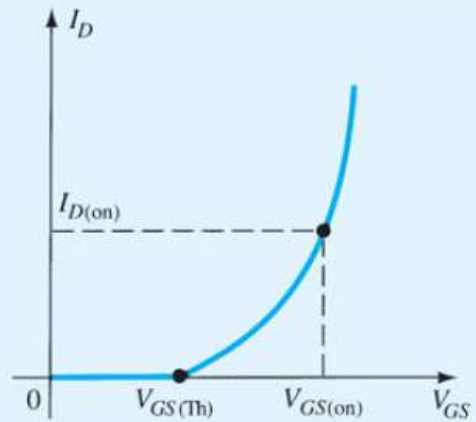


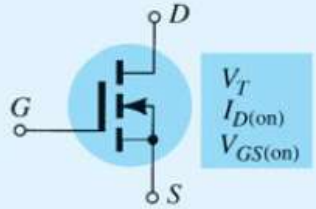
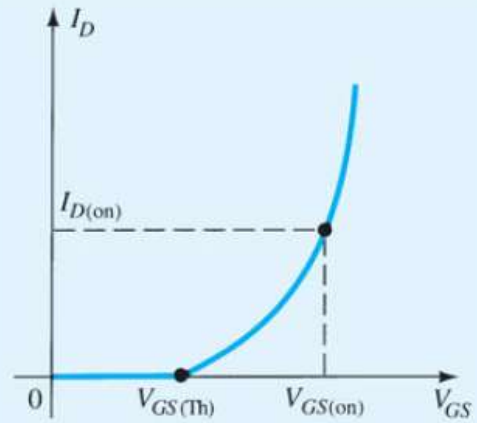
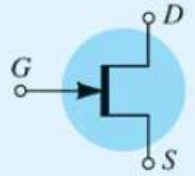
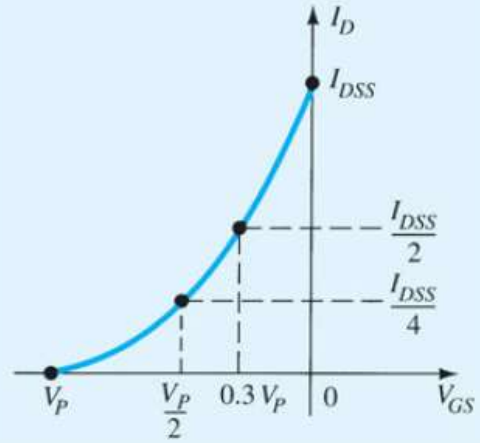
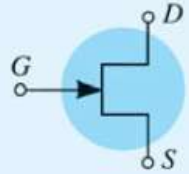
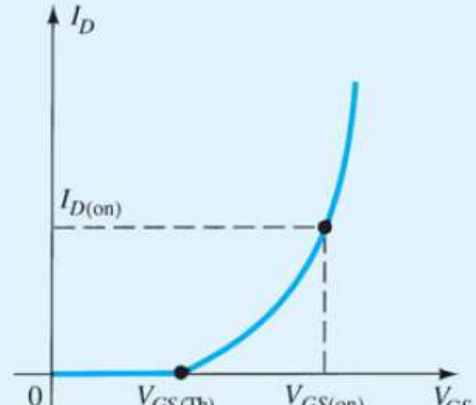
FIG. 6.46
Relative resistance levels for $V_i = 5$ V (1-state).

TABLE 6.3

Field Effect Transistors

Type	Symbol and Basic Relationships	Transfer Curve	Input Resistance and Capacitance
JFET (*n*-channel)	$I_G = 0\ \text{A}, I_D = I_S$  $I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_P}\right)^2$	 	$R_i > 100\ \text{M}\Omega$ C_i: (1 − 10) pF
MOSFET depletion type (*n*-channel)	$I_G = 0\ \text{A}, I_D = I_S$  $I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_P}\right)^2$	 	$R_i > 10^{10}\ \Omega$ C_i: (1 − 10) pF

MOSFET enhancement type (*n*-channel)	$I_G = 0\ \text{A}, I_D = I_S$  $I_D = k\,(V_{GS} - V_{GS\,(Th)})^2$ $k = \dfrac{I_{D(on)}}{(V_{GS(on)} - V_{GS\,(Th)})^2}$	 	$R_i > 10^{10}\ \Omega$ C_i: (1 − 10) pF

MOSFET enhancement type (*n*-channel)	$I_G = 0$ A, $I_D = I_S$  $I_D = k\,(V_{GS} - V_{GS\,(Th)})^2$ $k = \dfrac{I_{D(on)}}{(V_{GS(on)} - V_{GS\,(Th)})^2}$		$R_i > 10^{10}\ \Omega$ C_i: (1 − 10) pF
MESFET depletion type (*n*-channel)	$I_G = 0$ A, $I_D = I_S$  $I_D = I_{DSS}\left(1 - \dfrac{V_{GS}}{V_P}\right)^2$ $I_G = 0$ A, $I_D = I_S$		$R_i > 10^{12}\ \Omega$ C_i: (1 − 5) pF
MESFET enhancement type (*n*-channel)	 $I_D = k\,(V_{GS} - V_{GS\,(Th)})^2$ $k = \dfrac{I_{D(on)}}{(V_{GS(on)} - V_{GS\,(Th)})^2}$		$R_i > 10^{12}\ \Omega$ C_i: (1 − 5) pF

Equations

JFET:

$$I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_P}\right)^2$$

$$I_D = I_{DSS}\big|_{V_{GS}=0\ V}, \quad I_D = 0\ \text{mA}\big|_{V_{GS}=V_P}, \quad I_D = \left.\frac{I_{DSS}}{4}\right|_{V_{GS}=V_P/2}, \quad V_{GS} \cong 0.3V_P\big|_{I_D=I_{DSS}/2}$$

$$V_{GS} = V_P\left(1 - \sqrt{\frac{I_D}{I_{DSS}}}\right)$$

$$P_D = V_{DS}I_D$$

$$r_d = \frac{r_o}{(1 - V_{GS}/V_P)^2}$$

MOSFET (enhancement):

$$I_D = k(V_{GS} - V_T)^2$$

$$k = \frac{I_{D(\text{on})}}{(V_{GS(\text{on})} - V_T)^2}$$

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Chapter **6: FET Biasing**

Lecture 1

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CHAPTER OBJECTIVES

- Be able to perform a dc analysis of JFET, MOSFET
- Become proficient in the use of load-line analysis to examine FET networks.

7.1 INTRODUCTION

In Chapter 4 we found that the biasing levels for a silicon transistor configuration can be obtained using the approximate characteristic equations $V_{BE} = 0.7\ \text{V}$, $I_C = \beta I_B$, and $I_C \cong I_E$. The link between input and output variables is provided by β, which is assumed to be fixed in magnitude for the analysis to be performed. The fact that beta is a constant establishes a *linear* relationship between I_C and I_B. Doubling the value of I_B will double the level of I_C, and so on.

For the field-effect transistor, the relationship between input and output quantities is *nonlinear* due to the squared term in Shockley's equation. Linear relationships result in straight lines when plotted on a graph of one variable versus the other, whereas nonlinear functions result in curves as obtained for the transfer characteristics of a JFET. The nonlinear relationship between I_D and V_{GS} can complicate the mathematical approach to the dc analysis of FET configurations. A graphical approach may limit solutions to tenths-place accuracy, but it is a quicker method for most FET amplifiers. Since the graphical approach is in general the most popular, the analysis of this chapter will have graphical solutions rather than mathematical solutions.

$$I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_P}\right)^2$$

MOSFET (enhancement):

$$I_D = k(V_{GS} - V_T)^2$$

$$k = \frac{I_{D(on)}}{(V_{GS(on)} - V_T)^2}$$

Another distinct difference between the analysis of BJT and FET transistors is that:

The controlling variable for a BJT transistor is a current level, whereas for the FET a voltage is the controlling variable.

In both cases, however, the controlled variable on the output side is a current level that also defines the important voltage levels of the output circuit.

The general relationships that can be applied to the dc analysis of all FET amplifiers are

$$I_G \cong 0\ \text{A} \tag{7.1}$$

and

$$I_D = I_S \tag{7.2}$$

For JFETs and depletion-type MOSFETs , Shockley's equation is applied to relate the input and output quantities:

$$I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_P}\right)^2 \tag{7.3}$$

For enhancement-type MOSFETs , the following equation is applicable:

$$I_D = k(V_{GS} - V_T)^2 \tag{7.4}$$

It is particularly important to realize that all of the equations above are for the *field-effect transistor only*! They do not change with each network configuration so long as the device is in the active region. The network simply defines the level of current and voltage associated with the operating point through its own set of equations. In reality, the dc solution of BJT and FET networks is the solution of simultaneous equations established by the device and the network. The solution can be determined using a mathematical or graphical approach—a fact to be demonstrated by the first few networks to be analyzed. However, as noted earlier, the graphical approach is the most popular for FET networks and is employed in this book.

7.2 FIXED-BIAS CONFIGURATION

The simplest of biasing arrangements for the *n*-channel JFET appears in Fig. 7.1. Referred to as the fixed-bias configuration, it is one of the few FET configurations that can be solved just as directly using either a mathematical or a graphical approach. Both methods are included in this section to demonstrate the difference between the two methods and also to establish the fact that the same solution can be obtained using either approach.

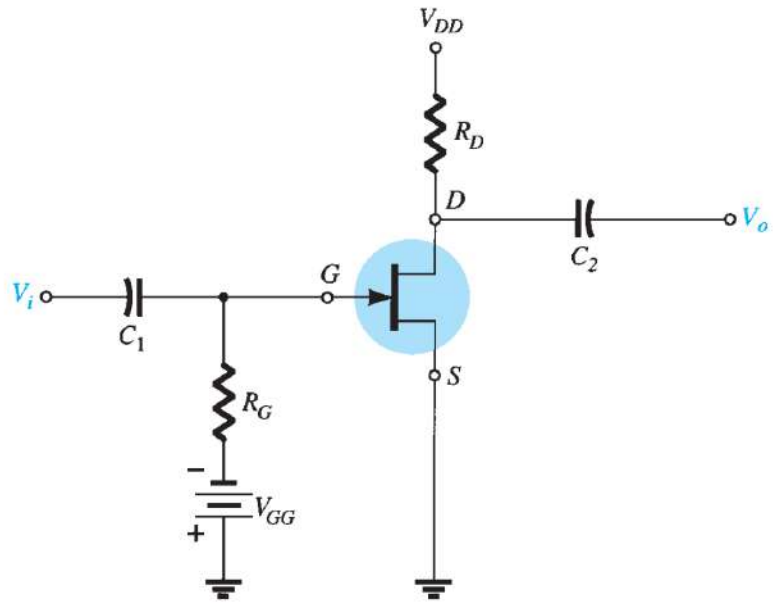


FIG. 7.1
Fixed-bias configuration.

The configuration of Fig. 7.1 includes the ac levels V_i and V_o and the coupling capacitors (C_1 and C_2). Recall that the coupling capacitors are "open circuits" for the dc analysis and low impedances (essentially short circuits) for the ac analysis. The resistor R_G is present to ensure that V_i appears at the input to the FET amplifier for the ac analysis (Chapter 8). For the dc analysis,

$$I_G \cong 0 \text{ A}$$

and

$$V_{R_G} = I_G R_G = (0 \text{ A}) R_G = 0 \text{ V}$$

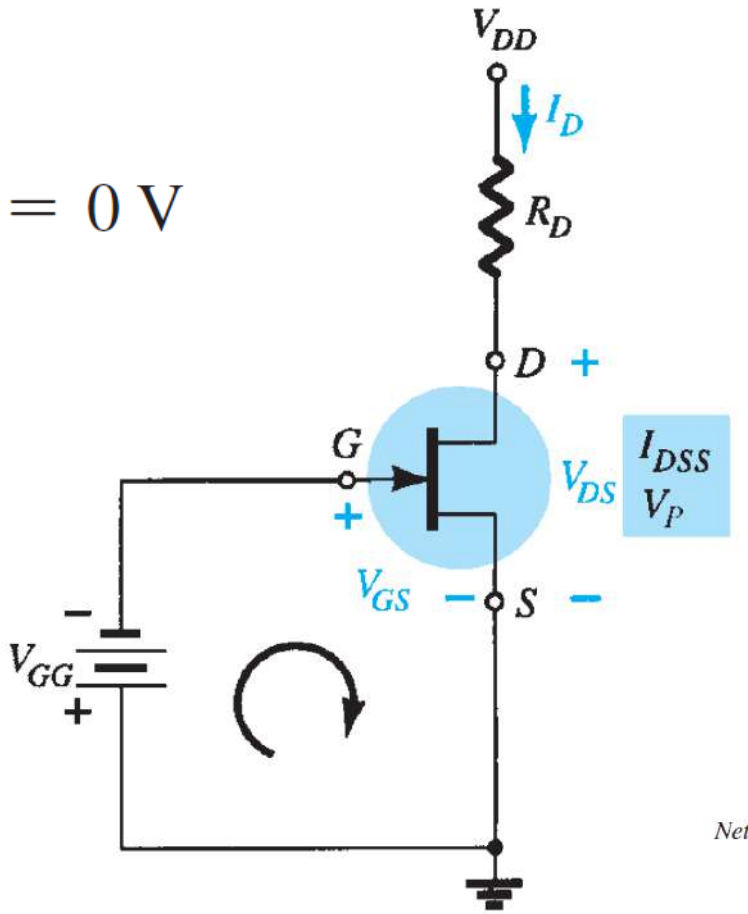


FIG. 7.2
Network for dc analysis.

The zero-volt drop across R_G permits replacing R_G by a short-circuit equivalent, as appearing in the network of Fig. 7.2, specifically redrawn for the dc analysis.

The fact that the negative terminal of the battery is connected directly to the defined positive potential of V_{GS} clearly reveals that the polarity of V_{GS} is directly opposite to that of V_{GG}. Applying Kirchhoff's voltage law in the clockwise direction of the indicated loop of Fig. 7.2 results in

$$-V_{GG} - V_{GS} = 0$$

and

$$\boxed{V_{GS} = -V_{GG}}$$

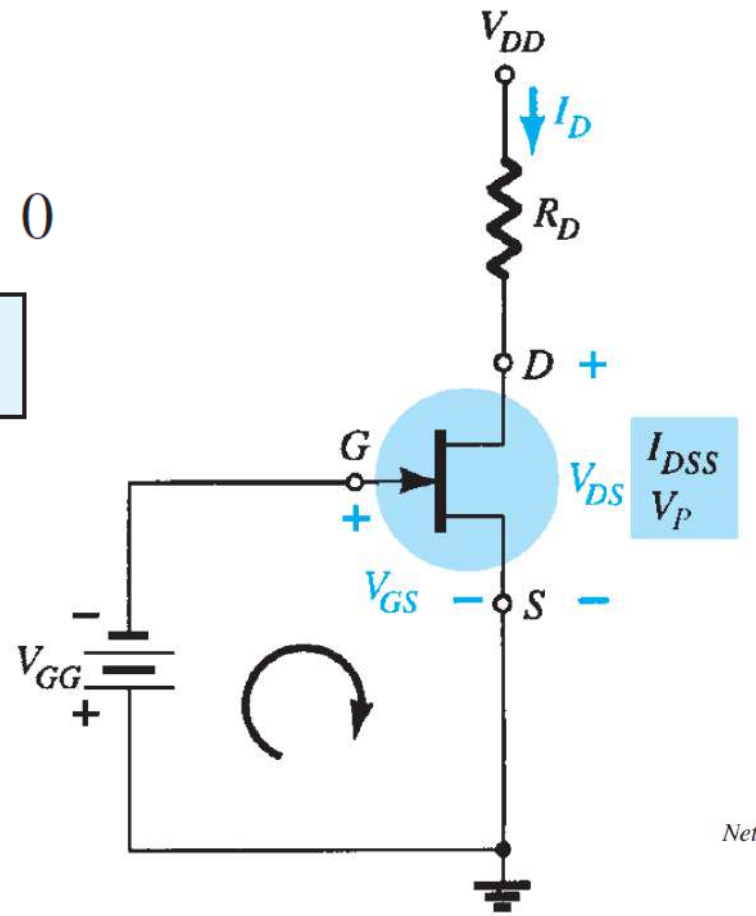


FIG. 7.2
Network for dc analysis.

Since V_{GG} is a fixed dc supply, the voltage V_{GS} is fixed in magnitude, resulting in the designation "fixed-bias configuration."

The resulting level of drain current I_D is now controlled by Shockley's equation:

$$I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_P}\right)^2$$

Since V_{GS} is a fixed quantity for this configuration, its magnitude and sign can simply be substituted into Shockley's equation and the resulting level of I_D calculated. This is one of the few instances in which a mathematical solution to a FET configuration is quite direct.

A graphical analysis would require a plot of Shockley's equation as shown in Fig. 7.3. Recall that choosing $V_{GS} = V_P/2$ will result in a drain current of $I_{DSS}/4$ when plotting the equation. For the analysis of this chapter, the three points defined by I_{DSS}, V_P, and the intersection just described will be sufficient for plotting the curve.

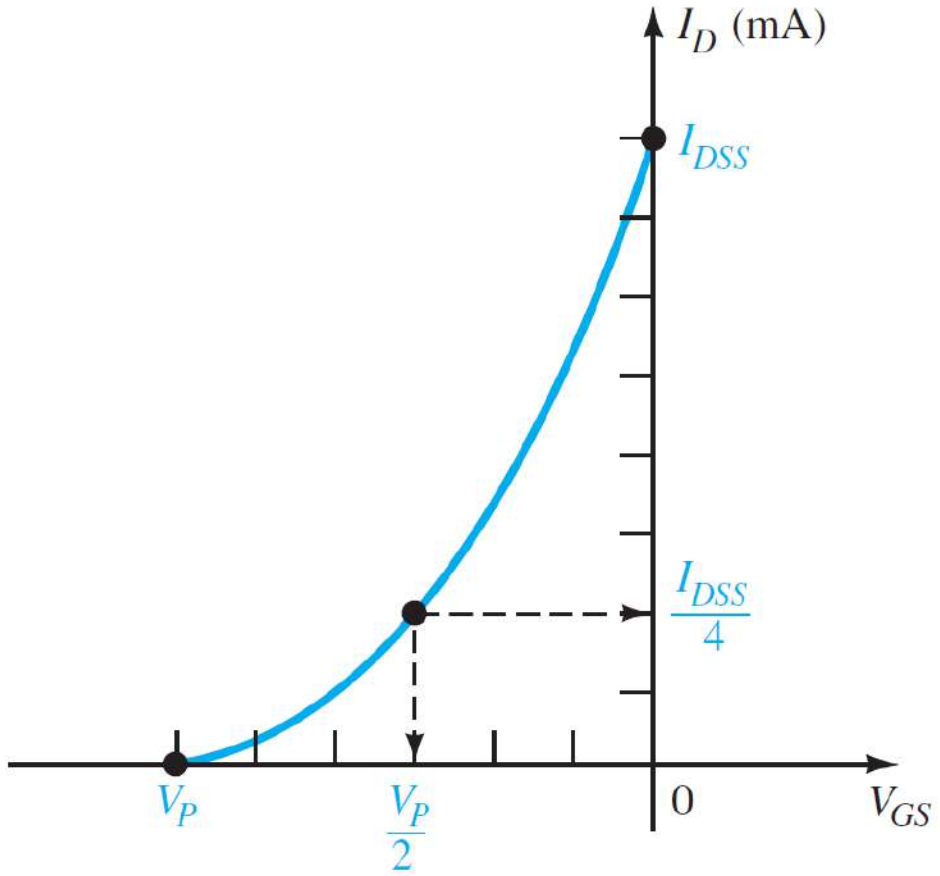


FIG. 7.3
Plotting Shockley's equation.

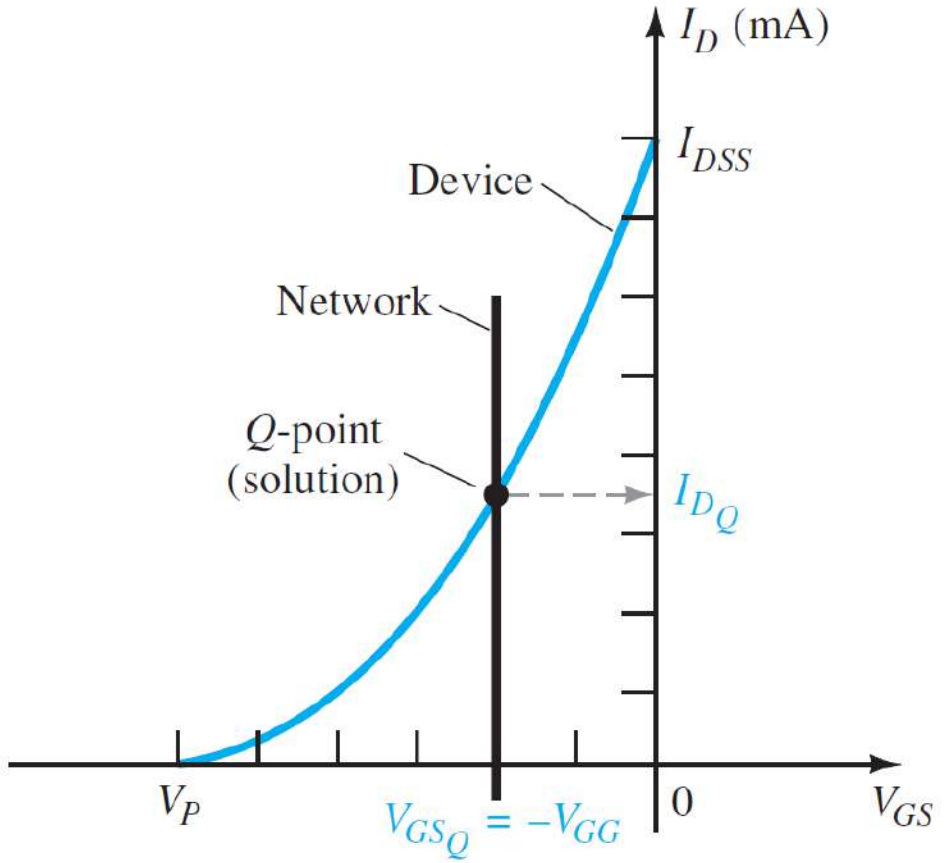


FIG. 7.4
Finding the solution for the fixed-bias configuration.

In Fig. 7.4, the fixed level of V_{GS} has been superimposed as a vertical line at $V_{GS} = -V_{GG}$. At any point on the vertical line, the level of V_{GS} is $-V_{GG}$—the level of I_D must simply be determined on this vertical line. The point where the two curves intersect is the common solution to the configuration—commonly referred to as the *quiescent* or *operating point.*

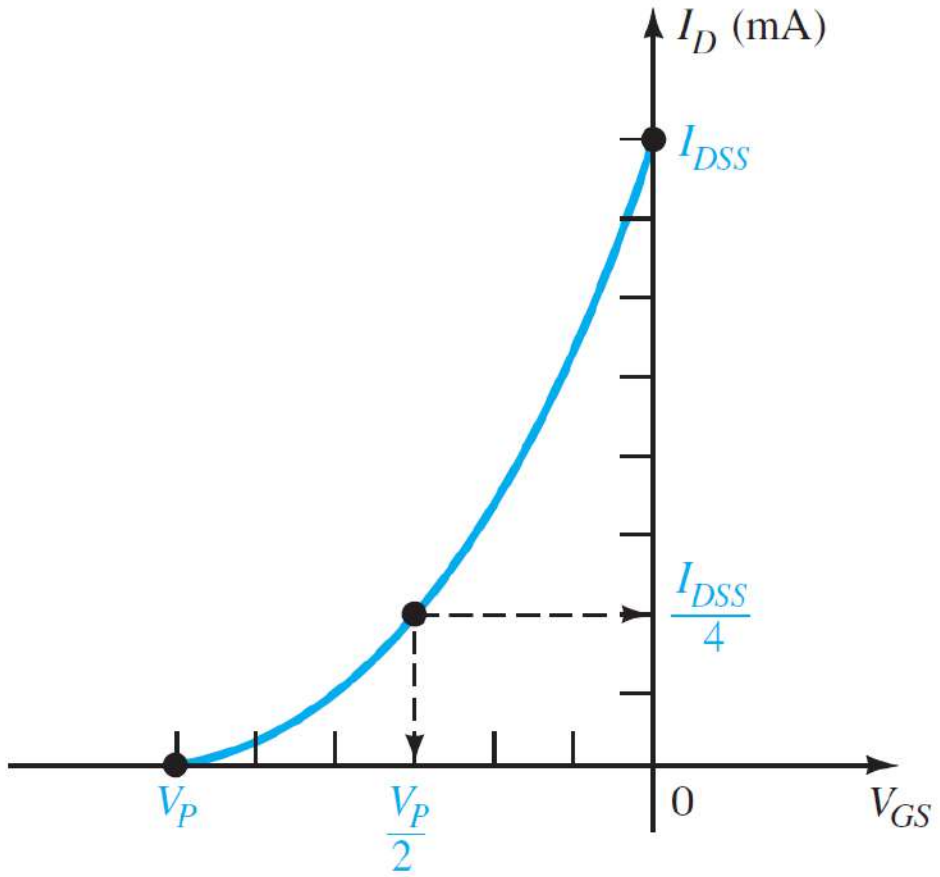


FIG. 7.3
Plotting Shockley's equation.

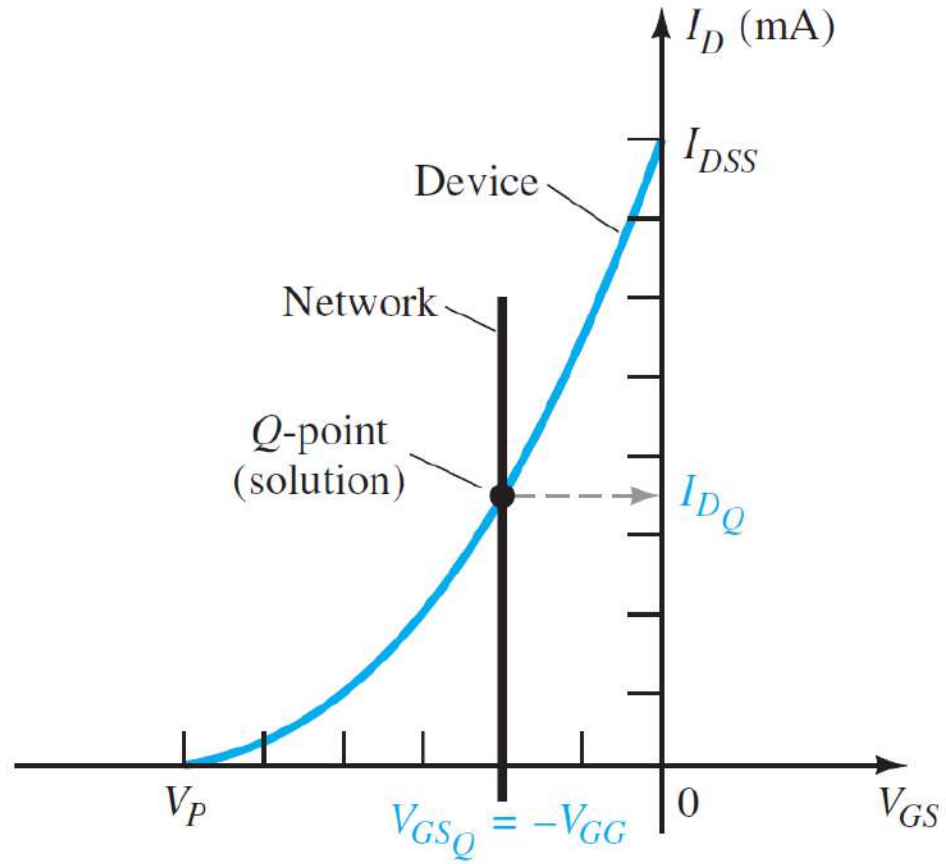


FIG. 7.4
Finding the solution for the fixed-bias configuration.

that once the network of Fig. 7.1 is constructed and operating, the dc levels of I_D and V_{GS} that will be measured by the meters of Fig. 7.5 are the quiescent values defined by Fig. 7.4.

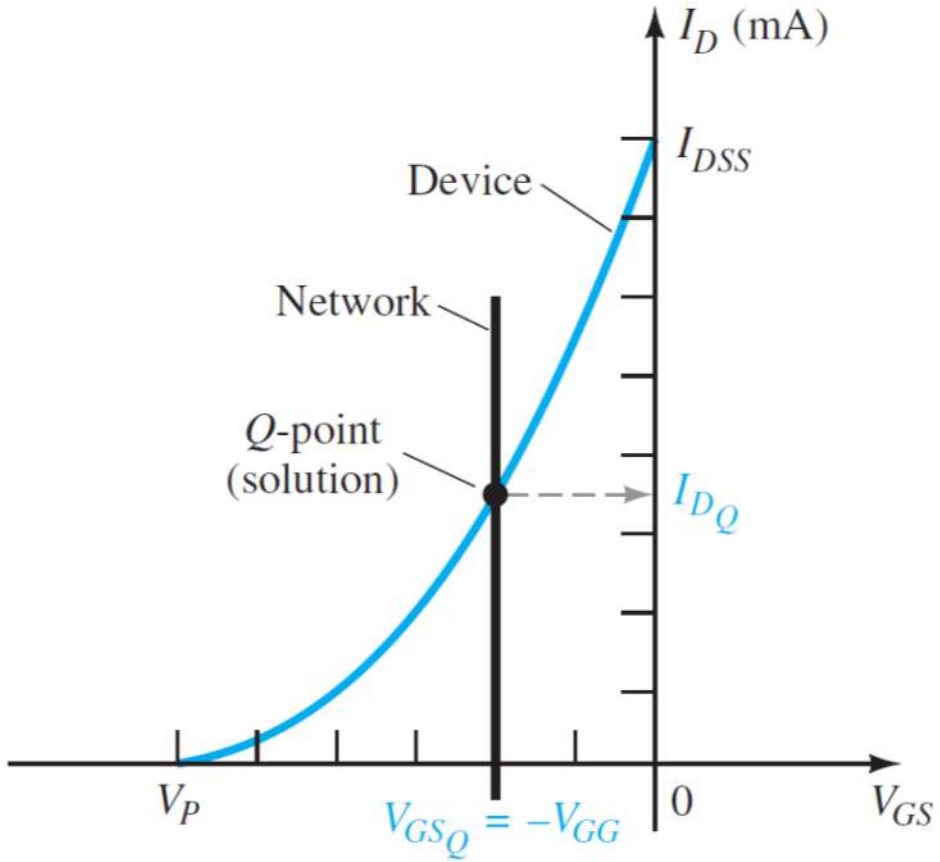


FIG. 7.4

Finding the solution for the fixed-bias configuration.

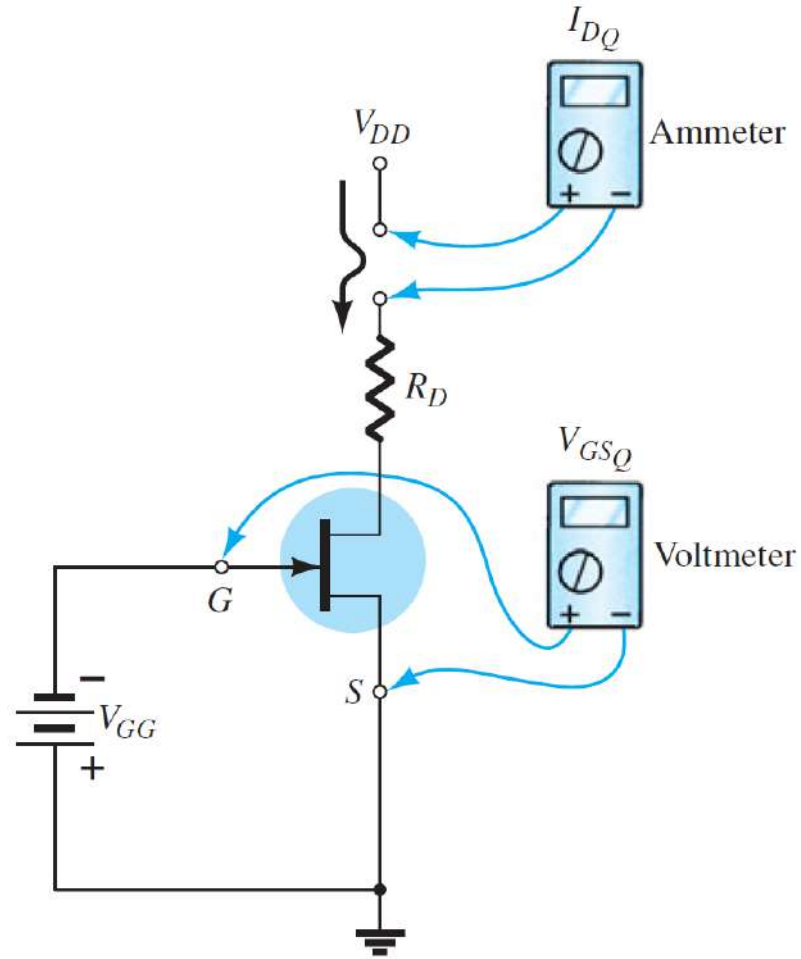


FIG. 7.5

Measuring the quiescent values of I_D and V_{GS}.

The drain-to-source voltage of the output section can be determined by applying Kirchhoff's voltage law as follows:

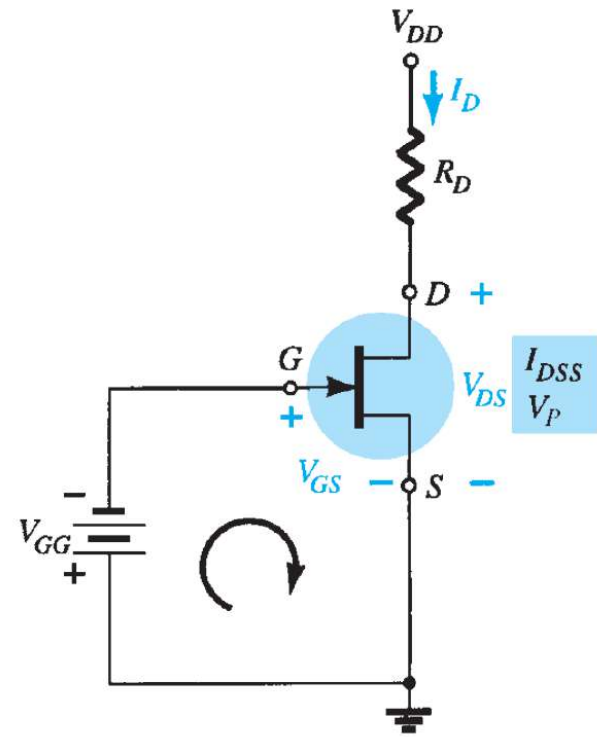
$$+V_{DS} + I_D R_D - V_{DD} = 0$$

and

$$\boxed{V_{DS} = V_{DD} - I_D R_D} \tag{7.6}$$

Recall that single-subscript voltages refer to the voltage at a point with respect to ground. For the configuration of Fig. 7.2,

$$\boxed{V_S = 0\,\text{V}}$$



Using double-subscript notation, we have

$$V_{DS} = V_D - V_S$$

The fact that $V_D = V_{DS}$ and $V_G = V_{GS}$ is fairly obvious from the fact that $V_S = 0\,\text{V}$,

EXAMPLE 7.1 Determine the following for the network of Fig. 7.6:

a. V_{GS_Q}.
b. I_{D_Q}.
c. V_{DS}.
d. V_D.
e. V_G.
f. V_S.

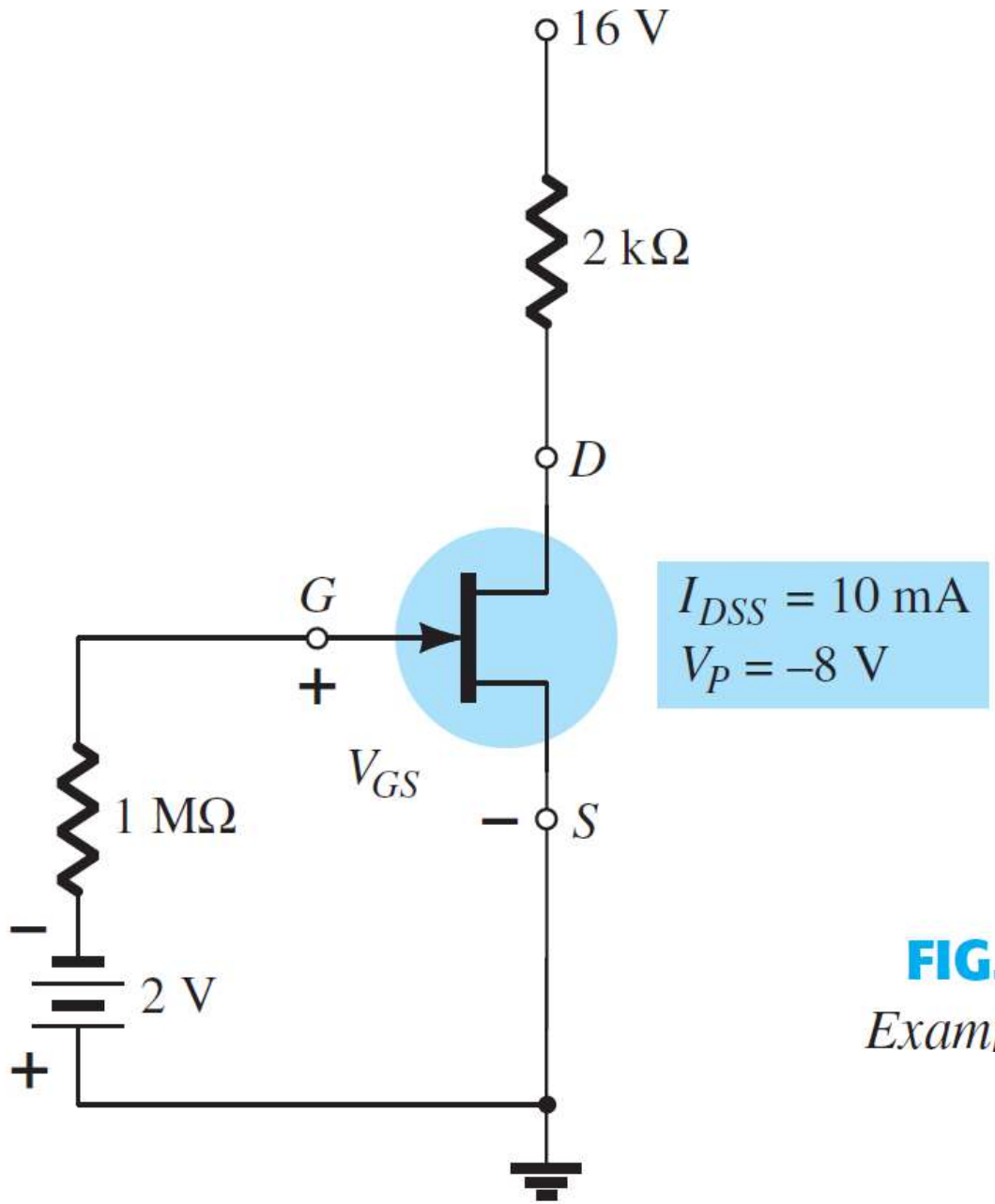


FIG. 7.6
Example 7.1.

Solution:

Mathematical Approach

a. $V_{GS_Q} = -V_{GG} = \mathbf{-2\ V}$

b. $$I_{D_Q} = I_{DSS}\left(1 - \frac{V_{GS}}{V_P}\right)^2 = 10\ \text{mA}\left(1 - \frac{-2\ \text{V}}{-8\ \text{V}}\right)^2$$
$$= 10\ \text{mA}(1 - 0.25)^2 = 10\ \text{mA}(0.75)^2 = 10\ \text{mA}(0.5625)$$
$$= \mathbf{5.625\ mA}$$

c. $$V_{DS} = V_{DD} - I_D R_D = 16\ \text{V} - (5.625\ \text{mA})(2\ \text{k}\Omega)$$
$$= 16\ \text{V} - 11.25\ \text{V} = \mathbf{4.75\ V}$$

d. $V_D = V_{DS} = \mathbf{4.75\ V}$

e. $V_G = V_{GS} = \mathbf{-2\ V}$

f. $V_S = \mathbf{0\ V}$

Graphical Approach The resulting Shockley curve and the vertical line at $V_{GS} = -2$ V are provided in Fig. 7.7. It is certainly difficult to read beyond the second place without

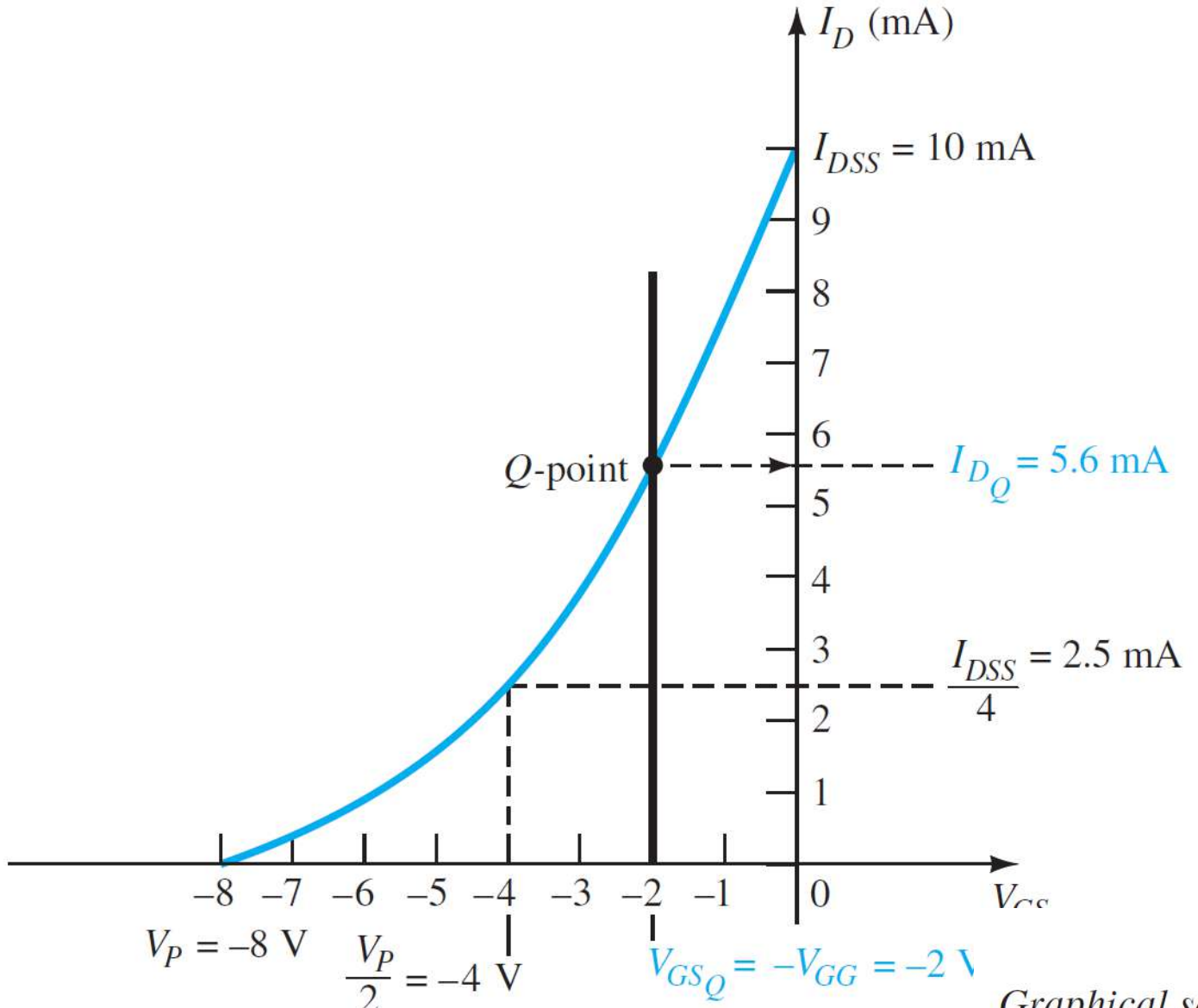


FIG. 7.7
Graphical solution for the network of Fig. 7.6.

significantly increasing the size of the figure, but a solution of 5.6 mA from the graph of Fig. 7.7 is quite acceptable.

a. Therefore,

$$V_{GS_Q} = -V_{GG} = \mathbf{-2\ V}$$

b. $I_{D_Q} = \mathbf{5.6\ mA}$

c. $V_{DS} = V_{DD} - I_D R_D = 16\ \text{V} - (5.6\ \text{mA})(2\ \text{k}\Omega)$
$= 16\ \text{V} - 11.2\ \text{V} = \mathbf{4.8\ V}$

d. $V_D = V_{DS} = \mathbf{4.8\ V}$

e. $V_G = V_{GS} = \mathbf{-2\ V}$

f. $V_S = \mathbf{0\ V}$

The results clearly confirm the fact that the mathematical and graphical approaches generate solutions that are quite close.

7.3 SELF-BIAS CONFIGURATION

The self-bias configuration eliminates the need for two dc supplies. The controlling gate-to-source voltage is now determined by the voltage across a resistor R_S introduced in the source leg of the configuration as shown in Fig. 7.8.

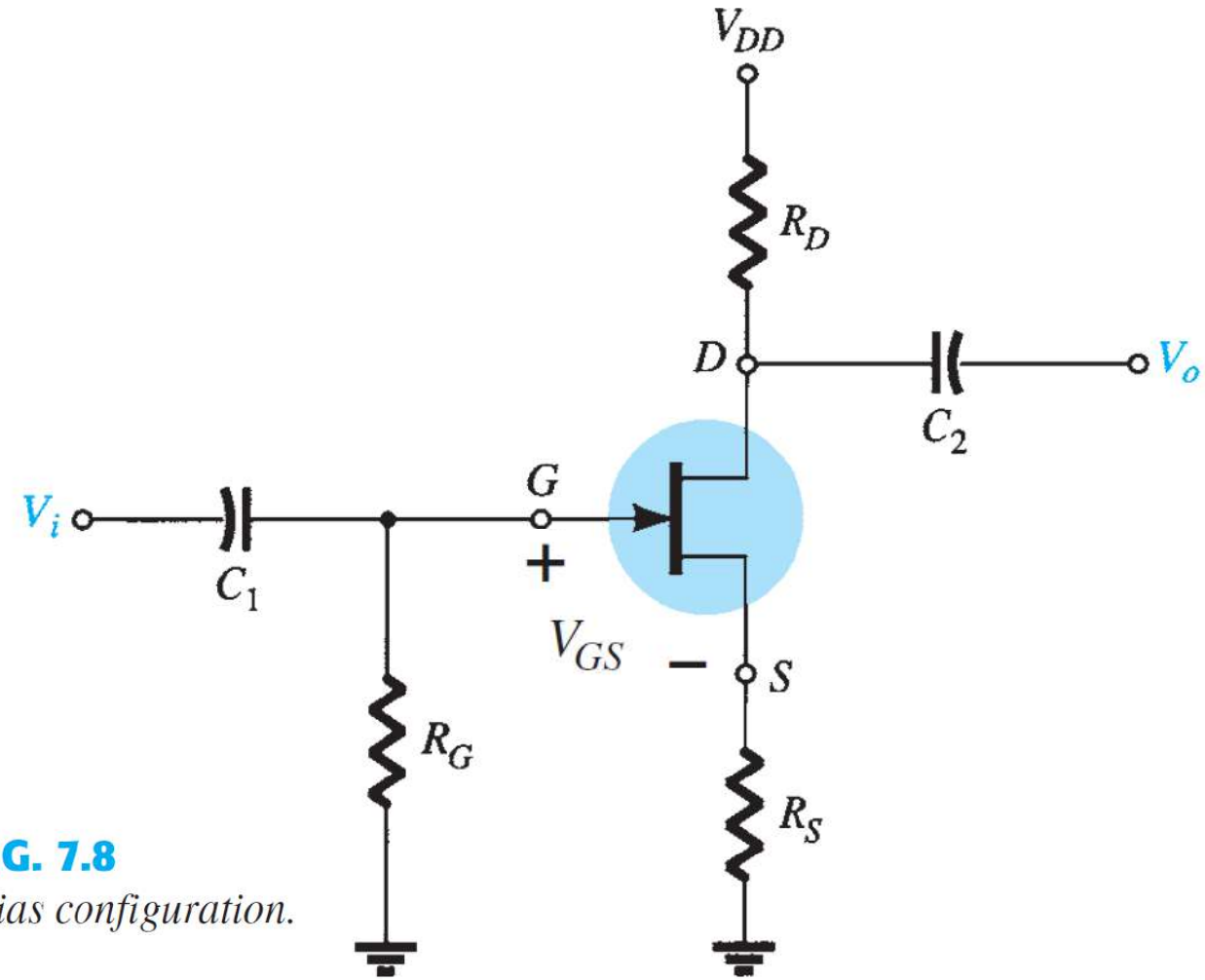


FIG. 7.8
JFET self-bias configuration.

For the dc analysis, the capacitors can again be replaced by "open circuits" and the resistor R_G replaced by a short-circuit equivalent since $I_G = 0$ A. The result is the network of Fig. 7.9 for the important dc analysis.

The current through R_S is the source current I_S, but $I_S = I_D$ and

$$V_{R_S} = I_D R_S$$

For the indicated closed loop of Fig. 7.9, we find that

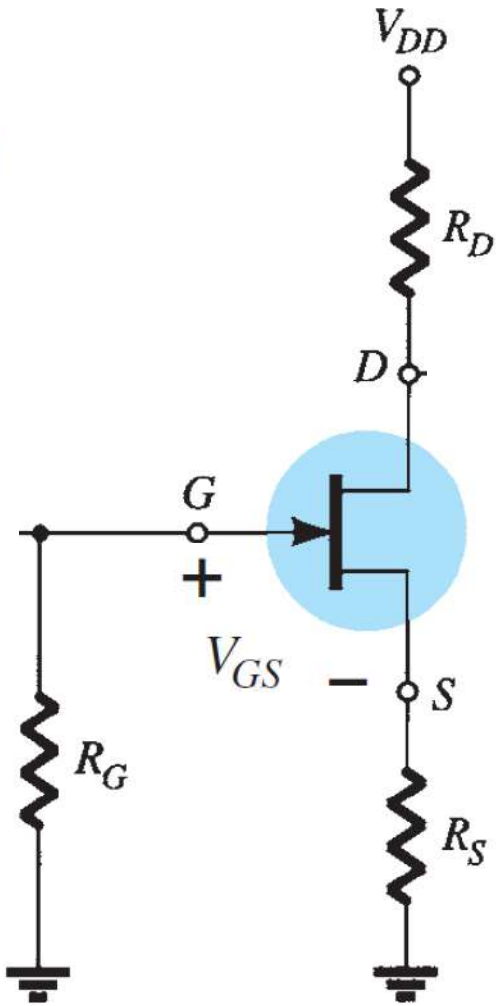
$$-V_{GS} - V_{R_S} = 0$$

and

$$V_{GS} = -V_{R_S}$$

or

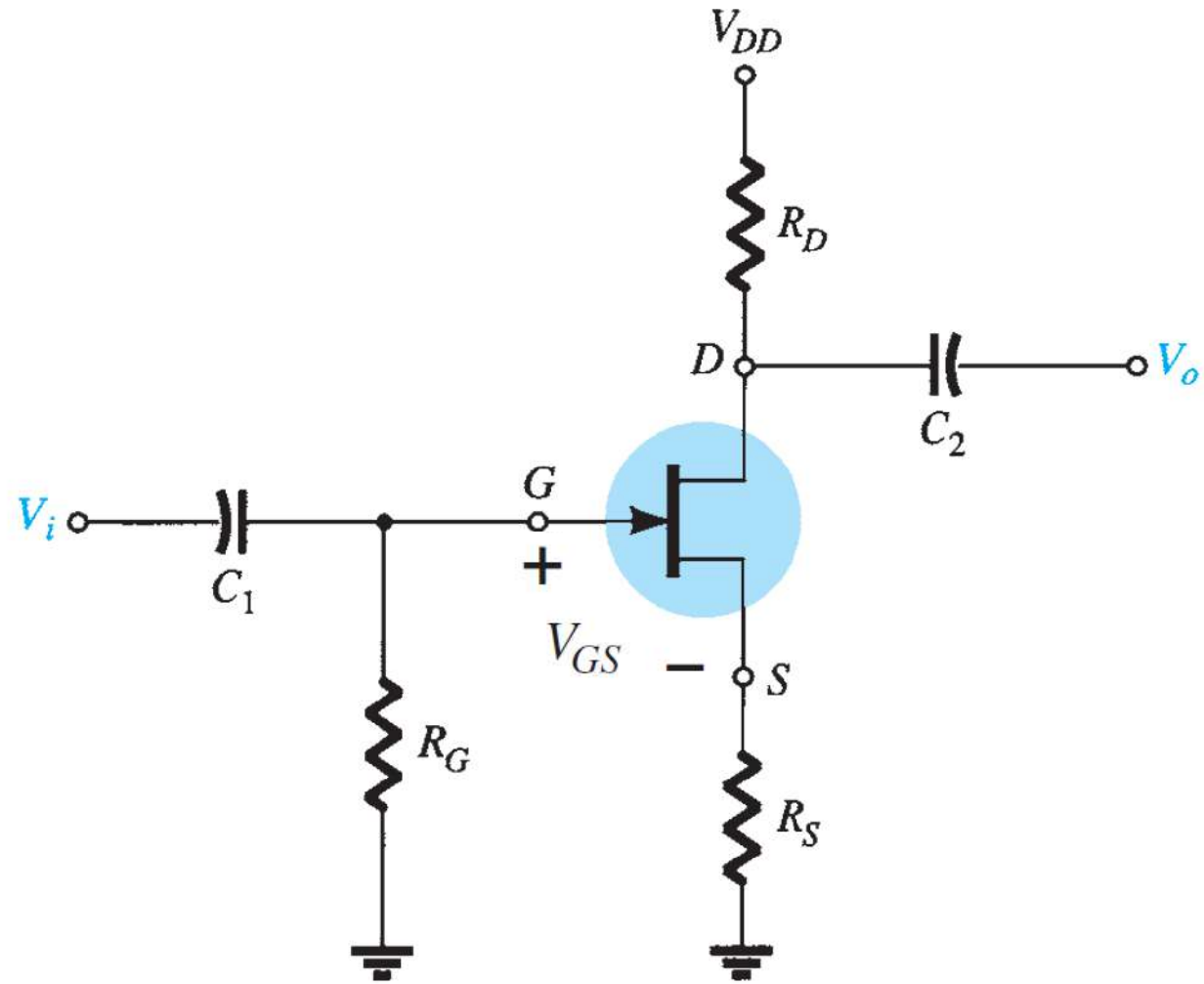
$$\boxed{V_{GS} = -I_D R_S}$$



Note in this case that V_{GS} is a function of the output current I_D and not fixed in magnitude as occurred for the fixed-bias configuration.

Equation (7.10) is defined by the network configuration, and Shockley's equation relates the input and output quantities of the device. Both equations relate the same two variables, I_D and V_{GS}, permitting either a mathematical or a graphical solution.

$$V_{GS} = -I_D R_S \qquad \textbf{(7.10)}$$



A mathematical solution could be obtained simply by substituting Eq. (7.10) into Shockley's equation as follows:

$$I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_P}\right)^2$$

$$= I_{DSS}\left(1 - \frac{-I_D R_S}{V_P}\right)^2$$

or

$$I_D = I_{DSS}\left(1 + \frac{I_D R_S}{V_P}\right)^2$$

By performing the squaring process indicated and rearranging terms, we obtain an equation of the following form:

$$I_D^2 + K_1 I_D + K_2 = 0$$

The quadratic equation can then be solved for the appropriate solution for I_D.

The sequence above defines the mathematical approach. The graphical approach requires that we first establish the device transfer characteristics as shown in Fig. 7.10. Since Eq. (7.10) defines a straight line on the same graph, let us now identify two points on the graph that are on the line and simply draw a straight line between the two points. The most obvious condition to apply is $I_D = 0$ A since it results in $V_{GS} = -I_D R_S = (0\text{ A})R_S = 0$ V. For Eq. (7.10), therefore, one point on the straight line is defined by $I_D = 0$ A and $V_{GS} = 0$ V, as appearing on Fig. 7.10.

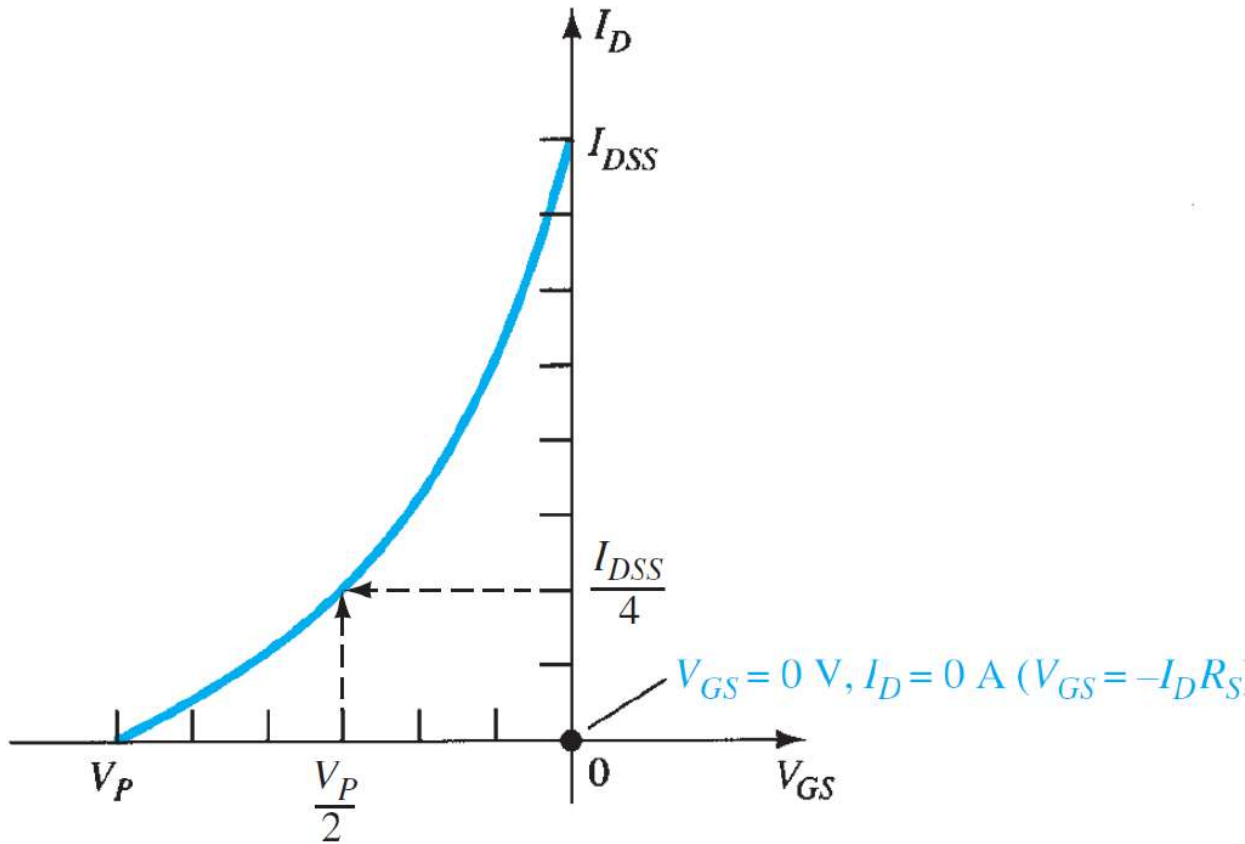


FIG. 7.10

Defining a point on the self-bias line.

The second point for Eq. (7.10) requires that a level of V_{GS} or I_D be chosen and the corresponding level of the other quantity be determined using Eq. (7.10). The resulting levels of I_D and V_{GS} will then define another point on the straight line and permit the drawing of the straight line. Suppose, for example, that we choose a level of I_D equal to one-half the saturation level. That is,

$$I_D = \frac{I_{DSS}}{2}$$

Then

$$V_{GS} = -I_D R_S = -\frac{I_{DSS} R_S}{2}$$

The result is a second point for the straight-line plot as shown in Fig. 7.11. The straight line as defined by Eq. (7.10) is then drawn and the quiescent point obtained at the intersection of the straight-line plot and the device characteristic curve. The quiescent values of I_D and V_{GS} can then be determined and used to find the other quantities of interest.

The level of V_{DS} can be determined by applying Kirchhoff's voltage law to the output circuit, with the result that

$$V_{R_S} + V_{DS} + V_{R_D} - V_{DD} = 0$$

$$V_{DS} = V_{DD} - V_{R_S} - V_{R_D} = V_{DD} - I_S R_S - I_D R_D$$

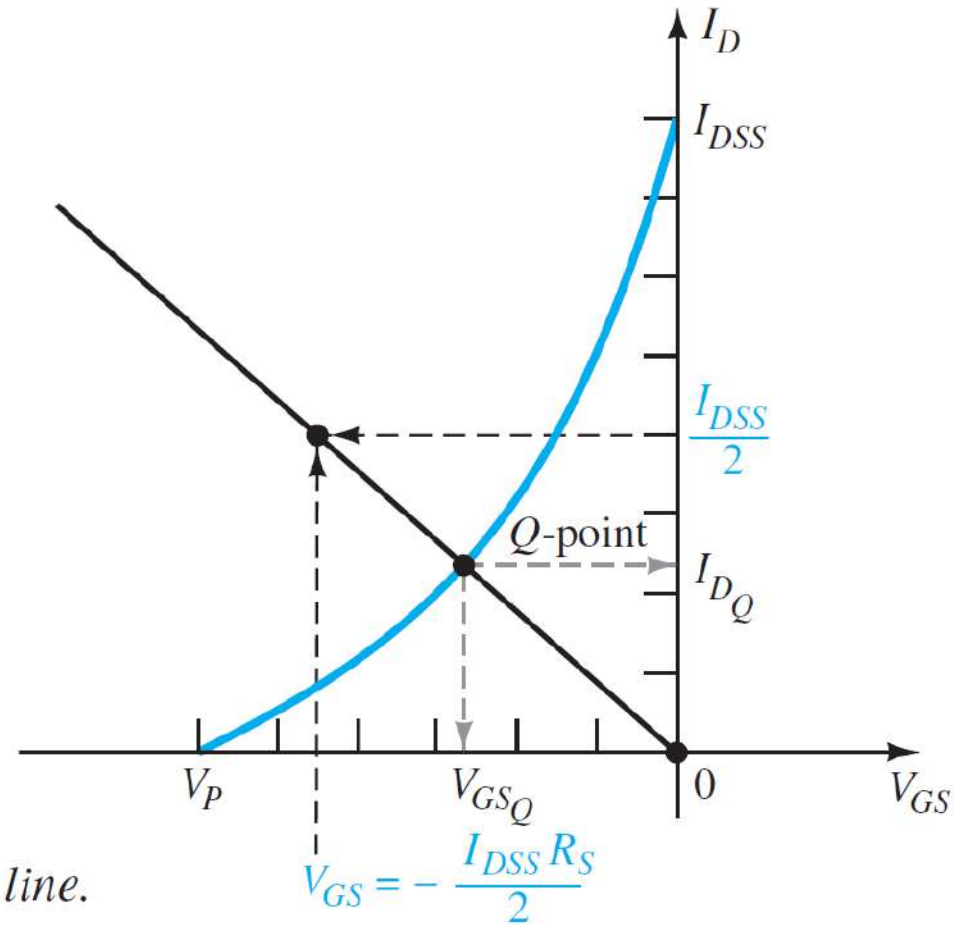


FIG. 7.11
Sketching the self-bias line.

but $$I_D = I_S$$

and $$\boxed{V_{DS} = V_{DD} - I_D(R_S + R_D)} \tag{7.11}$$

In addition,

$$\boxed{V_S = I_D R_S} \tag{7.12}$$

$$\boxed{V_G = 0\,\text{V}} \tag{7.13}$$

and $$\boxed{V_D = V_{DS} + V_S = V_{DD} - V_{R_D}} \tag{7.14}$$

EXAMPLE 7.2 Determine the following for the network of Fig. 7.12:

a. V_{GS_Q}.
b. I_{D_Q}.
c. V_{DS}.
d. V_S.
e. V_G.
f. V_D.

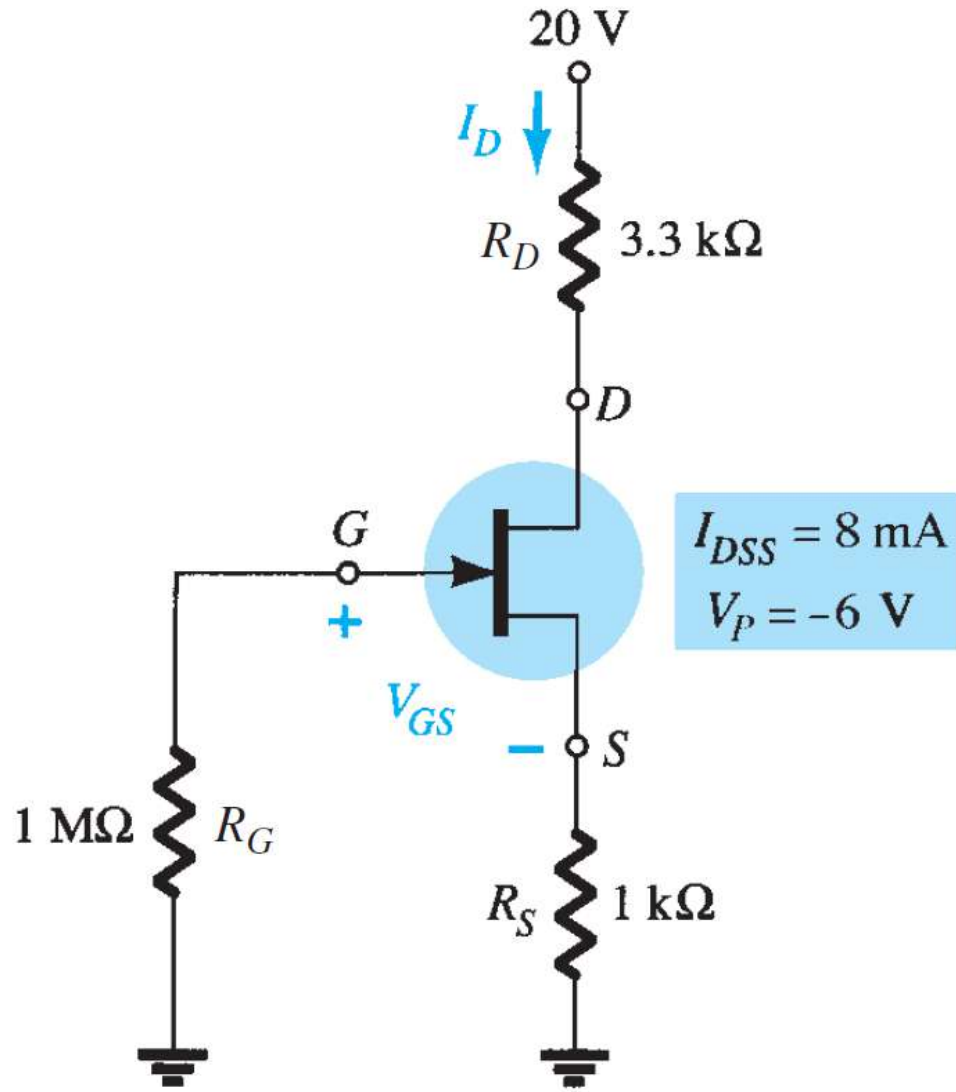


FIG. 7.12
Example 7.2.

Solution:

a. The gate-to-source voltage is determined by

$$V_{GS} = -I_D R_S$$

Choosing $I_D = 4$ mA, we obtain

$$V_{GS} = -(4\text{ mA})(1\text{ k}\Omega) = -4\text{ V}$$

The result is the plot of Fig. 7.13 as defined by the network.

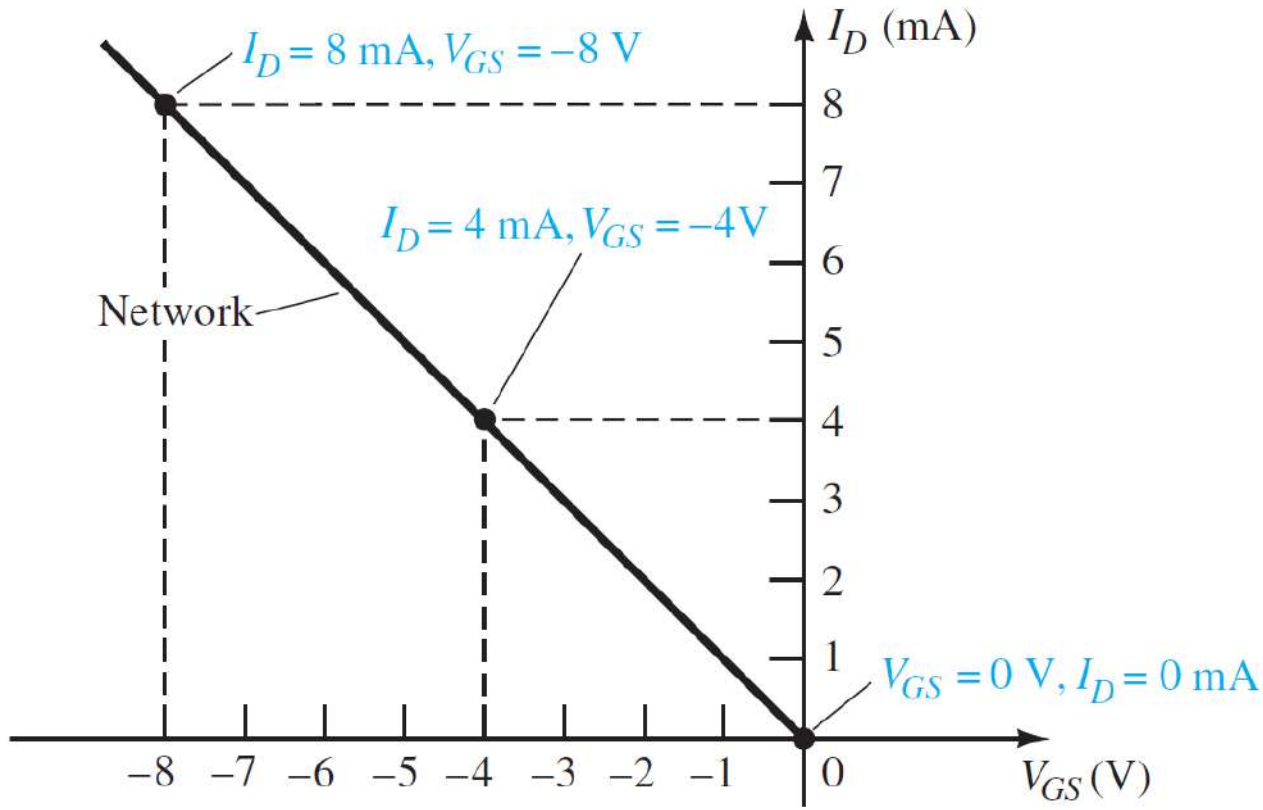


FIG. 7.13

Sketching the self-bias line for the network of Fig. 7.12.

For Shockley's equation, if we choose $V_{GS} = V_P/2 = -3\,\text{V}$, we find that $I_D = I_{DSS}/4 = 8\,\text{mA}/4 = 2\,\text{mA}$, and the plot of Fig. 7.14 will result, representing the characteristics of the device. The solution is obtained by superimposing the network characteristics defined by Fig. 7.13 on the device characteristics of Fig. 7.14

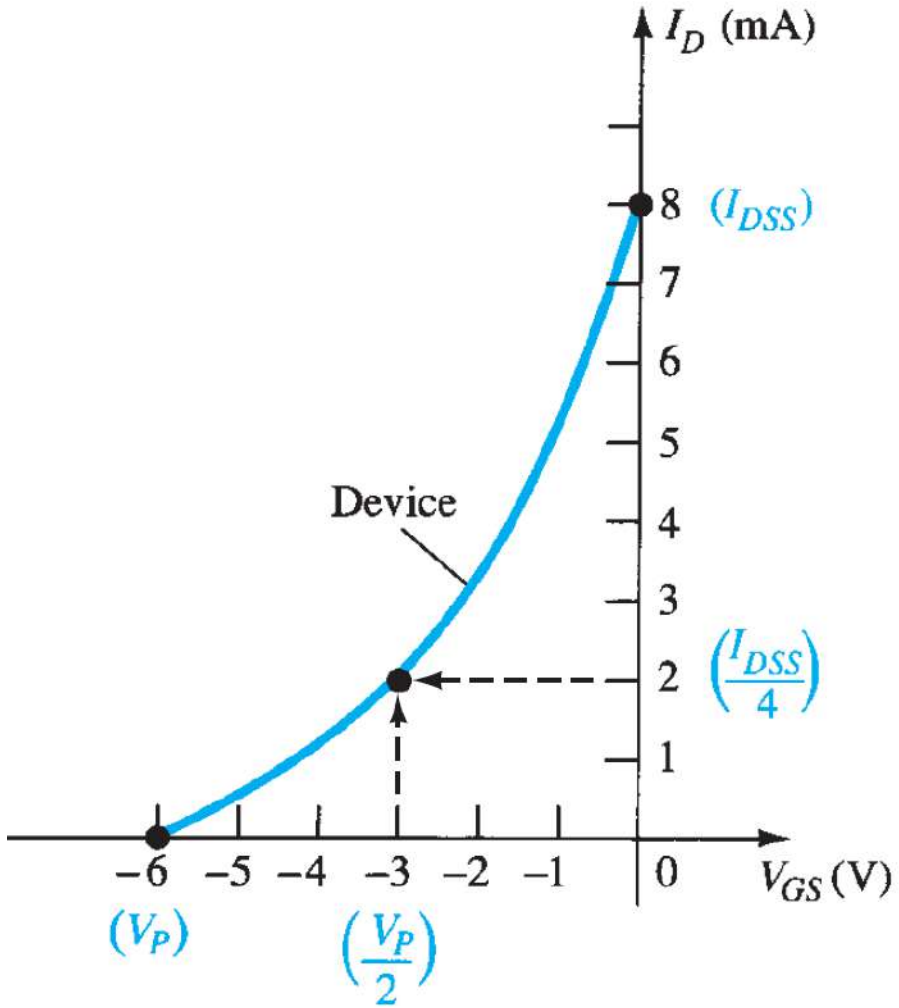


FIG. 7.14
Sketching the device characteristics for the JFET of Fig. 7.12.

finding
the point of intersection of the two as indicated on Fig. 7.15. The resulting operating point results in a quiescent value of gate-to-source voltage of

$$V_{GS_Q} = -2.6\ \text{V}$$

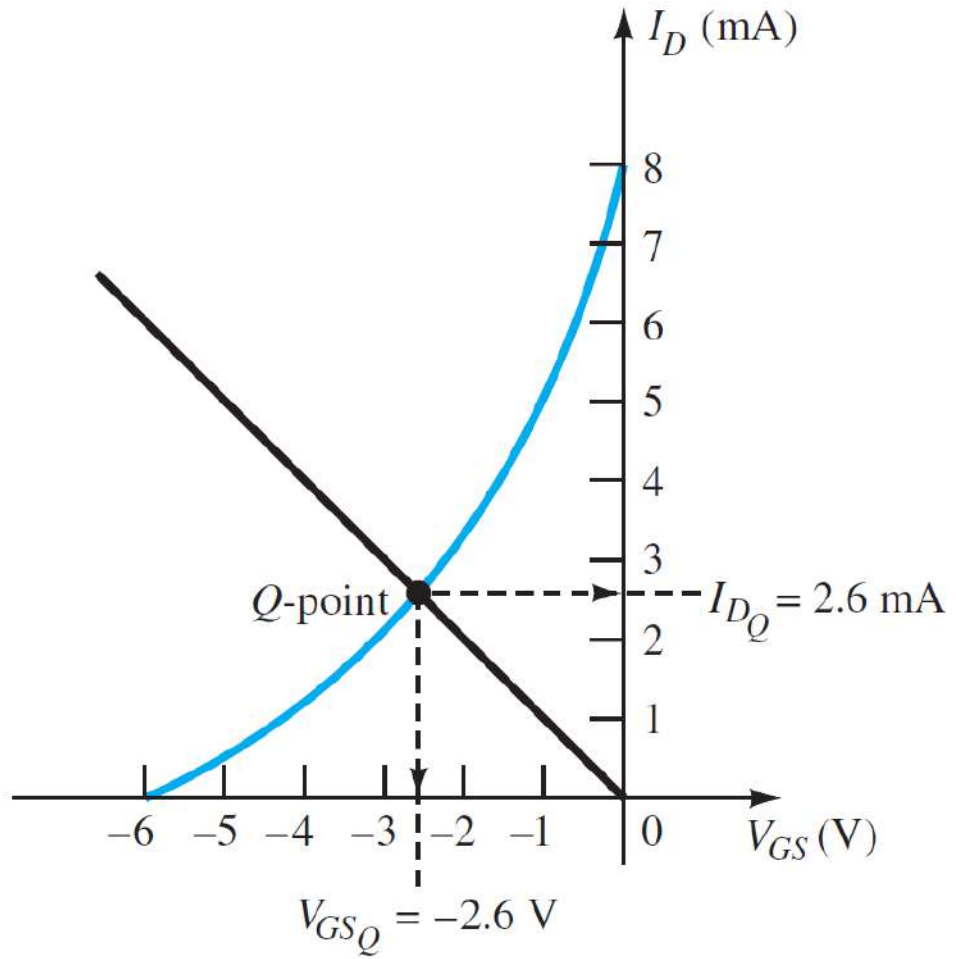


FIG. 7.15
Determining the Q-point for the network of Fig. 7.12.

b. At the quiescent point

$$I_{D_Q} = \mathbf{2.6\ mA}$$

c. Eq. (7.11): $V_{DS} = V_{DD} - I_D(R_S + R_D)$

$$= 20\ \text{V} - (2.6\ \text{mA})(1\ \text{k}\Omega + 3.3\ \text{k}\Omega)$$

$$= 20\ \text{V} - 11.18\ \text{V}$$

$$= \mathbf{8.82\ V}$$

d. Eq. (7.12): $V_S = I_D R_S$

$= (2.6\,\text{mA})(1\,\text{k}\Omega)$

$= \mathbf{2.6\,V}$

e. Eq. (7.13): $V_G = \mathbf{0\,V}$

f. Eq. (7.14): $V_D = V_{DS} + V_S = 8.82\,\text{V} + 2.6\,\text{V} = \mathbf{11.42\,V}$

or $V_D = V_{DD} - I_D R_D = 20\,\text{V} - (2.6\,\text{mA})(3.3\,\text{k}\Omega) = \mathbf{11.42\,V}$

EXAMPLE 7.3 Find the quiescent point for the network of Fig. 7.12 if:

a. $R_S = 100\ \Omega$.
b. $R_S = 10\ \text{k}\Omega$.

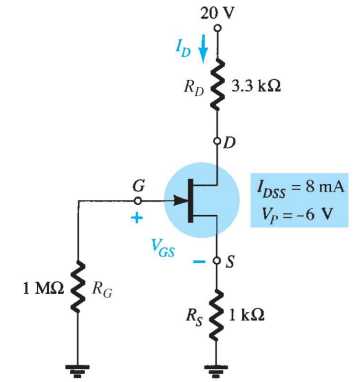


FIG. 7.12
Example 7.2.

Solution: Both $R_S = 100\ \Omega$ and $R_S = 10\ \text{k}\Omega$ are plotted on Fig. 7.16.

a. For $R_S = 100\ \Omega$:

$$I_{D_Q} \cong \mathbf{6.4\ mA}$$

and from Eq. (7.10),

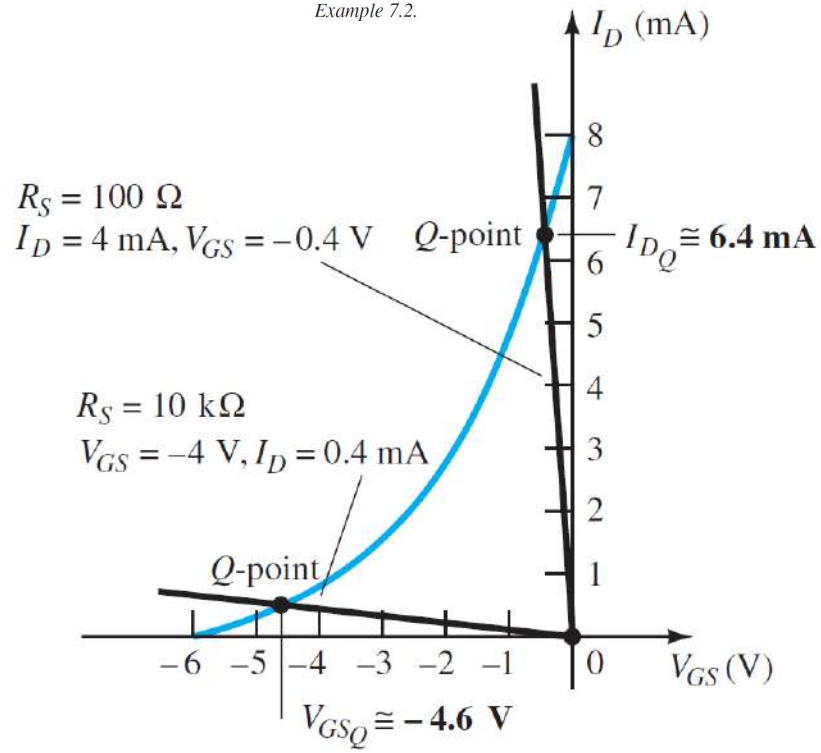
$$V_{GS_Q} \cong \mathbf{-0.64\ V}$$

b. For $R_S = 10\ \text{k}\Omega$

$$V_{GS_Q} \cong \mathbf{-4.6\ V}$$

and from Eq. (7.10),

$$I_{D_Q} \cong \mathbf{0.46\ mA}$$



In particular, note how lower levels of R_S bring the load line of the network closer to the I_D axis, whereas increasing levels of R_S bring the load line closer to the V_{GS} axis.

7.4 VOLTAGE-DIVIDER BIASING

The voltage-divider bias arrangement applied to BJT transistor amplifiers is also applied to FET amplifiers as demonstrated by Fig. 7.17. The basic construction is exactly the same, but the dc analysis of each is quite different. $I_G = 0$ A for FET amplifiers, but the magnitude of I_B for common-emitter BJT amplifiers can affect the dc levels of current and voltage in both the input and output circuits. Recall that I_B provides the link between input and output circuits for the BJT voltage-divider configuration, whereas V_{GS} does the same for the FET configuration.

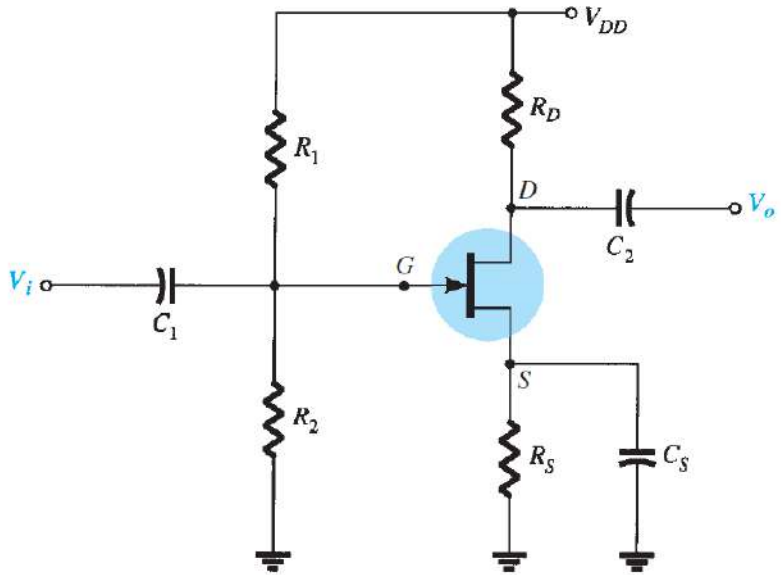


FIG. 7.17
Voltage-divider bias arrangement.

The network of Fig. 7.17 is redrawn as shown in Fig. 7.18 for the dc analysis. Note that all the capacitors, including the bypass capacitor C_S, have been replaced by an "open-circuit" equivalent in Fig. 7.18b. In addition, the source V_{DD} was separated into two equivalent sources to permit a further separation of the input and output regions of the network. Since $I_G = 0$ A, Kirchhoff's current law requires that $I_{R_1} = I_{R_2}$,

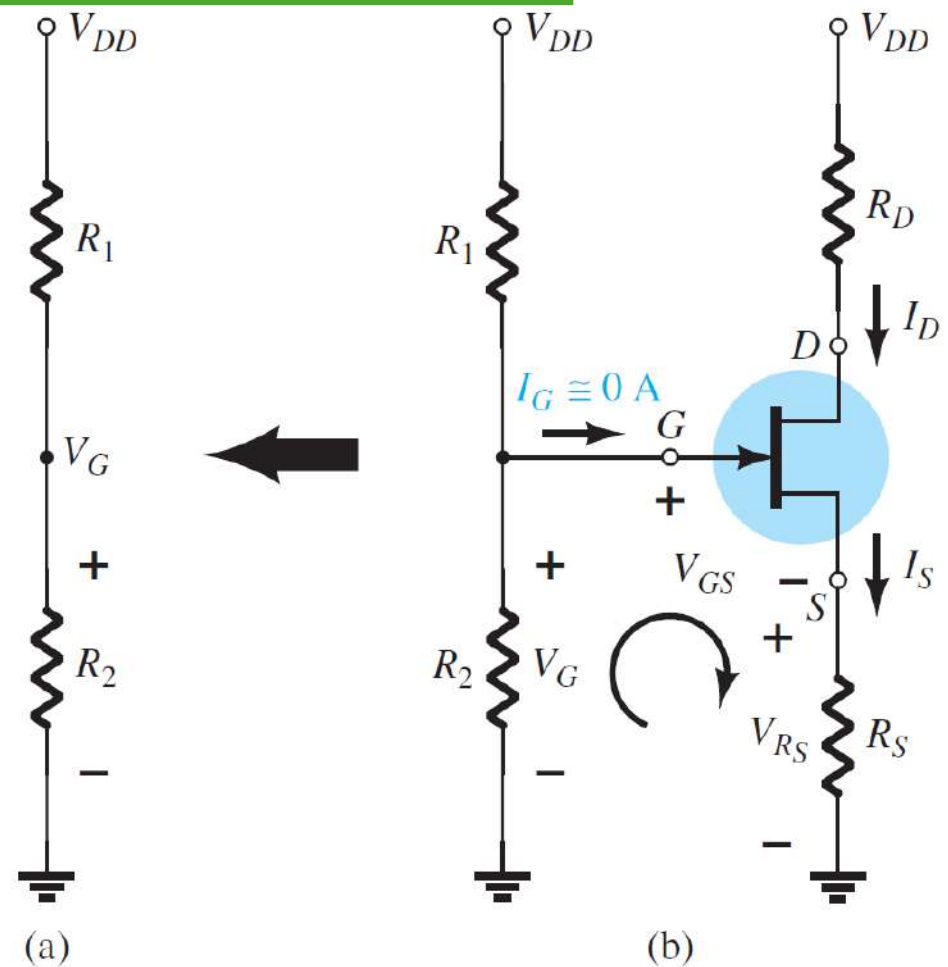


FIG. 7.18
Redrawn network of Fig. 7.17 for dc analysis.

The voltage V_G, equal to the voltage across R_2, can be found using the voltage-divider rule and Fig. 7.18a as follows:

$$V_G = \frac{R_2 V_{DD}}{R_1 + R_2} \tag{7.15}$$

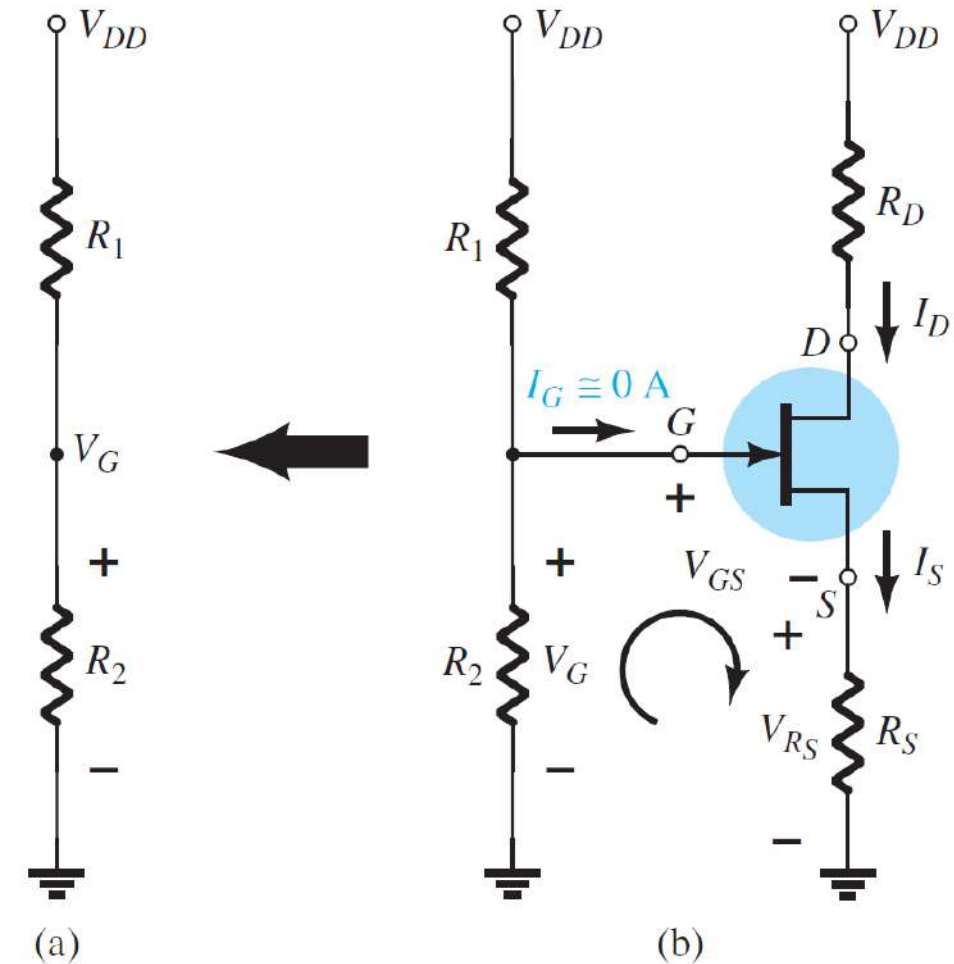


FIG. 7.18
Redrawn network of Fig. 7.17 for dc analysis.

Applying Kirchhoff's voltage law in the clockwise direction to the indicated loop of Fig. 7.18 results in

$$V_G - V_{GS} - V_{R_S} = 0$$

and

$$V_{GS} = V_G - V_{R_S}$$

Substituting $V_{R_S} = I_S R_S = I_D R_S$, we have

$$\boxed{V_{GS} = V_G - I_D R_S} \quad (7.16)$$

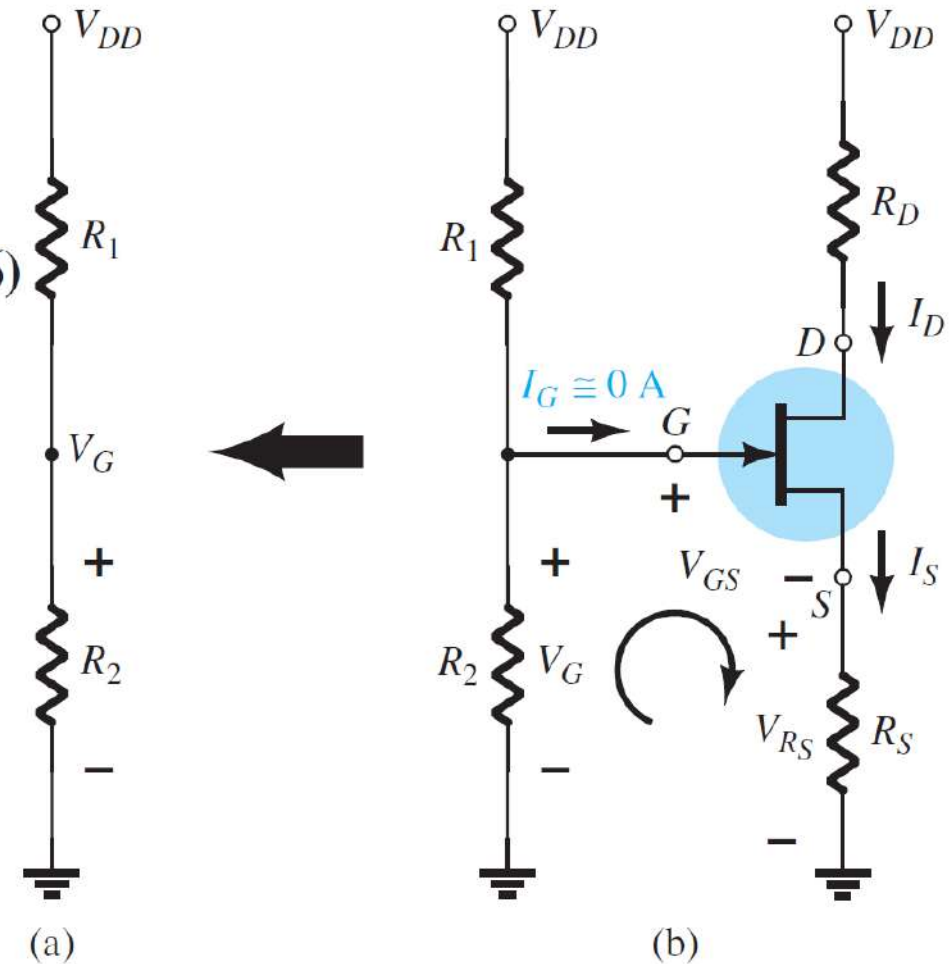


FIG. 7.18
Redrawn network of Fig. 7.17 for dc analysis.

The result is an equation that continues to include the same two variables appearing in Shockley's equation: V_{GS} and I_D. The quantities V_G and R_S are fixed by the network construction. Equation (7.16) is still the equation for a straight line, but the origin is no longer a point in the plotting of the line. The procedure for plotting Eq. (7.16) is not a difficult one and will proceed as follows. Since any straight line requires two points to be defined, let us first use the fact that anywhere on the horizontal axis of Fig. 7.19 the current $I_D = 0$ mA. If we therefore select I_D to be 0 mA, we are in essence stating that we are somewhere on the horizontal axis.

Substituting $V_{R_S} = I_S R_S = I_D R_S$, we have

$$\boxed{V_{GS} = V_G - I_D R_S} \qquad \textbf{(7.16)}$$

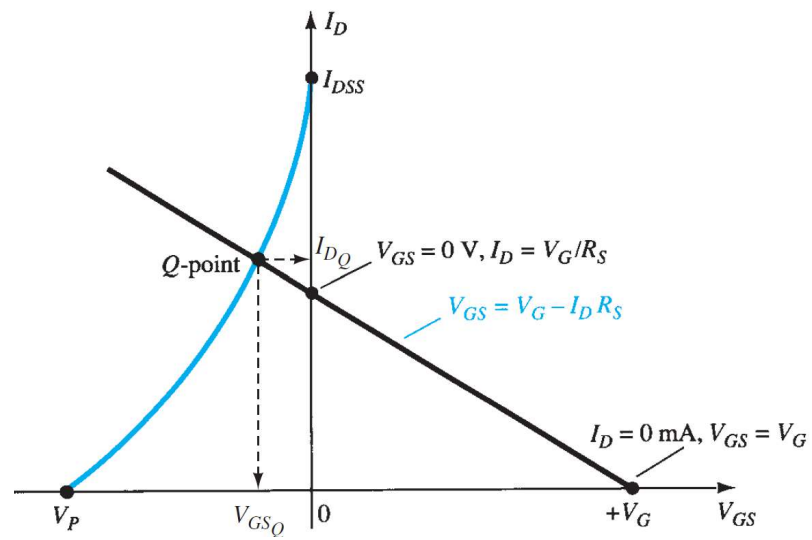
The exact location can be determined simply by substituting $I_D = 0\,\text{mA}$ into Eq. (7.16) and finding the resulting value of V_{GS} as follows:

$$V_{GS} = V_G - I_D R_S$$
$$= V_G - (0\,\text{mA})R_S$$

and

$$\boxed{V_{GS} = V_G|_{I_D=0\,\text{mA}}} \tag{7.17}$$

The result specifies that whenever we plot Eq. (7.16), if we choose $I_D = 0\,\text{mA}$, the value of V_{GS} for the plot will be V_G volts. The point just determined appears in Fig. 7.19.



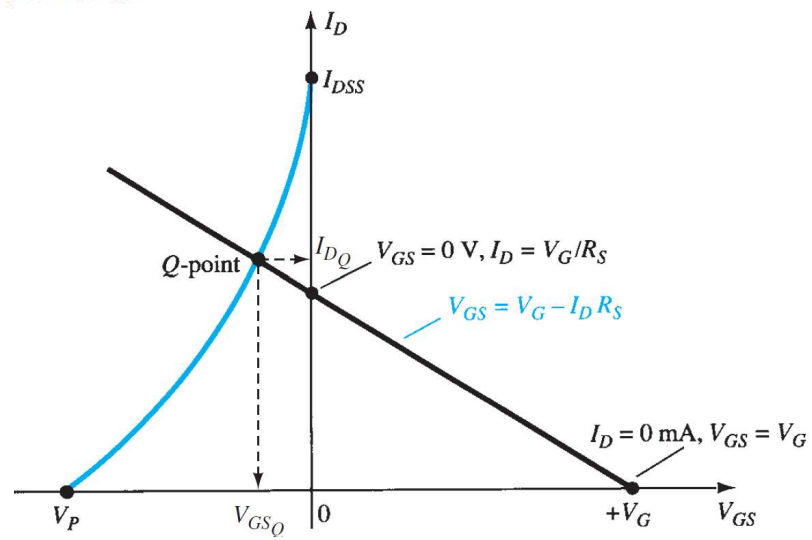
For the other point, let us now employ the fact that at any point on the vertical axis $V_{GS} = 0$ V and solve for the resulting value of I_D:

$$V_{GS} = V_G - I_D R_S$$
$$0\text{ V} = V_G - I_D R_S$$

and

$$I_D = \left.\frac{V_G}{R_S}\right|_{V_{GS}=0\text{ V}} \tag{7.18}$$

The result specifies that whenever we plot Eq. (7.16), if $V_{GS} = 0$ V, the level of I_D is determined by Eq. (7.18). This intersection also appears on Fig. 7.19.



$$V_{GS} = V_G - I_D R_S$$
$$= V_G - (0\,\text{mA})R_S$$

$$\boxed{V_{GS} = V_G|_{I_D=0\,\text{mA}}}$$

$$V_{GS} = V_G - I_D R_S$$
$$0\,\text{V} = V_G - I_D R_S$$

$$\boxed{I_D = \left.\frac{V_G}{R_S}\right|_{V_{GS}=0\,\text{V}}}$$

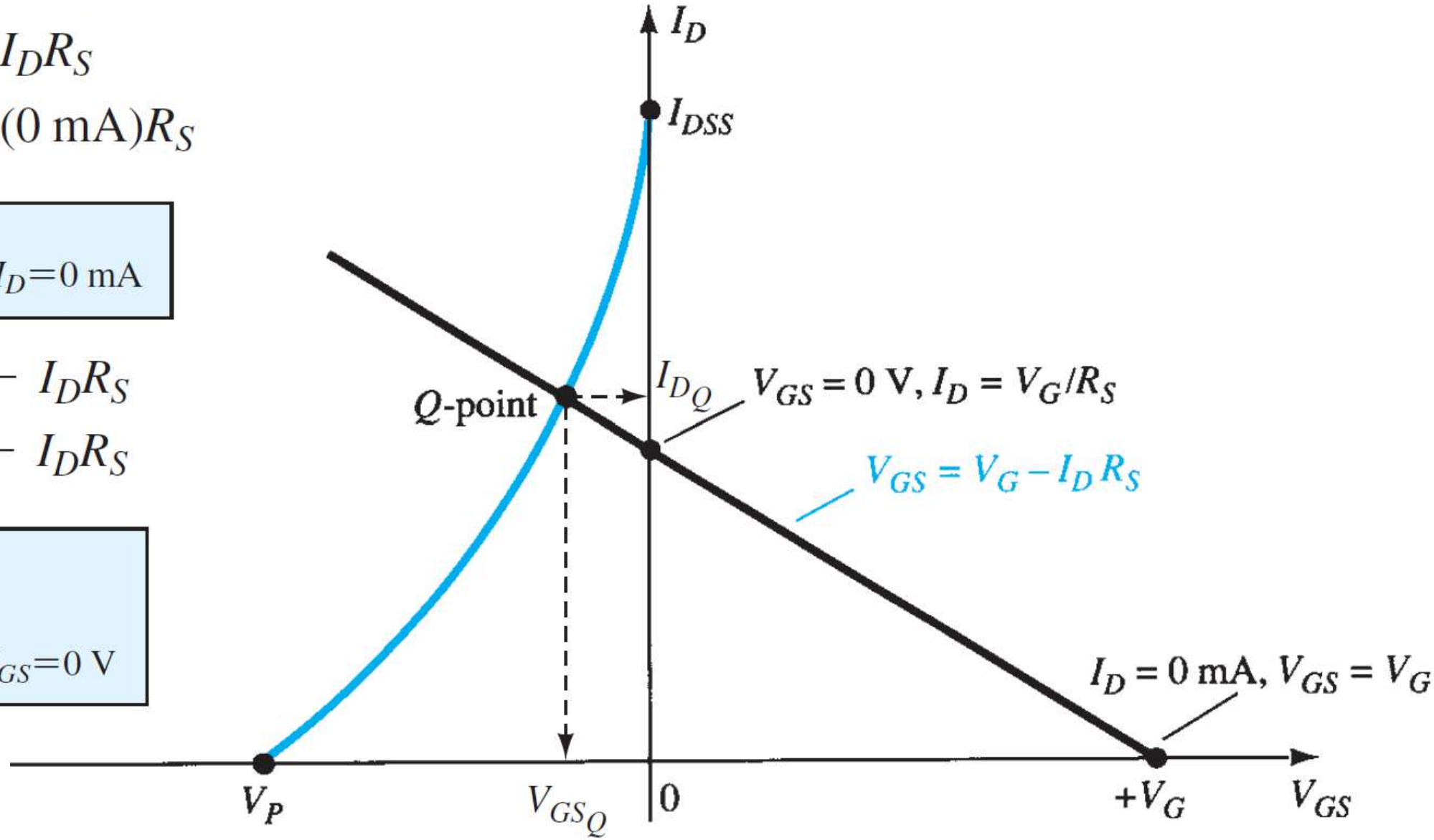


FIG. 7.19

Sketching the network equation for the voltage-divider configuration.

The two points defined above permit the drawing of a straight line to represent Eq. (7.16). The intersection of the straight line with the transfer curve in the region to the left of the vertical axis will define the operating point and the corresponding levels of I_D and V_{GS}.

Since the intersection on the vertical axis is determined by $I_D = V_G/R_S$ and V_G is fixed by the input network, increasing values of R_S will reduce the level of the I_D intersection as

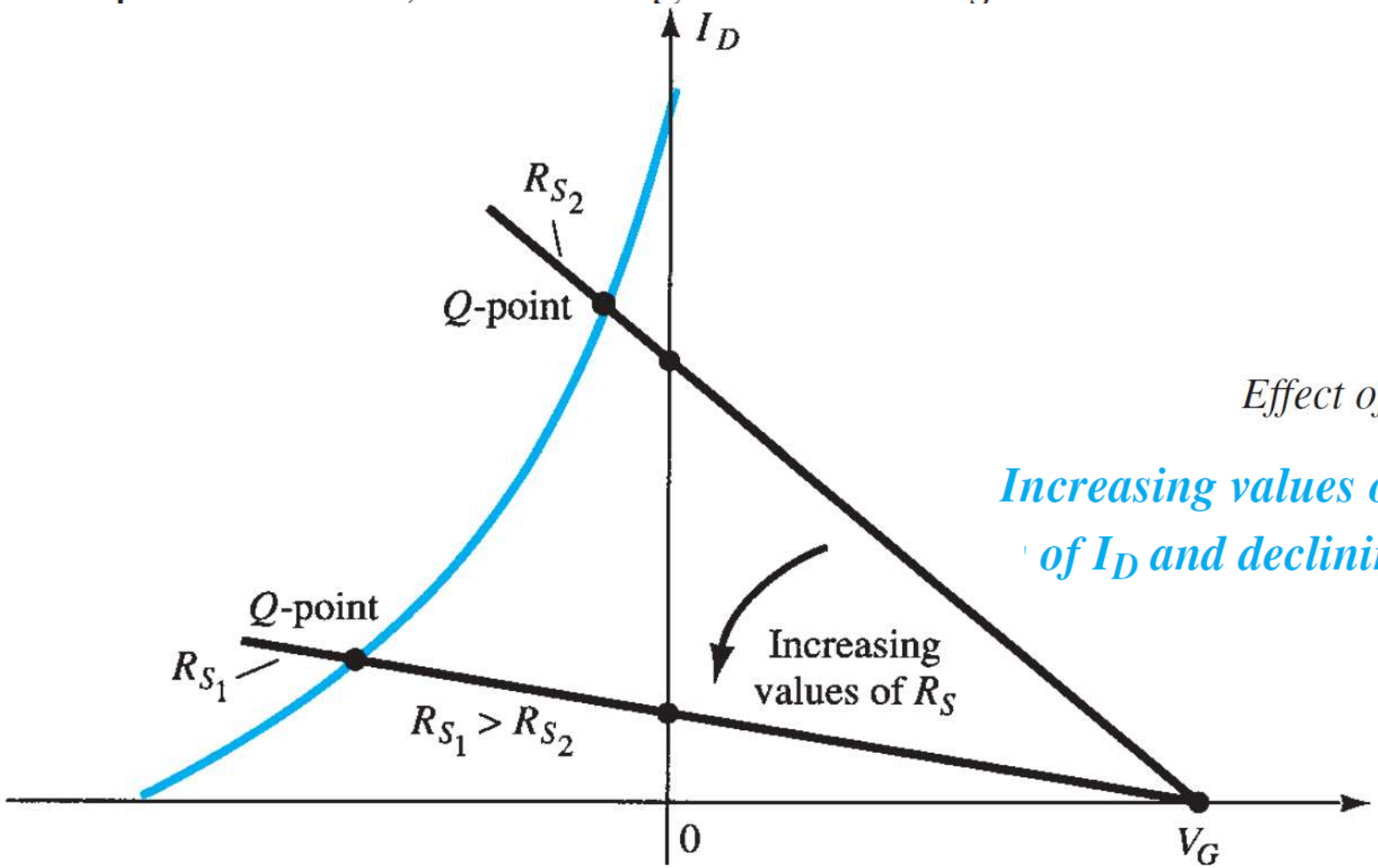


FIG. 7.20
Effect of R_S on the resulting Q-point.

Increasing values of R_S result in lower quiescent values of I_D and declining values of V_{GS}.

Once the quiescent values of I_{D_Q} and V_{GS_Q} are determined, the remaining network analysis can be performed in the usual manner. That is,

$$V_{DS} = V_{DD} - I_D(R_D + R_S) \tag{7.19}$$

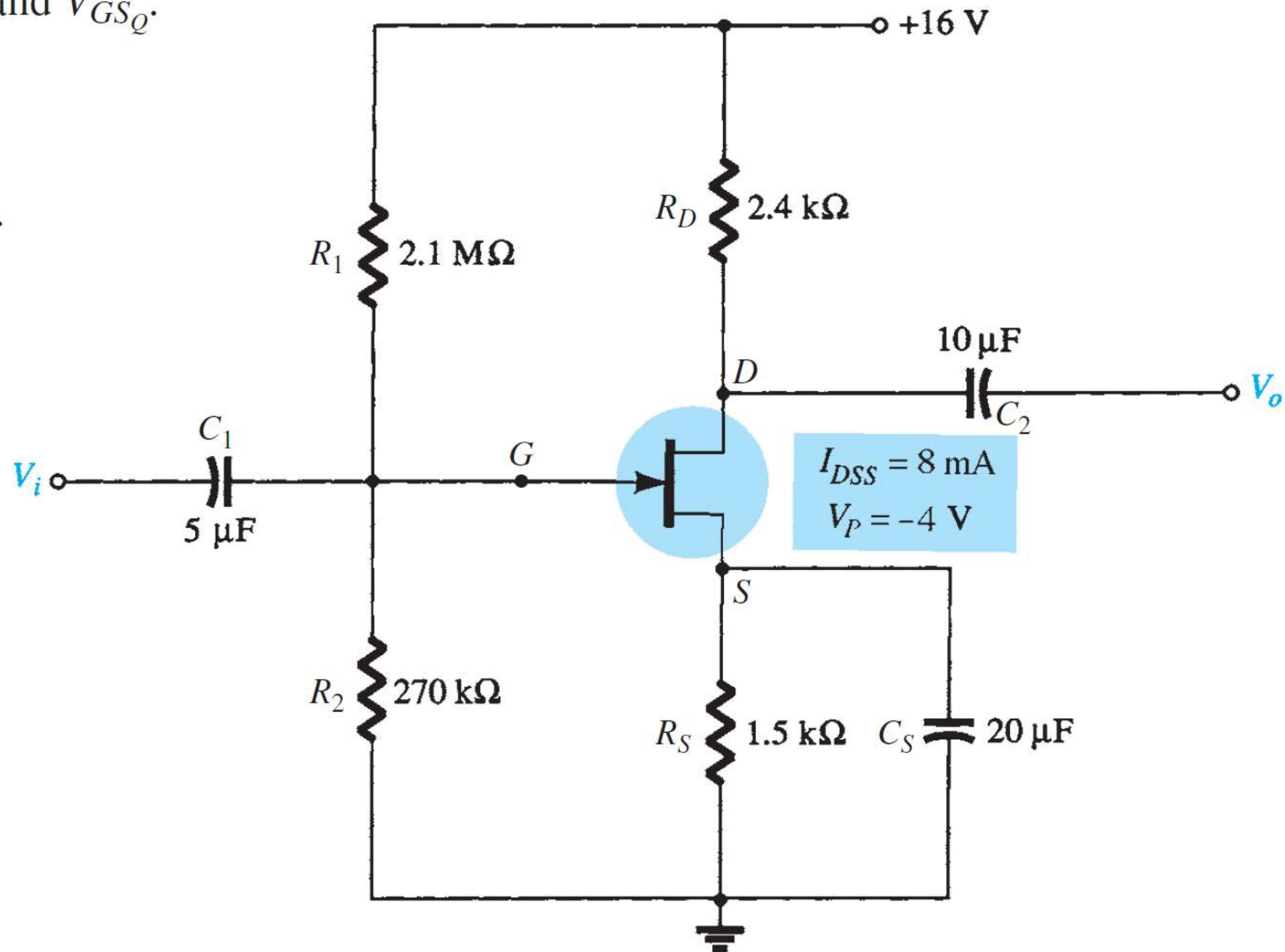
$$V_D = V_{DD} - I_D R_D \tag{7.20}$$

$$V_S = I_D R_S \tag{7.21}$$

$$I_{R_1} = I_{R_2} = \frac{V_{DD}}{R_1 + R_2} \tag{7.22}$$

EXAMPLE 7.4 Determine the following for the network of Fig. 7.21:

a. I_{D_Q} and V_{GS_Q}.
b. V_D.
c. V_S.
d. V_{DS}.
e. V_{DG}.



Solution:

a. For the transfer characteristics, if $I_D = I_{DSS}/4 = 8\,\text{mA}/4 = 2\,\text{mA}$, then $V_{GS} = V_P/2 = -4\,\text{V}/2 = -2\,\text{V}$. The resulting curve representing Shockley's equation appears in Fig. 7.22. The network equation is defined by

$$V_G = \frac{R_2 V_{DD}}{R_1 + R_2}$$

$$= \frac{(270\,\text{k}\Omega)(16\,\text{V})}{2.1\,\text{M}\Omega + 0.27\,\text{M}\Omega}$$

$$= 1.82\,\text{V}$$

and

$$V_{GS} = V_G - I_D R_S$$

$$= 1.82\,\text{V} - I_D(1.5\,\text{k}\Omega)$$

When $I_D = 0\,\text{mA}$,

$$V_{GS} = +1.82\,\text{V}$$

When $V_{GS} = 0\,\text{V}$,

$$I_D = \frac{1.82\,\text{V}}{1.5\,\text{k}\Omega} = 1.21\,\text{mA}$$

The resulting bias line appears on Fig. 7.22 with quiescent values of

$$I_{D_Q} = \mathbf{2.4\ mA}$$

and

$$V_{GS_Q} = \mathbf{-1.8\ V}$$

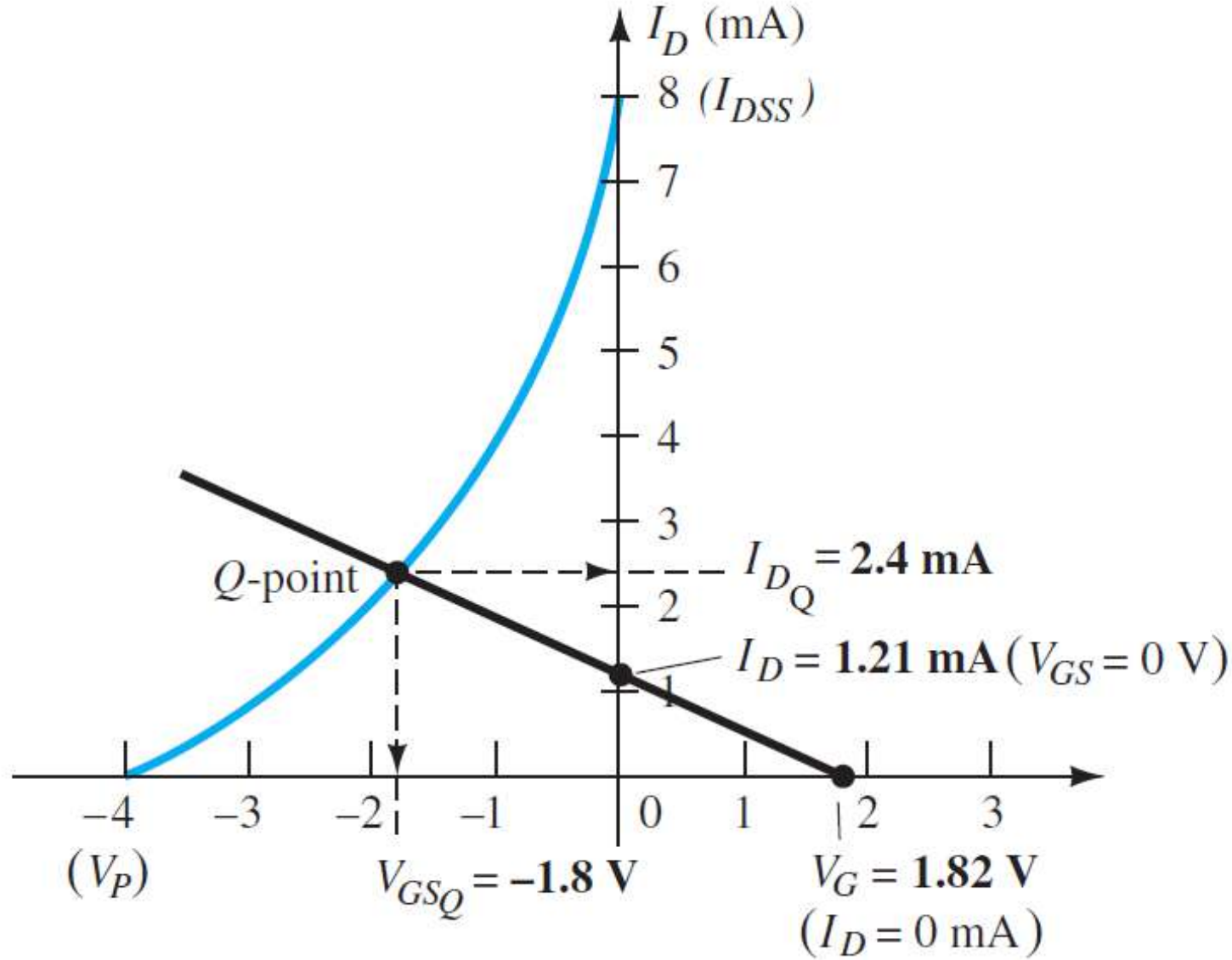


FIG. 7.22

Determining the Q-point for the network of Fig. 7.21.

b. $V_D = V_{DD} - I_D R_D$

$= 16\text{ V} - (2.4\text{ mA})(2.4\text{ k}\Omega)$

$= \mathbf{10.24\text{ V}}$

c. $V_S = I_D R_S = (2.4\text{ mA})(1.5\text{ k}\Omega)$

$= \mathbf{3.6\text{ V}}$

d. $V_{DS} = V_{DD} - I_D(R_D + R_S)$

$= 16\text{ V} - (2.4\text{ mA})(2.4\text{ k}\Omega + 1.5\text{ k}\Omega$

$= \mathbf{6.64\text{ V}}$

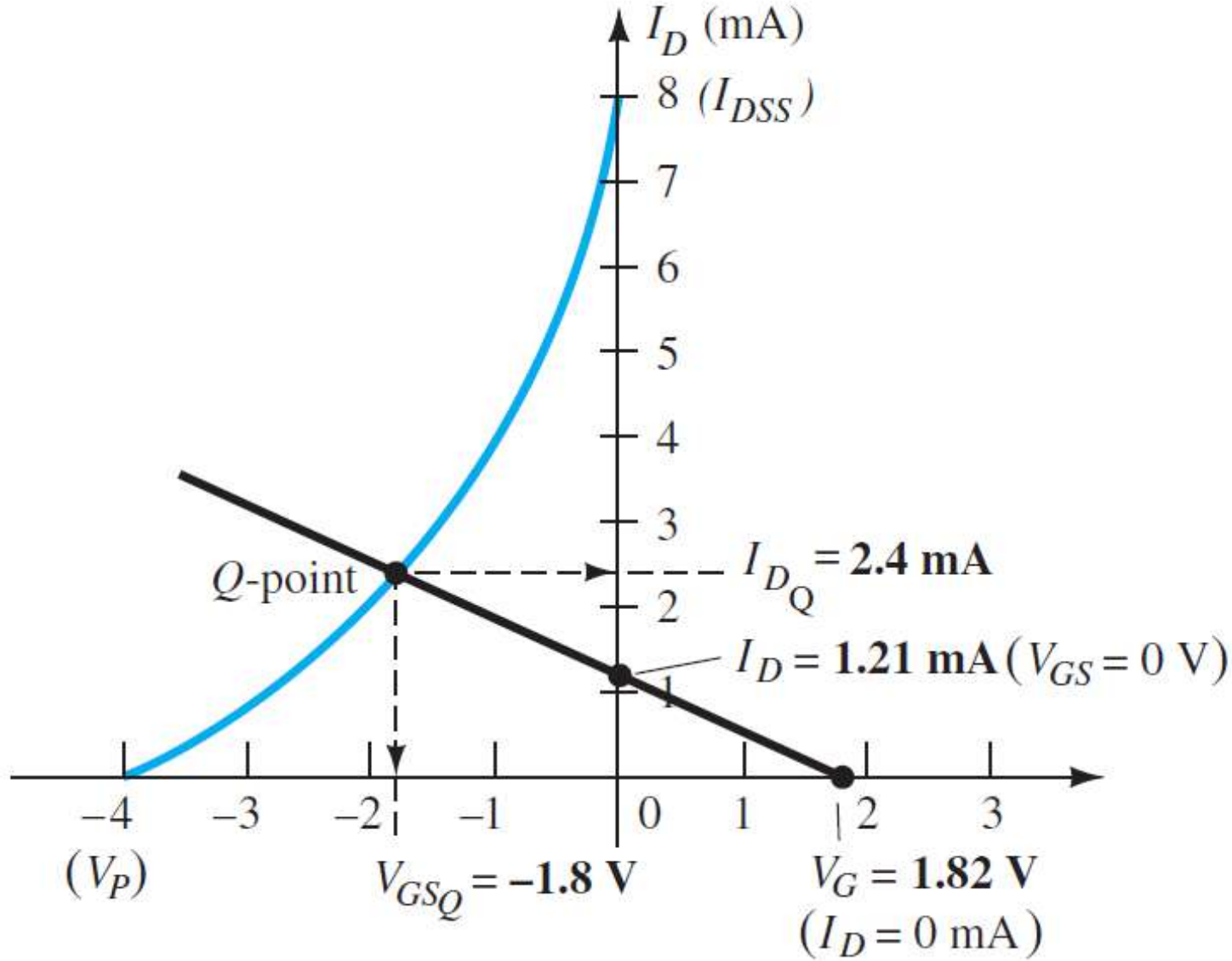
or $V_{DS} = V_D - V_S = 10.24\text{ V} - 3.6\text{ V}$

$= \mathbf{6.64\text{ V}}$

e. $V_{DG} = V_D - V_G$

$= 10.24\text{ V} - 1.82\text{ V}$

$= \mathbf{8.42\text{ V}}$



7.5 COMMON-GATE CONFIGURATION

The next configuration is one in which the gate terminal is grounded and the input signal typically applied to the source terminal and the output signal obtained at the drain terminal as shown in Fig. 7.23a. The network can also be drawn as shown in Fig. 7.23b.

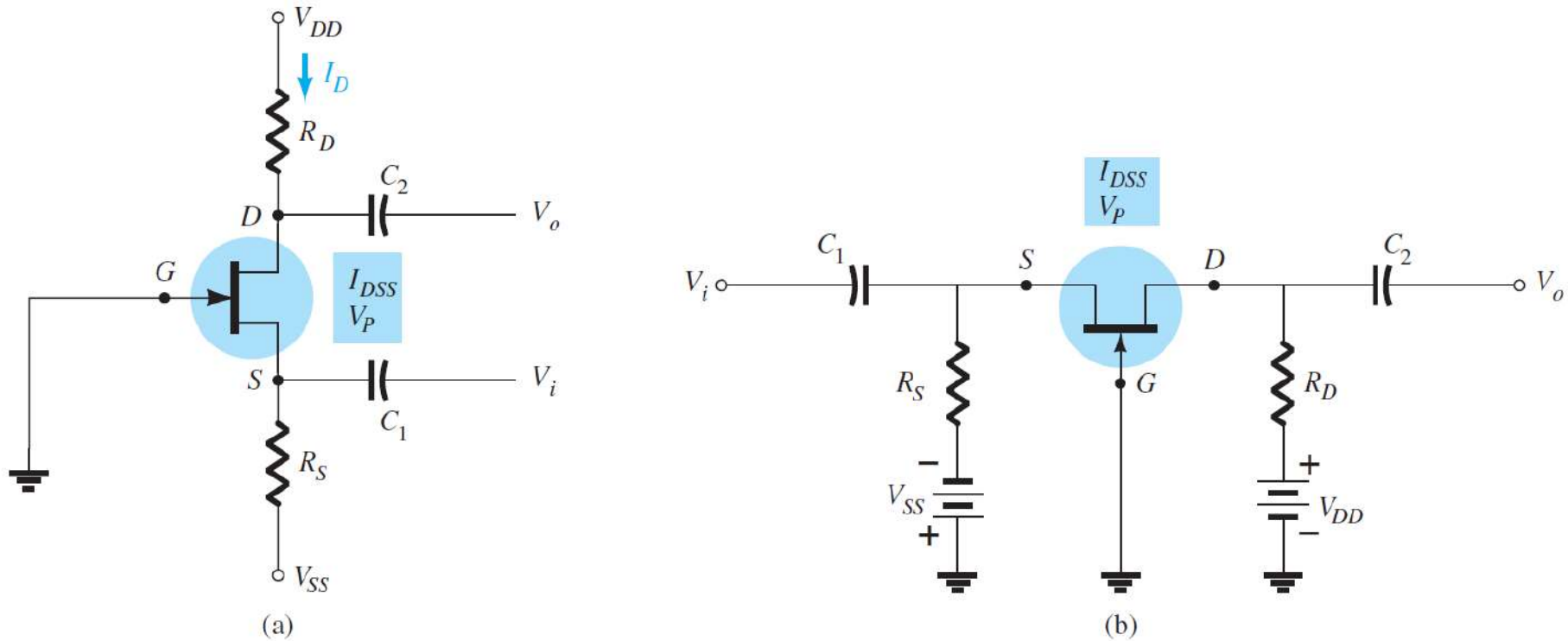


FIG. 7.23

Two versions of the common-gate configuration.

The network equation can be determined using Fig. 7.24.

Applying Kirchhoff's voltage law in the direction shown in Fig. 7.24 will result in

$$-V_{GS} - I_S R_S + V_{SS} = 0$$

and

$$V_{GS} = V_{SS} - I_S R_S$$

but

$$I_S = I_D$$

so

$$\boxed{V_{GS} = V_{SS} - I_D R_S} \tag{7.25}$$

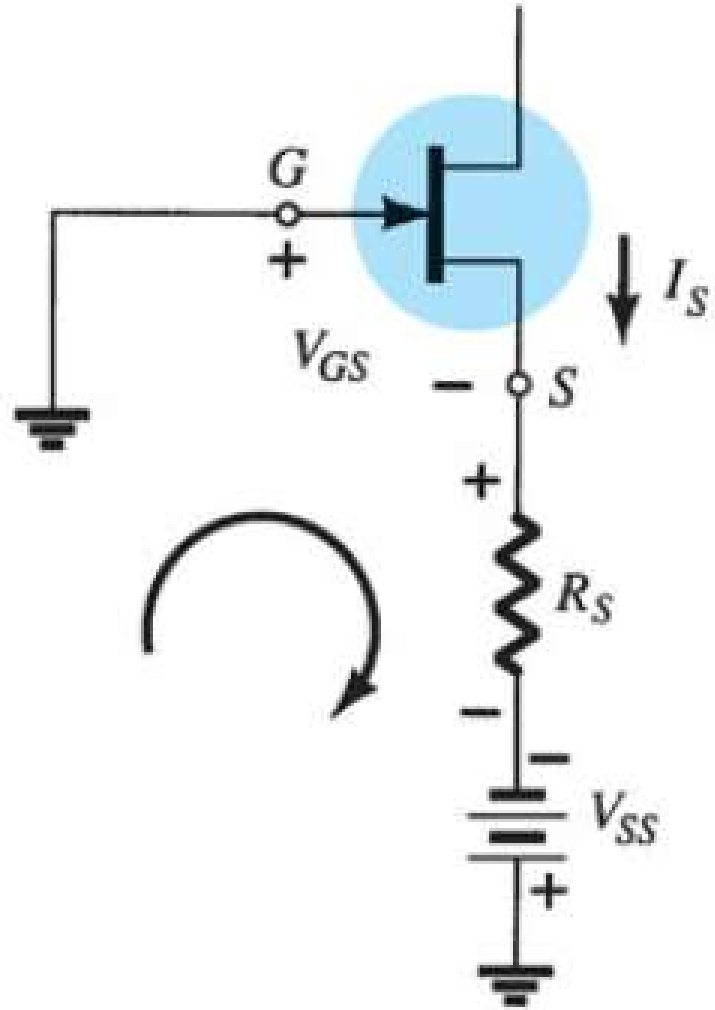


FIG. 7.24

Applying the condition $I_D = 0$ mA to Eq. 7.23 will result in

$$V_{GS} = V_{SS} - (0)R_S$$

and

$$\boxed{V_{GS} = V_{SS}|_{I_D=0\text{mA}}}$$

Applying the condition $V_{GS} = 0$ V to Eq. 7.23 will result in

$$0 = V_{SS} - I_D R_S$$

and

$$\boxed{I_D = \left.\frac{V_{SS}}{R_S}\right|_{V_{GS}=0\text{ V}}}$$

The resulting load-line appears in Fig. 7.25 intersecting the transfer curve for the JFET as shown in the figure.

The resulting intersection defines the operating current I_{D_Q} and voltage V_{D_Q} for the network as also indicated in the network.

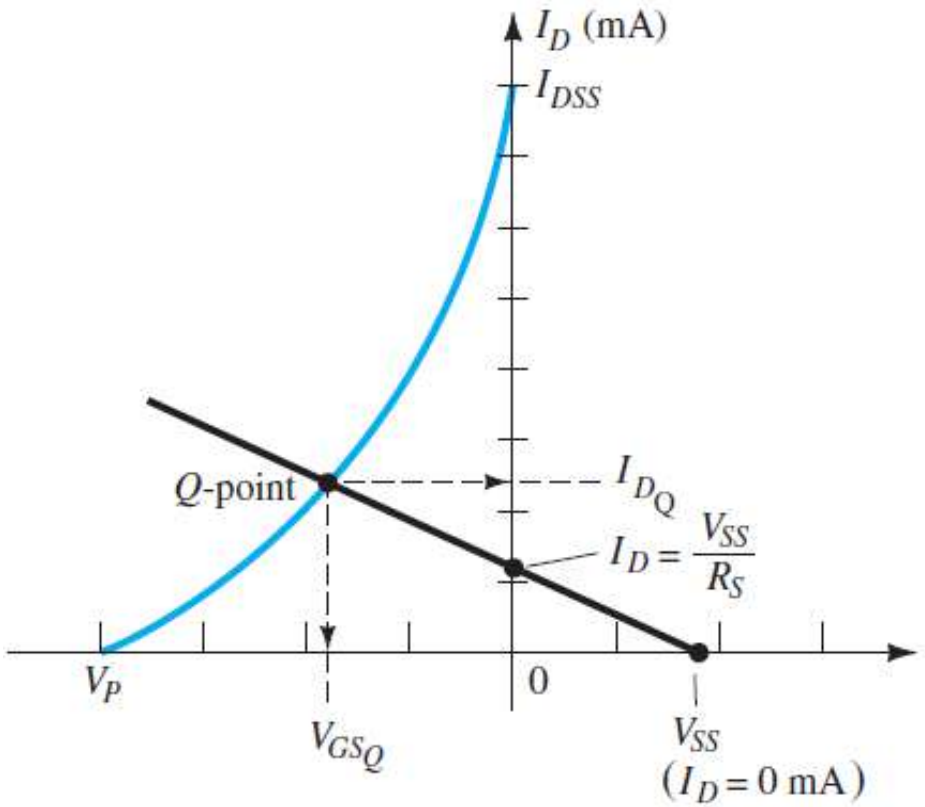


FIG. 7.25

Determining the Q-point for the network of Fig. 7.24.

Applying Kirchhoff's voltage law around the loop containing the two sources, the JFET and the resistors R_D and R_S in Fig. 7.23a and Fig. 7.23b will result in

$$+V_{DD} - I_D R_D - V_{DS} - I_S R_S + V_{SS} = 0$$

Substituting $I_S = I_D$ we have

$$+V_{DD} + V_{SS} - V_{DS} - I_D(R_D + R_S) = 0$$

so that

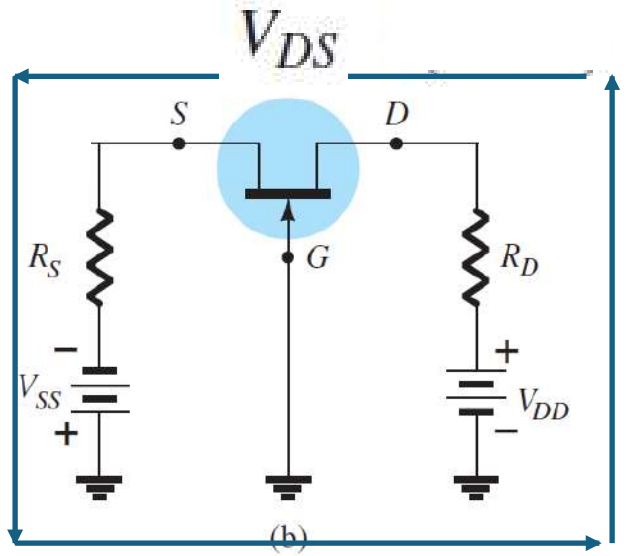
$$\boxed{V_{DS} = V_{DD} + V_{SS} - I_D(R_D + R_S)}$$

with

$$\boxed{V_D = V_{DD} - I_D R_D}$$

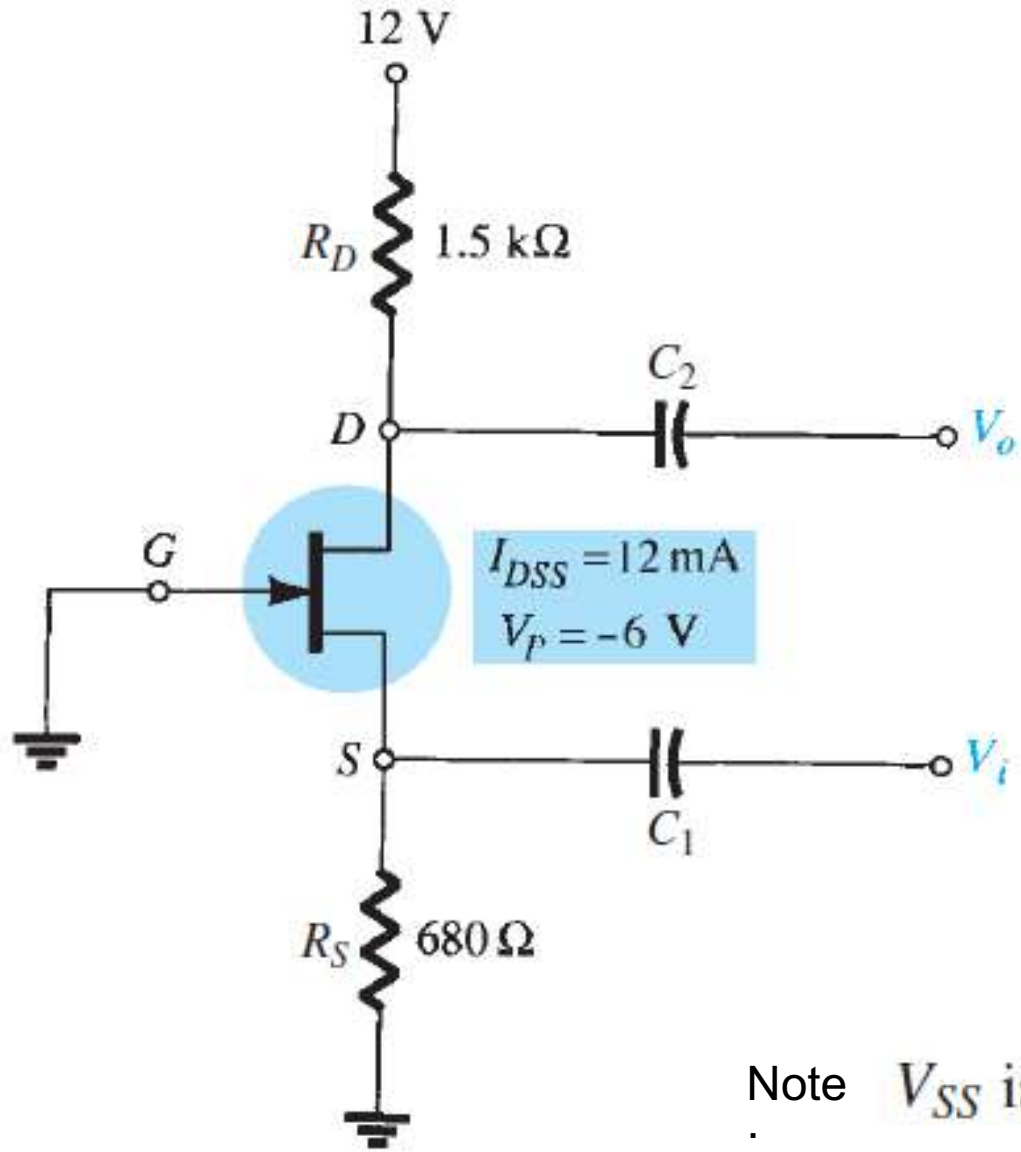
and

$$\boxed{V_S = -V_{SS} + I_D R_S}$$



EXAMPLE 7.5 Determine the following for the common-gate configuration of Fig. 7.26:

a. V_{GS_Q}
b. I_{D_Q}
c. V_D
d. V_G
e. V_S
f. V_{DS}



Note: V_{SS} is not present

FIG. 7.26

Solution: Even though V_{SS} is not present in this common-gate configuration the equations derived above can still be used by simply substituting $V_{SS} = 0$ V into each equation in which it appears.

a. For the transfer characteristics Eq. 7.23 becomes

$$V_{GS} = 0 - I_D R_S$$

and

$$V_{GS} = -I_D R_S$$

For this equation the origin is one point on the load line while the other must be determined at some arbitrary point. Choosing $I_D = 6$ mA and solving for V_{GS} will result in the following:

$$V_{GS} = -I_D R_S = -(6\text{ mA})(680\ \Omega) = -4.08\text{ V}$$

as shown in Fig. 7.27.

The device transfer curve is sketched using

$$I_D = \frac{I_{DSS}}{4} = \frac{12\text{ mA}}{4} = 3\text{ mA}(\text{at } V_P/2)$$

and $$V_{GS} \cong 0.3V_P = 0.3(-6\text{ V}) = -1.8\text{ V}\ (\text{at } I_D = I_{DSS}/2)$$

The resulting solution is:

$$V_{GS_Q} \cong \mathbf{-2.6\text{ V}}$$

b. From Fig. 7.27,

$$I_{D_Q} \cong \mathbf{3.8\text{ mA}}$$

c. $V_D = V_{DD} - I_D R_D$
$= 12\text{ V} - (3.8\text{ mA})(1.5\text{ k}\Omega) = 12\text{ V} - 5.7\text{ V}$
$= \mathbf{6.3\text{ V}}$

d. $V_G = \mathbf{0\text{ V}}$

e. $V_S = I_D R_S = (3.8\text{ mA})(680\ \Omega)$
$= \mathbf{2.58\text{ V}}$

f. $V_{DS} = V_D - V_S$
$= 6.3\text{ V} - 2.58\text{ V}$
$= \mathbf{3.72\text{ V}}$

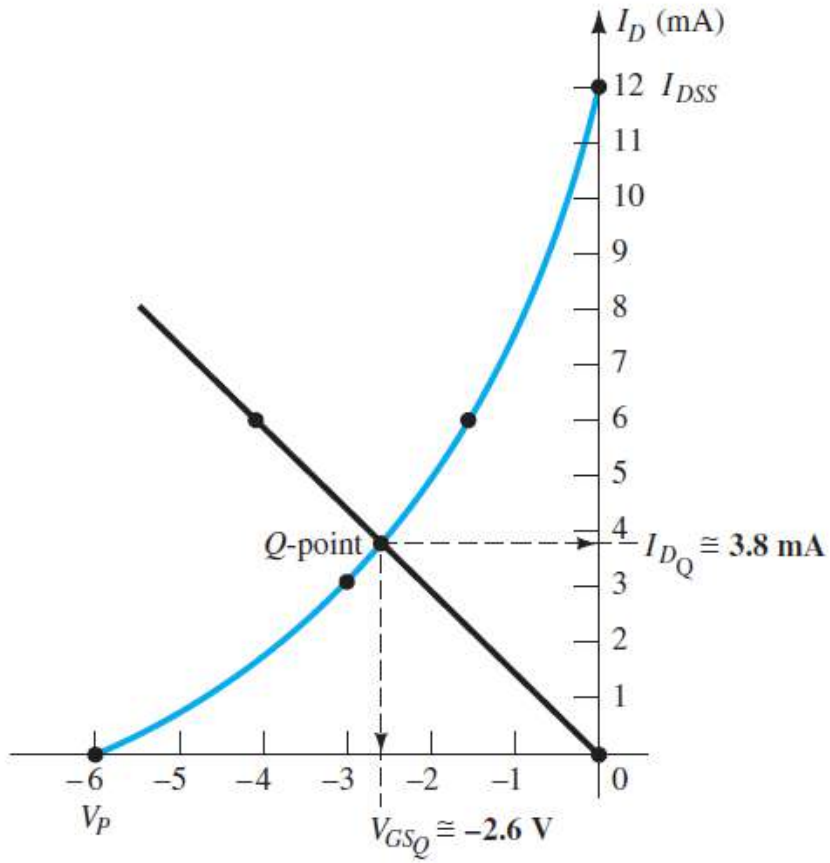


FIG. 7.27
Determining the Q-point for the network of Fig. 7.26.

7.6 SPECIAL CASE: $V_{GS_Q} = 0$ V

A network of recurring practical value because of its relative simplicity is the configuration of Fig. 7.28. Note that direct connection of the gate and source terminals to ground resulting in $V_{GS} = 0$ V. It specifies that for any dc condition the gate to source voltage must be zero volts. This will result in a vertical load line at $V_{GS_Q} = 0$ V as shown in Fig. 7.29.

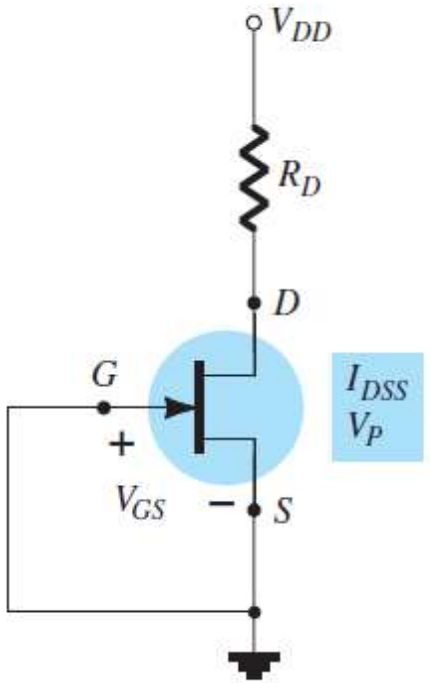


FIG. 7.28
Special case $V_{GS_Q} = 0$ V configuration.

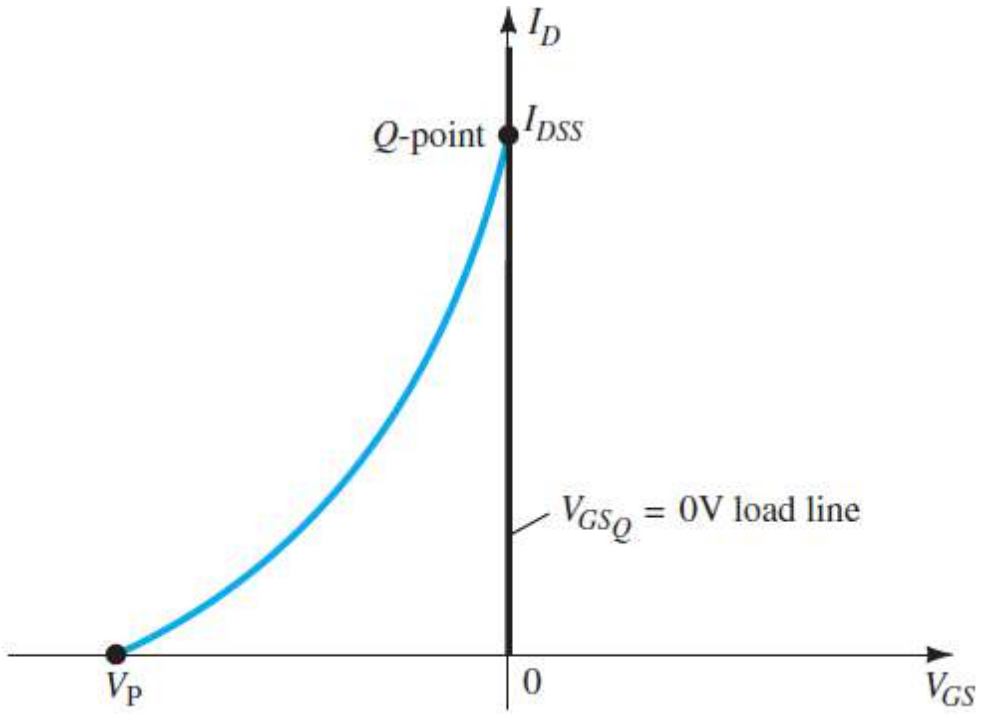


FIG. 7.29
Finding the Q-point for the network of Fig. 7.28.

Since the transfer curve of a JFET will cross the vertical axis at I_{DSS} the drain current for the network is set at that level.

Therefore,

$$I_{D_Q} = I_{DSS} \tag{7.29}$$

Applying Kirchhoff's voltage law:

$$V_{DD} - I_D R_D - V_{DS} = 0$$

and

$$V_{DS} = V_{DD} - I_D R_D$$

with

$$V_D = V_{DS}$$

and

$$V_S = 0\,\text{V}$$

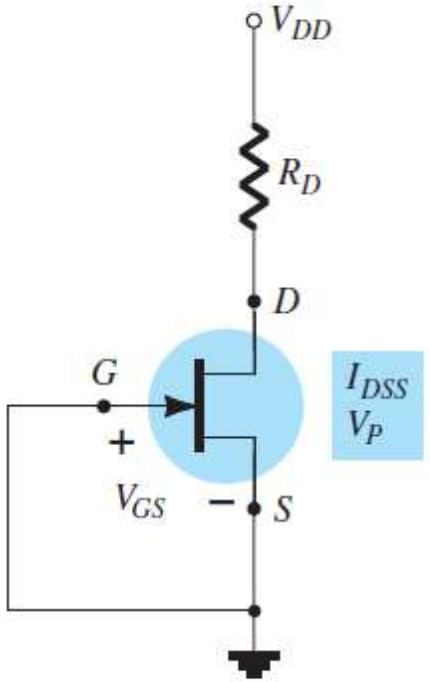


FIG. 7.28
Special case $V_{GS_Q} = 0$ V configuration.

7.7 DEPLETION-TYPE MOSFETs

The similarities in appearance between the transfer curves of JFETs and depletion-type MOSFETs permit a similar analysis of each in the dc domain. The primary difference between the two is the fact that depletion-type MOSFETs permit operating points with positive values of V_{GS} and levels of I_D that exceed I_{DSS}. In fact, for all the configurations discussed thus far, the analysis is the same if the JFET is replaced by a depletion-type MOSFET.

The only undefined part of the analysis is how to plot Shockley's equation for positive values of V_{GS}. How far into the region of positive values of V_{GS} and values of I_D greater than I_{DSS} does the transfer curve have to extend? For most situations, this required range will be fairly well defined by the MOSFET parameters and the resulting bias line of the network. A few examples will reveal the effect of the change in device on the resulting analysis.

EXAMPLE 7.6 For the *n*-channel depletion-type MOSFET of Fig. 7.30, determine:

a. I_{D_Q} and V_{GS_Q}.

b. V_{DS}.

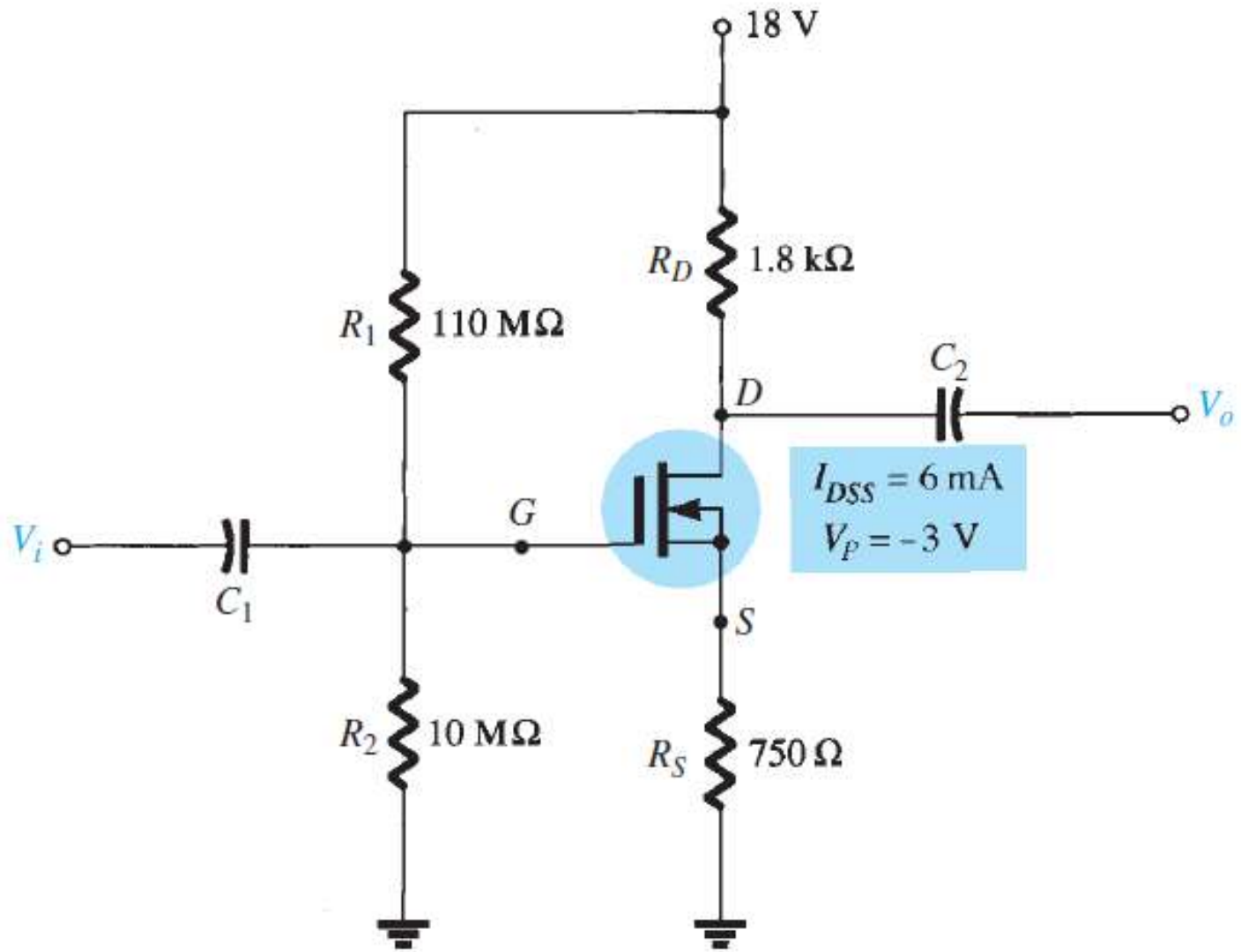


FIG. 7.30

Solution:

a. For the transfer characteristics, a plot point is defined by $I_D = I_{DSS}/4 = 6\text{ mA}/4 = 1.5\text{ mA}$ and $V_{GS} = V_P/2 = -3\text{ V}/2 = -1.5\text{ V}$. Considering the level of V_P and the fact that Shockley's equation defines a curve that rises more rapidly as V_{GS} becomes more positive, a plot point will be defined at $V_{GS} = +1\text{ V}$. Substituting into Shockley's equation yields

$$I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_P}\right)^2$$

$$= 6\text{ mA}\left(1 - \frac{+1\text{ V}}{-3\text{ V}}\right)^2 = 6\text{ mA}\left(1 + \frac{1}{3}\right)^2 = 6\text{ mA}\,(1.778)$$

$$= 10.67\text{ mA}$$

The resulting transfer curve appears in Fig. 7.31. Proceeding as described for JFETs, we have

$$V_G = \frac{10\,\text{M}\Omega(18\text{ V})}{10\,\text{M}\Omega + 110\,\text{M}\Omega} = 1.5\text{ V}$$

$$V_{GS} = V_G - I_D R_S = 1.5\text{ V} - I_D(750\ \Omega)$$

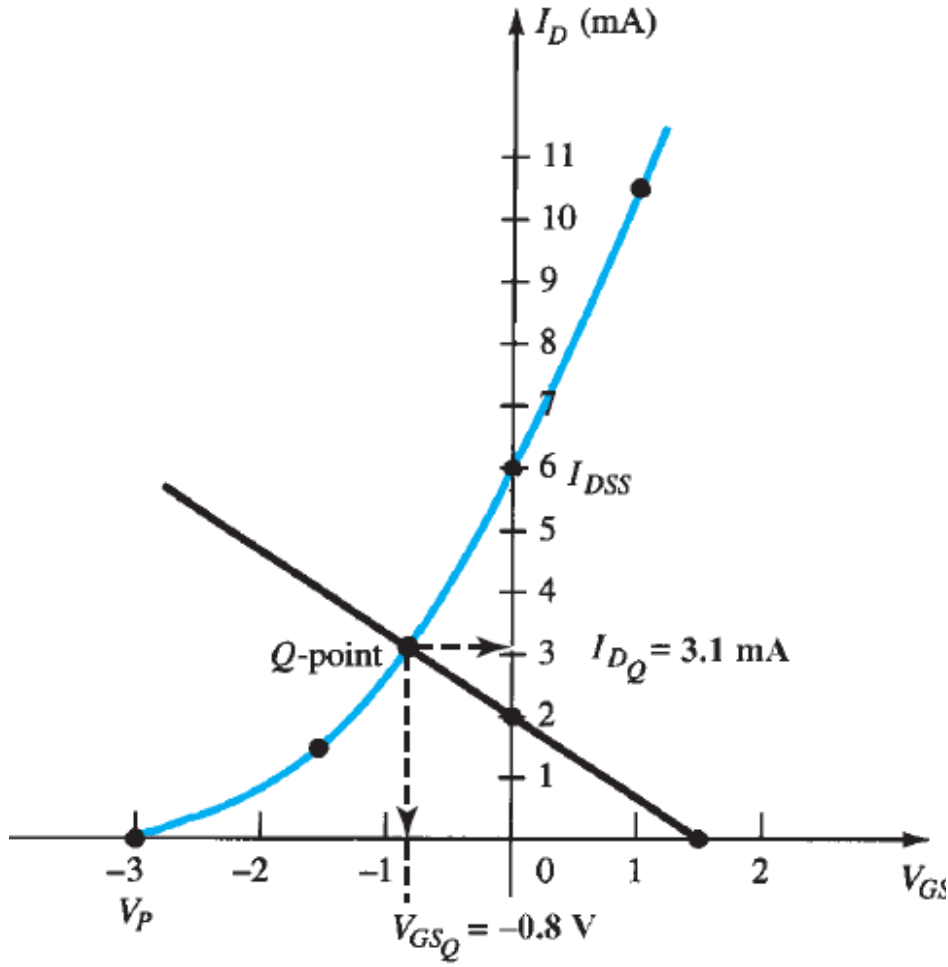


FIG. 7.31

Setting $I_D = 0$ mA results in

$$V_{GS} = V_G = 1.5\text{ V}$$

Setting $V_{GS} = 0$ V yields

$$I_D = \frac{V_G}{R_S} = \frac{1.5\text{ V}}{750\ \Omega} = 2\text{ mA}$$

The plot points and resulting bias line appear in Fig. 7.31. The resulting operating point is given by

$$I_{D_Q} = \mathbf{3.1\text{ mA}}$$
$$V_{GS_Q} = \mathbf{-0.8\text{ V}}$$

b. Eq. (7.19):

$$V_{DS} = V_{DD} - I_D(R_D + R_S)$$
$$= 18\text{ V} - (3.1\text{ mA})(1.8\text{ k}\Omega + 750\ \Omega)$$
$$\cong \mathbf{10.1\text{ V}}$$

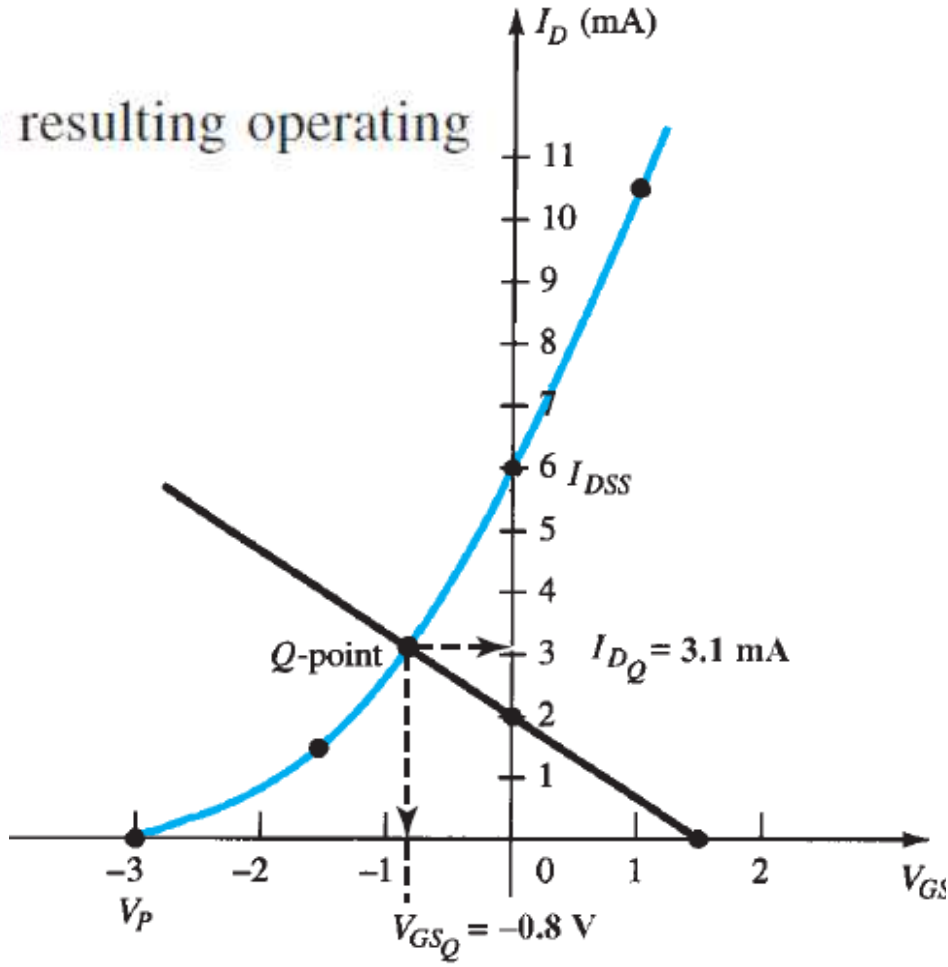


FIG. 7.31

EXAMPLE 7.7 Repeat Example 7.6 with $R_S = 150\ \Omega$.

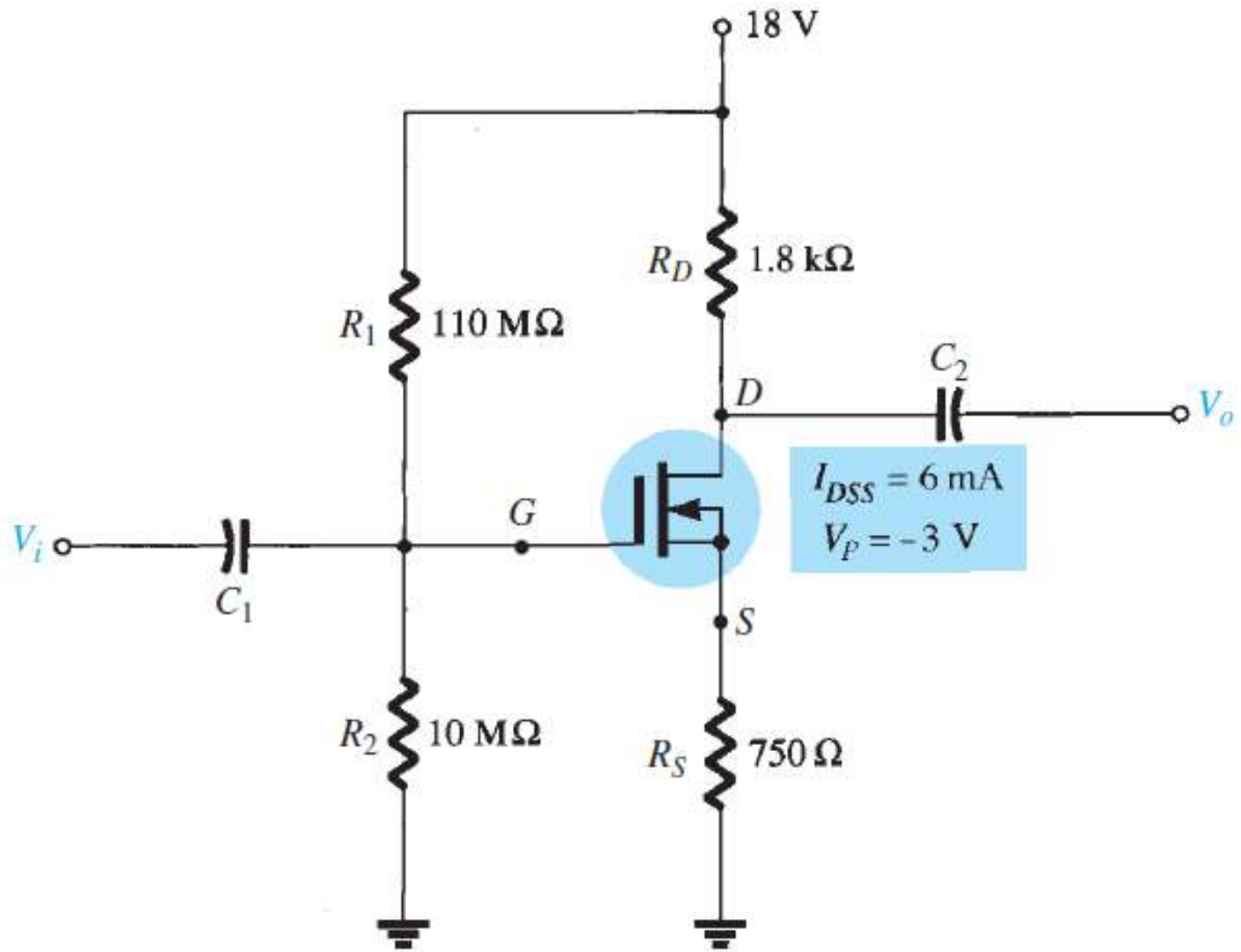


FIG. 7.30

Solution:

a. The plot points are the same for the transfer curve as shown in Fig. 7.32. For the bias line,

$$V_{GS} = V_G - I_D R_S = 1.5\text{ V} - I_D(150\ \Omega)$$

Setting $I_D = 0$ mA results in

$$V_{GS} = 1.5\text{ V}$$

Setting $V_{GS} = 0$ V yields

$$I_D = \frac{V_G}{R_S} = \frac{1.5\text{ V}}{150\ \Omega} = 10\text{ mA}$$

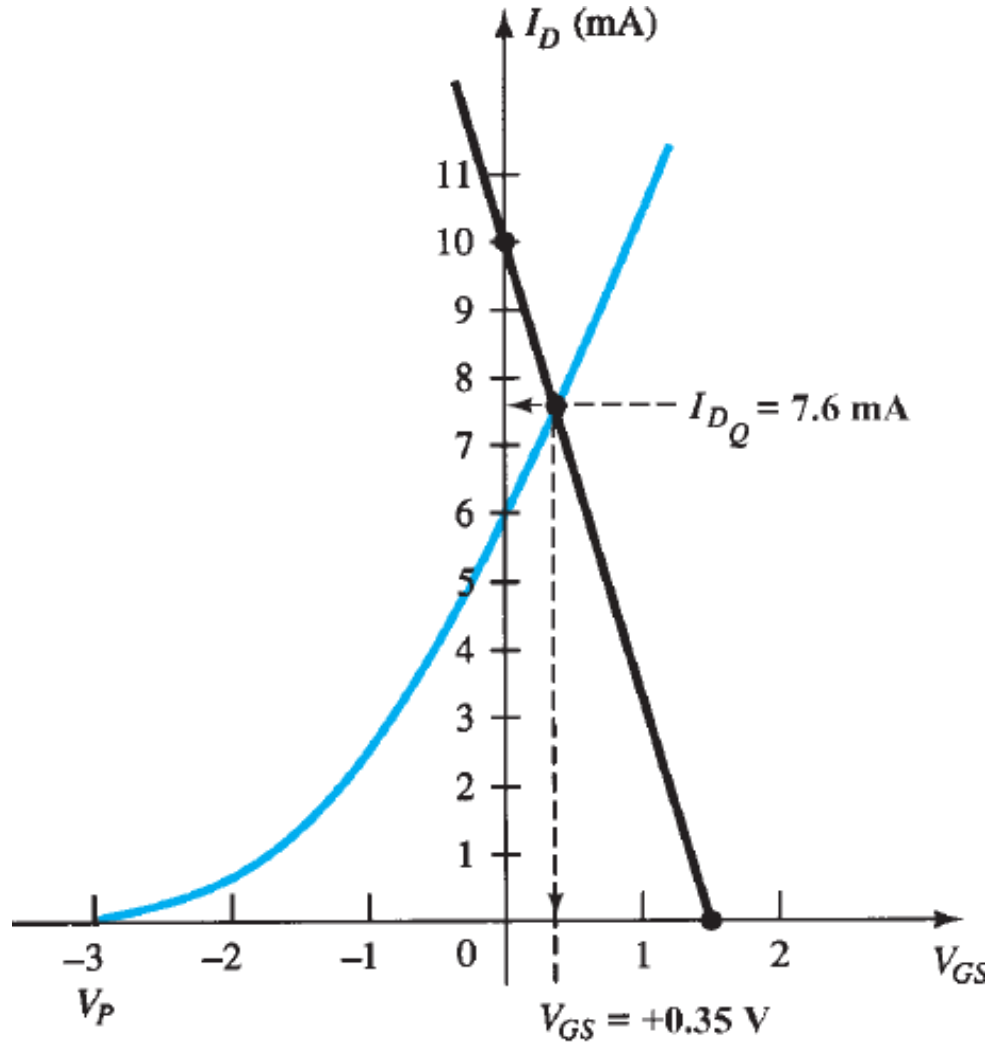


FIG. 7.32

The bias line is included on Fig. 7.32. Note in this case that the quiescent point results in a drain current that exceeds I_{DSS}, with a positive value for V_{GS}. The result is

$$I_{D_Q} = \mathbf{7.6\ mA}$$
$$V_{GS_Q} = \mathbf{+0.35\ V}$$

b. Eq. (7.19):

$$V_{DS} = V_{DD} - I_D(R_D + R_S)$$
$$= 18\ \text{V} - (7.6\ \text{mA})(1.8\ \text{k}\Omega + 150\ \Omega)$$
$$= \mathbf{3.18\ V}$$

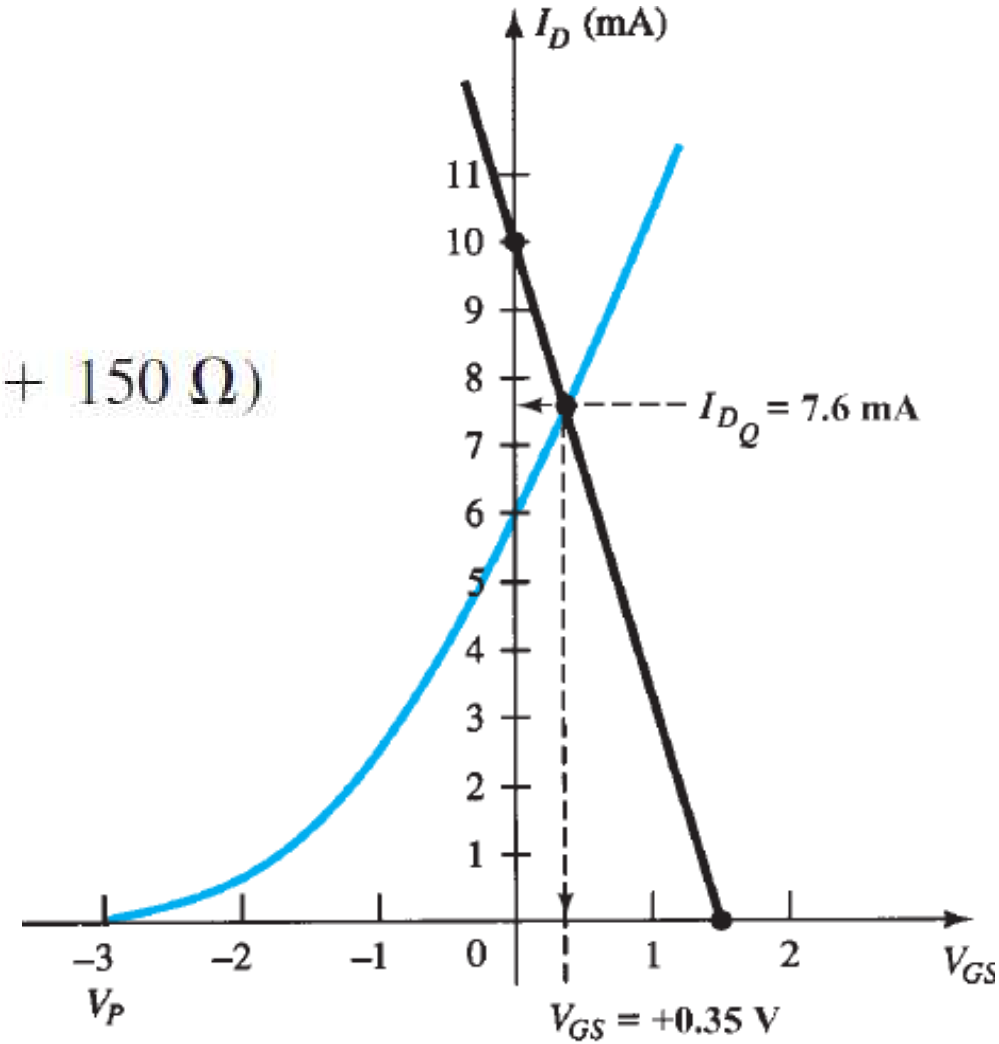


FIG. 7.32

EXAMPLE 7.8 Determine the following for the network of Fig. 7.33:

a. I_{D_Q} and V_{GS_Q}.
b. V_D.

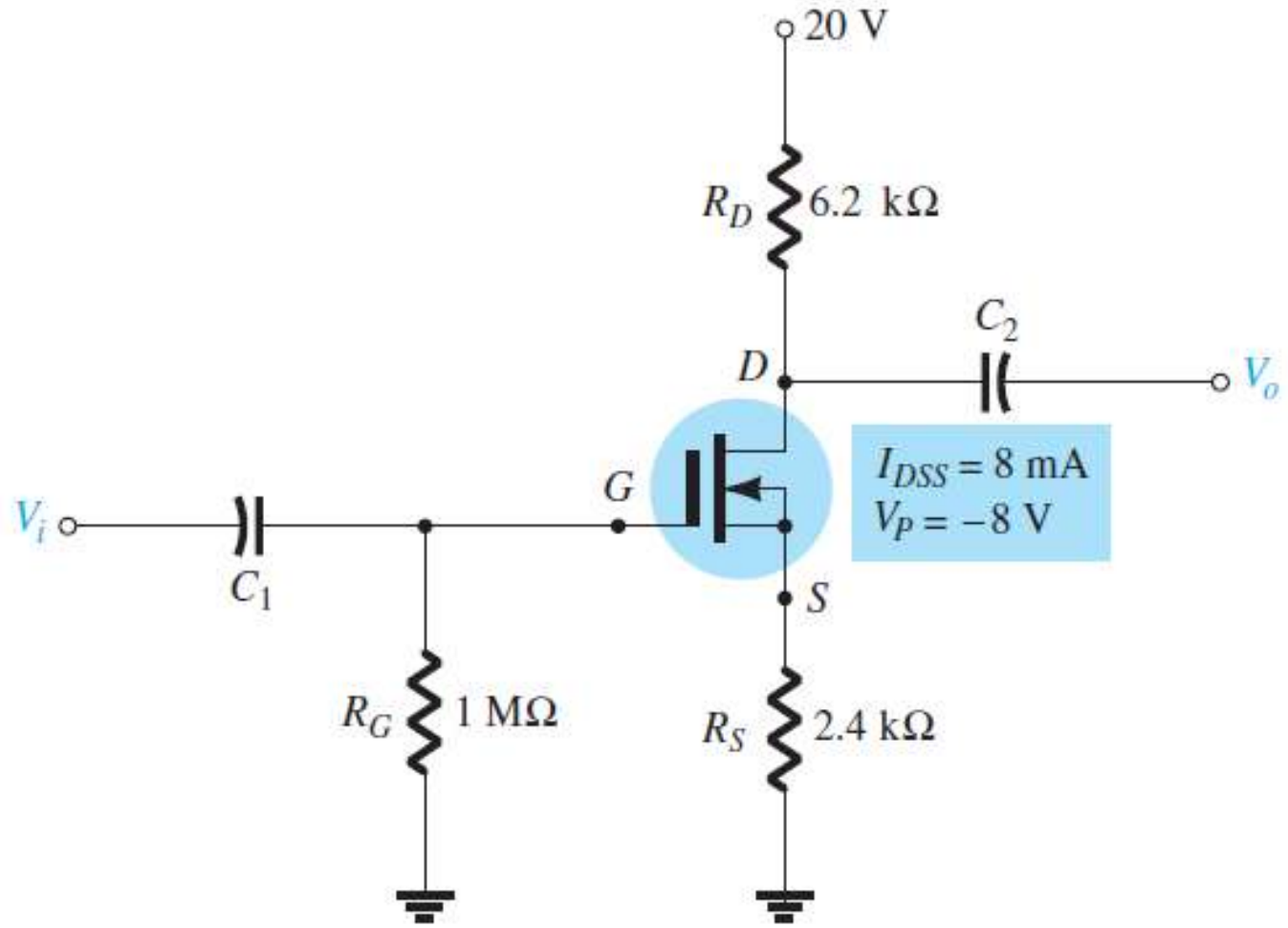


FIG. 7.33

Solution:

a. The self-bias configuration results in

$$V_{GS} = -I_D R_S$$

as obtained for the JFET configuration, establishing the fact that V_{GS} must be less than 0 V. There is therefore no requirement to plot the transfer curve for positive values of V_{GS}, although it was done on this occasion to complete the transfer characteristics. A plot point for the transfer characteristics for $V_{GS} < 0$ V is

$$I_D = \frac{I_{DSS}}{4} = \frac{8 \text{ mA}}{4} = 2 \text{ mA}$$

and

$$V_{GS} = \frac{V_P}{2} = \frac{-8 \text{ V}}{2} = -4 \text{ V}$$

and for $V_{GS} > 0$ V, since $V_P = -8$ V, we will choose

$$V_{GS} = +2 \text{ V}$$

and

$$I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_P}\right)^2 = 8 \text{ mA}\left(1 - \frac{+2 \text{ V}}{-8 \text{ V}}\right)^2$$

$$= 12.5 \text{ mA}$$

The resulting transfer curve appears in Fig. 7.34. For the network bias line, at $V_{GS} = 0\text{ V}, I_D = 0\text{ mA}$. Choosing $V_{GS} = -6\text{ V}$ gives

$$I_D = -\frac{V_{GS}}{R_S} = -\frac{-6\text{ V}}{2.4\text{ k}\Omega} = 2.5\text{ mA}$$

The resulting Q-point is given by

$$I_{D_Q} = \mathbf{1.7\text{ mA}}$$
$$V_{GS_Q} = \mathbf{-4.3\text{ V}}$$

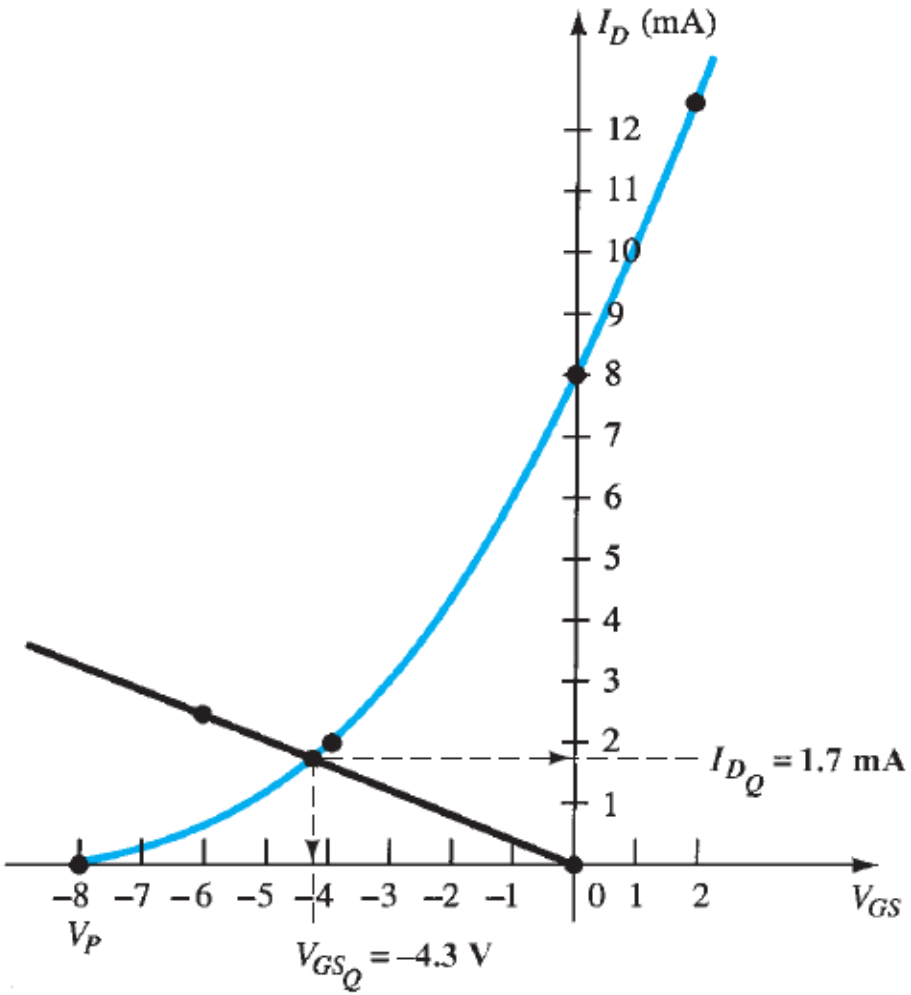


FIG. 7.34

EXAMPLE 7.9 Determine V_{DS} for the network of Fig. 7.35.

Solution: The direct connection between the gate and source terminals requires that

$$V_{GS} = 0\text{ V}$$

Since V_{GS} is fixed at 0 V, the drain current must be I_{DSS} (by definition). In other words,

$$V_{GS_Q} = \mathbf{0\text{ V}}$$

and

$$I_{D_Q} = \mathbf{10\text{ mA}}$$

There is therefore no need to draw the transfer curve, and

$$V_D = V_{DD} - I_D R_D = 20\text{ V} - (10\text{ mA})(1.5\text{ k}\Omega)$$
$$= 20\text{ V} - 15\text{ V}$$
$$= \mathbf{5\text{ V}}$$

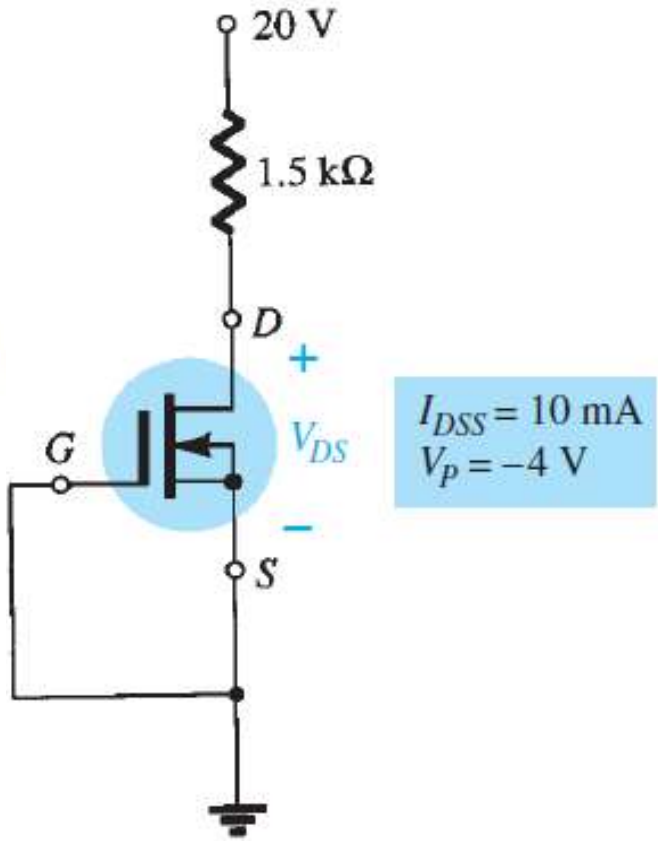


FIG. 7.35

7.8 ENHANCEMENT-TYPE MOSFETs

The transfer characteristics of the enhancement-type MOSFET are quite different from those encountered for the JFET and depletion-type MOSFETs, resulting in a graphical solution quite different from those of the preceding sections.

First and foremost, recall that for the *n* -channel enhancement-type MOSFET, the drain current is zero for levels of gate to-source voltage less than the threshold level $V_{GS(Th)}$, as shown in Fig. 7.36 . For levels of V_{GS} greater than $V_{GS(Th)}$, the drain current is defined by

$$I_D = k(V_{GS} - V_{GS(Th)})^2 \qquad (7.33)$$

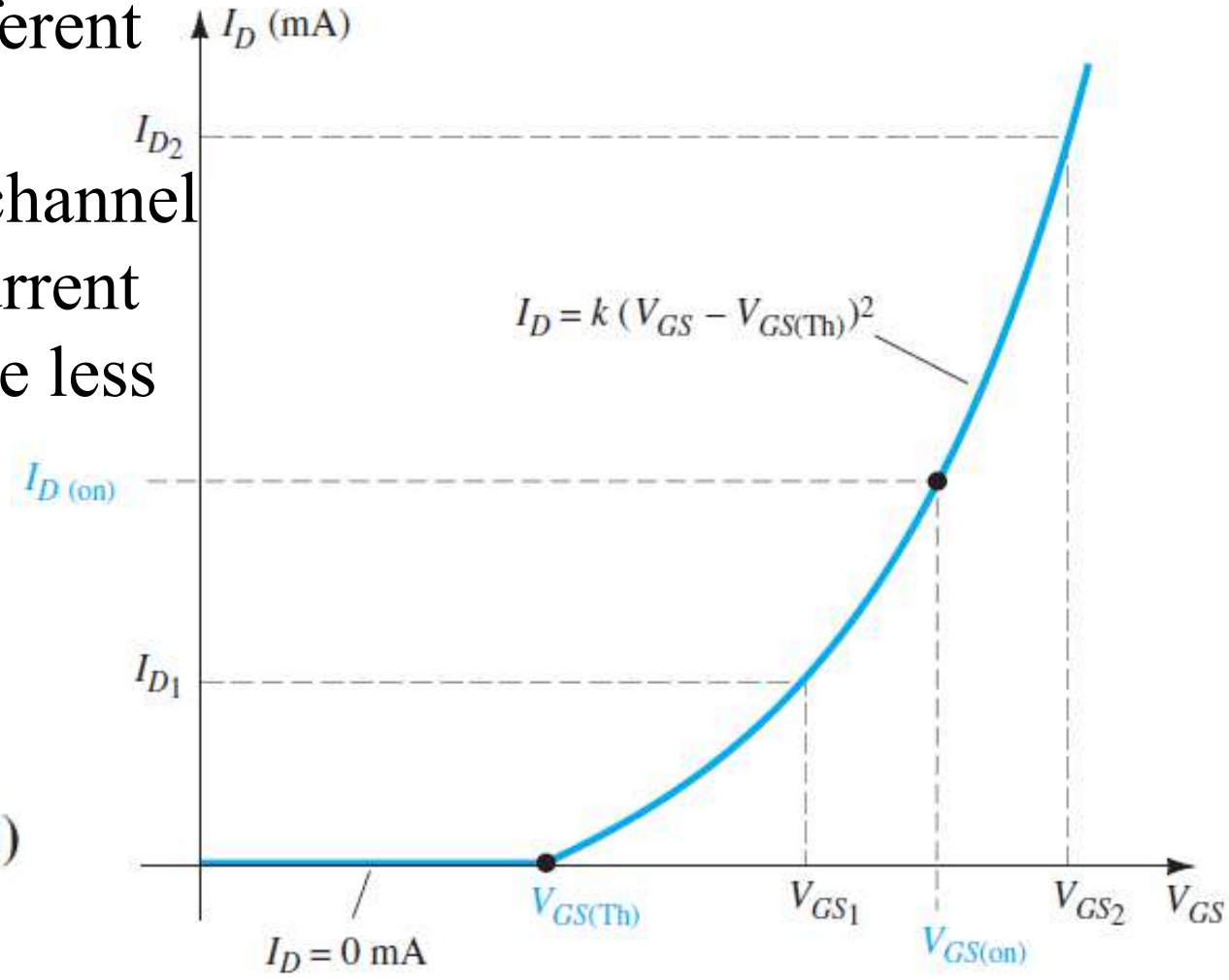


FIG. 7.36

Since specification sheets typically provide the threshold voltage and a level of drain current ($I_{D(on)}$) and its corresponding level of $V_{GS(on)}$, two points are defined immediately as shown in Fig. 7.36. To complete the curve, the constant k of Eq. (7.33) must be determined from the specification sheet data by substituting into Eq. (7.33) and solving for k as follows:

$$I_D = k(V_{GS} - V_{GS(Th)})^2$$
$$I_{D(on)} = k(V_{GS(on)} - V_{GS(Th)})^2$$

and

$$k = \frac{I_{D(on)}}{(V_{GS(on)} - V_{GS(Th)})^2}$$

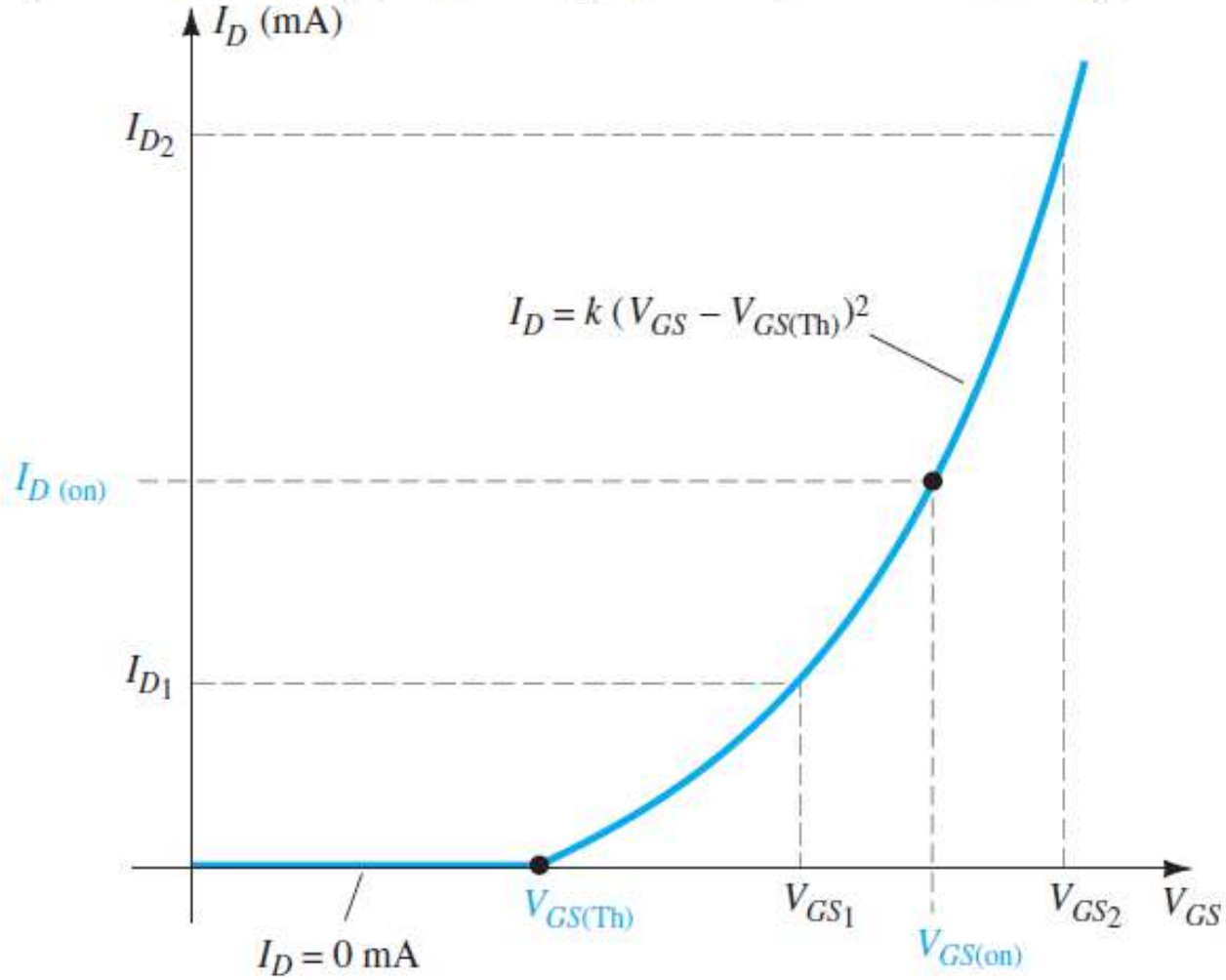


FIG. 7.36

Once k is defined, other levels of I_D can be determined for chosen values of V_{GS}. Typically, a point between $V_{GS(\text{Th})}$ and $V_{GS(\text{on})}$ and one just greater than $V_{GS(\text{on})}$ will provide a sufficient number of points to plot Eq. (7.33) (note I_{D_1} and I_{D_2} on Fig. 7.36).

$$I_D = k(V_{GS} - V_{GS(\text{Th})})^2$$
$$I_{D(\text{on})} = k(V_{GS(\text{on})} - V_{GS(\text{Th})})^2$$

and

$$k = \frac{I_{D(\text{on})}}{(V_{GS(\text{on})} - V_{GS(\text{Th})})^2}$$

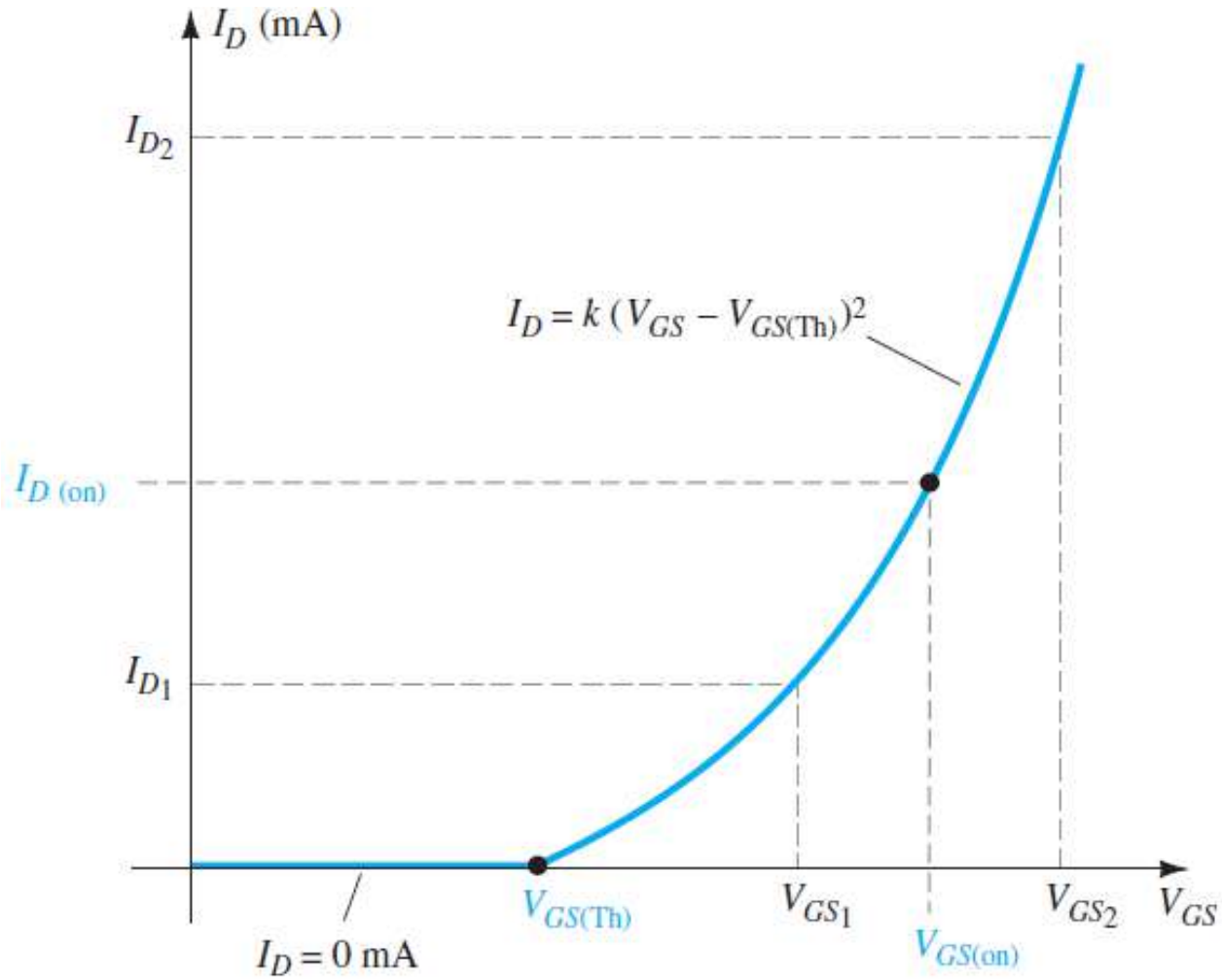


FIG. 7.36

Feedback Biasing Arrangement

A popular biasing arrangement for enhancement-type MOSFETs is provided in Fig. 7.37. The resistor R_G brings a suitably large voltage to the gate to drive the MOSFET "on." Since $I_G = 0$ mA, $V_{R_G} = 0$ V and the dc equivalent network appears as shown in Fig. 7.38.

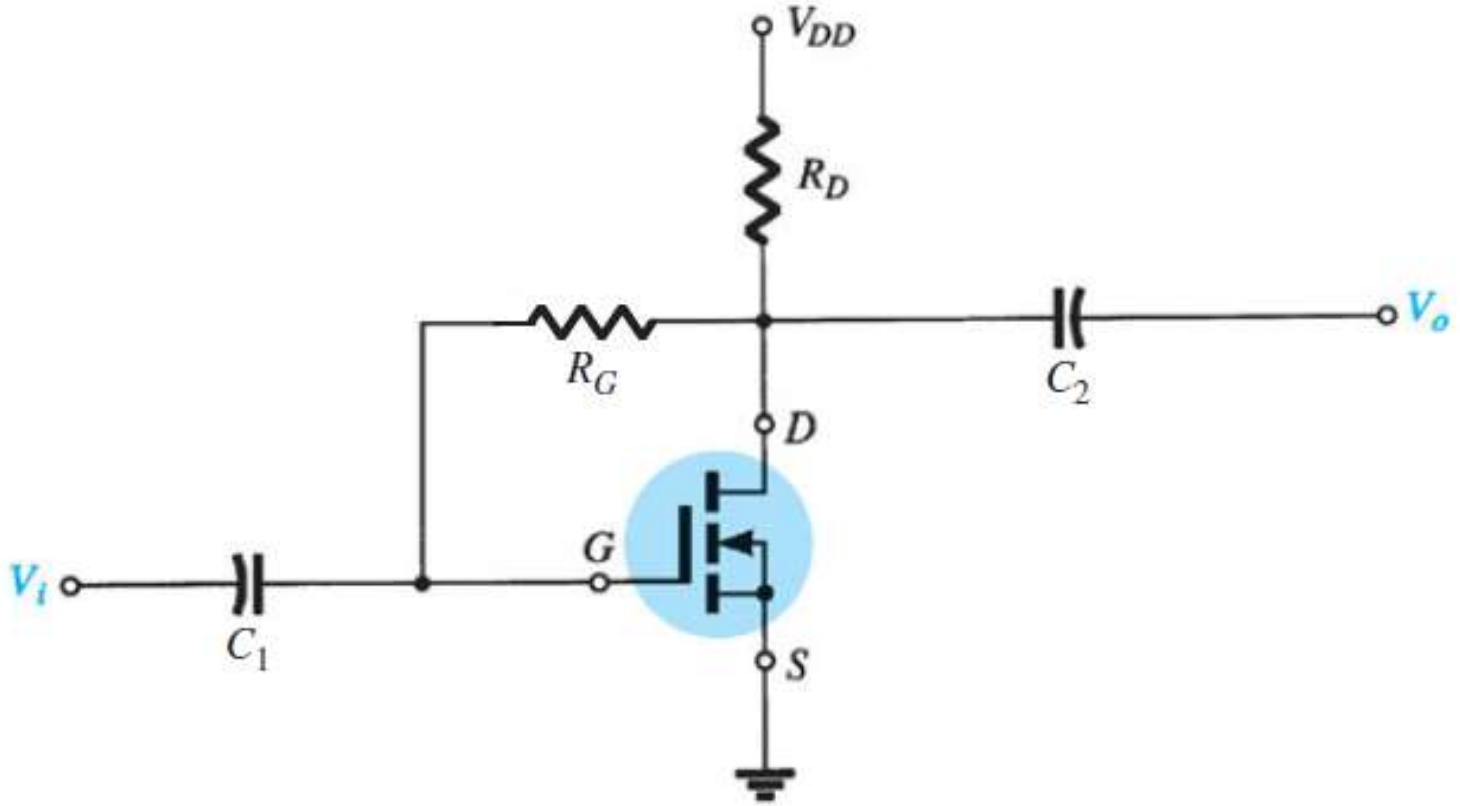


FIG. 7.37
Feedback biasing arrangement.

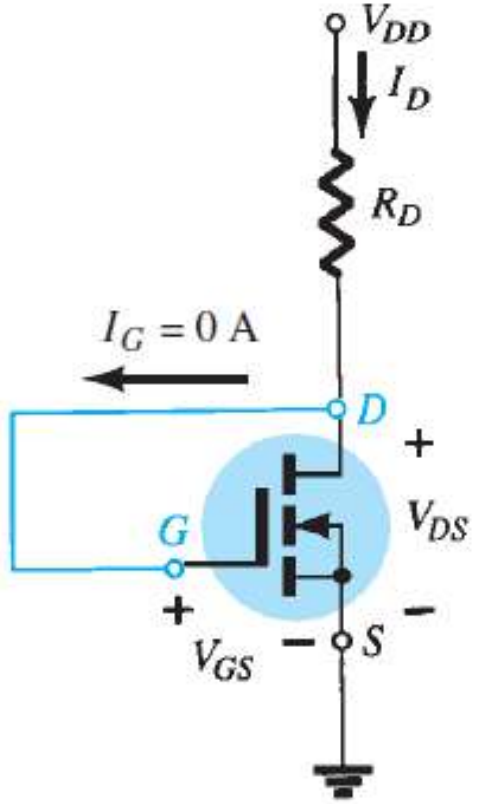


FIG. 7.38
DC equivalent of the network of Fig. 7.37.

A direct connection now exists between drain and gate, resulting in

$$V_D = V_G$$

and

$$\boxed{V_{DS} = V_{GS}}$$

For the output circuit,

$$V_{DS} = V_{DD} - I_D R_D$$

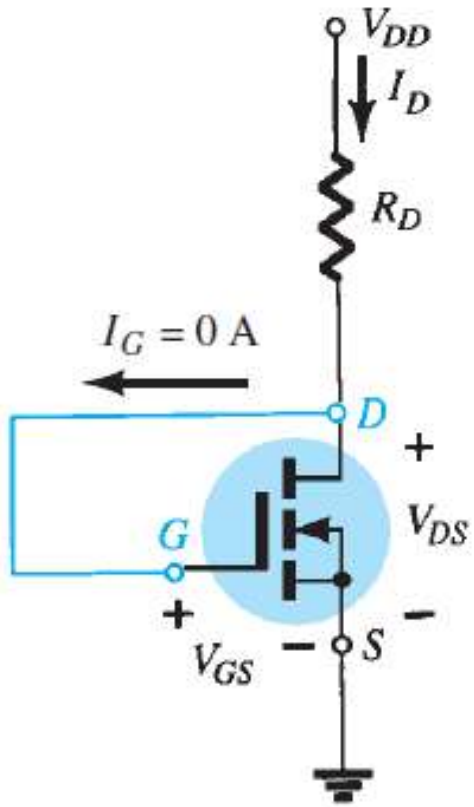


FIG. 7.38
DC equivalent of the network of Fig. 7.37.

which becomes the following after substituting Eq. (7.27):

$$V_{GS} = V_{DD} - I_D R_D \tag{7.36}$$

The result is an equation that relates I_D to V_{GS}, permitting the plot of both on the same set of axes.

Since Eq. (7.36) is that of a straight line, the same procedure described earlier can be employed to determine the two points that will define the plot on the graph. Substituting $I_D = 0$ mA into Eq. (7.36) gives

$$V_{GS} = V_{DD}\big|_{I_D = 0\text{ mA}}$$

Substituting $V_{GS} = 0$ V into Eq. (7.36), we have

$$I_D = \frac{V_{DD}}{R_D}\bigg|_{V_{GS} = 0\text{ V}}$$

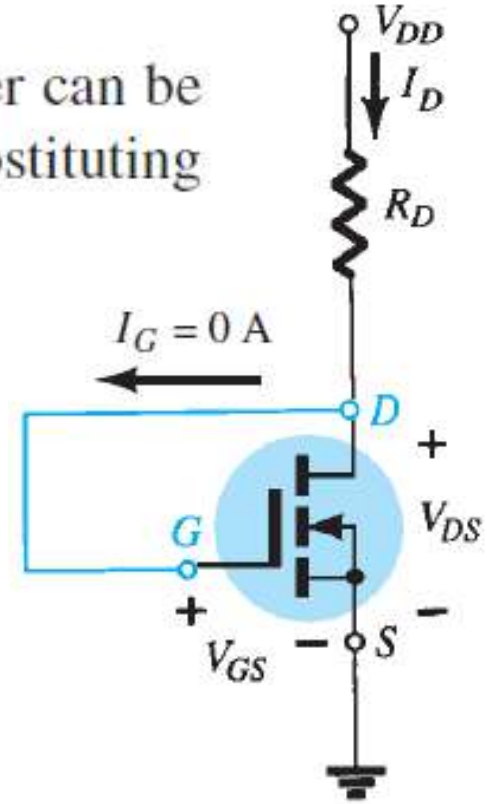


FIG. 7.38
DC equivalent of the network of Fig. 7.37.

The plots defined by Eqs. (7.33) and (7.36) appear in Fig. 7.39 with the resulting operating point.

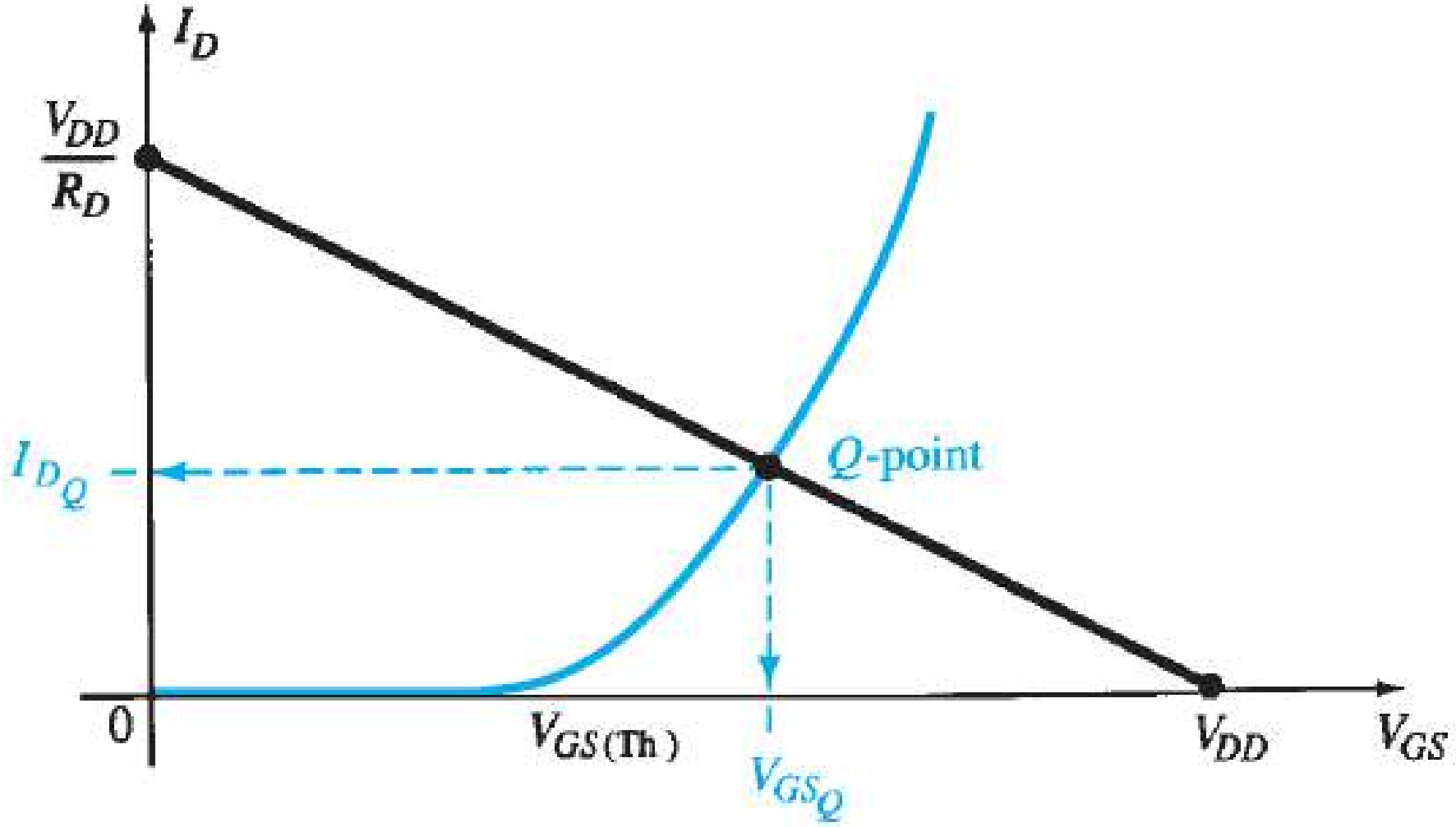


FIG. 7.39

Determining the Q-point for the network of Fig. 7.37.

EXAMPLE 7.10 Determine I_{D_Q} and V_{DS_Q} for the enhancement-type MOSFET of Fig. 7.40

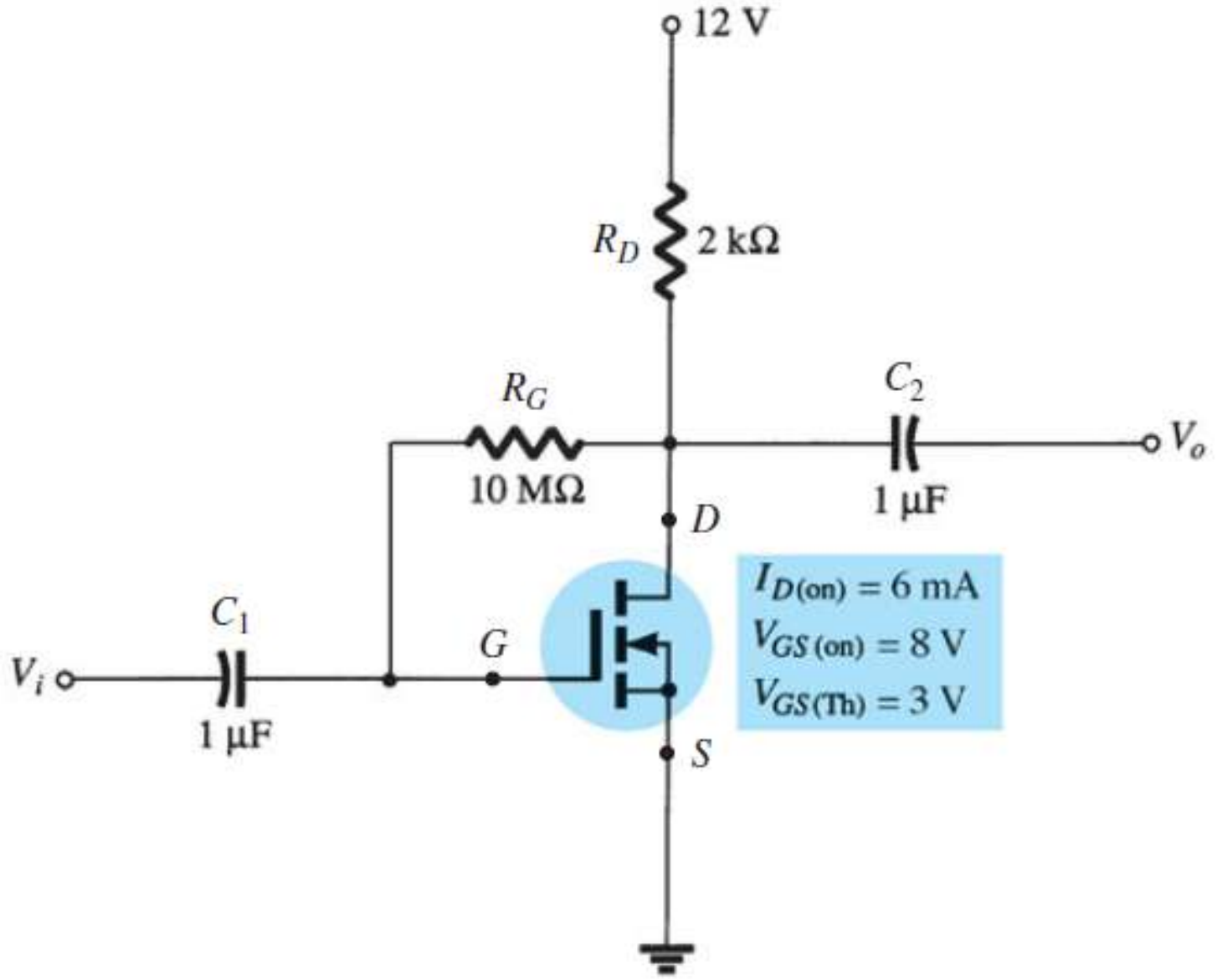


FIG. 7.40

Solution:

Plotting the Transfer Curve Two points are defined immediately as shown in Fig. 7.41.

Solving for k, we obtain

$$\text{Eq. (7.34):} \quad k = \frac{I_{D(\text{on})}}{(V_{GS(\text{on})} - V_{GS(\text{Th})})^2}$$

$$= \frac{6\ \text{mA}}{(8\ \text{V} - 3\ \text{V})^2} = \frac{6 \times 10^{-3}}{25}\ \text{A/V}^2$$

$$= \mathbf{0.24 \times 10^{-3}\ A/V^2}$$

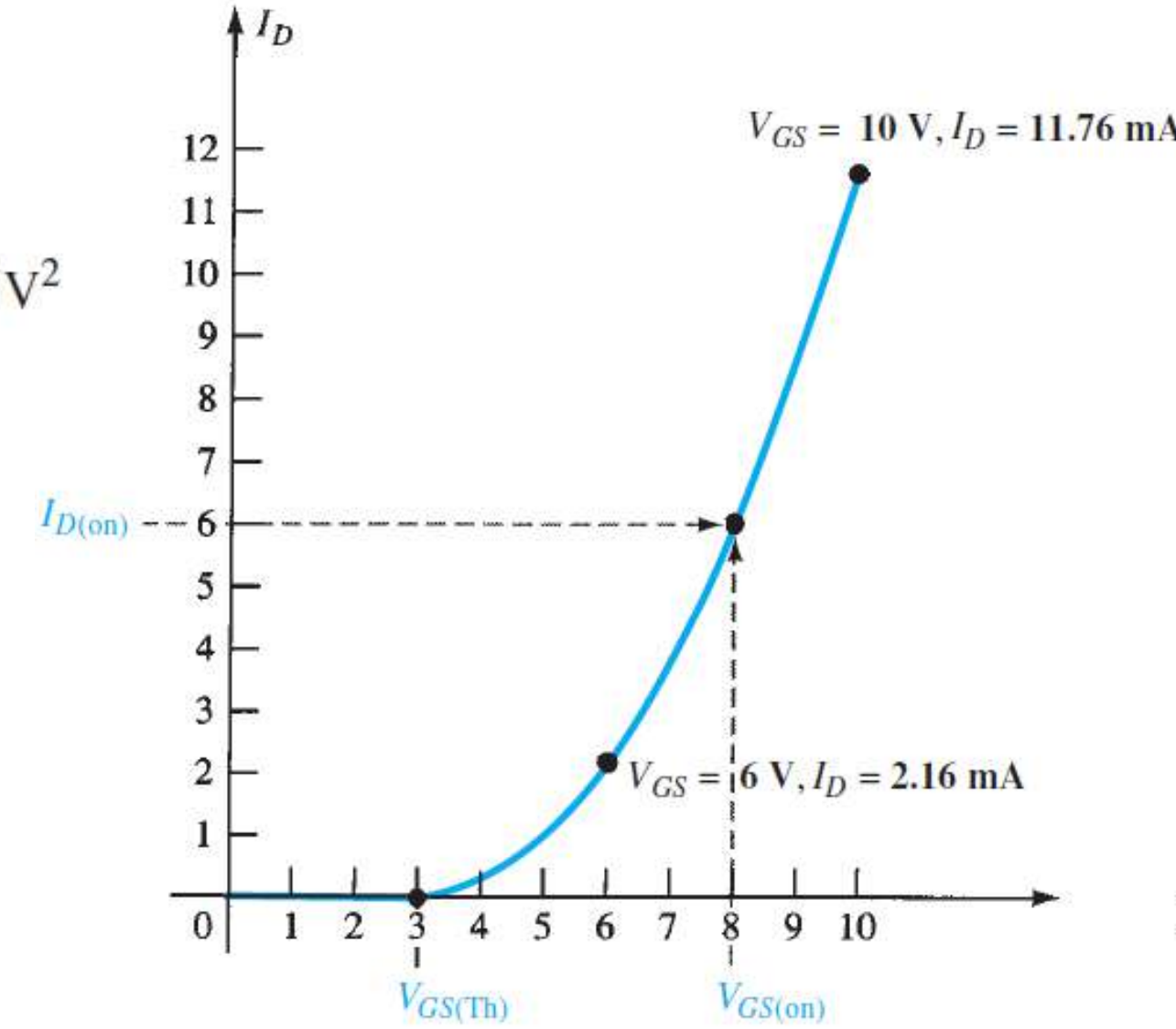


FIG. 7.41

For $V_{GS} = 6$ V (between 3 and 8 V):

$$I_D = 0.24 \times 10^{-3}(6\text{ V} - 3\text{ V})^2 = 0.24 \times 10^{-3}(9)$$
$$= 2.16\text{ mA}$$

on Fig. 7.41. For $V_{GS} = 10$ V (slightly greater than $V_{GS(\text{Th})}$),

$$I_D = 0.24 \times 10^{-3}(10\text{ V} - 3\text{ V})^2 = 0.24 \times 10^{-3}(49)$$
$$= 11.76\text{ mA}$$

as also appearing on Fig. 7.41. The four points are sufficient to plot the full curve for the range of interest as shown in Fig. 7.41.

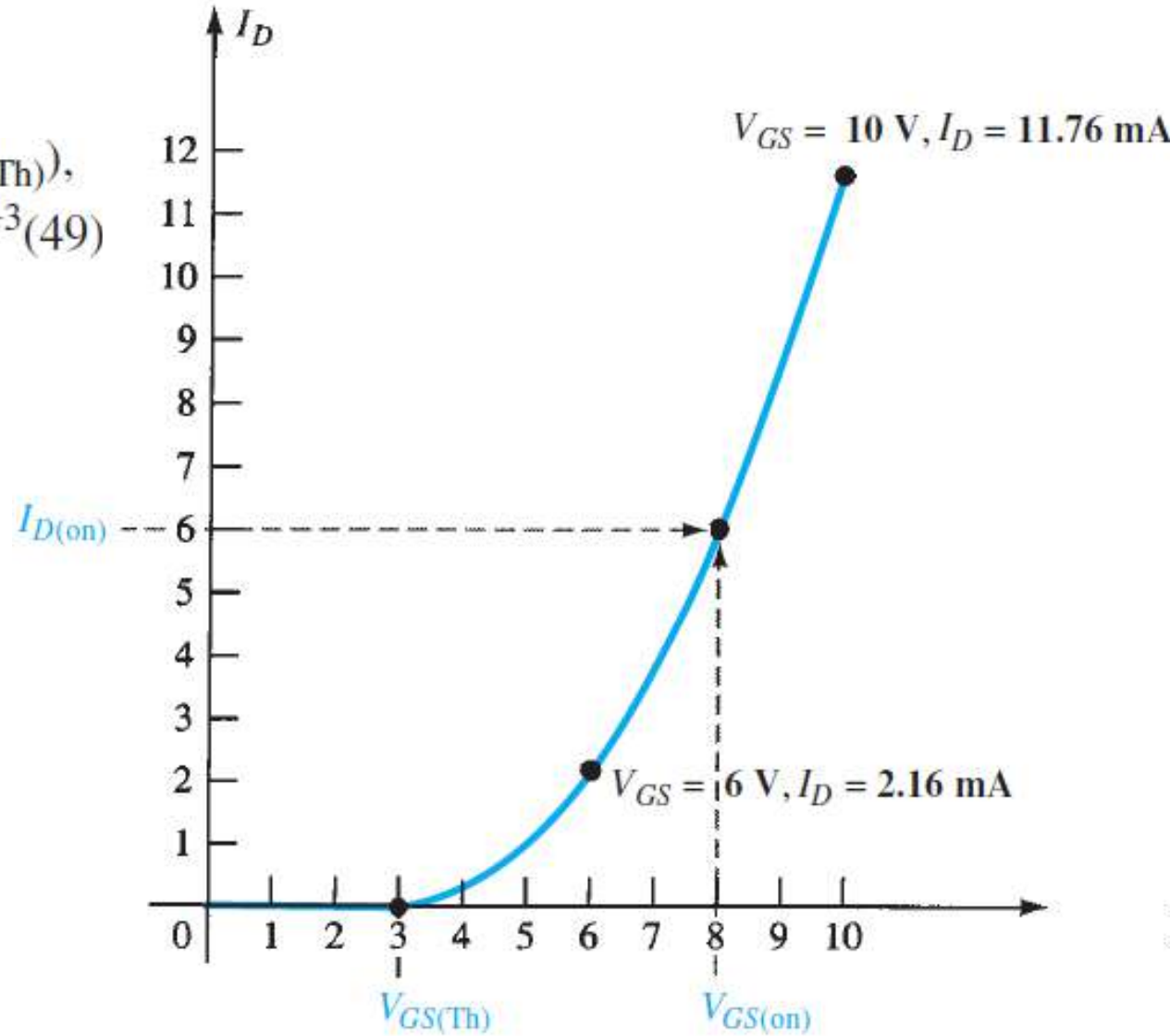


FIG. 7.41

For the Network Bias Line

$$V_{GS} = V_{DD} - I_D R_D$$
$$= 12\text{ V} - I_D(2\text{ k}\Omega)$$

Eq. (7.37): $$V_{GS} = V_{DD} = 12\text{ V}\big|_{I_D=0\text{ mA}}$$

Eq. (7.38): $$I_D = \frac{V_{DD}}{R_D} = \frac{12\text{ V}}{2\text{ k}\Omega} = 6\text{ mA}\big|_{V_{GS}=0\text{ V}}$$

The resulting bias line appears in Fig. 7.42.

At the operating point,

$$I_{D_Q} = \mathbf{2.75\text{ mA}}$$

and $$V_{GS_Q} = 6.4\text{ V}$$

with $$V_{DS_Q} = V_{GS_Q} = \mathbf{6.4\text{ V}}$$

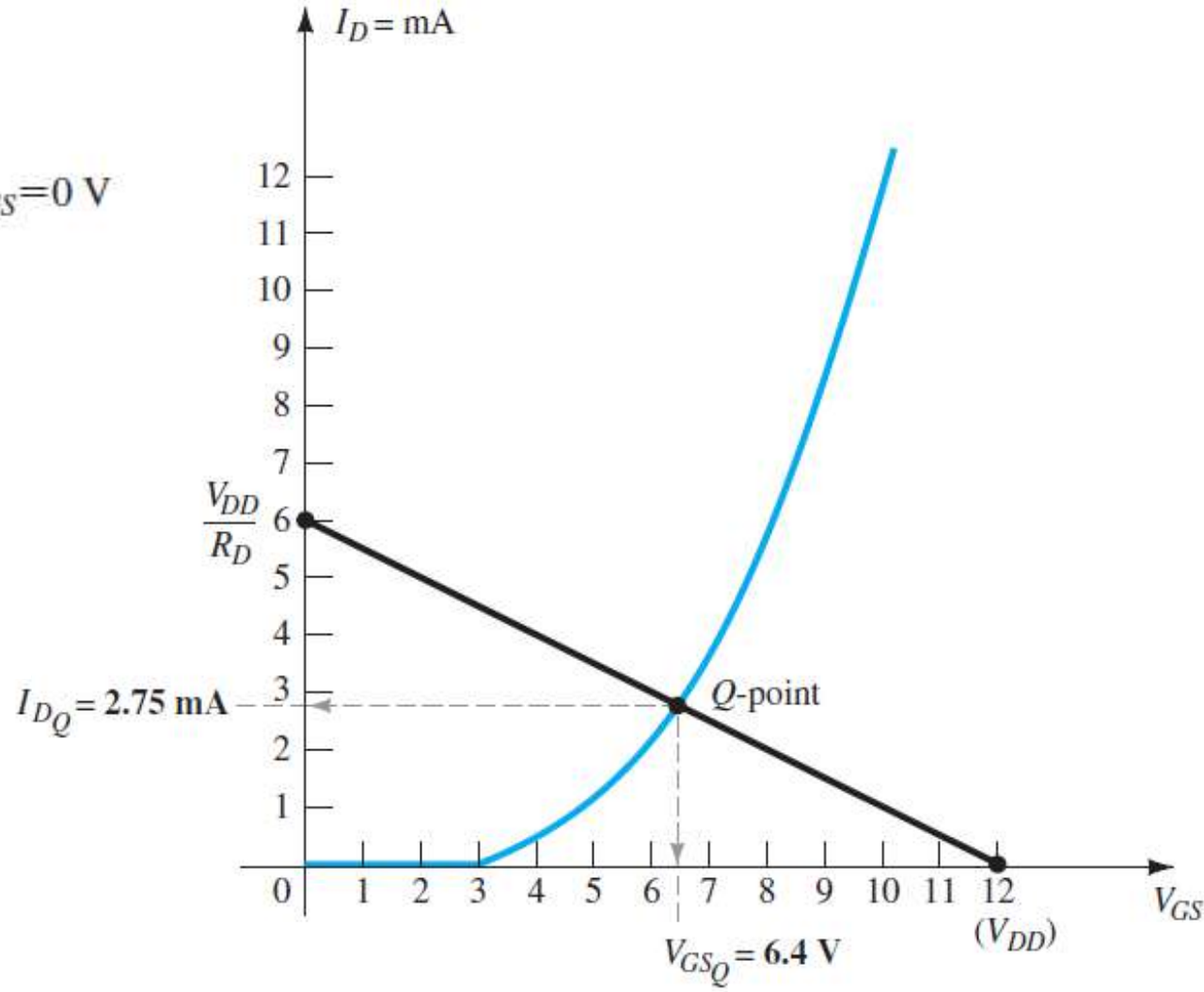


FIG. 7.42

Voltage-Divider Biasing Arrangement

A second popular biasing arrangement for the enhancement-type MOSFET appears in Fig. 7.43. The fact that $I_G = 0$ mA results in the following equation for V_{GG} as derived from an application of the voltage-divider rule:

$$V_G = \frac{R_2 V_{DD}}{R_1 + R_2}$$

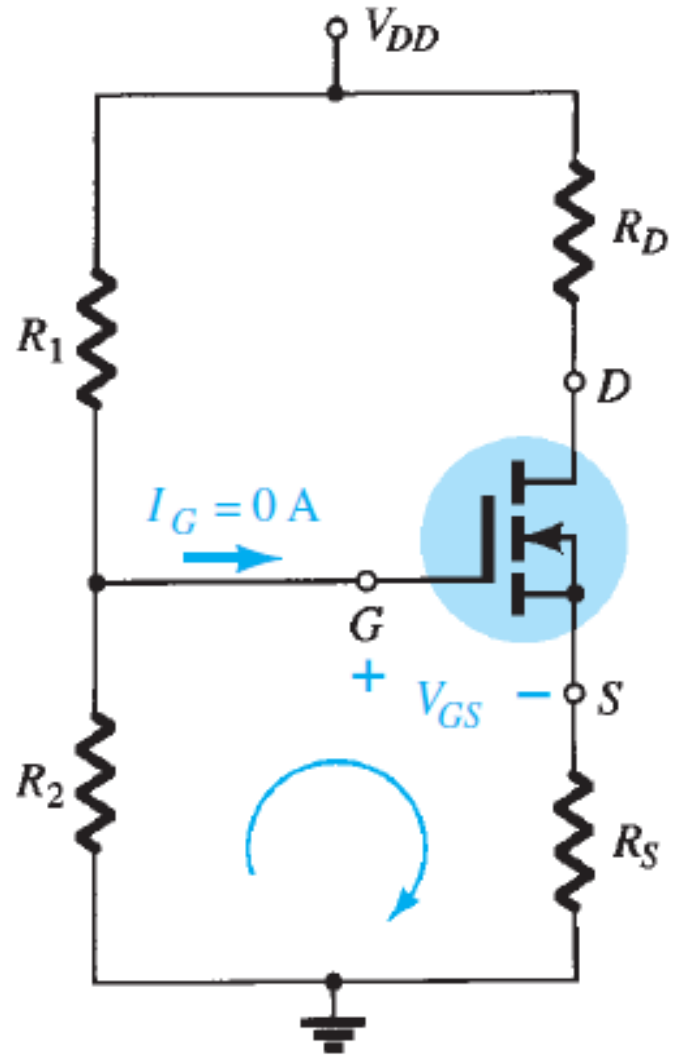


FIG. 7.43

Applying Kirchhoff's voltage law around the indicated loop of Fig. 7.43 results in

$$+V_G - V_{GS} - V_{R_S} = 0$$

and

$$V_{GS} = V_G - V_{R_S}$$

or

$$\boxed{V_{GS} = V_G - I_D R_S}$$

For the output section,

$$V_{R_S} + V_{DS} + V_{R_D} - V_{DD} = 0$$

and

$$V_{DS} = V_{DD} - V_{R_S} - V_{R_D}$$

or

$$\boxed{V_{DS} = V_{DD} - I_D(R_S + R_D)}$$

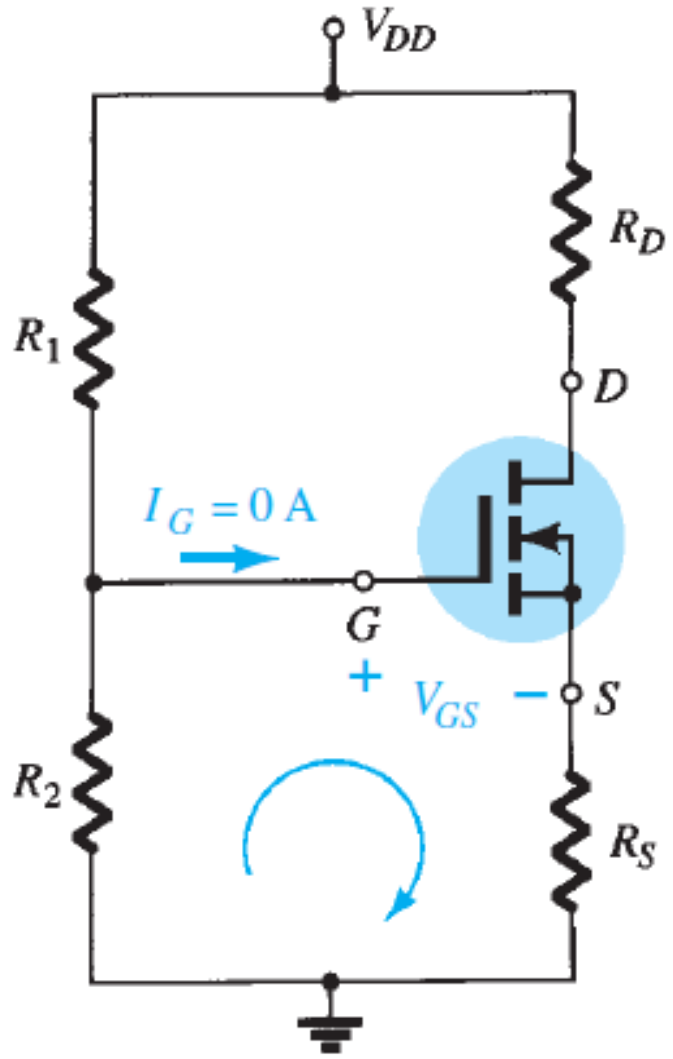


FIG. 7.43

Since the characteristics are a plot of I_D versus V_{GS} and Eq. (7.40) relates the same two variables, the two curves can be plotted on the same graph and a solution determined at their intersection. Once I_{D_Q} and V_{GS_Q} are known, all the remaining quantities of the network such as V_{DS}, V_D, and V_S can be determined.

$$V_{GS} = V_G - I_D R_S \tag{7.40}$$

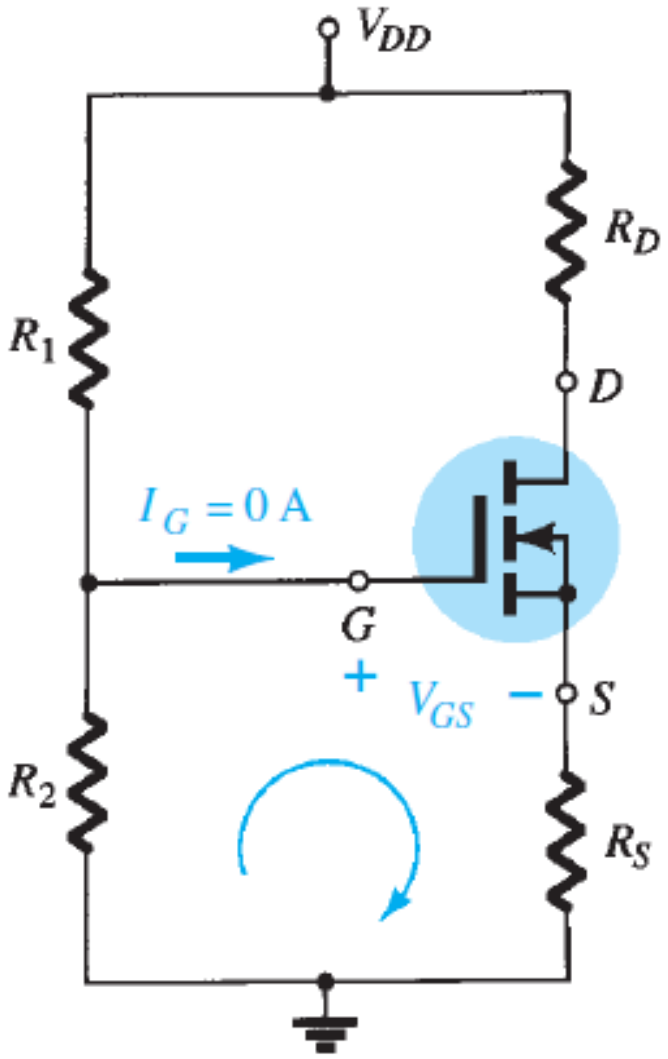


FIG. 7.43

EXAMPLE 7.11 Determine I_{D_Q}, V_{GS_Q}, and V_{DS} for the network of Fig. 7.44.

Solution:

Network

$$V_G = \frac{R_2 V_{DD}}{R_1 + R_2} = \frac{(18\,\text{M}\Omega)(40\,\text{V})}{22\,\text{M}\Omega + 18\,\text{M}\Omega} = 18\,\text{V}$$

$$V_{GS} = V_G - I_D R_S = 18\,\text{V} - I_D(0.82\,\text{k}\Omega)$$

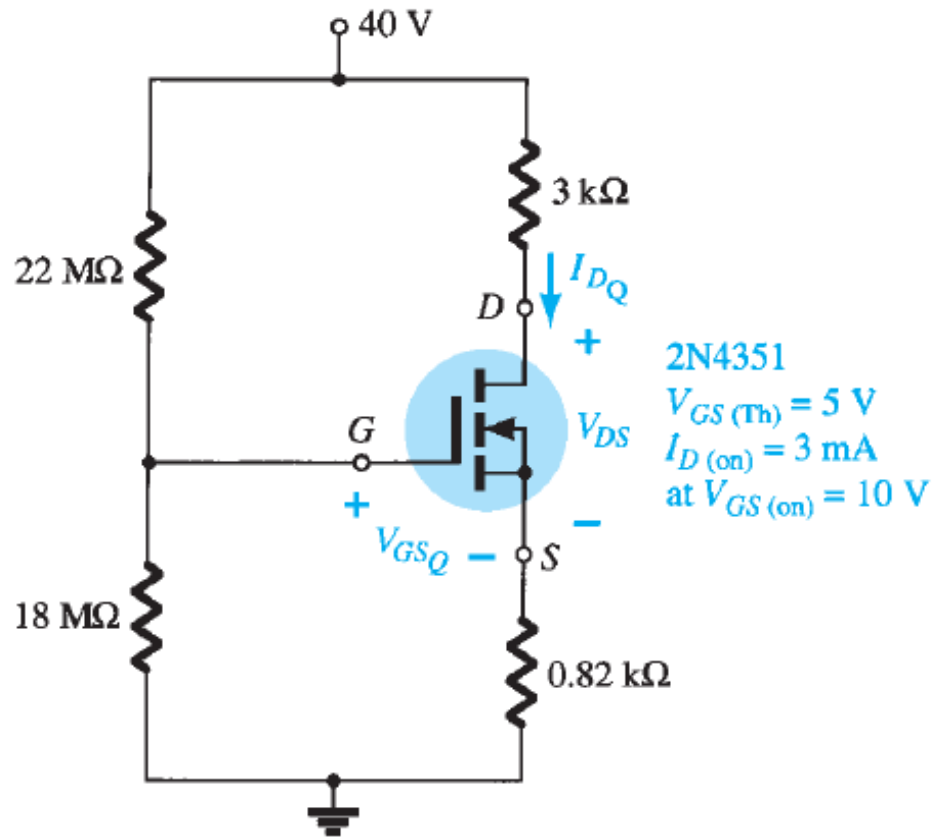


FIG. 7.44

When $I_D = 0$ mA,

$$V_{GS} = 18\text{ V} - (0\text{ mA})(0.82\text{ k}\Omega) = 18\text{ V}$$

as appearing on Fig. 7.45. When $V_{GS} = 0$ V,

$$V_{GS} = 18\text{ V} - I_D(0.82\text{ k}\Omega)$$

$$0 = 18\text{ V} - I_D(0.82\text{ k}\Omega)$$

$$I_D = \frac{18\text{ V}}{0.82\text{ k}\Omega} = 21.95\text{ mA}$$

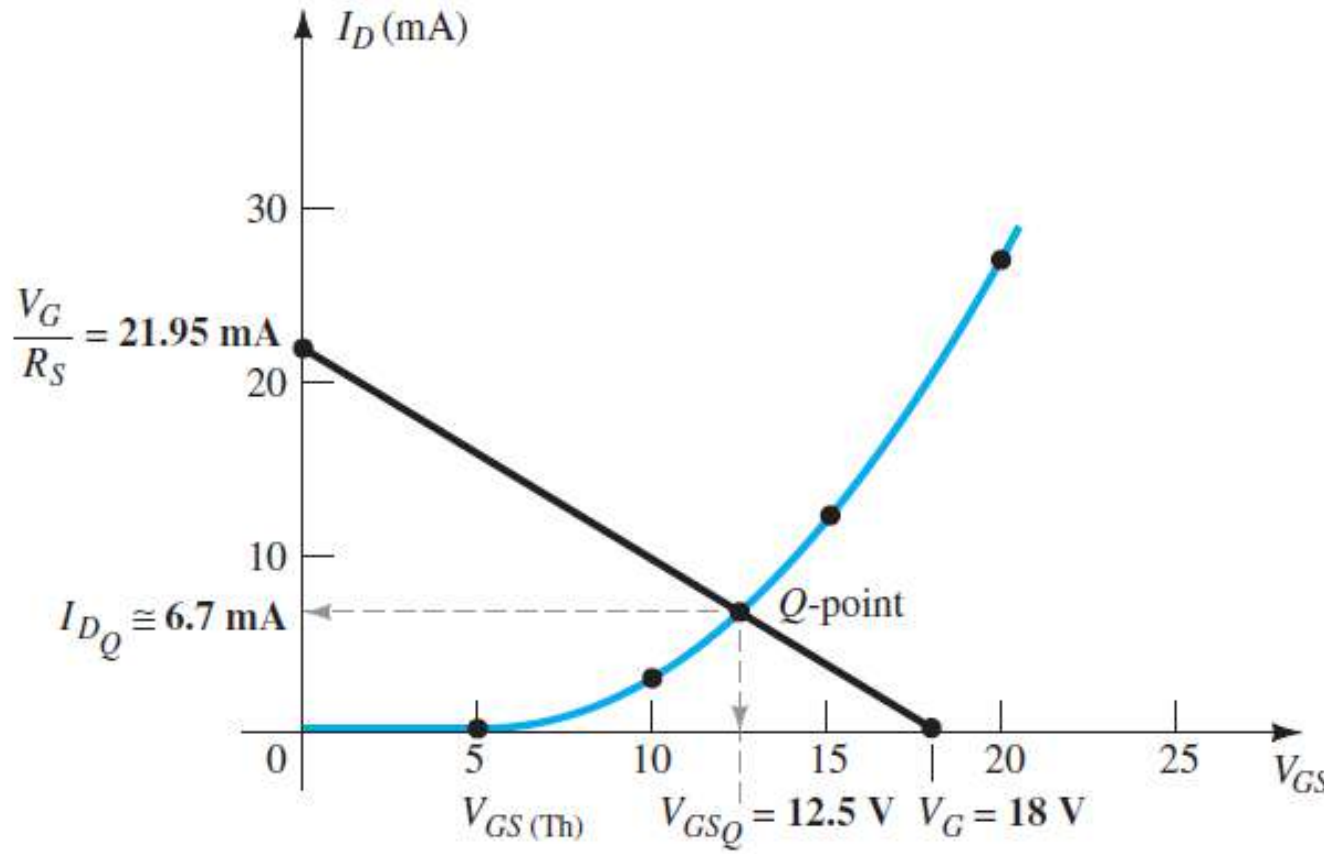


FIG. 7.45

Device

$$V_{GS(\text{Th})} = 5\text{ V}, \quad I_{D(\text{on})} = 3\text{ mA with } V_{GS(\text{on})} = 10\text{ V}$$

$$\text{Eq. (7.34):} \quad k = \frac{I_{D(\text{on})}}{(V_{GS(\text{on})} - V_{GS(\text{Th})})^2}$$

$$= \frac{3\text{ mA}}{(10\text{ V} - 5\text{ V})^2} = 0.12 \times 10^{-3}\text{ A/V}^2$$

and

$$I_D = k(V_{GS} - V_{GS(\text{Th})})^2$$

$$= 0.12 \times 10^{-3}(V_{GS} - 5)^2$$

which is plotted on the same graph (Fig. 7.45). From Fig. 7.45,

$$I_{D_Q} \cong \mathbf{6.7\text{ mA}}$$

$$V_{GS_Q} = \mathbf{12.5\text{ V}}$$

$$\text{Eq. (7.41):} \quad V_{DS} = V_{DD} - I_D(R_S + R_D)$$

$$= 40\text{ V} - (6.7\text{ mA})(0.82\text{ k}\Omega + 3.0\text{ k}\Omega)$$

$$= 40\text{ V} - 25.6\text{ V}$$

$$= \mathbf{14.4\text{ V}}$$

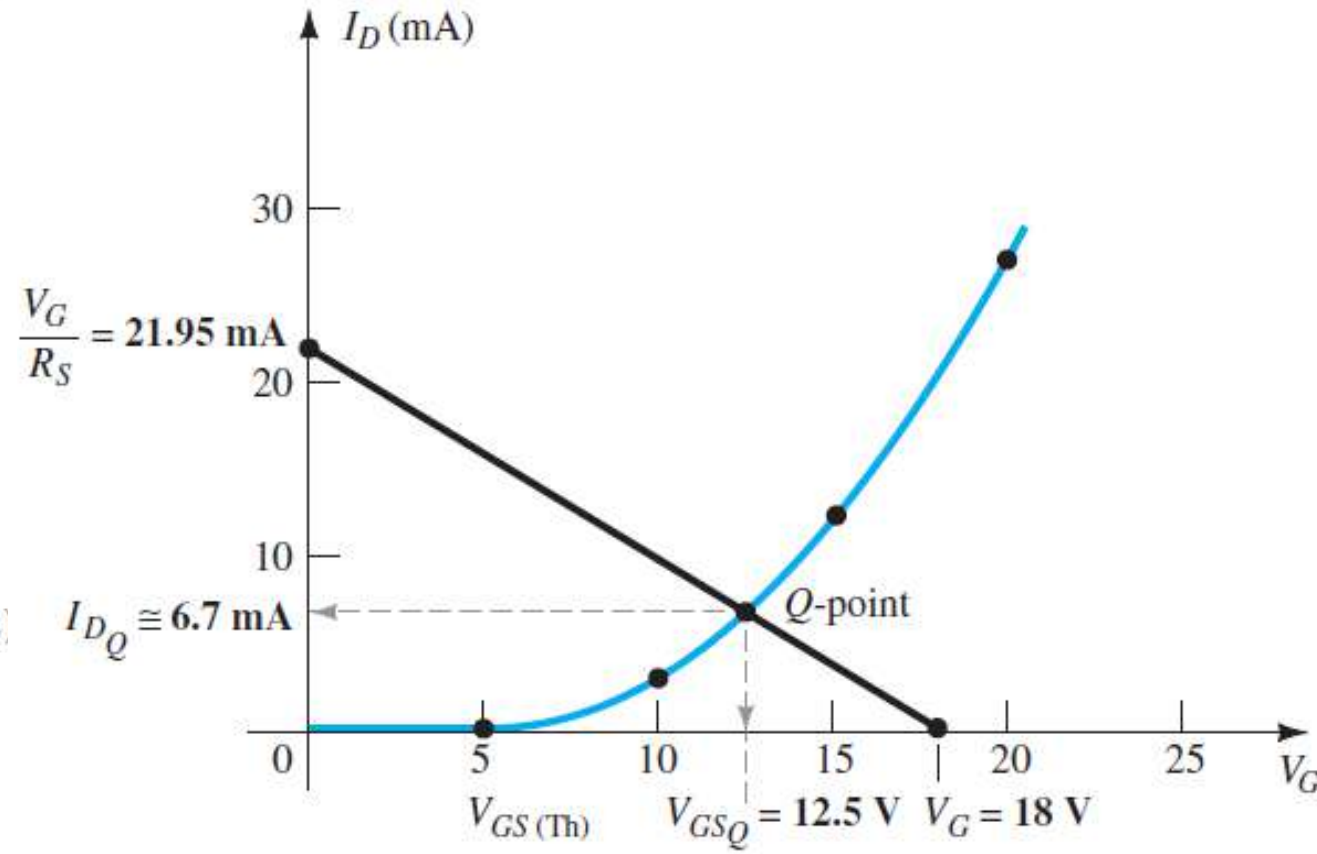


FIG. 7.45

FET Bias Configurations

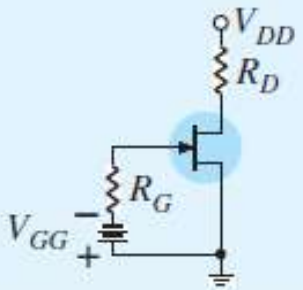
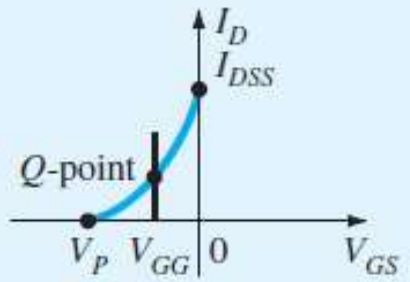
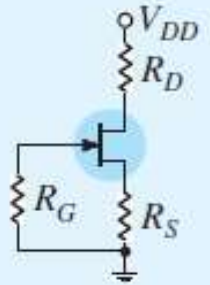
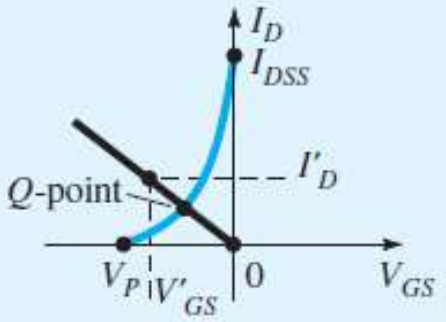
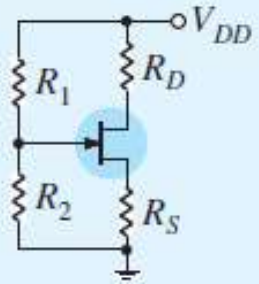
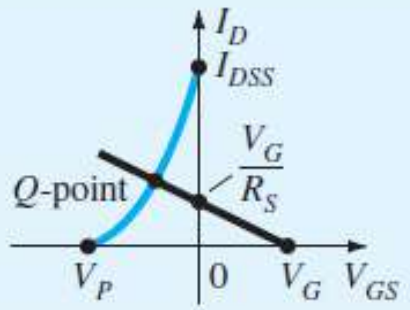
Type	Configuration	Pertinent Equations	Graphical Solution
JFET Fixed-bias		$V_{GS_Q} = -V_{GG}$ $V_{DS} = V_{DD} - I_D R_S$	
JFET Self-bias		$V_{GS} = -I_D R_S$ $V_{DS} = V_{DD} - I_D(R_D + R_S)$	
JFET Voltage-divider bias		$V_G = \dfrac{R_2 V_{DD}}{R_1 + R_2}$ $V_{GS} = V_G - I_D R_S$ $V_{DS} = V_{DD} - I_D(R_D + R_S)$	

TABLE 7.1

FET Bias Configurations

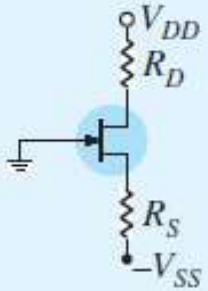
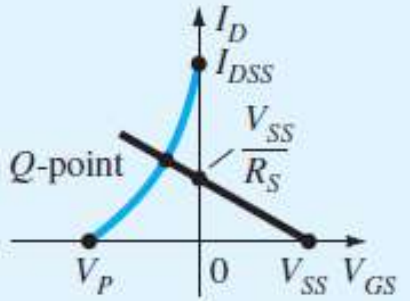
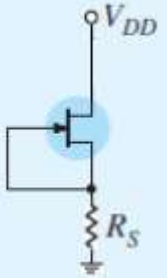
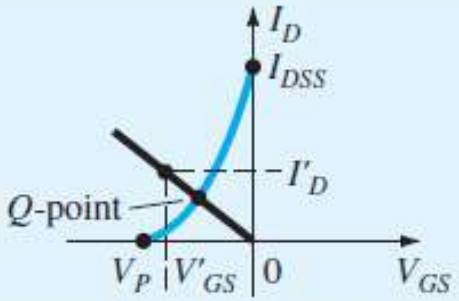
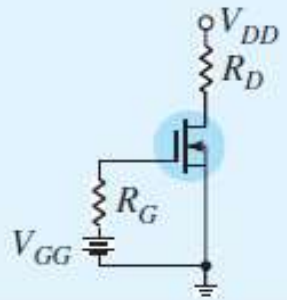
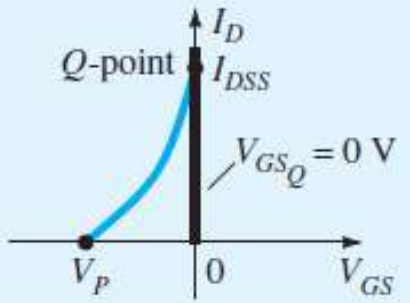
Type	Configuration	Pertinent Equations	Graphical Solution
JFET Common-gate		$V_{GS} = V_{SS} - I_D R_S$ $V_{DS} = V_{DD} + V_{SS} - I_D(R_D + R_S)$	
JFET ($R_D = 0\ \Omega$)		$V_{GS} = -I_D R_S$ $V_D = V_{DD}$ $V_S = I_D R_S$ $V_{DS} = V_{DD} - I_S R_S$	
JFET Special case ($V_{GS_Q} = 0$ V)		$V_{GS_Q} = 0$ V $I_{D_Q} = I_{DSS}$	

TABLE 7.1

FET Bias Configurations

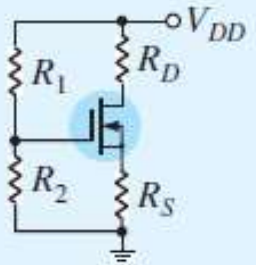
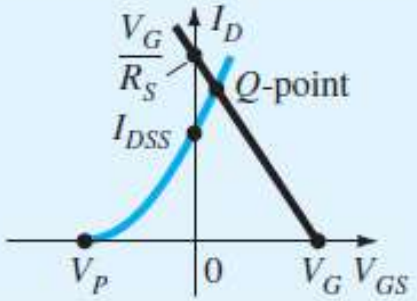
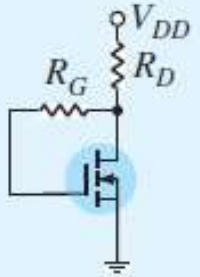
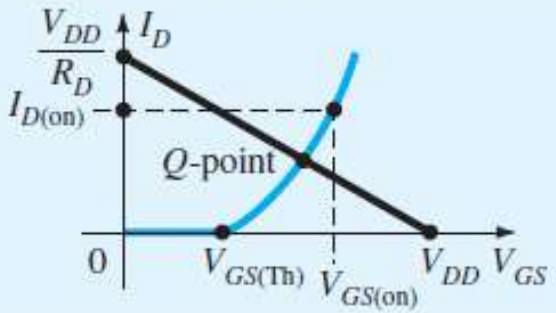
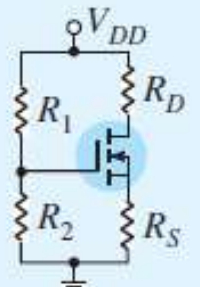
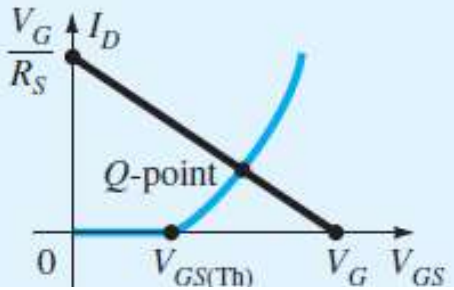
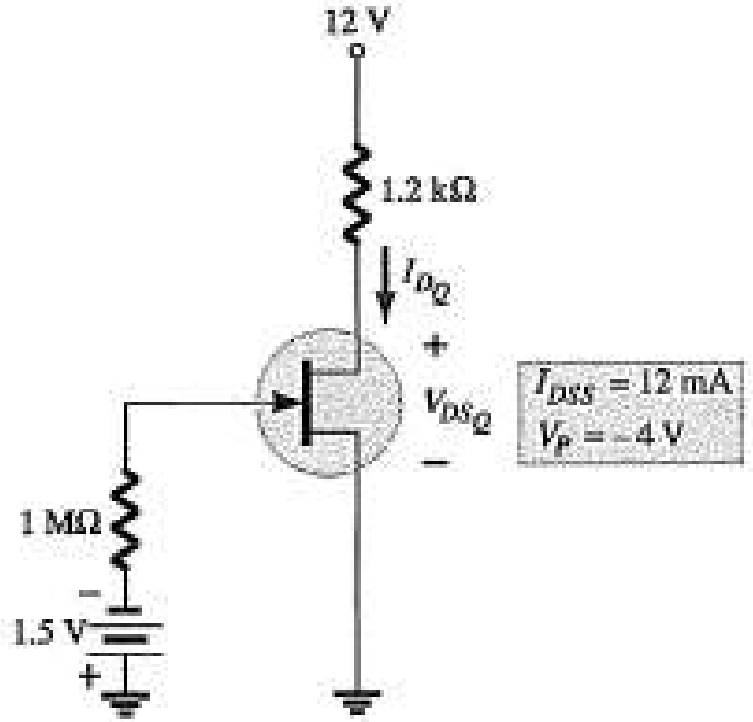
Type	Configuration	Pertinent Equations	Graphical Solution
Depletion-type MOSFET Voltage-divider bias (and MESFETs)		$V_G = \frac{R_2 V_{DD}}{R_1 + R_2}$ $V_{GS} = V_G - I_S R_S$ $V_{DS} = V_{DD} - I_D(R_D + R_S)$	

TABLE 7.1

FET Bias Configurations

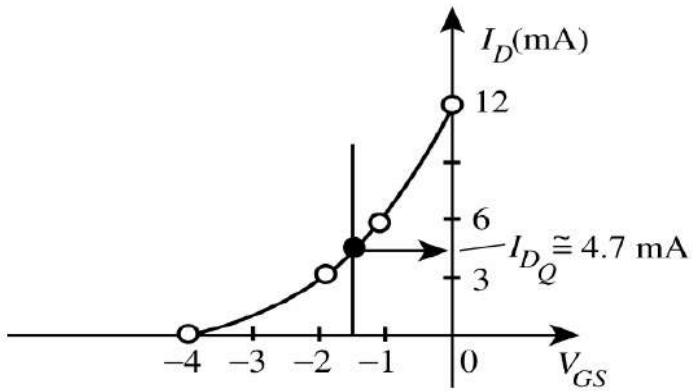
Type	Configuration	Pertinent Equations	Graphical Solution
Enhancement type MOSFET Feedback configuration (and MESFETs)		$V_{GS} = V_{DS}$ $V_{GS} = V_{DD} - I_D R_D$	
Enhancement type MOSFET Voltage-divider bias (and MESFETs)		$V_G = \frac{R_2 V_{DD}}{R_1 + R_2}$ $V_{GS} = V_G - I_D R_S$	

1. For the fixed-bias configuration of Fig. 7.75:
   a. Sketch the transfer characteristics of the device.
   b. Superimpose the network equation on the same graph.
   c. Determine I_{D_Q} and V_{DS_Q}.
   d. Using Shockley's equation, solve for I_{D_Q} and then find V_{DS_Q}. Compare with the solutions of part (c).



1. (a) $V_{GS} = 0$ V, $I_D = I_{DSS} = 12$ mA
$V_{GS} = V_P = -4$ V, $I_D = 0$ mA
$V_{GS} = V_P/2 = -2$ V, $I_D = I_{DSS}/4 = 3$ mA
$V_{GS} = 0.3V_P = -1.2$ V, $I_D = I_{DSS}/2 = 6$ mA

(b)



(c) $I_{D_Q} \cong$ **4.7 mA**

$V_{DS_Q} = V_{DD} - I_{D_Q}R_D = 12\text{ V} - (4.7\text{ mA})(1.2\text{ k}\Omega)$

$=$ **6.36 V**

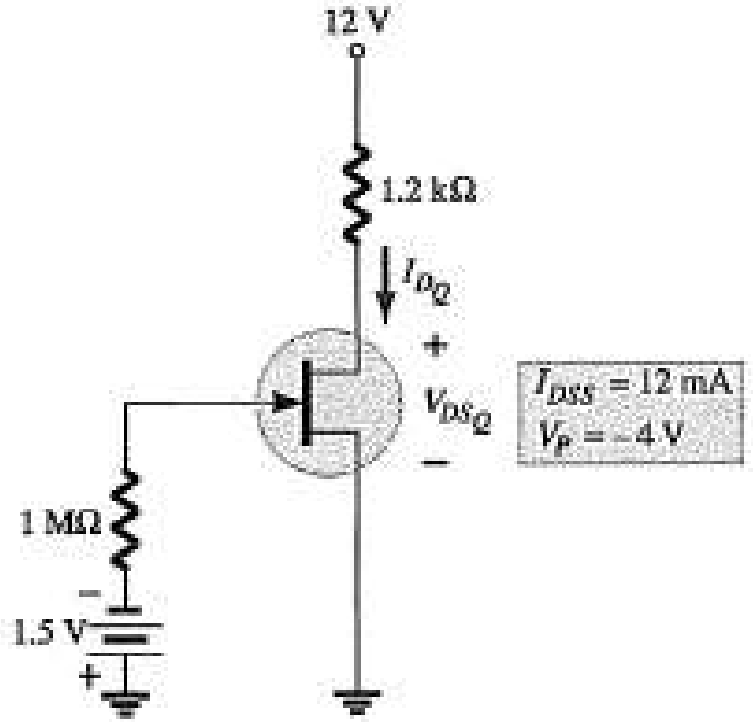
(d) $I_{D_Q} = I_{DSS}(1 - V_{GS}/V_P)^2 = 12\text{ mA}(1 - (-1.5\text{ V})/(-4\text{ V}))^2$

$=$ **4.69 mA**

$V_{DS_Q} = V_{DD} - I_{D_Q}R_D = 12\text{ V} - (4.69\text{ mA})(1.2\text{ k}\Omega)$

$=$ **6.37 V**

excellent comparison

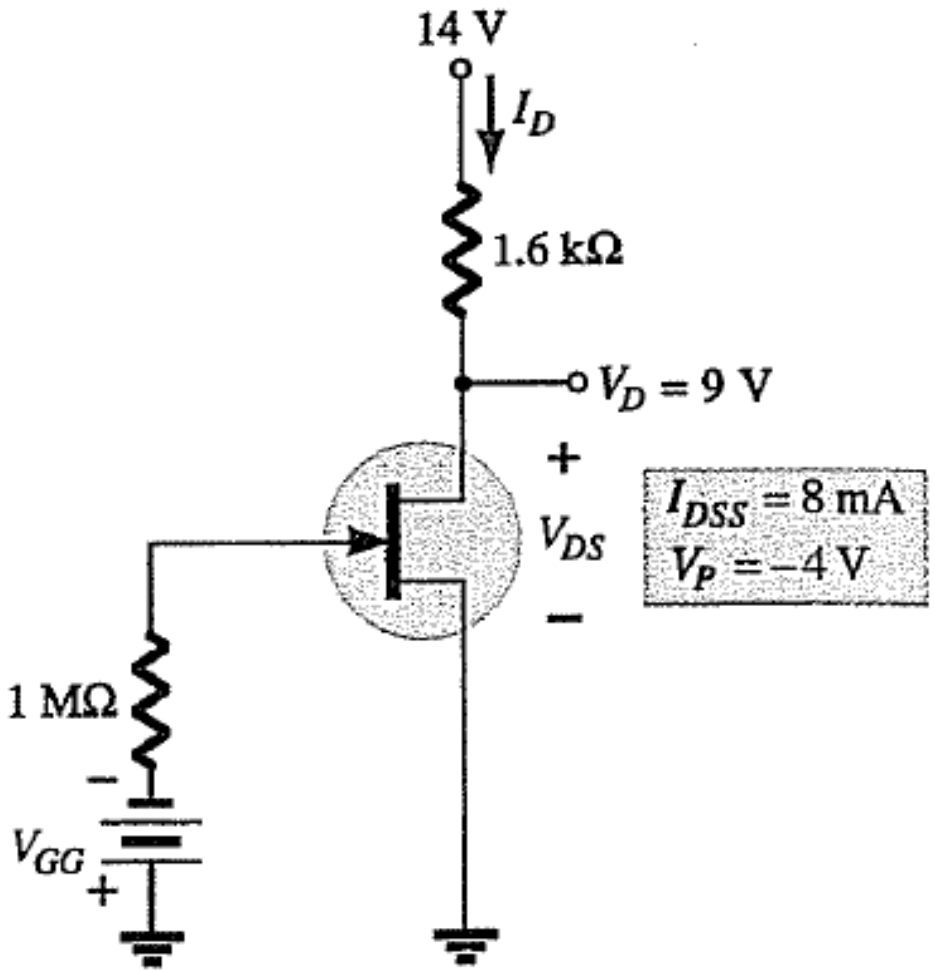


3. Given the measured value of V_D in Fig. 7.77, determine:

a. I_D.

b. V_{DS}.

c. V_{GG}.



(a) $$I_{D_Q} = \frac{V_{DD} - V_D}{R_D} = \frac{14\text{ V} - 9\text{ V}}{1.6\text{ k}\Omega} = \mathbf{3.125\text{ mA}}$$

(b) $$V_{DS} = V_D - V_S = 9\text{ V} - 0\text{ V} = \mathbf{9\text{ V}}$$

(c) $$I_D = I_{DSS}(1 - V_{GS}/V_P)^2 \Rightarrow V_{GS} = V_P\left(1 - \sqrt{\frac{I_D}{I_{DSS}}}\right)$$

$$V_{GS} = (-4\text{ V})\left(1 - \sqrt{\frac{3.125\text{ mA}}{8\text{ mA}}}\right)$$

$$= -1.5\text{ V}$$

$$\therefore V_{GG} = \mathbf{1.5\text{ V}}$$

6. For the self-bias configuration of Fig. 7.80:

a. Sketch the transfer curve for the device.

b. Superimpose the network equation on the same graph.

c. Determine I_{D_Q} and V_{GS_Q}.

d. Calculate V_{DS}, V_D, V_G, and V_S.

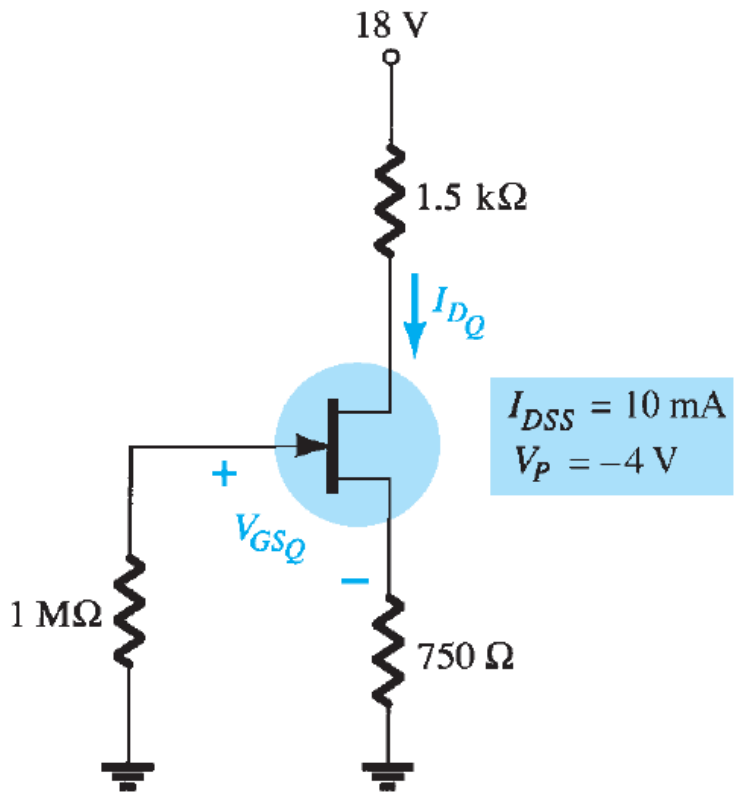


FIG. 7.80

6. (a)(b) $V_{GS} = 0$ V, $I_D = 10$ mA

$V_{GS} = V_P = -4$ V, $I_D = 0$ mA

$V_{GS} = \dfrac{V_P}{2} = -2$ V, $I_D = 2.5$ mA

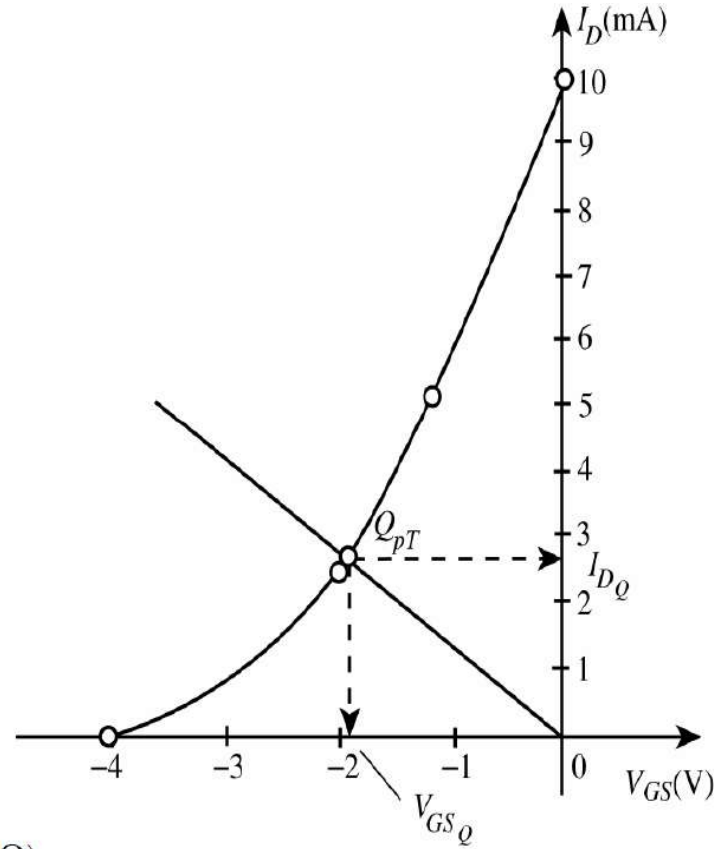
$V_{GS} = 0.3V_P = -1.2$ V, $I_D = 5$ mA

$V_{GS} = -I_D R_S$

$I_D = 5$ mA:

$V_{GS} = -(5\text{ mA})(0.75\text{ k}\Omega)$

$= -3.75$ V



(c) $I_{D_Q} \cong$ **2.7 mA**

$V_{GS_Q} \cong$ **−1.9 V**

(d) $V_{DS} = V_{DD} - I_D(R_D + R_S)$

$= 18\text{ V} - (2.7\text{ mA})(1.5\text{ k}\Omega + 0.75\text{ k}\Omega)$

$=$ **11.93 V**

$V_D = V_{DD} - I_D R_D$

$= 18\text{ V} - (2.7\text{ mA})(1.5\text{ k}\Omega)$

$=$ **13.95 V**

$V_G =$ **0 V**

$V_S = I_S R_S = I_D R_S$

$= (2.7\text{ mA})(0.75\text{ k}\Omega)$

$=$ **2.03 V**

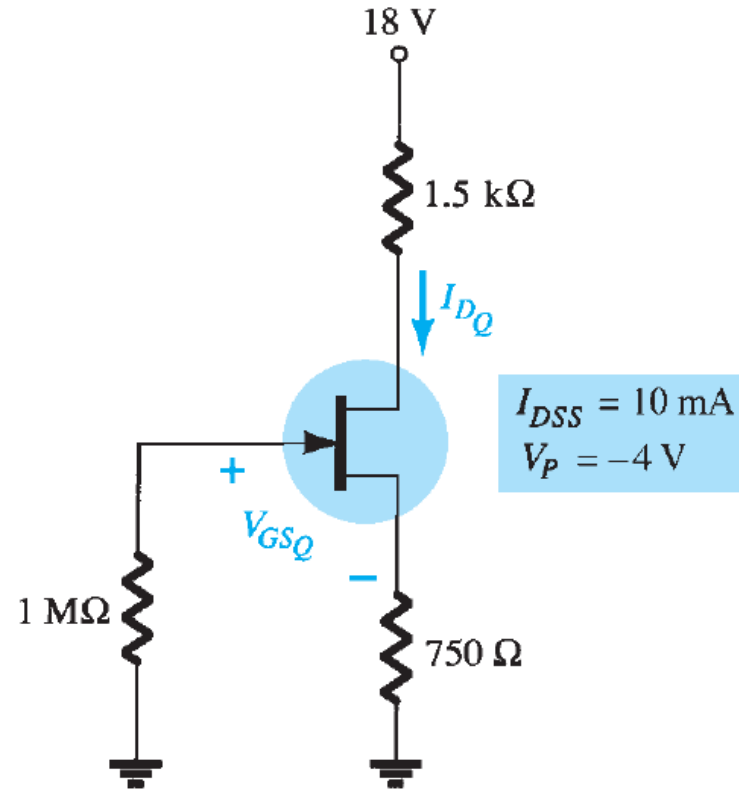


FIG. 7.80

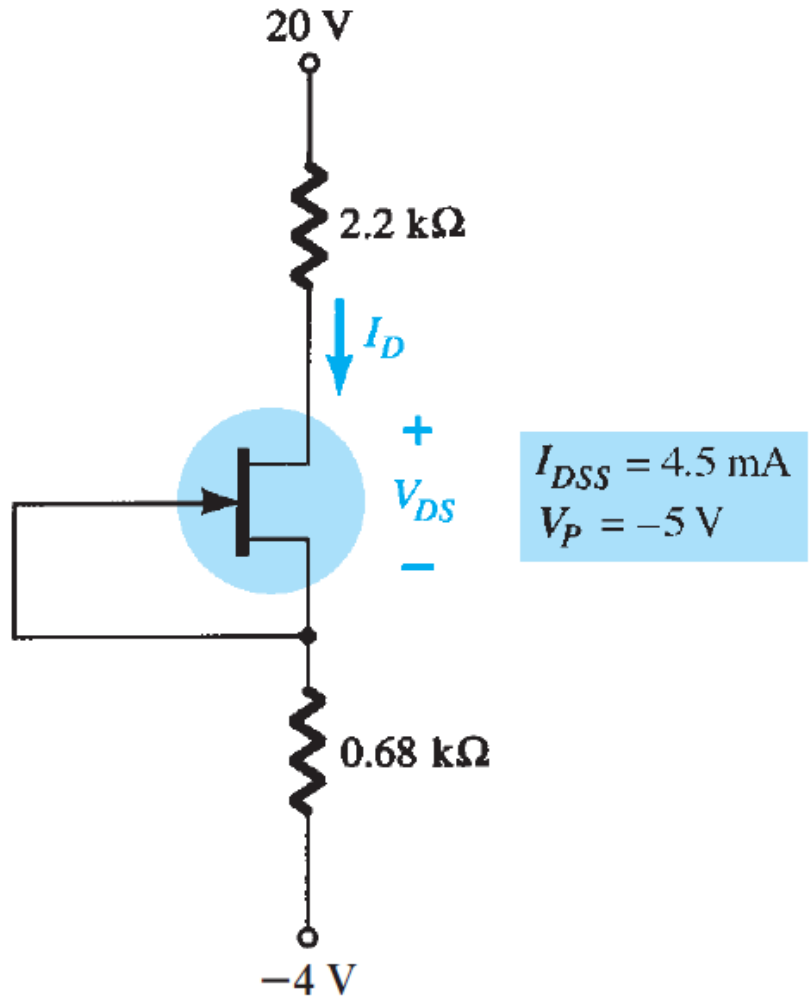
***10.** For the network of Fig. 7.83, determine:

a. I_D.

b. V_{DS}.

c. V_D.

d. V_S.



First for Vss =0

(a) $V_{GS} = 0\ \text{V}$

$\therefore I_D = I_{DSS} = \mathbf{4.5\ mA}$

(b) $V_{DS} = V_{DD} - I_D(R_D + R_S)$

$= 20\ \text{V} - (4.5\ \text{mA})(2.2\ \text{k}\Omega + 0.68\ \text{k}\Omega)$

$= 20\ \text{V} - 12.96$

$= \mathbf{7.04\ V}$

(c) $V_D = V_{DD} - I_D R_D$

$= 20\ \text{V} - (4.5\ \text{mA})(2.2\ \text{k}\Omega)$

$= \mathbf{10.1\ V}$

(d) $V_S = I_S R_S = I_D R_S$

$= (4.5\ \text{mA})(0.68\ \text{k}\Omega)$

$= \mathbf{3.06\ V}$

$V_S = I_D * R_S - 4 = -0.94\ \text{V}$

***11.** Find V_S for the network of Fig. 7.84.

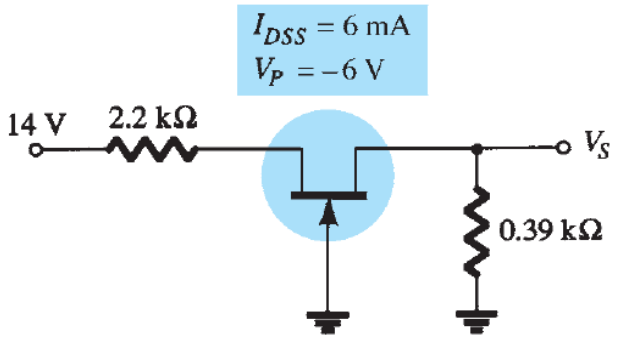
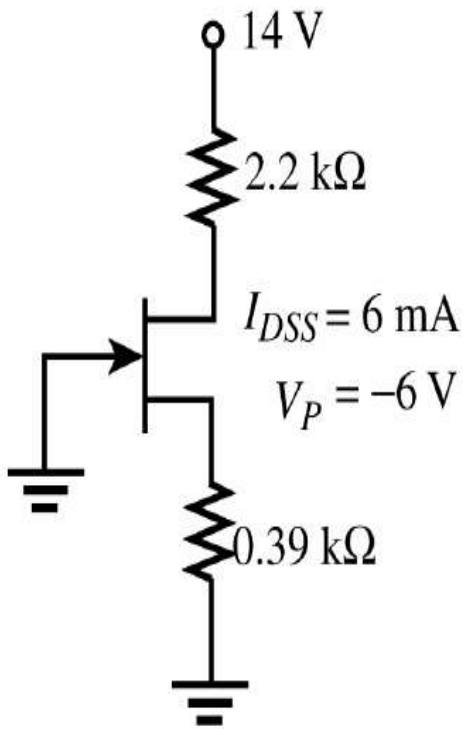


FIG. 7.84

Network redrawn:



$V_{GS} = 0\text{ V}, I_D = I_{DSS} = 6\text{ mA}$

$V_{GS} = V_P = -6\text{ V}, I_D = 0\text{ mA}$

$$V_{GS} = \frac{V_P}{2} = -3\text{ V}, I_D = 1.5\text{ mA}$$

$V_{GS} = 0.3V_P = -1.8\text{ V}, I_D = 3\text{ mA}$

$V_{GS} = -I_D R_S = -I_D(0.39\text{ k}\Omega)$

For $I_D = 5\text{ mA}$, $V_{GS} = -1.95\text{ V}$

***11.** Find V_S for the network of Fig. 7.84.

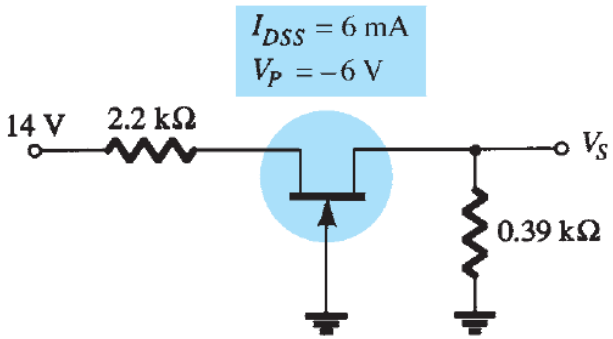


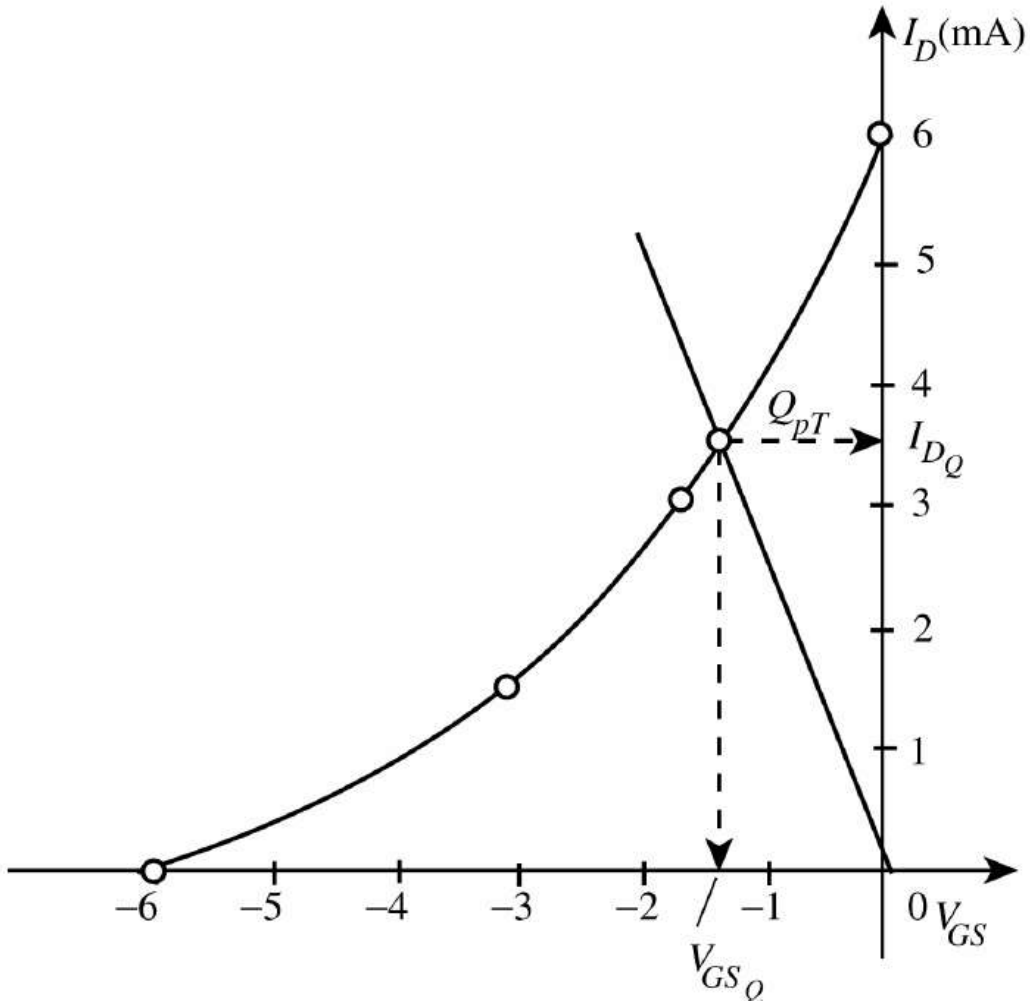
FIG. 7.84

From graph $I_{D_Q} \cong$ **3.55 mA**

$V_{GS_Q} \cong$ **−1.4 V**

$V_S = -\left(V_{GS_Q}\right) = -(-1.4\text{ V})$

$= +1.4$ **V**



12. For the network of Fig. 7.85, determine:

a. V_G.

b. I_{D_Q} and V_{GS_Q}.

c. V_D and V_S.

d. V_{DS_Q}.

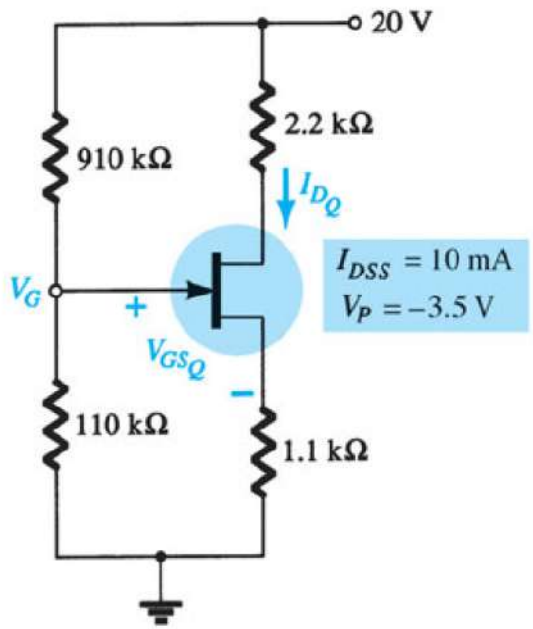


FIG. 7.85

(a) $V_G = \dfrac{R_2}{R_1 + R_2} V_{DD} = \dfrac{110\ \text{k}\Omega(20\ \text{V})}{910\ \text{k}\Omega + 110\ \text{k}\Omega}$

$= \mathbf{2.16\ V}$

$V_{GS} = 0\ \text{V}, I_D = I_{DSS} = 10\ \text{mA}$

$V_{GS} = V_P = -3.5\ \text{V}, I_D = 0\ \text{mA}$

$V_{GS} = \dfrac{V_P}{2} = -1.75\ \text{V}, I_D = 2.5\ \text{mA}$

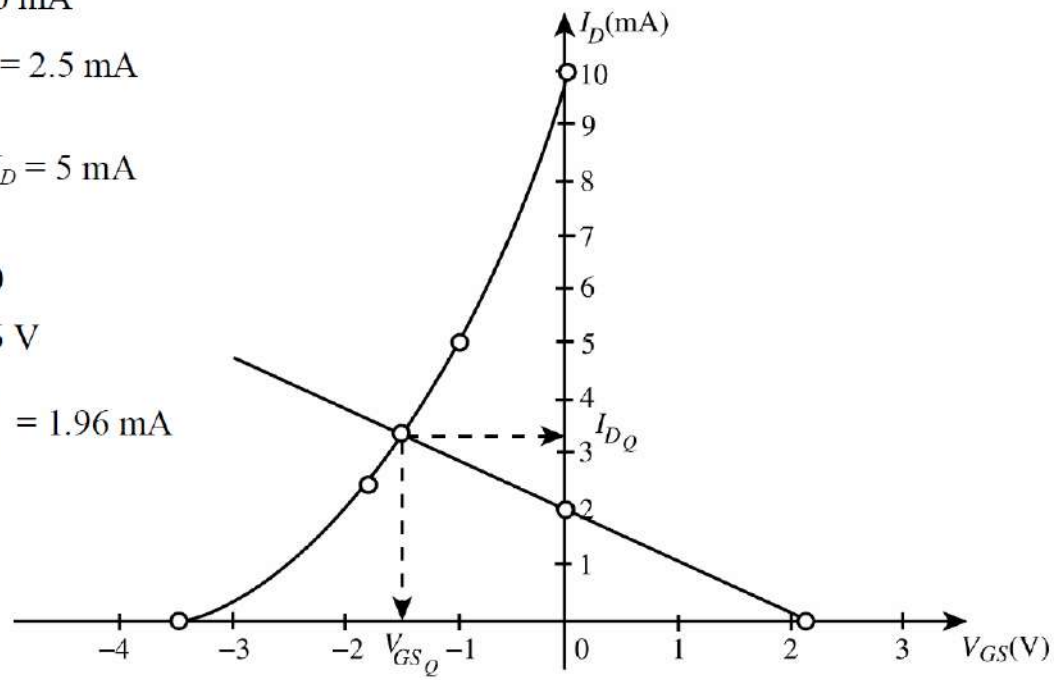
$V_{GS} = 0.3V_P = -1.05\ \text{V}, I_D = 5\ \text{mA}$

$V_{GS_Q} = V_G - I_D R_S$

$V_{GS_Q} = 2.16 - I_D(1.1\ \text{k}\Omega)$

$I_D = 0$: $V_{GS_Q} = V_G = 2.16\ \text{V}$

$V_{GS_Q} = 0\ \text{V}, I_D = \dfrac{2.16\ \text{V}}{1.1\ \text{k}\Omega} = 1.96\ \text{mA}$



(b) $I_{D_Q} \cong \mathbf{3.3\ mA}$

$V_{GS_Q} \cong \mathbf{-1.5\ V}$

(c) $V_D = V_{DD} - I_{D_Q} R_D$

$= 20\ \text{V} - (3.3\ \text{mA})(2.2\ \text{k}\Omega)$

$= \mathbf{12.74\ V}$

$V_S = I_S R_S = I_D R_S$

$= (3.3\ \text{mA})(1.1\ \text{k}\Omega)$

$= \mathbf{3.63\ V}$

(d) $V_{DS_Q} = V_{DD} - I_{D_Q}(R_D + R_S)$

$= 20\ \text{V} - (3.3\ \text{mA})(2.2\ \text{k}\Omega + 1.1\ \text{k}\Omega)$

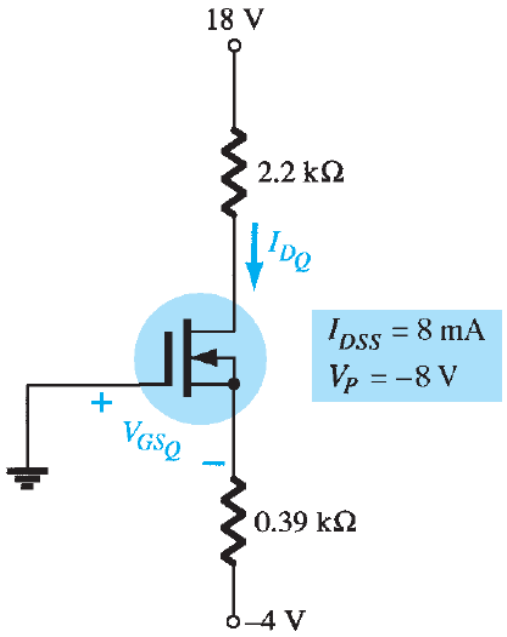
$= 20\ \text{V} - 10.89\ \text{V}$

$= \mathbf{9.11\ V}$

***21.** For the network of Fig. 7.93, determine:

a. I_{D_Q} and V_{GS_Q}.

b. V_{DS} and V_S.



(a) $V_{GS} = 0\text{ V}, I_D = I_{DSS} = 8\text{ mA}$

$V_{GS} = V_P = -8\text{ V}, I_D = 0\text{ mA}$

$V_{GS} = \dfrac{V_P}{2} = -4\text{ V}, I_D = 2\text{ mA}$

$V_{GS} = 0.3V_P = -2.4\text{ V}, I_D = 4\text{ mA}$

$V_{GS} = +1\text{ V}, I_D = 10.125\text{ mA}$

$V_{GS} = +2\text{ V}, I_D = 12.5\text{ mA}$

$V_{GS} = -V_{SS} - I_D R_S$

$= -(-4\text{ V}) - I_D(0.39\text{ k}\Omega)$

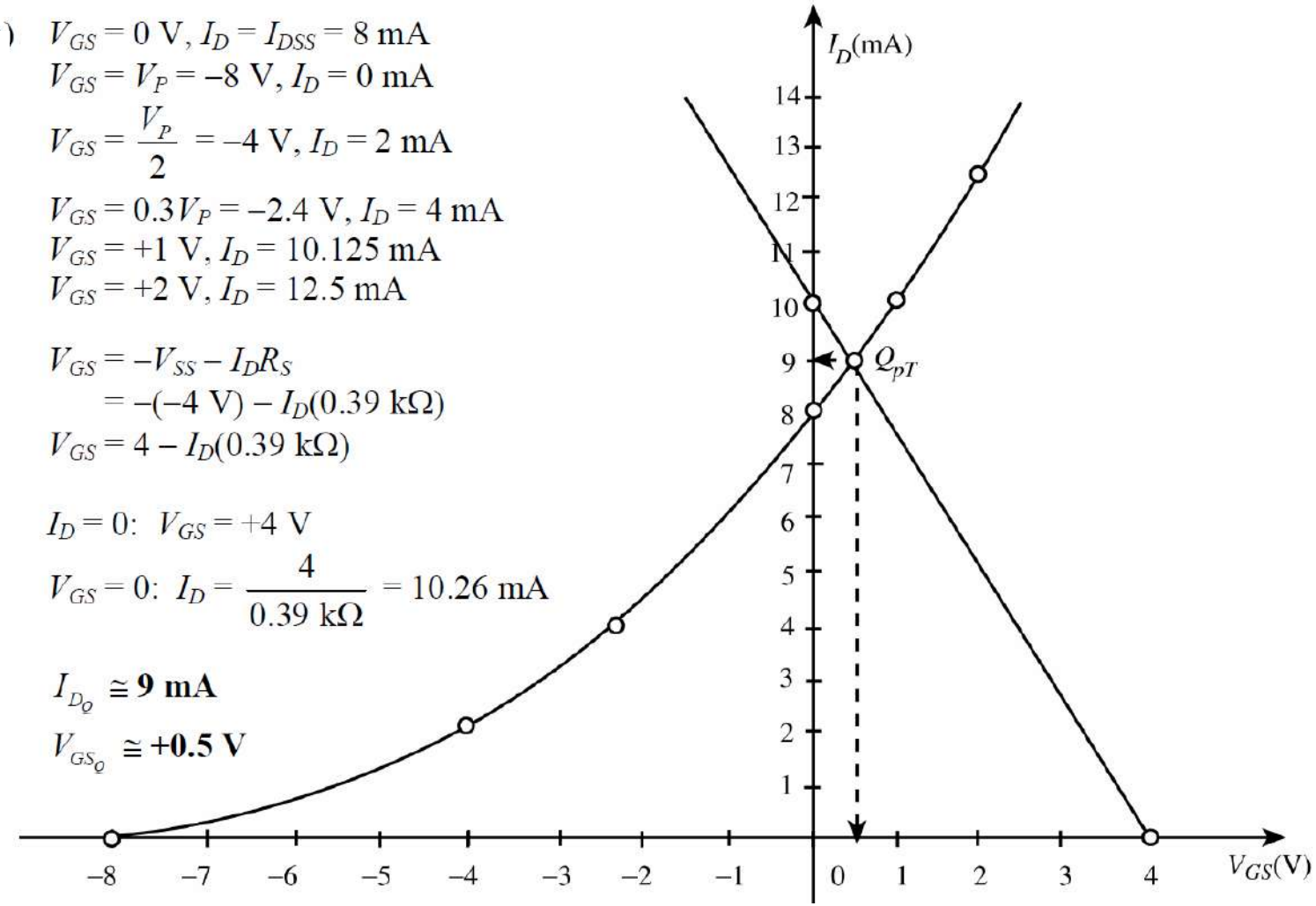
$V_{GS} = 4 - I_D(0.39\text{ k}\Omega)$

$I_D = 0$: $V_{GS} = +4\text{ V}$

$V_{GS} = 0$: $I_D = \dfrac{4}{0.39\text{ k}\Omega} = 10.26\text{ mA}$

$I_{D_Q} \cong \mathbf{9\text{ mA}}$

$V_{GS_Q} \cong \mathbf{+0.5\text{ V}}$



7.8 Enhancement-Type MOSFETs

22. For the network of Fig. 7.94, determine:

a. I_{D_Q}.

b. V_{GS_Q} and V_{DS_Q}.

c. V_D and V_S.

d. V_{DS}.

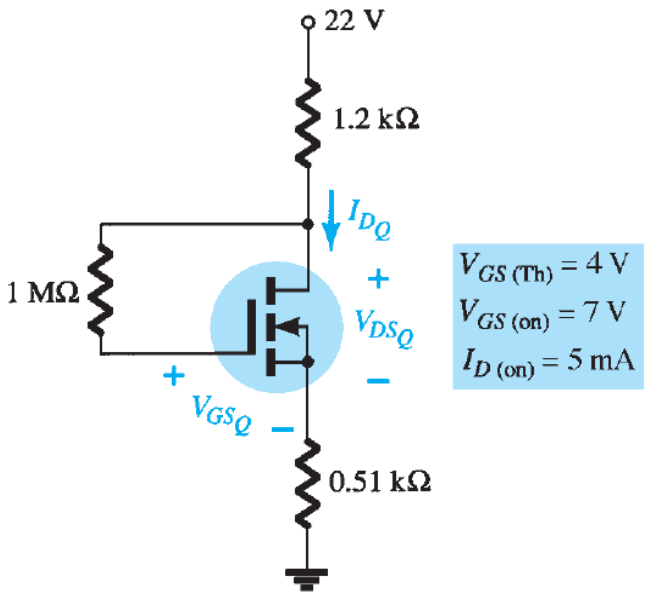


FIG. 7.94

$$I_D = k(V_{GS} - V_T)^2$$

$$k = \frac{I_{D(on)}}{(V_{GS(on)} - V_{Th})^2} = \frac{5\text{ mA}}{(7\text{ V} - 4\text{ V})^2} = \frac{5\text{ mA}}{9\text{ V}^2}$$

$$K = 0.556 \times 10^{-3}\text{ A/V}^2$$

and $$I_D = \mathbf{0.556 \times 10^{-3}(V_{GS} - 4\text{ V})^2}$$

$$V_{DS} = V_{DD} - I_D(R_D + R_S)$$

$V_{DS} = 0$ V; $I_D = \dfrac{V_{DD}}{R_D + R_S}$

$$= \frac{22\text{ V}}{1.2\text{ k}\Omega + 0.51\text{ k}\Omega}$$

$$= 12.87\text{ mA}$$

$I_D = 0$ mA, $V_{DS} = V_{DD}$

$$= 22\text{ V}$$

(c) $V_D = V_{DD} - I_D R_D$

$$= 22\text{ V} - (8.25\text{ mA})(1.2\text{ k}\Omega)$$

$$= \mathbf{12.1\text{ V}}$$

$V_S = I_S R_S = I_D R_S$

$$= (8.25\text{ mA})(0.51\text{ k}\Omega)$$

$$= \mathbf{4.21\text{ V}}$$

7.8 Enhancement-Type MOSFETs

22. For the network of Fig. 7.94, determine:

a. I_{D_Q}.

b. V_{GS_Q} and V_{DS_Q}.

c. V_D and V_S.

d. V_{DS}.

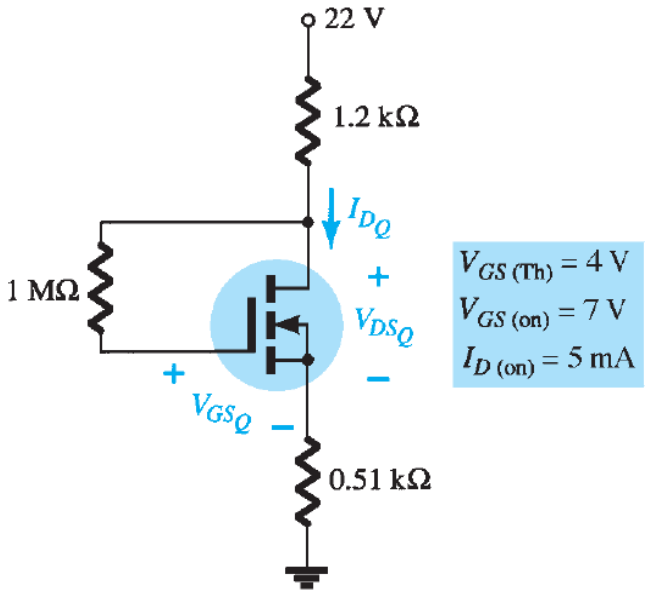
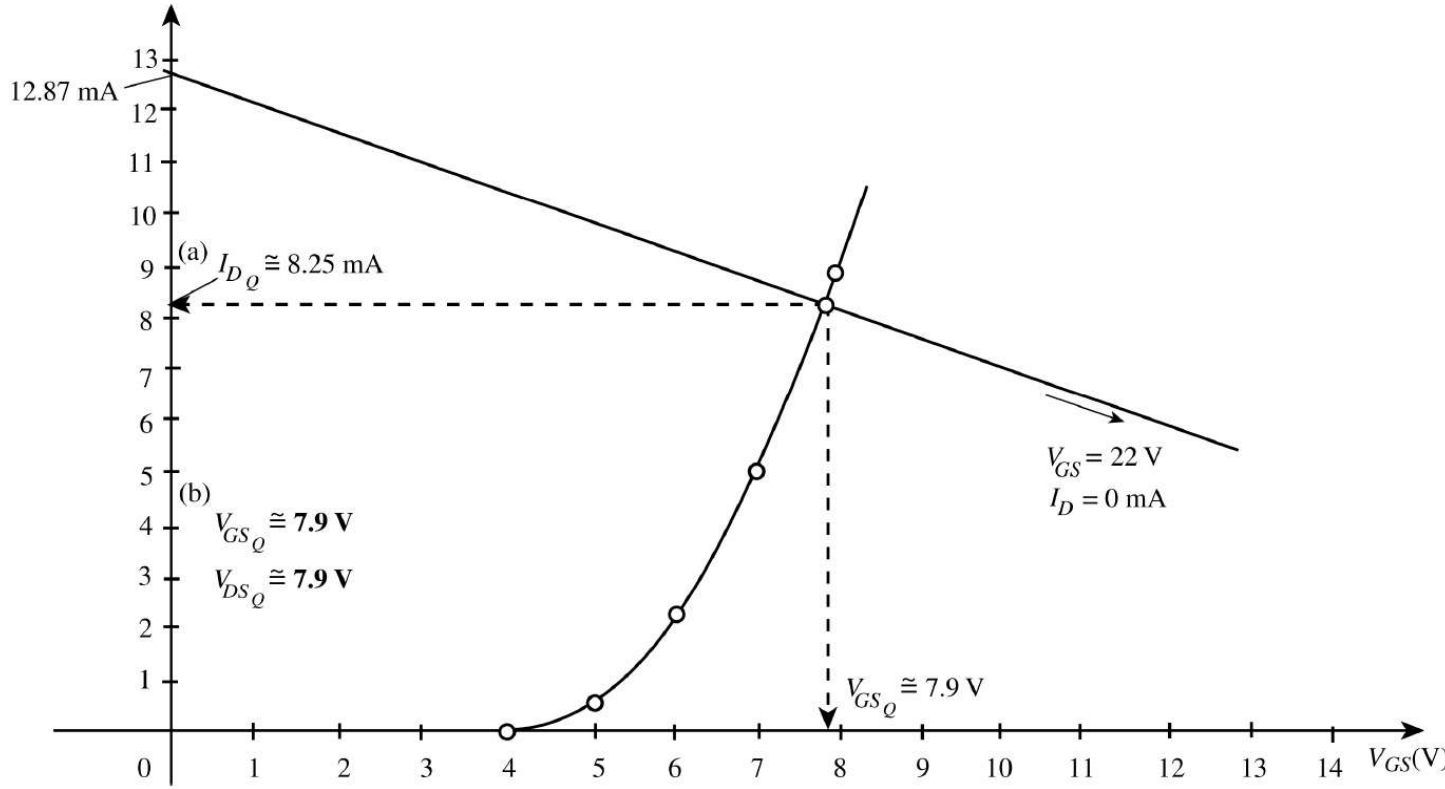


FIG. 7.94



(c) $V_D = V_{DD} - I_D R_D$

$= 22\text{ V} - (8.25\text{ mA})(1.2\text{ k}\Omega)$

$= \mathbf{12.1\text{ V}}$

$V_S = I_S R_S = I_D R_S$

$= (8.25\text{ mA})(0.51\text{ k}\Omega)$

$= \mathbf{4.21\text{ V}}$

23. For the voltage-divider configuration of Fig. 7.95, determine:

a. I_{D_Q} and V_{GS_Q}.

b. V_D and V_S.

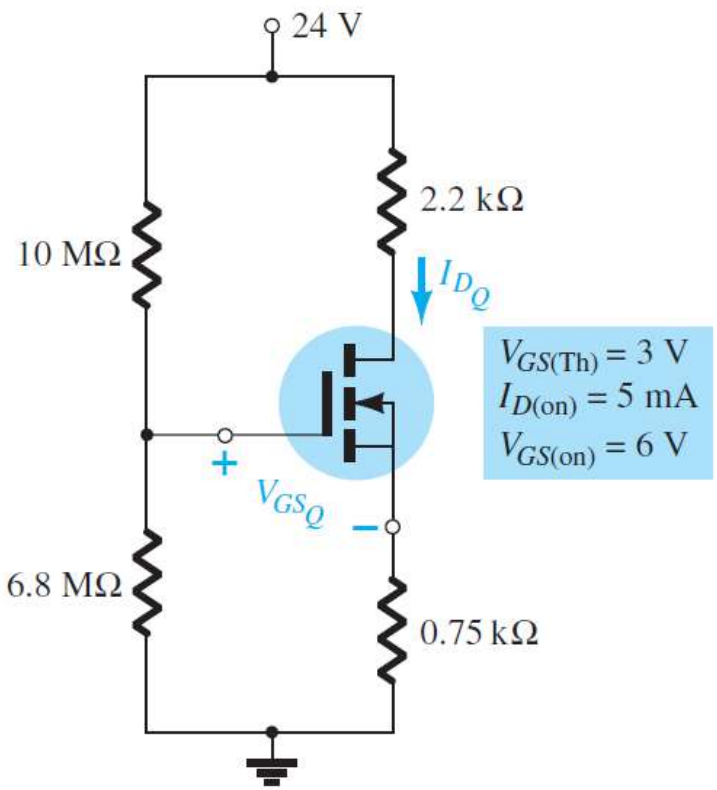


FIG. 7.95

$$I_D = k(V_{GS} - V_T)^2$$

$$k = \frac{I_{D(on)}}{\left(V_{GS(on)} - V_{Th}\right)^2} = \frac{5\text{ mA}}{(7\text{ V} - 4\text{ V})^2} = \frac{5\text{ mA}}{9\text{ V}^2}$$

$$K = 0.556 \times 10^{-3}\text{ A/V}^2$$

and $$I_D = \mathbf{0.556 \times 10^{-3}(V_{GS} - 4\text{ V})^2}$$

$$V_{DS} = V_{DD} - I_D(R_D + R_S)$$

$$V_{DS} = 0\text{ V}; I_D = \frac{V_{DD}}{R_D + R_S}$$

$$= \frac{22\text{ V}}{1.2\text{ k}\Omega + 0.51\text{ k}\Omega}$$

$$= 12.87\text{ mA}$$

$$I_D = 0\text{ mA}, V_{DS} = V_{DD}$$

$$= 22\text{ V}$$

BJT AC Analysis

CHAPTER OBJECTIVES

- Become familiar with the r_e, hybrid, and hybrid π models for the BJT transistor.
- Learn to use the equivalent model to find the important ac parameters for an amplifier.
- Understand the effects of a source resistance and load resistor on the overall gain and characteristics of an amplifier.
- Become aware of the general ac characteristics of a variety of important BJT configurations.
- Begin to understand the advantages associated with the two-port systems approach to single- and multistage amplifiers.
- Develop some skill in troubleshooting ac amplifier networks.

5.1 INTRODUCTION

The basic construction, appearance, and characteristics of the transistor were introduced in Chapter 3. The dc biasing of the device was then examined in detail in Chapter 4. We now begin to examine the ac response of the BJT amplifier by reviewing the *models* most frequently used to represent the transistor in the sinusoidal ac domain.

One of our first concerns in the sinusoidal ac analysis of transistor networks is the magnitude of the input signal. It will determine whether *small-signal* or *large-signal* techniques should be applied. There is no set dividing line between the two, but the application—and the magnitude of the variables of interest relative to the scales of the device characteristics—will usually make it quite clear which method is appropriate. The small-signal technique is introduced in this chapter, and large-signal applications are examined in Chapter 12.

There are three models commonly used in the small-signal ac analysis of transistor networks: the r_e model, the hybrid π model, and the hybrid equivalent model. This chapter introduces all three but emphasizes the r_e model.

5.2 AMPLIFICATION IN THE AC DOMAIN

It was demonstrated in Chapter 3 that the transistor can be employed as an amplifying device. That is, the output sinusoidal signal is greater than the input sinusoidal signal, or, stated another way, the output ac power is greater than the input ac power. The question then arises as to how the ac power output can be greater than the input ac power. Conservation of energy dictates that over time the total power output, P_o, of a system cannot be greater than its power

input, P_i, and that the efficiency defined by $\eta = P_o/P_i$ cannot be greater than 1. The factor missing from the discussion above that permits an ac power output greater than the input ac power is the applied dc power. It is the principal contributor to the total output power even though part of it is dissipated by the device and resistive elements. In other words, there is an "exchange" of dc power to the ac domain that permits establishing a higher output ac power. In fact, a *conversion efficiency* is defined by $\eta = P_{o(\text{ac})}/P_{i(\text{dc})}$, where $P_{o(\text{ac})}$ is the ac power to the load and $P_{i(\text{dc})}$ is the dc power supplied.

Perhaps the role of the dc supply can best be described by first considering the simple dc network of Fig. 5.1. The resulting direction of flow is indicated in the figure with a plot of the current i versus time. Let us now insert a control mechanism such as that shown in Fig. 5.2. The control mechanism is such that the application of a relatively small signal to the control mechanism can result in a substantial oscillation in the output circuit.

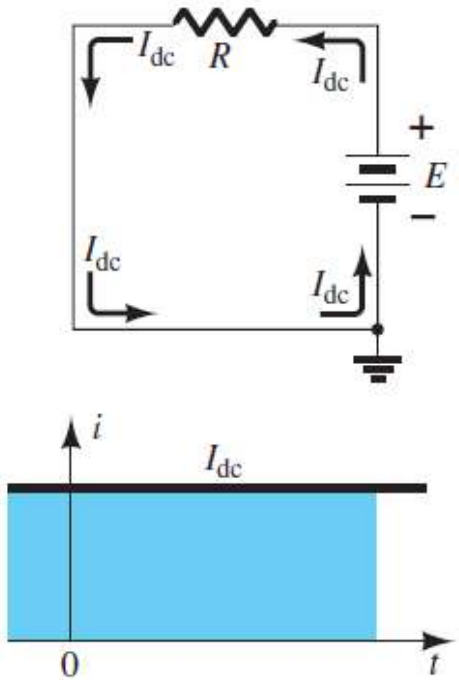


FIG. 5.1
Steady current established by a dc supply.

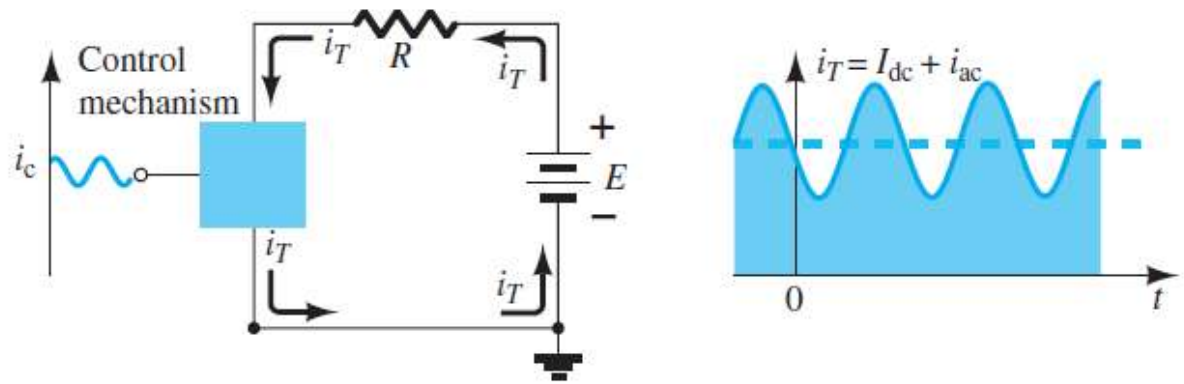


FIG. 5.2
Effect of a control element on the steady-state flow of the electrical system of Fig. 5.1.

That is, for this example,

$$i_{ac(p\text{-}p)} \gg i_{c(p\text{-}p)}$$

and amplification in the ac domain has been established. The peak-to-peak value of the output current far exceeds that of the control current.

For the system of Fig. 5.2, the peak value of the oscillation in the output circuit is controlled by the established dc level. Any attempt to exceed the limit set by the dc level will result in a "clipping" (flattening) of the peak region at the high and low end of the output signal. In general, therefore, proper amplification design requires that the dc and ac components be sensitive to each other's requirements and limitations.

However, it is extremely helpful to realize that:

The superposition theorem is applicable for the analysis and design of the dc and ac components of a BJT network, permitting the separation of the analysis of the dc and ac responses of the system.

In other words, one can make a complete dc analysis of a system before considering the ac response. Once the dc analysis is complete, the ac response can be determined using a completely ac analysis. It happens, however, that one of the components appearing in the ac analysis of BJT networks will be determined by the dc conditions, so there is still an important link between the two types of analysis.

5.3 BJT TRANSISTOR MODELING

The key to transistor small-signal analysis is the use of the equivalent circuits (models) to be introduced in this chapter.

A model is a combination of circuit elements, properly chosen, that best approximates the actual behavior of a semiconductor device under specific operating conditions.

Once the ac equivalent circuit is determined, the schematic symbol for the device can be replaced by this equivalent circuit and the basic methods of circuit analysis applied to determine the desired quantities of the network.

In the formative years of transistor network analysis the *hybrid equivalent network* was employed the most frequently. Specification sheets included the parameters in their listing, and analysis was simply a matter of inserting the equivalent circuit with the listed values.

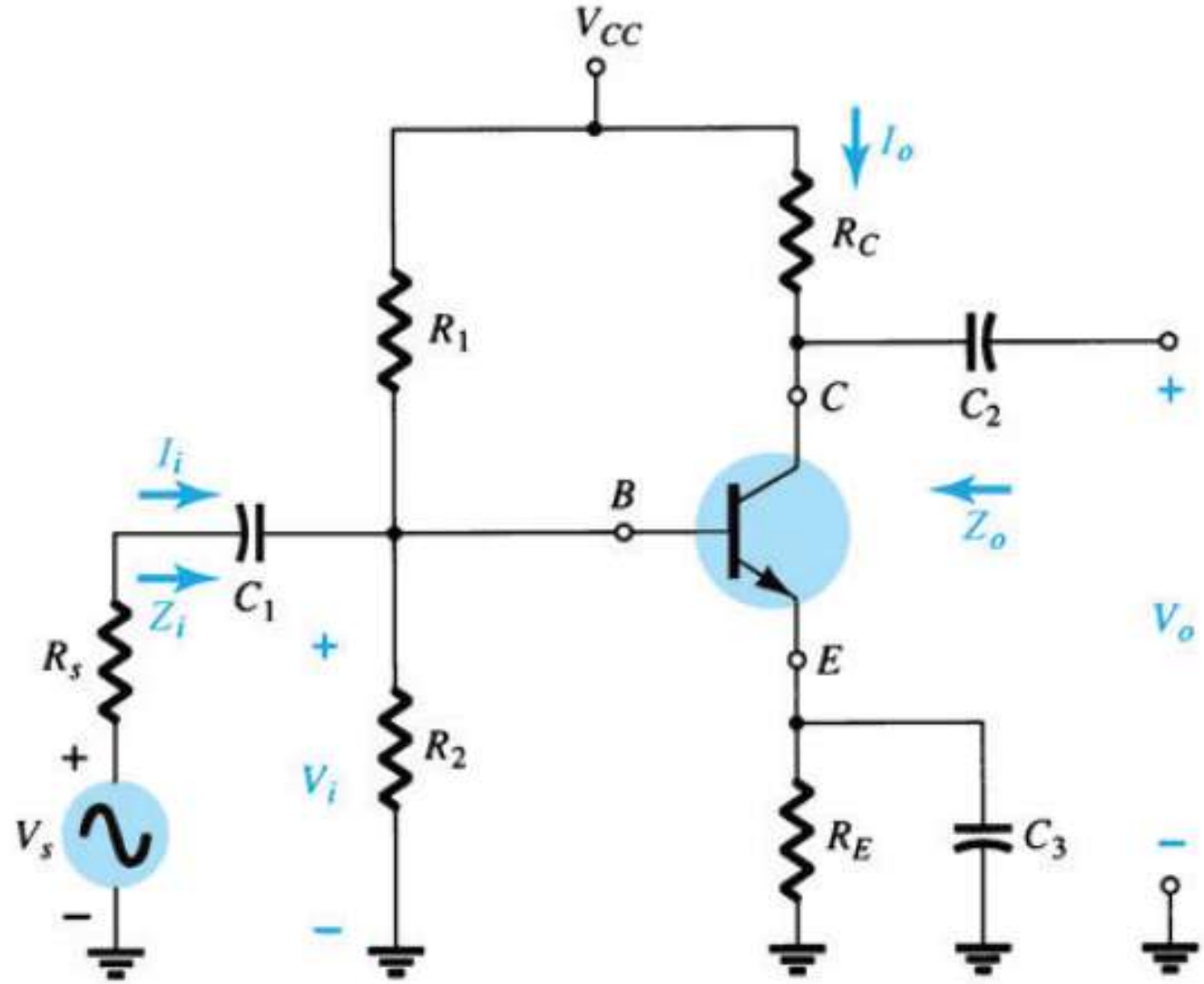


FIG. 5.3

Transistor circuit under examination in this introductory discussion.

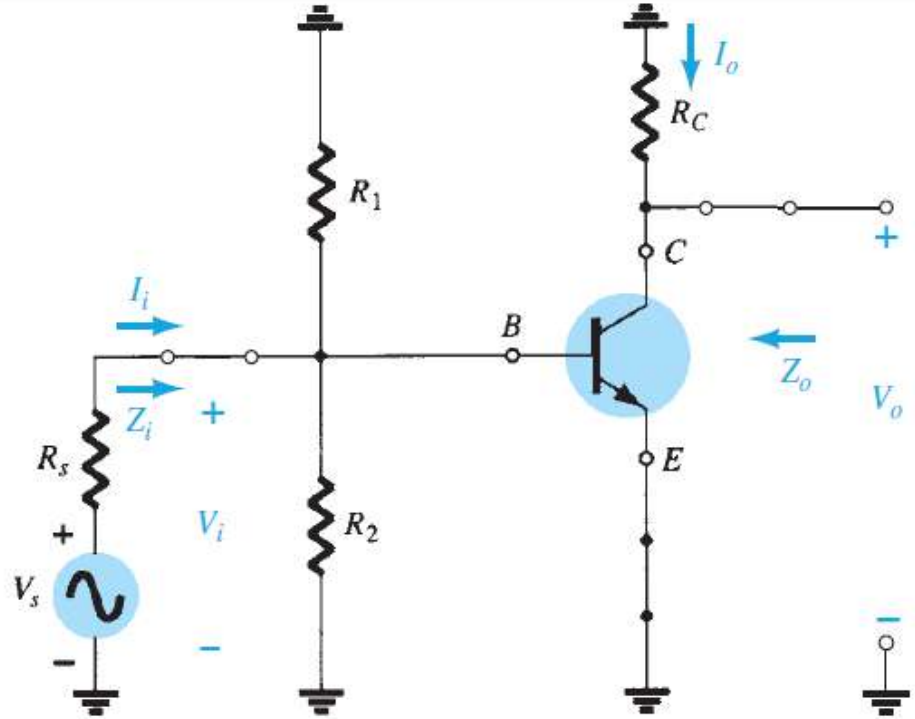


FIG. 5.4

The network of Fig. 5.3 following removal of the dc supply and insertion of the short-circuit equivalent for the capacitors.

It is important as you progress through the modifications of the network to define the ac equivalent that the parameters of interest such as Z_i, Z_o, I_i, and I_o as defined by Fig. 5.5 be carried through properly. Even though the network appearance may change, you want to be sure the quantities you find in the reduced network are the same as defined by the original network. In both networks the input impedance is defined from base to ground, the input current as the base current of the transistor, the output voltage as the voltage from collector to ground, and the output current as the current through the load resistor R_C.

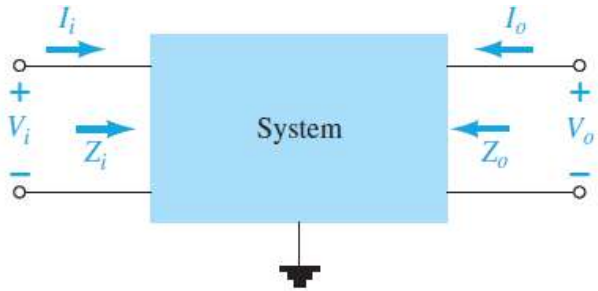


FIG. 5.5
Defining the important parameters of any system.

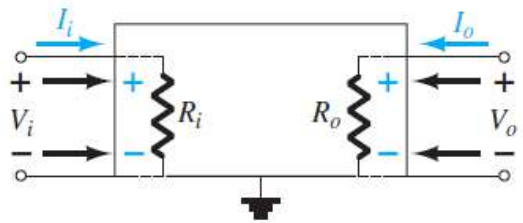


FIG. 5.6
Demonstrating the reason for the defined directions and polarities.

The parameters of Fig. 5.5 can be applied to any system whether it has one or a thousand components. For all the analysis to follow in this text, the directions of the currents, the polarities of the voltages, and the direction of interest for the impedance levels are as appearing in Fig. 5.5. In other words, the input current I_i and output current I_o are, by definition, defined to enter the system. If, in a particular example, the output current is leaving the system rather than entering the system as shown in Fig. 5.5, a minus sign must be applied. The defined polarities for the input and output voltages are also as appearing in Fig. 5.5. If V_o has the opposite polarity, the minus sign must be applied. Note that Z_i is the impedance "looking into" the system, whereas Z_o is the impedance "looking back into" the system from the output side. By choosing the defined directions for the currents and voltages as appearing in Fig. 5.5, both the input impedance and output impedance are defined as having positive values. For example, in Fig. 5.6 the input and output impedances for a particular system are both resistive. For the direction of I_i and I_o the resulting voltage across the resistive elements will have the same polarity as V_i and V_o, respectively. If I_o had been defined as the opposite direction in Fig. 5.5 a minus sign would have to be applied. For each case $Z_i = V_i/I_i$ and $Z_o = V_o/I_o$ with positive results if they all have the defined directions and polarity of Fig. 5.5. If the output current of an actual system has a direction opposite to that

of Fig. 5.5 a minus sign must be applied to the result because V_o must be defined as appearing in Fig. 5.5. Keep Fig. 5.5 in mind as you analyze the BJT networks in this chapter. It is an important introduction to "System Analysis," which is becoming so important with the expanded use of packaged IC systems.

If we establish a common ground and rearrange the elements of Fig. 5.4, R_1 and R_2 will be in parallel, and R_C will appear from collector to emitter as shown in Fig. 5.7. Because the components of the transistor equivalent circuit appearing in Fig. 5.7 employ familiar components such as resistors and independent controlled sources, analysis techniques such as superposition, Thévenin's theorem, and so on, can be applied to determine the desired quantities.

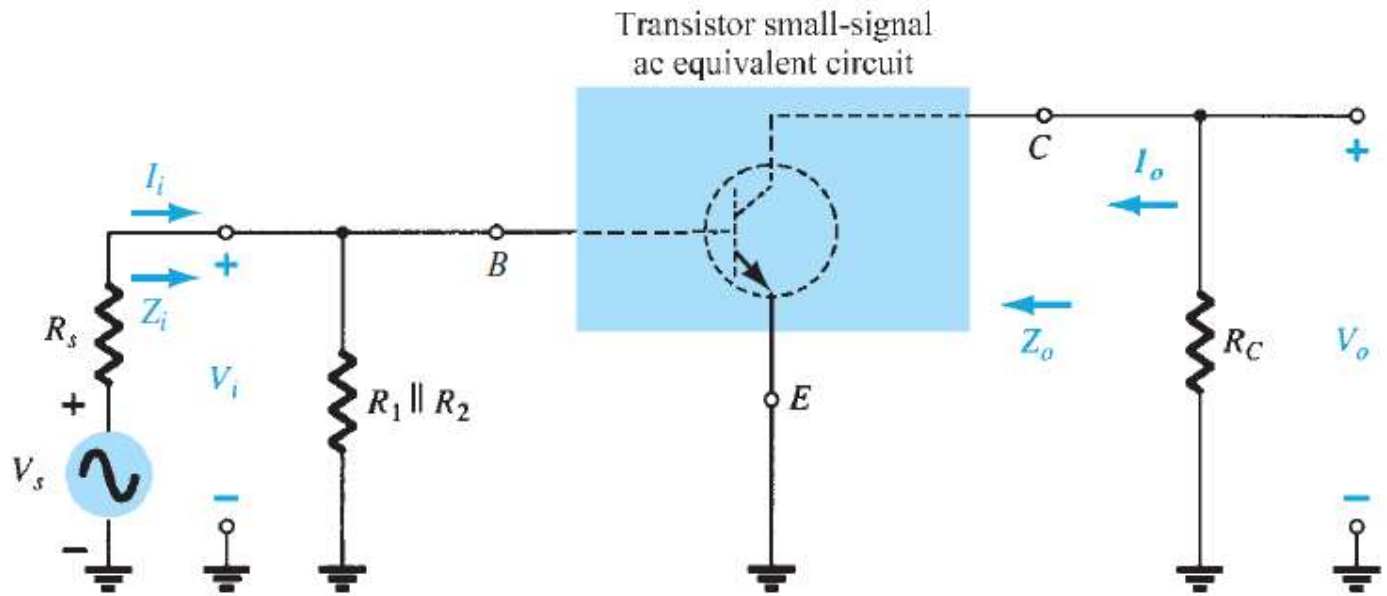


FIG. 5.7

Circuit of Fig. 5.4 redrawn for small-signal ac analysis.

Let us further examine Fig. 5.7 and identify the important quantities to be determined for the system. Because we know that the transistor is an amplifying device, we would expect some indication of how the output voltage V_o is related to the input voltage V_i—the *voltage gain.* Note in Fig. 5.7 for this configuration that the *current gain* is defined by $A_i = I_o/I_i$.

In summary, therefore, the ac equivalent of a transistor network is obtained by:

1. *Setting all dc sources to zero and replacing them by a short-circuit equivalent*
2. *Replacing all capacitors by a short-circuit equivalent*
3. *Removing all elements bypassed by the short-circuit equivalents introduced by steps 1 and 2*
4. *Redrawing the network in a more convenient and logical form*

In the sections to follow, a transistor equivalent model will be introduced to complete the ac analysis of the network of Fig. 5.7.

5.4 THE r_e TRANSISTOR MODEL

The r_e model for the CE, CB, and CC BJT transistor configurations will now be introduced with a short description of why each is a good approximation to the actual behavior of a BJT transistor.

Common-Emitter Configuration

The equivalent circuit for the common-emitter configuration will be constructed using the device characteristics and a number of approximations. Starting with the input side, we find the applied voltage V_i is equal to the voltage V_{be} with the input current being the base current I_b as shown in Fig. 5.8.

Recall from Chapter 3 that because the current through the forward-biased junction of the transistor is I_E, the characteristics for the input side appear as shown in Fig. 5.9a for various levels of V_{BE}. Taking the average value for the curves of Fig. 5.9a will result in the single curve of Fig. 5.9b, which is simply that of a forward-biased diode.

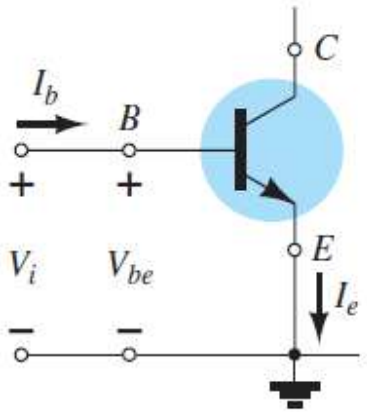


FIG. 5.8
Finding the input equivalent circuit for a BJT transistor.

5.19 THE HYBRID EQUIVALENT MODEL

The hybrid equivalent model was mentioned in the earlier sections of this chapter as one that was used in the early years before the popularity of the r_e model developed. Today there is a mix of usage depending on the level and direction of the investigation.

The r_e model has the advantage that the parameters are defined by the actual operating conditions,

whereas

the parameters of the hybrid equivalent circuit are defined in general terms for any operating conditions.

In other words, the hybrid parameters may not reflect the actual operating conditions but simply provide an indication of the level of each parameter to expect for general use. The r_e model suffers from the fact that parameters such as the output impedance and the feedback elements are not available, whereas the hybrid parameters provide the entire set on the specification sheet. In most cases, if the r_e model is employed, the investigator will simply examine the specification sheet to have some idea of what the additional elements might be. This section will show how one can go from one model to the other and how the parameters are related. Because all specification sheets provide the hybrid parameters and the model is still extensively used, it is important to be aware of both models. The hybrid parameters as shown in Fig. 5.92 are derived from the specification sheet for the 2N4400 transistor described in Chapter 3. The values are provided at a dc collector current of 1 mA and a collector-to-emitter voltage of 10 V. In addition, a range of values is provided for each parameter for guidance in the initial design or analysis of a system. One obvious advantage of the specification sheet listing is the immediate knowledge of typical levels for the parameters of the device as compared to other transistors.

The description of the hybrid equivalent model will begin with the general two-port system of Fig. 5.93. The following set of equations (5.131) and (5.132) is only one of a number of ways in which the four variables of Fig. 5.93 can be related. It is the most frequently employed in transistor circuit analysis, however, and therefore is discussed in detail in this chapter.

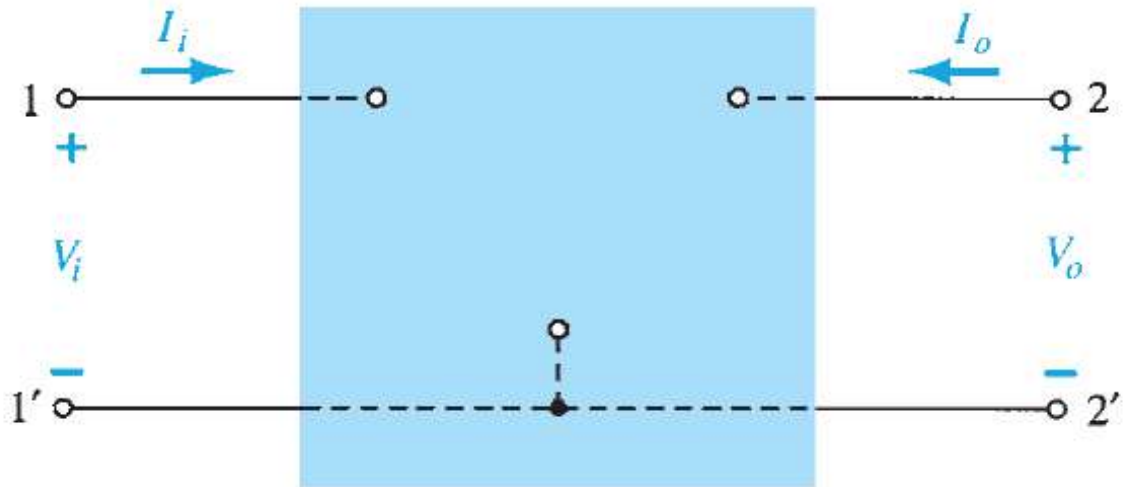
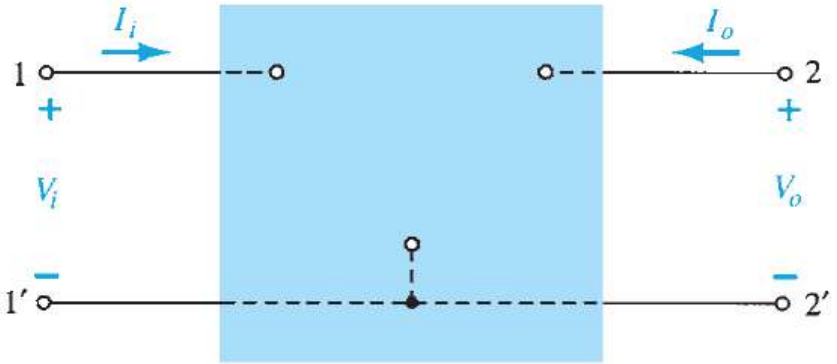


FIG. 5.93
Two-port system.



$$V_i = h_{11}I_i + h_{12}V_o \tag{5.133}$$

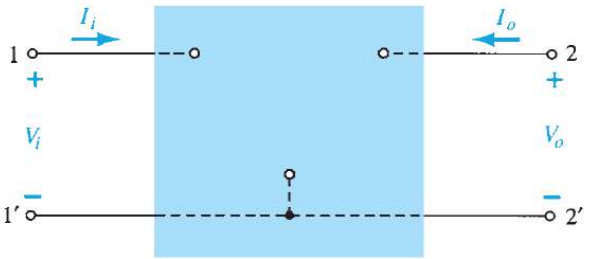
$$I_o = h_{21}I_i + h_{22}V_o \tag{5.134}$$

The parameters relating the four variables are called *h-parameters*, from the word "hybrid." The term *hybrid* was chosen because the mixture of variables (V and I) in each equation results in a "hybrid" set of units of measurement for the h-parameters. A clearer understanding of what the various h-parameters represent and how we can determine their magnitude can be developed by isolating each and examining the resulting relationship.

h_{11} If we arbitrarily set $V_o = 0$ (short circuit the output terminals) and solve for h_{11} in Eq. (5.133), we find

$$h_{11} = \left.\frac{V_i}{I_i}\right|_{V_o=0} \quad \text{ohms} \tag{5.135}$$

The ratio indicates that the parameter h_{11} is an impedance parameter with the units of ohms. Because it is the ratio of the *input* voltage to the *input* current with the output terminals *shorted*, it is called the *short-circuit input-impedance parameter*. The subscript 11 of h_{11} refers to the fact that the parameter is determined by a ratio of quantities measured at the input terminals.



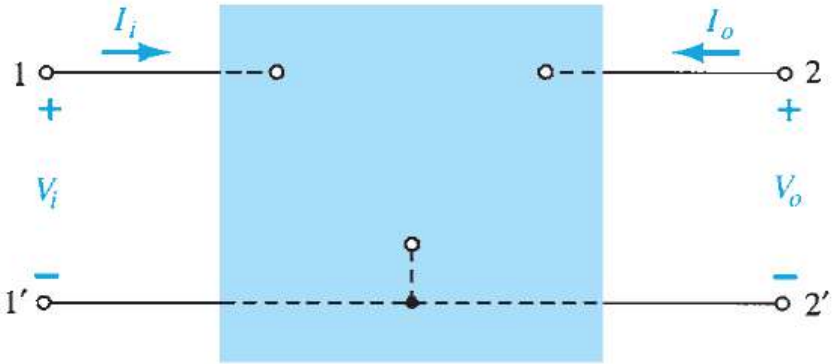
h_{12} If I_i is set equal to zero by opening the input leads, the following results for h_{12}:

$$h_{12} = \left.\frac{V_i}{V_o}\right|_{I_i=0} \qquad \text{unitless} \tag{5.136}$$

The parameter h_{12}, therefore, is the ratio of the input voltage to the output voltage with the input current equal to zero. It has no units because it is a ratio of voltage levels and is called the *open-circuit reverse transfer voltage ratio parameter*. The subscript 12 of h_{12} indicates that the parameter is a transfer quantity determined by a ratio of input (1) to output (2) measurements. The first integer of the subscript defines the measured quantity to appear in the numerator; the second integer defines the source of the quantity to appear in the denominator. The term *reverse* is included because the ratio is an input voltage over an output voltage rather than the reverse ratio typically of interest.

h_{21} If in Eq. (5.134) V_o is set equal to zero by again shorting the output terminals, the following results for h_{21}:

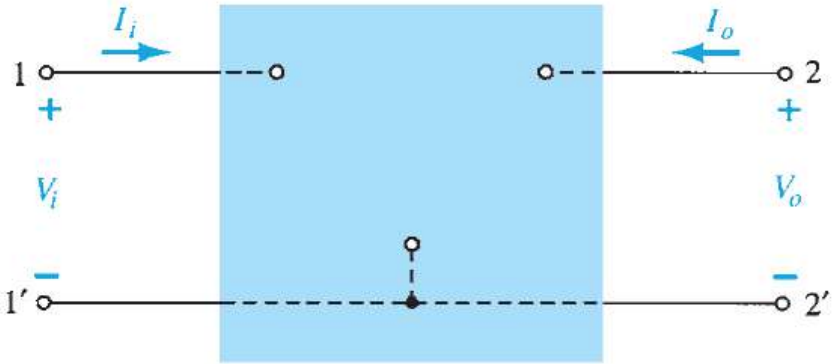
$$h_{21} = \left.\frac{I_o}{I_i}\right|_{V_o=0} \qquad \text{unitless} \tag{5.137}$$



h_{21} If in Eq. (5.134) V_o is set equal to zero by again shorting the output terminals, the following results for h_{21}:

$$h_{21} = \left.\frac{I_o}{I_i}\right|_{V_o=0} \qquad \text{unitless} \qquad \textbf{(5.137)}$$

Note that we now have the ratio of an output quantity to an input quantity. The term *forward* will now be used rather than *reverse* as indicated for h_{12}. The parameter h_{21} is the ratio of the output current to the input current with the output terminals shorted. This parameter, like h_{12}, has no units because it is the ratio of current levels. It is formally called the *short-circuit forward transfer current ratio parameter.* The subscript 21 again indicates that it is a transfer parameter with the output quantity (2) in the numerator and the input quantity (1) in the denominator.



h_{22} The last parameter, h_{22}, can be found by again opening the input leads to set $I_1 = 0$ and solving for h_{22} in Eq. (5.134):

$$h_{22} = \left.\frac{I_o}{V_o}\right|_{I_i=0} \quad \text{siemens} \tag{5.138}$$

Because it is the ratio of the output current to the output voltage, it is the output conductance parameter, and it is measured in siemens (S). It is called the *open-circuit output admittance parameter*. The subscript 22 indicates that it is determined by a ratio of output quantities.

The complete "ac" equivalent circuit for the basic three-terminal linear device is indicated in Fig. 5.96 with a new set of subscripts for the *h*-parameters. The notation of Fig. 5.96 is of a more practical nature because it relates the *h*-parameters to the resulting ratio obtained in the last few paragraphs. The choice of letters is obvious from the following listing:

$h_{11} \rightarrow$ *i*nput resistance $\rightarrow h_i$

$h_{12} \rightarrow$ *r*everse transfer voltage ratio $\rightarrow h_r$

$h_{21} \rightarrow$ *f*orward transfer current ratio $\rightarrow h_f$

$h_{22} \rightarrow$ *o*utput conductance $\rightarrow h_o$

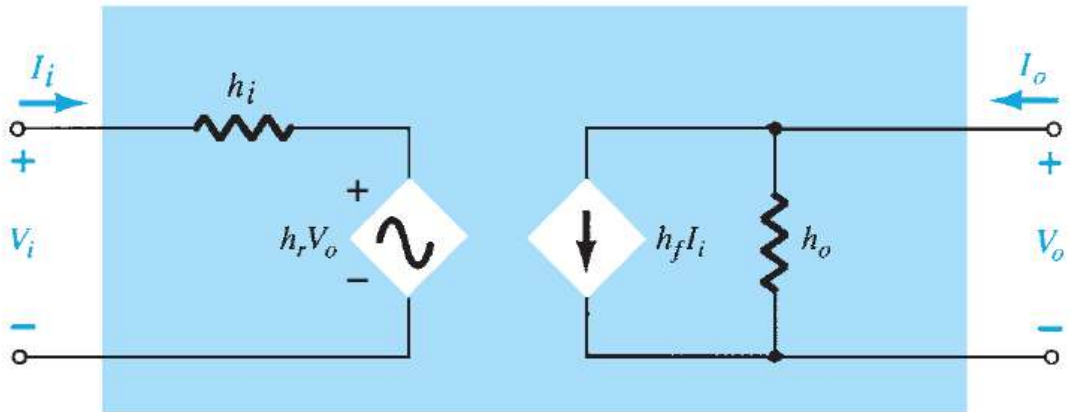


FIG. 5.96
Complete hybrid equivalent circuit.

FIG. 5.94
Hybrid input equivalent circuit.

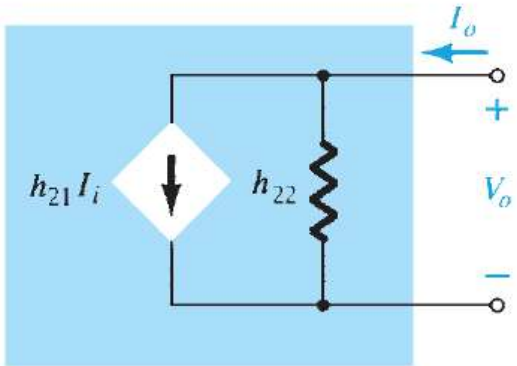


FIG. 5.95
Hybrid output equivalent circuit.

The circuit of Fig. 5.96 is applicable to any linear three-terminal electronic device or system with no internal independent sources. For the transistor, therefore, even though it has three basic configurations, *they are all three-terminal configurations*, so that the resulting equivalent circuit will have the same format as shown in Fig. 5.96. In each case, the bottom of the input and output sections of the network of Fig. 5.96 can be connected as shown in Fig. 5.97 because the potential level is the same. Essentially, therefore, the transistor model is a three-terminal two-port system. The *h*-parameters, however, will change with each configuration. To distinguish which parameter has been used or which is available, a second

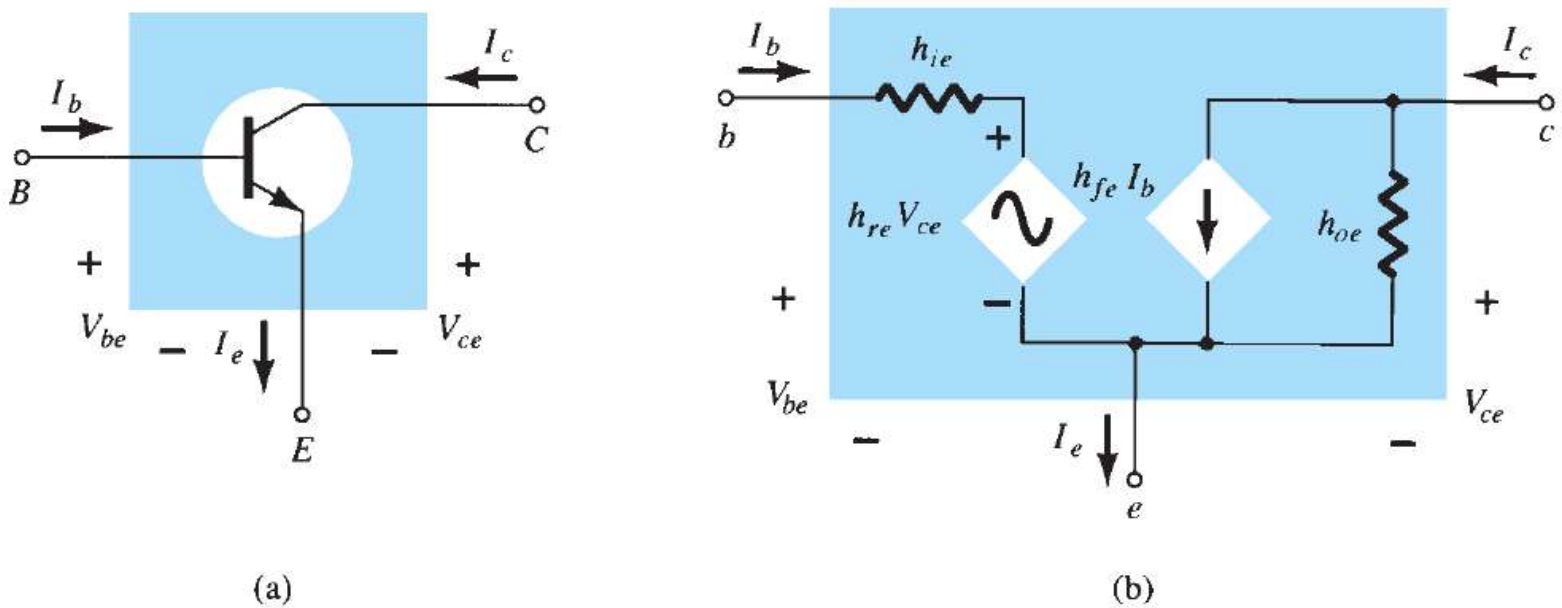


FIG. 5.97

Common-emitter configuration: (a) graphical symbol; (b) hybrid equivalent circuit.

subscript has been added to the h-parameter notation. For the common-base configuration, the lowercase letter b was added, whereas for the common-emitter and common-collector configurations, the letters e and c were added, respectively. The hybrid equivalent network for the common-emitter configuration appears with the standard notation in Fig. 5.97. Note that $I_i = I_b$, $I_o = I_c$, and, through an application of Kirchhoff's current law, $I_e = I_b + I_c$. The input voltage is now V_{be}, with the output voltage V_{ce}. For the common-base configuration of Fig. 5.98, $I_i = I_e$, $I_o = I_c$ with $V_{eb} = V_i$ and $V_{cb} = V_o$. The networks of Figs. 5.97 and 5.98 are applicable for *pnp* or *npn* transistors.

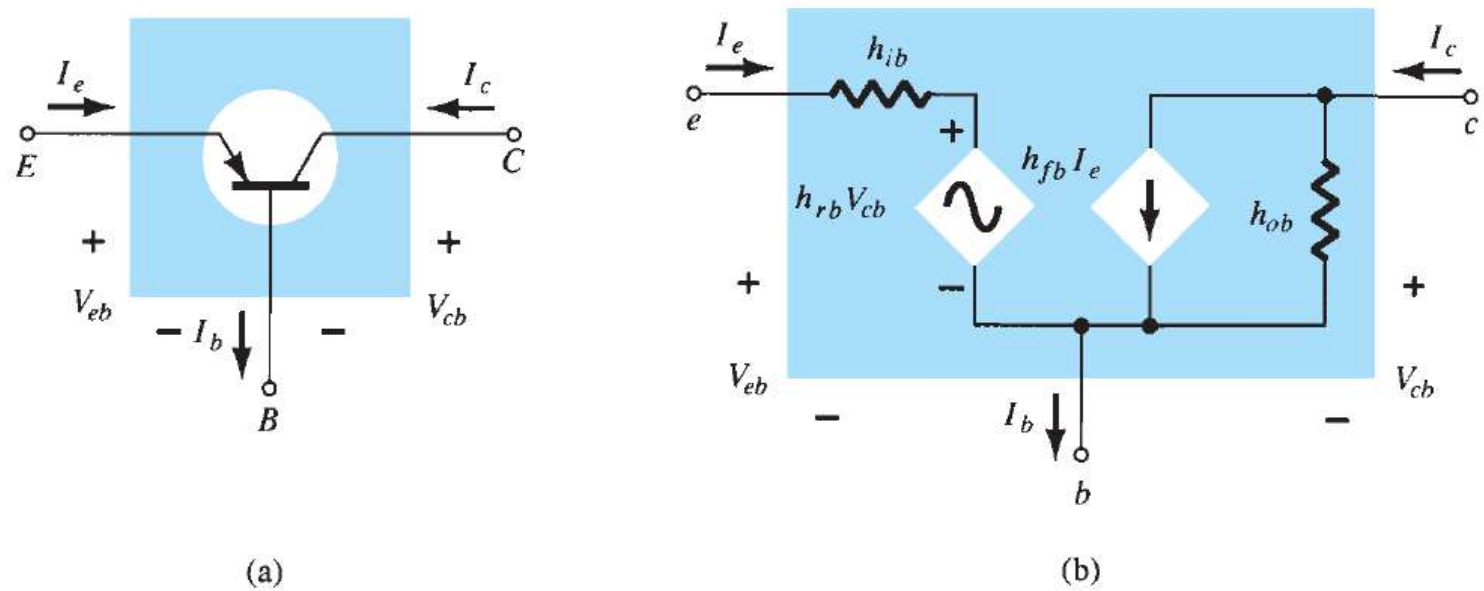


FIG. 5.98

Common-base configuration: (a) graphical symbol; (b) hybrid equivalent circuit.

The resulting equivalent of Fig. 5.100 is quite similar to the general structure of the common-base and common-emitter equivalent circuits obtained with the r_e model. In fact,

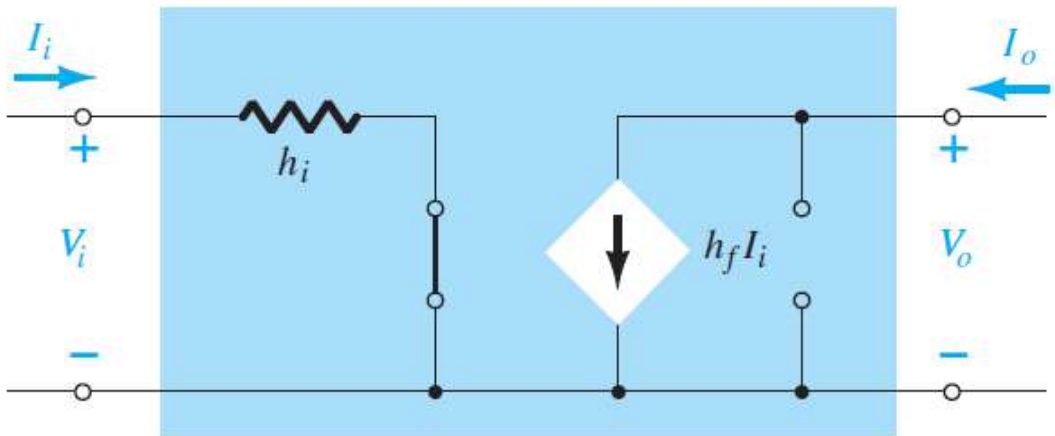


FIG. 5.99

Effect of removing h_{re} and h_{oe} from the hybird equivalent circuit.

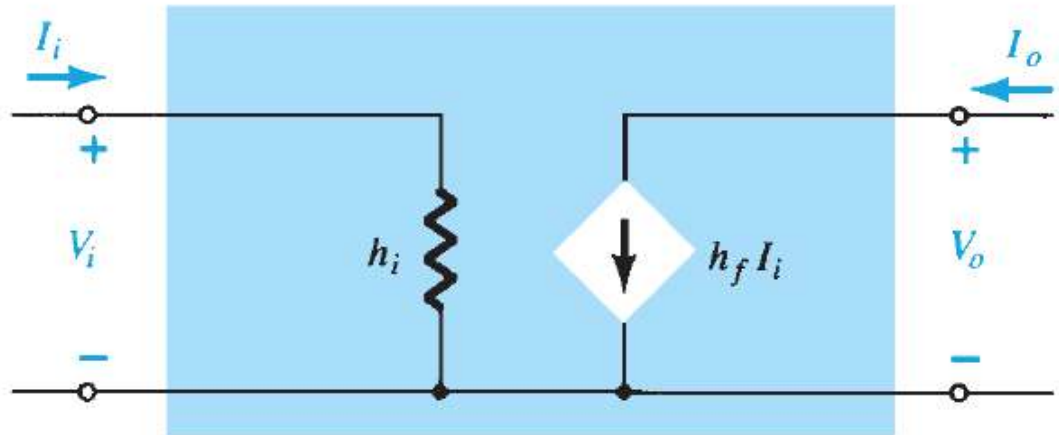


FIG. 5.100

Approximate hybrid equivalent model.

the hybrid equivalent and the r_e models for each configuration are repeated in Fig. 5.101 for comparison. It should be reasonably clear from Fig. 5.101a that

$$h_{ie} = \beta r_e \tag{5.139}$$

and

$$h_{fe} = \beta_{ac} \tag{5.140}$$

From Fig. 5.101b,

$$h_{ib} = r_e \tag{5.141}$$

and

$$h_{fb} = -\alpha \cong -1 \tag{5.142}$$

In particular, note that the minus sign in Eq. (5.142) accounts for the fact that the current source of the standard hybrid equivalent circuit is pointing down rather than in the actual direction as shown in the r_e model of Fig. 5.101b.

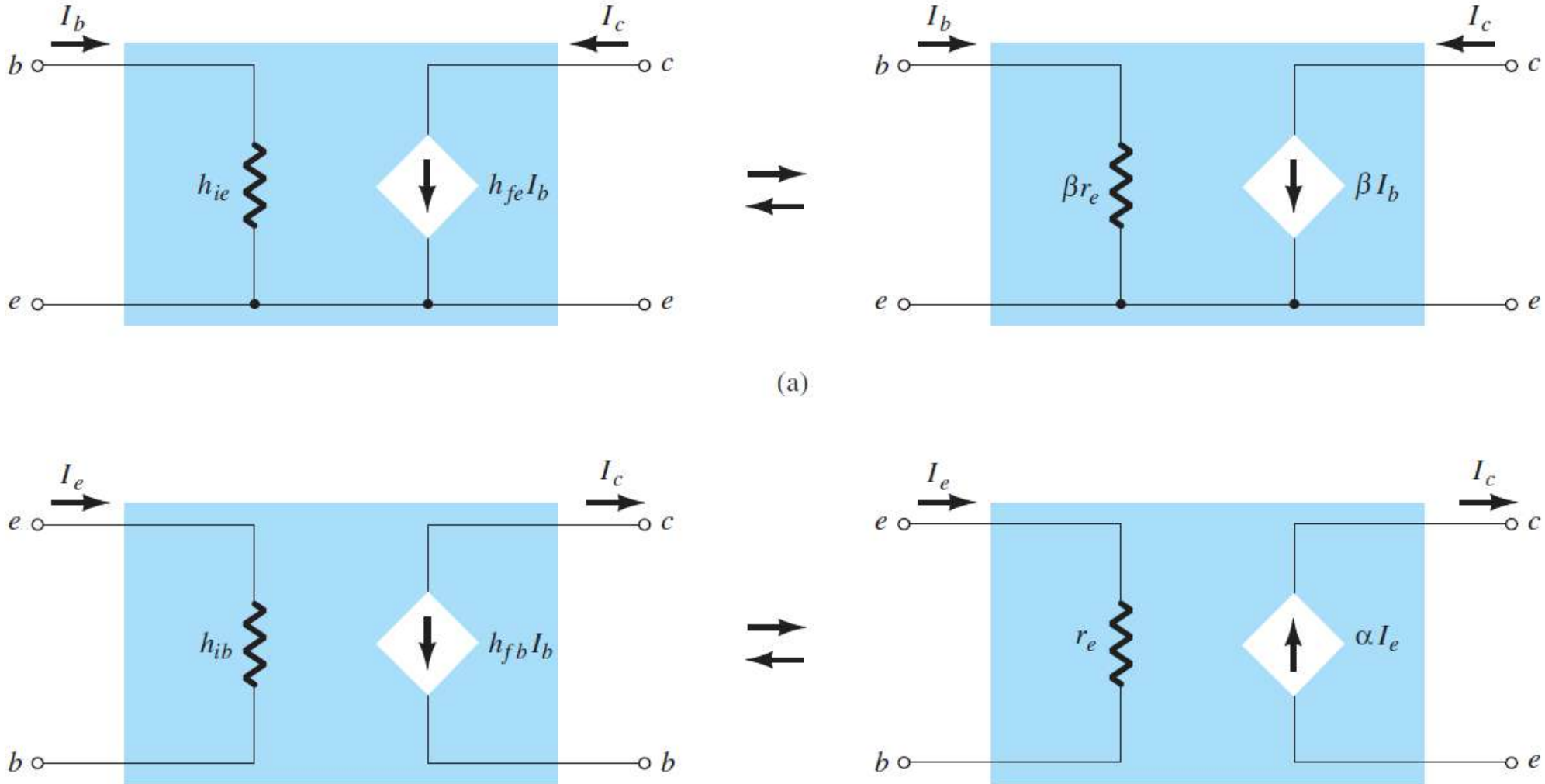


FIG. 5.101

Hybrid versus r_e model: (a) common-emitter configuration; (b) common-base configuration.

EXAMPLE 5.19 Given $I_E = 2.5$ mA, $h_{fe} = 140$, $h_{oe} = 20\ \mu$S (μmho), and $h_{ob} = 0.5\ \mu$S, determine:

a. The common-emitter hybrid equivalent circuit.
b. The common-base r_e model.

Solution:

a. $$r_e = \frac{26\text{ mV}}{I_E} = \frac{26\text{ mV}}{2.5\text{ mA}} = \mathbf{10.4\ \Omega}$$

$$h_{ie} = \beta r_e = (140)(10.4\ \Omega) = \mathbf{1.456\ k\Omega}$$

$$r_o = \frac{1}{h_{oe}} = \frac{1}{20\ \mu\text{S}} = \mathbf{50\ k\Omega}$$

Note Fig. 5.102.

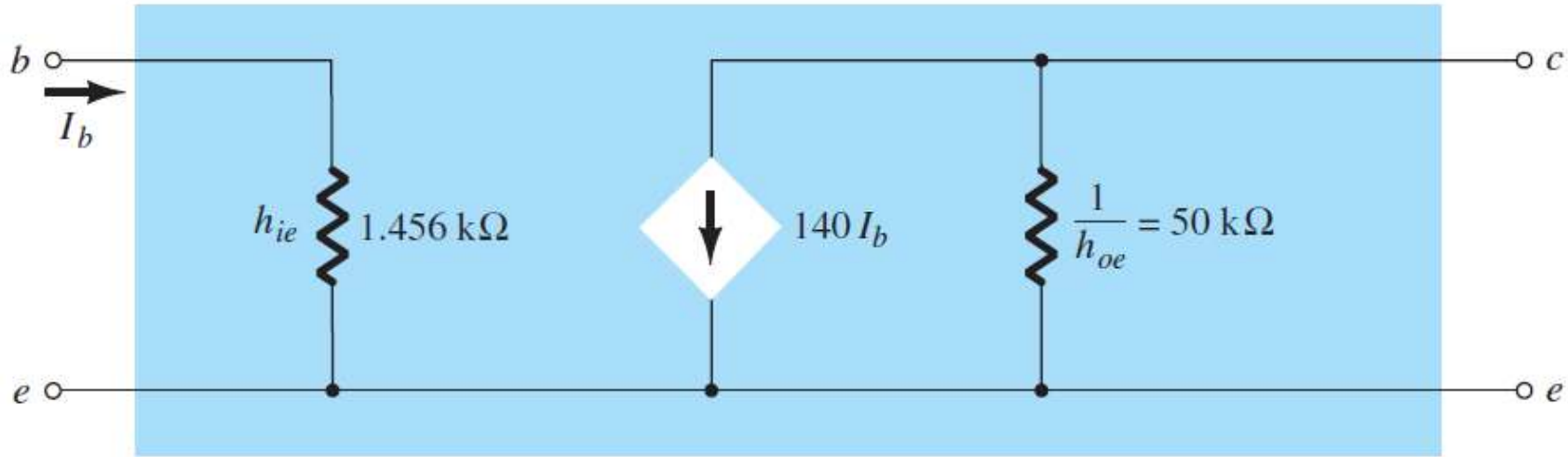


FIG. 5.102

Common-emitter hybrid equivalent circuit for the parameters of Example 5.19.

b. $r_e = \mathbf{10.4\ \Omega}$

$$\alpha \cong 1, \qquad r_o = \frac{1}{h_{ob}} = \frac{1}{0.5\ \mu\text{S}} = \mathbf{2\ M\Omega}$$

Note Fig. 5.103.

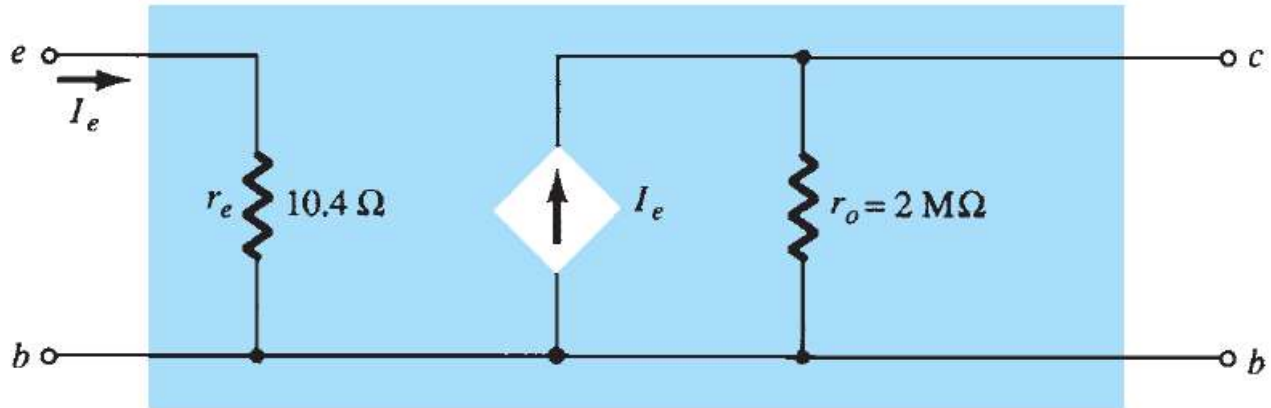


FIG. 5.103

Common-base r_e model for the parameters of Example 5.19.

A series of equations relating the parameters of each configuration for the hybrid equivalent circuit is provided in Appendix B. In Section 5.23 it is demonstrated that the hybrid parameter h_{fe} (β_{ac}) is the least sensitive of the hybrid parameters to a change in collector current. Assuming, therefore, that $h_{fe} = \beta$ is a constant for the range of interest, is a fairly good approximation. It is $h_{ie} = \beta r_e$ that will vary significantly with I_C and should be determined at operating levels because it can have a real effect on the gain levels of a transistor amplifier.

5.20 APPROXIMATE HYBRID EQUIVALENT CIRCUIT

The analysis using the approximate hybrid equivalent circuit of Fig. 5.104 for the common-emitter configuration and of Fig. 5.105 for the common-base configuration is very similar to that just performed using the r_e model. A brief overview of some of the most important configurations will be included in this section to demonstrate the similarities in approach and the resulting equations.

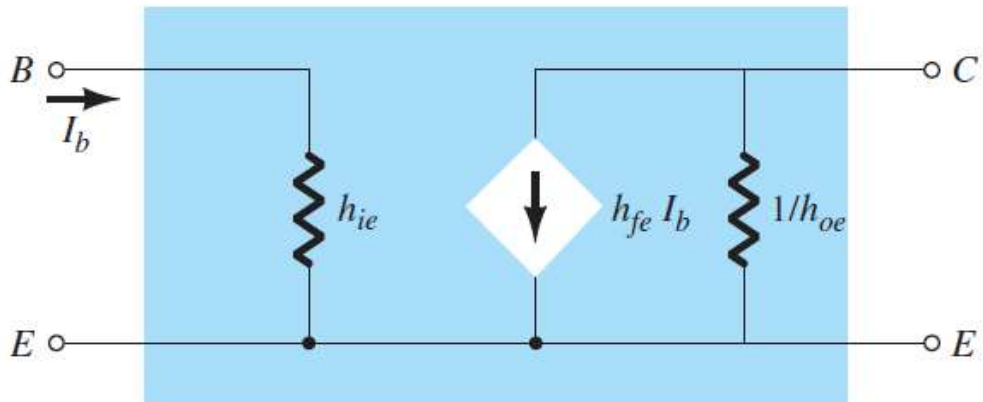


FIG. 5.104
Approximate common-emitter hybrid equivalent circuit.

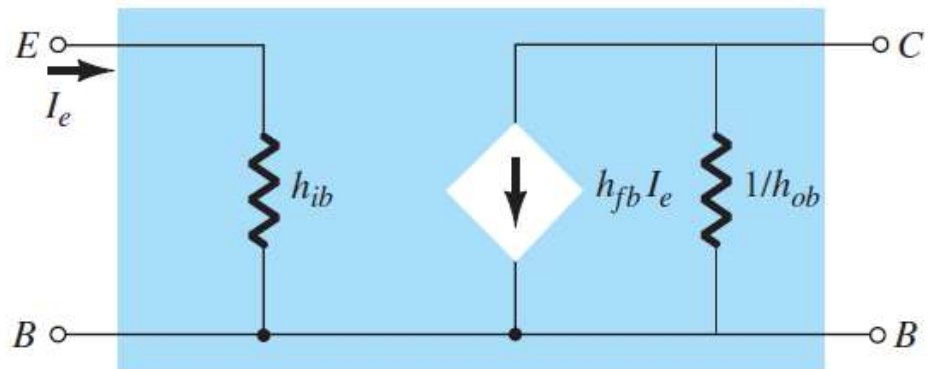


FIG. 5.105
Approximate common-base hybrid equivalent circuit.